

Polynomial–Logarithmic Transseries

Algebra, Composition, Series Reversal, and the Lambert W Archetype

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Abstract

This volume is a self-contained operational treatment of the smallest transseries scale that is closed under the operations one actually performs when solving an implicit equation with a logarithmic term: the polynomial–logarithmic scale generated by $t = X^{-1}$ and $L = \log X$ as $X \rightarrow +\infty$. Its elements are formal series $\sum_{n \geq n_0} p_n(L)t^n$ whose coefficient blocks are polynomials in L .

We develop the four ambient classes $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} \subset \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} \subset \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} \subset \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ together with the valuation, leading data, and truncation calculus that makes every computation finite; the arithmetic (multiplication, the exact unit criterion, reciprocals and division, powers, logarithms, exponentials, and the closure boundary); the differential calculus of the distinguished derivation $D_X = t\partial_L - t^2\partial_t$, including the repeated-derivative operator and the resonance that governs antiderivatives; composition, the exact formal Taylor formula $F \circ (X + H) = \sum_{k \geq 0} H^k D_X^k F / k!$, and the near-identity composition group; and series reversal at infinity by four independent routes – triangular correction, functional iteration, Newton’s method with its precision-doubling law, and a Lagrange–Bürmann formula proved directly in the formal setting.

The Lambert W function is treated as the archetype rather than as an application. The inverse of $Y \mapsto Y + \log Y$ is Wright’s ω , whose expansion coefficients are the canonical Lambert polynomials $\mathbb{L}_n^{\text{Lam}}$; we prove their operator definition, recurrence, Stirling-number closed form, generating equation, and integral transform, and derive the branch expansions of W_k from them. Throughout, a formal identity and an analytic estimate are kept rigorously apart: the transfer from a formal residual to an error bound on an actual inverse branch is stated as a theorem with its hypotheses printed, and no convergence claim is made that is not proved.

Two further parts cover the passage from formal transseries to Poincaré asymptotics – including power–log series near a finite point and the Dulac conventions – and the algorithms, residual certificates, and diagnostics needed to compute with these objects reliably. Appendices collect the Bell-polynomial bookkeeping, closed-form coefficient extraction, Lambert polynomial tables, an operational formula sheet, exercises with solutions, a concordance with the corpus notation catalogue, and the provenance of the six absorbed sources.

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1 Reader’s guide and the structure of this volume

1.1 The eight parts

The volume is arranged so that each part depends only on those before it. One of those dependencies is lighter than the ordering suggests: a reader who wants only the Lambert expansion may pass from Part I directly to Parts IV and V, because Part V uses Parts II and III only through the two statements quoted in reading path 1 below.

Part	Subject	What it establishes
I	The scale	The generators $t = X^{-1}$, $L = \log X$; dominance and the rank-two valuation; the rings $\mathcal{PL}_{\text{pow},\mathbb{K}}$, $\mathcal{PL}_{\text{poly},\mathbb{K}}$, $\mathcal{PL}_{\text{rat},\mathbb{K}}$, $\mathcal{PL}_{\text{itLaur},\mathbb{K}}$; coefficient support, leading block, leading monomial and leading scalar; exclusive truncation $\text{Trunc}_{<N} F$; t -adic completeness and formal summability.
II	Arithmetic and differential calculus	Cauchy convolution block by block; the exact unit criterion and the reciprocal and quotient recurrences; the distinguished derivation $D_X = t\partial_L - t^2\partial_t$ and the Euler operator ϑ ; repeated derivatives $\Delta_{n,k}$; antidifferentiation and the resonant block; powers, logarithms and exponentials of small series.
III	Composition	Admissible inner arguments; substitution on the generators; the exact formal Taylor formula and its formal summability; finite coefficient formulas; associativity and the identity; the near-identity group $\mathcal{G}_{\text{poly},\mathbb{K}}$; the exact points at which composition leaves the class.
IV	Series reversal	Existence and uniqueness of $F^{(-1)\circ}$ on $\mathcal{G}_{\text{poly},\mathbb{K}}$; four algorithms — contractive fixed point, triangular coefficient matching, Newton iteration with precision doubling, and Lagrange–Bürmann at infinity; residual certificates $\text{Err}_{\text{rev}}(F, H)$; reversion with a nontrivial leading monomial.
V	Wright omega, the Lambert polynomials, and Lambert W	The reversion of $X + L$; the canonical zero-based family $\mathbb{L}_n^{\text{Lam}}$ and the scaled integer family $\mathbb{L}_n^{\text{Lam,sc}}$; recurrence, generating equation, operator formula, closed form through unsigned Stirling cycle numbers, degree law, Laplace transform; all complex branches, the lower real branch W_{-1} , and the square-root branch point.
VI	From formal transseries to analytic asymptotics	What a formal identity does and does not give; block-wise Poincaré expansions with the logarithmic degree printed; transport of asymptotics through the algebraic operations; from a formal residual to an analytic inverse error; convergent versus divergent expansions; branch consistency in sectors.
VII	Algorithms, certificates, diagnostics	Sparse data structures for $n \mapsto p_n(L)$; truncated primitives and their cost; reversion strategies compared; verification invariants, residual certificates, and a decision procedure for the recurring failure modes.
VIII	Extensions and the wider landscape	Puiseux and Hahn exponent groups; rational-log and Laurent-log coefficients; iterated logarithms and multi-series; power-log and Dulac classes at zero; the relation to the logarithmic-exponential transseries field and to Hardy fields [20, 13, 33, 1].

1.2 Appendices

- *Bell polynomials and Faà di Bruno bookkeeping.* The partial and complete exponential Bell polynomials $\mathcal{B}_{n,k}^{\text{exp}}$ and $\mathcal{B}_n^{\text{exp}}$ in exactly the form used by the composition and reversion formulas of Parts III and IV.

- *Closed-form coefficient extraction.* Explicit formulas for $[t^n]F$ and $[t^n L^j]F$ after each operation, with the conventions of [Theorem 1.1](#) in force.
- *Lambert coefficient tables.* $\mathbb{L}_n^{\text{Lam}}$ and $\mathbb{L}_n^{\text{Lam,sc}}$ through a stated order, with Stirling diagonals $\begin{bmatrix} n \\ k \end{bmatrix}$ and a reproducible residual check for each row.
- *Operational formula sheet.* One entry per operation: hypotheses, formula, cost, and the certificate that validates a computed truncation.
- *Exercises with solutions.*
- *Notation concordance.* The map from each source draft’s local letters to the normative catalogue commands.
- *Provenance.* Which source contributed which result, and the editorial repairs applied to it.

1.3 Reading paths

1. *Only the Lambert W expansion.* Read Part I $\rightarrow$ Part IV $\rightarrow$ Part V. Part I supplies the scale, the ring and the valuation; Part IV supplies the one theorem this path needs, namely that every element of $\mathcal{G}_{\text{poly},\mathbb{K}}$ has a compositional inverse in $\mathcal{G}_{\text{poly},\mathbb{K}}$ computable to any order by a finite calculation; Part V performs the reversion of $X + L$ and reads off every branch. The only imports from Parts II and III along this path are the block derivative (14) and the formal Taylor formula, both quoted with full statements where used.
2. *The full calculus.* Read Parts I–IV in order. This is the self-contained algebraic core.
3. *Analytic use.* After Part I, read Part VI. A reader who intends to convert a formal truncation into a numerical bound should read Part VI before computing anything, because none of the formal machinery by itself licenses an analytic claim.
4. *Implementation.* Read Part I, then the recurrences of Parts II–IV, then Part VII with the formula-sheet appendix.
5. *Context.* Read Part VIII for the embedding of this class into Hahn series, Dulac classes, Hardy fields, and the full logarithmic-exponential transseries field.

Structural checkpoint

After Part I a reader should be able to: write an element of $\mathcal{PL}_{\text{poly},\mathbb{K}}$ in the normal form (13); compute ν_t , lm , lc , deg_L and the log-degree profile d_F of a given series; decide from the leading block alone whether the series is a unit; say which of $\mathcal{PL}_{\text{pow},\mathbb{K}}$, $\mathcal{PL}_{\text{poly},\mathbb{K}}$, $\mathcal{PL}_{\text{rat},\mathbb{K}}$, $\mathcal{PL}_{\text{itLaur},\mathbb{K}}$ a given expression belongs to; and state, for a displayed truncation, whether its tail is a formal valuation tail or an analytic remainder.

1.4 Standing conventions and normalizations

Remark 1.1 (Normalizations adopted, and how they differ from the source drafts). This volume merges six independently written treatments. Where their conventions collided, the notation catalogue of the corpus is the arbiter, and the following choices are in force everywhere.

1. *Regime and field.* Unless a statement says otherwise, $X \rightarrow +\infty$ along the positive real ray, $\mathbb{K}$ is a field of characteristic zero, and $t = X^{-1}$, $L = \log X$. Complex work names its sector and its branch of log. Local letters used by individual sources – λ , ℓ , x – are renamings of L or X and are not additional generators.
2. *Truncation is exclusive.* $\text{Trunc}_{<N} F = \sum_{n < N} p_n(L)t^n$ keeps every logarithmic monomial in every retained block; “through outer order t^{N-1} ” means the exclusive cutoff $< N$. One source uses the

- inclusive $[F]_{\leq N}$; the conversion is $[F]_{\leq N} = \text{Trunc}_{<N+1} F$, a one-block difference that must be applied whenever a formula is transported from that source.
3. *Coefficient extraction.* $[t^n]F = p_n(L)$ and $[t^n L^j]F = a_{n,j}$, with the series written to the right of the bracket; an absent coefficient is 0. The same bracket glyph also denotes unsigned Stirling cycle numbers $\begin{bmatrix} n \\ k \end{bmatrix}$ with two integer indices; the roles are separated by argument type and never abbreviated to the bare glyph in prose.
 4. *Lambert polynomials.* The canonical zero-based family L_n^{Lam} of (10) is used throughout, with $L_0^{\text{Lam}}(u) = -u$. The degree law $\deg L_n^{\text{Lam}} = n$ is asserted only for $n \geq 1$. The scaled family is $L_n^{\text{Lam,sc}} = n! L_n^{\text{Lam}} \in \mathbb{Z}[u]$ for $n \geq 1$; it is a storage normalization, not a different coefficient sequence. The bare symbol P_n is not reserved for this family and is not used for it.
 5. *The three O-roles* are kept apart exactly as in the warning box of Section 2.8.
 6. *Two inverses.* F^{-1} is the multiplicative reciprocal and $F^{\langle -1 \rangle \circ}$ the compositional inverse. They are unrelated operations, and the ambiguous notation F^{-1} is used for neither.
 7. *Filtration, not ideal.* In $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ the generator t is a unit, so the principal ideal generated by t^N is all of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, and $t^N \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ is an additive valuation-filtration tail rather than an ideal. Agreement below outer order N is therefore phrased as $\nu_t(F - G) \geq N$, as $\mathcal{O}_t^{\text{form}}(t^N)$, or as equality of exclusive truncations — never as a congruence modulo an ideal. Genuine quotient-ring language is legitimate only in $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, where (t^N) is a proper ideal.
 8. *Coefficient blocks are polynomials.* In $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ every p_n lies in $\mathbb{K}[L]$, so logarithmic exponents are nonnegative and each block is finite. Membership implies no uniform bound on $\deg_L p_n$; any such bound is a hypothesis or a proved invariant, never part of the definition.

Part I

The polynomial–logarithmic scale

2 Why powers and logarithms belong in one expansion

This volume is about a single asymptotic scale and the exact algebra it supports. The scale is generated by two symbols,

$$t := X^{-1}, \quad L := \log X, \quad X \rightarrow +\infty, \quad (1)$$

and its monomials are the products $t^n L^j$. A reader who has met only power series may reasonably ask why two generators are needed at all and, if two, why exactly these two. This section answers both questions. It exhibits the smallest problem that forces the scale – inverting $y \mapsto y + \log y$ – and then proves three separate minimality statements: a scale of pure powers stalls after one term, a scale of pure logarithms cannot even begin, and the ring generated by t and L is the smallest subring of germs at $+\infty$ that contains a logarithm and is closed under differentiation.

Throughout this section X is a real variable tending to $+\infty$, all logarithms are the real logarithm on $(0, \infty)$, and $f \sim g$ means $f/g \rightarrow 1$. Complex branches are introduced only in Part V, where every branch of W and every choice of $\log$ is printed with the statement it belongs to.

2.1 The inversion problem that starts everything

The whole subject is visible in one implicit equation with one logarithmic term.

Proposition 2.1 (The basic inverse and its elementary envelope). *The map $\Phi(y) = y + \log y$ is a strictly increasing bijection from $(0, \infty)$ onto $\mathbb{R}$. Denote its inverse by ω , so that*

$$\omega(X) + \log \omega(X) = X \quad (X \in \mathbb{R}). \quad (2)$$

Then ω is real analytic and strictly increasing, $\omega(X) \rightarrow +\infty$ as $X \rightarrow +\infty$, and for every $X \geq 1$, writing $L = \log X$,

$$X - L \leq \omega(X) \leq X. \quad (3)$$

Moreover $W_0(z) = \omega(\log z)$ for every $z > 0$, where W_0 is the principal real branch of Lambert’s function, defined by $W_0(z)e^{W_0(z)} = z$.

Proof. On $(0, \infty)$ we have $\Phi'(y) = 1 + 1/y > 0$, so Φ is strictly increasing and real analytic with nowhere vanishing derivative; $\Phi(y) \rightarrow -\infty$ as $y \downarrow 0$ and $\Phi(y) \rightarrow +\infty$ as $y \rightarrow +\infty$, so Φ is a bijection onto $\mathbb{R}$. The inverse function theorem gives that ω is real analytic and strictly increasing, and $\omega(X) \rightarrow +\infty$ as $X \rightarrow +\infty$.

Fix $X \geq 1$ and put $y = \omega(X)$. If $y < 1$ then $\log y < 0$ and hence $X = y + \log y < 1$, a contradiction; so $y \geq 1$ and $\log y \geq 0$. From $y = X - \log y$ we get $y \leq X$; then $\log y \leq \log X = L$, whence $y = X - \log y \geq X - L$. This is (3).

For the last claim let $z > 0$ and put $w = \omega(\log z)$. Then $w > 0$ and $w + \log w = \log z$, so $\log(we^w) = \log z$ and therefore $we^w = z$. Since $v \mapsto ve^v$ is strictly increasing on $(-1, \infty)$ and $w > 0 > -1$, the value w is the principal branch $W_0(z)$. $\square$

The envelope (3) already contains the message of this volume in miniature. It says that $\omega(X)$ equals X up to a correction of size at most $\log X$; so the answer to a question posed purely in powers of X is not expressible purely in powers of X . The next subsection makes the correction explicit to two further orders.

Remark 2.2 (Names). The function ω is Wright’s omega function [7, 5]. Working with ω rather than with W removes one layer of logarithms from every display: the two nested logarithms $\log z$ and $\log \log z$ of the classical Lambert expansion become the single large variable X and the single logarithmic coefficient variable L . Every result about W in Part V is obtained from a result about ω by substituting a branch coordinate for X .

2.2 Two orders by hand

The following computation is the motivating one. It is short enough to do on paper, and it already displays every structural feature of the scale: the corrections are inverse powers of X ; their coefficients are polynomials in $\log X$; and the logarithmic degree grows with the inverse power.

Proposition 2.3 (Two orders of the basic inverse). *With $L = \log X$ and ω as in [Theorem 2.1](#), as $X \rightarrow +\infty$ along the real axis,*

$$\omega(X) = X - L + \frac{L}{X} + \frac{1}{X^2} \left(\frac{L^2}{2} - L \right) + \mathcal{O}_{X \rightarrow +\infty} \left(\frac{L^3}{X^3} \right). \quad (4)$$

Moreover there is $X_0 \geq 4$ such that for $X \geq X_0$

$$\frac{1}{2} \cdot \frac{L}{X} \leq \omega(X) - X + L \leq 2 \cdot \frac{L}{X}. \quad (5)$$

In particular $\omega(X) - X \sim -L$ and $\omega(X) - X + L \sim L/X$.

Proof. Choose $X_0 \geq 4$ so large that $L \geq 1$ and $\varepsilon := L/X \leq \frac{1}{2}$ for $X \geq X_0$, and assume $X \geq X_0$ throughout. Write

$$r := \omega(X) - X + L, \quad u := \frac{L - r}{X},$$

so that $\omega(X) = X(1 - u)$. By (3) we have $0 \leq r \leq L$, hence

$$0 \leq u \leq \varepsilon \leq \frac{1}{2}. \quad (6)$$

Since $\omega(X) > 0$, we may take logarithms in $\omega(X) = X(1 - u)$: $\log \omega(X) = L + \log(1 - u)$. Substituting this into (2) and solving for $\omega(X)$ gives $\omega(X) = X - L - \log(1 - u)$, that is,

$$r = -\log(1 - u) = \sum_{k \geq 1} \frac{u^k}{k}, \quad (7)$$

the series converging because $0 \leq u \leq \frac{1}{2}$.

First estimate. For $0 \leq u \leq \frac{1}{2}$ one has $u \leq -\log(1 - u) \leq 2u$: the quotient $-\log(1 - u)/u = \sum_{k \geq 1} u^{k-1}/k$ is nondecreasing on $[0, \frac{1}{2}]$, equals 1 at $u = 0$ and equals $2 \log 2 < 2$ at $u = \frac{1}{2}$. Hence by (7) and (6), $u \leq r \leq 2u \leq 2\varepsilon$. Feeding $r \leq 2\varepsilon$ back into the definition of u gives $u = (L - r)/X \geq (L - 2L/X)/X \geq \frac{1}{2}\varepsilon$, because $X \geq 4$. Therefore

$$\frac{1}{2}\varepsilon \leq u \leq \varepsilon, \quad \frac{1}{2}\varepsilon \leq r \leq 2\varepsilon, \quad (8)$$

which is (5).

Second estimate. From (7) and $0 \leq u \leq \frac{1}{2}$ we get the two tail bounds

$$\left| r - u - \frac{u^2}{2} \right| = \left| \sum_{k \geq 3} \frac{u^k}{k} \right| \leq \frac{u^3}{3} \sum_{k \geq 0} u^k \leq \frac{2}{3} u^3 \leq \frac{2}{3} \varepsilon^3,$$

$$|r - u| \leq \frac{u^2}{2} \cdot \frac{1}{1 - u} \leq u^2 \leq \varepsilon^2.$$

Next, $u = \varepsilon - r/X$ with $0 \leq r \leq 2\varepsilon$, so $|u - \varepsilon| \leq 2\varepsilon/X \leq 2\varepsilon^2$, where the last step uses $1/X \leq L/X = \varepsilon$, valid because $L \geq 1$. Combining this with the second tail bound gives $r = \varepsilon + \mathcal{O}_{X \rightarrow +\infty}(\varepsilon^2)$, whence $r/X = \varepsilon/X + \mathcal{O}_{X \rightarrow +\infty}(\varepsilon^2/X)$ and consequently

$$u = \varepsilon - \frac{r}{X} = \varepsilon - \frac{\varepsilon}{X} + \mathcal{O}_{X \rightarrow +\infty} \left(\frac{\varepsilon^2}{X} \right). \quad (9)$$

Squaring (9) and using $r = \mathcal{O}_{X \rightarrow +\infty}(\varepsilon)$,

$$u^2 = \left(\varepsilon - \frac{r}{X}\right)^2 = \varepsilon^2 - \frac{2\varepsilon r}{X} + \frac{r^2}{X^2} = \varepsilon^2 + \mathcal{O}_{X \rightarrow +\infty}\left(\frac{\varepsilon^2}{X}\right).$$

Adding the three contributions,

$$r = u + \frac{u^2}{2} + \mathcal{O}_{X \rightarrow +\infty}(\varepsilon^3) = \varepsilon - \frac{\varepsilon}{X} + \frac{\varepsilon^2}{2} + \mathcal{O}_{X \rightarrow +\infty}\left(\frac{\varepsilon^2}{X}\right) + \mathcal{O}_{X \rightarrow +\infty}(\varepsilon^3).$$

Substituting $\varepsilon = L/X$, the three main terms are L/X , $-L/X^2$ and $L^2/(2X^2)$, while the two error terms are $\mathcal{O}_{X \rightarrow +\infty}(L^2/X^3)$ and $\mathcal{O}_{X \rightarrow +\infty}(L^3/X^3)$; both are $\mathcal{O}_{X \rightarrow +\infty}(L^3/X^3)$ because $L \geq 1$. Since $\omega(X) = X - L + r$, this is exactly (4).

The two asymptotic equivalences follow: $\omega(X) - X = -L + r$ with $0 \leq r \leq 2L/X = o_{X \rightarrow +\infty}(L)$, and $\omega(X) - X + L = r \sim \varepsilon = L/X$. $\square$

Remark 2.4 (What the two coefficients are). The three coefficient blocks produced by the computation are the members $n = 0, 1, 2$ of the canonical zero-based Lambert family; the next two members, recorded here for orientation but derived only in Part V, continue the pattern:

$$\begin{aligned} \mathsf{L}_0^{\text{Lam}}(u) &= -u, & \mathsf{L}_1^{\text{Lam}}(u) &= u, & \mathsf{L}_2^{\text{Lam}}(u) &= \frac{u^2}{2} - u, \\ \mathsf{L}_3^{\text{Lam}}(u) &= \frac{u^3}{3} - \frac{3}{2}u^2 + u, & \mathsf{L}_4^{\text{Lam}}(u) &= \frac{u^4}{4} - \frac{11}{6}u^3 + 3u^2 - u, \end{aligned} \quad (10)$$

defined and studied in Part V. In that normalization (4) reads

$$\omega(X) = X + \sum_{n=0}^2 \mathsf{L}_n^{\text{Lam}}(L) t^n + \mathcal{O}_{X \rightarrow +\infty}(L^3 t^3),$$

the block $n = 0$ being the term $-L$. The degree law $\deg \mathsf{L}_n^{\text{Lam}} = n$ holds for $n \geq 1$ only: $\mathsf{L}_0^{\text{Lam}}(u) = -u$ has degree one, so a blanket statement “ $\mathsf{L}_n^{\text{Lam}}$ has degree n ” over $n \in \mathbb{N}_0$ is false and is not made anywhere in this volume.

Remark 2.5 (The same computation for W). Substituting $X = \log z$ into (4) and using $W_0(z) = \omega(\log z)$ from Theorem 2.1 gives, with the iterated logarithms $L_1 = \log z$ and $L_2 = \log L_1$,

$$W_0(z) = L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2^2/2 - L_2}{L_1^2} + \mathcal{O}_{z \rightarrow +\infty}\left(\frac{L_2^3}{L_1^3}\right), \quad (11)$$

the classical large-argument expansion [5, 11]. This is the sense in which the algebra of this volume sits one level below Lambert’s function: the two nested logarithms of z become the single pair (X, L) . Here L_1 and L_2 are iterated logarithms and never Lambert polynomials; the Lambert family is always written $\mathsf{L}_n^{\text{Lam}}$.

2.3 Asymptotic scales, dominance, and blocks

Before showing that no one-generator scale works, we fix what a scale is and prove the one comparison on which everything rests.

Definition 2.6 (Asymptotic scale and expansion; working definition). An *asymptotic scale* at $+\infty$ is a set $\mathfrak{M}$ of functions, each positive on some half-line, such that for all distinct $m, m' \in \mathfrak{M}$ either $m/m' \rightarrow 0$ or $m'/m \rightarrow 0$ as $X \rightarrow +\infty$. A function f defined near $+\infty$ has the *asymptotic expansion* $\sum_{i \geq 1} c_i m_i$ with respect to $\mathfrak{M}$ if $c_i \in \mathbb{R} \setminus \{0\}$, the $m_i \in \mathfrak{M}$ satisfy $m_{i+1}/m_i \rightarrow 0$, and for every index N present in the (finite or infinite) list,

$$f - \sum_{i \leq N} c_i m_i = o_{X \rightarrow +\infty}(m_N).$$

A finite such expansion is *maximal* when it admits no further term, that is, when the final remainder is not asymptotically equivalent to a constant multiple of any element of $\mathfrak{M}$.

This is the coarse blockwise notion adequate for a motivating discussion. Part VI replaces it by the sharper statements needed for analytic work, including remainder estimates with the logarithmic degree printed and the exact sense in which a truncated polynomial-logarithmic series approximates a function.

Lemma 2.7 (Lexicographic dominance of power-logarithmic monomials). *Let $(a, b), (a', b') \in \mathbb{R}^2$. As $X \rightarrow +\infty$,*

$$\frac{X^a(\log X)^b}{X^{a'}(\log X)^{b'}} \longrightarrow \begin{cases} 0, & \text{if } a < a', \text{ or } a = a' \text{ and } b < b', \\ 1, & \text{if } (a, b) = (a', b'), \\ +\infty, & \text{if } a > a', \text{ or } a = a' \text{ and } b > b'. \end{cases}$$

Consequently $\{X^a(\log X)^b : (a, b) \in \mathbb{R}^2\}$ is an asymptotic scale in the sense of [Theorem 2.6](#), totally ordered by the lexicographic order on (a, b) ; and in the generators (1), for $n, m, j, k \in \mathbb{Z}$,

$$t^n L^j \succ t^m L^k \iff n < m, \text{ or } (n = m \text{ and } j > k). \quad (12)$$

Proof. Put $s = \log X$, so $s \rightarrow +\infty$, and write the ratio as

$$X^{a-a'}(\log X)^{b-b'} = e^{(a-a')s} s^{b-b'} = \exp((a-a')s + (b-b') \log s).$$

If $a < a'$ then $(a-a')s + (b-b') \log s \rightarrow -\infty$, because $\log s = o_{s \rightarrow +\infty}(s)$ and $a-a' < 0$; hence the ratio tends to 0. The case $a > a'$ is the reciprocal statement. If $a = a'$ the ratio is $s^{b-b'}$, which tends to 0, equals 1, or tends to $+\infty$ according as $b < b'$, $b = b'$, or $b > b'$. Since the trichotomy is exhaustive, distinct exponent pairs give ratios tending to 0 or to $+\infty$, which is exactly the scale condition of [Theorem 2.6](#).

For (12), apply the trichotomy with $(a, b) = (-n, j)$ and $(a', b') = (-m, k)$: the monomial $t^n L^j$ dominates $t^m L^k$ exactly when $-n > -m$, i.e. $n < m$, or when $n = m$ and $j > k$. $\square$

The dominance order is not a convention chosen to make bookkeeping convenient. The grouping of monomials into inverse-power blocks is intrinsic.

Proposition 2.8 (Blocks are the logarithmic-comparability classes). *Let $\mu = t^n L^j$ and $\nu = t^m L^k$ with $n, m, j, k \in \mathbb{Z}$. The following are equivalent.*

1. $n = m$.
2. There is an integer $N \geq 0$ with $(\log X)^{-N} \leq \mu/\nu \leq (\log X)^N$ for all sufficiently large X .

Hence “differ by at most a fixed power of $\log X$ ” is an equivalence relation on the group of monomials $t^n L^j$, its classes are exactly the sets $\{t^n L^j : j \in \mathbb{Z}\}$ for fixed n , and $\succ$ descends to the total order of $-n$ on the classes.

Proof. (i) $\Rightarrow$ (ii): if $n = m$ then $\mu/\nu = L^{j-k}$, so $N = |j-k|$ works, with equality on one side.

(ii) $\Rightarrow$ (i): suppose $n \neq m$. If $n > m$ then, with $s = \log X$, we have $\mu/\nu = e^{(m-n)s} s^{j-k}$ and, for every N ,

$$\frac{\mu/\nu}{s^{-N}} = e^{(m-n)s} s^{j-k+N} \longrightarrow 0$$

by [Theorem 2.7](#), since $m-n < 0$; so the lower bound in (ii) fails for every N . If $n < m$, the same computation applied to ν/μ shows that the upper bound fails.

For the last sentence: the relation is reflexive ($N = 0$) and symmetric (replace N by N), and transitive with N replaced by the sum of the two constants; by the equivalence just proved, its classes are the fibres of $t^n L^j \mapsto n$. That $\succ$ induces the order of $-n$ on classes is (12). $\square$

Theorem 2.8 is the structural justification for the normal form used throughout this volume: a polynomial–logarithmic transseries is written

$$F = \sum_{n \geq n_0} p_n(L) t^n, \quad p_n \in \mathbb{K}[L], \quad n_0 \in \mathbb{Z}, \quad (13)$$

one finite polynomial per comparability class, the classes indexed by the integer n and ordered by $-n$. The formal ring housing (13) is $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} = \mathbb{K}[L]((t))$, with power-series subring $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} = \mathbb{K}[L][[t]]$; their construction, valuation, leading data and truncation topology occupy the rest of Part I.

Remark 2.9 (Where the class ends). Three failures of closure mark the exact boundary of (13), and each names its own minimal enlargement.

- $(L + 1)^{-1}$ is not a polynomial in L . Unrestricted division needs the coefficient field $\mathbb{K}(L)$, giving $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}} = \mathbb{K}(L)((t))$; expanding those rational coefficients at $L = \infty$ gives instead $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}} = \mathbb{K}((L^{-1}))((t))$. These are different objects: $1/(L + 1)$ is exact in the first and becomes $L^{-1} - L^{-2} + L^{-3} - \dots$ in the second.
- $\log L$ arises from composing with an inner argument whose leading behaviour is logarithmic, and requires a second logarithmic level.
- $X^{1/2}$ arises from inverting a leading behaviour that is not linear: a leading power X^k with $k \neq 1$ inverts to one with exponents in $k^{-1}\mathbb{Z}$. This requires a Puiseux or Hahn exponent group.

Negative powers of L do not belong to $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. A display that mixes L^{-1} into a coefficient block has already moved to $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ or $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ and must say so. These enlargements are the subject of Part VIII.

2.4 Why one generator is never enough

We now show that neither obvious one-generator scale can carry the expansion (4). The two failures are of different kinds and deserve different names: the power scale *stalls*, the logarithmic scale is *blind*.

Proposition 2.10 (Both one-generator scales fail). *Set $\psi_1(X) = \omega(X) - X$ and $\psi_2(X) = \omega(X) - X + \log X$, and let*

$$\mathfrak{M}_{\text{pow}} = \{X^a : a \in \mathbb{R}\}, \quad \mathfrak{M}_{\log} = \{(\log X)^b : b \in \mathbb{R}\},$$

both asymptotic scales by Theorem 2.7.

1. (The power scale stalls.) *With respect to $\mathfrak{M}_{\text{pow}}$, the function ω has the one-term expansion X , and that expansion is maximal. Indeed $\psi_1 \sim -\log X$, so $X^a = o_{X \rightarrow +\infty}(|\psi_1|)$ for every $a \leq 0$ while $\psi_1 = o_{X \rightarrow +\infty}(X^a)$ for every $a > 0$; no element of $\mathfrak{M}_{\text{pow}}$ is asymptotically proportional to ψ_1 .*
2. (The logarithmic scale is blind.) *With respect to $\mathfrak{M}_{\log}$, the function ω has no expansion at all, because $\omega(X)/(\log X)^b \rightarrow +\infty$ for every $b \in \mathbb{R}$. Moreover the residual ψ_2 is strictly positive for large X and satisfies $\psi_2 = o_{X \rightarrow +\infty}((\log X)^b)$ for every $b \in \mathbb{R}$; so $\mathfrak{M}_{\log}$ assigns ψ_2 the empty expansion and cannot distinguish ψ_2 from 0.*
3. (One mixed monomial repairs both failures.) $\psi_2(X) \sim X^{-1} \log X$.

Proof. (i) By Theorem 2.3, $\psi_1 \sim -\log X$. For $a \leq 0$ we have $X^a \leq 1$ for $X \geq 1$ while $|\psi_1| \rightarrow +\infty$, so $X^a/|\psi_1| \rightarrow 0$; for $a > 0$, $\psi_1/X^a \rightarrow 0$ by Theorem 2.7 applied to $\log X$ and X^a . Hence there are no $a \in \mathbb{R}$ and $c \neq 0$ with $\psi_1 \sim cX^a$: for $a > 0$ the quotient ψ_1/X^a tends to 0, and for $a \leq 0$ it is unbounded. Finally $\omega \sim X$ by (3), and $\omega - X = \psi_1 = o_{X \rightarrow +\infty}(X)$, so X is a legitimate first – and, by the above, last – term.

(ii) By (3), $\omega(X) \geq X - \log X$, and $(X - \log X)/(\log X)^b = X(\log X)^{-b} - (\log X)^{1-b} \rightarrow +\infty$ by Theorem 2.7, since the first summand dominates the second. So no leading term exists. For the

second claim, (5) gives $\frac{1}{2} \log X/X \leq \psi_2(X) \leq 2 \log X/X$ for $X \geq X_0$; in particular $\psi_2 > 0$. For any b , $\psi_2(X)(\log X)^{-b} \leq 2(\log X)^{1-b}/X \rightarrow 0$, again by [Theorem 2.7](#).

(iii) This is the last assertion of [Theorem 2.3](#). $\square$

Remark 2.11 (The two failures are not symmetric). Part (i) is a *gap*: the true size of the correction, $\log X$, lies strictly between X^0 and X^a for every $a > 0$, so the power scale simply runs out of monomials at the place where one is needed. Part (ii) is a *loss of resolution*: the logarithmic scale has monomials on both sides of ψ_2 , but every one of them is too coarse, so an expansion exists and carries no information. A treatment that reports only “a pure power series is impossible” has stated the weaker of the two obstructions.

2.5 The smallest closed class

The negative results say what fails. The positive statement is an exact minimality theorem, and it is where the pair (t, L) is singled out. Let $\mathcal{G}$ denote the ring of germs at $+\infty$ of real functions smooth on some half-line, with the derivation $D_X = d/dX$.

Theorem 2.12 (The two-generator ring is the smallest differentially closed class containing a logarithm). *Inside $\mathcal{G}$, with $t = X^{-1}$ and $L = \log X$:*

1. *The smallest D_X -stable subring of $\mathcal{G}$ containing $\mathbb{Q}$ and X is $\mathbb{Q}[X]$. Every unbounded element of it grows at least as fast as X ; in particular it contains no element asymptotic to a nonzero multiple of $\log X$.*
2. *The smallest D_X -stable subring of $\mathcal{G}$ containing $\mathbb{Q}$ and L is $\mathbb{Q}[t, L]$.*
3. *The smallest D_X -stable subring of $\mathcal{G}$ containing $\mathbb{Q}$, X and L is $\mathbb{Q}[X, t, L] = \mathbb{Q}[t, t^{-1}][L]$, the ring of Laurent polynomials in t with coefficients polynomial in L .*
4. *t and L are algebraically independent over $\mathbb{R}$. Hence $\mathbb{R}[t, L]$ is a genuine two-variable polynomial ring, and each of its elements has a unique representation $\sum_{n \geq 0} p_n(L)t^n$ with $p_n \in \mathbb{R}[L]$ and only finitely many $p_n \neq 0$.*
5. *For $n \in \mathbb{Z}$ and $p \in \mathbb{Q}[L]$,*

$$D_X(t^n p(L)) = t^{n+1}(p'(L) - np(L)). \quad (14)$$

The same statements hold with $\mathbb{Q}$ replaced by any subfield of $\mathbb{R}$. Abstractly, for a field $\mathbb{K}$ of characteristic zero, the ring in (iii) is the Laurent polynomial ring $\mathbb{K}[t, t^{-1}][L]$ carrying the unique derivation with $D_X t = -t^2$ and $D_X L = t$.

Proof. We first verify (v). As germs, $D_X t = -X^{-2} = -t^2$ and $D_X L = X^{-1} = t$; by the product and chain rules,

$$D_X(t^n p(L)) = nt^{n-1}(-t^2)p(L) + t^n p'(L)t = t^{n+1}(p'(L) - np(L)).$$

(i) $\mathbb{Q}[X]$ contains $\mathbb{Q}$ and X and is D_X -stable, since $D_X X = 1$. Conversely any D_X -stable subring containing $\mathbb{Q}$ and X contains all powers of X and all rational combinations of them, hence contains $\mathbb{Q}[X]$. A nonzero element of $\mathbb{Q}[X]$ is a polynomial of some degree d ; if $d = 0$ it is a constant, and if $d \geq 1$ it is asymptotic to a nonzero multiple of X^d . Since $\log X/X^d \rightarrow 0$ for $d \geq 1$ and $\log X \rightarrow \infty$, no element is asymptotic to a nonzero multiple of $\log X$.

(ii) Put $R = \mathbb{Q}[t, L]$. It contains $\mathbb{Q}$ and L . It is D_X -stable: a derivation is determined on a finitely generated ring by its values on generators, and $D_X t = -t^2 \in R$, $D_X L = t \in R$, so $D_X R \subseteq R$. Conversely let S be D_X -stable with $\mathbb{Q} \cup \{L\} \subseteq S$. Then $t = D_X L \in S$, so S contains the subring generated by $\mathbb{Q}$, t and L , namely R .

(iii) Put $R' = \mathbb{Q}[X, t, L]$. Since $Xt = 1$, R' is the ring of Laurent polynomials in t with coefficients in $\mathbb{Q}[L]$. It is D_X -stable by $D_X X = 1$, $D_X t = -t^2$, $D_X L = t$. Any D_X -stable subring containing $\mathbb{Q}$, X and L contains $D_X L = t$, hence contains R' .

(iv) Suppose $P \in \mathbb{R}[T, \Lambda]$ is nonzero with $P(t, L) = 0$ as a germ, i.e. $P(X^{-1}, \log X) = 0$ for all large X . Write $P = \sum_{n \geq 0} T^n q_n(\Lambda)$ with $q_n \in \mathbb{R}[\Lambda]$, and let n_0 be minimal with $q_{n_0} \neq 0$. Multiplying the relation by X^{n_0} gives, for all large X ,

$$q_{n_0}(\log X) = - \sum_{n > n_0} X^{-(n-n_0)} q_n(\log X).$$

Each summand on the right is X^{-k} with $k \geq 1$ times a polynomial in $\log X$, hence tends to 0 by [Theorem 2.7](#); the sum is finite, so the right side tends to 0. The left side is a nonzero polynomial evaluated along $\log X \rightarrow +\infty$, so it tends to $\pm\infty$ or is a nonzero constant. This contradiction proves algebraic independence; uniqueness of the representation follows, since two representations of the same germ would differ by a nontrivial polynomial relation.

For the abstract version, $\mathbb{K}[t, t^{-1}][L]$ is the localization at t of the polynomial ring $\mathbb{K}[t, L]$. A derivation of a polynomial ring is determined freely by its values on the variables, and it extends uniquely to any localization by the quotient rule; taking $D_X t = -t^2$, $D_X L = t$ gives the derivation in question. $\square$

Remark 2.13 (Reading the theorem). Part (i) says that the pure power world is closed but expressively poor: a logarithm can never be produced from X by ring operations and differentiation, so if a problem's answer contains a logarithm, that logarithm must enter through an operation – inversion, antidifferentiation, composition – that leaves the class. Part (ii) says the converse asymmetry does not exist: once $\log X$ is admitted, X^{-1} is forced immediately, and then all of $\mathbb{Q}[t, L]$. There is no “logarithms only” differential algebra. This is the precise sense in which powers and logarithms belong in one expansion. The ambient object $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} = \mathbb{K}[L]((t))$ of Part I is obtained from the Laurent ring of (iii) by completing its *polynomial* subring $\mathbb{K}[L][t]$ in the t -adic topology, which gives $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} = \mathbb{K}[L][[t]]$, and then inverting t . Equivalently, $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is the completion of $\mathbb{K}[t, t^{-1}][L]$ for the valuation ν_t . One may not complete $\mathbb{K}[t, t^{-1}][L]$ t -adically: there t is a unit, so (t^N) is the whole ring for every N and the t -adic completion is trivial.

Corollary 2.14 (Both exponent directions are forced by ω). *Let $S \subseteq \mathbb{R}^2$ and put $\mathfrak{M}_S = \{X^a(\log X)^b : (a, b) \in S\}$, an asymptotic scale by [Theorem 2.7](#). If ω has an asymptotic expansion with respect to $\mathfrak{M}_S$ with at least three terms, then*

$$(1, 0) \in S, \quad (0, 1) \in S, \quad (-1, 1) \in S,$$

with coefficients 1, -1 , 1 respectively. Consequently the subgroup of $(\mathbb{R}^2, +)$ generated by S contains $\mathbb{Z}^2$; in particular S is contained neither in $\mathbb{R} \times \{0\}$ (a scale of pure powers) nor in $\{0\} \times \mathbb{R}$ (a scale of pure logarithms).

Proof. By [Theorem 2.7](#), a monomial $X^a(\log X)^b$ can be asymptotically equivalent to $cX^{a'}(\log X)^{b'}$ with $c \in (0, \infty)$ only if $(a, b) = (a', b')$ and $c = 1$, since otherwise the ratio tends to 0 or to $+\infty$.

By [Theorem 2.6](#), the first term satisfies $\omega \sim c_1 m_1$. By (3), $\omega \sim X = X^1(\log X)^0$, so $m_1 \sim c_1^{-1} X$; by the previous paragraph $m_1 = X$ and $c_1 = 1$, whence $(1, 0) \in S$. The first partial sum is therefore exactly X , and the second term satisfies $\omega - X \sim c_2 m_2$. By [Theorem 2.3](#), $\omega - X \sim -\log X$; as $m_2 > 0$, this forces $c_2 < 0$ and $m_2 \sim |c_2|^{-1} \log X$, hence $m_2 = \log X$ and $c_2 = -1$, so $(0, 1) \in S$. The second partial sum is exactly $X - \log X$, and the third term satisfies $\omega - X + \log X \sim c_3 m_3$; by [Theorem 2.3](#) this residual is $\sim X^{-1} \log X$, so $m_3 = X^{-1} \log X$, $c_3 = 1$ and $(-1, 1) \in S$. Finally $(1, 0)$ and $(0, 1)$ generate $\mathbb{Z}^2$. $\square$

Remark 2.15 (Scope of the corollary). The corollary is stated for monomial scales in the two generators, and the restriction is not cosmetic. Nothing above rules out an exotic scale built from other generators – for instance one containing $X + X^{1/2}$ and $X^{1/2}$ – in which the first terms of ω look different.

What is ruled out, by [Theorem 2.10](#), is any scale of powers alone or of logarithms alone, whatever the exponents; and what is asserted by [Theorem 2.12](#) is that among differentially closed rings the two-generator ring is minimal. Together these are the honest content of the phrase “smallest closed class”. A stronger uniqueness statement — that $\mathcal{PL}_{\text{poly}, \mathbb{K}}$ is the *unique* minimal differential ring in which the reversion of $X + L$ exists — is not proved in this volume and should not be inferred from it.

2.6 Where a logarithm is created

[Theorem 2.12](#)(i) says a logarithm cannot be manufactured from powers by differentiation. It can be manufactured by *antidifferentiation*, and by the discrete analogue of antidifferentiation, summation. These two mechanisms account for nearly every occurrence of the scale in practice, so we isolate them.

Lemma 2.16 (Block antidifferentiation and the resonant block). *Let $\mathbb{K}$ be a field of characteristic zero and work in the differential ring $\mathbb{K}[t, t^{-1}][L]$ of [Theorem 2.12](#)(iii). Let $c \in \mathbb{K}$ and consider the $\mathbb{K}$ -linear operator $\partial_L - c$ on $\mathbb{K}[L]$.*

1. *If $c \neq 0$, then $\partial_L - c$ is a degree-preserving bijection of $\mathbb{K}[L]$ onto itself. Consequently, for $n \neq 1$ and $p \in \mathbb{K}[L]$, the element $t^n p(L)$ has a unique antiderivative of the shape $t^{n-1} q(L)$ with $q \in \mathbb{K}[L]$, and $\deg q = \deg p$.*
2. *If $c = 0$, then ∂_L is surjective on $\mathbb{K}[L]$ with kernel $\mathbb{K}$, and $\deg q = \deg(\partial_L q) + 1$ for nonconstant q . Consequently $t p(L)$ has antiderivative $q(L)$ with $q' = p$, unique up to an additive constant, and $\deg q = \deg p + 1$.*

In particular $t = X^{-1}$ integrates to $L = \log X$. The outer level $n = 1$ is the unique block at which antidifferentiation raises logarithmic degree.

Proof. (i) Write $q = \sum_{i=0}^d q_i L^i$ with $q_d \neq 0$. Then $(\partial_L - c)q = -cq_d L^d + (\text{terms of degree} < d)$, so the operator preserves degree, hence is injective and maps the finite-dimensional space of polynomials of degree at most d into itself injectively, therefore bijectively; taking the union over d gives bijectivity on $\mathbb{K}[L]$. For the antiderivative statement, apply (14) with $n - 1$ in place of n : $D_X(t^{n-1}q) = t^n(q' - (n-1)q) = t^n(\partial_L - c)q$ with $c = n - 1 \neq 0$; solve $(\partial_L - c)q = p$, which has a unique solution of the same degree.

(ii) ∂_L is surjective on $\mathbb{K}[L]$ because $\partial_L(L^{i+1}/(i+1)) = L^i$, which uses $\text{char } \mathbb{K} = 0$; its kernel is $\mathbb{K}$; and it lowers degree by exactly one on nonconstant polynomials. Taking $n = 1$ in (14), with $n - 1 = 0$, gives $D_X q(L) = t q'(L)$, so the antiderivatives of $t p(L)$ are exactly the q with $q' = p$. The case $p = 1, q = L$ is $\int X^{-1} dX = \log X$. $\square$

Lemma 2.17 (The discrete counterpart: a harmonic increment makes a logarithm). *Let $(w_n)_{n \geq n_1}$ be real with $w_n \rightarrow +\infty$ and suppose that for constants $c_0 > 0$ and $c_1 \in \mathbb{R}$,*

$$w_{n+1} = w_n + c_0 + \frac{c_1}{w_n} + \mathcal{O}_{n \rightarrow \infty}(w_n^{-2}). \quad (15)$$

Then there is a constant C with

$$w_n = c_0 n + \frac{c_1}{c_0} \log n + C + o_{n \rightarrow \infty}(1). \quad (16)$$

Proof. Since $w_n \rightarrow +\infty$, the last two terms of (15) tend to 0, so $w_{n+1} - w_n \rightarrow c_0$. By the Stolz–Cesàro theorem applied to the increments, $w_n/n \rightarrow c_0$; in particular $w_n \sim c_0 n$ and $1/w_n = \mathcal{O}_{n \rightarrow \infty}(n^{-1})$.

Put $v_n = w_n - c_0 n$. Then $v_{n+1} - v_n = c_1/w_n + \mathcal{O}_{n \rightarrow \infty}(w_n^{-2}) = \mathcal{O}_{n \rightarrow \infty}(n^{-1})$; summing, $v_n = \mathcal{O}_{n \rightarrow \infty}(\log n)$.

Now bootstrap. Since $v_n = \mathcal{O}_{n \rightarrow \infty}(\log n) = o_{n \rightarrow \infty}(n)$, we have $|v_n/(c_0 n)| \leq \frac{1}{2}$ for large n , and

$$\frac{1}{w_n} = \frac{1}{c_0 n} \cdot \frac{1}{1 + v_n/(c_0 n)} = \frac{1}{c_0 n} - \frac{v_n}{c_0^2 n^2} + \mathcal{O}_{n \rightarrow \infty}\left(\frac{v_n^2}{n^3}\right) = \frac{1}{c_0 n} + \mathcal{O}_{n \rightarrow \infty}\left(\frac{\log n}{n^2}\right).$$

Hence $v_{n+1} - v_n = \frac{c_1}{c_0 n} + \mathcal{O}_{n \rightarrow \infty} \left(\frac{\log n}{n^2} \right)$. Summing from n_1 to $n - 1$, using $\sum_{k=n_1}^{n-1} k^{-1} = \log n + \gamma - \sum_{k < n_1} k^{-1} + o_{n \rightarrow \infty}(1)$ and the absolute convergence of $\sum_k k^{-2} \log k$ (so that its tail is $o_{n \rightarrow \infty}(1)$), we obtain (16) for a suitable constant C . $\square$

Example 2.18 (Iterating the sine). Let $x_0 \in (0, \pi)$ and $x_{n+1} = \sin x_n$. Since $0 < \sin x \leq \min(x, 1)$ for $x \in (0, \pi)$, the sequence stays in $(0, 1] \subset (0, \pi)$ and decreases; its limit ℓ satisfies $\ell = \sin \ell$, so $x_n \downarrow 0$. Put $u_n = x_n^{-2}$. From $\sin x = x(1 - x^2/6 + x^4/120 + \mathcal{O}_{x \rightarrow 0}(x^6))$ and $(1 + a)^{-2} = 1 - 2a + 3a^2 + \mathcal{O}_{a \rightarrow 0}(a^3)$ with $a = -x^2/6 + x^4/120$,

$$\left(\frac{\sin x}{x} \right)^{-2} = 1 + \frac{x^2}{3} + \left(\frac{3}{36} - \frac{1}{60} \right) x^4 + \mathcal{O}_{x \rightarrow 0}(x^6) = 1 + \frac{x^2}{3} + \frac{x^4}{15} + \mathcal{O}_{x \rightarrow 0}(x^6).$$

Multiplying by u_n and using $u_n x_n^2 = 1$,

$$u_{n+1} = u_n + \frac{1}{3} + \frac{1}{15u_n} + \mathcal{O}_{n \rightarrow \infty}(u_n^{-2}).$$

Theorem 2.17 with $c_0 = 1/3$, $c_1 = 1/15$ gives $u_n = n/3 + \frac{1}{5} \log n + C + o_{n \rightarrow \infty}(1)$, and therefore

$$x_n = \sqrt{\frac{3}{n}} \left(1 - \frac{3}{10} \cdot \frac{\log n}{n} + \mathcal{O}_{n \rightarrow \infty} \left(\frac{1}{n} \right) \right). \quad (17)$$

The coefficient of n^{-1} is a polynomial of degree one in $\log n$, and the whole expression is a Puiseux-logarithmic monomial times a polynomial-logarithmic factor: with $X = n$, $t = X^{-1}$, $L = \log X$, the right side of (17) is $\sqrt{3} t^{1/2} (1 - \frac{3}{10} L t + \dots)$, an element of the Puiseux enlargement $\mathbb{K}[L]((t^{1/2}))$. Continuing the same bootstrap produces further blocks, each again a polynomial in $\log n$. This is de Bruijn's example [10, Ch. 8].

Example 2.19 (Iterating a parabolic germ). Let $f(z) = z - z^2 + az^3 + \mathcal{O}_{z \rightarrow 0}(z^4)$ be an analytic germ at 0 tangent to the identity, and let $z_n = f^{\circ n}(z_0)$ for $z_0 > 0$ small enough that $z_n \downarrow 0$. With $w_n = 1/z_n$, the expansion $(1 - z + az^2 + \mathcal{O}_{z \rightarrow 0}(z^3))^{-1} = 1 + z + (1 - a)z^2 + \mathcal{O}_{z \rightarrow 0}(z^3)$ gives

$$w_{n+1} = w_n (1 + z_n + (1 - a)z_n^2 + \mathcal{O}_{n \rightarrow \infty}(z_n^3)) = w_n + 1 + \frac{1 - a}{w_n} + \mathcal{O}_{n \rightarrow \infty}(w_n^{-2}),$$

so **Theorem 2.17** yields $w_n = n + (1 - a) \log n + C + o_{n \rightarrow \infty}(1)$ and hence

$$z_n = \frac{1}{n} - \frac{(1 - a) \log n}{n^2} + \mathcal{O}_{n \rightarrow \infty} \left(\frac{1}{n^2} \right).$$

The same logarithm appears in the Fatou coordinate Φ of f , the solution of $\Phi \circ f = \Phi + 1$, whose leading behaviour as $z \downarrow 0$ is $\Phi(z) = 1/z - (1 - a) \log(1/z) + \text{const} + o_{z \downarrow 0}(1)$; one checks directly that this shape reproduces the increment 1 to the displayed order. Normal-form and conjugacy theory for such germs is one of the classical homes of this scale.

Section summary

Two mechanisms create the logarithm, and they are one mechanism in continuous and in discrete clothing. Antidifferentiation preserves logarithmic degree on every outer block except t^1 , where it raises it by one (**Theorem 2.16**); summation of increments is bounded except for the harmonic one, which produces $\log n$ (**Theorem 2.17**). Once one logarithm is present, **Theorem 2.12(ii)** forces the whole ring $\mathbb{K}[t, L]$.

2.7 Provenance: where these expansions arise

2.7.1 Implicit equations carrying a logarithmic term

The generic source is a relation in which the unknown occurs both linearly and inside a logarithm,

$$y + \alpha \log y + \frac{p_1(\log y)}{y} + \frac{p_2(\log y)}{y^2} + \cdots = X, \quad \alpha \in \mathbb{K}, \quad p_i \in \mathbb{K}[\cdot], \quad (18)$$

whose two-term case $y + \alpha \log y = X$ is equivalent to the exponentiated form $y^\alpha e^y = e^X$. Dominant balance gives $y \sim X$; substituting that into the logarithm creates $\log X$; iterating creates powers of X^{-1} carrying polynomials in $\log X$. Part IV turns this heuristic into a theorem: read as a transseries in the variable y , the left side of (18) lies in the near-identity class $\mathcal{G}_{\text{poly}, \mathbb{K}}$, and $\mathcal{G}_{\text{poly}, \mathbb{K}}$ is a group under composition, so the solution is again of the same shape.

A classical instance outside the Lambert family is the size of the n -th prime. Inverting $\pi(p_n) = n$ against the logarithmic-integral approximation gives Cipolla's expansion

$$\frac{p_n}{n} = \log n + \log \log n - 1 + \frac{\log \log n - 2}{\log n} + \mathcal{O}_{n \rightarrow \infty} \left(\frac{(\log \log n)^2}{(\log n)^2} \right). \quad (19)$$

Read with $X = \log n$ and $L = \log \log n$, the right side of (19) is a truncation of an element of $X + \mathbb{K}[L][[t]] = \mathcal{G}_{\text{poly}, \mathbb{K}}$: the very class that houses ω . Quantile asymptotics for laws with exponential-type tails have the same shape, for the same reason — one inverts a relation in which the unknown appears both linearly and logarithmically.

2.7.2 Dulac maps and first-return maps of hyperbolic sectors

Let v be a real analytic planar vector field with a hyperbolic saddle at the origin, and let D be the transition (Dulac, or corner) map from a transversal on one separatrix to a transversal on the other, written in coordinates $x \downarrow 0$ on the transversals. Then D has an asymptotic expansion in the power-logarithmic scale at zero,

$$D(x) \sim \sum_{i \geq 0} x^{\lambda_i} P_i(\log(1/x)), \quad 0 < \lambda_0 < \lambda_1 < \cdots \rightarrow +\infty, \quad P_i \in \mathbb{R}[\cdot], \quad (20)$$

where λ_0 is the hyperbolicity ratio of the saddle. The logarithms are forced precisely by resonances among the exponents: for a nonresonant saddle the P_i are constants. Composing finitely many such maps with regular transition maps produces the first-return map of a hyperbolic polycycle, whose asymptotic expansion is again of the form (20). The finiteness of the number of limit cycles accumulating on such a polycycle — Dulac's problem, settled independently by Écalle [12] and by Ilyashenko — turns on how much of this formal expansion can be promoted to an analytic statement. Normal forms for the associated germs are studied in [23, 27], the ε -neighbourhood length of an orbit of a Dulac map is computed in the same scale [24], and closure properties of the ambient logarithmic transseries algebras are in [26].

The change of variable $X = 1/x$, $L = \log X = \log(1/x)$ carries (20) to $\sum_i t^{\lambda_i} P_i(L)$, which is the normal form (13) with a real exponent set; Part VIII treats that Hahn enlargement. The core of this volume keeps the integer lattice, where the algorithms are exact and finite.

A common trap

Part of the dynamical literature uses $-1/\log x$ as its small logarithmic generator, so that the generator tends to 0 with x , rather than $\log(1/x)$, which tends to $+\infty$. The two conventions differ by an inversion of the coefficient variable, and a formula transported between them without applying that inversion will be wrong. In this volume $L = \log X$ always tends to $+\infty$.

2.7.3 Iteration, normal forms, and recursions

Theorems 2.18 and **2.19** are the model cases: whenever the increment of a slowly diverging quantity carries a term of size $1/w$, the partial sums manufacture a logarithm, and the subsequent expansion is organized in inverse powers of n with polynomial coefficients in $\log n$. In the analytic classification of parabolic germs the same logarithm is the logarithmic term of the Fatou coordinate; in the formal classification of maps tangent to the identity at infinity the conjugating series is an element of $\mathcal{G}_{\text{poly}, \mathbb{K}}$. Symbolic algorithms for computing such expansions – and for the broader exp-log asymptotic inversion problem – are due to Salvy and Shackell [29, 30]; Part VII specializes their framework to the present class.

2.7.4 The Fabius–Rvachev endpoint, where this corpus meets the scale

The corpus in which this volume sits studies the Fabius function F_{bd} and the Rvachev function up. Their contact with the polynomial-logarithmic scale is at the endpoints, and it is structural rather than decorative: the endpoint problem is an inversion problem of exactly the type analysed above, and its solution needs two generators.

The corpus proves, and has verified in Lean, the effective flatness family

$$F_{\text{bd}}(x) \leq 2^{\binom{n+1}{2}} x^n, \quad F_{\text{bd}}(x) \leq \frac{2^{\binom{n+1}{2}}}{n!} x^n, \quad (21)$$

valid for all $n \in \mathbb{N}_0$, $x \geq 0$ with $2^n x \leq 1$.

Optimizing the free index n in the sharper of the two turns the family into a quadratic-in-log decay law.

Proposition 2.20 (A two-term endpoint upper bound from the flatness family). *Write $\ell = \log(1/x)$ and $a = \log 2$, and set $C = \frac{1}{2} + 3/a$. For all $0 < x \leq \frac{1}{4}$,*

$$\log F_{\text{bd}}(x) \leq -\frac{\ell^2}{2a} - \frac{\ell \log \ell}{a} + C\ell. \quad (22)$$

Proof. Put $m = \ell/a$; since $x \leq \frac{1}{4}$ we have $\ell \geq 2a$, hence $m \geq 2$. Let $n = \lfloor m \rfloor \geq 2$ and $\theta = m - n \in [0, 1)$. The hypothesis $2^n x \leq 1$ is equivalent to $na \leq \ell$, i.e. to $n \leq m$, which holds. Taking logarithms in the second bound of (21) and using $\log x = -\ell$,

$$\log F_{\text{bd}}(x) \leq A_1 + A_2, \quad A_1 := \frac{an(n+1)}{2} - n\ell, \quad A_2 := -\log n!.$$

The quadratic part. Using $\ell = am$ and $n = m - \theta$,

$$A_1 = \frac{an}{2}(n+1-2m) = -\frac{a}{2}(m-\theta)(m+\theta-1) = -\frac{a}{2}(m^2 - m + \theta - \theta^2).$$

Since $\theta - \theta^2 \geq 0$ on $[0, 1)$, and since $am^2/2 = \ell^2/(2a)$ and $am/2 = \ell/2$,

$$A_1 \leq -\frac{\ell^2}{2a} + \frac{\ell}{2}.$$

The factorial part. From $e^n \geq n^n/n!$ we get $n! \geq n^n e^{-n}$, hence $A_2 \leq -n \log n + n$. As $m \geq 2$ and $0 \leq \theta < 1$, we have $0 \leq \theta/m \leq \frac{1}{2}$, so $\log(1 - \theta/m) \geq -2\theta/m \geq -2/m$ and therefore

$$n \log n = (m - \theta)(\log m + \log(1 - \theta/m)) \geq (m - \theta) \log m - 2.$$

Using $0 \leq \theta < 1$, $\log m \geq \log 2 > 0$ and $n \leq m$, this gives $A_2 \leq -m \log m + \log m + 2 + m$. Now

$$m \log m = \frac{\ell}{a}(\log \ell - \log a) = \frac{\ell \log \ell}{a} - \frac{\ell \log a}{a},$$

and $a = \log 2 < 1$ gives $\log a < 0$, so $-m \log m = -\ell \log \ell/a + \ell \log a/a \leq -\ell \log \ell/a$. Finally $\log m \leq m = \ell/a$ and $2 \leq m = \ell/a$, so

$$A_2 \leq -\frac{\ell \log \ell}{a} + \frac{3\ell}{a}.$$

Adding the two estimates gives (22) with $C = \frac{1}{2} + 3/a$. $\square$

Corollary 2.21 (The inverse endpoint lives on a Puiseux-logarithmic scale). *Let Q_F be the inverse of the strictly increasing restriction $F_{\text{bd}}|_{[0,1]}$, and put $B = \log(1/y)$. Then, as $y \downarrow 0$,*

$$\log \frac{1}{Q_F(y)} \leq \sqrt{2B \log 2} + \mathcal{O}_{y \downarrow 0}(1). \quad (23)$$

Proof. Let $x = Q_F(y)$, so that $y = F_{\text{bd}}(x)$ and $x \downarrow 0$ as $y \downarrow 0$; for y small, $x \leq \frac{1}{4}$. Put $\ell = \log(1/x)$ and $a = \log 2$. Since $\ell \geq 2a > 1$ we have $\log \ell > 0$, so Theorem 2.20 gives

$$B = \log \frac{1}{y} = -\log F_{\text{bd}}(x) \geq \frac{\ell^2}{2a} + \frac{\ell \log \ell}{a} - C\ell \geq \frac{\ell^2}{2a} - C\ell.$$

Hence $\ell^2 - 2aC\ell - 2aB \leq 0$, so $\ell \leq aC + \sqrt{a^2C^2 + 2aB} \leq 2aC + \sqrt{2aB}$, using $\sqrt{p+q} \leq \sqrt{p} + \sqrt{q}$. As $\ell = \log(1/Q_F(y))$, this is (23). $\square$

Remark 2.22 (What the corpus adds, and what must not be inferred). Theorems 2.20 and 2.21 are one-sided: they rest on a family of upper bounds, and an upper bound for F_{bd} yields only an upper bound for $\log(1/Q_F(y))$. The corpus obtains the matching lower bounds by a lattice computation along the dyadic scale ladder, giving the two-sided laws

$$\begin{aligned} \log F_{\text{bd}}(x) &= -\frac{\ell^2}{2 \log 2} - \frac{\ell \log \ell}{\log 2} + \mathcal{O}_{x \downarrow 0}(\ell), \\ \log \frac{1}{Q_F(y)} &= \sqrt{2B \log 2} - \frac{1}{2} \log B + \mathcal{O}_{y \downarrow 0}(1), \end{aligned}$$

with $\ell = \log(1/x)$ and $B = \log(1/y)$. Those results are quoted here, not proved; their proofs belong to the endpoint volumes of the corpus.

Two structural points connect them to the present volume. First, the inverse law is not a polynomial-logarithmic series in the core sense. Read with $X = B$, $t = X^{-1}$ and $L = \log X$, its displayed terms are $\sqrt{2 \log 2} X^{1/2} - \frac{1}{2} L + \mathcal{O}_{X \rightarrow +\infty}(1)$, an element of the Puiseux enlargement $\mathbb{K}[L]((t^{1/2}))$ of Part VIII, not of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. The half-integer exponent is the signature of inverting a leading behaviour that is quadratic rather than linear, a situation treated in Part IV under reversion beyond the near-identity class. Second, the corpus records that the refined dyadic endpoint expansions carry *log-periodic* corrections – bounded one-periodic functions of $\ell/\log 2$ – and that the natural coordinate for the associated saddle equations is the lower real branch W_{-1} . A bounded oscillating function of $\ell/\log 2$ is not a monomial of any power-logarithmic scale: by Theorem 2.7 it is not asymptotic to a constant multiple of any $t^n L^j$, so it cannot appear as a coefficient in (13). Its presence is a genuine departure from the class studied here, and the algebra of this volume applies to such an expansion only after the periodic factor has been separated out. The appearance of W_{-1} , by contrast, is squarely inside the volume: that branch is expanded in Part V.

2.8 Formal, asymptotic, convergent

The material above has used three logically independent kinds of statement, and the rest of the volume depends on never confusing them.

A common trap

A displayed infinite expansion can have any of three statuses.

1. A *formal identity*: an equality of coefficient arrays in a series ring. Written with the formal tail $\mathcal{O}_t^{\text{form}}(t^N)$, which means $\nu_t(F - G) \geq N$ and asserts no numerical bound, no convergence, and no uniformity in L .
2. An *asymptotic expansion*: a hierarchy of remainder estimates for an actual function on a stated domain. Written $\mathcal{O}_{X \rightarrow +\infty}(A(X))$, or $\mathcal{O}_{X \rightarrow \infty, X \in S}(A(X))$ in a sector, with the limit and the domain printed.
3. A *convergent expansion*: an actual infinite sum, with its domain of convergence stated.

A fourth use of the same glyph, algorithmic complexity $\mathcal{O}(N^2)$, is none of the three. No one of these four meanings may be inferred from another; in particular a bare $\mathcal{O}$ is never acceptable for an asymptotic claim in this volume.

For the Lambert expansion all three statuses are in fact available on suitable domains, which is exceptional rather than typical: the large-argument Lambert series is not only asymptotic but convergent in appropriate real and complex regions, a fact requiring separate analytic work [5, 11]. What is elementary, and worth isolating, is a statement about the *coefficients*: the formal series underlying (11), namely $L_1 + \sum_{n \geq 0} \mathbb{L}_n^{\text{Lam}}(L_2) L_1^{-n}$, collapses to its leading term at $z = e$, where $L_1 = 1$ and $L_2 = 0$. Indeed $\mathbb{L}_n^{\text{Lam}}(0) = 0$ for every $n \in \mathbb{N}_0$: in the defining formula $\mathbb{L}_n^{\text{Lam}}(u) = ((-1)^{n+1}/(n+1)!) [\prod_{j=0}^{n-1} (\partial_u - j)] u^{n+1}$ each of the n factors either differentiates or rescales, so every monomial of the result has degree at least $(n+1) - n = 1$ and $\mathbb{L}_n^{\text{Lam}}$ has no constant term. Hence every correction block vanishes at $z = e$ and the series returns the correct value $W_0(e) = 1$. This is an identity between coefficients, not a convergence statement: a convergence threshold is a different assertion and is not proved in this volume; Part VI states what is proved and cites what is not.

Section summary

The inverse of $y \mapsto y + \log y$ cannot be expanded in powers of X alone — the correction $-\log X$ falls in a gap of that scale — nor in powers of $\log X$ alone, which cannot even see the correction $X^{-1} \log X$. It can be expanded in the two-generator scale $t^n L^j$, and that scale is forced: the smallest differentially closed ring containing a logarithm is $\mathbb{K}[t, L]$. Dominance is lexicographic, one inverse power of X defeating every fixed power of $\log X$, so a series is organized into inverse-power blocks, each a polynomial in L . The same scale organizes implicit equations with a logarithmic term, Dulac and first-return maps, parabolic normal forms, recursions with a harmonic increment, and the endpoint asymptotics of the Fabius–Rvachev system.

3 The formal polynomial–logarithmic rings

3.1 Generators, coefficients, and what “formal” means here

Throughout this volume $\mathbb{K}$ is a field of characteristic zero. The default analytic regime is the positive ray $X \rightarrow +\infty$; for complex work a sector and a branch of $\log$ must be named before any analytic statement is made, and the algebra developed in this part is branch-neutral until such a choice is imposed.

Definition 3.1 (Generators). Set

$$t := X^{-1}, \quad L := \log X. \quad (24)$$

In the formal algebra t and L are two *independent* generators: the underlying additive structure of every class below is a space of families of scalars indexed by exponent pairs, and no algebraic relation between t and L is imposed.

Independence is a statement about the ring operations only, and it is exactly as strong as it needs to be.

Remark 3.2 (The two-variable illusion). For addition and multiplication, treating t and L as unrelated indeterminates is harmless and is what makes the coefficient bookkeeping finite. For differentiation and for composition it is false. The analytic relation $L = \log X$, $t = X^{-1}$ re-enters through the distinguished derivation

$$D_X = t \partial_L - t^2 \partial_t, \quad D_X t = -t^2, \quad D_X L = t,$$

studied in Part II, and through the substitution rule $t \mapsto G^{-1}$, $L \mapsto \log G$ of Part III. Every “shifted” operator $\partial_L - n$ that appears later is a trace of this coupling. A calculation that silently uses $\partial_L L = 1$ together with $\partial_t L = 0$ inside a differentiation is using the coupling, and must say so.

Formal is not analytic

Every statement in this part is an identity between formal objects: two elements are equal when all their coefficients agree. No such statement asserts that substituting a numerical value of X makes an infinite sum converge, nor that any function has the displayed expansion. The passage from a formal identity to an asymptotic or convergent statement is a separate theorem with separately printed hypotheses.

3.2 The four ambient classes

Definition 3.3 (The four classes). Define

$$\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} := \mathbb{K}[L][[t]] = \left\{ \sum_{n \geq 0} p_n(L) t^n : p_n \in \mathbb{K}[L] \right\}, \quad (25)$$

$$\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} := \mathbb{K}[L]((t)) = \left\{ \sum_{n \geq n_0} p_n(L) t^n : n_0 \in \mathbb{Z}, p_n \in \mathbb{K}[L] \right\}, \quad (26)$$

$$\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} := \mathbb{K}(L)((t)), \quad (27)$$

$$\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}} := \mathbb{K}((L^{-1}))((t)). \quad (28)$$

In each case the outer construction is a formal Laurent (or power) series in t ; the coefficient object at level n is called the *block* at t^n . Addition is blockwise and multiplication is the Cauchy convolution

$$\left(\sum_i p_i t^i \right) \left(\sum_k q_k t^k \right) = \sum_n \left(\sum_{i+k=n} p_i q_k \right) t^n,$$

in which every inner sum is finite because the outer supports are bounded below.

Two facts are definitional but deserve to be said once, because almost every later argument uses them. First, each block is a *single* finite polynomial (resp. rational function, resp. Laurent series in L^{-1}); no uniform bound on its degree is implied. Second, only finitely many negative powers of t , that is finitely many positive powers of X , may occur.

Proposition 3.4 (Localization). $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} = \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}[t^{-1}]$: the natural map from the localization of $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ at the multiplicative set $\{t^m : m \geq 0\}$ to $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is a ring isomorphism. The same holds with $\mathbb{K}[L]$ replaced by $\mathbb{K}(L)$ or $\mathbb{K}((L^{-1}))$.

Proof. The map sends F/t^m to $t^{-m}F$; it is well defined and a ring homomorphism. It is surjective because an element of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ with $\nu_t \geq -m$ is t^{-m} times an element of $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. It is injective because t^{-m} is invertible in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, so $t^{-m}F = 0$ forces $F = 0$, whence $F/t^m = 0$ in the localization. $\square$

Proposition 3.5 (Strict inclusions). *There are inclusions of $\mathbb{K}$ -algebras*

$$\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} \subsetneq \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} \subsetneq \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} \subsetneq \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}, \quad (29)$$

each given by the identity on coefficients except the last, which is the coefficientwise Laurent expansion at $L = \infty$. All three inclusions are strict, and:

- (i) $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ are integral domains and are not fields;
- (ii) $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ and $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ are fields.

Proof. The first two inclusions. The first is the inclusion $E[[t]] \subset E((t))$ for $E = \mathbb{K}[L]$, a subring inclusion by Theorem 3.3; the second is the blockwise extension of the coefficient inclusion $\mathbb{K}[L] \subset \mathbb{K}(L)$, hence an injective ring homomorphism. Strictness: $t^{-1} \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} \setminus \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, and $L^{-1} \in \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} \setminus \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ because $L^{-1} \notin \mathbb{K}[L]$.

The third inclusion. The polynomial ring $\mathbb{K}[L]$ sits inside $\mathbb{K}((L^{-1}))$ (a polynomial has finitely many positive and no negative powers of L). Every nonzero $p = aL^d + \dots \in \mathbb{K}[L]$, $a \in \mathbb{K}^\times$, is a unit there: writing $u = L^{-1}$ we have $p = au^{-d}(1+w)$ with $w \in u\mathbb{K}[u]$, and $1+w$ is invertible in $\mathbb{K}[[u]]$ by the geometric series $\sum_{r \geq 0} (-w)^r$, which is u -adically summable (Theorem 5.23 with coefficient ring $\mathbb{K}$ and u in place of t). By the universal property of the field of fractions, the inclusion $\mathbb{K}[L] \hookrightarrow \mathbb{K}((L^{-1}))$ extends uniquely to a field embedding $\mathbb{K}(L) \hookrightarrow \mathbb{K}((L^{-1}))$. Extending blockwise gives $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} \hookrightarrow \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$. Strictness is Theorem 3.7.

(i). A Laurent series ring $E((t))$ over an integral domain E is an integral domain: if $F, G \neq 0$, then by Theorem 4.2 (i) the block of FG at level $\nu_t(F) + \nu_t(G)$ is the product of two nonzero elements of E , hence nonzero. Neither $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ nor $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is a field: if $LG = 1$ with $G = \sum_n q_n(L)t^n$, comparison of the blocks at t^0 gives $Lq_0(L) = 1$ in $\mathbb{K}[L]$, which is impossible on degree grounds.

(ii). Let E be a field and $F \in E((t))$ nonzero, with $\nu = \nu_t(F)$ and block $c = p_\nu \in E^\times$. Then $F = ct^\nu(1+U)$ with $U = \sum_{n \geq 1} (p_{\nu+n}/c)t^n \in tE[[t]]$, and $\sum_{r \geq 0} (-U)^r$ is t -adically summable (Theorem 5.23), so F is invertible with $F^{-1} = c^{-1}t^{-\nu} \sum_{r \geq 0} (-U)^r$. Apply this with $E = \mathbb{K}(L)$ and $E = \mathbb{K}((L^{-1}))$. $\square$

Corollary 3.6 (Domains). *All four classes are integral domains, and ν_t and the rank-two valuation of Theorem 4.6 are defined on each of them.*

Proof. Immediate from Theorem 3.5, since a subring of a field is a domain and $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} \subset \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} \subset \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$. $\square$

Proposition 3.7 (The iterated field is strictly larger than the rational one). *The element*

$$\Phi := \sum_{k \geq 0} L^{-k^2} \in \mathbb{K}((L^{-1})) \subset \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}} \quad (30)$$

does not lie in the image of $\mathbb{K}(L)$; consequently $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} \subsetneq \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$.

Proof. Put $u = L^{-1}$, so that $\mathbb{K}((L^{-1})) = \mathbb{K}((u))$, the image of $\mathbb{K}(L)$ is the subfield $\mathbb{K}(u)$, and $\Phi = \sum_{k \geq 0} c_k u^k \in \mathbb{K}[[u]]$ with $c_k = 1$ if k is a perfect square and $c_k = 0$ otherwise. Suppose $\Phi \in \mathbb{K}(u)$. Write $\Phi = A/B$ in lowest terms with $A, B \in \mathbb{K}[u]$. If $B(0) = 0$, then $u \mid B$; since $\Phi \in \mathbb{K}[[u]]$ we get $A = B\Phi \in u\mathbb{K}[[u]] \cap \mathbb{K}[u]$, so $u \mid A$, contradicting coprimality. Hence $B(0) = b_0 \neq 0$. Let $B = \sum_{i=0}^m b_i u^i$ and $D = \deg A$. Comparing coefficients of u^k in $B\Phi = A$ gives, for every $k > D$,

$$b_0 c_k = - \sum_{i=1}^m b_i c_{k-i}.$$

Therefore, if $k > D + m$ and $c_{k-1} = \dots = c_{k-m} = 0$, then $c_k = 0$, and by induction $c_l = 0$ for all $l \geq k$. Consecutive squares differ by $(k+1)^2 - k^2 = 2k+1$, so the sequence (c_k) has runs of zeros of every length; choose one of length m beginning beyond $D + m$. Then $c_l = 0$ for all large l , contradicting $c_{k^2} = 1$ for arbitrarily large k . $\square$

Proposition 3.8 (The coefficient extensions are not localizations). *Let $S = \mathbb{K}[L] \setminus \{0\}$. Then the localization $S^{-1} \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is a proper subring of $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$. Explicitly,*

$$\Phi_0 := \sum_{n \geq 0} \frac{t^n}{L+n} \in \mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}} \quad (31)$$

admits no representation F/q with $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and $q \in S$.

Proof. If $\Phi_0 = F/q$ then $q\Phi_0 = F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, so for every $n \geq 0$ the block $q/(L+n)$ lies in $\mathbb{K}[L]$, that is, $L+n$ divides q . Since $\text{char } \mathbb{K} = 0$, the polynomials $L+n$, $n \geq 0$, are pairwise non-associate irreducibles, so q would have infinitely many pairwise non-associate irreducible factors; hence $q = 0$, a contradiction. $\square$

Remark 3.9 (What each step buys, and what it costs). [Table 1](#) records the trade-offs. The two extensions on the right are not interchangeable: $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ keeps rational dependence on L *exact* but does not resolve the secondary asymptotic scale, while $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ resolves that scale but replaces an exact coefficient such as $(L+1)^{-1}$ by the infinite tail $L^{-1} - L^{-2} + L^{-3} - \dots$ and therefore needs a second, inner cutoff before anything can be stored or printed.

Table 1: The four ambient classes. “Finite record” means the data a computer must hold to determine the object up to a declared precision.

Class	Buys	Costs
$\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} = \mathbb{K}[L][[t]]$	$t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is a genuine ideal, so residues modulo t^N form a ring; complete and separated; local finiteness is automatic	does not contain X ; not stable under multiplication by t^{-1}
$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} = \mathbb{K}[L]((t))$	smallest of the four containing X ; still exact polynomial blocks; finite record $n \mapsto p_n$	not a field; t is a unit, so $t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is only a filtration tail, not an ideal
$\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}} = \mathbb{K}(L)((t))$	a field; exact rational dependence on L ; unrestricted division of blocks	not a localization of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ (Theorem 3.8); a block has no intrinsic asymptotic expansion until one is chosen
$\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}} = \mathbb{K}((L^{-1}))((t))$	a field; the Hahn field of the rank-two value group (Theorem 3.19); the secondary scale in L^{-1} is resolved	blocks are infinite objects, so a finite record needs an inner L^{-1} -cutoff as well as the outer t -cutoff; the embedding of $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ destroys exactness

Remark 3.10 (The classes are not local rings). The subset $t \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is a prime ideal of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, with $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}/t \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} \cong \mathbb{K}[L]$; since $\mathbb{K}[L]$ is not a field, $t \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is not maximal and $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is not a local ring. In $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ the set $t \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is not an ideal at all ([Theorem 5.11](#)). By contrast $\mathbb{K}(L)[[t]]$ and $\mathbb{K}((L^{-1}))[[t]]$ are discrete valuation rings with residue fields $\mathbb{K}(L)$ and $\mathbb{K}((L^{-1}))$.

3.3 Coefficient support

Expanding every block into monomials gives the double form

$$F = \sum_{n \geq n_0} p_n(L) t^n = \sum_{\substack{n \in \mathbb{Z} \\ j \in \mathbb{N}_0}} a_{n,j} t^n L^j, \quad p_n(L) = \sum_{j \geq 0} a_{n,j} L^j. \quad (32)$$

Definition 3.11 (Coefficient support). For $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ written as in (32),

$$\text{supp}_{\text{coeff}}(F) := \{(n, j) \in \mathbb{Z} \times \mathbb{N}_0 : a_{n,j} \neq 0\}.$$

The zero series has empty coefficient support. This is support in an exponent-index lattice; it is unrelated to the pointwise support of a function.

Definition 3.12 (Dominance). For $(n, j), (m, k) \in \mathbb{Z} \times \mathbb{Z}$,

$$t^n L^j \succ t^m L^k \iff n < m, \text{ or } (n = m \text{ and } j > k). \quad (33)$$

We write $\prec, \preceq$ and $\succsim$ accordingly.

Lemma 3.13 (Analytic justification of dominance). *Let $\alpha, \alpha', \beta, \beta'$ be fixed reals with either $\alpha > \alpha'$, or $\alpha = \alpha'$ and $\beta > \beta'$. Then, on the part $X > e$ of the positive ray, where $\log X > 1$ and the quotient below is positive,*

$$\frac{X^\alpha (\log X)^\beta}{X^{\alpha'} (\log X)^{\beta'}} \longrightarrow +\infty \quad (X \rightarrow +\infty).$$

Consequently (33) orders the monomials $t^n L^j = X^{-n} (\log X)^j$ by decreasing eventual size.

Proof. For $X > e$ we have $\log X > 1$, so the quotient is positive and

$$\log \frac{X^\alpha (\log X)^\beta}{X^{\alpha'} (\log X)^{\beta'}} = (\alpha - \alpha') \log X + (\beta - \beta') \log \log X.$$

In the first case, since $\log \log X = o_{X \rightarrow +\infty}(\log X)$ and $\alpha - \alpha' > 0$, the right-hand side tends to $+\infty$. In the second case, $\alpha = \alpha'$ and $\beta > \beta'$, so the expression reduces to $(\beta - \beta') \log \log X \rightarrow +\infty$. $\square$

Remark 3.14 (Fixed exponents only). **Theorem 3.13** is the two-generator fragment of Hardy’s comparison of orders of infinity [20]: it compares two monomials with *fixed* exponents. It says nothing about a family in which the logarithmic exponent grows with the inverse-power level; see **Theorem 5.9** for what this costs.

Theorem 3.15 (Characterization of admissible supports). *For a subset $S \subseteq \mathbb{Z} \times \mathbb{N}_0$ the following are equivalent.*

- (a) $S = \text{supp}_{\text{coeff}}(F)$ for some $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$.
- (b) The projection $\{n : (n, j) \in S \text{ for some } j\}$ is bounded below, and for each n the fiber $S_n = \{j : (n, j) \in S\}$ is finite.
- (c) Every nonempty subset of S has a $\succ$ -greatest element (reverse well-ordering).
- (d) S can be enumerated as a strictly $\succ$ -decreasing sequence $s_0 \succ s_1 \succ s_2 \succ \dots$, finite or of order type ω , exhausting S .

Proof. (a) $\Leftrightarrow$ (b). By **Theorem 3.3** an element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is exactly a family $(a_{n,j})$ whose blocks $p_n = \sum_j a_{n,j} L^j$ are polynomials — that is, have finite support in j — and which vanishes for $n < n_0$. Conversely any S with property (b) is the support of $\sum_{(n,j) \in S} t^n L^j$.

(b) $\Rightarrow$ (d). If $S = \emptyset$ there is nothing to prove. Otherwise let $n_0 = \min \{n : S_n \neq \emptyset\}$, which exists because the projection is a nonempty subset of $\mathbb{Z}$ bounded below. List first the finitely many elements of $\{n_0\} \times S_{n_0}$ in order of decreasing j , then those of $\{n_0 + 1\} \times S_{n_0+1}$, and so on. By (33) the resulting list is strictly $\succ$ -decreasing, every element of S occupies a finite position, and every element occurs.

(d) $\Rightarrow$ (c). Given $\emptyset \neq T \subseteq S$, let $k = \min \{i : s_i \in T\}$; then $s_k \succ s_i$ for every other $s_i \in T$, so s_k is the greatest element of T .

(c) $\Rightarrow$ (b). If the projection of S were unbounded below, then S itself would have no greatest element, since any $(n, j) \in S$ is dominated by some $(n', j') \in S$ with $n' < n$. If some fiber S_n were infinite, then $\{n\} \times S_n$ would be a nonempty subset with no greatest element, because an infinite subset of $\mathbb{N}_0$ is unbounded above. $\square$

Remark 3.16 (Order type). Condition (d) is the sharp form of the “well-based support” hypothesis of Hahn’s construction [19, 13] in the present rank-two, integer-lattice case: the support of an element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is not merely reverse well-ordered, it has order type at most ω . The same is *not* true in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$, where fibers may be infinite downwards; see **Theorem 3.19**.

Example 3.17 (A support lattice). The table below marks with $\bullet$ the pairs (n, j) of a support whose log-degree profile is $d_n = 1, 2, 3, 3, 4, 5, 5$ for $n = 0, \dots, 6$. Each column is finite; the columns start at $n = 0$; the greatest monomial of any nonempty sub-diagram is found by moving as far left as possible and then as far up as possible.

$j = 5$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\bullet$	$\bullet$
$j = 4$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\bullet$	$\bullet$	$\bullet$
$j = 3$	$\cdot$	$\cdot$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$
$j = 2$	$\cdot$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$
$j = 1$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$
$j = 0$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$
	$n = 0$	1	2	3	4	5	6

3.4 The Hahn-series viewpoint

Let $\Gamma = \mathbb{Z} \times \mathbb{Z}$ be the additive group of exponent pairs of the monomials $t^n L^j$, $n, j \in \mathbb{Z}$, ordered by the rank-two valuation

$$v(t^n L^j) = (n, -j), \quad (34)$$

with Γ lexicographically ordered. Thus a smaller value means a more dominant monomial, and (34) is exactly [Theorem 3.12](#) rewritten. The general framework — an ordered monomial group, a coefficient field, and a support condition making convolution locally finite — is Hahn’s [19]; its transseries specialization is developed in [13, 33, 1]. The present rank-two case is the smallest nontrivial instance, and the whole of [Section 3.3](#) is the statement that its support condition can be checked coordinatewise.

Lemma 3.18 (Rank two). *The convex subgroups of the ordered abelian group Γ are precisely $0 \subset \{0\} \times \mathbb{Z} \subset \Gamma$. Hence Γ has rank two, and v is a valuation of rank two.*

Proof. $\{0\} \times \mathbb{Z}$ is convex: if $(0, b) < (a, c) < (0, b')$ then $a = 0$ by the lexicographic rule. Let H be a convex subgroup containing some (a, b) with $a \neq 0$; replacing (a, b) by its negative we may assume $a > 0$. For any $(c, d) \in \Gamma$ choose N with $Na > |c|$; then $-N(a, b) < (c, d) < N(a, b)$ and both bounds lie in H , so $(c, d) \in H$ by convexity. Thus $H = \Gamma$. A convex subgroup contained in $\{0\} \times \mathbb{Z}$ and containing some $(0, b) \neq 0$ is all of $\{0\} \times \mathbb{Z}$ by the same argument applied inside that subgroup. $\square$

Theorem 3.19 (The iterated field is the Hahn field of Γ). *Let $\mathbb{K}[[t^{\mathbb{Z}} L^{\mathbb{Z}}]]$ denote the Hahn field of formal sums $\sum_{(n,j)} a_{n,j} t^n L^j$ with coefficients in $\mathbb{K}$ whose support is reverse well ordered for $\succ$, with the convolution product. Then*

$$\mathbb{K}[[t^{\mathbb{Z}} L^{\mathbb{Z}}]] = \mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$$

as $\mathbb{K}$ -algebras: a family $(a_{n,j})_{(n,j) \in \Gamma}$ has reverse well-ordered support if and only if its n -projection is bounded below and every fiber S_n is bounded above in j ; such families are exactly the elements of $\mathbb{K}((L^{-1}))((t))$, and the two products agree.

Proof. Support condition. Suppose the support S is reverse well ordered. If its n -projection were unbounded below, S would have no $\succ$ -greatest element, as in the proof of [Theorem 3.15](#). If some fiber S_n were unbounded above in j , then $\{n\} \times S_n$ would have no greatest element. Conversely, assume the two conditions and let $\emptyset \neq T \subseteq S$. The n -projection of T is a nonempty subset of $\mathbb{Z}$ bounded below, so has a least element n_1 ; the fiber T_{n_1} is a nonempty subset of $\mathbb{Z}$ bounded above, so has a greatest element j_1 ; and (n_1, j_1) is $\succ$ -greatest in T .

Identification of the underlying sets. A family whose n -projection is bounded below by n_0 and whose fibers are bounded above is precisely a family $(p_n)_{n \geq n_0}$ with $p_n = \sum_j a_{n,j} L^j \in \mathbb{K}((L^{-1}))$, that is, an element of $\mathbb{K}((L^{-1}))((t))$; and conversely.

Products. In $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ the coefficient of t^n in a product is the finite sum $\sum_{i+k=n} p_i q_k$ (finite because both outer supports are bounded below), and the coefficient of L^j in $p_i q_k$ is the finite sum

$\sum_{j_1+j_2=j} a_{i,j_1} b_{k,j_2}$ (finite because both inner supports are bounded above). This is exactly the Hahn convolution $\sum_{(n_1,j_1)+(n_2,j_2)=(n,j)} a_{n_1,j_1} b_{n_2,j_2}$, whose local finiteness has just been verified. $\square$

Corollary 3.20 (Where the smaller classes sit). *Under the identification of Theorem 3.19: $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is the subring of elements with finite fibers contained in $\mathbb{N}_0$; $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ is the further subring with n -projection contained in $\mathbb{N}_0$; and $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ is the set of elements of $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ every block of which lies in the subfield $\mathbb{K}(L) \subset \mathbb{K}((L^{-1}))$.*

Proof. The first two statements are Theorem 3.15 (b) together with Theorem 3.3. The third is the definition of $\mathbb{K}(L)((t))$ transported along the blockwise embedding of Theorem 3.5. $\square$

Remark 3.21. The third description cannot be improved to “the subring generated by $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ and the inverses of the nonzero elements of $\mathbb{K}[L]$ ”: every element of that subring is a finite sum of finite products and therefore has a *common* denominator across all blocks, whereas Φ_0 of (31) has none, by Theorem 3.8.

3.5 Why the iteration order is part of the type

Both $\mathbb{K}((L^{-1}))((t))$ and $\mathbb{K}((t))((L^{-1}))$ are fields of formal sums of the monomials $t^n L^j$, and both are naturally sets of families $(a_{n,j}) \in \mathbb{K}^\Gamma$. They impose *different* support conditions, and neither is contained in the other.

Proposition 3.22 (Incomparability). *Regard both fields as sets of families $(a_{n,j})_{(n,j) \in \Gamma}$. Then*

$$\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}} = \{(a_{n,j}) : n \text{ bounded below; for each } n, j \text{ bounded above}\}, \quad (35)$$

$$\mathbb{K}((t))((L^{-1})) = \{(a_{n,j}) : j \text{ bounded above; for each } j, n \text{ bounded below}\}. \quad (36)$$

Neither set contains the other. Explicit witnesses:

$$A = \sum_{n \geq 0} t^n L^n \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} \subset \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}, \quad B = \sum_{r \geq 0} t^{-r} L^{-r} \in \mathbb{K}((t))((L^{-1})), \quad (37)$$

with $A \notin \mathbb{K}((t))((L^{-1}))$ and $B \notin \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$.

Proof. The two displayed descriptions are the definitions of an outer Laurent series whose coefficients are inner Laurent series, read in the two possible orders. For A : its support is $\{(n, n) : n \geq 0\}$, whose j -coordinate is unbounded above, so A violates the first condition in (36); but its n -coordinate is bounded below and each fiber is a single point, so $A \in \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ — indeed $A \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, since its block at t^n is the polynomial L^n . For B : its support is $\{(-r, -r) : r \geq 0\}$, whose n -coordinate is unbounded below, so $B \notin \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$; but its j -coordinate is bounded above by 0 and each of its fibers over a fixed $j = -r$ is the single point $n = -r$, so B satisfies (35). $\square$

Remark 3.23 (The common part, and why we choose t outermost). Intersecting (35) and (36) gives the families supported in a quadrant $\{n \geq n_0\} \times \{j \leq j_0\}$. These form a subring of both fields, but not a field: $1 + tL$ lies in it and its inverse $\sum_{r \geq 0} (-1)^r t^r L^r$ does not. We take t outermost because the intended hierarchy is $t \prec L^{-m} \prec 1 \prec L^m \prec t^{-1}$ for every fixed $m > 0$; this is the formal counterpart of $X^{-1} = o_{X \rightarrow +\infty}((\log X)^{-m})$, and it is exactly the statement that the convex subgroup $\{0\} \times \mathbb{Z}$ of Theorem 3.18 is the *smaller* one. The same phenomenon — that an iterated series field remembers the order in which its scales were adjoined — is standard in the general theory [14, 33]; what Theorem 3.22 adds is that here the two orders are *incomparable*, not merely distinct.

3.6 The Puiseux-log extension

Definition 3.24 (Puiseux-log ring). For $s \in \mathbb{N}_{>0}$ let

$$\mathbb{K}[L]((t^{1/s})) = \left\{ \sum_{r \geq r_0} p_r(L) t^{r/s} : r_0 \in \mathbb{Z}, p_r \in \mathbb{K}[L] \right\}, \quad (38)$$

the polynomial-logarithmic Laurent ring in the generator $t^{1/s}$. Its monomials are $t^\alpha L^j$ with $\alpha \in s^{-1}\mathbb{Z}$ and $j \in \mathbb{N}_0$.

Proposition 3.25 (Ramified tower). For $s \mid s'$ the substitution $t^{1/s} \mapsto (t^{1/s'})^{s'/s}$ is an injective ring homomorphism $\mathbb{K}[L]((t^{1/s})) \hookrightarrow \mathbb{K}[L]((t^{1/s'}))$, and these maps are compatible, so

$$\mathbb{K}[L]((t^{1/\infty})) := \bigcup_{s \in \mathbb{N}_{>0}} \mathbb{K}[L]((t^{1/s}))$$

is a ring, containing $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} = \mathbb{K}[L]((t^{1/1}))$. Every element of the union has all its exponents in a single $s^{-1}\mathbb{Z}$; consequently the union is strictly smaller than the Hahn ring of well-ordered supports in $\mathbb{Q} \times \mathbb{N}_0$.

Proof. The substitution is injective because it is injective on exponents and the identity on coefficients, and it is a ring homomorphism because it is induced by a monoid homomorphism of monomials; compatibility is immediate. A directed union of rings along injective homomorphisms is a ring. For strictness, consider $\sum_{k \geq 1} t^{1-1/k}$, whose support $\{1 - 1/k : k \geq 1\} \subset \mathbb{Q}$ is well ordered in the direction of increasing smallness and has no bounded denominator; it therefore lies in the Hahn ring but in no $\mathbb{K}[L]((t^{1/s}))$. $\square$

Remark 3.26 (What ramification costs). Formally, nothing in [Section 3.3](#) changes: the support is a subset of $s^{-1}\mathbb{Z} \times \mathbb{N}_0$ with finite fibers and lower-bounded projection, and [Theorem 3.15](#) applies verbatim after rescaling. Analytically a branch of $X^{1/s}$ must be chosen, and that choice is data, not a convention. Fractional exponents arise in this volume only where they are forced – at a degenerate leading derivative, and at the square-root branch point of W . The corresponding classes near a finite point, where one writes $\sum_{\lambda \in S} z^\lambda p_\lambda(\log z)$ as $z \rightarrow 0^+$, are the power–log and Dulac series of local dynamics [\[23, 24, 27, 26\]](#); the change of variable relating them to the present classes is carried out in Part V.

4 Valuation, leading data, and the degree profile

Throughout this section R denotes any one of the four classes of [Theorem 3.3](#), with coefficient ring $E \in \{\mathbb{K}[L], \mathbb{K}(L), \mathbb{K}((L^{-1}))\}$, and $F = \sum_{n \geq n_0} p_n t^n$ denotes the block representation of an element of R .

4.1 The outer valuation

Definition 4.1 (Outer valuation). For nonzero $F \in R$ set

$$\nu_t(F) := \min \{n \in \mathbb{Z} : p_n \neq 0\}, \quad \nu_t(0) := +\infty. \quad (39)$$

The symbol $+\infty$ obeys $+\infty + m = +\infty$ and $+\infty > m$ for every $m \in \mathbb{Z}$.

The minimum exists because the set of occurring n is a nonempty subset of $\mathbb{Z}$ bounded below. The convention $\nu_t(0) = +\infty$ is not cosmetic: it is what makes the laws below hold without case distinctions, and it is what makes $\{F : \nu_t(F) \geq N\}$ an additive subgroup.

Theorem 4.2 (Valuation laws). Let $F, G \in R$. Then

- (i) $\nu_t(FG) = \nu_t(F) + \nu_t(G)$;
- (ii) $\nu_t(F + G) \geq \min \{\nu_t(F), \nu_t(G)\}$;
- (iii) $\nu_t(F) = +\infty$ if and only if $F = 0$, and $\nu_t(-F) = \nu_t(F)$.

Moreover the inequality in (ii) is strict if and only if

$$\nu_t(F) = \nu_t(G) < +\infty \quad \text{and} \quad p_{\nu_t(F)} + q_{\nu_t(G)} = 0, \quad (40)$$

where q_n denotes the blocks of G . In particular ν_t satisfies the axioms of a valuation with values in $\mathbb{Z} \cup \{+\infty\}$ on each of the four classes, and $\{F \in R : \nu_t(F) \geq 0\} = E[[t]]$. When E is a field — that is, for $R = \mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ and $R = \mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ — R is a field (Theorem 3.5 (ii)), ν_t is surjective onto $\mathbb{Z}$, and ν_t is a discrete valuation of R in the usual sense, with valuation ring $E[[t]]$. For $R = \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $R = \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ the same laws hold, but R is not a field and the phrase “valuation ring” is not available.

Proof. (iii) is immediate from (39).

(i) If $F = 0$ or $G = 0$ both sides are $+\infty$. Otherwise put $m = \nu_t(F)$, $r = \nu_t(G)$. For $n < m + r$ the block of FG at t^n is $\sum_{i+k=n} p_i q_k$; in each summand either $i < m$ or $k < r$ (else $i + k \geq m + r > n$), so every summand vanishes. At $n = m + r$ the only surviving pair is (m, r) , giving the block $p_m q_r$, which is nonzero because E is an integral domain. Hence $\nu_t(FG) = m + r$.

(ii) and the strictness criterion. Write $\mu = \min \{\nu_t F, \nu_t G\}$. For $n < \mu$ both p_n and q_n vanish, so the block of $F + G$ at t^n vanishes; this is (ii). For the criterion, suppose first $\nu_t F \neq \nu_t G$, say $\nu_t F < \nu_t G$. Then the block of $F + G$ at $t^{\nu_t F}$ is $p_{\nu_t F} + q_{\nu_t F} = p_{\nu_t F} \neq 0$, so equality holds. If $\nu_t F = \nu_t G = \mu < +\infty$, the block of $F + G$ at t^μ is $p_\mu + q_\mu$, and the inequality is strict exactly when this vanishes. If $\mu = +\infty$ then $F = G = 0$ and both sides are $+\infty$.

Finally, (i)–(iii) are the axioms of a valuation with values in $\mathbb{Z} \cup \{+\infty\}$; the set $\{\nu_t \geq 0\}$ is, by (39), the set of series with no negative blocks, namely $E[[t]]$. That $E((t))$ is a field when E is one — so that ν_t is then a discrete valuation of a field with valuation ring $E[[t]]$ — is Theorem 3.5 (ii), whose proof uses only Theorem 5.23 and not the present theorem. The argument used about E only that it is an integral domain, so the same laws hold in $E[[t]]$ and $E((t))$ for every integral domain E ; this is the form in which they are applied in Theorem 3.5. $\square$

Example 4.3 (Strictness needs equal valuations and full cancellation). With $F = (L^2 + L)t^3 + t^4$ and $G = (-L^2 - L)t^3 + t^5$ we have $\nu_t F = \nu_t G = 3$ but $\nu_t(F + G) = 4$. Cancelling only part of a block is not enough: $F' = (L^2 + L)t^3$ and $G' = (-L^2 + 2)t^3$ give $F' + G' = (L + 2)t^3$, still of valuation 3. And with $F'' = t^3$, $G'' = -t^4$ the valuations differ, so $\nu_t(F'' + G'') = 3$ whatever the blocks are.

4.2 The rank-two valuation and the three leading objects

Definition 4.4 (Leading data). Let $F \in R$ be nonzero and $\nu = \nu_t(F)$. Write the block p_ν in its L^{-1} -expansion inside $\mathbb{K}((L^{-1}))$ — for $E = \mathbb{K}[L]$ this is the polynomial itself, and for $E = \mathbb{K}(L)$ it is the image under the embedding of Theorem 3.5 — so that $p_\nu = aL^d + (\text{strictly lower powers of } L)$ with $a \in \mathbb{K}^\times$ and $d = \deg_L p_\nu$ as in Theorem 4.6. Then:

- the *leading coefficient block* of F is the whole coefficient p_ν ;
- the *leading monomial* is $\text{lm}(F) = t^\nu L^d$;
- the *scalar leading coefficient* is $\text{lc}(F) = a$.

For $F = 0$ none of the three is defined.

Remark 4.5 (Leading block, leading monomial, leading scalar). p_ν is a polynomial (or rational function, or Laurent series in L^{-1}); $\text{lm}(F)$ is a monomial; $\text{lc}(F)$ is a scalar. The distinction is not pedantic. For $F = t(L + 1)$ we have $\text{lc}(F) = 1 \in \mathbb{K}^\times$ and $\text{lm}(F) = tL$, yet $t^{-1}F = L + 1$ is not invertible in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$: what governs invertibility is the leading *block*, not the leading scalar. Every “the leading coefficient is 1, so we may divide” argument must name which of the three objects it means.

Definition 4.6 (Rank-two valuation). For nonzero $F \in R$ set

$$v(F) := (\nu_t(F), -\deg_L p_{\nu_t(F)}) \in \Gamma = \mathbb{Z} \times \mathbb{Z}, \quad (41)$$

lexicographically ordered, and $v(0) := \infty_\Gamma$, a symbol larger than every element of Γ and absorbing under addition. Here $\deg_L$ of a nonzero block means the largest j with a nonzero coefficient of L^j ; for $E = \mathbb{K}((L^{-1}))$ this exists because inner supports are bounded above (Theorem 3.19), and for $E = \mathbb{K}(L)$ it is $\deg$ numerator minus $\deg$ denominator, computed in $\mathbb{K}((L^{-1}))$ under the embedding of Theorem 3.5. Thus $v(t^n L^j) = (n, -j)$, matching (34), and ν_t is the coarsening of v obtained by projecting to the first coordinate.

Theorem 4.7 (Rank-two valuation laws and leading-term arithmetic). For $F, G \in R$,

- (i) $v(FG) = v(F) + v(G) -$ equivalently, $\text{lm}(FG) = \text{lm}(F) \text{lm}(G)$ for nonzero F, G — and, in addition,

$$\text{lc}(FG) = \text{lc}(F) \text{lc}(G) \quad (F, G \neq 0);$$

- (ii) $v(F + G) \geq \min\{v(F), v(G)\}$, with equality unless $v(F) = v(G) < \infty_\Gamma$ and $\text{lc}(F) + \text{lc}(G) = 0$;

- (iii) if $\text{lm}(F) \neq \text{lm}(G)$, then $\text{lm}(F + G)$ is the $\succ$ -greater of $\text{lm}(F)$ and $\text{lm}(G)$, and $\text{lc}(F + G)$ is the corresponding leading scalar.

Consequently v is a valuation on R with values in the lexicographically ordered group Γ . Its image is all of Γ exactly when E is $\mathbb{K}(L)$ or $\mathbb{K}((L^{-1}))$; on $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ the image is $\{(n, -d) : n \in \mathbb{Z}, d \in \mathbb{N}_0\}$ and on $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ it is $\{(n, -d) : n, d \in \mathbb{N}_0\}$. In every case the image generates Γ , so the associated value group is Γ , which has rank two by Theorem 3.18.

Proof. (i) The cases involving 0 are handled by the absorbing convention. Let $F, G \neq 0$, $m = \nu_t F$, $r = \nu_t G$. By Theorem 4.2 (i) the leading block of FG is $p_m q_r$. In E — a polynomial ring, a rational function field, or a field of Laurent series in L^{-1} — the top L -degree is additive and the top coefficients multiply, because $\mathbb{K}$ is a domain: $\deg_L(p_m q_r) = \deg_L p_m + \deg_L q_r$ and the coefficient of the top power is the product of the two top coefficients. Hence $v(FG) = (m+r, -\deg_L p_m - \deg_L q_r) = v(F) + v(G)$, which is the displayed statement about lm and lc .

(ii)–(iii) If $\nu_t F < \nu_t G$, then $v(F) < v(G)$ lexicographically and the block of $F + G$ at $t^{\nu_t F}$ equals $p_{\nu_t F}$, so $v(F + G) = v(F) = \min$. If $\nu_t F = \nu_t G = \mu$ but $\deg_L p_\mu > \deg_L q_\mu$, then again $v(F) < v(G)$, and the top L -term of $p_\mu + q_\mu$ is that of p_μ , so $v(F + G) = v(F)$. These two cases cover $\text{lm}(F) \neq \text{lm}(G)$ and give (iii). In the remaining case $\text{lm}(F) = \text{lm}(G) = t^\mu L^d$, the coefficient of L^d in $p_\mu + q_\mu$ is $\text{lc}(F) + \text{lc}(G)$; if this is nonzero then $v(F + G) = v(F) = v(G)$, and if it vanishes then $v(F + G) > v(F)$.

The image of v . $v(t^n L^d) = (n, -d)$, and $t^n L^d$ lies in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ for $n, d \geq 0$, in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ for $n \in \mathbb{Z}, d \geq 0$, and in $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}} \subset \mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ for all $n, d \in \mathbb{Z}$; conversely $\deg_L$ of a block of an element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is a nonnegative integer and $\nu_t \geq 0$ on $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, which gives the reverse inclusions. Since $(1, 0)$ and $(0, -1)$ lie in every one of these images, each generates Γ . Rank two is then Theorem 3.18. $\square$

Remark 4.8 (What ν_t forgets). ν_t is deliberately coarser than v : it ignores the internal L -ordering of a block. That is precisely what makes it usable as a computational filtration — the set $\{\nu_t \geq N\}$ is a *finitely describable* tail, whereas $\{v \geq (N, -d)\}$ cuts a block in half and is not stable under the operations of Part II. All truncation, completeness and summability statements in this volume are for ν_t ; the finer v is used only to compare individual monomials.

4.3 Uniqueness of the block representation

The block form (32) is unique by construction if one *defines* an element to be its family of coefficients. What actually needs proof is that an element presented as a formally summable sum of monomials —

which is how every later formula presents its output – has uniquely determined blocks, independently of the order of summation. The proof uses the summability calculus of [Section 5.7](#); we state it here because it belongs to the leading-data discussion.

Theorem 4.9 (Uniqueness of the block representation). *Let $(a_{n,j})$ be a family of scalars indexed by $\mathbb{Z} \times \mathbb{N}_0$ whose index set satisfies [Theorem 3.15 \(b\)](#). Then the family of monomial terms $(a_{n,j}t^n L^j)$ is formally summable in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, its sum F is independent of the enumeration, and*

$$[t^n]F = \sum_j a_{n,j} L^j, \quad [t^n L^j]F = a_{n,j} \quad \text{for all } (n, j).$$

In particular, if $\sum a_{n,j}t^n L^j = \sum b_{n,j}t^n L^j$ with both index sets admissible, then $a_{n,j} = b_{n,j}$ for all (n, j) : the monomials $t^n L^j$ are linearly independent over $\mathbb{K}$ in the strong, summable sense.

Proof. Summability: for $N \in \mathbb{Z}$, a term $a_{n,j}t^n L^j$ has $\nu_t = n$, and the terms with $n < N$ are indexed by $\bigcup_{n_0 \leq n < N} \{n\} \times S_n$, a finite union of finite sets by [Theorem 3.15 \(b\)](#). Hence only finitely many terms have $\nu_t < N$, which is [Theorem 5.21](#). By [Theorem 5.22 \(ii\)–\(iii\)](#) the sum exists, is independent of the enumeration, and satisfies $[t^n]F = \sum_{(n',j)} [t^n](a_{n',j}t^{n'} L^j) = \sum_j a_{n,j} L^j$, a finite sum. Extracting the coefficient of L^j from this polynomial gives $[t^n L^j]F = a_{n,j}$. The last assertion follows because F determines all its coefficients. $\square$

Proposition 4.10 (Leading-block factorization). *Let E be a field and $F \in E((t))$ nonzero. Then there are unique $\nu \in \mathbb{Z}$, $c \in E^\times$ and $U \in tE[[t]]$ with*

$$F = ct^\nu (1 + U), \tag{42}$$

namely $\nu = \nu_t(F)$, $c = p_\nu$, and $U = \sum_{n \geq 1} (p_{\nu+n}/c)t^n$. The factorization is available in $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ and in $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$; over $E = \mathbb{K}[L]$ it is available with $c = p_\nu$ and $U \in t\mathbb{K}(L)[[t]]$, but in general not with $U \in t\mathbb{K}[L][[t]]$.

Proof. Existence over a field is the displayed formula, which makes sense because $c = p_\nu \in E^\times$. Uniqueness: applying ν_t to (42) and using [Theorem 4.2 \(i\)](#) with $\nu_t(1 + U) = 0$ gives $\nu = \nu_t(F)$; comparing blocks at t^ν gives $c = p_\nu$; and then $U = c^{-1}t^{-\nu}F - 1$ is determined. For the last assertion take $F = L + t \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. A factorization $F = L(1 + U)$ with $U = \sum_{n \geq 1} u_n(L)t^n \in t\mathbb{K}[L][[t]]$ would force, at t^1 , $1 = Lu_1(L)$ in $\mathbb{K}[L]$, which is impossible; and c must equal L by the uniqueness just proved. $\square$

4.4 The log-degree profile

Definition 4.11 (Log-degree profile). For $F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ define $d_F : \mathbb{Z} \rightarrow \mathbb{N}_0 \cup \{-\infty\}$ by

$$d_F(n) = \begin{cases} \deg_L p_n, & p_n \neq 0, \\ -\infty, & p_n = 0, \end{cases} \tag{43}$$

with the conventions $-\infty + m = -\infty$ and $\max(-\infty, m) = m$. Absent blocks are extended by zero, so $d_F(n) = -\infty$ for $n < \nu_t(F)$.

The profile is genuinely extra data: $\nu_t(F)$ is the first n at which $d_F(n) > -\infty$, and nothing more about d_F can be recovered from it.

Proposition 4.12 (Profile under the ring operations). *For $F, G \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ and every $n \in \mathbb{Z}$,*

$$d_{F+G}(n) \leq \max\{d_F(n), d_G(n)\}, \tag{44}$$

$$d_{FG}(n) \leq \max_{i+k=n} (d_F(i) + d_G(k)), \tag{45}$$

where in (45) the maximum is over the finitely many pairs with $i \geq \nu_t F$ and $k \geq \nu_t G$, and is $-\infty$ if there are none. Equality holds in (44) unless the top L -terms cancel, and in (45) unless the sum of the products of top coefficients over the maximizing pairs vanishes.

Proof. For a sum of polynomials the degree is at most the maximum of the degrees, with equality unless the coefficients of that maximal power cancel; apply this to $p_n + q_n$ for (44) and to $\sum_{i+k=n} p_i q_k$ for (45), using $\deg_L(p_i q_k) = \deg_L p_i + \deg_L q_k$ (valid with the $-\infty$ convention). The index set is finite because $i \geq \nu_t F$ and $k = n - i \geq \nu_t G$ force $\nu_t F \leq i \leq n - \nu_t G$. $\square$

Definition 4.13 (Series of bounded slope). For $\sigma \geq 0$ let

$$\mathcal{D}_\sigma := \{F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} : \exists C \in \mathbb{R} \forall n, d_F(n) \leq \sigma n + C\}.$$

Proposition 4.14 (Slope transport). Let $F, G \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ be nonzero with $m = \nu_t F$, $r = \nu_t G$, and suppose

$$d_F(n) \leq A + \alpha n \quad (n \geq m), \quad d_G(n) \leq B + \beta n \quad (n \geq r),$$

with $\alpha \geq \beta \geq 0$. Then

$$d_{FG}(n) \leq (A + B - (\alpha - \beta)r) + \alpha n \quad (n \geq m + r). \quad (46)$$

In particular each $\mathcal{D}_\sigma$ is a subring of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ containing $\mathbb{K}[L]$ and $t^{\pm 1}$, and $\mathcal{D}_\sigma \subseteq \mathcal{D}_{\sigma'}$ for $\sigma \leq \sigma'$. If moreover $r \geq 0$ – in particular if $F, G \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ – then (46) implies the simpler bound $d_{FG}(n) \leq A + B + \alpha n$.

Proof. By (45), for $i + k = n$ with $i \geq m$, $k \geq r$,

$$d_F(i) + d_G(k) \leq A + B + \alpha i + \beta k = A + B + \beta n + (\alpha - \beta)i \leq A + B + \beta n + (\alpha - \beta)(n - r),$$

using $i = n - k \leq n - r$ and $\alpha - \beta \geq 0$; this is (46). For the subring claim, closure under subtraction follows from (44), and closure under multiplication from (46), applied with $\alpha = \beta = \sigma$ (so that the correction term $-(\alpha - \beta)r$ vanishes); the constants A, B are fixed for given F, G . For the monotonicity, let $\sigma \leq \sigma'$ and $F \in \mathcal{D}_\sigma$, say $d_F(n) \leq \sigma n + C$ for all n , and put $m = \nu_t F$. For $n < m$ the left-hand side is $-\infty$ and nothing is to be proved; for $n \geq m$ one has $\sigma n + C = \sigma' n + C + (\sigma - \sigma')n \leq \sigma' n + C + (\sigma - \sigma')m$, because $\sigma - \sigma' \leq 0$. Hence $F \in \mathcal{D}_{\sigma'}$ with the constant $C + (\sigma - \sigma')m$. $\square$

Example 4.15 (The naive slope bound fails on the Laurent ring). Take $F = \sum_{n \geq 0} L^n t^n$ with $A = 0$, $\alpha = 1$, $m = 0$, and $G = t^{-2}$ with $B = 0$, $\beta = 0$, $r = -2$. Then $FG = \sum_{n \geq -2} L^{n+2} t^n$, so $d_{FG}(n) = n + 2$. The naive bound $A + B + \max(\alpha, \beta)n = n$ is violated for every n , while (46) gives $0 + 0 - (1 - 0)(-2) + n = n + 2$, attained.

Remark 4.16 (Degree bounds are hypotheses, never definitions). No class in Theorem 3.3 imposes any bound on d_F . A statement of the form $d_F(n) \leq \sigma n + C$ is either an explicit hypothesis, or an invariant proved for the particular construction at hand – as it is for the Lambert polynomials, where $\deg \mathbb{L}_n^{\text{Lam}} = n$ for $n \geq 1$. It is never available merely because the object lies in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. Two consequences are worth isolating. First, membership in $\mathcal{D}_\sigma$ must be re-proved after every operation that is not covered by Theorem 4.14; the subring property gives sums and products only. Second, a formal tail $\mathcal{O}_t^{\text{form}}(t^N)$ becomes an analytic estimate only after a degree bound on the discarded blocks is supplied; see Theorem 5.9.

Example 4.17 (Leading data and profile of a typical element). Let

$$F = X^2 + 3XL + 7 + (L^3 - 2L)t + \sum_{n \geq 2} (L^n + 1)t^n = t^{-2} + 3Lt^{-1} + 7 + (L^3 - 2L)t + \sum_{n \geq 2} (L^n + 1)t^n.$$

Then $\nu_t(F) = -2$; the leading block is the constant polynomial 1; $\text{lm}(F) = t^{-2} = X^2$; $\text{lc}(F) = 1$; and $v(F) = (-2, 0)$. The profile is

$$d_F(-2) = 0, \quad d_F(-1) = 1, \quad d_F(0) = 0, \quad d_F(1) = 3, \quad d_F(n) = n \quad (n \geq 2),$$

and $-\infty$ for $n < -2$. It is not monotone, and the smallest affine bound of slope 1 is $d_F(n) \leq n + 2$; thus $F \in \mathcal{D}_1$. Here the leading block happens to be a nonzero scalar, so F is invertible in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ by the unit criterion of Part II. Delete the term X^2 : the leading block of $F - t^{-2}$ becomes $3L$, whose scalar leading coefficient is still $3 \in \mathbb{K}^\times$ – and yet $3L$ is not a unit of $\mathbb{K}[L]$, so the reciprocal of $F - t^{-2}$ leaves $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. This is Theorem 4.5 in a single example.

5 Truncation, formal tails, and summability

5.1 The exclusive truncation operator

Definition 5.1 (Exclusive outer truncation). For $N \in \mathbb{Z}$ and $F = \sum_{n \geq n_0} p_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ put

$$\text{Trunc}_{<N} F := \sum_{n < N} p_n(L)t^n. \quad (47)$$

The sum is finite because $n_0 \leq n < N$; it is empty, hence 0, when $N \leq n_0$, and $\text{Trunc}_{<N} 0 = 0$. Every retained block is retained *whole*: all powers of L at a retained level survive. The phrase “through outer order t^{N-1} ” means the exclusive cutoff $< N$.

Remark 5.2 (Reconciliation with the inclusive convention). Source S6 writes $[F]_{\leq N} = \sum_{n \leq N} p_n(L)t^n$, which is *inclusive*. The translation is

$$[F]_{\leq N} = \text{Trunc}_{<N+1} F, \quad \text{Trunc}_{<N} F = [F]_{\leq N-1}. \quad (48)$$

Any statement imported from S6 — in particular its residual and stopping criteria — must absorb this one-block shift. We use (47) exclusively.

Proposition 5.3 (Elementary properties). For $F, G \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, $c \in \mathbb{K}$ and $M, N \in \mathbb{Z}$:

- (i) $\text{Trunc}_{<N} \cdot$ is $\mathbb{K}$ -linear and idempotent, and $\text{Trunc}_{<N}(\text{Trunc}_{<M} F) = \text{Trunc}_{<\min(M, N)} F$;
- (ii) $\nu_t(F - \text{Trunc}_{<N} F) \geq N$, and $\nu_t(\text{Trunc}_{<N} F) \geq \nu_t(F)$;
- (iii) $\text{Trunc}_{<N} F = \text{Trunc}_{<N} G$ if and only if $\nu_t(F - G) \geq N$;
- (iv) $F = G$ if and only if $\text{Trunc}_{<N} F = \text{Trunc}_{<N} G$ for every $N \in \mathbb{Z}$;
- (v) $\text{Trunc}_{<N}(cF + G) = c \text{Trunc}_{<N} F + \text{Trunc}_{<N} G$.

Proof. All five are read off blockwise from (47). For (iii), $F - G$ has all blocks below N equal to $p_n - q_n$, which vanish for all $n < N$ exactly when the two truncations agree; and $\nu_t(F - G) \geq N$ says precisely that those blocks vanish. For (iv), if the truncations agree for all N then $\nu_t(F - G) \geq N$ for all N , so $F - G = 0$ by Theorem 4.2 (iii). $\square$

Proposition 5.4 (Truncated product). Let $F, G \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ be nonzero, $m = \nu_t F$, $r = \nu_t G$, and $N \in \mathbb{Z}$. Then

$$\text{Trunc}_{<N}(FG) = \text{Trunc}_{<N}(\text{Trunc}_{<N-r} F \cdot \text{Trunc}_{<N-m} G). \quad (49)$$

If in addition $m \geq 0$ and $r \geq 0$ — in particular if $F, G \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ — then also

$$\text{Trunc}_{<N}(FG) = \text{Trunc}_{<N}(\text{Trunc}_{<N} F \cdot \text{Trunc}_{<N} G). \quad (50)$$

Proof. Write $F = \text{Trunc}_{<N-r} F + F_1$ and $G = \text{Trunc}_{<N-m} G + G_1$, so that $\nu_t F_1 \geq N - r$ and $\nu_t G_1 \geq N - m$ by Theorem 5.3 (ii). Then

$$FG - \text{Trunc}_{<N-r} F \text{Trunc}_{<N-m} G = F_1 G + \text{Trunc}_{<N-r} F G_1.$$

By Theorem 4.2, $\nu_t(F_1 G) \geq (N - r) + r = N$ and $\nu_t(\text{Trunc}_{<N-r} F G_1) \geq m + (N - m) = N$, using $\nu_t(\text{Trunc}_{<N-r} F) \geq m$. Hence the difference has $\nu_t \geq N$, and (49) follows from Theorem 5.3 (iii). When $m, r \geq 0$ the same computation with N in place of $N - r$ and $N - m$ gives $\nu_t(F_1 G) \geq N + r \geq N$ and $\nu_t(\text{Trunc}_{<N} F G_1) \geq m + N \geq N$, which is (50). $\square$

Example 5.5 (The symmetric law fails on the Laurent ring). Let $F = t^{-1} + 1$, $G = t^{-1}$ and $N = 0$. Then $FG = t^{-2} + t^{-1}$, so $\text{Trunc}_{<0}(FG) = t^{-2} + t^{-1}$. On the other hand $\text{Trunc}_{<0} F = t^{-1}$ and $\text{Trunc}_{<0} G = t^{-1}$, whose product truncates to t^{-2} . The block t^{-1} has been lost, because the term 1 of F , discarded at cutoff 0, meets the pole t^{-1} of G . Formula (49) keeps it: $m = -1$, $r = -1$, so one must retain F below $N - r = 1$ and G below $N - m = 1$, giving $(t^{-1} + 1)t^{-1} = t^{-2} + t^{-1}$.

Remark 5.6 (Precision accounting). [Theorem 5.4](#) is the prototype of every precision contract in this volume: an operation must declare how much input precision it consumes. Multiplication consumes $-\nu_t$ of the *other* factor; equivalently, each negative power of t in one factor costs one block of precision in the other. Composition with an inner argument of negative outer order amplifies the requested precision by a factor, and is analysed in Part III.

5.2 The formal tail token

Definition 5.7 (Formal big-O). For $F, G \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and $N \in \mathbb{Z}$, the statement

$$F = G + \mathcal{O}_t^{\text{form}}(t^N) \quad (51)$$

means any of the following four equivalent things:

$$\nu_t(F - G) \geq N; \quad F - G \in t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}; \quad \text{Trunc}_{<N} F = \text{Trunc}_{<N} G; \quad [t^n](F - G) = 0 \text{ for all } n < N.$$

The token $\mathcal{O}^{\text{form}}$ is the same operator with its arguments left implicit.

Proposition 5.8 (Equivalence, and the calculus of the token). *The four conditions in [Theorem 5.7](#) are equivalent. Moreover, for $F = G + \mathcal{O}_t^{\text{form}}(t^N)$ and any $H, H' \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and $M \in \mathbb{Z}$ — where, in (ii) and (iii), a tail exponent that evaluates to $+\infty$ (which happens only through a vanishing factor) is to be read as the assertion that the difference in question is 0:*

- (i) $F + H = G + H + \mathcal{O}_t^{\text{form}}(t^N)$;
- (ii) $HF = HG + \mathcal{O}_t^{\text{form}}(t^{N+\nu_t H})$; in particular multiplication by H with $\nu_t H < 0$ loses $|\nu_t H|$ blocks of precision;
- (iii) if also $H = H' + \mathcal{O}_t^{\text{form}}(t^M)$, then $FH = GH' + \mathcal{O}_t^{\text{form}}(t^K)$ with $K = \min\{N + \nu_t H, M + \nu_t G\}$;
- (iv) $\mathcal{O}_t^{\text{form}}(t^N) + \mathcal{O}_t^{\text{form}}(t^M) = \mathcal{O}_t^{\text{form}}(t^{\min(N, M)})$ as a statement about the set $t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} + t^M \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} = t^{\min(N, M)} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.

Proof. Equivalence: $\nu_t(F - G) \geq N$ says all blocks of $F - G$ below N vanish, which is the fourth condition and, by [Theorem 5.3](#) (iii), the third; and a series with $\nu_t \geq N$ is t^N times a series with $\nu_t \geq 0$, that is, an element of $t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and conversely.

(i) is [Theorem 5.3](#) (v). (ii): $H(F - G)$ has $\nu_t = \nu_t H + \nu_t(F - G) \geq \nu_t H + N$ by [Theorem 4.2](#) (i). (iii): write $FH - GH' = H(F - G) + G(H - H')$ and apply (ii) twice together with [Theorem 4.2](#) (ii). (iv): the inclusion $t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} + t^M \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} \subseteq t^{\min(N, M)} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is clear, and the reverse holds because one of the two summands already equals $t^{\min(N, M)} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. $\square$

Remark 5.9 (What the token does not assert). $F = G + \mathcal{O}_t^{\text{form}}(t^N)$ asserts *coefficient agreement below outer order N* and nothing else. It does not assert a numerical bound, a rate, convergence, uniformity in L , or the existence of any function with the displayed expansion. Concretely, the element

$$T = \sum_{n \geq N} L^{n^2} t^n$$

satisfies $T = \mathcal{O}_t^{\text{form}}(t^N)$, yet at $X = e^{10}$, so that $L = 10$, its very first term is $e^{-10N} 10^{N^2}$, which for $N = 5$ exceeds 1.9×10^3 and grows without bound in N . The formal tail is small in the filtration, not in size. An estimate of size requires, in addition, a bound on d_T on the discarded range — see [Theorem 4.16](#) — and then a separate analytic theorem.

Remark 5.10 (Contrast with the blockwise analytic statement). The analytic counterpart, proved for actual functions in Part V, reads as follows. Let S be an unbounded subset of the positive ray (or, for complex work, a sector on which a branch of $\log$ has been fixed), let $f : S \rightarrow \mathbb{C}$, and let

$\sum_{n \geq n_0} p_n(L)t^n \in \mathcal{PL}_{\text{poly}, \mathbb{K}}$ be a candidate expansion. One says that f has this blockwise asymptotic expansion on S if for every $N > n_0$ there is an exponent $M_N \in \mathbb{N}_0$ with

$$f(X) - \sum_{n_0 \leq n < N} p_n(\log X)X^{-n} = \mathcal{O}_{X \rightarrow +\infty, X \in S}(X^{-N}(\log X)^{M_N}).$$

The exponent M_N is part of the statement; it is exactly the data that [Theorem 5.9](#) shows the formal token cannot supply.

5.3 Why the Laurent tail is not an ideal

Proposition 5.11 (The tail is a filtration, not an ideal). *Fix $N \in \mathbb{Z}$ and set $\mathfrak{F}_N = t^N \mathcal{PL}_{\text{pow}, \mathbb{K}} \subseteq \mathcal{PL}_{\text{poly}, \mathbb{K}}$.*

- (i) *t is a unit of $\mathcal{PL}_{\text{poly}, \mathbb{K}}$, and the ideal of $\mathcal{PL}_{\text{poly}, \mathbb{K}}$ generated by t^N is all of $\mathcal{PL}_{\text{poly}, \mathbb{K}}$: $(t^N) = \mathcal{PL}_{\text{poly}, \mathbb{K}}$.*
- (ii) *$\mathfrak{F}_N$ is an additive subgroup and a $\mathcal{PL}_{\text{pow}, \mathbb{K}}$ -submodule of $\mathcal{PL}_{\text{poly}, \mathbb{K}}$, but for no N is it an ideal of $\mathcal{PL}_{\text{poly}, \mathbb{K}}$; indeed $t^{-1} \cdot t^N = t^{N-1} \notin \mathfrak{F}_N$.*
- (iii) *$\mathfrak{F}_N = \{F \in \mathcal{PL}_{\text{poly}, \mathbb{K}} : \nu_t F \geq N\}$, the chain $(\mathfrak{F}_N)_{N \in \mathbb{Z}}$ is decreasing, exhaustive ($\bigcup_N \mathfrak{F}_N = \mathcal{PL}_{\text{poly}, \mathbb{K}}$) and separated ($\bigcap_N \mathfrak{F}_N = 0$).*
- (iv) *In $\mathcal{PL}_{\text{pow}, \mathbb{K}}$, by contrast, $t^N \mathcal{PL}_{\text{pow}, \mathbb{K}}$ for $N \geq 0$ is a genuine ideal and*

$$\mathcal{PL}_{\text{pow}, \mathbb{K}}/t^N \mathcal{PL}_{\text{pow}, \mathbb{K}} \cong \mathbb{K}[L][t]/(t^N)$$

as $\mathbb{K}[L]$ -algebras.

Proof. (i) $t \cdot t^{-1} = 1$ in $\mathcal{PL}_{\text{poly}, \mathbb{K}}$, so $t \in \mathcal{PL}_{\text{poly}, \mathbb{K}}^\times$ and hence $t^N \in \mathcal{PL}_{\text{poly}, \mathbb{K}}^\times$; an ideal containing a unit is the whole ring.

(ii) $\mathfrak{F}_N$ is closed under addition and under multiplication by elements of $\mathcal{PL}_{\text{pow}, \mathbb{K}}$ because $\mathcal{PL}_{\text{pow}, \mathbb{K}}$ is a ring and $t^N \mathcal{PL}_{\text{pow}, \mathbb{K}} \cdot \mathcal{PL}_{\text{pow}, \mathbb{K}} = t^N \mathcal{PL}_{\text{pow}, \mathbb{K}}$. It is not an ideal of $\mathcal{PL}_{\text{poly}, \mathbb{K}}$: $t^N \in \mathfrak{F}_N$, $t^{-1} \in \mathcal{PL}_{\text{poly}, \mathbb{K}}$, and $\nu_t(t^{N-1}) = N-1 < N$, so $t^{N-1} \notin \mathfrak{F}_N$ by (iii).

(iii) $F \in t^N \mathcal{PL}_{\text{pow}, \mathbb{K}}$ means $F = t^N G$ with $\nu_t G \geq 0$, i.e. $\nu_t F \geq N$ by [Theorem 4.2](#) (i); conversely $t^{-N} F \in \mathcal{PL}_{\text{pow}, \mathbb{K}}$ whenever $\nu_t F \geq N$. The chain is decreasing since $\nu_t \geq N+1$ implies $\nu_t \geq N$; exhaustive because every element has finite ν_t or is 0; separated because $\nu_t F \geq N$ for all N forces $\nu_t F = +\infty$, i.e. $F = 0$.

(iv) For $N \geq 0$, $t^N \mathcal{PL}_{\text{pow}, \mathbb{K}}$ is the image of $\mathcal{PL}_{\text{pow}, \mathbb{K}}$ under multiplication by t^N , hence an ideal of $\mathcal{PL}_{\text{pow}, \mathbb{K}}$. The inclusion $\mathbb{K}[L][t] \hookrightarrow \mathcal{PL}_{\text{pow}, \mathbb{K}}$ induces a $\mathbb{K}[L]$ -algebra homomorphism $\varphi : \mathbb{K}[L][t]/(t^N) \rightarrow \mathcal{PL}_{\text{pow}, \mathbb{K}}/t^N \mathcal{PL}_{\text{pow}, \mathbb{K}}$. It is surjective because $F \equiv \text{Trunc}_{<N} F \pmod{t^N \mathcal{PL}_{\text{pow}, \mathbb{K}}}$ and $\text{Trunc}_{<N} F \in \mathbb{K}[L][t]$ for $F \in \mathcal{PL}_{\text{pow}, \mathbb{K}}$. It is injective because a polynomial of t -degree $< N$ lying in $t^N \mathcal{PL}_{\text{pow}, \mathbb{K}}$ has all blocks below N zero, hence is 0. $\square$

The three legitimate replacements

On $\mathcal{PL}_{\text{poly}, \mathbb{K}}$, never write “modulo t^N ” or use quotient-ring language. Use instead, in decreasing order of explicitness:

1. the valuation statement $\nu_t(F - G) \geq N$;
2. the formal tail token $F = G + \mathcal{O}_t^{\text{form}}(t^N)$ of [Theorem 5.7](#);
3. equality of exclusive truncations $\text{Trunc}_{<N} F = \text{Trunc}_{<N} G$.

These three are equivalent by [Theorem 5.8](#). Quotient-algebra language becomes legitimate only after the finite Laurent principal part has been removed — that is, after multiplying by $t^{-\nu_t F}$ and working in $\mathcal{PL}_{\text{pow}, \mathbb{K}}$, where [Theorem 5.11](#) (iv) applies.

5.4 Coefficient extraction

Definition 5.12 (Extraction brackets). For $F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ written as in (32),

$$[t^n]F := p_n(L) \in \mathbb{K}[L], \quad [t^n L^j]F := a_{n,j} \in \mathbb{K}. \quad (52)$$

The monomial is written to the *left* of the bracket and the series to the right. An absent coefficient is 0.

Proposition 5.13 (Extraction is linear and filtration-continuous). *Each map $[t^n]$ and $[t^n L^j]$ is $\mathbb{K}$ -linear, and $[t^n]F = 0$ for every $F \in \mathfrak{F}_N$ with $N > n$. Consequently, if $F = G + \mathcal{O}_t^{\text{form}}(t^N)$ then $[t^n]F = [t^n]G$ for all $n < N$, and F is determined by the family $([t^n L^j]F)_{(n,j)}$.*

Proof. Linearity is blockwise addition. If $\nu_t F \geq N > n$ then $p_n = 0$. The remaining statements follow from [Theorem 5.7](#) and [Theorem 4.9](#). $\square$

Remark 5.14 (Bracket collisions). The glyph $[\cdot]$ carries four unrelated meanings in this corpus: coefficient extraction $[t^n]F$; the inclusive truncation $[F]_{\leq N}$ of S6, banished here by [Theorem 5.2](#); the unsigned Stirling cycle number $\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]$; and the Gaussian binomial coefficient. They are told apart by argument type, and we always print the semantic form.

5.5 The separate logarithmic cutoff

A log-degree cap is not a truncation in the dominance order

Bounding $\deg_L p_n$ is an additional, independent approximation. It is not part of the t -adic topology, and — this is the essential point — inside a fixed block the discarded high powers of L are the *dominant* terms, not the negligible ones: by [Theorem 3.12](#), $t^n L^j \succ t^n L^k$ whenever $j > k$. Discarding them therefore removes the largest part of the block.

Example 5.15 (A log cap destroys a retained block). Let $F = G = 1 + L^3 t \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. Then $FG = 1 + 2L^3 t + L^6 t^2$, so $\text{Trunc}_{<2}(FG) = 1 + 2L^3 t$. Capping the logarithmic degree at 2 *before* multiplying replaces both factors by 1 and returns $\text{Trunc}_{<2}(FG) = 1$: an entire retained t -block has been lost, not merely perturbed. Capping *after* multiplying returns 1 as well. No t -adic precision statement detects the error, because both answers agree modulo $\mathcal{O}_t^{\text{form}}(t^1)$ and disagree at the retained level t^1 .

Remark 5.16 (When an inner cutoff is legitimate). In $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ the situation reverses: a block is a Laurent series in L^{-1} whose discarded terms L^{-j} with j large *are* the small ones, so an inner cutoff is a genuine approximation in the rank-two order of [Theorem 4.6](#). A finite record in $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ therefore needs a two-dimensional precision region: an outer bound $n < N$ and a level-dependent inner bound $j > J(n)$, with the staircase J chosen stable under the intended convolutions. An outer cutoff alone never makes an infinite Laurent-log block finite. In $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, by contrast, a log-degree cap is legitimate only when a proved bound on d_F says that the capped powers are absent.

5.6 Completeness and separatedness

Definition 5.17 (The t -adic topology). Give $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ the topology defined by the filtration $(\mathfrak{F}_N)_{N \in \mathbb{Z}}$ of [Theorem 5.11](#), equivalently by the ultrametric

$$d(F, G) = 2^{-\nu_t(F-G)}, \quad 2^{-\infty} := 0. \quad (53)$$

A sequence $(F_r)_{r \geq 0}$ is *Cauchy* if for every N there is R with $\nu_t(F_r - F_s) \geq N$ for all $r, s \geq R$; it *converges* to F if $\nu_t(F_r - F) \rightarrow +\infty$. We say (F_r) *stabilizes coefficientwise* if for every n the block $[t^n]F_r$ is eventually independent of r .

That (53) is an ultrametric – in particular $d(F, H) \leq \max \{d(F, G), d(G, H)\}$ – is [Theorem 4.2](#) (ii) restated, and $d(F, G) = 0$ iff $F = G$ is [Theorem 4.2](#) (iii).

Theorem 5.18 (Completeness and separatedness). (i) Separatedness. $\bigcap_{N \geq 0} t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} = 0$ in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and $\bigcap_{N \in \mathbb{Z}} \mathfrak{F}_N = 0$ in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$.

(ii) Cauchy = stabilizing truncations. (F_r) is Cauchy if and only if for every N the truncations $\text{Trunc}_{<N} F_r$ are eventually independent of r .

(iii) Completeness. $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ are complete: every Cauchy sequence converges, to a unique limit. The same holds for $E[[t]]$ and $E((t))$ for any coefficient ring E ; in particular for $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ and $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$.

(iv) Coefficientwise stabilization is weaker on the Laurent ring. Every Cauchy sequence stabilizes coefficientwise. On $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ the converse holds. On $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ it fails; it is restored exactly under the additional hypothesis that $\inf_r \nu_t(F_r) > -\infty$.

Proof. (i) If $F \in \mathfrak{F}_N$ for every N then $\nu_t F \geq N$ for every N by [Theorem 5.11](#) (iii), so $\nu_t F = +\infty$ and $F = 0$.

(ii) Immediate from [Theorem 5.3](#) (iii): the condition $\nu_t(F_r - F_s) \geq N$ is exactly $\text{Trunc}_{<N} F_r = \text{Trunc}_{<N} F_s$.

(iii) Let (F_r) be Cauchy. Apply (ii) with $N = 0$: there is R_0 with $\text{Trunc}_{<0} F_r = \text{Trunc}_{<0} F_{R_0}$ for $r \geq R_0$. Set

$$n_0 = \min \{0, \nu_t(\text{Trunc}_{<0} F_{R_0})\} \in \mathbb{Z},$$

a genuine integer (equal to 0 if the truncation vanishes). For $r \geq R_0$ all blocks of F_r below n_0 vanish, since they agree with those of $\text{Trunc}_{<0} F_{R_0}$ below 0 and there are none below n_0 ; hence $\nu_t(F_r) \geq n_0$. This is the step at which the Cauchy hypothesis, and not merely coefficientwise stabilization, is used: it supplies a *common* lower support bound.

Next, for each $n \geq n_0$, choosing $N = n+1$ in (ii) gives R_n with $[t^n]F_r = [t^n]F_{R_n} =: p_n$ for $r \geq R_n$. Since $\nu_t F_r \geq n_0$ for $r \geq R_0$, we have $p_n = 0$ for $n < n_0$, so $F := \sum_{n \geq n_0} p_n t^n$ is a well-defined element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ (and of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ if $n_0 \geq 0$, which is the case when all $F_r \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$). Given $N \geq n_0$, take $R = \max \{R_0, R_{n_0}, \dots, R_{N-1}\}$; then $\text{Trunc}_{<N} F_r = \text{Trunc}_{<N} F$ for $r \geq R$, i.e. $\nu_t(F_r - F) \geq N$. Hence $F_r \rightarrow F$. Uniqueness of the limit is (i). The argument used nothing about $\mathbb{K}[L]$ beyond its being a ring, so it applies verbatim to $E[[t]]$ and $E((t))$.

(iv) If $F_r \rightarrow F$ then $[t^n]F_r = [t^n]F$ for r large, by [Theorem 5.13](#); so Cauchy sequences stabilize coefficientwise. On $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ suppose conversely that (F_r) stabilizes coefficientwise. Given $N \geq 0$, the finitely many blocks $n = 0, 1, \dots, N-1$ all stabilize, say beyond R ; then $\text{Trunc}_{<N} F_r$ is constant for $r \geq R$, and (ii) gives Cauchy. The same argument works on $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ as soon as $\nu_t(F_r) \geq n_1$ for all r , because then only the finitely many blocks $n_1 \leq n < N$ matter. Failure without that hypothesis is [Theorem 5.19](#). That the hypothesis is not merely sufficient but necessary – which is what “exactly” asserts – follows from (iii): a Cauchy sequence satisfies $\nu_t(F_r) \geq n_0$ for all $r \geq R_0$, and the finitely many terms with $r < R_0$ each have $\nu_t(F_r) \in \mathbb{Z} \cup \{+\infty\}$, so $\inf_r \nu_t(F_r) > -\infty$. $\square$

Example 5.19 (Coefficientwise stabilization without convergence). Let

$$F_r = \sum_{n=-r}^{-1} t^n \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} \quad (r \geq 0),$$

so $F_0 = 0$. For each fixed n the block $[t^n]F_r$ equals 1 for all $r \geq -n$ when $n < 0$, and 0 for all r when $n \geq 0$: the sequence stabilizes coefficientwise. But $\nu_t(F_{r+1} - F_r) = -(r+1) \rightarrow -\infty$, so (F_r) is not Cauchy, and there is no $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ with $[t^n]F = 1$ for every $n < 0$, since such an F would have infinitely many negative blocks. The obstruction is exactly the missing common lower bound: $\inf_r \nu_t(F_r) = -\infty$.

Remark 5.20 (Separatedness is what makes residuals certificates). Separatedness says that an element with $\nu_t = +\infty$ is 0. Every verification scheme in this volume rests on it: to prove $F = G$ it suffices to prove $\text{Trunc}_{<N}(F - G) = 0$ for every N , which is a sequence of finite computations. Completeness says the converse construction is available: a consistent prescription of blocks defines an element. Together they are the formal substitute for “the iteration converges”.

5.7 Formal summability

Definition 5.21 (Formally summable family). Let I be an index set. A family $(S_i)_{i \in I}$ in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is *formally summable* if

$$\text{for every } N \in \mathbb{Z}, \quad \{i \in I : \nu_t(S_i) < N\} \text{ is finite.} \quad (54)$$

For a sequence indexed by $\mathbb{N}_0$ this is the condition $\nu_t(S_r) \rightarrow +\infty$. Its sum is defined blockwise by

$$[t^n] \sum_{i \in I} S_i := \sum_{i \in I} [t^n] S_i, \quad (55)$$

in which all but finitely many terms vanish.

Theorem 5.22 (The summability calculus). *Let $(S_i)_{i \in I}$ be formally summable in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. Then:*

- (i) (55) defines an element $S = \sum_{i \in I} S_i \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, with $\nu_t(S) \geq \inf_i \nu_t(S_i) > -\infty$;
- (ii) for every N , $\text{Trunc}_{<N} S = \sum_{i \in I_N} \text{Trunc}_{<N} S_i$ where $I_N = \{i : \nu_t(S_i) < N\}$ is finite; equivalently S is the limit of the net of finite partial sums, and for $I = \mathbb{N}_0$, $\sum_{r \leq R} S_r \rightarrow S$ in the t -adic topology;
- (iii) the sum does not depend on the enumeration of I : for any bijection $\pi : I \rightarrow I$, $\sum_i S_{\pi(i)} = S$;
- (iv) summable families form a $\mathbb{K}$ -module and $\sum$ is $\mathbb{K}$ -linear on it;
- (v) for $H \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, the family (HS_i) is summable and $\sum_i HS_i = H \sum_i S_i$;
- (vi) if $(T_k)_{k \in J}$ is also summable, then $(S_i T_k)_{(i,k) \in I \times J}$ is summable and $\sum_{(i,k)} S_i T_k = (\sum_i S_i) (\sum_k T_k)$.

Proof. (i) Fix n . Terms with $[t^n] S_i \neq 0$ satisfy $\nu_t S_i \leq n$, and by (54) with $N = n + 1$ there are finitely many such i ; so (55) is a finite sum in $\mathbb{K}[L]$ and defines a block P_n . Applying (54) with $N = 0$ shows that $\{i : \nu_t S_i < 0\}$ is finite, hence

$$\mu := \inf_i \nu_t(S_i) \geq \min \left\{ 0, \min_{i: \nu_t S_i < 0} \nu_t S_i \right\} > -\infty,$$

with the convention that the inner minimum is 0 when its index set is empty. Thus $\mu \in \mathbb{Z}$ unless every $S_i = 0$, in which case $\mu = +\infty$. For $n < \mu$ every $[t^n] S_i$ vanishes, so $P_n = 0$; thus $S = \sum_{n \geq \mu} P_n t^n \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and $\nu_t S \geq \mu$.

(ii) For $n < N$ and $i \notin I_N$ we have $\nu_t S_i \geq N > n$, so $[t^n] S_i = 0$; hence $[t^n] S = \sum_{i \in I_N} [t^n] S_i$ for all $n < N$, which is the displayed identity. Consequently, for the partial sums $\Sigma_R = \sum_{r \leq R} S_r$ with $I = \mathbb{N}_0$, choosing R with $I_N \subseteq \{0, \dots, R\}$ gives $\text{Trunc}_{<N} \Sigma_{R'} = \text{Trunc}_{<N} S$ for all $R' \geq R$, i.e. $\nu_t(\Sigma_{R'} - S) \geq N$.

(iii) The defining formula (55) is a sum over a finite subset of I determined by n alone, and a finite sum in $\mathbb{K}[L]$ is independent of order; and π preserves the summability condition.

(iv) If (S_i) and (S'_i) are summable then $\nu_t(cS_i + S'_i) \geq \min\{\nu_t S_i, \nu_t S'_i\}$, so $\{i : \nu_t(cS_i + S'_i) < N\} \subseteq \{i : \nu_t S_i < N\} \cup \{i : \nu_t S'_i < N\}$ is finite. Linearity of $\sum$ is linearity of (55) blockwise.

(v) $\nu_t(HS_i) = \nu_t H + \nu_t S_i$, so $\{i : \nu_t(HS_i) < N\} = \{i : \nu_t S_i < N - \nu_t H\}$ is finite (if $H = 0$ the claim is trivial). For the identity, fix n and compute

$$[t^n] \sum_i HS_i = \sum_i \sum_{a+b=n} [t^a] H [t^b] S_i = \sum_{a+b=n} [t^a] H \sum_i [t^b] S_i = [t^n] \left(H \sum_i S_i \right),$$

where the interchange is legitimate because every sum involved is finite: the outer index a ranges over $\nu_t H \leq a \leq n - \mu$, and for each b only finitely many i contribute.

(vi) $\nu_t(S_i T_k) = \nu_t S_i + \nu_t T_k$. Let $\mu_S = \inf_i \nu_t S_i$ and $\mu_T = \inf_k \nu_t T_k$, both integers by (i). If $\nu_t S_i + \nu_t T_k < N$, then $\nu_t S_i < N - \mu_T$ and $\nu_t T_k < N - \mu_S$; so

$$\{(i, k) : \nu_t(S_i T_k) < N\} \subseteq \{i : \nu_t S_i < N - \mu_T\} \times \{k : \nu_t T_k < N - \mu_S\},$$

a product of two finite sets. For the value, apply (v) with $H = S_i$ to get $\sum_k S_i T_k = S_i \sum_k T_k$ for each i ; then [Theorem 5.26](#), whose proof uses only (i) and (ii) above, and (v) again with $H = \sum_k T_k$, give $\sum_{(i,k)} S_i T_k = \sum_i (S_i \sum_k T_k) = (\sum_i S_i) (\sum_k T_k)$. $\square$

Corollary 5.23 (Geometric summability). *Let E be a commutative ring with 1 and let $U \in tE[[t]]$, i.e. $\nu_t U \geq 1$. Then $(U^r)_{r \geq 0}$ is formally summable in $E((t))$, $\sum_{r \geq 0} U^r \in E[[t]]$, and*

$$(1 - U) \sum_{r \geq 0} U^r = 1.$$

This applies in particular to $E = \mathbb{K}, \mathbb{K}[L], \mathbb{K}(L)$ and $\mathbb{K}((L^{-1}))$, and — reading L^{-1} for t — inside $\mathbb{K}[[L^{-1}]] \subset \mathbb{K}((L^{-1}))$.

Proof. By [Theorem 5.27](#) the summability calculus of [Theorem 5.22](#) is available in $E((t))$ for every commutative ring E . The Cauchy product gives the subadditivity $\nu_t(FG) \geq \nu_t F + \nu_t G$ there — this is the first half of the computation in the proof of [Theorem 4.2](#) (i), which needs no domain hypothesis — so $\nu_t(U^r) \geq r \nu_t U \geq r \rightarrow +\infty$ and (54) holds. Write $V = \sum_{r \geq 0} U^r$. By [Theorem 5.22](#) (iv),(v), $V - UV = \sum_{r \geq 0} U^r - \sum_{r \geq 0} U^{r+1} = \sum_{r \geq 0} (U^r - U^{r+1})$. Fix $N \geq 1$; for $N \leq 0$ both $\text{Trunc}_{<N}(V - UV)$ and $\text{Trunc}_{<N} 1$ vanish, since $V - UV$ and 1 lie in $E[[t]]$. Only the indices $r < N$ contribute to $\text{Trunc}_{<N}$, because $\nu_t(U^r - U^{r+1}) \geq r$ ([Theorem 5.22](#) (ii)), and the resulting finite sum telescopes:

$$\text{Trunc}_{<N}(V - UV) = \text{Trunc}_{<N}\left(\sum_{0 \leq r < N} (U^r - U^{r+1})\right) = \text{Trunc}_{<N}(1 - U^N) = \text{Trunc}_{<N} 1,$$

the last step because $\nu_t(U^N) \geq N$. Since this holds for every N , [Theorem 5.3](#) (iv) gives $V - UV = 1$. $\square$

Theorem 5.24 (Three formulations of summability). *Let $(S_i)_{i \in I}$ be a family in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. Consider:*

- (a) Valuation form. *For every N , $\{i : \nu_t S_i < N\}$ is finite ([Theorem 5.21](#)).*
- (b) Hahn form. $\bigcup_i \text{supp}_{\text{coeff}}(S_i)$ satisfies [Theorem 3.15](#) (b), and every monomial $t^n L^j$ lies in $\text{supp}_{\text{coeff}}(S_i)$ for only finitely many i .
- (c) Block-local finiteness with a common bound. *For every n the set $\{i : [t^n] S_i \neq 0\}$ is finite, and $\inf_i \nu_t(S_i) > -\infty$.*

Then (a), (b) and (c) are equivalent. If the family lies in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, the lower-bound clause in (c) is automatic and may be dropped. On $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ it may not: block-local finiteness alone does not imply summability.

Proof. (a) $\Rightarrow$ (c). $\{i : [t^n] S_i \neq 0\} \subseteq \{i : \nu_t S_i \leq n\}$, finite by (a) with $N = n + 1$; and the lower bound was produced in the proof of [Theorem 5.22](#) (i).

(c) $\Rightarrow$ (a). Let $\mu = \inf_i \nu_t S_i \in \mathbb{Z}$. For $N \leq \mu$ the set in (a) is empty. For $N > \mu$,

$$\{i : \nu_t S_i < N\} \subseteq \bigcup_{n=\mu}^{N-1} \{i : [t^n] S_i \neq 0\},$$

a finite union of finite sets.

(a) $\Rightarrow$ (b). The union of supports has n -projection bounded below by μ . Its fiber over n is the union of the fibers of the finitely many S_i with $\nu_t S_i \leq n$, each finite, hence finite. The monomial condition follows from $\{i : (n, j) \in \text{supp}_{\text{coeff}}(S_i)\} \subseteq \{i : \nu_t S_i \leq n\}$.

(b) $\Rightarrow$ (a). Let n_0 bound the n -projection of the union from below and let U_n be its (finite) fiber over n . Given N ,

$$\{i : \nu_t S_i < N\} \subseteq \bigcup_{n=n_0}^{N-1} \bigcup_{j \in U_n} \{i : (n, j) \in \text{supp}_{\text{coeff}}(S_i)\},$$

a finite union — finitely many n , finitely many j for each — of finite sets.

The $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ case. There $\nu_t S_i \geq 0$ for all i , so $\inf_i \nu_t S_i \geq 0$. $\square$

Example 5.25 (Two families that just fail). *Local finiteness without a common lower bound.* Let $S_r = t^{-r}$, $r \geq 0$, in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. Every block index n is hit by at most one member, so block-local finiteness holds; but $\nu_t S_r = -r \rightarrow -\infty$, the family is not summable, and the would-be sum $\sum_{n \leq 0} t^n$ has an infinite principal part and lies in no Laurent class. This is exactly the extra hypothesis that [Theorem 5.24](#) (c) adds on $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$.

Monomial finiteness without block finiteness. Let $S_r = tL^r$, $r \geq 0$, in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Each monomial tL^r occurs in exactly one member, yet $\nu_t S_r = 1$ for every r , the family is not summable, and the would-be block $\sum_{r \geq 0} L^r$ is not a polynomial. Note that the union of supports, $\{(1, r) : r \geq 0\}$, has an infinite fiber and so violates the Hahn clause of [Theorem 5.24](#) (b). Requiring only that each *monomial* occur finitely often is therefore not enough, even on $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$; the well-based clause is doing real work.

Proposition 5.26 (Fubini for summable families). *Let $(S_{i,k})_{(i,k) \in I \times J}$ be formally summable. Then for each i the family $(S_{i,k})_{k \in J}$ is summable with sum T_i ; the family $(T_i)_{i \in I}$ is summable; and*

$$\sum_{i \in I} T_i = \sum_{(i,k) \in I \times J} S_{i,k}.$$

The same holds with the roles of I and J exchanged, so iterated summation may be performed in either order.

Proof. Summability of each slice is inherited, since $\{k : \nu_t S_{i,k} < N\}$ injects into $\{(i', k) : \nu_t S_{i',k} < N\}$. If $\nu_t S_{i,k} \geq N$ for all $k \in J$, then every block of T_i below N vanishes by (55), so $\nu_t T_i \geq N$; hence $\{i : \nu_t T_i < N\} \subseteq \{i : \exists k, \nu_t S_{i,k} < N\}$, which is finite. Finally, for each n ,

$$[t^n] \sum_i T_i = \sum_i [t^n] T_i = \sum_i \sum_k [t^n] S_{i,k} = \sum_{(i,k)} [t^n] S_{i,k} = [t^n] \sum_{(i,k)} S_{i,k},$$

all sums being finite, so ordinary finite-sum associativity applies. $\square$

Remark 5.27 (Summability on the other three classes). [Theorem 5.21](#) and [Theorem 5.22](#) were stated for $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ but use only the valuation laws and the fact that each block lies in a fixed additive group; they hold verbatim in $E[[t]]$ and $E((t))$ for every commutative ring E — in particular in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ and $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$. The domain hypothesis on E was used only for the equalities $\nu_t(HS_i) = \nu_t H + \nu_t S_i$ and $\nu_t(S_i T_k) = \nu_t S_i + \nu_t T_k$ in parts (v) and (vi) of that theorem, and there the inequality $\geq -$ valid over any commutative ring — is all that the finiteness arguments need, the displayed set identities becoming inclusions. In $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$, however, (55) requires an additional check: the finitely many contributing blocks are Laurent series in L^{-1} , and their sum is again one, so nothing is lost. What does *not* transfer is the *monomialwise* version: in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ a single block already involves infinitely many monomials, so summability must be tested on blocks, never on monomials.

5.8 The three $\mathcal{O}$ -roles

Never infer one $\mathcal{O}$ -role from another

Three unrelated meanings share the letter $\mathcal{O}$ in this subject, and the corpus keeps them typographically distinct.

Printed form	Means	Does <i>not</i> mean
$\mathcal{O}_t^{\text{form}}(t^N)$	formal valuation tail: $\nu_t(F - G) \geq N$ (Theorem 5.7)	any numerical bound, rate, uniformity in L , convergence, or the existence of a function
$\mathcal{O}_{X \rightarrow +\infty, X \in S}(A(X))$	analytic estimate: $ R(X) \leq C A(X) $ on a stated late part of a stated domain, with branches fixed	anything formal; and it is void unless the regime and domain are printed
$\mathcal{O}(N^2)$	algorithmic cost in a declared model (coefficient-ring operations, bit operations, ...)	a statement about the size or accuracy of the computed object

Remark 5.28 (The bridge, and its price). Passing from the first column to the second is a theorem, never a notation change. It needs, at minimum: a function f on a stated domain; a proof that f has the blockwise expansion in the sense of [Theorem 5.10](#); and, for each cutoff N , an exponent M_N bounding the logarithmic degree of the discarded material. A bare $\mathcal{O}$ with no printed regime is never acceptable for an asymptotic claim.

Section summary

The formal ambient is $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} = \mathbb{K}[L]((t))$: supports are lattice sets with finite fibers and lower-bounded projection, hence reverse well ordered of order type at most ω . The outer valuation ν_t is the computational filtration; the rank-two valuation $v(t^n L^j) = (n, -j)$ is the monomial comparison, and $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ is exactly the Hahn field of its value group. $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is complete and separated and $t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is a genuine ideal there; in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ the same set is only a filtration tail, because t is a unit. All infinite formulas in the volume are legitimated by one criterion: a family is summable when $\nu_t(S_i) \rightarrow +\infty$, which on the Laurent ring is strictly stronger than block-local finiteness. Leading block, leading monomial and leading scalar are three different objects; log-degree bounds are always hypotheses or proved invariants; and a formal tail carries no size information.

Part II

Arithmetic and differential calculus

6 Multiplication: convolution of logarithmic blocks

Multiplication is the operation on which every later construction rests: reciprocals, powers, composition, and reversion are all built from it by triangular recurrences. Its whole content is that the Cauchy rule in the outer variable t is *coefficientwise finite*, so that each output block is an exact finite expression in finitely many input blocks. Nothing here is an analytic rearrangement, and nothing here requires convergence.

Throughout this section $\mathbb{K}$ is a field of characteristic zero, X is the large variable, the regime is $X \rightarrow +\infty$ on the positive ray unless a different domain is announced, and

$$t := X^{-1}, \quad L := \log X. \quad (56)$$

In the formal algebra t and L are independent generators; their analytic coupling is recorded only by the distinguished derivation, which plays no role in this section. The four ambient classes are

class	coefficient ring	elements
$\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$	$\mathbb{K}[L]$	$\sum_{n \geq 0} p_n(L) t^n$
$\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$	$\mathbb{K}[L]$	$\sum_{n \geq n_0} p_n(L) t^n, n_0 \in \mathbb{Z}$
$\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$	$\mathbb{K}(L)$	$\sum_{n \geq n_0} p_n(L) t^n, p_n \text{ rational in } L$
$\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$	$\mathbb{K}((L^{-1}))$	$\sum_{n \geq n_0} t^n \sum_{j \geq j_n} a_{n,j} L^{-j}$

All truncations below are *exclusive*: $\text{Trunc}_{<N} F$ keeps every logarithmic monomial in every block t^n with $n < N$, and discards the rest.

Remark 6.1 (Reconciliation of the source conventions). Two of the merged reports use an inclusive outer cutoff $[F]_{\leq N}$. The translation is $[F]_{\leq N} = \text{Trunc}_{<N+1} F$, and every dependency range, algorithm bound and error exponent quoted from those reports has been shifted by one block accordingly. Likewise, one report writes the coefficient bracket as $[F]_M$; here the series always stands to the right of the bracket, $[t^n]F = p_n(L)$ and $[t^n L^j]F = a_{n,j}$.

6.1 The block Cauchy product

It is economical to prove the algebra of all four classes at once. What the three coefficient rings $\mathbb{K}[L]$, $\mathbb{K}(L)$, $\mathbb{K}((L^{-1}))$ have in common is a degree function in L together with a coefficient-extraction map obeying the usual convolution rule.

Definition 6.2 (Logarithmically graded domain). A *logarithmically graded domain* over $\mathbb{K}$ is a commutative $\mathbb{K}$ -algebra R together with $\mathbb{K}$ -linear maps $\kappa_d: R \rightarrow \mathbb{K}$, one for each $d \in \mathbb{Z}$, such that for every nonzero $f \in R$ the set $\{d : \kappa_d(f) \neq 0\}$ is nonempty and bounded above, and:

- (G1) $f = 0$ if and only if $\kappa_d(f) = 0$ for every d ;
- (G2) $\kappa_d(fg) = \sum_{i+j=d} \kappa_i(f)\kappa_j(g)$ for all $f, g \in R$ and all $d \in \mathbb{Z}$, the sum having only finitely many nonzero terms;
- (G3) $\kappa_0(1) = 1$ and $\kappa_d(1) = 0$ for $d \neq 0$.

For $f \neq 0$ put $\deg_L f := \max \{d : \kappa_d(f) \neq 0\}$ and $\text{lc}_L(f) := \kappa_{\deg_L f}(f) \in \mathbb{K}^\times$; set $\deg_L 0 := -\infty$ and $\text{lc}_L(0) := 0$.

Here lc_L is the leading coefficient *inside one block*; it is not the same object as the leading scalar lc of a series, defined in [Theorem 6.18](#) below, although the two agree on the leading block.

Lemma 6.3 (Elementary consequences). *Let R be a logarithmically graded domain over $\mathbb{K}$. For all $f, g \in R$:*

$$\deg_L(fg) = \deg_L f + \deg_L g, \quad \text{lc}_L(fg) = \text{lc}_L(f) \text{lc}_L(g), \quad \deg_L(f+g) \leq \max(\deg_L f, \deg_L g). \quad (57)$$

Here $-\infty + c = -\infty$ and $\max(-\infty, c) = c$. In particular R is an integral domain. Moreover, if $\deg_L f \leq d$ and $\deg_L g \leq e$, then $\kappa_{d+e}(fg) = \kappa_d(f) \kappa_e(g)$.

Proof. If $f = 0$ or $g = 0$ every assertion is trivial under the stated conventions. The third assertion in (57) is immediate from $\mathbb{K}$ -linearity of each κ_d . For the first two, assume $f, g \neq 0$ and set $d = \deg_L f$, $e = \deg_L g$. If $c > d + e$ and $i + j = c$, then $i > d$ or $j > e$, so every term of (G2) vanishes and $\kappa_c(fg) = 0$. If $c = d + e$, the only possibly nonvanishing term is $i = d, j = e$, whence

$$\kappa_{d+e}(fg) = \kappa_d(f) \kappa_e(g) = \text{lc}_L(f) \text{lc}_L(g) \neq 0,$$

because $\mathbb{K}$ is a field and both factors are nonzero. This proves $\deg_L(fg) = d + e$, $\text{lc}_L(fg) = \text{lc}_L(f) \text{lc}_L(g)$, and, by (G1), $fg \neq 0$; so R is a domain. The last sentence is the same computation with d, e merely upper bounds, all other terms of (G2) again vanishing. $\square$

Lemma 6.4 (The three coefficient rings). *Each of*

$$\mathbb{K}[L], \quad \mathbb{K}(L), \quad \mathbb{K}((L^{-1}))$$

is a logarithmically graded domain over $\mathbb{K}$, with $\deg_L$ equal respectively to the polynomial degree, to $\deg_L(p/q) = \deg p - \deg q$, and to the largest exponent occurring in the Laurent expansion. There are injective $\mathbb{K}$ -algebra homomorphisms

$$\mathbb{K}[L] \hookrightarrow \mathbb{K}(L) \hookrightarrow \mathbb{K}((L^{-1})) \quad (58)$$

that preserve $\deg_L$ and lc_L .

Proof. For $\mathbb{K}[L]$ take $\kappa_d(f)$ to be the coefficient of L^d in f for $d \geq 0$ and $\kappa_d = 0$ for $d < 0$; (G1)–(G3) are the definition of polynomial multiplication. For $\mathbb{K}((L^{-1}))$, an element is $f = \sum_{j \leq d} a_j L^j$ with only finitely many $j > 0$ allowed but arbitrarily negative j permitted; put $\kappa_j(f) = a_j$. Then (G1) and (G3) are clear, and in (G2) the constraint $i + j = c$ with $i \leq \deg_L f, j \leq \deg_L g$ forces $i \geq c - \deg_L g$, so the sum is finite; it is the definition of the Laurent product. The support of f is bounded above by construction.

The inclusion $\mathbb{K}[L] \subset \mathbb{K}((L^{-1}))$ is a $\mathbb{K}$ -algebra map, and every nonzero element of $\mathbb{K}((L^{-1}))$ is invertible: writing $f \neq 0$ as $f = a_d L^d (1 + u)$ with $d = \deg_L f$ and $u \in L^{-1} \mathbb{K}[[L^{-1}]]$, the family $((-u)^r)_{r \geq 0}$ is L^{-1} -adically summable and its sum inverts $1 + u$, so $f^{-1} = a_d^{-1} L^{-d} \sum_{r \geq 0} (-u)^r$. Hence every nonzero polynomial becomes invertible, and the universal property of the field of fractions supplies a unique $\mathbb{K}$ -algebra homomorphism $\iota: \mathbb{K}(L) \rightarrow \mathbb{K}((L^{-1}))$ extending the inclusion; it is injective because $\mathbb{K}(L)$ is a field and $\iota(1) = 1$. Pulling the maps κ_d back along ι makes $\mathbb{K}(L)$ a logarithmically graded domain, and Theorem 6.3 applied to $\iota(p) = \iota(q)\iota(p/q)$ gives $\deg_L(p/q) = \deg p - \deg q$ and $\text{lc}_L(p/q) = \text{lc}_L(p) \text{lc}_L(q)^{-1}$. Both maps in (58) preserve κ_d by construction, hence preserve $\deg_L$ and lc_L . $\square$

Definition 6.5 (The block Cauchy product). Let R be a logarithmically graded domain over $\mathbb{K}$ and let

$$F = \sum_{n \geq a} p_n t^n, \quad G = \sum_{m \geq b} q_m t^m, \quad p_n, q_m \in R, \quad (59)$$

be elements of $R((t))$, the ring of formal Laurent series in t over R . Their product is

$$FG := \sum_{N \geq a+b} C_N t^N, \quad C_N := \sum_{\substack{n+m=N \\ n \geq a, m \geq b}} p_n q_m = \sum_{n=a}^{N-b} p_n q_{N-n}. \quad (60)$$

Lemma 6.6 (Finiteness of every fiber). *For fixed N the index set in (60) is $\{n \in \mathbb{Z} : a \leq n \leq N - b\}$, of cardinality $\max(N - a - b + 1, 0)$. In particular it is finite, and it is empty exactly when $N < a + b$.*

Proof. The constraints $n \geq a$ and $m = N - n \geq b$ are equivalent to $a \leq n \leq N - b$. An interval of integers $[a, N - b]$ has $N - b - a + 1$ elements when $N - b \geq a$ and is empty otherwise. $\square$

This is the entire mechanism. There is no limit, no rearrangement, and no hypothesis of numerical convergence: each C_N is a finite sum of products in R .

Theorem 6.7 (Closure of the four ambient classes). *Let R be a logarithmically graded domain over $\mathbb{K}$. Then (60) makes $R((t))$ a commutative $\mathbb{K}$ -algebra with unit 1, containing $R[[t]]$ as a subalgebra. Consequently each of*

$$\begin{aligned} \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} &= \mathbb{K}[L][[t]], & \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} &= \mathbb{K}[L]((t)), \\ \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} &= \mathbb{K}(L)((t)), & \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}} &= \mathbb{K}((L^{-1}))((t)) \end{aligned}$$

is closed under multiplication, and the chain of inclusions

$$\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} \subset \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} \subset \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} \subset \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}} \quad (61)$$

consists of injective $\mathbb{K}$ -algebra homomorphisms.

Proof. By Theorem 6.6 each C_N is a well-defined element of R , and $C_N = 0$ for $N < a + b$; so FG again has support bounded below and lies in $R((t))$. Commutativity is the symmetry of the index set under $(n, m) \mapsto (m, n)$; bilinearity over $\mathbb{K}$ is clear; and $1 \cdot F = F$ because the only admissible pair for $1 = 1 \cdot t^0$ at level N is $(0, N)$. For associativity let $H = \sum_{k \geq c} h_k t^k$. Then

$$[t^N]((FG)H) = \sum_{r+k=N} \left(\sum_{n+m=r} p_n q_m \right) h_k = \sum_{n+m+k=N} p_n q_m h_k,$$

the last sum ranging over the finite set of triples with $n \geq a, m \geq b, k \geq c$: indeed $n + m + k = N$ with the three lower bounds confines each index to a finite interval, so the regrouping is a finite rearrangement. The same computation with the other bracketing gives the same finite sum, whence $(FG)H = F(GH)$. Distributivity is likewise a finite regrouping.

If $\nu_t F \geq 0$ and $\nu_t G \geq 0$, then $C_N = 0$ for $N < 0$, so $R[[t]]$ is closed under the product; it is a subalgebra. Taking $R = \mathbb{K}[L]$ gives the first two classes, $R = \mathbb{K}(L)$ the third, $R = \mathbb{K}((L^{-1}))$ the fourth, using Theorem 6.4. Finally, a $\mathbb{K}$ -algebra homomorphism $\varphi: R \rightarrow R'$ induces a $\mathbb{K}$ -algebra homomorphism $R((t)) \rightarrow R'((t))$ by $\sum p_n t^n \mapsto \sum \varphi(p_n) t^n$, because φ commutes with the finite sums (60); injectivity is inherited blockwise. Applying this to (58) yields (61). $\square$

Proposition 6.8 (The monomial-level convolution and the support bound). *Let $F = \sum_{n,j} a_{n,j} t^n L^j$ and $G = \sum_{m,k} b_{m,k} t^m L^k$ lie in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ (so $j, k \in \mathbb{N}_0$, and $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ is the case $\nu_t \geq 0$) or in $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ (so $j, k \in \mathbb{Z}$, bounded above at each fixed outer level). Then*

$$[t^N L^d](FG) = \sum_{n+m=N} \sum_{i+j=d} a_{n,i} b_{m,j}, \quad (62)$$

a finite sum, and

$$\text{supp}_{\text{coeff}}(FG) \subseteq \text{supp}_{\text{coeff}}(F) + \text{supp}_{\text{coeff}}(G), \quad (63)$$

the Minkowski sum in $\mathbb{Z}^2$.

Proof. Formula (62) is (G2) applied to (60). Finiteness holds because the outer sum is finite by Theorem 6.6 and, for each of its terms, the inner sum is finite by (G2). For (63): if $(N, d) \notin \text{supp}_{\text{coeff}}(F) + \text{supp}_{\text{coeff}}(G)$, then every summand of (62) has $a_{n,i} = 0$ or $b_{m,j} = 0$, so $[t^N L^d](FG) = 0$. $\square$

Remark 6.9 ($\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ has no monomial support of its own). [Theorem 6.8](#) is deliberately not stated for $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$. An element of $\mathbb{K}(L)((t))$ has coefficient blocks that are rational functions of L , and a rational function is not a (finite or infinite) combination of powers of L until one chooses an expansion point. The coefficient support $\text{supp}_{\text{coeff}}(\cdot)$ becomes defined only after applying the embedding $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} \hookrightarrow \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ of [Theorem 6.7](#), which expands each block at $L = \infty$ and turns the exact coefficient $1/(L+1)$ into the infinite series $L^{-1} - L^{-2} + L^{-3} - \dots$. Exact rational dependence on L and a monomial support in L are different pieces of data; only the second is what (63) constrains.

Remark 6.10 (Why multiplication is a two-level, not a flat, operation). Although (62) is a genuine two-dimensional convolution, it should not be implemented as one. The outer index is the asymptotic hierarchy and the inner index is not: at fixed N the inner sum is bounded, whereas a cutoff imposed jointly on $|N| + |d|$ does not respect dominance. See [Theorem 6.17](#).

Remark 6.11 (Hahn supports). Nothing in [Theorem 6.6](#) uses that the exponents are integers, only that the two supports are well ordered in the direction of increasing smallness. If $S, T \subseteq \Gamma$ are reverse-well-ordered subsets of an ordered abelian group, then $S + T$ is reverse-well-ordered and each $\gamma \in S + T$ has only finitely many decompositions $\gamma = s + t$ with $s \in S, t \in T$; this is the Hahn support theorem [19, 13]. Everything in this section that does not mention the specific value group $\mathbb{Z}$ transfers verbatim to the Hahn power-log extension $\sum_{\alpha \in S} p_\alpha(L)t^\alpha$ and, in particular, to the Puiseux-log extensions $\mathbb{K}[L]((t^{1/s}))$ used later for fractional powers. We do not repeat the statements.

6.2 Finite determination and precision bookkeeping

The content of [Theorem 6.6](#) deserves to be recorded as a dependency statement, because that is the form in which it is used: it says exactly how much of the input must be known in order to know a given part of the output, and it says that no more is needed and no less will do.

Theorem 6.12 (Finite determination, with exact dependency range). *Let R be a logarithmically graded domain over $\mathbb{K}$, let $a, b, N \in \mathbb{Z}$, and let $F, G \in R((t))$ satisfy $\nu_t F \geq a$ and $\nu_t G \geq b$. Then*

$$[t^N](FG) = \sum_{n=a}^{N-b} p_n q_{N-n}, \quad (64)$$

so that $[t^N](FG)$ is a $\mathbb{K}$ -bilinear function of the two finite tuples

$$(p_a, p_{a+1}, \dots, p_{N-b}) \quad \text{and} \quad (q_b, q_{b+1}, \dots, q_{N-a}),$$

equivalently of $\text{Trunc}_{<N-b+1} F$ and $\text{Trunc}_{<N-a+1} G$. Both ranges are exact: for every n_0 with $a \leq n_0 \leq N-b$ there are $F, F' \in R((t))$ with $\nu_t \geq a$ that differ only in the block of index n_0 , and a G with $\nu_t G \geq b$, such that $[t^N](FG) \neq [t^N](F'G)$; and symmetrically in the second argument.

Proof. Equation (64) and the bilinearity are [Theorem 6.5](#) together with [Theorem 6.6](#); the blocks listed are precisely those occurring in the sum. For exactness fix $n_0 \in [a, N-b]$ and put $m_0 = N - n_0$, so that $m_0 \geq b$. Take $G = t^{m_0}$, which has $\nu_t G = m_0 \geq b$. If $n_0 > a$ set $F = t^a + t^{n_0}$ and $F' = t^a$; if $n_0 = a$ set $F = t^a$ and $F' = 2t^a$ (here $2 \neq 0$ because $\text{char } \mathbb{K} = 0$). In both cases $\nu_t F = \nu_t F' = a$, the two series agree outside index n_0 , and $[t^N](FG) - [t^N](F'G)$ equals 1 respectively -1 . The symmetric statement follows by commutativity. $\square$

Structural checkpoint

The precision rule. If F is known through $\text{Trunc}_{<P} F$ and G through $\text{Trunc}_{<Q} G$, and $\nu_t F = a$, $\nu_t G = b$, then FG is known through

$$\text{Trunc}_{<\min(P+b, Q+a)} FG$$

and, in the worst case, no further. The binding entry is contributed by whichever factor is paired with the *more singular* partner: a factor known to P blocks is degraded by the valuation b of the other factor. In particular, multiplying by a series with a pole of order r , that is with valuation $-r$, costs exactly r blocks of precision.

Corollary 6.13 (Truncated product over a general coefficient ring, at arbitrary sufficient cutoffs). *Let $F, G \in R((t))$ be nonzero with $a = \nu_t F$, $b = \nu_t G$, and let $N \in \mathbb{Z}$. Then for all integers $P \geq N - b$ and $Q \geq N - a$,*

$$\text{Trunc}_{<N} FG = \text{Trunc}_{<N} (\text{Trunc}_{<P} F) (\text{Trunc}_{<Q} G); \quad (65)$$

and if $N > a + b$ the pair $(P, Q) = (N - b, N - a)$ is optimal: neither entry can be lowered by one, that is, there exist F, G with these valuations for which $P = N - b - 1$, respectively $Q = N - a - 1$, falsifies (65). If $F, G \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $N \geq 0$, then $(P, Q) = (N, N)$ is admissible, so the cruder law

$$\text{Trunc}_{<N} FG = \text{Trunc}_{<N} (\text{Trunc}_{<N} F) (\text{Trunc}_{<N} G) \quad (N \geq 0) \quad (66)$$

holds there.

Proof. Write $F = \text{Trunc}_{<P} F + E$ and $G = \text{Trunc}_{<Q} G + H$, so that $\nu_t E \geq P$ and $\nu_t H \geq Q$, while $\nu_t \text{Trunc}_{<P} F \geq a$ and $\nu_t \text{Trunc}_{<Q} G \geq b$. Then

$$FG - (\text{Trunc}_{<P} F) (\text{Trunc}_{<Q} G) = E \text{Trunc}_{<Q} G + \text{Trunc}_{<P} F H + EH,$$

whose ν_t is at least $P + b \geq N$, $a + Q \geq N$ and $P + b \geq N$ respectively; for the third term note that $H = G - \text{Trunc}_{<Q} G$ has $\nu_t H \geq b$, every block of H being a block of G , so $\nu_t(EH) \geq P + b$. Hence the two products agree below t^N , which is (65).

For the optimality and the sharpness, assume $N > a + b$, so that $N - 1 \geq a + b$. By Theorem 6.12 the block $[t^{N-1}](FG)$ already depends on p_{N-1-b} and on q_{N-1-a} , so no pair below $(N - b, N - a)$ can serve for all F, G ; the following pair of series shows that neither entry may be lowered by one. If $N - b - 1 > a$ put $F = t^a + t^{N-b-1}$; if $N - b - 1 = a$ put $F = t^a$. In both cases take $G = t^b$. Then $[t^{N-1}](FG) = 1$, whereas $\text{Trunc}_{<N-b-1} F$ omits the block t^{N-b-1} , so $\text{Trunc}_{<N} (\text{Trunc}_{<N-b-1} F) G$ has zero block at t^{N-1} . (In the first case $N - 1 > a + b$, so the surviving term t^{a+b} does not sit at level $N - 1$.) The second cutoff is symmetric.

For (66), note that $a, b \geq 0$ gives $N \geq N - b$ and $N \geq N - a$, so $(P, Q) = (N, N)$ is admissible. $\square$

A common trap

The crude law (66) is *false* on the Laurent ring. Take $F = t^{-1}$, $G = 1 + t$, $N = 1$. Then $FG = t^{-1} + 1$, so $\text{Trunc}_{<1} FG = t^{-1} + 1$, whereas $\text{Trunc}_{<1} F \text{Trunc}_{<1} G = t^{-1} \cdot 1 = t^{-1}$. The correct cutoff for G is $N - \nu_t F = 2$, and indeed $t^{-1}(1 + t) = t^{-1} + 1$. Implementations that “normalize every series to start at t^0 ” silently commit this error whenever a genuine principal part is present.

Corollary 6.14 (Error rule for approximate factors). *Let $F, G \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ with $\nu_t F = a$, $\nu_t G = b$, and suppose $\tilde{F}, \tilde{G} \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ satisfy*

$$F = \tilde{F} + \mathcal{O}_t^{\text{form}}(t^p), \quad G = \tilde{G} + \mathcal{O}_t^{\text{form}}(t^q).$$

Then

$$FG = \tilde{F}\tilde{G} + \mathcal{O}_t^{\text{form}}\left(t^{\min(p+b, q+a, p+q)}\right). \quad (67)$$

If in addition $p \geq a$ and $q \geq b$ — that is, if neither approximation is coarser than the leading order of the quantity it approximates — then the third entry is redundant and

$$FG = \tilde{F}\tilde{G} + \mathcal{O}_t^{\text{form}}\left(t^{\min(p+b, q+a)}\right). \quad (68)$$

Proof. Write $E = F - \tilde{F}$ and $H = G - \tilde{G}$, so $\nu_t E \geq p$ and $\nu_t H \geq q$. Expanding,

$$FG - \tilde{F}\tilde{G} = E\tilde{G} + \tilde{F}H + EH. \quad (69)$$

Now $\tilde{G} = G - H$ gives $\nu_t \tilde{G} \geq \min(\nu_t G, \nu_t H) \geq \min(b, q)$, and likewise $\nu_t \tilde{F} \geq \min(a, p)$. Since ν_t is additive on products (Theorem 6.21 below, or directly Theorem 6.6) and satisfies $\nu_t(A + B) \geq \min(\nu_t A, \nu_t B)$,

$$\nu_t(FG - \tilde{F}\tilde{G}) \geq \min(p + \min(b, q), \min(a, p) + q, p + q) = \min(p + b, q + a, p + q),$$

which is (67). If $p \geq a$ and $q \geq b$, then $p + q \geq a + q$ and $p + q \geq p + b$, so $p + q \geq \min(p + b, q + a)$ and the third entry may be dropped. $\square$

Remark 6.15 (What (67) does and does not say). The symbol $\mathcal{O}_t^{\text{form}}(t^N)$ asserts coefficient agreement below outer order N and nothing else: no numerical bound, no convergence, no uniformity in L . It is also not a statement about a quotient ring. In $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ the element t is a unit, so the ideal generated by t^N is the whole ring; the subset $t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is an additive piece of the valuation filtration, not an ideal. Congruences of the form (67) must therefore be read as valuation inequalities $\nu_t(F - G) \geq N$, or equivalently as equalities of exclusive truncations. Quotient-algebra language becomes legitimate only after the finite principal part has been removed and one works inside $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, where (t^N) is a genuine ideal.

The corresponding algorithm is the schoolbook convolution restricted to the window supplied by Theorem 6.12.

Listing 1: Truncated block product

```
# F, G sparse maps exponent -> coefficient in R; a = ord(F), b = ord(G)
# returns the blocks of Trunc_{<N}(F*G)
function block_product(F, G, a, b, N):
    C := empty map
    for (n, Pn) in F with a <= n <= N-b-1:
        for (m, Qm) in G with b <= m <= N-n-1:
            C[n+m] := C.get(n+m, 0) + Pn*Qm
    normalize every coefficient of C # collect in L before zero-testing
    delete the blocks of C that are exactly 0
    return C
```

Remark 6.16 (Cost, and what actually dominates it). With dense consecutive support the loop performs $\mathcal{O}((N - a - b)^2)$ coefficient products; with sparse supports of sizes s_F, s_G it performs at most $s_F s_G$. If the retained blocks have logarithmic degree $\mathcal{O}(d)$, each coefficient product costs $\mathcal{O}(d^2)$ scalar operations by schoolbook polynomial arithmetic, so the total is $\mathcal{O}((N - a - b)^2 d^2)$; fast convolution in t and fast polynomial multiplication in L reduce both factors. These are complexity $\mathcal{O}$'s and carry no analytic meaning. In practice, at the moderate orders typical of hand asymptotics, exact coefficient normalization – expansion, collection, and gcd reduction if the coefficients are rational functions of L – costs more than the convolution itself, and the decisive implementation choice is to never form a product whose outer order exceeds the requested cutoff.

Remark 6.17 (A rectangular cutoff is not a truncation). Discarding all monomials $t^n L^j$ with $|n| + |j|$ or $|j|$ above a threshold does not respect dominance. With a cap $|j| \leq 10$ one deletes $t^2 L^{100}$ while retaining $t^3 L^{-100}$, although $t^3 L^{-100} \prec t^2 L^{100}$ by an enormous margin. Bounding $\deg_L p_n$ is a second, independent approximation; it is legitimate only when accompanied by a proved degree bound such as Theorem 6.29.

6.3 Leading data and the leading-term law

Definition 6.18 (Leading data over a logarithmically graded domain). Let R be a logarithmically graded domain over $\mathbb{K}$ and let $F = \sum_{n \geq n_0} p_n t^n \in R((t))$ be nonzero with $\nu_t F = \min \{n : p_n \neq 0\}$; set $\nu_t 0 = +\infty$. The *leading block* of F is the whole coefficient $p_{\nu_t F} \in R$, the *leading monomial* is

$$\text{lm}(F) := t^{\nu_t F} L^{\deg_L p_{\nu_t F}}, \quad (70)$$

and the *leading scalar* is $\text{lc}(F) := \text{lc}_L(p_{\nu_t F}) \in \mathbb{K}^\times$. The *rank-two valuation* of F is

$$v(F) := (\nu_t F, -\deg_L p_{\nu_t F}) \in \mathbb{Z}^2, \quad (71)$$

with $\mathbb{Z}^2$ lexicographically ordered and $v(0) = +\infty$. These are three different objects and none is determined by the coarse integer $\nu_t F$ alone.

Remark 6.19 (These are the objects of the ring chapter, over a general coefficient ring). [Theorem 6.18](#) introduces no new object. It repeats [Theorem 4.4](#) and [Theorem 4.6](#) with the three coefficient rings $\mathbb{K}[L]$, $\mathbb{K}(L)$, $\mathbb{K}((L^{-1}))$ replaced by an arbitrary logarithmically graded domain R , so that the leading-term law can be proved once for all four ambient classes simultaneously; on those four classes the two readings agree. The same applies to [Theorem 6.24](#) against [Theorem 4.11](#) — whose codomain is $\mathbb{N}_0 \cup \{-\infty\}$ because there the blocks are polynomials, whereas over $\mathbb{K}(L)$ and $\mathbb{K}((L^{-1}))$ the value $\deg_L p_n$ may be negative — and to [Theorem 6.13](#) against [Theorem 5.4](#), which the corollary strengthens to arbitrary sufficient cutoffs (P, Q) and supplements with optimality and sharpness. Nothing below re-proves a ring-chapter result for its own sake; the restatements exist only to make the general coefficient ring available.

Lemma 6.20 (Dominance is growth). *On the positive ray, for integers n, m and j, k ,*

$$t^n L^j \succ t^m L^k \iff \lim_{X \rightarrow +\infty} \frac{X^{-n} (\log X)^j}{X^{-m} (\log X)^k} = +\infty \iff n < m, \text{ or } (n = m \text{ and } j > k).$$

Consequently, for F in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ or in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$, $\text{lm}(F)$ is the $\succ$ -largest element of $\text{supp}_{\text{coeff}}(F)$. (For $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ the statement is meaningful only after the expansion of [Theorem 6.9](#).)

Proof. By [Theorem 3.12](#) the relation $\succ$ is defined by the right-hand condition, so the first equivalence is a definition and only the second requires an argument; it is [Theorem 3.13](#) specialized to integer exponents, and we repeat the two lines. The ratio equals $X^{m-n} (\log X)^{j-k}$. If $m > n$ it tends to $+\infty$ because $\log X = o_{X \rightarrow +\infty}(X^\varepsilon)$ for every $\varepsilon > 0$; if $m = n$ it is $(\log X)^{j-k}$, which tends to $+\infty$ exactly when $j > k$; if $m < n$ it tends to 0 for the same reason as the first case. For the last sentence: a monomial $t^n L^j$ in the support has $n \geq \nu_t F$, with equality attained; among those with $n = \nu_t F$ the largest j with $\kappa_j(p_{\nu_t F}) \neq 0$ is by definition $\deg_L p_{\nu_t F}$. $\square$

Theorem 6.21 (Leading-term law). *Let R be a logarithmically graded domain over $\mathbb{K}$ and let $F, G \in R((t))$ be nonzero. Then $FG \neq 0$ and*

$$\nu_t(FG) = \nu_t F + \nu_t G, \quad (72)$$

$$\text{lm}(FG) = \text{lm}(F) \text{lm}(G), \quad (73)$$

$$\text{lc}(FG) = \text{lc}(F) \text{lc}(G), \quad (74)$$

$$v(FG) = v(F) + v(G). \quad (75)$$

Moreover $\nu_t(F + G) \geq \min(\nu_t F, \nu_t G)$ and $v(F + G) \geq \min(v(F), v(G))$ in the lexicographic order. Each is an equality unless cancellation occurs at the common level: whole leading blocks cancel in the first case, top logarithmic coefficients in the second.

Proof. Put $a = \nu_t F$, $b = \nu_t G$. By [Theorem 6.6](#) the coefficient C_N of FG vanishes for $N < a + b$, and for $N = a + b$ the index set $[a, N - b] = \{a\}$ is a single point, so $C_{a+b} = p_a q_b$. Since R is a domain ([Theorem 6.3](#)) and $p_a, q_b \neq 0$, we get $C_{a+b} \neq 0$; hence $FG \neq 0$ and (72) holds. Applying (57) to $C_{a+b} = p_a q_b$ gives

$$\deg_L C_{a+b} = \deg_L p_a + \deg_L q_b, \quad \text{lc}_L(C_{a+b}) = \text{lc}_L(p_a) \text{lc}_L(q_b),$$

which are exactly (73) and (74); (75) is the same pair of identities rewritten in the notation (71).

For the sum, let $c = \min(a, b)$. Every block of $F + G$ of index $< c$ vanishes, so $\nu_t(F + G) \geq c$. If $a < b$ then $[t^c](F + G) = p_a \neq 0$ and the leading data of $F + G$ and F coincide, so $v(F + G) = v(F) = \min(v(F), v(G))$ – note that $a < b$ makes $v(F) < v(G)$ lexicographically. If $a = b = c$ and $p_c + q_c \neq 0$, then $\nu_t(F + G) = c$ and $\deg_L(p_c + q_c) \leq \max(\deg_L p_c, \deg_L q_c)$ by (57), i.e. $-\deg_L(p_c + q_c) \geq \min(-\deg_L p_c, -\deg_L q_c)$, which is $v(F + G) \geq \min(v(F), v(G))$. If $p_c + q_c = 0$, then $\nu_t(F + G) > c$ and $v(F + G) > \min(v(F), v(G))$ outright. $\square$

Corollary 6.22 (Integrality and fields). *Each of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$, $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ is an integral domain. The map v of (71) is a rank-two valuation on $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ and on $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$, with value group $\mathbb{Z}^2$ lexicographically ordered; on $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ it takes values in the sub-semigroup $\mathbb{Z} \times \mathbb{Z}_{\leq 0}$. The classes $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ and $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ are fields; $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ are not.*

Proof. Integrality is [Theorem 6.21](#), and the valuation axioms are its last two displays together with $v(F) = +\infty$ iff $F = 0$. Surjectivity onto $\mathbb{Z}^2$ for $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ and $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ follows from $v(t^n L^{-j}) = (n, j)$, legitimate there because L^{-j} lies in $\mathbb{K}(L)$ and in $\mathbb{K}((L^{-1}))$; the group $\mathbb{Z}^2$ with the lexicographic order has exactly one proper nontrivial convex subgroup, $\{0\} \times \mathbb{Z}$, so the valuation has rank two. On $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ the second coordinate is $-\deg_L p_{\nu_t F} \leq 0$ because polynomial degrees are nonnegative.

A Laurent-series ring $R((t))$ over a field R is a field, which covers $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ and $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$: given $F \neq 0$ with $\nu = \nu_t F$ and leading block $p_\nu \in R^\times$, write $F = p_\nu t^\nu (1 + U)$ with $U = \sum_{n > \nu} (p_n/p_\nu) t^{n-\nu} \in tR[[t]]$; the family $((-U)^r)_{r \geq 0}$ is summable because $\nu_t(-U)^r \geq r$, and by [Theorem 6.34\(2\)](#) its sum V satisfies

$$(1 + U)V = \sum_{r \geq 0} (-U)^r - \sum_{r \geq 0} (-U)^{r+1} = (-U)^0 = 1,$$

so $F^{-1} = p_\nu^{-1} t^{-\nu} V$. ([Theorem 6.34](#) is proved in [Section 6.5](#) from [Theorem 6.21](#) alone, so there is no circularity.) Finally $L \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is not invertible in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$: if $LH = 1$ then (73) gives $L \cdot \text{lm}(H) = 1$, forcing $\deg_L$ of the leading block of H to be -1 , impossible for a polynomial. Since $L \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} \subseteq \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, an inverse of L inside $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ would in particular be an inverse inside $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$; so L is not invertible in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ either, and neither class is a field. $\square$

Corollary 6.23 (Necessity half of the unit criterion). *If $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is invertible in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, then its leading block is a nonzero scalar: $p_{\nu_t F} \in \mathbb{K}^\times$. The same statement holds in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with the additional conclusion $\nu_t F = 0$.*

Proof. If $FH = 1$ then (72) gives $\nu_t F + \nu_t H = 0$ and (73) gives $\deg_L p_{\nu_t F} + \deg_L h_{\nu_t H} = 0$. Both degrees are nonnegative integers, hence both vanish, so $p_{\nu_t F} \in \mathbb{K}^\times$. In $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ both orders are ≥ 0 and sum to 0, so both are 0. $\square$

The converse – that a scalar leading block suffices – is a division statement and is proved with the reciprocal recurrence; it is not a multiplication fact and is not repeated here.

6.4 Degree and support bounds

Definition 6.24 (Log-degree profile on the four classes). For $F = \sum_n p_n t^n$ in one of the four classes, the *log-degree profile* is the function $d_F : \mathbb{Z} \rightarrow \mathbb{Z} \cup \{-\infty\}$,

$$d_F(n) := \begin{cases} \deg_L p_n, & p_n \neq 0, \\ -\infty, & p_n = 0. \end{cases} \quad (76)$$

The profile is data additional to $\nu_t F$; it is not recoverable from it. For two profiles define the *max-plus convolution*

$$(d_F * d_G)(N) := \max_{n+m=N} (d_F(n) + d_G(m)), \quad (77)$$

with the conventions $-\infty + c = -\infty$ and $\max \emptyset = -\infty$. Only the finitely many pairs of [Theorem 6.6](#) can contribute a value other than $-\infty$, so the maximum is effectively over a finite set.

Theorem 6.25 (Degree bound and exact criterion for attainment). *Let $F, G \in R((t))$ with R a logarithmically graded domain over $\mathbb{K}$, let $N \in \mathbb{Z}$, and set $M := (d_F * d_G)(N)$. Let*

$$P_N := \{(n, m) : n + m = N, p_n \neq 0, q_m \neq 0, d_F(n) + d_G(m) = M\}$$

be the (finite, possibly empty) set of maximizing pairs. Then

$$d_{FG}(N) \leq M, \quad (78)$$

If $P_N = \emptyset$, equivalently $M = -\infty$, then $[t^N](FG) = 0$ and (78) holds with equality. If $P_N \neq \emptyset$, so that $M \in \mathbb{Z}$, then the coefficient of L^M in the block $[t^N](FG)$ is exactly

$$\kappa_M([t^N](FG)) = \sum_{(n,m) \in P_N} \text{lc}_L(p_n) \text{lc}_L(q_m), \quad (79)$$

and equality holds in (78) if and only if the sum (79) is nonzero.

Proof. Each summand of $C_N = \sum_{n+m=N} p_n q_m$ has $\deg_L(p_n q_m) = d_F(n) + d_G(m) \leq M$ by (57), with the convention that a vanishing factor gives $-\infty$. A finite sum of elements of $\deg_L \leq M$ has $\deg_L \leq M$, which is (78). If $M = -\infty$ then every summand $p_n q_m$ vanishes, so $[t^N](FG) = 0$ and both sides of (78) equal $-\infty$; assume from now on $P_N \neq \emptyset$, so that $M \in \mathbb{Z}$. By $\mathbb{K}$ -linearity of κ_M ,

$$\kappa_M(C_N) = \sum_{n+m=N} \kappa_M(p_n q_m).$$

If $(n, m) \notin P_N$ then either a factor vanishes or $\deg_L(p_n q_m) < M$, and in both cases $\kappa_M(p_n q_m) = 0$. If $(n, m) \in P_N$ then $\deg_L(p_n q_m) = M$ and $\kappa_M(p_n q_m) = \text{lc}_L(p_n q_m) = \text{lc}_L(p_n) \text{lc}_L(q_m)$ by [Theorem 6.3](#). This proves (79), and equality in (78) means precisely $\kappa_M(C_N) \neq 0$. $\square$

Corollary 6.26 (Three sufficient conditions for equality). *In the situation of [Theorem 6.25](#), equality holds in (78) in each of the following cases.*

1. P_N is a singleton.
2. $N = \nu_t F + \nu_t G$ (both factors nonzero); then P_N is the singleton $\{(\nu_t F, \nu_t G)\}$ and one recovers (73).
3. $P_N \neq \emptyset$, $\mathbb{K}$ is an ordered field, and $\text{lc}_L(p_n) > 0, \text{lc}_L(q_m) > 0$ for every $(n, m) \in P_N$.

Proof. (1) The sum (79) is a single product of two elements of $\mathbb{K}^\times$, hence nonzero. (2) By [Theorem 6.6](#) the only admissible pair at $N = \nu_t F + \nu_t G$ is $(\nu_t F, \nu_t G)$, and both blocks are nonzero, so P_N is that singleton; apply (1). (3) The sum is a nonempty sum of strictly positive elements of an ordered field. $\square$

Example 6.27 (The bound is attained, at every order and by every pair). Over $\mathbb{K} = \mathbb{Q}$ let

$$F = G = \sum_{n \geq 0} L^n t^n \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}, \quad \text{so} \quad d_F(n) = d_G(n) = n.$$

Then $(d_F * d_G)(N) = \max_{n+m=N} (n + m) = N$, and every one of the $N + 1$ admissible pairs is maximizing, so P_N is as large as it can be and [Theorem 6.26\(1\)](#) does not apply. Nevertheless

$$[t^N](FG) = \sum_{n=0}^N L^n \cdot L^{N-n} = (N + 1)L^N,$$

so $d_{FG}(N) = N$ for every N : the bound (78) is attained at every order. Formula (79) predicts exactly the top coefficient $N + 1 = \sum_{(n,m) \in P_N} 1 \cdot 1$.

A common trap

[Theorem 6.27](#) is the place where characteristic zero is used. Over a field of characteristic $p > 0$ the same computation gives $[t^{p-1}](FG) = pL^{p-1} = 0$, so the block vanishes entirely and $d_{FG}(p - 1) = -\infty$ although the bound is $p - 1$. Every statement in this section is asserted over a field of characteristic zero, and this is not a formality.

Example 6.28 (The bound can fail by an arbitrary amount). Let $F = 1 + L^d t$ and $G = 1 - L^d t$ in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, $d \geq 1$. Then $(d_F * d_G)(1) = \max(0 + d, d + 0) = d$, while $[t^1](FG) = -L^d + L^d = 0$, so $d_{FG}(1) = -\infty$. Here $P_1 = \{(0, 1), (1, 0)\}$ and the sum (79) is $1 \cdot (-1) + 1 \cdot 1 = 0$, as it must be. A drop by exactly one occurs for $F = 1 + (L^2 + L)t$, $G = 1 - (L^2 - L)t$, where the bound at $N = 1$ is 2 while $[t^1](FG) = 2L$ has degree 1: here the top coefficients cancel but the next ones do not.

Proposition 6.29 (Affine profiles and the profile subalgebra). Let $\alpha, \alpha' \geq 0$ and $\beta, \gamma \in \mathbb{R}$. Let $F, G \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ be nonzero with $a = \nu_t F$, $b = \nu_t G$ and

$$d_F(n) \leq \alpha n + \beta \quad (n \geq a), \quad d_G(m) \leq \alpha' m + \gamma \quad (m \geq b).$$

Put $\mu = \max(\alpha, \alpha')$. Then for all N ,

$$d_{FG}(N) \leq \mu N + \beta + \gamma - (\mu - \alpha)a - (\mu - \alpha')b. \quad (80)$$

In particular, if $a, b \geq 0$ then $d_{FG}(N) \leq \mu N + \beta + \gamma$. Consequently, for each $\alpha \geq 0$ the set

$$\mathfrak{A}_\alpha := \{F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} : \exists \beta \in \mathbb{R}, d_F(n) \leq \alpha n + \beta \text{ for all } n\}$$

is a $\mathbb{K}$ -subspace with $\mathfrak{A}_\alpha \cdot \mathfrak{A}_{\alpha'} \subseteq \mathfrak{A}_{\max(\alpha, \alpha')}$, and $\mathfrak{A} := \bigcup_{\alpha \geq 0} \mathfrak{A}_\alpha$ is a $\mathbb{K}$ -subalgebra of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ containing $\mathbb{K}[L][t, t^{-1}]$.

Proof. Fix N and $n + m = N$ with $n \geq a$, $m \geq b$. Then

$$\alpha n + \alpha' m = \mu(n + m) - (\mu - \alpha)n - (\mu - \alpha')m \leq \mu N - (\mu - \alpha)a - (\mu - \alpha')b,$$

because $\mu - \alpha \geq 0$, $\mu - \alpha' \geq 0$, $n \geq a$ and $m \geq b$. Adding $\beta + \gamma$ and taking the maximum over the finite index set gives (80) via (78). If $a, b \geq 0$ then the two subtracted terms are ≥ 0 , so they may be dropped.

For the algebra statements: $\mathfrak{A}_\alpha$ is a subspace because $d_{F+G}(n) \leq \max(d_F(n), d_G(n)) \leq \alpha n + \max(\beta, \gamma)$ and scalars do not change profiles. The product statement is (80), whose right-hand side is $\mu N + \text{const}$. For $\alpha \leq \alpha''$ one has $\mathfrak{A}_\alpha \subseteq \mathfrak{A}_{\alpha''}$. Indeed, let $F \neq 0$ satisfy $d_F(n) \leq \alpha n + \beta$ for all n , and put $a = \nu_t F$. For $n \geq 0$ we have $\alpha n + \beta \leq \alpha'' n + \beta$; for $a \leq n < 0$,

$$\alpha n + \beta = \alpha'' n + (\alpha'' - \alpha)(-n) + \beta \leq \alpha'' n + \beta + (\alpha'' - \alpha) \max(0, -a);$$

and $d_F(n) = -\infty$ for $n < a$. Hence $\beta'' = \beta + (\alpha'' - \alpha) \max(0, -a)$ witnesses $F \in \mathfrak{A}_{\alpha''}$, and $0 \in \mathfrak{A}_{\alpha''}$ trivially. Consequently the union $\mathfrak{A}$ is closed under addition and multiplication and contains 1; every Laurent polynomial $\sum_{n=n_1}^{n_2} p_n(L) t^n$ lies in $\mathfrak{A}_0$ with $\beta = \max_n \deg_L p_n$. $\square$

Remark 6.30 ($\mathfrak{A}$ is not t -adically closed). The affine-profile condition is preserved by finite sums and products but not by t -adic limits. Take $F_r = t^r L^{r^2}$ for $r \geq 1$; each F_r lies in $\mathfrak{A}_0$, and $\nu_t F_r = r \rightarrow \infty$, so (F_r) is summable, yet the sum $S = \sum_{r \geq 1} t^r L^{r^2}$ has $d_S(r) = r^2$, which is bounded by no affine function. Hence $S \notin \mathfrak{A}$. A degree bound must therefore be re-derived, not inherited, whenever an infinite construction is performed.

Proposition 6.31 (Profile of a power). *Let $F \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $d_F(n) \leq \alpha n + \beta$ for all $n \geq 0$, where $\alpha \geq 0$ and $\beta \geq 0$. Then for every integer $k \geq 0$,*

$$d_{F^k}(N) \leq \alpha N + k\beta \quad (N \geq 0). \quad (81)$$

Proof. Induction on k . For $k = 0$ we have $F^0 = 1$, so $d_{F^0}(0) = 0 \leq 0$ and $d_{F^0}(N) = -\infty$ for $N > 0$. Assume the bound for k . By (78) applied to $F^{k+1} = F^k \cdot F$,

$$d_{F^{k+1}}(N) \leq \max_{n+m=N, n, m \geq 0} ((\alpha n + k\beta) + (\alpha m + \beta)) = \alpha N + (k+1)\beta.$$

□

Remark 6.32 (The Lambert profile, corrected). Two of the sources record the degree law of the canonical Lambert family as $\deg \mathbb{L}_n^{\text{Lam}} = n$ without restriction, and use it to assert that products of Lambert-type expansions have profile $\leq N$. The law $\deg \mathbb{L}_n^{\text{Lam}} = n$ holds only for $n \geq 1$: the normalization $\mathbb{L}_0^{\text{Lam}}(u) = -u$ has degree 1, not 0. The correct hypothesis for a series $F = \sum_{n \geq 0} \mathbb{L}_n^{\text{Lam}}(L)t^n$ is therefore $d_F(n) \leq n + 1$ for all $n \geq 0$, i.e. $\alpha = 1, \beta = 1$ in Theorem 6.29; a product of two such series has $d_{FG}(N) \leq N + 2$, and a k -th power has $d_{F^k}(N) \leq N + k$ by Theorem 6.31. The unrestricted claim $d_{FG}(N) \leq N$ is false already at $N = 0$, where $[t^0](FG) = \mathbb{L}_0^{\text{Lam}}(L)^2 = L^2$ has degree 2.

6.5 Powers and binomial powers

Infinite sums enter through powers, so we first record the exact sense in which they are legitimate.

Definition 6.33 (Summable family over a logarithmically graded domain). Let R be a logarithmically graded domain over $\mathbb{K}$. A family $(S_i)_{i \in I}$ in $R((t))$ is *summable* if for every $N \in \mathbb{Z}$ the set $\{i \in I : \nu_t S_i < N\}$ is finite. Its sum is defined blockwise: $[t^n] \sum_i S_i := \sum_i [t^n] S_i$, a finite sum in the coefficient ring.

Lemma 6.34 (Summable families are well behaved). *Let $(S_i)_{i \in I}$ and $(T_j)_{j \in J}$ be summable families in $R((t))$.*

1. $\inf_i \nu_t S_i > -\infty$, so $\sum_i S_i$ is a genuine element of $R((t))$ (its support is bounded below).
2. The family $(S_i T_j)_{(i,j) \in I \times J}$ is summable and $(\sum_i S_i)(\sum_j T_j) = \sum_{(i,j)} S_i T_j$.

Proof. (1) Taking $N = 0$ in Theorem 6.33, only finitely many i have $\nu_t S_i < 0$, and each of those has $\nu_t S_i \in \mathbb{Z}$ (a member of that finite set is in particular nonzero). Let λ be the minimum of that finite set of integers, or $\lambda = 0$ if the set is empty. Then $\nu_t S_i \geq \min(0, \lambda)$ for all i , and every block of $\sum_i S_i$ below that bound vanishes.

(2) Let $\alpha = \inf_i \nu_t S_i$ and $\beta = \inf_j \nu_t T_j$, both $> -\infty$ by (1). If $\alpha = +\infty$ or $\beta = +\infty$ then every S_i , respectively every T_j , is zero, so every $S_i T_j$ is zero and both assertions are trivial; assume therefore $\alpha, \beta \in \mathbb{Z}$. Fix N . If $\nu_t(S_i T_j) = \nu_t S_i + \nu_t T_j < N$ then $\nu_t S_i < N - \beta$ and $\nu_t T_j < N - \alpha$, and each of those conditions holds for only finitely many indices; so the family is summable. For the identity, fix n and compute in the coefficient ring, all sums having only finitely many nonzero terms:

$$[t^n] \left(\sum_i S_i \sum_j T_j \right) = \sum_{p+q=n} \left(\sum_i [t^p] S_i \right) \left(\sum_j [t^q] T_j \right) = \sum_{i,j} \sum_{p+q=n} [t^p] S_i [t^q] T_j = \sum_{i,j} [t^n] (S_i T_j).$$

□

Remark 6.35. Part (1) is worth isolating. For a *sequence* in a Laurent ring, t -adic Cauchyness alone does not guarantee a limit with a finite principal part; a common eventual lower support bound must be imposed. For a *summable family* no such extra hypothesis is needed, because the finiteness condition at $N = 0$ already forces a common lower bound.

Proposition 6.36 (Coefficients of an integer power). *Let $F = \sum_{n \geq 0} p_n t^n \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $k \geq 0$. Then*

$$[t^N]F^k = \sum_{\substack{n_1, \dots, n_k \geq 0 \\ n_1 + \dots + n_k = N}} p_{n_1} \cdots p_{n_k} := C_{N,k}(p_0, p_1, \dots), \quad (82)$$

with $C_{0,0} = 1$ and $C_{N,0} = 0$ for $N > 0$. These convolution polynomials satisfy

$$C_{N,k+1} = \sum_{n=0}^N p_n C_{N-n,k}, \quad (83)$$

which computes the whole triangular table in $\mathcal{O}(N^2 k)$ coefficient products. If $p_0 = 0$, then $C_{N,k} = 0$ for $N < k$, and, writing $F = \sum_{n \geq 1} p_n t^n$,

$$[t^N]F^k = \frac{k!}{N!} \mathcal{B}_{N,k}^{\text{exp}}(1!p_1, 2!p_2, \dots, (N-k+1)!p_{N-k+1}). \quad (84)$$

Proof. Formula (82) follows by induction on k from [Theorem 6.5](#): the case $k = 0$ is the convention, and

$$[t^N]F^{k+1} = \sum_{n+m=N} p_n [t^m]F^k = \sum_{n=0}^N p_n \sum_{\substack{n_1, \dots, n_k \geq 0 \\ n_1 + \dots + n_k = N-n}} p_{n_1} \cdots p_{n_k},$$

which is both (83) and the composition sum for $k + 1$. All sums are finite because every $n_i \geq 0$ and they sum to N . If $p_0 = 0$, every n_i in a nonvanishing term is ≥ 1 , so $N \geq k$.

For (84), recall the defining generating identity of the exponential partial Bell polynomials [4, 3]: in $R[[z]]$ over a commutative $\mathbb{Q}$ -algebra R ,

$$\frac{1}{k!} \left(\sum_{j \geq 1} x_j \frac{z^j}{j!} \right)^k = \sum_{N \geq k} \mathcal{B}_{N,k}^{\text{exp}}(x_1, x_2, \dots) \frac{z^N}{N!}.$$

Apply it with $R = \mathbb{K}[L]$ (a $\mathbb{Q}$ -algebra, as $\text{char } \mathbb{K} = 0$), $z = t$ and $x_j = j!p_j$, so that $\sum_{j \geq 1} x_j z^j / j! = F$. Comparing coefficients of t^N gives $\frac{1}{k!}[t^N]F^k = \frac{1}{N!}\mathcal{B}_{N,k}^{\text{exp}}(1!p_1, 2!p_2, \dots)$, which is (84); only the arguments $x_1, \dots, x_{N-k+1}$ occur because a term of $\mathcal{B}_{N,k}^{\text{exp}}$ is a product of k variables with index sum N , so no index can exceed $N - k + 1$. $\square$

Theorem 6.37 (Binomial powers and their finite determination). *Let $U \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $\nu_t U \geq e$ for an integer $e \geq 1$, write $u_n = [t^n]U$, and let $\sigma \in \mathbb{K}$. Put $\binom{\sigma}{r} = \sigma(\sigma-1) \cdots (\sigma-r+1)/r! \in \mathbb{K}$, with $\binom{\sigma}{0} = 1$. Then:*

1. The family $(\binom{\sigma}{r} U^r)_{r \geq 0}$ is summable, so

$$(1+U)^\sigma := \sum_{r \geq 0} \binom{\sigma}{r} U^r \quad (85)$$

is a well-defined element of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with constant block 1.

2. (Finite determination.) For $N \geq 0$,

$$[t^N](1+U)^\sigma = \sum_{r=0}^{\lfloor N/e \rfloor} \binom{\sigma}{r} [t^N]U^r, \quad (86)$$

and this depends only on the blocks $u_e, u_{e+1}, \dots, u_N$, that is, only on $\text{Trunc}_{<N+1} U$. The range of r is exact: for $U = t^e$ with $e \mid N$ the term $r = N/e$ contributes $\binom{\sigma}{N/e}$, nonzero whenever $\sigma \notin \{0, 1, \dots, N/e - 1\}$. The top block enters exactly linearly: for $N \geq 1$,

$$[t^N](1+U)^\sigma = \sigma u_N + \Psi_N(u_e, \dots, u_{N-e}), \quad (87)$$

where Ψ_N is a universal polynomial with coefficients in $\mathbb{Q}[\sigma]$, with the convention that $\Psi_N = 0$ when the argument list is empty. Hence for $\sigma \neq 0$ the block u_N cannot be omitted from the dependency range.

3. (Exponent law.) For $\sigma, \tau \in \mathbb{K}$,

$$(1+U)^\sigma(1+U)^\tau = (1+U)^{\sigma+\tau}, \quad (1+U)^0 = 1, \quad (88)$$

and for $k \in \mathbb{N}_0$ the series $(1+U)^k$ agrees with the k -fold product. Consequently $(1+U)^\sigma$ is a unit of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with inverse $(1+U)^{-\sigma}$, and $\sigma \mapsto (1+U)^\sigma$ is a homomorphism from $(\mathbb{K}, +)$ to the units of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.

4. (Profile.) If $d_U(n) \leq \alpha n + \beta$ for all $n \geq e$, with $\alpha \geq 0$, then, writing $\beta^+ = \max(\beta, 0)$,

$$d_{(1+U)^\sigma}(N) \leq \left(\alpha + \frac{\beta^+}{e}\right)N \quad (N \geq 0). \quad (89)$$

In particular $\mathfrak{A} \cap \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is stable under $U \mapsto (1+U)^\sigma$ whenever $\nu_t U \geq 1$.

Proof. (1) By (72), $\nu_t(U^r) \geq re$, which tends to $+\infty$; so for each N only the finitely many $r \leq N/e$ can have $\nu_t(\binom{\sigma}{r}U^r) < N$. Theorem 6.34(1) gives a well-defined sum, and $\nu_t U \geq e \geq 1$ makes the $r = 0$ term the only contributor to t^0 .

(2) Blockwise, $[t^N](1+U)^\sigma = \sum_{r \geq 0} \binom{\sigma}{r}[t^N]U^r$, and $[t^N]U^r = 0$ once $re > N$; this is (86). By Theorem 6.12 applied $r-1$ times, $[t^N]U^r$ depends only on the blocks u_n with $e \leq n \leq N - (r-1)e$, and for $r \geq 1$ this range is contained in $[e, N]$. For the sharpness of the r -range take $U = t^e$: then $U^r = t^{re}$ and the only contribution to t^N comes from $r = N/e$, with value $\binom{\sigma}{N/e}$, which vanishes exactly when σ is one of $0, 1, \dots, N/e - 1$. For (87), the term $r = 0$ contributes nothing to t^N for $N \geq 1$, the term $r = 1$ contributes σu_N , and each term with $r \geq 2$ involves only u_n with $n \leq N - (r-1)e \leq N - e$.

(3) By Theorem 6.34(2),

$$(1+U)^\sigma(1+U)^\tau = \sum_{r,s \geq 0} \binom{\sigma}{r} \binom{\tau}{s} U^{r+s} = \sum_{m \geq 0} \left(\sum_{r+s=m} \binom{\sigma}{r} \binom{\tau}{s} \right) U^m,$$

the regrouping being legitimate because for each fixed t -block only finitely many pairs (r, s) contribute. Vandermonde's convolution [18]

$$\sum_{r+s=m} \binom{\sigma}{r} \binom{\tau}{s} = \binom{\sigma+\tau}{m}$$

holds for all $\sigma, \tau \in \mathbb{K}$: both sides are polynomial expressions in (σ, τ) with coefficients in $\mathbb{Q}$; they agree for all pairs of nonnegative integers by comparing coefficients in $(1+z)^\sigma(1+z)^\tau = (1+z)^{\sigma+\tau}$ in $\mathbb{Q}[z]$; a polynomial in two variables over the infinite domain $\mathbb{Q}$ vanishing on $\mathbb{N}_0^2$ is the zero polynomial; and $\mathbb{Q} \subseteq \mathbb{K}$ because $\text{char } \mathbb{K} = 0$, so the identity persists in $\mathbb{K}$. This proves (88); $(1+U)^0 = 1$ is immediate. For $k \in \mathbb{N}_0$ we have $\binom{k}{r} = 0$ for $r > k$, so (85) is the finite binomial expansion of the k -fold product in the commutative ring $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Invertibility follows by taking $\tau = -\sigma$.

(4) By (86) it suffices to bound $\deg_L[t^N]U^r$ for $1 \leq r \leq \lfloor N/e \rfloor$ (the term $r = 0$ contributes only at $N = 0$, where the bound reads $0 \leq 0$). By (82) and (57),

$$\deg_L[t^N]U^r \leq \max_{\substack{n_1 + \dots + n_r = N \\ n_i \geq e}} \sum_{i=1}^r (\alpha n_i + \beta) = \alpha N + r\beta \leq \alpha N + \lfloor N/e \rfloor \beta^+ \leq \left(\alpha + \frac{\beta^+}{e}\right)N,$$

where the second inequality uses $r \leq \lfloor N/e \rfloor$ when $\beta \geq 0$ and $r\beta \leq 0 \leq \lfloor N/e \rfloor \beta^+$ when $\beta < 0$. Taking the maximum over r gives (89). $\square$

Corollary 6.38 (Formal roots). *With U as above and $p, q \in \mathbb{Z}, q \geq 1$,*

$$((1 + U)^{p/q})^q = (1 + U)^p,$$

the right-hand side being the ordinary p -th power (or the reciprocal of the $|p|$ -th power if $p < 0$). In particular $(1 + U)^{1/q}$ is a q -th root of $1 + U$ in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and it is the unique one with constant block 1.

Proof. Iterating (88) $q - 1$ times gives $((1 + U)^{p/q})^q = (1 + U)^{qp/q} = (1 + U)^p$, and Theorem 6.37(3) identifies the right-hand side with the ordinary power. For uniqueness, suppose $V, W \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ have constant block 1 and $V^q = W^q = 1 + U$. Then $(V - W) \sum_{i=0}^{q-1} V^i W^{q-1-i} = 0$; the second factor has constant block $q \neq 0$ (characteristic zero), hence is nonzero, and $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is a domain by Theorem 6.22, so $V = W$. $\square$

Remark 6.39 (Infinite products). If $(U_r)_{r \geq 1}$ lies in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $\nu_t U_r \rightarrow +\infty$, the partial products $P_R = \prod_{r \leq R} (1 + U_r)$ converge t -adically. Indeed, given N choose R_N with $\nu_t U_r \geq N$ for $r > R_N$; for $R > R_N$,

$$P_R - P_{R_N} = P_{R_N} \left(\prod_{R_N < r \leq R} (1 + U_r) - 1 \right),$$

and expanding the finite product shows that the bracket is a sum of products of at least one U_r with $r > R_N$, hence of $\nu_t \geq N$; since $\nu_t P_{R_N} \geq 0$ we get $\nu_t (P_R - P_{R_N}) \geq N$. Thus (P_R) is t -adically Cauchy in the complete ring $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $\prod_{r \geq 1} (1 + U_r)$ is well defined. On $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ the same argument applies once one notes that $\nu_t U_r \rightarrow +\infty$ forces all but finitely many factors to lie in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, so the partial products have a common lower support bound.

6.6 Three worked computations

Example 6.40 (A Laurent product through four blocks, with degree bookkeeping). Over $\mathbb{K} = \mathbb{Q}$ take

$$\begin{aligned} F &= 3t^{-2} + (L - 1)t^{-1} + (L^2 + 2) + 2Lt + (L^2 - L)t^2 + \mathcal{O}_t^{\text{form}}(t^3), \\ G &= t^{-1} + (2 - L) + (L + 1)t + L^2t^2 + \mathcal{O}_t^{\text{form}}(t^3). \end{aligned}$$

Here $\nu_t F = -2$ and $\nu_t G = -1$; F is known through $\text{Trunc}_{<3} F$ and G through $\text{Trunc}_{<3} G$. By the precision rule the product is known through

$$\text{Trunc}_{<\min(3+(-1), 3+(-2))} FG = \text{Trunc}_{<1} FG,$$

that is, through the block t^0 inclusive — four blocks in all, starting at t^{-3} . The extra block $[t^2]F$ is wasted: it can influence only $[t^1](FG)$ and beyond, for which $[t^3]G$ would also be needed.

Writing $A_n = [t^n]F$ and $B_m = [t^m]G$, the convolution (60) gives

$$\begin{aligned} C_{-3} &= A_{-2}B_{-1} = 3, \\ C_{-2} &= A_{-2}B_0 + A_{-1}B_{-1} = 3(2 - L) + (L - 1) = 5 - 2L, \\ C_{-1} &= A_{-2}B_1 + A_{-1}B_0 + A_0B_{-1} \\ &= 3(L + 1) + (-L^2 + 3L - 2) + (L^2 + 2) = 6L + 3, \\ C_0 &= A_{-2}B_2 + A_{-1}B_1 + A_0B_0 + A_1B_{-1} \\ &= 3L^2 + (L^2 - 1) + (-L^3 + 2L^2 - 2L + 4) + 2L = -L^3 + 6L^2 + 3. \end{aligned}$$

Hence

$$FG = 3t^{-3} + (5 - 2L)t^{-2} + (6L + 3)t^{-1} + (-L^3 + 6L^2 + 3) + \mathcal{O}_t^{\text{form}}(t^1). \quad (90)$$

Leading data: $\text{lm}(F) = t^{-2}$ with $\text{lc}(F) = 3$, $\text{lm}(G) = t^{-1}$ with $\text{lc}(G) = 1$; [Theorem 6.21](#) predicts $\text{lm}(FG) = t^{-3}$ and $\text{lc}(FG) = 3$, confirmed by $C_{-3} = 3$.

The degree bookkeeping is instructive. The profiles are $d_F(-2) = 0$, $d_F(-1) = 1$, $d_F(0) = 2$, $d_F(1) = 1$ and $d_G(-1) = 0$, $d_G(0) = 1$, $d_G(1) = 1$, $d_G(2) = 2$. Then:

N	maximizing pairs P_N	bound M	top coefficient (79)	$d_{FG}(N)$
-3	$(-2, -1)$	0	$3 \cdot 1 = 3$	0
-2	$(-2, 0), (-1, -1)$	1	$3(-1) + 1 \cdot 1 = -2$	1
-1	$(-1, 0), (0, -1)$	2	$1 \cdot (-1) + 1 \cdot 1 = 0$	$1 < 2$
0	$(0, 0)$	3	$1 \cdot (-1) = -1$	3

At $N = -2$ two pairs compete and their contributions -3 and $+1$ do not cancel, so the bound is attained, with top coefficient -2 : indeed $C_{-2} = -2L + 5$. At $N = -1$ two pairs compete and cancel exactly, so the degree drops from the bound 2 to the actual value 1: indeed $C_{-1} = 6L + 3$. At $N = 0$ the maximizing pair is unique, so [Theorem 6.26\(1\)](#) forces attainment, with top coefficient -1 : indeed $C_0 = -L^3 + \dots$. This is exactly the phenomenon [Theorem 6.25](#) isolates; no inspection of the finished polynomial is needed to predict it.

Example 6.41 (A square root through four blocks, verified by squaring). Let

$$U = 2Lt + (L^2 + 2)t^2 + 4Lt^3 + \mathcal{O}_t^{\text{form}}(t^4) \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}},$$

so $e = 1$ in [Theorem 6.37](#), and take $\sigma = \frac{1}{2}$, with

$$\binom{1/2}{0} = 1, \quad \binom{1/2}{1} = \frac{1}{2}, \quad \binom{1/2}{2} = -\frac{1}{8}, \quad \binom{1/2}{3} = \frac{1}{16}.$$

By (86) only $r \leq N$ contributes to t^N , so through t^3 we need

$$U^2 = 4L^2t^2 + (4L^3 + 8L)t^3 + \mathcal{O}_t^{\text{form}}(t^4), \quad U^3 = 8L^3t^3 + \mathcal{O}_t^{\text{form}}(t^4).$$

Block by block:

$$\begin{aligned} [t^1](1 + U)^{1/2} &= \frac{1}{2}(2L) = L, \\ [t^2](1 + U)^{1/2} &= \frac{1}{2}(L^2 + 2) - \frac{1}{8}(4L^2) = \frac{L^2}{2} + 1 - \frac{L^2}{2} = 1, \\ [t^3](1 + U)^{1/2} &= \frac{1}{2}(4L) - \frac{1}{8}(4L^3 + 8L) + \frac{1}{16}(8L^3) = 2L - \frac{L^3}{2} - L + \frac{L^3}{2} = L. \end{aligned}$$

Hence

$$(1 + U)^{1/2} = 1 + Lt + t^2 + Lt^3 + \mathcal{O}_t^{\text{form}}(t^4). \quad (91)$$

Two logarithmic cancellations occur, one at each of t^2 and t^3 ; the degree bound (89) with $\alpha = 1$, $\beta^+ = 0$, $e = 1$ predicts only $d_{(1+U)^{1/2}}(N) \leq N$, and the actual profile $0, 1, 0, 1$ is strictly smaller at $N = 2, 3$. A degree bound is an upper bound; it is not a prediction.

Verification. [Theorem 6.38](#) says the square of (91) must reproduce $1 + U$ through t^3 . Squaring by (60):

$$[t^0] = 1, \quad [t^1] = 2L, \quad [t^2] = 2 \cdot 1 + L^2 = L^2 + 2, \quad [t^3] = 2L + 2L = 4L,$$

which are exactly the blocks $1, u_1, u_2, u_3$ of $1 + U$. Note the precision accounting: squaring a series known through $\text{Trunc}_{<4} \cdot$ with $\nu_t = 0$ returns a result known through $\text{Trunc}_{<4} \cdot$, so the check is available at every block computed and at none beyond.

Example 6.42 (A product in the iterated field). In $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ the coefficient convolution is finite at the inner level too. Take the two blocks

$$f = \frac{1}{L+1} = L^{-1} - L^{-2} + L^{-3} - \dots, \quad g = \frac{1}{L-1} = L^{-1} + L^{-2} + L^{-3} + \dots$$

in $\mathbb{K}((L^{-1}))$. For $r \geq 2$, the coefficient of L^{-r} in fg is a sum over the finitely many decompositions $r = i + j$ with $i, j \geq 1$:

$$\kappa_{-r}(fg) = \sum_{i=1}^{r-1} (-1)^{i+1} \cdot 1 = \begin{cases} 1, & r \text{ even,} \\ 0, & r \text{ odd,} \end{cases}$$

so $fg = L^{-2} + L^{-4} + L^{-6} + \dots = (L^2 - 1)^{-1}$, as it must be. The degree law is visible: $\deg_L f = \deg_L g = -1$, $\deg_L(fg) = -2$, and $\text{lc}_L(f) = \text{lc}_L(g) = \text{lc}_L(fg) = 1$. Consequently, for $F = f t^{-1} + \dots$ and $G = g + \dots$ in $\mathcal{PL}_{\text{itLaur}, \mathbb{K}}$ one gets $\text{lm}(FG) = t^{-1}L^{-2}$ and $\text{lc}(FG) = 1$ by [Theorem 6.21](#), with no need to compute any further block. Observe that lm here carries a *negative* power of L : this is legitimate in $\mathcal{PL}_{\text{rat}, \mathbb{K}}$ and $\mathcal{PL}_{\text{itLaur}, \mathbb{K}}$ and impossible in $\mathcal{PL}_{\text{poly}, \mathbb{K}}$, which is exactly the content of [Theorem 6.22](#).

6.7 The analytic product rule

Everything above is formal algebra. We now record the exact circumstances under which the block convolution computes the asymptotic expansion of an actual product of functions. Nothing in this subsection asserts convergence: an asymptotic expansion is a statement about remainders, and the formal product theorem is used as an input, not replaced.

Definition 6.43 (Polynomial-logarithmic asymptotic expansion). Let $\mathbb{K} \subseteq \mathbb{C}$, let $X_0 > 1$, and let $S = [X_0, +\infty)$ with the real branch $L = \log X$. A function $f: S \rightarrow \mathbb{C}$ is *PL-asymptotic* to $F \in \mathcal{PL}_{\text{poly}, \mathbb{K}}$, written $f \approx F$, if for every $N \in \mathbb{Z}$ there exist $M_N \in \mathbb{N}_0$, $C_N > 0$ and $X_N \geq X_0$ such that

$$|f(X) - (\text{Trunc}_{<N} F)(X)| \leq C_N X^{-N} (\log X)^{M_N} \quad (X \geq X_N), \quad (92)$$

where $(\text{Trunc}_{<N} F)(X) = \sum_{\nu_t F \leq n < N} p_n(\log X) X^{-n}$ is the actual finite sum obtained by substituting the generators. Equivalently, $f - \text{Trunc}_{<N} F = \mathcal{O}_{X \rightarrow +\infty, X \in S}(X^{-N} (\log X)^{M_N})$.

Lemma 6.44 (Basic properties of the expansion relation $\approx$). Let $f, g: S \rightarrow \mathbb{C}$ and $F, G \in \mathcal{PL}_{\text{poly}, \mathbb{K}}$.

1. If (92) holds for arbitrarily large N , it holds for every N ; so it suffices to verify it along a cofinal set of cutoffs.
2. $f \approx F$ and $f \approx F'$ imply $F = F'$.
3. If $f \approx F$ and $g \approx G$, then $f + g \approx F + G$ and $\lambda f \approx \lambda F$ for $\lambda \in \mathbb{K}$.
4. $f \approx 0$ if and only if $f = \mathcal{O}_{X \rightarrow +\infty}(X^{-N})$ for every N .

Proof. (1) Let $N' < N$ and suppose (92) holds at N . Then

$$f - \text{Trunc}_{<N'} F = (f - \text{Trunc}_{<N} F) + \sum_{N' \leq n < N} p_n(\log X) X^{-n},$$

a finite sum; the first term is $\mathcal{O}_{X \rightarrow +\infty}(X^{-N} (\log X)^{M_N}) \subseteq \mathcal{O}_{X \rightarrow +\infty}(X^{-N'} (\log X)^{M_N})$ since $N \geq N'$, and each term of the sum is $\mathcal{O}_{X \rightarrow +\infty}(X^{-N'} (\log X)^{\deg_L p_n})$.

(2) Suppose $D = F - F' \neq 0$, with $n_1 = \nu_t D$ and leading block p of degree d and leading coefficient $c \neq 0$. Take $N = n_1 + 1$ and subtract the two instances of (92):

$$(\text{Trunc}_{<n_1+1} D)(X) = p(\log X) X^{-n_1} = \mathcal{O}_{X \rightarrow +\infty}(X^{-n_1-1} (\log X)^M)$$

for some M . But $|p(\log X)| \sim |c|(\log X)^d$, so the left side divided by $X^{-n_1-1} (\log X)^M$ is $\sim |c|X(\log X)^{d-M} \rightarrow +\infty$, a contradiction.

(3) Immediate from $\text{Trunc}_{<N} F + G = \text{Trunc}_{<N} F + \text{Trunc}_{<N} G$ and the triangle inequality.

(4) If $f \approx 0$ then $\text{Trunc}_{<N} 0 = 0$ and $f = \mathcal{O}_{X \rightarrow +\infty}(X^{-N} (\log X)^{M_N})$ for every N ; since $(\log X)^{M_N} = o_{X \rightarrow +\infty}(X)$, this gives $f = \mathcal{O}_{X \rightarrow +\infty}(X^{-N+1})$ for every N , hence $f = \mathcal{O}_{X \rightarrow +\infty}(X^{-N})$ for every N . The converse is trivial. $\square$

Theorem 6.45 (Analytic product rule). *Let $\mathbb{K} \subseteq \mathbb{C}$, $S = [X_0, +\infty)$ with the real logarithm, and let $f, g: S \rightarrow \mathbb{C}$ with $f \approx F$ and $g \approx G$ for $F, G \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. Then*

$$fg \approx FG,$$

where FG is the formal Cauchy product (60). Consequently the set $\mathcal{A}(S) = \{f : \exists F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}, f \approx F\}$ is a $\mathbb{K}$ -algebra, the map $f \mapsto F$ is a well-defined $\mathbb{K}$ -algebra homomorphism $\mathcal{A}(S) \rightarrow \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, and its kernel is the ideal of functions that are $\mathcal{O}_{X \rightarrow +\infty}(X^{-N})$ for every N .

Proof. First dispose of the degenerate case. Any g with $g \approx G$ satisfies $g = \mathcal{O}_{X \rightarrow +\infty}(X^c(\log X)^m)$ for some $c \in \mathbb{Z}$ and $m \in \mathbb{N}_0$: take $N = 1$ in (92), note that $\text{Trunc}_{<1} G$ is a finite sum of monomials $t^n L^j$, and set $c = \max(-\nu_t G, -1)$ (with $c = -1$ if $G = 0$). Hence, if $F = 0$, then Theorem 6.44(4) gives $f = \mathcal{O}_{X \rightarrow +\infty}(X^{-P})$ for every P , so $fg = \mathcal{O}_{X \rightarrow +\infty}(X^{-P+c}(\log X)^m)$ for every P , whence $fg \approx 0 = FG$ by Theorem 6.44(4) again; the case $G = 0$ is symmetric. Assume from now on $F, G \neq 0$ and set $a = \nu_t F$, $b = \nu_t G$.

Fix $N \in \mathbb{Z}$ and put

$$P = \max(N - b, a), \quad Q = \max(N - a, b).$$

Write $f_P = (\text{Trunc}_{<P} F)(X)$, $r = f - f_P$, $g_Q = (\text{Trunc}_{<Q} G)(X)$, $s = g - g_Q$. By hypothesis $r = \mathcal{O}_{X \rightarrow +\infty}(X^{-P}(\log X)^{m_1})$ and $s = \mathcal{O}_{X \rightarrow +\infty}(X^{-Q}(\log X)^{m_2})$ for suitable exponents. Since $\text{Trunc}_{<P} F$ is a finite sum of blocks with outer exponents $\geq a$,

$$f_P = \mathcal{O}_{X \rightarrow +\infty}(X^{-a}(\log X)^{m_3}), \quad g_Q = \mathcal{O}_{X \rightarrow +\infty}(X^{-b}(\log X)^{m_4})$$

with $m_3 = \max_{a \leq n < P} \deg_L p_n$ and similarly for m_4 (the estimates being vacuously true if the sums are empty). Therefore

$$fg - f_P g_Q = f_P s + r g_Q + r s$$

is, term by term,

$$\begin{aligned} & \mathcal{O}_{X \rightarrow +\infty}(X^{-(a+Q)}(\log X)^{m_3+m_2}) + \mathcal{O}_{X \rightarrow +\infty}(X^{-(P+b)}(\log X)^{m_1+m_4}) \\ & + \mathcal{O}_{X \rightarrow +\infty}(X^{-(P+Q)}(\log X)^{m_1+m_2}). \end{aligned}$$

Now $a + Q \geq a + (N - a) = N$, $P + b \geq (N - b) + b = N$, and $P + Q \geq a + (N - a) = N$; hence

$$fg - f_P g_Q = \mathcal{O}_{X \rightarrow +\infty}(X^{-N}(\log X)^{M'}) \tag{93}$$

for some M' .

Next, $f_P g_Q$ is the value at X of the finite polynomial–logarithmic expression $(\text{Trunc}_{<P} F)(\text{Trunc}_{<Q} G) \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, because substitution of the generators $t = X^{-1}$, $L = \log X$ into a finite sum of monomials is a ring homomorphism into functions on S . Since $P \geq N - b$ and $Q \geq N - a$, Theorem 6.13 gives

$$\text{Trunc}_{<N}(\text{Trunc}_{<P} F)(\text{Trunc}_{<Q} G) = \text{Trunc}_{<N} FG.$$

Hence $(\text{Trunc}_{<P} F)(\text{Trunc}_{<Q} G) - \text{Trunc}_{<N} FG$ is a finite sum of monomials $c t^n L^j$ with $n \geq N$, so its value at X is $\mathcal{O}_{X \rightarrow +\infty}(X^{-N}(\log X)^{M''})$ with M'' the largest j occurring. Combining with (93),

$$fg - (\text{Trunc}_{<N} FG)(X) = \mathcal{O}_{X \rightarrow +\infty}(X^{-N}(\log X)^{\max(M', M'')}),$$

which is (92) for fg and FG . As N was arbitrary, $fg \approx FG$.

The algebra statements now follow: $\mathcal{A}(S)$ is closed under $+$, scalars and $\cdot$ by Theorem 6.44(3) and the above; the assignment $f \mapsto F$ is well defined by Theorem 6.44(2) and is a homomorphism by the same two statements; its kernel is described by Theorem 6.44(4). $\square$

Remark 6.46 (Which hypotheses are actually used, and the sectorial version). The proof uses only three things: that f and g have remainders bounded by X^{-N} times *some* power of $\log X$; that finitely many blocks are involved at each stage; and the formal finite-determination theorem [Theorem 6.12](#), which is what allows the two truncation cutoffs P and Q to be chosen independently of the shape of F and G . No convergence and no uniformity in L is used or obtained.

The same proof works verbatim on an unbounded set $S \subseteq \{z : |\arg z| \leq \theta\}$ with $0 < \theta < \pi$ and the principal logarithm, with X^{-N} replaced by $|X|^{-N}$ and $(\log X)^M$ by $(\log |X|)^M$; the only additional input is $|\log X| \leq \log |X| + \theta \leq 2 \log |X|$ for $|X| \geq e^\theta$, so that a bound in $|\log X|$ and a bound in $\log |X|$ differ only by a constant factor. Uniformity claims on a sector must be stated for a *closed* subsector; the branch of log is part of the statement and cannot be recovered from the formal data.

A common trap

Three cautions, each corresponding to an overstatement found in the sources.

1. *A formal identity is not an analytic theorem.* [Theorem 6.45](#) presupposes that f and g *already* have expansions; it does not produce one. The homomorphism $f \mapsto F$ has a large kernel — $e^{-X} \approx 0$ — so a formal computation determines an actual function at best up to a flat error.
2. *The formal $\mathcal{O}_t^{\text{form}}(t^N)$ carries no uniformity in L .* Asserting $F = G + \mathcal{O}_t^{\text{form}}(t^N)$ says only that the blocks agree below outer order N . In particular the constant C_N and the exponent M_N of (92) are not supplied by any formal statement and must be produced separately.
3. *The three $\mathcal{O}$ -roles are distinct.* The formal tail $\mathcal{O}_t^{\text{form}}(t^N)$, the analytic estimate $\mathcal{O}_{X \rightarrow +\infty}(X^{-N}(\log X)^M)$ of (92), and the complexity bound $\mathcal{O}((N - a - b)^2 d^2)$ of [Theorem 6.16](#) are three different assertions and none follows from the others. A bare $\mathcal{O}$ for an asymptotic claim, with the regime left to context, is never acceptable.

Section summary

Multiplication is Cauchy convolution in $t = X^{-1}$ with exact coefficient arithmetic in $L = \log X$. Every output block below a cutoff is a finite bilinear expression in finitely many input blocks, with the exact range $a \leq n \leq N - b$, $b \leq m \leq N - a$; the resulting precision rule is $\text{Trunc}_{<\min(P+b, Q+a)} FG$ and it cannot be improved. All four ambient classes are closed and are integral domains, the leading monomial and leading scalar are multiplicative, and the rank-two valuation $v(F) = (\nu_t F, -\deg_L p_{\nu_t F})$ is additive. The log-degree profile obeys the max-plus bound $d_{FG} \leq d_F * d_G$, with equality exactly when the maximizing pairs do not cancel; affine profiles are stable under products, powers, and binomial powers, but not under t -adic limits. Binomial powers $(1 + U)^\sigma$ are defined for every $\sigma \in \mathbb{K}$ by a summable family, are finitely determined at every order, and satisfy the exponent law. Under printed remainder hypotheses on a stated domain and branch, the formal product computes the asymptotic expansion of the analytic product — and nothing more.

7 Units, reciprocals, and division

Addition and multiplication never leave $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} = \mathbb{K}[L]((t))$. Division is the first operation that does. This section settles exactly when it does not — the unit criterion — and then follows the three escape routes that the criterion forces: an accidental exact quotient inside $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, the rational-log field $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}} = \mathbb{K}(L)((t))$, and the iterated Laurent-log field $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}} = \mathbb{K}((L^{-1}))((t))$. Each of the three is a genuinely different object, and conflating them is the most frequent error in informal polynomial–logarithmic computation.

Throughout, $\mathbb{K}$ is a field of characteristic zero, $t := X^{-1}$ and $L := \log X$ are independent formal

generators, and a nonzero element of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is written

$$F = \sum_{n \geq n_0} p_n(L)t^n, \quad p_n \in \mathbb{K}[L], \quad p_{n_0} \neq 0, \quad (94)$$

so that $\nu_t F = n_0$. We call $p_{\nu_t F}$ the *leading block* of F ; it is a polynomial, not a scalar, and it must be distinguished from the leading monomial $\text{lm } F = t^{n_0} L^d$ and from the scalar leading coefficient $\text{lc } F \in \mathbb{K}^\times$, where $d = \deg_L p_{n_0}$. Truncation is always the exclusive operator $\text{Trunc}_{<N} F = \sum_{n < N} p_n(L)t^n$, and $F = G + \mathcal{O}_t^{\text{form}}(t^N)$ abbreviates $\nu_t(F - G) \geq N$; it carries no analytic content whatever.

7.1 The exact unit criterion

Two elementary facts do all the work. Both are standard, but the second is the step that three of the six sources compress into a single clause, and it is precisely the step that decides the nonunit direction of the criterion.

Lemma 7.1 (Units of the coefficient ring). *Let $\mathbb{K}$ be a field. Then $\mathbb{K}[L]^\times = \mathbb{K}^\times$: a polynomial is invertible in $\mathbb{K}[L]$ if and only if it is a nonzero constant.*

Proof. A nonzero constant c has inverse c^{-1} . Conversely, suppose $pq = 1$ with $p, q \in \mathbb{K}[L]$. Since $\mathbb{K}$ is a field, $\mathbb{K}[L]$ is an integral domain and $\deg_L(pq) = \deg_L p + \deg_L q$. Hence $\deg_L p + \deg_L q = \deg_L 1 = 0$, and as both degrees are nonnegative integers, $\deg_L p = \deg_L q = 0$. So $p \in \mathbb{K}$, and $p \neq 0$ because $pq = 1$. $\square$

Lemma 7.2 (The Laurent ring is a domain and ν_t is additive). *Let $F, G \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ be nonzero, with leading blocks $p_{\nu_t F}$ and $q_{\nu_t G}$. Then $FG \neq 0$,*

$$\nu_t(FG) = \nu_t F + \nu_t G, \quad (95)$$

and the leading block of FG is $p_{\nu_t F} q_{\nu_t G}$. In particular $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is an integral domain, and the same statements hold in $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, in $\mathbb{K}(L)((t))$, and in $E((t))$ for any integral domain E .

Proof. Write $m = \nu_t F$, $r = \nu_t G$. The Cauchy convolution gives $[t^k](FG) = \sum_{i+j=k} p_i q_j$, a finite sum because $i \geq m$ and $j \geq r$ force $m \leq i \leq k - r$. For $k < m + r$ every term has $i < m$ or $j < r$ and therefore vanishes. For $k = m + r$ the only surviving pair is $(i, j) = (m, r)$, so $[t^{m+r}](FG) = p_m q_r$, which is nonzero because $\mathbb{K}[L]$ is an integral domain. Hence $FG \neq 0$ and both assertions hold. $\square$

Theorem 7.3 (Unit criterion). *Let $F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ be nonzero, written as in Equation (94). The following are equivalent.*

1. F is a unit of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$.
2. The leading block p_{n_0} is a nonzero constant, that is $p_{n_0} \in \mathbb{K}^\times$.
3. $F = c t^{n_0}(1 + U)$ for some $c \in \mathbb{K}^\times$ and some $U \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$.

For $F \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ with $n_0 = 0$, these are equivalent to $p_0 \in \mathbb{K}^\times$, and F^{-1} then lies in $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$.

Proof. (2) $\Leftrightarrow$ (3). If $p_{n_0} = c \in \mathbb{K}^\times$, put $U := c^{-1}t^{-n_0}F - 1$; every block of $c^{-1}t^{-n_0}F$ of index ≥ 1 is a polynomial and the index-0 block is 1, so $U \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. Conversely, in $F = ct^{n_0}(1 + U)$ with $U \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ the block of index n_0 is c .

(3) $\Rightarrow$ (1). Set

$$G := c^{-1}t^{-n_0} \sum_{k \geq 0} (-U)^k. \quad (96)$$

Because $\nu_t(U^k) \geq k$, for each N only the terms with $k < N$ contribute to blocks of index $< N$; the family is therefore formally summable and $G \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, with polynomial blocks because $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ is closed under multiplication. For every $N \geq 0$,

$$\begin{aligned} \text{Trunc}_{<N} \left((1+U) \sum_{k \geq 0} (-U)^k \right) &= \text{Trunc}_{<N} \left((1+U) \sum_{k < N} (-U)^k \right) \\ &= \text{Trunc}_{<N} (1 - (-U)^N) = \text{Trunc}_{<N} 1, \end{aligned}$$

using the telescoping identity $(1+U) \sum_{k < N} (-U)^k = 1 - (-U)^N$ and $\nu_t((-U)^N) \geq N$. Since two elements of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ with equal exclusive truncations at every order are equal, $(1+U) \sum_{k \geq 0} (-U)^k = 1$, whence $FG = 1$.

(1) $\Rightarrow$ (2). Suppose $FG = 1$ with $G \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. Then $G \neq 0$, and [Theorem 7.2](#) gives $\nu_t F + \nu_t G = \nu_t 1 = 0$ together with $p_{\nu_t F} q_{\nu_t G} = 1$ in $\mathbb{K}[L]$, where $q_{\nu_t G}$ is the leading block of G . By [Theorem 7.1](#), $p_{\nu_t F} \in \mathbb{K}^\times$.

The statement for $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ is the case $n_0 = 0$ of the above, together with the observation that the G constructed in [Equation \(96\)](#) lies in $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ when $n_0 = 0$. $\square$

Corollary 7.4 (The unit group). *The group of units of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is*

$$\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}^\times = \{ c t^m (1+U) : m \in \mathbb{Z}, c \in \mathbb{K}^\times, U \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} \}, \quad (97)$$

and the factorisation is unique, so that $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}^\times \cong \mathbb{Z} \times \mathbb{K}^\times \times (1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}})$ as abelian groups. Moreover $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}^\times = \mathbb{K}^\times \cdot (1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}})$.

Proof. The description of the set is [Theorem 7.3](#). Uniqueness: in $F = ct^m(1+U)$ one has $m = \nu_t F$ by [Theorem 7.2](#), then c is the leading block, then $U = c^{-1}t^{-m}F - 1$. The set $1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ is a subgroup of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}^\times$: it is closed under multiplication, and by [Theorem 7.3](#) the inverse of $1+U$ is a unit of $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ with leading block 1, hence again of the form $1+U'$ with $U' \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. Since $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}^\times$ is abelian,

$$\mu : \mathbb{Z} \times \mathbb{K}^\times \times (1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}) \longrightarrow \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}^\times, \quad \mu(m, c, 1+U) = ct^m(1+U),$$

is a group homomorphism. It is surjective by [Theorem 7.3](#) and injective because $\mu(m, c, 1+U) = 1 = 1 \cdot t^0 \cdot (1+0)$ forces $m = 0$, $c = 1$ and $U = 0$ by the uniqueness just proved. Hence μ is an isomorphism. $\square$

Example 7.5 (A nonzero nonunit). The element $L+1 \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ is nonzero, but it is not a unit of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. Indeed $\nu_t(L+1) = 0$ and its leading block is $L+1$, which is not constant; [Theorem 7.3](#) applies. Unwinding the proof gives the direct argument: if $(L+1)G = 1$ with $G = \sum_{n \geq n_1} q_n(L)t^n$ nonzero, then $\nu_t G = 0$ and $(L+1)q_0 = 1$ in $\mathbb{K}[L]$, which is impossible because $\deg_L((L+1)q_0) = 1 + \deg_L q_0 \geq 1$. The same argument shows that $L, L+t, L^2+t^3$ and $t^{-5}(L-7)$ are nonzero nonunits. Their reciprocals exist in $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$:

$$\frac{1}{L+t} = \frac{1}{L} - \frac{t}{L^2} + \frac{t^2}{L^3} - \cdots \in \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} \setminus \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}. \quad (98)$$

Remark 7.6 (The dominant-monomial fallacy). Factoring out the dominant monomial is *not* enough to invert. Writing $L+t = L(1+t/L)$ and expanding the second factor geometrically already commits one to the coefficient ring $\mathbb{K}(L)$ or $\mathbb{K}((L^{-1}))$: the step that leaves $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is the inversion of the scalar-free factor L , not the infinite t -expansion. [Equation \(98\)](#) makes the point in one line – the failure is visible already in the block of index 0.

7.2 Existence, uniqueness, and the reciprocal recurrence

[Equation \(96\)](#) proves existence but is a poor algorithm. The practical construction is triangular.

Theorem 7.7 (Reciprocal of a unit). *Let $B = \sum_{n \geq 0} B_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $B_0 = c \in \mathbb{K}^\times$. Then there is exactly one $C \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $BC = 1$, namely $C = B^{-1} = \sum_{n \geq 0} C_n(L)t^n$ with*

$$C_0 = c^{-1}, \quad (99)$$

$$C_n = -c^{-1} \sum_{j=1}^n B_j C_{n-j} \quad (n \geq 1), \quad (100)$$

and every C_n lies in $\mathbb{K}[L]$. Moreover C is the unique reciprocal of B in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, and for a Laurent unit $F = t^m B$ one has $F^{-1} = t^{-m} C$.

Proof. Existence. Define $C_n \in \mathbb{K}[L]$ by [Equations \(99\) and \(100\)](#); the right-hand side of [Equation \(100\)](#) involves only $C_0, \dots, C_{n-1}$, so the definition is legitimate by strong induction, and each C_n is a $\mathbb{K}$ -linear combination of products of polynomials, hence a polynomial. Put $C := \sum_{n \geq 0} C_n t^n \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. For $n \geq 1$,

$$[t^n](BC) = \sum_{j=0}^n B_j C_{n-j} = cC_n + \sum_{j=1}^n B_j C_{n-j} = 0$$

by [Equation \(100\)](#), and $[t^0](BC) = cC_0 = 1$. Hence $BC = 1$.

Uniqueness. If $BC = \tilde{B}\tilde{C} = 1$ with $C, \tilde{C} \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, then $C = C(\tilde{B}\tilde{C}) = (C\tilde{B})\tilde{C} = \tilde{C}$. This argument uses only associativity and commutativity, so uniqueness holds in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ – and in any commutative ring – not merely in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.

Laurent case. t^m is a unit with inverse t^{-m} , so $(t^m B)(t^{-m} C) = BC = 1$.

Generality. Nothing above uses more about the coefficient ring than that it is commutative and that B_0 is invertible in it. Hence for any commutative ring R and any $B \in R[[t]]$ with $B_0 \in R^\times$, the same recurrence produces the unique reciprocal of B in $R[[t]]$, and that reciprocal remains the unique one in $R((t))$. $\square$

Corollary 7.8 (Quotient recurrence). *Let E be a field, $A = \sum_{n \geq 0} A_n t^n$ and $B = \sum_{n \geq 0} B_n t^n$ in $E[[t]]$ with $B_0 \neq 0$. Then $Q = A/B = \sum_{n \geq 0} Q_n t^n \in E[[t]]$ is the unique solution of $BQ = A$, and*

$$Q_n = B_0^{-1} \left(A_n - \sum_{j=1}^n B_j Q_{n-j} \right), \quad n \geq 0, \quad (101)$$

the sum being empty when $n = 0$. When $E = \mathbb{K}[L]$ is used instead of a field, [Equation \(101\)](#) still produces polynomial blocks provided $B_0 \in \mathbb{K}^\times$, and then $Q \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.

Proof. Existence and uniqueness follow from [Theorem 7.7](#) applied over the coefficient ring E (or $\mathbb{K}[L]$, where $B_0 \in \mathbb{K}^\times$ is invertible): $Q := A B^{-1}$ solves $BQ = A$, and if $BQ = BQ'$ then $Q = Q'$ after multiplying by B^{-1} . Comparing coefficients of t^n in $BQ = A$ gives $B_0 Q_n + \sum_{j=1}^n B_j Q_{n-j} = A_n$, which is [Equation \(101\)](#). Since [Equation \(101\)](#) determines Q_n from $Q_0, \dots, Q_{n-1}$, it also proves uniqueness directly. $\square$

The next statement is the one an implementation actually needs: it says that a finite computation with truncated inputs returns exactly the truncation of the true quotient, and that the truncated quotient is characterised without reference to the recurrence at all.

Theorem 7.9 (Correctness of truncated division). *Let $B \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $B_0 \in \mathbb{K}^\times$, let $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and let $N \geq 0$. Let $Q_0, \dots, Q_{N-1}$ be produced by [Equation \(101\)](#) and set $P := \sum_{n < N} Q_n t^n$. Then*

$$BP = A + \mathcal{O}_t^{\text{form}}(t^N), \quad (102)$$

and P is the only element of $\mathbb{K}[L][t]$ of t -degree $< N$ satisfying Equation (102). Consequently $P = \text{Trunc}_{<N}(A/B)$.

Proof. For $n < N$ the coefficient $[t^n](BP) = \sum_{j=0}^n B_j Q_{n-j}$ involves only $Q_0, \dots, Q_n$, all of which are among the computed blocks, and it equals A_n by Equation (101). This is Equation (102).

Uniqueness: if P' is another such polynomial, put $D = P - P'$, a polynomial in t of degree $< N$ with $\nu_t(BD) \geq N$. If $D \neq 0$ then Theorem 7.2 gives $\nu_t(BD) = \nu_t B + \nu_t D = \nu_t D < N$, a contradiction; so $D = 0$.

Finally, $B(A/B - P) = A - BP$ has $\nu_t \geq N$, and $\nu_t B = 0$, so $\nu_t(A/B - P) \geq N$ by Theorem 7.2. As P has t -degree $< N$, this says exactly $P = \text{Trunc}_{<N}(A/B)$. $\square$

Remark 7.10 (Truncation convention). Report S6 states Theorem 7.9 in the inclusive form

$$B \sum_{n \leq N} Q_n t^n = A + \mathcal{O}_t^{\text{form}}(t^{N+1}).$$

Because $[F]_{\leq N} = \text{Trunc}_{<N+1} F$, that statement is the case $N \mapsto N + 1$ of the one above. Anyone transferring a precision budget between the two conventions must account for the one-block shift; it is the commonest silent off-by-one in this subject.

Proposition 7.11 (Exact input requirement: relative precision is preserved). *Let E be a field, and let $A, B \in E((t))$ with $A \neq 0 \neq B$ (the case $A = 0$ being trivial). Put $a = \nu_t A$ and $b = \nu_t B$, so that $\nu_t(A/B) = a - b$. For every $M \geq 0$, the first M blocks of A/B , namely the coefficients of $t^{a-b}, \dots, t^{a-b+M-1}$, are determined by the first M blocks of A and the first M blocks of B , and by nothing else. Equivalently, in absolute cutoffs,*

$$\text{Trunc}_{<N}(A/B) \text{ is determined by } \text{Trunc}_{<N+b} A \text{ and } \text{Trunc}_{<N+2b-a} B. \quad (103)$$

Proof. Write $\tilde{B} = t^{-b} B$, so $\nu_t \tilde{B} = 0$ and $\tilde{B}_j = B_{j+b}$. Then $A/B = t^{-b} (A/\tilde{B})$, and $G := A/\tilde{B}$ has $\nu_t G = a$. Applying Equation (101) to G after shifting indices by a , the block G_{a+i} is a function of $A_a, \dots, A_{a+i}$ and $\tilde{B}_0, \dots, \tilde{B}_i$ alone. Taking $i < M$ gives the first assertion, since $\tilde{B}_0, \dots, \tilde{B}_{M-1}$ are the first M blocks of B . For Equation (103), note first that $\text{Trunc}_{<N}(A/B) = 0$ when $N \leq a - b$, so assume $N > a - b$. The requested blocks of A/B are then those of index $< N$, that is the first $M = N - (a - b)$ blocks; these need A through index $a + M - 1 = N + b - 1$ and B through index $b + M - 1 = N + 2b - a - 1$. $\square$

Remark 7.12 (Guard precision). Theorem 7.11 is the precise form of the informal rule that a quotient must be computed “beyond the requested block”. The guard is $b = \nu_t B$ blocks on the numerator and $2b - a$ on the denominator; it is zero exactly when $a = b = 0$, and it is what a symbolic system must propagate through an expression tree instead of a single global cutoff. A negative guard means that fewer input blocks suffice. On the numerator this happens exactly when $b < 0$, that is when the denominator has a Laurent principal part; on the denominator it happens exactly when $a > 2b$, for instance when a numerator of high t -order is divided by a unit.

Example 7.13 (A reciprocal computed to five blocks). Let

$$B = 1 + (L + 1)t + L^2 t^2 + (L - 2)t^3 \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}. \quad (104)$$

Here $B_0 = 1 \in \mathbb{K}^\times$ and $B_n = 0$ for $n \geq 4$. Equations (99) and (100) give

$$\begin{aligned} C_0 &= 1, \\ C_1 &= -(L + 1), \\ C_2 &= -((L + 1)C_1 + L^2 C_0) = 2L + 1, \\ C_3 &= -((L + 1)C_2 + L^2 C_1 + (L - 2)C_0) = L^3 - L^2 - 4L + 1, \\ C_4 &= -((L + 1)C_3 + L^2 C_2 + (L - 2)C_1) = -L^4 - 2L^3 + 5L^2 + 2L - 3, \end{aligned}$$

so that

$$B^{-1} = 1 - (L+1)t + (2L+1)t^2 + (L^3 - L^2 - 4L + 1)t^3 + (-L^4 - 2L^3 + 5L^2 + 2L - 3)t^4 + \mathcal{O}_t^{\text{form}}(t^5). \quad (105)$$

Multiplying Equation (104) by Equation (105) and truncating at t^5 returns 1; by Theorem 7.9 this residual check certifies the five blocks independently of the method used to produce them.

Example 7.14 (A quotient with polynomial blocks). With

$$A = 1 + (L+1)t + (2L^2 - L)t^2 + \mathcal{O}_t^{\text{form}}(t^3), \quad B = 1 + (2-L)t + (L^2 + 3)t^2 + \mathcal{O}_t^{\text{form}}(t^3),$$

Equation (101) gives $Q_0 = 1$, $Q_1 = (L+1) - (2-L) = 2L-1$, and

$$Q_2 = (2L^2 - L) - (2-L)(2L-1) - (L^2 + 3) = 3L^2 - 6L - 1,$$

so $A/B = 1 + (2L-1)t + (3L^2 - 6L - 1)t^2 + \mathcal{O}_t^{\text{form}}(t^3)$. The block Q_3 is *not* determined by the displayed data: Theorem 7.11 with $a = b = 0$ says three input blocks determine exactly three output blocks. If one additionally posits $A_3 = B_3 = 0$, then $Q_3 = L^3 - 11L^2 + 5L + 5$.

Do not read a block that the data does not contain

An input written with a formal tail $\mathcal{O}_t^{\text{form}}(t^3)$ determines the quotient only through t^2 . Continuing the recurrence past that point silently substitutes the assumption “the omitted blocks are zero”. Both S1 and S6 print one extra block of a worked example on exactly this unstated convention; when it is intended it must be said.

Proposition 7.15 (Degree profile of a reciprocal and of a quotient). *Let $\alpha \geq 0$ and $\beta \geq 0$ be real. Suppose $B \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ has $B_0 \in \mathbb{K}^\times$ and $\deg_L B_n \leq \alpha n$ for all $n \geq 1$. Then the blocks of B^{-1} satisfy*

$$\deg_L C_n \leq \alpha n \quad (n \geq 0). \quad (106)$$

If moreover $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ has $\deg_L A_n \leq \alpha n + \beta$, then the quotient blocks satisfy $\deg_L Q_n \leq \alpha n + \beta$.

Proof. Strong induction on n . For $n = 0$, $C_0 = B_0^{-1} \in \mathbb{K}$ has degree $0 \leq 0$. For $n \geq 1$, each summand of Equation (100) obeys $\deg_L(B_j C_{n-j}) \leq \alpha j + \alpha(n-j) = \alpha n$ by the inductive hypothesis and the additivity of $\deg_L$ on products in $\mathbb{K}[L]$; a finite sum cannot raise the bound, and multiplying by the scalar $-B_0^{-1}$ changes nothing. The quotient case is identical, with the extra term A_n of degree $\leq \alpha n + \beta$ and $\deg_L(B_j Q_{n-j}) \leq \alpha j + \alpha(n-j) + \beta$. $\square$

Remark 7.16. The bound in Equation (106) may be strict: cancellation can lower a block’s degree, and it does so in Theorem 7.13, where $\deg_L B_2 = 2$ and $\deg_L C_2 = 1$. An affine profile $d_F(n) \leq \alpha n + \beta$ is a hypothesis or a proved invariant, never part of the definition of the ring; the Lambert expansion is the archetype with $\alpha = 1$, $\beta = 0$.

7.3 Geometric inversion and two-level summability

The geometric formula Equation (96) is the conceptual construction, and in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ its justification is purely t -adic: $\nu_t(U^k) \geq k \rightarrow \infty$. Over a completed logarithmic coefficient field this reason is no longer available, and a second level of the argument is needed. Several sources normalise a divisor in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ as $B = b t^\nu L^\mu (1 + S)$ with $S \prec 1$ and then assert summability of $\sum_k (-S)^k$ “by the support conditions”. That is true but not for the t -adic reason: S may have $\nu_t S = 0$, so $\nu_t(S^k) = 0$ for every k . The correct statement is the following.

Lemma 7.17 (Two-level summability). *For $s \in \mathbb{K}((L^{-1}))$ let $\omega(s)$ denote the L^{-1} -adic order, so that $s \in L^{-\omega(s)}\mathbb{K}[[L^{-1}]]$ and $\omega(0) = +\infty$; ω is additive on products (Theorem 7.2 with L^{-1} in place of t and $E = \mathbb{K}$) and satisfies $\omega(s + s') \geq \min\{\omega(s), \omega(s')\}$. Let $S = \sum_{n \geq 0} s_n t^n \in \mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ satisfy*

$$\nu_t S \geq 0 \quad \text{and} \quad \omega(s_0) \geq 1, \quad (107)$$

which is exactly the condition $S \prec 1$ in the dominance order. Fix $n \geq 0$ and put

$$c_n := \max\{0, -\omega(s_1), \dots, -\omega(s_n)\} \in \mathbb{N}_0.$$

Then for every $k \geq 0$,

$$\omega([t^n]S^k) \geq k - n(1 + c_n). \quad (108)$$

Consequently, for each fixed monomial $t^n L^{-j}$ only the finitely many indices $k \leq j + n(1 + c_n)$ contribute to $\sum_{k \geq 0} (-S)^k$; the family is summable in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$, and $(1 + S) \sum_{k \geq 0} (-S)^k = 1$.

Proof. Because $\nu_t S \geq 0$,

$$[t^n]S^k = \sum_{\substack{(n_1, \dots, n_k) \in \mathbb{N}_0^k \\ n_1 + \dots + n_k = n}} s_{n_1} \cdots s_{n_k},$$

a finite sum in which every index satisfies $n_i \leq n$. Fix one tuple and let r be the number of indices with $n_i \geq 1$; then $r \leq n$, since each such index contributes at least 1 to the total n . The $k - r$ factors with $n_i = 0$ have $\omega(s_0) \geq 1$, and the r remaining factors have $\omega(s_{n_i}) \geq -c_n$ by definition of c_n . Additivity of ω on products gives

$$\omega(s_{n_1} \cdots s_{n_k}) \geq (k - r) \cdot 1 + r \cdot (-c_n) \geq k - n - nc_n,$$

using $r \leq n$ and $c_n \geq 0$. Taking the minimum over the finitely many tuples yields Equation (108).

A power L^{-j} can occur in an element of order ω only when $j \geq \omega$; hence $[t^n L^{-j}]S^k = 0$ as soon as $k - n(1 + c_n) > j$, which proves the finiteness assertion.

Summability in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ needs one more line. The outer support of every S^k lies in $n \geq 0$ because $\nu_t S \geq 0$, so the outer support of the sum is bounded below. For a fixed n , Equation (108) with $k \geq 0$ gives $\omega([t^n]S^k) \geq -n(1 + c_n)$ for every k , so each contribution to the n -th block is supported in $j \geq -n(1 + c_n)$; that block is therefore a well-defined element of $\mathbb{K}((L^{-1}))$, and the sum lies in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$.

For the identity, fix (n, j) and choose K larger than $j + n(1 + c_n)$. Telescoping gives $(1 + S) \sum_{k \leq K} (-S)^k = 1 - (-S)^{K+1}$, and by the finiteness just proved the coefficient of $t^n L^{-j}$ in $(-S)^{K+1}$ vanishes and the coefficient of $t^n L^{-j}$ in $(1 + S) \sum_{k \leq K} (-S)^k$ already equals that in $(1 + S) \sum_{k \geq 0} (-S)^k$. Hence the two sides agree at every monomial. $\square$

Remark 7.18. Theorem 7.17 is why the catalogue insists that a finite representation in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ needs an outer t -cutoff and an inner L^{-1} -cutoff, or one explicitly declared staircase. A single outer cutoff does not bound the work: the coefficient of $t^0 L^{-j}$ still receives contributions from $j + 1$ powers of S .

7.4 Why unrestricted division needs the rational-log field

Theorem 7.19 (Laurent series over a field). *Let E be a field. Then $E((t))$ is a field. In particular $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}} = \mathbb{K}(L)((t))$ and $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}} = \mathbb{K}((L^{-1}))((t))$ are fields.*

Proof. $E((t))$ is a commutative ring with 1, and it is an integral domain by Theorem 7.2. Let $F \neq 0$ with $m = \nu_t F$ and leading coefficient $f_m \in E^\times$. Put $U := f_m^{-1} t^{-m} F - 1 \in tE[[t]]$. Exactly as in the proof of Theorem 7.3, $\nu_t(U^k) \geq k$ makes $\sum_{k \geq 0} (-U)^k$ formally summable in $E[[t]]$, and telescoping gives $(1 + U) \sum_{k \geq 0} (-U)^k = 1$. Hence $F \cdot f_m^{-1} t^{-m} \sum_{k \geq 0} (-U)^k = 1$. Finally $\mathbb{K}(L)$ and $\mathbb{K}((L^{-1}))$ are fields: the first is the fraction field of the integral domain $\mathbb{K}[L]$; the second is the Laurent series field over $\mathbb{K}$ in the variable L^{-1} , a field by the case already proved. $\square$

The next theorem says that $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ is not merely a field containing $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, but the smallest one of the shape “Laurent series in t over a coefficient ring between $\mathbb{K}[L]$ and $\mathbb{K}(L)$ ”. This is the precise sense in which unrestricted division of *coefficient blocks* forces $\mathbb{K}(L)$.

Theorem 7.20 (Minimality of the rational-log coefficient field). *Let E be a subring of $\mathbb{K}(L)$ with $\mathbb{K}[L] \subseteq E$. Then $E((t))$ is a field if and only if $E = \mathbb{K}(L)$. Consequently $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ is the unique smallest field of the form $E((t))$ containing $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ with $E \subseteq \mathbb{K}(L)$.*

Proof. If $E = \mathbb{K}(L)$, apply [Theorem 7.19](#). Conversely suppose $E((t))$ is a field and let $e \in E$, $e \neq 0$. Its inverse $G = \sum_n G_n t^n \in E((t))$ satisfies $eG = 1$, i.e. $eG_0 = 1$ and $eG_n = 0$ for $n \neq 0$. Since $E \subseteq \mathbb{K}(L)$ is an integral domain and $e \neq 0$, we get $G_n = 0$ for $n \neq 0$ and $G_0 = e^{-1} \in E$. Hence E is a field. A subfield of $\mathbb{K}(L)$ containing $\mathbb{K}[L]$ contains p^{-1} for every nonzero $p \in \mathbb{K}[L]$ and hence every quotient p/q , so $E = \mathbb{K}(L)$. $\square$

7.5 An integral domain that is not a field, made precise

“ $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is an integral domain but not a field” is the standard slogan. It is worth saying exactly how far from a field it is, and – a point none of the six sources records – that its field of fractions is *strictly smaller* than $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$. Passing to $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ is therefore not the same operation as “forming the fraction field”.

Lemma 7.21 (Denominator lemma). *Let $A, B \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ with $B \neq 0$, and let $b_0 \in \mathbb{K}[L]$ be the leading block of B . Then, computed in $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$,*

$$\frac{A}{B} \in \mathbb{K}[L][b_0^{-1}]((t)), \quad (109)$$

where $\mathbb{K}[L][b_0^{-1}] = \{p/b_0^k : p \in \mathbb{K}[L], k \geq 0\} \subseteq \mathbb{K}(L)$. In particular every element of the fraction field of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ has all of its coefficient blocks in a single localisation $\mathbb{K}[L][q^{-1}]$ of $\mathbb{K}[L]$ at one nonzero polynomial q .

Proof. Write $m = \nu_t B$ and $S := \mathbb{K}[L][b_0^{-1}]$, a subring of $\mathbb{K}(L)$. Then $t^{-m}B = \sum_{k \geq 0} b_k t^k$ with $b_k \in \mathbb{K}[L]$, so $b_0^{-1}t^{-m}B = 1 + U$ with $U := \sum_{k \geq 1} (b_k/b_0)t^k \in tS[[t]]$. By the proof of [Theorem 7.19](#) applied to the ring S – only summability and telescoping are used, not that S is a field – $(1 + U)^{-1} = \sum_{k \geq 0} (-U)^k \in S[[t]]$. Hence $B^{-1} = t^{-m}b_0^{-1}(1 + U)^{-1} \in S((t))$ and, since $A \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} \subseteq S((t))$, also $A/B \in S((t))$. The last sentence is this statement with $q = b_0$. $\square$

Theorem 7.22 (The fraction field of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is strictly smaller than $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$). *Let $\mathbb{K}$ have characteristic zero. Then*

$$\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} \subsetneq \text{Frac}(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}) \subsetneq \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}. \quad (110)$$

An explicit witness for the second strict inclusion is

$$F := \sum_{n \geq 0} \frac{t^n}{L - n} \in \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}. \quad (111)$$

Proof. The first inclusion is strict because L is a nonzero nonunit ([Theorem 7.5](#)). For the second, $\text{Frac}(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}})$ embeds in the field $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ because $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ contains $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, so we must only show $F \notin \text{Frac}(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}})$. Certainly $F \in \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$: its blocks $(L - n)^{-1}$ lie in $\mathbb{K}(L)$ and its support is bounded below by 0.

Suppose $F = A/B$ with $A, B \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, $B \neq 0$. By [Theorem 7.21](#) there is a nonzero $q \in \mathbb{K}[L]$ with every block of F in $\mathbb{K}[L][q^{-1}]$. Fix $n \geq 0$. Then $(L - n)^{-1} = p/q^k$ for some $p \in \mathbb{K}[L]$ and $k \geq 0$, i.e. $q^k = (L - n)p$. Since $\mathbb{K}$ has characteristic zero, the elements $0, 1, 2, \dots$ of $\mathbb{K}$ are pairwise distinct, so $L - n$ is a nonassociate irreducible of the unique factorisation domain $\mathbb{K}[L]$ for each n , and $(L - n) \mid q^k$ forces $(L - n) \mid q$. As this holds for every $n \geq 0$, the polynomial q has infinitely many roots, hence $q = 0$, a contradiction. $\square$

Corollary 7.23 (The precise structure). $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is an integral domain whose unit group is [Equation \(97\)](#). It is not a field, and its nonunits do not form an ideal, so it is not a local ring: L and $1 - L$ are nonunits whose sum is the unit 1. The obstruction to division is therefore not concentrated at one maximal ideal but is spread over the primes of $\mathbb{K}[L]$: for each monic irreducible $\pi \in \mathbb{K}[L]$, the set $\pi \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is a proper nonzero ideal, and $\bigcap_{\pi} \pi \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} = 0$.

Proof. The first two sentences have been proved. For the last, $\pi \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is an ideal; it is nonzero and proper because π is a nonzero nonunit ([Theorem 7.3](#)). If $F \neq 0$ lay in $\pi \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ for every π , its leading block $p_{\nu_t F}$ would be divisible in $\mathbb{K}[L]$ by every monic irreducible – by [Theorem 7.2](#) the leading block of πG is π times the leading block of G – and $\mathbb{K}[L]$ has infinitely many pairwise nonassociate irreducibles (Euclid’s argument applies verbatim), forcing $p_{\nu_t F} = 0$. $\square$

7.6 Conditional divisibility

A nonunit divisor still divides some elements exactly. The following proposition makes the test explicit; it also shows that “being divisible” is a genuinely global condition, so that no unconditional division theorem in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is available.

Proposition 7.24 (Coefficientwise divisibility test). *Let $B \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ be nonzero with $\nu_t B = 0$ and leading block $B_0 \in \mathbb{K}[L]$, and let $A \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. Let $Q = A/B \in \mathbb{K}(L)[[t]]$ have blocks Q_n given by [Equation \(101\)](#) over $E = \mathbb{K}(L)$. Then:*

1. $A/B \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ if and only if $Q_n \in \mathbb{K}[L]$ for every $n \geq 0$;
2. given $Q_0, \dots, Q_{n-1} \in \mathbb{K}[L]$, the block Q_n lies in $\mathbb{K}[L]$ if and only if B_0 divides $A_n - \sum_{j=1}^n B_j Q_{n-j}$ in $\mathbb{K}[L]$;
3. the set $\{A \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} : A/B \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}\}$ is exactly the principal ideal $B \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, which is nonzero, and proper as soon as B is not a unit.

Proof. (1) is the definition of $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ as the set of series with polynomial blocks, together with the uniqueness in [Theorem 7.8](#): the blocks of A/B computed over $\mathbb{K}(L)$ are the only candidates. (2) is immediate from [Equation \(101\)](#), since $Q_n = B_0^{-1} R_n$ with $R_n = A_n - \sum_{j=1}^n B_j Q_{n-j} \in \mathbb{K}[L]$ under the stated hypothesis. For (3): $A/B \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ means $A = B \cdot (A/B)$ with $A/B \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, i.e. $A \in B \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$; conversely $A = BC$ with $C \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ gives $A/B = C$. The ideal $B \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ contains $B \neq 0$, and it equals $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ exactly when $1 \in B \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, i.e. when B is a unit. $\square$

Example 7.25 (Exact division by a nonunit, and a near miss). Take $B = L + t$, a nonzero nonunit. Then

$$\frac{L^2 - t^2}{L + t} = L - t \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}, \quad (112)$$

verified by the single multiplication $(L + t)(L - t) = L^2 - t^2$. So a nonunit can divide exactly. Change one sign and the divisibility fails at the third step: with $A = L^2 + t^2$, the test of [Theorem 7.24](#) (2) gives

$$Q_0 = \frac{L^2}{L} = L, \quad Q_1 = \frac{0 - 1 \cdot L}{L} = -1, \quad Q_2 = \frac{1 - 1 \cdot (-1)}{L} = \frac{2}{L} \notin \mathbb{K}[L],$$

consistently with the exact identity $\frac{L^2 + t^2}{L + t} = L - t + \frac{2t^2}{L + t}$. Two successful steps prove nothing about the third.

Remark 7.26 (The test is a semi-decision procedure). [Theorem 7.24](#) (2) refutes divisibility in finite time whenever it fails at some finite order. It never confirms divisibility in finite time on its own, because the condition quantifies over all n . A positive answer needs a finite certificate: either a candidate quotient $C \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ with only finitely many blocks – so that $BC = A$ is a finite identity, as in [Equation \(112\)](#) – or an a priori bound reducing the infinitely many conditions to finitely many, or a structural argument. Absent such a certificate, a computation that “keeps coming out polynomial” is evidence, not proof.

Structural checkpoint

Three closure questions must be kept apart.

1. Does the quotient exist in *some* field? Always, in $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$.
2. Does it have rational blocks and nothing worse? Always, again in $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$; this is [Theorem 7.19](#).
3. Does it have polynomial blocks? Only under the unit criterion, or by a certified accident as in [Theorem 7.25](#).

Answering (1) and reporting it as an answer to (3) is the standard failure mode.

7.7 Newton iteration and the exact precision-doubling law

The reciprocal recurrence produces one block per step. Newton’s method for the equation $C^{-1} - B = 0$ produces the map

$$C \mapsto C(2 - BC), \quad (113)$$

which doubles the number of correct blocks. Every source states [Equation \(113\)](#) and the identity $1 - BC(2 - BC) = (1 - BC)^2$; none states which truncations of B and of C suffice to compute the doubled output, which is precisely what makes the scheme an algorithm rather than an identity. Part (4) below supplies that statement.

Theorem 7.27 (Precision doubling). *Let $B \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ with $B_0 \in \mathbb{K}^\times$, let $C \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, let $N \geq 1$, and suppose*

$$\nu_t(1 - BC) \geq N. \quad (114)$$

Put $C' := C(2 - BC)$. Then:

1. $1 - BC' = (1 - BC)^2$, hence $\nu_t(1 - BC') \geq 2N$;
2. $\nu_t(B^{-1} - C') = \nu_t(1 - BC') \geq 2N$, and more precisely $\nu_t(B^{-1} - C) = \nu_t(1 - BC)$;
3. $\nu_t(1 - B \text{Trunc}_{<2N} C') \geq 2N$, so truncating the output at order $2N$ preserves the invariant;
4. $\text{Trunc}_{<2N} C'$ depends on B and C only through $\text{Trunc}_{<2N} B$ and $\text{Trunc}_{<N} C$. Explicitly, if $\bar{B} \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and $\bar{C} \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ satisfy $\nu_t(B - \bar{B}) \geq 2N$ and $\nu_t(C - \bar{C}) \geq N$, then $\nu_t(C' - \bar{C}(2 - \bar{B}\bar{C})) \geq 2N$.

Proof. (1) $BC' = BC(2 - BC) = 2BC - (BC)^2$, so $1 - BC' = 1 - 2BC + (BC)^2 = (1 - BC)^2$. Valuations add ([Theorem 7.2](#)), so $\nu_t(1 - BC') \geq 2N$.

(2) B is a unit by [Theorem 7.3](#) and $\nu_t(B^{-1}) = -\nu_t B = 0$. From $B^{-1} - C' = B^{-1}(1 - BC')$ and additivity of ν_t we get $\nu_t(B^{-1} - C') = \nu_t(1 - BC')$. The same computation with C in place of C' gives the second identity.

(3) $C' - \text{Trunc}_{<2N} C' \in t^{2N} \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, so $B(C' - \text{Trunc}_{<2N} C') \in t^{2N} \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ because $\nu_t B = 0$; adding this to (1) gives $\nu_t(1 - B \text{Trunc}_{<2N} C') \geq 2N$.

(4) Put $\delta = C - \bar{C}$ and $\varepsilon = B - \bar{B}$, so $\nu_t \delta \geq N$ and $\nu_t \varepsilon \geq 2N$. Since $C' = 2C - BC^2$ and $\bar{C}(2 - \bar{B}\bar{C}) = 2\bar{C} - \bar{B}\bar{C}^2$,

$$C' - \bar{C}(2 - \bar{B}\bar{C}) = 2\delta - (BC^2 - \bar{B}\bar{C}^2) = 2\delta - \varepsilon C^2 - \bar{B}(C + \bar{C})\delta = \delta(2 - \bar{B}(C + \bar{C})) - \varepsilon C^2,$$

using $BC^2 - \bar{B}\bar{C}^2 = \varepsilon C^2 + \bar{B}(C^2 - \bar{C}^2)$ and $C^2 - \bar{C}^2 = (C + \bar{C})\delta$. The second term has $\nu_t \geq 2N$. For the first,

$$2 - \bar{B}(C + \bar{C}) = (1 - \bar{B}C) + (1 - \bar{B}\bar{C}),$$

and $1 - \bar{B}C = (1 - BC) + \varepsilon C$ has $\nu_t \geq N$, while $1 - \bar{B}\bar{C} = (1 - BC) + \varepsilon C + B\delta - \varepsilon\delta$ also has $\nu_t \geq N$. Hence $\nu_t(\delta(2 - \bar{B}(C + \bar{C}))) \geq N + N = 2N$, and the whole difference has $\nu_t \geq 2N$. $\square$

Algorithm 7.28 (Newton reciprocal to exclusive order N). Given an integer $N \geq 1$ and $B \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $B_0 \in \mathbb{K}^\times$, known through order t^{N-1} :

```

function newton_reciprocal(B, N):
  require B[0] is a nonzero scalar in K
  C := 1/B[0]                # exact through order t^0
  m := 1
  while m < N:
    m2 := min(2*m, N)
    E := Trunc(1 - B*C, m2)  # valuation >= m by the invariant
    C := Trunc(C*(1 + E), m2)
    m := m2
  return C                    # equals Trunc_{<N} of the reciprocal

```

Corollary 7.29 (The iterates are exact truncations). *Let*

$$C^{(0)} = B_0^{-1}, \quad C^{(k+1)} = \text{Trunc}_{<2^{k+1}}(C^{(k)}(2 - BC^{(k)})).$$

Then for every $k \geq 0$,

$$C^{(k)} = \text{Trunc}_{<2^k} B^{-1}. \quad (115)$$

In particular [Theorem 7.28](#) returns $\text{Trunc}_{<N} B^{-1}$, and the intermediate steps require no blocks of B beyond index $2^{k+1} - 1$ at step k .

Proof. Induction on k . For $k = 0$, $1 - BC^{(0)} = 1 - B_0^{-1}B$ has zero constant block, so $\nu_t(1 - BC^{(0)}) \geq 1$, and by [Theorem 7.27](#) (2) $\nu_t(B^{-1} - C^{(0)}) \geq 1$; as $C^{(0)}$ has t -degree $0 < 1$, it equals $\text{Trunc}_{<1} B^{-1}$. Assume $C^{(k)} = \text{Trunc}_{<2^k} B^{-1}$; then $\nu_t(B^{-1} - C^{(k)}) \geq 2^k$, so $\nu_t(1 - BC^{(k)}) \geq 2^k$ by [Theorem 7.27](#) (2). Parts (1) and (3) give $\nu_t(1 - BC^{(k+1)}) \geq 2^{k+1}$, hence $\nu_t(B^{-1} - C^{(k+1)}) \geq 2^{k+1}$ by (2) again; since $C^{(k+1)}$ has t -degree $< 2^{k+1}$, it is $\text{Trunc}_{<2^{k+1}} B^{-1}$.

The capped step. In [Theorem 7.28](#) the new precision is $m' = \min\{2m, N\} \leq 2m$. Suppose $C = \text{Trunc}_{<m} B^{-1}$ at the top of the loop, so $\nu_t(B^{-1} - C) \geq m$ and hence $\nu_t(1 - BC) \geq m$ by [Theorem 7.27](#) (2). Part (1) gives $\nu_t(1 - BC') \geq 2m \geq m'$ for $C' = C(2 - BC)$, and the computation in the proof of part (3), run with m' in place of $2N$, gives $\nu_t(1 - B \text{Trunc}_{<m'} C') \geq m'$; by (2) once more $\nu_t(B^{-1} - \text{Trunc}_{<m'} C') \geq m'$, and $\text{Trunc}_{<m'} C'$ has t -degree $< m'$, so it equals $\text{Trunc}_{<m'} B^{-1}$. The invariant holds initially with $m = 1$ and the loop terminates with $m = N$, so the returned value is $\text{Trunc}_{<N} B^{-1}$ for every $N \geq 1$, not only for N a power of two. The statement about required blocks of B is [Theorem 7.27](#) (4). $\square$

Remark 7.30 (A wording to avoid). Writing $E = 1 - BC$, the step $\nu_t(E^2) \geq 2N$ holds because ν_t is *additive* on products ([Theorem 7.2](#)), not because it is multiplicative; ν_t is a homomorphism from the multiplicative monoid of nonzero elements to $(\mathbb{Z}, +)$.

Remark 7.31 (Where the iteration lives). [Theorem 7.27](#) used only that ν_t is additive and that B has an invertible leading block; it therefore holds verbatim over the coefficient rings $\mathbb{K}(L)$ and $\mathbb{K}((L^{-1}))$, and in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ after normalising the leading Laurent-log coefficient. It is an exact algebraic identity, not an approximation: unlike numerical Newton iteration, there is no convergence hypothesis and no loss of significance – by [Theorem 7.29](#) each iterate is literally a truncation of the answer, so the logarithmic degrees appearing in the iterates are exactly those of B^{-1} and cannot swell beyond them, provided one truncates at every step.

7.8 Cost: the recurrence against the doubling scheme

We count multiplications in the coefficient ring $\mathbb{K}[L]$: every product of two coefficient blocks costs one *block multiplication*, including those products one of whose factors happens to be the scalar block $C_0 = B_0^{-1}$. Additions are free, and so is the single outer normalisation by B_0^{-1} that [Equations \(100\)](#)

and (101) apply once per step. Let $M(N)$ denote the number of block multiplications used by whatever routine computes $\text{Trunc}_{<N}(PQ)$ from $\text{Trunc}_{<N} P$ and $\text{Trunc}_{<N} Q$.

Proposition 7.32 (Cost of the recurrence). *Computing $\text{Trunc}_{<N} B^{-1}$ by Equations (99) and (100) uses exactly*

$$\binom{N}{2} = \frac{N(N-1)}{2} \quad (116)$$

block multiplications, and computing $\text{Trunc}_{<N}(A/B)$ by Equation (101) uses the same number.

Proof. Step n forms the products $B_j C_{n-j}$ for $j = 1, \dots, n$, that is n block multiplications, for $n = 0, 1, \dots, N-1$. The total is $\sum_{n=0}^{N-1} n = N(N-1)/2$. $\square$

Proposition 7.33 (Cost of the doubling scheme). *Let $N = 2^K$.*

1. *If M is nondecreasing and satisfies $2M(n) \leq M(2n)$, then Theorem 7.28 uses at most $4M(N)$ block multiplications.*
2. *With schoolbook truncated multiplication, and exploiting the invariant $\nu_t(1 - BC) \geq m$ so that the first m blocks of BC are known in advance, Theorem 7.28 uses exactly*

$$\frac{N^2 + N - 2}{2} \quad (117)$$

block multiplications, which exceeds Equation (116) by exactly $N - 1$.

Proof. (1) The step from precision m to $2m$ forms two truncated products at order $2m$, so costs at most $2M(2m)$. Summing over $m = 1, 2, 4, \dots, 2^{K-1}$ gives at most $\sum_{j=1}^K 2M(2^j)$. The hypothesis $2M(n) \leq M(2n)$ gives $M(2^j) \leq M(2^K)/2^{K-j}$, so the sum is at most $2M(N) \sum_{j=1}^K 2^{j-K} \leq 4M(N)$.

(2) Consider the step from m to $2m$, with $C = \text{Trunc}_{<m} B^{-1}$ supported on indices $0, \dots, m-1$. Forming $E = 1 - BC$ modulo t^{2m} . By the invariant, the blocks of BC of index $< m$ are known to be $1, 0, \dots, 0$; only indices $m, \dots, 2m-1$ must be computed. The needed products are $B_j C_i$ with $0 \leq i < m$ and $m \leq i+j < 2m$, and for each of the m values of i there are exactly m admissible j . Cost: m^2 . Forming $C(1 + E)$ modulo t^{2m} . E is supported on indices $m, \dots, 2m-1$, and C on $0, \dots, m-1$; the products $C_i E_j$ with $i+j < 2m$ number $\sum_{j=m}^{2m-1} (2m-j) = \sum_{s=1}^m s = m(m+1)/2$. Total for the step: $m^2 + m(m+1)/2 = (3m^2 + m)/2$. Summing over $m = 2^k, k = 0, \dots, K-1$,

$$\sum_{k=0}^{K-1} \frac{3 \cdot 4^k + 2^k}{2} = \frac{3}{2} \cdot \frac{4^K - 1}{3} + \frac{2^K - 1}{2} = \frac{N^2 - 1}{2} + \frac{N - 1}{2} = \frac{N^2 + N - 2}{2}.$$

Subtracting Equation (116) gives $(2N - 2)/2 = N - 1$.

How fragile the gap is. Suppose instead that a product one of whose factors is the scalar $C_0 = B_0^{-1}$ is charged nothing. That removes one product per step $n = 1, \dots, N-1$ from the recurrence, hence $N-1$ in all, and $2m$ from the doubling step at precision m (the m products $B_j C_0$ and the m products $C_0 E_j$), hence $\sum_{k=0}^{K-1} 2 \cdot 2^k = 2N - 2$ in all. Both counts then drop to $\binom{N-1}{2}$ and the two schemes cost exactly the same. The $N-1$ gap of part (2) is therefore an artefact of the accounting; what is robust in the comparison is part (1). $\square$

Corollary 7.34 (When doubling pays). *Which of the two schemes wins is decided by the growth of M , and the two counts settle the two extremes. Precisely: if $M(N)/N^2 \rightarrow 0$ as $N \rightarrow \infty$ (with M as in Theorem 7.33 (1)), then the doubling scheme uses $o(N^2)$ block multiplications against $\Theta(N^2)$ for the recurrence; whereas with schoolbook block multiplication it uses exactly $N-1$ more block multiplications than the recurrence in the accounting fixed at the head of Section 7.8, and exactly as many under the alternative accounting described at the end of the proof of Theorem 7.33. Under neither accounting is it faster.*

Proof. Combine [Theorems 7.32](#) and [7.33](#); the first claim is part (1) with the stated growth hypothesis, the second is part (2). $\square$

Remark 7.35 (Scalar cost under an affine degree profile). Block multiplications are not unit-cost. Assume the hypothesis of [Theorem 7.15](#) with $\alpha > 0$ an integer, so $\deg_L B_n \leq \alpha n$ and $\deg_L C_n \leq \alpha n$, and count $\mathbb{K}$ -multiplications with schoolbook polynomial arithmetic in L . The recurrence then costs at most

$$\sum_{n=1}^{N-1} \sum_{j=1}^n (\alpha j + 1)(\alpha(n-j) + 1) = \Theta(\alpha^2 N^4), \quad \text{with leading term } \frac{\alpha^2 N^4}{24},$$

because $\sum_{j=1}^n j(n-j) = \binom{n+1}{3} \sim n^3/6$ and $\sum_{n < N} \binom{n+1}{3} = \binom{N+1}{4} \sim N^4/24$. Quoting a bound $\mathcal{O}(N^2 d^2)$ with a fixed d therefore understates the cost whenever the log-degree profile grows; substituting the honest $d \asymp \alpha N$ recovers $\alpha^2 N^4$. This quartic behaviour, not the quadratic block count, is what makes truncating after *every* multiplication the decisive implementation discipline; see [\[33\]](#) for the relaxed and fast-multiplication variants.

7.9 The embedding $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} \hookrightarrow \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$

The third response to a nonconstant leading block is to expand each rational coefficient at $L = \infty$. This resolves a finer asymptotic scale, and it is the natural home of a multiseries implementation [\[29\]](#); it is also the step most likely to be taken without noticing that the object has changed.

Definition 7.36 (Iterated Laurent-log field). $\mathbb{K}((L^{-1}))$ is the field of formal Laurent series in the variable L^{-1} : elements $\sum_{j \geq j_0} a_j L^{-j}$ with $j_0 \in \mathbb{Z}$ and $a_j \in \mathbb{K}$, so that only finitely many positive powers of L occur but arbitrarily many negative ones may. Then $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}} := \mathbb{K}((L^{-1}))((t))$, whose elements are $\sum_{n \geq n_0} t^n \sum_{j \geq j_n} a_{n,j} L^{-j}$.

Theorem 7.37 (The expansion map). *There is exactly one $\mathbb{K}$ -algebra homomorphism $\iota_0 : \mathbb{K}(L) \rightarrow \mathbb{K}((L^{-1}))$ with $\iota_0(L) = L$, namely Laurent expansion at $L = \infty$, and it is injective. Applying it blockwise defines an injective ring homomorphism $\iota : \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} \rightarrow \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ that restricts to the identity on $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. The map ι is not surjective; its image is therefore a proper subfield of $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$.*

Proof. Existence on coefficients. A polynomial $\sum_{i \leq d} c_i L^i$ already lies in $\mathbb{K}((L^{-1}))$ (as $\sum_{j \geq -d} c_{-j} L^{-j}$), giving an injective ring map $\mathbb{K}[L] \hookrightarrow \mathbb{K}((L^{-1}))$. Since $\mathbb{K}((L^{-1}))$ is a field ([Theorem 7.19](#)), every nonzero polynomial maps to a unit, so by the universal property of the localisation $\mathbb{K}(L) = \text{Frac } \mathbb{K}[L]$ the map extends uniquely to a ring homomorphism $\iota_0 : \mathbb{K}(L) \rightarrow \mathbb{K}((L^{-1}))$; a ring homomorphism out of a field is injective. Concretely, for $q = q_d L^d (1 + w)$ with $q_d \in \mathbb{K}^\times$ and $w \in L^{-1} \mathbb{K}[L^{-1}]$, $\iota_0(1/q) = q_d^{-1} L^{-d} \sum_{k \geq 0} (-w)^k$, the series being summable because w has positive L^{-1} -order.

Uniqueness of ι_0 is the uniqueness clause in that universal property: a $\mathbb{K}$ -algebra map on $\mathbb{K}(L)$ is determined by the image of L .

Extension to Laurent series. Define $\iota(\sum_{n \geq n_0} f_n t^n) = \sum_{n \geq n_0} \iota_0(f_n) t^n$. The support stays bounded below, so the image is in $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$. Applying a ring homomorphism blockwise to Laurent series is again a ring homomorphism, because each block of a sum or a Cauchy product is a finite algebraic expression in the blocks of the arguments; and ι is injective because ι_0 is. On $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ every block is a polynomial and ι_0 is the inclusion, so ι is the identity there.

Not surjective. Since $\text{char } \mathbb{K} = 0$, the binomial series $\sigma := \sum_{k \geq 0} \binom{1/2}{k} L^{-k} \in \mathbb{K}[[L^{-1}]]$ is defined and satisfies $\sigma^2 = 1 + L^{-1}$, the identity holding coefficientwise because it holds in $\mathbb{Q}[[u]]$ for $u = L^{-1}$. Suppose $\sigma = \iota_0(f)$ for some $f \in \mathbb{K}(L)$. Then $f^2 = (L+1)/L$, so $(Lf)^2 = L(L+1)$ with $Lf \in \mathbb{K}(L)$. Writing $Lf = u/v$ in lowest terms gives $u^2 = L(L+1)v^2$; comparing multiplicities of the irreducible L on the two sides of this equation in the unique factorisation domain $\mathbb{K}[L]$ yields an even number on the left and an odd one on the right, a contradiction. Hence σ is not in the image of ι_0 , and σ regarded as a t^0 -block is not in the image of ι . $\square$

Example 7.38 (The exact coefficient becomes an infinite series).

$$\iota\left(\frac{1}{L+1}\right) = L^{-1} - L^{-2} + L^{-3} - \dots = \sum_{j \geq 1} (-1)^{j-1} L^{-j}, \quad \iota\left(\frac{1}{L+t}\right) = \sum_{n \geq 0} (-1)^n L^{-n-1} t^n. \quad (118)$$

The left-hand sides are single finite data; the right-hand sides are infinite.

Proposition 7.39 (The iteration order is part of the type). *Regard both $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}} = \mathbb{K}((L^{-1}))((t))$ and $\mathbb{K}((t))((L^{-1}))$ as sets of families $(a_{n,j})$ of scalars indexed by $(n, j) \in \mathbb{Z}^2$, representing $\sum a_{n,j} t^n L^{-j}$. For the first, the support condition is: the set of n carrying a nonzero entry is bounded below, and for each such n the set of j with $a_{n,j} \neq 0$ is bounded below. For the second, the roles of n and j are exchanged. Then neither set contains the other. Witnesses:*

$$\sum_{n \geq 0} L^n t^n \in \mathbb{K}((L^{-1}))((t)) \setminus \mathbb{K}((t))((L^{-1})), \quad \sum_{j \geq 0} L^{-j} t^{-j} \in \mathbb{K}((t))((L^{-1})) \setminus \mathbb{K}((L^{-1}))((t)). \quad (119)$$

Proof. The first element has $a_{n,-n} = 1$ for $n \geq 0$ and all other entries zero. Its outer index n is bounded below by 0, and for each n exactly one j occurs, so the first support condition holds; indeed the element lies already in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. But for the second condition one must read the family by inner index: the set of j carrying a nonzero entry is $\{0, -1, -2, \dots\}$, which is not bounded below, so the element is not in $\mathbb{K}((t))((L^{-1}))$.

The second element has $a_{-j,j} = 1$ for $j \geq 0$ and all other entries zero. Its inner index j is bounded below by 0, and for each j exactly one n occurs, so it lies in $\mathbb{K}((t))((L^{-1}))$. But the set of n carrying a nonzero entry is $\{0, -1, -2, \dots\}$, so it has infinitely many positive powers of X and is excluded from $\mathbb{K}((L^{-1}))((t))$. $\square$

Expanding at $L = \infty$ changes the object

Three distinct things go wrong if ι is applied silently.

1. *Finiteness.* An exact block such as $1/(L+1)$ becomes an infinite series. A finite representation in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ therefore needs an outer t -cutoff and an inner L^{-1} -cutoff, or one declared staircase in the lexicographic monomial order. An outer cutoff alone does not make the data finite.
2. *Algebra.* ι is injective but not surjective (Theorem 7.37), so $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ sits inside $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ as a proper subfield: an identity established in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ need not descend to $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$, and an element of $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ need not be a rational function of L at all. Moreover the leading data change meaning – in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ each block has a finite $\deg_L$, while in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ the relevant invariant is the L^{-1} -adic order.
3. *Asymptotics.* Let $\mathbb{K} \subseteq \mathbb{C}$ and let $f \in \mathbb{K}(L)$ have largest pole modulus $\rho(f)$, with $\rho(f) = 0$ when f is a polynomial and $\iota_0(f)$ a finite sum. The numerical series $\iota_0(f)$ converges for $|L| > \rho(f)$, that is for $X > e^{\rho(f)}$ on the positive ray, and for nonpolynomial f diverges for $|L| < \rho(f)$. There is in general no threshold valid for all blocks at once: for $F = \sum_{n \geq 0} t^n / (L - n) \in \mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ the n -th block requires $X > e^n$. Hence “ F equals its L^{-1} -expansion for large X ” is not a theorem about F ; it is a statement about one block at a time, and any uniform claim needs its own proof with the regime printed as $\mathcal{O}_{X \rightarrow +\infty}(\cdot)$.

Remark 7.40 (When to expand and when not to). Keeping $1/(L+1)$ unexpanded is exact, finite, and preserves cancellation; expanding it resolves the finer scale $t \prec \dots \prec L^{-2} \prec L^{-1} \prec 1$ and is required as soon as one must compare two blocks of the same t -order. The decision is a modelling decision and should be recorded, not inferred: a system that expands rational coefficients automatically will destroy exact cancellations and swell expressions, while one that never expands cannot answer questions about the inner scale. The same tension appears in every multiseries implementation [29, 30].

Section summary

An element of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is invertible there exactly when its leading coefficient block is a nonzero scalar; the leading *block* is a polynomial, and factoring out the dominant monomial does not by itself invert. Reciprocals of units exist, are unique, and are computed by a triangular recurrence whose truncated output $P = \text{Trunc}_{<N}(A/B)$ is certified by the residual $BP = A + \mathcal{O}_t^{\text{form}}(t^N)$ alone; relative precision is exactly preserved, so M leading input blocks give M leading output blocks. Newton’s map $C \mapsto C(2 - BC)$ doubles the correct order and, with the input requirement of [Theorem 7.27](#) (4), yields exact truncations of the reciprocal; it beats the recurrence asymptotically when block multiplication is subquadratic, and does not beat it at all when multiplication is schoolbook. Unrestricted division of coefficient blocks forces $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$, the smallest Laurent field over an intermediate coefficient ring, which strictly contains the fraction field of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. Expanding each rational coefficient at $L = \infty$ embeds $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ into $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ injectively but not surjectively, and converts one exact datum into an infinite series with no uniform analytic threshold.

8 Powers, logarithms, exponentials, and the closure boundary

Addition, multiplication and division were settled by finite coefficient algebra. The operations of this section – powers with an arbitrary exponent, logarithms, exponentials, and more generally the substitution of a series into a scalar power series – are the first for which the answer to “does the result stay in the class?” is neither always yes nor always no. The purpose of the section is to replace the informal closure tables of the sources by statements with hypotheses and proofs, and in particular to prove the two negative results that fix the boundary of the polynomial–logarithmic scale: a logarithm leaves the scale exactly when the leading coefficient block is nonconstant, and an exponential leaves *every* power–log scale exactly when its argument is not of nonnegative outer order with an affine t^0 -block.

Throughout, $\mathbb{K}$ is a field of characteristic zero, $t := X^{-1}$ and $L := \log X$ are independent formal generators, the default analytic regime is $X \rightarrow +\infty$ on the positive ray, and a nonzero $F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} = \mathbb{K}[L](t)$ is written

$$F = \sum_{n \geq n_0} p_n(L)t^n, \quad p_n \in \mathbb{K}[L], \quad p_{n_0} \neq 0, \quad n_0 = \nu_t F. \quad (120)$$

We write $p_{\nu_t F}$ for the leading block, $\deg_L p_{\nu_t F}$ for its degree, and $t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ for the elements of $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} = \mathbb{K}[L][[t]]$ of strictly positive outer order. Truncation is always the exclusive operator $\text{Trunc}_{<N} F = \sum_{n < N} p_n(L)t^n$, and $F = G + \mathcal{O}_t^{\text{form}}(t^N)$ abbreviates $\nu_t(F - G) \geq N$, a statement about coefficients only.

Two facts proved elsewhere in the volume are used freely: the unit criterion ([Theorem 7.3](#)), namely that F is a unit of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ if and only if its leading block is a nonzero scalar; and the expansion embedding $\iota: \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} \hookrightarrow \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ ([Theorem 7.37](#)), Laurent expansion of each coefficient at $L = \infty$. On $\mathbb{K}((L^{-1}))$ we use the L^{-1} -adic order ω of [Theorem 7.17](#): $\omega(\sum_{j \geq j_0} a_j L^{-j}) = \min \{j : a_j \neq 0\}$, so that $\omega(p) = -\deg_L p$ for a nonzero polynomial p , ω is additive on products, and $\omega(h + h') \geq \min \{\omega(h), \omega(h')\}$.

The distinguished derivation $D_X = t\partial_L - t^2\partial_t$ appears in the hypotheses of several statements below; it is constructed, and all of its properties used here are proved, in [Section 9](#), which is independent of the present section. The only facts borrowed are the block formula $D_X(t^n p) = t^{n+1}(p' - np)$, the resulting bound $\nu_t(D_X F) \geq \nu_t F + 1$, and $\ker D_X = \mathbb{K}$.

8.1 Substituting a small series into a scalar power series

Every construction in this section rests on one mechanism: a series of positive outer order may be substituted into an arbitrary formal power series over $\mathbb{K}$, because the outer order of the r -th

power grows with r . All six sources state this; we record it once, in the form that also delivers the finite-determination statement that a computation actually needs.

Lemma 8.1 (Summable sequences and blockwise assembly). *Let $(F_r)_{r \geq 0}$ be a family in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ with $\nu_t F_r \rightarrow +\infty$ and $\inf_r \nu_t F_r > -\infty$. Then for each $n \in \mathbb{Z}$ only finitely many r have $[t^n]F_r \neq 0$, the element*

$$\sum_{r \geq 0} F_r := \sum_{n \in \mathbb{Z}} \left(\sum_{r \geq 0} [t^n]F_r \right) t^n$$

is a well-defined member of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, and the assignment is $\mathbb{K}$ -linear and compatible with multiplication by a fixed element: $G \sum_r F_r = \sum_r G F_r$. The same statements hold in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, in $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ and in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$, with the coefficient ring changed accordingly.

Proof. Fix n . Since $\nu_t F_r \rightarrow +\infty$, the set $R_n = \{r : \nu_t F_r \leq n\}$ is finite, and $[t^n]F_r = 0$ for $r \notin R_n$; so the inner sum is a finite sum in the coefficient ring. The common lower bound $m = \inf_r \nu_t F_r$ is finite and gives $[t^n]F_r = 0$ for all $n < m$, so the support of the result is bounded below and the result lies in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. Linearity is immediate. For the last claim, $[t^n](G \sum_r F_r) = \sum_{i+j=n} [t^i]G \sum_r [t^j]F_r$, a finite double sum that may be reordered. No step used that the coefficients are polynomials. $\square$

Proposition 8.2 (Substitution of a small series into a scalar power series). *Let $U \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $q := \nu_t U \geq 1$. For $\Phi = \sum_{r \geq 0} c_r z^r \in \mathbb{K}[[z]]$ put*

$$\Phi(U) := \sum_{r \geq 0} c_r U^r. \quad (121)$$

Then the sum is defined by Theorem 8.1, $\Phi(U) \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $\nu_t \Phi(U) \geq q \cdot \text{ord}_z \Phi$, and

$$\Phi \mapsto \Phi(U)$$

is a $\mathbb{K}$ -algebra homomorphism $\mathbb{K}[[z]] \rightarrow \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ sending z to U . It is the unique such homomorphism that is continuous for the z -adic and t -adic topologies. The same holds with $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ replaced by $\mathbb{K}(L)[[t]]$ or $\mathbb{K}((L^{-1}))[[t]]$.

Proof. $\nu_t(U^r) \geq rq \rightarrow +\infty$ and all orders are ≥ 0 , so Theorem 8.1 applies and gives $\Phi(U) \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$; if $c_r = 0$ for $r < r_0$ then every surviving term has order $\geq r_0 q$. Additivity is clear. For multiplicativity, let $\Psi = \sum_s d_s z^s$. Then

$$\Phi(U)\Psi(U) = \sum_r c_r U^r \Psi(U) = \sum_r \sum_s c_r d_s U^{r+s},$$

where the first step is the last clause of Theorem 8.1 and the second applies it again inside each term; the resulting double family is summable because $\nu_t(U^{r+s}) \geq (r+s)q$, so it may be summed by the antidiagonals $r+s=m$, giving $\sum_m (\sum_{r+s=m} c_r d_s) U^m = (\Phi\Psi)(U)$. Uniqueness: a continuous homomorphism is determined on the dense subring $\mathbb{K}[z]$, where it is determined by the image of z . $\square$

The next statement is what a computation uses. The sources give only its first half – that few powers contribute – and leave implicit which blocks of U must be known; both halves are needed to certify a truncated result.

Theorem 8.3 (Finite determination of a substituted scalar series). *Let $U \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $\nu_t U = q \geq 1$ and $\Phi = \sum_r c_r z^r \in \mathbb{K}[[z]]$. For $N \leq 0$, $\text{Trunc}_{<N} \Phi(U) = 0$. For $N \geq 1$, put $m = \lfloor (N-1)/q \rfloor$. Then*

$$\text{Trunc}_{<N} \Phi(U) = \text{Trunc}_{<N} \sum_{r=0}^m c_r (\text{Trunc}_{<N} U)^r. \quad (122)$$

In particular $\text{Trunc}_{<N} \Phi(U)$ depends only on the $m+1$ scalars $c_0, \dots, c_m$ and on the blocks $[t^n]U$ with $n < N$, and $[t^N]\Phi(U)$ receives contributions only from the powers U^r with $r \leq \lfloor N/q \rfloor$.

Proof. Both sides lie in $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, so the case $N \leq 0$ is trivial. Let $N \geq 1$. If $r > m$ then $r \geq m+1 > (N-1)/q$, hence $rq > N-1$ and $rq \geq N$; since $\nu_t(c_r U^r) \geq rq$, such terms do not affect $\text{Trunc}_{<N}$, which proves that only $r \leq m$ matter. For the replacement of U by $\text{Trunc}_{<N} U$, write $E = U - \text{Trunc}_{<N} U$, so $\nu_t E \geq N$. For $r \geq 1$,

$$U^r - (\text{Trunc}_{<N} U)^r = E \sum_{i=0}^{r-1} U^i (\text{Trunc}_{<N} U)^{r-1-i}$$

has outer order $\geq N$, because every factor in the sum has nonnegative order. Hence the two truncations agree. The last clause is the same computation with N replaced by $N+1$: $rq > N$ forces $[t^N](c_r U^r) = 0$. $\square$

Corollary 8.4 (The elementary series). *Let $U \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and $\alpha \in \mathbb{K}$. Then*

$$(1+U)^\alpha := \sum_{r \geq 0} \binom{\alpha}{r} U^r \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}, \quad (123)$$

$$\log(1+U) := \sum_{r \geq 1} \frac{(-1)^{r+1}}{r} U^r \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}, \quad (124)$$

$$\exp U := \sum_{r \geq 0} \frac{U^r}{r!} \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} \quad (125)$$

are well defined, and every identity between formal power series over $\mathbb{K}$ in one or two variables that involves only these operations transfers. In particular, for $U, V \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$,

$$(1+U)^\alpha (1+U)^\beta = (1+U)^{\alpha+\beta}, \quad ((1+U)^\alpha)^n = (1+U)^{n\alpha} \quad (n \in \mathbb{Z}), \quad (126)$$

$$\exp(U+V) = (\exp U)(\exp V), \quad \exp(\log(1+U)) = 1+U, \quad \log(\exp U) = U, \quad (127)$$

$$\log((1+U)(1+V)) = \log(1+U) + \log(1+V), \quad (1+U)^\alpha = \exp(\alpha \log(1+U)). \quad (128)$$

Proof. Each of Equations (123) to (125) is $\Phi(U)$ for a $\Phi \in \mathbb{K}[[z]]$ (here $\text{char } \mathbb{K} = 0$ is used to divide by r and $r!$), so Theorem 8.2 applies; the order statements come from $\text{ord}_z \Phi$. For the identities, note that a one-variable identity $\Phi_1 = \Phi_2$ in $\mathbb{K}[[z]]$ is carried to $\Phi_1(U) = \Phi_2(U)$ by the homomorphism. For the two-variable identities, the same argument applies to the unique continuous $\mathbb{K}$ -algebra homomorphism $\mathbb{K}[[z_1, z_2]] \rightarrow \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ with $z_i \mapsto U, V$, which exists by the proof of Theorem 8.2 verbatim; the required identities $\exp(z_1+z_2) = \exp z_1 \exp z_2$ and so on are classical in $\mathbb{Q}[[z_1, z_2]]$. The two identities in Equation (127) that involve a composition use one further step: if $\Psi \in \mathbb{K}[[z]]$ has $\Psi(0) = 0$, then $(\Phi \circ \Psi)(U) = \Phi(\Psi(U))$ for every Φ . Indeed both $\Phi \mapsto (\Phi \circ \Psi)(U)$ and $\Phi \mapsto \Phi(\Psi(U))$ are continuous $\mathbb{K}$ -algebra homomorphisms $\mathbb{K}[[z]] \rightarrow \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ – the first as the composite of $\Phi \mapsto \Phi \circ \Psi$, which is Theorem 8.2 applied inside $\mathbb{K}[[z]]$, with evaluation at U ; the second directly, since $\nu_t \Psi(U) \geq 1$ – and both send $z \mapsto \Psi(U)$, so they agree by the uniqueness clause. Applying this to the classical identities $\exp(\log(1+z)) = 1+z$ and $\log(\exp z) = z$ in $\mathbb{Q}[[z]]$ gives the two displayed identities. $\square$

For explicit coefficients there are two standard devices, a recurrence and a closed form; each is preferable in a different situation.

Proposition 8.5 (Coefficients of a substituted series). *Let $U = \sum_{n \geq 1} u_n t^n \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and $\Phi = \sum_{r \geq 0} c_r z^r \in \mathbb{K}[[z]]$. Then $[t^0]\Phi(U) = c_0$ and, for $N \geq 1$,*

$$[t^N]\Phi(U) = \sum_{k=1}^N c_k \frac{k!}{N!} \mathcal{B}_{N,k}^{\exp}(1! u_1, 2! u_2, \dots, (N-k+1)! u_{N-k+1}). \quad (129)$$

In particular, with $E = \exp U$ and $M = \log(1 + U)$,

$$[t^N]E = \frac{1}{N!} \mathcal{B}_N^{\exp}(1! u_1, \dots, N! u_N), \quad (130)$$

$$[t^N]M = \frac{1}{N!} \sum_{k=1}^N (-1)^{k-1} (k-1)! \mathcal{B}_{N,k}^{\exp}(1! u_1, \dots, (N-k+1)! u_{N-k+1}). \quad (131)$$

Moreover E and M satisfy the recurrences

$$N[t^N]E = \sum_{m=1}^N m u_m [t^{N-m}]E, \quad N[t^N]M = N u_N - \sum_{m=1}^{N-1} m [t^m]M u_{N-m}, \quad (132)$$

which compute all blocks through t^N with $\mathcal{O}(N^2)$ coefficient multiplications.

Proof. The partial exponential Bell polynomials are defined [4] by the identity

$$\frac{1}{k!} \left(\sum_{m \geq 1} a_m \frac{z^m}{m!} \right)^k = \sum_{n \geq k} \mathcal{B}_{n,k}^{\exp}(a_1, a_2, \dots) \frac{z^n}{n!}$$

in $\mathbb{Q}[a_1, a_2, \dots][[z]]$; since it is an identity with universal coefficients it holds after any specialisation into a commutative $\mathbb{Q}$ -algebra. Specialise $a_m = m! u_m \in \mathbb{K}[L]$ and $z = t$; then $\sum_m a_m z^m / m! = U$, and comparing blocks gives $[t^n]U^k = (k!/n!) \mathcal{B}_{n,k}^{\exp}(1! u_1, 2! u_2, \dots)$, the arguments beyond index $n - k + 1$ being irrelevant. Summing over k and using $[t^N]U^k = 0$ for $k > N$ gives Equation (129). Equations (130) and (131) are the cases $c_k = 1/k!$ and $c_k = (-1)^{k-1}/k$, together with $\mathcal{B}_N^{\exp} = \sum_{k=1}^N \mathcal{B}_{N,k}^{\exp}$ for $N \geq 1$.

For the recurrences, let ∂_t be the formal partial derivative holding L fixed. It is a derivation of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $\nu_t(\partial_t F) \geq \nu_t F - 1$, hence commutes with summable sums, so applying it to Equation (125) termwise gives $\partial_t E = (\partial_t U)E$. Comparing the coefficient of t^{N-1} and using $[t^{N-1}]\partial_t E = N[t^N]E$ and $[t^{N-1}]\partial_t U = m u_m$ yields the first recurrence. Likewise $\partial_t M = (\partial_t U)(1 + U)^{-1}$, that is $(1 + U)\partial_t M = \partial_t U$; comparing coefficients of t^{N-1} gives $N[t^N]M + \sum_{m=1}^{N-1} m[t^m]M u_{N-m} = N u_N$. Each recurrence performs $\mathcal{O}(N)$ coefficient multiplications per block. $\square$

Example 8.6 (A finite logarithm). Let $U = (L + 1)t + L^2 t^2$, so $q = 1$. By Theorem 8.3 with $N = 4$, only $r \leq 3$ contribute below t^4 , and

$$\begin{aligned} \log(1 + U) &= U - \frac{1}{2}U^2 + \frac{1}{3}U^3 + \mathcal{O}_t^{\text{form}}(t^4) \\ &= (L + 1)t + \left(L^2 - \frac{(L+1)^2}{2} \right) t^2 + \left(\frac{(L+1)^3}{3} - L^2(L + 1) \right) t^3 + \mathcal{O}_t^{\text{form}}(t^4). \end{aligned}$$

Every step is a finite polynomial computation, and the certificate $\exp(\log(1+U)) - (1+U) = \mathcal{O}_t^{\text{form}}(t^4)$ checks it.

8.2 The ambient exponential extension

The exponential of an element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ need not lie in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, so a statement such as “ $\exp F$ leaves the class” is meaningless until an ambient object has been named in which $\exp F$ exists. We therefore fix one once and for all. Nothing below depends on which ambient is chosen: every negative result is proved from the axioms alone, so it holds in all of them simultaneously.

Definition 8.7 (Exponential extension). An *exponential extension* of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is a triple $(\mathcal{T}, \partial, \exp)$ in which

1. $\mathcal{T}$ is an integral domain containing $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ as a subring, and ∂ is a derivation of $\mathcal{T}$ restricting to D_X on $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$;

2. $\exp: (\mathcal{T}, +) \rightarrow (\mathcal{T}^\times, \cdot)$ is a group homomorphism with $\partial(\exp F) = (\partial F) \exp F$ for all $F \in \mathcal{T}$;
3. (normalisation) $\exp L = t^{-1}$ and $\exp(t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}) \subseteq 1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$.

We write $\text{Const}(\mathcal{T}) = \ker \partial$ for the constants, say that $G \in \mathcal{T}$ is a *logarithm of F* when $\exp G = F$, and for $\alpha \in \mathbb{K}$ and F admitting a logarithm G we set $F^\alpha := \exp(\alpha G)$, a value that depends on the choice of G .

Remark 8.8 (Realisations). For $\mathbb{K} = \mathbb{R}$ the field $\mathbb{T}$ of logarithmic–exponential transseries is such an extension, with ∂ the transseries derivation and $\exp$ its exponential [1, 33, 13]; so is any Hardy field containing the germs at $+\infty$ of $X \mapsto X$ and $X \mapsto \log X$ and closed under $\exp$, for instance the germs of the log–exp-definable functions [32, 20]. Condition (iii) is a normalisation, not a restriction: it merely says that the ambient exponential extends the relation $L = \log X$ and does not multiply small exponentials by a stray constant. [Theorem 8.9](#) shows that (iii) already forces $\exp$ to agree on $t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ with the series of [Equation \(125\)](#).

Lemma 8.9 (The ambient exponential is the series exponential). *Let $(\mathcal{T}, \partial, \exp)$ be an exponential extension. For every $U \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$,*

$$\exp U = \sum_{r \geq 0} \frac{U^r}{r!},$$

the right-hand side being [Equation \(125\)](#). Consequently, for $V \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ the element $\log V$ of [Equation \(124\)](#) is a logarithm of V in $\mathcal{T}$.

Proof. Write $S = \sum_{r \geq 0} U^r / r! \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, a unit of $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. Since D_X is a derivation with $\nu_t(D_X F) \geq \nu_t F + 1$, it commutes with summable sums, so $D_X S = \sum_{r \geq 1} U^{r-1} (D_X U) / (r-1)! = (D_X U) S$. Hence $Z := (\exp U) S^{-1}$ lies in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ by (iii), and

$$\partial Z = (\partial \exp U) S^{-1} - (\exp U) S^{-2} D_X S = (D_X U) Z - (D_X U) Z = 0.$$

Thus $Z \in \ker D_X \cap \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} = \mathbb{K}$ ([Theorem 9.11](#)), while $Z \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ by (iii); the only scalar in $1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ is 1. For the last sentence, apply the first with $U = \log V$ and use [Equation \(127\)](#). $\square$

Lemma 8.10 (Logarithmic derivative). *Let $(\mathcal{T}, \partial, \exp)$ be an exponential extension and let $F \in \mathcal{T}$ admit a logarithm G . Then $F \in \mathcal{T}^\times$ and*

$$\partial G = \frac{\partial F}{F}. \quad (133)$$

Conversely, if G_1, G_2 are logarithms of F then $G_1 - G_2 \in \text{Const}(\mathcal{T})$, and if $H \in \mathcal{T}$ satisfies $\partial H = \partial F / F$ then $H - G \in \text{Const}(\mathcal{T})$.

Proof. $F = \exp G$ is a unit by (ii). Differentiating $F = \exp G$ gives $\partial F = (\partial G) F$, which is [Equation \(133\)](#). If $\exp G_1 = \exp G_2$ then $\exp(G_1 - G_2) = 1$, so $\partial(G_1 - G_2) = \partial(1)/1 = 0$. The last claim is immediate from [Equation \(133\)](#). $\square$

8.3 Power–log scales

To say that an operation leaves the polynomial–logarithmic class *and every reasonable enlargement of it* we must quantify over the enlargements. The following definition collects them: fractional or real exponents of X , rational or Laurent coefficients in L , and any enlargement of the scalars.

Definition 8.11 (Power–log scale). Let $\mathbb{K}'$ be a field of characteristic zero containing $\mathbb{K}$, let Γ be an ordered additive subgroup of $\mathbb{R}$ with $\mathbb{Z} \subseteq \Gamma$, fixed together with an embedding $\Gamma \hookrightarrow \mathbb{K}'$ of additive groups extending $\mathbb{Z} \subseteq \mathbb{K}'$ (so that the products γp_γ below are defined), and let C be one of $\mathbb{K}'[L]$,

$\mathbb{K}'(L)$, $\mathbb{K}'((L^{-1}))$, each equipped with the derivation ∂_L . The associated *power–log scale* is the Hahn series ring

$$\mathcal{H}(C, \Gamma) := \left\{ F = \sum_{\gamma \in S} p_\gamma t^\gamma : p_\gamma \in C, S \subseteq \Gamma \text{ well ordered} \right\},$$

with the Hahn product [19], the valuation $\nu_t F = \min \{\gamma : p_\gamma \neq 0\}$, and the operator

$$D_X F := \sum_{\gamma \in S} t^{\gamma+1} (\partial_L p_\gamma - \gamma p_\gamma). \quad (134)$$

Thus $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} = \mathcal{H}(\mathbb{K}[L], \mathbb{Z})$, $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}} = \mathcal{H}(\mathbb{K}(L), \mathbb{Z})$, $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}} = \mathcal{H}(\mathbb{K}((L^{-1})), \mathbb{Z})$, and the Puiseux-log ring $\mathbb{K}[L]((t^{1/s}))$ is $\mathcal{H}(\mathbb{K}[L], s^{-1}\mathbb{Z})$.

Lemma 8.12 (Basic properties of a power–log scale). *Let $\mathcal{H} = \mathcal{H}(C, \Gamma)$ be a power–log scale.*

1. $\mathcal{H}$ is an integral domain, ν_t is additive on nonzero products, and D_X is a derivation of $\mathcal{H}$.
2. $\omega(\partial_L h) \geq \omega(h) + 1$ for every $h \in C$, where ω denotes the L^{-1} -adic order computed in $\mathbb{K}'((L^{-1}))$ after the expansion embedding (Theorem 7.37). Consequently, for $h \neq 0$, $\omega(\partial_L h/h) \geq 1$, and $\partial_L h - \gamma h \neq 0$ for every $\gamma \in \Gamma \setminus \{0\}$.
3. $\nu_t(D_X F) \geq \nu_t F + 1$ for every $F \in \mathcal{H}$, and $\ker D_X = \mathbb{K}'$.
4. If $F \in \mathcal{H}$ is nonzero and F^{-1} exists in the fraction field of $\mathcal{H}$, then $\nu_t(D_X F/F) \geq 1$, and

$$[t^1] \left(\frac{D_X F}{F} \right) = \frac{\partial_L b}{b} - e, \quad e = \nu_t F, \quad b = [t^e] F. \quad (135)$$

Proof. (i) Hahn multiplication is well defined and makes $\mathcal{H}$ a domain when C is a domain, and ν_t is additive, because the minimum of the Minkowski sum of two well-ordered sets is the sum of the minima and the corresponding coefficient is the product of the leading coefficients [19]. That D_X is additive is clear; for the Leibniz rule it suffices, by Theorem 8.1 applied in the Hahn setting, to check it on monomials:

$$D_X(t^\alpha p \cdot t^\beta q) = t^{\alpha+\beta+1} ((pq)' - (\alpha + \beta)pq) = (D_X(t^\alpha p))t^\beta q + t^\alpha p D_X(t^\beta q).$$

(ii) For $h = \sum_{j \geq j_0} a_j L^{-j}$ with $a_{j_0} \neq 0$, we have $\partial_L h = \sum_j (-j) a_j L^{-j-1}$, whose terms all have order $\geq j_0 + 1$. The case $C = \mathbb{K}'(L)$ reduces to this one because the expansion embedding ι_0 of Theorem 7.37 commutes with ∂_L : it is a ring homomorphism fixing L , hence fixing polynomials and their derivatives, and applying it to $\partial_L(u/v) = (u'v - uv')/v^2$ for $u, v \in \mathbb{K}'[L]$ gives $\iota_0(\partial_L(u/v)) = \partial_L(\iota_0(u/v))$ by the quotient rule in $\mathbb{K}'((L^{-1}))$. Then $\omega(\partial_L h/h) = \omega(\partial_L h) - \omega(h) \geq 1$. If $\gamma \neq 0$ then $\omega(\gamma h) = \omega(h) < \omega(\partial_L h)$, so no cancellation is possible and $\partial_L h - \gamma h \neq 0$.

(iii) The bound is immediate from Equation (134). If $D_X F = 0$ then $\partial_L p_\gamma = \gamma p_\gamma$ for every γ ; by (ii) this forces $p_\gamma = 0$ for $\gamma \neq 0$. For $\gamma = 0$ we get $\partial_L p_0 = 0$; writing $p_0 = \sum_{j \geq j_0} a_j L^{-j}$ in $\mathbb{K}'((L^{-1}))$, this says $-j a_j = 0$ for all j , hence $a_j = 0$ for $j \neq 0$ and $p_0 = a_0 \in \mathbb{K}'$.

(iv) $\nu_t(D_X F/F) = \nu_t(D_X F) - \nu_t F \geq 1$ by (iii) and additivity. Write $F = t^e b(1 + S)$ with $\nu_t S > 0$, which is possible in the fraction field. Then, D_X being a derivation,

$$\frac{D_X F}{F} = \frac{D_X(t^e)}{t^e} + \frac{D_X b}{b} + \frac{D_X S}{1 + S} = -et + t \frac{\partial_L b}{b} + \frac{D_X S}{1 + S},$$

using $D_X(t^e) = -et^{e+1}$ and $D_X b = t \partial_L b$. The last term has order $\geq \nu_t S + 1 > 1$, so it does not contribute to the block at t^1 . $\square$

Remark 8.13 (A common ambient scale). Given two power–log scales $\mathcal{H}(C_1, \Gamma_1)$ and $\mathcal{H}(C_2, \Gamma_2)$ inside one exponential extension, both embed D_X -compatibly in $\mathcal{H}(\mathbb{K}'''((L^{-1})), \Gamma_1 + \Gamma_2)$, where $\mathbb{K}'''$ is the compositum of the two scalar fields: on coefficients this is Theorem 7.37, and on exponents it is the inclusion of well-ordered supports. Hence in the theorems below one may always assume that the scale containing the input also contains the output.

8.4 Exponentials: the exact boundary

The mechanism is the logarithmic derivative. If $E = \exp F$ lies in a power–log scale, then $D_X F = D_X E/E$ is a logarithmic derivative of an element of that scale, and [Theorem 8.12](#) constrains such an object twice: its outer order is at least 1, and its block at t^1 has positive L^{-1} -order. Those two constraints are the whole boundary.

Proposition 8.14 (Necessary conditions). *Let $(\mathcal{T}, \partial, \exp)$ be an exponential extension, let $\mathcal{H} = \mathcal{H}(C, \Gamma) \subseteq \mathcal{T}$ be a power–log scale, and let $F \in \mathcal{H}$ be such that $E := \exp F$ lies in $\mathcal{H}$. Put $e = \nu_t E \in \Gamma$ and $b = [t^e]E \neq 0$. Then*

$$\nu_t F \geq 0 \quad \text{and} \quad \partial_L([t^0]F) + e = \frac{\partial_L b}{b}. \quad (136)$$

In particular $\omega(\partial_L([t^0]F) + e) \geq 1$; and if $[t^0]F$ is a polynomial then

$$\partial_L([t^0]F) = -e, \quad \text{that is} \quad [t^0]F = a - eL \quad \text{with } a \in \mathbb{K}'. \quad (137)$$

Proof. E is nonzero and invertible in the fraction field of $\mathcal{H}$, to which ν_t extends with additivity. By [Theorem 8.10](#), $D_X F = D_X E/E$, so [Theorem 8.12\(iv\)](#) gives $\nu_t(D_X F) \geq 1$ and

$$[t^1]D_X F = \frac{\partial_L b}{b} - e.$$

Write $F = \sum_{\gamma} p_{\gamma} t^{\gamma}$. By [Equation \(134\)](#), $\nu_t(D_X F) \geq 1$ forces $\partial_L p_{\gamma} - \gamma p_{\gamma} = 0$ for every $\gamma < 0$, and [Theorem 8.12\(ii\)](#) then gives $p_{\gamma} = 0$ for $\gamma < 0$; that is, $\nu_t F \geq 0$. Again by [Equation \(134\)](#), the block of $D_X F$ at t^1 comes only from $\gamma = 0$ and equals $\partial_L p_0$, which proves [Equation \(136\)](#). Its right-hand side has $\omega \geq 1$ by [Theorem 8.12\(ii\)](#). If p_0 is a polynomial then $\partial_L p_0 + e$ is a polynomial, and a nonzero polynomial h has $\omega(h) = -\deg_L h \leq 0$; so $\partial_L p_0 + e = 0$, which is [Equation \(137\)](#). $\square$

Corollary 8.15 (The three ways out). *Let $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$.*

1. *If $\nu_t F < 0$ then $\exp F$ lies in no power–log scale whatsoever.*
2. *If $\nu_t F \geq 0$ and $\deg_L [t^0]F \geq 2$ then $\exp F$ lies in no power–log scale.*
3. *If $\nu_t F \geq 0$ and $[t^0]F = a + mL$, then $\exp F$ lies in no power–log scale whose exponent group omits m ; in particular it lies in no Puiseux–log ring at all when m is irrational.*

Proof. By [Theorem 8.13](#) we may work inside a single scale $\mathcal{H}(C, \Gamma)$ containing F , and $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ has polynomial coefficients, so [Equation \(137\)](#) applies. (i) If $\nu_t F < 0$ this contradicts the first half of [Equation \(136\)](#). (ii) Here $\partial_L [t^0]F$ is a nonzero polynomial of degree $\deg_L [t^0]F - 1 \geq 1$, so it is not the constant $-e$, contradicting [Equation \(137\)](#). (iii) Now $\partial_L [t^0]F = m$, so [Equation \(137\)](#) forces $m = -e \in \Gamma$. $\square$

Theorem 8.16 (Exponential closure boundary). *Let $(\mathcal{T}, \partial, \exp)$ be an exponential extension and $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. Then $\exp F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ if and only if*

$$\nu_t F \geq 0, \quad [t^0]F = a + mL \quad \text{with } m \in \mathbb{Z}, a \in \mathbb{K}, \exp a \in \mathbb{K}. \quad (138)$$

In that case

$$\exp F = (\exp a) t^{-m} \exp(F - [t^0]F), \quad \exp(F - [t^0]F) \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}. \quad (139)$$

Proof. Sufficiency. Put $V = F - [t^0]F \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Since $\exp$ is a homomorphism and $\exp L = t^{-1}$, for $m \in \mathbb{Z}$ we get $\exp(mL) = (\exp L)^m = t^{-m}$, so $\exp F = (\exp a) t^{-m} \exp V$; and $\exp V \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ by [Theorem 8.9](#). All three factors lie in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$.

Necessity. [Theorem 8.14](#) with $\mathcal{H} = \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, $C = \mathbb{K}[L]$, $\Gamma = \mathbb{Z}$ gives $\nu_t F \geq 0$ and $[t^0]F = a - eL$ with $a \in \mathbb{K}$ and $e = \nu_t(\exp F) \in \mathbb{Z}$; put $m = -e$. Then [Equation \(139\)](#) holds by the sufficiency computation, and solving it for $\exp a$ exhibits $\exp a$ as a product of elements of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, so $\exp a \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. Finally $\partial(\exp a) = (\partial a) \exp a = 0$, so $\exp a \in \ker D_X \cap \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} = \mathbb{K}$ by [Theorem 9.11](#). $\square$

Corollary 8.17 (Fractional and irrational slopes). *Let $\mathcal{H} = \mathcal{H}(\mathbb{K}'[L], \Gamma)$ be a power–log scale with polynomial coefficients on which the ambient exponential is normalised, in the sense that $\exp V \in 1 + \{W \in \mathcal{H} : \nu_t W > 0\}$ whenever $V \in \mathcal{H}$ has $\nu_t V > 0$. For $F \in \mathcal{H}$, $\exp F \in \mathcal{H}$ if and only if $\nu_t F \geq 0$, $[t^0]F = a + mL$ with $m \in \Gamma$, and the constant $\kappa := \exp(a + mL)t^m$ lies in $\mathbb{K}'$. Thus $\exp(\frac{1}{2}L) = X^{1/2}$ lies in the Puiseux–log ring $\mathbb{K}[L]((t^{1/2}))$ but not in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, and $\exp(\pi L) = X^\pi$ requires a Hahn scale with $\pi \in \Gamma$.*

Proof. That κ is a constant is the computation $\partial \kappa = m\kappa - m\kappa = 0$, using $D_X(t^m) = -mt^{m+1}$. Sufficiency: $\exp F = \kappa t^{-m} \exp(F - [t^0]F)$ with all factors in $\mathcal{H}$. Necessity: Equation (137) gives $\nu_t F \geq 0$ and $[t^0]F = a + mL$ with $m = -e \in \Gamma$, and then $\kappa = \exp(F) \exp(-(F - [t^0]F))t^m \in \mathcal{H} \cap \text{Const}(\mathcal{T}) = \mathbb{K}'$ by Theorem 8.12(iii). $\square$

Remark 8.18 (Laurent coefficients admit more exponentials). The affine criterion Equation (137) is a statement about polynomial coefficients. It genuinely fails for Laurent coefficients: $F = L^{-1}$ has $\nu_t F = 0$ and a t^0 -block that is not affine in L , yet $\exp(L^{-1}) = \sum_{k \geq 0} L^{-k}/k! \in \mathbb{K}((L^{-1})) \subseteq \mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$. There is no contradiction: Equation (136) is satisfied with $e = 0$ and $b = \exp(L^{-1})$, because $\partial_L b/b = -L^{-2}$ does have positive L^{-1} -order. The exact criterion in a scale with Laurent coefficients is precisely Equation (136): the t^0 -block of F , shifted by $-eL$, must be a logarithm inside C of an element of C .

Example 8.19 (The three witnesses). $\exp(t^{-1}) = e^X$ and $\exp(t^{-1}L) = X^X$ fail (i); $\exp(L^2) = X^{\log X}$ fails (ii); $\exp(\frac{1}{2}L) = X^{1/2}$ fails (iii) for $\Gamma = \mathbb{Z}$ but succeeds in the Puiseux–log ring. By contrast $\exp(3L + t) = X^3 e^{1/X} \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, and $\exp(Lt) = 1 + Lt + \frac{1}{2}L^2 t^2 + \dots \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$: smallness is measured by the dominance order, not by the size of the coefficient polynomials.

Remark 8.20 (The exponential hierarchy). Theorem 8.15(i) is the structural reason for the transseries hierarchy. The elements of negative outer order are exactly the *purely large* elements of the scale, and their exponentials are new transmonomials that no enlargement of the exponent group or of the coefficient field can express: enlarging Γ adds monomials t^γ , all of which have logarithmic derivative $-\gamma t$ of outer order 1, whereas e^X has logarithmic derivative 1. Closing under $\exp$ and $\log$ therefore requires a monomial group built by transfinite iteration, which is precisely the construction of the logarithmic–exponential transseries [1, 33, 13, 8]. The polynomial–logarithmic scale is the fragment of that field met after a dominant logarithmic change of variable, and its virtue is that all closure questions have the finite answers proved here.

Corollary 8.21 (Which elements are exponentials). *Let $\mathbb{K} = \mathbb{R}$ and let $\mathcal{T}$ be an exponential extension with $\exp(\mathbb{R}) \subseteq \mathbb{R}$ equal to the real exponential. Then*

$$\mathcal{E} := \{F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} : \exp F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}\} = \mathbb{R} \oplus \mathbb{Z}L \oplus t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}},$$

and $\exp$ restricts to a group isomorphism from $(\mathcal{E}, +)$ onto the subgroup of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}^\times$ consisting of the units whose scalar leading coefficient is positive.

Proof. The description of $\mathcal{E}$ is Theorem 8.16 with $\mathbb{K} = \mathbb{R}$, since $\exp a \in \mathbb{R}$ for every $a \in \mathbb{R}$. Equation (139) shows that $\exp$ maps $\mathcal{E}$ into the units with positive leading coefficient e^a . It is onto: a unit is $ct^{-m}(1+S)$ with $c > 0$, $m \in \mathbb{Z}$, $S \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ (Theorem 7.3), and $F = \log c + mL + \log(1+S) \in \mathcal{E}$ has $\exp F = ct^{-m}(1+S)$ by Equation (127) and Theorem 8.9. It is injective: if $\exp F = 1$ with $F \in \mathcal{E}$ then $D_X F = 0$ by Theorem 8.10, so $F \in \mathbb{K} = \mathbb{R}$ by Theorem 9.11, and $e^F = 1$ gives $F = 0$. $\square$

8.5 Logarithms: the leading block decides

We now prove the counterpart for logarithms. One elementary observation about Laurent coefficients does all the work in the negative direction, and it will reappear in Section 9 as the antiderivative obstruction; we state it here in the form needed.

Definition 8.22 (Logarithmic residue at $L = \infty$). For $h = \sum_{j \geq j_0} a_j L^{-j} \in \mathbb{K}'((L^{-1}))$ put $\text{res}(h) := a_1$, the coefficient of L^{-1} . For $h \in \mathbb{K}'(L)$, $\text{res}(h)$ means res of its expansion at $L = \infty$; for $h \in \mathbb{K}'[L]$, $\text{res}(h) = 0$. In the classical convention $\text{Res}_{L=\infty}(h \, dL) = -\text{res}(h)$.

Lemma 8.23 (Derivatives have no residue). $\text{res}(\partial_L h) = 0$ for every $h \in \mathbb{K}'((L^{-1}))$.

Proof. $\partial_L(\sum_{j \geq j_0} a_j L^{-j}) = \sum_{j \geq j_0} (-j) a_j L^{-j-1}$, and the exponent $-j-1$ equals -1 only for $j = 0$, whose coefficient $-0 \cdot a_0$ is 0. $\square$

Lemma 8.24 (Image and kernel of ∂_L). Let $g \in \mathbb{K}'((L^{-1}))$. There exists $h \in \mathbb{K}'((L^{-1}))$ with $\partial_L h = g$ if and only if $\text{res}(g) = 0$, and in that case h is unique up to an additive constant. Equivalently,

$$\ker \partial_L = \mathbb{K}', \quad \text{im } \partial_L = \{g \in \mathbb{K}'((L^{-1})) : \text{res}(g) = 0\},$$

so res induces an isomorphism $\mathbb{K}'((L^{-1}))/\text{im } \partial_L \cong \mathbb{K}'$.

Proof. Write $h = \sum_{j \geq j_0} a_j L^{-j}$ and $g = \sum_{k \geq k_0} b_k L^{-k}$. Then $\partial_L h = \sum_{j \geq j_0} (-j) a_j L^{-j-1}$, so $\partial_L h = g$ holds if and only if

$$-j a_j = b_{j+1} \quad \text{for every } j. \quad (140)$$

For $j \neq 0$ this determines $a_j = -b_{j+1}/j$ uniquely, and characteristic zero is what makes the division legitimate. For $j = 0$ the left side is 0, so (140) is solvable exactly when $b_1 = \text{res}(g) = 0$, and then a_0 is unconstrained. This gives both the criterion and uniqueness up to the additive constant a_0 .

The candidate h so defined is a legitimate element of $\mathbb{K}'((L^{-1}))$: if $b_k = 0$ for $k < k_0$ then $a_j = 0$ for $j < k_0 - 1$, so the support of h is bounded below by $k_0 - 1$. Finally $\ker \partial_L = \mathbb{K}'$ is the case $g = 0$ of the same computation. $\square$

Corollary 8.25 (Residue of a logarithmic derivative). For nonzero $h \in \mathbb{K}'((L^{-1}))$, $\text{res}(\partial_L h/h) = -\omega(h)$. In particular, for a nonzero polynomial p , $\text{res}(\partial_L p/p) = \deg_L p$, and for $p = a \prod_i (L - \lambda_i)^{m_i}$ over $\overline{\mathbb{K}'}$ the residue of $\partial_L p/p$ at the finite point λ_i is $m_i \neq 0$.

Proof. Write $h = a_{j_0} L^{-j_0} (1+w)$ with $\omega(w) \geq 1$ and $j_0 = \omega(h)$. Then $\partial_L h/h = -j_0 L^{-1} + \partial_L w/(1+w)$ by the Leibniz rule, and $\omega(\partial_L w/(1+w)) \geq \omega(w) + 1 \geq 2$ by Theorem 8.12(ii); so the coefficient of L^{-1} is $-j_0$. For a polynomial, $\omega(p) = -\deg_L p$. The last claim is the partial-fraction identity $\partial_L p/p = \sum_i m_i/(L - \lambda_i)$, and $m_i \neq 0$ in $\mathbb{K}'$ because $\text{char } \mathbb{K}' = 0$. $\square$

Theorem 8.26 (Logarithm boundary). Let $(\mathcal{T}, \partial, \exp)$ be an exponential extension, let $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ be nonzero with $n_0 = \nu_t F$, leading block $p = p_{n_0}$, scalar leading coefficient $a \in \mathbb{K}^\times$ and $d = \deg_L p$, and write $F = t^{n_0} p(1+S)$ with $S \in t\mathbb{K}(L)[[t]]$. Assume F has a logarithm G in $\mathcal{T}$ and that L has a logarithm Λ in $\mathcal{T}$. Then:

1. G is determined modulo $\text{Const}(\mathcal{T})$ by

$$G \equiv -n_0 L + d \Lambda + \log\left(\frac{p}{aL^d}\right) + \log(1+S) \pmod{\text{Const}(\mathcal{T})}, \quad (141)$$

where $\log(p/(aL^d)) = \sum_{r \geq 1} \frac{(-1)^{r+1}}{r} w^r$ with $w = p/(aL^d) - 1 \in L^{-1}\mathbb{K}[L^{-1}]$, and $\log(1+S)$ is the series Equation (124) with coefficients in $\mathbb{K}(L)$. All the displayed terms except $d\Lambda$ lie in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$.

2. G lies in some power–log scale if and only if $d = 0$, that is (Theorem 7.3) if and only if F is a unit of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. When $d = 0$ one may take $G = c - n_0 L + \log(1+S) \in \mathbb{K}(c)[L](\langle t \rangle)$ for any $c \in \mathcal{T}$ with $\exp c = a$.

Proof. (i) Call the right-hand side of Equation (141) H . By Theorem 8.10 it suffices to prove $\partial H = D_X F/F$, for then $G - H$ is a constant. Now $\partial \Lambda = D_X L/L = t/L$ by Theorem 8.10 applied to L , and D_X applied to the two logarithmic series termwise (legitimate by Theorem 8.1, the families being summable in the L^{-1} -adic and t -adic order respectively) gives $D_X \log(1+w) = D_X w/(1+w) = t \partial_L w/(1+w)$ and $D_X \log(1+S) = D_X S/(1+S)$. Hence

$$\partial H = -n_0 t + \frac{dt}{L} + t \frac{\partial_L w}{1+w} + \frac{D_X S}{1+S} = -n_0 t + t \frac{\partial_L p}{p} + \frac{D_X S}{1+S},$$

because $p = aL^d(1+w)$ gives $\partial_L p/p = d/L + \partial_L w/(1+w)$. On the other side, $F = t^{n_0} p(1+S)$ and the Leibniz rule give exactly the same expression for $D_X F/F$. The membership claims are clear: $w \in L^{-1}\mathbb{K}[L^{-1}]$ makes $\log(1+w) \in L^{-1}\mathbb{K}[[L^{-1}]]$, and $\log(1+S) \in t\mathbb{K}(L)[[t]] \subseteq \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ after Theorem 7.37.

(ii) *Sufficiency.* If $d = 0$ then $p = a$, $w = 0$, and Equation (141) exhibits a logarithm in $\mathbb{K}(c)[L]((t))$ once the constant is fixed as c ; note $S \in t\mathbb{K}[L][[t]]$ in this case, since $F = at^{n_0}(1+S)$ with $F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$.

Necessity. Suppose $G \in \mathcal{H}(C, \Gamma)$ and $d \geq 1$. By Theorem 8.10 and the computation in (i),

$$D_X G = \frac{D_X F}{F}, \quad [t^1]D_X G = \frac{\partial_L p}{p} - n_0.$$

On the other hand Equation (134) gives $[t^1]D_X G = \partial_L g_0$ with $g_0 = [t^0]G \in C$. Hence $\partial_L(g_0 + n_0 L) = \partial_L p/p$ with $g_0 + n_0 L \in C$. Apply res: the left side vanishes by Theorem 8.23, the right side equals $d \neq 0$ by Theorem 8.25 – a contradiction. The argument covers all three coefficient rings at once, since each embeds ∂_L -compatibly in $\mathbb{K}'((L^{-1}))$ (Theorem 8.12(ii)); for $C = \mathbb{K}'[L]$ one may argue even more directly, since $\partial_L p/p$ has a pole when $d \geq 1$ and is therefore not a polynomial. $\square$

Example 8.27 (A forced nested logarithm). Let $F = L^2 t^{-3}(1 + (L+1)t)$, so $n_0 = -3$, $p = L^2$, $a = 1$, $d = 2$, $w = 0$. Then

$$\log F \equiv 3L + 2\Lambda + (L+1)t - \frac{1}{2}(L+1)^2 t^2 + \frac{1}{3}(L+1)^3 t^3 + \mathcal{O}_t^{\text{form}}(t^4).$$

The new generator $\Lambda = \log \log X$ is forced by the factor L^2 , not by the small correction, and it enters with coefficient exactly $d = 2$.

Corollary 8.28 (Logarithms inside the class). *Let $F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ be nonzero. Then F has a logarithm in $\mathbb{K}'[L]((t))$ for some field $\mathbb{K}' \supseteq \mathbb{K}$ if and only if F is a unit of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$; one may take $\mathbb{K}' = \mathbb{K}(c)$ with $\exp c$ the scalar leading coefficient of F . If F is not a unit, a logarithm exists in $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}} \oplus \mathbb{K}\Lambda \oplus \text{Const}(\mathcal{T})$ and in no power–log scale.*

Proof. Immediate from Theorem 8.26; the second sentence is Equation (141), whose terms other than $d\Lambda$ lie in $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$. $\square$

8.6 Powers

Proposition 8.29 (General powers). *Let $F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ be a unit, $F = at^{n_0}(1+S)$ with $a \in \mathbb{K}^\times$, $n_0 \in \mathbb{Z}$, $S \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, let $\alpha \in \mathbb{K}$, fix $c \in \mathcal{T}$ with $\exp c = a$, and let $G = c - n_0 L + \log(1+S)$ be the corresponding logarithm of F (Theorem 8.26). Then*

$$F^\alpha = \exp(\alpha G) = \mu t^{\alpha n_0} (1+S)^\alpha, \quad (1+S)^\alpha = \sum_{r \geq 0} \binom{\alpha}{r} S^r \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}, \quad (142)$$

where $\mu := \exp(\alpha c - \alpha n_0 L) t^{\alpha n_0}$ is a constant of $\mathcal{T}$, equal to $\exp(\alpha c)$ whenever $\alpha n_0 \in \mathbb{Z}$. Consequently:

1. if $\alpha n_0 \in \mathbb{Z}$ and $\exp(\alpha c) \in \mathbb{K}'$, then $F^\alpha \in \mathbb{K}'[L]((t))$;

2. if $\alpha n_0 \in s^{-1}\mathbb{Z} \setminus \mathbb{Z}$, then F^α lies in the Puiseux-log ring $\mathbb{K}'[L]((t^{1/s}))$ and in no scale with $\alpha n_0 \notin \Gamma$; for αn_0 irrational a Hahn scale is required;
3. if F is not a unit, with leading block p of degree $d \geq 1$, then F^α lies in a power–log scale only if $\alpha d \in \mathbb{Z}$.

Proof. Since $\exp$ is a homomorphism, $\exp(\alpha G) = \exp(\alpha c - \alpha n_0 L) \exp(\alpha \log(1 + S))$, and the second factor is $(1 + S)^\alpha$ by the last identity of Equation (128) together with Theorem 8.9; inserting $t^{\alpha n_0} t^{-\alpha n_0}$ gives Equation (142). That μ is a constant is the computation of Theorem 8.17, and for $\alpha n_0 \in \mathbb{Z}$ the normalisation $\exp L = t^{-1}$ gives $\exp(-\alpha n_0 L) = t^{\alpha n_0}$, whence $\mu = \exp(\alpha c)$. Items (i) and (ii) follow, the negative half of (ii) from Theorem 8.15(iii) applied to αG . For (iii), Equation (141) shows that αG contains the term $\alpha d \Lambda$; equivalently, $D_X(F^\alpha)/F^\alpha = \alpha D_X F/F$ has $\text{res} = \alpha d$ by Theorem 8.25, while by that same corollary the logarithmic derivative of a nonzero element of any power–log scale has residue $-\omega(b) \in \mathbb{Z}$ when the coefficients are polynomial, and in general an integer whenever the leading coefficient lies in $\mathbb{K}'((L^{-1}))$. Hence $\alpha d \in \mathbb{Z}$. $\square$

Remark 8.30 (Branches). Over $\mathbb{C}$ the scalars $\exp c = a^1$ and $\exp \alpha c$ both depend on the choice of c , and different choices differ by $\exp(2\pi i k \alpha)$; over $\mathbb{R}$ with $a > 0$ the principal choice $c = \text{Log } a$ is available. A formula for F^α is incomplete until the branch is named, and the choice propagates into every later block. This is a genuine mathematical datum, not a notational nuisance: the two square roots of $X^2(1 + t)$ are $\pm X(1 + \frac{1}{2}t - \frac{1}{8}t^2 + \dots)$, and only the choice of sign makes the expansion an asymptotic expansion of a given branch as $X \rightarrow +\infty$.

8.7 The closure boundary

We can now state the closure boundary as a theorem rather than a table. The first part says that the polynomial–logarithmic class is generated from its two generators by one construction, Laurent assembly; the second says that a long list of further operations does not enlarge it; the third exhibits, for each operation that does enlarge it, a witness and the theorem that proves non-membership.

Theorem 8.31 (The generated class and its boundary). *Let $(\mathcal{T}, \partial, \exp)$ be an exponential extension.*

1. $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is the smallest subring $\mathcal{A} \subseteq \mathcal{T}$ that contains $\mathbb{K} \cup \{t, t^{-1}, L\}$ and is closed under Laurent assembly: whenever $p_n \in \mathbb{K}[L]$ for $n \geq n_0$ and every $p_n t^n$ lies in $\mathcal{A}$, the element $\sum_{n \geq n_0} p_n t^n \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ lies in $\mathcal{A}$.
2. $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is closed under: the ring operations; inversion of its units; $\Phi(U)$ for $\Phi \in \mathbb{K}[[z]]$ and $\nu_t U \geq 1$; the derivation D_X and the Euler derivation ϑ ; antidifferentiation; the exponential of any element of $\mathbb{K}_0 \oplus \mathbb{Z}L \oplus t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, where $\mathbb{K}_0 = \{a \in \mathbb{K} : \exp a \in \mathbb{K}\}$; and the logarithm of any unit whose scalar leading coefficient has a logarithm in $\mathbb{K}$.
3. Each of the following leaves $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, and none of the enlargements it forces can be avoided: reciprocal of a nonunit; a power with $\alpha n_0 \notin \mathbb{Z}$; the logarithm of a nonunit; the exponential of an element of negative outer order or with $\deg_L[t^0]F \geq 2$. Witnesses and minimal enlargements are given in the proof and collected in Table 2.

Proof. (i) The intersection of any family of subrings closed under Laurent assembly is again one, so a smallest $\mathcal{A}$ exists. $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is a subring of $\mathcal{T}$ containing the generators and is closed under Laurent assembly by definition, so $\mathcal{A} \subseteq \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. Conversely $\mathcal{A}$ contains $\mathbb{K}[L][t, t^{-1}]$, hence every $p_n t^n$, hence by assembly every element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$.

(ii) Ring operations and inversion of units are Theorem 7.3; $\Phi(U)$ is Theorem 8.2; D_X and ϑ are Theorem 9.12; antidifferentiation is Theorem 9.17; the exponential and logarithm clauses are Theorem 8.16 and Theorem 8.28.

(iii) The witnesses and their proofs are: $1/(L + 1) \notin \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ (Theorem 7.22), with minimal enlargement $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$, or $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ after expansion; $t^{1/2} = X^{-1/2}$ for $F = t$, $\alpha = 1/2$ (Theorem 8.29), with minimal enlargement a Puiseux-log or Hahn scale; $\log L = \Lambda$ for $F = L$ (Theorem 8.26),

with enlargement the adjunction of the new generator Λ ; e^X and $X^{\log X}$ (Theorem 8.15), with enlargement a genuinely exponential transseries field. In each case the cited statement proves non-membership in every power–log scale of the relevant type, so no enlargement of the coefficient field or of the exponent group alone suffices. $\square$

Table 2: Closure of the polynomial–logarithmic scale under the elementary operations. “Scale” means any power–log scale $\mathcal{H}(C, \Gamma)$ in the sense of the definition in this section.

Operation	Hypothesis	Result
$F + G, FG$	none	$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$
$1/F$	leading block in $\mathbb{K}^\times$	$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$
$1/F$	leading block of degree ≥ 1	$\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$; after expansion $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$
$\Phi(U), \Phi \in \mathbb{K}[[z]]$	$\nu_t U \geq 1$	$\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ (Theorem 8.2)
$D_X F, \vartheta F$	none	same class (Theorem 9.12)
antiderivative	none	$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ (Theorem 9.17)
antiderivative	in $\text{res } F = 0$	$\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$; otherwise adjoin Λ (Theorem 9.20)
$\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$		
F^α	F a unit, $\alpha \nu_t F \in \mathbb{Z}, a^\alpha \in \mathbb{K}$	$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$
F^α	$\alpha \nu_t F \notin \mathbb{Z}$	Puiseux-log or Hahn scale
$\log F$	F a unit, $\log a \in \mathbb{K}$	$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ (Theorem 8.28)
$\log F$	leading block of degree $d \geq 1$	$\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}} \oplus \mathbb{K}\Lambda$; no scale (Theorem 8.26)
$\exp F$	$\nu_t F \geq 0, [t^0]F = a + mL, m \in \mathbb{Z}, e^a \in \mathbb{K}$	$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ (Theorem 8.16)
$\exp F$	$\nu_t F \geq 0, [t^0]F = a + mL, m \notin \mathbb{Z}$	Hahn scale with $m \in \Gamma$
$\exp F$	$\deg_L [t^0]F \geq 2$, or $\nu_t F < 0$	no scale; exponential transseries (Theorem 8.15)

A common trap

Three errors recur in the sources and in informal practice.

- “Small” is not about the coefficients. Lt is small and L^{-1} is not, in the sense that decides summability of $\sum_r c_r U^r$: the criterion is $\nu_t U \geq 1$, not the numerical size of the blocks. For $\nu_t U = 0$ the series $\log(1 + U)$ need not be summable at all, and a normalisation must precede the expansion.
- A closure table is not a theorem. “ $\exp F$ may require new monomials” is not a statement one can act on. What is true is Theorem 8.15: for $\nu_t F < 0$ no enlargement of the coefficient field or of the exponent group ever suffices.
- Formal identities are not analytic ones. Every statement in this section is an identity between formal objects. That $\exp(\log(1 + U)) = 1 + U$ holds coefficientwise says nothing about the convergence of either series, and an equality $F = G + \mathcal{O}_t^{\text{form}}(t^N)$ carries no bound of the form $\mathcal{O}_{X \rightarrow +\infty}(X^{-N})$ without a separate realisation theorem.

Section summary

Substitution of a series of positive outer order into a formal power series is unconditional and finitely determined: $\text{Trunc}_{<N} \Phi(U)$ depends on $c_0, \dots, c_{\lfloor (N-1)/q \rfloor}$ and on $\text{Trunc}_{<N} U$ alone. Hence $(1 + U)^\alpha$, $\log(1 + U)$ and $\exp U$ stay in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Beyond that, the leading data decide. A logarithm stays in the scale precisely when the leading block is a scalar; otherwise the generator $\Lambda = \log L$ appears with coefficient $\deg_L p$, and the coefficients must be expanded at $L = \infty$. An exponential stays in a power–log scale precisely when $\nu_t F \geq 0$ and the t^0 -block is affine in L with

slope in the exponent group; otherwise the object is a new transmonomial and the exponential hierarchy begins.

9 Differentiation and antiderivatives

Differentiation is the operation for which the polynomial–logarithmic scale behaves best: it is unconditionally closed, it raises the outer order by exactly one except in one identifiable case, and – unlike almost every other restricted asymptotic scale – so does its inverse. This section proves those statements, in the sharp form, and then determines exactly which elements of each ambient class have an antiderivative inside that class. The answer is a residue: for $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ there is no obstruction at all, for $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ the obstruction is a single linear functional, and for $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ there is one linear condition at every pole of every coefficient block.

9.1 The distinguished derivation

In the formal algebra t and L are independent generators. The analytic relation $t = X^{-1}$, $L = \log X$ enters the algebra in exactly one place: it selects a derivation.

Theorem 9.1 (Existence and uniqueness). *Let ∂_L and ∂_t be the formal partial derivatives of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, each holding the other generator fixed.*

1. *The operator*

$$D_X := t \partial_L - t^2 \partial_t \quad (143)$$

is a $\mathbb{K}$ -derivation of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ with $D_X t = -t^2$ and $D_X L = t$, and it maps $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ into $t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$.

2. *D_X is the unique $\mathbb{K}$ -derivation δ of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ with $\delta t = -t^2$, $\delta L = t$ and $\delta(\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}) \subseteq \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. The last hypothesis may be replaced by t -adic continuity of δ .*

3. *Without any such hypothesis, uniqueness still holds on the subring $\mathbb{K}[L][t, t^{-1}]$.*

Proof. (i) Both ∂_L and ∂_t are $\mathbb{K}$ -derivations of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. Indeed each is defined blockwise or termwise, $\partial_L(\sum p_n t^n) = \sum p'_n t^n$ and $\partial_t(\sum p_n t^n) = \sum n p_n t^{n-1}$, both supports being bounded below; and the Leibniz rule holds because every block of a product is the finite sum $[t^k](FG) = \sum_{i+j=k} p_i q_j$, to which the Leibniz rule for polynomials, respectively the classical computation $k \sum_{i+j=k} p_i q_j = \sum_{i+j=k} (i p_i q_j + j p_i q_j)$, applies. A left $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ -linear combination of derivations is a derivation, so D_X is one. The values on the generators are immediate, and the block formula Equation (144) below gives $D_X(\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}) \subseteq t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$.

(ii) Let δ be as stated and put $\eta = \delta - D_X$, a $\mathbb{K}$ -derivation with $\eta t = \eta L = 0$, hence $\eta(t^{-1}) = -t^{-2}\eta(t) = 0$ and $\eta = 0$ on $\mathbb{K}[L][t, t^{-1}]$. Let $F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ and $N > \nu_t F$. Write $F = \text{Trunc}_{<N} F + t^N R$ with $\text{Trunc}_{<N} F \in \mathbb{K}[L][t, t^{-1}]$ and $R \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. Then

$$\eta F = \eta(t^N R) = N t^{N-1} \eta(t) R + t^N \eta R = t^N \eta R.$$

By hypothesis $\delta R \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and $D_X R \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, so $\eta R \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and $\nu_t(\eta F) \geq N$. As N was arbitrary, $\eta F = 0$. If instead η is continuous, choose N_0 with $\nu_t(\eta G) \geq 0$ whenever $\nu_t G \geq N_0$; applying the display with $N + N_0$ in place of N gives $\eta F = t^N \eta(t^{N_0} R)$ of order $\geq N$, and the same conclusion.

(iii) Contained in the first two sentences of (ii). □

Remark 9.2 (The hypothesis is not decorative). The sources, and the notation catalogue, call D_X the unique derivation with $D_X t = -t^2$ and $D_X L = t$. That is correct on the dense subring $\mathbb{K}[L][t, t^{-1}]$ and correct on $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ under the boundedness hypothesis of (ii), which every derivation arising in

practice satisfies; but the passage to the completion genuinely uses a hypothesis, since a derivation of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is not a priori bounded on $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. We have stated the hypothesis rather than suppressing it.

9.2 One block, and the Euler derivation

Proposition 9.3 (Block formula). *For $n \in \mathbb{Z}$ and $p \in \mathbb{K}[L]$,*

$$D_X(t^n p(L)) = t^{n+1}(\partial_L - n)p(L), \quad (144)$$

and hence, for $F = \sum_{n \geq n_0} p_n t^n \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$,

$$D_X F = \sum_{n \geq n_0} t^{n+1}(p'_n - n p_n), \quad \text{that is} \quad [t^{n+1}]D_X F = (\partial_L - n)[t^n]F. \quad (145)$$

The same formula holds with p in $\mathbb{K}(L)$ or $\mathbb{K}((L^{-1}))$ and with n replaced by any exponent α of a Puiseux-log or Hahn scale: $D_X(t^\alpha p) = t^{\alpha+1}(p' - \alpha p)$. Each output block receives a contribution from exactly one input block.

Proof. $\partial_L(t^n p) = t^n p'$ and $\partial_t(t^n p) = n t^{n-1} p$, so $D_X(t^n p) = t^{n+1} p' - n t^{n+1} p$. The series form follows because D_X is additive and commutes with summable sums, being continuous by Equation (144). For the general scale this is Equation (134), which was verified to be a derivation in Theorem 8.12(i). $\square$

Proposition 9.4 (The Euler derivation). *Put $\vartheta := t^{-1} D_X$, the formal counterpart of $X \, d/dX$. Then ϑ is a $\mathbb{K}$ -derivation of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ with $\vartheta t = -t$ and $\vartheta L = 1$, it preserves each outer block,*

$$\vartheta(t^n p(L)) = t^n (\partial_L - n)p(L), \quad (146)$$

and therefore $\nu_t(\vartheta F) \geq \nu_t F$ for all F . On the block $t^n \mathbb{K}[L]$ the operator ϑ acts as $\partial_L - n$, whose only eigenvalue is $-n$; it is invertible on that block if and only if $n \neq 0$.

Proof. A derivation multiplied by a ring element is again a derivation, and $t^{-1} \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$; the values on the generators and Equation (146) follow from Equation (144). On $t^n \mathbb{K}[L]$, the operator $\partial_L - n$ is, in the basis $1, L, \dots, L^D$ of polynomials of degree $\leq D$, triangular with every diagonal entry $-n$, because $\partial_L L^i = i L^{i-1}$ lowers the degree. Hence its determinant on that space is $(-n)^{D+1}$, which vanishes exactly when $n = 0$; and ∂_L itself is not injective on $\mathbb{K}[L]$. $\square$

Remark 9.5 (Resonance is a zero eigenvalue). Theorem 9.4 is the conceptual source of everything that follows. The scale decomposes into the ϑ -invariant blocks $t^n \mathbb{K}[L]$; on block n the Euler derivation is the scalar $-n$ plus a nilpotent, so it is invertible exactly when $n \neq 0$. The block $n = 0$ is *resonant*: there ϑ is the nilpotent ∂_L , with kernel $\mathbb{K}$ and cokernel 0 on $\mathbb{K}[L]$ – but with a one-dimensional cokernel on $\mathbb{K}((L^{-1}))$. Every statement below about constants of integration, logarithms and residues is a statement about that single block.

9.3 Repeated derivatives

Definition 9.6 (Repeated-derivative operator). For n in the exponent group and $k \in \mathbb{N}_0$ put

$$\Delta_{n,k} := \prod_{j=0}^{k-1} (\partial_L - n - j), \quad \Delta_{n,0} := 1. \quad (147)$$

The $k = 0$ product is empty and acts as the identity; the factors commute, each being a polynomial in ∂_L .

Two of the sources abbreviate the single factor $\partial_L - n$ by Δ_n . In the notation of this volume that operator is $\Delta_{n,1}$: the second index counts derivatives and is never optional, since $\Delta_{n,k}$ is *not* the k -th power of $\Delta_{n,1}$.

Theorem 9.7 (Repeated derivative formula). *For every $k \in \mathbb{N}_0$, every n , and every coefficient p ,*

$$D_X^k(t^n p(L)) = t^{n+k} \Delta_{n,k} p(L), \quad (148)$$

and $\Delta_{n,k+1} = \Delta_{n+1,k} \circ (\partial_L - n) = (\partial_L - n - k) \circ \Delta_{n,k}$.

Proof. Induction on k . For $k = 0$ both sides are $t^n p$, the empty product being the identity – the convention that the sources state and then, in Equation (148)’s applications, occasionally forget. Assume Equation (148) for k . Applying Equation (144) at outer exponent $n + k$ to the coefficient $\Delta_{n,k} p$ gives

$$D_X^{k+1}(t^n p) = t^{n+k+1} (\partial_L - n - k) \Delta_{n,k} p = t^{n+k+1} \Delta_{n,k+1} p,$$

the last step because the factors of Equation (147) commute. $\square$

Corollary 9.8 (Two specialisations). *For $k \geq 0$:*

$$D_X^k(t^n) = (-1)^k (n)^{\bar{k}} t^{n+k}, \quad (149)$$

with $(n)^{\bar{k}} = n(n+1) \cdots (n+k-1)$ and $(n)^{\bar{0}} = 1$; and, for $p \in \mathbb{K}[L]$,

$$\vartheta^k p(L) = t^{-k} D_X^k p(L) = \Delta_{0,k} p = \sum_{j=0}^k s(k, j) \partial_L^j p, \quad (150)$$

where $s(k, j)$ are the signed Stirling numbers of the first kind. More generally,

$$\Delta_{n,k} = \sum_{j=0}^k s(k, j) (\partial_L - n)^j.$$

Proof. For Equation (149), $\Delta_{n,k} 1 = \prod_{j=0}^{k-1} (-n - j) = (-1)^k \prod_{j=0}^{k-1} (n + j)$. For Equation (150), the generating identity of the signed Stirling numbers of the first kind is $(z)^{\bar{k}} = \prod_{j=0}^{k-1} (z - j) = \sum_j s(k, j) z^j$ [4, 18]; substituting the commuting operator $z = \partial_L$, respectively $z = \partial_L - n$, gives both displays, and $\vartheta^k = t^{-k} D_X^k$ holds on the block $n = 0$ by Equation (146). $\square$

Corollary 9.9 (Derivative budget). $\nu_t(D_X^k F) \geq \nu_t F + k$ for every $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and every $k \geq 0$. Consequently, in a Taylor-type family $\sum_{k \geq 0} H^k D_X^k F / k!$ with $\nu_t H \geq r \geq 0$, the k -th term has outer order at least $\nu_t F + (r + 1)k$; the family is summable as soon as $r \geq 0$, and the k -th term is invisible below t^N once $(r + 1)k > N - 1 - \nu_t F$.

Proof. Iterate Theorem 8.12(iii), or read off Equation (148). $\square$

9.4 The valuation of a derivative, exactly

The sources state $\nu_t(D_X F) \geq \nu_t F + 1$ and then describe the failure of equality by three cases. The three cases are correct, but they leave the actual value of $\nu_t(D_X F)$ undetermined in the third. It is determined, and the sharp statement is no harder.

Theorem 9.10 (Sharp valuation). *Let $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and write $a_{0,0} = [t^0 L^0] F \in \mathbb{K}$ for its constant term.*

1. $\nu_t(D_X F) \geq \nu_t F + 1$ always.

2. If $F \notin \mathbb{K}$, then

$$\nu_t(D_X F) = \nu_t(F - a_{0,0}) + 1. \quad (151)$$

3. Equality $\nu_t(D_X F) = \nu_t F + 1$ holds if and only if F is not of the form $a_{0,0} + \tilde{F}$ with $a_{0,0} \neq 0$ and $\nu_t \tilde{F} \geq 1$. Explicitly, with $n_0 = \nu_t F$, $p = p_{n_0}$, $d = \deg_L p$ and a the scalar leading coefficient:

- if $n_0 \neq 0$, the leading term of $D_X F$ is $-n_0 a t^{n_0+1} L^d$ and equality holds;
- if $n_0 = 0$ and $d > 0$, the leading term of $D_X F$ is $da t L^{d-1}$ and equality holds;
- if $n_0 = d = 0$, the leading block is annihilated and $\nu_t(D_X F) = \nu_t(F - a) + 1 \geq 2$.

The same statements hold in $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ and $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ with d replaced by $-\omega(p_{n_0})$, and in any power–log scale, except that in the third case the bound reads $\nu_t(D_X F) > 1$ when the exponent group is not $\mathbb{Z}$.

Proof. (i) is Theorem 8.12(iii). For (iii), apply Equation (144) to the leading block: the coefficient of t^{n_0+1} in $D_X F$ is $(\partial_L - n_0)p$. If $n_0 \neq 0$ this is a polynomial of degree d with leading coefficient $-n_0 a \neq 0$, so it is nonzero. If $n_0 = 0$ it is p' , of degree $d - 1$ with leading coefficient da , nonzero exactly when $d > 0$. If $n_0 = d = 0$ then $p = a$ and $p' = 0$, so the block vanishes.

For (ii), let $\tilde{F} = F - a_{0,0}$, which is nonzero since $F \notin \mathbb{K}$, and note $D_X F = D_X \tilde{F}$. It suffices to show that $\tilde{F}$ falls under one of the first two cases of (iii). Let $m = \nu_t \tilde{F}$. If $m \neq 0$ we are in the first case. If $m = 0$ then $[t^0] \tilde{F} = [t^0] F - a_{0,0}$ is a nonzero polynomial with vanishing constant term, hence of degree ≥ 1 , and we are in the second case. In both cases $\nu_t(D_X \tilde{F}) = m + 1$. The version for the larger classes is the same computation with ω in place of $-\deg_L$, using Theorem 8.12(ii). $\square$

Corollary 9.11 (Kernel). $\ker D_X = \mathbb{K}$ in $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$, $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$, and $\ker D_X = \mathbb{K}'$ in every power–log scale $\mathcal{H}(C, \Gamma)$. Consequently $\ker \vartheta = \mathbb{K}$ as well.

Proof. Theorem 8.12(iii); alternatively, if $F \notin \mathbb{K}$ then Equation (151) gives $\nu_t(D_X F) < +\infty$, so $D_X F \neq 0$. For $\vartheta = t^{-1} D_X$, multiplication by the unit t^{-1} does not change the kernel. $\square$

9.5 Closure of the ambient classes

Proposition 9.12 (Differential closure). D_X and ϑ map each of $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$, $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$, $\mathbb{K}[L]((t^{1/s}))$ and every power–log scale into itself. Moreover $D_X(\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}) = t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and $D_X(t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}) = t^2\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, so $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and its powers $t^k\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ form a filtration by D_X -stable subgroups, and each of $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$, $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ is a differential field.

Proof. Closure under D_X follows from Theorem 9.3, because the operator $\partial_L - n$ maps each of the three coefficient rings into itself. For ϑ multiply by t^{-1} , a unit of every class listed except $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. The two inclusions

$$D_X(\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}) \subseteq t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}, \quad D_X(t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}) \subseteq t^2\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$$

are immediate from Equation (144). Conversely, run the construction of Theorem 9.17 below on an F with $\nu_t F \geq 1$: it produces blocks only at indices $n \geq 0$, so the antiderivative lies in $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. On an F with $\nu_t F \geq 2$ it meets only nonresonant indices $n \geq 1$, so the antiderivative lies in $t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. $\square$

Remark 9.13 (Iterated logarithms are not polynomially closed). Set $L_1 = L$ and $L_{j+1} = \log L_j$. Then, by Theorem 8.10,

$$D_X L_j = \frac{D_X L_{j-1}}{L_{j-1}}, \quad \text{hence} \quad D_X L_j = \frac{t}{L_1 L_2 \cdots L_{j-1}} \quad (j \geq 1), \quad (152)$$

by induction from $D_X L_1 = t$. Consequently a polynomial ring $\mathbb{K}[L_1, \dots, L_r]((t))$ with $r \geq 2$ is *not* closed under D_X : already $D_X L_2 = t/L_1$ leaves it. The smallest natural differential extension replaces the coefficient ring by $\mathbb{K}(L_1, \dots, L_r)$, or by iterated Laurent coefficients, and the chain rule then reads

$$D_X(a(L_1, \dots, L_r)t^n) = t^{n+1} \left(\partial_{L_1} a + \frac{1}{L_1} \partial_{L_2} a + \dots + \frac{1}{L_1 \dots L_{r-1}} \partial_{L_r} a - na \right).$$

This is the exact sense in which one logarithmic level is special: it is the only depth at which polynomial coefficients are differentially closed.

A common trap

The formal partial derivative ∂_t , which extracts coefficients, is not the derivative with respect to X . A system that treats L as a symbol independent of X will compute $D_X(p(L)t^n) = -np(L)t^{n+1}$, losing the term $t^{n+1}p'(L)$ – exactly the term that carries all logarithmic information. Conversely, differentiating after the substitution $L = -\log t$ is a different operation again. Print the differentiated variable.

A separate trap is analytic. Termwise differentiation of an *asymptotic* expansion is not automatic: a remainder that is $o_{X \rightarrow +\infty}(X^{-N})$ may have a derivative that is not $o_{X \rightarrow +\infty}(X^{-N-1})$. Formal closure of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ under D_X is an algebraic fact and licenses no analytic conclusion without a realisation theorem – analyticity in a sector, or monotonicity plus a derivative estimate [25, 20].

9.6 The block equation and the resonant block

Antidifferentiation is, by Equation (145), a block-by-block problem: $D_X G = F$ with $G = \sum q_n t^n$ is the system

$$(\partial_L - n)q_n = p_{n+1}, \quad n \in \mathbb{Z}, \quad (153)$$

one independent equation per block, with the outer index shifted down by one. Everything therefore reduces to the solvability of one linear ordinary equation in one polynomial or coefficient unknown.

Lemma 9.14 (Nonresonant blocks are triangular). *Let $a \in \mathbb{K} \setminus \{0\}$ and $q \in \mathbb{K}[L]$ of degree D . Then $(\partial_L - a)p = q$ has the unique polynomial solution*

$$p = - \sum_{j=0}^D \frac{q^{(j)}}{a^{j+1}}, \quad (154)$$

a finite sum, and $\deg_L p = D$ with leading coefficient $-a^{-1}$ times that of q . The same formula solves the equation uniquely in $\mathbb{K}((L^{-1}))$ for any $q \in \mathbb{K}((L^{-1}))$, the sum then being convergent in the L^{-1} -adic order; hence $\partial_L - a$ is bijective on $\mathbb{K}[L]$ and on $\mathbb{K}((L^{-1}))$.

Proof. In the basis $1, L, \dots, L^D$, the operator $\partial_L - a$ is triangular with diagonal $-a \neq 0$, hence bijective on polynomials of degree $\leq D$; this proves existence, uniqueness and the degree statement. For the closed form, apply $\partial_L - a$ to the right-hand side of Equation (154):

$$(\partial_L - a) \left(- \sum_{j=0}^D \frac{q^{(j)}}{a^{j+1}} \right) = - \sum_{j \geq 0} \frac{q^{(j+1)}}{a^{j+1}} + \sum_{j \geq 0} \frac{q^{(j)}}{a^j} = q,$$

the sums telescoping and $q^{(j)} = 0$ for $j > D$. In $\mathbb{K}((L^{-1}))$ the same computation is valid because $\omega(q^{(j)}) \geq \omega(q) + j \rightarrow +\infty$ by Theorem 8.12(ii), so the family is summable and may be differentiated termwise; injectivity is Theorem 8.12(ii). $\square$

Lemma 9.15 (The resonant block). *On $\mathbb{K}[L]$, ∂_L is surjective with kernel $\mathbb{K}$: every $q \in \mathbb{K}[L]$ of degree D has the antiderivatives $p = c + \sum_{i=0}^D \frac{q_i}{i+1} L^{i+1}$, of degree $D+1$, one for each $c \in \mathbb{K}$. On $\mathbb{K}((L^{-1}))$, ∂_L has kernel $\mathbb{K}$ and image $\{h : \text{res}(h) = 0\}$, a subspace of codimension one.*

Proof. The polynomial statement uses $\text{char } \mathbb{K} = 0$ to divide by $i + 1$; that $\deg_L p = D + 1$ is clear from the leading term, and $p' = 0$ forces $p \in \mathbb{K}$. The statement for $\mathbb{K}((L^{-1}))$ is [Theorem 8.24](#), and res is onto $\mathbb{K}$ because $\text{res}(L^{-1}) = 1$. $\square$

Remark 9.16 (The same dichotomy as in the motivation part). [Theorem 9.14](#) and [Theorem 9.15](#) are the nonresonant and resonant halves of [Theorem 2.16](#), sharpened in the two ways the antiderivative theorem needs: the nonresonant inverse is given in closed form by [Equation \(154\)](#), and both halves are proved on $\mathbb{K}((L^{-1}))$ as well as on $\mathbb{K}[L]$, where the resonant image acquires the codimension-one residue obstruction. Nothing about the mechanism is new: the unique resonant outer level is still $n = 0$ for $\partial_L - n$, equivalently the level $n = 1$ of the series being antidifferentiated, and it is still the only level at which the logarithmic degree can rise.

9.7 Antiderivatives in the polynomial–logarithmic class

Theorem 9.17 (D_X is onto $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$). *The sequence*

$$0 \longrightarrow \mathbb{K} \longrightarrow \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} \xrightarrow{D_X} \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} \longrightarrow 0 \quad (155)$$

is exact. That is: every $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ has an antiderivative $G \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, and G is unique up to an additive constant of $\mathbb{K}$. Explicitly, with $F = \sum_{n \geq n_0} p_n t^n$, the antiderivatives are

$$G = c + \int p_1 \, dL + \sum_{\substack{n \geq n_0 - 1 \\ n \neq 0}} t^n \left(- \sum_{j \geq 0} \frac{p_{n+1}^{(j)}}{n^{j+1}} \right), \quad c \in \mathbb{K}, \quad (156)$$

where $\int p_1 \, dL$ is the polynomial antiderivative with zero constant term. The log-degree profile of G is

$$d_G(n) = d_F(n + 1) \quad (n \neq 0), \quad d_G(0) = d_F(1) + 1 \quad (157)$$

for the choice $c = 0$, with the convention $-\infty + 1 = -\infty$. The same holds in every Puiseux-log ring and in every power-log scale with polynomial coefficients.

Proof. Exactness at the left and middle is [Theorem 9.11](#). For surjectivity, solve [Equation \(153\)](#) block by block: for $n \neq 0$ use [Theorem 9.14](#) with $a = n$, obtaining the unique $q_n \in \mathbb{K}[L]$ displayed in [Equation \(156\)](#); for $n = 0$ use [Theorem 9.15](#), obtaining $q_0 = c + \int p_1 \, dL$. The resulting family $(q_n t^n)$ has support bounded below by $n_0 - 1$, so [Theorem 8.1](#) assembles it into an element $G \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, and [Equation \(145\)](#) gives $D_X G = F$. Uniqueness up to $\mathbb{K}$ is again [Theorem 9.11](#). The profile statement is the degree bookkeeping of [Theorems 9.14](#) and [9.15](#): the nonresonant solution has the same degree as its right-hand side, the resonant one has degree one higher. Nothing in the argument used $\Gamma = \mathbb{Z}$; for a general scale with polynomial coefficients the only resonant exponent is still $\gamma = 0$. $\square$

Example 9.18 (Both blocks). Take $F = L^2 t^3 + L^2 t$. The first term is nonresonant with $n = 2$: [Equation \(154\)](#) with $a = 2$, $q = L^2$ gives $q_2 = -\frac{1}{2}L^2 - \frac{1}{2}L - \frac{1}{4}$. The second is the resonant block $n = 0$: $q_0 = \frac{1}{3}L^3 + c$. Hence

$$\int \left(\frac{(\log X)^2}{X^3} + \frac{(\log X)^2}{X} \right) dX = -\frac{2L^2 + 2L + 1}{4X^2} + \frac{L^3}{3} + c.$$

Differentiation checks both blocks. Note the degree growth in the resonant block only, as [Equation \(157\)](#) predicts.

Remark 9.19 (Why the classical obstruction is invisible here). [Theorem 9.17](#) may look too strong, since $\int dX/(X \log X) = \log \log X$ is the standard example of an integral that leaves an elementary scale. There is no conflict: the integrand is t/L , which is not in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ at all. The obstruction is invisible in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ precisely because a polynomial has no poles and no residue; it becomes visible, and is then exactly one linear condition, as soon as the coefficient ring is enlarged. That is the content of the next subsection.

9.8 The exact obstruction in the larger coefficient classes

Theorem 9.20 (Antiderivatives in $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$). *Define $\text{res}: \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}} \rightarrow \mathbb{K}$ by*

$$\text{res}(F) := [t^1 L^{-1}]F = \text{res}([t^1]F), \quad (158)$$

the coefficient of the single monomial tL^{-1} . Then the sequence

$$0 \longrightarrow \mathbb{K} \longrightarrow \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}} \xrightarrow{D_X} \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}} \xrightarrow{\text{res}} \mathbb{K} \longrightarrow 0 \quad (159)$$

is exact. That is: $F \in \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ has an antiderivative in $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ if and only if $\text{res}(F) = 0$; the antiderivative is then unique up to $\mathbb{K}$. In general, if Λ is a logarithm of L in an exponential extension, then

$$F - \text{res}(F) \frac{t}{L} \in D_X(\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}), \quad (160)$$

so that every $F \in \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ has an antiderivative in $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}} \oplus \mathbb{K}\Lambda$.

Proof. Exactness at $\mathbb{K}$ and at the first $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ is [Theorem 9.11](#). The system [Equation \(153\)](#) has, by [Theorem 9.14](#), a unique solution $q_n \in \mathbb{K}((L^{-1}))$ for every $n \neq 0$, and by [Theorem 9.15](#) a solution q_0 if and only if $\text{res}(p_1) = 0$, in which case it is unique up to $\mathbb{K}$. Since $\text{res}(p_1) = \text{res}(F)$ by definition, and since the assembled family has support bounded below by $\nu_t F - 1$ ([Theorem 8.1](#)), this proves exactness at the second $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$. Surjectivity of res follows from $\text{res}(t/L) = 1$. For [Equation \(160\)](#), subtracting $\text{res}(F)t/L$ makes the residue vanish, and $D_X \Lambda = D_X L/L = t/L$ by [Theorem 8.10](#). $\square$

Corollary 9.21 (Where the second logarithm comes from). *$F = t/L = 1/(X \log X)$ has $\text{res}(F) = 1$, so it has no antiderivative in $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$, hence none in $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ or $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$; it has the antiderivative $\Lambda = \log \log X$. More generally, by [Theorem 8.25](#), for nonzero F in $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ with leading coefficient b ,*

$$\text{res}\left(\frac{D_X F}{F}\right) = -\omega(b) = \deg_L b \text{ when } b \text{ is a polynomial}, \quad (161)$$

so the logarithmic derivative of F has an antiderivative in $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ if and only if $\omega(b) = 0$. This is [Theorem 8.26\(ii\)](#) again: $\log F = \int D_X F/F$, and the two obstructions are the same functional.

Proof. Combine [Theorem 9.20](#) with [Theorem 8.12\(iv\)](#) and [Theorem 8.25](#): $[t^1](D_X F/F) = \partial_L b/b - e$, whose residue is $-\omega(b)$, the constant e contributing nothing. $\square$

The rational-coefficient field behaves differently. The sources record, all of them correctly, that $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ is closed under differentiation, and they prove closure under antiderivatives for the polynomial class only; none of them remarks that the antiderivative statement fails after the enlargement to $\mathbb{K}(L)$ – an inference their closure tables invite. It does fail, and not only at the resonant block: the obstruction is a linear condition at every pole of every coefficient block.

Lemma 9.22 (Twisted residues). *Let $a \in \mathbb{K}$ and $q \in \mathbb{K}(L)$. For $\lambda \in \overline{\mathbb{K}}$ define the twisted residue*

$$\rho_{a,\lambda}(q) := \text{Res}_{L=\lambda}\left(e^{-a(L-\lambda)} q\right) = \sum_{k \geq 0} \frac{(-a)^k}{k!} \text{Res}_{L=\lambda}((L-\lambda)^k q), \quad (162)$$

a finite sum, and note $\rho_{0,\lambda} = \text{Res}_\lambda$. Then

$$(\partial_L - a)\mathbb{K}(L) = \{q \in \mathbb{K}(L) : \rho_{a,\lambda}(q) = 0 \text{ for every } \lambda \in \overline{\mathbb{K}}\}. \quad (163)$$

In particular $\mathbb{K}[L] \subseteq (\partial_L - a)\mathbb{K}(L)$, while for $a \neq 0$ neither $1/L$ nor $1/L^2$ lies in the image.

Proof. Work first over $\overline{\mathbb{K}}$ and fix λ ; write $u = L - \lambda$ and expand in $\overline{\mathbb{K}}((u))$. The multiplier $e^{-au} = \sum_k (-au)^k/k!$ is a unit of $\overline{\mathbb{K}}[[u]]$, the displayed sum is finite because $u^k q$ is regular for k at least the pole order, and

$$e^{-au}(\partial_L - a)p = \partial_L(e^{-au}p)$$

for every $p \in \overline{\mathbb{K}}((u))$. Since the residue of a derivative of a Laurent series vanishes, $\rho_{a,\lambda}$ kills the image; this proves the inclusion $\subseteq$, the value at a point where q is regular being 0.

For $\supseteq$, decompose $q = Q + \sum_\lambda q_\lambda$ into a polynomial and its polar parts over $\overline{\mathbb{K}}$; $\rho_{a,\lambda}(q) = \rho_{a,\lambda}(q_\lambda)$ because the other summands are regular at λ . The polynomial part is in the image by [Theorem 9.14](#) (for $a \neq 0$) or [Theorem 9.15](#) (for $a = 0$). Fix λ and let M be the pole order of q_λ . If p has a pole of order $m \geq 1$ at λ , then $(\partial_L - a)p$ has a pole of order exactly $m + 1$ there, since the top coefficient is $-mc_m \neq 0$. Hence the linear map sending polar parts of order $\leq M - 1$ to polar parts of order $\leq M$ is injective, so its image has codimension one in the M -dimensional space of polar parts of order $\leq M$; the functional $\rho_{a,\lambda}$ vanishes on that image and is not identically zero on the target (it takes the value d_1 on $d_1 u^{-1}$). Therefore the image is exactly $\ker \rho_{a,\lambda}$, and q_λ lies in it. Summing the local solutions and the polynomial solution gives $p \in \overline{\mathbb{K}}(L)$ with $(\partial_L - a)p = q$.

Finally, descend to $\mathbb{K}$. If $a \neq 0$, p is unique by [Theorem 9.14](#), hence fixed by every automorphism of $\overline{\mathbb{K}}$ over $\mathbb{K}$ (which fixes q , a and L), so $p \in \mathbb{K}(L)$. If $a = 0$, p is unique up to a constant; choosing a finite Galois extension $E/\mathbb{K}$ with $p \in E(L)$ and replacing p by $[E : \mathbb{K}]^{-1} \sum_{\sigma \in \text{Gal}(E/\mathbb{K})} \sigma(p)$, which is again an antiderivative because each $\sigma(p)$ is one, gives an element of $\mathbb{K}(L)$. The examples: $1/L$ has $\rho_{a,0} = 1$; $1/L^2$ has $\rho_{a,0} = -a$. $\square$

Theorem 9.23 (Antiderivatives in $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$). *Let $F = \sum_{n \geq n_0} p_n t^n \in \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$. Then F has an antiderivative in $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ if and only if*

$$\rho_{n-1,\lambda}(p_n) = 0 \quad \text{for every } n \in \mathbb{Z} \text{ and every } \lambda \in \overline{\mathbb{K}}, \quad (164)$$

and the antiderivative is then unique up to $\mathbb{K}$. The condition at $n = 1$ is the vanishing of all residues of $[t^1]F$; the conditions at $n \neq 1$ are the twisted conditions with weight $e^{-(n-1)(L-\lambda)}$. Consequently:

1. *every $F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ satisfies [Equation \(164\)](#) (polynomials have no poles), recovering [Theorem 9.17](#);*
2. *$t^2/L = 1/(X^2 \log X)$ has no antiderivative in $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$; it does have one in $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$, namely $t \sum_{j \geq 0} (-1)^{j+1} j! L^{-j-1}$;*
3. *$\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ is a differential field that is not closed under antidifferentiation, whereas $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is a differential ring, not a field, that is closed under antidifferentiation.*

Proof. The system [Equation \(153\)](#) is solvable in $\mathbb{K}(L)$ block by block if and only if $p_{n+1} \in (\partial_L - n)\mathbb{K}(L)$ for every n , which by [Theorem 9.22](#) is [Equation \(164\)](#) after the index shift $n \mapsto n - 1$. Solutions assemble by [Theorem 8.1](#) because the support is bounded below, and the ambiguity is $\ker D_X = \mathbb{K}$ by [Theorem 9.11](#). (i) is immediate. For (ii), $n = 2$ and $\lambda = 0$ give $\rho_{1,0}(1/L) = 1 \neq 0$, so the criterion fails; the element displayed is the unique solution of $(\partial_L - 1)q_1 = L^{-1}$ provided by [Theorem 9.14](#) in $\mathbb{K}((L^{-1}))$, namely $q_1 = -\sum_{j \geq 0} (L^{-1})^{(j)} = -\sum_{j \geq 0} (-1)^j j! L^{-j-1}$. (iii) collects [Theorem 9.12](#), [Theorem 9.17](#) and part (ii); that $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is not a field is [Theorem 7.23](#). $\square$

Remark 9.24 (A divergent formal antiderivative). The series $t \sum_{j \geq 0} (-1)^{j+1} j! L^{-j-1}$ of [Theorem 9.23\(ii\)](#) is a perfectly good element of $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$: its defining sum is L^{-1} -adically summable, and its blocks are determined by finitely many operations each. It is not, and does not claim to be, a convergent series; the factorial growth of its coefficients is the algebraic shadow of the fact that $\int dX/(X^2 \log X)$ is not elementary, and the analytic content of the expansion is a matter of summation theory rather than of the formal algebra [12, 8]. Nothing in [Theorems 9.20](#) and [9.23](#) asserts convergence; the statements are exact statements about coefficients.

Table 3: The differential calculus of the four ambient classes. Here $\text{res } F = [t^1 L^{-1}]F$ and $\rho_{a,\lambda}$ are the twisted residues defined in this section.

Class	D_X -closed	$\ker D_X$	antiderivative exists in the class
$\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ $\mathbb{K}[L][[t]]$	= yes, $t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$	into $\mathbb{K}$	iff $\nu_t F \geq 1$ (Theorem 9.12)
$\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ $\mathbb{K}[L]((t))$	= yes	$\mathbb{K}$	always (Theorem 9.17)
$\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ $\mathbb{K}(L)((t))$	= yes	$\mathbb{K}$	iff all $\rho_{n-1,\lambda}([t^n]F) = 0$ (Theorem 9.23)
$\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ $\mathbb{K}((L^{-1}))((t))$	= yes	$\mathbb{K}$	iff $\text{res } F = 0$ (Theorem 9.20)

Structural checkpoint

The whole of Section 9 turns on one line: D_X sends the outer block n to the block $n + 1$ by the operator $\partial_L - n$. Everything else is a statement about that operator. It is injective for $n \neq 0$ and has kernel $\mathbb{K}$ for $n = 0$; it is surjective on $\mathbb{K}[L]$ for every n , surjective on $\mathbb{K}((L^{-1}))$ for $n \neq 0$ and of corank one for $n = 0$, and of infinite corank on $\mathbb{K}(L)$ for every n as soon as poles are admitted. Reading the four ambient classes off from that single operator is the fastest route through the material.

Section summary

The distinguished derivation $D_X = t\partial_L - t^2\partial_t$ is the unique bounded $\mathbb{K}$ -derivation with $D_X t = -t^2$ and $D_X L = t$; on one block it is $D_X(t^n p) = t^{n+1}(\partial_L - n)p$, and k derivatives give $t^{n+k}\Delta_{n,k}p$ with the empty-product convention. The Euler derivation $\vartheta = t^{-1}D_X$ preserves each block and acts there as $\partial_L - n$, whose invertibility fails exactly at the resonant block $n = 0$. Valuation rises by exactly one except when the leading block is a nonzero constant, and then $\nu_t(D_X F) = \nu_t(F - a_{0,0}) + 1$. All four ambient classes are closed under D_X . For antiderivatives the answer is a residue: D_X is onto $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ with kernel $\mathbb{K}$; on $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ the single functional $\text{res } F = [t^1 L^{-1}]F$ is the entire obstruction, and it is repaired by adjoining $\Lambda = \log L$; on $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ there is one twisted-residue condition at every pole of every block, so exact rational coefficients are the one enlargement that costs closure under integration.

Part III

Composition

10 Admissible composition and substitution

Throughout this part $\mathbb{K}$ is a field of characteristic zero, X is the large variable, and

$$t := X^{-1}, \quad L := \log X$$

are the two generators. Unless a sector is named, the analytic regime is the positive real ray $X \rightarrow +\infty$ and every logarithm is the real logarithm there. We write $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ for $\mathbb{K}[L][[t]]$, $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ for $\mathbb{K}[L]((t))$, ν_t for the outer valuation, $\text{Trunc}_{<N} F$ for the *exclusive* truncation $\sum_{n < N} p_n(L)t^n$, and $D_X = t\partial_L - t^2\partial_t$ for the distinguished derivation. Recall the repeated-derivative operator

$$\Delta_{n,k} := \prod_{j=0}^{k-1} (\partial_L - n - j), \quad \Delta_{n,0} = 1, \quad D_X^k(t^n p(L)) = t^{n+k} \Delta_{n,k} p(L). \quad (165)$$

The empty product at $k = 0$ is the identity operator; this convention is used without further comment.

Composition is the first operation for which the analytic relation between the two generators is essential. Addition and multiplication hold L fixed and merely recombine supports that are already present. Substitution of $X \mapsto G(X)$ moves both generators at once, and it can move them out of the ring: fractional powers, a second logarithmic level, or exponential monomials may appear. The whole content of this section is a set of hypotheses under which the substitution is defined, closed, associative, and computable block by block, together with a careful separation of that formal statement from the analytic statement that it computes the asymptotics of an actual composite function.

10.1 What substitution has to mean

For a polynomial in t and L , substitution is finite algebra and needs no theory. For an infinite series $F = \sum_{n \geq n_0} p_n(L)t^n$ one wants to declare

$$F \circ G := \sum_{n \geq n_0} p_n(L \circ G) (t \circ G)^n,$$

and three separate things must be checked.

1. *The generators must land in a named ring.* Both $t \circ G$ and $L \circ G$ must be expansible in the chosen monomial scale.
2. *The outer sum must be summable.* For each output monomial, only finitely many indices n may contribute; otherwise a coefficient is an undefined infinite sum in $\mathbb{K}$.
3. *The result must have admissible support.* In a Laurent setting this means finitely many negative outer exponents; in a Hahn setting it means a well-ordered exponent set.

Each requirement fails for some perfectly meaningful G .

Example 10.1 (Four failures of naive substitution). 1. $F = t$, $G = X^{1/2}$: here $t \circ G = t^{1/2} \notin \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, and a Puiseux extension is forced.

2. $F = t$, $G = \log X$: here $t \circ G = L^{-1} \notin \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, and negative powers of L are forced, that is, a Laurent-log coefficient field.

3. $F = L$, $G = X \log X$: here $L \circ G = L + \log L$, and a second logarithmic level is forced.

4. $F = L, G = e^X$: here $t \circ G = e^{-X}$, smaller than every power of t , and an exponential monomial is forced.

Failure of summability is separate and more insidious. Let $F = \sum_{n \geq 0} t^n$ and let $G = c$ be a nonzero constant, so that $t \circ G = c^{-1}$ has outer valuation 0. Every summand c^{-n} then lands in the block t^0 , and the coefficient of t^0 in the putative composite is $\sum_{n \geq 0} c^{-n}$, which is not an element of $\mathbb{K}$. The same collapse occurs for any G whose leading exponent is 0; this is why the hypothesis $\rho > 0$ below is not decorative.

Remark 10.2. Nothing in [Theorem 10.1](#) says that the composite function does not exist. Closure is a statement about a chosen representation class, not about existence. When a substitution leaves the class, the correct response is to name the enlargement it demands, not to suppress the offending symbol.

We record the summability criterion once, in the generality needed later. Let $\Gamma \subseteq \mathbb{R}$ be an ordered additive subgroup and let

$$\mathbb{K}[L][[t^\Gamma]] := \left\{ \sum_{\alpha \in S} p_\alpha(L) t^\alpha : p_\alpha \in \mathbb{K}[L], S \subseteq \Gamma \text{ well ordered} \right\}$$

be the associated polynomial–logarithmic Hahn ring. For $F \neq 0$ write $E(F) := \{\alpha \in \Gamma : [t^\alpha]F \neq 0\}$ for its exponent support, so that $\nu_t(F) = \min E(F)$ and $\text{supp}_{\text{coeff}}(F)$ is the finer set of pairs (α, j) with $a_{\alpha,j} \neq 0$. For $\Gamma = \mathbb{Z}$ this ring is exactly $\mathcal{PL}_{\text{poly}, \mathbb{K}}$, and for $\Gamma = s^{-1}\mathbb{Z}$ it is the Puiseux–log ring $\mathbb{K}[L]((t^{1/s}))$.

Lemma 10.3 (Summable families in a polynomial–logarithmic Hahn ring). *Let $\mathcal{A} = \mathbb{K}[L][[t^\Gamma]]$ and let $(S_i)_{i \in I}$ be a family in $\mathcal{A}$ such that*

$$\{i \in I : \nu_t(S_i) \leq \beta\} \quad \text{is finite for every } \beta \in \Gamma. \quad (166)$$

Then $\bigcup_i E(S_i)$ is a well-ordered subset of Γ , every exponent occurs in only finitely many S_i , and

$$\sum_{i \in I} S_i := \sum_{\alpha} \left(\sum_i [t^\alpha] S_i \right) t^\alpha$$

is a well-defined element of $\mathcal{A}$. The sum is unchanged by any regrouping or reindexing of I ; in particular a doubly indexed summable family may be summed in either order. Finally, call a $\mathbb{K}$ -linear map $\Psi : \mathcal{A} \rightarrow \mathcal{B}$ between two such rings compatible with summable families when there is a non-decreasing unbounded φ with $\nu_t(\Psi a) \geq \varphi(\nu_t a)$ for all a and $\Psi(\sum_i S_i) = \sum_i \Psi S_i$ for every summable family; the first condition alone already forces (ΨS_i) to satisfy (166), and a composite of two compatible maps is compatible.

Proof. Let $T \subseteq \bigcup_i E(S_i)$ be nonempty and pick $\beta \in T$. By (166) the index set $I_\beta = \{i : \nu_t(S_i) \leq \beta\}$ is finite, and every element of T that is $\leq \beta$ lies in $\bigcup_{i \in I_\beta} E(S_i)$, a finite union of well-ordered sets and hence well ordered. Therefore $T \cap (-\infty, \beta]$ has a least element, which is the least element of T . So the union is well ordered. Taking $\beta = \alpha$ shows that only the finitely many $i \in I_\alpha$ can contribute to the exponent α , so each inner sum is finite and the displayed series has well-ordered support. Regrouping and reindexing change none of the finitely many terms contributing to a fixed α . For the last assertion, let φ be non-decreasing and unbounded with $\nu_t(\Psi a) \geq \varphi(\nu_t a)$, and fix β . Choose $\beta_0 \in \Gamma$ with $\varphi(\gamma) > \beta$ for all $\gamma > \beta_0$. Then $\{i : \nu_t(\Psi S_i) \leq \beta\} \subseteq \{i : \nu_t(S_i) \leq \beta_0\}$, which is finite. $\square$

Remark 10.4. Condition (166) at $\beta = 0$ already forces a common lower support bound: only finitely many members have valuation ≤ 0 , and each of those has finite valuation. A t -adically Cauchy sequence in a Laurent ring carries no such guarantee, which is why the family formulation is the one used throughout.

10.2 Formal exponential and logarithm of a one-unit

Write $\mathfrak{m} := t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ for the maximal ideal of $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$; in a Hahn ring $\mathfrak{m}$ denotes the ideal of elements of strictly positive valuation. For $Z \in \mathfrak{m}$ put

$$e^Z := \sum_{k \geq 0} \frac{Z^k}{k!}, \quad \log(1 + Z) := \sum_{r \geq 1} \frac{(-1)^{r-1}}{r} Z^r. \quad (167)$$

Both families satisfy (166) because $\nu_t(Z^k) \geq k\nu_t(Z) \rightarrow +\infty$.

Lemma 10.5 (Formal exponential and logarithm). *The maps (167) are mutually inverse bijections between $\mathfrak{m}$ and $1 + \mathfrak{m}$, and*

$$e^{Z_1+Z_2} = e^{Z_1}e^{Z_2}, \quad \log(P_1P_2) = \log P_1 + \log P_2, \quad \log(P^n) = n \log P$$

for $Z_1, Z_2 \in \mathfrak{m}$, $P, P_1, P_2 \in 1 + \mathfrak{m}$ and $n \in \mathbb{Z}$. Moreover, if Ψ is a $\mathbb{K}$ -algebra homomorphism into another such ring, compatible with summable families and carrying elements of positive valuation to elements of positive valuation, then $\Psi(e^Z) = e^{\Psi Z}$ and $\Psi(\log(1 + Z)) = \log(1 + \Psi Z)$.

Proof. In $\mathbb{Q}[[z_1, z_2]]$ one has the classical identities $\exp(z_1 + z_2) = \exp z_1 \exp z_2$, $\log((1+z_1)(1+z_2)) = \log(1 + z_1) + \log(1 + z_2)$, $\log \exp z_1 = z_1$ and $\exp \log(1 + z_1) = 1 + z_1$. Because $\nu_t(Z_i) > 0$, the monomial family $(a_{k_1 k_2} Z_1^{k_1} Z_2^{k_2})_{k_1, k_2}$ satisfies (166) for every choice of coefficients $a_{k_1 k_2} \in \mathbb{Q}$, so $z_i \mapsto Z_i$ extends to a $\mathbb{Q}$ -algebra homomorphism $\mathbb{Q}[[z_1, z_2]] \rightarrow \mathcal{A}$ that is compatible with summable families (Theorem 10.3); applying it to the four identities gives the first three assertions for $n \geq 0$, and $\log(P^{-1}) = -\log P$ follows from $\log(PP^{-1}) = \log 1 = 0$. The last sentence is termwise application of Ψ to (167). $\square$

10.3 Monomial-leading inner arguments and the substitution homomorphism

Definition 10.6 (Admissible monomial-leading argument). An *admissible monomial-leading inner argument* over $\mathbb{K}$ is a quadruple $(c, \log c, \rho, U)$ consisting of

$$c \in \mathbb{K}^\times, \quad \text{a chosen value } \log c \in \mathbb{K}, \quad \rho \in \mathbb{N}_{>0}, \quad U \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}},$$

presented as the inner argument

$$G(X) = cX^\rho(1 + U(X)) = ct^{-\rho}(1 + U) \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}. \quad (168)$$

We put

$$V := \log(1 + U) \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}. \quad (169)$$

On the real ray we additionally require $c > 0$ and take $\log c$ real, so that $G(X) \rightarrow +\infty$.

The datum $\log c$ is part of the inner argument, not a consequence of it. If $\mathbb{K}$ does not already contain a logarithm of c , it must first be enlarged; and if $\mathbb{K} \subseteq \mathbb{C}$ contains $2\pi i$, then c admits several admissible logarithms, which give genuinely different substitutions (Theorem 10.9).

Theorem 10.7 (Substitution homomorphism of a monomial-leading argument). *Let G be as in Theorem 10.6. For $F = \sum_{n \geq n_0} p_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ the family*

$$(p_n(\rho L + \log c + V) c^{-n} t^{\rho n} (1 + U)^{-n})_{n \geq n_0} \quad (170)$$

is summable in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, so that

$$F \circ G := \sum_{n \geq n_0} p_n(\rho L + \log c + V) c^{-n} t^{\rho n} (1 + U)^{-n} \quad (171)$$

is a well-defined element of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. The resulting map $\Phi_G : F \mapsto F \circ G$ is

1. a $\mathbb{K}$ -algebra homomorphism $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} \rightarrow \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, compatible with summable families;
2. determined by its values on the generators,

$$\Phi_G(t) = c^{-1}t^\rho(1+U)^{-1}, \quad \Phi_G(L) = \rho L + \log c + V; \quad (172)$$

3. exactly valuation-multiplying: $\nu_t(F \circ G) = \rho \nu_t(F)$ for all F . In particular Φ_G is injective, and $\Phi_G(\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}) \subseteq \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$.

Proof. Since $\nu_t(U) \geq 1$, [Theorem 10.5](#) gives $V \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, and the binomial family $((\binom{-n}{r}U^r)_{r \geq 0})$ is summable for every $n \in \mathbb{Z}$, so $(1+U)^{-n} \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. Because p_n is a polynomial of some finite degree d_n ,

$$p_n(\rho L + \log c + V) = \sum_{j=0}^{d_n} \frac{p_n^{(j)}(\rho L + \log c)}{j!} V^j \quad (173)$$

is a finite sum, each summand lying in $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$; here $p_n^{(j)}$ is the j -th derivative of p_n in its own variable and the polynomial is evaluated at $\rho L + \log c$. Hence the n -th member of (170) lies in $t^{\rho n} \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, and its valuation is at least ρn . Since $\rho \geq 1$ and $n \geq n_0$, condition (166) holds, and [Theorem 10.3](#) gives the sum (171).

For the algebra structure, note that (171) is exactly "substitute (172) into the expansion of F ". Concretely, let $\Psi_0 : \mathbb{K}[L][t, t^{-1}] \rightarrow \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ be the unique $\mathbb{K}$ -algebra homomorphism with $\Psi_0(L) = \rho L + \log c + V$ and $\Psi_0(t) = c^{-1}t^\rho(1+U)^{-1}$; it exists because L and t are algebraically independent over $\mathbb{K}$ and $\Psi_0(t)$ is a unit of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, the required inverse being $c t^{-\rho}(1+U)$. Then $\Phi_G(p_n(L)t^n) = \Psi_0(p_n(L)t^n)$ for every n and every p_n , and $\Phi_G(F) = \sum_n \Psi_0(p_n(L)t^n)$. Additivity of Φ_G is clear. For multiplicativity, let $F = \sum_n p_n t^n$ and $F' = \sum_m q_m t^m$. The double family $(\Psi_0(p_n t^n) \Psi_0(q_m t^m))_{n,m}$ satisfies (166), because its (n, m) member has valuation at least $\rho(n+m)$; summing it in either order ([Theorem 10.3](#)) gives

$$\Phi_G(F) \Phi_G(F') = \sum_{n,m} \Psi_0(p_n q_m t^{n+m}) = \sum_r \Psi_0\left(\sum_{n+m=r} p_n q_m t^r\right) = \Phi_G(F F').$$

It remains to check compatibility with summable families in the sense of [Theorem 10.3](#). The valuation half holds with $\varphi(x) = \rho x$, which is non-decreasing and unbounded. For the interchange, let $(S_i)_{i \in I}$ be summable with sum S and write $S_i = \sum_n p_{i,n}(L)t^n$. By [Theorem 10.4](#) there is n_1 with $\nu_t(S_i) \geq n_1$ for all i , so $p_{i,n} = 0$ for $n < n_1$, and for each fixed n only the finitely many i with $\nu_t(S_i) \leq n$ can have $p_{i,n} \neq 0$; consequently $[t^n]S = \sum_i p_{i,n}$ is a finite sum. The doubly indexed family $(\Psi_0(p_{i,n} t^n))_{i,n}$ has (i, n) member of valuation at least $\rho n \geq \rho n_1$ and therefore satisfies (166). Summing it with i innermost gives $\sum_n \Psi_0([t^n]S \cdot t^n) = \Phi_G(S)$, and summing it with n innermost gives $\sum_i \Phi_G(S_i)$; the two agree by [Theorem 10.3](#). This proves (i) and (ii).

For (iii), let $n_0 = \nu_t(F)$. Every summand with $n > n_0$ has valuation at least $\rho n > \rho n_0$, and the summand with $n = n_0$ equals $c^{-n_0} t^{\rho n_0} (p_{n_0}(\rho L + \log c) + W)$ with $\nu_t(W) \geq 1$, by (173) and $(1+U)^{-n_0} \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. Since $\rho \neq 0$, the substitution $L \mapsto \rho L + \log c$ is a $\mathbb{K}$ -algebra automorphism of $\mathbb{K}[L]$, so $p_{n_0}(\rho L + \log c) \neq 0$. Hence $[t^{\rho n_0}](F \circ G) = c^{-n_0} p_{n_0}(\rho L + \log c) \neq 0$ and $\nu_t(F \circ G) = \rho n_0$. Injectivity and preservation of $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ follow at once. $\square$

Corollary 10.8 (Scaling and pure power substitution). *With $U = 0$,*

$$F(cX^\rho) = \sum_{n \geq n_0} c^{-n} t^{\rho n} p_n(\rho L + \log c).$$

Thus a pure scaling acts on coefficient blocks by the affine change of logarithmic variable $L \mapsto \rho L + \log c$ and on outer exponents by $n \mapsto \rho n$; no new monomials are created.

Proof. Immediate from (171) with $U = V = 0$. $\square$

Remark 10.9 (The choice of logarithm is part of the data). Take $\mathbb{K} = \mathbb{C}$, $c = 1$, $\rho = 1$, $U = 0$, so that $G = X$ is the identity map. Choosing $\log c = 0$ gives $\Phi_G = \text{id}$. Choosing $\log c = 2\pi i$ gives the different $\mathbb{K}$ -algebra automorphism $t \mapsto t$, $L \mapsto L + 2\pi i$, which is (171) for the very same G . Therefore " $F \circ G$ " is shorthand for $\Phi_{(c, \log c, \rho, U)}(F)$, and a symbolic implementation must carry $\log c$ as data rather than replace it by a principal value. All statements below that compose two inner arguments carry an explicit compatibility hypothesis on the chosen logarithms.

10.4 Finite determination

The practical content of [Theorem 10.7](#) is that each output block is a finite algebraic expression in finitely many input blocks, with an explicit and *triangular* precision budget.

Theorem 10.10 (Finite determination of a composition). *Let G be as in [Theorem 10.6](#), let $F = \sum_{n \geq a} p_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ with $a = \nu_t(F)$, and let $N \in \mathbb{Z}$. Then*

$$[t^N](F \circ G) = \sum_{\substack{a \leq n \\ \rho n \leq N}} c^{-n} \sum_{\substack{j \geq 0, r \geq 0 \\ j \leq \deg p_n}} \frac{\binom{-n}{r}}{j!} [t^{N-\rho n}](V^j U^r) p_n^{(j)}(\rho L + \log c), \quad (174)$$

and in the inner sum only the pairs with $j + r \leq N - \rho n$ contribute. Both index ranges are finite, so every output block below any cutoff is a finite sum. Consequently $\text{Trunc}_{<N} F \circ G$ is a finite algebraic expression in $c^{\pm 1}$, $\log c$, ρ , the blocks p_n with $a \leq n < N/\rho$, and the blocks of U below outer order $N - \rho a$.

Proof. Insert (173) and the binomial expansion $(1 + U)^{-n} = \sum_{r \geq 0} \binom{-n}{r} U^r$ into (171) and extract the coefficient of t^N ; the triple family is summable because its (n, j, r) member has valuation at least $\rho n + j + r$, so [Theorem 10.3](#) licenses the rearrangement. The outer index is restricted by $a \leq n$ and $\rho n \leq N$, that is $n \leq \lfloor N/\rho \rfloor$, and the inner indices by $j + r \leq N - \rho n$, because $\nu_t(V^j U^r) \geq j + r$; both are finite ranges. Finally, $\text{Trunc}_{<N} F \circ G$ collects (174) for all $N' < N$; the constraint $\rho n < N$ is $n < N/\rho$, and every occurrence of U or V sits inside a factor $t^{\rho n}$ with $\rho n \geq \rho a$, so only the blocks of U below outer order $N - \rho a$ can be reached. $\square$

The last clause is worth isolating, because it is the difference between a usable and an unusable implementation: the required precision of the inner data is not uniform across outer blocks.

Corollary 10.11 (Triangular precision budget). *In (171) the n -th summand needs V and $(1 + U)^{-n}$ only through outer order $N - \rho n - 1$. Computing all of them to the single global precision $N - \rho a - 1$ is correct but wasteful.*

Proof. The n -th summand carries the prefactor $t^{\rho n}$, so anything of outer order $\geq N - \rho n$ inside it contributes only to $\mathcal{O}_t^{\text{form}}(t^N)$. $\square$

Finite determination also yields the two stability estimates that make composition continuous in each argument. They are used repeatedly in the reversion theory.

Proposition 10.12 (Stability in the outer and inner arguments). *Let G, G_1, G_2 be admissible monomial-leading arguments with the same $c, \log c, \rho$ and with $U_1, U_2 \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.*

1. *If $\nu_t(F_1 - F_2) \geq M$ then $\nu_t(F_1 \circ G - F_2 \circ G) \geq \rho M$.*
2. *If $\nu_t(U_1 - U_2) \geq M \geq 1$ and $a = \nu_t(F)$, then $\nu_t(F \circ G_1 - F \circ G_2) \geq \rho a + M$.*

Proof. (i) is [Theorem 10.7\(iii\)](#) applied to $F_1 - F_2$ together with linearity of Φ_G .

For (ii) write $V_i = \log(1 + U_i)$ and $D = U_1 - U_2$. From $U_1^r - U_2^r = D \sum_{i=0}^{r-1} U_1^i U_2^{r-1-i}$ and $\nu_t(U_i) \geq 1$ we get $\nu_t(U_1^r - U_2^r) \geq M + r - 1$, whence $\nu_t(V_1 - V_2) \geq M$ by (167) and [Theorem 10.3](#). The same computation applied to $(1 + U_i)^{-n} = \sum_r \binom{-n}{r} U_i^r$ gives $\nu_t((1 + U_1)^{-n} - (1 + U_2)^{-n}) \geq M$. For a polynomial $p \in \mathbb{K}[L]$ there is a two-variable polynomial q over $\mathbb{K}$ with $p(y_1) - p(y_2) =$

$(y_1 - y_2)q(y_1, y_2)$; substituting $y_i = \rho L + \log c + V_i$, whose values have nonnegative valuation, gives $\nu_t(p_n(\rho L + \log c + V_1) - p_n(\rho L + \log c + V_2)) \geq M$. Writing the n -th summands of (171) as $A_1 B_1 - A_2 B_2 = (A_1 - A_2)B_1 + A_2(B_1 - B_2)$ with $A_i = p_n(\rho L + \log c + V_i)$ and $B_i = c^{-n} t^{\rho n} (1 + U_i)^{-n}$, all factors having valuation at least 0 resp. ρn , we obtain a valuation at least $\rho n + M \geq \rho a + M$ for every n . $\square$

10.5 The shifted-Taylor form of the composition formula

Formula (171) separates the logarithmic shift from the power prefactor. The two can be packaged into a single shifted operator, which is both more compact and better adapted to coefficient recurrences.

Lemma 10.13 (Shifted-Taylor identity). *Let $p \in \mathbb{K}[L]$, let $n \in \mathbb{Z}$, and let W be an element of positive valuation in a polynomial-logarithmic Hahn ring. Then*

$$e^{-nW} p(z + W) = \sum_{k \geq 0} \frac{W^k}{k!} ((\partial_z - n)^k p)(z), \quad (175)$$

where z is any element of the ring, ∂_z denotes differentiation of the polynomial p in its own variable, and the operator is applied to p before evaluation at z . The family on the right is summable; for $p \neq 0$ it terminates exactly when $n = 0$, since for $n \neq 0$ the polynomial $(\partial_z - n)^k p$ has the same degree as p and leading coefficient $(-n)^k$ times that of p .

Proof. Both sides are $\mathbb{K}$ -linear in p , so it suffices to expand. On the left,

$$e^{-nW} p(z + W) = \left(\sum_{i \geq 0} \frac{(-n)^i W^i}{i!} \right) \left(\sum_{j \geq 0} \frac{W^j}{j!} p^{(j)}(z) \right),$$

the second factor being the finite Taylor expansion of the polynomial p . Collecting the total power $k = i + j$ gives the coefficient $\sum_{j=0}^k \frac{(-n)^{k-j}}{(k-j)! j!} p^{(j)}(z)$. On the right, the binomial theorem for the commuting operators ∂_z and $-n$ gives

$$\frac{1}{k!} (\partial_z - n)^k p = \frac{1}{k!} \sum_{j=0}^k \binom{k}{j} (-n)^{k-j} p^{(j)} = \sum_{j=0}^k \frac{(-n)^{k-j}}{(k-j)! j!} p^{(j)}.$$

The two coefficients agree. Summability holds because $\nu_t(W^k) \geq k\nu_t(W) \rightarrow +\infty$, and the rearrangement of the double family is licensed by Theorem 10.3. $\square$

Corollary 10.14 (Operator form of the composition formula). *With G as in Theorem 10.6,*

$$F \circ G = \sum_{n \geq n_0} c^{-n} t^{\rho n} E_n, \quad E_n := \sum_{k \geq 0} \frac{V^k}{k!} ((\partial_z - n)^k p_n)(\rho L + \log c). \quad (176)$$

Proof. By Theorem 10.6 and Theorem 10.5, $(1 + U)^{-n} = e^{-nV}$. Apply Theorem 10.13 with $W = V$ and $z = \rho L + \log c$, and substitute into (171). $\square$

A common trap

Two ordering traps. First, *normal ordering*. The operator in (176) may be written suggestively as $:e^{V(\partial_z - n)}:$, the colons meaning that every power of V stands to the left and is never differentiated. It is *not* the operator exponential $\exp(M_V(\partial_L - n))$, where M_V is multiplication by V : iterating $M_V \partial_L$ would let ∂_L act on V , which depends on L . The two agree only when $\partial_L V = 0$. Second, *which variable is differentiated*. In (176) the operator ∂_z acts on the coefficient polynomial p_n in its own variable, and the result is evaluated at $\rho L + \log c$ afterwards. Writing ∂_L instead is

harmless only when $\rho = 1$. Indeed the chain rule gives

$$(\partial_L - n)^k [p_n(\rho L + \log c)] = ((\rho \partial_z - n)^k p_n)(\rho L + \log c),$$

so the misprint replaces the operator $(\partial_z - n)^k$ by $(\rho \partial_z - n)^k$. These two agree only for $\rho = 1$, and they are not even proportional: the factor ρ^k relates them in the single case $n = 0$. For $\rho = 2$, $n = 1$, $\log c = 0$, $p_n(z) = z$ and $k = 1$, the correct coefficient of V is $1 - 2L$ whereas the misprint produces $2 - 2L$.

The operator form makes the block coefficients of a single outer level satisfy an exact first-order recurrence. We state it for arbitrary ρ ; the case $\rho = 1$ is the form found in the sources.

Proposition 10.15 (Exact block recurrence). *Assume $\rho \in \mathbb{K}^\times$. With E_n as in (176), write $E_n = \sum_{r \geq 0} E_{n,r}(L)t^r$ and*

$$\theta := \frac{\partial_t V}{\rho + \partial_L V} = \sum_{q \geq 0} \theta_q(L)t^q, \quad (177)$$

the denominator being a unit of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Then

$$\partial_t E_n = \theta (\partial_L - n\rho) E_n, \quad (178)$$

and therefore

$$E_{n,0} = p_n(\rho L + \log c), \quad r E_{n,r} = \sum_{q=0}^{r-1} \theta_q (\partial_L - n\rho) E_{n,r-1-q} \quad (r \geq 1). \quad (179)$$

Proof. By Theorem 10.14 and Theorem 10.13, $E_n = e^{-nV} p_n(\rho L + \log c + V)$. Write $P = p_n$ and $w = \rho L + \log c + V$. Since ∂_t and ∂_L are formal partial derivatives and $\partial_t w = \partial_t V$, $\partial_L w = \rho + \partial_L V$,

$$\begin{aligned} \partial_t E_n &= e^{-nV} \partial_t V (P' - nP)(w), \\ \partial_L E_n &= e^{-nV} \left[-n(\partial_L V) P(w) + (\rho + \partial_L V) P'(w) \right]. \end{aligned}$$

Subtracting $n\rho E_n = e^{-nV} n\rho P(w)$ from the second line gives

$$(\partial_L - n\rho) E_n = e^{-nV} (\rho + \partial_L V) (P' - nP)(w),$$

because $-n(\partial_L V) - n\rho = -n(\rho + \partial_L V)$. Dividing the two displays yields (178); the denominator $\rho + \partial_L V$ is a unit because $\rho \in \mathbb{K}^\times$ and $\partial_L V \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Comparing coefficients of t^{r-1} in (178) gives (179); and $E_{n,0} = [t^0] E_n = p_n(\rho L + \log c)$ because $\nu_t(V^k) \geq k$, so only $k = 0$ contributes to the block t^0 in (176). $\square$

Remark 10.16. The naive substitute $\theta = \partial_t V$ is wrong. It ignores that the logarithmic shift itself depends on L , and it already fails at $r = 2$ whenever $\partial_L V \neq 0$. The correction factor $(\rho + \partial_L V)^{-1}$ is exactly the Jacobian of the change of logarithmic coordinate.

10.6 Rational and real leading exponents

Nothing in the proof of Theorem 10.7 used integrality of ρ except to keep the output in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$.

Proposition 10.17 (Puisseux-log and Hahn extensions). *Let $F = \sum_{n \geq a} p_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, let $c \in \mathbb{K}^\times$ with a chosen $\log c \in \mathbb{K}$, let $U \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and let $\rho > 0$ be real with $\rho \in \mathbb{K}$. Define $G = cX^\rho(1 + U)$ and $F \circ G$ by (171).*

1. *If $\rho = r/s$ with $r, s \in \mathbb{N}_{>0}$, then $F \circ G \in \mathbb{K}[L]((t^{1/s}))$, the Puiseux-log ring obtained by adjoining $t^{1/s}$.*

2. For arbitrary real $\rho > 0$, the family (170) is summable in $\mathbb{K}[L][[t^\Gamma]]$ with $\Gamma = \mathbb{Z} + \rho\mathbb{Z}$, and

$$\text{supp}_{\text{coeff}}(F \circ G) \subseteq \{(\rho n + m, j) : n \geq a, m \in \mathbb{N}_0, j \in \mathbb{N}_0\}.$$

The exponent set $\{\rho n + m : n \geq a, m \geq 0\}$ is locally finite: only finitely many of its elements lie below any real bound.

3. In both cases the conclusions of [Theorem 10.7](#) and [Theorem 10.10](#) hold with $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ replaced as target by $\mathbb{K}[L]((t^{1/s}))$ resp. $\mathbb{K}[L][[t^\Gamma]]$, with N ranging over Γ and $\lfloor N/\rho \rfloor$ read as the largest admissible n .

Proof. Only the support statement needs an argument. Fix a real bound B . Since $\rho > 0$, the inequality $\rho n + m \leq B$ with $n \geq a, m \geq 0$ forces $a \leq n \leq B/\rho$, finitely many values, and for each such n it forces $0 \leq m \leq B - \rho n$, again finitely many. Hence the exponent set is locally finite, in particular well ordered, and it is contained in the group Γ . In particular the n -th member of (170) has valuation at least ρn and $\{n \geq a : \rho n \leq \beta\}$ is finite for every real β , which is (166); so [Theorem 10.3](#) applies and every proof above goes through word for word. When $\rho = r/s$ we have $\Gamma \subseteq s^{-1}\mathbb{Z}$ and local finiteness upgrades to "finitely many negative exponents", which is the Puiseux-log condition. $\square$

Remark 10.18. The union over all s of the rings $\mathbb{K}[L]((t^{1/s}))$ admits rational exponents but no single denominator bound; it is a field, but a fixed element of it does not carry a uniform s . Real exponents genuinely require the Hahn setting, where the support condition, not the exponent group, does the work. Complex exponents require in addition a declared tie-breaking order, since ordering by real part alone is not a total order [19, 33].

Example 10.19 (Where the one-logarithm class ceases to be closed). The following table is a diagnostic; each row is proved by writing down $t \circ G$ and $L \circ G$.

Leading behavior of G	New object created	Minimal enlargement
$cX^\rho, \rho \in \mathbb{N}_{>0}$	none	$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$
$cX^{r/s}$	$t^{1/s}$	$\mathbb{K}[L]((t^{1/s}))$
cX^ρ, ρ real	t^ρ	$\mathbb{K}[L][[t^\Gamma]]$
$cX^\rho L^\sigma, \sigma \neq 0$	$\log L$	one further iterated-log level
$cL^\sigma, \sigma > 0$	$L^{-\sigma}$, and $\log L$ if some $p_n \notin \mathbb{K}$	completion order reversed
e^X	e^{-nX}	logarithmic-exponential transseries

For the fourth row, $\log G = \rho L + \sigma \log L + \log c + \log(1 + U)$, so a single outer coefficient of positive degree in its argument already produces $\log \log X$. The fifth row is the most drastic: $t \circ G = c^{-1}L^{-\sigma}$ has outer valuation 0, so the composite is not a series in t at all and the roles of the two completions are exchanged; since $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}} = \mathbb{K}((L^{-1}))((t)) \neq \mathbb{K}((t))((L^{-1}))$, naming the target field is part of the statement. For the sixth, $t \circ e^X = e^{-X}$, which is smaller than every power of t ; such height-changing substitutions belong to the field of logarithmic-exponential transseries [13, 33, 1].

Example 10.20 (A scale-changing substitution computed exactly). Let $F(Y) = Y + \log Y + Y^{-1}$ and $G(X) = 3X^2L^{-1}$ on the ray $X \rightarrow +\infty$. The inner argument is *not* admissible in the sense of [Theorem 10.6](#), because of the factor L^{-1} ; this is exactly why a new scale appears. Substituting the generators,

$$F \circ G = 3X^2L^{-1} + (\log 3 + 2L - \log L) + \frac{1}{3}Lt^2.$$

The second logarithmic level appears exactly, not as a remainder; the result lies in a two-logarithm class and in no smaller one.

10.7 Generator substitution for a near-identity argument

The case $c = 1$, $\log c = 0$, $\rho = 1$ is the one that governs reversion.

Definition 10.21 (Near-identity argument). For $H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ set

$$G = X + H = X(1 + tH),$$

so that $U = tH \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, $c = 1$, $\log c = 0$, $\rho = 1$. We write $\mathcal{G}_{\text{poly}, \mathbb{K}} = \{X + H : H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}\}$.

Proposition 10.22 (Exact generator substitution). For $G = X + H$ with $H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$,

$$t \circ G = \frac{t}{1 + tH}, \quad L \circ G = L + \log(1 + tH), \quad (180)$$

and consequently, for $F = \sum_{n \geq n_0} p_n(L)t^n$,

$$F \circ (X + H) = \sum_{n \geq n_0} t^n (1 + tH)^{-n} p_n(L + \log(1 + tH)). \quad (181)$$

Moreover $\nu_t(F \circ (X + H)) = \nu_t(F)$, so Φ_{X+H} preserves the valuation exactly and maps $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ injectively into $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.

Proof. Specialize (172) and (171) to $c = 1$, $\log c = 0$, $\rho = 1$, $U = tH$; the valuation statement is Theorem 10.7(iii) with $\rho = 1$. $\square$

A common trap

Both generators move, and neither moves the way it looks. Let $H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $G = X + H$.

Wrong rule 1: $t \mapsto t + H$. The symbol t denotes X^{-1} , not X . Adding H to t models $X^{-1} \mapsto X^{-1} + H$, which is not $(X + H)^{-1}$. The mistake is visible at once from valuations: for $H \notin t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ one has $\nu_t(t + H) = 0$, whereas $\nu_t(1/G) = 1$ because $G \rightarrow \infty$.

Wrong rule 2: update L , hold t fixed. Take $F = tL = \log X/X$ and $H = L$, so $G = X + \log X$. The correct value, from (181), is

$$F \circ G = (t - Lt^2 + L^2t^3 - \dots)(L + Lt - \frac{1}{2}L^2t^2 + \dots) = tL + (L - L^2)t^2 + \left(L^3 - \frac{3}{2}L^2\right)t^3 + \mathcal{O}_t^{\text{form}}(t^4).$$

Substituting only $L \mapsto L + \log(1 + Lt)$ returns $tL + Lt^2 - \frac{1}{2}L^2t^3 + \mathcal{O}_t^{\text{form}}(t^4)$, which misses the term $-L^2t^2$ entirely: the answer is already wrong in the first corrected block. Substituting only $t \mapsto t/(1 + Lt)$ returns $tL - L^2t^2 + L^3t^3 + \mathcal{O}_t^{\text{form}}(t^4)$ and misses $+Lt^2$. A correct implementation substitutes the two generators *simultaneously*: the replacement for L contains t , and the replacement for t contains L , so no sequential substitution is legitimate.

Proposition 10.23 (The near-identity monoid). For $A, B \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$,

$$(X + A) \circ (X + B) = X + B + A \circ (X + B), \quad (182)$$

and $A \circ (X + B) \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Hence $\mathcal{G}_{\text{poly}, \mathbb{K}}$ is a monoid under composition, with identity X , and the map $G \mapsto \Phi_G$ is an anti-homomorphism of monoids (Theorem 10.27).

Proof. Apply Φ_{X+B} , a $\mathbb{K}$ -algebra homomorphism, to $X + A = t^{-1} + A$: $\Phi_{X+B}(t^{-1}) = (\Phi_{X+B}t)^{-1} = X + B$, and $\Phi_{X+B}(A) = A \circ (X + B)$, which lies in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ by Theorem 10.22. $\square$

Proposition 10.24 (Filtration continuity, with a one-block gain on the right). Let $F, F_1, F_2 \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and $H, K \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Then

$$\nu_t(F_1 \circ (X + H) - F_2 \circ (X + H)) = \nu_t(F_1 - F_2), \quad (183)$$

$$\nu_t(F \circ (X + H) - F \circ (X + K)) \geq \nu_t(F) + 1 + \nu_t(H - K). \quad (184)$$

Proof. (183) is [Theorem 10.22](#) applied to $F_1 - F_2$. For (184) use the Taylor formula (195), proved independently in [Section 11](#): the difference equals $\sum_{k \geq 1} \frac{H^k - K^k}{k!} D_X^k F$. Since $H^k - K^k = (H - K) \sum_{i=0}^{k-1} H^i K^{k-1-i}$ and all factors lie in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, $\nu_t(H^k - K^k) \geq \nu_t(H - K)$; and $\nu_t(D_X^k F) \geq \nu_t(F) + k \geq \nu_t(F) + 1$ for $k \geq 1$. $\square$

Example 10.25 (A worked near-identity composition). Let

$$F = L^2 + (L + 1)t + (L^2 - L)t^2 + (2L + 1)t^3, \quad G = X + L.$$

Here $U = Lt$ and, through outer order 3,

$$t \circ G = t - Lt^2 + L^2t^3 + \mathcal{O}_t^{\text{form}}(t^4), \quad L \circ G = L + Lt - \frac{1}{2}L^2t^2 + \frac{1}{3}L^3t^3 + \mathcal{O}_t^{\text{form}}(t^4).$$

Simultaneous substitution and collection of blocks give

$$F \circ G = L^2 + (2L^2 + L + 1)t + (-L^3 + L^2 - L)t^2 + \left(\frac{2}{3}L^4 - 2L^3 + \frac{7}{2}L^2 + L + 1\right)t^3 + \mathcal{O}_t^{\text{form}}(t^4). \quad (185)$$

Example 10.26 (Increasing logarithmic degree is not an artifact). Let $F = L + L^2t$ and $G = X + L$. Then

$$F \circ G = L + (L^2 + L)t + \left(-L^3 + \frac{3}{2}L^2\right)t^2 + \left(L^4 - \frac{8}{3}L^3 + L^2\right)t^3 + \left(-L^5 + \frac{41}{12}L^4 - 2L^3\right)t^4 + \mathcal{O}_t^{\text{form}}(t^5).$$

The log-degree profile $d_{F \circ G}$ grows by one per block: every extra power of $\log(1 + Lt)$ supplies another factor L . No uniform bound on $\deg_L$ survives composition, which is why the ring is defined without one.

10.8 The associativity domain

Associativity is a theorem, not a formality: it holds on the nose only when the composite inner argument is itself admissible *and* the chosen logarithms are compatible.

Theorem 10.27 (Associativity). *Let*

$$G = c_G X^\rho (1 + U), \quad H = c_H X^\sigma (1 + W)$$

be admissible monomial-leading arguments with chosen logarithms $\log c_G, \log c_H \in \mathbb{K}$, with $\rho, \sigma \in \mathbb{N}_{>0}$ and $U, W \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Put

$$c := c_G c_H^\rho, \quad \log c := \log c_G + \rho \log c_H, \quad 1 + U^* := (1 + W)^\rho (1 + \Phi_H U). \quad (186)$$

Then $U^ \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, the quadruple $(c, \log c, \rho\sigma, U^*)$ is an admissible monomial-leading argument representing $G \circ H$, and for every $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$*

$$(F \circ G) \circ H = F \circ (G \circ H), \quad \text{equivalently} \quad \Phi_H \circ \Phi_G = \Phi_{G \circ H}. \quad (187)$$

The identity X is a two-sided unit, and $\mathcal{G}_{\text{poly}, \mathbb{K}}$ is closed under composition with the canonical choices $c = 1, \log c = 0$.

Proof. First, admissibility. Φ_H is a $\mathbb{K}$ -algebra homomorphism compatible with summable families ([Theorem 10.7](#)), and $\nu_t(\Phi_H U) = \sigma \nu_t(U) \geq \sigma \geq 1$; also $\nu_t((1 + W)^\rho - 1) \geq 1$, since ρ is a positive integer and $(1 + W)^\rho - 1$ is a sum of positive powers of W . Hence $U^* = (1 + W)^\rho (1 + \Phi_H U) - 1 \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Furthermore

$$G \circ H = \Phi_H(G) = c_G (\Phi_H t)^{-\rho} (1 + \Phi_H U) = c_G c_H^\rho t^{-\rho\sigma} (1 + W)^\rho (1 + \Phi_H U) = c t^{-\rho\sigma} (1 + U^*),$$

which is (186).

Second, the identity. Both $\Phi_H \circ \Phi_G$ and $\Phi_{G \circ H}$ are $\mathbb{K}$ -algebra homomorphisms $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} \rightarrow \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ compatible with summable families, the first because a composite of two such maps is one (Theorem 10.3). By the definition (171) of Φ , a homomorphism with this compatibility is determined by its values on t and L : indeed it maps $p_n(L)t^n$ to $p_n(\text{image of } L) \cdot (\text{image of } t)^n$ and then sums. So it suffices to compare the generators.

For t :

$$\Phi_H(\Phi_G t) = \Phi_H(c_G^{-1} t^\rho (1+U)^{-1}) = c_G^{-1} (c_H^{-1} t^\sigma (1+W)^{-1})^\rho (1+\Phi_H U)^{-1} = c^{-1} t^{\rho\sigma} (1+U^*)^{-1} = \Phi_{G \circ H}(t).$$

For L , using Theorem 10.5 twice,

$$\begin{aligned} \Phi_H(\Phi_G L) &= \Phi_H(\rho L + \log c_G + \log(1+U)) \\ &= \rho(\sigma L + \log c_H + \log(1+W)) + \log c_G + \log(1+\Phi_H U) \\ &= \rho\sigma L + (\log c_G + \rho \log c_H) + \log((1+W)^\rho (1+\Phi_H U)) \\ &= \rho\sigma L + \log c + \log(1+U^*) = \Phi_{G \circ H}(L). \end{aligned}$$

Hence the two homomorphisms agree. Taking $G = X$ with $c = 1$, $\log c = 0$, $\rho = 1$, $U = 0$ gives $\Phi_X = \text{id}$, which is the two-sided identity statement; closure of $\mathcal{G}_{\text{poly}, \mathbb{K}}$ is Theorem 10.23. $\square$

Remark 10.28 (Right substitution reverses the apparent order). Equation (187) reads $\Phi_H \circ \Phi_G = \Phi_{G \circ H}$: the operator that acts *second* is the one indexed by the inner argument that acts *first*. Substitution on the right is a contravariant operation, so $G \mapsto \Phi_G$ is an anti-homomorphism. Implementations of nested composition should test exactly this identity; reversing it silently produces correct-looking output for commuting pairs and wrong output otherwise.

Remark 10.29 (The compatibility hypothesis on logarithms cannot be dropped). Suppose $\mathbb{K} = \mathbb{C}$ and one chooses for $G \circ H$ the logarithm $\log c + 2\pi i$ instead of $\log c_G + \rho \log c_H$. Then $\Phi_{G \circ H}(L)$ differs from $\Phi_H(\Phi_G L)$ by $2\pi i$, and (187) fails for every F with a coefficient block of positive degree. Associativity is a statement about admissible *data*, not about the underlying maps alone.

Remark 10.30 (Rational and real exponents). Theorem 10.27 was stated for $\rho, \sigma \in \mathbb{N}_{>0}$ so that every displayed object stays in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. For $\rho, \sigma > 0$ lying in $\mathbb{K}$, the same proof works verbatim inside the Hahn ring $\mathbb{K}[L][[t^\Gamma]]$ with $\Gamma = \mathbb{Z} + \rho\mathbb{Z} + \sigma\mathbb{Z} + \rho\sigma\mathbb{Z}$, provided one *defines* $(1+W)^\rho := e^{\rho \log(1+W)}$, which is legitimate by Theorem 10.5 and agrees with the integer power when $\rho \in \mathbb{Z}$. The exponent of the composite is $\rho\sigma$, so the class of admissible exponents must be closed under multiplication; $\mathbb{N}_{>0}$ and $\mathbb{Q}_{>0}$ are, a fixed denominator lattice $s^{-1}\mathbb{Z}$ is not.

A common trap

Formal closure is not a license. Theorem 10.27 asserts $(F \circ G) \circ H = F \circ (G \circ H)$ only when every displayed composition exists in the selected support class. A syntactically meaningful nested expression in a larger Hahn field need not have summable support; support theorems for general transseries composition are genuinely harder and carry their own hypotheses [14, 33].

10.9 The chain rule

Theorem 10.31 (Chain rule). *Let G be an admissible monomial-leading argument with $\rho > 0$. Then for every $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$,*

$$D_X(F \circ G) = (D_X F) \circ G \cdot D_X G. \quad (188)$$

Proof. Set $\delta_1(F) = D_X(\Phi_G F)$ and $\delta_2(F) = \Phi_G(D_X F) \cdot D_X G$. Both are $\mathbb{K}$ -linear, both are compatible with summable families (each is a composite of maps that transform valuations by a non-decreasing unbounded function, so Theorem 10.3 applies), and both satisfy the twisted Leibniz rule $\delta(F_1 F_2) = \delta(F_1)\Phi_G(F_2) + \Phi_G(F_1)\delta(F_2)$: for δ_1 because D_X is a derivation and Φ_G a homomorphism, for δ_2 for

the same reason. It therefore suffices to check (188) for $F = t$ and $F = L$; the twisted Leibniz rule then propagates it to $\mathbb{K}[L][t]$, the relation $\delta(t \cdot t^{-1}) = 0$ propagates it to t^{-1} , and compatibility with summable families finishes.

Write $G = ct^{-\rho}(1 + U)$. From $D_X(t^\alpha p(L)) = t^{\alpha+1}(p' - \alpha p)$ we get $D_X(t^{-\rho}) = \rho t^{1-\rho}$, so

$$\frac{D_X G}{G} = \frac{c\rho t^{1-\rho}(1 + U) + ct^{-\rho}D_X U}{ct^{-\rho}(1 + U)} = \rho t + \frac{D_X U}{1 + U}. \quad (189)$$

For $F = t$: $\delta_1(t) = D_X(G^{-1}) = -G^{-2}D_X G$, while $\delta_2(t) = \Phi_G(-t^2)D_X G = -G^{-2}D_X G$. For $F = L$: $\delta_2(L) = \Phi_G(t)D_X G = D_X G/G$, while

$$\delta_1(L) = D_X(\rho L + \log c + \log(1 + U)) = \rho t + \sum_{r \geq 1} (-1)^{r-1} U^{r-1} D_X U = \rho t + \frac{D_X U}{1 + U},$$

termwise differentiation being legitimate by Theorem 10.3. By (189) the two agree. $\square$

10.10 Analytic admissibility

Everything above is an identity between formal objects. It becomes a statement about functions only under additional hypotheses, which we now make explicit. Take $\mathbb{K} \subseteq \mathbb{C}$.

Definition 10.32 (Asymptotic realization). Let S be a ray $[X_0, \infty)$ or a closed subsector of a sector at infinity on which a branch of $\log$ has been fixed. A function $f : S \rightarrow \mathbb{C}$ realizes $F = \sum_{n \geq a} p_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ on S , written $f \sim_S F$, if for every $N \in \mathbb{Z}$

$$f(X) - (\text{Trunc}_{<N} F)(X) = o_{X \rightarrow \infty, X \in S}(X^{1-N}), \quad (190)$$

where $(\text{Trunc}_{<N} F)(X) = \sum_{a \leq n < N} p_n(\log X)X^{-n}$ is a finite sum evaluated with the fixed branch.

Lemma 10.33 (Equivalent forms and uniqueness). *$f \sim_S F$ holds if and only if for every N and every $\varepsilon > 0$, $f - \text{Trunc}_{<N} F = \mathcal{O}_{X \rightarrow \infty, X \in S}(X^{-N+\varepsilon})$. A function realizes at most one F . The stronger relation $f - \text{Trunc}_{<N} F = \mathcal{O}_{X \rightarrow \infty, X \in S}(X^{-N})$ fails for a realization whenever $\deg_L p_N > 0$, since the first omitted block is then not $\mathcal{O}_{X \rightarrow \infty}(X^{-N})$; the ε may not be dropped.*

Proof. If (190) holds for all N , apply it at $N + 1$ and add $p_N(\log X)X^{-N} = \mathcal{O}_{X \rightarrow \infty}(X^{-N+\varepsilon})$. Conversely the ε -form at N with $\varepsilon < 1$ implies (190). For uniqueness, if $F \neq \tilde{F}$ first differ in block M by a nonzero $R \in \mathbb{K}[L]$, subtracting the two relations (190) at $N = M + 1$ gives $R(\log X)X^{-M} = o_{X \rightarrow \infty}(X^{-M})$, so $R(\log X) \rightarrow 0$; a nonzero polynomial does not tend to 0 along $\log X \rightarrow \infty$. The last sentence is the observation that $X^{-N}(\log X)^d$ is not $\mathcal{O}_{X \rightarrow \infty}(X^{-N})$ for $d > 0$. $\square$

Lemma 10.34 (Realizations form a ring). *If $f \sim_S F$ and $g \sim_S \tilde{F}$ then $f + g \sim_S F + \tilde{F}$ and $fg \sim_S F\tilde{F}$.*

Proof. Addition is clear. For the product put $a = \nu_t F$, $\tilde{a} = \nu_t \tilde{F}$ and fix N and $\varepsilon > 0$. Since $\text{Trunc}_{<N'} F\tilde{F} - \text{Trunc}_{<N} F\tilde{F}$ is, for $N' \geq N$, a finite sum of terms $q(\log X)X^{-m}$ with $m \geq N$ and hence $\mathcal{O}_{X \rightarrow \infty}(X^{-N+\varepsilon})$, the assertion at a cutoff N' implies the assertion at every smaller cutoff; we may therefore assume $N \geq a + \tilde{a}$. Write $f = \text{Trunc}_{<N-\tilde{a}} F + R$, $g = \text{Trunc}_{<N-a} \tilde{F} + \tilde{R}$. By Theorem 10.33, $R = \mathcal{O}_{X \rightarrow \infty}(X^{-N+\tilde{a}+\varepsilon/4})$ and $\tilde{R} = \mathcal{O}_{X \rightarrow \infty}(X^{-N+a+\varepsilon/4})$, while $\text{Trunc}_{<N-\tilde{a}} F = \mathcal{O}_{X \rightarrow \infty}(X^{-a+\varepsilon/4})$ and similarly for $\tilde{F}$. Hence each of the three cross terms is $\mathcal{O}_{X \rightarrow \infty}(X^{-N+\varepsilon})$; for the term $R\tilde{R}$, whose exponent is $-2N+a+\tilde{a}+\varepsilon/2$, this is where $N \geq a+\tilde{a}$ is used. The remaining product of two finite sums is a finite sum of terms $q(\log X)X^{-m}$; those with $m < N$ constitute $\text{Trunc}_{<N} F\tilde{F}$ by the definition of the Cauchy product, and those with $m \geq N$ are $\mathcal{O}_{X \rightarrow \infty}(X^{-N+\varepsilon})$. $\square$

Theorem 10.35 (Analytic transfer under composition). *Let $S' = [X_1, \infty)$ and $S = [Y_0, \infty)$ be rays, let $f \sim_S F$ with $F = \sum_{n \geq a} p_n(L)t^n$, and let $g : S' \rightarrow \mathbb{R}$ satisfy the following three conditions.*

- (A1) Regime. $g(X) \rightarrow +\infty$ and $g(X) \in S$ for all sufficiently large $X \in S'$.
- (A2) Monomial-leading realization. There are $c > 0, \rho > 0$ real and $U \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with real coefficients such that $u := c^{-1}X^{-\rho}g(X) - 1$ satisfies $u \sim_{S'} U$ and $u(X) \rightarrow 0$.
- (A3) Branch compatibility. $\log g(X) = \rho \log X + \log c + \log(1 + u(X))$ for all large $X \in S'$, with the principal real logarithm on both sides.

Then $f \circ g$ realizes $F \circ G$ on S' , where $G = cX^\rho(1 + U)$ and $F \circ G$ is (171) in the Hahn ring of Theorem 10.17, whose support is locally finite so that $\text{Trunc}_{<\beta} \cdot$ at a real cutoff β is again a finite sum; that is, for every real β ,

$$f(g(X)) - (\text{Trunc}_{<\beta} F \circ G)(X) = o_{X \rightarrow +\infty}(X^{-\beta+1}).$$

Proof. Fix β and $\varepsilon \in (0, 1)$. Choose an integer $N > (\beta+1)/\rho$ with $N > a$, and split $f = \text{Trunc}_{<N} F + R_N$. By Theorem 10.33, $R_N(Y) = \mathcal{O}_{Y \rightarrow +\infty}(Y^{-N+\varepsilon})$. By (A1) and (A2), $g(X) = cX^\rho(1 + u(X))$ with $u \rightarrow 0$, so $g(X) \asymp X^\rho$ and

$$R_N(g(X)) = \mathcal{O}_{X \rightarrow +\infty}(X^{-\rho(N-\varepsilon)}) = o_{X \rightarrow +\infty}(X^{-\beta}) \quad (191)$$

for ε small, since $\rho N > \beta + 1$. This is the *remainder transport* step; note that it consumes ρ^{-1} times as many outer orders of the outer expansion as it produces.

It remains to expand the finite sum $\text{Trunc}_{<N} F \circ g$. By (A3) and Theorem 10.33, $\log(1 + u(X))$ realizes $V = \log(1 + U)$ on S' : indeed for any M , $\log(1 + u) = \sum_{r=1}^M \frac{(-1)^{r-1}}{r} u^r + \mathcal{O}_{X \rightarrow +\infty}(|u|^{M+1})$ by Taylor's theorem with remainder for $\log(1 + \cdot)$ at 0, applicable because $u \rightarrow 0$; each u^r realizes U^r by Theorem 10.34, and $|u|^{M+1} = \mathcal{O}_{X \rightarrow +\infty}(X^{-M-1+\varepsilon})$. The same argument with the binomial series shows that $(1 + u(X))^{-n}$ realizes $(1 + U)^{-n}$ for each n . Consequently, for each $n < N$,

$$p_n(\log g(X))g(X)^{-n} = p_n(\rho \log X + \log c + \log(1 + u(X)))c^{-n}X^{-\rho n}(1 + u(X))^{-n},$$

which by (A3), the finite expansion (173) and Theorem 10.34 realizes $p_n(\rho L + \log c + V)c^{-n}t^{\rho n}(1 + U)^{-n}$ on S' . Summing the finitely many $n < N$ and using (191) gives the claim, because by Theorem 10.10 the blocks of $F \circ G$ of exponent $< \beta$ come only from those n with $\rho n < \beta \leq \rho N$. $\square$

Example 10.36 (Each hypothesis is used). 1. (A1)/(A2) with $\rho > 0$ are not decorative. Let $f(Y) = e^{-Y}$. Then $f \sim_S 0$ on any ray: the zero series is a correct realization to all orders. Substituting $g(X) = \log X$, which has no positive power of X as leading monomial, gives $f(g(X)) = X^{-1}$, which realizes $t \neq 0$. Substituting $g(X) = X^{-1} \rightarrow 0$, which violates (A1), gives $f(g(X)) \rightarrow 1$, which realizes the series $1 \neq 0$. So composition is not even well defined on realizations without (A1)–(A2); what makes the theorem work is that $\rho > 0$ transports flatness to flatness.

2. (A3) is a hypothesis, not a consequence. On a complex sector containing $\arg X$ close to π , the principal logarithm satisfies $\text{Log } g(X) = \rho \text{Log } X + \log c + \text{Log}(1 + u)$ only while $|\rho \arg X + \arg(1 + u)| < \pi$; crossing that threshold changes the answer by $2\pi i$ in every coefficient block of positive degree. On the positive real ray with $c > 0$, (A3) is automatic.
3. Precision transport is lossy for $\rho < 1$. By (191), an output valid through exponent β requires the outer expansion through outer order $N > \beta/\rho$. For $\rho = 1/2$ one must double the outer precision; for $\rho = 2$ one gains.

A common trap

Three separate obligations. A formal identity $F \circ G$ established by Theorem 10.7, Theorem 10.10 or Theorem 10.27 carries no analytic content. To claim an asymptotic statement one must additionally exhibit (A1) a domain on which the inner map lands in the outer expansion's regime, (A2) a

monomial-leading realization of the inner map with a positive exponent, and (A3) a branch identity for every logarithm and fractional power used. Differentiating a realization requires yet more: a Poincaré expansion cannot in general be differentiated termwise without sectorial analyticity or direct derivative estimates [25, 10]. The formal derivation D_X is always defined; its analytic interpretation is not.

11 The formal Taylor formula

For near-identity inner arguments there is a second description of composition that looks exactly like the classical Taylor expansion but is an exact identity of formal series. We prove it from scratch, because the summability and the multiplicativity of the Taylor operator are both nontrivial and are the two points at which the sources are thin.

11.1 The normal-ordered Taylor operator

Definition 11.1 (Taylor operator). For $H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ define, for $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$,

$$\mathcal{T}_H(F) := \sum_{k \geq 0} \frac{H^k}{k!} D_X^k F. \quad (192)$$

Lemma 11.2 (Summability, with an explicit derivative budget). *Let $H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ with $a = \nu_t(F)$. Then*

$$\nu_t\left(\frac{H^k}{k!} D_X^k F\right) \geq a + k + k \nu_t(H) \geq a + k, \quad (193)$$

so the family in (192) is summable and $\mathcal{T}_H(F) \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ with $\nu_t(\mathcal{T}_H F) \geq a$. Consequently, for every $N \in \mathbb{Z}$,

$$\text{Trunc}_{<N} \mathcal{T}_H(F) = \text{Trunc}_{<N} \left(\sum_{k=0}^{N-a-1} \frac{H^k}{k!} D_X^k F \right), \quad (194)$$

that is, only the indices $k \leq N - a - 1$ can affect any block below t^N , and only $k \leq N - a$ can affect the block t^N itself. If moreover $\nu_t(H) \geq 1$, the budget halves: only $k \leq (N - a)/2$ can affect t^N .

Proof. Write $F = \sum_{n \geq a} p_n(L)t^n$. By (165) and Theorem 10.3 applied to the continuous operator D_X , $D_X^k F = \sum_{n \geq a} t^{n+k} \Delta_{n,k} p_n$, so $\nu_t(D_X^k F) \geq a + k$; the inequality may be strict, and is so exactly when the operators $\Delta_{n,k}$ annihilate the leading blocks, but the bound is all we need. Multiplying by H^k , whose valuation is $k \nu_t(H) \geq 0$, gives (193). The right-hand side tends to $+\infty$ with k , which is (166), so Theorem 10.3 applies. A summand contributes to the block t^N only if $a + k \leq N$, and to a block strictly below t^N only if $a + k \leq N - 1$. If $\nu_t(H) \geq 1$, (193) improves to $a + 2k \leq N$. $\square$

Lemma 11.3 (The Taylor operator is a homomorphism). *For every $H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, $\mathcal{T}_H : \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} \rightarrow \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is a $\mathbb{K}$ -algebra homomorphism compatible with summable families.*

Proof. $\mathbb{K}$ -linearity is clear from (192), and $\mathcal{T}_H(1) = 1$ because $D_X 1 = 0$.

For compatibility with summable families, the valuation half holds with $\varphi = \text{id}$, since $\nu_t(\mathcal{T}_H F) \geq \nu_t(F)$ by (193). For the interchange, observe first that D_X itself is compatible with summable families: it is $\mathbb{K}$ -linear, raises valuation by at least 1, and by (165) the block $[t^{n+1}]D_X F$ depends $\mathbb{K}$ -linearly on the single block $[t^n]F$, so $D_X(\sum_i S_i) = \sum_i D_X S_i$ for every summable family; the same then holds for each D_X^k . Now let $(S_i)_{i \in I}$ be summable with sum S . By Theorem 10.4 the valuations $\nu_t(S_i)$ are bounded below, so the doubly indexed family $(\frac{H^k}{k!} D_X^k S_i)_{i,k}$, whose (i, k) member has valuation

at least $\nu_t(S_i) + k$ by (193), satisfies (166). Summing it with i innermost gives $\mathcal{T}_H(S)$ and with k innermost gives $\sum_i \mathcal{T}_H(S_i)$.

For multiplicativity, first record the Leibniz rule for the derivation D_X : by induction on m , $D_X^m(F_1 F_2) = \sum_{i=0}^m \binom{m}{i} D_X^i F_1 D_X^{m-i} F_2$. Hence

$$\mathcal{T}_H(F_1 F_2) = \sum_{m \geq 0} \frac{H^m}{m!} \sum_{i=0}^m \binom{m}{i} D_X^i F_1 D_X^{m-i} F_2 = \sum_{m \geq 0} \sum_{i=0}^m \frac{H^i D_X^i F_1}{i!} \cdot \frac{H^{m-i} D_X^{m-i} F_2}{(m-i)!}.$$

The doubly indexed family $(\frac{H^i D_X^i F_1}{i!} \cdot \frac{H^j D_X^j F_2}{j!})_{i,j \geq 0}$ has $\nu_t \geq \nu_t(F_1) + \nu_t(F_2) + i + j$ by (193), hence is summable; regrouping it by $i + j = m$ recovers the display, and regrouping it as a product of two summable families gives $\mathcal{T}_H(F_1) \mathcal{T}_H(F_2)$. Both regroupings are licensed by Theorem 10.3. $\square$

Remark 11.4. Theorem 11.3 is not a formality. The operator $\mathcal{T}_H$ is the normal-ordered exponential $:e^{H D_X}:$, in which every power of H is placed to the left and is never differentiated. It is *not* the operator exponential $\exp(M_H D_X)$, since $(M_H D_X)^2 = M_H M_{D_X H} D_X + M_{H^2} D_X^2$. The latter is also an automorphism, but a different one. What makes (192) multiplicative is the Leibniz rule combined with the elementary identity $\binom{m}{i}/m! = 1/(i!(m-i)!)$, not any operator exponential formalism.

11.2 The formal Taylor formula

Lemma 11.5 (Generators determine such a homomorphism). *Let $\mathcal{A}$ be a polynomial-logarithmic Hahn ring and let $\Psi_1, \Psi_2 : \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} \rightarrow \mathcal{A}$ be $\mathbb{K}$ -algebra homomorphisms compatible with summable families. If $\Psi_1(t) = \Psi_2(t)$ and $\Psi_1(L) = \Psi_2(L)$, then $\Psi_1 = \Psi_2$.*

Proof. Both fix $\mathbb{K}$ and agree on L , hence agree on $\mathbb{K}[L]$; they agree on t , hence on t^n for $n \geq 0$, and on t^{-1} because $\Psi_i(t) \Psi_i(t^{-1}) = 1$ forces $\Psi_i(t^{-1}) = \Psi_i(t)^{-1}$. So they agree on every $p_n(L) t^n$. For $F = \sum_n p_n t^n$ the family $(p_n(L) t^n)_n$ is summable, so $\Psi_i(F) = \sum_n \Psi_i(p_n t^n)$, and the two sums coincide. $\square$

Theorem 11.6 (Formal Taylor formula). *For every $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and every $H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$,*

$$F \circ (X + H) = \sum_{k \geq 0} \frac{H^k}{k!} D_X^k F. \quad (195)$$

This is an exact identity in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, not an approximation: both sides have the same coefficient in every block. If $a = \nu_t(F)$, the coefficient of t^N on the right receives contributions only from $k \leq N - a$.

Proof. By Theorem 10.22 the map Φ_{X+H} is a $\mathbb{K}$ -algebra homomorphism compatible with summable families, and by Theorem 11.3 so is $\mathcal{T}_H$. By Theorem 11.5 it suffices to compare them on t and L .

For $F = t$: from $D_X t = -t^2$ and (165) with $n = 1, p = 1$, one gets $D_X^k t = t^{1+k} \prod_{j=0}^{k-1} (-1 - j) = (-1)^k k! t^{k+1}$. Hence

$$\mathcal{T}_H(t) = \sum_{k \geq 0} \frac{H^k}{k!} (-1)^k k! t^{k+1} = t \sum_{k \geq 0} (-tH)^k = \frac{t}{1 + tH} = \Phi_{X+H}(t),$$

the geometric family being summable because $\nu_t(tH) \geq 1$.

For $F = L$: $D_X L = t$, so $D_X^k L = D_X^{k-1} t = (-1)^{k-1} (k-1)! t^k$ for $k \geq 1$. Hence

$$\mathcal{T}_H(L) = L + \sum_{k \geq 1} \frac{H^k}{k!} (-1)^{k-1} (k-1)! t^k = L + \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} (tH)^k = L + \log(1 + tH) = \Phi_{X+H}(L),$$

by (167). The derivative budget is Theorem 11.2. $\square$

Remark 11.7 (The small parameter is H/X , not H). No smallness of H is assumed, and none is available: H may be L^{100} , which tends to infinity. What the proof uses is $H/X = tH \prec 1$, which is automatic for $H \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ because every polynomial in $\log X$ is $o_{X \rightarrow +\infty}(X)$. The mechanism is that D_X raises the outer order by one while multiplication by H does not lower it. The base point is the asymptotic variable itself, and the conclusion is an identity, not an estimate; there is no analytic differentiability hypothesis anywhere.

Remark 11.8 (Where the mechanism fails). In a larger transseries field, " Δ small compared with X " is not enough. Take $F(X) = e^{e^X}$ and $\Delta = e^{-X/2}$, so $\Delta \prec X$. Then

$$\Delta D_X F = e^{-X/2} e^X e^{e^X} = e^{X/2} F \succ F,$$

so the $k = 1$ Taylor term dominates the $k = 0$ term and the family is not summable. The quantity that must be small is not Δ but $\Delta \cdot D_X \mathfrak{b}/\mathfrak{b}$, where $\mathfrak{b}$ is the dominant monomial of F : the increment multiplied by the logarithmic derivative of the outer scale. In the polynomial-logarithmic class this is automatically small, because for $\mathfrak{b} = t^n L^j$ one has $D_X \mathfrak{b}/\mathfrak{b} = t(jL^{-1} - n) \prec 1$, whereas for $\mathfrak{b} = e^{e^X}$ one has $D_X \mathfrak{b}/\mathfrak{b} = e^X \succ 1$. Sharper Taylor approximation theorems for larger classes require exactly such a logarithmic-derivative hypothesis [14, 22].

Remark 11.9 (On a tempting but circular proof). One sometimes sees the Taylor formula deduced by introducing a scalar parameter s , forming $\Phi(s) = F(X + sH)$ "with H held constant", and expanding Φ in powers of s at $s = 0$. This is not a proof: the symbol $F(X + sH)$ has no meaning prior to the composition one is trying to compute, and the claim $\partial_s^k \Phi(0) = H^k D_X^k F$ is the chain rule for that same composition. One can rescue the argument by working in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}[[s]]$ and defining $\Phi(s) = \Phi_{X+sH}(F)$, but then one must still prove that $\partial_s \Phi = H \cdot \Phi_{X+sH}(D_X F)$, which is [Theorem 10.31](#). The route through [Theorem 11.3](#) and [Theorem 11.5](#) is shorter and self-contained.

11.3 The finite coefficient formula

Theorem 11.10 (Finite coefficient formula). *Let $F = \sum_{n \geq a} p_n(L) t^n \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ and $H \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, and write*

$$H^k = \sum_{m \geq 0} h_{k,m}(L) t^m, \quad h_{0,0} = 1, \quad h_{0,m} = 0 \ (m > 0). \quad (196)$$

Then for every $r \in \mathbb{Z}$

$$[t^r](F \circ (X + H)) = \sum_{\substack{n \geq a, k \geq 0, m \geq 0 \\ n+k+m=r}} \frac{h_{k,m}(L)}{k!} \Delta_{n,k} p_n(L), \quad (197)$$

a sum with exactly $\binom{r-a+2}{2}$ index triples when $r \geq a$, and empty when $r < a$. If $\nu_t(H) \geq 1$, the constraint $m \geq k$ applies as well, so $n + 2k \leq r$ and roughly half the triples drop out.

Proof. By [Theorem 11.6](#) and (165),

$$F \circ (X + H) = \sum_{k \geq 0} \frac{H^k}{k!} \sum_{n \geq a} t^{n+k} \Delta_{n,k} p_n = \sum_{k \geq 0} \sum_{n \geq a} \sum_{m \geq 0} \frac{h_{k,m}}{k!} \Delta_{n,k} p_n t^{n+k+m},$$

the triple family being summable because its (n, k, m) member has valuation at least $n + k + m$. Extracting $[t^r]$ gives (197). The affine constraint $n + k + m = r$ with $n \geq a$ and $k, m \geq 0$ has exactly $\binom{r-a+2}{2}$ solutions. If $\nu_t(H) \geq 1$ then $\nu_t(H^k) \geq k$, so $h_{k,m} = 0$ for $m < k$. $\square$

Remark 11.11 (Three separable pieces of work). Formula (197) factors the computation into (a) the powers H^k , computed once by truncated multiplication and reused for every outer series; (b) the shifted differential operators $\Delta_{n,k}$ acting on coefficient polynomials, which depend only on F ; and (c) a finite index convolution. It is therefore the formula of choice when one inner correction H is composed with many outer series, and the generator form (181) is the formula of choice when a single sparse F is composed once.

11.4 Convolution coefficients

The blocks $h_{k,m}$ of (196) deserve a name and a recurrence. Write

$$H = \sum_{j \geq 0} h_j(L) t^j, \quad h_j \in \mathbb{K}[L]. \quad (198)$$

Definition 11.12 (Convolution coefficients). For $m, r \in \mathbb{N}_0$ put

$$C_{m,r} := \sum_{\substack{j_1, \dots, j_r \geq 0 \\ j_1 + \dots + j_r = m}} h_{j_1} \cdots h_{j_r} \in \mathbb{K}[L]. \quad (199)$$

The value at $r = 0$ is an empty product summed over the empty composition, so

$$C_{0,0} = 1, \quad C_{m,0} = 0 \quad (m > 0). \quad (200)$$

A common trap

Index order. With (196) and (199) one has

$$h_{k,m} = C_{m,k} :$$

in $h_{k,m}$ the *first* index is the exponent of H and the second is the outer order; in $C_{m,r}$ it is the other way round. Both notations occur in the literature. Any transfer between them must transpose.

Proposition 11.13 (Structure of the convolution coefficients). *With the conventions (200):*

1. $H^r = \sum_{m \geq 0} C_{m,r} t^m$ for every $r \in \mathbb{N}_0$.
2. $C_{m,1} = h_m$, and for all $m, r \geq 0$

$$C_{m,r+1} = \sum_{j=0}^m h_j C_{m-j,r}. \quad (201)$$

3. If $h_0 = 0$, then $C_{m,r} = 0$ whenever $m < r$, and $C_{r,r} = h_1^r$.
4. If $h_0 = 0$, then for $m \geq r \geq 1$

$$C_{m,r} = \frac{r!}{m!} \mathcal{B}_{m,r}^{\exp}(1! h_1, 2! h_2, \dots, (m-r+1)! h_{m-r+1}). \quad (202)$$

Proof. (i) Expanding $H^r = \left(\sum_j h_j t^j\right)^r$ and collecting the total outer order m is exactly (199); for $r = 0$ the statement is $H^0 = 1$, which is (200). The rearrangement is legitimate because the family indexed by $(j_1, \dots, j_r)$ has valuation $j_1 + \dots + j_r$ and is summable.

(ii) Take $r = 1$ in (i). For the recurrence, write $H^{r+1} = H \cdot H^r$ and read off the coefficient of t^m using (i); the inner index is bounded by m because $C_{m',r}$ is 0 for $m' < 0$. At $r = 0$ the recurrence returns $C_{m,1} = h_m C_{0,0} = h_m$, consistent with (200).

(iii) If $h_0 = 0$ then $\nu_t(H) \geq 1$, so $\nu_t(H^r) \geq r$ and (i) forces $C_{m,r} = 0$ for $m < r$. For $m = r$ the only composition of r into r positive parts is $(1, \dots, 1)$, giving h_1^r .

(iv) With $h_0 = 0$, set $\hat{h}_j = j! h_j$. The partial exponential Bell polynomials are defined by the generating identity

$$\sum_{m \geq r} \mathcal{B}_{m,r}^{\exp}(\hat{h}_1, \hat{h}_2, \dots) \frac{z^m}{m!} = \frac{1}{r!} \left(\sum_{j \geq 1} \hat{h}_j \frac{z^j}{j!} \right)^r = \frac{1}{r!} \left(\sum_{j \geq 1} h_j z^j \right)^r$$

[4]. Substituting $z \mapsto t$, which is legitimate coefficientwise since both sides are formal power series in one variable with coefficients in $\mathbb{K}[L]$, and comparing with (i) gives $\frac{C_{m,r}}{r!} = \frac{1}{m!} \mathcal{B}_{m,r}^{\exp}(\hat{h})$, which is (202). $\square$

Example 11.14 (The first convolution coefficients). For $H = h_0 + h_1t + h_2t^2 + h_3t^3 + \mathcal{O}_t^{\text{form}}(t^4)$:

	$m = 0$	$m = 1$	$m = 2$	$m = 3$
$r = 0$	1	0	0	0
$r = 1$	h_0	h_1	h_2	h_3
$r = 2$	h_0^2	$2h_0h_1$	$h_1^2 + 2h_0h_2$	$2h_0h_3 + 2h_1h_2$
$r = 3$	h_0^3	$3h_0^2h_1$	$3h_0^2h_2 + 3h_0h_1^2$	$h_1^3 + 6h_0h_1h_2 + 3h_0^2h_3$

Each entry is obtained from the row above by (201). Setting $h_0 = 0$ empties the lower-left triangle, as Theorem 11.13(iii) predicts.

11.5 Low-order composition laws

We now write out (197) for the case that governs reversion: both maps tangent to the identity, through four outer blocks.

Proposition 11.15 (Composition through four blocks). *Let*

$$F = X + f_0(L) + f_1(L)t + f_2(L)t^2 + f_3(L)t^3 + \mathcal{O}_t^{\text{form}}(t^4), \quad (203)$$

$$G = X + g_0(L) + g_1(L)t + g_2(L)t^2 + g_3(L)t^3 + \mathcal{O}_t^{\text{form}}(t^4), \quad (204)$$

with all $f_j, g_j \in \mathbb{K}[L]$, and write $F \circ G = X + h_0 + h_1t + h_2t^2 + h_3t^3 + \mathcal{O}_t^{\text{form}}(t^4)$. Primes denote ∂_L . Then

$$h_0 = f_0 + g_0, \quad (205)$$

$$h_1 = f_1 + g_1 + g_0f'_0, \quad (206)$$

$$h_2 = f_2 + g_2 + g_1f'_0 + g_0(f'_1 - f_1) + \frac{g_0^2}{2}(f''_0 - f'_0), \quad (207)$$

$$\begin{aligned} h_3 = & f_3 + g_3 + g_2f'_0 + g_1(f'_1 - f_1) + g_0(f'_2 - 2f_2) + g_0g_1(f''_0 - f'_0) \\ & + \frac{g_0^2}{2}(f''_1 - 3f'_1 + 2f_1) + \frac{g_0^3}{6}(f'''_0 - 3f''_0 + 2f'_0). \end{aligned} \quad (208)$$

Proof. Put $H = G - X = g_0 + g_1t + g_2t^2 + g_3t^3 + \mathcal{O}_t^{\text{form}}(t^4)$, so $H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $\nu_t(F) = -1$. By Theorem 11.2 only $k \leq 3 - (-1) = 4$ can reach the block t^3 ; but $D_X^k X = 0$ for $k \geq 2$ and $\nu_t(D_X^k(F - X)) \geq k$, so in fact only $k \leq 3$ matter. We need the first three derivatives of F , which (165) supplies blockwise with $\Delta_{n,k} = \prod_{j=0}^{k-1}(\partial_L - n - j)$:

$$D_X F = 1 + tf'_0 + t^2(f'_1 - f_1) + t^3(f'_2 - 2f_2) + \mathcal{O}_t^{\text{form}}(t^4),$$

$$D_X^2 F = t^2(f''_0 - f'_0) + t^3(f''_1 - 3f'_1 + 2f_1) + \mathcal{O}_t^{\text{form}}(t^4),$$

$$D_X^3 F = t^3(f'''_0 - 3f''_0 + 2f'_0) + \mathcal{O}_t^{\text{form}}(t^4),$$

using $\Delta_{0,1} = \partial_L$, $\Delta_{1,1} = \partial_L - 1$, $\Delta_{2,1} = \partial_L - 2$, $\Delta_{0,2} = \partial_L(\partial_L - 1)$, $\Delta_{1,2} = (\partial_L - 1)(\partial_L - 2)$ and $\Delta_{0,3} = \partial_L(\partial_L - 1)(\partial_L - 2)$. Also $H^2 = g_0^2 + 2g_0g_1t + \mathcal{O}_t^{\text{form}}(t^2)$ and $H^3 = g_0^3 + \mathcal{O}_t^{\text{form}}(t)$. Substituting into (195),

$$\begin{aligned} F \circ G = & (X + f_0 + f_1t + f_2t^2 + f_3t^3) \\ & + (g_0 + g_1t + g_2t^2 + g_3t^3)(1 + tf'_0 + t^2(f'_1 - f_1) + t^3(f'_2 - 2f_2)) \\ & + \frac{1}{2}(g_0^2 + 2g_0g_1t)(t^2(f''_0 - f'_0) + t^3(f''_1 - 3f'_1 + 2f_1)) \\ & + \frac{1}{6}g_0^3t^3(f'''_0 - 3f''_0 + 2f'_0) + \mathcal{O}_t^{\text{form}}(t^4). \end{aligned}$$

Collecting the blocks $t^0, \dots, t^3$ gives (205)–(208). □

Remark 11.16 (Which derivatives can occur, and with which coefficients). The formulas are triangular: derivatives fall only on the *outer* coefficients f_j , never on the inner g_j , because it is the outer logarithmic argument that is Taylor-expanded while the inner coefficients only supply the increment. At block t^r the highest derivative of f_n that occurs is of order $r - n$, and the operator applied to f_n at Taylor order k is exactly $\Delta_{n,k}$. The integers appearing in (207)–(208) are therefore signed Stirling numbers of the first kind: since $\Delta_{n,k} = \prod_{j=0}^{k-1} ((\partial_L - n) - j) = ((\partial_L - n))^k$,

$$\Delta_{n,k} = \sum_{i=0}^k s(k, i) (\partial_L - n)^i. \quad (209)$$

For $n = 0, k = 2$ this gives $\partial_L^2 - \partial_L$; for $n = 1, k = 2$, $\partial_L^2 - 3\partial_L + 2$; for $n = 0, k = 3$, $\partial_L^3 - 3\partial_L^2 + 2\partial_L$, which are precisely the three brackets displayed above.

Example 11.17 (Consistency check: a constant shift). Take $G = X + c$ with $c \in \mathbb{K}$, so $g_0 = c$ and $g_1 = g_2 = g_3 = 0$. Then (205)–(207) reduce to

$$h_0 = f_0 + c, \quad h_1 = f_1 + cf'_0, \quad h_2 = f_2 + c(f'_1 - f_1) + \frac{c^2}{2}(f''_0 - f'_0),$$

which is (195) read off directly, since $D_X f_0(L) = tf'_0$ and $D_X^2 f_0(L) = t^2(f''_0 - f'_0)$.

Corollary 11.18 (First commutator block). Let $F = X + f(L)$ and $G = X + g(L)$ with $f, g \in \mathbb{K}[L]$. Then

$$F \circ G - G \circ F = (gf' - fg')t + \mathcal{O}_t^{\text{form}}(t^2). \quad (210)$$

In particular composition is associative (Theorem 10.27) but not commutative, and the leading obstruction is the Lie bracket of the one-dimensional vector fields $f\partial_L$ and $g\partial_L$.

Proof. Apply (205)–(206) with $f_0 = f$, $g_0 = g$ and all other blocks zero: $h_0 = f + g$ is symmetric and $h_1 = gf'$; exchanging the roles of F and G gives fg' . $\square$

Example 11.19 (Explicit noncommutativity). For $F = X + L$ and $G = X + L^2$,

$$\begin{aligned} F \circ G &= X + L^2 + L + L^2t - \frac{1}{2}L^4t^2 + \mathcal{O}_t^{\text{form}}(t^3), \\ G \circ F &= X + L + L^2 + 2L^2t + (L^2 - L^3)t^2 + \mathcal{O}_t^{\text{form}}(t^3), \end{aligned}$$

so $F \circ G - G \circ F = -L^2t + \mathcal{O}_t^{\text{form}}(t^2)$, in agreement with (210): $gf' - fg' = L^2 - 2L^2 = -L^2$.

Example 11.20 (A four-block composition with nontrivial coefficients). Let F and G be the *exact* finite series

$$F(Y) = Y + ((\log Y)^2 - 1) + \frac{2 \log Y}{Y} + \frac{1}{Y^2}, \quad G(X) = X + (1 - L) + \frac{L^2}{X},$$

so that $f_0 = L^2 - 1$, $f_1 = 2L$, $f_2 = 1$, $f_3 = 0$, and $g_0 = 1 - L$, $g_1 = L^2$, $g_2 = g_3 = 0$. Then Theorem 11.15 gives

$$h_0 = L^2 - L, \quad h_1 = -L^2 + 4L, \quad h_2 = L^3 + 5L^2 - 7L + 4, \quad h_3 = \frac{4}{3}L^4 - L^3 - 8L^2 + \frac{41}{3}L - 6,$$

that is,

$$F \circ G = X + (L^2 - L) + \frac{-L^2 + 4L}{X} + \frac{L^3 + 5L^2 - 7L + 4}{X^2} + \frac{\frac{4}{3}L^4 - L^3 - 8L^2 + \frac{41}{3}L - 6}{X^3} + \mathcal{O}_t^{\text{form}}(t^4).$$

Had F and G been specified only modulo Y^{-3} and t^2 , the block h_3 would not have been determined, since it contains $f_3 + g_3 + g_2f'_0$.

Example 11.21 (One step of a reversion). Let $F(Y) = Y + \log Y$ and $G(X) = X - L + LX^{-1}$, so $f_0 = L$, $f_1 = f_2 = f_3 = 0$, $g_0 = -L$, $g_1 = L$, $g_2 = g_3 = 0$. Then $h_0 = -L + L = 0$ and $h_1 = f_1 + g_1 + g_0 f'_0 = 0 + L - L = 0$, while

$$h_2 = f_2 + g_2 + g_1 f'_0 + g_0(f'_1 - f_1) + \frac{g_0^2}{2}(f''_0 - f'_0) = 0 + 0 + L + 0 + \frac{L^2}{2}(0 - 1) = L - \frac{L^2}{2}.$$

Adding $L_2^{\text{Lam}}(L) = \frac{1}{2}L^2 - L$ to the X^{-2} block of G cancels h_2 , because that block enters h_2 additively through g_2 . The canonical Lambert polynomial L_2^{Lam} is thus recovered as the unique second-order correction, exactly as the reversion theory predicts.

11.6 Composing with an outer analytic function

Two further compositions occur constantly and are governed by Bell polynomials: substituting a polynomial-logarithmic series of positive valuation into a scalar power series, and differentiating a composite repeatedly.

Remark 11.22 (Substitution into a scalar power series: where it is proved). Let $\varphi(z) = \sum_{k \geq 0} \frac{\varphi_k}{k!} z^k \in \mathbb{K}[[z]]$ and let $W = \sum_{j \geq 1} w_j(L) t^j \in t \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, so that $\nu_t W \geq 1$. Everything used below about $\varphi(W) := \sum_{k \geq 0} \frac{\varphi_k}{k!} W^k$ is proved in the elementary-series part, applied to $\Phi = \sum_{r \geq 0} c_r z^r$ with $c_r = \varphi_r/r!$ and $U = W$. The family is summable, $\varphi(W) \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and $\varphi \mapsto \varphi(W)$ is the unique continuous $\mathbb{K}$ -algebra homomorphism $\mathbb{K}[[z]] \rightarrow \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ sending z to W ([Theorem 8.2](#)); only $\varphi_0, \dots, \varphi_r$ are needed for the block t^r ([Theorem 8.3](#)); and

$$[t^0]\varphi(W) = \varphi_0, \quad [t^r]\varphi(W) = \frac{1}{r!} \sum_{k=1}^r \varphi_k \mathcal{B}_{r,k}^{\text{exp}}(1!w_1, 2!w_2, \dots, (r-k+1)!w_{r-k+1}) \quad (r \geq 1),$$

which is (129) of [Theorem 8.5](#) after that normalisation. Two additions to what is proved there. First, if φ is analytic at z_0 with Taylor coefficients in $\mathbb{K}$ and $W - z_0 \in t \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, then the same three statements hold after translation: apply them to $\psi(z) := \varphi(z_0 + z) \in \mathbb{K}[[z]]$ and to $W - z_0$. Second, the coefficient formula also follows from the convolution structure established in this part, without the generating identity for the partial Bell polynomials: [Theorem 11.13\(iv\)](#), applied to W in place of H (whose block w_0 vanishes), gives $[t^r]W^k = \frac{k!}{r!} \mathcal{B}_{r,k}^{\text{exp}}(1!w_1, \dots)$ for $r \geq k \geq 1$ and 0 for $r < k$; multiplying by $\varphi_k/k!$ and summing over $k \leq r$ reproduces the display.

Example 11.23. Taking $\varphi(z) = (1+z)^{-n}$, $\varphi(z) = \log(1+z)$ and $\varphi(z) = e^z$ in [Theorem 11.22](#) recovers, respectively, the binomial factor $(1+U)^{-n}$, the logarithmic increment $V = \log(1+U)$, and $e^V = 1+U$ used throughout [Section 10](#). For the exponential one may replace the partial by the complete Bell polynomial: $e^W = \sum_{r \geq 0} \frac{1}{r!} \mathcal{B}_r^{\text{exp}}(1!w_1, \dots, r!w_r) t^r$, with $\mathcal{B}_0^{\text{exp}} = 1$.

For repeated differentiation of a composite the relevant identity is Faà di Bruno's. We prove it in the exact generality needed, from a recurrence for the partial Bell polynomials which we also prove.

Lemma 11.24 (Shift recurrence for partial Bell polynomials). *Let $\mathcal{D} = \sum_{i \geq 1} x_{i+1} \partial_{x_i}$ act on polynomials in the indeterminates $x_1, x_2, \dots$ over $\mathbb{Q}$. Then for all $n \geq 0$ and all $k \geq 1$, with the convention $\mathcal{B}_{n,k}^{\text{exp}} = 0$ for $k > n$,*

$$\mathcal{B}_{n+1,k}^{\text{exp}} = \mathcal{D} \mathcal{B}_{n,k}^{\text{exp}} + x_1 \mathcal{B}_{n,k-1}^{\text{exp}}, \quad (211)$$

with $\mathcal{B}_{n,0}^{\text{exp}} = 0$ for $n \geq 1$ and $\mathcal{B}_{0,0}^{\text{exp}} = 1$.

Proof. Let $\Xi(z) = \sum_{m \geq 1} x_m z^m / m!$, so that $\sum_{n \geq k} \mathcal{B}_{n,k}^{\text{exp}} z^n / n! = \Xi^k / k!$ is the defining generating identity [4]. Applying $\mathcal{D}$ coefficientwise, $\mathcal{D}\Xi = \sum_{i \geq 1} x_{i+1} z^i / i! = \Xi'(z) - x_1$, because $\Xi'(z) = \sum_{i \geq 0} x_{i+1} z^i / i!$. Since $\mathcal{D}$ is a derivation that does not act on z ,

$$\mathcal{D} \left(\frac{\Xi^k}{k!} \right) = \frac{\Xi^{k-1}}{(k-1)!} \mathcal{D}\Xi = \frac{\Xi^{k-1}}{(k-1)!} \Xi' - x_1 \frac{\Xi^{k-1}}{(k-1)!} = \frac{d}{dz} \left(\frac{\Xi^k}{k!} \right) - x_1 \frac{\Xi^{k-1}}{(k-1)!}.$$

Comparing coefficients of $z^n/n!$ and using $\frac{d}{dz} \sum_n a_n z^n/n! = \sum_n a_{n+1} z^n/n!$ gives $\mathcal{D}\mathcal{B}_{n,k}^{\text{exp}} = \mathcal{B}_{n+1,k}^{\text{exp}} - x_1 \mathcal{B}_{n,k-1}^{\text{exp}}$ for all $n \geq 0$ and $k \geq 1$; when $k > n+1$ both sides vanish, and when $k = n+1$ the identity reads $0 = x_1^{n+1} - x_1 \cdot x_1^n$. $\square$

Proposition 11.25 (Faà di Bruno for admissible composition). *Let G be an admissible monomial-leading argument with $\rho > 0$, and let $\gamma := D_X G$. Then for every $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and every $n \geq 1$,*

$$D_X^n (F \circ G) = \sum_{k=1}^n (D_X^k F) \circ G \mathcal{B}_{n,k}^{\text{exp}}(\gamma, D_X \gamma, \dots, D_X^{n-k} \gamma). \quad (212)$$

Proof. Induction on n . For $n = 1$ this is [Theorem 10.31](#), since $\mathcal{B}_{1,1}^{\text{exp}} = x_1$. Assume (212) for n . Apply D_X , using the product rule, the chain rule (188) in the form $D_X((D_X^k F) \circ G) = (D_X^{k+1} F) \circ G \cdot \gamma$, and the fact that $D_X[\mathcal{B}_{n,k}^{\text{exp}}(\gamma, D_X \gamma, \dots)] = (\mathcal{D}\mathcal{B}_{n,k}^{\text{exp}})(\gamma, D_X \gamma, \dots)$, which is the ordinary chain rule for a polynomial in the entries $x_i = D_X^{i-1} \gamma$. This gives

$$D_X^{n+1} (F \circ G) = \sum_{k \geq 1} (D_X^k F) \circ G \left[\gamma \mathcal{B}_{n,k-1}^{\text{exp}} + \mathcal{D}\mathcal{B}_{n,k}^{\text{exp}} \right](\gamma, D_X \gamma, \dots),$$

after re-indexing the first sum. By [Theorem 11.24](#) with $x_1 = \gamma$ the bracket equals $\mathcal{B}_{n+1,k}^{\text{exp}}$, completing the induction. All rearrangements involve finitely many terms for each n . $\square$

Remark 11.26 (Valuation bookkeeping). For a near-identity $G = X + H = t^{-1} + H$ one has $D_X(t^{-1}) = t^0(0+1) = 1$, hence $\gamma = D_X G = 1 + D_X H$ with $\nu_t(D_X H) \geq 1$, and $D_X^i \gamma = D_X^{i+1} H$ has valuation at least $i+1$ for $i \geq 1$. Hence $\mathcal{B}_{n,k}^{\text{exp}}(\gamma, D_X \gamma, \dots)$ has valuation at least $n-k$, and the k -th term of (212) has valuation at least $\nu_t(F) + k + (n-k) = \nu_t(F) + n$, consistent with $\nu_t(D_X^n (F \circ G)) \geq \nu_t(F) + n$. In this class the size of (212) is a combinatorial, not an analytic, difficulty.

11.7 Which engine to use

The generator form (181) and the Taylor form (195) have identical formal content, by [Theorem 11.6](#), and different computational profiles.

	Generator substitution	Taylor composition
Core data	$(1+U)^{-1}$ and $\log(1+U)$	$D_X^k F$ and $H^k/k!$
Reused across	many outer blocks of one F	many outer series with one H
Truncation index	powers of U	derivative order $k \leq N - \nu_t F$
Valid for	any admissible $\rho > 0$	$\rho = 1, c = 1$ only
Typical error	updating one generator only	applying it when $H \notin \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$

Structural checkpoint

Before composing, verify in order: (1) the inner argument has a leading monomial cX^ρ with $\rho > 0$ and a chosen $\log c$ in the coefficient field; (2) the residual factor is $1+U$ with $U \prec 1$; (3) the images $1/G$ and $\log G$ stay in the declared exponent group and logarithmic depth; (4) both generators are updated simultaneously; (5) the precision of U and V is set blockwise by [Theorem 10.11](#), not globally; (6) if an asymptotic statement is intended, hypotheses (A1)–(A3) of [Theorem 10.35](#) are separately verified.

Section summary

For an admissible monomial-leading $G = cX^\rho(1+U)$ with a chosen $\log c \in \mathbb{K}$, composition is exact simultaneous substitution of the two generators, $t \mapsto c^{-1}t^\rho(1+U)^{-1}$ and $L \mapsto \rho L + \log c + \log(1+U)$. It is a $\mathbb{K}$ -algebra homomorphism, multiplies valuations by ρ , determines every output block

by a finite sum with a triangular precision budget, and is associative on the nose once the chosen logarithms are compatible; right substitution reverses the apparent operator order. For $\rho = 1$, $c = 1$ it coincides with the exact formal Taylor formula $F \circ (X + H) = \sum_{k \geq 0} H^k D_X^k F / k!$, whose summability comes from the fact that D_X raises outer order while multiplication by $H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ does not lower it, so that only $k \leq N - \nu_t(F)$ reach the block t^N . None of this is an analytic statement; asymptotic validity requires the regime, branch and remainder-transport hypotheses separately.

12 The near-identity composition group

Throughout this section $\mathbb{K}$ is a field of characteristic zero, the large variable is X , and

$$t := X^{-1}, \quad L := \log X, \quad \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} = \mathbb{K}[L][[t]], \quad \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} = \mathbb{K}[L]((t)).$$

The regime is $X \rightarrow +\infty$ along the positive real ray; in a complex sector a branch of $\log$ must be fixed once and for all, and every statement below is then read with that branch. The outer valuation is ν_t , the distinguished derivation is $D_X = t\partial_L - t^2\partial_t$, and $\Delta_{n,k} = \prod_{j=0}^{k-1} (\partial_L - n - j)$ with $\Delta_{n,0} = 1$, so that

$$D_X^k(t^n p(L)) = t^{n+k} \Delta_{n,k} p(L), \quad n \in \mathbb{Z}, p \in \mathbb{K}[L]. \quad (213)$$

All truncations are the exclusive ones, $\text{Trunc}_{<N} F = \sum_{n < N} p_n(L) t^n$.

12.1 The set, and why its correction term is unrestricted

Definition 12.1 (The near-identity class).

$$\mathcal{G}_{\text{poly}, \mathbb{K}} := \{X + A : A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}\} \subseteq \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}.$$

Membership imposes no smallness on A itself: the correction may be L^{100} , or any polynomial block of any degree. What is small is the *relative* correction

$$\frac{X + A}{X} = 1 + tA \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}, \quad (214)$$

and every element of $1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ arises exactly once this way. This is the whole content of the phrase “tangent to the identity” in the present class: $tA \prec 1$ because $t^n L^j \prec 1$ for every $n \geq 1$ and every $j \geq 0$ in the dominance order of the positive ray.

A common trap

$1 + tA = 1 + \mathcal{O}_t^{\text{form}}(t)$ is a statement about coefficient support only. It asserts no numerical bound, no uniformity in L , and no analytic realization. The corresponding analytic statement, $G(X)/X \rightarrow 1$ as $X \rightarrow +\infty$, is a separate hypothesis about a chosen realization of the series.

We shall use the formal exponential and logarithm on $1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ constantly.

Remark 12.2 (Formal exponential and logarithm: where they are proved). The assignments $u \mapsto \exp u$ and $1 + u \mapsto \log(1 + u)$, given by the usual series, are mutually inverse bijections between $t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and they carry products to sums. This is [Theorem 10.5](#) applied to the Hahn ring $\mathcal{A} = \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ with $\Gamma = \mathbb{Z}$ and $\mathfrak{m} = t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$; every reference to the present remark below is a reference to that lemma, whose extra clauses — additivity of $\exp$, $\log(P^n) = n \log P$ for $n \in \mathbb{Z}$, and commutation with a homomorphism compatible with summable families — are used here as well.

12.2 Substitution, and the exact Taylor formula

Definition 12.3 (Substitution by a near-identity map). Let $G = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ and put

$$R := (1 + tA)^{-1} \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}, \quad V := \log(1 + tA) \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}.$$

For $F = \sum_{n \geq n_0} p_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ define

$$\Phi_G(F) = F \circ G := \sum_{n \geq n_0} t^n R^n p_n(L + V). \quad (215)$$

The two generator images encoded in (215) are

$$t \circ G = \frac{t}{1 + tA}, \quad L \circ G = L + \log(1 + tA). \quad (216)$$

A common trap

The substitution is $t \mapsto t(1 + tA)^{-1}$, never $t \mapsto t + A$: the symbol t denotes X^{-1} , not X . Updating L while leaving t fixed, or conversely, is equally wrong; the two generators must be replaced simultaneously.

Proposition 12.4 (Substitution operator). Let $G = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$. Then

1. the family in (215) is formally summable, so $\Phi_G : \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} \rightarrow \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is well defined, and $\text{Trunc}_{<N}(F \circ G)$ depends only on $\text{Trunc}_{<N} F$ and on $\text{Trunc}_{<N-n_0} A$;
2. Φ_G is a $\mathbb{K}$ -algebra endomorphism of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$;
3. $\Phi_G(t) = t(1 + tA)^{-1}$, $\Phi_G(L) = L + \log(1 + tA)$, and $\Phi_G(X) = G$;
4. $\nu_t(\Phi_G(F)) = \nu_t(F)$ for every F , and $\Phi_G(t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}) \subseteq t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ for every $N \in \mathbb{Z}$; in particular Φ_G is injective and t -adically continuous;
5. $\Phi_G(\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}) \subseteq \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $\Phi_G(\log(1 + u)) = \log(1 + \Phi_G(u))$ for $u \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.

Proof. (1) Since $R \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $V \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and since p_n is a polynomial, $p_n(L + V) = \sum_{j \leq \deg_L p_n} \frac{p_n^{(j)}(L)}{j!} V^j$ is a finite sum lying in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$; likewise $R^n \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ for $n \geq 0$ and $R^n = (1 + tA)^{-n} \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ for $n < 0$. The n -th summand of (215) therefore lies in $t^n \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and ν_t of the n -th summand tends to $+\infty$; the family is summable and n_0 bounds the support below. Only the summands with $n < N$ meet outer order $< N$, and within such a summand only R and V through order t^{N-n-1} are needed, whence the stated dependency.

(2) Additivity is clear. For multiplicativity write $F = \sum_n p_n t^n$, $\tilde{F} = \sum_m q_m t^m$. Evaluation $p \mapsto p(L + V)$ is a $\mathbb{K}$ -algebra homomorphism $\mathbb{K}[L] \rightarrow \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, so

$$\Phi_G(F\tilde{F}) = \sum_r t^r R^r \left(\sum_{n+m=r} p_n q_m \right) (L+V) = \sum_{n,m} t^{n+m} R^{n+m} p_n(L+V) q_m(L+V) = \Phi_G(F) \Phi_G(\tilde{F}),$$

all rearrangements being legitimate because the families are summable.

(3) Take $F = t$, $F = L$, $F = t^{-1}$ in (215); the last gives $t^{-1} R^{-1} = t^{-1}(1 + tA) = X + A = G$.

(4) The coefficient of t^{n_0} in $\Phi_G(F)$ is $p_{n_0}(L)$, because $R^{n_0} \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $p_{n_0}(L + V) \in p_{n_0}(L) + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$; no other summand reaches outer order n_0 . Hence $\nu_t(\Phi_G F) = n_0 = \nu_t(F)$, and the displayed inclusion is immediate from (215).

(5) The inclusion is the case $n_0 \geq 0$. For the last identity, Φ_G is multiplicative and continuous, and $\Phi_G(u) \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ by (4), so $\Phi_G(\log(1 + u)) = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} \Phi_G(u)^k = \log(1 + \Phi_G(u))$. $\square$

Remark 12.5 (Exact formal Taylor formula). For every $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and every $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ the family $(A^k D_X^k F / k!)_{k \geq 0}$ is formally summable and $F \circ (X + A) = \sum_{k \geq 0} A^k D_X^k F / k!$; if $a = \nu_t(F)$, only the indices $k \leq N - a$ affect the coefficient of t^N . This is [Theorem 11.6](#), equation (195), proved there; it is an identity of formal series and carries no analytic content. Every citation of the present remark is a citation of that theorem. Two features of its proof are used repeatedly below: the operator $T_A = \sum_{k \geq 0} \frac{A^k}{k!} D_X^k$ is a *continuous* $\mathbb{K}$ -algebra endomorphism of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ ([Theorem 11.3](#)), and two such endomorphisms agreeing on t and L are equal ([Theorem 11.5](#)).

The single most useful consequence is a gain of one outer block whenever a near-identity argument is perturbed.

Corollary 12.6 (Filtered gain, and the Lipschitz contraction). *Let $A, B, C \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Then*

$$\nu_t(A \circ (X + B) - A \circ (X + C)) \geq \nu_t(A) + \nu_t(B - C) + 1, \quad (217)$$

$$\nu_t(A \circ (X + B) - A) \geq \nu_t(A) + \nu_t(B) + 1, \quad (218)$$

and, more precisely,

$$A \circ (X + B) - A \equiv B D_X A \pmod{t^{\nu_t(A) + 2\nu_t(B) + 2} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}}. \quad (219)$$

Proof. By [Theorem 12.5](#), $A \circ (X + B) - A \circ (X + C) = \sum_{k \geq 1} \frac{B^k - C^k}{k!} D_X^k A$. Since $B^k - C^k = (B - C) \sum_{i=0}^{k-1} B^i C^{k-1-i}$ and all factors have nonnegative valuation, $\nu_t(B^k - C^k) \geq \nu_t(B - C)$; and $\nu_t(D_X^k A) \geq \nu_t(A) + k$. Hence the k -th summand has $\nu_t \geq \nu_t(B - C) + \nu_t(A) + k$, and the minimum over $k \geq 1$ is attained at $k = 1$; this gives (217). Taking $C = 0$ and using $\nu_t(B^k) \geq k \nu_t(B)$ gives, for the k -th term, the bound $\nu_t(A) + k(1 + \nu_t(B))$, which is $\geq \nu_t(A) + \nu_t(B) + 1$ for $k \geq 1$ and $\geq \nu_t(A) + 2\nu_t(B) + 2$ for $k \geq 2$; this is (218) and (219). $\square$

12.3 The group theorem

Theorem 12.7 (The near-identity composition group). $\mathcal{G}_{\text{poly}, \mathbb{K}}$ is a group under composition. Explicitly:

1. (Closure) for $F = X + A$ and $G = X + B$ in $\mathcal{G}_{\text{poly}, \mathbb{K}}$,

$$F \circ G = X + B + A \circ G \in \mathcal{G}_{\text{poly}, \mathbb{K}}; \quad (220)$$

2. (Associativity) $\Phi_H \circ \Phi_G = \Phi_{G \circ H}$ for $G, H \in \mathcal{G}_{\text{poly}, \mathbb{K}}$; consequently $(F \circ G) \circ H = F \circ (G \circ H)$ for every $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and $G, H \in \mathcal{G}_{\text{poly}, \mathbb{K}}$;
3. (Identity) $X \circ G = G$ and $F \circ X = F$;
4. (Inverses) every $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ has a unique two-sided compositional inverse $F^{(-1)\circ} = X + B \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, and $\nu_t(B) = \nu_t(A)$.

The group is not abelian.

Proof. (1) $\Phi_G(X + A) = \Phi_G(X) + \Phi_G(A) = G + A \circ G$ by [Theorem 12.4](#)(2),(3), and $A \circ G \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ by part (5) of the same proposition; so $F \circ G = X + (B + A \circ G)$ with correction in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.

(2) Both $\Phi_H \circ \Phi_G$ and $\Phi_{G \circ H}$ are continuous $\mathbb{K}$ -algebra endomorphisms of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, so by [Theorem 11.5](#) it suffices to check them on t and L . Write $G = X + A$, $H = X + B$. Since Φ_H is a ring homomorphism with $\Phi_H(X) = H$ and $\Phi_H(A) = A \circ H$,

$$\Phi_H(G) = H + A \circ H = G \circ H. \quad (221)$$

Hence $\Phi_H(\Phi_G(t)) = \Phi_H(G^{-1}) = \Phi_H(G)^{-1} = (G \circ H)^{-1} = \Phi_{G \circ H}(t)$. For L , using [Theorem 12.4](#)(5) and [Theorem 12.2](#),

$$\begin{aligned} \Phi_H(\Phi_G(L)) &= \Phi_H(L + \log(1 + tA)) = L + \log(1 + tB) + \log\left(1 + \frac{t}{1+tB}(A \circ H)\right) \\ &= L + \log\left((1 + tB)\left(1 + \frac{t(A \circ H)}{1+tB}\right)\right) = L + \log(1 + t(B + A \circ H)) = \Phi_{G \circ H}(L), \end{aligned}$$

the last equality because the correction of $G \circ H$ is $B + A \circ H$ by (220). Applying the resulting operator identity to F gives $(F \circ G) \circ H = F \circ (G \circ H)$.

(3) $\Phi_G(X) = G$ is [Theorem 12.4\(3\)](#), and $\Phi_X = \text{id}$ because $A = 0$ makes $R = 1$, $V = 0$ in (215).

(4) By (1)–(3), $(\mathcal{G}_{\text{poly}, \mathbb{K}}, \circ)$ is a monoid with identity X , and by (220) an element $G = X + B$ of $\mathcal{G}_{\text{poly}, \mathbb{K}}$ satisfies $F \circ G = X$ if and only if

$$B = -A \circ (X + B). \quad (222)$$

Theorem 14.11 – proved from items (1)–(3) above, [Theorem 12.6](#), and the t -adic completeness of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, so that the present item may cite it – says that (222) has exactly one solution $B \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, that the resulting $G = X + B$ is the unique element of $\mathcal{G}_{\text{poly}, \mathbb{K}}$ with $F \circ G = X$, that it satisfies $G \circ F = X$ as well and is the unique element of $\mathcal{G}_{\text{poly}, \mathbb{K}}$ doing so, and that $\nu_t(B) = \nu_t(A)$ with $[t^{\nu_t(A)}]B = -[t^{\nu_t(A)}]A$. Writing $F^{(-1)\circ} = F^{(-1)\circ} = G$ gives (4).

Noncommutativity is [Theorem 12.8](#). □

Example 12.8 (Two maps that do not commute). Let $F = X + L$ and $G = X + 1$. Then $G \circ F = F + 1 = X + L + 1$, whereas

$$F \circ G = G + L \circ G = X + 1 + \log(X + 1) = X + 1 + L + \log(1 + t) = X + 1 + L + t - \frac{1}{2}t^2 + \mathcal{O}_t^{\text{form}}(t^3).$$

$$\text{Hence } F \circ G - G \circ F = t - \frac{1}{2}t^2 + \mathcal{O}_t^{\text{form}}(t^3) \neq 0.$$

Remark 12.9 (Direction of the dependency on Part IV). Items (1)–(3) of [Theorem 12.7](#) are proved here from [Theorem 12.4](#) and [Theorem 11.5](#) alone, and [Theorem 12.6](#) needs nothing beyond [Theorem 12.5](#). Item (4) is different: the existence, uniqueness, two-sidedness and leading data of $F^{(-1)\circ}$ are proved once in the volume, in [Theorem 14.11](#), and item (4) cites that theorem. The dependency runs in one direction only. Part IV takes items (1)–(3) and [Theorem 12.6](#) as given – they are the imports recorded in [Theorem 14.3](#) – and from them constructs the solution of (222) by Picard iteration with the sharp rate $\nu_t(B_{j+1} - B_j) \geq (j+1)\alpha + j$, $\alpha = \nu_t(A)$; it adds the first-order form $F^{(-1)\circ} = X - A + \mathcal{O}_t^{\text{form}}(t^{2\alpha+1})$ ([Theorem 14.12](#)), the residual certificates, the Lagrange–Bürmann coefficient formula and the algorithms that compute it. A reader who wants only the group law therefore needs, beyond this section, the existence half of Part IV and nothing else.

Remark 12.10 (Substitution reverses order). By [Theorem 12.7\(2\)](#), $G \mapsto \Phi_G$ is an injective *anti*-homomorphism from $\mathcal{G}_{\text{poly}, \mathbb{K}}$ into the group of continuous $\mathbb{K}$ -algebra automorphisms of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$; injectivity holds because $\Phi_G(X) = G$. Every sign discrepancy between group-theoretic and vector-field formulas below can be traced to this order reversal.

12.4 What the group is not

A common trap

$\mathcal{G}_{\text{poly}, \mathbb{K}}$ is *not* a multiplicative group of coefficient series. Four distinct statements are involved, and conflating them is the commonest error in this subject.

- No element of $\mathcal{G}_{\text{poly}, \mathbb{K}}$ lies in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$: each has a t^{-1} block. The elements of $\mathcal{G}_{\text{poly}, \mathbb{K}}$ are maps, not coefficient series.
- The bijection $X + A \mapsto 1 + tA$ of (214) onto the multiplicative group $1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is *not* a group homomorphism. Indeed $1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is abelian and, by [Theorem 12.8](#), $\mathcal{G}_{\text{poly}, \mathbb{K}}$ is not.
- $F^{(-1)\circ} \neq F^{-1}$. For $F = X + L$, $F^{-1} = t(1 + tL)^{-1} = t - Lt^2 + L^2t^3 - \dots$, whereas $F^{(-1)\circ} = X - L + Lt + \mathcal{O}_t^{\text{form}}(t^2)$.
- $\mathcal{G}_{\text{poly}, \mathbb{K}}$ is not the additive group of corrections either: $F \circ G = X + A + B + (A \circ G - A)$, and the last bracket vanishes only in the associated graded object ([Theorem 12.15](#)).

Note also that $t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is not an ideal of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, because t is a unit there; all filtration

statements below are statements about the valuation ν_t , not about quotient rings.

12.5 The valuation filtration and its normal subgroups

Definition 12.11 (Contact filtration). For $r \in \mathbb{N}_0$ put

$$\mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)} := \{X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}} : \nu_t(A) \geq r\}, \quad \mathcal{G}_{\text{poly}, \mathbb{K}}^{(0)} = \mathcal{G}_{\text{poly}, \mathbb{K}}.$$

Write $F \equiv_r F'$ when the corrections of $F, F' \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ differ by an element of $t^r \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, i.e. when $F - F' = \mathcal{O}_t^{\text{form}}(t^r)$; one says that F and F' have *contact of order r* .

Proposition 12.12 (The filtration is by normal subgroups). *For every $r \geq 0$, $\equiv_r$ is a congruence for composition: if $F \equiv_r F'$ and $G \equiv_r G'$ then $F \circ G \equiv_r F' \circ G'$. Consequently $\mathcal{G}_{\text{poly}, \mathbb{K}} / \equiv_r$ is a group, the quotient map is a surjective homomorphism, and $\mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)}$ is a normal subgroup of $\mathcal{G}_{\text{poly}, \mathbb{K}}$ with $\mathcal{G}_{\text{poly}, \mathbb{K}} / \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)} \cong \mathcal{G}_{\text{poly}, \mathbb{K}} / \equiv_r$.*

Proof. Write $F = X + A, F' = X + A', G = X + B, G' = X + B'$. Then

$$F \circ G - F' \circ G = (A - A') \circ G,$$

which has $\nu_t = \nu_t(A - A') \geq r$ by [Theorem 12.4\(4\)](#). Next

$$F' \circ G - F' \circ G' = (B - B') + (A' \circ G - A' \circ G'),$$

whose first term has $\nu_t \geq r$ and whose second has $\nu_t \geq r + 1$ by (217). Adding gives $F \circ G \equiv_r F' \circ G'$. Since $\equiv_r$ is a congruence on a group, the quotient is a group and the class of X , which is exactly $\mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)}$, is a normal subgroup. $\square$

Proposition 12.13 (Separation and completeness). $\bigcap_{r \geq 0} \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)} = \{X\}$, and the natural map

$$\mathcal{G}_{\text{poly}, \mathbb{K}} \longrightarrow \varprojlim_r \mathcal{G}_{\text{poly}, \mathbb{K}} / \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)}$$

is an isomorphism of groups.

Proof. If $X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)}$ for all r then $\nu_t(A) = +\infty$, so $A = 0$; this is injectivity as well. For surjectivity, let $(F_r \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)})_r$ be a compatible family. Compatibility says $F_{r+1} \equiv_r F_r$, so the coefficient blocks of the corrections stabilise: $[t^n]A_{F_r}$ is independent of r once $r > n$. Define A by $[t^n]A := [t^n]A_{F_{n+1}}$; then $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $X + A \equiv_r F_r$ for every r . $\square$

Proposition 12.14 (Commutators drop one further block). *For $r, s \geq 0$,*

$$[\mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)}, \mathcal{G}_{\text{poly}, \mathbb{K}}^{(s)}] \subseteq \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r+s+1)}, \quad \text{where } [F, G] = F^{(-1) \circ} \circ G^{(-1) \circ} \circ F \circ G.$$

More precisely, if $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)}$ and $G = X + B \in \mathcal{G}_{\text{poly}, \mathbb{K}}^{(s)}$ then the correction of $[F, G]$ satisfies

$$[F, G] - X \equiv B D_X A - A D_X B \pmod{t^{r+s+2} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}}. \quad (223)$$

Consequently the lower central series obeys $\gamma_k(\mathcal{G}_{\text{poly}, \mathbb{K}}) \subseteq \mathcal{G}_{\text{poly}, \mathbb{K}}^{(k-1)}$, each quotient $\mathcal{G}_{\text{poly}, \mathbb{K}} / \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)}$ is nilpotent of class at most r , and $\mathcal{G}_{\text{poly}, \mathbb{K}}$ is residually nilpotent.

Proof. Put $P = F \circ G = X + A_P$ and $Q = G \circ F = X + A_Q$. By (220) and (219),

$$A_P = B + A \circ G \equiv A + B + B D_X A, \quad A_Q = A + B \circ F \equiv A + B + A D_X B$$

modulo $t^{r+s+2} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$; note that the error terms in (219) are of order t^{r+2s+2} and t^{s+2r+2} , both at least t^{r+s+2} . Hence

$$A_P - A_Q \equiv B D_X A - A D_X B \pmod{t^{r+s+2} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}}, \quad \nu_t(A_P - A_Q) \geq r + s + 1. \quad (224)$$

Now $[F, G] = Q^{(-1)^\circ} \circ P$. Writing $Q^{(-1)^\circ} = X + B_Q$ and using $X = Q^{(-1)^\circ} \circ Q = X + A_Q + B_Q \circ Q$, i.e. $B_Q \circ Q = -A_Q$, we get

$$[F, G] = P + B_Q \circ P = X + (A_P - A_Q) + (B_Q \circ P - B_Q \circ Q).$$

The last bracket has $\nu_t \geq \nu_t(A_P - A_Q) + 1 \geq r + s + 2$ by (217). With (224) this proves both displayed claims. The statement about γ_k follows by induction: $\gamma_1 = \mathcal{G}_{\text{poly}, \mathbb{K}} = \mathcal{G}_{\text{poly}, \mathbb{K}}^{(0)}$ and $\gamma_{k+1} = [\gamma_k, \mathcal{G}_{\text{poly}, \mathbb{K}}] \subseteq [\mathcal{G}_{\text{poly}, \mathbb{K}}^{(k-1)}, \mathcal{G}_{\text{poly}, \mathbb{K}}^{(0)}] \subseteq \mathcal{G}_{\text{poly}, \mathbb{K}}^{(k)}$. $\square$

12.6 The associated graded object

Proposition 12.15 (Graded pieces). *For every $r \geq 0$ the map*

$$\mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)} / \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r+1)} \longrightarrow (\mathbb{K}[L], +), \quad (X + A) \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r+1)} \longmapsto [t^r] A,$$

is a group isomorphism. In particular each graded piece is abelian and uniquely divisible.

Proof. If $F = X + A$ and $G = X + B$ lie in $\mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)}$, then by (220) and (218) $F \circ G = X + A + B + (A \circ G - A)$ with $\nu_t(A \circ G - A) \geq r + r + 1 \geq r + 1$; hence $[t^r](A_{F \circ G}) = [t^r]A + [t^r]B$, so the map is a homomorphism. It is surjective (take $A = p(L)t^r$) and its kernel is $\mathcal{G}_{\text{poly}, \mathbb{K}}^{(r+1)}$. $\square$

Theorem 12.16 (The graded Lie algebra). *Let*

$$\mathcal{V} := \mathbb{K}[L][t] D_X,$$

the $\mathbb{K}$ -span of the derivations $a(L)t^r D_X$ with $a \in \mathbb{K}[L]$ and $r \in \mathbb{N}_0$, a Lie algebra under $[\alpha D_X, \beta D_X] = (\alpha D_X \beta - \beta D_X \alpha) D_X$, graded by $\deg(a(L)t^r D_X) := r + 1$; explicitly

$$[a t^r D_X, b t^s D_X] = t^{r+s+1} (a(b' - sb) - b(a' - ra)) D_X, \quad ' = \partial_L. \quad (225)$$

Then the associated graded group $\text{gr } \mathcal{G}_{\text{poly}, \mathbb{K}} = \bigoplus_{r \geq 0} \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)} / \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r+1)}$, equipped with the bracket induced by the group commutator, is a graded Lie algebra, and

$$\theta : \text{gr } \mathcal{G}_{\text{poly}, \mathbb{K}} \longrightarrow \mathcal{V}, \quad \theta((X + A) \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r+1)}) = -([t^r] A) t^r D_X$$

is an isomorphism of graded Lie algebras.

Proof. That $\mathcal{V}$ is a Lie algebra needs no verification beyond closure: for $\alpha, \beta \in \mathbb{K}[L][t]$, $\alpha D_X \beta - \beta D_X \alpha \in \mathbb{K}[L][t]$ by (213), and the bracket of two derivations of an associative algebra is a derivation, so the Jacobi identity is inherited. Formula (225) is (213) applied twice, and it shows that the bracket adds the degrees: $(r + 1) + (s + 1) = (r + s + 1) + 1$.

By Theorem 12.14 the group commutator induces a well-defined biadditive map $\mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)} / \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r+1)} \times \mathcal{G}_{\text{poly}, \mathbb{K}}^{(s)} / \mathcal{G}_{\text{poly}, \mathbb{K}}^{(s+1)} \rightarrow \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r+s+1)} / \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r+s+2)}$: replacing A by $A + \tilde{A}$ with $\nu_t \tilde{A} \geq r + 1$ changes $B D_X A -$

$AD_X B$ by an element of $t^{r+s+2} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and biadditivity follows from (223) together with [Theorem 12.15](#). Writing $A = a(L)t^r + \dots$ and $B = b(L)t^s + \dots$, (213) gives

$$[t^{r+s+1}](B D_X A - A D_X B) = b(a' - ra) - a(b' - sb) = (ba' - ab') + (s - r)ab.$$

On the other hand (225) gives

$$[-a t^r D_X, -b t^s D_X] = t^{r+s+1} \left(a(b' - sb) - b(a' - ra) \right) D_X = -t^{r+s+1} \left((ba' - ab') + (s - r)ab \right) D_X,$$

which is exactly θ applied to the previous display. Thus θ intertwines the two brackets; it is bijective and degree preserving by [Theorem 12.15](#). Since $\mathcal{V}$ satisfies the Jacobi identity, so does $\text{gr } \mathcal{G}_{\text{poly}, \mathbb{K}}$. $\square$

Remark 12.17. The sign in θ is forced by [Theorem 12.10](#): composition and operator composition run in opposite directions, so the induced bracket is the opposite of the vector-field bracket. For $r = s = 0$, that is for $F = X + f(L)$, $G = X + g(L)$, (223) reads $F \circ G - G \circ F = t(gf' - fg') + \mathcal{O}_t^{\text{form}}(t^2)$, the classical bracket of one-dimensional vector fields in the coordinate $L = \log X$.

12.7 The exponential of a vector field, and iteration

The Taylor operator $T_A = \sum_k \frac{A^k}{k!} D_X^k$ of [Theorem 12.5](#) is *normally ordered*: every power of A stands to the left of every D_X . It must not be confused with the operator exponential $\exp(AD_X)$, in which D_X also differentiates A . Both are automorphisms, and both are substitutions – but by different maps.

Theorem 12.18 (Exponential correspondence). *For $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ the operator $\exp(AD_X) = \sum_{k \geq 0} \frac{1}{k!} (AD_X)^k$ is a well-defined continuous $\mathbb{K}$ -algebra automorphism of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, and*

$$\exp(AD_X) = \Phi_{\mathcal{E}(A)}, \quad \mathcal{E}(A) := \exp(AD_X)(X) = X + A + \frac{1}{2} A D_X A + \dots \in \mathcal{G}_{\text{poly}, \mathbb{K}}. \quad (226)$$

The map $\mathcal{E} : \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} \rightarrow \mathcal{G}_{\text{poly}, \mathbb{K}}$ is a bijection, and $\mathcal{E}(A) \equiv X + A \pmod{t^{2\nu_t(A)+1} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}}$.

Proof. Well-definedness. $\delta := AD_X$ is a derivation with $\delta(t^n \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}) \subseteq t^{n+1} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, so for $F \in t^{-m} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ the family $(\delta^k F / k!)_k$ is summable and $\exp(\delta)F \in t^{-m} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$; continuity and $\mathbb{K}$ -linearity are clear. Multiplicativity follows from the Leibniz rule exactly as for T_A , and $\exp(-\delta)$ is a two-sided inverse because the identity $\sum_k \frac{x^k}{k!} \sum_j \frac{(-x)^j}{j!} = 1$ holds coefficientwise.

Identification with a substitution. Let $\Psi_s = \exp(s\delta)$; for each n , $[t^n]\Psi_s(F)$ is a polynomial in s with coefficients in $\mathbb{K}[L]$, because $\nu_t(\delta^k F) \geq \nu_t(F) + k$. Put

$$u(s) = \Psi_s(L), \quad v(s) = \log(\Psi_s(X)) := L + \log(1 + t(\Psi_s(X) - X)),$$

the latter legitimate because $\Psi_s(X) - X \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, so that $t(\Psi_s(X) - X) \in t \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and [Theorem 12.2](#) applies. Differentiating in s coefficientwise, and using $\frac{d}{ds} \Psi_s = \Psi_s \circ \delta$,

$$u'(s) = \Psi_s(A D_X L) = \Psi_s(At), \quad v'(s) = \frac{\Psi_s(A D_X X)}{\Psi_s(X)} = \Psi_s(A) \Psi_s(t) = \Psi_s(At),$$

where $\Psi_s(X)^{-1} = \Psi_s(t)$ because Ψ_s is a ring automorphism. Since $u(0) = v(0) = L$ and $\mathbb{K}$ has characteristic zero, $u = v$. Setting $s = 1$ and writing $G = \mathcal{E}(A) = \Psi_1(X)$, we get $\Psi_1(L) = \log G = \Phi_G(L)$ and $\Psi_1(t) = \Psi_1(X)^{-1} = G^{-1} = \Phi_G(t)$; both maps are continuous algebra homomorphisms, so $\Psi_1 = \Phi_G$.

Bijection. Write $\mathcal{E}(A) = X + A + \mathcal{R}(A)$ with $\mathcal{R}(A) = \sum_{k \geq 2} \frac{1}{k!} (AD_X)^k A$. Each term of $\mathcal{R}(A)$ is a sum of products of k factors drawn from A and its D_X -derivatives, carrying $k - 1$ derivatives in total, so $\nu_t \geq k \nu_t(A) + k - 1 \geq 2\nu_t(A) + 1$; this proves the last assertion. Replacing A by A' one factor at a time in each such product shows, for $A, A' \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$,

$$\nu_t(\mathcal{R}(A) - \mathcal{R}(A')) \geq \nu_t(A - A') + 1,$$

because each single-factor difference contributes $\nu_t(A - A')$ and the remaining $k - 1 \geq 1$ derivatives contribute at least 1. Hence $\mathcal{E}(A) - \mathcal{E}(A') \equiv A - A'$ modulo $t^{\nu_t(A-A')+1} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Injectivity is immediate. For surjectivity, let $G = X + A_* \in \mathcal{G}_{\text{poly}, \mathbb{K}}$; construct $A^{(r)}$ with $\mathcal{E}(A^{(r)}) \equiv_r G$ by setting $A^{(0)} = 0$ and $A^{(r+1)} = A^{(r)} - e_r t^r$, where $e_r t^r$ is the outer block of order r of $\mathcal{E}(A^{(r)}) - G$. The sequence is Cauchy and its limit A satisfies $\mathcal{E}(A) = G$ by continuity. $\square$

Corollary 12.19 (Unique divisibility and continuous iteration). *For $F \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ let $A = \mathcal{E}^{-1}(F)$ and define*

$$F^{[s]} := \exp(sAD_X)(X) = \mathcal{E}(sA), \quad s \in \mathbb{K}.$$

Then $F^{[1]} = F$, $F^{[0]} = X$, $F^{[s]} \circ F^{[s']} = F^{[s+s']}$, $F^{[n]}$ is the n -fold composite for $n \in \mathbb{N}_{>0}$, and $F^{[-1]} = F^{(-1)\circ}$. In particular $\mathcal{G}_{\text{poly}, \mathbb{K}}$ is uniquely divisible: for each $n \in \mathbb{N}_{>0}$ there is exactly one $H \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ with $H^{[n]} = F$, namely $H = \mathcal{E}(A/n)$. Moreover $F^{[s]} \equiv X + sA \pmod{t^{2\nu_t(A)+1} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}}$.

Proof. By Theorem 12.18 and Theorem 12.7(2), $\Phi_{F^{[s']} \circ F^{[s]}} = \Phi_{F^{[s]}} \Phi_{F^{[s']}} = \exp(sAD_X) \exp(s'AD_X) = \exp((s+s')AD_X) = \Phi_{F^{[s+s'()]}}$; apply both sides to X and use injectivity of $G \mapsto \Phi_G$. The remaining assertions are immediate, uniqueness of n -th roots because $H^{[n]} = \mathcal{E}(nB)$ for $H = \mathcal{E}(B)$ and $\mathcal{E}$ is injective. $\square$

Remark 12.20 (Normal ordering versus the flow). $T_A \neq \exp(AD_X)$ in general. $T_A = \Phi_{X+A}$ substitutes the map $X + A$; $\exp(AD_X) = \Phi_{\mathcal{E}(A)}$ is the time-one flow of the vector field AD_X , and $\mathcal{E}(A) = X + A + \frac{1}{2}AD_X A + \dots$. For $A = L$, say, $\mathcal{E}(L) = X + L + \frac{1}{2}Lt + \mathcal{O}_t^{\text{form}}(t^2) \neq X + L$. The two operators agree only when $AD_X A = 0$ to all relevant orders, in particular when $A \in \mathbb{K}$. Confusing them turns an exact composition formula into an approximate one.

Section summary

$\mathcal{G}_{\text{poly}, \mathbb{K}} = X + \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is a group under composition, with identity X and with inverses produced by a fixed-point iteration that gains one outer block per step. Its contact filtration $(\mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)})$ consists of normal subgroups with $[\mathcal{G}_{\text{poly}, \mathbb{K}}^{(r)}, \mathcal{G}_{\text{poly}, \mathbb{K}}^{(s)}] \subseteq \mathcal{G}_{\text{poly}, \mathbb{K}}^{(r+s+1)}$; the quotients are nilpotent, the group is their inverse limit, and the associated graded object is the graded Lie algebra $\mathbb{K}[L][t]D_X$ of polynomial vector fields. Exponentiating those vector fields makes $\mathcal{G}_{\text{poly}, \mathbb{K}}$ uniquely divisible and gives a continuous iteration $F^{[s]}$. None of this is a statement about multiplication of coefficient series.

13 Scale-changing composition and deeper logarithms

Section 12 treated inner maps that perturb the scale. This section treats inner maps that *change* it. The governing principle is the same in both cases and is worth isolating before any case analysis.

13.1 The substitution principle

Every $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is built from the two generators $t = X^{-1}$ and $L = \log X$ by sums, products and t -adic limits. Therefore a substitution $X \mapsto G$ is completely determined by the two images

$$t \circ G = G^{-1}, \quad L \circ G = \log G, \quad (227)$$

and the only question is which ambient class contains them and is closed enough to receive the resulting sums. Answering that question *first* is the difference between a correct computation and a silently truncated one: computing in a class that lacks a monomial forced by (227) does not make that monomial disappear, it makes the retained terms wrong.

Definition 13.1 (Nested logarithms, monomial groups, Hahn ambients). Put

$$L_0 := X, \quad L_{j+1} := \log L_j,$$

so that $L_1 = L = \log X$ and $L_2 = \log \log X$. For $r \geq 1$ and an ordered subgroup $\Gamma \subseteq \mathbb{R}$ let

$$\mathcal{M}_r(\Gamma) := \{L_0^{\alpha_0} L_1^{\alpha_1} \cdots L_r^{\alpha_r} : \alpha_0, \dots, \alpha_r \in \Gamma\},$$

ordered by declaring one monomial dominant over another when, at the least index j at which the exponents differ, the first exponent is the larger. This is a totally ordered abelian group, and we write $\mathcal{H}_r(\mathbb{K}, \Gamma)$ for the Hahn field of series $\sum_{m \in S} c_m m$ with $c_m \in \mathbb{K}$ and $S \subseteq \mathcal{M}_r(\Gamma)$ well ordered in the direction of increasing smallness [19, 13]. On the positive ray this order is the asymptotic order as $X \rightarrow +\infty$. One has

$$\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} \subset \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} \subset \mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}} \subset \mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}} \subset \mathcal{H}_1(\mathbb{K}, \mathbb{Z}), \quad \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} \subset \mathbb{K}[L]((t^{1/s})) \subset \mathcal{H}_1(\mathbb{K}, \tfrac{1}{s}\mathbb{Z}).$$

A common trap

$\mathcal{H}_r(\mathbb{K}, \Gamma)$ is a bookkeeping device, not a licence. Membership is a statement about support; it says nothing about convergence, and for non-integer exponents it presupposes a branch of X^{α_0} and of every logarithm involved. All statements below are for $X \rightarrow +\infty$ on the positive real ray with the real positive branches; in a sector, each power and each logarithm must be assigned a branch before the classification is applied.

13.2 Power–logarithm inner maps: the classification

Theorem 13.2 (Classification of power–log scale changes). *Let $\rho \in \mathbb{R}_{>0}$, $\sigma \in \mathbb{R}$, $c \in \mathbb{K}^\times$ with a chosen $\log c \in \mathbb{K}$, and $U \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Put*

$$G = c X^\rho L^\sigma (1 + U),$$

and let $\Gamma_0 = \rho\mathbb{Z} + \mathbb{Z}$, $\Gamma_1 = \sigma\mathbb{Z} + \mathbb{Z}$. Then for every $F = \sum_{n \geq n_0} p_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ the family

$$F \circ G = \sum_{n \geq n_0} c^{-n} t^{\rho n} L^{-\sigma n} (1 + U)^{-n} p_n(\rho L + \sigma L_2 + \log c + \log(1 + U)) \quad (228)$$

is well based and locally finite, so $F \circ G$ is a well-defined element of $\mathcal{H}_2(\mathbb{K}, \Gamma_0 + \Gamma_1)$, with L_0 -exponents in $-\Gamma_0$, L_1 -exponents in $-\sigma\mathbb{Z} + \mathbb{N}_0$ and L_2 -exponents in $\mathbb{N}_0$; and $\Phi_G : F \mapsto F \circ G$ is an injective $\mathbb{K}$ -algebra homomorphism. Moreover:

1. *if $\sigma = 0$ and $\rho \in \mathbb{N}_{>0}$, then $\Phi_G(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}) \subseteq \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$;*
2. *if $\sigma = 0$ and $\rho = r/s \in \mathbb{Q}_{>0}$ in lowest terms, then $\Phi_G(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}) \subseteq \mathbb{K}[L]((t^{1/s}))$; for $s > 1$ this is sharp, in that $\Phi_G(t) \notin \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and the subgroup of $\mathbb{R}$ generated by the L_0 -exponents occurring in $\Phi_G(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}})$ is exactly $\frac{1}{s}\mathbb{Z}$;*
3. *if $\sigma = 0$ and $\rho \notin \mathbb{Q}$, then no Puiseux-log extension $\mathbb{K}[L]((t^{1/s}))$ contains $\Phi_G(t)$, and Γ_0 is dense in $\mathbb{R}$;*
4. *if $\sigma \neq 0$, then L_2 occurs in $F \circ G$ if and only if $p_n \notin \mathbb{K}$ for some n ; and if $p_n \neq 0$ with $\sigma n \notin -\mathbb{N}_0$ then an L_1 -exponent outside $\mathbb{N}_0$ occurs. In particular $\Phi_G(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}})$ is contained in no depth-one class as soon as some coefficient block of some F is nonconstant.*

Proof. Formula (228) is (227) substituted into $F = \sum p_n(L)t^n$, using

$$G^{-1} = c^{-1} t^\rho L^{-\sigma} (1 + U)^{-1}, \quad \log G = \log c + \rho L_1 + \sigma L_2 + \log(1 + U)$$

with the chosen branches, together with the finite Taylor expansion $p_n(a+w) = \sum_{j \leq \deg_L p_n} \frac{p_n^{(j)}(a)}{j!} w^j$ at $a = \rho L + \sigma L_2 + \log c$, $w = \log(1+U) \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$.

Support. The n -th summand has L_0 -exponents in $-\rho n - \mathbb{N}_0$, since $(1+U)^{-n}$ and $\log(1+U)$ lie in $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and p_n is a polynomial. Because $\rho > 0$, for each real M only finitely many pairs $(n, k) \in \mathbb{Z}_{\geq n_0} \times \mathbb{N}_0$ satisfy $\rho n + k \leq M$. Hence the set of L_0 -exponents occurring is well ordered in the direction of increasing smallness, each is reached by only finitely many pairs, and each contribution carries only finitely many monomials in L_1, L_2 . The family is therefore summable in $\mathcal{H}_2(\mathbb{K}, \Gamma_0 + \Gamma_1)$, and the stated exponent ranges are read off the display.

Homomorphism and injectivity. Multiplicativity is proved exactly as in [Theorem 12.4\(2\)](#), evaluation of polynomials being a ring homomorphism. For injectivity let $F \neq 0$ with $\nu_t F = n_0$. Since $\rho \neq 0$, the map $Y \mapsto \rho L + \sigma L_2 + \log c$ induces an injection $\mathbb{K}[Y] \hookrightarrow \mathbb{K}[L_1, L_2]$, so $p_{n_0}(\rho L + \sigma L_2 + \log c) \neq 0$; its dominant monomial times $c^{-n_0} X^{-\rho n_0} L^{-\sigma n_0}$ is the dominant monomial of the whole sum, because every other summand and every U -correction has strictly smaller L_0 -exponent. Hence $F \circ G \neq 0$.

(1) With $\sigma = 0$ and $\rho \in \mathbb{N}_{>0}$ all L_0 -exponents lie in $\mathbb{Z}$, all L_1 -exponents in $\mathbb{N}_0$, no L_2 occurs, and only finitely many L_0 -exponents exceed $-\rho n_0$; this is exactly membership of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$.

(2) All exponents lie in $\frac{1}{s}\mathbb{Z}$. Taking $U = 0$ and $F = t^k$ gives $\Phi_G(t^k) = c^{-k} t^{kr/s}$; as k ranges over $\mathbb{Z}$ these exponents generate $\frac{r}{s}\mathbb{Z} + \mathbb{Z} = \frac{1}{s}\mathbb{Z}$, because $\gcd(r, s) = 1$. For $s > 1$ we have $\rho \notin \mathbb{Z}$, so $\Phi_G(t)$ has L_0 -exponent $-\rho \notin \mathbb{Z}$ and cannot lie in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, all of whose L_0 -exponents are integers.

(3) $\Phi_G(t)$ has L_0 -exponent $-\rho \notin \mathbb{Q}$, while every element of $\mathbb{K}[L]((t^{1/s}))$ has L_0 -exponents in $\frac{1}{s}\mathbb{Z} \subseteq \mathbb{Q}$. Density of $\rho\mathbb{Z} + \mathbb{Z}$ for irrational ρ is the classical equidistribution-free statement that a nondiscrete subgroup of $\mathbb{R}$ is dense.

(4) If every p_n lies in $\mathbb{K}$, then (228) reduces to $\sum_n c^{-n} p_n t^{\rho n} L^{-\sigma n} (1+U)^{-n}$, in which L_2 does not occur. Conversely, let n_1 be least with $\deg_L p_{n_1} \geq 1$. Summands with $n < n_1$ have constant p_n and contribute no L_2 , so the coefficient of $X^{-\rho n_1}$ receives L_2 only from $n = n_1$, where it equals $c^{-n_1} L^{-\sigma n_1} p_{n_1}(\rho L + \sigma L_2 + \log c)$; its L_2 -degree is $\deg_L p_{n_1} \geq 1$, with leading coefficient $\sigma^{\deg_L p_{n_1}}$ times that of p_{n_1} , hence nonzero because $\sigma \neq 0$. The claim about L_1 -exponents is read off the same summand, whose L_1 -exponents are $-\sigma n + \mathbb{N}_0$. $\square$

Corollary 13.3 (The three closure regimes for a pure power). *For $G = X^\rho$ with $\rho > 0$ one has $t \circ G = t^\rho$, $L \circ G = \rho L$, and*

$$\left(\sum_{n \geq n_0} p_n(L) t^n \right) \circ X^\rho = \sum_{n \geq n_0} p_n(\rho L) t^{\rho n}. \quad (229)$$

The composite stays in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ for $\rho \in \mathbb{N}_{>0}$; requires exactly the Puiseux-log extension $\mathbb{K}[L]((t^{1/s}))$ for $\rho = r/s \in \mathbb{Q}_{>0}$ in lowest terms; and requires a Hahn exponent group containing $\rho\mathbb{Z} + \mathbb{Z}$ for irrational ρ . For $\rho \in \mathbb{N}_{>0}$, Φ_{X^ρ} is an automorphism of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ if and only if $\rho = 1$.

Proof. All but the last sentence is [Theorem 13.2\(1\)–\(3\)](#) with $c = 1$, $\sigma = 0$, $U = 0$. For $\rho \geq 2$, (229) shows every L_0 -exponent of the image lies in $\rho\mathbb{Z}$, so t is not in the image; for $\rho = 1$, Φ_X is the identity. $\square$

Proposition 13.4 (Dilations and the semidirect structure). *Let $C \subseteq \mathbb{K}^\times$ be a subgroup carrying a fixed group homomorphism $\lg : C \rightarrow (\mathbb{K}, +)$, a multiplicative branch choice; an analytic realisation additionally requires $e^{\lg(c)} = c$. For $c \in C$ write $m_c(X) = cX$. Then*

$$t \circ m_c = c^{-1} t, \quad L \circ m_c = L + \lg(c),$$

Φ_{m_c} is an automorphism of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ preserving $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, the dilations normalise $\mathcal{G}_{\text{poly},\mathbb{K}}$ by

$$m_c^{(-1)\circ} \circ (X + A) \circ m_c = X + c^{-1} (A \circ m_c) \in \mathcal{G}_{\text{poly},\mathbb{K}},$$

and $\mathcal{G}_{\text{lin}} = \{cX + A : c \in C, A \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}\}$ is a group under composition; it is the semidirect product of C acting on the normal subgroup $\mathcal{G}_{\text{poly},\mathbb{K}}$ by the displayed conjugation.

Proof. The generator images are immediate from (227), and they show that Φ_{m_c} maps $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ into $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ with two-sided inverse $\Phi_{m_{c^{-1}}}$, using $\lg(c^{-1}) = -\lg(c)$. Next, $(X + A) \circ m_c = cX + A \circ m_c$, and composing with $m_{c^{-1}}$ on the outside multiplies by c^{-1} ; since $A \circ m_c \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, the conjugate lies in $\mathcal{G}_{\text{poly},\mathbb{K}}$. Finally $cX + A = m_c \circ (X + c^{-1}A)$ writes $\mathcal{G}_{\text{lin}}$ as the product of the subgroup $\{m_c : c \in C\} \cong C$ with the normal subgroup $\mathcal{G}_{\text{poly},\mathbb{K}}$, the two intersecting trivially; the group axioms follow from Theorem 12.7 together with the displayed normalisation. $\square$

Remark 13.5 (Why the branch constant must be carried as data). For $\mathbb{K} = \mathbb{R}$ and $C = \mathbb{R}_{>0}$ the ordinary logarithm is a homomorphism and Theorem 13.4 applies verbatim. For $\mathbb{K} = \mathbb{C}$ no multiplicative branch choice exists on all of $\mathbb{C}^\times$: a homomorphism $\lg : \mathbb{C}^\times \rightarrow (\mathbb{C}, +)$ splitting $\exp$ would exhibit $\ker \exp = 2\pi i\mathbb{Z}$ as a direct summand of $(\mathbb{C}, +)$; but the latter is a $\mathbb{Q}$ -vector space, hence divisible, every direct summand of a divisible group is divisible, and $\mathbb{Z}$ is not. Consequently a symbolic implementation must carry the chosen value of $\log c$ as data attached to the composition rather than recompute it from a principal branch: changing $\text{Log } c$ by $2\pi i j$ changes the constant block of $L \circ G$ and can move the result onto a different branch of a multivalued inverse.

13.3 Two boundary cases: a logarithmic shift and a logarithmic factor

The maps $X \mapsto X + c \log X$ and $X \mapsto X \log X$ look similar and behave completely differently.

Proposition 13.6 (A logarithmic shift stays inside, at a cost). *Let $c \in \mathbb{K}^\times$ and $G = X + cL$. Then $G \in \mathcal{G}_{\text{poly},\mathbb{K}}$; in particular $\Phi_G(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}) \subseteq \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ and G is invertible in $\mathcal{G}_{\text{poly},\mathbb{K}}$. The logarithmic degrees of the composite are nevertheless unbounded:*

$$t \circ G = \frac{t}{1 + cLt} = \sum_{n \geq 1} (-c)^{n-1} L^{n-1} t^n, \quad L \circ G = L + \sum_{n \geq 1} \frac{(-1)^{n+1} c^n}{n} L^n t^n,$$

so that, in the log-degree profile notation, $d_{(t \circ G)}(n) = n - 1$ and $d_{(L \circ G)}(n) = n$ for $n \geq 1$.

Proof. $cL \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, so $G \in \mathcal{G}_{\text{poly},\mathbb{K}}$ and Theorem 12.4 and Theorem 12.7 apply. The two displays are the geometric and logarithmic series for $u = cLt \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, legitimate by Theorem 12.2; each coefficient block is a single monomial in L , so its degree is read off directly. $\square$

Remark 13.7. Theorem 13.6 is the reason why the definition of $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ imposes no uniform bound on $\deg_L p_n$: the smallest class closed under composition by $X + cL$ must already contain blocks whose degree grows linearly in n . A linear bound $d_F(n) \leq an + b$ is therefore a hypothesis or a proved invariant, never part of the ring definition. The compositional inverse of $X + cL$ is the Wright-omega object whose coefficient blocks are the Lambert polynomials L_n^{Lam} ; see Theorem 14.11 and the surrounding material in Part IV.

Proposition 13.8 (A logarithmic factor forces depth two). *Let $G = XL = X \log X$. Then $t \circ G = tL^{-1}$, $L \circ G = L + L_2$, and for $F = \sum_{n \geq n_0} p_n(L) t^n \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$,*

$$F \circ (XL) = \sum_{n \geq n_0} t^n L^{-n} p_n(L + L_2) \in \mathbb{K}[L_2][L, L^{-1}]((t)). \quad (230)$$

The symbol L_2 occurs if and only if some p_n is nonconstant; a negative power of L occurs if and only if $p_n \neq 0$ for some $n > 0$. Hence

$$F \circ (XL) \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} \iff F \in \mathbb{K}[X] = \mathbb{K}[t^{-1}].$$

Proof. Apply Theorem 13.2 with $c = 1$, $\rho = 1$, $\sigma = 1$, $U = 0$; (228) becomes (230). The L_2 criterion is part (4) of that theorem. For the powers of L : the coefficient of t^n is $L^{-n} p_n(L + L_2)$, whose L -exponents run over $\{-n, \dots, -n + \deg_L p_n\}$, so a negative exponent occurs precisely when some

$n > 0$ has $p_n \neq 0$ – note that the top L_2 -degree part of $p_n(L + L_2)$ has L -degree 0, so no cancellation can remove the exponent $-n$. Both defects are excluded in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, whose L -exponents lie in $\mathbb{N}_0$ and which contains no L_2 . Membership in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ therefore forces every p_n to be a constant and $p_n = 0$ for $n > 0$, that is, $F \in \mathbb{K}[t^{-1}] = \mathbb{K}[X]$; conversely $X^k \circ (XL) = t^{-k}L^k \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ for $k \geq 0$. $\square$

Example 13.9 (Both enlargements at once). Let $F(X) = X + \log X + X^{-1}$ and $G(X) = 3X^2L^{-1}$, with $\log 3 \in \mathbb{K}$. Then

$$F \circ G = 3X^2L^{-1} + \log(3X^2L^{-1}) + \frac{L}{3X^2} = 3X^2L^{-1} + 2L - L_2 + \log 3 + \frac{1}{3}L t^2.$$

Here $\rho = 2$ and $\sigma = -1$: the leading block carries L^{-1} and the second logarithm L_2 appears *exactly*, not as a remainder. The ambient predicted by [Theorem 13.2](#) with $\Gamma_0 = \Gamma_1 = \mathbb{Z}$ is $\mathbb{K}[L_2][L, L^{-1}]((t))$, which is where the answer lies.

13.4 The two extreme scale changes: logarithm and exponential

Proposition 13.10 (Composition with log is a level shift). *Let $\Sigma(F) := F \circ \log$. Then*

$$t \circ \log = L_1^{-1}, \quad L \circ \log = L_2, \quad \Sigma\left(\sum_{n \geq n_0} p_n(L)t^n\right) = \sum_{n \geq n_0} p_n(L_2) L_1^{-n},$$

and Σ is an isomorphism of ordered rings from $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} = \mathbb{K}[L_1]((L_0^{-1}))$ onto $\mathbb{K}[L_2]((L_1^{-1}))$, carrying the valuation ν_t to the L_1^{-1} -adic valuation and the dominance order to the dominance order. It is a differential isomorphism for the derivation D_{L_1} of the target, i.e. $\Sigma \circ D_X = D_{L_1} \circ \Sigma$; it is not compatible with D_X on the target, where instead

$$D_X \circ \Sigma = t (\Sigma \circ D_X). \quad (231)$$

In particular $\mathbb{K}[L_2]((L_1^{-1}))$ is not stable under D_X .

Proof. The generator images are (227) with $G = \log X$. Substituting them into $F = \sum p_n(L)t^n$ gives the displayed formula, which is a relabelling of indeterminates: it sends the topological generators $t^{\pm 1}, L_1$ of the source bijectively to the topological generators $L_1^{\mp 1}, L_2$ of the target, preserving sums, products, limits, the lower-bounded integer support, and the lexicographic dominance order. Hence Σ is an isomorphism of ordered rings, and $\Sigma \circ D_X = D_{L_1} \circ \Sigma$ because both sides are continuous $\mathbb{K}$ -linear maps satisfying the same Leibniz rule and agreeing on t and L_1 . Since $D_X = \frac{dL_1}{dX} D_{L_1} = t D_{L_1}$ on the target, formula (231) follows. Finally $D_X(L_1^{-1}) = -tL_1^{-2}$ involves t , which does not occur in $\mathbb{K}[L_2]((L_1^{-1}))$. $\square$

Proposition 13.11 (Composition with exp leaves every logarithmic class). *One has $t \circ e^X = e^{-X}$ and $L \circ e^X = X$, so that*

$$\left(\sum_{n \geq n_0} p_n(L)t^n\right) \circ e^X = \sum_{n \geq n_0} p_n(X) e^{-nX}. \quad (232)$$

The monomial e^{-X} belongs to no $\mathcal{M}_r(\Gamma)$ of [Theorem 13.1](#): for $\mathfrak{m} = \prod_{j \leq r} L_j^{\alpha_j}$ one has $\log \mathfrak{m} = \mathcal{O}_{X \rightarrow +\infty}(L_1)$, whereas $\log e^{-X} = -X$ and $X/L_1 \rightarrow +\infty$. Moreover the loss is total on the analytic side: if f is a function on some $[X_1, +\infty)$ admitting F as an asymptotic expansion in the scale $\{X^{-n}(\log X)^j\}$ with $a = \nu_t F \geq 1$ and $d = \deg_L p_a$, then

$$f(e^X) = \mathcal{O}_{X \rightarrow +\infty}(e^{-aX} X^d) = o_{X \rightarrow +\infty}(X^{-N}) \quad \text{for every } N, \quad (233)$$

so the polynomial–logarithmic expansion of $X \mapsto f(e^X)$ is identically zero although the function itself need not be.

Proof. The generator images are (227) with $G = e^X$, and (232) is their substitution. For the monomials, $\log m = \alpha_0 L_1 + \alpha_1 L_2 + \cdots + \alpha_{r-1} L_r + \alpha_r \log L_r$, each summand being $\mathcal{O}_{X \rightarrow +\infty}(L_1)$. For the last claim, the hypothesis gives $f(Y) = p_a(\log Y)Y^{-a} + \mathcal{O}_{Y \rightarrow +\infty}(Y^{-a-1}(\log Y)^{d'})$ for a suitable d' , hence $|f(Y)| \leq CY^{-a}(\log Y)^d$ for Y large; put $Y = e^X$. Since $a \geq 1$, $e^{-aX} X^d$ is smaller than every power X^{-N} . $\square$

Remark 13.12. (232) is the reason a formal exponential change of scale must be met by enlarging the *monomial* group, not the coefficient ring: the objects $e^{-nX} X^j$ are new transmonomials, and the resulting field is the logarithmic–exponential transseries field of positive exponential height [1, 13, 33]. The polynomial–logarithmic class is exactly its height-zero, depth-one corner.

13.5 The table of scale changes

Table 4: Scale changes and the resulting ambient class. Throughout $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, $U \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, $c \in \mathbb{K}^\times$ with a chosen $\log c \in \mathbb{K}$, and the regime is $X \rightarrow +\infty$ on the positive ray.

Inner map G	$t \circ G$; $L \circ G$	Smallest ambient class for $F \circ G$
$X + A$	$t(1 + tA)^{-1}$; $L + \log(1 + tA)$	$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$; a group (Theorem 12.7)
cX	$c^{-1}t$; $L + \log c$	$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ (Theorem 13.4)
$X + cL$	$t(1 + cLt)^{-1}$; $L + \log(1 + cLt)$	$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$; $\deg_L$ grows linearly (Theorem 13.6)
$cX^\rho(1 + U)$, $\rho \in \mathbb{N}_{>0}$	$c^{-1}t^\rho(1 + U)^{-1}$; $\rho L + \log c + \log(1 + U)$	$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ (Theorem 13.2(1))
$X^{r/s}$, $\gcd(r, s) = 1$, $s > 1$	$t^{r/s}$; $(r/s)L$	Puiseux-log $\mathbb{K}[L]((t^{1/s}))$; not $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$
X^ρ , $\rho > 0$ irrational	t^ρ ; ρL	Hahn $\mathcal{H}_1(\mathbb{K}, \rho\mathbb{Z} + \mathbb{Z})$; no Puiseux extension suffices
XL^σ , $\sigma \neq 0$	$tL^{-\sigma}$; $L + \sigma L_2$	depth two: $\mathcal{H}_2$; for $\sigma \in \mathbb{Z}$, $\mathbb{K}[L_2][L, L^{-1}]((t))$
$cX^\rho L^\sigma(1 + U)$, $\rho > 0$, $\sigma \neq 0$	$c^{-1}t^\rho L^{-\sigma}(1 + U)^{-1}$; $\rho L + \sigma L_2 + \log c + \log(1 + U)$	$\mathcal{H}_2(\mathbb{K}, \Gamma_0 + \Gamma_1)$ (Theorem 13.2)
$\log X$	L^{-1} ; L_2	$\mathbb{K}[L_2]((L^{-1}))$: the level shift Σ (Theorem 13.10)
e^X	e^{-X} ; X	outside every $\mathcal{H}_r$: logarithmic–exponential transseries (Theorem 13.11)
cX^ρ , $\rho \leq 0$	–	inadmissible: G does not tend to $+\infty$
cX , $c < 0$, $\mathbb{K} \subseteq \mathbb{R}$	–	inadmissible on the positive ray: $\log c \notin \mathbb{R}$; extend $\mathbb{K}$ and fix a sector
$X = 1/x$ (regime change)	$t \mapsto x$; $L \mapsto -\log x$	$\mathbb{K}[\Lambda]((x))$, $\Lambda = -\log x$: an isomorphism, not an endomorphism (Theorem 13.18)

A common trap

The last three rows are not failures of a formula but failures of a *hypothesis*. An expression may be perfectly meaningful while lying outside $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$; and conversely a formally computed composite is meaningless if the inner map does not carry the outer expansion’s regime. Closure is a statement about a chosen representation class, never about existence of the composed function. In particular the whole table is directional: that $F \circ G$ is defined and lies in a given class says nothing about $G \circ F$.

13.6 The nested logarithmic scale

Proposition 13.13 (Derivation of the logarithmic tower). *With $L_0 = X$ and $L_{j+1} = \log L_j$,*

$$D_X L_j = \frac{t}{L_1 L_2 \cdots L_{j-1}} \quad (j \geq 1), \quad (234)$$

the product being empty for $j = 1$. Put

$$\mathcal{D}_r := \partial_{L_1} + \sum_{j=2}^r \frac{1}{L_1 L_2 \cdots L_{j-1}} \partial_{L_j}, \quad \mathcal{L}_r := \mathbb{K}[L_1^{\pm 1}, \dots, L_{r-1}^{\pm 1}, L_r].$$

Then $\mathcal{D}_r(\mathcal{L}_r) \subseteq \mathcal{L}_r$, and for $n \in \mathbb{Z}, p \in \mathcal{L}_r$,

$$D_X(t^n p) = t^{n+1}(\mathcal{D}_r - n)p. \quad (235)$$

By contrast, for $r \geq 2$ the polynomial ring $\mathbb{K}[L_1, \dots, L_r]$ is not stable under $\mathcal{D}_r$: $\mathcal{D}_r L_2 = L_1^{-1}$.

Proof. (234) holds for $j = 1$ since $D_X L_1 = t$, and inductively $D_X L_{j+1} = (D_X L_j)/L_j$. Stability: ∂_{L_1} maps $\mathcal{L}_r$ into $\mathcal{L}_r$ because the ring is Laurent in L_1 ; for $2 \leq j \leq r$, ∂_{L_j} lowers the L_j -degree and the prefactor $(L_1 \cdots L_{j-1})^{-1}$ lies in $\mathcal{L}_r$ — here it matters that only $L_1, \dots, L_{r-1}$ need be inverted, since ∂_{L_r} is multiplied by $(L_1 \cdots L_{r-1})^{-1}$. Formula (235) follows from $D_X t^n = -n t^{n+1}$ and (234) by the Leibniz rule. $\square$

Theorem 13.14 (The near-identity group at logarithmic depth r). *Let $r \geq 1$ and*

$$\mathcal{N}_r := \{X + A : A \in \mathcal{L}_r[[t]]\}, \quad \mathcal{N}_1 = \mathcal{G}_{\text{poly}, \mathbb{K}}.$$

Then $\mathcal{N}_r$ is a group under composition, and every statement of Section 12 holds verbatim with $\mathbb{K}[L]$ replaced by $\mathcal{L}_r$, ∂_L by $\mathcal{D}_r$, and $\Delta_{n,k}$ by $\prod_{j=0}^{k-1} (\mathcal{D}_r - n - j)$. The corresponding statement for $\{X + A : A \in \mathbb{K}[L_1, \dots, L_r][[t]]\}$ is false for $r \geq 2$.

Proof. Fix $G = X + A$ with $A \in \mathcal{L}_r[[t]]$ and define Φ_G on the topological generators $t^{\pm 1}, L_1^{\pm 1}, \dots, L_{r-1}^{\pm 1}, L_r$ of $\mathcal{L}_r((t))$ by $\Phi_G(t) = t(1 + tA)^{-1}$, $\Phi_G(L_{j+1}) = \log(\Phi_G(L_j))$, starting from $\Phi_G(L_0) = X(1 + tA)$. Set $w_0 = tA$ and, for $j \geq 1$,

$$w_j := L_j^{-1} \log(1 + w_{j-1}). \quad (236)$$

An induction using Theorem 12.2 gives $w_j \in t\mathcal{L}_r[[t]]$ and

$$\Phi_G(L_j) = L_j(1 + w_j) \quad (0 \leq j \leq r-1), \quad \Phi_G(L_r) = L_r + \log(1 + w_{r-1}),$$

since $\log(L_j(1 + w_j)) = L_{j+1} + \log(1 + w_j) = L_{j+1}(1 + L_{j+1}^{-1} \log(1 + w_j))$. The construction inverts exactly $L_1, \dots, L_{r-1}$ and never L_r ; this is what forces the coefficient ring $\mathcal{L}_r$ and no smaller one. Extending Φ_G multiplicatively and t -adically reproduces Theorem 12.4 word for word, and (235) reproduces $D_X(\mathcal{L}_r[[t]]) \subseteq t\mathcal{L}_r[[t]]$, which is the only property of D_X used in Theorem 12.5 and Theorem 12.6.

Two points genuinely need rechecking, and both go through. First, the *gain estimate*: if $\nu_t(B - C) = m$, then comparing (236) for $G = X + B$ and $G' = X + C$ gives $\nu_t(w_0 - w'_0) \geq m + 1$ and, inductively, $\nu_t(w_j - w'_j) \geq m + 1$, because $\log(1 + w) - \log(1 + w') = \log(1 + \frac{w-w'}{1+w'})$ and L_j^{-1} is a unit of $\mathcal{L}_r$ for $j \leq r-1$. Hence every generator except X itself has Φ_G - and $\Phi_{G'}$ -images differing by an element of $t^{m+1}\mathcal{L}_r[[t]]$, and expanding $\Phi_G(a_n t^n) - \Phi_{G'}(a_n t^n)$ by $PQ - P'Q' = (P - P')Q + P'(Q - Q')$ yields $\nu_t(A \circ G - A \circ G') \geq m + 1$. Second, *associativity*: $\Phi_H(\Phi_G(L_0)) = \Phi_H(G) = G \circ H = \Phi_{G \circ H}(L_0)$ as in (221), and if the identity $\Phi_H \Phi_G(L_j) = \Phi_{G \circ H}(L_j)$ holds then applying the continuous algebra homomorphism Φ_H to $\Phi_G(L_{j+1}) = L_{j+1} + \log(1 + w_j)$ and using $\Phi_H(\log(1 + u)) = \log(1 + \Phi_H(u))$ together with $\log(uv) = \log u + \log v$ gives the identity at level $j + 1$; the case $j + 1 = 1$ is the computation displayed in the proof of Theorem 12.7(2). With these, the fixed-point construction of

inverses ([Theorem 14.11](#), whose proof uses only the gain estimate [Theorem 12.6](#) and t -adic completeness, both available verbatim in $\mathcal{L}_r[[t]]$) and the monoid argument are unchanged.

For the final assertion take $r = 2$, $G = X + 1 \in \mathcal{N}_2$ and $F = L_2 \in \mathbb{K}[L_1, L_2]$. Then

$$F \circ G = \log \log(X + 1) = L_2 + \log\left(1 + L_1^{-1} \log(1 + t)\right) = L_2 + L_1^{-1}t + \mathcal{O}_t^{\text{form}}(t^2),$$

whose block of outer order 1 is $L_1^{-1} \notin \mathbb{K}[L_1, L_2]$. Hence $\{X + A : A \in \mathbb{K}[L_1, L_2][[t]]\}$ is not closed under composition. $\square$

Proposition 13.15 (The unit criterion changes shape with depth). $\mathcal{L}_r^\times = \{c L_1^{a_1} \cdots L_{r-1}^{a_{r-1}} : c \in \mathbb{K}^\times, a_i \in \mathbb{Z}\}$. Consequently a nonzero $F = \sum_{n \geq n_0} p_n t^n \in \mathcal{L}_r((t))$ is a unit if and only if its leading block p_{n_0} is a nonzero scalar times a Laurent monomial in $L_1, \dots, L_{r-1}$. For $r = 1$ this is the classical criterion “ $p_{n_0} \in \mathbb{K}^\times$ ”; for $r \geq 2$ strictly more elements are invertible, for instance $L_1^{-1} + \mathcal{O}_t^{\text{form}}(t)$.

Proof. Write $\mathcal{L}_r = R[L_r]$ with $R = \mathbb{K}[L_1^{\pm 1}, \dots, L_{r-1}^{\pm 1}]$. Since R is an integral domain, $(R[L_r])^\times = R^\times$; and by induction on the number of variables, the units of a Laurent polynomial ring over a domain are the units of the base times monomials. The criterion for $\mathcal{L}_r((t))$ is the standard one for a Laurent series ring over a commutative ring: if p_{n_0} is a unit, $F = t^{n_0} p_{n_0} (1 + v)$ with $v \in t \mathcal{L}_r[[t]]$, and $(1 + v)^{-1}$ is given by the geometric series; conversely $FF^{-1} = 1$ forces $p_{n_0} [t^{-n_0}] F^{-1} = 1$. $\square$

13.7 Lifting a one-logarithm result, and where the lift fails

Proposition 13.16 (Transport by the level shift). Let $\mathcal{S}$ be any statement about the structure $(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}; t, L, \nu_t, \preceq, D_X)$ expressed purely in that language. Then $\mathcal{S}$ holds if and only if the corresponding statement holds for $(\mathbb{K}[L_2]((L_1^{-1})); L_1^{-1}, L_2, \nu_{L_1^{-1}}, \preceq, D_{L_1})$. In particular

$$\{L_1 + B : B \in \mathbb{K}[L_2][[L_1^{-1}]]\}$$

is a group under composition in the variable L_1 , with all the structure of [Theorem 12.7](#), [Theorem 12.12](#) and [Theorem 12.16](#).

Proof. [Theorem 13.10](#) exhibits Σ as an isomorphism of the two structures. $\square$

Remark 13.17 (The correct fast variable). This is the precise content of the practical rule that a nested expansion should be read in the right variable. A series $\sum_{n \geq 0} p_n (\log \log X) (\log X)^{-n}$, which is a depth-two object in X , is a *one-logarithm* object in the fast variable $T = L_1 = \log X$, with $\Lambda = L_2 = \log T$. Choosing the wrong fast variable makes such a problem look as though it fell outside the theory.

A common trap

Three things do *not* lift.

- *Statements about D_X .* By (231), Σ intertwines D_X with D_{L_1} , not with D_X ; and $\mathbb{K}[L_2]((L_1^{-1}))$ is not even stable under D_X . A result whose statement mentions the derivation of the *original* variable must be re-derived, not transported.
- *Genuinely mixed depth.* Σ reaches only the shifted copy. A series such as $\sum_n p_n(L_1, L_2) t^n$ with genuinely bivariate blocks lies in neither $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ nor $\Sigma(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}})$, and no change of fast variable makes it single-logarithmic; one must work in $\mathcal{L}_r((t))$ and accept [Theorem 13.13](#).
- *The unit criterion.* Its shape changes at depth ≥ 2 ([Theorem 13.15](#)), so any argument that used “leading block a nonzero scalar” must be restated as “leading block a unit of the coefficient ring”.

13.8 From infinity to a finite point

Proposition 13.18 (The infinity–zero dictionary). *Put $x = X^{-1}$ and*

$$\Lambda := \log \frac{1}{x} = -\log x = \log X = L.$$

Then $F \mapsto F(1/x)$ is an isomorphism of ordered rings

$$\iota : \mathbb{K}[L]((t)) \longrightarrow \mathbb{K}[\Lambda]((x)), \quad t \longmapsto x, \quad L \longmapsto \Lambda,$$

matching the regime $X \rightarrow +\infty$ with $x \rightarrow 0^+$ and the dominance order with the order of decreasing size as $x \rightarrow 0^+$. On the differential side

$$\frac{d}{dx} = -X^2 D_X, \quad \frac{d}{dx} = \partial_x - \frac{1}{x} \partial_\Lambda, \quad \frac{d}{dx} (x^n P(\Lambda)) = x^{n-1} (nP - P'), \quad (237)$$

the exact mirror of (213). Under ι the group $\mathcal{G}_{\text{poly}, \mathbb{K}}$ corresponds to the group of germs

$$\left\{ g(x) = \frac{x}{1 + x a(x)} : a \in \mathbb{K}[\Lambda][[x]] \right\}$$

with the composition law of $\mathcal{G}_{\text{poly}, \mathbb{K}}$, the isomorphism being $G \mapsto j \circ G \circ j$ with $j(x) = 1/x$.

Proof. The map ι is the relabelling $t \mapsto x$, $L \mapsto \Lambda$; it is a ring isomorphism, and it preserves the order because $t^n L^j \succ t^m L^k$ and $x^n \Lambda^j \succ x^m \Lambda^k$ are governed by the same inequalities on (n, j) and (m, k) . From $x = X^{-1}$, $\frac{dx}{dX} = -X^{-2}$, so $D_X = -x^2 \frac{d}{dx}$, which is the first identity; the second is the chain rule with $\partial_x \Lambda = -1/x$; the third follows: $\frac{d}{dx} (x^n P) = nx^{n-1} P - x^n \frac{1}{x} P'$. Finally j is an involution, so $G \mapsto j \circ G \circ j$ is a group isomorphism, and for $G = X + A$, $j \circ G \circ j = (x^{-1} + A(1/x))^{-1} = x(1 + x \iota(A))^{-1}$. $\square$

A common trap

The dictionary is $L = -\log x$, never $L = \log x$. Substituting the wrong sign reverses ∂_L and turns the block operator $\Delta_{n,k} = \prod_{j=0}^{k-1} (\partial_L - n - j)$ into a different operator; all reversion coefficients then come out with wrong signs while remaining formally consistent, which makes the error hard to detect.

Remark 13.19 (What is deferred). **Theorem 13.18** is only the integer-exponent dictionary. Local dynamics at a finite point uses *Dulac* classes: real or well-ordered exponent sets x^α , the generator $\ell = -1/\log x$ in place of Λ , and their iterated versions, for which composition and normal forms require the machinery of [23, 24, 27, 26]. That development, including the change between Λ and ℓ and the consequences of admitting real exponents, is carried out systematically in [Section 24](#). Note also that the other natural passage to a finite point, $X = -\log x$, is the exponential change of scale of [Theorem 13.11](#) and leaves the class.

Section summary

A substitution is decided by the two generator images G^{-1} and $\log G$, and by nothing else. For $G = cX^\rho L^\sigma(1 + U)$ with $\rho > 0$: integral ρ and $\sigma = 0$ keep the composite in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$; rational $\rho = r/s$ forces exactly the Puiseux-log extension $\mathbb{K}[L]((t^{1/s}))$; irrational ρ forces a Hahn exponent group; $\sigma \neq 0$ forces the second logarithm L_2 precisely when some coefficient block is nonconstant, together with L -exponents outside $\mathbb{N}_0$. Composition with $\log$ is an isomorphism onto the level-shifted copy – the formal statement of the “correct fast variable” rule – while composition with $\exp$ leaves every logarithmic class and annihilates the polynomial–logarithmic expansion entirely. At depth r the near-identity group survives, but only over the coefficient ring $\mathbb{K}[L_1^{\pm 1}, \dots, L_{r-1}^{\pm 1}, L_r]$

that the derivation and the generator substitutions force.

Part IV

Series reversal at infinity

14 Existence and uniqueness of the near-identity inverse

14.1 Conventions, and what is imported from the composition part

Throughout this part $\mathbb{K}$ is a field of characteristic zero, the large variable is X with

$$t := X^{-1}, \quad L := \log X,$$

and the default analytic regime, whenever one is invoked at all, is $X \rightarrow +\infty$ on the positive real ray. We work in

$$\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} = \mathbb{K}[L][[t]], \quad \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} = \mathbb{K}[L]((t)), \quad \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} = \mathbb{K}(L)((t)),$$

with the outer valuation ν_t and the exclusive truncation $\text{Trunc}_{<N} F = \sum_{n < N} p_n(L)t^n$. The near-identity group is

$$\mathcal{G}_{\text{poly},\mathbb{K}} := \{X + A : A \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}\} = \{F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} : \nu_t(F) \geq -1 \text{ and } [t^{-1}]F = 1\}. \quad (238)$$

Definition 14.1 (Truncation and blocks of a near-identity map). For $G = X + B \in \mathcal{G}_{\text{poly},\mathbb{K}}$ with $B = \sum_{n \geq 0} b_n(L)t^n$ we call b_n the n -th inverse block of G and set

$$\text{Trunc}_{<N} G := X + \text{Trunc}_{<N} B = X + \sum_{n < N} b_n(L)t^n.$$

For $F \in \mathcal{G}_{\text{poly},\mathbb{K}}$, the phrase “ G is correct through outer block N ” means

$$\text{Trunc}_{<N+1} G = \text{Trunc}_{<N+1} F^{(-1)\circ},$$

that is, agreement in the $N + 1$ blocks $b_0, \dots, b_N$.

Remark 14.2 (Inclusive versus exclusive cutoffs). Report 6 writes its truncations inclusively as $[F]_{\leq N}$. The translation is $[F]_{\leq N} = \text{Trunc}_{<N+1} F$, so every cutoff index borrowed from that source has been raised by one below. The place where this matters most is the finite certificate ([Theorem 14.16](#)): correctness *through* block N is the exclusive cutoff $< N + 1$, and the certifying residual is a $\mathcal{O}_t^{\text{form}}(t^{N+1})$, not a $\mathcal{O}_t^{\text{form}}(t^N)$.

Remark 14.3 (Standing facts imported from the composition part). Four facts about $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, D_X and $\circ$ are used constantly. They are proved elsewhere in this volume; we record them in the exact form needed here and prove nothing about them except the completeness statement, whose precise shape is used below.

(I1) *Block derivative* ([Theorem 9.7](#)). For $n \in \mathbb{Z}$ and $p \in \mathbb{K}[L]$,

$$D_X(t^n p(L)) = t^{n+1}(\partial_L - n)p(L), \quad D_X^k(t^n p(L)) = t^{n+k} \Delta_{n,k} p(L), \quad (239)$$

where $\Delta_{n,k} = \prod_{j=0}^{k-1} (\partial_L - n - j)$ and $\Delta_{n,0} = 1$ is the empty product.

(I2) *Exact formal Taylor formula* ([Theorem 11.6](#)). For $S \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ and $P \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ the family $(P^k D_X^k S / k!)_{k \geq 0}$ is formally summable and

$$S \circ (X + P) = \sum_{k \geq 0} \frac{P^k}{k!} D_X^k S. \quad (240)$$

This is an identity of formal series; it carries no analytic content.

(I3) *Closure, associativity and chain rule* (Theorem 12.7 (1)–(3), Theorem 10.31). For $F = X + A$ and $G = X + B$ in $\mathcal{G}_{\text{poly}, \mathbb{K}}$,

$$F \circ G = X + B + A \circ G \in \mathcal{G}_{\text{poly}, \mathbb{K}}; \quad (241)$$

$(S \circ H_1) \circ H_2 = S \circ (H_1 \circ H_2)$ whenever all three displayed composites exist in the ambient class; this covers near-identity inner arguments and monomial inner arguments cX for which a value of $\log c$ lies in $\mathbb{K}$. The series X is a two-sided identity for $\circ$, and

$$D_X(S \circ H) = ((D_X S) \circ H) \cdot D_X H. \quad (242)$$

(I4) *Completeness* (Theorem 5.18). $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is complete for the t -adic filtration: if (F_j) is Cauchy then for each n the block $[t^n]F_j \in \mathbb{K}[L]$ is eventually constant, say p_n ; the series $F = \sum_{n \geq 0} p_n t^n$ lies in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ – no uniform bound on $\deg_L p_n$ is required – and $F_j \rightarrow F$. On $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ the same argument needs the extra hypothesis that the $\nu_t(F_j)$ are bounded below, without which a Cauchy sequence need not define a Laurent series with a finite principal part.

Lemma 14.4 (Valuation gain of the derivation). *For every $S \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and every $k \in \mathbb{N}_0$, $\nu_t(D_X^k S) \geq \nu_t(S) + k$.*

Proof. By (239), D_X^k sends the block $t^n p$ to $t^{n+k} \Delta_{n,k} p$, hence maps $t^m \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ into $t^{m+k} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ for every $m \in \mathbb{Z}$. Apply this with $m = \nu_t(S)$. Equality may fail, since $\Delta_{n,k} p$ can vanish; for instance $D_X(t^0 \cdot 1) = 0$. $\square$

Lemma 14.5 (Right composition with a near-identity preserves leading data). *Let $S \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, $P \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and $m = \nu_t(S)$. Then*

$$\nu_t(S \circ (X + P)) = m, \quad [t^m](S \circ (X + P)) = [t^m]S.$$

In particular $S \circ (X + P) = 0$ if and only if $S = 0$, and right composition by an element of $\mathcal{G}_{\text{poly}, \mathbb{K}}$ carries units of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ to units of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.

Proof. In (240) the term $k = 0$ is S . For $k \geq 1$, $\nu_t(P^k D_X^k S / k!) \geq k \nu_t(P) + m + k \geq m + k > m$, using Theorem 14.4 and $\nu_t(P) \geq 0$. So no earlier block is created and the block at level m is unchanged. $\square$

The next lemma re-expands a composition around a shifted base point without inverting anything. It may therefore be used inside the existence proof, and it is the engine of every estimate in this part.

Lemma 14.6 (Two-point Taylor expansion). *Let $S \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and $P, Q \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Then the family $(Q^j (D_X^j S) \circ (X + P) / j!)_{j \geq 0}$ is formally summable and*

$$S \circ (X + P + Q) = \sum_{j \geq 0} \frac{Q^j}{j!} (D_X^j S) \circ (X + P). \quad (243)$$

Proof. Summability: by Theorems 14.4 and 14.5, $\nu_t(Q^j (D_X^j S) \circ (X + P) / j!) \geq j \nu_t(Q) + \nu_t(S) + j \rightarrow \infty$.

Apply (240) with $P + Q \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ in place of P :

$$S \circ (X + P + Q) = \sum_{k \geq 0} \frac{(P + Q)^k}{k!} D_X^k S = \sum_{k \geq 0} \sum_{i+j=k} \frac{P^i}{i!} \frac{Q^j}{j!} D_X^{i+j} S.$$

Each term has $\nu_t \geq \nu_t(S) + i + j$, so the double family is summable and may be summed in either order. Grouping by j and applying (240) once more, now to $D_X^j S$,

$$\sum_{j \geq 0} \frac{Q^j}{j!} \sum_{i \geq 0} \frac{P^i}{i!} D_X^i (D_X^j S) = \sum_{j \geq 0} \frac{Q^j}{j!} (D_X^j S) \circ (X + P),$$

which is (243). $\square$

14.2 The inverse equation as a fixed point

Proposition 14.7 (The reversion equation). *Let $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ and $G = X + B \in \mathcal{G}_{\text{poly}, \mathbb{K}}$. Then $F \circ G = X$ if and only if $B = -A \circ (X + B)$.*

Proof. By (241), $F \circ G = X + B + A \circ G$, which equals X exactly when $B + A \circ (X + B) = 0$. $\square$

Thus B is a fixed point of

$$\Phi: \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} \rightarrow \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}, \quad \Phi(B) := -A \circ (X + B). \quad (244)$$

The map Φ is neither linear nor defined on all of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$: its argument must be an admissible near-identity displacement.

Remark 14.8 (Variable names). Reports 3 and 6 write the reversion equation in a second letter, $B(Y) = -A(Y + B(Y))$, to separate visually the argument of the inverse from the argument of F . This is a typographic device only: X is a free generator of the formal algebra and the renaming is the identity map. We use X throughout, in agreement with the notation catalogue.

Remark 14.9 (Filtered Lipschitz gain: the fifth standing import). Let $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $\alpha := \nu_t(A)$, and let $B, C \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Then

$$\nu_t(A \circ (X + B) - A \circ (X + C)) \geq \alpha + \nu_t(B - C) + 1, \quad (245)$$

which is (217) of [Theorem 12.6](#), proved there. In particular, for the map Φ of (244), $\nu_t(\Phi(B) - \Phi(C)) \geq \alpha + \nu_t(B - C) + 1$. The reader who prefers an argument internal to this part may instead put $S = A$, $P = C$, $Q = B - C$ in [Theorem 14.6](#) and bound the j -th term by $\alpha + j(\nu_t(Q) + 1)$ using [Theorems 14.4](#) and [14.5](#); the minimum over $j \geq 1$ is at $j = 1$.

Remark 14.10 (What “a gain of one block” means, precisely). All six reports describe (245) as a contraction. Three points must be kept apart.

1. *It is a statement about ν_t alone.* If B and C agree in the blocks $0, \dots, r - 1$, then $\Phi(B)$ and $\Phi(C)$ agree in the blocks $0, \dots, r$: one further block of the output is pinned down. Nothing whatever is asserted about the size, the coefficients, or the logarithmic degree of the blocks that still differ; $\deg_L$ is not contracted and in general grows.
2. *It is a genuine metric contraction for the standard t -adic absolute value.* Setting $|F|_t = 2^{-\nu_t(F)}$ and $|0|_t = 0$, inequality (245) says exactly that Φ is Lipschitz with constant $2^{-1-\alpha} \leq \frac{1}{2}$. The space $(\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}, |\cdot|_t)$ is complete by (I4), so the Banach fixed-point theorem applies verbatim. We nevertheless give the coefficient-stabilization proof below, because that is what an implementation performs and because it yields the sharp iteration rate.
3. *The gain is at least one, and in general exactly one.* Take $A = L$, $B = 0$ and $C = ct^r$ with $c \in \mathbb{K}^\times$, $r \geq 0$. Then $A \circ (X + C) - A \circ (X + B) = \log(1 + ct^{r+1}) = ct^{r+1} + \mathcal{O}_t^{\text{form}}(t^{r+2})$, and (245) is an equality. The summand $\alpha = \nu_t(A)$ in (245) does not appear in any of the six sources; it is what makes the sharp Newton estimate of [Theorem 15.11](#) come out.

14.3 Existence, uniqueness, and the two-sided property

Theorem 14.11 (Near-identity inverse theorem). *Let $\mathbb{K}$ have characteristic zero and let $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ with $\alpha = \nu_t(A)$. Then:*

1. *the Picard iterates $B_0 = 0$, $B_{j+1} = \Phi(B_j) = -A \circ (X + B_j)$ satisfy*

$$\nu_t(B_{j+1} - B_j) \geq (j + 1)\alpha + j \quad (j \geq 0), \quad (246)$$

and converge t -adically to some $B \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with

$$\nu_t(B - B_j) \geq (j + 1)\alpha + j; \quad (247)$$

2. B is the unique element of $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ with $B = -A \circ (X + B)$, and $G := X + B$ is the unique element of $\mathcal{G}_{\text{poly},\mathbb{K}}$ with $F \circ G = X$;
3. $G \circ F = X$ as well, and G is also the unique element of $\mathcal{G}_{\text{poly},\mathbb{K}}$ with that property; we write $G = F^{(-1)\circ}$;
4. $\nu_t(B) = \alpha$ and $[t^\alpha]B = -[t^\alpha]A$.

Proof. (1) For $j = 0$, $B_1 - B_0 = -A \circ X = -A$ has $\nu_t \geq \alpha = (0 + 1)\alpha + 0$. If (246) holds for $j - 1$, then by Theorem 14.9

$$\nu_t(B_{j+1} - B_j) = \nu_t(\Phi(B_j) - \Phi(B_{j-1})) \geq \alpha + (j\alpha + j - 1) + 1 = (j + 1)\alpha + j.$$

The bound tends to $+\infty$, so (B_j) is Cauchy and converges to some $B \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ by (I4); moreover $\nu_t(B - B_j) \geq \min_{i \geq j} \nu_t(B_{i+1} - B_i) \geq (j + 1)\alpha + j$, which is (247).

That B solves $B = \Phi(B)$ needs no separate continuity argument. For every j , $B - \Phi(B) = (B - B_{j+1}) + (\Phi(B_j) - \Phi(B))$. The first summand has $\nu_t \geq (j + 2)\alpha + j + 1$ by (247); the second has $\nu_t \geq \alpha + \nu_t(B_j - B) + 1 \geq (j + 2)\alpha + j + 1$ by Theorem 14.9. Hence $\nu_t(B - \Phi(B)) \geq (j + 2)\alpha + j + 1$ for every j , which exceeds every fixed integer as $j \rightarrow \infty$; so $B = \Phi(B)$.

(2) If B and $\tilde{B}$ are fixed points with $r = \nu_t(B - \tilde{B}) < \infty$, then Theorem 14.9 gives $r = \nu_t(\Phi(B) - \Phi(\tilde{B})) \geq \alpha + r + 1 > r$, a contradiction; so $B = \tilde{B}$. By Theorem 14.7, $G = X + B$ is then the unique right inverse of F inside $\mathcal{G}_{\text{poly},\mathbb{K}}$.

(3) Apply (1)–(2) to $G \in \mathcal{G}_{\text{poly},\mathbb{K}}$: there is $H \in \mathcal{G}_{\text{poly},\mathbb{K}}$ with $G \circ H = X$. Using (I3),

$$F = F \circ X = F \circ (G \circ H) = (F \circ G) \circ H = X \circ H = H,$$

so $G \circ F = G \circ H = X$. If $G' \in \mathcal{G}_{\text{poly},\mathbb{K}}$ also satisfies $G' \circ F = X$, then $G' = G' \circ (F \circ G) = (G' \circ F) \circ G = X \circ G = G$.

(4) From $B = -A \circ (X + B)$ and Theorem 14.5, $\nu_t(B) = \nu_t(A) = \alpha$ and $[t^\alpha]B = -[t^\alpha]A$. $\square$

Together with closure and associativity, which are (I3), part (3) of Theorem 14.11 supplies the missing clause of Theorem 12.7: $(\mathcal{G}_{\text{poly},\mathbb{K}}, \circ)$ is a group with identity X , and $F \mapsto F^{(-1)\circ}$ is its inversion map. This is the only place in the volume where the inverse is constructed.

Corollary 14.12 (First-order form of the inverse). *For $F = X + A \in \mathcal{G}_{\text{poly},\mathbb{K}}$ with $\alpha = \nu_t(A)$, $F^{(-1)\circ} = X - A + \mathcal{O}_t^{\text{form}}(t^{2\alpha+1})$.*

Proof. Take $j = 1$ in (247): $B_1 = -A$ and $\nu_t(B - B_1) \geq 2\alpha + 1$. $\square$

Remark 14.13 (A second proof that a one-sided inverse is two-sided). Report 1 argues differently, and the argument is worth keeping because it localizes the only arithmetic input. Let G be the right inverse and put $J = G \circ F = X + R \in \mathcal{G}_{\text{poly},\mathbb{K}}$. By (I3), $J \circ J = G \circ (F \circ G) \circ F = G \circ F = J$, so J is a composition idempotent. On the other hand, by (240),

$$J \circ J = (X + R) \circ (X + R) = X + R + R \circ (X + R) = X + 2R + \sum_{k \geq 1} \frac{R^k}{k!} D_X^k R,$$

and the last sum has $\nu_t \geq 2\nu_t(R) + 1$ by Theorem 14.4. If $R \neq 0$ with $\rho = \nu_t(R)$, comparing the t^ρ blocks in $J \circ J = J$ gives $2[t^\rho]R = [t^\rho]R$, hence $[t^\rho]R = 0$, a contradiction. So $R = 0$. This proof does not construct a right inverse for G , but it does compare two different integer multiples of one block; the monoid argument in Theorem 14.11(3) uses nothing beyond associativity.

Section summary

For $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ the equation $B = -A \circ (X + B)$ has exactly one solution, because $D_X A$ has strictly positive outer valuation: changing the guess in block r changes $-A \circ (X + \cdot)$ only from block $r + 1 + \nu_t(A)$ onward. Iteration from $B_0 = 0$ therefore freezes at least one further block per step, and t -adic completeness turns that gain into existence. Uniqueness, the group property of $\mathcal{G}_{\text{poly}, \mathbb{K}}$, and two-sidedness of the inverse all follow.

14.4 The reversal residual and the finite certificate

Definition 14.14 (Reversal residual). For $F, H \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ define

$$\text{Err}_{\text{rev}}(F, H) := F \circ H - X \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}. \quad (248)$$

We call $\text{Err}_{\text{rev}}(F, H)$ the *right residual* of the candidate inverse H for the map F , and $H \circ F - X$ the *left residual*.

A common trap

The residual has *two* arguments and is not determined by the candidate H alone. Several source reports abbreviate it by a one-argument symbol in which the ambient map F is pushed into a subscript or into the surrounding prose. In a document that reverses more than one map — for instance $X + \log X$ and its perturbation $X + \log X + c/X$, both of which occur below — such a symbol is ambiguous, and the ambiguity is invisible in the typeset formula. Always print both arguments, as in $\text{Err}_{\text{rev}}(F, H)$. The same warning applies to the left residual, which is a different function of the same pair.

The next proposition is what makes residuals usable as certificates. It is stated in none of the six sources, although several of them silently pass from “the residual has order m ” to “ m blocks are correct”.

Proposition 14.15 (The residual measures the error exactly, on both sides). *Let $F \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, $G = F^{(-1)\circ}$, $H \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, and $E = H - G \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Then*

$$\nu_t(\text{Err}_{\text{rev}}(F, H)) = \nu_t(H \circ F - X) = \nu_t(E), \quad (249)$$

and, if $E \neq 0$ with $e = \nu_t(E)$, all three series have the same leading coefficient block:

$$[t^e] \text{Err}_{\text{rev}}(F, H) = [t^e](H \circ F - X) = [t^e]E.$$

Proof. Right residual. Write $F = X + A$, $G = X + B$, $H = X + B + E$, and apply [Theorem 14.6](#) with $S = F$, $P = B$, $Q = E$:

$$F \circ H = F \circ G + ((D_X F) \circ G)E + \sum_{j \geq 2} \frac{E^j}{j!} (D_X^j F) \circ G.$$

Here $F \circ G = X$. Since $D_X F = 1 + D_X A$ with $\nu_t(D_X A) \geq 1$, [Theorem 14.5](#) gives, for $J := (D_X F) \circ G$, that $\nu_t(J) = 0$ and $[t^0]J = 1$. For $j \geq 2$, $D_X^j F = D_X^j A$ has $\nu_t \geq j$, so the j -th tail term has $\nu_t \geq je + j \geq 2e + 2$. Therefore

$$\text{Err}_{\text{rev}}(F, H) = JE + \mathcal{O}_t^{\text{form}}(t^{2e+2}), \quad \nu_t(J) = 0, \quad [t^0]J = 1. \quad (250)$$

Since $e \geq 0$ we have $2e + 2 > e$, while $\nu_t(JE) = e$ and $[t^e](JE) = [t^0]J \cdot [t^e]E = [t^e]E$.

Left residual. Right composition by F is substitution, hence a ring homomorphism, so, using $G \circ F = X$,

$$H \circ F - X = H \circ F - G \circ F = (H - G) \circ F = E \circ F,$$

and [Theorem 14.5](#) gives $\nu_t(E \circ F) = e$ with $[t^e](E \circ F) = [t^e]E$. □

Corollary 14.16 (Finite certificate). *Let $F \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, $N \geq 0$, and $G_N = X + \sum_{j=0}^N b_j(L)t^j$ with $b_j \in \mathbb{K}[L]$. The following are equivalent.*

1. $G_N = \text{Trunc}_{<N+1} F^{(-1)\circ}$, that is, G_N is correct through outer block N .
2. $\text{Err}_{\text{rev}}(F, G_N) = F \circ G_N - X = \mathcal{O}_t^{\text{form}}(t^{N+1})$ (right certificate).
3. $G_N \circ F - X = \mathcal{O}_t^{\text{form}}(t^{N+1})$ (left certificate).

Proof. By [Theorem 14.15](#), each of (2) and (3) is equivalent to $\nu_t(G_N - F^{(-1)\circ}) \geq N + 1$. Since G_N has no block beyond t^N , that is exactly $\text{Trunc}_{<N+1} G_N = \text{Trunc}_{<N+1} F^{(-1)\circ}$, i.e. (1). $\square$

Proposition 14.17 (The residual reads off the next inverse block). *Let $F \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, write $F^{(-1)\circ} = X + \sum_{n \geq 0} b_n(L)t^n$, and let $G_{N-1} = \text{Trunc}_{<N} F^{(-1)\circ}$ for $N \geq 0$. Then*

$$\text{Err}_{\text{rev}}(F, G_{N-1}) = -b_N(L)t^N + \mathcal{O}_t^{\text{form}}(t^{N+1}). \quad (251)$$

Proof. Here $E = G_{N-1} - F^{(-1)\circ} = -\sum_{n \geq N} b_n t^n$, so $\nu_t(E) \geq N$ and $[t^N]E = -b_N$. By (250) the residual has $\nu_t \geq N$ and $[t^N] \text{Err}_{\text{rev}}(F, G_{N-1}) = [t^N]E = -b_N$. $\square$

Remark 14.18 (Why both certificates are computed, and what they really test). [Theorem 14.15](#) gives more than the equivalence of the two certificates: for an exact $G_N \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ the right and left residuals have the same valuation *and* the same leading coefficient block. Report 1’s claim that at finite order the two “need not have identical first omitted blocks” is therefore incorrect as stated. What is true, and is the genuine reason to compute both, is this.

- The two are computed along different code paths: $F \circ G_N$ substitutes the candidate into the map, $G_N \circ F$ substitutes the map into the candidate. A transposed argument order, a wrong update of $L \circ G = L + \log(1 + t(G - X))$, or a routine that substitutes only the t -part of the inner series will usually break exactly one of them.
- As soon as the inputs are themselves truncated – F known only to some block, or G_N stored with different tails in the two calls – the two computed residuals are the residuals of *different* exact elements, and then their leading blocks may indeed differ. That is a statement about inconsistent precision management, not about the algebra of $\mathcal{G}_{\text{poly}, \mathbb{K}}$, and it is quantified by [Theorem 15.14](#) and [Theorem 15.15](#).

Proposition 14.19 (The derivative certificate is one level weaker). *Let $F, H \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, $G = F^{(-1)\circ}$, $E = H - G$, $e = \nu_t(E)$. Then*

$$\nu_t(D_X H - ((D_X F) \circ H)^{-1}) \geq e + 1, \quad (252)$$

with equality whenever $e \geq 1$. For $e = 0$ the bound need not be attained: if $E = c \in \mathbb{K}^\times$ then $D_X E = 0$ and the left-hand side has valuation at least 2.

Proof. Differentiating $F \circ G = X$ by the chain rule (242) gives $((D_X F) \circ G) D_X G = 1$, so $D_X G = ((D_X F) \circ G)^{-1}$. Hence

$$D_X H - ((D_X F) \circ H)^{-1} = D_X E + \left(((D_X F) \circ G)^{-1} - ((D_X F) \circ H)^{-1} \right).$$

By [Theorem 14.6](#) applied to $S = D_X F$, $(D_X F) \circ H - (D_X F) \circ G = \sum_{j \geq 1} \frac{E^j}{j!} (D_X^{j+1} A) \circ G$, of valuation $\geq e + 2$ because $\nu_t(D_X^{j+1} A) \geq j + 1 \geq 2$. Both reciprocals have valuation 0, so the bracket has valuation $\geq e + 2$. Finally $\nu_t(D_X E) \geq e + 1$, with equality when $e \geq 1$: if $[t^e]E = c \neq 0$ then $[t^{e+1}]D_X E = (\partial_L - e)c$, and $(\partial_L - e)c = 0$ forces $c' = ec$, impossible for a nonzero polynomial when $e \neq 0$. For $e = 0$ the operator ∂_L annihilates constants. $\square$

Remark 14.20. Report 1 recommends the derivative identity $D_X G = ((D_X F) \circ G)^{-1}$ as a check that “often detects a wrong block one order earlier”. [Theorem 14.19](#) shows the opposite: the derivative residual first becomes nonzero at least one level *later* than $\text{Err}_{\text{rev}}(F, H)$, and it under-reports by two levels an error whose leading block is a nonzero constant at level 0. It remains a useful *independent* check — it exercises D_X and the reciprocal rather than the composition engine — but as a detector of the first wrong block it is strictly weaker.

14.5 What the theorem does not say

A common trap

[Theorem 14.11](#) is a theorem about formal series. It asserts nothing about the existence of an analytic inverse branch, about convergence of $\sum_n b_n(L)t^n$ for any value of X , or about any constant uniform in L . A formal certificate $\text{Err}_{\text{rev}}(F, G_N) = \mathcal{O}_t^{\text{form}}(t^{N+1})$ means coefficient agreement below outer order $N + 1$ and nothing more; in particular it does not imply $f(g_N(X)) - X = \mathcal{O}_{X \rightarrow +\infty}(X^{-N-1})$ for any realization f, g_N , since the discarded blocks carry unbounded powers of $\log X$.

Proposition 14.21 (Analytic residual transfer). *Let $I \subseteq \mathbb{R}$ be an interval, let $f: I \rightarrow \mathbb{R}$ be continuously differentiable and injective, let $y \in f(I)$, and let $x_* = f^{(-1)\circ}(y)$. Let $x_N \in I$ and suppose $|f'(x)| \geq m > 0$ for all x between x_N and x_* . Then*

$$|x_N - x_*| \leq \frac{|f(x_N) - y|}{m}.$$

Proof. By the mean value theorem there is ξ between x_N and x_* with $f(x_N) - y = f(x_N) - f(x_*) = f'(\xi)(x_N - x_*)$, whence $|f(x_N) - y| \geq m |x_N - x_*|$. $\square$

Remark 14.22. Both hypotheses are essential and neither is supplied by the formal theory. For $f(x) = x + \log x$ on $(1, \infty)$ one may take $m = 1$, because $f'(x) = 1 + 1/x > 1$, so there a numerical residual does control the inverse error — but that is a statement about one realization on one ray, proved analytically, and it is not a corollary of [Theorem 14.16](#). Transferring a *formal* residual to an asymptotic estimate additionally requires a realization theorem for the class, treated separately in this volume.

15 Triangular reversal, Newton iteration, and precision doubling

15.1 The triangular correction

Lemma 15.1 (Block perturbation). *Let $F \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, $G \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, $N \in \mathbb{N}_0$, $c \in \mathbb{K}[L]$. Then*

$$F \circ (G + c(L)t^N) = F \circ G + c(L)t^N + \mathcal{O}_t^{\text{form}}(t^{N+1}). \quad (253)$$

Proof. Apply [Theorem 14.6](#) with $S = F$, $P = G - X$, $Q = \delta := ct^N$:

$$F \circ (G + \delta) - F \circ G = ((D_X F) \circ G)\delta + \sum_{j \geq 2} \frac{\delta^j}{j!} (D_X^j F) \circ G.$$

By [Theorem 14.5](#), $(D_X F) \circ G = 1 + \mathcal{O}_t^{\text{form}}(t)$, so the linear term is $ct^N + \mathcal{O}_t^{\text{form}}(t^{N+1})$; and for $j \geq 2$ the j -th term has $\nu_t \geq jN + j \geq 2N + 2 > N$ since $N \geq 0$. $\square$

Theorem 15.2 (Triangular reversal). *Let $F \in \mathcal{G}_{\text{poly}, \mathbb{K}}$. Put $G_{-1} = X$ and, for $n = 0, 1, 2, \dots$,*

$$r_n(L) := [t^n] \text{Err}_{\text{rev}}(F, G_{n-1}), \quad G_n := G_{n-1} - r_n(L)t^n. \quad (254)$$

Then for every $n \geq -1$,

$$\text{Err}_{\text{rev}}(F, G_n) = \mathcal{O}_t^{\text{form}}(t^{n+1}), \quad G_n = \text{Trunc}_{< n+1} F^{(-1)\circ},$$

and $r_n = -b_n$, where $F^{(-1)\circ} = X + \sum_{j \geq 0} b_j(L)t^j$.

Proof. Induction on n . For $n = -1$, $\text{Err}_{\text{rev}}(F, X) = F - X = A$ has $\nu_t \geq 0$, which is the assertion $\mathcal{O}_t^{\text{form}}(t^0)$, and $\text{Trunc}_{< 0} F^{(-1)\circ} = X$. Assume $\text{Err}_{\text{rev}}(F, G_{n-1}) = \mathcal{O}_t^{\text{form}}(t^n)$. Then r_n is its coefficient at level n , and by [Theorem 15.1](#) with $c = -r_n$,

$$\text{Err}_{\text{rev}}(F, G_n) = \text{Err}_{\text{rev}}(F, G_{n-1}) - r_n t^n + \mathcal{O}_t^{\text{form}}(t^{n+1}) = \mathcal{O}_t^{\text{form}}(t^{n+1}),$$

since the inherited residual and the correction contribute $r_n t^n$ and $-r_n t^n$ at level n and nothing below. As G_n has no block beyond t^n , [Theorem 14.16](#) identifies $G_n = \text{Trunc}_{< n+1} F^{(-1)\circ}$. Finally [Theorem 14.17](#), applied to $G_{n-1} = \text{Trunc}_{< n} F^{(-1)\circ}$, gives $r_n = -b_n$. $\square$

Algorithm 15.3. Residual-cancellation reversal. *Input:* $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ with A known below t^{N+1} ; a target block $N \geq 0$. *Output:* $\text{Trunc}_{< N+1} F^{(-1)\circ}$.

1. $G \leftarrow X$.
2. For $n = 0, 1, \dots, N$: compute $R \leftarrow \text{Trunc}_{< N+1}(F \circ G - X)$; set $r_n \leftarrow [t^n]R$; set $G \leftarrow G - r_n(L)t^n$.
3. Return G together with the certificate $\text{Trunc}_{< N+1}(F \circ G - X) = 0$.

Every intermediate composition may be truncated below t^{N+1} : by [Theorem 14.9](#) a change of G at order $\geq N+1$ changes $F \circ G$ only at order $\geq N+1$, so a discarded block never returns.

Step (254) divides the residual block by the leading block of $D_X F$, which is 1 for $F \in \mathcal{G}_{\text{poly}, \mathbb{K}}$. Outside $\mathcal{G}_{\text{poly}, \mathbb{K}}$ this is exactly where the unit criterion of the coefficient ring enters, and the sources treat it too briefly.

Proposition 15.4 (Reversal and triangular correction at a general leading slope). *Let $a \in \mathbb{K}^\times$ and fix a value $\log a \in \mathbb{K}$. Let θ be the $\mathbb{K}$ -algebra automorphism of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ induced by the substitution $X \mapsto a^{-1}X$, that is $\theta(t) = at$ and $\theta(L) = L - \log a$; equivalently $\theta(S) = S \circ (a^{-1}X)$. Let $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, $F = aX + A$, $G_0 := X + a^{-1}A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, and $\mathcal{G}_a := a^{-1}X + \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} = \theta(\mathcal{G}_{\text{poly}, \mathbb{K}})$. Then:*

1. θ preserves ν_t , maps $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ onto $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and satisfies $\theta(S \circ H) = S \circ \theta(H)$ whenever the composites exist;
2. $F = aG_0$ (a product, not a composite), and F has exactly one compositional inverse in $\mathcal{G}_a$, namely $F^{(-1)\circ} = \theta(G_0^{(-1)\circ})$, which is two-sided;
3. for every $G \in \mathcal{G}_a$, $c \in \mathbb{K}[L]$ and $N \geq 0$,

$$F \circ (G + c(L)t^N) = F \circ G + a c(L)t^N + \mathcal{O}_t^{\text{form}}(t^{N+1}), \quad (255)$$

so a residual block r_N is cancelled by the correction $-r_N/a$, which stays in $\mathbb{K}[L]$.

Proof. (1) $\theta(t^n p(L)) = a^n t^n p(L - \log a)$, so θ is a bijective ring homomorphism preserving ν_t and $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$; it is well defined because $\log a \in \mathbb{K}$. The intertwining relation is (I3): $\theta(S \circ H) = (S \circ H) \circ (a^{-1}X) = S \circ (H \circ (a^{-1}X)) = S \circ \theta(H)$.

(2) $F = aX + A = a(X + a^{-1}A) = aG_0$. Put $H = G_0^{(-1)\circ}$. Then, by (1), $F \circ \theta(H) = \theta(F \circ H) = \theta(a(G_0 \circ H)) = \theta(aX) = a \cdot a^{-1}X = X$. On the other side, $\theta(H) \circ F = H \circ (a^{-1}X) \circ F = H \circ (a^{-1}F) = H \circ G_0 = X$. For uniqueness, let $G' \in \mathcal{G}_a$ with $F \circ G' = X$ and write $G' = \theta(\tilde{G})$ with $\tilde{G} \in \mathcal{G}_{\text{poly}, \mathbb{K}}$. Then $\theta(F \circ \tilde{G}) = X$, so $F \circ \tilde{G} = \theta^{-1}(X) = aX$, i.e. $G_0 \circ \tilde{G} = X$, so $\tilde{G} = G_0^{(-1)\circ}$ by [Theorem 14.11](#)(2).

(3) Write $G = \theta(\tilde{G})$ with $\tilde{G} \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ and $\delta = ct^N$, $\eta = \theta^{-1}(\delta) = a^{-N}c(L + \log a)t^N \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Since θ is additive, $G + \delta = \theta(\tilde{G} + \eta)$, so by (1) and $F = aG_0$,

$$F \circ (G + \delta) = \theta(F \circ (\tilde{G} + \eta)) = \theta(aG_0 \circ (\tilde{G} + \eta)).$$

By [Theorem 15.1](#), $G_0 \circ (\tilde{G} + \eta) = G_0 \circ \tilde{G} + \eta + \mathcal{O}_t^{\text{form}}(t^{N+1})$. Applying $a\theta(\cdot)$ and using $\theta(\eta) = \delta$ together with the fact that θ preserves ν_t gives (255). $\square$

Remark 15.5 (Two repairs to the general-slope rule). (i) *The scaling reduction needs $\log a \in \mathbb{K}$.* Report 3 performs the reduction $F = aG_0$, $F^{(-1)\circ}(X) = G_0^{(-1)\circ}(X/a)$ and notes in passing that the substitution “shifts the logarithm by $-\log c$ ”; report 4 states the division rule with no hypothesis at all. As [Theorem 15.4](#)(1) makes explicit, the substitution $X \mapsto a^{-1}X$ is an automorphism of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ only if a value of $\log a$ lies in $\mathbb{K}$; otherwise $\mathbb{K}$ must first be extended, and the inverse of a map with coefficients in $\mathbb{K}$ acquires coefficients in $\mathbb{K}(\log a)$. Composition with F itself is likewise undefined without that value.

(ii) *The divisor must be a unit of the coefficient ring, not merely nonzero.* What is really being inverted in (255) is $\lambda := [t^0](D_X F)$, the leading block of the derivative. For $F = aX + A$ it equals $a \in \mathbb{K}^\times$ and all is well. But λ can be a nonconstant polynomial: for $F = X \log X = t^{-1}L$ one has $D_X F = (\partial_L + 1)L = L + 1$, so $\lambda = L + 1$, which is not a unit of $\mathbb{K}[L]$. Division by λ then leaves $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ for $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$. This is not a removable inconvenience: $X \log X$ has no compositional inverse in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ at all. Indeed, for any $G \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ the composite $(X + B) \log(X + B)$ has $\nu_t = -1$ with t^{-1} -block $L \neq 1$, so it cannot equal X ; and $\log(X \log X) = L + \log L$ forces the second logarithm into the scale. A leading slope that is not a unit is a reliable symptom that the inverse leaves the one-logarithm class.

15.2 Coefficient stabilization by functional iteration

Algorithm 15.6. Coefficient-stabilization reversal. *Input:* $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, target block $N \geq 0$. *Output:* $\text{Trunc}_{<N+1} F^{(-1)\circ}$.

1. $B \leftarrow 0$.
2. Repeat $N + 1$ times: $B \leftarrow \text{Trunc}_{<N+1}(-A \circ (X + B))$.
3. Return $X + B$ with the certificate $\text{Trunc}_{<N+1}(F \circ (X + B) - X) = 0$.

One may stop as soon as two consecutive iterates agree below t^{N+1} . By [Theorem 14.11](#)(1) that happens after at most $N + 1$ steps, and after at most $\lceil (N + 1 - \alpha)/(\alpha + 1) \rceil$ steps when $\alpha = \nu_t(A)$.

Proposition 15.7 (Truncation is safe). *Let $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, $M \geq 0$, and $B, \tilde{B} \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $\nu_t(B - \tilde{B}) \geq M$. Then $\text{Trunc}_{<M} \Phi(B) = \text{Trunc}_{<M} \Phi(\tilde{B})$. Consequently, truncating the iterate below t^M after every step of [Theorem 15.6](#) alters no block below t^M of any later iterate, and the algorithm returns the same answer as its untruncated form.*

Proof. [Theorem 14.9](#) gives $\nu_t(\Phi(B) - \Phi(\tilde{B})) \geq \nu_t(A) + M + 1 > M$, which is the first claim. For the second, let B_j be the exact iterates and $\hat{B}_j$ the truncated ones, and suppose $\nu_t(B_j - \hat{B}_j) \geq M$. Then $\nu_t(\Phi(B_j) - \Phi(\hat{B}_j)) \geq M + 1$, and truncating $\Phi(\hat{B}_j)$ below t^M introduces an error of order $\geq M$; hence $\nu_t(B_{j+1} - \hat{B}_{j+1}) \geq M$, and the invariant persists. $\square$

Remark 15.8 (An algebraically equivalent iteration need not be order-improving). An existence proof by iteration depends on the chosen normal form, not only on the equation. For $F = X + \log X$ the inverse $\omega = F^{(-1)\circ}$ satisfies $\omega + \log \omega = X$. The rearrangement

$$\omega = X - \log \omega \tag{256}$$

is order-improving: if $\omega_1, \omega_2 \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ and $\omega_1 - \omega_2 = \mathcal{O}_t^{\text{form}}(t^r)$, then $\omega_1 \omega_2^{-1} = 1 + \mathcal{O}_t^{\text{form}}(t^{r+1})$, hence $\log \omega_1 - \log \omega_2 = \mathcal{O}_t^{\text{form}}(t^{r+1})$; this is [Theorem 14.9](#) again. The algebraically equivalent rearrangement

$$\omega = e^{X-\omega} \quad (257)$$

fails for a sharper reason than slow convergence: e^X is not an element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, so the right-hand side of (257) does not define a map of the class and no truncation of it is meaningful. Report 6 makes the same point with an exponential rearrangement of the Lambert equation, attributing the observation to Edgar [13]: a fixed-point scheme must increase asymptotic precision, and algebraic equivalence is no evidence that it does. The test is supplied by [Theorem 14.9](#): the iteration map must send a difference of valuation r to one of valuation $> r$, uniformly in r .

15.3 Newton iteration

Lemma 15.9 (The Newton denominator is a unit). *Let $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ and $G \in \mathcal{G}_{\text{poly}, \mathbb{K}}$. Then $J := (D_X F) \circ G$ satisfies $J = 1 + \mathcal{O}_t^{\text{form}}(t)$ and is a unit of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, with $J^{-1} = \sum_{k \geq 0} (1 - J)^k$.*

Proof. $D_X F = 1 + D_X A$ and $\nu_t(D_X A) \geq \nu_t(A) + 1 \geq 1$ by [Theorem 14.4](#). By [Theorem 14.5](#), $\nu_t(J) = 0$ and $[t^0]J = 1$, so $W := 1 - J$ has $\nu_t(W) \geq 1$; the family $(W^k)_{k \geq 0}$ is summable and $J \sum_{k \geq 0} W^k = (1 - W) \sum_{k \geq 0} W^k = 1$. (This is the unit criterion in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$: U is a unit exactly when $[t^0]U \in \mathbb{K}^\times$.) $\square$

Definition 15.10 (Newton reversion map). For $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ and $G \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ put

$$\mathcal{N}_F(G) := G - \frac{\text{Err}_{\text{rev}}(F, G)}{(D_X F) \circ G} \in \mathcal{G}_{\text{poly}, \mathbb{K}}. \quad (258)$$

Theorem 15.11 (Precision doubling, exact arithmetic). *Let $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, $\alpha = \nu_t(A)$, $G \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, and $m = \nu_t(\text{Err}_{\text{rev}}(F, G))$. Then*

$$\nu_t(\text{Err}_{\text{rev}}(F, \mathcal{N}_F(G))) \geq 2m + 2 + \alpha. \quad (259)$$

Equivalently, by [Theorem 14.15](#): if G is correct through outer block $m - 1$, then $\mathcal{N}_F(G)$ is correct through outer block $2m + 1 + \alpha$; the number of correct blocks passes from m to $2m + 2 + \alpha$.

Proof. Put $R = \text{Err}_{\text{rev}}(F, G)$, $J = (D_X F) \circ G$, and $C = -RJ^{-1}$, so $\mathcal{N}_F(G) = G + C$. By [Theorem 15.9](#), $\nu_t(C) = \nu_t(R) = m$. By [Theorem 14.6](#) with $S = F$, $P = G - X$, $Q = C$,

$$\text{Err}_{\text{rev}}(F, G + C) = \underbrace{R + JC}_{=0} + \sum_{j \geq 2} \frac{C^j}{j!} (D_X^j F) \circ G.$$

For $j \geq 2$, $D_X^j F = D_X^j A$, and [Theorems 14.4](#) and [14.5](#) give $\nu_t((D_X^j A) \circ G) \geq \alpha + j$. Hence the j -th term has $\nu_t \geq jm + \alpha + j = \alpha + j(m + 1)$, minimal at $j = 2$ with value $2m + 2 + \alpha$. $\square$

Remark 15.12 (Diagnosis of the six sources). The reports state the Newton estimate in five different ways.

- Reports 1, 5 and 6 assert the exponent $2m + 2$, which is (259) with the term α omitted; their proofs are correct as far as they go.
- Report 2 states $2N$ under the extra hypothesis $N \geq 1$. The hypothesis is never used and is not needed; the case $N = 0$ is legitimate and is precisely the situation at the first Newton step from $G = X$, where the residual is A itself. The argument given there in fact yields $2N + 2$.
- Report 3 states $2m$ and adds only that “in this particular scale the order is often even higher”. The improvement is not occasional but automatic and equals $2m + 2 + \nu_t(A)$.

- Report 4 is internally inconsistent: it announces t^{2N+1} in the text and proves t^{2N+2} two lines later, bridging the gap with the phrase “allowing coarse bounds and Laurent shifts”. There are no Laurent shifts here: every object in (258) lies in $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, and the estimate is exact.

A second and subtler slide occurs in reports 1 and 5, which pass from “the residual has order $2m + 2$ ” to “the number of correct blocks doubles” without argument. That step is legitimate, but only because of [Theorem 14.15](#), which identifies the residual valuation with $\nu_t(G - F^{(-1)\circ})$; no source proves or even states that identification. Without it, a small residual is a priori compatible with a large coefficient error — exactly as it would be for a map whose derivative were not a unit.

Remark 15.13 (Sharpness). The exponent in (259) cannot be improved in general. For $F = X + \log X$ (so $\alpha = 0$) and $G = X - L(1+t)^{-1}$ one has $m = 2$, and after one exact Newton step the residual is $-\frac{1}{8}L^4t^6 + \mathcal{O}_t^{\text{form}}(t^7)$, of order precisely $2m + 2 = 6$. See [Theorem 15.17](#).

15.4 Precision doubling with truncated intermediates

In an implementation none of R , J or the quotient is known exactly, and the exponent $2m + 2 + \alpha$ is attained only if the truncations are deep enough. Reports 1, 2, 4, 5 and 6 omit the hypothesis; report 3 gestures at it (“computed consistently to sufficient precision”). The following two results make it exact.

Theorem 15.14 (Newton step with an inexact correction). *Let $F = X + A \in \mathcal{G}_{\text{poly},\mathbb{K}}$, $\alpha = \nu_t(A)$, $G \in \mathcal{G}_{\text{poly},\mathbb{K}}$, $R = \text{Err}_{\text{rev}}(F, G)$, $m = \nu_t(R)$, $J = (D_X F) \circ G$, and let $C = -RJ^{-1}$ be the exact Newton correction. Let $\hat{C} \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ satisfy $\nu_t(\hat{C} - C) \geq \sigma$ with $\sigma \geq m$. Then*

$$\nu_t\left(\text{Err}_{\text{rev}}\left(F, G + \hat{C}\right)\right) \geq \min\{2m + 2 + \alpha, \sigma\}. \quad (260)$$

Proof. Since $\nu_t(C) = m$ and $\nu_t(\hat{C} - C) \geq \sigma \geq m$, we get $\nu_t(\hat{C}) \geq m$. By [Theorem 14.6](#) with $S = F$, $P = G - X$, $Q = \hat{C}$,

$$\text{Err}_{\text{rev}}\left(F, G + \hat{C}\right) = R + J\hat{C} + \sum_{j \geq 2} \frac{\hat{C}^j}{j!} (D_X^j A) \circ G.$$

The tail has $\nu_t \geq 2m + 2 + \alpha$ exactly as in [Theorem 15.11](#). For the head, $R + JC = 0$ gives $R + J\hat{C} = J(\hat{C} - C)$, and $\nu_t(J) = 0$, so $\nu_t(R + J\hat{C}) = \nu_t(\hat{C} - C) \geq \sigma$. $\square$

Lemma 15.15 (Precision propagation through the Newton step). *Keep the notation of [Theorem 15.14](#) and let $\hat{R}, \hat{K} \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and $q \in \mathbb{N}_0$ satisfy*

$$\nu_t(\hat{R} - R) \geq \sigma_R \geq m, \quad \nu_t(\hat{K} - J^{-1}) \geq \sigma_K \geq 0.$$

Put $\hat{C} = -\text{Trunc}_{<q}(\hat{R}\hat{K})$. Then $\nu_t(\hat{C} - C) \geq \min\{q, \sigma_R, m + \sigma_K\}$. Consequently, to certify $\nu_t(\text{Err}_{\text{rev}}(F, G + \hat{C})) \geq q$ for a target $q \leq 2m + 2 + \alpha$, it suffices to take $\sigma_R \geq q$ and $\sigma_K \geq q - m$. Moreover:

1. $\hat{K}$ with $\sigma_K = s$ is obtained from $\text{Trunc}_{<s} J$, since for units $J, \hat{J}$ with $[t^0]J = [t^0]\hat{J} = 1$ one has $\hat{J}^{-1} - J^{-1} = (J - \hat{J})\hat{J}^{-1}J^{-1}$;
2. $\text{Trunc}_{<s} J$ is determined by $\text{Trunc}_{<s-1} A$ and $\text{Trunc}_{<s-2} G$, because changing G by $\mathcal{O}_t^{\text{form}}(t^u)$ changes J only by $\mathcal{O}_t^{\text{form}}(t^{u+2})$;
3. $\text{Trunc}_{<q} R$ is determined by $\text{Trunc}_{<q} A$ and $\text{Trunc}_{<q} G$, and no coarser knowledge of G suffices: for $\epsilon \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$,

$$\text{Err}_{\text{rev}}(F, G + \epsilon) - \text{Err}_{\text{rev}}(F, G) = J\epsilon + \mathcal{O}_t^{\text{form}}\left(t^{2\nu_t(\epsilon)+2}\right),$$

whose valuation is exactly $\nu_t(\epsilon)$. Thus a block of G at level $u < q$ moves R at level u , and if G is stored with fewer than q blocks its zero padding is part of the definition of the element being corrected and must be used consistently throughout the step.

Proof. Write $\widehat{C} - C = -(\text{Trunc}_{<q}(\widehat{R}\widehat{K}) - \widehat{R}\widehat{K}) - (\widehat{R}\widehat{K} - RJ^{-1})$. The first bracket has $\nu_t \geq q$. For the second,

$$\widehat{R}\widehat{K} - RJ^{-1} = (\widehat{R} - R)\widehat{K} + R(\widehat{K} - J^{-1}),$$

and $\nu_t(\widehat{K}) \geq \min\{0, \sigma_K\} = 0$ because $\nu_t(J^{-1}) = 0$; so its valuation is at least $\min\{\sigma_R, m + \sigma_K\}$. With $\sigma_R \geq q$, $\sigma_K \geq q - m$ the bound becomes q , and [Theorem 15.14](#) applies since $q \geq m$.

For (1), the displayed identity gives $\nu_t(\widehat{J}^{-1} - J^{-1}) = \nu_t(J - \widehat{J}) \geq s$. For (2), [Theorem 14.6](#) applied to $S = D_X F$ gives $(D_X F) \circ (G + \epsilon) - (D_X F) \circ G = \sum_{j \geq 1} \frac{\epsilon^j}{j!} (D_X^{j+1} A) \circ G$, whose valuation is at least $\nu_t(\epsilon) + 2$ because $\nu_t(D_X^{j+1} A) \geq j + 1 \geq 2$. For (3), [Theorem 14.6](#) with $S = F$, $P = G - X$, $Q = \epsilon$ gives $\text{Err}_{\text{rev}}(F, G + \epsilon) - \text{Err}_{\text{rev}}(F, G) = J\epsilon + \sum_{j \geq 2} \frac{\epsilon^j}{j!} (D_X^j A) \circ G$, and the tail has valuation at least $2\nu_t(\epsilon) + 2 > \nu_t(\epsilon)$ while $\nu_t(J\epsilon) = \nu_t(\epsilon)$. $\square$

Corollary 15.16 (Doubling schedule). *Let $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ and let $N \geq 0$ be a target block. Put $H_0 = X$, $q_0 = \nu_t(A)$, and for $i \geq 0$*

$$q_{i+1} = \min\{N + 1, 2q_i + 2\}, \quad H_{i+1} = \text{Trunc}_{<q_{i+1}} \left(H_i - \text{Trunc}_{<q_{i+1}} R_i \cdot \text{Trunc}_{<q_{i+1}-q_i} J_i^{-1} \right),$$

where $R_i = \text{Err}_{\text{rev}}(F, H_i)$ and $J_i = (D_X F) \circ H_i$. Then $\nu_t(\text{Err}_{\text{rev}}(F, H_i)) \geq q_i$ for every i , and $H_i = \text{Trunc}_{<N+1} F^{(-1)\circ}$ as soon as $q_i = N + 1$. The number of steps is $\mathcal{O}(\log N)$.

Proof. Induction on i . The case $i = 0$ is $\text{Err}_{\text{rev}}(F, X) = A$. Assume $\nu_t(R_i) \geq q_i$. Apply [Theorem 15.15](#) with $m = q_i$, $q = q_{i+1}$, $\sigma_R = q_{i+1}$ and $\sigma_K = q_{i+1} - q_i$, which is ≥ 1 because $q_{i+1} \geq \min\{N + 1, 2q_i + 2\} > q_i$ while $q_i < N + 1$; the final outer truncation $\text{Trunc}_{<q_{i+1}}$ contributes an error of valuation $\geq q_{i+1}$ and is absorbed in the same bound. Then [Theorem 15.14](#) gives $\nu_t(R_{i+1}) \geq \min\{2q_i + 2 + \nu_t(A), q_{i+1}\} = q_{i+1}$. The identification with $\text{Trunc}_{<N+1} F^{(-1)\circ}$ is [Theorem 14.16](#). Finally $q_{i+1} + 1 \geq 2(q_i + 1)$ until the cap is reached, so $q_i + 1 \geq 2^i$. $\square$

A common trap

A Newton step at target q treats the stored G as an exact element of $\mathcal{G}_{\text{poly}, \mathbb{K}}$ — zero-padded beyond its last stored block — and must form $R = \text{Err}_{\text{rev}}(F, G)$ below t^q from *that same* element, even though only $m < q$ of its blocks are known to be correct. The blocks between levels m and $q - 1$ are the guard blocks: they are not required to be correct, but they are required to be fixed and used identically in the residual, in the Jacobian factor and in the update. Computing J to the final cutoff while computing R only to the current accuracy m , or recomputing R from a differently truncated copy of G , destroys the doubling: by (260) the guaranteed order collapses to σ , and the collapse is invisible in the printed series.

Example 15.17 (Doubling, and its failure, for $F = X + \log X$). Let $F = X + \log X = X + L$, so $A = L$, $\alpha = 0$, and write $\omega = F^{(-1)\circ}$. Here $D_X F = 1 + t$, and for $G = X + B$

$$\text{Err}_{\text{rev}}(F, G) = B + L + \log(1 + Bt), \quad J = (D_X F) \circ G = 1 + G^{-1} = 1 + \frac{t}{1 + Bt}.$$

First step. From $H_0 = X$: $R_0 = L$, $m = 0$, $J_0 = 1 + t$, so

$$H_1 = X - \frac{L}{1 + t} = X - L + Lt - Lt^2 + Lt^3 - \dots$$

[Theorem 15.11](#) predicts $\nu_t(R_1) \geq 2$, and indeed

$$R_1 = -\frac{1}{2}L^2t^2 + (L^2 - \frac{1}{3}L^3)t^3 + (-\frac{3}{2}L^2 + L^3 - \frac{1}{4}L^4)t^4 + \mathcal{O}_t^{\text{form}}(t^5),$$

so $m = 2$ exactly. By [Theorem 14.15](#) the error must have the same leading block, and $H_1 - \omega = -\frac{1}{2}L^2t^2 + \mathcal{O}_t^{\text{form}}(t^3)$.

Exact second step. With $q = 2m + 2 = 6$ and all data carried below t^6 ,

$$\text{Err}_{\text{rev}}(F, \mathcal{N}_F(H_1)) = -\frac{1}{8}L^4t^6 + \mathcal{O}_t^{\text{form}}(t^7),$$

of order exactly 6: blocks $0, \dots, 5$ of ω are correct, six blocks in place of two.

Guard-starved second step. Suppose the residual is truncated one doubling too early, $\widehat{R} = \text{Trunc}_{<4} R_1$, while J and the division are still carried below t^6 . Then $\sigma_R = 4$ and [Theorem 15.15](#) guarantees only order 4 – and the bound is attained. With $\rho_4 := [t^4]R_1 = -\frac{1}{4}L^4 + L^3 - \frac{3}{2}L^2$,

$$\widehat{C} - C = (R_1 - \widehat{R})J^{-1} = \rho_4 t^4 + \mathcal{O}_t^{\text{form}}(t^5),$$

hence $\text{Err}_{\text{rev}}(F, H_1 + \widehat{C}) = \rho_4 t^4 + \mathcal{O}_t^{\text{form}}(t^5)$ and, by [Theorem 14.15](#), the computed inverse first deviates from ω in block 4, by exactly ρ_4 . Four correct blocks in place of six: the loss equals the two blocks of residual that were discarded, and no amount of extra precision in J or in the division repairs it.

15.5 Four reversal strategies compared

Write $C(n)$ for the cost of one composition $F \circ G$ truncated below t^n , $M(n)$ for a truncated multiplication, and $I(n)$ for the reciprocal of a unit of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. The counts below are counts of these operations, not bit complexities: each block is a polynomial in L whose degree grows with the block index – linearly when $\deg_L[t^0]A = 1$, and like dn when $\deg_L[t^0]A = d$ – so a further degree-dependent factor is hidden inside C , M and I .

Strategy	Operation count to block N	Strength	Failure mode
Triangular, Theorem 15.2	$N + 1$ residual evaluations; naively $\sum_{n \leq N} C(n+1)$, less with incremental residual updating	one exact new block per step, each of them printable; no division beyond the leading slope; the residual is itself the certificate	fails when the leading block of $D_X F$ is not a unit (Theorem 15.5); strictly sequential, so it cannot exploit fast arithmetic
Functional iteration, Theorem 15.6	$N + 1$ compositions below t^{N+1} : $(N+1)C(N+1)$	shortest existence proof; needs composition only; insensitive to the initial guess	gains only $\nu_t(A) + 1$ blocks per step and recomputes low blocks each time; an algebraically equivalent rearrangement may gain nothing (Theorem 15.8)
Newton, Theorem 15.16	$\mathcal{O}(\log N)$ steps at precisions q_i with $q_i + 1 \geq 2^i$; total $\sum_i (2C(q_i) + M(q_i) + I(q_i))$	residual valuation more than doubles each step; fastest at high order	needs $D_X F$, a reciprocal and a composition per step; silently loses blocks when the guard precisions of Theorem 15.15 are not respected
Lagrange–Bürmann, Section 16	powers A^r , $r \leq N + 1$: N multiplications, plus $\mathcal{O}(N^2)$ applications of $\Delta_{n,k}$	closed universal formula; no recursion; the natural route to coefficient identities and degree laws	expression swell in A^r and in the repeated derivatives; logarithmic degrees reach $\deg_L \approx d(N+1)$ although only N blocks are wanted

Remark 15.18 (On the cost claims). The assertion that Newton is asymptotically superior, made in reports 1, 3, 4 and 6, is a statement about the cost model: it holds when C , M and I are superadditive, so that the geometric schedule $q_0 < q_1 < \dots < N + 1$ sums to a constant multiple of the cost at the final precision. It is not a theorem about the polynomial–logarithmic class, and it can fail if a composition routine carries a large fixed overhead, or if the coefficient degrees rather than the block count dominate the work. We assert only what is proved here: the number of Newton steps is $\mathcal{O}(\log N)$, and every step is subject to [Theorem 15.15](#). Practical schemes seed with two or three triangular blocks, switch to Newton, and certify with an independent Lagrange–Bürmann coefficient; algorithms for the wider exp–log setting are in [29, 30, 33].

15.6 Two worked reversions with certificates

Example 15.19 (The inverse of $X + \log X$, through block 4). Take $F = X + \log X = X + L \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, so $A = L$ and $\alpha = 0$. For $G = X + B$,

$$\text{Err}_{\text{rev}}(F, G) = B + L + \log(1 + Bt), \quad (261)$$

since $F \circ G = G + \log G = X + B + L + \log(1 + Bt)$. Run [Theorem 15.2](#).

Stage –1. $G_{-1} = X$, $B = 0$, so $\text{Err}_{\text{rev}}(F, G_{-1}) = L$; $r_0 = L$ and $b_0 = -L$.

Stage 0. $G_0 = X - L$. Now $B + L = 0$ and (261) collapses to a closed form,

$$\text{Err}_{\text{rev}}(F, G_0) = \log(1 - Lt) = -Lt - \frac{1}{2}L^2t^2 - \frac{1}{3}L^3t^3 - \frac{1}{4}L^4t^4 - \dots,$$

so $r_1 = -L$ and $b_1 = L$.

Stage 1. $G_1 = X - L + Lt$, and

$$\text{Err}_{\text{rev}}(F, G_1) = (L - \frac{1}{2}L^2)t^2 + (L^2 - \frac{1}{3}L^3)t^3 + (-\frac{1}{2}L^2 + L^3 - \frac{1}{4}L^4)t^4 + \mathcal{O}_t^{\text{form}}(t^5),$$

so $r_2 = L - \frac{1}{2}L^2$ and $b_2 = \frac{1}{2}L^2 - L$.

Stage 2. $G_2 = G_1 + (\frac{1}{2}L^2 - L)t^2$, and

$$\text{Err}_{\text{rev}}(F, G_2) = (-\frac{1}{3}L^3 + \frac{3}{2}L^2 - L)t^3 + (-\frac{1}{4}L^4 + \frac{3}{2}L^3 - \frac{3}{2}L^2)t^4 + \mathcal{O}_t^{\text{form}}(t^5),$$

so $b_3 = \frac{1}{3}L^3 - \frac{3}{2}L^2 + L$.

Stage 3. $G_3 = G_2 + (\frac{1}{3}L^3 - \frac{3}{2}L^2 + L)t^3$, and

$$\text{Err}_{\text{rev}}(F, G_3) = (-\frac{1}{4}L^4 + \frac{11}{6}L^3 - 3L^2 + L)t^4 + \mathcal{O}_t^{\text{form}}(t^5),$$

so $b_4 = \frac{1}{4}L^4 - \frac{11}{6}L^3 + 3L^2 - L$.

Stage 4. $G_4 = G_3 + b_4t^4$ and $\text{Err}_{\text{rev}}(F, G_4) = \mathcal{O}_t^{\text{form}}(t^5)$, which is the certificate of [Theorem 14.16](#) for $N = 4$. Collecting,

$$(X + \log X)^{(-1)^\circ} = X - L + Lt + \left(\frac{L^2}{2} - L\right)t^2 + \left(\frac{L^3}{3} - \frac{3L^2}{2} + L\right)t^3 + \left(\frac{L^4}{4} - \frac{11L^3}{6} + 3L^2 - L\right)t^4 + \mathcal{O}_t^{\text{form}}(t^5). \quad (262)$$

The blocks are the canonical zero-based Lambert polynomials, $b_n = \mathbb{L}_n^{\text{Lam}}(L)$ for $0 \leq n \leq 4$, with $\mathbb{L}_0^{\text{Lam}}(u) = -u$, $\mathbb{L}_1^{\text{Lam}}(u) = u$, $\mathbb{L}_2^{\text{Lam}}(u) = \frac{1}{2}u^2 - u$, $\mathbb{L}_3^{\text{Lam}}(u) = \frac{1}{3}u^3 - \frac{3}{2}u^2 + u$ and $\mathbb{L}_4^{\text{Lam}}(u) = \frac{1}{4}u^4 - \frac{11}{6}u^3 + 3u^2 - u$. The map $(X + \log X)^{(-1)^\circ}$ is Wright’s omega function ω [5, 7]; its polynomials, the degree law $\deg \mathbb{L}_n^{\text{Lam}} = n$ for $n \geq 1$ – which fails at $n = 0$, where $\deg \mathbb{L}_0^{\text{Lam}} = 1$ – and the Stirling-number formula are developed in the Lambert part of this volume.

Remark 15.20 (The residual is minus the next Lambert polynomial). Every residual displayed in [Theorem 15.19](#) has leading block $-\mathbb{L}_n^{\text{Lam}}(L)$ at level n . This is not an accident of the computation but an instance of [Theorem 14.17](#): for $G_{n-1} = \text{Trunc}_{<n} \omega$,

$$\text{Err}_{\text{rev}}(X + \log X, \text{Trunc}_{<n} \omega) = -\mathbb{L}_n^{\text{Lam}}(L)t^n + \mathcal{O}_t^{\text{form}}(t^{n+1}).$$

This is a one-line certificate for any published table of Lambert polynomials, and it is independent of whatever recurrence generated the table.

Example 15.21 (A two-parameter logarithmic-rational perturbation, through block 4). Let $a, b \in \mathbb{K}$ and

$$F(X) = X + a \log X + \frac{b}{X} = X + aL + bt \in \mathcal{G}_{\text{poly}, \mathbb{K}}.$$

For $G = X + B$,

$$\text{Err}_{\text{rev}}(F, G) = B + aL + a \log(1 + Bt) + \frac{bt}{1 + Bt}. \quad (263)$$

Running [Theorem 15.2](#) on (263) gives $r_n = -b_n$ with

$$b_0 = -aL, \quad (264)$$

$$b_1 = a^2L - b, \quad (265)$$

$$b_2 = a^3 \left(\frac{L^2}{2} - L \right) + ab(1 - L), \quad (266)$$

$$b_3 = a^4 \left(\frac{L^3}{3} - \frac{3L^2}{2} + L \right) + a^2b(-L^2 + 3L - 1) - b^2, \quad (267)$$

$$b_4 = a^5 \left(\frac{L^4}{4} - \frac{11L^3}{6} + 3L^2 - L \right) + a^3b \left(-L^3 + \frac{11L^2}{2} - 6L + 1 \right) + ab^2 \left(\frac{5}{2} - 3L \right). \quad (268)$$

The certificates exhibited at the successive stages are

$$\begin{aligned} \text{Err}_{\text{rev}}(F, G_{-1}) &= aL + bt, \\ \text{Err}_{\text{rev}}(F, G_0) &= (b - a^2L)t + \mathcal{O}_t^{\text{form}}(t^2), \\ \text{Err}_{\text{rev}}(F, G_1) &= \left[a^3 \left(L - \frac{L^2}{2} \right) + ab(L - 1) \right] t^2 + \mathcal{O}_t^{\text{form}}(t^3), \\ \text{Err}_{\text{rev}}(F, G_2) &= -b_3 t^3 + \mathcal{O}_t^{\text{form}}(t^4), \\ \text{Err}_{\text{rev}}(F, G_3) &= -b_4 t^4 + \mathcal{O}_t^{\text{form}}(t^5), \\ \text{Err}_{\text{rev}}(F, G_4) &= \mathcal{O}_t^{\text{form}}(t^5), \end{aligned}$$

their valuations being 0, 1, 2, 3, 4 and ≥ 5 respectively (for $a \neq 0$). Three checks come free.

- $b = 0$. Then $b_n = a^{n+1} \mathbb{L}_n^{\text{Lam}}(L)$ for $0 \leq n \leq 4$; at $a = 1$ this reproduces (262). The pattern persists for every n , since in the Lagrange–Bürmann formula of [Section 16](#) only the index $r = n + 1$ contributes to block n when $A = aL$, giving $a^{n+1} \frac{(-1)^{n+1}}{(n+1)!} \Delta_{0,n} L^{n+1} = a^{n+1} \mathbb{L}_n^{\text{Lam}}(L)$.
- $a = 0$. Then $b_0 = b_2 = b_4 = 0$ and $b_1 = -b, b_3 = -b^2$. The vanishing of the even blocks is forced by parity: $F = X + bX^{-1}$ satisfies $F(-X) = -F(X)$, so by the uniqueness in [Theorem 14.11](#) its inverse is odd as well, and $F^{(-1)^\circ}(-X) = -F^{(-1)^\circ}(X)$ reads $b_n(-1)^n = -b_n$. See [Theorem 15.22](#).
- *Degrees*. For $a \neq 0$, $\deg_L b_n = n$ for $1 \leq n \leq 4$ while $\deg_L b_0 = 1$, matching the Lambert degree law and its exception at $n = 0$.

Formulas (264)–(267) agree with the specializations recorded in reports 1, 2, 3, 4, 5 and 6, which we have checked term by term against each other and against the triangular recursion; (268) appears in none of them.

Example 15.22 (An exact benchmark: the Catalan inverse). The case $a = 0$ of [Theorem 15.21](#) has a closed form and is therefore the sharpest available test of signs and of the branch convention. Let $c \in \mathbb{K}^\times$ and $F = X + ct$. Since $\mathbb{K}$ has characteristic zero,

$$S := (1 - 4ct^2)^{1/2} = \sum_{k \geq 0} \binom{1/2}{k} (-4c)^k t^{2k} \in \mathbb{K}[[t]]$$

is well defined, and $\binom{1/2}{k} (-4)^k = -\frac{2}{k} \binom{2k-2}{k-1}$ for $k \geq 1$, i.e. -2 times the Catalan number C_{k-1} . Put $G = \frac{1}{2}X(1 + S)$. Then $\nu_t(G - X) \geq 1$, so $G \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, and

$$G^2 - XG = \frac{1}{4}X^2(1 + S)[(1 + S) - 2] = \frac{1}{4}X^2(S^2 - 1) = -cX^2t^2 = -c,$$

so $G^2 - XG + c = 0$. Dividing by G , a unit of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ because $\nu_t(G) = -1$ with leading block 1, gives $G + cG^{-1} = X$, that is $F \circ G = X$. By [Theorem 14.11](#) this G is $F^{(-1)\circ}$, and expanding,

$$(X + cX^{-1})^{(-1)\circ} = X - \sum_{k \geq 1} C_{k-1} c^k t^{2k-1} = X - \frac{c}{X} - \frac{c^2}{X^3} - \frac{2c^3}{X^5} - \frac{5c^4}{X^7} - \cdots,$$

with Catalan coefficients $1, 1, 2, 5, 14, \dots$ [4, 18]. Blocks 0, 1 and 3 reproduce (264), (265) and (267) at $a = 0$. Note that the second root $\frac{1}{2}X(1 - S)$ has $\nu_t = 1$ and does not lie in $\mathcal{G}_{\text{poly},\mathbb{K}}$: the near-identity normalization is exactly what selects the branch, and no coefficient comparison is needed to discard the other one.

Structural checkpoint

Before accepting a computed reversal, record five items: the coefficient field $\mathbb{K}$, including any logarithm of a scaling constant that had to be adjoined; the exclusive cutoff $N + 1$ of the answer; the guard precisions actually used for R , for J and for the quotient; the valuation of the right residual; and the valuation of the left residual. A result whose residual valuation is not *strictly* greater than the advertised block index is not certified, however plausible its coefficients look.

15.7 Exercises

Exercise 15.23. Let $F = X + A \in \mathcal{G}_{\text{poly},\mathbb{K}}$ with $\nu_t(A) = \alpha$. Show that [Theorem 15.6](#) reaches block N after at most $\lceil (N + 1 - \alpha)/(\alpha + 1) \rceil$ iterations, and exhibit an A with $\alpha = 0$ for which $N + 1$ iterations are necessary.

Exercise 15.24. Carry out the guard-starved Newton step of [Theorem 15.17](#) in the other direction: keep $q = \sigma_R = 6$ but under-resolve the Jacobian, taking $\sigma_K \in \{2, 3, 4\}$. Predict the resulting residual order from [Theorem 15.15](#), and verify that in each case the bound $\min\{q, \sigma_R, m + \sigma_K\}$ is attained exactly.

Exercise 15.25. Let $F = X \log X$. Show that $\nu_t(D_X F) = 0$ with leading block $L + 1$, and that the first triangular correction already requires $(L + 1)^{-1}$. Show that F has no compositional inverse in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, identify the leading monomial of the true inverse, and locate the first block in which $\log \log X$ appears.

Exercise 15.26. Let $F = X + A \in \mathcal{G}_{\text{poly},\mathbb{K}}$ and $\alpha = \nu_t(A)$. Using [Theorem 14.15](#) and [Theorem 14.12](#), show that $H \in \mathcal{G}_{\text{poly},\mathbb{K}}$ satisfies $\text{Err}_{\text{rev}}(F, H) = \mathcal{O}_t^{\text{form}}(t^{2\alpha+1})$ if and only if $H = X - A + \mathcal{O}_t^{\text{form}}(t^{2\alpha+1})$. Deduce that $X - \text{Trunc}_{<2\alpha+1} A$ is the unique element of $\mathcal{G}_{\text{poly},\mathbb{K}}$ whose correction is supported below $t^{2\alpha+1}$ and whose residual has valuation at least $2\alpha + 1$.

16 The Lagrange–Bürmann formula at infinity

Fixed-point and Newton reversion produce the inverse of a near-identity map one filtration block at a time. The Lagrange–Bürmann formula produces all blocks at once, as a single explicit family of differential monomials in the perturbation. This section states that formula, proves it *formally* – no analytic Lagrange inversion theorem is invoked and no convergence is used or claimed – derives the finite coefficient formula in the polynomial–logarithmic basis together with a sharp determination of its index range, proves the transport formula for a composed observable, and establishes the degree law that governs the growth of logarithmic degrees under reversion.

Throughout, $\mathbb{K}$ is a field of characteristic zero, X is the large variable, and

$$t := X^{-1}, \quad L := \log X.$$

The default analytic regime, whenever one is mentioned at all, is $X \rightarrow +\infty$ along the positive real ray. Truncation is always *exclusive*: $\text{Trunc}_{<N} F = \sum_{n < N} p_n(L) t^n$. Coefficient extraction is written with the series to the right of the bracket, $[t^n]F = p_n(L)$ and $[t^n L^j]F = a_{n,j}$.

16.1 Reversion frames

Every argument below uses only four features of $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$: it is a characteristic-zero algebra, it carries a valuation, it is complete for that valuation, and its distinguished derivation strictly increases the valuation. Isolating these makes the theorems of this section available verbatim in the Puiseux–log, iterated Laurent-log and nested-logarithmic enlargements that [Section 17](#) is forced to introduce. The cost of the abstraction is one definition.

Definition 16.1 (Reversion frame). Let $\mathbb{K}$ be a field of characteristic zero and let $\Gamma \subseteq \mathbb{R}$ be an ordered subgroup with $1 \in \Gamma$. A *reversion frame* over $\mathbb{K}$ with value group Γ is a quadruple $(\mathfrak{A}, \nu, \partial, X)$ in which $\mathfrak{A}$ is a commutative $\mathbb{K}$ -algebra, ∂ is a $\mathbb{K}$ -derivation of $\mathfrak{A}$, $X \in \mathfrak{A}$, and $\nu: \mathfrak{A} \rightarrow \Gamma \cup \{+\infty\}$ satisfies the following.

- (F1) $\nu(f) = +\infty$ exactly when $f = 0$; $\nu(f + g) \geq \min\{\nu(f), \nu(g)\}$; $\nu(fg) = \nu(f) + \nu(g)$; and $\nu(\lambda) = 0$ for every $\lambda \in \mathbb{K}^\times$.
- (F2) Call a family $(f_i)_{i \in I}$ in $\mathfrak{A}$ *summable* when $\{i \in I : \nu(f_i) < \gamma\}$ is finite for every $\gamma \in \Gamma$. Every summable family has a sum $\sum_i f_i \in \mathfrak{A}$; summation is $\mathbb{K}$ -linear, satisfies $\nu(\sum_i f_i) \geq \inf_i \nu(f_i)$, is unchanged by regrouping the index set into disjoint blocks, and satisfies $(\sum_i f_i)(\sum_j g_j) = \sum_{i,j} f_i g_j$ for summable (f_i) and (g_j) .
- (F3) $\nu(\partial f) \geq \nu(f) + 1$ for every $f \in \mathfrak{A}$.
- (F4) $\partial X = 1$.

We write

$$\mathfrak{A}_{>-1} := \{f \in \mathfrak{A} : \nu(f) > -1\},$$

the set of *admissible perturbations*. It is closed under addition and under multiplication by $\{\nu \geq 0\}$, and contains 0; it is not in general a subring, since $\nu(f^2) = 2\nu(f)$ may drop below -1 .

Remark 16.2. The threshold is -1 , not 0, because the summability of every family below is governed by the *gain*

$$\delta := 1 + \nu(A) > 0 \tag{269}$$

of one derivative against one factor of the perturbation. In the polynomial–logarithmic frame the value group is $\mathbb{Z}$, so $\nu_t(A) > -1$ is the same as $A \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and $\mathcal{G}(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}) = \mathcal{G}_{\text{poly},\mathbb{K}}$; the distinction matters only in the Puiseux–log frames of [Section 17](#), where a perturbation of outer order $-\frac{1}{2}$ is admissible and is exactly what the reduction of a ramified reversion produces ([Theorem 17.11](#)).

Condition (F3) forces ∂ to be ν -continuous, so it commutes with sums of summable families: if (f_i) is summable then so is (∂f_i) , and $\partial \sum_i f_i = \sum_i \partial f_i$, because the two sides agree modulo $\{\nu \geq \gamma\}$ for every γ after discarding the finitely many indices with $\nu(f_i) < \gamma$.

Lemma 16.3 (The polynomial–logarithmic frame). $(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}, \nu_t, D_X, t^{-1})$ is a reversion frame over $\mathbb{K}$ with $\Gamma = \mathbb{Z}$.

Proof. $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} = \mathbb{K}[L]((t))$ is a domain because $\mathbb{K}[L]$ is, and (F1) is the standard outer valuation together with multiplicativity of leading blocks. For (F2), let (f_i) be summable. Taking $\gamma = n + 1$ shows that only finitely many i have $\nu_t(f_i) \leq n$, so $[t^n] \sum_i f_i := \sum_i [t^n] f_i$ is a finite sum of elements of $\mathbb{K}[L]$; taking $\gamma = 0$ shows that $\inf_i \nu_t(f_i)$ is finite, so the resulting series has only finitely many negative powers of t and lies in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. Linearity, the valuation bound, regrouping and the product rule are then finite-sum statements at each fixed n . For (F3), the block formula $D_X(t^n p(L)) = t^{n+1}(\partial_L - n)p(L)$ gives $\nu_t(D_X f) \geq \nu_t(f) + 1$. Finally $D_X(t^{-1}) = D_X X = 1$, which is (F4). $\square$

Remark 16.4. The same verification, word for word, applies to the Puiseux–log ring $\mathbb{K}[L]((t^{1/s}))$ with $\Gamma = s^{-1}\mathbb{Z}$, to the iterated Laurent-log field $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ with $\Gamma = \mathbb{Z}$, and to the nested-logarithmic ring $\mathbb{K}((L^{-1}))[\log L]((t))$, on which $D_X(t^n p) = t^{n+1}(\partial_L p + L^{-1} \partial_{\log L} p - n p)$. All four are used in [Section 17](#). What is *not* covered is a value group of rank two, such as the refinement of ν_t by logarithmic degree; see [Theorem 17.15](#).

Lemma 16.5 (Substitution calculus). *Let $(\mathfrak{A}, \nu, \partial, X)$ be a reversion frame and let $H \in \mathfrak{A}_{>-1}$. Define*

$$T_H(f) := \sum_{k \geq 0} \frac{H^k}{k!} \partial^k f, \quad f \in \mathfrak{A}, \quad (270)$$

and write $f \circ (X + H) := T_H(f)$. Then:

1. *the family in Equation (270) is summable, its k -th member having $\nu \geq \nu(f) + k(1 + \nu(H))$, and $\nu(T_H f) \geq \nu(f)$;*
2. *T_H is a $\mathbb{K}$ -algebra endomorphism of $\mathfrak{A}$, continuous for ν , with $T_0 = \text{id}$ and $T_H(X) = X + H$;*
3. *$\partial \circ T_H = (1 + \partial H) \cdot (T_H \circ \partial)$;*
4. *for $H, H' \in \mathfrak{A}_{>-1}$, $T_{H'} \circ T_H = T_{H''}$ with $H'' = H' + T_{H'}(H) \in \mathfrak{A}_{>-1}$;*
5. *$\mathcal{G}(\mathfrak{A}) := \{X + H : H \in \mathfrak{A}_{>-1}\}$ is a group under $(X + H) \circ (X + H') := T_{H'}(X + H) = X + H' + T_{H'}(H)$, and T is an anti-homomorphism from it to the monoid of endomorphisms of $\mathfrak{A}$. In the polynomial–logarithmic frame, $\mathcal{G}(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}) = \mathcal{G}_{\text{poly}, \mathbb{K}}$.*

Proof. (1) By (F1) and (F3), $\nu(H^k \partial^k f / k!) \geq k\nu(H) + \nu(f) + k = \nu(f) + k(1 + \nu(H))$, which tends to $+\infty$ because $1 + \nu(H) > 0$; summability and the valuation bound follow from (F2), the minimum over $k \geq 0$ being attained at $k = 0$.

(2) Linearity is clear. For multiplicativity, expand $\partial^k(fg)$ by Leibniz and regroup, which (F2) permits:

$$T_H(fg) = \sum_{k \geq 0} \frac{H^k}{k!} \sum_{i+j=k} \binom{k}{i} \partial^i f \partial^j g = \sum_{i,j \geq 0} \frac{H^i \partial^i f}{i!} \cdot \frac{H^j \partial^j g}{j!} = T_H(f) T_H(g).$$

$T_H(1) = 1$ and $T_H(\lambda) = \lambda$. $T_H(X) = X + H$ because $\partial X = 1$ and $\partial^k X = 0$ for $k \geq 2$. Continuity follows from $\nu(T_H f) \geq \nu(f)$ and linearity.

(3) Differentiate Equation (270) term by term, which is legitimate because ∂ commutes with summable sums:

$$\partial T_H(f) = \sum_{k \geq 1} \frac{k H^{k-1} (\partial H)}{k!} \partial^k f + \sum_{k \geq 0} \frac{H^k}{k!} \partial^{k+1} f = (\partial H) T_H(\partial f) + T_H(\partial f).$$

(4) Let $\mathfrak{A}\langle z \rangle_a$ denote the set of formal series $P = \sum_{m \geq 0} P_m z^m$ with $P_m \in \mathfrak{A}$ and $\nu(P_m) \geq m - a$, and $\mathfrak{A}\langle z \rangle = \bigcup_{a \in \Gamma} \mathfrak{A}\langle z \rangle_a$. By (F1) and (F2) this is a subring of $\mathfrak{A}[[z]]$ with $\mathfrak{A}\langle z \rangle_a \mathfrak{A}\langle z \rangle_b \subseteq \mathfrak{A}\langle z \rangle_{a+b}$, and for each $g \in \mathfrak{A}_{>-1}$ the evaluation $\text{ev}_g(P) = \sum_m P_m g^m$ is a well-defined $\mathbb{K}$ -algebra homomorphism $\mathfrak{A}\langle z \rangle \rightarrow \mathfrak{A}$, because $\nu(P_m g^m) \geq m(1 + \nu(g)) - a \rightarrow +\infty$ makes the family summable and (F2) makes the product rule a rearrangement of a summable double family. Put $\tau(f) = \sum_k (\partial^k f) z^k / k!$; by the computation in (1), $\tau(f) \in \mathfrak{A}\langle z \rangle_{-\nu(f)}$, by the Leibniz argument in (2) τ is a ring homomorphism, and $\partial_z \tau = \tau \partial$, $\text{ev}_0 \tau = \text{id}$, $T_g = \text{ev}_g \circ \tau$.

Fix $f \in \mathfrak{A}$ and consider the doubly indexed family

$$w_{j,k} := \frac{\partial^k f}{j! (k-j)!} \tau(H)^j z^{k-j} \in \mathfrak{A}\langle z \rangle, \quad 0 \leq j \leq k.$$

Its coefficient of z^m is nonzero only when $k - j \leq m$, and $\nu([z^m] w_{j,k}) \geq (\nu(f) + k) + (j\nu(H) + m - k + j) = \nu(f) + m + j(1 + \nu(H))$. Since $k \leq m + j$ and $1 + \nu(H) > 0$, for each fixed m and each γ only finitely many pairs (j, k) give valuation below γ ; the family is therefore summable in every z -degree, and may be regrouped freely. Summing first over k and then over j gives $\sum_j \tau(H)^j \partial_z^j \tau(f) / j!$, which by $\partial_z^j \tau = \tau \partial^j$ and multiplicativity of τ equals $\tau(T_H f)$; summing first over j and then over k , and using the binomial theorem, gives $\sum_k (\partial^k f / k!) (z + \tau(H))^k$. Hence

$$\tau(T_H f) = \sum_{k \geq 0} \frac{\partial^k f}{k!} (z + \tau(H))^k.$$

Applying the ring homomorphism $\text{ev}_{H'}$, which likewise commutes with the regrouping of this summable family, and using $\text{ev}_{H'}\tau(H) = T_{H'}(H)$ and $\text{ev}_{H'}(z) = H'$, gives $T_{H'}(T_H f) = \sum_k (\partial^k f)(H')(H' + T_{H'}H)^k/k! = T_{H'}(f)$. Finally $\nu(H'') \geq \min\{\nu(H'), \nu(T_{H'}H)\} \geq \min\{\nu(H'), \nu(H)\} > -1$.

(5) Part (4) says exactly that $\circ$ is associative on $\mathcal{G}(\mathfrak{A})$ and that $T_{F \circ G} = T_G \circ T_F$; the element X is a two-sided identity. By [Theorem 16.6](#) below every element has a right inverse, and a monoid in which every element has a right inverse is a group: if $ab = X$ and $bc = X$ then $ba = ba(bc) = b(ab)c = bc = X$. There is no circularity, because the proof of [Theorem 16.6](#) uses only parts (1) and (2). $\square$

Proposition 16.6 (Right inverse: existence, uniqueness, characterisation). *Let $(\mathfrak{A}, \nu, \partial, X)$ be a reversion frame, $A \in \mathfrak{A}_{>-1}$, and $F = X + A$. There is exactly one $B \in \mathfrak{A}_{>-1}$ with*

$$F \circ (X + B) = X, \quad \text{equivalently} \quad B = -T_B(A) = -A \circ (X + B). \quad (271)$$

It satisfies $\nu(B) \geq \nu(A)$. We write $F^{(-1)\circ} = X + B$.

Proof. The two displayed conditions are the same because $F \circ (X + B) = T_B(X + A) = X + B + T_B(A)$.

A contraction estimate. Let $B, B' \in \mathfrak{A}_{>-1}$ and put $m = \min\{\nu(B), \nu(B')\} > -1$. Then

$$\nu(T_{B'}(A) - T_B(A)) \geq \nu(B' - B) + \delta, \quad \delta = 1 + \nu(A) > 0. \quad (272)$$

Indeed $T_{B'}(A) - T_B(A) = \sum_{k \geq 1} (B'^k - B^k) \partial^k A / k!$, the $k = 0$ terms cancelling, and for $k \geq 1$ $B'^k - B^k = (B' - B) \sum_{i=0}^{k-1} B'^i B^{k-1-i}$ has valuation at least $\nu(B' - B) + (k-1)m$, while $\nu(\partial^k A) \geq \nu(A) + k$. The k -th term therefore has valuation at least $\nu(B' - B) + \nu(A) + k(1+m) - m$, which is increasing in k because $1+m > 0$; its value at $k = 1$ is $\nu(B' - B) + \nu(A) + 1$.

Uniqueness. If B, B' both solve [Equation \(271\)](#), then [Equation \(272\)](#) gives $\nu(B - B') \geq \nu(B' - B) + \delta$ with $\delta > 0$, forcing $\nu(B - B') = +\infty$.

Existence. Set $B_0 = 0$ and $B_{n+1} = -T_{B_n}(A)$; by [Theorem 16.5\(1\)](#), $\nu(B_{n+1}) \geq \nu(A) > -1$, so every B_n lies in $\mathfrak{A}_{>-1}$ and [Equation \(272\)](#) applies with $m \geq \min\{0, \nu(A)\}$. Hence $\nu(B_{n+1} - B_n) \geq \nu(B_n - B_{n-1}) + \delta$, so $\nu(B_{n+1} - B_n) \geq \nu(A) + n\delta$. The family $(B_{n+1} - B_n)_{n \geq 0}$ is summable, and $B := \sum_{n \geq 0} (B_{n+1} - B_n)$ satisfies $\nu(B - B_n) \geq \nu(A) + n\delta$. Applying [Equation \(272\)](#) once more, $\nu(T_B(A) - T_{B_n}(A)) \geq \nu(A) + (n+1)\delta$, so $\nu(B + T_B(A)) \geq \min\{\nu(B - B_{n+1}), \nu(T_{B_n}A - T_BA)\} \rightarrow +\infty$ and $B = -T_B(A)$. Finally $\nu(B) = \nu(T_B(A)) \geq \nu(A)$. $\square$

Remark 16.7. The characterisation [Equation \(271\)](#) is the only property of $F^{(-1)\circ}$ used below; the two-sidedness supplied by [Theorem 16.5\(5\)](#) is never needed in the proof of the Lagrange–Bürmann formula and is recorded only so that “the” inverse is unambiguous.

16.2 Formal summability

Lemma 16.8 (Order of the r -th Lagrange term). *Let $(\mathfrak{A}, \nu, \partial, X)$ be a reversion frame and $A \in \mathfrak{A}$. For every $r \geq 1$,*

$$\nu\left(\partial^{r-1}(A^r)\right) \geq r\nu(A) + r - 1 = r\delta - 1, \quad \delta = 1 + \nu(A). \quad (273)$$

If $A \in \mathfrak{A}_{>-1}$, so that $\delta > 0$, the family $((-1)^r \partial^{r-1}(A^r)/r!)_{r \geq 1}$ is summable, and its sum S satisfies $\nu(S) \geq \nu(A) > -1$, whence $S \in \mathfrak{A}_{>-1}$.

Proof. $\nu(A^r) = r\nu(A)$ by (F1) and $r - 1$ applications of (F3) add at least $r - 1$. If $\delta > 0$ the right-hand side of [Equation \(273\)](#) tends to $+\infty$, so the family is summable by (F2); its minimum over $r \geq 1$ is attained at $r = 1$ and equals $\nu(A)$, which bounds $\nu(S)$. $\square$

The hypothesis $\nu(A) > -1$ is not merely convenient: it is exactly the summability condition, in the polynomial–logarithmic frame and in its Puiseux–log enlargements alike.

Proposition 16.9 (Sharp summability criterion). *Fix $s \in \mathbb{N}_{>0}$ and let $A \in \mathbb{K}[L]((t^{1/s}))$ be nonzero, with $\alpha := \nu_t(A) \in s^{-1}\mathbb{Z}$. The family $((-1)^r D_X^{r-1}(A^r)/r!)_{r \geq 1}$ is summable if and only if $\alpha > -1$. Indeed, if $\alpha \leq -1$ then*

$$\nu_t\left(D_X^{r-1}(A^r)\right) = r(\alpha + 1) - 1 \quad \text{for every } r \geq 1, \quad (274)$$

which is nonincreasing in r and hence never tends to $+\infty$. For $s = 1$, that is in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, the value group is $\mathbb{Z}$ and the criterion reads $\nu_t(A) \geq 0$, i.e. $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.

Proof. Sufficiency is [Theorem 16.8](#). For necessity assume $\alpha \leq -1$ and write $A = t^\alpha p(L) + A_1$ with $p \neq 0$ and $\nu_t(A_1) > \alpha$. Then $A^r = t^{r\alpha} p^r + (\text{terms of } \nu_t > r\alpha)$. Applying the repeated-derivative formula $D_X^{r-1}(t^n p(L)) = t^{n+r-1} \Delta_{n, r-1} p(L)$ with $n = r\alpha$,

$$D_X^{r-1}(A^r) = t^{r\alpha+r-1} \prod_{j=0}^{r-2} (\partial_L - r\alpha - j) p(L)^r + (\text{terms of } \nu_t > r\alpha + r - 1).$$

For $0 \leq j \leq r-2$ one has $-r\alpha - j \geq r - (r-2) = 2 > 0$, so each factor $\partial_L - (r\alpha + j)$ multiplies the leading coefficient of a nonzero polynomial by the nonzero scalar $-r\alpha - j$ and in particular is injective on $\mathbb{K}[L] \setminus \{0\}$. Hence the displayed leading block is nonzero and [Equation \(274\)](#) holds. Since $\alpha + 1 \leq 0$, the exponent $r(\alpha + 1) - 1$ is at most -1 for all r , so $\{r : \nu_t < 0\}$ is infinite and the family is not summable. $\square$

Remark 16.10. [Theorem 16.9](#) makes precise what “near identity” means here, and it is *not* that A is bounded. For $A = L = \log X$ one has $A \rightarrow +\infty$ yet $\nu_t(A) = 0 > -1$, and the theory applies; for $A = X/L$ one has $A = o_{X \rightarrow +\infty}(X)$ yet $\nu_t(A) = -1$, and it does not. The decisive quantity is the outer order measured against the threshold $-1 = \nu_t(X)$, equivalently the dominance comparison $A \prec X$ refined to the outer filtration: logarithmic gains are invisible to ν_t , and a perturbation of the same outer order as X itself is never admissible however many logarithms separate it from X .

16.3 The parameter identity

The classical Lagrange–Bürmann theorem is a statement about a formal parameter. We prove the version we need – the *shifted*, or reversion, form – from scratch, over an arbitrary commutative $\mathbb{Q}$ -algebra and with no invertibility hypothesis whatsoever.

Lemma 16.11 (Lagrange reversion identity). *Let C be a commutative $\mathbb{Q}$ -algebra and let $\psi \in C[[z]]$. For $f \in C[[z]]$ and $w \in \varepsilon C[[z]][[\varepsilon]]$ write $f_{[w]} := \sum_{k \geq 0} (\partial_z^k f) w^k / k!$. There is exactly one $u \in \varepsilon C[[z]][[\varepsilon]]$ with*

$$u = \varepsilon \psi_{[u]}, \quad (275)$$

and for every $\gamma \in C[[z]]$ and every $m \geq 1$,

$$[\varepsilon^m] \gamma_{[u]} = \frac{1}{m!} \partial_z^{m-1} (\gamma' \psi^m), \quad \gamma' := \partial_z \gamma. \quad (276)$$

Proof. Existence and uniqueness of u : comparing coefficients of ε^m in [Equation \(275\)](#) determines $[\varepsilon^m]u$ from $[\varepsilon^1]u, \dots, [\varepsilon^{m-1}]u$, because w^k contributes to ε^m only for $k \leq m$ and then only through those coefficients.

For fixed u , the map $f \mapsto f_{[u]}$ is a C -algebra homomorphism of $C[[z]]$ into $C[[z]][[\varepsilon]]$ (Leibniz, exactly as in [Theorem 16.5\(2\)](#)), and differentiating its defining series term by term gives the two chain rules

$$\partial_\varepsilon(f_{[u]}) = f'_{[u]} \partial_\varepsilon u, \quad \partial_z(f_{[u]}) = f'_{[u]} (1 + \partial_z u). \quad (277)$$

Applying [Equation \(277\)](#) to [Equation \(275\)](#),

$$\partial_\varepsilon u = \psi'_{[u]} + \varepsilon \psi'_{[u]} \partial_\varepsilon u, \quad \partial_z u = \varepsilon \psi'_{[u]} (1 + \partial_z u).$$

Since $\varepsilon\psi'_{[u]} \in \varepsilon C[[z]][[\varepsilon]]$, the element $1 - \varepsilon\psi'_{[u]}$ is a unit; solving the second relation gives $1 + \partial_z u = (1 - \varepsilon\psi'_{[u]})^{-1}$, and substituting into the first gives the key identity

$$\partial_\varepsilon u = \psi_{[u]}(1 + \partial_z u). \quad (278)$$

Now let $\gamma \in C[[z]]$ be arbitrary and let $\Gamma \in C[[z]]$ be the formal antiderivative of $\gamma'\psi$ with $\Gamma(0) = 0$, which exists because C is a $\mathbb{Q}$ -algebra. Using Equation (277), Equation (278) and multiplicativity of $f \mapsto f_{[u]}$,

$$\partial_\varepsilon(\gamma_{[u]}) = \gamma'_{[u]}\psi_{[u]}(1 + \partial_z u) = (\gamma'\psi)_{[u]}(1 + \partial_z u) = \Gamma'_{[u]}(1 + \partial_z u) = \partial_z(\Gamma_{[u]}).$$

Extracting $[\varepsilon^{m-1}]$ from both ends and using $[\varepsilon^{m-1}]\partial_\varepsilon = m[\varepsilon^m]$,

$$m[\varepsilon^m]\gamma_{[u]} = \partial_z([\varepsilon^{m-1}]\Gamma_{[u]}), \quad m \geq 1. \quad (279)$$

We prove Equation (276) by induction on m , the statement being quantified over all $\gamma \in C[[z]]$. For $m = 1$, $[\varepsilon^0]\Gamma_{[u]} = \Gamma$ because $u \in \varepsilon C[[z]][[\varepsilon]]$, so Equation (279) gives $[\varepsilon^1]\gamma_{[u]} = \Gamma' = \gamma'\psi$, which is Equation (276) with $m = 1$. For $m \geq 2$, the inductive hypothesis applied to Γ gives $[\varepsilon^{m-1}]\Gamma_{[u]} = \partial_z^{m-2}(\Gamma'\psi^{m-1})/(m-1)! = \partial_z^{m-2}(\gamma'\psi^m)/(m-1)!$, and Equation (279) yields $m[\varepsilon^m]\gamma_{[u]} = \partial_z^{m-1}(\gamma'\psi^m)/(m-1)!$. $\square$

Remark 16.12. A second argument runs through formal residues. Writing res_ε for the coefficient of ε^{-1} in a formal Laurent series, one extracts $[\varepsilon^m]$ as a residue and substitutes $\varepsilon = z/\psi(z)$. That change of variables is legitimate only when $\psi(0)$ is invertible, so the general case must be recovered by performing the computation in the universal polynomial ring generated by the coefficients of ψ with the constant coefficient inverted, observing that both sides are polynomial in finitely many of those generators at each fixed m , and specialising. The route is instructive, but the ∂_ε – ∂_z exchange Equation (278) used above needs no localisation, no residue theory, and no hypothesis on $\psi(0)$. Classical accounts of formal Lagrange inversion are in [4, 18].

16.4 Specialising the bookkeeping parameter

Every source we merge performs the step “now set $\varepsilon = 1$ ” and justifies it by the sentence that only finitely many terms contribute below any fixed outer order. That sentence is true but it is not a proof: evaluation at $\varepsilon = 1$ has to be a *ring homomorphism* defined on the subalgebra in which the computation takes place, and the fixed-point equation Equation (275) contains an infinite sum that must be shown to survive it. The following lemma supplies the missing object.

Lemma 16.13 (Weighted deformation algebra and evaluation at one). *Let $(\mathfrak{A}, \nu, \partial, X)$ be a reversion frame and fix a weight $\delta \in \Gamma_{>0}$. For $a \in \Gamma$ put*

$$\mathfrak{A}\langle\varepsilon\rangle_a := \left\{ u = \sum_{m \geq 0} u_m \varepsilon^m \in \mathfrak{A}[[\varepsilon]] : \nu(u_m) \geq m\delta - a \text{ for all } m \right\}, \quad \mathfrak{A}\langle\varepsilon\rangle = \bigcup_{a \in \Gamma} \mathfrak{A}\langle\varepsilon\rangle_a.$$

Then:

1. $\mathfrak{A}\langle\varepsilon\rangle$ is a subring of $\mathfrak{A}[[\varepsilon]]$ containing $\mathfrak{A}$ and ε , with $\mathfrak{A}\langle\varepsilon\rangle_a \mathfrak{A}\langle\varepsilon\rangle_b \subseteq \mathfrak{A}\langle\varepsilon\rangle_{a+b}$, and it is stable under the coefficientwise extension of ∂ ;
2. $\text{ev}: \mathfrak{A}\langle\varepsilon\rangle \rightarrow \mathfrak{A}$, $\text{ev}(u) = \sum_{m \geq 0} u_m$, is a well-defined $\mathbb{K}$ -algebra homomorphism with $\text{ev} \circ \partial = \partial \circ \text{ev}$, $\text{ev}(\varepsilon) = 1$, and $\nu(\text{ev}(u)) \geq -a$ for $u \in \mathfrak{A}\langle\varepsilon\rangle_a$;
3. if $(f^{(k)})_{k \geq 0}$ is a family in $\mathfrak{A}\langle\varepsilon\rangle_a$ with $f^{(k)} \in \varepsilon^k \mathfrak{A}[[\varepsilon]]$, then $\sum_k f^{(k)} \in \mathfrak{A}\langle\varepsilon\rangle_a$, the family $(\text{ev } f^{(k)})_k$ is summable in $\mathfrak{A}$, and $\text{ev}(\sum_k f^{(k)}) = \sum_k \text{ev}(f^{(k)})$.

Proof. (1) If $\nu(u_m) \geq m\delta - a$ and $\nu(v_m) \geq m\delta - b$ then $\nu(\sum_{i+j=m} u_i v_j) \geq \min_{i+j=m} (i\delta - a + j\delta - b) = m\delta - a - b$. Stability under ∂ follows from (F3), which gives $\nu(\partial u_m) \geq m\delta - (a - 1)$.

(2) For $u \in \mathfrak{A}\langle \varepsilon \rangle_a$ the family $(u_m)_{m \geq 0}$ is summable, since $\nu(u_m) \geq m\delta - a \rightarrow +\infty$; the sum exists by (F2) and has $\nu \geq \inf_m (m\delta - a) = -a$. Additivity is clear, and multiplicativity is the product rule of (F2): $\text{ev}(uv) = \sum_m \sum_{i+j=m} u_i v_j = \sum_{i,j} u_i v_j = (\sum_i u_i)(\sum_j v_j)$, the regrouping being licensed because $\nu(u_i v_j) \geq (i + j)\delta - a - b$ makes the double family summable. Commutation with ∂ is the continuity of ∂ .

(3) Since $f^{(k)} \in \varepsilon^k \mathfrak{A}[[\varepsilon]]$, for each m only the indices $k \leq m$ contribute to $[\varepsilon^m] \sum_k f^{(k)}$, so the sum is defined coefficientwise and lies in $\mathfrak{A}\langle \varepsilon \rangle_a$. The doubly indexed family $([\varepsilon^m] f^{(k)})_{k,m}$ is summable: it vanishes for $k > m$, and $\nu([\varepsilon^m] f^{(k)}) \geq m\delta - a$, so $\{(k, m) : \nu < \gamma\} \subseteq \{(k, m) : m\delta < \gamma + a, k \leq m\}$ is finite. Both iterated sums therefore equal the unordered sum, by the regrouping clause of (F2). $\square$

16.5 The formula and the transport formula

Theorem 16.14 (Lagrange–Bürmann reversion at infinity). *Let $(\mathfrak{A}, \nu, \partial, X)$ be a reversion frame and let $A \in \mathfrak{A}_{>-1}$, $F = X + A$. Then the family $((-1)^r \partial^{r-1}(A^r)/r!)_{r \geq 1}$ is summable and*

$$F^{(-1)^\circ} = X + \sum_{r \geq 1} \frac{(-1)^r}{r!} \partial^{r-1}(A^r). \quad (280)$$

In the polynomial–logarithmic frame this reads

$$(X + A)^{(-1)^\circ}(X) = X + \sum_{r \geq 1} \frac{(-1)^r}{r!} D_X^{r-1}(A(X)^r), \quad A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}, \quad (281)$$

and the r -th summand has $\nu_t \geq r - 1$.

Proof. Summability is Theorem 16.8; call the sum S , so $\nu(S) \geq \nu(A) > -1$. By Theorem 16.6 it suffices to prove $S = -T_S(A)$. Write $\delta = 1 + \nu(A) > 0$ and use this δ as the weight in Theorem 16.13.

Take $C = \mathfrak{A}$ in Theorem 16.11 and let $\tau : \mathfrak{A} \rightarrow \mathfrak{A}[[z]]$, $\tau(f) = \sum_k (\partial^k f) z^k / k!$, be the Taylor-shift homomorphism of Theorem 16.5(4); recall $\partial_z \tau = \tau \partial$ and $\text{ev}_{z=0} \tau = \text{id}$. Put

$$\psi := \tau(-A) \in \mathfrak{A}[[z]],$$

and let $u \in \varepsilon \mathfrak{A}[[z]][[\varepsilon]]$ be the unique solution of Equation (275). Apply Equation (276) with $\gamma = \tau(X) = X + z$, so that $\gamma' = 1 = \tau(1)$:

$$[\varepsilon^m] \gamma_{[u]} = \frac{1}{m!} \partial_z^{m-1}(\psi^m) = \frac{(-1)^m}{m!} \partial_z^{m-1} \tau(A^m) = \frac{(-1)^m}{m!} \tau(\partial^{m-1}(A^m)),$$

using that τ is a ring homomorphism intertwining ∂_z with ∂ . Let $\text{ev}_{z=0} : \mathfrak{A}[[z]][[\varepsilon]] \rightarrow \mathfrak{A}[[\varepsilon]]$ act coefficientwise in ε ; it is a ring homomorphism. Setting $U := \text{ev}_{z=0}(u) \in \varepsilon \mathfrak{A}[[\varepsilon]]$ and using $\gamma_{[u]} = \sum_k (\partial_z^k \gamma) u^k / k! = (X + z) + u$, we obtain

$$U = \sum_{m \geq 1} \frac{(-1)^m}{m!} \partial^{m-1}(A^m) \varepsilon^m. \quad (282)$$

Likewise, applying $\text{ev}_{z=0}$ to Equation (275) and using $\psi_{[u]} = \sum_k (\partial_z^k \psi) u^k / k!$ together with $\text{ev}_{z=0} \partial_z^k \tau(-A) = \partial^k(-A)$,

$$U = -\varepsilon \sum_{k \geq 0} \frac{\partial^k A}{k!} U^k. \quad (283)$$

Now specialise. By Equation (282) and Theorem 16.8, $\nu([\varepsilon^m] U) \geq m\delta - 1$, so $U \in \mathfrak{A}\langle \varepsilon \rangle_1$ and $\text{ev}(U) = S$ by construction. Since U has zero constant term, $[\varepsilon^m](U^k) = 0$ for $m < k$ and,

splitting $m = m_1 + \dots + m_k$ with every $m_i \geq 1$, $\nu([\varepsilon^m](U^k)) \geq \sum_i (m_i \delta - 1) = m\delta - k$; with $\nu(\partial^k A) \geq \nu(A) + k \geq k - 1 + \delta \geq k \cdot \min\{1, \delta\} -$ more simply, with $\nu(\partial^k A) \geq \nu(A) + k -$ this gives

$$\nu([\varepsilon^m] \left(\frac{\partial^k A}{k!} U^k \right)) \geq (\nu(A) + k) + (m\delta - k) = m\delta + \nu(A), \quad \frac{\partial^k A}{k!} U^k \in \varepsilon^k \mathfrak{A}[[\varepsilon]].$$

So [Theorem 16.13\(3\)](#) applies to $f^{(k)} = \partial^k A U^k / k!$ with $a = -\nu(A)$, and applying ev to [Equation \(283\)](#) gives

$$S = \text{ev}(U) = - \sum_{k \geq 0} \frac{\partial^k A}{k!} \text{ev}(U)^k = - \sum_{k \geq 0} \frac{S^k}{k!} \partial^k A = -T_S(A).$$

By [Theorem 16.6](#), $X + S = F^{(-1)\circ}$. □

The same construction, run with a general observable in place of X , produces the expansion of any transseries along the inverse map without first building the inverse.

Theorem 16.15 (Transport formula for a composed observable). *Let $(\mathfrak{A}, \nu, \partial, X)$ be a reversion frame, let $A \in \mathfrak{A}_{\geq 0}$, $F = X + A$, and let $\varphi \in \mathfrak{A}$ be arbitrary. Then the family $((-1)^r \partial^{r-1}((\partial\varphi)A^r)/r!)_{r \geq 1}$ is summable, the r -th member having*

$$\nu\left(\partial^{r-1}((\partial\varphi)A^r)\right) \geq \nu(\varphi) + r(1 + \nu(A)) \geq \nu(\varphi) + r, \quad (284)$$

and

$$\varphi \circ F^{(-1)\circ} = \varphi + \sum_{r \geq 1} \frac{(-1)^r}{r!} \partial^{r-1}((\partial\varphi)A^r). \quad (285)$$

In particular $[t^N](\varphi \circ F^{(-1)\circ})$ in the polynomial–logarithmic frame receives contributions only from $r \leq N - \nu_t(\varphi)$.

Proof. For [Equation \(284\)](#): $\nu(\partial\varphi) \geq \nu(\varphi) + 1$ by (F3), $\nu(A^r) = r\nu(A)$ by (F1), and $r - 1$ further derivatives add at least $r - 1$; the total is $\nu(\varphi) + 1 + r\nu(A) + r - 1 = \nu(\varphi) + r(1 + \nu(A))$, which is at least $\nu(\varphi) + r \rightarrow +\infty$. Summability follows from (F2).

Run the proof of [Theorem 16.14](#) with $\gamma := \tau(\varphi)$ in [Equation \(276\)](#). Since $\gamma' = \partial_z \tau(\varphi) = \tau(\partial\varphi)$ and τ is a ring homomorphism intertwining ∂_z with ∂ ,

$$[\varepsilon^m] \gamma_{[u]} = \frac{1}{m!} \partial_z^{m-1} (\tau(\partial\varphi) \tau(-A)^m) = \frac{(-1)^m}{m!} \tau\left(\partial^{m-1}((\partial\varphi)A^m)\right).$$

Applying $\text{ev}_{z=0}$ and writing $c_m := (-1)^m \partial^{m-1}((\partial\varphi)A^m)/m!$,

$$\sum_{k \geq 0} \frac{\partial^k \varphi}{k!} U^k = \varphi + \sum_{m \geq 1} c_m \varepsilon^m \quad \text{in } \mathfrak{A}[[\varepsilon]], \quad (286)$$

where U is as in [Equation \(282\)](#). By [Equation \(284\)](#), $\nu(c_m) \geq \nu(\varphi) + m$, so the right-hand side of [Equation \(286\)](#) lies in $\mathfrak{A}\langle\varepsilon\rangle_{-\nu(\varphi)}$. On the left, $\nu([\varepsilon^m](\partial^k \varphi U^k/k!)) \geq (\nu(\varphi) + k) + (m - k) = \nu(\varphi) + m$ and $\partial^k \varphi U^k/k! \in \varepsilon^k \mathfrak{A}[[\varepsilon]]$, so [Theorem 16.13\(3\)](#) applies with $a = -\nu(\varphi)$. Applying ev to [Equation \(286\)](#) and using $\text{ev}(U) = S$ with $X + S = F^{(-1)\circ}$ gives $T_S(\varphi) = \varphi + \sum_{m \geq 1} c_m$, which is [Equation \(285\)](#). □

Corollary 16.16. *With $F = X + A$, $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$:*

$$F^{(-1)\circ} = X + \sum_{r \geq 1} \frac{(-1)^r}{r!} D_X^{r-1}(A^r) \quad (\varphi = X), \quad (287)$$

$$\log F^{(-1)\circ} = L + \sum_{r \geq 1} \frac{(-1)^r}{r!} D_X^{r-1}(t A^r) \quad (\varphi = L), \quad (288)$$

$$(F^{(-1)\circ})^{-q} = t^q - q \sum_{r \geq 1} \frac{(-1)^r}{r!} D_X^{r-1}(t^{q+1} A^r) \quad (\varphi = t^q, q \in \mathbb{Z}). \quad (289)$$

Proof. Substitute $\varphi = X = t^{-1}$, $\varphi = L$ and $\varphi = t^q$ in Equation (285), using $D_X X = 1$, $D_X L = t$ and $D_X t^q = -q t^{q+1}$. In Equation (288) the left-hand side means the observable L evaluated along $F^{(-1)\circ}$, that is $L \circ F^{(-1)\circ} = \log(F^{(-1)\circ})$ in the sense of the generator substitution $L \circ (X + B) = L + \log(1 + tB)$. $\square$

Example 16.17. For $A = L$, Equation (288) gives, with $\omega = (X + L)^{(-1)\circ}$,

$$\log \omega = L - tL + t^2 \left(L - \frac{L^2}{2} \right) + t^3 \left(-\frac{L^3}{3} + \frac{3}{2} L^2 - L \right) + \mathcal{O}_t^{\text{form}}(t^4),$$

which is exactly $X - \omega$ read off from the defining relation $\omega + \log \omega = X$, and therefore a nontrivial check of both Theorem 16.15 and the coefficient table for $\mathbb{L}_n^{\text{Lam}}$.

16.6 The finite coefficient formula and its exact index range

Theorem 16.18 (Coefficient formula in the polynomial–logarithmic basis). *Let $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, $F = X + A$, and write*

$$A = \sum_{m \geq 0} a_m(L) t^m, \quad A^r = \sum_{m \geq 0} A_{r,m}(L) t^m, \quad F^{(-1)\circ} = X + \sum_{N \geq 0} b_N(L) t^N.$$

Then for every $N \in \mathbb{N}_0$,

$$b_N(L) = \sum_{r=1}^{N+1} \frac{(-1)^r}{r!} \left[\prod_{s=N-r+1}^{N-1} (\partial_L - s) \right] A_{r, N-r+1}(L), \quad (290)$$

where a product whose lower bound exceeds its upper bound is empty and acts as the identity; this occurs exactly at $r = 1$.

Proof. By Theorem 16.14 and the summability of the family, the coefficient of t^N may be computed term by term. For fixed $r \geq 1$ and $m \geq 0$, the repeated-derivative formula gives

$$D_X^{r-1}(A_{r,m}(L)t^m) = t^{m+r-1} \Delta_{m,r-1} A_{r,m}(L), \quad \Delta_{m,r-1} = \prod_{j=0}^{r-2} (\partial_L - m - j).$$

This contributes to t^N exactly when $m = N - r + 1$, and then the parameters $m + j$ with $0 \leq j \leq r - 2$ run over the integers $N - r + 1, N - r + 2, \dots, N - 1$, so $\Delta_{N-r+1, r-1} = \prod_{s=N-r+1}^{N-1} (\partial_L - s)$. The constraint $m \geq 0$ is $r \leq N + 1$. For $r = 1$ the operator is $\Delta_{N,0} = 1$, the empty product. $\square$

Proposition 16.19 (The range $r \leq N + 1$ is exactly right). *Fix $N \in \mathbb{N}_0$ and let $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ be nonzero.*

1. *No summand with $r > N + 1$ contributes to b_N : indeed $\nu_t(D_X^{r-1}(A^r)) \geq r - 1 > N$.*
2. *The bound cannot be lowered. For $N \geq 1$, the summand $r = N + 1$ contributes a nonzero polynomial to Equation (290) if and only if $\deg_L a_0 \geq 1$; for $N = 0$ the only summand is $r = 1$ and $b_0 = -a_0 \neq 0$. In particular, for $A = L$ every $N \geq 0$ attains the bound, so no uniform improvement of the range is possible.*

Proof. (1) is Theorem 16.8. For (2), the $r = N + 1$ summand of Equation (290) is $\frac{(-1)^{N+1}}{(N+1)!} [\prod_{s=0}^{N-1} (\partial_L - s)] a_0^{N+1}$, because $A_{N+1,0} = a_0^{N+1}$. Write $q = a_0^{N+1}$; since $\mathbb{K}[L]$ is a domain, $q \neq 0$ and $\deg_L q = (N + 1) \deg_L a_0$. If $\deg_L a_0 = 0$ then q is a nonzero constant and $\partial_L q = 0$, so the product vanishes. If $\deg_L a_0 \geq 1$ then $\partial_L q \neq 0$, and for $s \geq 1$ the operator $\partial_L - s$ multiplies the leading coefficient of any nonzero polynomial by $-s \neq 0$, hence is injective; so the whole product is nonzero. For $A = L$ one has $a_0 = L$ and $\deg_L a_0 = 1$. $\square$

Remark 16.20. [Theorem 16.19](#) is the exact stopping rule for computation: $\text{Trunc}_{<N+1} F^{(-1)\circ}$, that is, the inverse through outer order t^N inclusive, needs A^r only for $r \leq N+1$ and only through outer order t^{N-r+1} . In the inclusive notation of report 6 this is $[F^{(-1)\circ}]_{\leq N} = \text{Trunc}_{<N+1} F^{(-1)\circ}$; the one-block difference between the two conventions is a frequent source of off-by-one errors when tables are transferred between reports.

16.7 The degree law

Reversion is finite at every outer block, but the *logarithmic* degrees inside those blocks grow. Recall the log-degree profile $d_F(n) = \deg_L p_n$, with $d_F(n) = -\infty$ when $p_n = 0$.

Theorem 16.21 (Degree law for linearly bounded profiles). *Let $A \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, $F = X + A$, $F^{(-1)\circ} = X + B$. Suppose there are real $\alpha, \beta \geq 0$ with*

$$d_A(n) \leq \alpha n + \beta \quad \text{for all } n \in \mathbb{N}_0. \quad (291)$$

Then

$$d_B(N) \leq N \max\{\alpha, \beta\} + \beta \quad \text{for all } N \in \mathbb{N}_0. \quad (292)$$

More precisely, $d_B(0) \leq \beta$ and, for $N \geq 1$,

$$d_B(N) \leq \max\left\{(N+1)\beta - 1, \max_{1 \leq r \leq N} [\alpha(N-r+1) + r\beta]\right\}. \quad (293)$$

In particular the class of perturbations with linearly bounded log-degree profile is stable under reversion.

Proof. Every operator $\partial_L - s$ satisfies $\deg_L((\partial_L - s)p) \leq \deg_L p$, with the strict bound $\deg_L(\partial_L p) \leq \deg_L p - 1$ when $s = 0$. In [Equation \(290\)](#) the operator attached to index r is $\prod_{s=N-r+1}^{N-1} (\partial_L - s)$; the value $s = 0$ occurs in that range precisely when $N-r+1 \leq 0 \leq N-1$, that is (given $1 \leq r \leq N+1$) when $N \geq 1$ and $r = N+1$.

Next, $A_{r,m} = \sum_{m_1+\dots+m_r=m} a_{m_1} \cdots a_{m_r}$, so [Equation \(291\)](#) gives $\deg_L A_{r,m} \leq \sum_i (\alpha m_i + \beta) = \alpha m + r\beta$. Hence the r -th summand of [Equation \(290\)](#) has degree at most $\alpha(N-r+1) + r\beta$, improved by one when $r = N+1$ and $N \geq 1$, where $m = 0$ and the bound is $(N+1)\beta - 1$. Taking the maximum over $1 \leq r \leq N+1$ gives [Equation \(293\)](#), and $d_B(0) = \deg_L(-a_0) \leq \beta$.

For [Equation \(292\)](#), note $\alpha(N-r+1) + r\beta = \alpha N + \alpha + r(\beta - \alpha)$ is monotone in r , so on $1 \leq r \leq N$ it is at most $\max\{\alpha N + \beta, \alpha + N\beta\} \leq N \max\{\alpha, \beta\} + \beta$, while $(N+1)\beta - 1 < N\beta + \beta \leq N \max\{\alpha, \beta\} + \beta$. $\square$

The bound is attained, with an exact leading coefficient, in the situation the sources treat.

Theorem 16.22 (Exact degree and leading coefficient). *Let $A = \sum_{n \geq 0} a_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ with*

$$\deg_L a_0 = d \geq 1, \quad a_d := [L^d]a_0 \neq 0, \quad \deg_L a_n \leq d-1 \quad \text{for all } n \geq 1.$$

Then $b_0 = -a_0$ has degree d , and for every $N \geq 1$

$$\deg_L b_N = d(N+1) - 1, \quad [L^{d(N+1)-1}]b_N = \frac{d}{N} a_d^{N+1}. \quad (294)$$

Thus $d_B(N) = dN + (d-1)$ for $N \geq 1$: the logarithmic degree of the leading block of the perturbation becomes the slope of the inverse's log-degree profile.

Proof. Consider the summands of [Equation \(290\)](#) for $N \geq 1$.

The summand $r = N+1$. It equals $\frac{(-1)^{N+1}}{(N+1)!} \partial_L(\partial_L - 1) \cdots (\partial_L - N + 1) a_0^{N+1}$. The polynomial a_0^{N+1} has degree $d(N+1)$ and leading coefficient a_d^{N+1} . The factor ∂_L lowers the degree by one and multiplies the leading coefficient by $d(N+1)$; each factor $\partial_L - s$ with $1 \leq s \leq N-1$ preserves the

degree and multiplies the leading coefficient by $-s$. Hence this summand has degree $d(N+1) - 1$ and leading coefficient

$$\frac{(-1)^{N+1}}{(N+1)!} d(N+1) a_d^{N+1} (-1)^{N-1} (N-1)! = \frac{d(N+1)(N-1)!}{(N+1)!} a_d^{N+1} = \frac{d}{N} a_d^{N+1},$$

which is nonzero because $\mathbb{K}$ has characteristic zero.

The summands $1 \leq r \leq N$. Here $m = N - r + 1 \geq 1$, so in every product $a_{m_1} \cdots a_{m_r}$ occurring in $A_{r,m}$ at least one index m_i is ≥ 1 and contributes degree at most $d-1$, the remaining $r-1$ factors contributing at most d each. Therefore $\deg_L A_{r,N-r+1} \leq rd - 1 \leq Nd - 1 < d(N+1) - 1$, and the attached operator does not raise degrees.

Adding the summands, the term of degree $d(N+1) - 1$ survives with the stated coefficient. $\square$

Corollary 16.23 (Pure polynomial–logarithmic perturbations). *Let $P \in \mathbb{K}[L]$ with $\deg_L P = d \geq 1$ and leading coefficient a_d , and let $F = X + P(L)$. Then only the summand $r = N+1$ contributes to b_N , so*

$$F^{(-1)\circ} = X - P(L) + \sum_{N \geq 1} b_N(L) t^N, \quad b_N(L) = \frac{(-1)^{N+1}}{(N+1)!} \left[\prod_{s=0}^{N-1} (\partial_L - s) \right] P(L)^{N+1}, \quad (295)$$

and $\deg_L b_N = d(N+1) - 1$ with $[L^{d(N+1)-1}] b_N = (d/N) a_d^{N+1}$ for $N \geq 1$. For $P = aL$ this specialises to $b_N = a^{N+1} \mathbb{L}_N^{\text{Lam}}(L)$, so

$$(X + a \log X)^{(-1)\circ}(Y) = Y + \sum_{N \geq 0} a^{N+1} \mathbb{L}_N^{\text{Lam}}(\log Y) Y^{-N}, \quad a \in \mathbb{K}. \quad (296)$$

Proof. For $A = P(L)$ one has $A_{r,m} = P^r \delta_{m,0}$, so Equation (290) retains only $r = N+1$, giving Equation (295); the degree statement is Theorem 16.22 with $a_n = 0$ for $n \geq 1$. For $P = aL$, Equation (295) reads $b_N = a^{N+1} \frac{(-1)^{N+1}}{(N+1)!} [\prod_{s=0}^{N-1} (\partial_L - s)] L^{N+1}$, which is the defining formula of the canonical zero-based Lambert polynomial $\mathbb{L}_N^{\text{Lam}}$; the case $N = 0$ gives $\mathbb{L}_0^{\text{Lam}}(L) = -L$, so the displayed sum may start at $N = 0$. $\square$

Remark 16.24 (Expression swell, stated correctly). Theorem 16.22 is a statement about *sizes of exact symbolic objects*, not about an algorithm. It says that computing the inverse through outer order t^N requires manipulating polynomials in L of degree about dN ; it says nothing about the number of arithmetic operations, and nothing whatever about the analytic size of the discarded tail. The three big- O roles must be kept apart: the tail statement $F^{(-1)\circ} - \text{Trunc}_{<N} F^{(-1)\circ} = \mathcal{O}_t^{\text{form}}(t^N)$ is a formal valuation statement with no analytic content.

16.8 Worked extraction of the first coefficients

Writing a dot for ∂_L , Equation (290) gives the first four universal coefficients of the inverse of $F = X + \sum_{n \geq 0} a_n(L) t^n$ directly. Grouping by the Lagrange index r :

$$b_0 = -a_0, \quad (297)$$

$$b_1 = -a_1 + a_0 \dot{a}_0, \quad (298)$$

$$b_2 = -a_2 + [\dot{a}_0 a_1 + a_0 \dot{a}_1 - a_0 a_1] + \left[-a_0 \dot{a}_0^2 + \frac{1}{2} a_0^2 \dot{a}_0 - \frac{1}{2} a_0^2 \ddot{a}_0 \right], \quad (299)$$

$$\begin{aligned} b_3 = & -a_3 + [\dot{a}_0 a_2 + a_0 \dot{a}_2 + a_1 \dot{a}_1 - 2a_0 a_2 - a_1^2] \\ & + \left[3a_0 \dot{a}_0 a_1 - a_0^2 \dot{a}_1 - \dot{a}_0^2 a_1 - a_0 \ddot{a}_0 a_1 - 2a_0 \dot{a}_0 \dot{a}_1 + \frac{3}{2} a_0^2 \dot{a}_1 - \frac{1}{2} a_0^2 \ddot{a}_1 \right] \\ & + \left[a_0 \dot{a}_0^3 - \frac{3}{2} a_0^2 \dot{a}_0^2 + \frac{1}{3} a_0^3 \dot{a}_0 + \frac{3}{2} a_0^2 \dot{a}_0 \ddot{a}_0 - \frac{1}{2} a_0^3 \ddot{a}_0 + \frac{1}{6} a_0^3 \ddot{a}_0 \right]. \end{aligned} \quad (300)$$

In each line the successive bracketed groups are the contributions of $r = 2, 3, 4$; the leading $-a_N$ is the contribution of $r = 1$.

Derivation. Apply Equation (290). For $N = 0$ only $r = 1$ occurs and the operator is empty. For $N = 1$, $r = 1$ gives $-A_{1,1} = -a_1$ and $r = 2$ gives $\frac{1}{2}\partial_L(a_0^2) = a_0\dot{a}_0$. For $N = 2$: $r = 1$ gives $-a_2$; $r = 2$ gives $\frac{1}{2}(\partial_L - 1)A_{2,1} = \frac{1}{2}(\partial_L - 1)(2a_0a_1)$, which is the first bracket; $r = 3$ gives $-\frac{1}{6}\partial_L(\partial_L - 1)a_0^3$, and with $\partial_L(\partial_L - 1)q = \ddot{q} - \dot{q}$ applied to $q = a_0^3$, $\dot{q} = 3a_0^2\dot{a}_0$ and $\ddot{q} = 6a_0\dot{a}_0^2 + 3a_0^2\ddot{a}_0$, this is the second bracket. For $N = 3$: $r = 1$ gives $-a_3$; $r = 2$ gives $\frac{1}{2}(\partial_L - 2)A_{2,2}$ with $A_{2,2} = 2a_0a_2 + a_1^2$; $r = 3$ gives $-\frac{1}{6}(\partial_L - 1)(\partial_L - 2)A_{3,1}$ with $A_{3,1} = 3a_0^2a_1$ and $(\partial_L - 1)(\partial_L - 2)q = \ddot{q} - 3\dot{q} + 2q$; $r = 4$ gives $\frac{1}{24}\partial_L(\partial_L - 1)(\partial_L - 2)a_0^4$ with $\partial_L(\partial_L - 1)(\partial_L - 2)q = \ddot{q} - 3\ddot{q} + 2\dot{q}$, and $\dot{q} = 4a_0^3\dot{a}_0$, $\ddot{q} = 12a_0^2\dot{a}_0^2 + 4a_0^3\ddot{a}_0$, $\ddot{q} = 24a_0\dot{a}_0^3 + 36a_0^2\dot{a}_0\ddot{a}_0 + 4a_0^3\ddot{a}_0$. Collecting gives the displayed brackets. $\square$

Example 16.25 (A fully mixed perturbation). Let

$$F(X) = X + 2\log X + 1 + \frac{3\log X - 1}{X} + \frac{(\log X)^2 + 2}{X^2},$$

so $a_0 = 2L + 1$, $a_1 = 3L - 1$, $a_2 = L^2 + 2$, $a_n = 0$ for $n \geq 3$. Then Equation (297)–Equation (300) give

$$\begin{aligned} F^{(-1)\circ}(Y) = & Y - 2M - 1 + \frac{M + 3}{Y} + \frac{-3M^2 + 7M - 3}{Y^2} \\ & + \frac{-\frac{32}{3}M^3 + 25M^2 - 6M - \frac{59}{6}}{Y^3} + \mathcal{O}_t^{\text{form}}(t^4), \quad M = \log Y. \end{aligned} \quad (301)$$

Grouped by Lagrange index, the contributions to the block t^3 are $-4M^3 - 5M^2 + 9M - 4$ from $r = 2$, $-12M^3 + 46M^2 - 11M - \frac{17}{2}$ from $r = 3$, and $\frac{16}{3}M^3 - 16M^2 - 4M + \frac{8}{3}$ from $r = 4$; nothing comes from $r = 1$. A summand that is “later” in the Lagrange index therefore still reaches the same outer block: the stopping rule is the filtration bound of Theorem 16.19, never the number of displayed summands.

Example 16.26 (A logarithmic–rational family). For $F(X) = X + a\log X + bX^{-1} + cX^{-2}$, that is $a_0 = aL$, $a_1 = b$, $a_2 = c$,

$$\begin{aligned} F^{(-1)\circ}(Y) = & Y - aM + \frac{a^2M - b}{Y} + \frac{a^3(\frac{1}{2}M^2 - M) - abM + ab - c}{Y^2} \\ & + \frac{a^4(\frac{1}{3}M^3 - \frac{3}{2}M^2 + M) + a^2b(-M^2 + 3M - 1) - 2acM + ac - b^2}{Y^3} + \mathcal{O}_t^{\text{form}}(t^4). \end{aligned} \quad (302)$$

Setting $b = c = 0$ recovers Equation (296); setting $a = c = 0$ recovers Theorem 16.27 with $\kappa = b$. The parameter c , which enters F at outer order t^2 , first perturbs the inverse at the same order, and the product b^2 first appears at t^3 , where two rational perturbations can meet.

Example 16.27 (An exact check: Catalan numbers). For $F(X) = X + \kappa X^{-1}$, Equation (281) with $A = \kappa t$ and $D_X^{r-1}(t^r) = t^{2r-1} \prod_{j=0}^{r-2} (-r - j) = (-1)^{r-1} \frac{(2r-2)!}{(r-1)!} t^{2r-1}$ gives

$$F^{(-1)\circ}(Y) = Y - \sum_{r \geq 1} \frac{(2r-2)!}{r!(r-1)!} \kappa^r Y^{-(2r-1)} = Y - \frac{\kappa}{Y} - \frac{\kappa^2}{Y^3} - \frac{2\kappa^3}{Y^5} - \frac{5\kappa^4}{Y^7} - \dots,$$

the coefficients $(2r-2)!/(r!(r-1)!)$ being the Catalan numbers. This matches the branch $\frac{1}{2}(Y + \sqrt{Y^2 - 4\kappa})$ of $Z + \kappa Z^{-1} = Y$ that is asymptotic to Y , and simultaneously checks the signs, the derivative order $r-1$, and the branch selection implicit in the near-identity normalisation.

The coefficients produced by [Theorem 16.18](#) are certified by a finite computation, and the certificate is the one already established in the reversion part: for $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ and $G_N = X + \sum_{j=0}^N g_j(L)t^j$, the three conditions $G_N = \text{Trunc}_{<N+1} F^{(-1)\circ}$, $\text{Err}_{\text{rev}}(F, G_N) = \mathcal{O}_t^{\text{form}}(t^{N+1})$ and $G_N \circ F - X = \mathcal{O}_t^{\text{form}}(t^{N+1})$ are equivalent ([Theorem 14.16](#)); moreover the order of either residual equals $\nu_t(G_N - F^{(-1)\circ})$ exactly, with the same leading block ([Theorem 14.15](#)). Nothing in that argument uses the Lagrange–Bürmann representation, so it certifies coefficients obtained by any of the routes of this part.

Example 16.28. Substituting [Equation \(301\)](#) back into the F of [Theorem 16.25](#), using the exact generator substitutions $t \circ G = t/(1+U)$ and $L \circ G = L + \log(1+U)$ with $U = t(G-Y)$, gives

$$F \circ G - Y = \left(28M^4 - \frac{211}{3}M^3 - 15M^2 + 91M - \frac{59}{3}\right)t^4 + \mathcal{O}_t^{\text{form}}(t^5).$$

The vanishing through t^3 certifies every coefficient claimed in [Equation \(301\)](#); the nonzero t^4 block is expected, being the first block not represented in G . Computing one residual block beyond the target is the cheapest available check, and it detects exactly the errors that a second run of the same code would not.

What the formula does and does not say

For $F = X + A$ with $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, $F^{(-1)\circ} = X + \sum_{r \geq 1} \frac{(-1)^r}{r!} D_X^{r-1}(A^r)$, and the block t^N is a finite sum over $1 \leq r \leq N+1$. Summability holds precisely because $\nu_t(A) \geq 0$, and *only* then ([Theorem 16.9](#)). The same construction transports any observable $\varphi \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ along the inverse without building it ([Theorem 16.15](#)). Logarithmic degrees grow linearly, with slope $\deg_L a_0$ ([Theorem 16.22](#)). None of this asserts convergence, an analytic inverse branch, or any bound uniform in L .

17 Reversion beyond the identity scale

[Section 16](#) inverts $F = X + A$ with $\nu_t(A) \geq 0$. Two things can go wrong outside that class: the leading exponent of F need not be 1, and the leading coefficient block need not be a scalar. This section shows that these are the *only* two obstructions, proves a criterion that decides in advance which enlargement of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ the inverse will require, turns dominant balance into an algorithm with a proof attached, and exhibits the way a careless scale change silently corrupts an otherwise correct near-identity computation.

Throughout this section, composition means composition of maps: whenever we write $F \circ G$ for germs at $+\infty$, associativity and the functorial identity

$$(F \circ G)^{(-1)\circ} = G^{(-1)\circ} \circ F^{(-1)\circ} \quad (303)$$

are the ordinary group-theoretic facts, valid in any group of invertible maps, and are used without further comment for genuine germs. For formal series they hold in the near-identity group of any reversion frame by [Theorem 16.5\(5\)](#), and more generally whenever all displayed composites are admissible in the chosen support class.

17.1 Monomial-leading maps and the closure criterion

Definition 17.1 (Monomial-leading map). Let $\Gamma \subseteq \mathbb{Q}$ be an exponent group. A series G is *monomial-leading of exponent μ* if

$$G = dX^\mu(1+V), \quad d \in \mathbb{K}^\times, \quad \mu \in \Gamma_{>0}, \quad \nu_t(V) > 0, \quad (304)$$

with $\log d \in \mathbb{K}$. This is the class for which the generator substitutions

$$t \circ G = d^{-1}t^\mu(1+V)^{-1}, \quad L \circ G = \mu L + \log d + \log(1+V) \quad (305)$$

stay inside the ambient coefficient ring, and hence the class for which composition with an outer polynomial–logarithmic series is defined.

Remark 17.2. The scalar hypothesis $d \in \mathbb{K}^\times$ cannot be relaxed to a polynomial leading block. If $G = X^\mu q(L)(1+V)$ with $q \in \mathbb{K}[L]$ of degree $e \geq 1$ and leading coefficient q_e , then $L \circ G = \mu L + \log q(L) + \log(1+V)$ and $\log q(L) = e \log L + \log q_e + \mathcal{O}_{X \rightarrow +\infty}(L^{-1})$, so a new generator $\log L$ appears. That generator is genuinely new: if $\log L = p(L) + R$ with $p \in \mathbb{K}[L]$ and $R \rightarrow 0$ as $X \rightarrow +\infty$, then $\log L - p(L) \rightarrow 0$, which is impossible, since for constant p the difference tends to $+\infty$ and for $\deg_L p \geq 1$ it tends to $\pm\infty$. This is the same argument used in [Theorem 17.4](#).

Theorem 17.3 (Closure criterion for reversion in the one-logarithm ring). *Let $\Gamma \subseteq \mathbb{Q}$ be an exponent group containing $\mathbb{Z}$, let $\mathfrak{R}$ be the corresponding Puiseux–log ring (so $\mathfrak{R} = \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ when $\Gamma = \mathbb{Z}$, and $\mathfrak{R} = \mathbb{K}[L]((t^{1/s}))$ when $\Gamma = s^{-1}\mathbb{Z}$), and let*

$$F = cX^\rho L^\sigma(1+U), \quad c \in \mathbb{K}^\times, \quad \rho \in \Gamma_{>0}, \quad \sigma \in \mathbb{N}_0, \quad \nu_t(U) > 0. \quad (306)$$

Then F admits a monomial-leading right inverse G in the sense of [Theorem 17.1](#), over $\mathfrak{R}$ or over any extension of $\mathfrak{R}$ obtained by enlarging the coefficient field alone, if and only if

$$\rho^{-1} \in \Gamma \quad \text{and} \quad \sigma = 0. \quad (307)$$

When [Equation \(307\)](#) holds, the exponent of G is $\mu = \rho^{-1}$ and its leading scalar is any d with $cd^\rho = 1$, so the construction requires a field $\mathbb{K}' \supseteq \mathbb{K}$ containing such a d together with $\log d$. In particular:

1. *in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ itself ($\Gamma = \mathbb{Z}$), the criterion is $\rho = 1$ and $\sigma = 0$, that is, $c^{-1}F \in \mathcal{G}_{\text{poly},\mathbb{K}}(\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}})$;*
2. *enlarging the exponent group to $\rho^{-1}\mathbb{Z}$ removes the exponent obstruction but never the obstruction $\sigma \geq 1$.*

Proof. Write $F(Y) = \sum_{n \geq -\rho} p_n(\log Y)Y^{-n}$ with $p_{-\rho}(\lambda) = c\lambda^\sigma$ and $\nu_t(F - cX^\rho L^\sigma) > -\rho$, and let $G = dX^\mu(1+V)$ be monomial-leading. By [Equation \(305\)](#) and the monomial-leading composition formula,

$$F \circ G = \sum_{n \geq -\rho} p_n(\mu L + \log d + \log(1+V))d^{-n}t^{\mu n}(1+V)^{-n}.$$

Since $\nu_t(V) > 0$ and $\nu_t \log(1+V) > 0$, the term $n = -\rho$ contributes

$$cd^\rho(\mu L + \log d)^\sigma t^{-\mu\rho} + (\text{terms of } \nu_t > -\mu\rho),$$

and every term with $n > -\rho$ has $\nu_t \geq -\mu n > -\mu\rho$. Hence $\nu_t(F \circ G) = -\mu\rho$ and the leading block of $F \circ G$ is $cd^\rho(\mu L + \log d)^\sigma$.

Requiring $F \circ G = X = t^{-1}$ forces $\mu\rho = 1$, so $\mu = \rho^{-1}$ must lie in Γ , and forces

$$cd^\rho(\mu L + \log d)^\sigma = 1. \quad (308)$$

If $\sigma \geq 1$, the left side of [Equation \(308\)](#) is a polynomial in L of degree $\sigma \geq 1$ with leading coefficient $cd^\rho\mu^\sigma \neq 0$, so it cannot be the constant 1; hence $\sigma = 0$, and then [Equation \(308\)](#) reads $cd^\rho = 1$.

Conversely, suppose $\rho^{-1} \in \Gamma$ and $\sigma = 0$. Choose d with $cd^\rho = 1$ in a finite extension $\mathbb{K}' \supseteq \mathbb{K}$ also containing $\log d$, and set $F_0 = cX^\rho$, $F_0^{(-1)^\circ} = dX^{1/\rho}$. Then $\Phi := F \circ F_0^{(-1)^\circ}$ satisfies $\Phi = cd^\rho X(1+U \circ F_0^{(-1)^\circ}) = X(1+\tilde{U})$ with $\nu_t(\tilde{U}) > 0$, because $\nu_t(U \circ F_0^{(-1)^\circ}) = \rho^{-1}\nu_t(U) > 0$. Hence $\Phi \in \mathcal{G}_{\text{poly},\mathbb{K}}$ of the Puiseux–log frame $\mathbb{K}'[L]((t^{1/s}))$, s a common denominator of ρ^{-1} and the exponents of $\tilde{U}$, and [Theorem 16.14](#) produces $\Phi^{(-1)^\circ}$. By [Equation \(303\)](#), $F^{(-1)^\circ} = F_0^{(-1)^\circ} \circ \Phi^{(-1)^\circ}$, which is monomial-leading of exponent ρ^{-1} . $\square$

Proposition 17.4 (A leading logarithm forces a nested logarithm). *Let $F = cX^\rho L^\sigma(1 + U)$ as in Equation (306) with $\sigma \geq 1$, and let G be a germ at $+\infty$ with $F(G(Y)) = Y$ for all large Y . Put $S = \log Y$ and $u = \log G(Y)$. Then*

$$u = \frac{1}{\rho} \left(S - \sigma \log S + \sigma \log \rho - \log c \right) + \mathcal{O}_{S \rightarrow +\infty} \left(\frac{\log S}{S} \right), \quad (309)$$

so $\log G$ contains the term $-(\sigma/\rho) \log \log Y$ with nonzero coefficient. Consequently G lies in no Puiseux–log ring $\mathbb{K}'[S]((Y^{-1/s}))$ with $\mathbb{K}' \supseteq \mathbb{K}$ and $s \in \mathbb{N}_{>0}$: a second logarithm is unavoidable.

Proof. Taking logarithms in $F(G(Y)) = Y$ and writing $u = \log G(Y)$,

$$S = \log c + \rho u + \sigma \log u + \log(1 + U(G(Y))). \quad (310)$$

Since $\nu_t(U) > 0$ there is $\delta > 0$ with $U(X) = \mathcal{O}_{X \rightarrow +\infty}(X^{-\delta})$, so the last term of Equation (310) is $\mathcal{O}_{Y \rightarrow +\infty}(G(Y)^{-\delta})$; since $G(Y) \rightarrow +\infty$ like a positive power of Y , it is smaller than every power of $1/S$. From Equation (310), $\rho u \sim S$, hence $\log u = \log S - \log \rho + o_{S \rightarrow +\infty}(1)$; substituting back, $\rho u = S - \log c - \sigma \log S + \sigma \log \rho + o_{S \rightarrow +\infty}(1)$, which is Equation (309).

Suppose $G \in \mathbb{K}'[S]((Y^{-1/s}))$. Then $G = q(S)Y^{-\lambda}(1 + o_{Y \rightarrow +\infty}(1))$ for some $\lambda \in s^{-1}\mathbb{Z}$ and some nonzero $q \in \mathbb{K}'[S]$, so

$$u = \log G = -\lambda S + \log q(S) + o_{Y \rightarrow +\infty}(1) = -\lambda S + (\deg q) \log S + \log(\text{lc } q) + o_{Y \rightarrow +\infty}(1).$$

Comparing with Equation (309), the coefficients of S give $\lambda = -1/\rho$ and those of $\log S$ give $\deg q = -\sigma/\rho < 0$, which is impossible for a polynomial. $\square$

Corollary 17.5 (Which enlargement is needed). *For F as in Equation (306):*

leading data	obstruction	smallest ambient for $F^{(-1)\circ}$
$\rho = 1, \sigma = 0$	none	$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ over $\mathbb{K}(c^{-1}, \log c)$
$\rho \in \mathbb{Q}_{>0}, \sigma = 0$	exponent	$\mathbb{K}'[L]((t^{1/s}))$, s a denominator of ρ^{-1}
$\rho \notin \mathbb{Q}, \sigma = 0$	exponent	Hahn power-log with $\rho^{-1}\mathbb{Z} \subseteq \Gamma$
$\sigma \geq 1$	logarithm	nested-logarithmic; $\log \log X$ required

Proof. Rows one to three are Theorem 17.3 together with the explicit construction in its converse half; row four is Theorem 17.4 together with Theorem 17.7 below, which produces the inverse in the nested-logarithmic frame. $\square$

17.2 Dominant balance as a theorem

The workflow “guess the leading behaviour, normalise, iterate” becomes a theorem once one names the object that does the normalising.

Theorem 17.6 (Reduction to the near-identity case). *Let $\mathfrak{G}$ be a group of invertible germs at $+\infty$ (or of formal maps in a fixed support class, with all displayed composites admissible), let $F \in \mathfrak{G}$, and let $F_0 \in \mathfrak{G}$ be any leading model, that is, any element whose inverse is known. Put*

$$\Phi := F \circ F_0^{(-1)\circ}. \quad (311)$$

Then $F = \Phi \circ F_0$ and

$$F^{(-1)\circ} = F_0^{(-1)\circ} \circ \Phi^{(-1)\circ}. \quad (312)$$

If moreover $\Phi = X + A$ with A in the nonnegative part of some reversion frame, then $\Phi^{(-1)\circ}$ is given by Equation (280), and

$$F^{(-1)\circ} = F_0^{(-1)\circ} \circ \left(X + \sum_{r \geq 1} \frac{(-1)^r}{r!} \partial^{r-1}(A^r) \right). \quad (313)$$

The dual normalisation $\Psi := F_0^{(-1)\circ} \circ F$ gives $F = F_0 \circ \Psi$ and $F^{(-1)\circ} = \Psi^{(-1)\circ} \circ F_0^{(-1)\circ}$.

Proof. $\Phi \circ F_0 = F \circ F_0^{(-1)\circ} \circ F_0 = F$. Applying Equation (303) to $F = \Phi \circ F_0$ gives $F^{(-1)\circ} = F_0^{(-1)\circ} \circ \Phi^{(-1)\circ}$. The last two statements are the same computation with the factors exchanged, and Equation (313) is Theorem 16.14 substituted into Equation (312). $\square$

Theorem 17.6 is content-free until a leading model is produced. For a monomial-leading F the model is supplied by one canonical change of scale, and this is where dominant balance actually happens.

Theorem 17.7 (Logarithmic normal form of a power–log leading term). *Let $c \in \mathbb{K}^\times$ with $\log c \in \mathbb{K}$, let $\rho \in \mathbb{Q}_{>0}$ with $\log \rho \in \mathbb{K}$, let $\sigma \in \mathbb{N}_0$, and set $F_0(X) = cX^\rho(\log X)^\sigma$. Write $E(X) = e^X$ and $m_\rho(X) = \rho X$. Then*

$$G := m_\rho^{(-1)\circ} \circ \log \circ F_0 \circ E = u + \frac{\sigma}{\rho} \log u + \frac{\log c}{\rho} \quad (314)$$

as a map in the large variable u ; in particular G lies in the near-identity group of the reversion frame $\mathbb{K}[\log u][[u^{-1}]]$ with perturbation $A(u) = (\sigma/\rho) \log u + (\log c)/\rho$, and

$$F_0^{(-1)\circ} = E \circ G^{(-1)\circ} \circ m_{1/\rho} \circ \log. \quad (315)$$

Consequently, with $S = \log Y$ and $\Lambda = \log S$,

$$\begin{aligned} \log F_0^{(-1)\circ}(Y) &= \frac{1}{\rho} \left[S - \sigma \Lambda + \sigma \log \rho - \log c \right. \\ &\quad \left. + \frac{\sigma^2 \Lambda + \sigma \log c - \sigma^2 \log \rho}{S} + \mathcal{O}_{S^{-1}}^{\text{form}}(S^{-2}) \right], \end{aligned} \quad (316)$$

$$F_0^{(-1)\circ}(Y) = c^{-1/\rho} \rho^{\sigma/\rho} Y^{1/\rho} S^{-\sigma/\rho} \left[1 + \frac{\sigma^2 \Lambda + \sigma \log c - \sigma^2 \log \rho}{\rho S} + \mathcal{O}_{S^{-1}}^{\text{form}}(S^{-2}) \right]. \quad (317)$$

Proof. For $X = e^u$, $\log F_0(e^u) = \log c + \rho u + \sigma \log u$. Post-composing with $m_\rho^{(-1)\circ} = m_{1/\rho}$ divides by ρ and gives Equation (314). Since $\log u \in \mathbb{K}[\log u][[u^{-1}]]$ has zero outer order in the variable u , the perturbation A has $\nu(A) = 0$, so $G \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ of that frame. Reading Equation (314) as $G = m_{1/\rho} \circ \log \circ F_0 \circ E$ and inverting with Equation (303), using $E^{(-1)\circ} = \log$ and $(\log)^{(-1)\circ} = E$,

$$G^{(-1)\circ} = E^{(-1)\circ} \circ F_0^{(-1)\circ} \circ (\log)^{(-1)\circ} \circ (m_{1/\rho})^{(-1)\circ} = \log \circ F_0^{(-1)\circ} \circ E \circ m_\rho,$$

which rearranges to Equation (315).

For the expansions, apply Equation (297) and Equation (298) to $A(u) = a_0(\log u)$ with $a_0(\lambda) = (\sigma/\rho)\lambda + (\log c)/\rho$ and $a_n = 0$ for $n \geq 1$:

$$b_0 = -a_0, \quad b_1 = a_0 \dot{a}_0 = \frac{\sigma}{\rho} \left(\frac{\sigma}{\rho} \lambda + \frac{\log c}{\rho} \right) = \frac{\sigma}{\rho^2} (\sigma \lambda + \log c).$$

Hence, writing v for the large variable of $G^{(-1)\circ}$ and $\lambda = \log v$,

$$G^{(-1)\circ}(v) = v - \frac{\sigma \lambda + \log c}{\rho} + \frac{\sigma(\sigma \lambda + \log c)}{\rho^2 v} + \mathcal{O}_{v^{-1}}^{\text{form}}(v^{-2}).$$

By Equation (315), $\log F_0^{(-1)\circ}(Y) = G^{(-1)\circ}(S/\rho)$. Substituting $v = S/\rho$, so $\lambda = \Lambda - \log \rho$ and $v^{-1} = \rho/S$,

$$\log F_0^{(-1)\circ}(Y) = \frac{S}{\rho} - \frac{\sigma(\Lambda - \log \rho) + \log c}{\rho} + \frac{\sigma(\sigma(\Lambda - \log \rho) + \log c)}{\rho S} + \mathcal{O}_{S^{-1}}^{\text{form}}(S^{-2}),$$

which is Equation (316). Exponentiating and using $e^{S/\rho} = Y^{1/\rho}$, $e^{-\sigma \Lambda/\rho} = S^{-\sigma/\rho}$, and $e^{(\sigma \log \rho - \log c)/\rho} = \rho^{\sigma/\rho} c^{-1/\rho}$, gives Equation (317), the exponential of the S^{-1} block being expanded as $1 + w + \mathcal{O}_{S^{-1}}^{\text{form}}(S^{-2})$. $\square$

Remark 17.8. Equation (314) explains the “one level down” phenomenon: after conjugation by e , the expansion parameter is $1/\log Y$ and the coefficients are polynomials in $\log \log Y$. The polynomial–logarithmic calculus is applied in the frame of the variable $u = \log X$, not of X . Note also that Equation (314) is exact only for $U = 0$; for $F = F_0 \cdot (1 + U)$ with $\nu_t(U) > 0$ the conjugated map carries an extra term $\rho^{-1} \log(1 + U(e^u))$, which is smaller than every power of u^{-1} and therefore invisible in the frame $\mathbb{K}[\log u][[u^{-1}]]$. That term is not negligible after exponentiating back, and it is precisely what the second stage Equation (311) of Theorem 17.6 handles: with $\Phi = F \circ F_0^{(-1)\circ} = X(1 + U \circ F_0^{(-1)\circ})$, the two scales $1/\log Y$ and $Y^{-1/\rho}$ are separated rather than mixed.

Algorithm 17.9 (Reversion with a nonidentity leading term). Input: a germ F at $+\infty$ presented as $F = cX^\rho L^\sigma(1 + U)$ with $c \in \mathbb{K}^\times$, $\rho \in \mathbb{Q}_{>0}$, $\sigma \in \mathbb{N}_0$, $\nu_t(U) > 0$; a target outer order N .

1. **Extend the coefficient field.** Replace $\mathbb{K}$ by a field containing $\log c$, $\log \rho$ and $c^{-1/\rho}$. Record the branch of every root and logarithm chosen.
2. **Fix the ambient.** If $\sigma = 0$ and $\rho = 1$, the ambient is $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. If $\sigma = 0$ and $\rho = p/q$ in lowest terms, the ambient is $\mathbb{K}[L]((t^{1/p}))$. If $\sigma \geq 1$, the ambient is the nested-logarithmic frame of Theorem 17.7. (Theorem 17.3 and Theorem 17.5.)
3. **Invert the leading model.** Set $F_0 = cX^\rho L^\sigma$. If $\sigma = 0$, $F_0^{(-1)\circ} = c^{-1/\rho} X^{1/\rho}$. If $\sigma \geq 1$, obtain $F_0^{(-1)\circ}$ from Equation (315): revert $G(u) = u + (\sigma/\rho) \log u + (\log c)/\rho$ by Theorem 16.14 in the u -frame to the required order, then evaluate at $u = S/\rho$ and exponentiate.
4. **Normalise.** Form $\Phi = F \circ F_0^{(-1)\circ}$. By construction $\Phi = X(1 + U \circ F_0^{(-1)\circ})$, which is near-identity in the ambient of step 2.
5. **Revert the near-identity factor.** Apply Theorem 16.14 or the coefficient formula Equation (290) to Φ , through the order fixed by Theorem 16.19.
6. **Recompose.** Return $F^{(-1)\circ} = F_0^{(-1)\circ} \circ \Phi^{(-1)\circ}$, as in Equation (312). Both generators must be substituted: t and L change together, by Equation (305).
7. **Certify.** Verify $F \circ F^{(-1)\circ} - X = \mathcal{O}_t^{\text{form}}(t^{N+1})$ and, independently, $F^{(-1)\circ} \circ F - X = \mathcal{O}_t^{\text{form}}(t^{N+1})$, as in Theorem 14.16. The two residuals detect different substitution-order errors.

17.3 Worked examples

Example 17.10 (Leading behaviour $X + c \log X$: inside the theory). The map $F(X) = X + a \log X + b$ with $a \neq 0$ has $A = aL + b \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, so $\nu_t(A) = 0$ and no enlargement is needed, even though $A \rightarrow +\infty$. By Theorem 16.23 with $P = aL + b$, the inverse is

$$\begin{aligned} F^{(-1)\circ}(Y) = Y - aM - b + \frac{a(aM + b)}{Y} + \frac{a(aM + b)(aM + b - 2a)}{2Y^2} \\ + \frac{a(aM + b)}{6Y^3} (2a^2M^2 - 9a^2M + 4abM + 6a^2 - 9ab + 2b^2) + \mathcal{O}_t^{\text{form}}(t^4), \end{aligned} \quad (318)$$

with $M = \log Y$, and $\deg_L b_N = 2(N + 1) - 1$ when $\deg P = 1 \dots$ no: with $d = \deg_L P = 1$, Equation (294) gives $\deg_L b_N = N$ and $[M^N]b_N = a^{N+1}/N$ for $N \geq 1$. The family also has an exact closed form: $Y = X + a \log X + b$ is equivalent to $\frac{X}{a} e^{X/a} = \frac{1}{a} e^{(Y-b)/a}$, whence

$$F^{(-1)\circ}(Y) = a W_k\left(\frac{1}{a} e^{(Y-b)/a}\right), \quad (319)$$

with the branch k selected by the intended real interval or sector. Equation (318) is the large- Y expansion of Equation (319); the formal calculation does not choose k [5, 11, 7].

Example 17.11 (Leading behaviour cX^ρ : Puiseux). Let $F(X) = X^2 + X$. Here $\rho = 2$, $\sigma = 0$, $c = 1$, so [Theorem 17.3](#) forbids an inverse in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ and [Theorem 17.5](#) prescribes $\mathbb{K}[L]((t^{1/2}))$. [Theorem 17.9](#) gives $F_0 = X^2$, $F_0^{(-1)\circ} = X^{1/2}$, $\Phi = F \circ F_0^{(-1)\circ} = X + X^{1/2}$, so $A = X^{1/2} = t^{-1/2}$ has $\nu_t(A) = -\frac{1}{2} < 0$ and step 4 fails: the normalisation must be redone with $F_0 = X^2 + X$ treated exactly, or, more simply, with the substitution $X = Z - \frac{1}{2}$. Doing the latter, $F = Z^2 - \frac{1}{4}$, and

$$F^{(-1)\circ}(Y) = \left(Y + \frac{1}{4}\right)^{1/2} - \frac{1}{2} = Y^{1/2} - \frac{1}{2} + \frac{1}{8}Y^{-1/2} - \frac{1}{128}Y^{-3/2} + \dots,$$

which is the branch $\frac{1}{2}(-1 + \sqrt{1 + 4Y})$ asymptotic to $Y^{1/2}$. No integer-power ansatz can reproduce it.

Remark 17.12. [Theorem 17.11](#) shows that the normalisation step of [Theorem 17.9](#) must use the *full* leading monomial $cX^\rho L^\sigma$ of F , not merely a convenient power. In [Theorem 17.3](#) the identity $\Phi = X(1 + U \circ F_0^{(-1)\circ})$ holds because $F = F_0(1 + U)$ exactly; if $F - F_0$ is not $o_{X \rightarrow +\infty}(F_0)$, the step produces a Φ outside $\mathcal{G}_{\text{poly},\mathbb{K}}$ and [Theorem 16.9](#) then says that [Equation \(280\)](#) is not available.

Example 17.13 (Leading behaviour $X \log X$: nested logarithm). Let $F(X) = X \log X$, so $\rho = 1$, $\sigma = 1$, $c = 1$. By [Theorem 17.3](#) there is no monomial-leading inverse in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, and by [Theorem 17.4](#) none in any Puiseux–log ring. [Theorem 17.7](#) gives the leading model in logarithmic coordinates: $G(u) = u + \log u$, whose inverse is Wright omega, so

$$F^{(-1)\circ}(Y) = e^{\omega(S)} = \frac{Y}{W(Y)}, \quad S = \log Y, \quad (320)$$

the second equality because $Z \log Z = Y$ with $Z = e^w$ gives $we^w = Y$. Expanding $W(Y) = S - \Lambda + \sum_{n \geq 1} \mathbb{L}_n^{\text{Lam}}(\Lambda) S^{-n}$ with $\Lambda = \log S$ and inverting the resulting series,

$$F^{(-1)\circ}(Y) = \frac{Y}{S} \left(1 + \frac{\Lambda}{S} + \frac{\Lambda^2 - \Lambda}{S^2} + \mathcal{O}_{Y \rightarrow +\infty}\left(\frac{\Lambda^3}{S^3}\right) \right). \quad (321)$$

Every coefficient is a polynomial in $\Lambda = \log \log Y$; a one-logarithm ansatz, however many terms it carries, cannot represent [Equation \(321\)](#). The same mechanism produces depth two in the inverses of Dulac transseries with a logarithm in the leading monomial [\[24, 27\]](#).

Example 17.14 (General power–log monomial). For $Y = cX^\rho(\log X)^\sigma$ with $c, \rho > 0$, [Equation \(317\)](#) gives, in fully explicit form,

$$X = c^{-1/\rho} \rho^{\sigma/\rho} Y^{1/\rho} (\log Y)^{-\sigma/\rho} \left[1 + \frac{\sigma^2 \log \log Y + \sigma \log c - \sigma^2 \log \rho}{\rho \log Y} + \mathcal{O}_{Y \rightarrow +\infty}\left(\frac{(\log \log Y)^2}{(\log Y)^2}\right) \right].$$

An exact Lambert form is also available when $\sigma \neq 0$: putting $s = \log X$, the equation becomes $(Y/c)^{1/\sigma} = se^{(\rho/\sigma)s}$, so

$$X = \exp \left[\frac{\sigma}{\rho} W_k \left(\frac{\rho}{\sigma} \left(\frac{Y}{c} \right)^{1/\sigma} \right) \right]. \quad (322)$$

Because the argument of W contains a power of Y , the expansion of [Equation \(322\)](#) contains $\log Y$ and $\log \log Y$: a second logarithm is forced whenever $\sigma \neq 0$, in agreement with [Theorem 17.4](#).

A scale change that silently corrupts a correct series

Let $F(X) = 2X + \log X$. This is not near-identity, but it is affine in the leading scale, so [Theorem 17.6](#) applies with the model $F_0 = m_2$, $m_2(Z) = 2Z$. Writing $G(X) = X + \frac{1}{2} \log X$, one has $F = m_2 \circ G$ and therefore

$$F^{(-1)\circ}(Y) = G^{(-1)\circ}(Y/2). \quad (323)$$

By Equation (296) with $a = \frac{1}{2}$, $G^{(-1)\circ}(Z) = Z + \sum_{N \geq 0} 2^{-(N+1)} \mathbf{L}_N^{\text{Lam}}(\log Z) Z^{-N}$. Substituting $Z = Y/2$ changes *both* generators: $Z^{-N} = 2^N Y^{-N}$ and $\log Z = \log Y - \log 2$. Hence

$$F^{(-1)\circ}(Y) = \frac{Y}{2} + \frac{1}{2} \sum_{N \geq 0} \mathbf{L}_N^{\text{Lam}}(\log Y - \log 2) Y^{-N}. \quad (324)$$

The silent failure is to substitute only the power, leaving $\log Z$ as $\log Y$. The result, $\frac{Y}{2} + \frac{1}{2} \sum_N \mathbf{L}_N^{\text{Lam}}(\log Y) Y^{-N}$, is a perfectly well formed element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$; every block has the right shape, the right degree, and the right sign pattern, and no formal check inside the near-identity calculation detects anything. It is wrong by a constant. Numerically, with $Y = 20 + \log 10 = 22.302585\dots$, for which $F^{(-1)\circ}(Y) = 10$ exactly, the truncations through Y^{-3} give

$$\text{Equation (324)} : 10.000004\dots, \quad \text{naive substitution} : 9.67026\dots,$$

the leading discrepancy being $\frac{1}{2} \log 2 = 0.34657\dots$. Two further observations belong with this example.

- The coefficient field must contain $\log 2$. Over $\mathbb{K} = \mathbb{Q}$ the identity Equation (324) is not even a statement about $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$; the field has to be extended first (Theorem 17.9, step 1).
- The residual certificate of Theorem 14.16 does catch the error, because it is computed against F , not against G . This is why step 7 of Theorem 17.9 certifies against the original map and never against the normalised one.

Remark 17.15 (Sufficient, not necessary). Theorem 16.9 settles summability for the outer valuation. It does not say that Equation (280) fails whenever $\nu_t(A) < 0$; it says the theorem as proved does not apply. Take $F(X) = X + X/L = X(1 + 1/L)$, so $A = t^{-1}L^{-1} \in \mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ with $\nu_t(A) = -1$, while $A \prec X$. Here every Lagrange summand contributes to the single outer block t^{-1} :

$$-A = -XL^{-1}, \quad \frac{1}{2}D_X(A^2) = XL^{-2} - XL^{-3}, \quad -\frac{1}{6}D_X^2(A^3) = -XL^{-3} + \frac{5}{2}XL^{-4} - 2XL^{-5},$$

and so on, so no finite number of summands determines any block. The family is nevertheless summable for the L^{-1} -adic topology of the coefficient field $\mathbb{K}((L^{-1}))$ at each fixed outer level, and the resulting series $Y(1 - M^{-1} + M^{-2} - 2M^{-3} + \dots)$, $M = \log Y$, does satisfy $F \circ F^{(-1)\circ} - Y = \mathcal{O}_{Y \rightarrow +\infty}(YM^{-4})$, as direct substitution confirms. A theorem covering this case needs a rank-two value group and Hahn-style summability, in which $\{i : \nu(f_i) < \gamma\}$ is replaced by a well-ordered-support condition. We have not carried that out, and we do not claim it: the statement “Equation (280) holds whenever $A \prec X$ in the rank-two order, with summation in the refined filtration” is left as an open problem.

Remark 17.16 (What does not belong here). Reversion at a critical point is a different problem in a different regime. If $f(x) - f(x_0) = c_m(x - x_0)^m + \dots$ with $c_m \neq 0$ and $m \geq 2$, the local inverse begins $x - x_0 \sim c_m^{-1/m}(y - f(x_0))^{1/m}$ and has m branches; Lambert W near $-e^{-1}$ is the case $m = 2$ [5]. This is a Puiseux expansion at a finite point, not an expansion in the scale $X \rightarrow +\infty$, and none of the machinery of this section applies to it. In the at-infinity setting the derivative never vanishes on the relevant class: for $F = X + A$ with $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, $D_X F = 1 + \mathcal{O}_t^{\text{form}}(t)$ is a unit of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.

Beyond the identity scale

A monomial-leading $F = cX^\rho L^\sigma(1 + U)$ has a monomial-leading inverse inside a Puiseux–log ring exactly when ρ^{-1} lies in the exponent group and $\sigma = 0$ (Theorem 17.3). A noninteger ρ is repaired by enlarging the exponent group; $\sigma \geq 1$ cannot be repaired that way and forces $\log \log X$ (Theorem 17.4). In every case reversion factors as $F^{(-1)\circ} = F_0^{(-1)\circ} \circ \Phi^{(-1)\circ}$ with $\Phi = F \circ F_0^{(-1)\circ}$ near-identity (Theorem 17.6), and the leading model F_0 is inverted by conjugating with e and dividing by ρ (Theorem 17.7). Every scale change substitutes *both* generators; substituting only the

power silently produces a plausible and wrong series ([Section 17.3](#)).

Part V

Wright omega, the Lambert polynomials, and Lambert W

18 Wright omega: the canonical logarithmic inverse

Every result of Parts I–IV was stated for a general polynomial–logarithmic series. This part specialises all of it to a single map,

$$F(Y) = Y + \log Y,$$

and shows that this one reversion carries the whole Lambert circle of problems. The correction blocks of $F^{(-1)\circ}$ are the Lambert polynomials L_n^{Lam} ; they are defined and studied in [Section 19](#), and the present section explains where they come from and exactly what is being asserted when one writes them down.

18.1 Standing conventions for this part

Throughout, $\mathbb{K}$ is a field of characteristic zero, so that $\mathbb{Q} \subseteq \mathbb{K}$. On the formal side we work in the rings of [Theorem 3.3](#) with the large variable X , the small generator $t = X^{-1}$, and the logarithmic generator $L = \log X$; the near-identity class is $\mathcal{G}_{\text{poly}, \mathbb{K}} = X + \mathbb{K}[L][[t]]$ of [Theorem 12.1](#), the derivation is $D_X = t \partial_L - t^2 \partial_t$, and truncation is always the *exclusive* $\text{Trunc}_{<N} F = \text{Trunc}_{\leq N} F$.

On the analytic side the regime is $x \rightarrow +\infty$ along the positive real ray, with the real logarithm; every deviation from that regime – a complex sector, a nonprincipal branch, a limit at a finite point – is announced explicitly.

Two symbols carry a variable role and must be read locally. In this section L is the generator $\log X$ of the coefficient ring. In [Section 19](#) the Lambert polynomials are treated as polynomials in an indeterminate u , and s is the formal variable of their generating series; in [Theorem 19.14](#) the letter s is, instead, a complex parameter of an honest integral. The two roles of s are unrelated and never appear in the same statement.

A common trap

Three inverses meet in this part and must not be conflated. For a series F , the multiplicative inverse is F^{-1} ([Theorem 7.7](#)) and the compositional inverse is $F^{(-1)\circ}$ ([Theorem 14.11](#)); for the Lambert function, the subscript in W_{-1} is a *branch label* and is neither of the two. The glyph -1 is the same in all three cases and the objects are different.

18.2 The real bijection and its inverse

Proposition 18.1 (The real logarithmic bijection). *The map*

$$\phi: (0, \infty) \longrightarrow \mathbb{R}, \quad \phi(Y) = Y + \log Y,$$

is a strictly increasing real-analytic bijection. Its inverse $\omega: \mathbb{R} \rightarrow (0, \infty)$ is strictly increasing and real-analytic, and satisfies, for every $x \in \mathbb{R}$,

$$\omega(x) + \log \omega(x) = x, \quad \omega'(x) = \frac{\omega(x)}{1 + \omega(x)}, \quad \omega(x) = W_0(e^x). \quad (325)$$

Here W_0 is the branch of the inverse of $w \mapsto we^w$ taking values in $[-1, \infty)$. Moreover $\omega(1) = 1$, and for every x with $\omega(x) > 1$,

$$x - \log x < \omega(x) < x. \quad (326)$$

Proof. On $(0, \infty)$ the function ϕ is real-analytic with $\phi'(Y) = 1 + Y^{-1} > 1 > 0$, so it is strictly increasing and injective. As $Y \rightarrow 0^+$ we have $\log Y \rightarrow -\infty$ and $Y \rightarrow 0$, so $\phi(Y) \rightarrow -\infty$; as $Y \rightarrow +\infty$, $\phi(Y) \geq Y \rightarrow +\infty$. By the intermediate value theorem ϕ is onto $\mathbb{R}$. Hence ϕ is a bijection and $\omega := \phi^{(-1)\circ}$ is defined on all of $\mathbb{R}$, strictly increasing, and real-analytic by the inverse function theorem, ϕ' being nowhere zero.

Differentiating $\omega + \log \omega = x$ gives $\omega'(1 + \omega^{-1}) = 1$, i.e. the displayed derivative formula. For the Lambert identity fix $x \in \mathbb{R}$ and put $w = \omega(x) > 0$. Then $\log(we^w) = \log w + w = x$, so $we^w = e^x$. The map $w \mapsto we^w$ is strictly increasing on $[-1, \infty)$ with value $-e^{-1}$ at $w = -1$, hence injective there; since $w > 0 > -1$, w is the value of the principal branch, i.e. $w = W_0(e^x)$.

Finally $\phi(1) = 1 + \log 1 = 1$, so $\omega(1) = 1$. If $Y = \omega(x) > 1$ then $\log Y > 0$, so $Y = x - \log Y < x$; and then $\log Y < \log x$, whence $Y = x - \log Y > x - \log x$. $\square$

Remark 18.2 (What is not repeated here). The elementary two-sided envelope for ω and the sharper two-order statement were established in [Theorem 2.1](#) and [Theorem 2.3](#); [Equation \(326\)](#) is recorded only because the proof of [Theorem 18.17](#) uses the crude lower bound $\omega(x) > x - \log x$ directly. In particular $\omega(x) \rightarrow +\infty$ and $\omega(x)/x \rightarrow 1$ as $x \rightarrow +\infty$.

Remark 18.3 (Totality versus asymptotics). ω is a genuine function on all of $\mathbb{R}$; it is not defined only near $+\infty$. The polynomial–logarithmic expansion studied below describes it *only* as $x \rightarrow +\infty$, and says nothing about, say, $x \leq 0$, where $\log x$ is not even real. Several of the six source drafts introduce ω with the phrase “for x sufficiently large”; that restriction belongs to the expansion, not to the function.

18.3 Every Lambert problem is this one reversion

The point of isolating ω is that a one-parameter family of logarithmic inversion problems collapses onto it exactly, with no new coefficients. We give the analytic reduction first and the formal one second.

Lemma 18.4 (Exact scaling reduction, analytic form). *Let $a > 0$ be real. Then $y \mapsto y + a \log y$ is a strictly increasing bijection of $(0, \infty)$ onto $\mathbb{R}$, and its inverse is*

$$y = a \omega\left(\frac{x}{a} - \log a\right), \quad x \in \mathbb{R}. \quad (327)$$

Equivalently, for $\alpha > 0$ and $X > 0$, the unique $y > 0$ with $y^\alpha e^y = X$ is $y = \alpha \omega(\alpha^{-1} \log X - \log \alpha)$.

Proof. Write $y = az$ with $z > 0$. Then $y + a \log y = az + a \log a + a \log z = a(z + \log z) + a \log a$, so $y + a \log y = x$ is equivalent to $z + \log z = x/a - \log a$, that is to $z = \omega(x/a - \log a)$ by [Theorem 18.1](#). Monotonicity and bijectivity are inherited from ϕ through the same substitution. For the second form, take logarithms in $y^\alpha e^y = X$: since $y > 0$ and $X > 0$, this is equivalent to $y + \alpha \log y = \log X$, and [\(327\)](#) applies with $a = \alpha$, $x = \log X$. $\square$

The formal counterpart is stronger: it holds over any characteristic-zero $\mathbb{K}$, for every scalar a , and it exhibits the *same* polynomials with only a scalar reweighting.

Theorem 18.5 (The one-parameter family of logarithmic reversions). *Let $a \in \mathbb{K}$ and let $F_a := X + aL \in \mathcal{G}_{\text{poly}, \mathbb{K}}$. Then $F_a^{(-1)\circ} \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ exists, is unique, and equals*

$$F_a^{(-1)\circ}(X) = X + \sum_{n \geq 0} a^{n+1} \mathbb{L}_n^{\text{Lam}}(L) t^n = X - aL + \sum_{n \geq 1} a^{n+1} \mathbb{L}_n^{\text{Lam}}(L) t^n, \quad (328)$$

with $\mathbb{L}_n^{\text{Lam}}$ as in [Theorem 19.1](#). In particular $F_1^{(-1)\circ} = \omega$ and

$$(X - L)^{(-1)\circ}(X) = X + L + \sum_{n \geq 1} (-1)^{n+1} \mathbb{L}_n^{\text{Lam}}(L) t^n. \quad (329)$$

Proof. Existence and uniqueness are [Theorem 14.11](#), since $aL \in \mathbb{K}[L][[t]]$. For the formula, write $\mathcal{P}(t, L) := \sum_{n \geq 0} \mathbb{L}_n^{\text{Lam}}(L) t^n \in \mathbb{K}[L][[t]]$. By [Theorem 18.9](#),

$$\mathcal{P} + L + \log(1 + t\mathcal{P}) = 0. \quad (330)$$

Substituting $t \mapsto at$ into (330) – a legitimate $\mathbb{K}$ -algebra endomorphism of $\mathbb{K}[L][[t]]$, continuous for the t -adic filtration, hence commuting with the formal logarithm – gives

$$\mathcal{P}(at, L) + L + \log(1 + at\mathcal{P}(at, L)) = 0.$$

Put $g := a\mathcal{P}(at, L) \in \mathbb{K}[L][[t]]$ and multiply by a :

$$g + aL + a\log(1 + tg) = 0. \quad (331)$$

Now let $G = X + g$. By generator substitution ([Theorem 10.22](#)), $L \circ G = L + \log(1 + tg)$, so

$$F_a \circ G = G + a(L \circ G) = X + g + aL + a\log(1 + tg) = X$$

by (331). Thus G is a right inverse of F_a in $\mathcal{G}_{\text{poly}, \mathbb{K}}$, hence the two-sided inverse by uniqueness ([Theorem 14.11\(2\),\(3\)](#)). Finally $g = a \sum_{n \geq 0} \mathbb{L}_n^{\text{Lam}}(L) a^n t^n = \sum_{n \geq 0} a^{n+1} \mathbb{L}_n^{\text{Lam}}(L) t^n$, which is (328); and $a^{n+1}|_{a=-1} = (-1)^{n+1}$, with $(-1)^1 \mathbb{L}_0^{\text{Lam}}(L) = L$, gives (329). $\square$

Remark 18.6 (Why no “second Lambert family” exists). The inverse of $X - \log X$ is often presented as a separate expansion with its own alternating coefficients. Equation (329) settles the matter: the blocks are $(-1)^{n+1} \mathbb{L}_n^{\text{Lam}}$, the same polynomials with a sign. The frequently quoted weight $(-1)^n$ is correct for $-(X - L)^{(-1)^\circ}$, not for $(X - L)^{(-1)^\circ}$ itself; that sign flip is exactly the substitution $w \mapsto -w$ used to reach the lower real Lambert branch, and it is the source of a recurring off-by-one-sign error in the source drafts. Concretely, for $x \rightarrow 0^-$, putting $T = \log(1/(-x))$ and $\ell = \log T$, the substitution $W_{-1}(x) = -v$ turns $we^w = x$ into $v - \log v = T$, so (329) gives the block pattern

$$W_{-1}(x) \text{ blocks} = -T - \ell + \sum_{n \geq 1} (-1)^n \mathbb{L}_n^{\text{Lam}}(\ell) T^{-n}.$$

The formal identity is (329); the asymptotic assertion about W_{-1} needs the analytic transfer of [Section 18.6](#) applied on the branch in question and is not proved here.

Remark 18.7 (Branch coordinates). Taking a compatible logarithm in $W(z)e^{W(z)} = z$ gives $W_k(z) + \log W_k(z) = \xi_k$ with $\xi_k = \text{Log } z + 2\pi i k$. Thus every large branch of W is the *same* reversion evaluated at a different large variable, and the formal expansion of W_k is obtained from [Theorem 18.10](#) by the substitutions $X \mapsto \xi_k$, $L \mapsto \ell_k = \log_{\mathcal{B}} \xi_k$ for a declared branch $\mathcal{B}$. The polynomials $\mathbb{L}_n^{\text{Lam}}$ are therefore branch-independent; the coordinates ξ_k, ℓ_k , the branch $\mathcal{B}$, the sector, and the resulting analytic estimates are not. We do not develop the complex branch analysis here.

18.4 The formal expansion

The reversion machinery of Part IV applies verbatim, and in this special case it degenerates in a useful way: each Lagrange–Bürmann summand produces exactly one block, so no cross-order collection is ever needed. We first record the operator identity that makes this visible.

Lemma 18.8 (Repeated derivative of a coefficient block). *Let $n \in \mathbb{Z}$, $k \in \mathbb{N}_0$ and $p \in \mathbb{K}[L]$. Then*

$$D_X^k(t^n p(L)) = t^{n+k} \Delta_{n,k} p(L), \quad \Delta_{n,k} = \prod_{j=0}^{k-1} (\partial_L - n - j), \quad (332)$$

the empty product at $k = 0$ acting as the identity.

Proof. Induction on k . The case $k = 0$ is trivial. Assume (332) for k and write $q = \Delta_{n,k}p \in \mathbb{K}[L]$. Since $D_X t = -t^2$ and $D_X L = t$, Leibniz and the chain rule for the derivation D_X give

$$D_X(t^{n+k}q(L)) = (n+k)t^{n+k-1}(-t^2)q(L) + t^{n+k}q'(L)t = t^{n+k+1}(\partial_L - (n+k))q(L).$$

The factors of $\Delta_{n,k}$ are polynomials in ∂_L and therefore commute, so $(\partial_L - (n+k))\Delta_{n,k} = \Delta_{n,k+1}$, which is the statement for $k+1$. $\square$

Proposition 18.9 (The implicit equation for the correction). *Let $\mathcal{T}: \mathbb{K}[L][[t]] \rightarrow \mathbb{K}[L][[t]]$ be $\mathcal{T}(U) = -L - \log(1+tU)$, the logarithm being the formal one of Theorem 10.5. Then $\mathcal{T}$ has a unique fixed point $\mathcal{P} \in \mathbb{K}[L][[t]]$; equivalently, the equation*

$$\mathcal{P} + L + \log(1+t\mathcal{P}) = 0 \tag{333}$$

has exactly one solution in $\mathbb{K}[L][[t]]$. That solution lies in $-L + t\mathbb{K}[L][[t]]$, satisfies $\mathcal{P}(t, 0) = 0$, and

$$\omega = (X + L)^{\langle -1 \rangle^\circ} = X + \mathcal{P}. \tag{334}$$

Proof. For $U \in \mathbb{K}[L][[t]]$ we have $tU \in t\mathbb{K}[L][[t]]$, so the family $((-1)^{j-1}(tU)^j/j)_{j \geq 1}$ is formally summable and $\log(1+tU) \in t\mathbb{K}[L][[t]]$; hence $\mathcal{T}$ is well defined and maps into $-L + t\mathbb{K}[L][[t]]$. If $\nu_t(U - V) \geq N$, then, $1+tV$ being a unit of $\mathbb{K}[L][[t]]$,

$$\mathcal{T}(U) - \mathcal{T}(V) = -\log \frac{1+tU}{1+tV} = -\log \left(1 + \frac{t(U-V)}{1+tV} \right) \in t^{N+1}\mathbb{K}[L][[t]],$$

so $\nu_t(\mathcal{T}(U) - \mathcal{T}(V)) \geq \nu_t(U - V) + 1$. Thus $\mathcal{T}$ is a strict contraction for the t -adic filtration on the whole of $\mathbb{K}[L][[t]]$, which is separated and complete (Theorem 12.13). Two fixed points would satisfy $\nu_t(U - V) \geq \nu_t(U - V) + 1$, forcing $\nu_t(U - V) = +\infty$ and $U = V$; and the iterates $U_0 = 0$, $U_{r+1} = \mathcal{T}(U_r)$ form a Cauchy sequence whose limit is a fixed point. Its membership in $-L + t\mathbb{K}[L][[t]]$ is the image statement above.

Setting $L = 0$ in (333) – that is, applying the $\mathbb{K}$ -algebra map $\mathbb{K}[L][[t]] \rightarrow \mathbb{K}[[t]]$, $L \mapsto 0$, which is t -adically continuous and hence commutes with the formal logarithm – shows that $V := \mathcal{P}(t, 0)$ solves $V + \log(1+tV) = 0$ in $\mathbb{K}[[t]]$. The same contraction argument applied to $V \mapsto -\log(1+tV)$ gives a unique solution, and $V = 0$ is one; hence $\mathcal{P}(t, 0) = 0$.

Finally put $G = X + \mathcal{P} \in \mathcal{G}_{\text{poly}, \mathbb{K}}$. By Theorem 10.22, $(X + L) \circ G = G + L + \log(1+t\mathcal{P}) = X$ by (333), so G is a right inverse of $X + L$, hence $G = (X + L)^{\langle -1 \rangle^\circ}$ by Theorem 14.11(2),(3). That $(X + L)^{\langle -1 \rangle^\circ}$ is what the notation ω denotes formally is Theorem 18.1 together with Theorem 18.17 below; at this point (334) is a definition of the symbol on the formal side. $\square$

Theorem 18.10 (Polynomial–logarithmic expansion of ω). *In $\mathcal{G}_{\text{poly}, \mathbb{K}}$,*

$$\omega = X + \sum_{n \geq 0} \mathbb{L}_n^{\text{Lam}}(L) t^n = X - L + \sum_{n \geq 1} \mathbb{L}_n^{\text{Lam}}(L) t^n, \tag{335}$$

where $\mathbb{L}_n^{\text{Lam}}$ is the Lambert polynomial of Theorem 19.1. More precisely, the r -th Lagrange–Bürmann summand for $F = X + L$ is a single block:

$$\frac{(-1)^r}{r!} D_X^{r-1}(L^r) = \mathbb{L}_{r-1}^{\text{Lam}}(L) t^{r-1}, \quad r \geq 1. \tag{336}$$

Proof. Apply the Lagrange–Bürmann formula Theorem 16.14, in the form (281), to $F = X + A$ with $A = L$:

$$F^{\langle -1 \rangle^\circ}(X) = X + \sum_{r \geq 1} \frac{(-1)^r}{r!} D_X^{r-1}(L^r),$$

the family being formally summable because the r -th summand has t -order at least $r - 1$. By [Theorem 18.8](#) with $n = 0$, $k = r - 1$, $p = L^r$,

$$D_X^{r-1}(L^r) = t^{r-1} \Delta_{0,r-1} L^r = t^{r-1} \left[\prod_{j=0}^{r-2} (\partial_L - j) \right] L^r.$$

Setting $n = r - 1$, the scalar prefactor is $\frac{(-1)^r}{r!} = \frac{(-1)^{n+1}}{(n+1)!}$ and the operator is $\prod_{j=0}^{n-1} (\partial_L - j)$ applied to L^{n+1} ; by [Theorem 19.1](#) the product of the two is exactly $\mathbb{L}_n^{\text{Lam}}(L)$. This is (336). Since the summand of index r has t -order exactly $r - 1$ and no other summand meets that order, summing over $r \geq 1$ gives (335) with no collection of contributions; the $r = 1$ term is $-L = \mathbb{L}_0^{\text{Lam}}(L)$. $\square$

Remark 18.11 (Two normalisations of the same expansion). The two displays in (335) are the same series. The zero-based one is canonical: it is the form in which (336) says “one Lagrange index per block”, and the form in which [Theorem 18.5](#) needs no case distinction. The one-based form, which displays $-L$ separately, is the classical way of writing the Lambert expansion and is the one the source drafts mostly use. The price of the zero-based normalisation is that $\deg \mathbb{L}_0^{\text{Lam}} = 1 \neq 0$: the degree law $\deg \mathbb{L}_n^{\text{Lam}} = n$ holds for $n \geq 1$ only ([Theorem 19.7](#)).

18.5 The truncation residual

Definition 18.12 (Omega truncations). For $N \in \mathbb{N}_0$ set

$$\omega_N := X + \text{Trunc}_{<N+1} \mathcal{P} = X - L + \sum_{n=1}^N \mathbb{L}_n^{\text{Lam}}(L) t^n \in \mathcal{G}_{\text{poly}, \mathbb{K}}. \quad (337)$$

In the inclusive notation of source 6 this is $X + [\mathcal{P}]_{\leq N}$; the one-block difference $[F]_{\leq N} = \text{Trunc}_{<N+1} F$ is the only discrepancy between the two conventions and is a frequent source of apparent off-by-one disagreements between the drafts.

Theorem 18.13 (Exact leading residual). *For every $N \in \mathbb{N}_0$, in $\mathbb{K}[L][[t]]$,*

$$\text{Err}_{\text{rev}}(X + L, \omega_N) = \omega_N + \log \omega_N - X = -\mathbb{L}_{N+1}^{\text{Lam}}(L) t^{N+1} + \mathcal{O}_t^{\text{form}}(t^{N+2}). \quad (338)$$

In particular $\nu_t(\text{Err}_{\text{rev}}(X + L, \omega_N)) = N + 1$ exactly, for every $N \geq 0$.

Proof. Write $\mathcal{P}_N = \text{Trunc}_{<N+1} \mathcal{P}$ and $\Delta = \mathcal{P} - \mathcal{P}_N = \sum_{n \geq N+1} \mathbb{L}_n^{\text{Lam}}(L) t^n$, so that $\nu_t \Delta = N + 1$ and $[t^{N+1}] \Delta = \mathbb{L}_{N+1}^{\text{Lam}}(L)$. By generator substitution ([Theorem 10.22](#)),

$$\text{Err}_{\text{rev}}(X + L, \omega_N) = \mathcal{P}_N + L + \log(1 + t\mathcal{P}_N).$$

Using $\mathcal{P}_N = \mathcal{P} - \Delta$, the identity $1 + t\mathcal{P}_N = (1 + t\mathcal{P}) \left(1 - \frac{t\Delta}{1 + t\mathcal{P}}\right)$, and additivity of the formal logarithm on $1 + t\mathbb{K}[L][[t]]$ ([Theorem 10.5](#)), we get

$$\text{Err}_{\text{rev}}(X + L, \omega_N) = \underbrace{\mathcal{P} + L + \log(1 + t\mathcal{P})}_{=0 \text{ by (333)}} - \Delta + \log\left(1 - \frac{t\Delta}{1 + t\mathcal{P}}\right).$$

Since $1 + t\mathcal{P}$ is a unit of $\mathbb{K}[L][[t]]$ and $\nu_t \Delta = N + 1$, the argument of the last logarithm lies in $1 + t^{N+2} \mathbb{K}[L][[t]]$, so that logarithm lies in $t^{N+2} \mathbb{K}[L][[t]]$. Hence

$$\text{Err}_{\text{rev}}(X + L, \omega_N) = -\Delta + \mathcal{O}_t^{\text{form}}(t^{N+2}) = -\mathbb{L}_{N+1}^{\text{Lam}}(L) t^{N+1} + \mathcal{O}_t^{\text{form}}(t^{N+2}).$$

Exactness of the order follows because $\mathbb{L}_{N+1}^{\text{Lam}} \neq 0$ for every $N \geq 0$, by [Theorem 19.7](#). $\square$

Corollary 18.14 (Finite certificate for the Lambert polynomials). *Let $p_0, \dots, p_N \in \mathbb{K}[L]$ and set $H = X + \sum_{n=0}^N p_n(L)t^n$. Then $p_n = \mathbb{L}_n^{\text{Lam}}$ for $0 \leq n \leq N$ if and only if $\text{Err}_{\text{rev}}(X + L, H) = \mathcal{O}_t^{\text{form}}(t^{N+1})$.*

Proof. Necessity is [Theorem 18.13](#). Conversely, if the residual has t -order at least $N + 1$ then H is a right inverse of $X + L$ through block N in the sense of [Theorem 14.16](#), so $\text{Trunc}_{<N+1} H - X = \text{Trunc}_{<N+1} \mathcal{P}$, i.e. $p_n = \mathbb{L}_n^{\text{Lam}}$ for $n \leq N$. $\square$

Example 18.15 (The residual at $N = 2$). With $\mathcal{P}_2 = -L + Lt + (\frac{1}{2}L^2 - L)t^2$,

$$\mathcal{P}_2 + L + \log(1 + t\mathcal{P}_2) = -\left(\frac{1}{3}L^3 - \frac{3}{2}L^2 + L\right)t^3 + \mathcal{O}_t^{\text{form}}(t^4) = -\mathbb{L}_3^{\text{Lam}}(L)t^3 + \mathcal{O}_t^{\text{form}}(t^4),$$

which one checks by expanding $\log(1 + t\mathcal{P}_2) = t\mathcal{P}_2 - \frac{1}{2}t^2\mathcal{P}_2^2 + \frac{1}{3}t^3\mathcal{P}_2^3 - \dots$ and collecting t^0, t^1, t^2, t^3 . This single computation certifies $\mathbb{L}_0^{\text{Lam}}, \mathbb{L}_1^{\text{Lam}}, \mathbb{L}_2^{\text{Lam}}$ simultaneously and predicts $\mathbb{L}_3^{\text{Lam}}$; it is the cheapest available unit test for an implementation of the recurrence.

Example 18.16 (Every truncation is exact at $x = 1$). Because $\mathbb{L}_n^{\text{Lam}}(0) = 0$ for all n ([Theorem 19.2](#)), substituting $x = 1$, where $L = \log 1 = 0$, gives $\omega_N(1) = 1 - 0 + 0 = 1 = \omega(1)$ for every N . The same phenomenon in the Lambert coordinates is the exactness of the branch expansion at $z = e$, where $\xi_0 = 1$ and $\ell_0 = 0$. It is a consequence of the vanishing $\mathbb{L}_n^{\text{Lam}}(0) = 0$, not of any convergence property.

18.6 Formal statement, analytic statement

[Theorem 18.13](#) is an identity in $\mathbb{K}[L][[t]]$. It asserts coefficient agreement below a cutoff and nothing else: no numerical bound, no convergence, no uniformity in L , no claim that ω exists as a function. An analytic statement about the real function of [Theorem 18.1](#) needs four further ingredients, none of which is formal algebra:

1. a *regime*: here the ray $x \rightarrow +\infty$ with the real logarithm, so that $L = \log x$ is a real quantity that grows with x ;
2. an *analytic remainder* for the truncated logarithm, replacing the formal $\log(1 + t\mathcal{P}_N)$ by a finite sum plus an estimated tail;
3. a *lower bound on the derivative* of ϕ between the true and the approximate root, to convert a residual bound into an error bound;
4. an *admissibility check* that the approximation stays inside the domain of ϕ , here $\omega_N(x) > 0$.

The following theorem supplies all four. Note that the estimate is *not* uniform in N , and that the exponent of $\log x$ in the remainder is quadratic in N , not linear; this is an honest feature of the argument, not an artefact.

Theorem 18.17 (Asymptotic validity on the positive real ray). *Fix $N \in \mathbb{N}_0$ and put $K = (N + 2)^2$. Then, with $L = \log x$,*

$$\omega_N(x) + \log \omega_N(x) - x = -\frac{\mathbb{L}_{N+1}^{\text{Lam}}(L)}{x^{N+1}} + \mathcal{O}_{x \rightarrow +\infty}\left(\frac{L^K}{x^{N+2}}\right) \quad (339)$$

and

$$\omega(x) - \omega_N(x) = \frac{\mathbb{L}_{N+1}^{\text{Lam}}(L)}{x^{N+1}} + \mathcal{O}_{x \rightarrow +\infty}\left(\frac{L^K}{x^{N+2}}\right). \quad (340)$$

Consequently

$$\omega(x) - \omega_N(x) \sim \frac{(\log x)^{N+1}}{(N + 1)x^{N+1}}, \quad x \rightarrow +\infty, \quad (341)$$

so ω_N approximates ω to order exactly $L^{N+1}x^{-(N+1)}$, and $X + \sum_{n \geq 0} \mathbb{L}_n^{\text{Lam}}(L)t^n$ is an asymptotic expansion of ω on the scale $\{(\log x)^m x^{-n}\}$ in the sense of [Theorem 6.43](#).

Proof. Write $t = 1/x$, $L = \log x$, and $p = p(x) := -L + \sum_{n=1}^N \mathsf{L}_n^{\text{Lam}}(L)x^{-n}$, so $\omega_N(x) = x + p = x(1 + \varepsilon)$ with $\varepsilon = p/x$.

Step 1: ε is small. For $x \geq e$ we have $L \geq 1$, and $|\mathsf{L}_n^{\text{Lam}}(L)| \leq c_n L^n$ with $c_n = \sum_m |[u^m] \mathsf{L}_n^{\text{Lam}}|$. Hence, once $x \geq L$, $\sum_{n=1}^N |\mathsf{L}_n^{\text{Lam}}(L)| x^{-n} \leq \sum_{n=1}^N c_n (L/x)^n \leq C_N L/x$. Therefore $|p| \leq L + C_N L/x \leq 2L$ and $|\varepsilon| \leq 2L/x$ for all x beyond some x_N , which we enlarge so that $|\varepsilon| \leq \frac{1}{2}$. In particular $\omega_N(x) \geq x/2 > 0$, settling item (4) above, and $\log \omega_N(x) = L + \log(1 + \varepsilon)$ with the real logarithm.

Step 2: an analytic remainder for the logarithm. For $|\varepsilon| \leq \frac{1}{2}$,

$$\log(1 + \varepsilon) = \sum_{j=1}^{N+1} \frac{(-1)^{j-1}}{j} \varepsilon^j + \theta, \quad |\theta| \leq \sum_{j>N+1} \frac{|\varepsilon|^j}{j} \leq 2|\varepsilon|^{N+2} \leq 2 \left(\frac{2L}{x} \right)^{N+2}.$$

Consequently

$$\omega_N(x) + \log \omega_N(x) - x = \Pi(1/x, \log x) + \theta, \quad \Pi := \mathcal{P}_N + L + \sum_{j=1}^{N+1} \frac{(-1)^{j-1}}{j} (t\mathcal{P}_N)^j, \quad (342)$$

where $\Pi \in \mathbb{K}[L][t]$ is a *polynomial* in t and L , evaluated at the real numbers $1/x$ and $\log x$.

Step 3: identifying Π . In $\mathbb{K}[L][[t]]$ we have $\nu_t(t\mathcal{P}_N) \geq 1$, so $\log(1+t\mathcal{P}_N) - \sum_{j \leq N+1} \frac{(-1)^{j-1}}{j} (t\mathcal{P}_N)^j \in t^{N+2} \mathbb{K}[L][[t]]$. Combining this with [Theorem 18.13](#),

$$\Pi \equiv -\mathsf{L}_{N+1}^{\text{Lam}}(L) t^{N+1} \pmod{t^{N+2} \mathbb{K}[L][t]}.$$

Now $\mathcal{P}_N$ has t -degree $\leq N$ and L -degree $\leq \max(N, 1)$, so Π has t -degree $\leq (N+1)^2$ and L -degree $\leq (N+1) \max(N, 1) \leq K$. Writing $\Pi = -\mathsf{L}_{N+1}^{\text{Lam}}(L) t^{N+1} + \sum_{i=N+2}^{(N+1)^2} c_i(L) t^i$ with $\deg c_i \leq K$, we obtain, for $x \geq x_N$,

$$\left| \Pi(1/x, \log x) + \mathsf{L}_{N+1}^{\text{Lam}}(L) x^{-(N+1)} \right| \leq C'_N \frac{L^K}{x^{N+2}}.$$

Since also $|\theta| \leq 2^{N+3} L^{N+2} x^{-(N+2)}$ and $N+2 \leq K$, (339) follows from (342).

Step 4: residual to error. Denote the left side of (339) by $R_N(x)$, so $\phi(\omega_N(x)) - \phi(\omega(x)) = R_N(x)$ with $\phi(Y) = Y + \log Y$. Both $\omega_N(x)$ and $\omega(x)$ lie in $(0, \infty)$, and by [Equation \(326\)](#) and Step 1 both exceed $x/2$ for $x \geq x_N$ (enlarging x_N again). By the mean value theorem there is ζ between them, hence $\zeta > x/2$, with

$$\omega_N(x) - \omega(x) = \frac{R_N(x)}{\phi'(\zeta)} = \frac{R_N(x)}{1 + \zeta^{-1}}, \quad \frac{1}{1 + \zeta^{-1}} = 1 + \mathcal{O}_{x \rightarrow +\infty}(x^{-1}).$$

This is the general residual-transfer principle of [Theorem 14.21](#) in the concrete case at hand, with derivative lower bound $m = 1$. Multiplying (339) by $-(1 + \zeta^{-1})^{-1}$ and absorbing $|\mathsf{L}_{N+1}^{\text{Lam}}(L)| x^{-(N+1)}$. $\mathcal{O}_{x \rightarrow +\infty}(x^{-1}) = \mathcal{O}_{x \rightarrow +\infty}(L^{N+1} x^{-(N+2)})$ into the remainder gives (340).

Step 5. By [Theorem 19.7](#), $\mathsf{L}_{N+1}^{\text{Lam}}(L) = L^{N+1}/(N+1) + \mathcal{O}_{L \rightarrow +\infty}(L^N)$, and $L^K x^{-(N+2)} = \mathcal{O}_{x \rightarrow +\infty}(L^{N+1} x^{-(N+1)})$ because $L^{K-N-1}/x \rightarrow 0$. Hence (341). Since (341) holds for every N , the partial sums are an asymptotic expansion on the stated scale. $\square$

Remark 18.18 (The rate in (341) is only logarithmic). Statement (340) is the useful one: the error of ω_N equals the *next block* with a relative error $\mathcal{O}_{x \rightarrow +\infty}(L^K/x)$. The simplified form (341) discards the lower-degree part of $\mathsf{L}_{N+1}^{\text{Lam}}$, whose largest term is $-H_N L^N$; the relative error of (341) is therefore only $\mathcal{O}_{L \rightarrow \infty}(1/L)$, i.e. $\mathcal{O}_{x \rightarrow +\infty}(1/\log x)$. For $N = 4$ and $x = 5 \cdot 10^5$ the ratio of the true error to $\mathsf{L}_5^{\text{Lam}}(L) x^{-5}$ is $1.0000148\dots$, while its ratio to $L^5/(5x^5)$ is still far from 1. Quoting (341) where (340) is meant is a standard way to make a correct theorem look numerically wrong.

The next proposition isolates the one convergence statement that the formal theory does yield, and makes clear why it is not the statement one wants.

Proposition 18.19 (Convergence at frozen logarithm). *For each $u_0 \in \mathbb{C}$ there are $r > 0$ and $\rho > 0$ such that the series $\sum_{n \geq 1} \mathbb{L}_n^{\text{Lam}}(u) s^n$ converges absolutely for $|s| < r$, uniformly for $|u - u_0| \leq \rho$, and its sum $Q(s, u)$ is the unique holomorphic solution near $(0, u_0)$ of*

$$Q = -\log(1 - su + sQ), \quad Q(0, u) = 0. \quad (343)$$

Proof. Let $\Phi(s, u, Q) = Q + \text{Log}(1 - su + sQ)$, holomorphic on a neighbourhood of $\{(0, u, 0) : u \in \mathbb{C}\}$ because $1 - su + sQ = 1$ there. We have $\Phi(0, u, 0) = 0$ and $\partial_Q \Phi(0, u, 0) = 1 + \frac{s}{1 - su + sQ} \Big|_{s=0} = 1 \neq 0$. The holomorphic implicit function theorem gives $r, \rho > 0$ and a unique holomorphic Q on $|s| < r$, $|u - u_0| < \rho$, with $Q(0, u) = 0$ and $\Phi = 0$. Its Taylor coefficients in s satisfy the same triangular recursion as those of the formal solution of (343), which by Theorem 19.11 is $\sum_{n \geq 1} \mathbb{L}_n^{\text{Lam}}(u) s^n$; by uniqueness of Taylor coefficients the two agree. $\square$

Remark 18.20 (What is *not* proved here). Theorem 18.19 freezes u and lets $s \rightarrow 0$. The asymptotic problem does the opposite: it couples them by $u = \log(1/s)$, so that $u \rightarrow \infty$ as $s \rightarrow 0$, and Theorem 18.19 says nothing about it, because r is not claimed to be bounded below as $|u_0| \rightarrow \infty$. In fact the implicit function theorem fails on the locus $\partial_Q \Phi = 0$, which after eliminating Q from (343) is $u = s^{-1} + 1 - \log(-s)$; this suggests $r \asymp 1/|u|$ for large $|u|$, which would be compatible with genuine convergence of $\sum_n \mathbb{L}_n^{\text{Lam}}(\log x) x^{-n}$ for large x . We record that as a heuristic only. The literature asserts absolute convergence of the associated double series on explicit regions [5, 21, 11]; no proof of any such convergence statement is given here, and none follows from anything in Parts I–IV. Likewise, uniformity of (340) in a complex sector, in the branch index k of Theorem 18.7, or in N , is neither claimed nor proved.

19 The Lambert polynomials

This section is self-contained: the polynomials are defined by an operator formula, and every property – the recurrence, the closed Stirling form, the coefficient structure, the integrality, the generating equation, the Laplace transform – is derived from that definition without using Section 18. The two sections meet in Theorem 19.11, which reproves Theorem 18.10 without the reversion machinery.

Throughout, u is an indeterminate, ∂_u is d/du , and $\Delta_{0,n} = \prod_{j=0}^{n-1} (\partial_u - j)$ is the falling-factorial operator ∂_u^n , with the empty product at $n = 0$ acting as the identity.

19.1 Definition and first values

Definition 19.1 (Canonical zero-based Lambert polynomials). For $n \in \mathbb{N}_0$ put

$$\mathbb{L}_n^{\text{Lam}}(u) := \frac{(-1)^{n+1}}{(n+1)!} \left[\prod_{j=0}^{n-1} (\partial_u - j) \right] u^{n+1} \in \mathbb{Q}[u], \quad (344)$$

the product being empty at $n = 0$. Since $\text{char } \mathbb{K} = 0$ we regard $\mathbb{L}_n^{\text{Lam}} \in \mathbb{K}[u]$ whenever convenient.

At $n = 0$ the operator is the identity and the prefactor is $(-1)^1/1! = -1$, so $\mathbb{L}_0^{\text{Lam}}(u) = -u$. The next values are

$$\mathbb{L}_1^{\text{Lam}}(u) = u, \quad \mathbb{L}_2^{\text{Lam}}(u) = \frac{u^2}{2} - u, \quad \mathbb{L}_3^{\text{Lam}}(u) = \frac{u^3}{3} - \frac{3}{2}u^2 + u, \quad (345)$$

computed as follows. For $n = 1$, $\partial_u u^2 = 2u$ and $(-1)^2/2! = \frac{1}{2}$, giving u . For $n = 2$, $(\partial_u - 1)\partial_u u^3 = (\partial_u - 1)(3u^2) = 6u - 3u^2$ and $(-1)^3/3! = -\frac{1}{6}$, giving $\frac{1}{2}u^2 - u$. For $n = 3$, $(\partial_u - 2)(\partial_u - 1)\partial_u u^4 = (\partial_u - 2)(\partial_u - 1)(4u^3) = (\partial_u - 2)(12u^2 - 4u^3) = 24u - 12u^2 - 24u^2 + 8u^3$, and $(-1)^4/4! = \frac{1}{24}$, giving $\frac{1}{3}u^3 - \frac{3}{2}u^2 + u$.

A common trap

The degree law $\deg \mathbb{L}_n^{\text{Lam}} = n$ is asserted for $n \geq 1$ only. The zero-index member $\mathbb{L}_0^{\text{Lam}}(u) = -u$ has degree 1. Any statement quantified over $n \geq 0$ that mentions the degree, the leading coefficient, or a coefficient indexed relative to the top must exclude $n = 0$ explicitly.

19.2 The recurrence

Theorem 19.2 (Differential and integral recurrence). *For every $n \in \mathbb{N}_0$,*

$$\mathbb{L}_n^{\text{Lam}}(0) = 0, \quad (346)$$

$$\frac{d}{du} \mathbb{L}_{n+1}^{\text{Lam}}(u) = n \mathbb{L}_n^{\text{Lam}}(u) - \frac{d}{du} \mathbb{L}_n^{\text{Lam}}(u), \quad (347)$$

$$\mathbb{L}_{n+1}^{\text{Lam}}(u) = -\mathbb{L}_n^{\text{Lam}}(u) + n \int_0^u \mathbb{L}_n^{\text{Lam}}(v) \, dv. \quad (348)$$

Moreover, (348) for all $n \geq 0$ is equivalent to the conjunction of (347) for all $n \geq 0$ with (346) for all $n \geq 0$; and either version, together with $\mathbb{L}_0^{\text{Lam}}(u) = -u$, determines the whole family uniquely.

Proof. Boundary value. Expanding the operator in (344) as a $\mathbb{Q}$ -linear combination of ∂_u^k , $0 \leq k \leq n$, and applying it to u^{n+1} produces a linear combination of the monomials u^{n+1-k} with $n+1-k \geq 1$. Hence $\mathbb{L}_n^{\text{Lam}}$ has zero constant term, which is (346).

Differential recurrence. Write $A_n = \Delta_{0,n}$, so $A_{n+1} = (\partial_u - n)A_n$ and A_n commutes with ∂_u , all factors being polynomials in ∂_u . From (344),

$$\mathbb{L}_{n+1}^{\text{Lam}}(u) = \frac{(-1)^n}{(n+2)!} (\partial_u - n)A_n u^{n+2}.$$

Differentiate and commute ∂_u past $(\partial_u - n)A_n$:

$$\frac{d}{du} \mathbb{L}_{n+1}^{\text{Lam}}(u) = \frac{(-1)^n}{(n+2)!} (\partial_u - n)A_n \partial_u u^{n+2} = \frac{(-1)^n(n+2)}{(n+2)!} (\partial_u - n)A_n u^{n+1} = \frac{(-1)^n}{(n+1)!} (\partial_u - n)A_n u^{n+1}.$$

By (344) again, $A_n u^{n+1} = (-1)^{n+1}(n+1)! \mathbb{L}_n^{\text{Lam}}(u)$, so

$$\frac{d}{du} \mathbb{L}_{n+1}^{\text{Lam}}(u) = \frac{(-1)^n(-1)^{n+1}(n+1)!}{(n+1)!} (\partial_u - n) \mathbb{L}_n^{\text{Lam}}(u) = -(\partial_u - n) \mathbb{L}_n^{\text{Lam}}(u),$$

which is (347).

Equivalence of the two forms. Given (347) and (346), integrate (347) from 0 to u and use $\mathbb{L}_{n+1}^{\text{Lam}}(0) = \mathbb{L}_n^{\text{Lam}}(0) = 0$ to get (348). Conversely, differentiating (348) returns (347), and evaluating (348) at $u = 0$ gives $\mathbb{L}_{n+1}^{\text{Lam}}(0) = -\mathbb{L}_n^{\text{Lam}}(0)$; since $\mathbb{L}_0^{\text{Lam}}(0) = 0$, induction yields (346) for all n .

Uniqueness. (348) computes $\mathbb{L}_{n+1}^{\text{Lam}}$ from $\mathbb{L}_n^{\text{Lam}}$ by operations with no free constant, so the family is determined by $\mathbb{L}_0^{\text{Lam}} = -u$. $\square$

Remark 19.3 (Reconciliation with the one-based drafts). Sources 3 and 6 state the recurrence as starting from $\mathbb{L}_1^{\text{Lam}}(u) = u$ with $\mathbb{L}_n^{\text{Lam}}(0) = 0$ imposed. The zero-based normalisation subsumes this: with $\mathbb{L}_0^{\text{Lam}} = -u$, (348) at $n = 0$ reads $\mathbb{L}_1^{\text{Lam}} = -\mathbb{L}_0^{\text{Lam}} + 0 = u$. Source 4 derives (347) by “equating coefficients” in the first-order equation $T\mathcal{L}\partial_T\mathcal{L} + (1+T)\partial_T\mathcal{L} + \mathcal{L} = 0$ satisfied by the generating series; that equation is *quadratic* in $\mathcal{L}$ and equating coefficients in it yields a convolution recurrence, not the two-term linear one. The correct generating-function route runs through the *linear* equation (358) below, and is given in Theorem 19.11.

19.3 The closed Stirling form

We use the signed and unsigned Stirling numbers of the first kind in the catalogue normalisation:

$$(z)^{\underline{n}} = z(z-1)\cdots(z-n+1) = \sum_{k=0}^n s(n, k) z^k, \quad \left[\begin{matrix} n \\ k \end{matrix} \right] = (-1)^{n-k} s(n, k), \quad (349)$$

with $s(0, 0) = 1$ and $s(n, k) = 0$ for $k < 0$ or $k > n$.

Theorem 19.4 (Stirling-number formula). *For every $n \geq 1$,*

$$\mathbb{L}_n^{\text{Lam}}(u) = \sum_{m=1}^n (-1)^{n-m} \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] \frac{u^m}{m!}. \quad (350)$$

Equivalently, writing $\mathbb{L}_n^{\text{Lam}}(u) = \sum_{m \geq 0} \lambda_{n,m} u^m$,

$$\lambda_{n,m} = \frac{(-1)^{n-m}}{m!} \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] = \frac{(-1)^{n+1}}{m!} s(n, n-m+1), \quad 1 \leq m \leq n, \quad (351)$$

and $\lambda_{n,m} = 0$ for $m = 0$ and for $m > n$. If the upper limit in (350) is raised from n to $n+1$, the formula holds for $n = 0$ as well and returns $\mathbb{L}_0^{\text{Lam}}(u) = -u$.

Proof. Substituting $z \mapsto \partial_u$ in (349) – a substitution of a commuting operator into a polynomial identity, hence valid – gives $\Delta_{0,n} = \sum_{k=0}^n s(n, k) \partial_u^k$. Since $\partial_u^k u^{n+1} = \frac{(n+1)!}{(n+1-k)!} u^{n+1-k}$ for $0 \leq k \leq n+1$, (344) gives

$$\mathbb{L}_n^{\text{Lam}}(u) = \frac{(-1)^{n+1}}{(n+1)!} \sum_{k=0}^n s(n, k) \frac{(n+1)!}{(n+1-k)!} u^{n+1-k} = (-1)^{n+1} \sum_{k=0}^n s(n, k) \frac{u^{n+1-k}}{(n+1-k)!}.$$

Put $m = n+1-k$, so $k = n-m+1$ and m runs from 1 to $n+1$:

$$\mathbb{L}_n^{\text{Lam}}(u) = (-1)^{n+1} \sum_{m=1}^{n+1} s(n, n-m+1) \frac{u^m}{m!}.$$

This is the second equality of (351), and the first follows from $s(n, n-m+1) = (-1)^{n-(n-m+1)} \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] = (-1)^{m-1} \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right]$, so that $(-1)^{n+1}(-1)^{m-1} = (-1)^{n+m} = (-1)^{n-m}$. For $n \geq 1$, the term $m = n+1$ carries $s(n, 0) = 0$ and may be dropped, giving (350). For $n = 0$ the only term is $m = 1$, with $s(0, 0) = 1$ and prefactor $(-1)^1$, giving $-u$. $\square$

Remark 19.5 (A second proof, and where the Stirling numbers come from). Formula (350) can also be obtained from Theorem 19.2 alone, and that route explains the appearance of Stirling numbers structurally. Put $\sigma_{n,m} := m! \lambda_{n,m}$. The integral recurrence (348) reads, coefficientwise, $\lambda_{n+1,m} = -\lambda_{n,m} + \frac{n}{m} \lambda_{n,m-1}$, which in the σ normalisation becomes

$$\sigma_{n+1,m} = -\sigma_{n,m} + n \sigma_{n,m-1}, \quad \sigma_{0,1} = -1. \quad (352)$$

Setting $\sigma_{n,m} = (-1)^{n-m} \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right]$ and $k = n-m+2$ turns (352) into $\left[\begin{matrix} n+1 \\ k \end{matrix} \right] = \left[\begin{matrix} n \\ k-1 \end{matrix} \right] + n \left[\begin{matrix} n \\ k \end{matrix} \right]$, the defining recurrence of the unsigned Stirling numbers. Thus the Lambert recurrence is the Stirling recurrence, transported by the normalisation $u^m \mapsto u^m/m!$. Source 4 uses this induction; it is valid, but it presupposes the recurrence, whereas the proof above needs only the definition.

19.4 Coefficient structure

The four “edge” coefficients below all come from a single generating polynomial, so we prove the Stirling evaluations rather than quote them.

Lemma 19.6 (Four Stirling evaluations). *For $n \geq 1$ let $H_r = \sum_{j=1}^r j^{-1}$ and $H_r^{(2)} = \sum_{j=1}^r j^{-2}$, with $H_0 = H_0^{(2)} = 0$. Then*

$$\begin{bmatrix} n \\ 1 \end{bmatrix} = (n-1)!, \quad \begin{bmatrix} n \\ 2 \end{bmatrix} = (n-1)! H_{n-1}, \quad \begin{bmatrix} n \\ 3 \end{bmatrix} = \frac{(n-1)!}{2} (H_{n-1}^2 - H_{n-1}^{(2)}),$$

and for $n \geq 2$, $\begin{bmatrix} n \\ n-1 \end{bmatrix} = \binom{n}{2}$, while $\begin{bmatrix} n \\ n \end{bmatrix} = 1$.

Proof. Replacing z by $-z$ in (349) gives the unsigned form $(z)^{\bar{n}} = \sum_k \begin{bmatrix} n \\ k \end{bmatrix} z^k$, where $(z)^{\bar{n}} = z(z+1) \cdots (z+n-1)$. Since $\begin{bmatrix} n \\ 0 \end{bmatrix} = 0$ for $n \geq 1$, divide by z and set

$$f_n(z) := \prod_{j=1}^{n-1} (z+j) = \sum_{k \geq 1} \begin{bmatrix} n \\ k \end{bmatrix} z^{k-1}.$$

Then $\begin{bmatrix} n \\ 1 \end{bmatrix} = f_n(0) = (n-1)!$; f_n is monic of degree $n-1$, so $\begin{bmatrix} n \\ n \end{bmatrix} = 1$; and the coefficient of z^{n-2} in f_n is $\sum_{j=1}^{n-1} j = \binom{n}{2}$, i.e. $\begin{bmatrix} n \\ n-1 \end{bmatrix} = \binom{n}{2}$. For the remaining two, logarithmic differentiation gives $f'_n = f_n \sum_{j=1}^{n-1} (z+j)^{-1}$ and $f''_n = f_n [(\sum_{j=1}^{n-1} (z+j)^{-1})^2 - \sum_{j=1}^{n-1} (z+j)^{-2}]$; evaluating at $z = 0$, $\begin{bmatrix} n \\ 2 \end{bmatrix} = f'_n(0) = (n-1)! H_{n-1}$ and $\begin{bmatrix} n \\ 3 \end{bmatrix} = \frac{1}{2} f''_n(0) = \frac{1}{2} (n-1)! (H_{n-1}^2 - H_{n-1}^{(2)})$. $\square$

Corollary 19.7 (Coefficient structure of $\mathbb{L}_n^{\text{Lam}}$). *Let $n \geq 1$. Then:*

- (i) $\deg \mathbb{L}_n^{\text{Lam}} = n$, the constant term is 0, and $u \mid \mathbb{L}_n^{\text{Lam}}$;
- (ii) all n coefficients $\lambda_{n,1}, \dots, \lambda_{n,n}$ are nonzero and strictly alternate: $\text{sgn } \lambda_{n,m} = (-1)^{n-m}$;
- (iii) $[u^n] \mathbb{L}_n^{\text{Lam}} = 1/n$;
- (iv) $[u^{n-1}] \mathbb{L}_n^{\text{Lam}} = -H_{n-1}$ for $n \geq 2$;
- (v) $[u] \mathbb{L}_n^{\text{Lam}} = (-1)^{n-1}$, equivalently $\mathbb{L}_n^{\text{Lam}'}(0) = (-1)^{n-1}$;
- (vi) $[u^{n-2}] \mathbb{L}_n^{\text{Lam}} = \frac{n-1}{2} (H_{n-1}^2 - H_{n-1}^{(2)})$ for $n \geq 3$;
- (vii) $[u^2] \mathbb{L}_n^{\text{Lam}} = (-1)^n \frac{n(n-1)}{4}$ for $n \geq 2$.

In particular $\mathbb{L}_n^{\text{Lam}}(u) = \frac{u^n}{n} - H_{n-1} u^{n-1} + \cdots + (-1)^{n-1} u$.

Proof. All statements read off (351) together with Theorem 19.6. (i) and (ii): for $1 \leq m \leq n$ the index $k = n - m + 1$ satisfies $1 \leq k \leq n$, so $\begin{bmatrix} n \\ k \end{bmatrix} > 0$ (it counts the permutations of n letters with k cycles, and there is at least one); hence $\lambda_{n,m} \neq 0$ with sign $(-1)^{n-m}$, and $\lambda_{n,m} = 0$ outside $1 \leq m \leq n$. (iii): $m = n$ gives $k = 1$ and $\lambda_{n,n} = \begin{bmatrix} n \\ 1 \end{bmatrix} / n! = (n-1)! / n! = 1/n$. (iv): $m = n-1$ gives $k = 2$ and $\lambda_{n,n-1} = -\begin{bmatrix} n \\ 2 \end{bmatrix} / (n-1)! = -H_{n-1}$. (v): $m = 1$ gives $k = n$ and $\lambda_{n,1} = (-1)^{n-1} \begin{bmatrix} n \\ n \end{bmatrix} = (-1)^{n-1}$. (vi): $m = n-2$ gives $k = 3$ and $\lambda_{n,n-2} = \begin{bmatrix} n \\ 3 \end{bmatrix} / (n-2)! = \frac{(n-1)!}{2(n-2)!} (H_{n-1}^2 - H_{n-1}^{(2)})$. (vii): $m = 2$ gives $k = n-1$ and $\lambda_{n,2} = (-1)^{n-2} \begin{bmatrix} n \\ 2 \end{bmatrix} / 2! = (-1)^n n(n-1)/4$. $\square$

Remark 19.8 (The formulas overlap consistently). At small n several of the seven statements describe the same coefficient, and they agree. At $n = 3$, (iv) and (vii) both give $[u^2] \mathbb{L}_3^{\text{Lam}} = -\frac{3}{2}$, and (v) and (vi) both give $[u] \mathbb{L}_3^{\text{Lam}} = 1$. At $n = 4$, (vi) and (vii) both give $[u^2] \mathbb{L}_4^{\text{Lam}} = 3$, since $\frac{3}{2} (H_3^2 - H_3^{(2)}) = \frac{3}{2} (\frac{121}{36} - \frac{49}{36}) = 3$. Formula (vi) also returns the correct value 0 at $n = 2$, where $H_1^2 - H_1^{(2)} = 0$ and $[u^0] \mathbb{L}_2^{\text{Lam}} = 0$; it is stated for $n \geq 3$ only because u^{n-2} is not a monomial for $n < 2$.

19.5 Integrality and the scaled family

Corollary 19.9 (Integrality). *For $n \geq 1$ and $0 \leq m \leq n$,*

$$\sigma_{n,m} := m! [u^m] \mathbb{L}_n^{\text{Lam}}(u) = (-1)^{n-m} \begin{bmatrix} n \\ n-m+1 \end{bmatrix} \in \mathbb{Z}. \quad (353)$$

Consequently the scaled family

$$\mathbb{L}_n^{\text{Lam,sc}}(u) := n! \mathbb{L}_n^{\text{Lam}}(u) = \sum_{m=1}^n \frac{n!}{m!} \sigma_{n,m} u^m \quad (354)$$

lies in $\mathbb{Z}[u]$, is divisible by u , has degree n , leading coefficient $(n-1)!$ and linear coefficient $(-1)^{n-1}n!$, and satisfies

$$\mathbb{L}_{n+1}^{\text{Lam,sc}}(u) = -(n+1) \mathbb{L}_n^{\text{Lam,sc}}(u) + n(n+1) \int_0^u \mathbb{L}_n^{\text{Lam,sc}}(v) \, dv. \quad (355)$$

The convention $\mathbb{L}_0^{\text{Lam,sc}}(u) = 0! \mathbb{L}_0^{\text{Lam}}(u) = -u$ makes (355) valid at $n = 0$ as well.

Proof. (353) is (351) rewritten, and $\begin{bmatrix} n \\ k \end{bmatrix} \in \mathbb{Z}$. Then $n! \lambda_{n,m} = \frac{n!}{m!} \sigma_{n,m}$ is an integer because $m \leq n$. The listed data are Theorem 19.7(i), (iii), (v) multiplied by $n!$. Multiplying (348) by $(n+1)!$ and using $(n+1)! = (n+1) \cdot n!$ gives (355); at $n = 0$ both sides equal u . $\square$

A common trap

(355) is an identity in $\mathbb{Q}[u]$ and does *not* manifestly preserve $\mathbb{Z}[u]$: the antiderivative divides the coefficient of u^m by $m+1$, and $n(n+1)/(m+1)$ need not be an integer. Integrality of $\mathbb{L}_n^{\text{Lam,sc}}$ is a theorem (Theorem 19.9), proved through the Stirling form, not a visible feature of the recurrence. An implementation that wants to stay inside $\mathbb{Z}$ should iterate (352) on the integers $\sigma_{n,m}$ – equivalently, build the Stirling triangle – and divide by $m!$ only at output time.

The first scaled polynomials are

$$\begin{aligned} \mathbb{L}_0^{\text{Lam,sc}}(u) &= -u, & \mathbb{L}_1^{\text{Lam,sc}}(u) &= u, \\ \mathbb{L}_2^{\text{Lam,sc}}(u) &= u^2 - 2u, & \mathbb{L}_3^{\text{Lam,sc}}(u) &= 2u^3 - 9u^2 + 6u, \\ \mathbb{L}_4^{\text{Lam,sc}}(u) &= 6u^4 - 44u^3 + 72u^2 - 24u, & \mathbb{L}_5^{\text{Lam,sc}}(u) &= 24u^5 - 250u^4 + 700u^3 - 600u^2 + 120u, \end{aligned} \quad (356)$$

$$\mathbb{L}_6^{\text{Lam,sc}}(u) = 120u^6 - 1644u^5 + 6750u^4 - 10200u^3 + 5400u^2 - 720u. \quad (357)$$

Further values, and the corresponding $\mathbb{L}_n^{\text{Lam}}$, are tabulated in Section D.

19.6 The generating equation

Lemma 19.10 (The linear generating equation). *Let $\mathcal{P}(s, u) := \sum_{n \geq 0} \mathbb{L}_n^{\text{Lam}}(u) s^n \in \mathbb{K}[u][[s]]$. Then*

$$s^2 \partial_s \mathcal{P} = (1+s) \partial_u \mathcal{P} + 1. \quad (358)$$

Proof. Multiply (347) by s^{n+1} and sum over $n \geq 0$. The right-hand side gives $\sum_{n \geq 0} n \mathbb{L}_n^{\text{Lam}}(u) s^{n+1} = s \cdot s \partial_s \mathcal{P} = s^2 \partial_s \mathcal{P}$. The left-hand side gives

$$\sum_{n \geq 0} \left(\mathbb{L}_{n+1}^{\text{Lam}'}(u) + \mathbb{L}_n^{\text{Lam}'}(u) \right) s^{n+1} = (\partial_u \mathcal{P} - \mathbb{L}_0^{\text{Lam}'}(u)) + s \partial_u \mathcal{P} = (1+s) \partial_u \mathcal{P} + 1,$$

using $\mathbb{L}_0^{\text{Lam}'}(u) = -1$. $\square$

Theorem 19.11 (Implicit generating equation and formal uniqueness). *Let $Q(s, u) := \sum_{n \geq 1} \mathbb{L}_n^{\text{Lam}}(u) s^n \in \mathbb{K}[u][[s]]$. Then $Q(0, u) = 0$ and*

$$Q = -\log(1 - su + sQ), \quad (359)$$

the logarithm being the formal one on $1 + s\mathbb{K}[u][[s]]$. Conversely, (359) together with $Q(0, u) = 0$ has exactly one solution in $\mathbb{K}[u][[s]]$; hence it determines the family $(\mathbb{L}_n^{\text{Lam}})_{n \geq 1}$, and therefore also $\mathbb{L}_0^{\text{Lam}} = -u$ through (348), uniquely.

Proof. Write $\mathcal{P} = -u + Q$ as in Theorem 19.10. Since $s\mathcal{P} \in s\mathbb{K}[u][[s]]$, the element $E := \mathcal{P} + u + \log(1 + s\mathcal{P}) \in \mathbb{K}[u][[s]]$ is defined, and (359) is exactly $E = 0$.

Step 1: E satisfies the homogeneous equation. Put $D = 1 + s\mathcal{P}$, a unit of $\mathbb{K}[u][[s]]$. Then

$$\partial_u E = \partial_u \mathcal{P} + 1 + \frac{s \partial_u \mathcal{P}}{D}, \quad \partial_s E = \partial_s \mathcal{P} + \frac{\mathcal{P} + s \partial_s \mathcal{P}}{D}.$$

Using $s^2 \partial_s \mathcal{P} = (1 + s) \partial_u \mathcal{P} + 1$ from (358), hence $s^3 \partial_s \mathcal{P} = s(1 + s) \partial_u \mathcal{P} + s$, we get

$$s^2 \partial_s E - (1 + s) \partial_u E = [(1 + s) \partial_u \mathcal{P} + 1] - (1 + s) \partial_u \mathcal{P} - (1 + s) + \frac{s^2 \mathcal{P} + s(1 + s) \partial_u \mathcal{P} + s - s(1 + s) \partial_u \mathcal{P}}{D}.$$

The fraction is $\frac{s^2 \mathcal{P} + s}{D} = \frac{s(1 + s\mathcal{P})}{D} = s$, so the whole expression equals $1 - (1 + s) + s = 0$. Thus

$$s^2 \partial_s E = (1 + s) \partial_u E. \quad (360)$$

Step 2: E vanishes on $u = 0$. By (346), $\mathcal{P}(s, 0) = \sum_n \mathbb{L}_n^{\text{Lam}}(0) s^n = 0$. Applying the continuous $\mathbb{K}$ -algebra map $u \mapsto 0$ to E gives $E(s, 0) = 0 + 0 + \log(1 + 0) = 0$.

Step 3. Write $E = \sum_{n \geq 0} e_n(u) s^n$. Comparing coefficients of s^0 and s^{n+1} in (360) gives $e'_0 = 0$ and $e'_{n+1} = n e_n - e'_n$; Step 2 gives $e_n(0) = 0$ for all n . Hence e_0 is a constant with $e_0(0) = 0$, so $e_0 = 0$; and inductively, if $e_n = 0$ then $e'_{n+1} = 0$ and $e_{n+1}(0) = 0$, so $e_{n+1} = 0$. Thus $E = 0$, which is (359).

Uniqueness. Let $\mathcal{S}(Q) = -\log(1 - su + sQ)$. For $Q \in \mathbb{K}[u][[s]]$ the argument lies in $1 + s\mathbb{K}[u][[s]]$, so $\mathcal{S}(Q) \in s\mathbb{K}[u][[s]]$ and in particular $\mathcal{S}(Q)(0, u) = 0$. If $Q_1 - Q_2 \in s^N \mathbb{K}[u][[s]]$ then, $1 - su + sQ_2$ being a unit,

$$\mathcal{S}(Q_1) - \mathcal{S}(Q_2) = -\log\left(1 + \frac{s(Q_1 - Q_2)}{1 - su + sQ_2}\right) \in s^{N+1} \mathbb{K}[u][[s]].$$

So $\mathcal{S}$ is a strict contraction for the s -adic filtration on $\mathbb{K}[u][[s]]$, which is separated and complete; it therefore has exactly one fixed point, and that fixed point automatically satisfies $Q(0, u) = 0$. $\square$

Corollary 19.12 (Reversion-free proof of Theorem 18.10). *The generating equation (359) determines the expansion of ω on its own:*

$$(X + L)^{\langle -1 \rangle \circ} = X + \mathcal{P}(t, L), \quad \mathcal{P}(t, L) = \sum_{n \geq 0} \mathbb{L}_n^{\text{Lam}}(L) t^n,$$

so Theorem 18.10 holds without appeal to Lagrange–Bürmann and without Theorem 18.8.

Proof. Substitute $s \mapsto t$ and $u \mapsto L$ in (359), and add $u = L$ to both sides of $\mathcal{P} = -u + Q$. The result is exactly the assertion that $\mathcal{P}(t, L)$ satisfies the implicit equation (333). By the uniqueness half of Theorem 18.9, the solution of that equation in $\mathcal{G}_{\text{poly}, \mathbb{K}}$ is unique, so $(X + L)^{\langle -1 \rangle \circ} = X + \mathcal{P}(t, L)$.

The two ingredients used here are the existence and uniqueness of the fixed point of the generating equation and the uniqueness half of Theorem 18.9; neither depends on Theorem 16.14 or on the repeated-derivative computation, which is what makes this route independent. $\square$

Remark 19.13 (Which route to use). Three independent characterisations of the same family are now available: the operator formula (344), the recurrence (348), and the generating equation (359). The recurrence is the cheapest generator (one exact polynomial integration and one subtraction per index); the operator formula is the right tool for proving identities, since it makes the falling factorial visible; the generating equation is the right tool for uniqueness statements and for transporting the family to a new inversion problem, as in Theorem 18.5. Agreement of two of them on a computed table is a genuine check only if the two were implemented independently; (338) of Theorem 18.13 is a stronger test, because it exercises the composition code as well.

19.7 The Laplace transform and the exponential moments

The remaining result is of a different type: it is an assertion about convergent integrals of the real polynomial functions $\mathcal{L}_n^{\text{Lam}}$, not a formal identity, and its hypotheses have real content.

Theorem 19.14 (Laplace transform of $\mathcal{L}_n^{\text{Lam}}$). *Let $n \geq 1$ and let $s \in \mathbb{C}$ with $\Re s > 0$. Then the integral below converges absolutely and*

$$\int_0^\infty e^{-su} \mathcal{L}_n^{\text{Lam}}(u) \, du = \frac{1}{s^{n+1}} \prod_{j=1}^{n-1} (j-s) = \frac{(1-s)^{\overline{n-1}}}{s^{n+1}}, \quad (361)$$

the product being empty (and equal to 1) when $n = 1$.

Proof. Absolute convergence is clear: $\mathcal{L}_n^{\text{Lam}}$ is a polynomial and $|e^{-su}| = e^{-u\Re s}$ with $\Re s > 0$, so the integrand is $\mathcal{O}_{u \rightarrow +\infty}(e^{-u\Re s/2})$. Write $I_n(s) = \int_0^\infty e^{-su} \mathcal{L}_n^{\text{Lam}}(u) \, du$.

The integration by parts. For $0 < b < \infty$,

$$\int_0^b e^{-su} \mathcal{L}_n^{\text{Lam}'}(u) \, du = \left[e^{-su} \mathcal{L}_n^{\text{Lam}}(u) \right]_0^b + s \int_0^b e^{-su} \mathcal{L}_n^{\text{Lam}}(u) \, du.$$

The boundary term at $u = 0$ vanishes because $\mathcal{L}_n^{\text{Lam}}(0) = 0$ (Theorem 19.2); the boundary term at $u = b$ tends to 0 as $b \rightarrow \infty$ because $\Re s > 0$ and $\mathcal{L}_n^{\text{Lam}}$ is a polynomial. Both hypotheses are used, and neither can be dropped: without $\Re s > 0$ the integral does not converge, and without $\mathcal{L}_n^{\text{Lam}}(0) = 0$ an extra term $-\mathcal{L}_n^{\text{Lam}}(0)$ would survive. Letting $b \rightarrow \infty$,

$$\int_0^\infty e^{-su} \mathcal{L}_n^{\text{Lam}'}(u) \, du = s I_n(s). \quad (362)$$

The recursion. Apply (362) to both sides of (347), which is an identity of polynomials and may therefore be integrated against e^{-su} :

$$s I_{n+1}(s) = n I_n(s) - s I_n(s), \quad \text{i.e.} \quad I_{n+1}(s) = \frac{n-s}{s} I_n(s) \quad (n \geq 1).$$

The base case is $I_1(s) = \int_0^\infty e^{-su} u \, du = s^{-2}$. Iterating, $I_n(s) = s^{-2} \prod_{j=1}^{n-1} \frac{j-s}{s} = s^{-(n+1)} \prod_{j=1}^{n-1} (j-s)$. Finally $\prod_{j=1}^{n-1} (j-s) = (1-s)(2-s) \cdots (n-1-s) = (1-s)^{\overline{n-1}}$. $\square$

Corollary 19.15 (Exponential moment cancellations). *Let $n \geq 1$. For every integer r with $1 \leq r \leq n-1$,*

$$\int_0^\infty e^{-ru} \mathcal{L}_n^{\text{Lam}}(u) \, du = 0. \quad (363)$$

The range is empty for $n = 1$, so the first genuine cancellation is $\int_0^\infty e^{-u} \mathcal{L}_2^{\text{Lam}}(u) \, du = 0$.

Proof. Such an r is a positive real number, so $\Re r > 0$ and Theorem 19.14 applies with $s = r$. The factor $j = r$ occurs in $\prod_{j=1}^{n-1} (j-s)$ precisely because $1 \leq r \leq n-1$, and it vanishes at $s = r$. $\square$

Remark 19.16 (The transform is the falling factorial again). [Theorem 19.14](#) and [Theorem 19.4](#) are equivalent, and the equivalence explains the cancellations of [Theorem 19.15](#). Indeed $(s)^{\underline{n}} = s \prod_{j=1}^{n-1} (s-j) = (-1)^{n-1} s \prod_{j=1}^{n-1} (j-s)$, so by (349)

$$\prod_{j=1}^{n-1} (j-s) = (-1)^{n-1} \sum_{k \geq 1} s(n, k) s^{k-1} = \sum_{k \geq 1} (-1)^{k-1} \begin{bmatrix} n \\ k \end{bmatrix} s^{k-1},$$

whence $I_n(s) = \sum_{k=1}^n (-1)^{k-1} \begin{bmatrix} n \\ k \end{bmatrix} s^{-(n-k+2)}$. Comparing with the termwise transform $I_n(s) = \sum_{m=1}^n \lambda_{n,m} m! s^{-(m+1)}$ and matching powers via $m = n - k + 1$ returns exactly (351). Thus (363) says that the falling factorial $(s)^{\underline{n}}$ vanishes at $s = 1, \dots, n-1$ – the cancellations are the roots of the very operator that defines $\mathbb{L}_n^{\text{Lam}}$ in (344).

A common trap

The letter s carries two unrelated meanings in this section. In [Theorem 19.10](#) and [Theorem 19.11](#) it is a formal indeterminate and no analytic content is asserted. In [Theorem 19.14](#) it is a complex number constrained by $\Re s > 0$, and the identity is an equality of convergent integrals with analytic functions of s on the half-plane. A “Laplace transform of the generating series” mixing the two would be a different object and is not claimed anywhere here.

20 Lambert W : branches and expansions

Every result of Parts I–IV was formal: an identity between elements of $\mathcal{PL}_{\text{pow}, \mathbb{K}}$, $\mathcal{PL}_{\text{poly}, \mathbb{K}}$, $\mathcal{PL}_{\text{rat}, \mathbb{K}}$ or $\mathcal{PL}_{\text{itLaur}, \mathbb{K}}$, or between elements of the near-identity group $\mathcal{G}_{\text{poly}, \mathbb{K}}$. Lambert’s W is where that formal theory meets a genuine multivalued analytic function, and it is therefore the place where the boundary between an algebraic identity and an analytic theorem has to be drawn with care. The six source reports draw it in six different places, and none of them draws it entirely correctly; the present section states exactly what is proved, proves it, and marks what is not.

The organising principle is a single substitution. The equation $we^w = z$ is not itself a polynomial–logarithmic reversion problem: its natural variable z is exponentially larger than its solution. After one logarithm it becomes the Wright omega problem $w + \log w = \xi$, which is exactly the reversion of $X + L$ in $\mathcal{G}_{\text{poly}, \mathbb{K}}$, already solved in [Section 18](#). All of W – every branch, the lower real branch near zero, and the generalized Lambert families of [Section 21](#) – is that one reversion evaluated at different arguments. What changes from branch to branch is the argument, not the reversion.

20.1 The defining equation and the real branches

Definition 20.1 (Branches of W). For $k \in \mathbb{Z}$, a *branch* W_k of the Lambert function is a determination of the inverse of

$$g: \mathbb{C} \longrightarrow \mathbb{C}, \quad g(w) := we^w,$$

so that

$$W_k(z) e^{W_k(z)} = z. \tag{364}$$

The map g is entire and infinite-to-one, so [Equation \(364\)](#) alone determines nothing: a branch is the pair consisting of a solution and a rule for selecting it. Following the notation catalogue, W_0 is the *principal* branch, real and increasing on $[-e^{-1}, \infty)$ with $W_0(0) = 0$, analytic at 0, and cut along $(-\infty, -e^{-1}]$; W_{-1} is the *lower real* branch, real on $[-e^{-1}, 0)$ with values in $(-\infty, -1]$. A branchless symbol $W(z)$ is admissible only where the domain forces the real principal branch.

Proposition 20.2 (The two real branches). *Let $g(w) = we^w$ on $\mathbb{R}$. Then:*

1. g is strictly decreasing on $(-\infty, -1]$, with image $[-e^{-1}, 0)$, and strictly increasing on $[-1, \infty)$, with image $[-e^{-1}, \infty)$;

2. consequently g restricted to $[-1, \infty)$ has a continuous strictly increasing inverse $W_0: [-e^{-1}, \infty) \rightarrow [-1, \infty)$, and g restricted to $(-\infty, -1]$ has a continuous strictly decreasing inverse $W_{-1}: [-e^{-1}, 0) \rightarrow (-\infty, -1]$;
3. for $x \in (-e^{-1}, 0)$ the equation $g(w) = x$ has exactly the two real solutions $W_0(x) \in (-1, 0)$ and $W_{-1}(x) \in (-\infty, -1)$; for $x = -e^{-1}$ it has the single solution -1 ; for $x \geq 0$ it has exactly one; and for $x < -e^{-1}$ none;
4. $W_{-1}(x) \rightarrow -\infty$ as $x \uparrow 0$.

Proof. We have $g'(w) = (1+w)e^w$, which is negative for $w < -1$, zero at $w = -1$ and positive for $w > -1$; thus g is strictly decreasing on $(-\infty, -1]$ and strictly increasing on $[-1, \infty)$, with the single minimum $g(-1) = -e^{-1}$. On $(-\infty, -1]$ we have $g(w) < 0$ for $w < 0$ and $g(w) \rightarrow 0^-$ as $w \rightarrow -\infty$, because $|w|e^w \rightarrow 0$; hence g maps $(-\infty, -1]$ bijectively onto $[-e^{-1}, 0)$, decreasingly. On $[-1, \infty)$, $g(-1) = -e^{-1}$ and $g(w) \rightarrow +\infty$, so g maps $[-1, \infty)$ bijectively onto $[-e^{-1}, \infty)$, increasingly. Items (i)–(iii) follow, and (iv) is the statement that the decreasing inverse of $g|_{(-\infty, -1]}$ tends to $-\infty$ as its argument tends to the endpoint 0^- of the image. $\square$

Lemma 20.3 (Derivative of a branch). *Let $w = W_k(z)$ be analytic near z_0 with $z_0 \neq 0$ and $w \neq -1$. Then*

$$\frac{dW_k}{dz}(z) = \frac{W_k(z)}{z(1+W_k(z))}. \quad (365)$$

Proof. Differentiating $we^w = z$ gives $(1+w)e^w w' = 1$. Since $e^w = z/w$ whenever $w \neq 0$, this is $(1+w)(z/w)w' = 1$, which is Equation (365). The excluded cases $z = 0$ and $w = -1$ are exactly the points where the substitution $e^w = z/w$ or the division by $1+w$ is illegitimate; $w = -1$ is the branch point studied in Section 20.7. $\square$

20.2 Branch coordinates and the substitution to the omega problem

Definition 20.4 (Branch coordinates). Let $z \in \mathbb{C} \setminus \{0\}$ and $k \in \mathbb{Z}$. The *outer branch coordinate* is

$$\xi_k := \text{Log } z + 2\pi i k, \quad (366)$$

with Log the principal logarithm, so that $e^{\xi_k} = z$. Given in addition $\xi_k \neq 0$, an *inner branch coordinate* is any $\ell_k \in \mathbb{C}$ with

$$e^{\ell_k} = \xi_k, \quad (367)$$

written $\ell_k = \log_{\mathcal{B}_k} \xi_k$ when a particular determination $\mathcal{B}_k$ has been fixed. The pair (ξ_k, ℓ_k) is the *branch datum*. A *coordinate sector* is a connected open $S_k \subseteq \mathbb{C} \setminus \{0\}$ on which $z \mapsto \xi_k$ and $\xi_k \mapsto \ell_k$ admit single-valued analytic determinations and on which $|\xi_k| \rightarrow \infty$ along the limits considered.

Three data must be recorded and are routinely conflated: the choice of $\text{Log } z$ (equivalently, of the strip in which $\text{Arg } z$ is taken), the integer k , and the determination $\mathcal{B}_k$ of the second logarithm. Reports S1, S2 and S4 all write $\ell_k = \text{Log } \xi_k$, i.e. they silently force $\mathcal{B}_k$ to be principal. That is a legitimate normalisation and we use it in all numerical statements, but it is a hypothesis, not a convention-free fact: Theorem 20.15 shows exactly where it is used.

Proposition 20.5 (Reduction to the omega problem). *Let $z \neq 0$, $k \in \mathbb{Z}$, and let $w \in \mathbb{C} \setminus \{0\}$. The following are equivalent.*

1. $we^w = z$ and there is a logarithm λ of w (that is, $e^\lambda = w$) with $w + \lambda = \xi_k$.
2. $w + \lambda = \xi_k$ for some λ with $e^\lambda = w$.

Moreover, if $we^w = z$ and λ is any logarithm of w , then $w + \lambda = \xi_j$ for exactly one $j \in \mathbb{Z}$, and every j arises from a suitable choice of λ .

Proof. Assume (ii). Then $we^w = e^\lambda e^w = e^{w+\lambda} = e^{\xi_k} = z$, which is (i). Conversely (i) contains (ii).

For the last statement, let $we^w = z$ and $e^\lambda = w$. Then $e^{w+\lambda} = we^w = z = e^{\text{Log } z}$, so $w + \lambda - \text{Log } z \in 2\pi i\mathbb{Z}$, that is $w + \lambda = \xi_j$ for a unique j . Replacing λ by $\lambda + 2\pi im$ (still a logarithm of w) replaces j by $j + m$, so every integer occurs. $\square$

Remark 20.6 (What the substitution does and does not say). [Theorem 20.5](#) is the exact form of the informal step “take a logarithm of $we^w = z$ ”. Two things must be noticed. First, the branch index k is *defined* by the relation $w + \lambda = \xi_k$; it is not an independent piece of data that can be chosen after w and λ . Second, the equivalence is with the *additive* equation $w + \lambda = \xi_k$, in which λ is a logarithm of the unknown w . Whether λ is the principal logarithm of w is a separate question, settled for large $|\xi_k|$ by [Theorem 20.15](#). The discrepancy $\lambda - \text{Log } w$, divided by $2\pi i$, is the *unwinding number*; the identity $\omega(s) = W(e^s)$ asserted without qualification in S1 and S3 holds only where that unwinding number vanishes.

20.3 The formal expansion and branch independence

We first record the formal statement, which is an identity in the near-identity group and involves no analysis whatsoever. Throughout, $\mathbb{K}$ is a field of characteristic zero, X is the large variable, $t = X^{-1}$, $L = \log X$, and $\mathcal{G}_{\text{poly}, \mathbb{K}} = X + \mathbb{K}[L][[t]]$.

Theorem 20.7 (Formal affine-logarithmic reversion). *Let $a \in \mathbb{K}$ and let $F = X + aL \in \mathcal{G}_{\text{poly}, \mathbb{K}}$. Then the compositional inverse of F in $\mathcal{G}_{\text{poly}, \mathbb{K}}$ is*

$$F^{(-1)\circ} = X + a \sum_{n \geq 0} a^n \mathbb{L}_n^{\text{Lam}}(L) t^n = X - aL + \sum_{n \geq 1} a^{n+1} \mathbb{L}_n^{\text{Lam}}(L) t^n. \quad (368)$$

In particular the coefficient of t^n in $F^{(-1)\circ}$ is the fixed polynomial $\mathbb{L}_n^{\text{Lam}}$ multiplied by the scalar a^{n+1} ; no new polynomial family occurs for any value of a , and $a = 1$ recovers the Wright omega expansion of [Theorem 18.10](#).

Proof. Let $\mathcal{Q}(s, u) = \sum_{n \geq 1} \mathbb{L}_n^{\text{Lam}}(u) s^n \in \mathbb{K}[u][[s]]$ be the generating series of [Theorem 19.11](#), characterised there as the unique element of $s\mathbb{K}[u][[s]]$ with

$$\mathcal{Q} = -\log(1 - su + s\mathcal{Q}). \quad (369)$$

Because at has positive t -order, the assignment $s \mapsto at$, $u \mapsto L$ extends to a continuous $\mathbb{K}$ -algebra homomorphism $\mathbb{K}[u][[s]] \rightarrow \mathbb{K}[L][[t]]$ for the respective adic topologies. Write

$$\mathcal{Q}_a := \mathcal{Q}(at, L) = \sum_{n \geq 1} a^n \mathbb{L}_n^{\text{Lam}}(L) t^n \in t\mathbb{K}[L][[t]].$$

Applying the homomorphism to [Equation \(369\)](#) gives, in $\mathbb{K}[L][[t]]$,

$$\mathcal{Q}_a = -\log(1 - atL + at\mathcal{Q}_a), \quad (370)$$

the logarithm being the formal logarithm of [Theorem 10.5](#), legitimate because the argument lies in $1 + t\mathbb{K}[L][[t]]$.

Put $G := X - aL + a\mathcal{Q}_a$. Since $-aL + a\mathcal{Q}_a \in \mathbb{K}[L][[t]]$, we have $G \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, and

$$G = X \left(1 - at(L - \mathcal{Q}_a) \right),$$

whose second factor lies in $1 + t\mathbb{K}[L][[t]]$. By the exact generator substitution rule of [Theorem 10.22](#),

$$L \circ G = \log G = L + \log(1 - atL + at\mathcal{Q}_a) = L - \mathcal{Q}_a,$$

the last equality being [Equation \(370\)](#). Hence

$$F \circ G = G + a(L \circ G) = X - aL + aQ_a + aL - aQ_a = X.$$

So G is a right inverse of F inside $\mathcal{G}_{\text{poly}, \mathbb{K}}$. By [Theorem 14.11](#) the inverse in $\mathcal{G}_{\text{poly}, \mathbb{K}}$ exists and is unique and two-sided, so $G = F^{(-1)^\circ}$. Finally $aL_0^{\text{Lam}}(L) = -aL$, which converts the second display of [Equation \(368\)](#) into the first. $\square$

Remark 20.8 (Two proofs in the sources, and why this one is preferred). Reports S2 and S4 obtain [Equation \(368\)](#) by inspecting the Lagrange–Bürmann series and observing that its m -th term carries a factor a^m . That argument is correct but needs the summability of the Lagrange family ([Theorem 16.9](#)) and a separate identification of the m -th term with L_m^{Lam} . The proof above verifies a closed-form candidate against the defining equation and appeals only to uniqueness in $\mathcal{G}_{\text{poly}, \mathbb{K}}$; it is shorter, it is the same proof for every a including $a = -1$, and it is the one that transports without change to the analytic statement of [Theorem 20.12](#).

Theorem 20.9 (Formal branch expansion). *Let $k \in \mathbb{Z}$ and let (ξ_k, ℓ_k) be a branch datum in the sense of [Theorem 20.4](#). Substituting $X \mapsto \xi_k$, $L \mapsto \ell_k$, $t \mapsto \xi_k^{-1}$ in [Equation \(368\)](#) with $a = 1$ produces the formal series*

$$\xi_k + \sum_{n \geq 0} \frac{L_n^{\text{Lam}}(\ell_k)}{\xi_k^n} = \xi_k - \ell_k + \sum_{n \geq 1} \frac{L_n^{\text{Lam}}(\ell_k)}{\xi_k^n}, \quad (371)$$

equivalently, by [Theorem 19.4](#),

$$\xi_k - \ell_k + \sum_{n \geq 1} \frac{(-1)^n}{\xi_k^n} \sum_{m=1}^n \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] \frac{(-\ell_k)^m}{m!}. \quad (372)$$

The coefficient polynomials L_n^{Lam} do not depend on k , on the determination $\mathcal{B}_k$, or on the sector.

Proof. The first display is [Equation \(368\)](#) with $a = 1$ after the stated substitution, which is a substitution of symbols and requires no justification beyond $\xi_k \neq 0$. For the second, insert the closed coefficient formula $L_n^{\text{Lam}}(u) = \sum_{m=1}^n (-1)^{n-m} \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] u^m / m!$ of [Theorem 19.4](#) and use $(-1)^{n-m} = (-1)^n (-1)^m$, which converts u^m into $(-u)^m$ with an overall factor $(-1)^n$. Branch independence is the observation that [Theorem 20.7](#) was proved once, in $\mathbb{K}[L][[t]]$, with no branch data present. $\square$

The next theorem is the precise form of the catalogue’s assertion that “the polynomial family is branch-independent; the coordinates ξ_k, ℓ_k , their branches, and the analytic domain are not”. Its first half is the uniqueness of the coefficients, and its second half exhibits the genuine k -dependence.

Theorem 20.10 (Branch independence of the coefficients, dependence of the coordinates).

1. (Universality.) *There is exactly one sequence $(p_n)_{n \geq 0}$ in $\mathbb{K}[u]$ such that for every $a \in \mathbb{K}$ the map $X + aL \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ has inverse $X + a \sum_{n \geq 0} a^n p_n(L) t^n$; namely $p_n = L_n^{\text{Lam}}$. A fortiori, for each fixed k , the expansion [Equation \(371\)](#) determines the polynomials L_n^{Lam} uniquely, and the answer is the same for all k .*
2. (Uniqueness against the coordinates.) *Fix k and a coordinate sector S_k on which $|\xi_k| \rightarrow \infty$ and $\ell_k = \text{Log } \xi_k$. If $(q_n)_{n \geq 0}$ in $\mathbb{C}[u]$ satisfies $\sum_{n=0}^N q_n(\ell_k) \xi_k^{-n} = \mathcal{O}_{|\xi_k| \rightarrow \infty, \xi_k \in S_k} \left(\ell_k^{N+1} / \xi_k^{N+1} \right)$ for every N , then $q_n = 0$ for all n .*
3. (The coordinates really differ.) *For $j \neq k$ and z in a common domain, $\xi_j - \xi_k = 2\pi i(j-k) \neq 0$, and the corresponding solutions w_j, w_k of [Equation \(364\)](#) supplied by [Theorem 20.12](#) satisfy $w_j - w_k \rightarrow 2\pi i(j-k)$ as $|z| \rightarrow \infty$ with $\text{Arg } z$ bounded. In particular they are distinct functions.*

Proof. (i) Existence is [Theorem 20.7](#). For uniqueness take $a = 1$: if $X + \sum_{n \geq 0} p_n(L)t^n$ and $X + \sum_{n \geq 0} \mathbb{L}_n^{\text{Lam}}(L)t^n$ are both inverses of $X + L$ in $\mathcal{G}_{\text{poly}, \mathbb{K}}$, they are equal by the uniqueness clause of [Theorem 14.11](#), and equality in $\mathbb{K}[L][[t]]$ is coefficientwise by [Theorem 4.9](#). (Uniqueness already follows from the single value $a = 1$; no compatibility across different a is needed.)

(ii) Induct on n . Suppose $q_0 = \dots = q_{n-1} = 0$ (vacuous for $n = 0$). Taking $N = n$ in the hypothesis and multiplying by ξ_k^n gives $q_n(\ell_k) = \mathcal{O}_{|\xi_k| \rightarrow \infty, \xi_k \in S_k}(\ell_k^{n+1}/\xi_k)$. Suppose $q_n \neq 0$, say of degree $d \geq 0$ with leading coefficient $c \neq 0$. Since $\ell_k = \text{Log } \xi_k$ we have $|\ell_k| \leq \log |\xi_k| + \pi$ and $|\ell_k| \geq \log |\xi_k|$, so $|\ell_k| \rightarrow \infty$ and $|q_n(\ell_k)| \geq \frac{1}{2}|c||\ell_k|^d$ for $|\xi_k|$ large, while the right-hand side is at most $C(\log |\xi_k| + \pi)^{n+1}/|\xi_k| \rightarrow 0$. For $d \geq 1$ the left side tends to ∞ and for $d = 0$ it is the nonzero constant $|c|$; either way we contradict the bound. Hence $q_n = 0$.

(iii) The identity $\xi_j - \xi_k = 2\pi i(j - k)$ is immediate from [Equation \(366\)](#). By [Theorem 20.12](#) applied with base point ξ_j , respectively ξ_k , we have $w_j = \xi_j - \ell_j + \mathcal{Y}_j$ and $w_k = \xi_k - \ell_k + \mathcal{Y}_k$, where [Theorem 20.13](#) with $N = 0$ bounds $|\mathcal{Y}_j|, |\mathcal{Y}_k|$ by $10(1 + |\ell|)/|\xi|$, which tends to 0. Hence

$$w_j - w_k - 2\pi i(j - k) = -(\ell_j - \ell_k) + \mathcal{Y}_j - \mathcal{Y}_k.$$

Now $\ell_j - \ell_k = \text{Log}(\xi_j) - \text{Log}(\xi_k) \rightarrow 0$, because $\xi_j/\xi_k = 1 + 2\pi i(j - k)/\xi_k \rightarrow 1$ and both $\text{Arg } \xi_j$ and $\text{Arg } \xi_k$ tend to 0 when $\text{Arg } z$ stays bounded. All three terms on the right tend to 0, so $w_j - w_k \rightarrow 2\pi i(j - k) \neq 0$. $\square$

Universality is a statement about one problem, not about all of them

[Theorem 20.10](#) (i) says that the *same* polynomials serve every a and every branch. It does not say that $\mathbb{L}_n^{\text{Lam}}$ is the coefficient family of every logarithmic reversion. Even inside the affine-logarithmic family the scalar a^{n+1} is present, and as soon as the perturbation ceases to be aL – for instance for $X + aL^2$ with $a \neq 0$, whose t^n -coefficient has degree $2n + 1$ in L and leading coefficient $2a^{n+1}/n$ by [Theorem 16.22](#) – the coefficients are new polynomials. The universality of $\mathbb{L}_n^{\text{Lam}}$ is exactly the universality of the single map $X + L$.

20.4 The analytic status of the expansion

[Theorem 20.9](#) is a theorem about symbols. It asserts nothing about the function W_k . The sources at this point diverge sharply. S1 and S3 assert Poincaré validity “in an admissible sector” without defining the sector; S6 asserts convergence by an implicit-function argument that quantifies incorrectly (see [Theorem 20.17](#)); S1, S3 and S5 all repeat from the literature the much stronger claim that for real z the series converges for $z \geq e$. We prove here a self-contained analytic theorem which is strong enough for every use made of the expansion in this volume – it gives absolute convergence, an explicit region, and an explicit uniform remainder – and we mark the literature claim as unverified.

The whole analytic content sits in one two-variable lemma. Its point is that the two small quantities in the problem are $1/\xi_k$ and ℓ_k/ξ_k , and that the solution is jointly analytic in them near the origin. Separating them is what makes the estimate uniform in ℓ_k , which is precisely the step that no source performs.

Lemma 20.11 (The two-variable solution). *Let*

$$D := \{(\sigma, \tau) \in \mathbb{C}^2 : |\sigma| + |\tau| < \tfrac{1}{2}\}.$$

For $(\sigma, \tau) \in D$ the equation

$$y = -\text{Log}(1 - \tau + \sigma y) \tag{373}$$

has a unique solution $y = \mathcal{Y}(\sigma, \tau)$ in the closed unit disc $|y| \leq 1$, and it satisfies $|\mathcal{Y}(\sigma, \tau)| \leq \log 2$. The function $\mathcal{Y}$ is holomorphic on D ; its Taylor series $\mathcal{Y} = \sum_{a,b \geq 0} c_{ab} \sigma^a \tau^b$ converges absolutely at every point of D ; the coefficients satisfy

$$|c_{ab}| \leq 5^{a+b} \quad (a, b \geq 0); \tag{374}$$

and for every $u \in \mathbb{C}$ and every $n \geq 0$,

$$\sum_{a+b=n} c_{ab} u^b = \begin{cases} 0, & n = 0, \\ \mathcal{L}_n^{\text{Lam}}(u), & n \geq 1. \end{cases} \quad (375)$$

Consequently $c_{ab} = \lambda_{a+b,b}$ for $a + b \geq 1$, where $\mathcal{L}_n^{\text{Lam}}(u) = \sum_m \lambda_{n,m} u^m$, and for all $(\sigma, \tau) \in D$ and $n \geq 1$,

$$\left| \sum_{a+b=n} c_{ab} \sigma^a \tau^b \right| \leq (5(|\sigma| + |\tau|))^n. \quad (376)$$

Proof. Existence, uniqueness and the bound. Fix $(\sigma, \tau) \in D$ and put $\theta = |\sigma| + |\tau| < \frac{1}{2}$. For $|y| \leq 1$ we have $|\sigma y - \tau| \leq |\sigma| + |\tau| = \theta < \frac{1}{2}$, so $\zeta := 1 - \tau + \sigma y$ lies in the open disc of radius $\frac{1}{2}$ about 1, where the principal logarithm is holomorphic and, by the Maclaurin series $\text{Log}(1 + \nu) = \sum_{m \geq 1} (-1)^{m+1} \nu^m / m$,

$$|\text{Log}(1 + \nu)| \leq \sum_{m \geq 1} \frac{|\nu|^m}{m} = -\log(1 - |\nu|) \quad (|\nu| < 1). \quad (377)$$

Hence the map $\Phi(y) = -\text{Log}(1 - \tau + \sigma y)$ sends the closed unit disc into the closed disc of radius $-\log(1 - \theta) \leq \log 2 < 1$. Moreover

$$|\Phi'(y)| = \left| \frac{\sigma}{1 - \tau + \sigma y} \right| \leq \frac{|\sigma|}{1 - \theta} \leq \frac{\theta}{1 - \theta} < 1,$$

so Φ is a contraction of the complete metric space $\{|y| \leq 1\}$. Banach's fixed point theorem gives a unique fixed point $\mathcal{Y}(\sigma, \tau)$, and it lies in the image, so $|\mathcal{Y}| \leq -\log(1 - \theta) \leq \log 2$.

Holomorphy. Let $y_0 \equiv 0$ and $y_{j+1} = \Phi(y_j)$. Each y_j is holomorphic on D as a function of (σ, τ) , by induction: $1 - \tau + \sigma y_j$ is holomorphic with values in $\{|\zeta - 1| < \frac{1}{2}\}$, on which Log is holomorphic. On a compact $E \subset D$ we have $\sup_E \theta = \theta_E < \frac{1}{2}$, hence $|y_{j+1} - y_j| \leq (\theta_E / (1 - \theta_E))^j |y_1 - y_0|$ with $\theta_E / (1 - \theta_E) < 1$, so (y_j) converges uniformly on E to $\mathcal{Y}$. A locally uniform limit of holomorphic functions is holomorphic.

Taylor expansion and Cauchy bounds. Let $(\sigma_0, \tau_0) \in D$ and choose $\rho_1 > |\sigma_0|$, $\rho_2 > |\tau_0|$ with $\rho_1 + \rho_2 < \frac{1}{2}$. The closed polydisc of polyradius (ρ_1, ρ_2) is contained in D and $|\mathcal{Y}| \leq 1$ there, so the Cauchy integral formula on the distinguished boundary yields the absolutely and uniformly convergent expansion $\mathcal{Y} = \sum_{a,b} c_{ab} \sigma^a \tau^b$ on the open polydisc, with $|c_{ab}| \leq \rho_1^{-a} \rho_2^{-b}$. Since (σ_0, τ_0) was arbitrary in D , the series converges absolutely at every point of D , and the coefficients c_{ab} do not depend on the polydisc. Taking $\rho_1 = \rho_2 = \frac{1}{5}$, which satisfies $\rho_1 + \rho_2 = \frac{2}{5} < \frac{1}{2}$, gives Equation (374).

Identification of the diagonal sums. Fix $u \in \mathbb{C}$ and restrict to the line $\tau = \sigma u$; the restriction lies in D for $|\sigma| < (2(1 + |u|))^{-1}$. On that disc, $s \mapsto \mathcal{Y}(s, su)$ is holomorphic, vanishes at $s = 0$, and satisfies $\mathcal{Y} = -\text{Log}(1 - su + s\mathcal{Y})$; hence its Taylor coefficients form the unique element of $s\mathbb{C}[[s]]$ satisfying Equation (369) at the given u . By the formal uniqueness clause of Theorem 19.11, those coefficients are $\mathcal{L}_n^{\text{Lam}}(u)$. On the other hand, absolute convergence permits the rearrangement $\mathcal{Y}(s, su) = \sum_{n \geq 0} (\sum_{a+b=n} c_{ab} u^b) s^n$. Comparing coefficients gives Equation (375) for the given u ; as u ranges over $\mathbb{C}$, two polynomials agreeing everywhere are equal, and comparing coefficients of u^b gives $c_{ab} = \lambda_{a+b,b}$.

The block bound. For $x, y \geq 0$, $\sum_{a+b=n} x^a y^b \leq \sum_{a+b=n} \binom{n}{a} x^a y^b = (x + y)^n$. Combining this with Equation (374) gives Equation (376). $\square$

Theorem 20.12 (Convergent branch expansion from an arbitrary base point). *Let $\xi \in \mathbb{C}$. Let $K \in \mathbb{C} \setminus \{0\}$, let $\log K$ be any logarithm of K , and put*

$$M := K + \log K - \xi. \quad (378)$$

Assume the seating condition

$$2(1 + |M|) < |K|. \quad (379)$$

Then:

1. the series $\sum_{n \geq 0} \mathbb{L}_n^{\text{Lam}}(M) K^{-n}$ converges absolutely;
2. the number

$$w := K + \sum_{n \geq 0} \frac{\mathbb{L}_n^{\text{Lam}}(M)}{K^n} = K - M + \sum_{n \geq 1} \frac{\mathbb{L}_n^{\text{Lam}}(M)}{K^n}$$

satisfies $|w - (K - M)| \leq \log 2$ and $w \neq 0$;

3. with $\log_* w := \log K + \text{Log}(w/K)$, which is a logarithm of w ,

$$w + \log_* w = \xi, \quad \text{hence} \quad we^w = e^\xi.$$

Proof. Set $\sigma = 1/K$ and $\tau = M/K$. Then $|\sigma| + |\tau| = (1 + |M|)/|K| < \frac{1}{2}$ by Equation (379), so $(\sigma, \tau) \in D$. Let $\mathcal{Y} = \mathcal{Y}(\sigma, \tau)$ be as in Theorem 20.11. Applying Equation (375) with $u = M$ and using $\tau = \sigma M$,

$$\mathcal{Y} = \sum_{n \geq 1} \mathbb{L}_n^{\text{Lam}}(M) \sigma^n = \sum_{n \geq 1} \frac{\mathbb{L}_n^{\text{Lam}}(M)}{K^n},$$

the series converging absolutely by Theorem 20.11. Since $\mathbb{L}_0^{\text{Lam}}(M) = -M$, this is (i), and $w = K - M + \mathcal{Y}$ with $|\mathcal{Y}| \leq \log 2$, which is the first half of (ii). For $w \neq 0$: $|w| \geq |K| - |M| - \log 2 > 2 + |M| - \log 2 > 1$, using Equation (379).

For (iii), compute

$$\frac{w}{K} = \frac{K - M + \mathcal{Y}}{K} = 1 - \tau + \sigma \mathcal{Y},$$

which lies in the disc of radius $\frac{1}{2}$ about 1 by the proof of Theorem 20.11; hence $\text{Log}(w/K)$ is defined and, by Equation (373), $\text{Log}(w/K) = -\mathcal{Y}$. Therefore $\log_* w = \log K - \mathcal{Y}$, and $e^{\log_* w} = K \cdot (w/K) = w$, so $\log_* w$ is indeed a logarithm of w . Finally

$$w + \log_* w = (K - M + \mathcal{Y}) + (\log K - \mathcal{Y}) = K + \log K - M = \xi$$

by Equation (378), and exponentiating gives $we^w = e^{w + \log_* w} = e^\xi$. □

Theorem 20.13 (Explicit uniform remainder). *In the situation of Theorem 20.12, put*

$$\theta := \frac{5(1 + |M|)}{|K|} \tag{380}$$

and assume $\theta \leq \frac{1}{2}$, that is $|K| \geq 10(1 + |M|)$. Then for every $N \in \mathbb{N}_0$,

$$\left| w - K - \sum_{n=0}^N \frac{\mathbb{L}_n^{\text{Lam}}(M)}{K^n} \right| \leq 2\theta^{N+1} = 2 \left(\frac{5(1 + |M|)}{|K|} \right)^{N+1}. \tag{381}$$

The bound is uniform: its constants 5 and 2 are absolute, and no hypothesis beyond $|K| \geq 10(1 + |M|)$ is used.

Proof. With $\sigma = 1/K$, $\tau = M/K$ as above, $5(|\sigma| + |\tau|) = \theta$, so Equation (376) reads $|\mathbb{L}_n^{\text{Lam}}(M) K^{-n}| \leq \theta^n$ for $n \geq 1$. Since $\theta \leq \frac{1}{2}$,

$$\left| \sum_{n > N} \frac{\mathbb{L}_n^{\text{Lam}}(M)}{K^n} \right| \leq \sum_{n > N} \theta^n = \frac{\theta^{N+1}}{1 - \theta} \leq 2\theta^{N+1}. \quad \square$$

Specialising to the canonical seating $K = \xi_k$ gives the statement the notation catalogue asks for, now with every hypothesis printed and every constant explicit.

Corollary 20.14 (Canonical branch expansion). *Let $z \neq 0$, $k \in \mathbb{Z}$, and let (ξ_k, ℓ_k) be a branch datum with $\xi_k \neq 0$. Take $K = \xi_k$ and $\log K = \ell_k$ in [Theorem 20.12](#); then $M = \ell_k$. Consequently:*

1. *if $2(1 + |\ell_k|) < |\xi_k|$, the series*

$$w = \xi_k - \ell_k + \sum_{n \geq 1} \frac{\mathcal{L}_n^{\text{Lam}}(\ell_k)}{\xi_k^n}$$

converges absolutely and defines a solution of $w e^w = z$ with $\log_ w = \xi_k - w$;*

2. *if moreover $|\xi_k| \geq 10(1 + |\ell_k|)$, then for every N ,*

$$\left| w - \xi_k + \ell_k - \sum_{n=1}^N \frac{\mathcal{L}_n^{\text{Lam}}(\ell_k)}{\xi_k^n} \right| \leq 2 \left(\frac{5(1 + |\ell_k|)}{|\xi_k|} \right)^{N+1} = \mathcal{O}_{|\xi_k| \rightarrow \infty, |\ell_k| \geq 1} \left(\ell_k^{N+1} / \xi_k^{N+1} \right);$$

3. *in particular, if $\ell_k = \text{Log } \xi_k$, both hypotheses hold as soon as $|\xi_k| \geq 100$, and for real $z \geq e^6$ with $k = 0$ hypothesis (i) holds while for real $z \geq e^{50}$ hypothesis (ii) holds.*

Proof. (i) and (ii) are [Theorems 20.12](#) and [20.13](#) with $M = \xi_k + \ell_k - \xi_k = \ell_k$; the $\mathcal{O}(\cdot)$ form of the bound in (ii) follows because $(1 + |\ell_k|) \leq 2|\ell_k|$ once $|\ell_k| \geq 1$, so the displayed bound is at most $2 \cdot 10^{N+1} |\ell_k / \xi_k|^{N+1}$.

(iii) If $\ell_k = \text{Log } \xi_k$ then $|\ell_k| \leq \log |\xi_k| + \pi$. Put $R = |\xi_k|$. Hypothesis (ii) holds if $R \geq 10(1 + \pi + \log R)$; the function $R \mapsto R - 10(1 + \pi + \log R)$ has derivative $1 - 10/R > 0$ for $R > 10$ and is positive at $R = 100$ (indeed $10(1 + \pi + \log 100) < 87.5$), hence for all $R \geq 100$; and hypothesis (i) is weaker. For real $z > 1$ and $k = 0$, $\xi_0 = \log z > 0$ and $\ell_0 = \log \log z$ are real, so (i) requires $2(1 + |\log \xi_0|) < \xi_0$; at $\xi_0 = 6$ one has $2(1 + \log 6) < 5.59 < 6$, and $\xi_0 \mapsto \xi_0 - 2 - 2 \log \xi_0$ is increasing for $\xi_0 > 2$. Similarly (ii) requires $\xi_0 \geq 10(1 + \log \xi_0)$, which holds at $\xi_0 = 50$ ($10(1 + \log 50) < 49.2$) and hence for all $\xi_0 \geq 50$ since $\xi_0 \mapsto \xi_0 - 10 - 10 \log \xi_0$ is increasing for $\xi_0 > 10$. $\square$

Proposition 20.15 (Identification of the branch label). *Let $z \neq 0$, $k \in \mathbb{Z}$, and let (ξ_k, ℓ_k) be a branch datum with $2(1 + |\ell_k|) < |\xi_k|$. Let $w = w_k(z)$ be the solution produced by [Theorem 20.14](#) (i), and assume in addition*

$$|\text{Im } \ell_k| \leq \pi - \frac{3}{4}, \quad (382)$$

that is, ξ_k is at angular distance at least $\frac{3}{4}$ from the cut of the chosen determination of the second logarithm. Then $\log_ w = \text{Log } w$, so*

$$w + \text{Log } w = \xi_k, \quad \text{Im } w \in [2\pi k + \text{Arg } z - \pi, 2\pi k + \text{Arg } z + \pi). \quad (383)$$

In particular distinct k produce solutions in disjoint horizontal strips, so $k \mapsto w_k(z)$ is injective. Independently of [Equation \(382\)](#), $w_k(z)$ is the unique solution of $w e^w = z$ in the closed disc $|v - (\xi_k - \ell_k)| \leq 1$.

Proof. By the proof of [Theorem 20.12](#), $\log_* w = \ell_k - \mathcal{Y}$ with $|\mathcal{Y}| \leq \log 2 < \frac{3}{4}$. Hence $|\text{Im } \log_* w| \leq |\text{Im } \ell_k| + \log 2 < (\pi - \frac{3}{4}) + \frac{3}{4} = \pi$. A logarithm of w whose imaginary part lies in $(-\pi, \pi)$ is the principal one, so $\log_* w = \text{Log } w$. Then [Theorem 20.12](#) (iii) reads $w + \text{Log } w = \xi_k$; taking imaginary parts and using $\text{Im } \text{Log } w = \text{Arg } w \in (-\pi, \pi]$ and $\text{Im } \xi_k = \text{Arg } z + 2\pi k$ gives $\text{Im } w = \text{Arg } z + 2\pi k - \text{Arg } w$, which lies in the stated interval. Those intervals for distinct k are disjoint half-open intervals of length 2π , which gives injectivity.

For the disc uniqueness, let $v e^v = z$ with $|v - (\xi_k - \ell_k)| \leq 1$, and put $\sigma = 1/\xi_k$, $\tau = \ell_k/\xi_k$, so $(\sigma, \tau) \in D$ by the seating condition. Then $|v - \xi_k| \leq |\ell_k| + 1 \leq \frac{1}{2} |\xi_k|$, so $|v/\xi_k - 1| \leq \frac{1}{2}$ and $y := -\text{Log}(v/\xi_k)$ is defined with $|y| \leq \log 2$ by [Equation \(377\)](#). Now $\lambda := \ell_k - y$ satisfies $e^\lambda = \xi_k \cdot (v/\xi_k) = v$, so [Theorem 20.5](#) gives $v + \lambda = \xi_k + 2\pi i m$ for some $m \in \mathbb{Z}$, that is $v = \xi_k - \ell_k + y + 2\pi i m$. Since $|v - (\xi_k - \ell_k)| \leq 1$ and $|y| \leq \log 2$, we get $2\pi |m| \leq 1 + \log 2 < 2\pi$, hence $m = 0$ and $v = \xi_k - \ell_k + y$. Dividing by ξ_k gives $v/\xi_k = 1 - \tau + \sigma y$, so $y = -\text{Log}(1 - \tau + \sigma y)$: y is a fixed point of [Equation \(373\)](#) in $|y| \leq 1$. By the uniqueness in [Theorem 20.11](#), $y = \mathcal{Y}(\sigma, \tau)$, whence $v = w_k(z)$. $\square$

Remark 20.16 (No sector is needed for the estimate). The remainder bound Equation (381) holds pointwise on the explicitly described region $\{|\xi_k| \geq 10(1 + |\ell_k|)\}$, with no sector hypothesis at all. The sector enters for two other purposes, and it is worth separating them:

1. to make $z \mapsto \xi_k$ and $\xi_k \mapsto \ell_k$ single-valued analytic functions, so that w_k is an analytic function of z rather than a value attached to a branch datum;
2. to identify w_k with the function that the standard references call W_k ; this is where the distance-to-the-cut hypothesis Equation (382) is used.

Sources that write “in an admissible sector” without saying which are conflating these two roles with the estimate itself. With Theorems 20.14 and 20.15 one may take, for instance,

$$S_k = \left\{ z : |\operatorname{Arg} z| < \pi, |\xi_k| \geq 100, |\operatorname{Arg} \xi_k| \leq \pi - \frac{3}{4} \right\},$$

on which ξ_k and $\ell_k = \operatorname{Log} \xi_k$ are analytic, w_k is analytic, and all hypotheses hold.

Remark 20.17 (The gap in the convergence argument of S6). S6 argues: the implicit function theorem shows that for each fixed u the series $\sum_n L_n^{\operatorname{Lam}}(u) s^n$ converges for $|s|$ small; along the Lambert path $s = 1/\xi$, $u = \ell$, one has $|su| \rightarrow 0$; hence the series converges. The conclusion does not follow from the premises. The radius of convergence in s supplied by the implicit function theorem depends on u and shrinks as $|u| \rightarrow \infty$, so a statement about $|su|$ is not a statement about $|s|$ lying inside that radius. What repairs the argument is exactly the separation of variables in Theorem 20.11: one must apply the implicit function theorem in the two variables $(\sigma, \tau) = (1/\xi, \ell/\xi)$, both of which are small, rather than in s alone with u frozen. The frozen- u statement is true – it is Theorem 18.19 – but it is not sufficient.

What is proved here, and what is only reported

Theorem 20.14 proves absolute convergence of the canonical series for real $z \geq e^6 \approx 403$, and the explicit uniform remainder for $z \geq e^{50}$. Reports S1, S3 and S5 all state, citing [11, 5], that for the principal branch and real z the series converges for every $z \geq e$. That claim is far stronger than anything proved here and we have not verified it; it is recorded as a literature report, not as a theorem of this volume. Note that the check offered by S5 – “at $z = e$ one has $\ell_0 = 0$, so every term vanishes” – is no evidence for it at all: it shows only that the series is trivially convergent at the single point where all its terms are zero (Theorem 20.19). Similarly, S5’s report of the convergence domains $X > (\alpha e)^\alpha$ for $\alpha \geq 1$ and $X > e$ for $\alpha < 1$ in the generalized problem [21] is repeated in Section 21 as attribution only.

20.5 The principal real specialization

Corollary 20.18 (Principal real branch at infinity). *Let $z > 1$ be real, and take $k = 0$, so that $\xi_0 = \log z > 0$ and $\ell_0 = \log \log z$ are real. Write $L_1 = \log z$ and $L_2 = \log \log z$. Then for $z \geq e^{50}$,*

$$\begin{aligned} W_0(z) = & L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2(L_2 - 2)}{2L_1^2} + \frac{L_2(2L_2^2 - 9L_2 + 6)}{6L_1^3} \\ & + \frac{L_2(3L_2^3 - 22L_2^2 + 36L_2 - 12)}{12L_1^4} \\ & + \frac{L_2(12L_2^4 - 125L_2^3 + 350L_2^2 - 300L_2 + 60)}{60L_1^5} + \varrho_5, \end{aligned} \quad (384)$$

with the explicit bound $|\varrho_5| \leq 2(5(1 + L_2)/L_1)^6$, and the infinite series converges absolutely to $W_0(z)$ already for $z \geq e^6$.

Proof. For real $z > 1$ the datum $(\xi_0, \ell_0) = (\log z, \log \log z)$ satisfies Equation (382) with $\operatorname{Im} \ell_0 = 0$, so Theorem 20.15 applies and identifies the constructed solution with the real principal branch: $w_0(z)$

is real (the whole construction is real for real data, because $\mathcal{Y}(\sigma, \tau)$ is real for real $(\sigma, \tau) \in D$ by uniqueness of the fixed point in $\{|y| \leq 1\}$) and $w_0 > 1$, hence $w_0 = W_0(z)$ by [Theorem 20.2](#) (iii). The five displayed corrections are $\mathbb{L}_1^{\text{Lam}}, \dots, \mathbb{L}_5^{\text{Lam}}$ evaluated at L_2 , in the factored forms of [Section D](#), and the bound on ϱ_5 is [Theorem 20.14](#) (ii) with $N = 5$. Convergence for $z \geq e^6$ is [Theorem 20.14](#) (iii). $\square$

Proposition 20.19 (Exactness at $z = e$, and what it does not mean). *At $z = e$ one has $L_1 = 1, L_2 = 0$, and every truncation $\xi_0 - \ell_0 + \sum_{n=1}^N \mathbb{L}_n^{\text{Lam}}(\ell_0) \xi_0^{-n}$ equals 1, which is the exact value $W_0(e) = 1$. The reason is that $\mathbb{L}_n^{\text{Lam}}(0) = 0$ for every $n \in \mathbb{N}_0$. This gives no information whatever about convergence, because at that point every term of the series is 0; indeed the seating condition [Equation \(379\)](#) reads $2 < 1$ there and fails.*

Proof. $W_0(e) = 1$ because $1 \cdot e^1 = e$ and $1 > -1$, so [Theorem 20.2](#) identifies the branch. For $n \geq 1$, [Theorem 19.7](#) expresses $\mathbb{L}_n^{\text{Lam}}$ as a sum over $1 \leq m \leq n$ of multiples of u^m , so $\mathbb{L}_n^{\text{Lam}}(0) = 0$; and $\mathbb{L}_0^{\text{Lam}}(u) = -u$ gives $\mathbb{L}_0^{\text{Lam}}(0) = 0$. Hence every truncation is $1 - 0 + 0 = 1$. For the last sentence, $|K| = \xi_0 = 1$ and $|M| = |\ell_0| = 0$, so [Equation \(379\)](#) asserts $2 < 1$. $\square$

20.6 The lower real branch near zero

The catalogue reserves the symbol ε_{br} for the positive small argument of the lower branch; it is not an error tolerance and not an inverse iterated logarithm. We write ε for it in this subsection, where no other ε occurs.

Proposition 20.20 (The lower-branch equation). *The map $h(u) = u - \log u$ is a strictly increasing bijection from $(1, \infty)$ onto $(1, \infty)$. For $0 < \varepsilon < e^{-1}$, put*

$$S := \log(1/\varepsilon) > 1, \quad L := \log S. \quad (385)$$

Then $u := -W_{-1}(-\varepsilon)$ is the unique solution in $(1, \infty)$ of

$$u - \log u = S. \quad (386)$$

Proof. $h'(u) = 1 - 1/u > 0$ on $(1, \infty)$, $h(1) = 1$, and $h(u) \rightarrow \infty$; so h is a strictly increasing bijection onto $(1, \infty)$. By [Theorem 20.2](#), $W_{-1}(-\varepsilon) \in (-\infty, -1)$ for $0 < \varepsilon < e^{-1}$, so $u > 1$. Writing $w = -u$ in $we^w = -\varepsilon$ gives $ue^{-u} = \varepsilon$; taking the real logarithm, $\log u - u = \log \varepsilon = -S$, which is [Equation \(386\)](#). Uniqueness is the injectivity of h . $\square$

Theorem 20.21 (Expansion of the lower real branch). *With S, L as in [Equation \(385\)](#), and provided $2(1 + |L|) < S$, the series below converges absolutely and*

$$W_{-1}(-\varepsilon) = -S - L + \sum_{n \geq 1} (-1)^n \frac{\mathbb{L}_n^{\text{Lam}}(L)}{S^n}, \quad (387)$$

equivalently, in the zero-based normalisation and with $\mathbb{L}_0^{\text{Lam}}(L) = -L$,

$$W_{-1}(-\varepsilon) = -S + \sum_{n \geq 0} (-1)^n \frac{\mathbb{L}_n^{\text{Lam}}(L)}{S^n}. \quad (388)$$

If in addition $S \geq 10(1 + |L|)$, then for every N ,

$$\left| W_{-1}(-\varepsilon) + S + L - \sum_{n=1}^N (-1)^n \frac{\mathbb{L}_n^{\text{Lam}}(L)}{S^n} \right| \leq 2 \left(\frac{5(1 + |L|)}{S} \right)^{N+1}. \quad (389)$$

Explicitly,

$$\begin{aligned} W_{-1}(-\varepsilon) = & -S - L - \frac{L}{S} + \frac{L(L-2)}{2S^2} - \frac{L(2L^2-9L+6)}{6S^3} \\ & + \frac{L(3L^3-22L^2+36L-12)}{12S^4} + \dots \end{aligned} \quad (390)$$

Proof. This is the analytic affine-logarithmic theorem with $a = -1$. Concretely, apply [Theorem 20.11](#) with $\sigma = -1/S$ and $\tau = -L/S$; then $|\sigma| + |\tau| = (1 + |L|)/S < \frac{1}{2}$ by hypothesis, so $(\sigma, \tau) \in D$, and $\tau = \sigma L$, so [Equation \(375\)](#) gives

$$\mathcal{Y} = \sum_{n \geq 1} \mathbb{L}_n^{\text{Lam}}(L) \sigma^n = \sum_{n \geq 1} (-1)^n \frac{\mathbb{L}_n^{\text{Lam}}(L)}{S^n},$$

absolutely convergent. Put $v := S + L - \mathcal{Y}$. Then

$$\frac{v}{S} = 1 + \frac{L}{S} - \frac{\mathcal{Y}}{S} = 1 - \tau + \sigma \mathcal{Y},$$

which lies in the disc of radius $\frac{1}{2}$ about 1; hence $v > 0$ and $\text{Log}(v/S) = -\mathcal{Y}$ by [Equation \(373\)](#). All the data being real, $\mathcal{Y}$ is real and $\log(v/S) = -\mathcal{Y}$ is the real logarithm. Therefore

$$v - \log v = v - \log S - \log(v/S) = S + L - \mathcal{Y} - L + \mathcal{Y} = S.$$

Also $v > 1$: the hypothesis $2(1 + |L|) < S$ gives $|L| < S/2 - 1$ and $S > 2$, hence $v \geq S + L - |\mathcal{Y}| \geq S - |L| - \log 2 > S/2 + 1 - \log 2 > 2 - \log 2 > 1$. By the uniqueness in [Theorem 20.20](#), $v = u = -W_{-1}(-\varepsilon)$, which gives [Equation \(388\)](#) after using $\mathbb{L}_0^{\text{Lam}}(L) = -L$. The remainder bound is [Equation \(376\)](#) exactly as in [Theorem 20.13](#), with $5(|\sigma| + |\tau|) = 5(1 + |L|)/S \leq \frac{1}{2}$. The displayed first terms are $\mathbb{L}_1^{\text{Lam}}, \dots, \mathbb{L}_4^{\text{Lam}}$ with the alternating signs. $\square$

The alternating signs do not define a second Lambert family

Every source displays [Equation \(390\)](#) and remarks that “the same polynomials reappear with alternating block signs”. The statement deserves to be made exactly, because the alternation is sometimes absorbed into the polynomials, producing a spurious second family $\widetilde{\mathbb{L}}_n^{\text{Lam}} = (-1)^n \mathbb{L}_n^{\text{Lam}}$ with its own tables and its own recurrence. There is no second family. By [Theorem 20.7](#), the coefficient of X^{-n} in the inverse of $X + aL$ is $a^{n+1} \mathbb{L}_n^{\text{Lam}}(L)$, a scalar multiple of the one canonical polynomial; the lower real branch is the single value $a = -1$ of that scalar, so the coefficient of S^{-n} is $(-1)^{n+1} \mathbb{L}_n^{\text{Lam}}(L)$ in the equation $u - \log u = S$ and $(-1)^n \mathbb{L}_n^{\text{Lam}}(L)$ after the reflection $w = -u$. The sign is data about the argument, not about the polynomial. Concretely: the recurrence $\mathbb{L}_{n+1}^{\text{Lam}} = -\mathbb{L}_n^{\text{Lam}} + n \int_0^u \mathbb{L}_n^{\text{Lam}}$ of [Theorem 19.2](#), the Stirling formula of [Theorem 19.4](#) and the tables of [Section D](#) are used unchanged for the lower branch; only the evaluation point and the sign of the outer variable change.

Remark 20.22 (Why the lower branch is not literally an ω value). Setting $y = aY$ in $y + a \log y = X$ turns it into $Y + \log Y = X/a - \log a$, so for $a > 0$ the affine-logarithmic problem is a rescaled Wright omega evaluation. For $a = -1$ this substitution requires $\log(-1)$, and the resulting omega argument is $-X \mp \pi i$: the real lower branch corresponds to a complex point of ω . This is why [Theorem 20.21](#) is proved directly from [Theorem 20.11](#) rather than by quoting the omega expansion, and it is the precise content of the catalogue’s warning that ε_{br} is a branch-point-independent parameter for a problem of its own.

20.7 The square-root branch point

Lemma 20.23 (The critical point of we^w). *Let $g(w) = we^w$ and $w = -1 + q$. Then*

$$1 + e g(-1 + q) = \sum_{n \geq 2} \frac{(n-1)q^n}{n!} = \frac{q^2}{2} + \frac{q^3}{3} + \frac{q^4}{8} + \frac{q^5}{30} + \frac{q^6}{144} + \dots, \quad (391)$$

an entire function of q . In particular $g(-1) = -e^{-1}$, $g'(-1) = 0$ and $g''(-1) = e^{-1} \neq 0$, so -1 is a critical point of g of order exactly two.

Proof. $e g(-1 + q) = e(-1 + q)e^{-1+q} = (q - 1)e^q$. Now

$$(q - 1)e^q = \sum_{n \geq 0} \frac{q^{n+1}}{n!} - \sum_{n \geq 0} \frac{q^n}{n!} = -1 + \sum_{n \geq 1} q^n \left(\frac{1}{(n-1)!} - \frac{1}{n!} \right) = -1 + \sum_{n \geq 1} \frac{(n-1)q^n}{n!},$$

and the $n = 1$ summand vanishes. Adding 1 gives Equation (391). Comparing coefficients, $e g'(-1) = 0$ and $e g''(-1)/2! = \frac{1}{2}$, i.e. $g''(-1) = e^{-1}$. $\square$

Theorem 20.24 (Puiseux expansion at the branch point). *Put*

$$p := \sqrt{2(1 + ez)}. \quad (392)$$

There are $r > 0$ and a function φ holomorphic on $|p| < r$ with $\varphi(0) = 0$, $\varphi'(0) = 1$, such that for $|1 + ez| < r^2/2$ the two solutions of $we^w = z$ near $w = -1$ are exactly

$$w = -1 + \varphi(p) \quad \text{and} \quad w = -1 + \varphi(-p), \quad (393)$$

for either determination of the square root in Equation (392). Its expansion begins

$$\varphi(p) = p - \frac{p^2}{3} + \frac{11p^3}{72} - \frac{43p^4}{540} + \frac{769p^5}{17280} - \frac{221p^6}{8505} + \dots \quad (394)$$

In particular the leading behaviour is

$$w + 1 = \pm \sqrt{2(1 + ez)} + \mathcal{O}_{z \rightarrow -e^{-1}}(|1 + ez|). \quad (395)$$

For real z slightly larger than $-e^{-1}$ and $p > 0$, the branch $\varphi(p)$ is W_0 and $\varphi(-p)$ is W_{-1} ; the two are exchanged by $p \mapsto -p$.

Proof. Write Equation (391) as $1 + ez = \frac{1}{2}q^2 h(q)$ with

$$h(q) := \sum_{n \geq 2} \frac{2(n-1)}{n!} q^{n-2} = 1 + \frac{2q}{3} + \frac{q^2}{4} + \frac{q^3}{15} + \frac{q^4}{72} + \dots,$$

entire with $h(0) = 1$. Choose $r_0 > 0$ with $h \neq 0$ on $|q| < r_0$ and let $h^{1/2}$ be the holomorphic square root there with $h^{1/2}(0) = 1$. Put $\psi(q) = q h^{1/2}(q)$, so $\psi(0) = 0$, $\psi'(0) = 1$ and

$$\psi(q)^2 = q^2 h(q) = 2(1 + ez) \quad \text{whenever } z = g(-1 + q). \quad (396)$$

By the holomorphic inverse function theorem ψ is a biholomorphism of a disc $|q| < r_1 \leq r_0$ onto a neighbourhood of 0 containing a disc $|p| < r$; let $\varphi = \psi^{-1}$ on that disc, so $\varphi(0) = 0$ and $\varphi'(0) = 1/\psi'(0) = 1$.

Now fix z with $|1 + ez| < r^2/2$ and let p be either square root of $2(1 + ez)$, so $|p| < r$. If $w = -1 + q$ with $|q| < r_1$ solves $g(w) = z$, then Equation (396) gives $\psi(q)^2 = p^2$, hence $\psi(q) = \pm p$ and $q = \varphi(\pm p)$; conversely each of $q = \varphi(\pm p)$ satisfies $\frac{1}{2}q^2 h(q) = \frac{1}{2}\psi(q)^2 = \frac{1}{2}p^2 = 1 + ez$, i.e. $g(-1 + q) = z$ by Equation (391). Shrinking r so that $|\varphi| < r_1$ on $|p| < r$ makes this a bijection between the two square roots and the two nearby solutions; they are distinct for $p \neq 0$ because $\varphi(p) - \varphi(-p) = 2p + \mathcal{O}_{p \rightarrow 0}(p^3)$.

For the coefficients, write $\varphi(p) = \sum_{i \geq 1} a_i p^i$ with $a_1 = 1$ and determine $a_2, a_3, \dots$ by substituting into $\psi(\varphi(p)) = p$, where $\psi(q) = q + \frac{1}{3}q^2 + \frac{5}{72}q^3 + \frac{11}{1080}q^4 + \frac{59}{51840}q^5 + \dots$ is obtained by expanding $h^{1/2}$ from the displayed h . Comparing coefficients of $p^2, \dots, p^6$ gives in turn $a_2 = -\frac{1}{3}$, $a_3 = \frac{11}{72}$, $a_4 = -\frac{43}{540}$, $a_5 = \frac{769}{17280}$, $a_6 = -\frac{221}{8505}$, which is Equation (394). Since $\varphi(p) = p + \mathcal{O}_{p \rightarrow 0}(p^2)$ and $p^2 = 2(1 + ez)$, we get Equation (395).

Finally let $z \in (-e^{-1}, -e^{-1} + \delta)$ be real and take $p = +\sqrt{2(1 + ez)} > 0$ small. Then $\varphi(p) > 0$ and $\varphi(-p) < 0$, so $-1 + \varphi(p) > -1$ and $-1 + \varphi(-p) < -1$; by Theorem 20.2 (iii) these are $W_0(z)$ and $W_{-1}(z)$ respectively. $\square$

Corollary 20.25 (No power–logarithmic expansion at the branch point). *Let $z_0 = -e^{-1}$.*

1. *There is no single-valued holomorphic function F on a punctured disc $0 < |z - z_0| < \rho$ with $g(F(z)) = z$ and $F(z) \rightarrow -1$.*
2. *The local monodromy of the two-valued germ at z_0 has order exactly two.*
3. *Consequently no determination of the inverse near -1 has, on a slit disc, a representation*

$$F(z) = \sum_{j=0}^d A_j(z) (\log(z - z_0))^j$$

with $d \geq 1$, the A_j single-valued holomorphic on the punctured disc, and $A_d \neq 0$. That is, no logarithm of $z - z_0$ occurs at any power; the correct local scale is the Puiseux scale $\{(z - z_0)^{m/2}\}$, on which [Theorem 20.24](#) is an expansion.

Proof. (ii) Continuing $p = \sqrt{2e(z - z_0)}$ once counterclockwise around z_0 replaces p by $-p$, hence exchanges the two determinations $-1 + \varphi(\pm p)$ of [Theorem 20.24](#); continuing twice restores them. The exchange is nontrivial because the two determinations differ for $p \neq 0$. So the monodromy permutation has order two.

(i) A single-valued F would be invariant under one continuation, but the only two candidate values are exchanged; contradiction.

(iii) Write $\Lambda = \log(z - z_0)$ and suppose $F = \sum_{j=0}^d A_j \Lambda^j$ with $d \geq 1$ and $A_d \neq 0$. Continuation m times around z_0 replaces Λ by $\Lambda + 2\pi i m$ and hence F by $F_m = \sum_j A_j (\Lambda + 2\pi i m)^j$. Expanding, F_m is again of the same shape with top coefficient A_d and next coefficient $A_{d-1} + 2\pi i m d A_d$. Since the representation of a germ in the form $\sum_{j \leq d} A_j \Lambda^j$ with single-valued A_j is unique – the functions $1, \Lambda, \dots, \Lambda^d$ are linearly independent over the field of single-valued meromorphic germs, because Λ is transcendental over it – the germs $F_m, m \in \mathbb{Z}$, are pairwise distinct. So the germ would have infinitely many determinations, contradicting (ii). If instead F were given by a convergent series in integer powers of $z - z_0$, it would be single-valued, contradicting (i). $\square$

Remark 20.26 (The general principle). [Theorem 20.25](#) is the local instance of a rule used repeatedly in [Section 21](#): the fractional-power scale of an inverse is dictated by the order of vanishing of the derivative of the direct map. A simple zero of g' – equivalently, a critical point of order two – forces the exponent $1/2$ and monodromy of order two. A zero of order $m - 1$ would force exponent $1/m$; an essential degeneracy would force something else again. Logarithms in an inverse come from a different mechanism entirely, namely a logarithm in the direct map ([Theorem 17.4](#)), and the two mechanisms never substitute for one another.

20.8 Residual correction and one Newton step in branch coordinates

The formal theory of [Section 14](#) measures the quality of a truncated inverse by the reversal residual $\text{Err}_{\text{rev}}(F, H) = F \circ H - X$, and [Theorem 14.15](#) shows that its t -order is exactly the number of correct blocks. In branch coordinates the analytic counterpart is the *logarithmic residual*, and it is a much better numerical object than the multiplicative residual $ve^v - z$, which overflows as soon as z is large.

Definition 20.27 (Truncations and the logarithmic residual). Let (ξ, ℓ) be a branch datum with $|\xi| \geq 10(1 + |\ell|)$. For $N \in \mathbb{N}_0$ put

$$w_N := \xi - \ell + \sum_{n=1}^N \frac{\mathbb{L}_n^{\text{Lam}}(\ell)}{\xi^n}, \quad (397)$$

so that w_N is the value of the exclusive truncation $\text{Trunc}_{<N+1}(X + L)^{(-1)^\circ}$ at $(X, L) = (\xi, \ell)$. On the closed disc $\mathcal{D} := \{v : |v - (\xi - \ell)| \leq 1\}$ define $\log_* v := \ell + \text{Log}(v/\xi)$ and $F(v) := v + \log_* v$; set

$$R_N := F(w_N) - \xi = w_N + \log_* w_N - \xi, \quad \delta_N := -\frac{R_N}{1 + 1/w_N}. \quad (398)$$

Proposition 20.28 (The disc is admissible). *Under the hypothesis of Theorem 20.27: $\mathcal{D}$ is contained in the region where $\log_*$ is defined and holomorphic; for $v \in \mathcal{D}$ one has $|v| \geq \frac{9}{10} |\xi| \geq 9$, $\frac{8}{9} \leq |F'(v)| \leq \frac{10}{9}$, and $|F''(v)| \leq \frac{5}{4} |\xi|^{-2}$; and both w_N (for every N) and the exact solution w of Theorem 20.14 lie in $\mathcal{D}$.*

Proof. Write $\theta = 5(1 + |\ell|)/|\xi| \leq \frac{1}{2}$. For $v \in \mathcal{D}$, $|v - \xi| \leq |\ell| + 1 \leq |\xi|/10$, so $|v/\xi - 1| \leq \frac{1}{10} < 1$ and $\text{Log}(v/\xi)$ is defined and holomorphic there; also $|v| \geq \frac{9}{10} |\xi|$, and $|\xi| \geq 10(1 + |\ell|) \geq 10$ gives $|v| \geq 9$. Since $F'(v) = 1 + 1/v$ and $F''(v) = -1/v^2$, we get $|F'(v) - 1| \leq \frac{1}{9}$ and $|F''(v)| \leq (10/9)^2 |\xi|^{-2} \leq \frac{5}{4} |\xi|^{-2}$. Finally $|w_N - (\xi - \ell)| \leq \sum_{n \geq 1} \theta^n = \theta/(1 - \theta) \leq 1$ by Equation (376), and $|w - (\xi - \ell)| = |\mathcal{Y}| \leq \log 2 \leq 1$. $\square$

Theorem 20.29 (Residual order, residual-to-error transfer, and one Newton step). *In the situation of Theorem 20.27, put $\theta = 5(1 + |\ell|)/|\xi| \leq \frac{1}{2}$ and let w be the exact solution of Theorem 20.14. Then, with $e := w - w_N$:*

1. $|e| \leq 2\theta^{N+1}$ and $|R_N| \leq \frac{20}{9}\theta^{N+1}$; in particular

$$R_N = \mathcal{O}_{|\xi| \rightarrow \infty, |\ell| \geq 1}(\ell^{N+1}/\xi^{N+1}),$$

the same order as the error itself;

2. $|e| \leq \frac{5}{4} |R_N|$ (residual-to-error transfer, with an explicit constant);
3. $|e - \delta_N| \leq \frac{3}{2} \frac{|R_N|^2}{|\xi|^2}$ (one Newton step is quadratically better).

Proof. (i) The bound on e is Theorem 20.13. The segment $[w_N, w]$ lies in the convex set $\mathcal{D}$ by Theorem 20.28, and $F(w) = \xi$ by Theorem 20.12 (iii) – note that on $\mathcal{D}$ the function $\log_*$ of Theorem 20.27 agrees with the $\log_*$ of Theorem 20.12 – so $|R_N| = |F(w_N) - F(w)| \leq \sup_{\mathcal{D}} |F'| \cdot |e| \leq \frac{10}{9} \cdot 2\theta^{N+1}$. The $\mathcal{O}(\cdot)$ form follows as in Theorem 20.14 (ii).

(ii) and (iii). Taylor’s formula with integral remainder along $[w_N, w] \subset \mathcal{D}$ gives

$$0 = F(w) - \xi = R_N + F'(w_N)e + E, \quad E = \int_0^1 (1-s)F''(w_N + se)e^2 \, ds,$$

so $|E| \leq \frac{1}{2} \cdot \frac{5}{4} |\xi|^{-2} |e|^2 = \frac{5}{8} |e|^2 |\xi|^{-2}$. Since both points lie in $\mathcal{D}$ we have $|e| \leq 2$, and $|\xi| \geq 10$, so $|E| \leq \frac{5}{8} \cdot 2 |e|/100 = |e|/80$. Using $|F'(w_N)| \geq \frac{8}{9}$,

$$\frac{8}{9} |e| \leq |R_N| + \frac{1}{80} |e|, \quad \text{hence} \quad |e| \leq \frac{|R_N|}{8/9 - 1/80} \leq \frac{5}{4} |R_N|,$$

which is (ii). Dividing the Taylor identity by $F'(w_N) = 1 + 1/w_N$ gives $e = \delta_N - E/F'(w_N)$, so

$$|e - \delta_N| \leq \frac{9}{8} |E| \leq \frac{9}{8} \cdot \frac{5}{8} \cdot \frac{|e|^2}{|\xi|^2} \leq \frac{45}{64} \cdot \frac{25}{16} \frac{|R_N|^2}{|\xi|^2} \leq \frac{3}{2} \frac{|R_N|^2}{|\xi|^2}. \quad \square$$

Remark 20.30 (Where the transfer fails). Every constant above degrades as $|v|$ approaches 0, and the argument collapses entirely at $v = -1$, where $F'(v) = 1 + 1/v = 0$. That is exactly the branch point of Section 20.7: the residual-to-error transfer is a statement about a nonvanishing derivative, and at a critical point a small residual is compatible with an error of the order of its square root. This is the analytic shadow of the algebraic fact recorded in Theorem 15.9: Newton reversion needs the denominator to be a unit.

Remark 20.31 (Exact identities as independent checks). Two exact relations cost nothing and catch branch mistakes that coefficient checks do not. On any branch, $\log_* W_k(z) = \xi_k - W_k(z)$ and $e^{W_k(z)} = z/W_k(z)$. The first lets one expand $\log W_k$ without composing a logarithm with a series, and the second lets one check $W_k e^{W_k} = z$ by a division rather than an exponentiation. A truncation that satisfies the coefficient recurrences but violates the first identity has a branch mismatch, not a coefficient error.

20.9 Numerical behaviour of the truncations

All six sources tabulate truncation errors, and the tables disagree – not in their numbers but in what their column headings mean. S1 and S4 tabulate against $x = \log z$, S3 and S5 against z . Both families of numbers are correct, and each is a subset of the other’s grid; we tabulate once, in the intrinsic variable $\xi_0 = \log z$, and record the concordance.

Table 5: Absolute error $|w_N - W_0(e^{\xi_0})|$ of the truncation Equation (397) on the principal real branch, $\ell_0 = \log \xi_0$. Recomputed here to 60 significant digits and in agreement with the tables of S1, S3, S4 and S5 wherever those overlap.

N	$\xi_0 = 10$	$\xi_0 = 25$	$\xi_0 = 50$	$\xi_0 = 100$
0	$2.3201 \cdot 10^{-1}$	$1.3180 \cdot 10^{-1}$	$7.9742 \cdot 10^{-2}$	$4.6657 \cdot 10^{-2}$
1	$1.7467 \cdot 10^{-3}$	$3.0442 \cdot 10^{-3}$	$1.5017 \cdot 10^{-3}$	$6.0513 \cdot 10^{-4}$
2	$1.7370 \cdot 10^{-3}$	$9.4510 \cdot 10^{-5}$	$5.7352 \cdot 10^{-6}$	$5.2671 \cdot 10^{-6}$
3	$1.5606 \cdot 10^{-4}$	$1.7340 \cdot 10^{-5}$	$1.5654 \cdot 10^{-6}$	$8.1538 \cdot 10^{-8}$
5	$5.1336 \cdot 10^{-6}$	$2.7114 \cdot 10^{-8}$	$2.2625 \cdot 10^{-9}$	$1.5423 \cdot 10^{-10}$
8	$1.4208 \cdot 10^{-8}$	$1.0628 \cdot 10^{-11}$	$3.9030 \cdot 10^{-13}$	$2.9383 \cdot 10^{-15}$
12	$1.9269 \cdot 10^{-11}$	$2.7265 \cdot 10^{-15}$	$1.8339 \cdot 10^{-18}$	$2.1347 \cdot 10^{-21}$

Table 6: Absolute error of the truncation of Equation (387) for $W_{-1}(-\varepsilon)$, with $S = \log(1/\varepsilon)$, $L = \log S$.

N	$\varepsilon = 10^{-3}$ ($S = 6.908$)	$\varepsilon = 10^{-6}$ ($S = 13.816$)	$\varepsilon = 10^{-20}$ ($S = 46.052$)
0	$2.7761 \cdot 10^{-1}$	$1.8521 \cdot 10^{-1}$	$8.1518 \cdot 10^{-2}$
1	$2.1725 \cdot 10^{-3}$	$4.8547 \cdot 10^{-3}$	$1.6446 \cdot 10^{-3}$
2	$3.5365 \cdot 10^{-3}$	$5.5019 \cdot 10^{-4}$	$7.5505 \cdot 10^{-6}$
3	$2.9763 \cdot 10^{-4}$	$8.7527 \cdot 10^{-5}$	$1.8886 \cdot 10^{-6}$
5	$1.8100 \cdot 10^{-5}$	$1.0098 \cdot 10^{-6}$	$3.0375 \cdot 10^{-9}$
8	$1.8025 \cdot 10^{-8}$	$2.4269 \cdot 10^{-9}$	$6.2816 \cdot 10^{-13}$
12	$3.8085 \cdot 10^{-10}$	$2.5958 \cdot 10^{-13}$	$3.3774 \cdot 10^{-18}$

Remark 20.32 (Concordance of the source tables). For the reader of a particular source: the S1 columns headed $L = 10, 50, 100$ and the S4 columns headed $x = 10, 25, 100$ are columns of Table 5 (S4’s index N counts from the zero-based family, $\omega_N(x) = x + \sum_{n=0}^N \mathbb{L}_n^{\text{Lam}}(\log x)x^{-n}$, which is our w_N). The S3 and S5 columns headed $z = 10, 100, 10^6$ correspond to $\xi_0 = 2.3026, 4.6052, 13.8155$. For the first two the seating condition Equation (379) fails ($2(1+0.8340) = 3.67 \not\leq 2.30$ and $2(1+1.5272) = 5.05 \not\leq 4.61$), so nothing proved here applies to them; for the third it holds ($2(1+2.6259) = 7.25 < 13.82$), so the series converges there, but the remainder hypothesis $|\xi_0| \geq 10(1+|\ell_0|) = 36.3$ still fails and the explicit bound Equation (381) is unavailable. Those columns are therefore empirical illustrations, not instances of a theorem. Every numerical value in all four source tables was recomputed to 60 digits and found correct.

Remark 20.33 (Non-monotonicity is not a contradiction). At $\xi_0 = 10$ the error at $N = 2$ is essentially the same as at $N = 1$, and at $\varepsilon = 10^{-3}$ the error at $N = 2$ is *larger* than at $N = 1$. Nothing is wrong: Theorem 20.13 bounds the tail by $2\theta^{N+1}$ with $\theta = 5(1+|\ell|)/|\xi|$, and at $\xi_0 = 10$ one has $\theta \approx 1.65 > 1$, so the theorem says nothing there at all. Even inside the proved region the bound is a geometric envelope, not a term-by-term ordering: individual $\mathbb{L}_n^{\text{Lam}}(\ell)$ change sign (Theorem 19.7), so two consecutive corrections can nearly cancel. The residual R_N of Theorem 20.27, not the size of the last retained block, is the quantity that should decide where to truncate; by Theorem 20.29 (ii) it controls the error with an explicit constant.

Algorithm 20.34 (Branch-aware evaluation).

```

input : z != 0, branch index k, order N, tolerance tol
1  xi := Log(z) + 2*pi*i*k          # principal Log; record the choice
2  if xi = 0: fail (the coordinates are undefined)
3  ell := Log(xi)                   # record the determination B_k
4  if |xi| < 10*(1 + |ell|):
      warn: outside the region of Theorem lw-uniform-remainder
5  w := xi - ell
6  for n := 1 to N: w := w + P_n(ell) / xi^n    # tables, or the recurrence
7  R := w + ell + Log(w/xi) - xi          # the same branch data as line 3
8  while |R| > tol:
      w := w - R / (1 + 1/w)              # Newton in branch coordinates
      R := w + ell + Log(w/xi) - xi
9  return w, R

```

Line 7 evaluates $\log_* w$ as $\ell + \text{Log}(w/\xi)$, never as $\text{Log } w$: by [Theorem 20.15](#) the two agree only when the distance-to-the-cut hypothesis [Equation \(382\)](#) holds, and an implementation that silently uses $\text{Log } w$ will report a spurious residual of size 2π near the cut. The multiplicative residual $we^w - z$ must not be substituted for line 7: it overflows for $|z|$ large, whereas R never forms e^w .

Section summary

Lambert W is one reversion evaluated at many arguments. The substitution $we^w = z \rightsquigarrow w + \log_* w = \xi_k$ ([Theorem 20.5](#)) turns each branch into the reversion of $X + L$, so the coefficient polynomials L_n^{Lam} are the same for every branch ([Theorem 20.10](#)) while the coordinates ξ_k, ℓ_k carry all the branch dependence. The formal expansion is a theorem in $\mathcal{G}_{\text{poly}, \mathbb{K}}$ with no hypotheses; its analytic validity is a separate theorem with explicit ones ([Theorems 20.12](#) and [20.13](#)), and the frequently repeated claim of convergence for real $z \geq e$ is not among them. The lower real branch is the single scalar $a = -1$ of the affine-logarithmic family and introduces no new polynomials. At $z = -e^{-1}$ the derivative of we^w vanishes, the local monodromy has order two, and the correct scale is a square root, not a logarithm.

Exercise 20.35. Show that the seating condition [Equation \(379\)](#) holds for $K = \xi_0 = \log z$ and $M = \ell_0$ whenever $z \geq e^6$, but fails for every $z \leq e^5$; and find the exact threshold to four decimal places.

Exercise 20.36. Take $K = \xi_k - \ell_k$ in [Theorem 20.12](#) and show that $M = \log(1 - \ell_k/\xi_k)$, so that $|M| = \mathcal{O}_{|\xi_k| \rightarrow \infty}(|\ell_k/\xi_k|)$. Deduce that the seating condition for this base point is satisfied on a strictly larger region than for $K = \xi_k$, and compare with [Theorem 21.21](#).

Exercise 20.37. Using $\log_* W = \xi - W$ and the division rules of [Section 7](#), expand $W_k(z)/(1 + W_k(z))$ through ξ^{-3} and verify against [Equation \(365\)](#).

Exercise 20.38. Prove that the two determinations of [Theorem 20.24](#) satisfy $\varphi(p) + \varphi(-p) = -\frac{2}{3}p^2 - \frac{43}{270}p^4 + \mathcal{O}_{p \rightarrow 0}(p^6)$, and explain why this sum is an even function of p , hence a single-valued holomorphic function of $1 + ez$ near 0.

21 Generalized Lambert equations and reusable inverse templates

[Section 20](#) solved one equation, $w + \log w = \xi$, and obtained every branch of W from it. This section asks how far that leverage extends. The answer has three layers. First, an explicitly delimited family of equations reduces to $w + \log w = \xi$ by an exact algebraic change of variable, so that no new coefficient family is needed at all: this is the class $y^\alpha e^{\beta y} = x$ and its relatives ([Section 21.2](#)). Second, a much larger class does not reduce exactly but has the same *shape* of dominant balance — a monomial times a

power of a logarithm — and for that class there is a mechanical template producing the polynomial–logarithmic inverse, with the Lambert polynomials appearing as the leading part and computable corrections beyond (Section 21.3). Third, and equally important, there are equations that look similar and are not in the class; the last worked example is one of them, and the point of including it is that the reader should be able to recognise the boundary before spending a page computing on the wrong side of it.

21.1 The affine-logarithmic core, analytically

Theorem 20.7 is the formal statement. Its analytic counterpart is proved by exactly the argument of **Theorem 20.12**, with the pair (σ, τ) chosen to carry the parameter a .

Theorem 21.1 (Analytic affine-logarithmic reversion). *Let $a \in \mathbb{C} \setminus \{0\}$, $X \in \mathbb{C} \setminus \{0\}$, and let ℓ satisfy $e^\ell = X$. Assume*

$$2|a|(1 + |\ell|) < |X|. \quad (399)$$

Then the series below converges absolutely; the number

$$y := X + a \sum_{n \geq 0} \left(\frac{a}{X}\right)^n \mathsf{L}_n^{\text{Lam}}(\ell) = X - a\ell + \sum_{n \geq 1} a^{n+1} \frac{\mathsf{L}_n^{\text{Lam}}(\ell)}{X^n} \quad (400)$$

satisfies $|y/X - 1| \leq \frac{1}{2}$, hence $y \neq 0$; and with $\log_ y := \ell + \text{Log}(y/X)$, which is a logarithm of y ,*

$$y + a \log_* y = X. \quad (401)$$

If moreover $10|a|(1 + |\ell|) \leq |X|$, then with $\theta_a := 5|a|(1 + |\ell|)/|X| \leq \frac{1}{2}$ and every $N \in \mathbb{N}_0$,

$$\left| y - X - a \sum_{n=0}^N \left(\frac{a}{X}\right)^n \mathsf{L}_n^{\text{Lam}}(\ell) \right| \leq 2|a| \theta_a^{N+1}. \quad (402)$$

All data being real, y is real.

Proof. Put $\sigma = a/X$ and $\tau = a\ell/X$, so that $\tau = \sigma\ell$ and $|\sigma| + |\tau| = |a|(1 + |\ell|)/|X| < \frac{1}{2}$ by Equation (399); thus $(\sigma, \tau) \in D$. Let $\mathcal{Y} = \mathcal{Y}(\sigma, \tau)$ be as in **Theorem 20.11**. By Equation (375) with $u = \ell$,

$$\mathcal{Y} = \sum_{n \geq 1} \mathsf{L}_n^{\text{Lam}}(\ell) \sigma^n = \sum_{n \geq 1} a^n \frac{\mathsf{L}_n^{\text{Lam}}(\ell)}{X^n},$$

absolutely convergent, and since $\mathsf{L}_0^{\text{Lam}}(\ell) = -\ell$ the number y of Equation (400) equals $X - a\ell + a\mathcal{Y}$. Then

$$\frac{y}{X} = 1 - \frac{a\ell}{X} + \frac{a\mathcal{Y}}{X} = 1 - \tau + \sigma\mathcal{Y},$$

which lies in the disc of radius $\frac{1}{2}$ about 1, so $|y/X - 1| \leq \frac{1}{2}$, $y \neq 0$, $\text{Log}(y/X) = -\mathcal{Y}$ by Equation (373), and $e^{\log_* y} = X \cdot (y/X) = y$. Therefore

$$y + a \log_* y = (X - a\ell + a\mathcal{Y}) + a(\ell - \mathcal{Y}) = X.$$

The remainder bound is Equation (376) with $5(|\sigma| + |\tau|) = \theta_a$, multiplied by the outer factor $|a|$. Reality follows from uniqueness of the fixed point in **Theorem 20.11**: for real (σ, τ) the contraction Φ preserves the real interval $[-1, 1]$, so its fixed point is real. $\square$

21.2 The class $y^\alpha e^{\beta y} = x$

Proposition 21.2 (Exact reduction, with the branch bookkeeping made explicit). *Let $\alpha, \beta \in \mathbb{C} \setminus \{0\}$ and $x \in \mathbb{C} \setminus \{0\}$; fix a logarithm ξ of x and a logarithm γ of α/β . Then the map*

$$(y, \lambda) \mapsto (u, \mu) := \left(\frac{\beta}{\alpha} y, \lambda - \gamma \right) \quad (403)$$

is a bijection between

$$\left\{ (y, \lambda) : e^\lambda = y, \beta y + \alpha \lambda = \xi \right\} \quad \text{and} \quad \left\{ (u, \mu) : e^\mu = u, u + \mu = \zeta \right\}, \quad \zeta := \frac{\xi}{\alpha} - \gamma.$$

Consequently, writing $x^{1/\alpha} := e^{\xi/\alpha}$, every solution of $y^\alpha e^{\beta y} = x$ compatible with the chosen logarithms has the form

$$y = \frac{\alpha}{\beta} u, \quad u e^u = e^\zeta = \frac{\beta}{\alpha} x^{1/\alpha}, \quad (404)$$

that is, $y = (\alpha/\beta) W_k((\beta/\alpha) x^{1/\alpha})$ for the branch label k determined by [Theorem 20.5](#).

Proof. Let $e^\lambda = y$ and $\beta y + \alpha \lambda = \xi$; set $u = (\beta/\alpha)y$ and $\mu = \lambda - \gamma$. Then $e^\mu = e^\lambda e^{-\gamma} = y \cdot (\beta/\alpha) = u$, and $u + \mu = (\beta/\alpha)y + \lambda - \gamma = \alpha^{-1}(\beta y + \alpha \lambda) - \gamma = \xi/\alpha - \gamma = \zeta$. The inverse map is $(u, \mu) \mapsto ((\alpha/\beta)u, \mu + \gamma)$, so the correspondence is bijective. For the last statement, $e^\mu = u$ and $u + \mu = \zeta$ give $u e^u = e^{u+\mu} = e^\zeta$ by [Theorem 20.5](#), and $e^\zeta = e^{\xi/\alpha} e^{-\gamma} = (\beta/\alpha) x^{1/\alpha}$. Finally $y^\alpha e^{\beta y}$ means $e^{\alpha\lambda + \beta y}$ once the logarithm λ of y has been chosen, so the equation $y^\alpha e^{\beta y} = x$ is exactly $\beta y + \alpha \lambda = \xi$ modulo $2\pi i\mathbb{Z}$, i.e. exactly the left-hand set after adjusting ξ by a multiple of $2\pi i$. $\square$

“With branches chosen consistently” is three choices, not a disclaimer

Reports S3, S4, S5 and S6 all display $y = (\alpha/\beta) W_k((\beta/\alpha) x^{1/\alpha})$ followed by a phrase such as “with powers and branches chosen consistently”. [Theorem 21.2](#) says what the phrase must mean: one chooses (a) a logarithm ξ of x , which fixes the meaning of $x^{1/\alpha}$ as $e^{\xi/\alpha}$ and is *not* determined by x alone when $\alpha \notin \mathbb{Z}$; (b) a logarithm γ of α/β ; (c) the branch label k , which by [Theorem 20.5](#) is then determined by the solution, not free. Different choices at (a) or (b) give genuinely different solutions y of the same equation, not different names for one.

Theorem 21.3 (Generalized Lambert expansion). *In the situation of [Theorem 21.2](#), put*

$$a := \frac{\alpha}{\beta}, \quad X := \frac{\xi}{\beta}, \quad (405)$$

and let ℓ satisfy $e^\ell = X$. If $2|a|(1 + |\ell|) < |X|$, then

$$y = X + a \sum_{n \geq 0} \left(\frac{a}{X} \right)^n \mathcal{L}_n^{\text{Lam}}(\ell) = \frac{\xi}{\beta} - \frac{\alpha}{\beta} \ell + \sum_{n \geq 1} \frac{\alpha^{n+1}}{\beta} \frac{\mathcal{L}_n^{\text{Lam}}(\ell)}{\xi^n} \quad (406)$$

converges absolutely and solves $y^\alpha e^{\beta y} = x$ with $\lambda = \ell + \text{Log}(y/X)$ as the logarithm of y ; the remainder after N terms is bounded by [Equation \(402\)](#). In particular, for $\beta = 1$, real $x \rightarrow +\infty$, $L_1 = \log x$ and $L_2 = \log \log x$,

$$y = L_1 - \alpha L_2 + \sum_{n \geq 1} \alpha^{n+1} \frac{\mathcal{L}_n^{\text{Lam}}(L_2)}{L_1^n}. \quad (407)$$

Proof. By [Theorem 21.2](#) the equation is $\beta y + \alpha \lambda = \xi$ with $e^\lambda = y$; dividing by β gives $y + a\lambda = X$. [Theorem 21.1](#) with this a and X produces y as in [Equation \(400\)](#), which is the first equality in [Equation \(406\)](#); the second is $a(a/X)^n = \alpha^{n+1} \beta^{-n-1} \cdot \beta^n \xi^{-n} = \alpha^{n+1} \beta^{-1} \xi^{-n}$. For $\beta = 1$ we have $X = \xi = L_1$, $\ell = L_2$, $a = \alpha$, and $a \cdot a^n \mathcal{L}_n^{\text{Lam}}(L_2) L_1^{-n} = \alpha^{n+1} \mathcal{L}_n^{\text{Lam}}(L_2) L_1^{-n}$. $\square$

Remark 21.4 (Comparison with the sources, and with the literature). Equation (407) is S5’s generalized Lambert expansion and S2’s α -expansion; both are correct and both are proved there only formally. What is added here is the analytic half: a printed seating condition and an explicit remainder. The inversion of $y^\alpha e^y$ is classical: Comtet treated it together with $y \log y$ [3], and Jeffrey, Corless, Hare and Knuth expressed the coefficients through associated Stirling numbers and studied the convergence of the resulting series [21]. S5 reports their sufficient domains as $x > (\alpha e)^\alpha$ for $\alpha \geq 1$ and $x > e$ for $\alpha < 1$. Those domains are much larger than the region $2|\alpha|(1 + |L_2|) < L_1$ proved here; they are recorded as attribution and are not theorems of this volume. See Section 21.5 for the structural reason why a different regrouping converges on a larger domain, and for what this volume can prove about it.

Proposition 21.5 (The r -Lambert equation: the same expansion to all orders). *Fix $r \in \mathbb{R}$. There is $z_r > 0$ such that for real $z > z_r$ the equation*

$$we^w + rw = z \quad (408)$$

has a unique large positive solution $w = \mathcal{W}_r(z)$, it tends to $+\infty$, and

$$|\mathcal{W}_r(z) - W_0(z)| \leq \frac{3|r|\log z}{z} \quad (z \rightarrow +\infty). \quad (409)$$

Consequently $\mathcal{W}_r$ and W_0 have the same polynomial–logarithmic asymptotic expansion to all orders in the scale $\{\ell_0^j \xi_0^{-n}\}$: the parameter r is invisible beyond all orders.

Proof. Let $f(w) = we^w + rw$. Then $f'(w) = (1+w)e^w + r$, and $(1+w)e^w \rightarrow \infty$, so there is $w_r \geq 1$ with $f' > 0$ on $[w_r, \infty)$; since $f \rightarrow +\infty$, f is a strictly increasing bijection from $[w_r, \infty)$ onto $[f(w_r), \infty)$. Put $z_r = \max\{f(w_r), e^2\}$; for $z > z_r$ let $w > w_r$ be the unique preimage. As $z \rightarrow \infty$, $w \rightarrow \infty$.

From Equation (408), $we^w = z - rw$, so

$$w + \log w = \log(z - rw) = \xi_0 + R, \quad R := \log\left(1 - \frac{rw}{z}\right), \quad \xi_0 = \log z, \quad (410)$$

valid once $z - rw > 0$. To see that this holds for z large, note first that $we^w = z - rw \leq z + |r|w$; since $w \rightarrow \infty$ we may assume $e^w \geq 2|r|$, and then $2|r|w \leq we^w \leq z + |r|w$ forces $|r|w \leq z$ and hence $we^w \leq 2z$. Since $W_0(y) \leq \log y$ for $y \geq e$, this gives $w \leq W_0(2z) \leq \log(2z) \leq \frac{3}{2} \log z$ for $z \geq 4$. For z large, $|rw/z| \leq \frac{1}{2}$ and therefore $|R| \leq 2|r|w/z \leq 3|r|\log z/z$, using $|\log(1 - \nu)| \leq 2|\nu|$ for $|\nu| \leq \frac{1}{2}$ and $w \leq \log(2z) \leq \frac{3}{2} \log z$.

Let $F(v) = v + \log v$ on $(0, \infty)$; then $F'(v) = 1 + 1/v > 1$, so $|F(v_1) - F(v_2)| \geq |v_1 - v_2|$. The principal branch satisfies $F(W_0(z)) = \xi_0$ (take the real logarithm of $W_0 e^{W_0} = z$, legitimate since $W_0(z) > 0$), and $F(w) = \xi_0 + R$; hence $|w - W_0(z)| \leq |R|$, which is Equation (409).

Finally $\log z/z = \xi_0 e^{-\xi_0}$, which is $o_{\xi_0 \rightarrow \infty}(\xi_0^{-n})$ for every n ; so subtracting $\mathcal{W}_r$ from W_0 changes no coefficient of the expansion. $\square$

Remark 21.6 (Why this proposition is here). None of the six sources treats the r -Lambert family, so nothing is consolidated in Theorem 21.5; it is included because the family is the standard first generalisation of W in the literature and because the answer is instructive. It exhibits, in the simplest possible setting, the limitation that every result of this volume shares: a polynomial–logarithmic expansion sees a function only modulo quantities that are smaller than every power of the large variable, and a one-parameter family of genuinely different functions can share one expansion. The same phenomenon is revisited in Theorem 21.18.

21.3 The template

Definition 21.7 (Monomial–logarithmic inverse datum). Let $\mathbb{K}$ be a field of characteristic zero. An inverse problem $\Phi(y) = x$ is of *monomial–logarithmic type* with data (ρ, μ, β, B) if, after a logarithm

ξ of x and a logarithm Z of y have been chosen, it is equivalent to

$$\rho Z + \mu \log Z + \beta + B(Z) = \xi, \quad (411)$$

where $\rho \in \mathbb{K}^\times$, $\mu, \beta \in \mathbb{K}$, and, writing $t = Z^{-1}$ and $L = \log Z$ for the generators of the logarithmic chart,

$$B = \sum_{m \geq 1} b_m(L) t^m \in t\mathbb{K}[L][[t]], \quad b_m \in \mathbb{K}[L]. \quad (412)$$

Exponentiating, this says $\Phi(y) = e^\beta y^\rho (\log y)^\mu e^{B(\log y)}$, all powers being defined through the chosen logarithms; the leading balance is the monomial $e^\beta y^\rho (\log y)^\mu$.

The three hypotheses are worth naming, because each fails in a recognisable way.

- (H1) $\rho \neq 0$: there is a genuine monomial in the balance. If $\rho = 0$ the inverse changes exponential level and leaves every polynomial–logarithmic class ([Theorem 13.11](#)); see [Theorem 21.17](#).
- (H2) μ is a *constant*, so that the logarithmic part of the balance is $\mu \log Z$ and not something with a variable exponent.
- (H3) B has strictly positive t -order and coefficients *polynomial* in L . If B has t -order 0 or its coefficients are not polynomial, the reversion leaves $\mathcal{G}_{\text{poly}, \mathbb{K}}$; see [Theorem 21.15](#).

Theorem 21.8 (Reusable inverse template). *Let $\Phi(y) = x$ be of monomial–logarithmic type with data (ρ, μ, β, B) as in [Theorem 21.7](#). Put*

$$a := \frac{\mu}{\rho}, \quad X := \frac{\xi - \beta}{\rho}, \quad t = X^{-1}, \quad \ell = \log X. \quad (413)$$

Then:

1. $F := X + aL + \rho^{-1}B$ lies in $\mathcal{G}_{\text{poly}, \mathbb{K}}$, and [Equation \(411\)](#) is equivalent to $F(Z) = X$; hence there is a unique $Z \in X + \mathbb{K}[\ell][[t]]$ solving it, namely $Z = F^{(-1)^\circ}$ evaluated at X ;
2. writing $F^{(-1)^\circ} = X + \sum_{n \geq 0} q_n(L) t^n$, one has $q_0 = -aL$, and for every $N \geq 0$ the polynomial q_N is a polynomial function of a and of the coefficients of $b_1, \dots, b_N$ only, given explicitly by the coefficient formula of [Theorem 16.18](#);
3. $q_N = a^{N+1} \mathcal{L}_N^{\text{Lam}} + E_N$, where E_N depends only on $a, b_1, \dots, b_N$ and vanishes identically when $b_1 = \dots = b_N = 0$; in particular $q_N = a^{N+1} \mathcal{L}_N^{\text{Lam}}$ for all N when $B = 0$;
4. re-exponentiating,

$$y = e^Z = \left(\frac{x}{e^\beta} \right)^{1/\rho} X^{-\mu/\rho} \exp \left(\sum_{n \geq 1} q_n(\ell) X^{-n} \right), \quad (414)$$

with $x^{1/\rho} := e^{\xi/\rho}$; the last factor lies in $1 + t\mathbb{K}[\ell][[t]]$. In particular the leading behaviour is

$$y \sim e^{-\beta/\rho} x^{1/\rho} \left(\frac{\xi}{\rho} \right)^{-\mu/\rho}. \quad (415)$$

When $1/\rho$ or μ/ρ is not an integer, [Equation \(414\)](#) lives in the ramified Puiseux–logarithmic tower of [Theorem 3.25](#), not in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$.

Proof. (i) Dividing [Equation \(411\)](#) by ρ and moving β/ρ to the right gives $Z + a \log Z + \rho^{-1}B(Z) = X$. The perturbation $A := aL + \rho^{-1}B$ lies in $\mathbb{K}[L][[t]]$ by [Equation \(412\)](#), so $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ ([Theorem 12.1](#)), and [Theorem 14.11](#) gives a unique inverse in $\mathcal{G}_{\text{poly}, \mathbb{K}}$.

(ii) Write $A^r = \sum_{m \geq 0} A_{r,m}(L) t^m$. By [Theorem 16.18](#),

$$q_N(L) = \sum_{r=1}^{N+1} \frac{(-1)^r}{r!} \left[\prod_{s=N-r+1}^{N-1} (\partial_L - s) \right] A_{r,N-r+1}(L),$$

with the empty product acting as the identity. Now $A_{r,m}$ is a sum of products of r blocks $A_{j_1}, \dots, A_{j_r}$ with $j_1 + \dots + j_r = m$, so it depends only on $A_0, \dots, A_m$; since $m = N - r + 1 \leq N$, the whole formula involves only $A_0 = aL$ and $A_1 = \rho^{-1}b_1, \dots, A_N = \rho^{-1}b_N$, polynomially. The block q_0 is the $r = 1, m = 0$ term, namely $-A_{1,0} = -A_0 = -aL$.

(iii) Setting $b_1 = \dots = b_N = 0$ in the formula of (ii) leaves the coefficients of $(X + aL)^{\langle -1 \rangle_\circ}$, which are $a^{N+1} \mathbb{L}_N^{\text{Lam}}$ by [Theorem 20.7](#). Define E_N as the difference; by (ii) it is a polynomial function of $a, b_1, \dots, b_N$ vanishing when the b 's do.

(iv) From $Z = X + q_0(\ell) + \sum_{n \geq 1} q_n(\ell) X^{-n}$ and $q_0(\ell) = -a\ell$,

$$e^Z = e^X \cdot e^{-a\ell} \cdot \exp\left(\sum_{n \geq 1} q_n(\ell) X^{-n}\right).$$

Here $e^{-a\ell} = (e^\ell)^{-a} = X^{-a} = X^{-\mu/\rho}$ and $e^X = e^{(\xi-\beta)/\rho} = e^{\xi/\rho} e^{-\beta/\rho} = x^{1/\rho} e^{-\beta/\rho}$. The exponential of an element of $t\mathbb{K}[\ell][[t]]$ lies in $1 + t\mathbb{K}[\ell][[t]]$ by [Theorem 10.5](#), which also gives [Equation \(415\)](#) after replacing $X = (\xi - \beta)/\rho$ by ξ/ρ , a substitution absorbed by the $1 + t\mathbb{K}[\ell][[t]]$ factor. The final sentence is [Theorem 10.17](#) together with [Theorem 3.25](#). $\square$

Corollary 21.9 (Closed form when $B = 0$). *If $B = 0$, then*

$$Z = \frac{\xi - \beta}{\rho} + \frac{\mu}{\rho} \sum_{n \geq 0} \left(\frac{\mu}{\xi - \beta} \right)^n \mathbb{L}_n^{\text{Lam}}(\ell), \quad \ell = \log \frac{\xi - \beta}{\rho}, \quad (416)$$

and the series converges absolutely, with the remainder bound [Equation \(402\)](#), as soon as $2|\mu|(1 + |\ell|) < |\xi - \beta|$.

Proof. [Theorem 21.1](#) with $a = \mu/\rho$ and $X = (\xi - \beta)/\rho$; note $a/X = \mu/(\xi - \beta)$ and $a \cdot (a/X)^n$ has the prefactor μ/ρ shown. The seating condition $2|a|(1 + |\ell|) < |X|$ is $2|\mu|(1 + |\ell|) < |\xi - \beta|$ after multiplying by $|\rho|$. $\square$

Note the shape of [Equation \(416\)](#): the exponent ρ of the monomial appears only in the leading term and inside ℓ , while the exponent μ of the logarithm is the expansion parameter. This is why the generalized Lambert family is a one-parameter family and not a two-parameter one.

Algorithm 21.10 (The template, mechanically).

```

input : the direct map Phi, the target x, an order N
1  choose logarithms: xi with e^xi = x; Z will be a logarithm of y
2  rewrite Phi(y) = x as  rho*Z + mu*log Z + beta + B(Z) = xi
   # if this is impossible with constant rho != 0 and constant mu,
   # or with B of positive t-order and polynomial coefficients,
   # the template does not apply: see Example lw-failure
3  a := mu/rho ; X := (xi - beta)/rho ; ell := log X ; t := 1/X
4  A := a*L + B/rho                               # an element of K[L][[t]]
5  Z := reverse(X + A) to order N                   # triangular or Newton reversion
   # equivalently q_n = a^{n+1} P_n(ell) + E_n(b_1..b_n)
6  y := exp(Z)                                       # = (x e^{-beta})^{(1/rho)} * X^{-a} * unit
7  residual := (rho*Z + mu*log Z + beta + B(Z) - xi) # must be O(t^{N+1})

```

Step 5 is where the whole of [Section 15](#) is reused unchanged; step 7 is the finite certificate of [Theorem 14.16](#). Step 6 is the only place where the answer can leave $\mathcal{PL}_{\text{poly}, \mathbb{K}}$, and it does so exactly when $1/\rho$ or μ/ρ is not an integer.

21.4 Three worked examples

Example 21.11 (A fractional power forcing a Puiseux–logarithmic extension). Solve $Y = X^2 \log X$ for X as $Y \rightarrow +\infty$.

Here $\rho = 2$, $\mu = 1$, $\beta = 0$, $B = 0$, and $a = \mu/\rho = 1/2$. With $\xi = \log Y$, the template coordinates are $X_* = \xi/2$ and $\ell = \log(\xi/2)$, and $a/X_* = \mu/\xi = 1/\xi$. [Theorem 21.9](#) gives

$$\log X = \frac{\xi}{2} + \frac{1}{2} \sum_{n \geq 0} \frac{\mathbb{L}_n^{\text{Lam}}(\ell)}{\xi^n} = \frac{\xi}{2} - \frac{\ell}{2} + \frac{\ell}{2\xi} + \frac{\ell^2/2 - \ell}{2\xi^2} + \cdots, \quad (417)$$

convergent as soon as $2(1 + |\ell|) < \xi$. Exponentiating,

$$X = \sqrt{\frac{2Y}{\log Y}} \left(1 + \frac{\ell}{2\xi} + \frac{3\ell^2/8 - \ell/2}{\xi^2} + \mathcal{O}_{Y \rightarrow +\infty}(\ell^3/\xi^3) \right), \quad \xi = \log Y, \ell = \log \frac{\log Y}{2}. \quad (418)$$

Two features are structural rather than accidental. First, the answer contains $Y^{1/2}$: by [Theorem 21.8](#) the exponent is $1/\rho$, so any ρ with $1/\rho \notin \mathbb{Z}$ takes the answer out of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ into the ramified tower of [Theorem 3.25](#). Second, the answer contains $\log \log Y$ through ℓ , although the problem contained only one logarithm; this is [Theorem 17.4](#), and it is why the coefficient ring has to be $\mathbb{K}[\ell]$ and not $\mathbb{K}$. The relative errors of the three displayed truncations at $Y = 10^{20}$ are $3.4 \cdot 10^{-2}$, $9.8 \cdot 10^{-4}$ and $1.3 \cdot 10^{-5}$.

The general case $Y = cX^\rho(\log X)^\mu$ is [Equation \(415\)](#): $X \sim c^{-1/\rho} \rho^{\mu/\rho} Y^{1/\rho} (\log Y)^{-\mu/\rho}$, which is the leading form quoted in S3 and S6.

Example 21.12 (An iterated logarithm, and where the coefficient field breaks). Solve $Y = F(X) = X + \alpha \log X + \gamma \log \log X$ for X as $Y \rightarrow +\infty$, with $\alpha, \gamma \in \mathbb{K}$.

This is *not* covered by [Theorem 21.8](#) as stated. In the template’s variables the equation is $Z + \alpha \log Z + \gamma \log \log Z = Y$, so besides the permitted block $\mu \log Z$ with $\mu = \alpha$ there is a *second* block of t -order 0, namely $\gamma \log \log Z$, whose coefficient $M = \log L$ is not in $\mathbb{K}[L]$: hypothesis (H3) fails. Equivalently, in the reversion chart the perturbation is $A = \alpha L + \gamma M$ with $L = \log Y$, $M = \log L$, and $A \notin \mathbb{K}[L][[t]]$. The example is covered by the depth-two version: by [Theorem 13.14](#) the near-identity group at logarithmic depth 2 is a group over the coefficient ring generated by L and M , and the proof of [Theorem 21.8](#) goes through verbatim once $\mathbb{K}[L]$ is replaced by that ring. Writing

$$A = \alpha L + \gamma M, \quad \varpi := \alpha + \frac{\gamma}{L}, \quad \frac{dA}{dY} = \frac{\varpi}{Y}, \quad \frac{d^2 A}{dY^2} = -\frac{1}{Y^2} \left(\alpha + \frac{\gamma}{L} + \frac{\gamma}{L^2} \right), \quad (419)$$

the first three Lagrange–Bürmann terms of [Theorem 16.14](#) give

$$\begin{aligned} F^{(-1)\circ}(Y) &= Y - A + \frac{A\varpi}{Y} + \frac{1}{Y^2} \left[-A\varpi^2 + \frac{A^2}{2} \left(\alpha + \frac{\gamma}{L} + \frac{\gamma}{L^2} \right) \right] \\ &\quad + \mathcal{O}_{Y \rightarrow +\infty}(L^4/Y^3). \end{aligned} \quad (420)$$

The instructive point is where the coefficient ring first fails to be polynomial.

Proposition 21.13 (The first non-polynomial coefficient). *In [Equation \(420\)](#), expand the block of Y^{-1} :*

$$A\varpi = (\alpha L + \gamma M) \left(\alpha + \frac{\gamma}{L} \right) = \alpha^2 L + \alpha\gamma M + \alpha\gamma + \frac{\gamma^2 M}{L}. \quad (421)$$

Hence for every $\gamma \neq 0$ the coefficient of Y^{-1} already fails to lie in $\mathbb{K}[L, M]$ — one block earlier than S6 records — and the natural coefficient ring is $\mathbb{K}((L^{-1}))[M]$. For $\gamma = 0$ the problem degenerates to the affine-logarithmic case and every block is in $\mathbb{K}[L]$.

Proof. The displayed product is immediate. The functions $L = \log Y$ and $M = \log \log Y$ are algebraically independent over $\mathbb{K}$: if $P \in \mathbb{K}[U, V]$ were nonzero with $P(L, M) \equiv 0$ for large Y , then among the monomials $L^i M^j$ occurring in P there would be a unique dominant one by [Theorem 2.7](#) (the order being lexicographic in (i, j) , since $M = o_{Y \rightarrow +\infty}(L^\epsilon)$ for every $\epsilon > 0$), and a dominant term cannot cancel. Hence $\mathbb{K}[L, M]$ is a polynomial ring in two indeterminates, in which L is prime and does not divide M ; therefore $M/L \notin \mathbb{K}[L, M]$. Since $\alpha^2 L + \alpha \gamma M + \alpha \gamma \in \mathbb{K}[L, M]$ and $\gamma^2 \neq 0$, the sum [Equation \(421\)](#) lies outside $\mathbb{K}[L, M]$. For $\gamma = 0$, $A = \alpha L$ and [Theorem 20.7](#) applies, giving $q_n = \alpha^{n+1} \mathbf{L}_n^{\text{Lam}}(L) \in \mathbb{K}[L]$. $\square$

Remark 21.14 (Numerical check). With $\alpha = \gamma = 1$ and $Y = 10^{10}$, the absolute errors of the truncations of [Equation \(420\)](#) at orders Y^0, Y^{-1}, Y^{-2} are $2.73 \cdot 10^{-9}$, $3.29 \cdot 10^{-18}$ and $5.16 \cdot 10^{-27}$; the ratio of consecutive errors is about $Y^{-1}L$, as [Equation \(420\)](#) predicts.

Example 21.15 (An equation that fails the hypotheses). Solve

$$Y = X + \frac{X}{\log X} \quad (422)$$

for X as $Y \rightarrow +\infty$.

The map looks like a perturbation of the identity, and it is one analytically: $X/\log X = o_{X \rightarrow +\infty}(X)$. It is not one *in the sense of* $\mathcal{G}_{\text{poly}, \mathbb{K}}$. Writing $t = X^{-1}$ and $L = \log X$, the perturbation is $L^{-1}t^{-1}$; read inside $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ it has $\nu_t = -1$, the same outer order as X itself, and its coefficient L^{-1} is not in $\mathbb{K}[L]$. So hypothesis (H3) of [Theorem 21.7](#) fails twice over – wrong outer order and wrong coefficient ring – and [Theorem 14.11](#) does not apply.

The correct diagnosis is that the problem is a near-identity problem one level down. Put $\lambda = \log Y$ and $Z = \log X$; taking logarithms of [Equation \(422\)](#),

$$\lambda = Z + \log \left(1 + \frac{1}{Z} \right) = Z + \frac{1}{Z} - \frac{1}{2Z^2} + \frac{1}{3Z^3} - \cdots, \quad (423)$$

which is of monomial–logarithmic type with $\rho = 1$, $\mu = 0$, $\beta = 0$ and $B(Z) = \log(1 + 1/Z) \in Z^{-1}\mathbb{K}[[Z^{-1}]]$. [Theorem 21.8](#) therefore applies in the logarithmic chart and gives, with all q_n constants because $a = \mu/\rho = 0$ and every b_m is constant,

$$Z = \lambda - \frac{1}{\lambda} + \frac{1}{2\lambda^2} - \frac{4}{3\lambda^3} + \frac{7}{4\lambda^4} - \frac{121}{30\lambda^5} + \cdots. \quad (424)$$

Re-exponentiating, or equivalently using $X = Y(1 - 1/(Z + 1))$,

$$X = Y \left(1 - \frac{1}{\lambda} + \frac{1}{\lambda^2} - \frac{2}{\lambda^3} + \frac{7}{2\lambda^4} - \frac{22}{3\lambda^5} + \frac{179}{12\lambda^6} - \cdots \right), \quad \lambda = \log Y. \quad (425)$$

The relative errors of the first four truncations at $Y = 10^{40}$ are $1.09 \cdot 10^{-2}$, $1.17 \cdot 10^{-4}$, $2.54 \cdot 10^{-6}$, $4.81 \cdot 10^{-8}$.

Proposition 21.16 (The answer is not a polynomial–logarithmic series). *Let $X = X(Y)$ be the solution of [Equation \(422\)](#) for $Y \rightarrow +\infty$. There is no $G = \sum_{n \geq n_0} p_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ – with $t = Y^{-1}$, $L = \log Y$, coefficients $p_n \in \mathbb{K}[L]$ – that is an asymptotic expansion of X in the sense of [Theorem 6.43](#). The answer [Equation \(425\)](#) does lie in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}} = \mathbb{K}((L^{-1}))((t))$.*

Proof. Suppose such a G existed. Since $X/Y \rightarrow 1$, the outer order is $n_0 = -1$, and [Theorem 6.43](#) implies in particular that, for some $d \in \mathbb{N}_0$ and all large Y ,

$$X = p_{-1}(L)Y + \mathcal{O}_{Y \rightarrow +\infty}(L^d), \quad (426)$$

the error absorbing the block $p_0(L)$ together with the tail. Dividing by Y , Equation (425) gives

$$\frac{X}{Y} = 1 - \frac{1}{L} + \mathcal{O}_{Y \rightarrow +\infty}(L^{-2}),$$

whereas Equation (426) gives $X/Y = p_{-1}(L) + \mathcal{O}_{Y \rightarrow +\infty}(L^d/Y)$. Subtracting,

$$p_{-1}(L) - 1 + \frac{1}{L} = \mathcal{O}_{Y \rightarrow +\infty}(L^{-2}) + \mathcal{O}_{Y \rightarrow +\infty}(L^d/Y).$$

As $Y \rightarrow +\infty$ both right-hand terms tend to 0 (the second because $L^d/Y \rightarrow 0$ for every fixed d), while $1/L \rightarrow 0$; hence $p_{-1}(L) \rightarrow 1$ along $L \rightarrow +\infty$, which for a polynomial forces $p_{-1} \equiv 1$. But then the left side is exactly $1/L$, which is not $\mathcal{O}_{Y \rightarrow +\infty}(L^{-2}) + \mathcal{O}_{Y \rightarrow +\infty}(L^d/Y)$, because $L \cdot (1/L) = 1 \not\rightarrow 0$. Contradiction.

For the positive half, Equation (425) exhibits X as $c(L)t^{-1}$ with $c(L) = 1 - L^{-1} + L^{-2} - 2L^{-3} + \dots \in \mathbb{K}((L^{-1}))$, which is one block of an element of $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ (Theorem 3.3); the inclusion $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} \subsetneq \mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ and its strictness are Theorem 3.7. $\square$

Remark 21.17 (The other way to fail: no monomial at all). Hypothesis (H1) fails for $Y = (\log X)^2$, where Equation (411) reads $0 \cdot Z + 2 \log Z = \xi$. The solution is $X = \exp(e^{\xi/2}) = \exp(\sqrt{Y})$ – an exponential of a power of the large variable, hence outside every polynomial–logarithmic class by Theorem 13.11. No amount of coefficient-ring enlargement recovers it: the failure is a change of exponential level, not of coefficient field. The rule of thumb is that (H1) controls the *level*, (H2) the shape of the leading logarithm, and (H3) the coefficient field; a violation of each is repaired, if at all, in a different direction.

Remark 21.18 (Beyond all orders). A last failure mode passes every hypothesis and still leaves the answer undetermined. For $Y = X + e^{-X}$, the perturbation e^{-X} is not in $\mathbb{K}[L][[t]]$, but it is $\mathcal{O}_{X \rightarrow +\infty}(t^n)$ for every n ; the template applied to the truncated data returns $X = Y$, and every polynomial–logarithmic truncation of the true inverse is indeed Y . The expansion is correct and useless: it does not see e^{-Y} . The same phenomenon appears in Theorem 21.5, where an entire one-parameter family of functions shares one expansion. Recovering the invisible terms requires transseries with exponential monomials [12, 8, 1], which are outside the classes of Section 3.

Remark 21.19 (Two classical instances, recorded for reference). The template covers, with no new work, two examples that recur across the sources.

1. $Y = X \log X$. Put $u = \log X$; then $ue^u = Y$, so $X = e^{W(Y)} = Y/W(Y)$, and dividing by the branch expansion, using the reciprocal recurrence of Theorem 7.7,

$$X = \frac{Y}{L_1} \left(1 + \frac{L_2}{L_1} + \frac{L_2^2 - L_2}{L_1^2} + \frac{L_2^3 - \frac{5}{2}L_2^2 + L_2}{L_1^3} + \mathcal{O}_{Y \rightarrow +\infty}(L_2^4/L_1^4) \right),$$

with $L_1 = \log Y$, $L_2 = \log \log Y$. This is the form given in S3 and S6, and it is correct.

2. $X = y/\log y$. Setting $v = y/X$ turns this into $v - \log v = \log X$, which is the lower-branch equation of Theorem 20.20 with $S = T$, $T := \log X$; hence $y = -XW_{-1}(-1/X)$ and, with $\Lambda = \log T$,

$$y = X \left(T + \Lambda + \sum_{n \geq 1} (-1)^{n+1} \frac{\mathcal{L}_n^{\text{Lam}}(\Lambda)}{T^n} \right),$$

convergent whenever $2(1 + |\Lambda|) < T$ by Theorem 20.21. Note the sign: it is $(-1)^{n+1}$, not $(-1)^n$, because the reflection $y = Xv$ has already been undone.

Remark 21.20 (Reductions that are exact, and reductions that are not). It is worth separating two things the sources sometimes present together. Theorem 21.2 and Theorem 21.19 are *exact* reductions:

an algebraic change of variable turns the equation into $u + \log u = \zeta$ with no error term, so the whole of [Section 20](#) applies verbatim, convergence statements included. [Theorem 21.8](#) is not of that kind: it produces a formal inverse in $\mathcal{G}_{\text{poly}, \mathbb{K}}$ whose analytic status must be argued separately, either by [Theorem 10.35](#) (if the direct map realises its expansion) or by an inverse-function estimate as in [Theorem 14.21](#). S6’s example $Y = X + \alpha \log X + \beta + (\gamma \log X + \delta)/X + \cdots$, whose inverse is

$$X = Y - \alpha L - \beta + \frac{(\alpha^2 - \gamma)L + \alpha\beta - \delta}{Y} + \mathcal{O}_{Y \rightarrow +\infty}(L^2/Y^2), \quad L = \log Y,$$

is of the second kind. Its displayed coefficients are correct, and they are the case $\rho = 1$, $\mu = \alpha$, $b_1 = \gamma\ell + \delta$ of [Theorem 21.8](#): the template shifts the constant away, working in $X = Y - \beta$ and $\ell = \log(Y - \beta)$, where $q_1(\ell) = (\alpha^2 - \gamma)\ell - \delta$ with no constant term, and the extra $\alpha\beta$ in the display above is produced when ℓ is re-expanded as $L - \beta/Y + \mathcal{O}_{Y \rightarrow +\infty}(Y^{-2})$. The error term, however, is legitimate only under a hypothesis on the direct map, not merely on its formal expansion.

21.5 Reseating the expansion, and the associated Stirling numbers

[Theorem 20.12](#) was stated with an arbitrary base point K for a reason: the choice $K = \xi_k$ is convenient but not optimal, and the literature on Lambert W contains a whole *sequence* of series obtained by other choices [\[6\]](#), as well as rearrangements whose coefficients are associated Stirling numbers rather than ordinary ones [\[21, 3\]](#). This subsection isolates the mechanism, which is entirely elementary once the two-variable picture of [Theorem 20.11](#) is available: the coefficient array (c_{ab}) can be summed along its diagonals $a + b = n$, which gives the Lambert polynomials, or along its rows $a = \text{const}$, which gives something else – and the something else turns out to be the same family again, evaluated elsewhere.

Theorem 21.21 (Transverse resummation). *In $\mathbb{Q}[[\sigma, \tau]]$, let $\mathcal{Y}$ be the unique solution with $\mathcal{Y}(0, 0) = 0$ of $\mathcal{Y} = -\log(1 - \tau + \sigma\mathcal{Y})$. Then*

$$\mathcal{Y}(\sigma, \tau) = \sum_{a \geq 0} \mathbb{L}_a^{\text{Lam}}(\log(1 - \tau)) \left(\frac{\sigma}{1 - \tau} \right)^a, \quad (427)$$

where $\log(1 - \tau) = -\sum_{m \geq 1} \tau^m/m \in \tau\mathbb{Q}[[\tau]]$. Equivalently, writing $\mathbb{L}_n^{\text{Lam}}(u) = \sum_m \lambda_{n,m} u^m$, for all $a, b \geq 0$ with $a + b \geq 1$,

$$\lambda_{a+b,b} = [\tau^b] \left\{ \frac{\mathbb{L}_a^{\text{Lam}}(\log(1 - \tau))}{(1 - \tau)^a} \right\}. \quad (428)$$

Proof. Existence and uniqueness of $\mathcal{Y}$ in $\mathbb{Q}[[\sigma, \tau]]$ with $\mathcal{Y}(0, 0) = 0$ follow from the (σ, τ) -adic contraction argument, the map $y \mapsto -\log(1 - \tau + \sigma y)$ raising the order by one on the maximal ideal; the analytic realisation is [Theorem 20.11](#). Put $\zeta = 1 - \tau$, $\Lambda = -\log \zeta \in \tau\mathbb{Q}[[\tau]]$ and $\hat{\sigma} = \sigma/\zeta$. Then

$$\mathcal{Y} = -\log(\zeta + \sigma\mathcal{Y}) = -\log \zeta - \log(1 + \hat{\sigma}\mathcal{Y}) = \Lambda - \log(1 + \hat{\sigma}\mathcal{Y}).$$

Set $\mathcal{Q} := \mathcal{Y} - \Lambda$. Substituting $\mathcal{Y} = \Lambda + \mathcal{Q}$,

$$\mathcal{Q} = -\log(1 + \hat{\sigma}\Lambda + \hat{\sigma}\mathcal{Q}) = -\log(1 + \hat{\sigma}(-\Lambda) + \hat{\sigma}\mathcal{Q}),$$

which is precisely the generating equation [Equation \(369\)](#) with $s = \hat{\sigma}$ and $u = -\Lambda$. Moreover $\mathcal{Q}$ has positive order in $\hat{\sigma}$. By the uniqueness clause of [Theorem 19.11](#), transported along the continuous $\mathbb{Q}$ -algebra homomorphism $\mathbb{Q}[u][[s]] \rightarrow \mathbb{Q}[[\sigma, \tau]]$ sending $s \mapsto \hat{\sigma}$ and $u \mapsto -\Lambda$,

$$\mathcal{Q} = \sum_{a \geq 1} \mathbb{L}_a^{\text{Lam}}(-\Lambda) \hat{\sigma}^a.$$

Since $\mathbb{L}_0^{\text{Lam}}(-\Lambda) = \Lambda$, adding Λ gives $\mathcal{Y} = \sum_{a \geq 0} \mathbb{L}_a^{\text{Lam}}(-\Lambda) \hat{\sigma}^a$, which is [Equation \(427\)](#) because $-\Lambda = \log(1 - \tau)$.

For [Equation \(428\)](#), compare with the diagonal grouping. Restricting [Equation \(375\)](#) to $\tau = \sigma u$ identified $\sum_{a+b=n} c_{ab} u^b = \mathbb{L}_n^{\text{Lam}}(u)$, i.e. $c_{ab} = \lambda_{a+b,b}$ for $a + b \geq 1$. Extracting $\sigma^a \tau^b$ from [Equation \(427\)](#) gives $c_{ab} = [\tau^b] \{ \mathbb{L}_a^{\text{Lam}}(\log(1 - \tau)) (1 - \tau)^{-a} \}$. $\square$

Corollary 21.22 (The edge coefficients of the Lambert polynomials, twice). *Taking $a = 0$ and $a = 1$ in Equation (428):*

$$\lambda_{b,b} = [\tau^b](-\log(1 - \tau)) = \frac{1}{b} \quad (b \geq 1), \quad (429)$$

$$\lambda_{b+1,b} = [\tau^b] \frac{\log(1 - \tau)}{1 - \tau} = -H_b \quad (b \geq 0), \quad (430)$$

where $H_b = \sum_{j=1}^b 1/j$. These recover the leading coefficient $1/n$ and the next coefficient $-H_{n-1}$ of $\mathcal{L}_n^{\text{Lam}}$ recorded in Theorem 19.7, now as the Taylor coefficients of two elementary functions. Continuing to $a = 2$,

$$\frac{\mathcal{L}_2^{\text{Lam}}(\log(1 - \tau))}{(1 - \tau)^2} = \tau + 3\tau^2 + \frac{35}{6}\tau^3 + \frac{75}{8}\tau^4 + \frac{203}{15}\tau^5 + \dots,$$

whose coefficients are $\lambda_{3,1}, \lambda_{4,2}, \lambda_{5,3}, \lambda_{6,4}, \lambda_{7,5}$.

Proof. For $a = 0$, $\mathcal{L}_0^{\text{Lam}}(u) = -u$ gives $-\log(1 - \tau) = \sum_{b \geq 1} \tau^b/b$. For $a = 1$, $\mathcal{L}_1^{\text{Lam}}(u) = u$ gives $\log(1 - \tau)/(1 - \tau)$, and $\sum_{b \geq 1} H_b \tau^b = -\log(1 - \tau)/(1 - \tau)$ is the classical Cauchy-product identity $(\sum_{m \geq 1} \tau^m/m)(\sum_{j \geq 0} \tau^j) = \sum_{b \geq 1} H_b \tau^b$, with $H_0 = 0$. The listed values for $a = 2$ are obtained by the same expansion. $\square$

Corollary 21.23 (Reseating at the two-term approximation). *In Theorem 20.12, take the base point $\tilde{K} := K - M$ with $\log \tilde{K} := \log K + \text{Log}(1 - M/K)$, assuming $|M/K| < 1$. Then the associated parameter is*

$$\tilde{M} = \text{Log}\left(1 - \frac{M}{K}\right), \quad |\tilde{M}| \leq -\log\left(1 - \frac{|M|}{|K|}\right), \quad (431)$$

so that $\tilde{M} \rightarrow 0$ when $M/K \rightarrow 0$. If $2(1 + |\tilde{M}|) < |\tilde{K}|$, then $w = \tilde{K} + \sum_{a \geq 0} \mathcal{L}_a^{\text{Lam}}(\tilde{M})\tilde{K}^{-a}$ absolutely, with the remainder bound Equation (381) in the reseeded data. In the canonical case $K = \xi_k$, $M = \ell_k$, and under the hypotheses of Theorem 20.15, this reads

$$W_k(z) = (\xi_k - \ell_k) + \sum_{a \geq 0} \frac{\mathcal{L}_a^{\text{Lam}}(\text{Log}(1 - \ell_k/\xi_k))}{(\xi_k - \ell_k)^a}. \quad (432)$$

Proof. By Equation (378), the parameter attached to a base point $\tilde{K}$ with a chosen logarithm is $\tilde{M} = \tilde{K} + \log \tilde{K} - \xi$. With $\tilde{K} = K - M$ and the stated logarithm,

$$\tilde{M} = (K - M) + \log K + \text{Log}(1 - M/K) - \xi = (K + \log K - \xi) - M + \text{Log}(1 - M/K),$$

and the bracket is M , leaving $\tilde{M} = \text{Log}(1 - M/K)$. The stated modulus bound is Equation (377). Note $e^{\log \tilde{K}} = K(1 - M/K) = K - M = \tilde{K}$, so the chosen logarithm is legitimate. The rest is Theorems 20.12 and 20.13 applied to the reseeded data, and both base points produce the same w , namely the solution of $w + \log_* w = \xi$ in the relevant region. $\square$

Remark 21.24 (What this buys, quantitatively). The reseeded parameter $\tilde{M} = \text{Log}(1 - \ell_k/\xi_k)$ is $\mathcal{O}_{|\xi_k| \rightarrow \infty}(\ell_k/\xi_k)$, whereas $M = \ell_k \rightarrow \infty$. The seating condition Equation (379) therefore holds for the reseeded series on a strictly larger region: for real z it becomes $2(1 + |\log(1 - \ell_0/\xi_0)|) < \xi_0 - \ell_0$, which holds already at $\xi_0 = 5$, whereas the canonical condition needs $\xi_0 > 5.3569\dots$ Numerically, with $K = \xi_0$ and $\tilde{K} = \xi_0 - \ell_0$:

ξ_0	base point	M	$N = 2$	$N = 5$	$N = 10$
5	ξ_0	+1.6094	$6.4 \cdot 10^{-3}$	$3.0 \cdot 10^{-5}$	$1.4 \cdot 10^{-7}$
	$\xi_0 - \ell_0$	-0.3884	$1.1 \cdot 10^{-2}$	$1.1 \cdot 10^{-3}$	$4.9 \cdot 10^{-5}$
10	ξ_0	+2.3026	$1.7 \cdot 10^{-3}$	$5.1 \cdot 10^{-6}$	$6.4 \cdot 10^{-10}$
	$\xi_0 - \ell_0$	-0.2617	$6.9 \cdot 10^{-4}$	$4.3 \cdot 10^{-6}$	$1.7 \cdot 10^{-9}$
30	ξ_0	+3.4012	$4.1 \cdot 10^{-5}$	$1.5 \cdot 10^{-9}$	$8.2 \cdot 10^{-14}$
	$\xi_0 - \ell_0$	-0.1203	$7.3 \cdot 10^{-6}$	$6.9 \cdot 10^{-10}$	$2.2 \cdot 10^{-16}$

The honest reading: reseating enlarges the region on which convergence is *proved*, and it is numerically better at $\xi_0 = 10$ and 30; at $\xi_0 = 5$ the canonical series happens to be better even though nothing here proves it converges. A larger proved domain and a smaller error at a given point are different goods.

Remark 21.25 (Associated Stirling numbers). [Theorem 21.21](#) is the structural reason why a second family of combinatorial numbers appears in this problem. Summing the coefficient array (c_{ab}) along diagonals $a + b = n$ gives L_n^{Lam} , whose coefficients are ordinary Stirling cycle numbers ([Theorem 19.4](#)). Summing along rows $a = \text{const}$ gives the functions $L_a^{\text{Lam}}(\log(1 - \tau))(1 - \tau)^{-a}$, whose Taylor coefficients are a different array. Comtet inverted $y^\alpha e^y$ and $y \log y$ [3], and Jeffrey, Corless, Hare and Knuth identified the coefficients of the row-type rearrangements with *associated* Stirling numbers — those counting permutations all of whose cycles have length at least two — and used them to obtain series converging on larger domains [21]; Corless, Jeffrey and Knuth developed the resulting sequence of series systematically [6]. S5’s observation that the two small parameters are $\sigma = \alpha/L_1$ and $\tau = \alpha L_2/L_1$, and that associated Stirling numbers arise from “a different small-parameter geometry”, is correct and is made precise by [Theorems 21.21](#) and [21.23](#): the different geometry is the base-point change $K \rightsquigarrow K - M$. This volume does not prove the identification of the row coefficients with associated Stirling numbers; that identification is cited, not derived, and [Equation \(428\)](#) is what is proved here.

Section summary

Three layers. *Exact reduction*: $y^\alpha e^{\beta y} = x$ and its relatives become $u + \log u = \zeta$ by an algebraic substitution, so [Section 20](#) transfers verbatim, convergence included ([Theorems 21.2](#) and [21.3](#)); the only new content is bookkeeping, and L_n^{Lam} reappears with the scalar a^{n+1} , $a = \mu/\rho$. *Template*: any equation whose dominant balance is a monomial times a power of a logarithm, perturbed by a series of positive outer order with polynomial logarithmic coefficients, has a mechanically computable polynomial–logarithmic inverse ([Theorem 21.8](#)), whose leading blocks are the Lambert polynomials and whose later blocks differ from them by explicitly computable corrections. *Boundary*: the hypotheses are not decorative. A zero monomial exponent changes exponential level ([Theorem 21.17](#)); a perturbation of outer order zero forces the iterated Laurent coefficient field $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ ([Theorem 21.16](#)); an exponentially small perturbation is invisible to the whole apparatus ([Theorem 21.18](#)).

Exercise 21.26. Carry [Equation \(406\)](#) to $n = 3$ for $\alpha = \frac{1}{2}$, $\beta = 1$ and verify the seating condition at $x = e^{40}$.

Exercise 21.27. Apply [Theorem 21.8](#) to $Y = X + \alpha \log X + \beta + (\gamma \log X + \delta)/X$, in the shifted chart $X_* = Y - \beta$, $\ell = \log(Y - \beta)$, and check from [Theorem 16.18](#) that $q_1(\ell) = (\alpha^2 - \gamma)\ell - \delta$. Identify the part $a^2 L_1^{\text{Lam}}(\ell) = \alpha^2 \ell$ and the correction $E_1 = -\gamma \ell - \delta$. Then re-expand ℓ in $L = \log Y$ and recover the block $((\alpha^2 - \gamma)L + \alpha\beta - \delta)/Y$ displayed above.

Exercise 21.28. Show that $L_3^{\text{Lam}}(\log(1 - \tau))(1 - \tau)^{-3}$ has Taylor coefficients $0, -1, -5, -\frac{85}{6}, -\frac{245}{8}, \dots$, and identify them among the coefficients of $L_3^{\text{Lam}}, L_4^{\text{Lam}}, L_5^{\text{Lam}}, L_6^{\text{Lam}}, L_7^{\text{Lam}}$.

Exercise 21.29. Prove that the logarithmic integral satisfies $\text{li}(x) \sim x/\log x \cdot \sum_{j \geq 0} j! (\log x)^{-j}$ and that formal inversion of this relation is an instance of [Theorem 21.8](#) with $\rho = 1$, $\mu = -1$; deduce the shape of the first three blocks of the inverse, and explain why an analytic theorem about li — not the template — is what is needed to pass from the formal inverse to an asymptotic formula for the n th prime.

Part VI

From formal transseries to analytic asymptotics

22 Poincaré asymptotics on the polynomial–logarithmic scale

Everything proved so far about $\mathcal{PL}_{\text{pow},\mathbb{K}}$, $\mathcal{PL}_{\text{poly},\mathbb{K}}$, $\mathcal{PL}_{\text{rat},\mathbb{K}}$ and $\mathcal{PL}_{\text{itLaur},\mathbb{K}}$ is algebra: statements about coefficient arrays, the valuation ν_t , and the t -adic filtration. None of it mentions a function, a limit, or an inequality. This part supplies the bridge, and states its toll. The toll is real: four of the six sources assert asymptotic validity, uniformity in a parameter, or termwise differentiability on the strength of a formal identity alone, and each such assertion is marked with a % ed. comment and repaired where it occurs below.

Two relations between a function and a series have already been used in this volume: $f \approx F$ of [Theorem 6.43](#), introduced with a $\mathcal{O}(\cdot)$ remainder carrying an unspecified power of $\log X$, and $f \sim_S F$ of [Theorem 10.32](#), introduced with a $o(\cdot)$ remainder. They were introduced independently, in different parts, for different purposes. The first order of business is to prove that they are the same relation ([Theorem 22.6](#)), so that results proved about one may be applied to the other without comment. The second is to determine exactly how much of a function the relation remembers ([Theorem 22.13](#) and [Theorem 22.14](#)); the answer is: everything except a flat perturbation, and nothing less.

Throughout, $\mathbb{K} \subseteq \mathbb{C}$ is a field of characteristic zero, the regime is $X \rightarrow +\infty$ along the real ray $S = [X_0, +\infty)$ with $X_0 > 1$ and the real logarithm $L = \log X$, and $t = X^{-1}$. Where a statement holds verbatim on a closed subsector of a sector at infinity with a fixed branch of the logarithm, this is said explicitly.

22.1 What may be called a scale here

Poincaré’s definition of an asymptotic expansion presupposes an *enumerated* scale: a sequence $\phi_0, \phi_1, \dots$ of positive functions with $\phi_{k+1} = o_{X \rightarrow +\infty}(\phi_k)$. Two of the sources assert that the polynomial–logarithmic monomials $X^{-n}(\log X)^j$ form such a scale under the lexicographic dominance order of [Theorem 3.12](#). They do not, and the reason is structural rather than pedantic.

Proposition 22.1 (The full monomial family admits no decreasing enumeration). *Let*

$$\Gamma = \{X^{-n}(\log X)^j : n \in \mathbb{Z}, j \in \mathbb{N}_0\},$$

ordered by dominance $\prec$ as in [Theorem 3.12](#). Then for every $n \in \mathbb{Z}$ the set $\{\mu \in \Gamma : \mu \prec X^{-n}\}$ has no $\prec$ -greatest element. Consequently there is no sequence $(\phi_k)_{k \in \mathbb{N}_0}$ in Γ with $\phi_{k+1} \prec \phi_k$ whose terms exhaust Γ .

Proof. Fix n . A monomial $X^{-m}(\log X)^j$ is $\prec X^{-n} = X^{-n}(\log X)^0$ if and only if $m > n$, since for $m = n$ the only monomial below $(\log X)^0$ in the block would need a negative logarithmic exponent, and there is none in Γ . Given such a monomial, $X^{-m}(\log X)^{j+1}$ is again $\prec X^{-n}$ (still $m > n$) and strictly dominates it. So the set has no greatest element.

For the consequence, suppose (ϕ_k) were a strictly decreasing enumeration of Γ . The monomial X^{-n} occurs as some ϕ_k , and then $\phi_{k+1} \prec X^{-n}$ while every element of Γ below X^{-n} must occur later in the sequence, hence be $\preceq \phi_{k+1}$. That makes ϕ_{k+1} a greatest element of a set which has none. $\square$

Definition 22.2 (Admissible enumerated scale). Let $n_0 \in \mathbb{Z}$ and let $\delta: \mathbb{Z}_{\geq n_0} \rightarrow \mathbb{N}_0$ be any function. The *scale of profile δ* is the family

$$\Sigma_\delta = \{X^{-n}(\log X)^j : n \geq n_0, 0 \leq j \leq \delta(n)\},$$

enumerated by increasing n and, inside each block, by decreasing j .

Lemma 22.3 (A profile makes a genuine Poincaré scale). *The enumeration of Theorem 22.2 is strictly decreasing for dominance, and consecutive members satisfy $\phi_{k+1} = o_{X \rightarrow +\infty}(\phi_k)$.*

Proof. Each block contributes the finitely many indices $j = \delta(n), \dots, 0$, so the enumeration is well defined and of order type ω . Inside a block, $X^{-n}(\log X)^{j-1}/X^{-n}(\log X)^j = (\log X)^{-1} \rightarrow 0$. At a block boundary, the successor of X^{-n} is $X^{-n-1}(\log X)^{\delta(n+1)}$ and the ratio is $(\log X)^{\delta(n+1)}/X \rightarrow 0$. $\square$

22.2 The expansion relation, and the equivalent remainder conditions

Theorem 6.43 defines $f \approx F$ for $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. We record the extension to rational logarithmic blocks, which is forced on us later by division (Theorem 22.16) and is already needed to say anything at all about so simple a function as $1/\log X$ (Theorem 22.10).

Definition 22.4 (Polynomial–logarithmic asymptotic expansion over the rational-log field). Let $F = \sum_{n \geq n_0} p_n(L)t^n$ lie in $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$, so that every block p_n is a rational function of L over $\mathbb{K}$. Fix $N \in \mathbb{Z}$. Only the finitely many blocks with $n_0 \leq n < N$ are involved below, and each has finitely many poles, so for all sufficiently large X none of them has a pole at $\log X$; for such X put

$$(\text{Trunc}_{<N} F)(X) = \sum_{n_0 \leq n < N} p_n(\log X) X^{-n},$$

a finite sum of complex numbers. A function $f: S \rightarrow \mathbb{C}$ is *PL-asymptotic* to F , written $f \approx F$, if for every $N \in \mathbb{Z}$ there are $M_N \in \mathbb{N}_0$, $C_N > 0$ and $X_N \geq X_0$ with

$$|f(X) - (\text{Trunc}_{<N} F)(X)| \leq C_N X^{-N} (\log X)^{M_N} \quad (X \geq X_N). \quad (433)$$

For $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ this is exactly Theorem 6.43.

Remark 22.5 (Why the ambient is not $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$). The definition does not extend to $F \in \mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ as it stands: there a block p_n lies in $\mathbb{K}((L^{-1}))$, so it is an infinite series and not a function of $\log X$, and $(\text{Trunc}_{<N} F)(X)$ is not a number. The correct object there is a *doubly* indexed condition, in which one truncates first in t and then, inside each retained block, in L^{-1} ; the two limits do not commute, since t is smaller than every power of L^{-1} (Theorem 3.22). Two of the sources use $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ coefficients and an undecorated one-parameter remainder in the same formula. We do not develop the iterated notion here and no statement below uses it; it is recorded as an obligation in Section G.

Theorem 22.6 (Equivalent forms of the remainder condition). *Let $F = \sum_{n \geq n_0} p_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ and let $f: S \rightarrow \mathbb{C}$. Write $k_n \in \mathbb{Z}$ for the degree of p_n as a rational function, so that $p_n(\ell) \sim c_n \ell^{k_n}$ with $c_n \neq 0$ when $p_n \neq 0$. The following are equivalent.*

- (i) *For every N , $f - \text{Trunc}_{<N} F = o_{X \rightarrow +\infty}(X^{1-N})$.*
- (ii) *For every N and every $\varepsilon > 0$, $f - \text{Trunc}_{<N} F = \mathcal{O}_{X \rightarrow +\infty}(X^{-N+\varepsilon})$.*
- (iii) *$f \approx F$, that is, (433) holds for every N .*
- (iv) *For every N , $f - \text{Trunc}_{<N} F - p_N(\log X)X^{-N} = o_{X \rightarrow +\infty}(X^{-N})$.*
- (v) *There are $M_N \in \mathbb{N}_0$ with $f - \text{Trunc}_{<N+1} F = o_{X \rightarrow +\infty}(X^{-N}(\log X)^{M_N})$ for every N .*

If moreover $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, these are further equivalent to:

- (vi) *for some (equivalently, every) profile δ with $\delta(n) \geq \deg_L p_n$ for all n , the function f has the Poincaré expansion $\sum_k c_k \phi_k$ relative to the scale Σ_δ of Theorem 22.2, where c_k is the coefficient in F of the monomial ϕ_k ; that is, $f - \sum_{i \leq k} c_i \phi_i = o_{X \rightarrow +\infty}(\phi_k)$ for every k .*

The equivalence is between the families of conditions, and cannot be localised to one cutoff. Cutoff by cutoff only the implications (iv) $\Rightarrow$ (iii) $\Rightarrow$ (ii) $\Rightarrow$ (i) hold; the reverse implications need the whole family. For instance, with $f(X) = X^{-N+1/2}$ and $F = 0$, condition (i) holds at that N while (iii) fails at that N for every M_N .

Proof. Throughout we use that for any fixed $m \in \mathbb{Z}$, $d \geq 0$ and $\varepsilon > 0$,

$$X^{-m}(\log X)^d = o_{X \rightarrow +\infty}(X^{-m+\varepsilon}), \quad X^{-m-1}(\log X)^d = o_{X \rightarrow +\infty}(X^{-m}), \quad (434)$$

both because $(\log X)^d/X^\varepsilon \rightarrow 0$. We also use that a nonzero $p \in \mathbb{K}(L)$ satisfies $p(\log X) = \mathcal{O}_{X \rightarrow +\infty}((\log X)^{\max(k,0)})$ with k its degree, so each individual block of F obeys $p_n(\log X)X^{-n} = \mathcal{O}_{X \rightarrow +\infty}(X^{-n}(\log X)^{\max(k_n,0)})$.

(iii) $\Rightarrow$ (ii) $\Rightarrow$ (i). The first is (434); the second follows by taking $\varepsilon < 1$.

(i) $\Rightarrow$ (iv). Condition (i) at the cutoff $N+1$ reads $f - \text{Trunc}_{<N+1} F = o_{X \rightarrow +\infty}(X^{-N})$, and one has the identity $\text{Trunc}_{<N+1} F = \text{Trunc}_{<N} F + p_N(\log X)X^{-N}$.

(iv) $\Rightarrow$ (iii). Given N , (iv) at that N gives $f - \text{Trunc}_{<N} F = p_N(\log X)X^{-N} + o_{X \rightarrow +\infty}(X^{-N})$, which is $\mathcal{O}_{X \rightarrow +\infty}(X^{-N}(\log X)^{M_N})$ with $M_N = \max(k_N, 0)$.

(iv) $\Rightarrow$ (v) with $M_N = 0$, since (iv) at N is precisely $f - \text{Trunc}_{<N+1} F = o_{X \rightarrow +\infty}(X^{-N})$.

(v) $\Rightarrow$ (i). Apply (v) at index $N+1$: $f - \text{Trunc}_{<N+2} F = o_{X \rightarrow +\infty}(X^{-N-1}(\log X)^{M_{N+1}})$, which by (434) is $o_{X \rightarrow +\infty}(X^{-N})$. Adding the single block $p_{N+1}(\log X)X^{-N-1} = o_{X \rightarrow +\infty}(X^{-N})$ gives $f - \text{Trunc}_{<N+1} F = o_{X \rightarrow +\infty}(X^{-N})$, which is (i) at cutoff $N+1$. As N is arbitrary, (i) holds at every cutoff.

(iii) $\Rightarrow$ (vi), for $F \in \mathcal{PL}_{\text{poly}, \mathbb{K}}$. Fix an admissible profile δ , and let $\phi_k = X^{-n}(\log X)^j$. The partial sum $\sum_{i \leq k} c_i \phi_i$ equals $\text{Trunc}_{<n} F(X) + X^{-n} \sum_{i \geq j} a_{n,i}(\log X)^i$, where $p_n = \sum_i a_{n,i} L^i$. Hence

$$f - \sum_{i \leq k} c_i \phi_i = \left(f - \text{Trunc}_{<n+1} F \right) + X^{-n} \sum_{i < j} a_{n,i}(\log X)^i.$$

By (iii) at cutoff $n+1$ and (434), the first bracket is $o_{X \rightarrow +\infty}(X^{-n})$, hence $o_{X \rightarrow +\infty}(\phi_k)$ because $\phi_k \geq X^{-n}$; the second is $\mathcal{O}_{X \rightarrow +\infty}(X^{-n}(\log X)^{j-1}) = o_{X \rightarrow +\infty}(\phi_k)$ (empty, hence zero, if $j = 0$).

(vi) $\Rightarrow$ (i). Take $\phi_k = X^{-n}$, the last member of block n ; the corresponding partial sum is exactly $\text{Trunc}_{<n+1} F(X)$, so (vi) gives $f - \text{Trunc}_{<n+1} F = o_{X \rightarrow +\infty}(X^{-n})$ for every $n \geq n_0$, which is (i) at every cutoff $> n_0$; at cutoffs $\leq n_0$ the truncation is 0 and the estimate follows from the one at $n_0 + 1$. That the choice of admissible δ is immaterial is now clear, since the conclusion (i) does not mention δ .

Finally, the last assertion: for $f = X^{-N+1/2}$ and $F = 0$ one has $f = o_{X \rightarrow +\infty}(X^{1-N})$, while $X^{-N+1/2}/(X^{-N}(\log X)^M) = X^{1/2}(\log X)^{-M} \rightarrow +\infty$ for every M . $\square$

Remark 22.7 (Concordance with the six sources and with this volume). [Theorem 22.6](#) reconciles every convention in play. Form (iii) is [Theorem 6.43](#); form (i) is [Theorem 10.32](#); form (iv), written with the inclusive cutoff $\sum_{n \leq N}$, is the definition used by S4 and S5; form (v), again inclusive and with a logarithmic weight on the right, is the definition used by S6. All of them say the same thing, and consequently [Theorems 10.33 to 10.35](#) may be used interchangeably with [Theorems 6.44 and 6.45](#). What is *not* interchangeable is a single truncation order: the last clause of the theorem shows that a one-cutoff statement in one convention is strictly weaker or stronger than the corresponding one-cutoff statement in another, which is why a printed asymptotic formula must always say which convention and which cutoff it uses.

Proposition 22.8 (Uniqueness, with rational blocks). *A function $f: S \rightarrow \mathbb{C}$ is PL-asymptotic to at most one $F \in \mathcal{PL}_{\text{rat}, \mathbb{K}}$. The same holds on a closed subsector $\{z: |z| \geq R, |\arg z - \theta_0| \leq \theta\}$ of a sector at infinity carrying a fixed analytic branch of the logarithm.*

Proof. Suppose $f \approx F$ and $f \approx \tilde{F}$ with $D = F - \tilde{F} \neq 0$. Put $n_1 = \nu_t D$ and let $q \in \mathbb{K}(L)$ be its leading block, of degree $k \in \mathbb{Z}$ and leading coefficient $c \neq 0$, so $q(\ell)/\ell^k \rightarrow c$ as $|\ell| \rightarrow \infty$. Subtracting the two instances of (433) at $N = n_1 + 1$ and using $\text{Trunc}_{<n_1+1} D = q(L)t^{n_1}$ gives

$$q(\log X) X^{-n_1} = \mathcal{O}_{X \rightarrow +\infty}(X^{-n_1-1}(\log X)^M)$$

for some $M \in \mathbb{N}_0$. Dividing, $|q(\log X)| X (\log X)^{-M}$ would have to stay bounded; but it is asymptotic to $|c| X (\log X)^{k-M} \rightarrow +\infty$, a contradiction. On a subsector, $\log z = \log |z| + i \arg z$ has $|\arg z|$ bounded, so $|\log z| \sim \log |z|$ and the same computation applies with $|z|$ in place of X . $\square$

Remark 22.9 (Exactly what uniqueness gives, and what it does not). [Theorem 22.8](#) says the map $f \mapsto F$ is well defined where it is defined. It does *not* say:

1. that f is determined by F . It is not, by [Theorem 22.14](#); the fibre over F is an entire coset of the flat ideal.
2. that F determines f even up to a null set, or on any finite interval. The relation is a statement about germs at $+\infty$ only.
3. that F converges anywhere. Convergence is a separate theorem about one series and one domain; see [Theorem 18.19](#) and [Theorem 18.20](#) for the only case in this volume where the question is settled at all, and there only in a form that does not answer the asymptotic question.
4. that f is differentiable, or that $f' \approx D_X F$ if it is ([Theorem 22.20](#)).
5. that f exists. That every F is realized is a theorem ([Theorem 22.13](#)), and its proof constructs f by hand; no part of the formal algebra supplies it.
6. anything about the coefficients being computable from finitely many values of f . Each p_n is a limit of quotients along the whole ray.

Example 22.10 (A smooth positive function with no $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ expansion). Let $f(X) = 1/\log X$ on $[e, \infty)$. If $f \approx F$ with $F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, then $\log X \cdot f(X) = 1$ and $\log X \approx L$, so [Theorem 6.45](#) gives $LF = 1$ in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. But L is not a unit of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ by [Theorem 7.3](#), since its leading block L is not a unit of $\mathbb{K}[L]$; contradiction. On the other hand $f \approx L^{-1} \in \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ trivially, the remainder being zero at every cutoff $N \geq 1$. So the enlargement of [Theorem 22.4](#) from $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ to $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ is forced by an example this elementary, and is not a convenience.

22.3 The algebra of expansions and the flat ideal

Definition 22.11 (The expansion algebra and its flat ideal). Let $\mathcal{A}$ be the set of germs at $+\infty$ of functions $f: [X_f, \infty) \rightarrow \mathbb{C}$ for which there exists $F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ with $f \approx F$, and let

$$\mathcal{N} = \{f \in \mathcal{A} : f \approx 0\} = \{f : f = \mathcal{O}_{X \rightarrow +\infty}(X^{-N}) \text{ for every } N\}.$$

Members of $\mathcal{N}$ are called *flat*. Write $\exp: \mathcal{A} \rightarrow \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, $\exp(f) = F$, for the expansion map, well defined by [Theorem 22.8](#).

The second description of $\mathcal{N}$ is [Theorem 6.44\(4\)](#), and that $\mathcal{N}$ is an ideal of the $\mathbb{K}$ -algebra $\mathcal{A}$ is part of [Theorem 6.45](#): $\mathcal{N} = \ker \exp$ and $\exp$ is a $\mathbb{K}$ -algebra homomorphism. The next proposition records the three finer structural facts, which are not in the sources.

Proposition 22.12 (Structure of the flat ideal). $\mathcal{N}$ is a proper ideal of $\mathcal{A}$, and

1. $\mathcal{N}$ is prime but not maximal;
2. $\mathcal{N}$ is not stable under differentiation, even on the subalgebra of germs of entire functions;
3. $\mathcal{N}$ contains no minimal nonzero scale: if $f \in \mathcal{N}$ is eventually nonvanishing then $f^2 \in \mathcal{N}$ and $f^2 = \mathcal{O}_{X \rightarrow +\infty}(f)$.

Proof. Properness: $1 \approx 1 \neq 0$, so $1 \notin \mathcal{N}$.

(1) By [Theorem 22.13](#) below, $\exp$ is onto, so $\mathcal{A}/\mathcal{N} \cong \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. That ring is an integral domain ([Theorem 6.22](#)) and is not a field ([Theorem 7.23](#)); hence $\mathcal{N}$ is prime and not maximal. A witness for non-maximality is the germ of $\log X$, whose class is nonzero and not invertible: an inverse would force L to be a unit of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, refuted in [Theorem 22.10](#).

(2) See [Theorem 22.20](#), where $f \in \mathcal{N}$ is entire and $f' \notin \mathcal{A}$, a fortiori $f' \notin \mathcal{N}$.

(3) $\mathcal{N}$ is an ideal and $f \in \mathcal{N} \subseteq \mathcal{A}$, so $f^2 \in \mathcal{N}$. Taking $N = 1$ in the definition of $\mathcal{N}$ gives $f(X) \rightarrow 0$, so $f^2/f = f \rightarrow 0$ wherever $f \neq 0$, that is $f^2 = \mathcal{O}_{X \rightarrow +\infty}(f)$. $\square$

Theorem 22.13 (Borel–Ritt on the polynomial–logarithmic scale). *For every $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ there is a C^∞ function $f: (0, \infty) \rightarrow \mathbb{C}$, real valued if $\mathbb{K} \subseteq \mathbb{R}$, with $f \approx F$. Consequently*

$$0 \longrightarrow \mathcal{N} \longrightarrow \mathcal{A} \xrightarrow{\text{exp}} \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} \longrightarrow 0$$

is an exact sequence of $\mathbb{K}$ -algebras and $\mathcal{A}/\mathcal{N} \cong \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$.

Proof. Write $F = \sum_{n \geq n_0} p_n(L)t^n$. Fix once and for all a C^∞ function $\chi: \mathbb{R} \rightarrow [0, 1]$ with $\chi = 0$ on $(-\infty, 0]$ and $\chi = 1$ on $[1, \infty)$. Choose real numbers $X_{n_0} < X_{n_0+1} < \dots$ with $X_n \geq \max(e, X_{n-1} + 1)$, $X_n \rightarrow +\infty$, and

$$|p_n(\log X)| \leq 2^{-|n|} X \quad \text{for all } X \geq X_n; \quad (435)$$

this is possible because $p_n(\log X)/X \rightarrow 0$ for each fixed n . Set

$$f(X) = \sum_{n \geq n_0} \chi(X - X_n) p_n(\log X) X^{-n}.$$

On any bounded interval only the finitely many n with $X_n < \sup$ of that interval contribute, so the sum is locally finite and $f \in C^\infty((0, \infty))$.

By Theorem 6.44(1) it suffices to establish (433) for every $N > n_0$. Fix such an N and split

$$f - \text{Trunc}_{<N} F = \underbrace{\sum_{n_0 \leq n < N} (\chi(X - X_n) - 1) p_n(\log X) X^{-n}}_{=: \Sigma_1} + \underbrace{\sum_{n \geq N} \chi(X - X_n) p_n(\log X) X^{-n}}_{=: \Sigma_2}.$$

Σ_1 is a finite sum whose n -th term vanishes once $X \geq X_n + 1$; hence $\Sigma_1 = 0$ for $X \geq X_{N-1} + 1$. In Σ_2 , the term $n = N$ is bounded by $|p_N(\log X)| X^{-N} \leq C X^{-N} (\log X)^{M_N}$ with $M_N = \max(\deg_L p_N, 0)$. For $n \geq N + 1$ the n -th term vanishes unless $X \geq X_n$, and where it does not vanish (435) and $\chi \leq 1$ bound it by $2^{-|n|} X^{1-n} \leq 2^{-|n|} X^{-N}$ for $X \geq 1$; summing $\sum_{n \in \mathbb{Z}} 2^{-|n|} = 3$,

$$|\Sigma_2| \leq C X^{-N} (\log X)^{M_N} + 3 X^{-N}.$$

This is (433) at the cutoff N .

Exactness at $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is the surjectivity just proved, exactness at $\mathcal{A}$ is $\mathcal{N} = \ker \text{exp}$ (Theorem 6.45), and exp is a $\mathbb{K}$ -algebra map by the same theorem. $\square$

Corollary 22.14 (The converse fails, and by exactly one coset). *For $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ the set of $f \in \mathcal{A}$ with $f \approx F$ is nonempty, and equals $f_0 + \mathcal{N}$ for any one of its members f_0 . The ideal $\mathcal{N}$ is nonzero, since it contains e^{-X} , $e^{-\sqrt{X}}$ and $X^{-\log X}$; so infinitely many pairwise distinct germs share every polynomial–logarithmic expansion, and no property of a function that is not invariant under adding a flat germ can be read off from its expansion.*

Proof. Nonemptiness is Theorem 22.13; the fibre description is the statement that exp is additive with kernel $\mathcal{N}$. For membership in $\mathcal{N}$: $e^{-X} X^N \rightarrow 0$, $e^{-\sqrt{X}} X^N \rightarrow 0$ and $X^{-\log X} X^N = e^{-(\log X)^2 + N \log X} \rightarrow 0$ for every N . $\square$

Example 22.15 (Flatness has no bottom). Extending the dominance symbol of Theorem 3.12 from monomials to arbitrary positive germs by declaring $\phi \succ \psi$ to mean $\psi = o_{X \rightarrow +\infty}(\phi)$, the germs

$$X^{-\log X} \succ e^{-\sqrt{X}} \succ e^{-X} \succ e^{-e^X}$$

all lie in $\mathcal{N}$; iterating Theorem 22.12(3) produces from any one of them an infinite strictly decreasing chain inside $\mathcal{N}$. The expansion map cannot distinguish any of them from 0. This is the precise sense in which a polynomial–logarithmic expansion “remembers the whole power–log sector and nothing beyond it”; adjoining exponential monomials is the standard remedy and takes one out of the class studied here (Theorem 8.16).

Theorem 22.16 (Quotients). *Let $f \approx F$ and $g \approx G$ with $F, G \in \mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ and $G \neq 0$. Put $m = \nu_t G$ and let $q \in \mathbb{K}(L)$ be the leading block of G . Then g is eventually nonvanishing, and*

$$\frac{f}{g} \approx \frac{F}{G} \in \mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}.$$

If $F, G \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and $q \in \mathbb{K}^\times$, then $F/G \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$; if $q \notin \mathbb{K}$, then F/G need not lie in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, the quotient $F = 1, G = L$ being the smallest example (Theorem 22.10).

Proof. Write k for the degree of q and $c \neq 0$ for its leading coefficient. Applying (433) to g at cutoff $m + 1$,

$$g(X) = X^{-m} \left(q(\log X) + \mathcal{O}_{X \rightarrow +\infty}(X^{-1}(\log X)^M) \right),$$

and $|q(\log X)| \sim |c| (\log X)^k$ while $X^{-1}(\log X)^M = o_{X \rightarrow +\infty}((\log X)^k)$. Hence for large X

$$|g(X)| \geq \frac{1}{2} |c| X^{-m} (\log X)^k > 0, \quad (436)$$

so g is eventually nonvanishing and $|1/g(X)| \leq 2|c|^{-1} X^m (\log X)^{-k}$.

Since $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}} = \mathbb{K}(L)((t))$ is the field of Laurent series over the field $\mathbb{K}(L)$ (Theorem 7.19), $H := F/G \in \mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ exists and $F = GH$. Fix N . The truncation $\text{Trunc}_{<N} H$ is a finite polynomial–logarithmic expression, hence realizes itself: $\text{Trunc}_{<N} H \approx \text{Trunc}_{<N} H$, because for any cutoff N' the difference $\text{Trunc}_{<N} H - \text{Trunc}_{<N'} \text{Trunc}_{<N} H$ is a finite sum of blocks of index $\geq N'$. By Theorem 6.45 and linearity (Theorem 6.44(3)),

$$\phi := f - g \cdot \text{Trunc}_{<N} H \approx E := F - G \text{Trunc}_{<N} H = G(H - \text{Trunc}_{<N} H).$$

By additivity of ν_t (Theorem 4.2), $\nu_t E \geq m + N$, so $\text{Trunc}_{<m+N} E = 0$ and (433) applied to ϕ at cutoff $m + N$ gives $\phi = \mathcal{O}_{X \rightarrow +\infty}(X^{-(m+N)}(\log X)^{M'})$. Dividing by g and using the lower bound (436),

$$\frac{f}{g} - \text{Trunc}_{<N} H = \frac{\phi}{g} = \mathcal{O}_{X \rightarrow +\infty}(X^{-N}(\log X)^{\max(M'-k, 0)}).$$

As N was arbitrary, $f/g \approx H$. For the last sentence: F/G lies in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ for every $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ if and only if $t^{-m}G$ is a unit of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, which by Theorem 7.3 happens if and only if its leading block q lies in $\mathbb{K}^\times$, and then $1/G$ is given by the recurrence of Theorem 7.7. $\square$

Proposition 22.17 (Substitution into a function holomorphic at the origin). *Let $u: S \rightarrow \mathbb{C}$ satisfy $u \approx U$ with $U \in \mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$, $\nu_t U \geq 1$. Let $h(z) = \sum_{k \geq 0} h_k z^k$ be holomorphic on $|z| < r$ with coefficients in $\mathbb{K}$. Then $u(X) \rightarrow 0$, $h \circ u$ is defined for large X , and*

$$h \circ u \approx \sum_{k \geq 0} h_k U^k,$$

the right side being the formally summable substitution of Theorem 11.22.

Proof. Since $\nu_t U \geq 1$, $\text{Trunc}_{<1} U = 0$, so (433) at cutoff 1 gives $u = \mathcal{O}_{X \rightarrow +\infty}(X^{-1}(\log X)^{M_1})$, whence $u \rightarrow 0$ and, for each $k \geq 0$, $u^k = \mathcal{O}_{X \rightarrow +\infty}(X^{-k}(\log X)^{kM_1})$. In particular $|u| \leq r/2$ for large X , so $h \circ u$ is defined there.

Fix N and choose an integer $J \geq \max(N, 0)$. Write $h(z) = \sum_{k < J} h_k z^k + z^J \tilde{h}(z)$ with $\tilde{h}$ holomorphic on $|z| < r$; let $C_J = \max_{|z| \leq r/2} |\tilde{h}(z)|$. Then

$$h(u) - \sum_{k < J} h_k u^k = u^J \tilde{h}(u) = \mathcal{O}_{X \rightarrow +\infty}(X^{-J}(\log X)^{JM_1}) = \mathcal{O}_{X \rightarrow +\infty}(X^{-N}(\log X)^{JM_1}).$$

By Theorem 6.45 and linearity, $\sum_{k < J} h_k u^k \approx \sum_{k < J} h_k U^k$. Finally $\nu_t(U^k) \geq k \geq J \geq N$ for $k \geq J$, so $\text{Trunc}_{<N} \sum_{k < J} h_k U^k = \text{Trunc}_{<N} \sum_{k \geq 0} h_k U^k$. Combining the last three displays gives (433) for $h \circ u$ at cutoff N . $\square$

Corollary 22.18 (The three small-argument operations). *Let $u \approx U$ with $\nu_t U \geq 1$, and let $\alpha \in \mathbb{K}$. Then, with the real logarithm and the principal branch of the power for large real X ,*

$$\log(1 + u) \approx \log(1 + U), \quad (1 + u)^\alpha \approx (1 + U)^\alpha, \quad e^u \approx \exp U,$$

where the right-hand sides are the formal operations of [Theorem 10.5](#) and [Theorem 6.37](#). In particular the formal geometric, logarithmic and binomial series may be substituted termwise, with no hypothesis beyond $\nu_t U \geq 1$.

Proof. Apply [Theorem 22.17](#) to $h(z) = \log(1 + z)$ and $h(z) = (1 + z)^\alpha$, both holomorphic on $|z| < 1$, and to $h(z) = e^z$, entire; $u \rightarrow 0$ puts the argument in the disc for large X , and on the positive real ray the principal branches agree with the power series. The formal sums coincide with the named formal operations by [Theorem 11.22](#). $\square$

Remark 22.19 (Composition: the hypotheses are geometric, not algebraic). Composition with a general inner function is *not* covered by [Theorem 22.17](#) and is not free. The statement to use is [Theorem 10.35](#), whose three hypotheses (A1) regime, (A2) monomial-leading realization with a positive exponent, and (A3) branch compatibility are each shown to be necessary in [Theorem 10.36](#). All five sources that discuss composition list “uniformity of the outer remainder over the range of the inner map” as a requirement; that is (A1) together with the remainder transport step (191), and it costs ρ^{-1} outer orders per output order. None of the sources records that cost.

22.4 Termwise differentiation is not automatic

Example 22.20 (An entire flat function whose derivative has no expansion). Let

$$f(X) = e^{-X} \sin(e^X), \quad X \geq 1.$$

Then f is the restriction to the ray of an entire function, $|f(X)| \leq e^{-X}$, so $f \in \mathcal{N}$ and $f \approx 0$; the formal derivative of 0 is 0. But

$$f'(X) = \cos(e^X) - e^{-X} \sin(e^X),$$

and f' is not PL-asymptotic to *any* $F \in \mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$. Indeed $|f'| \leq 1 + e^{-1}$ is bounded, so if $f' \approx F$ then $\nu_t F \geq 0$ – otherwise (433) at cutoff $\nu_t F + 1$ would force $|f'| \sim |p_{\nu_t F}(\log X)| X^{-\nu_t F} \rightarrow \infty$ – and then (433) at cutoff 1 gives $f'(X) - p_0(\log X) \rightarrow 0$. Since $f'(X) = \cos(e^X) + o_{X \rightarrow +\infty}(1)$, this forces $p_0(\log X) \rightarrow 1$ along $X_k = \log(2k\pi)$ and $p_0(\log X) \rightarrow -1$ along $Y_k = \log((2k+1)\pi)$. Both sequences tend to $+\infty$, and a rational function has a single limit in $\mathbb{K} \cup \{\infty\}$ at $+\infty$; contradiction.

An expansion says nothing about the derivative

The example is entire, bounded, and flat. Real-analyticity is therefore not a sufficient hypothesis for termwise differentiation, nor is boundedness, nor is flatness. In particular the frequently seen inference

$$“f(X) = X + \mathcal{O}_{X \rightarrow +\infty}(\log X), \text{ hence } f'(X) = 1 + o_{X \rightarrow +\infty}(1)”$$

is invalid: $f(X) = X + \sin(e^X)$ satisfies the hypothesis and has $f'(X) = 1 + e^X \cos(e^X)$, which is unbounded. Two sufficient hypotheses that do work are given in [Theorem 22.25](#) and [Theorem 22.27](#).

22.5 Termwise integration is unconditional

The asymmetry between the two operations is complete: differentiation needs an extra hypothesis and integration needs none. The reason is visible in [Theorem 22.20](#): differentiation amplifies fast oscillation, while integration averages it away.

Lemma 22.21 (The elementary tail integral). *Let $N \geq 2$ and $M \in \mathbb{N}_0$. For $X \geq e$,*

$$\int_X^\infty s^{-N} (\log s)^M ds \leq C_{N,M} X^{1-N} (\log X)^M, \quad C_{N,M} = \frac{1}{N-1} \sum_{i=0}^M \frac{M!}{(M-i)!(N-1)^i}.$$

Proof. Let $I_M(X)$ denote the integral. Integrating by parts with $u = (\log s)^M$ and $dv = s^{-N} ds$, and using $s^{1-N} (\log s)^M \rightarrow 0$,

$$I_M(X) = \frac{X^{1-N} (\log X)^M}{N-1} + \frac{M}{N-1} I_{M-1}(X), \quad I_0(X) = \frac{X^{1-N}}{N-1}.$$

Unwinding the recursion gives $I_M(X) = \frac{X^{1-N}}{N-1} \sum_{i=0}^M \frac{M!}{(M-i)!(N-1)^i} (\log X)^{M-i}$, and $(\log X)^{M-i} \leq (\log X)^M$ for $X \geq e$. $\square$

Theorem 22.22 (Termwise integration). *Let $g: [X_0, \infty) \rightarrow \mathbb{C}$ be locally integrable with $g \approx G \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, and let $A \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ be any antiderivative of G , that is $D_X A = G$; such an A exists and is unique up to an additive constant of $\mathbb{K}$ by Theorem 9.17 and Theorem 9.11. Then there is a constant $c \in \mathbb{C}$ with*

$$\left(\int_{X_0}^X g(s) ds \right) - c \approx A,$$

equivalently $\int_{X_0}^X g \approx A + c$ once the coefficient field is enlarged from $\mathbb{K}$ to $\mathbb{K}(c)$. That constant is the only ambiguity: no hypothesis beyond $g \approx G$ is needed, and none of g beyond local integrability.

Proof. Write $G = \sum_n g_n(L)t^n$ and $A = \sum_n a_n(L)t^n$, and put $h(X) = \int_{X_0}^X g$. We first record the exact formal identity

$$\frac{d}{dX} \left[(\text{Trunc}_{<N-1} A)(X) \right] = (\text{Trunc}_{<N} D_X A)(X) = (\text{Trunc}_{<N} G)(X), \quad (437)$$

valid for every N . Indeed by Theorem 9.3 the block of $D_X A$ at index m is $(\partial_L - (m-1))a_{m-1}$, so the blocks of $D_X A$ of index $< N$ are exactly the images of the blocks of A of index $< N-1$; differentiating the finite sum $\sum_{n < N-1} a_n(\log X) X^{-n}$ term by term reproduces them.

Fix $N \geq 2$ and set $\psi_N = h - \text{Trunc}_{<N-1} A$. By (437), $\psi'_N = g - \text{Trunc}_{<N} G =: r_N$, and by hypothesis $|r_N(X)| \leq C_N X^{-N} (\log X)^{M_N}$ for $X \geq X_N$. By Theorem 22.21 this is integrable on $[X_N, \infty)$, so $c_N := \lim_{X \rightarrow \infty} \psi_N(X)$ exists and

$$|\psi_N(X) - c_N| = \left| \int_X^\infty r_N \right| \leq C_N C_{N,M_N} X^{1-N} (\log X)^{M_N}. \quad (438)$$

Moreover for $N \geq 2$,

$$c_{N+1} - c_N = \lim_{X \rightarrow \infty} (\text{Trunc}_{<N-1} A - \text{Trunc}_{<N} A)(X) = - \lim_{X \rightarrow \infty} a_{N-1}(\log X) X^{1-N} = 0,$$

since $N-1 \geq 1$. So all c_N with $N \geq 2$ coincide; call the common value c .

Then (438) says $h - c - \text{Trunc}_{<N-1} A = \mathcal{O}_{X \rightarrow +\infty}(X^{-(N-1)} (\log X)^{M_N})$ for every $N \geq 2$, that is, (433) holds for $h - c$ against A at every cutoff ≥ 1 . By Theorem 6.44(1) a cofinal family of cutoffs suffices, so $h - c \approx A$. $\square$

Corollary 22.23 (The tail integral of an integrable expansion). *If $g \approx G$ with $\nu_t G \geq 2$, then g is integrable near $+\infty$ and*

$$\int_X^\infty g(s) ds \approx -A,$$

where A is the unique antiderivative of G in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ with $\nu_t A \geq 1$.

Proof. Existence and uniqueness of A with $\nu_t A \geq 1$: writing $G = \sum_n g_n(L)t^n$ and $A = \sum_n a_n(L)t^n$, by [Theorem 9.3](#) the block of A at index $n \geq 1$ must satisfy $(\partial_L - n)a_n = g_{n+1}$, which for $n \neq 0$ has a unique solution in $\mathbb{K}[L]$ by [Theorem 9.14](#), and $\nu_t G \geq 2$ means only indices $n \geq 1$ occur; the resonant index $n = 0$, the one carrying the constant of [Theorem 9.11](#) and the extra logarithmic degree of (157), is not reached. Now apply [Theorem 22.22](#): with $h(X) = \int_{X_0}^X g$ there is $c \in \mathbb{C}$ with $h - c \approx A$. Since $\nu_t A \geq 1$ we have $(\text{Trunc}_{<1} A)(X) = 0$, so (433) at cutoff 1 gives $h(X) - c \rightarrow 0$; hence $\int_{X_0}^\infty g$ converges, with value c , and $\int_X^\infty g = c - h(X) = -(h - c) \approx -A$. $\square$

Remark 22.24 (Where the resonant block bites). If $\nu_t G = 1$, the antiderivative acquires a block at index 0 of logarithmic degree one higher than that of g_1 ((157)): integrating $p(\log s)s^{-1}$ produces a polynomial in $\log X$, not a constant. This is the analytic shadow of the resonance of [Theorem 9.15](#), and it is the reason [Theorem 22.23](#) needs $\nu_t G \geq 2$ and not merely $\nu_t G \geq 1$. In $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ the same mechanism at the block $L^{-1}t$ produces $\log \log X$ and leaves the class altogether ([Theorems 9.21](#) and [9.23](#)).

Theorem 22.25 (Termwise differentiation is rigid where it is possible). *Let f be differentiable on a ray with $f \approx F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. If $f' \approx G$ for some $G \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, then $G = D_X F$.*

Proof. First, f' is locally integrable on a tail. It is Borel measurable, being a pointwise limit of continuous difference quotients, and $f' \approx G$ applied at the cutoff $\nu_t G + 1$ exhibits f' on some $[X_1, \infty)$ as a continuous function plus a remainder bounded on compacts; hence f' is bounded on compact subsets of $[X_1, \infty)$. A function that is differentiable everywhere on a compact interval with Lebesgue integrable derivative satisfies the fundamental theorem of calculus there, so $f(X) - f(X_1) = \int_{X_1}^X f'$ for $X \geq X_1$.

Now [Theorem 22.22](#) applied to $g = f'$ gives, for any $A \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ with $D_X A = G$, a constant $c \in \mathbb{C}$ with $f - \gamma \approx A$, where $\gamma = c + f(X_1)$. Also $f \approx F$, so $f - \gamma \approx F - \gamma$. Both A and $F - \gamma$ lie in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ over the field $\mathbb{K}(\gamma) \subseteq \mathbb{C}$, so [Theorem 22.8](#), applied over that field, gives $F - \gamma = A$. Applying D_X and using $\ker D_X = \mathbb{K}(\gamma)$ ([Theorem 9.11](#)) gives $D_X F = D_X A = G$. $\square$

Remark 22.26 (The exact shape of the extra hypothesis). [Theorem 22.25](#) isolates the content of the folklore statement “an asymptotic expansion may not be differentiated termwise”. The formal answer is never wrong; what can fail is that f' has no expansion at all. So the extra hypothesis needed for $f \approx F \Rightarrow f' \approx D_X F$ is exactly:

f' admits some polynomial–logarithmic asymptotic expansion.

This is a hypothesis about f' , not about F , and no property of F implies it. [Theorem 22.27](#) gives the standard checkable sufficient condition; a Hardy-field or tame-germ hypothesis is another [[31](#), [32](#), [1](#)].

Theorem 22.27 (Sectorial expansions may be differentiated termwise). *Let $0 < \theta \leq \pi$ and $R > 1$, let $S(\theta, R) = \{z \in \mathbb{C} : |z| \geq R, |\arg z| < \theta\}$ with the principal logarithm, and let f be holomorphic on $S(\theta, R)$. Suppose $F \in \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ and that for every $\theta' < \theta$ and every N there are C, M, R' with*

$$|f(z) - (\text{Trunc}_{<N} F)(z)| \leq C |z|^{-N} (\log |z|)^M \quad (|z| \geq R', |\arg z| \leq \theta'). \quad (439)$$

Then f' satisfies (439) with F replaced by $D_X F$: the expansion may be differentiated termwise, on every closed subsector.

Proof. Fix $\theta' < \theta$ and N . Choose θ'' with $\theta' < \theta'' < \theta$ and put $\delta = \min(\frac{1}{2}, \sin(\theta'' - \theta'))$. For z with $|\arg z| \leq \theta'$, every w with $|w - z| \leq \delta |z|$ satisfies $|\arg w - \arg z| \leq \arcsin \delta \leq \theta'' - \theta'$, hence $|\arg w| \leq \theta''$, and $(1 - \delta)|z| \leq |w| \leq (1 + \delta)|z|$.

Apply (439) on the subsector θ'' at cutoff $N + 1$: setting $G = f - \text{Trunc}_{<N+1} F$, which is holomorphic there, there are C, M, R'' with

$$|G(w)| \leq C |w|^{-N-1} (\log |w|)^M \quad (|\arg w| \leq \theta'', |w| \geq R'').$$

For $|z| \geq 2R''$ with $|\arg z| \leq \theta'$, the circle $|w - z| = \delta|z|$ lies in that region, so Cauchy’s estimate gives

$$|G'(z)| \leq \frac{1}{\delta|z|} \max_{|w-z|=\delta|z|} |G(w)| \leq \frac{C(1-\delta)^{-N-1}(\log((1+\delta)|z|))^M}{\delta|z|^{N+2}} \leq C'|z|^{-(N+2)}(\log|z|)^M$$

for $|z|$ large. Finally, exactly as in (437), the derivative of $\text{Trunc}_{<N+1} F$ with respect to z equals $\text{Trunc}_{<N+2} D_X F$, so that

$$G' = f' - \text{Trunc}_{<N+2} D_X F.$$

The display above is therefore (439) for f' against $D_X F$, at the cutoff $N+2$. Since N was arbitrary and such cutoffs are cofinal, the conclusion holds at every cutoff by Theorem 6.44(1). $\square$

Remark 22.28 (The hypothesis is sectorial, not real-analytic). Theorem 22.20 shows why (439) must be required on a genuine two-dimensional sector. The function $e^{-z} \sin e^z$ is entire, and its restriction to the ray is flat; but on any sector $|\arg z| \leq \theta'$ with $\theta' > 0$ it is unbounded, so it fails (439) against $F = 0$ at every cutoff. “Analytic” alone is worth nothing here; “analytic with a uniform remainder bound on a sector of positive opening” is worth Cauchy’s estimate, which is the whole of the proof.

Section summary

On the polynomial–logarithmic scale, the four elementary transfers behave as follows. *Sums and products*: unconditional (Theorem 6.44, Theorem 6.45). *Quotients*: unconditional once the divisor’s leading block is nonzero, with the quotient living in $\mathcal{PL}_{\text{rat}, \mathbb{K}}$ unless that block is a constant (Theorem 22.16). *Substitution into a function holomorphic at 0*: unconditional (Theorem 22.17). *Composition with a general inner map*: three geometric hypotheses, each necessary (Theorem 10.35). *Integration*: unconditional (Theorem 22.22). *Differentiation*: not transferable at all without an extra hypothesis, but rigid once f' is known to have any expansion (Theorem 22.25, Theorem 22.27). Every function with an expansion is a member of a coset of the flat ideal, and every formal series is realized (Theorem 22.13, Theorem 22.14).

23 Residual-to-error transfer

A formal reversion produces a candidate G_N and a certificate: the residual $\text{Err}_{\text{rev}}(F, G_N)$ has t -order at least N (Theorem 14.16). A user of the answer wants a number: how far is $G_N(x)$ from the true solution of $F(y) = x$? The passage from the first to the second is the subject of this section. It has three steps, and only the first is trivial:

1. convert a residual *value* into an error bound (calculus);
2. convert the formal residual *order* into an analytic residual *size* (Theorem 23.5, which needs an analytic hypothesis on the map being reversed);
3. verify that the true solution exists, is unique, and lies where the derivative bound holds (Theorem 23.2).

Steps (2) and (3) are exactly what the sources omit.

23.1 The transfer inequality and its enclosure form

Remark 23.1 (What Theorem 14.21 does and does not assume). Theorem 14.21 states: if f is C^1 and injective on an interval I , $y \in f(I)$, $x_* = f^{(-1) \circ}(y)$, $x_N \in I$ and $|f'| \geq m > 0$ between x_N and x_* , then

$$|x_N - x_*| \leq \frac{|f(x_N) - y|}{m}. \quad (440)$$

Two observations about the hypotheses, neither of them in the sources. First, injectivity is used only to *name* x_* ; the mean-value argument itself needs only differentiability on the segment joining x_N to x_* . Second, and more seriously, the derivative bound is required on an interval whose right endpoint is the unknown. In an actual computation one knows x_N and $f(x_N) - y$ and does not know x_* ; a bound “between x_N and x_* ” is therefore not directly verifiable, and using (440) as stated is circular unless a separate argument locates x_* . **Theorem 23.2** removes the circularity by hypothesising the derivative bound on a *chosen* interval around x_N and deducing the existence of x_* inside it.

Theorem 23.2 (Residual enclosure: existence, uniqueness, and two-sided error). *Let $\rho > 0$, let $I = [x_N - \rho, x_N + \rho] \subseteq \mathbb{R}$, and let $f: I \rightarrow \mathbb{R}$ be continuously differentiable with*

$$m := \min_{x \in I} |f'(x)| > 0, \quad M := \max_{x \in I} |f'(x)|.$$

Let $y \in \mathbb{R}$ satisfy

$$|f(x_N) - y| \leq m\rho. \quad (441)$$

Then f is strictly monotone on I ; there is exactly one $x_ \in I$ with $f(x_*) = y$; and*

$$\frac{|f(x_N) - y|}{M} \leq |x_N - x_*| \leq \frac{|f(x_N) - y|}{m}. \quad (442)$$

No hypothesis of injectivity is imposed; it is a conclusion.

Proof. Since f' is continuous on the interval I and never zero, the intermediate value theorem applied to f' shows that f' has constant sign; replacing f by $-f$ and y by $-y$ if necessary, assume $f' \geq m > 0$. Then f is strictly increasing, hence injective.

By the mean value theorem, $f(x_N + \rho) - f(x_N) \geq m\rho \geq y - f(x_N)$ and $f(x_N) - f(x_N - \rho) \geq m\rho \geq f(x_N) - y$, using (441) in both. Hence $f(x_N - \rho) \leq y \leq f(x_N + \rho)$, and the intermediate value theorem applied to f produces $x_* \in I$ with $f(x_*) = y$; it is unique by injectivity.

Finally the mean value theorem gives ξ strictly between x_N and x_* (or $x_N = x_*$, in which case (442) is trivial) with $f(x_N) - y = f(x_N) - f(x_*) = f'(\xi)(x_N - x_*)$. Since $m \leq |f'(\xi)| \leq M$, both inequalities of (442) follow. $\square$

Theorem 23.3 (Complex enclosure by Rouché). *Let f be holomorphic on a neighbourhood of the closed disc $\overline{D} = \{z : |z - x_N| \leq \rho\} \subseteq \mathbb{C}$, put $\mu = |f'(x_N)|$, assume $\mu > 0$, and suppose*

$$|f'(z) - f'(x_N)| \leq \frac{\mu}{4} \quad (z \in \overline{D}), \quad |f(x_N) - y| < \frac{\mu\rho}{4}. \quad (443)$$

Then f is injective on $\overline{D}$, the equation $f(z) = y$ has exactly one solution x_ in D , and*

$$\frac{4|f(x_N) - y|}{5\mu} \leq |x_N - x_*| \leq \frac{4|f(x_N) - y|}{3\mu}. \quad (444)$$

Proof. Write $g(z) = f(z) - y$ and let $h(z) = (f(x_N) - y) + f'(x_N)(z - x_N)$ be its linearisation at x_N . For $z \in \overline{D}$, integrating $f' - f'(x_N)$ along the segment from x_N to z , which lies in the convex set $\overline{D}$, gives

$$|g(z) - h(z)| = \left| \int_0^1 (f'(x_N + s(z - x_N)) - f'(x_N))(z - x_N) \, ds \right| \leq \frac{\mu}{4} |z - x_N|. \quad (445)$$

On the circle $|z - x_N| = \rho$ this is $\leq \mu\rho/4$, while $|h(z)| \geq \mu\rho - |f(x_N) - y| > \mu\rho - \mu\rho/4 = 3\mu\rho/4$. Thus $|g - h| < |h|$ on the circle, and Rouché’s theorem gives that g and h have equally many zeros in D , counted with multiplicity. The only zero of h is $z_0 = x_N - (f(x_N) - y)/f'(x_N)$, and $|z_0 - x_N| < (\mu\rho/4)/\mu = \rho/4 < \rho$, so $z_0 \in D$ and g has exactly one zero $x_* \in D$.

For injectivity and the error bounds, note that for $z_1, z_2 \in \overline{D}$,

$$f(z_1) - f(z_2) = (z_1 - z_2) \int_0^1 f'(z_2 + s(z_1 - z_2)) ds,$$

and the integral differs from $f'(x_N)$ by at most $\mu/4$ by (443); hence its modulus lies in $[3\mu/4, 5\mu/4]$ because $|f'(x_N)| = \mu$. Injectivity follows from the lower end. Taking $z_1 = x_N$ and $z_2 = x_*$,

$$\frac{3\mu}{4} |x_N - x_*| \leq |f(x_N) - y| \leq \frac{5\mu}{4} |x_N - x_*|,$$

which is (444). $\square$

Remark 23.4 (Using a lower bound for $|f'(x_N)|$). If μ is only known to satisfy $0 < \mu \leq |f'(x_N)|$, the proof of Theorem 23.3 still gives injectivity, existence and uniqueness of $x_* \in D$, and the upper bound $|x_N - x_*| \leq \frac{4}{3\mu} |f(x_N) - y|$; only the left-hand inequality of (444) uses $|f'(x_N)| = \mu$. This is the asymmetry that produces the constant 5 on the left and 3 on the right.

A formal residual is not an inequality

The statement $\text{Err}_{\text{rev}}(F, G_N) = \mathcal{O}_t^{\text{form}}(t^N)$ is a membership in a filtration step of $\mathcal{PL}_{\text{pow}, \mathbb{K}}$ (Theorem 5.7); the statement $F(G_N(x)) - x = \mathcal{O}_{x \rightarrow +\infty}(x^{-N}(\log x)^M)$ is an inequality between real numbers valid beyond some x_N . Neither implies the other without a theorem. The direction one needs is supplied by Theorem 23.5, and its hypothesis is that the map being reversed *has* a polynomial–logarithmic asymptotic expansion – an analytic fact about a function, of the kind that Theorem 22.13 shows cannot be recovered from the coefficients.

23.2 From the formal residual order to an analytic residual size

Theorem 23.5 (Analytic size of a residual). *Let $Y_0 > 1$, let $\Phi = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, and let $\varphi: [Y_0, \infty) \rightarrow \mathbb{R}$ satisfy $\varphi \approx \Phi$ in the sense of Theorem 22.4, with the real logarithm and the regime $Y \rightarrow +\infty$. Let $H = X + B \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ with $B \in \mathbb{K}[L][t]$ a finite sum of blocks, and let $h(x) = x + B(\log x, 1/x)$ be its evaluation for large real x . Put $r = \nu_t(\text{Err}_{\text{rev}}(\Phi, H)) \in \mathbb{N}_0 \cup \{+\infty\}$. Then $h(x) \rightarrow +\infty$, $h(x)/x \rightarrow 1$, and for every finite $N \leq r$ there are M, C, x_1 with*

$$|\varphi(h(x)) - x| \leq C x^{-N} (\log x)^M \quad (x \geq x_1). \quad (446)$$

Proof. Since $\nu_t B \geq 0$ and B is a finite sum of blocks, $B(\log x, 1/x) = \mathcal{O}_{x \rightarrow +\infty}((\log x)^d)$ with $d = \max_n \deg_L b_n$; hence $h(x) = x(1 + u(x))$ with $u(x) = B(\log x, 1/x)/x \rightarrow 0$, and $h(x) \rightarrow +\infty$, $h(x)/x \rightarrow 1$. In particular $h(x) \geq Y_0$ for large x , so $\varphi \circ h$ is defined there.

We verify the three hypotheses of Theorem 10.35 for the outer function φ and the inner function h . (A1) is the previous paragraph. (A2) holds with $c = 1$, $\rho = 1$ and $U = tB \in t\mathcal{PL}_{\text{pow}, \mathbb{K}}$: the function u is the evaluation of the finite sum tB , so $u \approx U$ exactly (its remainder vanishes at every cutoff beyond the t -degree of tB), and $u \rightarrow 0$. (A3) holds automatically on the positive real ray with $c = 1 > 0$. By Theorem 22.6 the relation $\approx$ of Theorem 22.4 and the relation $\sim_S$ used in Theorem 10.35 coincide, so the theorem applies and yields

$$\varphi \circ h \approx \Phi \circ H.$$

The identity function realizes the series X , so by Theorem 6.44(3) the germ $x \mapsto \varphi(h(x)) - x$ is PL-asymptotic to $\Phi \circ H - X = \text{Err}_{\text{rev}}(\Phi, H)$. Let $N \leq r$ be finite. Since $\nu_t \text{Err}_{\text{rev}}(\Phi, H) = r \geq N$, we have $\text{Trunc}_{<N} \text{Err}_{\text{rev}}(\Phi, H) = 0$, and (433) at cutoff N is exactly (446). $\square$

Theorem 23.6 (Transfer of a formal inverse expansion to the analytic inverse). *Assume:*

- (H1) Regime and derivative bound. $\varphi: [Y_0, \infty) \rightarrow \mathbb{R}$ is C^1 with $\varphi'(Y) \geq m_0 > 0$ for all $Y \geq Y_0$, and $\varphi(Y) \rightarrow +\infty$. (Then φ is a strictly increasing bijection onto $[\varphi(Y_0), \infty)$; write ψ for its inverse.)

(H2) Analytic expansion of the map. $\varphi \approx \Phi$ for some $\Phi = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, with the real logarithm as $Y \rightarrow +\infty$.

Let $G = \Phi^{(-1)\circ} = X + B \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ be the formal compositional inverse supplied by [Theorem 14.11](#), and put $g_N(x) = x + (\text{Trunc}_{<N} B)(x)$. Then for every $N \in \mathbb{N}_0$ there are $M_N \in \mathbb{N}_0$, $C_N > 0$ and a threshold τ_N with

$$|\psi(x) - g_N(x)| \leq C_N x^{-N} (\log x)^{M_N} \quad (x \geq \tau_N), \quad (447)$$

and consequently $\psi \approx G$: the formal compositional inverse is the asymptotic expansion of the analytic inverse.

Proof. (H1) makes ψ well defined. $\varphi' \geq m_0 > 0$ forces φ strictly increasing, hence injective, and $\varphi(Y) \rightarrow +\infty$ with the intermediate value theorem makes it onto $[\varphi(Y_0), \infty)$; ψ is its inverse, $\psi(x) \geq Y_0$, and $\psi(x) \rightarrow +\infty$.

Formal order of the residual. Fix N and put $G_N = X + \text{Trunc}_{<N} B \in \mathcal{G}_{\text{poly}, \mathbb{K}}$, a finite sum of blocks. Then $G - G_N = B - \text{Trunc}_{<N} B$ has $\nu_t \geq N$, so by [Theorem 14.15](#)

$$\nu_t(\text{Err}_{\text{rev}}(\Phi, G_N)) = \nu_t(G_N - G) \geq N.$$

Analytic size of the residual. By (H2) and [Theorem 23.5](#) applied with $H = G_N$, there are M_N, C, x_1 with $|\varphi(g_N(x)) - x| \leq C x^{-N} (\log x)^{M_N}$ for $x \geq x_1$.

Transfer. For x large, both $g_N(x)$ and $\psi(x)$ lie in the interval $[Y_0, \infty)$, on all of which $|\varphi'| \geq m_0$. Applying [Theorem 14.21](#) on that interval with $f = \varphi$, argument value $y = x$, approximation $g_N(x)$, exact solution $x_* = \psi(x)$ and $m = m_0$ gives

$$|\psi(x) - g_N(x)| \leq \frac{|\varphi(g_N(x)) - x|}{m_0} \leq \frac{C}{m_0} x^{-N} (\log x)^{M_N},$$

which is (447). Since $g_N(x) = (\text{Trunc}_{<N} G)(x)$ for every $N \geq 0$, this is (433) at every cutoff $N \geq 0$, and cutoffs $N < 0$ follow from [Theorem 6.44\(1\)](#). Hence $\psi \approx G$. $\square$

Remark 23.7 (Which hypothesis does which job). (H1) is the only source of the constant m_0 , and it must be verified on the function, not deduced from an expansion: by [Theorem 22.20](#) the expansion $\Phi = X + A$ does not entail $\varphi' = 1 + o_{Y \rightarrow +\infty}(1)$, or indeed any statement about φ' whatever. (H2) is what turns a t -adic order into an inequality; it is an analytic hypothesis and no algebraic theorem supplies it. A third condition, the admissibility check $g_N(x) \geq Y_0$, is listed as a hypothesis in several of the sources and in the four-item list preceding [Theorem 18.17](#); it is *not* a hypothesis of [Theorem 23.6](#), because $g_N(x) = x + \mathcal{O}_{x \rightarrow +\infty}((\log x)^d) \rightarrow +\infty$ makes it automatic in the limit. It is nonetheless indispensable at any *specific* x at which one wants to use the resulting inequality: a truncated formal inverse can take values outside $[Y_0, \infty)$ for moderate x , and then $\varphi(g_N(x))$ is not even defined, so the numerical bound is vacuous while the asymptotic theorem remains true. Item (2) of the list preceding [Theorem 18.17](#) is the specialisation of (H2) to $\varphi(Y) = Y + \log Y$.

23.3 Explicit constants for the Wright omega truncations

For the archetype $\varphi(Y) = Y + \log Y$ the constants of [Theorem 23.2](#) can be written down exactly, with no asymptotic hypothesis at all. This is the strongest form in which a residual bound is available anywhere in this volume, and it is available because $\varphi'(Y) = 1 + Y^{-1}$ is monotone and explicitly bounded on $(0, \infty)$.

Proposition 23.8 (A computable two-sided error bound, valid for every $x > 1$). *Let $x > 1$, let $N \in \mathbb{N}_0$, let $\omega_N(x) = x - \log x + \sum_{n=1}^N \mathbb{L}_n^{\text{Lam}}(\log x) x^{-n}$ be the evaluation of the truncation of [Theorem 18.12](#), and suppose $\omega_N(x) > 0$. Put*

$$R_N(x) = \omega_N(x) + \log \omega_N(x) - x, \quad \mu_N(x) = \min \{ \omega_N(x), x - \log x \}.$$

Then $\mu_N(x) > 0$ and

$$\frac{|R_N(x)|}{1 + \mu_N(x)^{-1}} \leq |\omega(x) - \omega_N(x)| \leq |R_N(x)|, \quad (448)$$

and $\omega(x) - \omega_N(x)$ has the sign of $-R_N(x)$. Both bounds are computable from x and N alone; neither involves an implied constant, an unspecified threshold, or a limit.

Proof. By [Theorem 18.1](#), $\varphi(Y) = Y + \log Y$ is a strictly increasing bijection $(0, \infty) \rightarrow \mathbb{R}$ with $\varphi'(Y) = 1 + Y^{-1}$, and for $x > 1$ the envelope (326) gives $\omega(x) > x - \log x$. Also $x - \log x \geq 1 > 0$ for every $x > 0$, because $x \mapsto x - \log x$ has its minimum at $x = 1$. Hence both $\omega(x)$ and $\omega_N(x)$ lie in $[\mu_N(x), \infty)$, and so does the whole segment between them.

On $[\mu_N(x), \infty)$ the derivative satisfies $1 < \varphi'(Y) \leq 1 + \mu_N(x)^{-1}$. Since $\varphi(\omega(x)) = x$, the mean value theorem gives ζ in that segment with

$$R_N(x) = \varphi(\omega_N(x)) - \varphi(\omega(x)) = \varphi'(\zeta)(\omega_N(x) - \omega(x)),$$

whence the sign statement and, dividing by $\varphi'(\zeta) \in (1, 1 + \mu_N(x)^{-1}]$, the two inequalities (448). $\square$

Proposition 23.9 (The two-term approximation, in closed form). *For every $x \geq 1$, with $L = \log x$,*

$$R_0(x) = (x - L) + \log(x - L) - x = \log\left(1 - \frac{L}{x}\right) \leq 0, \quad (449)$$

and

$$\frac{L}{x} \leq |R_0(x)| \leq \frac{L}{x} + \frac{L^2}{x^2}. \quad (450)$$

Consequently $x - \log x$ always underestimates $\omega(x)$, and

$$\frac{L}{x} \cdot \frac{x - L}{x - L + 1} \leq \omega(x) - (x - \log x) \leq \frac{L}{x} + \frac{L^2}{x^2}. \quad (451)$$

Proof. For $x \geq 1$ we have $L \geq 0$ and $x - L \geq 1 > 0$, so $\log(x - L)$ is defined and $(x - L) + \log(x - L) - x = \log(x - L) - L = \log \frac{x-L}{x} = \log(1 - L/x)$, which is ≤ 0 . Set $u = L/x$. The function $x \mapsto (\log x)/x$ attains its maximum e^{-1} at $x = e$, so $0 \leq u \leq e^{-1} < \frac{1}{2}$. Then

$$-\log(1 - u) = \sum_{j \geq 1} \frac{u^j}{j} \geq u, \quad -\log(1 - u) \leq u + \frac{1}{2} \sum_{j \geq 2} u^j = u + \frac{u^2}{2(1 - u)} \leq u + u^2,$$

the last step because $u \leq \frac{1}{2}$. This is (450). Now apply [Theorem 23.8](#) with $N = 0$, where $\omega_0(x) = x - L$ and hence $\mu_0(x) = x - L$: the sign of $-R_0 \geq 0$ gives $\omega(x) \geq x - L$, the upper bound of (448) combined with the right half of (450) gives the right half of (451), and the lower bound of (448) combined with the left half of (450) gives $|R_0|/(1 + (x - L)^{-1}) \geq (L/x)(x - L)/(x - L + 1)$. $\square$

Remark 23.10 (What is *not* obtained). [Theorem 23.8](#) is uniform in N only because it does not evaluate R_N ; the closed form (449) is available at $N = 0$ and, by the same computation, at $N = 1$, but a constant C with $|R_N(x)| \leq C |\mathbf{L}_{N+1}^{\text{Lam}}(\log x)| x^{-N-1}$ uniformly in N is *not* obtained here and is not obtained in [Theorem 18.17](#) either: the constant there depends on N , and the logarithmic exponent $K = (N + 2)^2$ in its remainder is quadratic in N . Any statement in the literature of the form “the error is bounded by the first omitted term” for this expansion, with a constant independent of N , is stronger than anything proved here. We do not assert it.

23.4 A worked bound, to digits

Example 23.11 (The enclosure at $x = 100$, $N = 3$). Take $x = 100$, so that

$$L = \log 100 = 4.605170185988091368 \dots,$$

and use the Lambert polynomials of [Theorem 19.1](#),

$$\mathsf{L}_1^{\text{Lam}}(u) = u, \quad \mathsf{L}_2^{\text{Lam}}(u) = \frac{1}{2}u^2 - u, \quad \mathsf{L}_3^{\text{Lam}}(u) = \frac{1}{3}u^3 - \frac{3}{2}u^2 + u.$$

The three correction terms are

$$\begin{aligned} \mathsf{L}_1^{\text{Lam}}(L) x^{-1} &= 4.60517018598809137 \times 10^{-2}, \\ \mathsf{L}_2^{\text{Lam}}(L) x^{-2} &= 5.99862603496870465 \times 10^{-4}, \\ \mathsf{L}_3^{\text{Lam}}(L) x^{-3} &= 5.34863899981332245 \times 10^{-6}, \end{aligned}$$

so that

$$\omega_3(100) = 95.4414867271142862294 \dots$$

Substituting into $\varphi(Y) = Y + \log Y$ gives the residual

$$R_3(100) = \omega_3(100) + \log \omega_3(100) - 100 = 8.23927836460666826 \times 10^{-8}.$$

Here $x - \log x = 95.39482981401190863 \dots$ and $\omega_3(100) > x - \log x$, so $\mu_3 = 95.394829814 \dots$ and $1 + \mu_3^{-1} = 1.010482748404 \dots$. [Theorem 23.8](#) therefore yields the rigorous enclosure

$$8.153804088 \times 10^{-8} \leq |\omega(100) - \omega_3(100)| \leq 8.239278365 \times 10^{-8},$$

with $\omega(100) < \omega_3(100)$ because $R_3 > 0$. The true value, computed independently to 40 digits from $\omega(100) = W_0(e^{100}) = 95.44148664557583184017 \dots$, is

$$|\omega(100) - \omega_3(100)| = 8.153845439 \times 10^{-8},$$

inside the enclosure and within 5×10^{-6} relative of its lower end. The lower bound is close to sharp because $\varphi'(\zeta) = 1 + \zeta^{-1}$ is close to 1; the whole enclosure has relative width $\mu_3^{-1} \approx 1.05\%$.

For comparison, [Theorem 18.17](#) predicts the error $\mathsf{L}_4^{\text{Lam}}(L)x^{-4} = -7.593614686 \times 10^{-8}$ up to a relative correction $\mathcal{O}_{x \rightarrow +\infty}(L^K/x)$; the predicted and true errors agree to about 6.9%, which is the size of the ratio $\mathsf{L}_5^{\text{Lam}}(L)x^{-5}/\mathsf{L}_4^{\text{Lam}}(L)x^{-4} = 7.2 \times 10^{-2}$. The enclosure of [Theorem 23.8](#) has width 8.55×10^{-10} , about seven times smaller than the 5.60×10^{-9} by which the asymptotic prediction misses; and unlike the asymptotic statement it is a theorem about this one value of x , with no implied constant and no threshold.

Table 7: Rigorous enclosure of $|\omega(100) - \omega_N(100)|$ by [Theorem 23.8](#). All entries computed at 40 significant digits and rounded; $\mu_N = x - \log x = 95.394829814 \dots$ in every row.

N	$ R_N(100) $	lower bound	true error	upper bound
0	$4.714580382 \times 10^{-2}$	$4.665671324 \times 10^{-2}$	$4.665683156 \times 10^{-2}$	$4.714580382 \times 10^{-2}$
1	$6.114700456 \times 10^{-4}$	$6.051266551 \times 10^{-4}$	$6.051297040 \times 10^{-4}$	$6.114700456 \times 10^{-4}$
2	$5.322287245 \times 10^{-6}$	$5.267073836 \times 10^{-6}$	$5.267100545 \times 10^{-6}$	$5.322287245 \times 10^{-6}$
3	$8.239278365 \times 10^{-8}$	$8.153804088 \times 10^{-8}$	$8.153845439 \times 10^{-8}$	$8.239278365 \times 10^{-8}$
4	$5.661006395 \times 10^{-9}$	$5.602279113 \times 10^{-9}$	$5.602307524 \times 10^{-9}$	$5.661006395 \times 10^{-9}$
5	$1.558418770 \times 10^{-10}$	$1.542251733 \times 10^{-10}$	$1.542259554 \times 10^{-10}$	$1.558418770 \times 10^{-10}$

Remark 23.12 (Cancellation in the residual, with the digit count). [Table 7](#) is a table of *differences of numbers of size 95*. At $N = 5$ the residual is 1.56×10^{-10} , so forming it loses about $\log_{10}(95.44/1.56 \times 10^{-10}) \approx 11.8$ decimal digits. A double-precision evaluation of $R_N(x)$ carries roughly 16 digits and is therefore already worthless at $N = 6$, where $R_6 \approx 2.05 \times 10^{-12}$ and the loss is about 13.7 digits. The bound of [Theorem 23.8](#) is exact mathematics and useless numerics unless the residual is formed in extended precision. This is a property of the bound, not of the expansion: the expansion itself is evaluated by a stable Horner recurrence. One source advises “sufficient guard precision” when evaluating the coordinates; none quantifies the cancellation in the residual itself, although it is that cancellation, and not the coordinates, that decides how many digits are needed.

Remark 23.13 (The error is not monotone in N). At $x = 100$ the errors in [Table 7](#) decrease at every step, but this is a feature of that value of x , not a theorem. At $x = 10$, where $\omega(10) = 7.929420095019697348562\dots$, the successive errors $\omega(10) - \omega_N(10)$ are

$$\begin{array}{ll} N = 4 : & 1.9026 \times 10^{-5}, \quad N = 5 : \quad 5.1336 \times 10^{-6}, \\ N = 6 : & 3.8679 \times 10^{-8}, \quad N = 7 : \quad -1.2836 \times 10^{-7}, \\ N = 11 : & 4.8696 \times 10^{-12}, \quad N = 12 : \quad -1.9269 \times 10^{-11}, \end{array}$$

so the absolute error increases from $N = 6$ to $N = 7$ and again from $N = 11$ to $N = 12$. This is the reason to truncate by residual size rather than by formal order when a numerical answer is wanted.

23.5 An example where the hypotheses fail

Example 23.14 (The Lambert branch point: a residual of 10^{-8} with an error of 2.3×10^{-4}). Let $\varphi(Y) = Ye^Y$ on $\mathbb{R}$, so that $\varphi'(Y) = (1+Y)e^Y$ and $\varphi'(-1) = 0$; the two real solutions of $\varphi(Y) = z$ for $-e^{-1} < z < 0$ are $W_0(z) > -1$ and $W_{-1}(z) < -1$ [[5](#), [11](#)]. Take

$$\eta = 10^{-8}, \quad z = -e^{-1} + \eta = -0.36787943117144232159552\dots, \quad Y_{\text{app}} = -1.$$

The residual is exactly

$$\varphi(Y_{\text{app}}) - z = -e^{-1} - z = -\eta = -10^{-8},$$

as small as one could wish. The two true solutions are

$$W_0(z) = -0.99976685372178274818\dots, \quad W_{-1}(z) = -1.00023318252197543531\dots,$$

so that the actual error of Y_{app} is $2.3314627822 \times 10^{-4}$ or $2.3318252198 \times 10^{-4}$ according to which root is meant – larger than the residual by a factor of about 2.33×10^4 .

Every hypothesis fails, and each failure is instructive.

1. *No admissible m .* On the interval joining $Y_{\text{app}} = -1$ to either root, $\inf |\varphi'| = 0$, attained at the endpoint -1 . So there is no $m > 0$ satisfying the hypothesis of [Theorem 14.21](#), and [\(440\)](#) is not merely weak, it is unavailable.
2. *No enclosure.* In [Theorem 23.2](#) one needs $m = \min_I |\varphi'| > 0$ on an interval I around Y_{app} ; any such I contains -1 , where φ' vanishes, so $m = 0$ and [\(441\)](#) cannot be satisfied for any ρ .
3. *No uniqueness.* The residual cannot distinguish the two roots: both lie within 2.34×10^{-4} of Y_{app} , and the single number -10^{-8} certifies neither. A residual bound is a statement about a root, and at a critical point “a root” is not “the root”.
4. *The true law is a square root.* Since $\varphi(-1) = -e^{-1}$, $\varphi'(-1) = 0$ and $\varphi''(-1) = e^{-1}$, Taylor’s theorem gives $\varphi(Y) + e^{-1} = \frac{1}{2}e^{-1}(Y+1)^2 + \mathcal{O}_{Y \rightarrow -1}((Y+1)^3)$, so the error obeys $|Y+1| \approx \sqrt{2e|R|}$. With $|R| = \eta = 10^{-8}$ this predicts $\sqrt{2e\eta} = 2.3316439816 \times 10^{-4}$, which matches both true errors to four significant figures. A linear residual bound with any finite constant is false here for all small η , since $\sqrt{\eta}/\eta \rightarrow \infty$.

The repair is a change of chart, not a better constant: in the variable $p = \sqrt{2e(z + e^{-1})}$ the two branches are analytic and the transfer machinery applies again. That is the analytic reason the branch point carries a Puiseux, not a polynomial–logarithmic, expansion.

Remark 23.15 (The three ways the transfer fails, in general). [Theorem 23.14](#) exhibits the first and worst of three failure modes.

1. *A critical point of the map.* φ' vanishes between the approximation and the root. The residual then controls only a fractional power of the error, with the exponent $1/k$ at a zero of φ' of order $k - 1$. No choice of constants rescues the linear bound.
2. *A small derivative without a critical point.* φ' is bounded below by $m > 0$, but m is tiny. Then (442) is true and useless: the error bound is inflated by m^{-1} , and the enclosure hypothesis (441) demands a residual smaller than $m\rho$. This is the situation for W_{-1} at moderate arguments, where φ' is small but nonzero.
3. *An unverifiable analytic residual.* (H2) of [Theorem 23.6](#) fails or is not checked: the formal residual has high t -order and nothing is known about the size of $\varphi(g_N(x)) - x$. This mode is silent — no computation goes wrong, no inequality is violated, and the resulting statement is simply unproved. It is the mode that occurs in the sources.

Section summary

A residual is converted into an error bound by the mean value theorem, and that step is elementary. Everything else is not. The derivative lower bound must hold on an interval that contains the unknown root, which is circular unless one instead *encloses* the root ([Theorem 23.2](#), and [Theorem 23.3](#) by Rouché in the complex case). The analytic size of the residual does not follow from its formal t -order; it follows from the formal order *together with* an analytic expansion of the map being reversed ([Theorem 23.5](#)), and with that in hand the formal inverse is the asymptotic expansion of the analytic inverse ([Theorem 23.6](#)). For $\varphi(Y) = Y + \log Y$ all constants are explicit and the enclosure (448) holds at every single $x > 1$, with relative width $1/(x - \log x)$; at $x = 100$, $N = 3$ it brackets the true error 8.153845×10^{-8} between 8.153804×10^{-8} and 8.239278×10^{-8} . Where the derivative vanishes — the Lambert branch point — a residual of 10^{-8} coexists with an error of 2.3×10^{-4} , and no constant repairs the bound.

24 Power–log series near a finite point and the Dulac conventions

Every result proved so far has been stated for $X \rightarrow +\infty$. The literature that motivates the class — local dynamics of planar vector fields, normal forms of parabolic and hyperbolic germs, Dulac’s non-accumulation problem — works instead at a finite point, almost always $x \rightarrow 0^+$, and with generators, signs, orderings and truncation conventions that differ from ours in every one of those four respects. The purpose of this section is to make the passage exact.

[Theorem 13.18](#) already recorded the integer-exponent dictionary $x = X^{-1}$, $\Lambda = \log(1/x) = L$, and [Theorem 13.10](#) recorded the level shift $F \mapsto F \circ \log$. Those two propositions are the input here; the remark closing [Section 13](#) deferred to the present section three things they do not supply:

- (a) a statement of *which* class each of the two natural substitutions produces, and a proof that the two are not interchangeable;
- (b) the Dulac conventions themselves — what a Dulac series is, what a Dulac germ is, how the support condition compares with the one that defines $\mathcal{PL}_{\text{poly}, \mathbb{K}}$, and where the two genuinely differ — together with the reason the first-return map of a hyperbolic polycycle lands in the class;
- (c) the enlargements a finite point forces — real exponents in x , real exponents in the logarithm, and the two distinct mechanisms by which finiteness of the logarithmic degree fails.

The contribution of this section is precision rather than machinery. The sources are thin on proofs here and thick on convention: S2 devotes a chapter to the small-variable dictionary and states its results without proof; S1 and S6 each give half a page. Worse, S1, S2 and S6 do not use the same letter for the same object, and the discrepancy is invisible in any single formula. We therefore state definitions, prove transport theorems, and record as *checkable* propositions the eight ways a transported statement goes wrong.

24.1 Zero-side generators, and the two substitutions

Definition 24.1 (Local coordinate and zero-side generators). Let $a \in \mathbb{R}$. A *local coordinate at a from the right* is $x = z - a$; from the left it is $x = a - z$. In either case the regime is $x \rightarrow 0^+$, and we work on an interval $0 < x < x_0$ with $x_0 < e^{-1}$, so that the two zero-side generators

$$x, \quad \Lambda := \log \frac{1}{x} = -\log x \quad (452)$$

satisfy $0 < x < e^{-1} < 1 < \Lambda$ and $\Lambda \rightarrow +\infty$. We write $\Lambda_2 := \log \Lambda$ for the second zero-side logarithm and

$$\ell := \Lambda^{-1} = -\frac{1}{\log x} \in (0, 1), \quad \ell \rightarrow 0^+, \quad (453)$$

for the *small* logarithmic generator used in the normal-form literature. In the formal algebra x and Λ are independent indeterminates, and the four zero-side classes are

$$\mathbb{K}[\Lambda][[x]], \quad \mathbb{K}[\Lambda]((x)), \quad \mathbb{K}(\Lambda)((x)), \quad \mathbb{K}((\Lambda^{-1}))((x)) = \mathbb{K}((\ell))((x)),$$

carrying the x -adic valuation ν_x , the exclusive truncation $\text{Trunc}_{<N} f = \sum_{n < N} p_n(\Lambda) x^n$, and the derivation determined by

$$\frac{dx}{dx} = 1, \quad \frac{d\Lambda}{dx} = -\frac{1}{x}. \quad (454)$$

For a complex germ one replaces the interval by a sector $\{0 < |x| < x_0, |\arg x| < \theta\}$ and fixes a branch of $\log$; every statement below that mentions positivity or a real branch is then a hypothesis, not a consequence.

The letter ℓ means two reciprocal things

Source S2 writes $\ell = \log(1/x)$; sources S1 and S6, following [23, 24, 27, 26], write $\ell = -1/\log x$. These are reciprocal: $\ell_{S2} = \Lambda \rightarrow +\infty$ while $\ell_{S1,S6} = \Lambda^{-1} \rightarrow 0^+$. A monomial $x^\alpha \ell^m$ therefore means $x^\alpha \Lambda^m$ in one convention and $x^\alpha \Lambda^{-m}$ in the other; the exponent of the logarithm changes sign, and with it the direction in which the logarithmic part of the dominance order runs. Throughout this volume the large zero-side logarithm is Λ and the small one is written Λ^{-1} or, where the comparison with the normal-form literature is the point, ℓ with (453) recalled on the spot.

Definition 24.2 (The reciprocal and the exponential substitution). Let $\mathcal{M}_2(\mathbb{K}, \mathbb{R})$ and $\mathcal{H}_2(\mathbb{K}, \mathbb{R})$ be the depth-two monomial group and Hahn field of Theorem 13.1. Write $\mathcal{Z} := \iota(\mathcal{H}_2(\mathbb{K}, \mathbb{R}))$ for its zero-side copy under the relabelling

$$\iota : \quad L_0 = X \mapsto x^{-1}, \quad L_1 \mapsto \Lambda, \quad L_2 \mapsto \Lambda_2, \quad (455)$$

so that a monomial of $\mathcal{Z}$ is $x^a \Lambda^b \Lambda_2^c$ with $(a, b, c) \in \mathbb{R}^3$, and

$$x^a \Lambda^b \Lambda_2^c \succ x^{a'} \Lambda^{b'} \Lambda_2^{c'} \iff a < a', \text{ or } (a = a', b > b'), \text{ or } (a = a', b = b', c > c'). \quad (456)$$

This is (455) applied to the order of Theorem 13.1, since the exponent of $L_0 = x^{-1}$ is $-a$. The two substitutions are the $\mathbb{K}$ -algebra embeddings of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ into $\mathcal{Z}$ determined on generators by

$$\sigma : \quad t \mapsto x, \quad L \mapsto \Lambda \quad (\text{reciprocal: } X = 1/x), \quad (457)$$

$$\tau : \quad t \mapsto \Lambda^{-1}, \quad L \mapsto \Lambda_2 \quad (\text{exponential: } X = \log(1/x), \text{ i.e. } x = e^{-X}). \quad (458)$$

On germs, σ is $F \mapsto F(1/x)$, which is ι restricted to $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$; and τ is $F \mapsto F(\log(1/x))$, which is $\iota \circ \Sigma$ with $\Sigma = (\cdot) \circ \log$ the level shift of [Theorem 13.10](#). Throughout this section σ and τ denote these two substitutions and nothing else; elsewhere in the volume σ is a logarithmic exponent ([Theorem 13.2](#)) and τ is the Taylor-shift homomorphism used in the proof of [Theorem 16.14](#), whose role here is played by θ .

Theorem 24.3 (What each substitution does to each class). *Let $\mathbb{K}$ have characteristic zero.*

(i) σ is an isomorphism of ordered rings onto its image and restricts to isomorphisms

$$\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} \rightarrow \mathbb{K}[\Lambda][[x]], \quad \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} \rightarrow \mathbb{K}[\Lambda]((x)),$$

$$\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} \rightarrow \mathbb{K}(\Lambda)((x)), \quad \mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}} \rightarrow \mathbb{K}((\Lambda^{-1}))((x)),$$

and, for every $s \in \mathbb{N}_{>0}$ and every ordered subgroup $\Gamma \subseteq \mathbb{R}$, $\mathbb{K}[L]((t^{1/s})) \rightarrow \mathbb{K}[\Lambda]((x^{1/s}))$ and $\mathcal{H}_1(\mathbb{K}, \Gamma) \rightarrow \iota(\mathcal{H}_1(\mathbb{K}, \Gamma))$. It carries ν_t to ν_x , $\text{Trunc}_{<N} \cdot$ to $\text{Trunc}_{<N} \cdot$, $\deg_L$ to $\deg_\Lambda$, and the dominance order of [Theorem 3.12](#) to the order of decreasing eventual size as $x \rightarrow 0^+$. It does not carry D_X to d/dx ; instead

$$\sigma \circ D_X = \left(-x^2 \frac{d}{dx}\right) \circ \sigma. \quad (459)$$

(ii) τ is an isomorphism of ordered rings onto $\mathbb{K}[\Lambda_2]((\Lambda^{-1}))$, carrying ν_t to the Λ^{-1} -adic valuation and dominance to dominance. It intertwines D_X with the derivation D_Λ of the target determined by $D_\Lambda \Lambda = 1$, $D_\Lambda \Lambda_2 = \Lambda^{-1}$; it does not intertwine D_X with d/dx , and $\mathbb{K}[\Lambda_2]((\Lambda^{-1}))$ is not stable under d/dx , because

$$\frac{d}{dx} = -\frac{1}{x} D_\Lambda, \quad \frac{d}{dx} \Lambda^{-1} = \frac{1}{x \Lambda^2}. \quad (460)$$

(iii) Inside $\mathcal{Z}$,

$$\sigma(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}) \cap \tau(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}) = \mathbb{K}[\Lambda]. \quad (461)$$

More precisely, in the exponent coordinates (a, b, c) of (456), $\sigma(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}})$ consists of the elements of $\mathcal{Z}$ whose support lies in $\mathbb{Z} \times \mathbb{N}_0 \times \{0\}$ with bounded-below first coordinate, while $\tau(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}})$ consists of those whose support lies in $\{0\} \times \mathbb{Z} \times \mathbb{N}_0$ with bounded-above second coordinate.

Proof. (i) The map σ is the relabelling $t \mapsto x$, $L \mapsto \Lambda$ of two algebraically independent generators, extended continuously; it is therefore a ring isomorphism onto the correspondingly relabelled ring, and each of the six displayed restrictions is the same relabelling applied to the same support condition. Since σ does not move the exponent pair (n, j) of a monomial, it commutes with the operators $\nu_t \mapsto \nu_x$, $\text{Trunc}_{<N} \cdot$, $\deg_L \mapsto \deg_\Lambda$, which are defined from that pair alone; and it preserves the order because (33) and its zero-side counterpart are governed by the same inequalities on (n, j) . That the zero-side order is the order of decreasing eventual size is [Theorem 3.13](#) composed with $X = 1/x$. Finally (459) is (237), restated as an identity of operators.

(ii) Composing [Theorem 13.10](#) with the relabelling (455) gives at once that τ is an isomorphism of ordered rings of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} = \mathbb{K}[L_1]((L_0^{-1}))$ onto $\mathbb{K}[L_2]((L_1^{-1})) = \mathbb{K}[\Lambda_2]((\Lambda^{-1}))$ with the stated transport of valuation and order, and $\Sigma \circ D_X = D_{L_1} \circ \Sigma$ becomes $\tau \circ D_X = D_\Lambda \circ \tau$. For the negative statement, the chain rule and (454) give $d/dx = (d\Lambda/dx)D_\Lambda = -x^{-1}D_\Lambda$, whence $d\Lambda^{-1}/dx = -x^{-1} \cdot (-\Lambda^{-2}) = (x\Lambda^2)^{-1}$, which involves x and therefore does not lie in $\mathbb{K}[\Lambda_2]((\Lambda^{-1}))$; this is (231) written at zero.

(iii) Both images lie in $\mathcal{Z}$ and Hahn representations are unique, so it suffices to identify supports, which we record in the coordinates (a, b, c) of (456). An element of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is $\sum_{n \geq n_0} p_n(L)t^n$; applying σ gives support $\{(n, j, 0)\}$ with $n \geq n_0$ and $0 \leq j \leq \deg p_n$, while applying τ gives support $\{(0, -n, j)\}$ with the same ranges. An element of the intersection has support in $\{0\} \times \mathbb{Z} \times \{0\}$, i.e. it is a series in Λ alone; lying in $\sigma(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}})$ makes it a *polynomial* in Λ , and every polynomial in Λ does lie in $\tau(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}) = \mathbb{K}[\Lambda_2]((\Lambda^{-1}))$, since that ring admits finitely many negative powers of its outer generator Λ^{-1} . Hence the intersection is exactly $\mathbb{K}[\Lambda]$. $\square$

Remark 24.4 (The transport theorem is deliberately trivial as algebra). Item (i) is a relabelling and nothing more; this is the point. All of the difficulty in passing between infinity and a finite point is concentrated in the two places where the relabelling is *not* available: the derivation, by (459), and the analytic meaning of a monomial, by Theorem 6.43 versus Theorem 24.9 below. A transported statement that mentions neither is automatically correct; a transported statement that mentions either must be re-derived. The eight traps of Theorem 24.14 are all instances of this one sentence.

The two substitutions are not alternatives between which one may choose freely; they produce classes that are, apart from the polynomials in Λ , disjoint. The next proposition is the precise obstruction, and it also supplies the practical criterion.

Proposition 24.5 (The two scales at a finite point are incomparable). *Work on $0 < x < e^{-1}$ with the real branches, so that $\Lambda > 1$ and $\Lambda_2 > 0$.*

(i) *Every nonzero $f \in \tau(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}})$, evaluated at the generators, satisfies*

$$\log |f(x)| = \mathcal{O}_{x \rightarrow 0^+}(\Lambda_2) = o_{x \rightarrow 0^+}(\Lambda);$$

in particular, for every $\varepsilon > 0$, $x^\varepsilon \prec f \prec x^{-\varepsilon}$.

(ii) $\Lambda_2 = \log \log(1/x)$ *is not asymptotically equivalent to $c x^\alpha \Lambda^\beta$ for any $c \neq 0$ and any $\alpha, \beta \in \mathbb{R}$. Consequently Λ_2 admits no asymptotic expansion in the scale $\{x^\alpha \Lambda^\beta : \alpha, \beta \in \mathbb{R}\}$, and in particular lies in none of the four σ -images, nor in $\sigma(\mathcal{H}_1(\mathbb{K}, \mathbb{R}))$ realized on that scale.*

(iii) *Symmetrically, x is not asymptotically equivalent to $c \Lambda^\beta \Lambda_2^\gamma$ for any $c \neq 0$ and any $\beta, \gamma \in \mathbb{R}$; so x has no expansion in the scale produced by τ .*

Proof. (i) Let $c \Lambda^{-n} \Lambda_2^j$, with $c \neq 0$, be the leading term of f . Then $f/(c \Lambda^{-n} \Lambda_2^j) \rightarrow 1$, by Theorem 3.13 transported through τ . Hence

$$\log |f| = -n\Lambda_2 + j \log \Lambda_2 + \mathcal{O}_{x \rightarrow 0^+}(1) = \mathcal{O}_{x \rightarrow 0^+}(\Lambda_2),$$

and $\Lambda_2 = o_{x \rightarrow 0^+}(\Lambda)$ with $\log x = -\Lambda$ gives $\log |f| / \log x \rightarrow 0$, which is the second assertion.

(ii) Suppose $\Lambda_2 \sim c x^\alpha \Lambda^\beta$. Taking logarithms,

$$\log \Lambda_2 = \log |c| - \alpha \Lambda + \beta \Lambda_2 + o_{x \rightarrow 0^+}(1).$$

The left side is $o_{x \rightarrow 0^+}(\Lambda_2)$, hence $o_{x \rightarrow 0^+}(\Lambda)$. If $\alpha \neq 0$ the right side is $\asymp \Lambda$, a contradiction; so $\alpha = 0$. Then the right side is $\beta \Lambda_2 + \mathcal{O}_{x \rightarrow 0^+}(1)$ while the left side is $o_{x \rightarrow 0^+}(\Lambda_2)$, forcing $\beta = 0$; but then the right side is bounded and the left side tends to $+\infty$. For the consequence: if Λ_2 had an expansion in that scale with a nonzero leading term, that term would be such a $c x^\alpha \Lambda^\beta$, and if the expansion were 0 then $\Lambda_2 = \mathcal{O}_{x \rightarrow 0^+}(x^N)$ for some $N > 0$, which is false.

(iii) Suppose $x \sim c \Lambda^\beta \Lambda_2^\gamma$. Taking logarithms, $-\Lambda = \log |c| + \beta \Lambda_2 + \gamma \log \Lambda_2 + o_{x \rightarrow 0^+}(1)$. The right side is $\mathcal{O}_{x \rightarrow 0^+}(\Lambda_2) = o_{x \rightarrow 0^+}(\Lambda)$ while the left side is $-\Lambda$, a contradiction. $\square$

Remark 24.6 (Which substitution to use). Theorem 24.5 turns the choice into a measurement. Given a germ f at 0^+ , form $\log |f(x)| / \log x$.

- If it tends to a finite limit α and the corrections to f/x^α are powers of x times polynomials in Λ , the reciprocal substitution σ is the right one: f is a power–log germ and $f(1/X)$ is a candidate element of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ or one of its enlargements.
- If it tends to 0 while f itself is unbounded or tends to 0, the germ lives on the logarithmic scale and σ is useless: by Theorem 24.5(ii) even the depth-one Hahn class in x and Λ cannot express Λ_2 . Use τ , i.e. take $X = \log(1/x)$ as the large variable.
- If it tends to $\pm\infty$, neither substitution applies: the germ is beyond every logarithmic class, by Theorem 13.11 read through ι .

Theorem 24.7 is the standard germ for which the second case occurs, and for which the first case fails in an instructive way.

Example 24.7 (The lower Lambert branch at a finite point). For $0 < x < e^{-1}$ the branch $W_{-1}(-x)$ is the unique solution $w < -1$ of $we^w = -x$. Writing $w = -U$ with $U > 1$, the equation becomes $Ue^{-U} = x$, i.e.

$$U - \log U = \Lambda. \quad (462)$$

The map $g(u) = u - \log u$ has $g'(u) = 1 - 1/u > 0$ on $(1, +\infty)$, with $g(1) = 1$ and $g(u) \rightarrow +\infty$; it is therefore an increasing bijection $(1, +\infty) \rightarrow (1, +\infty)$, and $U = g^{(-1)^\circ}(\Lambda)$ is well defined for $\Lambda > 1$, i.e. for $0 < x < e^{-1}$.

Now apply τ . In the variables $X = \Lambda$, $L = \Lambda_2$, the equation (462) is exactly the reversion of $X \mapsto X - L$, which is the member $a = -1$ of the family of **Theorem 18.5**. Substituting (329) gives the expansion – with $\approx$ the relation of **Theorem 6.43** read in the large variable $X = \Lambda$, not the zero-side relation $\approx_0$ of **Theorem 24.9**, whose scale is powers of x –

$$W_{-1}(-x) \approx -\Lambda - \Lambda_2 - \sum_{n \geq 1} (-1)^{n+1} \frac{\mathbf{L}_n^{\text{Lam}}(\Lambda_2)}{\Lambda^n} = -\Lambda - \Lambda_2 - \frac{\Lambda_2}{\Lambda} + \frac{\Lambda_2^2/2 - \Lambda_2}{\Lambda^2} - \dots, \quad (463)$$

whose analytic validity is **Theorem 24.8**. Two points deserve emphasis. First, the expansion uses *no* power of x : it lies entirely in $\tau(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}})$, consistently with $\log |W_{-1}(-x)| / \log x \rightarrow 0$. Second, the reciprocal substitution genuinely fails here: by **Theorem 24.5(ii)** the term Λ_2 already lies outside every σ -image, so no amount of enlarging the coefficient field in Λ will produce (463). A second logarithm at infinity would be needed – and that is precisely the level shift that τ performs once and for all.

Proposition 24.8 (Asymptotic validity of the lower-branch expansion). *For $N \in \mathbb{N}_0$ put*

$$V_N(\Lambda) := \Lambda + \Lambda_2 + \sum_{n=1}^N (-1)^{n+1} \frac{\mathbf{L}_n^{\text{Lam}}(\Lambda_2)}{\Lambda^n}.$$

Then there are $K_N \in \mathbb{N}_0$, $C_N > 0$ and $\Lambda_N > 1$ such that, with $U = -W_{-1}(-x)$ and $\Lambda = \log(1/x)$,

$$|U - V_N(\Lambda)| \leq C_N \frac{\Lambda_2^{K_N}}{\Lambda^{N+1}} \quad (\Lambda \geq \Lambda_N). \quad (464)$$

In particular (463) is an asymptotic expansion of $W_{-1}(-x)$ as $x \rightarrow 0^+$ on the scale $\{\Lambda^{-n} \Lambda_2^j\}$, and

$$W_{-1}(-x) = -\Lambda - \Lambda_2 - \frac{\Lambda_2}{\Lambda} + \frac{\Lambda_2^2/2 - \Lambda_2}{\Lambda^2} + \mathcal{O}_{x \rightarrow 0^+} \left(\frac{\Lambda_2^3}{\Lambda^3} \right).$$

Proof. Throughout, $g(u) = u - \log u$ and $U = g^{(-1)^\circ}(\Lambda)$ as in **Theorem 24.7**.

Step 1: an a priori bound. From $U = \Lambda + \log U$ and $U > 1$ we get $U > \Lambda$. For $u \geq 8$ one has $\log u \leq u/2$, hence $g(u) \geq u/2$; so if $\Lambda \geq 4$ and $u > 2\Lambda \geq 8$ then $g(u) > \Lambda$, and monotonicity of g gives $U \leq 2\Lambda$. Therefore $\Lambda < U \leq 2\Lambda$ and $\Lambda_2 < \log U \leq \Lambda_2 + \log 2$ for $\Lambda \geq 4$.

Step 2: the formal identity. Work in $\mathbb{K}[L][[t]]$ and put $\Psi := L + \sum_{n \geq 1} (-1)^{n+1} \mathbf{L}_n^{\text{Lam}}(L) t^n$, so that (329) reads $(X - L)^{(-1)^\circ} = X + \Psi$. Applying $X - L$ to $X + \Psi$ and using generator substitution (**Theorem 10.22**), $L \circ (X + \Psi) = L + \log(1 + t\Psi)$, the defining relation becomes

$$\Psi = L + \log(1 + t\Psi). \quad (465)$$

Let $\Psi_N := \text{Trunc}_{<N+1} \Psi$. Since $\nu_t(\Psi - \Psi_N) \geq N + 1$ and $\log(1 + t \cdot)$ raises the outer order by at least one, $\nu_t(\log(1 + t\Psi) - \log(1 + t\Psi_N)) \geq N + 2$; subtracting (465) therefore gives

$$\nu_t(\Psi_N - L - \log(1 + t\Psi_N)) \geq N + 1. \quad (466)$$

Step 3: the analytic residual. Put $\rho_N(\Lambda) = \sum_{n=1}^N (-1)^{n+1} \mathbb{L}_n^{\text{Lam}}(\Lambda_2) \Lambda^{-n}$, so $V_N = \Lambda + \Lambda_2 + \rho_N = \Lambda(1 + w)$ with $w = (\Lambda_2 + \rho_N)/\Lambda$. For Λ large, $|w| \leq 2\Lambda_2/\Lambda \leq \frac{1}{2}$, so $V_N > \Lambda/2 > 1$ and

$$R_N := g(V_N) - \Lambda = V_N - \log V_N - \Lambda = \rho_N - \log(1 + w).$$

Expand $\log(1 + w) = \sum_{k=1}^{N+1} \frac{(-1)^{k-1}}{k} w^k + \theta$ with $|\theta| \leq 2|w|^{N+2} \leq 2^{N+3} \Lambda_2^{N+2} \Lambda^{-(N+2)}$, the bound on the tail being $\sum_{k \geq N+2} |w|^k / k \leq |w|^{N+2} / ((N+2)(1 - |w|))$. What remains,

$$P := \rho_N - \sum_{k=1}^{N+1} \frac{(-1)^{k-1}}{k} w^k,$$

is the value at $t = \Lambda^{-1}$, $L = \Lambda_2$ of the polynomial $\mathcal{P} = (\Psi_N - L) - \sum_{k=1}^{N+1} \frac{(-1)^{k-1}}{k} (t\Psi_N)^k \in \mathbb{K}[L][t]$, because both are the same finite algebraic combination of t , L and Ψ_N . Now

$$\mathcal{P} = \underbrace{\left(\Psi_N - L - \log(1 + t\Psi_N) \right)}_{\nu_t \geq N+1 \text{ by (466)}} + \underbrace{\sum_{k \geq N+2} \frac{(-1)^{k-1}}{k} (t\Psi_N)^k}_{\nu_t \geq N+2},$$

so $\nu_t \mathcal{P} \geq N+1$, i.e. $\mathcal{P} = t^{N+1} Q$ with $Q \in \mathbb{K}[L][t]$. Writing $K_N = \deg_L Q$ and evaluating at $t = \Lambda^{-1} \leq 1$, $L = \Lambda_2 \geq 1$ gives $|P| \leq C \Lambda_2^{K_N} \Lambda^{-(N+1)}$; enlarging K_N to absorb θ , $|R_N| \leq C'_N \Lambda_2^{K_N} \Lambda^{-(N+1)}$.

Step 4: inversion. We have $g(U) = \Lambda$ and $g(V_N) = \Lambda + R_N$. Both U and V_N exceed $\Lambda/2 \geq 2$ for Λ large, so on the interval between them $g'(u) = 1 - 1/u \geq 1/2$. The mean value theorem gives $|U - V_N| \leq 2|R_N|$, which is (464).

For the last display, apply (464) with $N = 3$ and add back the $n = 3$ term: $U - V_2 = \mathbb{L}_3^{\text{Lam}}(\Lambda_2) \Lambda^{-3} + \mathcal{O}_{x \rightarrow 0^+}(\Lambda_2^{K_3} \Lambda^{-4}) = \mathcal{O}_{x \rightarrow 0^+}(\Lambda_2^3 \Lambda^{-3})$, since $\deg \mathbb{L}_3^{\text{Lam}} = 3$ and $\Lambda_2^K \Lambda^{-4} = o_{x \rightarrow 0^+}(\Lambda_2^3 \Lambda^{-3})$ for every fixed K . $\square$

24.2 The transported dictionary, and the repeated derivative

The purely algebraic half of the dictionary is σ ; the analytic half needs its own definition, because Theorem 6.43 names the regime $X \rightarrow +\infty$ explicitly.

Definition 24.9 (Zero-side asymptotic expansion). Let $\mathbb{K} \subseteq \mathbb{C}$, let $0 < x_0 < e^{-1}$, and let $f: (0, x_0) \rightarrow \mathbb{C}$. Let $\hat{f}$ be a formal series in the zero-side generators with left-finite x -support (in the sense of Theorem 28.1(b)) and polynomial coefficient blocks. We write $f \approx_0 \hat{f}$ if for every $N \in \mathbb{R}$ there are $M_N \in \mathbb{N}_0$, $C_N > 0$ and $x_N \in (0, x_0]$ with

$$\left| f(x) - (\text{Trunc}_{<N} \hat{f})(x) \right| \leq C_N x^N \Lambda^{M_N} \quad (0 < x < x_N), \quad (467)$$

where $(\text{Trunc}_{<N} \hat{f})(x)$ is the finite sum obtained by substituting the generators.

Proposition 24.10 (The two asymptotic relations correspond). *Let $F \in \mathcal{PL}_{\text{poly}, \mathbb{K}}$ and let f be a function on $(0, x_0)$. Put $\tilde{f}(X) := f(1/X)$ on $(1/x_0, +\infty)$. Then $f \approx_0 \sigma(F)$ if and only if $\tilde{f} \approx F$ in the sense of Theorem 6.43. The same equivalence holds for the Puiseux-log and Hahn enlargements, with N ranging over the relevant exponent group.*

Proof. Substituting $X = 1/x$ turns $(\text{Trunc}_{<N} F)(X)$ into $(\text{Trunc}_{<N} \sigma F)(x)$ monomial by monomial, because σ commutes with $\text{Trunc}_{<N} \cdot$ (Theorem 24.3(i)); it turns $\log X$ into Λ and $X^{-N} (\log X)^M$ into $x^N \Lambda^M$; and it is a bijection between $(1/x_0, +\infty)$ and $(0, x_0)$ matching $X \geq X_N$ with $0 < x \leq 1/X_N$. Thus (92) and (467) are the same inequality read in two coordinates. $\square$

We can now transport a genuinely differential statement, and we do it in both directions, on the theorem whose zero-side form the Dulac literature uses most: the repeated-derivative formula. Recall $\Delta_{n,k} = \prod_{j=0}^{k-1} (\partial_L - n - j)$, with the empty product equal to the identity.

Theorem 24.11 (Repeated derivative at a finite point). *Let $p \in \mathbb{K}[\Lambda]$, $n \in \mathbb{Z}$ and $k \in \mathbb{N}_0$. Then, in $\mathbb{K}[\Lambda]((x))$,*

$$\frac{d^k}{dx^k}(x^n p(\Lambda)) = (-1)^k x^{n-k} \Delta_{n-k+1,k} p(\Lambda) = x^{n-k} \prod_{j=0}^{k-1} (n - j - \partial_\Lambda) p(\Lambda), \quad (468)$$

the factors of the product commuting. In particular $d(x^n p)/dx = x^{n-1}(np - p')$, and $\nu_x(df/dx) \geq \nu_x(f) - 1$ for every $f \in \mathbb{K}[\Lambda]((x))$, with equality unless $\nu_x(f) = 0$ and the block $[x^0]f$ is a nonzero constant.

Proof. By (459), $d/dx = -x^{-2} \sigma D_X \sigma^{-1}$ as operators on $\mathbb{K}[\Lambda]((x))$. Iterating,

$$\frac{d^k}{dx^k} = (-x^{-2} \sigma D_X \sigma^{-1})^k,$$

which is not a power of $\sigma D_X \sigma^{-1}$ because x^{-2} and $\sigma D_X \sigma^{-1}$ do not commute; so we argue directly by induction on k , using the case $k = 1$ supplied by (237), namely $d(x^m q)/dx = x^{m-1}(mq - q')$.

For $k = 0$ both sides of (468) are $x^n p$, the empty product being the identity. Assume (468) for k , in the second form. Applying the case $k = 1$ at outer exponent $n - k$ to the coefficient $q = \prod_{j=0}^{k-1} (n - j - \partial_\Lambda) p$ gives

$$\frac{d^{k+1}}{dx^{k+1}}(x^n p) = x^{n-k-1}((n - k)q - q') = x^{n-(k+1)}(n - k - \partial_\Lambda)q = x^{n-(k+1)} \prod_{j=0}^k (n - j - \partial_\Lambda) p,$$

which is the case $k + 1$; the factors commute because they are polynomials in the single operator ∂_Λ . Finally $\prod_{j=0}^{k-1} (n - j - \partial_\Lambda) = (-1)^k \prod_{j=0}^{k-1} (\partial_\Lambda - n + j)$, and as j runs from 0 to $k - 1$ the shifts $n - j$ run over $n, n - 1, \dots, n - k + 1$, the same set as the shifts $(n - k + 1) + i$ for $0 \leq i \leq k - 1$; hence the product is $(-1)^k \Delta_{n-k+1,k}$, which is the first form.

For the valuation statement, let $n_0 = \nu_x(f)$ and $p = [x^{n_0}]f$. The coefficient of x^{n_0-1} in df/dx is $n_0 p - p'$. If $n_0 \neq 0$ this has the same Λ -degree as p and leading coefficient n_0 times that of p , hence is nonzero; if $n_0 = 0$ it is $-p'$, nonzero exactly when $\deg_\Lambda p > 0$. $\square$

The shift index moves, and only by k

The infinity-side formula (148) reads $D_X^k(t^n p) = t^{n+k} \Delta_{n,k} p$: the outer exponent moves *up* by k and the shift index stays at n . The finite-point formula (468) reads $d^k(x^n p)/dx^k = (-1)^k x^{n-k} \Delta_{n-k+1,k} p$: the outer exponent moves *down* by k , a sign $(-1)^k$ appears, and the shift index moves to $n - k + 1$. Writing $\Delta_{n,k}$ on the zero side is correct for $k \leq 1$ and wrong for every $k \geq 2$.

Transporting in the other direction is not a formality either, because the statement the sources make at a finite point is stated for a *real* exponent, as Dulac exponents require, whereas Theorem 9.7 is stated over $\mathbb{Z}$. Carrying it back therefore yields a strengthening.

Proposition 24.12 (The return trip, and a real-exponent extension). *Let $\Gamma \subseteq \mathbb{R}$ be an ordered subgroup and let $\alpha \in \Gamma$, $p \in \mathbb{K}[L]$, $k \in \mathbb{N}_0$. In the Hahn power–log ring $\mathbb{K}[L][[t^\Gamma]]$, with the derivation determined by $D_X t^\alpha = -\alpha t^{\alpha+1}$ and $D_X L = t$,*

$$D_X^k(t^\alpha p(L)) = t^{\alpha+k} \Delta_{\alpha,k} p(L), \quad \Delta_{\alpha,k} = \prod_{j=0}^{k-1} (\partial_L - \alpha - j), \quad (469)$$

and applying σ returns (468) with n replaced by α . In particular the two transports are mutually inverse: starting from (148), carrying it to the finite point by [Theorem 24.11](#) and carrying the result back by (459) reproduces (148) verbatim, with the index substitutions $n \mapsto n$ and $n \mapsto n - k + 1$ composing to the identity on the pair (outer exponent, shift index).

Proof. The block rule $D_X(t^\alpha p) = t^{\alpha+1}(p' - \alpha p)$ holds for real α by the Leibniz rule applied to $D_X t^\alpha = -\alpha t^{\alpha+1}$ and $D_X L = t$; the induction proving [Theorem 9.7](#) uses nothing about n beyond this rule and the commutation of the factors, so it goes through verbatim with n replaced by α . This is (469).

Applying σ to (469) and using (459) k times gives $(-x^2 d/dx)^k(x^\alpha p(\Lambda)) = x^{\alpha+k} \Delta_{\alpha,k} p(\Lambda)$; and the induction of [Theorem 24.11](#), which again uses only the case $k = 1$, gives $d^k(x^\alpha p)/dx^k = (-1)^k x^{\alpha-k} \Delta_{\alpha-k+1,k} p$. These two are consistent. Indeed, one application of $-x^2 d/dx$ sends $x^m q$ to

$$-x^2 \cdot x^{m-1}(mq - q') = x^{m+1}(\partial_\Lambda - m)q,$$

so starting from $x^\alpha p$ the $(j+1)$ -st application acts at outer exponent $\alpha + j$ and contributes the factor $\partial_\Lambda - \alpha - j$; after k steps the result is $x^{\alpha+k} \prod_{j=0}^{k-1} (\partial_\Lambda - \alpha - j)p = x^{\alpha+k} \Delta_{\alpha,k} p$, which is exactly the σ -image of (469). Reading the same computation backwards recovers (148), the shift index of the composite being α again. $\square$

Remark 24.13 (What the sources say). S2 displays (468) in the second, product form, for real λ and with no proof and no empty-product convention. The formula is correct; what is missing is the identification of the operator, which is where the $k \geq 2$ index shift of [Section 24.2](#) becomes visible, and the observation that the real-exponent case is a strengthening of [Theorem 9.7](#) rather than an instance of it. S6 gives only the case $k = 1$.

24.3 Eight traps, each as a checkable statement

Proposition 24.14 (Transport traps). *Let $F = \sum_{n \geq n_0} p_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ and $f = \sigma F$. Each of the following is an identity or an equivalence, and each is the correct form of a step that is routinely taken in the reverse or the wrong direction.*

- (T1) Index sign. Writing F in powers of the variable, $F = \sum_{m \leq -n_0} q_m(L)X^m$ with $q_{-n} = p_n$, one has $f = \sum_{m \leq -n_0} q_m(\Lambda)x^{-m}$. Thus $\deg_X F = -\nu_t F = -\nu_x f$: the outer index used at infinity and the one used at zero are negatives of each other, while ν_t and ν_x agree.
- (T2) Dominance against the variable. $X^a \succ X^b$ if and only if $a > b$, whereas $x^a \succ x^b$ if and only if $a < b$. The dominance order read on the exponent of the variable therefore reverses; read on the exponent of t , respectively x , it is preserved.
- (T3) Leading term. The leading monomial of F is the one of largest X -exponent and that of f the one of smallest x -exponent, and σ matches them. Consequently a series written in increasing powers of X has its leading term last, while a series written in increasing powers of x has it first.
- (T4) Truncation bound. $\text{Trunc}_{<N} F$ retains exactly the monomials with X -exponent $> -N$ — a lower bound — and $\text{Trunc}_{<N} f$ retains exactly those with x -exponent $< N$ — an upper bound. The remainder $F - \text{Trunc}_{<N} F$ consists of the monomials of small X -exponent and of large x -exponent.
- (T5) The derivation is not transported. $\sigma D_X \sigma^{-1} = -x^2 d/dx$, not d/dx . On blocks, $\partial_L - n$ becomes $n - \partial_\Lambda$, and for k derivatives the shift index moves as in [Section 24.2](#).
- (T6) Differentiation changes the remainder in opposite directions. $\nu_t(D_X F) \geq \nu_t F + 1$ ([Theorem 9.10](#)), whereas $\nu_x(df/dx) \geq \nu_x(f) - 1$ ([Theorem 24.11](#)). Hence $\mathcal{O}_t^{\text{form}}(t^N)$ differentiates to $\mathcal{O}_t^{\text{form}}(t^{N+1})$ at infinity and $\mathcal{O}_x^{\text{form}}(x^N)$ differentiates to $\mathcal{O}_x^{\text{form}}(x^{N-1})$ at a finite point: at infinity a derivative improves a formal tail, at a finite point it costs one order.

- (T7) Sign of the logarithm. $\Lambda = -\log x$. A block written as a polynomial P in $\log x$ equals $p(\Lambda)$ with $p(u) = P(-u)$; degrees agree, and every coefficient of odd degree changes sign. Writing $\log x$ for Λ therefore leaves every formula formally consistent while reversing the sign of half the coefficients.
- (T8) Reciprocal logarithmic generator. With $\ell = \Lambda^{-1}$, a monomial $x^\alpha \ell^\beta$ is $x^\alpha \Lambda^{-\beta}$, i.e. $X^{-\alpha} L^{-\beta}$. The exponent of the logarithm changes sign; a family listed by increasing powers of ℓ is listed by decreasing size in the logarithmic direction, the opposite of a family listed by increasing powers of Λ .

Proof. (T1) $X^m = t^{-m}$ and $\sigma(t^{-m}) = x^{-m}$; the rest is the definition of ν_t and ν_x .

(T2) For $a > b$, $X^a/X^b = X^{a-b} \rightarrow +\infty$ as $X \rightarrow +\infty$, so $X^a \succ X^b$; and $x^a/x^b = x^{a-b} \rightarrow 0$ as $x \rightarrow 0^+$, so $x^a \prec x^b$, i.e. $x^a \succ x^b$ exactly when $a < b$. On the exponent of t , respectively x , (33) is literally the same inequality on both sides, which is [Theorem 24.3\(i\)](#).

(T3) Immediate from (T1) and (T2).

(T4) By [Theorem 5.1](#), $\text{Trunc}_{<N} F = \sum_{n < N} p_n t^n$; the monomial t^n is X^{-n} , and $n < N$ is $-n > -N$. On the zero side t^n becomes x^n and the condition stays $n < N$. The statement about the remainder is the negation of the two conditions.

(T5) (459) and [Theorem 24.11](#). The two operators are genuinely different: $\sigma D_X \sigma^{-1}$ raises ν_x by one and d/dx lowers it by one.

(T6) The two valuation statements are [Theorem 9.10\(i\)](#) and the last clause of [Theorem 24.11](#). The consequences for formal tails follow because $\mathcal{O}_t^{\text{form}}(t^N)$ means $\nu_t \geq N$ ([Theorem 5.7](#)) and likewise at zero.

(T7) $P(\log x) = P(-\Lambda) = p(\Lambda)$ with $p(u) = P(-u)$; the coefficient of u^j in p is $(-1)^j$ times that of P .

(T8) $\ell^\beta = \Lambda^{-\beta}$ by (453), and $\Lambda = L$, $x = t$ by (457). Since $\ell \rightarrow 0^+$ and $\Lambda \rightarrow +\infty$, $\ell^\beta \prec \ell^{\beta'}$ exactly when $\beta > \beta'$, the reverse of (33) read in Λ . $\square$

Three ways a transported theorem is wrong while looking right

The dangerous traps are (T5)–(T8), because each of them leaves a formula *formally consistent*. A calculation performed at a finite point with $\Delta_{n,k}$ in place of $\Delta_{n-k+1,k}$, or with $\log x$ in place of Λ , or with the reciprocal ℓ in place of Λ , produces a self-consistent array of coefficients that satisfies every internal check and is simply not the expansion of the intended function. The cheapest detection is a two-sided residual: substitute the computed inverse back and verify that the residual has the predicted order *and the predicted sign*, as in [Theorem 14.15](#). The second cheapest is a numerical spot check at one value of x , which $\Lambda \leftrightarrow \ell$ confusion fails immediately.

24.4 A statement that looks transportable and is not: reversion at a finite point

All three of S1, S2 and S6 assert that near-identity reversion and the Lagrange–Bürmann formula “have exactly the same form” at a finite point. The formulas are indeed the same; the reason is not that they are transports of one another, and asserting that they are is a mistake with consequences, because the summability mechanism runs in the opposite direction. The following proposition is the obstruction, stated so that it can be checked.

Proposition 24.15 (The reversion-frame axioms are not σ -invariant). *Let $\mathfrak{B} := \mathbb{K}[\Lambda]((x))$ and $\partial := d/dx$.*

- (i) σ carries the polynomial–logarithmic reversion frame $(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}, \nu_t, D_X, t^{-1})$ of [Theorem 16.3](#) to the frame $(\mathfrak{B}, \nu_x, -x^2\partial, x^{-1})$. Its derivation is $-x^2\partial$, not ∂ , and its distinguished element is x^{-1} , not x .
- (ii) The quadruple $(\mathfrak{B}, \nu_x, \partial, x)$ — the one that governs reversion of germs at 0^+ — satisfies (F1), (F2) and (F4) of [Theorem 16.1](#) with $\Gamma = \mathbb{Z}$, but not (F3): instead

$$\nu_x(\partial f) \geq \nu_x(f) - 1 \quad (f \in \mathfrak{B}), \quad (470)$$

and (F3) fails already at $f = x$, where $\nu_x(\partial x) = 0 < \nu_x(x) + 1 = 2$.

- (iii) No rescaling repairs this: for $c \in \mathbb{R}^\times$ the map $c\nu_x$ satisfies (F3) for ∂ only if $c \leq -1$, and for $c < 0$ it violates (F1), as witnessed by $f = x$, $g = -x + x^5$, for which $c\nu_x(f + g) = 5c < c = \min\{c\nu_x f, c\nu_x g\}$.

Consequently [Theorem 16.14](#) does not apply to $(\mathfrak{B}, \nu_x, \partial, x)$, and the near-zero Lagrange–Bürmann formula is a second theorem, with its own proof.

Proof. (i) σ is a ring isomorphism carrying ν_t to ν_x ([Theorem 24.3\(i\)](#)), D_X to $-x^2\partial$ ((459)) and $X = t^{-1}$ to x^{-1} . Since the frame axioms are formulated in terms of these four data, the image is a frame.

(ii) (F1): $\mathbb{K}[\Lambda]$ is a domain, so ν_x is a valuation with $\nu_x(\lambda) = 0$ for $\lambda \in \mathbb{K}^\times$. (F2): a family with $\{i : \nu_x(f_i) < \gamma\}$ finite for all γ has coefficientwise finite sums at each level and only finitely many levels below any bound, exactly as in the proof of [Theorem 16.3](#). (F4): $\partial x = 1$. Inequality (470) is the last clause of [Theorem 24.11](#), and $\partial x = 1$ gives the stated failure of (F3).

(iii) If $\nu = c\nu_x$ satisfies (F3), then applying it to $f = x^n$ with $\partial^n x^n = n!$ gives $0 = \nu(n!) \geq \nu(x^n) + n = n(c + 1)$ for every $n \geq 1$, so $c \leq -1$; in particular $c < 0$. The displayed witness then contradicts the ultrametric inequality in (F1). $\square$

The correct zero-side objects are these.

Definition 24.16 (The near-identity class at a finite point). Put

$$\mathfrak{m}_0 := x^2\mathbb{K}[\Lambda][[x]] = \{h \in \mathbb{K}[\Lambda]((x)) : \nu_x(h) \geq 2\}, \quad \mathcal{G}_0 := \{x + h : h \in \mathfrak{m}_0\}.$$

For $B \in \mathfrak{m}_0$ define the *shift operator*

$$T_B(f) := \sum_{k \geq 0} \frac{B^k}{k!} \frac{d^k f}{dx^k}, \quad f \in \mathbb{K}[\Lambda]((x)), \quad (471)$$

and write $f \circ (x + B) := T_B(f)$.

Lemma 24.17 (The shift operator is substitution of $x + B$). Let $B \in \mathfrak{m}_0$. Then:

- (i) the family in (471) is summable, its k -th member having $\nu_x \geq \nu_x(f) + k$; hence $\nu_x(T_B f) \geq \nu_x(f)$;
- (ii) T_B is a $\mathbb{K}$ -algebra endomorphism of $\mathbb{K}[\Lambda]((x))$, continuous for ν_x , with $T_0 = \text{id}$;
- (iii) on generators,

$$T_B(x) = x + B, \quad T_B(\Lambda) = \Lambda - \log\left(1 + \frac{B}{x}\right) = \log \frac{1}{x + B}; \quad (472)$$

- (iv) $T_{B'} \circ T_B = T_{B''}$ with $B'' = B' + T_{B'}(B) \in \mathfrak{m}_0$; consequently $\mathcal{G}_0$ is a monoid under $(x + B) \circ (x + B') := T_{B'}(x + B) = x + B''$, with neutral element x , and T is an anti-homomorphism from it into the monoid of endomorphisms of $\mathbb{K}[\Lambda]((x))$.

Proof. (i) $\nu_x(B^k) \geq 2k$ and, by (470), $\nu_x(d^k f / dx^k) \geq \nu_x(f) - k$; the product has $\nu_x \geq \nu_x(f) + k \rightarrow +\infty$, so (F2) for $(\mathfrak{B}, \nu_x)$ – established in [Theorem 24.15\(ii\)](#) – gives the sum, and the bound is the minimum at $k = 0$.

(ii) Linearity is clear. For multiplicativity, expand $d^k(fg)/dx^k$ by Leibniz and regroup; the double family $(B^i \partial^i f / i!) (B^j \partial^j g / j!)$ has $\nu_x \geq \nu_x(f) + \nu_x(g) + i + j$, so it is summable and the regrouping clause of (F2) applies:

$$T_B(fg) = \sum_k \frac{B^k}{k!} \sum_{i+j=k} \binom{k}{i} \partial^i f \partial^j g = \sum_{i,j} \frac{B^i \partial^i f}{i!} \cdot \frac{B^j \partial^j g}{j!} = T_B(f) T_B(g).$$

Continuity follows from (i).

(iii) $\partial x = 1$ and $\partial^k x = 0$ for $k \geq 2$, giving $T_B(x) = x + B$. For Λ : by (454), $\partial \Lambda = -x^{-1}$ and hence $\partial^k \Lambda = (-1)^k (k-1)! x^{-k}$ for $k \geq 1$, so

$$T_B(\Lambda) = \Lambda + \sum_{k \geq 1} \frac{B^k}{k!} (-1)^k (k-1)! x^{-k} = \Lambda + \sum_{k \geq 1} \frac{(-1)^k}{k} \left(\frac{B}{x}\right)^k = \Lambda - \log\left(1 + \frac{B}{x}\right),$$

the formal logarithm being defined because $\nu_x(B/x) \geq 1$. Since $\Lambda = \log(1/x)$ and $\log(1 + B/x) = \log((x+B)/x)$, this is $\log(1/(x+B))$.

(iv) Both $T_{B'} \circ T_B$ and $T_{B''}$ are continuous $\mathbb{K}$ -algebra endomorphisms, and $\mathbb{K}[\Lambda]((x))$ is topologically generated by $x^{\pm 1}$ and Λ ; so it suffices to compare them there. On x : $T_{B'}(T_B x) = T_{B'}(x+B) = x+B'+T_{B'}(B) = x+B''$, and $T_{B''}(x) = x+B''$. On x^{-1} : a ring homomorphism sends the inverse of a unit to the inverse of its image, and $x+B'$ is a unit of $\mathbb{K}[\Lambda]((x))$, its leading block being the scalar 1 (Theorem 7.3, transported by Theorem 24.3(i)); so the values on x^{-1} are determined by those on x . On Λ : $T_{B'}$ is a continuous ring homomorphism, hence $T_{B'}(\log(1+w)) = \log(1+T_{B'}(w))$ whenever $\nu_x(w) > 0$, and $T_{B'}(B/x) = T_{B'}(B)/(x+B')$. Therefore, by (iii),

$$\begin{aligned} T_{B'}(T_B \Lambda) &= \Lambda - \log\left(1 + \frac{B'}{x}\right) - \log\left(1 + \frac{T_{B'}(B)}{x+B'}\right) \\ &= \Lambda - \log\left(\frac{x+B'}{x} \cdot \frac{x+B'+T_{B'}(B)}{x+B'}\right) = \Lambda - \log\left(1 + \frac{B''}{x}\right), \end{aligned}$$

which is $T_{B''}(\Lambda)$. Finally $\nu_x(B'') \geq \min\{\nu_x(B'), \nu_x(B)\} \geq 2$ by (i), so $B'' \in \mathfrak{m}_0$ and $\mathcal{G}_0$ is closed; neutrality of x is $T_0 = \text{id}$, and associativity is (iv) itself. $\square$

Theorem 24.18 (Lagrange–Bürmann reversion at a finite point). *Let $h \in \mathfrak{m}_0$, i.e. $h \in \mathbb{K}[\Lambda][[x]]$ with $\nu_x(h) \geq 2$, and put $\delta := \nu_x(h) - 1 \geq 1$. Then the family*

$$\left(B_r\right)_{r \geq 1}, \quad B_r := \frac{(-1)^r}{r!} \frac{d^{r-1}}{dx^{r-1}}(h^r),$$

is summable, with $\nu_x(B_r) \geq r\delta + 1$; its sum B lies in $\mathfrak{m}_0$; and

$$(x+h)^{(-1)^\circ} = x + \sum_{r \geq 1} \frac{(-1)^r}{r!} \frac{d^{r-1}}{dx^{r-1}}(h^r) \quad (473)$$

is the two-sided compositional inverse of $x+h$ in $\mathcal{G}_0$. In particular $\mathcal{G}_0$ is a group. The same statement holds in $\mathbb{K}[\Lambda]((x^{1/s}))$ and in the Hahn enlargements, with $\mathfrak{m}_0$ replaced by $\{h : \nu_x(h) > 1\}$; the threshold $\nu_x(h) > 1$ is exactly the mirror of $\nu_t(A) > -1$ of Theorem 16.1, and $1 = \nu_x(x)$ is the mirror of $-1 = \nu_t(X)$.

Proof. Write $\partial = d/dx$ and $\mathfrak{B} = \mathbb{K}[\Lambda]((x))$.

Step 1: summability. $\nu_x(h^r) \geq r\nu_x(h)$ and ∂^{r-1} costs at most $r-1$ by (470), so $\nu_x(B_r) \geq r\nu_x(h) - (r-1) = r\delta + 1$, which tends to $+\infty$ since $\delta > 0$. Hence (B_r) is summable and $\nu_x(B) \geq \delta + 1 = \nu_x(h) \geq 2$; also $B \in \mathbb{K}[\Lambda][[x]]$, so $B \in \mathfrak{m}_0$ and $x+B \in \mathcal{G}_0$.

Step 2: what has to be proved. By Theorem 24.17(iv), $(x+h) \circ (x+B) = T_B(x+h) = x+B+T_B(h)$. So $x+B$ is a right inverse of $x+h$ in $\mathcal{G}_0$ if and only if

$$B = -T_B(h). \quad (474)$$

Granting (474), every element of the monoid $\mathcal{G}_0$ has a right inverse; and in a monoid in which every element has a right inverse, every element is invertible: if $ab = x$ and $bc = x$ then $a = a \circ (b \circ c) = (a \circ b) \circ c = c$, whence $ba = bc = x$. So $\mathcal{G}_0$ is a group and $x+B = (x+h)^{(-1)^\circ}$ is the two-sided inverse. It remains to prove (474).

Step 3: the weighted deformation algebra. Define $\mathfrak{B}\langle\varepsilon\rangle_a$ and $\mathfrak{B}\langle\varepsilon\rangle$ exactly as in [Theorem 16.13](#), with the present ν_x , the present ∂ and the weight δ . The proof of [Theorem 16.13](#) uses (F3) in one place only, to see that ∂ maps $\mathfrak{B}\langle\varepsilon\rangle_a$ into some $\mathfrak{B}\langle\varepsilon\rangle_{a'}$; by (470) that holds here with $a' = a + 1$ in place of $a - 1$. Everything else in that proof uses only (F1) and (F2), which hold by [Theorem 24.15\(ii\)](#). Thus all three conclusions of [Theorem 16.13](#) are available: $\mathfrak{B}\langle\varepsilon\rangle$ is a ∂ -stable subring of $\mathfrak{B}[[\varepsilon]]$; evaluation $\text{ev}(u) = \sum_m u_m$ is a $\mathbb{K}$ -algebra homomorphism with $\text{ev}(\varepsilon) = 1$; and ev commutes with sums of families $(f^{(k)})$ lying in a common $\mathfrak{B}\langle\varepsilon\rangle_a$ with $f^{(k)} \in \varepsilon^k \mathfrak{B}[[\varepsilon]]$.

Step 4: the parameter identity. [Theorem 16.11](#) is stated for an arbitrary commutative $\mathbb{Q}$ -algebra and uses no frame; take $C = \mathfrak{B}$. Let

$$\theta: \mathfrak{B} \rightarrow \mathfrak{B}[[z]], \quad \theta(f) = \sum_{k \geq 0} (\partial^k f) \frac{z^k}{k!}.$$

This is a $\mathbb{K}$ -algebra homomorphism (Leibniz, as in [Theorem 24.17\(ii\)](#), with no summability needed because $\mathfrak{B}[[z]]$ is a formal power-series ring), it satisfies $\partial_z \theta = \theta \partial$ by a term shift, and $\text{ev}_{z=0} \theta = \text{id}$. Put $\psi = \theta(-h)$ and $\gamma = \theta(x) = x + z$, so $\partial_z \gamma = 1$. Let $u \in \varepsilon \mathfrak{B}[[z]][[\varepsilon]]$ be the unique solution of $u = \varepsilon \psi_{[u]}$. Since $\gamma_{[u]} = x + z + u$, formula (276) gives, for $m \geq 1$,

$$[\varepsilon^m]u = \frac{1}{m!} \partial_z^{m-1} (\psi^m) = \frac{(-1)^m}{m!} \partial_z^{m-1} \theta(h^m) = \theta(B_m). \quad (475)$$

Step 5: evaluation at $z = 0$. The map $\text{ev}_{z=0}$, applied coefficientwise in ε , is a $\mathbb{K}$ -algebra homomorphism $\mathfrak{B}[[z]][[\varepsilon]] \rightarrow \mathfrak{B}[[\varepsilon]]$; each ε -coefficient of $\psi_{[u]} = \sum_k (\partial_z^k \psi) u^k / k!$ is a finite sum because $u \in \varepsilon(\cdot)$, so applying it to $u = \varepsilon \psi_{[u]}$ is legitimate. Write $\tilde{u} := \text{ev}_{z=0}(u) = \sum_{m \geq 1} B_m \varepsilon^m$ by (475), and note $\text{ev}_{z=0}(\partial_z^k \psi) = \text{ev}_{z=0} \theta(\partial^k(-h)) = -\partial^k h$. Hence

$$\tilde{u} = \varepsilon \sum_{k \geq 0} \frac{-\partial^k h}{k!} \tilde{u}^k \quad \text{in } \mathfrak{B}[[\varepsilon]]. \quad (476)$$

By Step 1, $\nu_x(B_m) \geq m\delta + 1$, so $\tilde{u} \in \mathfrak{B}\langle\varepsilon\rangle_0$ and $\text{ev}(\tilde{u}) = \sum_m B_m = B$.

Step 6: evaluation at $\varepsilon = 1$. Put $f^{(k)} := \varepsilon \frac{-\partial^k h}{k!} \tilde{u}^k \in \varepsilon^{k+1} \mathfrak{B}[[\varepsilon]]$. Its ε^m coefficient is a finite sum of terms $\frac{-\partial^k h}{k!} B_{m_1} \cdots B_{m_k}$ with $m_1 + \cdots + m_k = m - 1$ and every $m_i \geq 1$, so

$$\nu_x([\varepsilon^m]f^{(k)}) \geq (\nu_x(h) - k) + \sum_{i=1}^k (m_i \delta + 1) = \nu_x(h) + \delta(m - 1) = \delta m + 1 \geq \delta m,$$

the bound being uniform in k . Thus every $f^{(k)}$ lies in the same $\mathfrak{B}\langle\varepsilon\rangle_0$, and [Theorem 16.13\(3\)](#) applies to (476):

$$B = \text{ev}(\tilde{u}) = \sum_{k \geq 0} \text{ev}(f^{(k)}) = \sum_{k \geq 0} \frac{-\partial^k h}{k!} \text{ev}(\tilde{u})^k = - \sum_{k \geq 0} \frac{B^k}{k!} \partial^k h = -T_B(h),$$

using $\text{ev}(\varepsilon) = 1$ and multiplicativity. This is (474).

For the Puiseux-log and Hahn versions, every step used only $\delta = \nu_x(h) - 1 > 0$ and the properties (F1), (F2) of ν_x , which hold verbatim over the enlarged exponent group. $\square$

Proposition 24.19 (Logarithmic degree of a finite-point reversion). *Let $h = \sum_{n \geq 2} c_n(\Lambda) x^n \in \mathfrak{m}_0$ with $\deg_\Lambda c_n \leq d$ for all n . Then, for every $n \geq 2$,*

$$\deg_\Lambda [x^n](x + h)^{(-1)^\circ} \leq (n - 1) d, \quad (477)$$

and the bound is attained for $h = ax^2\Lambda$ with $a \neq 0$, $d = 1$.

Proof. Differentiation does not raise the logarithmic degree: by [Theorem 24.11](#) one has

$$\frac{d}{dx}(x^m p) = x^{m-1}(mp - p'), \quad \deg_\Lambda(mp - p') \leq \deg_\Lambda p.$$

Multiplication adds degrees, so every block of h^r has $\deg_\Lambda \leq rd$, and hence so does every block of $\partial^{r-1}(h^r)$. By Step 1 of [Theorem 24.18](#) with $\delta \geq 1$, $\nu_x(B_r) \geq r + 1$, so only the terms with $r \leq n - 1$ contribute to $[x^n](x + h)^{(-1)\circ}$; the largest degree among them is at most $(n - 1)d$. Attainment is [Theorem 24.20](#) below, where the x^n block has degree exactly $n - 1$ for $n = 2, 3, 4$; in general the $r = n - 1$ term of $h = ax^2\Lambda$ contributes $\frac{(-1)^{n-1}}{(n-1)!} \partial^{n-2}(a^{n-1}x^{2n-2}\Lambda^{n-1})$, whose Λ^{n-1} coefficient is $\frac{(-1)^{n-1}a^{n-1}}{(n-1)!} \prod_{j=0}^{n-3} (2n - 2 - j) \neq 0$ by (468) – the pure top-degree part of $\prod_j (m - j - \partial_\Lambda) \Lambda^{n-1}$ being $\prod_j (m - j) \Lambda^{n-1}$ with $m = 2n - 2$ – and no other r contributes to Λ^{n-1} at level x^n . $\square$

Example 24.20 (A Dulac reversion computed at zero and checked at infinity). Let $a \in \mathbb{K}^\times$ and $f(x) = x + ax^2\Lambda$, a tangent-to-identity Dulac germ. Here $h = ax^2\Lambda$, $\nu_x(h) = 2$, $\delta = 1$. Using (468) at each step:

$$\begin{aligned} B_1 &= -h = -ax^2\Lambda, \\ B_2 &= \frac{1}{2}\partial(a^2x^4\Lambda^2) = \frac{a^2}{2}x^3(4\Lambda^2 - 2\Lambda) = a^2x^3(2\Lambda^2 - \Lambda), \\ B_3 &= -\frac{1}{6}\partial^2(a^3x^6\Lambda^3) = -\frac{a^3}{6}x^4(\partial_\Lambda - 5)(\partial_\Lambda - 6)\Lambda^3 = -\frac{a^3}{6}x^4(30\Lambda^3 - 33\Lambda^2 + 6\Lambda), \end{aligned}$$

and $\nu_x(B_4) \geq 5$. Hence

$$f^{(-1)\circ}(x) = x - ax^2\Lambda + a^2x^3(2\Lambda^2 - \Lambda) + a^3x^4\left(-5\Lambda^3 + \frac{11}{2}\Lambda^2 - \Lambda\right) + \mathcal{O}_x^{\text{form}}(x^5). \quad (478)$$

Direct verification. Write $g = x + B$ with $B = -a\Lambda x^2 + a^2(2\Lambda^2 - \Lambda)x^3 + \beta_4x^4$, $w = B/x$, and use $\Lambda \circ g = \Lambda - \log(1 + w)$ from (472). Since g^2 has $\nu_x = 2$, only two terms of $\log(1 + w)$ are needed:

$$\begin{aligned} \log(1 + w) &= -a\Lambda x + a^2\left(\frac{3}{2}\Lambda^2 - \Lambda\right)x^2 + \mathcal{O}_x^{\text{form}}(x^3), \\ g^2 &= x^2 - 2a\Lambda x^3 + a^2(5\Lambda^2 - 2\Lambda)x^4 + \mathcal{O}_x^{\text{form}}(x^5). \end{aligned}$$

Multiplying,

$$ag^2(\Lambda \circ g) = a\Lambda x^2 + a^2(\Lambda - 2\Lambda^2)x^3 + a^3(5\Lambda^3 - \frac{11}{2}\Lambda^2 + \Lambda)x^4 + \mathcal{O}_x^{\text{form}}(x^5),$$

so that $f \circ g = g + ag^2(\Lambda \circ g) = x$ through order x^4 exactly when $\beta_4 = a^3(-5\Lambda^3 + \frac{11}{2}\Lambda^2 - \Lambda)$, which is (478).

The same germ at infinity. Conjugating by $j(x) = 1/x$ as in [Theorem 13.18](#) and writing $L = \Lambda$, the germ $F := j \circ f \circ j$ is $F(X) = 1/f(1/X)$:

$$F(X) = \frac{1}{X^{-1} + aX^{-2}L} = \frac{X}{1 + aL} = X - aL + a^2L^2t - a^3L^3t^2 + \cdots \in \mathcal{G}_{\text{poly}, \mathbb{K}}.$$

Thus the single zero-side block $ax^2\Lambda$ becomes an *infinite* perturbation $A = -aL + a^2L^2t - \cdots \in \mathcal{PL}_{\text{pow}, \mathbb{K}}$ at infinity: the map conjugation $f \mapsto j \circ f \circ j$ is not the coefficientwise dictionary σ , even though σ is a term-by-term relabelling on series. Reversion at infinity now proceeds by (281), with D_X and the shift operators $\Delta_{n,k}$ – a different computation with a different summability count, whose answer is $j \circ f^{(-1)\circ} \circ j$. This is the concrete form of [Theorem 24.15](#).

Remark 24.21 (The source formula for tangent-to-identity Dulac germs). S1 computes, for $f(x) = x + ax^\alpha \ell^m$ with $\ell = -1/\log x = \Lambda^{-1}$ and $\alpha > 1$,

$$f^{(-1)\circ}(x) = x - ax^\alpha \ell^m + a^2x^{2\alpha-1}(\alpha \ell^{2m} + m \ell^{2m+1}) + \cdots.$$

Here $h = ax^\alpha \ell^m$ has $\nu_x(h) = \alpha > 1$ but is not in $\mathbb{K}[\Lambda][[x]]$ when $m > 0$; the applicable form of [Theorem 24.18](#) is therefore its last clause, in the Hahn enlargement with real x -exponents and Laurent Λ -coefficients. Granting that, the formula agrees with (473) at $r \leq 2$: from $d\ell/dx = \ell^2/x$ one gets $h' = ax^{\alpha-1}(\alpha\ell^m + m\ell^{m+1})$ and $\frac{1}{2}\partial(h^2) = hh'$. In our generator the same statement reads $f^{(-1)\circ} = x - ax^\alpha \Lambda^{-m} + a^2 x^{2\alpha-1}(\alpha\Lambda^{-2m} + m\Lambda^{-2m-1}) + \dots$: the exponents of the logarithm are the *negatives* of S1's. Note also that the dominant logarithmic term of the $x^{2\alpha-1}$ block is $\alpha\ell^{2m}$ when $m > 0$, since $\ell \rightarrow 0$; in Λ it is $\alpha\Lambda^{-2m}$, again the dominant one, so the ordering statement survives the translation even though every exponent changes sign. This is [Theorem 24.14](#) (T8) in action.

24.5 Dulac series, Dulac germs, and how the support condition differs

Two distinct conventions travel under the name “power–log series near zero”. The older one, of Dulac and of the finiteness literature, keeps *polynomial* coefficients in the large logarithm and enlarges the exponents of x to a set of reals increasing to $+\infty$. The newer one, of the normal-form literature [23, 24, 27, 26], keeps x -exponents real and takes real exponents of the *small* logarithm $\ell = \Lambda^{-1}$. They are not the same class, they are not nested, and their intersection is small. We define both and prove the comparison.

Definition 24.22 (Dulac series). A set $S \subseteq \mathbb{R}$ is *left-finite* if $S \cap (-\infty, M)$ is finite for every $M \in \mathbb{R}$; equivalently ([Theorem 28.6\(i\)](#)), S is finite or can be enumerated as a strictly increasing sequence $\alpha_0 < \alpha_1 < \dots \rightarrow +\infty$. A *Dulac series* over $\mathbb{K}$ is a formal expression

$$\hat{f} = \sum_{\alpha \in S} x^\alpha p_\alpha(\Lambda), \quad S \subseteq \mathbb{R} \text{ left-finite}, \quad p_\alpha \in \mathbb{K}[\Lambda] \setminus \{0\}, \quad (479)$$

and $\mathfrak{D}(\mathbb{K})$ denotes the set of these. For $\hat{f} \neq 0$ and $N \in \mathbb{R}$ we write

$$\nu_x(\hat{f}) = \min S, \quad \text{Trunc}_{<N} \hat{f} = \sum_{\alpha < N} x^\alpha p_\alpha(\Lambda);$$

both are meaningful because S is left-finite. For a sub-semigroup $\Gamma \subseteq (\mathbb{R}, +)$ and $\alpha_0 \in \mathbb{R}$ we say that $\hat{f}$ is *carried by the grid* $\alpha_0 + \Gamma$ if $S \subseteq \alpha_0 + \Gamma$; this is [Theorem 28.1\(c\)](#). Finally $\mathfrak{D}_{>0}(\mathbb{K}) := \{\hat{f} \in \mathfrak{D}(\mathbb{K}) : \nu_x(\hat{f}) > 0\}$.

Lemma 24.23 (Finitely generated positive semigroups are left-finite). *Let $g_1, \dots, g_k \in \mathbb{R}_{>0}$ and let $\Gamma = \{n_1 g_1 + \dots + n_k g_k : n_i \in \mathbb{N}_0\}$. Then Γ is left-finite, and every $\gamma \in \Gamma$ has only finitely many representations. Consequently every grid $\alpha_0 + \Gamma$ is left-finite, and a sum of two such grids is again one.*

Proof. Put $g_{\min} = \min_i g_i > 0$. If $\sum n_i g_i \leq M$ then $g_{\min} \sum n_i \leq M$, so $\sum n_i \leq M/g_{\min}$; there are only finitely many tuples $(n_1, \dots, n_k) \in \mathbb{N}_0^k$ with bounded sum, which proves both assertions at once. Translation preserves left-finiteness, and $(\alpha_0 + \Gamma) + (\alpha'_0 + \Gamma') = (\alpha_0 + \alpha'_0) + \Gamma''$ with Γ'' generated by the union of the two generating sets. $\square$

Proposition 24.24 ($\mathfrak{D}(\mathbb{K})$ is a complete valued domain). *$\mathfrak{D}(\mathbb{K})$ is a commutative $\mathbb{K}$ -algebra under the operations*

$$\hat{f} + \hat{g} = \sum_{\alpha} x^\alpha (p_\alpha + q_\alpha), \quad \hat{f}\hat{g} = \sum_{\gamma} x^\gamma \sum_{\alpha+\beta=\gamma} p_\alpha q_\beta,$$

both well defined; it is an integral domain; ν_x is a valuation on it with value group $\mathbb{R}$; it is complete and separated for the filtration by $\{\nu_x \geq N\}$; and

$$\hat{f} \in \mathfrak{D}(\mathbb{K})^\times \iff p_{\nu_x(\hat{f})} \in \mathbb{K}^\times, \quad (480)$$

i.e. exactly when the leading coefficient block is a nonzero scalar — the verbatim mirror of [Theorem 7.3](#).

Proof. Well-definedness of the sum is clear (a union of two left-finite sets is left-finite). For the product, [Theorem 28.6\(iii\)](#) gives that $S + T$ is left-finite and that every fiber $\{(\alpha, \beta) \in S \times T : \alpha + \beta = \gamma\}$ is finite, so each inner sum is a finite sum in $\mathbb{K}[\Lambda]$. Associativity, distributivity and the $\mathbb{K}$ -algebra axioms are then verified fiberwise.

Leading blocks multiply: $[x^{\nu_x \hat{f} + \nu_x \hat{g}}](\hat{f}\hat{g}) = p_{\nu_x \hat{f}} q_{\nu_x \hat{g}} \neq 0$ because $\mathbb{K}[\Lambda]$ is a domain. This gives both integrality and $\nu_x(\hat{f}\hat{g}) = \nu_x \hat{f} + \nu_x \hat{g}$; the ultrametric inequality is immediate.

Separatedness is clear. For completeness, let $(\hat{f}_k)$ be Cauchy: for every M there is k_M with $\nu_x(\hat{f}_k - \hat{f}_l) \geq M$ for $k, l \geq k_M$. Then for each M the truncations $\text{Trunc}_{<M} \hat{f}_k$ are eventually constant; define $\hat{f}$ by these stable truncations. Its support below any M equals the support of $\text{Trunc}_{<M} \hat{f}_{k_M}$, which is finite, so the support of $\hat{f}$ is left-finite and $\hat{f} \in \mathfrak{D}(\mathbb{K})$, with $\hat{f}_k \rightarrow \hat{f}$.

For (480): if $\hat{f}\hat{g} = 1$ then the leading blocks multiply to 1 in $\mathbb{K}[\Lambda]$, whose units are $\mathbb{K}^\times$. Conversely if $p_{\alpha_0} = c \in \mathbb{K}^\times$ with $\alpha_0 = \nu_x \hat{f}$, write $\hat{f} = cx^{\alpha_0}(1 + u)$ with $\nu_x(u) > 0$; the support of u is left-finite and contained in $\mathbb{R}_{>0}$, so by [Theorem 28.6\(iv\)](#) the semigroup it generates is left-finite, the family $(-u)^k$ is summable, and $\hat{f}^{-1} = c^{-1}x^{-\alpha_0} \sum_{k \geq 0} (-u)^k \in \mathfrak{D}(\mathbb{K})$. $\square$

Theorem 24.25 (Dulac series against the classes of this volume). *Inside $\mathfrak{D}(\mathbb{K})$, with σ as in (457):*

- (i) $\sigma(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}) = \{\hat{f} \in \mathfrak{D}(\mathbb{K}) : S \subseteq \mathbb{Z}\}$ and $\sigma(\mathbb{K}[L]((t^{1/s}))) = \{\hat{f} : S \subseteq s^{-1}\mathbb{Z}\}$; the inclusions

$$\sigma(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}) \subsetneq \bigcup_{s \geq 1} \sigma(\mathbb{K}[L]((t^{1/s}))) \subsetneq \mathfrak{D}(\mathbb{K})$$

are strict, the second witnessed by $x^{\sqrt{2}}$.

- (ii) $\mathfrak{D}(\mathbb{K})$ is strictly contained in the set of elements of $\sigma(\mathcal{H}_1(\mathbb{K}, \mathbb{R}))$ all of whose coefficient blocks are polynomials in Λ : the Hahn condition on such an element is well-basedness of its x -support, which is strictly weaker than left-finiteness once the exponent group is dense. The series $\sum_{k \geq 1} x^{1-1/k}$ has well-based x -support of order type ω and is not a Dulac series.
- (iii) For $S \subseteq \mathbb{Z}$ – and more generally for $S \subseteq \Gamma$ with $\Gamma \subseteq \mathbb{R}$ a discrete subgroup – the three conditions “bounded below”, “left-finite” and “well-based” coincide, which is [Theorem 3.15](#). This is the exact sense in which the support condition defining $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and the one defining $\mathfrak{D}(\mathbb{K})$ agree, and (ii) is the exact sense in which they differ.

Proof. (i) By [Theorem 24.3\(i\)](#), σ sends $\sum_{n \geq n_0} p_n(L)t^n$ to $\sum_{n \geq n_0} p_n(\Lambda)x^n$, whose support is a subset of $\mathbb{Z}$ bounded below, hence left-finite; conversely a left-finite $S \subseteq \mathbb{Z}$ is bounded below and $\sum_{\alpha \in S} x^\alpha p_\alpha$ is the σ -image of the corresponding element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. The same argument with $s^{-1}\mathbb{Z}$ gives the Puiseux case. Strictness of the first inclusion: $x^{1/2}$. Of the second: $x^{\sqrt{2}}$ lies in $\mathfrak{D}(\mathbb{K})$ but its exponent lies in no $s^{-1}\mathbb{Z}$.

(ii) The displayed series has support $\{1 - 1/k : k \geq 1\}$, which is well ordered of order type ω and therefore admissible in a Hahn ring, but its intersection with $(-\infty, 1)$ is infinite, so it is not left-finite; this is [Theorem 28.6\(ii\)](#). Every left-finite set is well based, since a strictly decreasing sequence would have infinitely many terms below its first, so the inclusion holds and is strict.

(iii) A subset of a discrete subgroup $\Gamma \subseteq \mathbb{R}$ is bounded below if and only if it is left-finite, because Γ itself is left-finite above any bound; and bounded below is equivalent to well based in a discrete group for the same reason. $\square$

“Left-finite” is the Dulac condition, “well-based” is not

The condition $\alpha_j \rightarrow +\infty$ that every source imposes on a Dulac series is left-finiteness, not well-basedness. The distinction is invisible in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, where the two coincide by [Theorem 24.25\(iii\)](#), and is essential the moment real exponents enter: a Hahn series such as $\sum_k x^{1-1/k}$ has a perfectly legitimate support, a leading term, and a well-defined product, but has no finite truncation $\text{Trunc}_{<1}$ other than 0 – so every algorithm of [Section 25](#), every finite-determination theorem, and the very

notion of “computing the expansion to order N ” fails for it. Dulac theory needs left-finiteness precisely because it needs those.

Definition 24.26 (Dulac germ). Let $\mathbb{K} = \mathbb{R}$. A germ $f: (0, x_0) \rightarrow \mathbb{R}$ is a *Dulac germ* if there is $\hat{f} \in \mathfrak{D}_{>0}(\mathbb{R})$ with $f \approx_0 \hat{f}$ in the sense of [Theorem 24.9](#). We call $\hat{f}$ the *Dulac series* of f ; it is unique when it exists, and we write $\nu_x(f) := \nu_x(\hat{f}) > 0$ for its leading exponent.

Proposition 24.27 (Uniqueness, and the flat kernel). *Let f, g be germs at 0^+ and $\hat{f}, \hat{g} \in \mathfrak{D}(\mathbb{R})$.*

- (i) *If $f \approx_0 \hat{f}$ and $f \approx_0 \hat{g}$ then $\hat{f} = \hat{g}$.*
- (ii) *$f \approx_0 0$ if and only if $f(x) = \mathcal{O}_{x \rightarrow 0^+}(x^N)$ for every N . Such germs exist and are nonzero: $e^{-1/x} \approx_0 0$.*
- (iii) *If $f \approx_0 \hat{f}$ and $g \approx_0 \hat{g}$ then $f + g \approx_0 \hat{f} + \hat{g}$ and $fg \approx_0 \hat{f}\hat{g}$. Consequently the germs admitting a Dulac series form a ring, the map $f \mapsto \hat{f}$ is a ring homomorphism onto $\mathfrak{D}(\mathbb{R})$, and by (ii) its kernel — the flat germs — is a nonzero ideal. The map is therefore not injective.*

Proof. (i) Let $\hat{h} = \hat{f} - \hat{g} \neq 0$, with $\alpha_0 = \nu_x(\hat{h})$ and leading block $p \neq 0$. Because the support of $\hat{h}$ is left-finite, there is $\epsilon > 0$ with no support element in $(\alpha_0, \alpha_0 + \epsilon)$; put $N = \alpha_0 + \epsilon$, so that $\text{Trunc}_{<N} \hat{h} = x^{\alpha_0} p(\Lambda)$. Subtracting the two estimates (467) at this N and using linearity of $\text{Trunc}_{<N} \cdot$ gives $|\text{Trunc}_{<N} \hat{h}(x)| \leq Cx^N \Lambda^M$. But $|x^{\alpha_0} p(\Lambda)| \geq cx^{\alpha_0} \Lambda^{\deg p}$ for small x , with $c > 0$ half the absolute value of the leading coefficient of p , and $x^{\alpha_0} \Lambda^{\deg p} / (x^{\alpha_0 + \epsilon} \Lambda^M) = x^{-\epsilon} \Lambda^{\deg p - M} \rightarrow +\infty$. Contradiction.

(ii) If $\hat{f} = 0$ then $\text{Trunc}_{<N} \hat{f} = 0$ and (467) is exactly $|f| \leq C_N x^N \Lambda^{M_N}$, which for every N' is $\mathcal{O}_{x \rightarrow 0^+}(x^{N'})$ upon taking $N = N' + 1$; the converse is immediate. Finally $e^{-1/x} x^{-N} \rightarrow 0$ for every N .

(iii) Additivity is the triangle inequality applied to (467), since $\text{Trunc}_{<N} \cdot$ is linear: take C_N the sum and M_N the maximum of the two data.

For multiplicativity, first note that if $\hat{h} \in \mathfrak{D}(\mathbb{R})$ is nonzero with $b = \nu_x(\hat{h})$ and $h \approx_0 \hat{h}$, then $\text{Trunc}_{<N'} \hat{h}$ is, for each N' , a finite sum $\sum_{b \leq \gamma < N'} x^\gamma q_\gamma(\Lambda)$, whence

$$\left| (\text{Trunc}_{<N'} \hat{h})(x) \right| \leq C x^b \Lambda^M, \quad |h(x)| \leq C' x^b \Lambda^{M'} \quad (481)$$

for small x , the second by adding the remainder at $N' = b + 1$. Now let $a = \nu_x(\hat{f})$, $b = \nu_x(\hat{g})$, both finite, fix $N \in \mathbb{R}$ and put $N_1 = N - b$, $N_2 = N - a$. Then

$$fg - (\text{Trunc}_{<N_1} \hat{f})(\text{Trunc}_{<N_2} \hat{g}) = (f - \text{Trunc}_{<N_1} \hat{f})g + (\text{Trunc}_{<N_1} \hat{f})(g - \text{Trunc}_{<N_2} \hat{g}),$$

and by (467) together with (481) the two summands are $\mathcal{O}_{x \rightarrow 0^+}(x^{N_1+b} \Lambda^*)$ and $\mathcal{O}_{x \rightarrow 0^+}(x^{a+N_2} \Lambda^*)$, both $\mathcal{O}_{x \rightarrow 0^+}(x^N \Lambda^*)$. Finally $(\text{Trunc}_{<N_1} \hat{f})(\text{Trunc}_{<N_2} \hat{g}) - \text{Trunc}_{<N} \hat{f}\hat{g}$ is a finite sum of monomials $x^{\alpha+\beta} \Lambda^*$ with $\alpha+\beta \geq N$: a pair (α, β) with $\alpha+\beta < N$, $\alpha \geq a$, $\beta \geq b$ automatically has $\alpha < N - b = N_1$ and $\beta < N - a = N_2$, so no term of $\text{Trunc}_{<N} \hat{f}\hat{g}$ is missing from the product of truncations. Hence this difference is $\mathcal{O}_{x \rightarrow 0^+}(x^N \Lambda^*)$ as well, and $fg \approx_0 \hat{f}\hat{g}$. If $\hat{f} = 0$, then $f = \mathcal{O}_{x \rightarrow 0^+}(x^N)$ for every N by (ii) and $|g| \leq C' x^b \Lambda^{M'}$ by (481), so $fg \approx_0 0 = \hat{f}\hat{g}$. $\square$

A Dulac series is not a Dulac germ

In $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ the series is the object. At a finite point this is false in both directions. A Dulac series is typically divergent, so it defines no germ; and by [Theorem 24.27\(ii\)](#) a germ is not determined by its Dulac series, since exponentially flat germs are invisible to it. That the germs *arising from analytic vector fields* are nevertheless determined by their Dulac series — quasi-analyticity of the corresponding class — is a deep theorem, equivalent in strength to the non-accumulation of

limit cycles on a polycycle, and is the content of the resolution of Dulac’s problem by Écalle [12] and, independently, by Il’yashenko; see [27, 24] for the modern account and for the associated ε -neighbourhood invariants. No formal statement in this section, and no amount of algebra in $\mathfrak{D}(\mathbb{K})$, implies it.

Theorem 24.28 (Composition of Dulac series). *Let $\mathbb{K} = \mathbb{R}$. Let $\hat{f} = \sum_{\alpha \in S} x^\alpha p_\alpha(\Lambda) \in \mathfrak{D}(\mathbb{K})$ and let*

$$\hat{g} = c x^\beta (1 + u), \quad c > 0, \quad \beta > 0, \quad u \in \mathfrak{D}(\mathbb{K}), \quad \nu_x(u) > 0.$$

Define

$$\hat{f} \circ \hat{g} := \sum_{\alpha \in S} c^\alpha x^{\alpha\beta} (1 + u)^\alpha p_\alpha(\beta\Lambda - \log c - \log(1 + u)), \quad (482)$$

where $(1 + u)^\alpha = \sum_{k \geq 0} \binom{\alpha}{k} u^k$ and $\log(1 + u) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} u^k$. Then:

(i) *the family in (482) is summable and $\hat{f} \circ \hat{g} \in \mathfrak{D}(\mathbb{K})$, the set of x -exponents occurring in it satisfying*

$$\{x\text{-exponents of } \hat{f} \circ \hat{g}\} \subseteq \beta S + \Gamma_u, \quad (483)$$

Γ_u denoting the sub-semigroup of $\mathbb{R}_{\geq 0}$ generated by the x -support of u ;

- (ii) *if $S \subseteq \alpha_0 + \Gamma_1$ and the support of u lies in a finitely generated $\Gamma_2 \subseteq \mathbb{R}_{>0}$, with Γ_1 finitely generated, then $\hat{f} \circ \hat{g}$ is carried by the grid $\alpha_0\beta + \Gamma_3$, where Γ_3 is generated by $\beta \cdot$ (generators of Γ_1) together with the generators of Γ_2 ; in particular Γ_3 is finitely generated;*
- (iii) *$\nu_x(\hat{f} \circ \hat{g}) = \beta \nu_x(\hat{f})$, the leading block is $c^{\alpha_0} p_{\alpha_0}(\beta\Lambda - \log c)$ with $\alpha_0 = \nu_x(\hat{f})$, and $\deg_\Lambda$ of the leading block equals $\deg p_{\alpha_0}$;*
- (iv) *$\hat{f} \mapsto \hat{f} \circ \hat{g}$ is an injective $\mathbb{K}$ -algebra homomorphism of $\mathfrak{D}(\mathbb{K})$ into itself, continuous for the filtration.*

Proof. Write T for the support of u ; it is left-finite and contained in $\mathbb{R}_{>0}$, so by Theorem 28.6(iv) the semigroup $\Gamma_u = T^*$ is left-finite and every element of it has only finitely many representations as a sum of elements of T .

The individual summands. Since $\nu_x(u) > 0$, the families $(\binom{\alpha}{k} u^k)_{k \geq 0}$ and $(\frac{(-1)^{k-1}}{k} u^k)_{k \geq 1}$ are summable in $\mathfrak{D}(\mathbb{K}) - \nu_x(u^k) \geq k\nu_x(u) \rightarrow +\infty$ – so $(1 + u)^\alpha$ and $\log(1 + u)$ are defined, with supports contained in Γ_u . Substituting the element $\beta\Lambda - \log c - \log(1 + u)$ into the polynomial p_α is a finite algebraic operation, so $p_\alpha(\beta\Lambda - \log c - \log(1 + u))$ lies in $\mathfrak{D}(\mathbb{K})$ with support in Γ_u and with Λ -degree $\deg p_\alpha$ at the exponent 0. Hence the α -th summand of (482) has support in $\alpha\beta + \Gamma_u$.

Summability. Fix $M \in \mathbb{R}$. A summand can contribute below M only if $\alpha\beta \leq M$, i.e. $\alpha \leq M/\beta$; since S is left-finite and $\beta > 0$, only finitely many α qualify. Each of those contributes a left-finite set of exponents. Therefore the total support meets $(-\infty, M)$ in a finite set, the sum is defined coefficientwise, and (483) holds; the result lies in $\mathfrak{D}(\mathbb{K})$.

(ii) $\beta S + \Gamma_u \subseteq \beta\alpha_0 + \beta\Gamma_1 + \Gamma_2$, and $\beta\Gamma_1 + \Gamma_2 = \Gamma_3$ is generated by the displayed finite set; Theorem 24.23 makes it left-finite.

(iii) The exponent $\alpha_0\beta$ is attained only by the summand $\alpha = \alpha_0$ and, within it, only by the Γ_u -degree-zero part, which is $c^{\alpha_0} p_{\alpha_0}(\beta\Lambda - \log c)$; this is nonzero of the same Λ -degree as p_{α_0} because $\beta \neq 0$.

(iv) Additivity is clear from (482). For multiplicativity, write $\hat{g}^\alpha := c^\alpha x^{\alpha\beta} (1 + u)^\alpha$ and $\Lambda \circ \hat{g} := \beta\Lambda - \log c - \log(1 + u)$, so that (482) is the $\mathbb{K}$ -linear extension of

$$x^\alpha \Lambda^j \mapsto \hat{g}^\alpha (\Lambda \circ \hat{g})^j$$

over the (summable) monomial decomposition of $\hat{f}$. This assignment is multiplicative on monomials: $c^{\alpha+\alpha'} = c^\alpha c^{\alpha'}$, and $(1 + u)^\alpha (1 + u)^{\alpha'} = (1 + u)^{\alpha+\alpha'}$ because the coefficient of u^k in the product is

$\sum_{i+j=k} \binom{\alpha}{i} \binom{\alpha'}{j} = \binom{\alpha+\alpha'}{k}$ by the Chu–Vandermonde identity, which is a polynomial identity in α, α' . Multiplicativity on all of $\mathfrak{D}(\mathbb{K})$ then follows by expanding both sides as summable double families and regrouping, which the finiteness established above licenses. Continuity and injectivity follow from (iii), which gives $\nu_x(\hat{f} \circ \hat{g}) = \beta \nu_x(\hat{f})$. $\square$

Corollary 24.29 (The class does not depend on the analytic chart). *Let φ be a germ at 0 of a real-analytic diffeomorphism with $\varphi(0) = 0$ and $\varphi'(0) > 0$. Then $\hat{f} \mapsto \hat{f} \circ \hat{\varphi}$ maps $\mathfrak{D}(\mathbb{R})$ into itself, preserves ν_x and the leading Λ -degree, and carries a grid $\alpha_0 + \Gamma$ to the grid $\alpha_0 + \Gamma'$ with Γ' generated by the generators of Γ together with 1. Consequently the notions “Dulac series”, “leading exponent” and “leading logarithmic degree” do not depend on the analytic parametrisation of the transversal used to define them; the corresponding statement for Dulac germs follows from the analytic composition clause quoted in Theorem 24.32.*

Proof. Write $\varphi(x) = x\psi(x)$ with ψ analytic and $c := \psi(0) = \varphi'(0) > 0$. Then $\hat{\varphi} = cx^1(1+u)$ with $u = \psi(x)/c - 1 \in x\mathbb{R}[[x]]$, whose support lies in $\mathbb{N}_{>0}$; so $\beta = 1$ and Γ_2 is the semigroup generated by 1, and Theorem 24.28 applies, giving all the assertions. The last sentence follows because two analytic parametrisations of the same transversal differ by such a φ , possibly after replacing φ by $\varphi^{(-1)^\circ}$, which is again analytic with positive derivative. $\square$

Proposition 24.30 (The two logarithmic conventions). *Recall $\ell = \Lambda^{-1}$. Then, inside the Hahn ambient $\mathcal{Z}$:*

- (i) $\mathbb{K}[\Lambda] \cap \mathbb{K}[[\ell]] = \mathbb{K}$; the Dulac coefficient ring $\mathbb{K}[\Lambda]$ and the normal-form coefficient ring $\mathbb{K}[[\ell]]$ meet only in the constants.
- (ii) The smallest of the four classes of Theorem 24.1 containing both is $\mathbb{K}((\ell))((x)) = \sigma(\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}})$, the transported iterated Laurent-log field: its coefficient ring $\mathbb{K}((\Lambda^{-1}))$ contains $\mathbb{K}[\Lambda]$ and $\mathbb{K}[[\ell]]$, and any ring containing both contains their sums.
- (iii) Admitting real exponents of the logarithm, i.e. monomials $x^\alpha \ell^\gamma = x^\alpha \Lambda^{-\gamma}$ with $\gamma \in \mathbb{R}$ and well-based γ -support, produces the rank-two Hahn ring $\sigma(\mathcal{H}_1(\mathbb{K}, \mathbb{R}))$ refined by real logarithmic exponents; both conventions embed in it.
- (iv) In the dominance order (456), $\ell^\gamma \prec \ell^{\gamma'}$ if and only if $\gamma > \gamma'$. A list of monomials $x^\alpha \ell^m$ written by increasing m is therefore written by decreasing size, while a list of $x^\alpha \Lambda^j$ written by increasing j is written by increasing size.

Proof. (i) In $\mathcal{Z}$ a polynomial in Λ has support in $\{0\} \times \mathbb{N}_0 \times \{0\}$ and an element of $\mathbb{K}[[\ell]]$ has support in $\{0\} \times (-\mathbb{N}_0) \times \{0\}$; the intersection of the supports is $\{(0, 0, 0)\}$, and uniqueness of the Hahn representation gives the claim.

(ii) $\mathbb{K}((\Lambda^{-1}))$ consists of the series $\sum_{j \geq j_0} a_j \Lambda^{-j}$ with $j_0 \in \mathbb{Z}$; taking $j_0 \leq 0$ gives every polynomial in Λ , and taking $j_0 = 0$ gives $\mathbb{K}[[\ell]]$. Among $\mathbb{K}[\Lambda]$, $\mathbb{K}(\Lambda)$ and $\mathbb{K}((\Lambda^{-1}))$, the first does not contain ℓ , and the second does not contain $\sum_{j \geq 0} j! \ell^j$. Indeed, the Laurent expansion at $\Lambda = \infty$ of a rational function P/Q has coefficients satisfying the linear recurrence with constant coefficients determined by Q , whereas $(j!)$ satisfies none: if $\sum_{i=0}^d c_i(j+i)! = 0$ for all large j , division by $j!$ gives $\sum_{i=0}^d c_i(j+1)(j+2) \cdots (j+i) = 0$ for all large j , a polynomial identity in j whose leading coefficient is c_d , so $c_d = 0$ and, inductively, every $c_i = 0$. So $\mathbb{K}((\Lambda^{-1}))$ is the smallest of the three.

(iii) Immediate from (453).

(iv) $\ell^\gamma = \Lambda^{-\gamma}$, and by (456) $\Lambda^b \succ \Lambda^{b'}$ iff $b > b'$; put $b = -\gamma$. $\square$

24.6 The transition map of a hyperbolic sector

The reason the class of Theorem 24.22 is the right one is a theorem about planar vector fields. We state it precisely, we prove the model case, and we mark clearly which part is quoted.

Example 24.31 (The linear saddle: an exact power law, and where the logarithm hides). Consider $\dot{u} = \lambda_u u$, $\dot{v} = -\lambda_s v$ on $\mathbb{R}^2$ with $\lambda_u, \lambda_s > 0$, and put

$$r := \frac{\lambda_s}{\lambda_u} > 0$$

(the *hyperbolicity ratio*). Fix $\varepsilon > 0$ and take the transversals $\Sigma^{\text{in}} = \{v = \varepsilon, 0 < u < \varepsilon\}$ with coordinate $x = u$ and $\Sigma^{\text{out}} = \{u = \varepsilon, 0 < v < \varepsilon\}$ with coordinate $y = v$. The orbit through (x, ε) is $u(\theta) = x e^{\lambda_u \theta}$, $v(\theta) = \varepsilon e^{-\lambda_s \theta}$; it meets Σ^{out} at the transit time

$$T(x) = \frac{1}{\lambda_u} \log \frac{\varepsilon}{x} = \frac{1}{\lambda_u} (\Lambda + \log \varepsilon), \quad (484)$$

and the transition map is

$$D(x) = \varepsilon e^{-\lambda_s T(x)} = \varepsilon^{1-r} x^r, \quad (485)$$

an exact power law with no logarithm. The logarithm is nevertheless present: by (484) the time spent near the saddle is exactly Λ/λ_u up to a constant, and it is the exact exponential cancellation in (485) that removes it. For an analytic saddle that is not linearisable the cancellation is only approximate, and the surviving discrepancy is what produces the polynomials p_j in [Theorem 24.32](#).

Theorem 24.32 (Dulac; quoted, not proved here). *Let $\mathcal{X}$ be a real-analytic vector field on a neighbourhood of $0 \in \mathbb{R}^2$ with a hyperbolic saddle at 0, of eigenvalues $\lambda_u > 0 > -\lambda_s$ and hyperbolicity ratio $r = \lambda_s/\lambda_u > 0$. Let Σ^{in} and Σ^{out} be analytic arcs transverse to the stable and to the unstable separatrix, analytically parametrised by $x \geq 0$ and $y \geq 0$ with the separatrix at the parameter value 0. Then the transition map $D: (0, \varepsilon) \rightarrow (0, \infty)$ obtained by following the flow from Σ^{in} to Σ^{out} is a Dulac germ in the sense of [Theorem 24.26](#): there are $c > 0$, a left-finite set $\{r = \alpha_0 < \alpha_1 < \dots\} \subseteq \mathbb{R}_{>0}$ contained in a finitely generated sub-semigroup of $\mathbb{R}_{>0}$, and polynomials $p_j \in \mathbb{R}[\Lambda]$ with $p_0 = c$ constant, such that*

$$D \approx_0 \hat{D}, \quad \hat{D}(x) = c x^r + \sum_{j \geq 1} x^{\alpha_j} p_j(\Lambda). \quad (486)$$

Moreover the composition of two Dulac germs whose leading exponents are positive is again a Dulac germ, with Dulac series the formal composite (482).

Remark 24.33 (Provenance, and what we have and have not verified). [Theorem 24.32](#) is classical: it originates with Dulac and is proved in the modern finiteness literature; the accounts we have used are [27, 24], with the normal-form machinery that makes the exponent grid explicit in [23, 26]. Three points of hygiene.

- (a) We have *proved* the model case ([Theorem 24.31](#)), the closure of the formal class under the composition (482) ([Theorem 24.28](#)), and the chart-independence ([Theorem 24.29](#)). What is quoted is the existence of the asymptotic expansion for an analytic saddle and the analytic half of the composition statement, which requires uniform remainder control of the kind [Theorem 10.35](#) supplies at infinity and which we have not carried out at a finite point.
- (b) The sources record further structure that we quote but do not use: that the exponents may be taken in the sub-semigroup of $\mathbb{R}_{>0}$ generated by 1 and r , and that in the non-resonant case $r \notin \mathbb{Q}$ every p_j is constant, so that the expansion has no logarithms at all. We do not assert these; only the left-finiteness and the finite generation of the grid are used below, and both are consequences of the displayed statement.
- (c) The polynomials are usually printed as $P_j(\ln x)$ in that literature. The translation to our normalisation is $p_j(u) = P_j(-u)$, [Theorem 24.14](#) (T7): the degrees agree and every odd-degree coefficient changes sign.

Corollary 24.34 (The first-return map of a hyperbolic polycycle). *Let γ be a hyperbolic polycycle of a real-analytic planar vector field: a cyclically ordered finite family of hyperbolic saddles $s_1, \dots, s_k$, with ratios $r_1, \dots, r_k > 0$, joined by regular orbit arcs. Let Σ be an analytic transversal at a regular point of γ , analytically parametrised by $x \geq 0$, and let P be the first-return map on the side from which the orbits approach γ . Then P is a Dulac germ. Its Dulac series $\hat{P}$ is the formal composite*

$$\hat{P} = \hat{R}_k \circ \hat{D}_k \circ \dots \circ \hat{R}_1 \circ \hat{D}_1,$$

carried by a finitely generated grid, with

$$\nu_x(\hat{P}) = r := \prod_{i=1}^k r_i, \quad [x^r]\hat{P} \in \mathbb{R}_{>0} \text{ a constant}, \quad (487)$$

and $\hat{P}$ is independent of the analytic parametrisation of Σ in the sense of [Theorem 24.29](#). In particular, if $r \neq 1$ then $\hat{P}(x) - x$ has leading term $([x^r]\hat{P})x^r$ when $r < 1$ and $-x$ when $r > 1$; if $r = 1$ the leading block of $\hat{P} - x$ is not determined by the ratios alone and may carry a logarithm.

Proof. Each corner map D_i is a Dulac germ with leading exponent $r_i > 0$, by [Theorem 24.32](#); each regular transition R_i is a germ of an analytic diffeomorphism fixing 0 with positive derivative, hence a Dulac germ with $\hat{R}_i \in x\mathbb{R}[[x]]$, leading exponent 1 and support in $\mathbb{N}_{>0}$ (a convergent Taylor series is $\approx_0$ itself). That P is the displayed composite of germs is the definition of the return map. By the last clause of [Theorem 24.32](#) the composite is a Dulac germ with Dulac series the formal composite; by [Theorem 24.28\(i\),\(ii\)](#) the formal composite lies in $\mathfrak{D}(\mathbb{R})$ and is carried by a finitely generated grid, since each factor is; by [Theorem 24.28\(iii\)](#) applied $2k$ times the leading exponent multiplies, giving $\prod_i r_i \cdot 1^k = r$. The leading block is a positive constant by the same induction: at each stage the outer leading block is a positive constant κ and the inner leading coefficient a positive constant c , and [Theorem 24.28\(iii\)](#) gives the composite leading block $c^{\alpha_0}\kappa > 0$. Independence of the chart is [Theorem 24.29](#). The final sentence compares $\nu_x(\hat{P}) = r$ with $\nu_x(x) = 1$ in the dominance order (456). $\square$

Remark 24.35 (Dulac’s problem, and the exact place formal algebra stops). Limit cycles accumulating on γ correspond to zeros of $P(x) - x$ accumulating at 0. If $\hat{P}(x) \neq x$, the formal algebra of [Theorem 24.24](#) shows that $\hat{P}(x) - x$ has a leading block $x^\alpha p(\Lambda)$ with $p \neq 0$, and hence — if P were determined by $\hat{P}$ — that $P(x) - x$ has finitely many zeros near 0. The gap is exactly [Theorem 24.27](#): a germ whose Dulac series is x may still oscillate, and nothing in $\mathfrak{D}(\mathbb{R})$ forbids it. Closing that gap for germs of analytic origin is Dulac’s problem, solved by Écalle [12] and by Il’yashenko. The algebraic content of this section — closure of the class, the grid, the leading data — is what makes the statement of the problem precise; it is not, and cannot be, an ingredient of its solution. Compare the o-minimal route: the unary germs definable in $\mathbb{R}_{\text{an,exp}}$ form a Hardy field whose asymptotic expansions live in the field of logarithmic–exponential series [31, 32], so for a germ known to be definable the quasi-analyticity is automatic; we make no claim about the definability of a general Dulac germ.

24.7 Enlargements forced at a finite point

At infinity the enlargements of [Section 13](#) were forced by composition. At a finite point three further mechanisms operate, and the third is the one the sources treat least carefully.

Proposition 24.36 (Real and ramified exponents). *Let $r > 0$ be the hyperbolicity ratio of [Theorem 24.31](#) and let $D(x) = cx^r$ be the model transition map.*

- (i) *If $r \in \mathbb{N}_{>0}$ then $\hat{D} \in \sigma(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}})$.*
- (ii) *If $r = p/q \in \mathbb{Q}_{>0}$ in lowest terms, then $\hat{D} \in \sigma(\mathbb{K}[L]((t^{1/q})))$ and in no $\sigma(\mathbb{K}[L]((t^{1/s})))$ with $q \nmid s$; the Puiseux-log ramification index is exactly q .*

- (iii) If $r \notin \mathbb{Q}$ then $\hat{D}$ lies in no Puiseux-log ring at all. The grid produced by composing $\hat{D}$ with analytic charts, by [Theorem 24.29](#), is the semigroup generated by r and 1; it is left-finite by [Theorem 24.23](#), but the group $\mathbb{Z} + r\mathbb{Z}$ it generates is dense in $\mathbb{R}$, so no ramification index bounds it.

Proof. (i) and the membership in (ii) are [Theorem 24.25\(i\)](#). For the sharpness in (ii): the exponent p/q lies in $s^{-1}\mathbb{Z}$ if and only if $sp/q \in \mathbb{Z}$, i.e. if and only if $q \mid sp$, i.e. — since $\gcd(p, q) = 1$ — if and only if $q \mid s$. For (iii), an irrational r lies in no $s^{-1}\mathbb{Z}$; and $\mathbb{Z} + r\mathbb{Z}$ is a non-cyclic subgroup of $\mathbb{R}$, hence dense. This is [Theorem 13.2\(2\),\(3\)](#) read through σ . $\square$

Theorem 24.37 (Two ways the logarithmic degree stops being finite). *Work in a Hahn ambient admitting monomials $x^a \Lambda^b$ with $a, b \in \mathbb{R}$ and well-based supports.*

- (i) (Division.) $(x(\Lambda + 1))^{-1} = x^{-1} \sum_{k \geq 0} (-1)^k \Lambda^{-k-1}$. Its coefficient block is an infinite series in Λ^{-1} ; it lies in $\sigma(\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}})$ and in $\sigma(\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}})$, but in no class with polynomial blocks. The block has ceased to be finite while every exponent is still an integer.
- (ii) (Non-integer powers.) Let $\rho \in \mathbb{R} \setminus \mathbb{N}_0$. Then

$$(\Lambda + 1)^\rho = \Lambda^\rho \sum_{k \geq 0} \binom{\rho}{k} \Lambda^{-k}, \quad \binom{\rho}{k} \neq 0 \text{ for every } k \geq 0.$$

Consequently $(x(\Lambda + 1))^\rho = x^\rho \Lambda^\rho \sum_{k \geq 0} \binom{\rho}{k} \Lambda^{-k}$ has infinitely many nonzero logarithmic coefficients and, if $\rho \notin \mathbb{Z}$, a non-integer leading logarithmic exponent; it lies in none of $\sigma(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}})$, $\sigma(\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}})$, $\sigma(\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}})$, nor in any Puiseux-log or Dulac class, all of which have integer logarithmic exponents. The degree has ceased to be a natural number.

- (iii) Neither failure is exotic. Failure (i) is forced by a single division, since $\Lambda + 1$ is not a unit of $\mathbb{K}[\Lambda]$ ([Theorem 7.3](#) transported). Failure (ii) is forced by a single square root: the element $x(\Lambda + 1)$, a member of $\sigma(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}})$, has by (ii) with $\rho = 1/2$ the square root $x^{1/2} \Lambda^{1/2} \sum_{k \geq 0} \binom{1/2}{k} \Lambda^{-k}$, which lies in no Dulac, Puiseux-log or iterated Laurent-log class.
- (iv) Within $\sigma(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}})$ itself, where every block is a polynomial, the degrees need not be bounded: $\sum_{n \geq 1} x^n \Lambda^n$ has $\deg_\Lambda[x^n] = n$. No statement in this volume may assume a uniform logarithmic degree.

Proof. (i) $(\Lambda + 1)^{-1} = \Lambda^{-1}(1 + \Lambda^{-1})^{-1} = \sum_{k \geq 0} (-1)^k \Lambda^{-k-1}$, the geometric series being summable in the Λ^{-1} -adic topology. All coefficients are $\pm 1 \neq 0$, so the support is infinite; membership in $\sigma(\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}})$ is the definition of $\mathbb{K}((\Lambda^{-1}))$, and membership in $\sigma(\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}})$ holds because $(\Lambda + 1)^{-1} \in \mathbb{K}(\Lambda)$.

(ii) Write $\Lambda + 1 = \Lambda(1 + \Lambda^{-1})$; the binomial series converges Λ^{-1} -adically and $\binom{\rho}{k} = \rho(\rho - 1) \cdots (\rho - k + 1)/k!$ vanishes only if $\rho \in \{0, 1, \dots, k - 1\}$, which is excluded by $\rho \notin \mathbb{N}_0$. The leading monomial is $x^\rho \Lambda^\rho$; if $\rho \notin \mathbb{Z}$ its logarithmic exponent is not an integer, whereas every monomial of the listed classes has logarithmic exponent in $\mathbb{Z}$ — for $\sigma(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}})$ and the Dulac and Puiseux-log classes in $\mathbb{N}_0$, for $\sigma(\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}})$ and $\sigma(\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}})$ in $\mathbb{Z}$ — and Hahn representations are unique.

(iii) The reciprocal of $x(\Lambda + 1)$ is not in $\sigma(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}})$ because $\Lambda + 1$ is not a unit of $\mathbb{K}[\Lambda]$; this is [Theorem 7.3](#). The square-root assertion is (ii) with $\rho = 1/2$: the displayed element has leading monomial $x^{1/2} \Lambda^{1/2}$, whose logarithmic exponent $1/2$ is not an integer, and its logarithmic support is infinite because $\binom{1/2}{k} \neq 0$ for all k .

(iv) Immediate; the series lies in $\mathbb{K}[\Lambda][[x]]$ because each block is a single monomial. $\square$

Proposition 24.38 (A logarithm in the inner monomial forces Λ_2). Let $\hat{g} = c x^\beta \Lambda^\gamma (1 + u)$ with $c > 0$, $\beta > 0$, $\gamma \in \mathbb{R}$, $\nu_x(u) > 0$. Then

$$\Lambda \circ \hat{g} = \log \frac{1}{\hat{g}} = \beta \Lambda - \gamma \Lambda_2 - \log c - \log(1 + u), \quad (488)$$

so that for $\hat{f} = \sum_\alpha x^\alpha p_\alpha(\Lambda)$:

- (i) if $\gamma = 0$, [Theorem 24.28](#) applies and $\hat{f} \circ \hat{g}$ stays in $\mathfrak{D}(\mathbb{K})$;
- (ii) if $\gamma \neq 0$, the second logarithm Λ_2 occurs in $\hat{f} \circ \hat{g}$ if and only if $p_\alpha \notin \mathbb{K}$ for some α ; and for every α with $p_\alpha \neq 0$ and $\gamma\alpha \notin \mathbb{N}_0$ a Λ -exponent outside $\mathbb{N}_0$ occurs. In either case $\hat{f} \circ \hat{g}$ lies in no class of depth one at a finite point, that is, in no class whose monomials are $x^\alpha \Lambda^b$.

Proof. Formula (488) is $-\log$ of the product, using $\log x = -\Lambda$ and $\log \Lambda = \Lambda_2$. Item (i) is [Theorem 24.28](#).

For (ii), substitute the generator images into $\hat{f} = \sum_\alpha x^\alpha p_\alpha(\Lambda)$:

$$\hat{f} \circ \hat{g} = \sum_{\alpha \in S} c^\alpha x^{\alpha\beta} \Lambda^{\gamma\alpha} (1+u)^\alpha p_\alpha(\beta\Lambda - \gamma\Lambda_2 - \log c - \log(1+u)),$$

which is summable by the argument of [Theorem 24.28](#) verbatim, since the extra factor $\Lambda^{\gamma\alpha}$ affects no x -exponent. Because $\alpha \mapsto \alpha\beta$ is injective, distinct α contribute at distinct x -exponents and there is no cancellation between them; so it suffices to examine one summand. At the x -exponent $\alpha\beta$ exactly, the factors $(1+u)^\alpha$ and $\log(1+u)$ contribute only their constant terms 1 and 0, so that block is

$$c^\alpha \Lambda^{\gamma\alpha} p_\alpha(\beta\Lambda - \gamma\Lambda_2 - \log c). \quad (489)$$

Writing $d = \deg p_\alpha$ and $a_d \neq 0$ for its leading coefficient, the coefficient of Λ_2^d in (489) is $c^\alpha \Lambda^{\gamma\alpha} a_d (-\gamma)^d$, which is nonzero when $d \geq 1$ because $\gamma \neq 0$; and it is 0 for every α when every p_α is a constant, in which case (489) contains no Λ_2 at all. This proves the “if and only if”. For the second assertion, the Λ -exponents occurring in (489) form the set $\gamma\alpha + \{0, 1, \dots, d\}$, and the exponent $\gamma\alpha$ itself occurs with coefficient $c^\alpha p_\alpha(-\gamma\Lambda_2 - \log c) \neq 0$; so if $\gamma\alpha \notin \mathbb{N}_0$ a Λ -exponent outside $\mathbb{N}_0$ occurs. Finally, depth one at a finite point means support in $\{(a, b, 0)\}$ in the coordinates of (456), which a nonzero Λ_2 -exponent contradicts; and a non-natural Λ -exponent excludes every class with $b \in \mathbb{N}_0$. Read through σ^{-1} , this is [Theorem 13.2\(4\)](#). $\square$

Table 8: The enlargement ladder at a finite point. Every row names the smallest class containing the object produced, and the statement that forces it.

what forces it	smallest zero-side class	reference
nothing (integer exponents)	$\mathbb{K}[\Lambda]((x)) = \sigma(\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}})$	Theorem 24.3
division by a nonconstant block	$\mathbb{K}(\Lambda)((x))$, then $\mathbb{K}((\ell))((x))$	Theorem 24.37(i)
ratio $r = p/q \in \mathbb{Q}$	$\mathbb{K}[\Lambda]((x^{1/q}))$	Theorem 24.36(ii)
ratio $r \notin \mathbb{Q}$	$\mathfrak{D}(\mathbb{K})$, grid $\mathbb{N}_0 + r\mathbb{N}_0$	Theorem 24.36(iii)
non-integer power of a block	real Λ -exponents, rank-two Hahn	Theorem 24.37(ii)
inner monomial with Λ^γ , $\gamma \neq 0$	depth two: Λ_2 appears	Theorem 24.38
a logarithmic leading scale	the τ -image $\mathbb{K}[\Lambda_2]((\ell))$	Theorem 24.3(ii)
an exponentially flat germ	none: the expansion is 0	Theorem 24.27(ii)

Remark 24.39 (What is not claimed). Three questions are left open here, and none of them is answered by anything above.

- (a) Whether the logarithmic degrees $\deg p_j$ of the Dulac series of a hyperbolic-polycycle return map grow at most linearly in j . [Theorem 24.19](#) gives a linear bound for reversion of a perturbation of uniformly bounded logarithmic degree, and [Theorem 16.21](#) gives the corresponding statement at infinity; neither applies to a composite of transition maps, whose logarithmic degrees are not supplied by [Theorem 24.32](#) in any form we have verified.
- (b) Whether the analytic half of [Theorem 24.32](#) — that a composition of Dulac germs is a Dulac germ with the formally composed series — can be proved by transporting [Theorem 10.35](#) through [Theorem 24.10](#). The obstruction is that the latter is stated for integer exponents, and the uniform control needed for the outer expansion at real exponents has not been established here.

- (c) Whether every Dulac germ arising from an analytic saddle is definable in an o-minimal expansion of the real field; see [Theorem 24.35](#).

Section summary

At a finite point there are two substitutions, not one. The reciprocal substitution $\sigma: X = 1/x$ is a relabelling of $\mathcal{PL}_{\text{poly}, \mathbb{K}}$ onto $\mathbb{K}[\Lambda]((x))$ with $\Lambda = \log(1/x)$; it preserves valuation, truncation, leading data and dominance, and carries D_X to $-x^2 d/dx$, not to d/dx . The exponential substitution $\tau: X = \log(1/x)$ lands on the disjoint logarithmic scale $\mathbb{K}[\Lambda_2]((\ell))$ and is the only one that reaches germs such as $W_{-1}(-x)$. Every trap in transporting a theorem is an instance of one sentence: a statement mentioning neither the derivation nor the analytic meaning of a monomial transports automatically, and one mentioning either must be re-derived – as the near-zero Lagrange–Bürmann formula must, since the reversion-frame axiom $\nu(\partial f) \geq \nu(f) + 1$ becomes $\nu_x(\partial f) \geq \nu_x(f) - 1$. A Dulac series is an element of the Hahn ring in x and Λ whose x -support is *left-finite*, strictly more than the well-basedness a Hahn ring requires and exactly what finite truncation needs; over $\mathbb{Z}$ the two conditions coincide, which is why the distinction is invisible in $\mathcal{PL}_{\text{poly}, \mathbb{K}}$. The Dulac class is closed under composition with a positive-leading-exponent argument and under analytic changes of chart, which is what puts the first-return map of a hyperbolic polycycle in it; but the passage from a germ to its Dulac series has a nonzero kernel, and closing that gap is Dulac’s problem, not algebra.

Part VII

Algorithms, certificates, and diagnostics

25 Data structures and finite-order algorithms

Parts I–IV established what the polynomial–logarithmic calculus *is*: which sums, products, quotients, compositions and inverses exist, and which finitely many input blocks determine a given output block. This section turns those finiteness theorems into routines. Nothing here is a new closure result; everything here is a contract. The mathematical content is of three kinds: the exact statement of what a truncated object denotes, the exact statement of how much precision each primitive delivers, and the exact statement of what each primitive costs.

Two conventions are in force throughout, and both are consequences of the normative catalogue rather than choices of this section.

- Truncation is *exclusive*: $\text{Trunc}_{<N} F = \sum_{n < N} p_n(L)t^n$. A source that writes $[F]_{\leq N}$ is using the inclusive marker, and $[F]_{\leq N} = \text{Trunc}_{<N+1} F$. Every index bound below has been converted; the one-block discrepancy this conversion repairs is the subject of [Section 27](#).
- The three O -roles are never conflated. In this section $\mathcal{O}_t^{\text{form}}(t^N)$ is a statement about coefficients, $\mathcal{O}_{X \rightarrow +\infty}(\cdot)$ is a statement about functions, and $\mathcal{O}(\cdot)$ counts machine operations. A complexity bound is not an error bound and an error bound is not a complexity bound.

A common trap

The tail $t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is an additive subgroup of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and a filtration step, *not* an ideal of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$: in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} = \mathbb{K}[L]((t))$ the element t is a unit, so the ideal generated by t^N is the whole ring. Every statement below about “working modulo t^N ” is therefore a statement about cosets of an additive subgroup, and quotient-ring language is illegitimate on the Laurent ring. Reduction modulo (t^N) is a genuine ring quotient only after one restricts to $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.

25.1 What a computation stores

Definition 25.1 (Sparse block representation). Let R be a commutative $\mathbb{K}$ -algebra with a decidable zero test and a distinguished canonical form; the cases of interest are $R = \mathbb{K}[L]$, $R = \mathbb{K}(L)$, and $R = \mathbb{K}((L^{-1}))$ truncated at a declared inner cutoff, together with the tower of [Theorem 7.36](#). A *truncated representative* over R is a pair

$$\mathcal{T} = (v, N), \quad N \in \mathbb{Z} \cup \{+\infty\},$$

in which v is a finitely supported map

$$v: \mathbb{Z} \longrightarrow R, \quad v(n) = 0 \text{ for } n \geq N,$$

stored as the strictly increasing list of the pairs $(n, v(n))$ with $v(n) \neq 0$. The outer index n is the primary key and the internal representation of $v(n)$ – for $R = \mathbb{K}[L]$, a polynomial stored by logarithmic degree – is subordinate to it. The representative additionally carries the name of R and, when the ground field contains chosen values of logarithms or fractional powers, the branch data selecting them.

The two-level key is forced by the dominance order of [Theorem 2.7](#): $t^n L^j \succ t^m L^k$ compares n first and j only within a fixed n . Any storage scheme that flattens (n, j) into a single total degree destroys the hierarchy, because $L^D \succ 1$ while L^D has the larger total degree; truncating by total degree therefore discards *dominant* monomials. Blocks are finite ([Theorem 2.8](#)) but their degrees are unbounded across blocks ([Theorem 6.24](#)), so the second level must be a variable-length object, never a fixed second array dimension.

Definition 25.2 (Denotation). For a truncated representative $\mathcal{T} = (v, N)$ over $R = \mathbb{K}[L]$ put

$$\text{Den}(\mathcal{T}) := \{F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} : \text{Trunc}_{<N} F = v\}.$$

Lemma 25.3 (The denotation is a coset of a filtration step). *Let $\mathcal{T} = (v, N)$ with N finite. Then*

$$\text{Den}(\mathcal{T}) = v + t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}},$$

and every $F \in \text{Den}(\mathcal{T})$ satisfies $\nu_t(F) \geq m(\mathcal{T})$, where

$$m(\mathcal{T}) := \begin{cases} \nu_t(v), & v \neq 0, \\ N, & v = 0. \end{cases}$$

The bound $m(\mathcal{T})$ is attained: it equals $\min \{\nu_t(F) : F \in \text{Den}(\mathcal{T})\}$.

Proof. If $\text{Trunc}_{<N} F = v$ then every block of $F - v$ with index $< N$ vanishes, so $F - v \in t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$; conversely if $F = v + w$ with $w \in t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ then $\text{Trunc}_{<N} F = \text{Trunc}_{<N} v = v$ because v is supported below N . This proves the displayed equality. For the valuation, $\nu_t(F) \geq \min \{\nu_t(v), N\}$, and $\nu_t(v) \leq N - 1 < N$ when $v \neq 0$, which gives $\nu_t(F) \geq \nu_t(v) = m(\mathcal{T})$ with equality for $F = v$. When $v = 0$ the set is $t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, whose minimum valuation is N , attained at t^N . $\square$

Remark 25.4 (The order bound is derived, not stored). **Theorem 25.3** shows that the valuation bound $m(\mathcal{T})$ is determined by (v, N) . An implementation must therefore recompute it from the canonical form and must not carry it as an independent mutable field: a retained zero block at the nominal least exponent makes a stored bound stale, and a stale bound falsifies exactly the precision law of **Theorem 25.10**, where m occurs on the right-hand side.

Definition 25.5 (Relative precision). The *absolute precision* of $\mathcal{T} = (v, N)$ is N ; its *relative precision* is

$$\rho(\mathcal{T}) := N - m(\mathcal{T}) \in \mathbb{N}_0 \cup \{+\infty\}.$$

Thus ρ counts the number of outer blocks known from the leading one onwards, and $\rho(\mathcal{T}) = 0$ exactly when $v = 0$.

Definition 25.6 (Soundness and sharpness). Let $\varphi: \mathcal{D} \rightarrow \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ be a partial operation on k -tuples. A routine R that maps truncated representatives $\mathcal{T}_1, \dots, \mathcal{T}_k$ to a truncated representative $R(\mathcal{T}_1, \dots, \mathcal{T}_k) = (w, N_{\text{out}})$ is *sound* for φ if

$$\varphi(\text{Den}(\mathcal{T}_1) \times \dots \times \text{Den}(\mathcal{T}_k)) \subseteq \text{Den}(R(\mathcal{T}_1, \dots, \mathcal{T}_k))$$

whenever every tuple in the product lies in $\mathcal{D}$. It is *sharp* if in addition no larger value of N_{out} has this property, that is, if there exist two admissible tuples whose φ -images differ in the block of index N_{out} .

Soundness is the invariant that every routine must preserve. It is worth isolating the reason that it is enough to check it one primitive at a time.

Proposition 25.7 (Soundness composes). *Let R_1 be sound for φ_1 and R_2 sound for φ_2 , and suppose the declared output type of R_1 is an admissible input type of R_2 . Then $R_2 \circ R_1$ is sound for $\varphi_2 \circ \varphi_1$ on the locus where the latter is defined.*

Proof. Let $F \in \text{Den}(\mathcal{T})$ with $\varphi_2(\varphi_1(F))$ defined. Soundness of R_1 gives $\varphi_1(F) \in \text{Den}(R_1(\mathcal{T}))$. Applying soundness of R_2 to the representative $R_1(\mathcal{T})$ gives $\varphi_2(\varphi_1(F)) \in \text{Den}(R_2(R_1(\mathcal{T})))$, which is the assertion. The argument uses nothing about R_1 except that its output denotation contains $\varphi_1(F)$; in particular it is indifferent to how many blocks R_1 discarded. $\square$

Corollary 25.8 (Truncating early is free). *If every primitive in a straight-line computation is sound, then truncating each intermediate result to its own declared precision changes no block of the final result that the final declared precision advertises.*

Proof. Immediate from [Theorem 25.7](#): the truncation of an intermediate is the passage from an element of a denotation to the representative of that denotation, and soundness of the next routine is stated for the whole denotation. $\square$

[Theorem 25.8](#) is the theorem behind the universal advice of all six sources to truncate after every operation. The advice is not a heuristic trade of accuracy against speed: at the declared precision the answer is bit-for-bit the same.

Definition 25.9 (The routine invariant). A routine is *conforming* when it guarantees all of:

- (P1) *Soundness* in the sense of [Theorem 25.6](#), with the declared N_{out} .
- (P2) *Canonicity*: the returned map has strictly increasing keys, no key $\geq N_{\text{out}}$, no zero value, and every value in the canonical form of R .
- (P3) *Order honesty*: the order bound is recomputed from the returned canonical form ([Theorem 25.4](#)).
- (P4) *Ring honesty*: if the computation left the declared R – for instance by dividing by a nonunit block, [Theorem 7.3](#) – the enlarged R is part of the returned type and not a silent simplification.

25.2 The precision calculus

The following theorem is the arithmetic that (P1) requires. All six sources state fragments of it; none states it sharply, and the missing sharpness is the source of the guard-block failures catalogued in [Section 27](#).

Theorem 25.10 (Precision calculus, with sharpness). *Let $\mathcal{T}_i = (v_i, N_i)$ be truncated representatives with finite N_i , order bounds $m_i = m(\mathcal{T}_i)$ and relative precisions $\rho_i = N_i - m_i$. Then the following output precisions are sound, and each is sharp in the sense of [Theorem 25.6](#) – for every instance in cases 1–5, and for every instance satisfying the nondegeneracy hypothesis stated there in case 6.*

1. Sum. $N_{\text{out}} = \min \{N_1, N_2\}$.
2. Product. $N_{\text{out}} = \min \{N_1 + m_2, N_2 + m_1\}$, with output order bound $m_1 + m_2$.
3. Reciprocal. *If every $F \in \text{Den}(\mathcal{T}_1)$ is a unit of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, then $N_{\text{out}} = N_1 - 2m_1$, with output order bound $-m_1$.*
4. Quotient. *If every $G \in \text{Den}(\mathcal{T}_2)$ is a unit, then $N_{\text{out}} = \min \{N_1 - m_2, N_2 + m_1 - 2m_2\}$, with output order bound $m_1 - m_2$.*
5. Derivation. $N_{\text{out}} = N_1 + 1$ for D_X , and this is sharp without exception. The output order bound $m_1 + 1$ is always sound, and it is attained if and only if $(\partial_L - m_1)v_1(m_1) \neq 0$, that is (for $v_1 \neq 0$) unless $m_1 = 0$ and $v_1(0) \in \mathbb{K}^\times$; in that exceptional case the true order is strictly larger and must be recomputed from the returned canonical form ([Theorem 25.4](#)).
6. Near-identity composition. *Let $\mathcal{T}_1$ represent the outer series $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and let $\mathcal{T}_2 = (u, N_2)$ represent $H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, so that the inner map is $X + H \in \mathcal{G}_{\text{poly}, \mathbb{K}}$. Then*

$$N_{\text{out}} = \min \{N_1, N_2 + m_1 + 1\},$$

with output order bound m_1 . Sharpness holds for every instance in which $N_1 \leq N_2 + m_1 + 1$, and, when $N_2 + m_1 + 1 < N_1$, for every instance with $\nu_t(D_X v_1) = m_1 + 1$. The excluded case is real: for $v_1 = c \in \mathbb{K}^\times$ every admissible pair gives $F \circ (X + H) = c + a \circ (X + H)$ with $\nu_t(a) \geq N_1$, so the composite is in fact determined to the larger cutoff N_1 , and the declared $N_{\text{out}} = N_2 + 1$ is sound but pessimistic.

Proof. Write $F = v_1 + a$ and $G = v_2 + b$ with $\nu_t(a) \geq N_1$, $\nu_t(b) \geq N_2$; by [Theorem 25.3](#) this parametrises the denotations, and $\nu_t(v_i) \geq m_i$, $N_i \geq m_i$.

1. $(F + G) - (v_1 + v_2) = a + b$ has valuation at least $\min\{N_1, N_2\}$. For sharpness assume $N_1 \leq N_2$ (otherwise exchange the two arguments) and compare the admissible pairs (v_1, v_2) and $(v_1 + t^{N_1}, v_2)$: their sums differ by t^{N_1} , a nonzero block of index exactly $N_1 = N_{\text{out}}$. This witness exists for every instance, not only for $v_1 = v_2 = 0$.

2. $FG - v_1v_2 = v_1b + av_2 + ab$. The three summands have valuations at least $m_1 + N_2$, $N_1 + m_2$ and $N_1 + N_2$; since $N_i \geq m_i$ the third is not smaller than the first two, so $\nu_t(FG - v_1v_2) \geq N_{\text{out}}$. Additivity of ν_t ([Theorem 7.2](#)) gives the order bound. For sharpness one must exhibit, for the given instance, two admissible pairs whose products differ in the block of index *exactly* N_{out} . *Case* $N_1 + m_2 \leq N_2 + m_1$ with $v_2 \neq 0$. The admissible pairs (v_1, v_2) and $(v_1 + t^{N_1}, v_2)$ have products differing by $t^{N_1}v_2$, of valuation exactly $N_1 + \nu_t(v_2) = N_1 + m_2 = N_{\text{out}}$. *Case* $N_2 + m_1 \leq N_1 + m_2$ with $v_1 \neq 0$. Symmetrically the pairs (v_1, v_2) and $(v_1, v_2 + t^{N_2})$ have products differing by $t^{N_2}v_1$, of valuation exactly $N_2 + m_1 = N_{\text{out}}$. *Remaining cases.* If $v_2 = 0$ and $v_1 \neq 0$ then $m_2 = N_2$ while $m_1 \leq N_1 - 1$, so $N_2 + m_1 < N_1 + N_2 = N_1 + m_2$ and the second case applies; symmetrically if $v_1 = 0$ and $v_2 \neq 0$. If $v_1 = v_2 = 0$ then $N_{\text{out}} = N_1 + N_2$ and the pairs $(0, 0)$, (t^{N_1}, t^{N_2}) have products 0 and $t^{N_1+N_2}$. Every instance is covered.

3. Here $v_1 \neq 0$ and $m_1 = \nu_t(v_1) = \nu_t(F)$. From $F^{-1} - v_1^{-1} = (v_1 - F)/(Fv_1) = -a/(Fv_1)$ and additivity of ν_t ,

$$\nu_t(F^{-1} - v_1^{-1}) = \nu_t(a) - 2m_1 \geq N_1 - 2m_1.$$

Equality holds for $a = t^{N_1}$, which proves sharpness.

4. With $A = v_1 + a$ and $B = v_2 + b$,

$$\frac{A}{B} - \frac{v_1}{v_2} = \frac{av_2 - bv_1}{Bv_2},$$

whose valuation is at least $\min\{N_1 + m_2, N_2 + m_1\} - 2m_2$, which is the assertion. For sharpness note that $\nu_t(v_2) = m_2$ and $N_2 > m_2$, so $\nu_t(v_2(v_2 + t^{N_2})) = 2m_2$ exactly. Perturbing the numerator from v_1 to $v_1 + t^{N_1}$ at fixed denominator v_2 changes the quotient by t^{N_1}/v_2 , of valuation exactly $N_1 - m_2$. Perturbing the denominator from v_2 to $v_2 + t^{N_2}$ at a fixed admissible numerator A changes it by $-At^{N_2}/(v_2(v_2 + t^{N_2}))$, of valuation exactly $\nu_t(A) + N_2 - 2m_2$; take $A = v_1$ if $v_1 \neq 0$ and $A = t^{N_1}$ if $v_1 = 0$, so that $\nu_t(A) = m_1$ in both cases and the valuation is $m_1 + N_2 - 2m_2$. Whichever of the two competing bounds is the smaller, the corresponding witness differs in the block of index exactly N_{out} .

5. $D_X(F - v_1) = D_X a$ and $\nu_t(D_X a) \geq \nu_t(a) + 1 \geq N_1 + 1$ ([Theorem 14.4](#)). For sharpness take $a = t^{N_1}L$, which is admissible because $\nu_t(a) = N_1$. Then $D_X a = t^{N_1+1}(\partial_L - N_1)L = t^{N_1+1}(1 - N_1)L \neq 0$ is a nonzero block of index exactly $N_1 + 1 = N_{\text{out}}$, for every $N_1 \in \mathbb{Z}$, including $N_1 = 0$. For the order clause, the block of index $m_1 + 1$ of $D_X v_1$ is $(\partial_L - m_1)v_1(m_1)$ by the block formula; the operator $\partial_L - m_1$ is injective on $\mathbb{K}[L]$ for $m_1 \neq 0$ and has kernel exactly $\mathbb{K}$ for $m_1 = 0$, while $v_1(m_1) \neq 0$ whenever $v_1 \neq 0$. Hence that block vanishes precisely when $m_1 = 0$ and $v_1(0) \in \mathbb{K}^\times$. This – and not any failure of the cutoff law – is the algorithmic shadow of $\ker D_X = \mathbb{K}$ ([Theorem 9.11](#)).

6. Write $G = X + H$. Perturbing the outer argument by Φ with $\nu_t(\Phi) \geq N_1$ changes the value by $\Phi \circ G$, and

$$\nu_t(\Phi \circ G) = \nu_t(\Phi) \quad (\Phi \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}, G \in \mathcal{G}_{\text{poly}, \mathbb{K}}), \quad (490)$$

because the exact Taylor formula [Theorem 12.5](#) gives $\Phi \circ G = \sum_{k \geq 0} H^k D_X^k \Phi / k!$, whose $k = 0$ term is Φ and whose $k \geq 1$ terms have valuation at least $\nu_t(\Phi) + k > \nu_t(\Phi)$; this is [Theorem 14.5](#). Hence the outer argument contributes precision N_1 , sharply, by taking $\Phi = t^{N_1}$. Perturbing the inner correction by η with $\nu_t(\eta) \geq N_2$ changes the value by

$$F \circ (X + H + \eta) - F \circ (X + H) = \sum_{k \geq 1} \frac{(H + \eta)^k - H^k}{k!} D_X^k F.$$

Since $\nu_t(H) \geq 0$ we have $\nu_t((H + \eta)^k - H^k) \geq \nu_t(\eta) \geq N_2$, while $\nu_t(D_X^k F) \geq m_1 + k$; so the k -th summand has valuation at least $N_2 + m_1 + k$, and the minimum over $k \geq 1$ is $N_2 + m_1 + 1$. For sharpness in the regime $N_2 + m_1 + 1 < N_1$, assume $\nu_t(D_X v_1) = m_1 + 1$ and take $F = v_1$, $\eta = t^{N_2}$. The $k = 1$ summand of the displayed difference is exactly $\eta D_X F = t^{N_2} D_X v_1$, of valuation exactly $N_2 + m_1 + 1$; for $k \geq 2$ one has $\nu_t((H + \eta)^k - H^k) \geq N_2$ and $\nu_t(D_X^k F) \geq \nu_t(D_X F) + (k - 1) = m_1 + k \geq m_1 + 2$, so all later summands have valuation at least $N_2 + m_1 + 2$. The admissible inner data H and $H + \eta$ therefore give composites differing in the block of index exactly N_{out} . Finally the output order bound is m_1 by Equation (490). $\square$

Corollary 25.11 (Relative precision is the conserved quantity). *In the notation of Theorem 25.10, and writing $h = m_2 \geq 0$ for the inner order in part 6,*

$$\begin{aligned} \rho(FG) &= \min \{\rho_1, \rho_2\}, & \rho(F^{-1}) &= \rho_1, \\ \rho(A/B) &= \min \{\rho_1, \rho_2\}, & \rho(F \circ (X + H)) &= \min \{\rho_1, \rho_2 + h + 1\}. \end{aligned}$$

For the derivation, $\rho(D_X F) \leq \rho_1$, with equality if and only if $(\partial_L - m_1)v_1(m_1) \neq 0$; in the exceptional case the drop can be total, and is so for $v_1 = c \in \mathbb{K}^\times$. For the sum, no bound in terms of ρ_1, ρ_2 holds in either direction: ρ can drop to 1.

Proof. Each identity is $N_{\text{out}} - m_{\text{out}}$ computed from Theorem 25.10; for the product, the reciprocal, the quotient and the composition respectively,

$$\begin{aligned} \min \{N_1 + m_2, N_2 + m_1\} - (m_1 + m_2) &= \min \{N_1 - m_1, N_2 - m_2\}, \\ (N_1 - 2m_1) - (-m_1) &= N_1 - m_1, \\ \min \{N_1 - m_2, N_2 + m_1 - 2m_2\} - (m_1 - m_2) &= \min \{N_1 - m_1, N_2 - m_2\}, \\ \min \{N_1, N_2 + m_1 + 1\} - m_1 &= \min \{\rho_1, (N_2 - h) + h + 1\}. \end{aligned}$$

For the derivation, the output cutoff is $N_1 + 1$ and the output order is $m_{\text{out}} = \nu_t(D_X v_1) \geq m_1 + 1$, so $\rho(D_X F) = N_1 + 1 - m_{\text{out}} \leq N_1 - m_1 = \rho_1$, with equality exactly when $m_{\text{out}} = m_1 + 1$, which by Theorem 25.105 is the condition $(\partial_L - m_1)v_1(m_1) \neq 0$; for $v_1 = c \in \mathbb{K}^\times$ the returned representative is $(0, N_1 + 1)$, of relative precision 0. For the sum, take

$$F = \sum_{n=-1}^{N-1} t^n, \quad G = - \sum_{n=-1}^{N-2} t^n,$$

both with $m = -1$ and $N_i = N$, hence $\rho_i = N + 1$; their sum is t^{N-1} , with order bound $N - 1$ and absolute precision N , so $\rho = 1$. $\square$

Remark 25.12 (Addition, not multiplication, is the dangerous primitive). Theorem 25.11 inverts the usual folklore. Products, quotients and compositions transport relative precision unchanged or improved; the two primitives that can destroy it are addition, which destroys it exactly when leading blocks cancel – the formal counterpart of catastrophic cancellation – and, in the single exceptional case $(\partial_L - m_1)v_1(m_1) = 0$, the derivation, which destroys it for the same reason one block further in. A routine that reports only absolute cutoffs cannot see this loss; a routine that reports ρ sees it immediately, because ρ drops at the subtraction that caused it.

Example 25.13 (The one-block guard). Let $F = X = t^{-1}$, known exactly, and let $G \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ be known to exclusive precision N with $m_2 = 0$. Then Theorem 25.102 gives

$$N_{\text{out}} = \min \{\infty + 0, N + (-1)\} = N - 1 :$$

the product is determined only one block less far than the factor. The two admissible completions $G_1 = 1$ and $G_2 = 1 + t^N$ satisfy

$$\text{Trunc}_{<N} G_1 = \text{Trunc}_{<N} G_2 = 1, \quad FG_1 = t^{-1}, \quad FG_2 = t^{-1} + t^{N-1},$$

so the two products differ in the block of index $N - 1$. Conversely, to obtain the product through block $N - 1$ inclusively one must request G through block N , that is, one guard block beyond the naive expectation, because the block of index n of FG is the block of index $n + 1$ of G .

Remark 25.14 (Why the sharp law matters). The sound-but-blunt rule “both inputs modulo t^N give the output modulo t^N ” is correct only when both order bounds are ≥ 0 . On the Laurent ring it is false, by [Theorem 25.13](#), and the failure is silent: the routine returns a block it has no right to return. This is failure mode 27.2 of [Section 27](#).

25.3 Cost models

Definition 25.15 (Two cost models). In the *coefficient-operation model* one counts multiplications, additions, and inversions of units in the coefficient ring R , each at unit cost. In the *scalar model* one counts multiplications and additions in $\mathbb{K}$. Complexity statements below use $\mathcal{O}(\cdot)$ in the sense of operation counting only; the notation carries no analytic content.

The two models are not interchangeable, and the discrepancy between them is governed by the log-degree profile d_F of [Theorem 6.24](#), which is data attached to a series and is *not* bounded by any function of the outer cutoff.

Lemma 25.16 (Scalar cost of a block-bilinear routine). *Let a routine over $R = \mathbb{K}[L]$ perform a multiset $\mathcal{M}$ of coefficient multiplications $p \cdot q$, each carried out by schoolbook polynomial multiplication. Then it performs exactly*

$$\sum_{(p,q) \in \mathcal{M}} (\deg p + 1)(\deg q + 1)$$

scalar multiplications in $\mathbb{K}$, with the convention that a zero polynomial contributes the factor 0.

Proof. Schoolbook multiplication of polynomials of degrees d and e forms exactly the $(d + 1)(e + 1)$ products of pairs of coefficients. Summing over $\mathcal{M}$ gives the count. $\square$

Proposition 25.17 (Exact scalar cost of truncated block convolution). *Let $F, G \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $N \geq 1$. The schoolbook algorithm computing $\text{Trunc}_{<N} FG$ as $\sum_{i+j < N} p_i q_j$ performs exactly*

$$\sum_{\substack{i,j \geq 0 \\ i+j < N}} (d_F(i) + 1)(d_G(j) + 1)$$

scalar multiplications, where $d_F(n) + 1$ is read as 0 when the block p_n vanishes. In the coefficient-operation model the count is at most $\binom{N+1}{2}$ multiplications in $\mathbb{K}[L]$.

Proof. Immediate from [Theorem 25.16](#) together with [Theorem 6.5](#): the pairs contributing to blocks below N are exactly those with $i, j \geq 0$ and $i + j < N$, of which there are $\binom{N+1}{2}$. $\square$

Corollary 25.18 (The scalar cost is not $\mathcal{O}(N^2)$). 1. *If $d_F(n), d_G(n) \leq D$ for all $n < N$, the scalar cost is at most $(D + 1)^2 \binom{N+1}{2}$.*

2. *If $d_F(n) = d_G(n) = n$ for all $n < N$, the scalar cost is exactly $\binom{N+3}{4}$.*

3. *No bound of the form $\mathcal{O}(N^c)$, with an absolute implied constant, holds uniformly over $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$: already for $N = 1$ the cost of multiplying $F = G = L^M$ is $(M + 1)^2$, unbounded in M .*

Proof. Part (i) is immediate. For part (ii),

$$\sum_{i+j=r} (i+1)(j+1) = \sum_{k=1}^{r+1} k(r+2-k) = (r+2) \frac{(r+1)(r+2)}{2} - \frac{(r+1)(r+2)(2r+3)}{6} = \binom{r+3}{3},$$

using $\sum_{k=1}^M k = M(M+1)/2$ and $\sum_{k=1}^M k^2 = M(M+1)(2M+1)/6$ with $M = r+1$. Summing over $0 \leq r \leq N-1$ and applying the hockey-stick identity $\sum_{j=3}^{N+2} \binom{j}{3} = \binom{N+3}{4}$ gives the stated closed form. Part (iii) is the displayed example. $\square$

A common trap

The profile of case (ii) is not artificial: up to the single block of index 0 it is the profile of the compositional inverse of $X + L$, by [Theorem 27.4](#) below, for which the exact scalar count is $\binom{N+3}{4} + N^2 + N + 1$. So the natural output of the reversion algorithms already costs $N^4/24 + \mathcal{O}(N^3)$ scalar multiplications per truncated product: the exponent is four, not the naively expected two. Every complexity statement in this section is therefore stated in the coefficient-operation model and must be multiplied by the profile factor before it describes a machine.

25.4 Multiplication

Algorithm 25.19 (Truncated block product). **Input.** Truncated representatives $\mathcal{T}_1 = (v_1, N_1)$, $\mathcal{T}_2 = (v_2, N_2)$ over R , with derived order bounds m_1, m_2 .

Precondition. None beyond [Theorem 25.9](#).

Output. (w, N) with $N = \min\{N_1 + m_2, N_2 + m_1\}$ and $w = \text{Trunc}_{<N} v_1 v_2$.

Postcondition. $FG \in \text{Den}(w, N)$ for every $F \in \text{Den}(\mathcal{T}_1)$, $G \in \text{Den}(\mathcal{T}_2)$, and N is the largest cutoff with this property.

Steps.

1. Compute m_1, m_2 from the canonical forms and set $N \leftarrow \min\{N_1 + m_2, N_2 + m_1\}$. If $N \leq m_1 + m_2$ return the zero representative $(0, N)$.
2. Initialise an empty ordered map w .
3. For each stored pair $(i, v_1(i))$ and each stored pair $(j, v_2(j))$ with $i+j < N$, accumulate $w(i+j) \leftarrow w(i+j) + v_1(i)v_2(j)$.
4. Normalise every touched value in R and delete the zero values.

Cost. At most $\binom{N-m_1-m_2+1}{2}$ multiplications in R , and exactly $s_1 s_2$ minus the number of skipped pairs when the supports have s_1, s_2 stored blocks. In the scalar model the count is that of [Theorem 25.17](#).

Correctness. Soundness and sharpness of the declared N are [Theorem 25.102](#). That the loop computes $\text{Trunc}_{<N} v_1 v_2$ is [Theorem 6.5](#) together with the finiteness of every fiber ([Theorem 6.6](#)); no pair with $i+j \geq N$ can contribute to a retained block, so skipping such pairs is exact rather than approximate. $\square$

```
function multiply(T1, T2):
    m1 := order(T1); m2 := order(T2)
    N := min(T1.cutoff + m2, T2.cutoff + m1)
    w := empty ordered map
    for (i, a) in T1.blocks:
        if i >= N - m2: break          # keys are increasing
        for (j, b) in T2.blocks:
            if i + j >= N: break
            w[i+j] := normalize(w[i+j] + a*b)
    return canonicalize(w, N)
```

The two break statements are legitimate exactly because the keys are stored in increasing order; with a hash map they would be wrong, and the routine would also lose the determinism required in [Section 26.5](#).

25.5 Reciprocal, division, and powers

Algorithm 25.20 (Reciprocal by the linear recurrence). **Input.** $\mathcal{T} = (v, N_1)$ over R , with $m_1 = m(\mathcal{T})$. **Precondition.** $v \neq 0$ and the leading block $v(m_1)$ is a unit of R . For $R = \mathbb{K}[L]$ this means $v(m_1) \in \mathbb{K}^\times$, by [Theorems 7.1](#) and [7.3](#); if the test fails the routine must not proceed but must report the obstruction, per (P4) of [Theorem 25.9](#).

Output. $(w, N_1 - 2m_1)$ with $w = \text{Trunc}_{<N_1-2m_1} v^{-1}$.

Steps. Factor $v = t^{m_1} \tilde{v}$ with $\tilde{v} = b_0 + b_1 t + \dots$, b_0 a unit; set $c_0 \leftarrow b_0^{-1}$ and, for $n = 1, \dots, \rho - 1$ where $\rho = N_1 - m_1$,

$$c_n \leftarrow -b_0^{-1} \sum_{j=1}^n b_j c_{n-j};$$

return $t^{-m_1} \sum_{n < \rho} c_n t^n$.

Cost. Exactly $\binom{\rho}{2}$ products $b_j c_{n-j}$ in R , plus $\rho - 1$ multiplications by the precomputed b_0^{-1} and one inversion. In the scalar model, $\sum_{n=1}^{\rho-1} \sum_{j=1}^n (\deg b_j + 1)(\deg c_{n-j} + 1)$.

Correctness is [Theorem 7.7](#); the precision statement is [Theorem 25.103](#), and the coefficient-model count restates [Theorem 7.32](#) in the present bookkeeping.

Algorithm 25.21 (Reciprocal by Newton doubling). **Input.** As in [Theorem 25.20](#).

Precondition. As in [Theorem 25.20](#).

Output. The same representative as [Theorem 25.20](#).

Steps. With $\tilde{v}$ as above, set $C \leftarrow b_0^{-1}$, $\mu \leftarrow 1$; while $\mu < \rho$, set $\mu' \leftarrow \min \{2\mu, \rho\}$ and

$$C \leftarrow \text{Trunc}_{<\mu'} C (2 - \tilde{v}C), \quad \mu \leftarrow \mu'.$$

Return $t^{-m_1} C$.

Cost. $\lceil \log_2 \rho \rceil$ iterations, the i -th using two truncated products at cutoff 2^i .

Correctness. The update squares the residual: with $E = 1 - \tilde{v}C$,

$$1 - \tilde{v}C(2 - \tilde{v}C) = 1 - 2\tilde{v}C + (\tilde{v}C)^2 = E^2.$$

Hence $\nu_t(E) \geq \mu$ implies $\nu_t(E^2) \geq 2\mu$, and by [Theorem 25.103](#) the reciprocal is then known to precision 2μ . That the iterates are exactly the truncations of the reciprocal, so that no spurious blocks are produced, is [Theorem 7.29](#); the precision schedule and the crossover with [Theorem 25.20](#) are [Theorems 7.27](#), [7.33](#) and [7.34](#). $\square$

A common trap

Newton doubling is stated for the *reciprocal*, an operation on the multiplicative structure. It has nothing to do with the compositional inverse beyond a formal analogy, and the two must not share a symbol: this volume writes F^{-1} and $F^{(-1)\circ}$ for the two operations and never a bare superscript -1 .

Algorithm 25.22 (Truncated division). **Input.** $\mathcal{T}_1 = (v_1, N_1)$ (numerator) and $\mathcal{T}_2 = (v_2, N_2)$ (denominator).

Precondition. The leading block of v_2 is a unit of R .

Output. (w, N) with $N = \min \{N_1 - m_2, N_2 + m_1 - 2m_2\}$, equivalently the representative of relative precision $\min \{\rho_1, \rho_2\}$ and order $m_1 - m_2$.

Steps. Either compose [Theorem 25.20](#) or [Theorem 25.21](#) with [Theorem 25.19](#), or run the direct recurrence of [Theorem 7.8](#) on the normalised operands, which saves one temporary and ρ coefficient products.

Cost. $\binom{\rho}{2} + \mathcal{O}(\rho)$ coefficient multiplications for the direct recurrence, where $\rho = \min \{\rho_1, \rho_2\}$.

Correctness of the truncated recurrence is [Theorem 7.9](#); that the declared relative precision is exactly right, and in particular that the numerator need not be known to a larger absolute cutoff than the denominator, is [Theorem 7.11](#) together with [Theorem 25.11](#). The profile of the answer is [Theorem 7.15](#); it is in general *not* bounded by the profiles of the inputs, which is why the scalar cost of the recurrence cannot be read off from the inputs alone.

For powers, two routines are available and they are not interchangeable.

Proposition 25.23 (Power recurrence). *Let $\mathbb{K}$ have characteristic zero, let $B = \sum_{n \geq 0} b_n t^n \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with b_0 a unit of R , and let $\alpha \in \mathbb{K}$. Define $C = B^\alpha$ by [Theorem 6.37](#), that is, $C = b_0^\alpha (1 + U)^\alpha$ with $U = b_0^{-1} B - 1$ of positive order and a chosen value $b_0^\alpha \in R$. Then $C = \sum_{n \geq 0} c_n t^n$ satisfies $c_0 = b_0^\alpha$ and, for $n \geq 1$,*

$$n b_0 c_n = \sum_{j=1}^n ((\alpha + 1)j - n) b_j c_{n-j}. \quad (491)$$

Proof. Let ∂_t be the formal partial derivative in t with L held fixed; it is an R -linear derivation of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and it is continuous for the t -adic filtration because $\partial_t(t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}) \subseteq t^{N-1} \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Write $C = b_0^\alpha (1 + U)^\alpha = b_0^\alpha \sum_{k \geq 0} \binom{\alpha}{k} U^k$, a summable family since $\nu_t(U^k) \geq k$ ([Theorem 6.34](#)). Applying ∂_t termwise, which is legitimate for a summable family by continuity, and using $k \binom{\alpha}{k} = \alpha \binom{\alpha-1}{k-1}$,

$$\partial_t C = b_0^\alpha \sum_{k \geq 1} k \binom{\alpha}{k} U^{k-1} \partial_t U = \alpha b_0^\alpha (1 + U)^{\alpha-1} \partial_t U.$$

Multiplying by $B = b_0(1 + U)$ and using $\partial_t B = b_0 \partial_t U$ gives the differential relation

$$B \partial_t C = \alpha C \partial_t B. \quad (492)$$

Extracting the coefficient of t^{n-1} from [Equation \(492\)](#) yields

$$\sum_{j=0}^n (n-j) b_j c_{n-j} = \alpha \sum_{j=1}^n j b_j c_{n-j},$$

that is, $n b_0 c_n + \sum_{j=1}^n (n-j) b_j c_{n-j} = \alpha \sum_{j=1}^n j b_j c_{n-j}$, which rearranges to [Equation \(491\)](#). Division by n uses characteristic zero. $\square$

Algorithm 25.24 (Powers). **Input.** $\mathcal{T} = (v, N_1)$, an exponent α .

Precondition. For $\alpha \in \mathbb{N}_0$: none. For general $\alpha \in \mathbb{K}$: $m_1 = 0$, b_0 a unit of R , and a chosen value $b_0^\alpha \in R$ recorded as branch data.

Output. Relative precision ρ_1 , order αm_1 (an integer when $\alpha \in \mathbb{Z}$; otherwise the computation lives in a Puiseux extension and must be typed accordingly).

Steps. For $\alpha \in \mathbb{N}_0$ either binary powering by [Theorem 25.19](#) or [Equation \(491\)](#). For general α , [Equation \(491\)](#) only.

Cost. Binary powering: $\mathcal{O}(\log \alpha)$ truncated products, hence $\mathcal{O}(\rho^2 \log \alpha)$ coefficient multiplications. Recurrence [Equation \(491\)](#): exactly $\binom{\rho}{2}$ products $b_j c_{n-j}$, independent of α .

Remark 25.25. Binary powering is preferable only for very small α ; the recurrence is independent of α altogether and is the *only* option for $\alpha \notin \mathbb{N}_0$, including the formal roots of [Theorem 6.38](#). The profile of the answer is [Theorem 6.31](#).

25.6 Logarithm and exponential of a positive-order argument

Proposition 25.26 (Coefficient recurrences for log and exp). *Let $\mathbb{K}$ have characteristic zero and let $U = \sum_{n \geq 1} u_n t^n \in t \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.*

1. If $\Lambda = \log(1 + U) = \sum_{n \geq 1} \lambda_n t^n$, then for $n \geq 1$

$$\lambda_n = u_n - \frac{1}{n} \sum_{j=1}^{n-1} j \lambda_j u_{n-j}. \quad (493)$$

2. If $E = \exp(U) = \sum_{n \geq 0} e_n t^n$, then $e_0 = 1$ and for $n \geq 1$

$$e_n = \frac{1}{n} \sum_{j=1}^n j u_j e_{n-j}. \quad (494)$$

Both series lie in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, with $\nu_t(\Lambda) = \nu_t(U)$ and $E \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, by [Theorem 10.5](#).

Proof. Both families $\sum_{m \geq 1} (-1)^{m+1} U^m / m$ and $\sum_{k \geq 0} U^k / k!$ are summable, since $\nu_t(U^m) \geq m$ ([Theorem 6.34](#)), so their sums are defined and ∂_t may be applied termwise by the continuity noted in the proof of [Theorem 25.23](#). Termwise differentiation gives

$$\partial_t \Lambda = \left(\sum_{m \geq 1} (-1)^{m+1} U^{m-1} \right) \partial_t U = \frac{\partial_t U}{1 + U}, \quad \partial_t E = E \partial_t U,$$

the first because $\sum_{m \geq 1} (-1)^{m+1} U^{m-1}$ is the reciprocal of $1 + U$ ([Theorem 7.7](#)). Rewrite the first as $(1 + U) \partial_t \Lambda = \partial_t U$ and extract the coefficient of t^{n-1} :

$$n \lambda_n + \sum_{\substack{i \geq 1, j \geq 1 \\ i+j=n}} u_i j \lambda_j = n u_n,$$

which is [Equation \(493\)](#) after writing $i = n - j$. From the second, the coefficient of t^{n-1} is $n e_n = \sum_{j=1}^n j u_j e_{n-j}$, which is [Equation \(494\)](#). Division by n uses characteristic zero. $\square$

Algorithm 25.27 (Logarithm and exponential). **Input.** $\mathcal{T} = (u, N)$ representing U .

Precondition. $m(\mathcal{T}) \geq 1$. The routine must *refuse* an argument of order 0: $\exp$ of a nonconstant block leaves the ring ($\exp(L) = X$), and $\log$ of a unit $b_0(1 + U)$ with $b_0 \notin \mathbb{K}$, or of an element of nonzero order, produces $\log b_0$ or a multiple of L and must be handled by the generator formulas of [Theorem 10.22](#), not by these recurrences.

Output. Precision N in both cases; order ≥ 1 for $\log$, order 0 for $\exp$.

Steps. [Equation \(493\)](#) or [Equation \(494\)](#) for $n = 1, \dots, N - 1$.

Cost. Exactly $\binom{N-1}{2}$ coefficient multiplications $\lambda_j u_{n-j}$ for [Equation \(493\)](#) and exactly $\binom{N}{2}$ coefficient multiplications $u_j e_{n-j}$ for [Equation \(494\)](#), against $\mathcal{O}(N)$ truncated products, hence $\mathcal{O}(N^3)$ coefficient multiplications, for the naive summation of the defining series.

Precision. Perturb U by η with $\nu_t(\eta) \geq N$. Then $\log(1 + U + \eta) - \log(1 + U) = \log(1 + \eta(1 + U)^{-1})$ has order $\nu_t(\eta) \geq N$, and $\exp(U + \eta) - \exp(U) = \exp(U)(\exp(\eta) - 1)$ likewise has order $\nu_t(\eta) \geq N$. Both are sharp, as $\eta = t^N$ shows. $\square$

25.7 The derivation and its iterates

Algorithm 25.28 (The distinguished derivation). **Input.** $\mathcal{T} = (v, N)$ over $R = \mathbb{K}[L]$.

Precondition. R supports ∂_L .

Output. $(w, N + 1)$ with $w(n + 1) = (\partial_L - n)v(n)$ for every stored key n , and order bound $m + 1$.

Cost. One ∂_L and one scalar multiple per stored block: $\mathcal{O}(N)$ coefficient operations, $\mathcal{O}(\sum_n (d_F(n) + 1))$ scalar operations. No coefficient *multiplication* is required.

Correctness is the block formula $D_X(t^n p) = t^{n+1}(\partial_L - n)p$; the precision gain and its single exception are [Theorem 25.105](#). Iterates use the shifted operator $\Delta_{n,k} = \prod_{j=0}^{k-1}(\partial_L - n - j)$, with $\Delta_{n,0}$ the empty product and hence the identity:

$$D_X^k(t^n p(L)) = t^{n+k} \Delta_{n,k} p(L).$$

Proposition 25.29 (Cost of iterated differentiation). *Computing $D_X^k F$ from F by k applications of [Theorem 25.28](#) costs $\mathcal{O}(kN)$ coefficient operations, whereas applying $\Delta_{n,k}$ directly to a single block costs $\mathcal{O}(k)$. Moreover, for $p \in \mathbb{K}[L]$ and $k \geq 1$,*

$$\Delta_{n,k} p = 0 \iff p = 0, \quad \text{or} \quad p \in \mathbb{K} \text{ and } n \leq 0 \leq n + k - 1,$$

and $\deg \Delta_{n,k} p = \deg p$ whenever $n, n+1, \dots, n+k-1$ are all nonzero, while $\deg \Delta_{n,k} p = \deg p - 1$ when one of them vanishes and $\deg p \geq 1$.

Proof. The operation counts are immediate. Each factor $\partial_L - s$ with $s \neq 0$ maps a polynomial of degree $d \geq 0$ to one of degree d , its leading coefficient multiplied by $-s$; in particular it is injective, and being degree-preserving and $\mathbb{K}$ -linear on each finite-dimensional space of polynomials of degree $\leq d$ it is bijective on $\mathbb{K}[L]$. The factor ∂_L lowers the degree by one and has kernel $\mathbb{K}$. The factors commute, each being a polynomial in ∂_L . If none of $n, \dots, n+k-1$ vanishes, $\Delta_{n,k}$ is a composite of bijections, whence injective and degree-preserving. If exactly one of them vanishes — they are distinct consecutive integers, so at most one can — write $\Delta_{n,k} = \partial_L \circ Q$ with Q the product of the remaining factors, a degree-preserving bijection mapping $\mathbb{K}$ onto $\mathbb{K}$. Then $\Delta_{n,k} p = 0$ if and only if $Qp \in \mathbb{K}$, that is, if and only if $p \in \mathbb{K}$; and for $\deg p \geq 1$ the degree of $\Delta_{n,k} p$ is $\deg Qp - 1 = \deg p - 1$. $\square$

Antidifferentiation is the inverse problem and is needed both by integration-based algorithms and by the diagnosis of [Section 27.6](#).

Corollary 25.30 (Antiderivative). *The map $D_X: \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} \rightarrow t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ is surjective and $\mathbb{K}$ -linear with kernel $\mathbb{K}$. Concretely, given $G = \sum_{m \geq 1} g_m t^m$, the solutions $Y = \sum_{n \geq 0} y_n t^n$ of $D_X Y = G$ are obtained blockwise from $(\partial_L - n)y_n = g_{n+1}$; for $n \geq 1$ the solution is unique, and for $n = 0$ it is $y_0 = \int_0^L g_1 + c$ with $c \in \mathbb{K}$ arbitrary.*

Proof. The blockwise reduction is the block formula for D_X . For $n \neq 0$ the operator $\partial_L - n$ is bijective on $\mathbb{K}[L]$: it preserves the filtration by degree and acts on the $(d+1)$ -dimensional space of polynomials of degree $\leq d$ by a triangular matrix with all diagonal entries $-n \neq 0$. For $n = 0$ it is ∂_L , which is surjective on $\mathbb{K}[L]$ in characteristic zero with kernel $\mathbb{K}$. Assembling the blocks gives surjectivity onto $t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and kernel $\mathbb{K}$. This is the ring-level form of [Theorem 2.16](#). $\square$

A common trap

[Theorem 25.30](#) says that an antiderivative routine is *not* a function of its input. A routine that returns one antiderivative without recording how the resonant constant was chosen is nondeterministic in the sense of [Theorem 26.24](#), and two correct runs may disagree by an element of $\mathbb{K}$, which is asymptotically larger than every block of positive order.

25.8 Composition

Two engines are available. The Taylor engine of [Theorem 12.5](#) applies to a near-identity inner map; the generator engine of [Theorem 10.22](#) applies to every admissible monomial-leading inner argument of [Theorem 10.6](#). They compute the same thing ([Theorem 10.7](#)) and are the canonical pair of independent implementations for the differential test of [Section 26.4](#).

Proposition 25.31 (Sharp Taylor budget, exclusive convention). *Let $F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ with $\nu_t(F) \geq a$, let $H \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ with $\nu_t(H) \geq h \geq 0$, and let $N \in \mathbb{Z}$. Then*

$$\text{Trunc}_{<N} F \circ (X + H) = \text{Trunc}_{<N} \sum_{k=0}^{K-1} \frac{H^k}{k!} D_X^k F, \quad K = \left\lceil \frac{N-a}{h+1} \right\rceil,$$

and no smaller K works for every such pair F and H . Equivalently, the last retained index is $k = \lfloor (N-1-a)/(h+1) \rfloor$.

Proof. By Theorem 12.5 the identity $F \circ (X + H) = \sum_{k \geq 0} H^k D_X^k F / k!$ is exact, and the family is summable because $\nu_t(H^k D_X^k F) \geq kh + a + k = a + k(h+1)$ by Theorem 14.4. A summand can influence a block of index $< N$ only if $a + k(h+1) < N$, that is, $k < (N-a)/(h+1)$; the number of nonnegative integers k satisfying this is $\lceil (N-a)/(h+1) \rceil$ and the largest is $\lfloor (N-1-a)/(h+1) \rfloor$. Sharpness: take $F = t^a L$ and $H = t^h$, so that $\nu_t F = a$ and $\nu_t H = h$. Then $D_X^k F = t^{a+k} \Delta_{a,k} L$, and $\Delta_{a,k} L \neq 0$ for every $k \geq 0$ because $L \notin \mathbb{K}$ (Theorem 25.29). Hence $H^k D_X^k F / k!$ is a nonzero multiple of $t^{a+k(h+1)}$ for every k , and the summand with $k = K-1$ contributes a nonzero block of index $a + (K-1)(h+1) < N$, while all summands with $k \geq K$ have valuation at least N . Deleting the term $k = K-1$ therefore changes the truncation. $\square$

Algorithm 25.32 (Near-identity composition, Taylor engine). **Input.** $\mathcal{T}_F = (v_F, N_F)$, $\mathcal{T}_H = (v_H, N_H)$ with $m_H \geq 0$.

Precondition. $m_H \geq 0$, so that $X + H \in \mathcal{G}_{\text{poly},\mathbb{K}}$. An inner argument of negative order is *not* near-identity and must be routed to Theorem 25.34 after the dominant monomial has been normalised.

Output. Precision $N = \min\{N_F, N_H + m_F + 1\}$, order m_F .

Steps. Set $S \leftarrow 0$, $P \leftarrow 1$, $D \leftarrow v_F$, $k \leftarrow 0$. While $\nu_t(P) + \nu_t(D) < N$: set $S \leftarrow \text{Trunc}_{<N} S + PD/k!$; then $D \leftarrow D_X D$, $P \leftarrow \text{Trunc}_{<N-\nu_t(D)} PH$, $k \leftarrow k+1$. Return S .

Cost. $K = \lceil (N - m_F)/(m_H + 1) \rceil$ iterations, each one truncated product and one derivative: $\mathcal{O}(K\rho^2) = \mathcal{O}(\rho^3/(m_H + 1))$ coefficient multiplications, with $\rho = N - m_F$.

Precision and sharpness are Theorem 25.106; the stopping test is Theorem 25.31, implemented as the dynamic test $\nu_t(P) + \nu_t(D) \geq N$, which is never weaker than the static bound and is often much stronger because D_X may annihilate blocks. The cutoff $N - \nu_t(D)$ carried by the running power is the exact one demanded by Theorem 25.102: the discarded part of P has valuation at least $N - \nu_t(D)$, so its product with D has valuation at least N and cannot reach a retained block. For $m_F \geq 0$ this cutoff is at most N and the naive choice N is merely wasteful; for $m_F < 0$, which is the normal case when the outer series is Laurent, it is strictly larger than N and the naive choice is wrong.

For the generator engine one needs the elementary but useful observation that substituting into the logarithmic generator is a *finite* Taylor shift.

Proposition 25.33 (Polynomial shift). *Let $p \in \mathbb{K}[L]$ of degree d and let $V \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. Then, in $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$,*

$$p(L + V) = \sum_{k=0}^d \frac{(\partial_L^k p)(L)}{k!} V^k, \quad (495)$$

a sum of $d+1$ terms.

Proof. Both sides are $\mathbb{K}$ -linear in p , so it suffices to take $p = L^e$ with $e \leq d$. The binomial theorem gives $p(L + V) = \sum_{k=0}^e \binom{e}{k} L^{e-k} V^k$, and $\partial_L^k L^e / k! = \binom{e}{k} L^{e-k}$. Terms with $k > e$ vanish on both sides. Characteristic zero is used only to divide by $k!$. $\square$

Algorithm 25.34 (Composition by generator images). **Input.** $\mathcal{T}_F = (v_F, N_F)$; an inner argument $G = cX^\varrho(1 + U)$ with $c \in \mathbb{K}^\times$, $\varrho \in \mathbb{N}_{>0}$, $U \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ given by $\mathcal{T}_U = (v_U, N_U)$; a chosen value $\log c \in \mathbb{K}$.

Precondition. G admissible in the sense of [Theorem 10.6](#); $\log c \in \mathbb{K}$, failing which $\mathbb{K}$ must first be enlarged; $\varrho \in \mathbb{N}_{>0}$, failing which the output lives in a Puiseux or Hahn extension ([Theorem 10.17](#)) and the type must say so.

Output. $\text{Trunc}_{<N} F \circ G$ at the precision furnished by [Theorem 10.11](#).

Steps.

1. $V \leftarrow \log(1 + U)$ by [Theorem 25.27](#); $R \leftarrow (1 + U)^{-1}$ by [Theorem 25.20](#) or [Theorem 25.21](#).
2. Let $D = \max_n \deg v_F(n)$ over retained blocks. Precompute $V^k/k!$ for $1 \leq k \leq \min\{D, N-1\}$; the cap is legitimate because $\nu_t(V^k) \geq k$.
3. Precompute R^n incrementally for the retained outer indices n (for negative n , $R^n = (1 + U)^{|n|}$, an ordinary positive power).
4. Accumulate $\sum_n c^{-n} t^{\varrho n} R^n p_n(\varrho L + \log c + V)$ with the inner evaluation performed by [Equation \(495\)](#) applied to $p_n(\varrho L + \log c + \cdot)$, truncating after every product.

Cost. At most $\min\{D, N-1\}$ truncated products for the powers of V , at most N for the powers of R , and $\sum_n (\deg v_F(n) + 1)$ polynomial-by-series products for the shifts: in the coefficient-operation model $\mathcal{O}((N+D)N^2)$.

Remark 25.35 (Why the powers of V are shared). [Theorem 25.33](#) converts every one of the $\mathcal{O}(N)$ polynomial evaluations into a fixed linear combination of the *same* powers $V^k/k!$. This is the precise content of the caching advice given by all six sources. Horner evaluation in $L+V$, by contrast, performs $\deg p_n$ truncated products *per block* and does not share work between blocks; it is asymptotically worse by a factor N .

A common trap

Composition changes *both* generators. For $G = X + H = X(1 + tH)$,

$$t \circ G = \frac{t}{1 + tH}, \quad L \circ G = L + \log(1 + tH),$$

by [Theorem 10.22](#). Substituting $t \mapsto t + H$, or updating one generator and not the other, is not an approximation but a different and wrong operation; see [Section 27.3](#).

25.9 Reversion by the four strategies

Throughout this subsection $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ with $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, $h = \nu_t(A) \geq 0$, and $G = F^{(-1)\circ} = X + B$. All four strategies compute the same B , by uniqueness ([Theorem 14.11](#)). All four are precision-neutral in the following sense, so none of them needs guard blocks in the outer argument.

Proposition 25.36 (Reversion neither gains nor loses precision). *If $F_1 = X + A_1$ and $F_2 = X + A_2$ lie in $\mathcal{G}_{\text{poly}, \mathbb{K}}$ with $\nu_t(A_1 - A_2) \geq N$, then $\nu_t(F_1^{(-1)\circ} - F_2^{(-1)\circ}) \geq N$. Consequently a routine that receives A at precision N may return B at precision N , and no larger cutoff is sound.*

Proof. Put $G_2 = F_2^{(-1)\circ}$. Then $F_1 \circ G_2 - X = (F_2 \circ G_2 - X) + (A_1 - A_2) \circ G_2 = (A_1 - A_2) \circ G_2$, whose order equals $\nu_t(A_1 - A_2) \geq N$ by [Equation \(490\)](#). By [Theorem 26.5](#) below, applied to F_1 and the candidate G_2 , $\nu_t(G_2 - F_1^{(-1)\circ}) = \nu_t(F_1 \circ G_2 - X) \geq N$. Sharpness: take $A_1 = A_2 + t^N$; the same computation gives $\nu_t(G_2 - F_1^{(-1)\circ}) = N$ exactly. The statement is the block form of [Theorem 15.7](#). $\square$

Algorithm 25.37 (Strategy 1: Picard iteration). **Input.** A at precision N .

Precondition. $\nu_t(A) \geq 0$.

Output. B at precision N .

Steps. $B_0 \leftarrow 0$; $B_{r+1} \leftarrow \text{Trunc}_{<r+1} - A \circ (X + B_r)$ for $r = 0, \dots, N-1$; return B_N .

Cost. N compositions, the r -th at cutoff $r+1$.

Correctness is the fixed-point equation $B = -A \circ (X + B)$ of [Theorem 14.7](#) together with the one-block gain of [Theorem 14.9](#), which gives $\nu_t(B - B_r) \geq r$; the limit exists by completeness ([Theorem 12.13](#)). Picard is the cheapest strategy to implement and the most expensive to run; its virtue is that it needs nothing but composition.

Algorithm 25.38 (Strategy 2: triangular block solve). **Input.** A at precision N .

Precondition. $\nu_t(A) \geq 0$; the leading slope of F is 1. For leading slope $c \in \mathbb{K}^\times$ divide each correction by c ([Theorem 15.4](#)).

Output. B at precision N .

Steps. $G \leftarrow X$; for $n = 0, \dots, N - 1$: compute the block of index n of $\text{Err}_{\text{rev}}(F, G)$, call it $r_n(L)$, and set $G \leftarrow G - r_n(L)t^n$.

Cost. N residual evaluations; with the block-perturbation update of [Theorem 15.1](#) each step touches only the blocks that the new correction can reach, and the total is one composition-sized computation rather than N .

Correctness is [Theorem 15.2](#); that the residual block reads off the next inverse block is [Theorem 14.17](#). This is the strategy of choice when only a few blocks are wanted, when the answer is required in expanded differential-polynomial form, or when a lazy interface must produce blocks on demand.

Algorithm 25.39 (Strategy 3: Newton reversion with doubling). **Input.** A at precision N ; an initial G correct to precision $\mu_0 \geq 1$.

Precondition. $\nu_t(A) \geq 0$; $D_X F \circ G$ is a unit, which holds automatically since it equals $1 + \mathcal{O}_t^{\text{form}}(t)$ ([Theorem 15.9](#)).

Output. B at precision N .

Steps. While $\mu < N$: set $\mu' \leftarrow \min \{2\mu, N\}$ and

$$G \leftarrow \text{Trunc}_{<\mu'} G - \frac{F \circ G - X}{(D_X F) \circ G}, \quad \mu \leftarrow \mu'.$$

Recompute $\nu_t(\text{Err}_{\text{rev}}(F, G))$ after each step and abort if it is smaller than μ .

Cost. $\lceil \log_2(N/\mu_0) \rceil$ stages; each stage one composition, one derivative-composition and one division at the stage cutoff.

Correctness in exact arithmetic is [Theorem 15.11](#); the behaviour with inexact data, and hence the guard precision each stage must request, is [Theorem 15.14](#), and the schedule is [Theorem 15.16](#).

Algorithm 25.40 (Strategy 4: Lagrange–Bürmann streaming). **Input.** A at precision N .

Precondition. $\nu_t(A) \geq h \geq 0$.

Output. B at precision N .

Steps. $B \leftarrow 0, P \leftarrow 1$; for $r = 1, 2, \dots$: set $P \leftarrow \text{Trunc}_{<}. PA$, form $Q \leftarrow D_X^{r-1} P$ by $r-1$ applications of [Theorem 25.28](#), and accumulate $B \leftarrow \text{Trunc}_{<N} B + (-1)^r Q/r!$. Stop when $r(h+1) - 1 \geq N$, that is, after $R = \lceil (N+1)/(h+1) \rceil - 1$ terms.

Cost. R truncated products and $\mathcal{O}(R^2)$ derivative passes; memory $\mathcal{O}(N)$, since only the current power is retained.

Stopping rule. By [Theorem 16.14](#), $B = \sum_{r \geq 1} ((-1)^r/r!) D_X^{r-1}(A^r)$. Since $\nu_t(A^r) \geq rh$ and each D_X raises the order by at least one ([Theorem 14.4](#)), $\nu_t(D_X^{r-1}(A^r)) \geq rh + r - 1 = r(h+1) - 1$. A summand can meet a block of index $< N$ only if $r(h+1) - 1 < N$, i.e. $r < (N+1)/(h+1)$, which is the stated bound. For $h = 0$ it reads $r \leq N$; for the inclusive statement “through block N ” it reads $r \leq N+1$, which is the source of the apparent one-block discrepancy discussed in [Section 27.1](#). $\square$

Algorithm 25.41 (Hybrid driver). **Input.** A map F whose dominant part need not be X ; a target N . **Steps.**

1. Normalise: factor the dominant monomial of F and invert it exactly. If the leading exponent is not a positive integer, or if the leading derivative vanishes, stop: the problem is not near-identity reversion and the class must be changed (Section 27.9).
2. Run Theorem 25.38 for two or three blocks. This exposes the coefficient ring actually required and the correct leading slope before any expensive stage.
3. Run Theorem 25.39 with the doubling schedule to the target cutoff.
4. Compute *both* residuals of Theorem 26.5 and compare their orders.
5. If either order falls short of the target, enlarge the working cutoff and repeat from step 3; do not adjust the claimed precision instead.

Proposition 25.42 (Comparison of the four strategies). *Let $C(\nu)$ denote the cost of one composition at cutoff ν in the coefficient-operation model, and assume C is nondecreasing and satisfies $C(2\nu) \geq 2C(\nu)$. Then, to reach precision N from A of order $h \geq 0$:*

1. *Picard costs $\sum_{r \leq N} C(r)$ and uses only composition;*
2. *the triangular solve costs $\mathcal{O}(C(N))$ when the block-perturbation update is used, and $\sum_{n \leq N} C(n)$ when it is not;*
3. *Newton costs $\sum_{i \leq \log_2 N} (2C(2^i) + \mathcal{O}(4^i)) \leq 4C(N) + \mathcal{O}(N^2)$;*
4. *Lagrange–Bürmann costs $R = \lceil (N+1)/(h+1) \rceil - 1$ truncated products and $\mathcal{O}((N/(h+1))^2)$ derivative passes, and uses no composition at all.*

Proof. Parts (i)–(iii) are the step counts stated with the algorithms; the estimate in (iii) uses

$$\sum_{i \leq \log_2 N} C(2^i) \leq 2C(N),$$

which follows by summing the geometric majorisation implied by $C(2\nu) \geq 2C(\nu)$, and the division at stage i costs $\mathcal{O}(4^i)$ by Theorem 25.22. Part (iv) is the stopping rule of Theorem 25.40: the r -th term needs one product to update A^r and $r-1$ derivative passes, and $\sum_{r \leq R} (r-1) = \mathcal{O}(R^2)$. $\square$

Remark 25.43 (Which to use). Lagrange–Bürmann is the only strategy that never composes, and it is therefore the strategy of choice for theoretical derivations and for maps whose powers and derivatives simplify – the Lambert polynomials being the model case, as Theorem 27.4 shows. Its weakness is that its cost is insensitive to cancellation: it forms A^r for every r up to the stopping index even when most terms cancel. Newton dominates at high N , provided composition is itself efficient; Picard and the triangular solve dominate at low N and are the only ones that give blocks lazily.

Section summary

Every routine in this section returns a coset, not an element. Its contract is the triple (order bound, absolute cutoff, coefficient ring), its correctness obligation is soundness at the declared cutoff, and its cost must be quoted in a named model. Relative precision, not absolute cutoff, is the quantity that products, quotients and compositions conserve. Addition destroys it whenever leading blocks cancel; differentiation gains one block of absolute cutoff and conserves the relative precision, except in the single annihilating case $(\partial_L - m_1)v_1(m_1) = 0$, where it destroys that too.

26 Residual certificates and reproducible verification

Section 25 says what each routine must deliver. This section says how one finds out whether it did. The mechanism is uniform: every operation of this volume is characterised by an equation that its

output satisfies exactly, so a computed candidate can be substituted back into that equation and the valuation of the result compared with the declared cutoff. The mathematics needed to make this a proof rather than a habit is small but it is not empty, and it is exactly the following three points, each of which is made precise below.

- A residual certifies a *coset*, and only when the residual equation is *faithful* — when the valuation of the residual determines the valuation of the error. Every primitive of [Section 25](#) has a faithful equation with an explicit offset ([Table 9](#)), with exactly one exception, the antiderivative, and the exception is the resonant constant ([Theorem 26.18](#)).
- A residual test is a statement about coefficients in R , decided by comparison of canonical forms. Substituting numbers for L does not decide it ([Theorem 26.3](#)).
- A residual computed by the same code path that produced the answer, and in particular the residual that *drove* an iteration, certifies nothing at all ([Theorem 26.7](#)). This is the precise reason for computing both residuals of a reversion, and the reason is not that one of them is more accurate: at finite precision they certify exactly as far as each other ([Theorem 26.53](#)).

26.1 Faithful defining equations

Definition 26.1 (Defining equation, faithfulness, gauge). Let φ be a partial operation on the ambient class, defined on a set $\mathcal{D}$ of input tuples with values in a declared solution class $\mathcal{Y}$. A *defining equation* for φ is a map E that assigns to an input tuple $z \in \mathcal{D}$ and a candidate $Y \in \mathcal{Y}$ an element $E(Y; z)$ of the ambient class, computable from Y and z by finitely many conforming routines, such that

$$E(Y; z) = 0 \iff Y = \varphi(z).$$

The equation is *faithful with gauge* $\kappa \in \mathbb{Z}$ if moreover

$$\nu_t E(Y; z) = \nu_t (Y - \varphi(z)) + \kappa \quad \text{for all } z \in \mathcal{D}, Y \in \mathcal{Y}, \quad (496)$$

with the convention $\nu_t 0 = +\infty$. A *residual certificate* for a computed Y at cutoff M is the pair consisting of the computed residual $E(Y; z)$ and the verified assertion $\nu_t E(Y; z) \geq M + \kappa$.

Proposition 26.2 (What a faithful certificate certifies). *Let E be faithful with gauge κ and let $M \in \mathbb{Z}$. Then, for every $Y \in \mathcal{Y}$,*

$$\nu_t E(Y; z) \geq M + \kappa \iff \nu_t (Y - \varphi(z)) \geq M \iff \text{Trunc}_{<M} Y = \text{Trunc}_{<M} \varphi(z).$$

In particular the certificate is an equivalence, not an implication: a candidate that passes is correct below M , and a candidate that is correct below M passes.

Proof. The first equivalence is (496). The second is [Theorem 25.3](#): $\nu_t (Y - \varphi(z)) \geq M$ says exactly that $Y - \varphi(z) \in t^M \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, which says that Y and $\varphi(z)$ have the same blocks of index $< M$. $\square$

The gauge is not a decoration: it is the number of blocks by which the residual test must be run *beyond* the cutoff one wants to certify, and it is the commonest source of an apparently failing test on a correct answer. For the reciprocal of a series of order m it equals m , which is negative exactly when the series has a pole.

Proposition 26.3 (Exact zero testing). *Let $\mathbb{K}$ be infinite and let $R = \mathbb{K}[L]$. Let $s \geq 1$ and let $\lambda_1, \dots, \lambda_s \in \mathbb{K}$ be any finite list of sample points. Then there is a nonzero $p \in R$ of degree s with $p(\lambda_i) = 0$ for all i , namely $p = \prod_{i=1}^s (L - \lambda_i)$. Consequently no test that evaluates residual blocks at finitely many values of L can decide whether a residual block vanishes, and a residual test is sound only when it compares canonical forms in R .*

Proof. The displayed polynomial is monic of degree s , hence nonzero, and vanishes at each λ_i . A test that inspects only the values $p(\lambda_1), \dots, p(\lambda_s)$ cannot distinguish it from 0. $\square$

Remark 26.4 (The three coefficient rings test differently). For $R = \mathbb{K}[L]$ the canonical form is the coefficient vector and the test is a comparison with the zero vector. For $R = \mathbb{K}(L)$ one must normalise the fraction and test the numerator, since an unnormalised numerator and denominator may share a factor; testing a rational function at sample points is subject to [Theorem 26.3](#) twice over, once for the numerator and once for a spurious pole. For $R = \mathbb{K}((L^{-1}))$, a block is stored only to a declared *inner* cutoff, so the test decides vanishing only below that cutoff, and the inner cutoff is a second precision parameter that the type must carry and the certificate must quote ([Theorem 3.22](#)). A test that silently compares a polynomial block with a truncated Laurent block is comparing objects of two different types.

A common trap

A certificate is a statement about the pair (candidate, defining equation). It is not a statement about the routine that produced the candidate, and it is not a statement about the routine that computes the residual. If the residual is computed by the very operation whose correctness is in question, a passing test is consistent with that operation being arbitrarily wrong; see [Theorem 26.7](#) and the engine of [Theorem 26.8](#), which passes its own residual test identically while getting every block after the first one wrong.

26.2 The two residuals of a reversion

Throughout this subsection $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ with $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and $G \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ is a candidate for $F^{(-1)\circ}$. The two residuals are the right residual $\text{Err}_{\text{rev}}(F, G) = F \circ G - X$ and the left residual $G \circ F - X$ of [Theorem 14.14](#).

Theorem 26.5 (Two-sided residual certificate). *Let $F, G \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ and put $\Delta = G - F^{(-1)\circ} \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$; the letter E is reserved for defining equations, as in [Theorem 26.1](#).*

1. Both equations are faithful with gauge 0:

$$\nu_t(\text{Err}_{\text{rev}}(F, G)) = \nu_t(G \circ F - X) = \nu_t(\Delta),$$

and when $\Delta \neq 0$ all three series have the same leading block.

2. Certification. For every $M \in \mathbb{N}_0$, each of $\nu_t(\text{Err}_{\text{rev}}(F, G)) \geq M$ and $\nu_t(G \circ F - X) \geq M$ is equivalent to $\text{Trunc}_{<M} G - X = \text{Trunc}_{<M} F^{(-1)\circ} - X$.
3. Attainable certification cutoff. Suppose F and G are given by conforming representatives of the corrections A and $G - X$, with finite cutoffs N_A and N_B . Then each of the two residuals is determined by that data exactly to the cutoff $\min\{N_A, N_B\}$, and to no larger cutoff. Hence neither residual can certify G beyond block $\min\{N_A, N_B\} - 1$, and the two residuals certify exactly as far as each other.

Proof. Part 1 is [Theorem 14.15](#), recorded here in the form the certificate discipline uses; in the language of [Theorem 26.1](#) it says that $E(G; F) = \text{Err}_{\text{rev}}(F, G)$ and $E'(G; F) = G \circ F - X$ are defining equations for $\varphi = \cdot^{(-1)\circ}$ on $\mathcal{Y} = \mathcal{G}_{\text{poly}, \mathbb{K}}$, faithful with gauge $\kappa = 0$. Part 2 is then [Theorem 26.2](#); it contains [Theorem 14.16](#), which is the special case in which $G - X$ is a polynomial in t of degree $M - 1$.

For part 3, apply [Theorem 25.106](#) to $F \circ G$: the outer argument is F , whose representative has order bound $m_F = -1$ — the block $X = t^{-1}$ — and cutoff N_A ; the inner correction is $G - X$, with cutoff N_B . The declared output cutoff is therefore

$$\min\{N_A, N_B + m_F + 1\} = \min\{N_A, N_B\},$$

and subtracting the exactly known X changes neither cutoff nor determinacy. Sharpness in the outer-limited regime is automatic; in the inner-limited regime it needs $\nu_t(D_X v_F) = m_F + 1 = 0$,

which holds for every admissible F because $D_X X = 1$ and every other block of v_F has index ≥ 0 , so $\nu_t(D_X v_F) = 0$. The same computation with the roles of F and G exchanged gives $\min\{N_B, N_A\}$ for the left residual. Since certification below M requires the residual to be known to be zero on all blocks of index $< M$, and blocks of index $\geq \min\{N_A, N_B\}$ are not determined by the data, no certificate can reach beyond $\min\{N_A, N_B\} - 1$. $\square$

Remark 26.6 (Why compute both, then). [Theorem 26.5](#) removes two folklore reasons for computing both residuals and leaves one. It is *not* true that one residual is sharper than the other: by [1](#) they have the same order and even the same leading block, and by [3](#) they are determined to the same cutoff by the same data. It is *not* true that the left residual detects an error the right one cannot: by [1](#) either one detects an error exactly when there is one. What remains, and what is decisive in practice, is independence: the two residuals evaluate the composition routine at arguments in the opposite order, so a composition implemented correctly in one order and incorrectly in the other is caught, and – far more importantly – an iteration driven by one of them cannot be certified by that one ([Theorem 26.7](#)).

Proposition 26.7 (A driven residual certifies nothing). *Let $\odot$ be any binary operation on $\mathcal{G}_{\text{poly}, \mathbb{K}}$ -representatives whatsoever, and suppose the triangular loop of [Theorem 25.38](#) is run with $\odot$ in place of $\circ$, the block r_n at each step being read from $F \odot G - X$. If the loop terminates having driven the blocks of $F \odot G - X$ of index $< N$ to zero, then the assertion “ $\nu_t(F \odot G - X) \geq N$ ” holds by construction and is independent of whether $\odot = \circ$. Consequently it carries no information about G , and in particular it does not certify $\text{Trunc}_{<N} G - X = \text{Trunc}_{<N} F^{(-1)\circ} - X$.*

Proof. The loop’s exit condition is precisely the assertion in question: the loop terminates only when the blocks of $F \odot G - X$ of index $< N$ have been made to vanish, and each step changes G so as to make one more of them vanish. Nothing in the argument used any property of $\odot$. Hence the assertion is a theorem about the loop, not about G , and it holds equally for an operation $\odot \neq \circ$, for which the conclusion $\text{Trunc}_{<N} G - X = \text{Trunc}_{<N} F^{(-1)\circ} - X$ is false; an explicit such $\odot$ is exhibited in [Theorem 26.8](#). $\square$

Example 26.8 (An engine that passes its own test and is wrong from the second block on). Define $\Psi_G: \mathcal{PL}_{\text{poly}, \mathbb{K}} \rightarrow \mathcal{PL}_{\text{poly}, \mathbb{K}}$ as the $\mathbb{K}[L]$ -algebra homomorphism that substitutes the outer generator and leaves the logarithmic generator alone:

$$\Psi_G\left(\sum_n p_n(L)t^n\right) := \sum_n p_n(L)(G^{-1})^n, \quad G = X + B \in \mathcal{G}_{\text{poly}, \mathbb{K}}.$$

This is well defined: $G^{-1} = t/(1 + tB)$ has $\nu_t = 1$, so the family is summable ([Theorem 6.34](#)). It is the “substitute t , forget L ” engine of [Section 27.3](#). Take $F = X + L$, the map whose inverse is ω ([Theorem 18.10](#)). Then

$$F \odot G := \Psi_G(F) = \Psi_G(t^{-1}) + \Psi_G(L) = G + L,$$

so the triangular loop driven by $F \odot G - X = B + L$ sets $b_0 = -L$ and every later block to 0, terminating with

$$\tilde{G} = X - L \quad \text{and} \quad F \odot \tilde{G} - X = 0 \text{ identically.}$$

The true inverse is $\omega = X - L + Lt + (\frac{1}{2}L^2 - L)t^2 + \dots$, so $\tilde{G}$ is correct in the block of index 0 and wrong in every later block. The true right residual detects this at once:

$$\text{Err}_{\text{rev}}(F, \tilde{G}) = (X - L) + \log(X - L) - X = \log(1 - Lt) = -Lt - \frac{1}{2}L^2t^2 - \dots,$$

of order 1, and the true left residual is

$$\tilde{G} \circ F - X = (X + L) - \log(X + L) - X = -\log(1 + Lt) = -Lt + \frac{1}{2}L^2t^2 - \dots,$$

also of order 1, with the same leading block $-L$ as predicted by [Theorem 26.51](#).

The engine of [Theorem 26.8](#) is worth one more moment, because it shows which structural tests can and cannot see a wrong generator image.

Proposition 26.9 (The ring tests are blind to the generator; the chain rule is not). *Let $G = X + B \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ with $B \neq 0$, $h = \nu_t(B) \geq 0$, and let Ψ_G be as in [Theorem 26.8](#).*

1. Ψ_G is a $\mathbb{K}[L]$ -algebra homomorphism of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, continuous for the t -adic filtration. Hence it satisfies every identity built from addition, multiplication, reciprocals and $\mathbb{K}[L]$ -linearity – in particular $\Psi_G(F_1 F_2) = \Psi_G(F_1) \Psi_G(F_2)$ and $\Psi_G(F^{-1}) = \Psi_G(F)^{-1}$ – so no test built from those operations distinguishes it from composition.
2. Ψ_G fails the chain rule already on the single generator L :

$$D_X(\Psi_G(L)) - \Psi_G(D_X L) D_X G = t - G^{-1} D_X G,$$

and this element has valuation exactly $h + 2$, with block $-(\partial_L - h - 1)b_h$ there, where b_h is the leading block of B .

Proof. (i) Ψ_G is defined on generators by $t \mapsto G^{-1}$, $L \mapsto L$, and extended by the summable family in the definition; summability, since $\nu_t((G^{-1})^n) = n$, makes the extension a continuous $\mathbb{K}[L]$ -algebra map, exactly as in [Theorem 10.7](#) with the logarithmic generator held fixed. Multiplicativity and compatibility with reciprocals are formal consequences.

(ii) $\Psi_G(L) = L$ and $D_X L = t$, so the left side is $t - \Psi_G(t) D_X G = t - G^{-1} D_X G$. Now $G^{-1} = t(1 + tB)^{-1}$ and $D_X G = 1 + D_X B$, so

$$G^{-1} D_X G = t(1 + D_X B)(1 - tB + t^2 B^2 - \dots) = t[1 + (D_X B - tB) + W],$$

where W collects the products of at least two of the small terms $D_X B, tB$, each of valuation $\geq h + 1$; hence $\nu_t W \geq 2h + 2 \geq h + 2$. By the block formula the block of index $h + 1$ of $D_X B$ is $(\partial_L - h)b_h$ and that of tB is b_h , so $D_X B - tB$ has block $(\partial_L - h - 1)b_h$ at index $h + 1$. Since $h + 1 \neq 0$, the operator $\partial_L - h - 1$ is injective on $\mathbb{K}[L]$ ([Theorem 25.29](#)), so that block is nonzero and $\nu_t(D_X B - tB) = h + 1 < h + 2 \leq \nu_t W$. Therefore $t - G^{-1} D_X G = -t(D_X B - tB) - tW$ has valuation exactly $h + 2$ and block $-(\partial_L - h - 1)b_h$ there. $\square$

Remark 26.10. [Theorem 26.9](#) explains why the chain-rule test is listed as mandatory in [Section 26.4](#) even though it looks redundant next to the composition residual: the composition residual computed with a wrong engine is the object of [Theorem 26.7](#), whereas the chain rule couples the engine to the derivation, which is an independent primitive ([Theorem 25.28](#)) with an independent certificate.

26.3 Certificates for the algebraic primitives

Proposition 26.11 (Quotient and reciprocal certificates). *Let $B \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ be a unit, $m = \nu_t(B)$, and $A \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. For every candidate $Q \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$,*

$$\nu_t(BQ - A) = \nu_t\left(Q - \frac{A}{B}\right) + m.$$

Thus $E(Q) = BQ - A$ is a defining equation for the quotient, faithful with gauge m ; the case $A = 1$ is the reciprocal, with the same gauge.

Proof. $BQ - A = B(Q - A/B)$, and ν_t is additive on $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ ([Theorem 7.2](#)), with $\nu_t(B) = m$ finite because $B \neq 0$. $\square$

Remark 26.12 (The gauge is where the sources' division test goes wrong). Two of the six sources state the division certificate in the form “residual $\mathcal{O}_t^{\text{form}}(t^N)$ certifies the quotient through block $N - 1$ ”. By [Theorems 26.2](#) and [26.11](#) the correct statement is that such a residual certifies the quotient through block $N - m - 1$, where $m = \nu_t(B)$. The two agree exactly when the denominator is normalised to order 0, which is the case the sources have in mind and the case an implementation should enforce before testing – but the enforcement is then part of the contract and must be checked, not assumed.

Proposition 26.13 (Logarithm and exponential certificates). *Let $U \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and let ∂_t be the formal derivative in t at fixed L .*

1. *For every candidate $\Lambda \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, $\nu_t(\exp \Lambda - (1 + U)) = \nu_t(\Lambda - \log(1 + U))$: gauge 0.*
2. *For every candidate $\Lambda \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, $\nu_t((1 + U)\partial_t \Lambda - \partial_t U) = \nu_t(\Lambda - \log(1 + U)) - 1$: gauge -1 .*
3. *For every candidate $\mathcal{E} \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, $\nu_t(\log \mathcal{E} - U) = \nu_t(\mathcal{E} - \exp U)$: gauge 0.*

Proof. Write $\Delta = \Lambda - \log(1 + U) \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$.

(i) By [Theorem 10.5](#), $\exp \Lambda = \exp(\log(1 + U)) \exp \Delta = (1 + U) \exp \Delta$, so $\exp \Lambda - (1 + U) = (1 + U)(\exp \Delta - 1)$. Now $\exp \Delta - 1 = \sum_{k \geq 1} \Delta^k/k!$ and $\nu_t(\Delta^k) \geq k\nu_t \Delta > \nu_t \Delta$ for $k \geq 2$ because $\nu_t \Delta \geq 1$; hence $\nu_t(\exp \Delta - 1) = \nu_t \Delta$. Multiplying by the unit $1 + U$ of order 0 preserves the order.

(ii) Termwise differentiation of the summable family defining the logarithm gives $(1 + U)\partial_t \log(1 + U) = \partial_t U$ (proof of [Theorem 25.26](#)), so $(1 + U)\partial_t \Lambda - \partial_t U = (1 + U)\partial_t \Delta$. If $\Delta \neq 0$ has $\nu_t \Delta = n \geq 1$ and leading block $\delta_n \neq 0$, then $\partial_t \Delta$ has block $n\delta_n \neq 0$ at index $n - 1$, using characteristic zero, and no block below; so $\nu_t(\partial_t \Delta) = n - 1$. Multiplying by $1 + U$ preserves it.

(iii) With $\Theta = \mathcal{E} - \exp U$, one has $\log \mathcal{E} - U = \log(\mathcal{E}(\exp U)^{-1}) = \log(1 + \Theta(\exp U)^{-1})$, and $\exp U$ is a unit of order 0, so the argument of the logarithm lies in $1 + t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and $\nu_t \log(1 + \Xi) = \nu_t \Xi$ for $\Xi \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ by [Theorem 10.5](#). Hence the order equals $\nu_t \Theta$. $\square$

Remark 26.14 (The differential certificate is cheaper, not stronger). Part (ii) of [Theorem 26.13](#) detects an error one block earlier than part (i), and it costs one truncated product and two derivatives instead of an exponential. It does not, however, certify further. If the data are known to cutoff N , the residual of (i) is known to cutoff N and certifies $\nu_t \Delta \geq M$ for $M \leq N$; the residual of (ii) is known only to cutoff $N - 1$, because ∂_t shifts the filtration by one, and its threshold is $M - 1$, so it too certifies exactly $M \leq N$. The two reach the same block for opposite reasons.

Proposition 26.15 (Power and root certificates). *Let $B \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ with $\nu_t(B) = 0$ and leading scalar $b_0 \in \mathbb{K}^\times$, let $p \in \mathbb{Z}$ and $q \in \mathbb{N}_{>0}$, and let $C = B^{p/q}$ be the power determined by a chosen value $b_0^{p/q} \in \mathbb{K}^\times$ satisfying $(b_0^{p/q})^q = b_0^p$, the powers being those of [Theorem 6.37](#). Then $C^q = B^p$, and for every candidate $D \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ of order 0 whose leading scalar equals $b_0^{p/q}$,*

$$\nu_t(D^q - B^p) = \nu_t(D - C) :$$

gauge 0. The restriction on the leading scalar of D cannot be dropped, and the equation cannot replace it: if $\zeta \in \mathbb{K}^\times$ satisfies $\zeta^q = 1$ and $\zeta \neq 1$, then $D = \zeta C$ satisfies $D^q - B^p = 0$ identically while $D \neq C$. The branch datum is part of the specification and is certified only by comparing leading scalars.

Proof. The exponent law $(1 + U)^a(1 + U)^b = (1 + U)^{a+b}$ for binomial powers is the Vandermonde convolution $\sum_k \binom{a}{k} \binom{b}{n-k} = \binom{a+b}{n}$ [[18](#)], valid in any $\mathbb{Q}$ -algebra; iterating it q times with $a = b = \dots = p/q$ gives $(1 + U)^p$ where $U = b_0^{-1}B - 1$. Combined with the hypothesis $(b_0^{p/q})^q = b_0^p$ this yields $C^q = B^p$. Next,

$$D^q - C^q = (D - C) \sum_{i=0}^{q-1} D^i C^{q-1-i},$$

and each summand has order 0 with leading scalar $d_0^i c_0^{q-1-i}$, where $c_0 = b_0^{p/q}$ and d_0 is the leading scalar of D . Under the hypothesis $d_0 = c_0$ the sum has leading scalar $q c_0^{q-1} \neq 0$, because $\mathbb{K}$ has characteristic zero and $c_0 \neq 0$; hence the sum has order exactly 0 and the displayed identity follows from additivity of ν_t . For the final assertion, $(\zeta C)^q = \zeta^q C^q = C^q = B^p$ while $\zeta C - C = (\zeta - 1)C \neq 0$. $\square$

Proposition 26.16 (A faithful composition certificate). *Let $F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, $G \in \mathcal{G}_{\text{poly},\mathbb{K}}$, and let $H \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ be a candidate for $F \circ G$. Then*

$$\nu_t(H \circ G^{(-1)\circ} - F) = \nu_t(H - F \circ G) :$$

gauge 0.

Proof. Right composition with $G^{(-1)\circ} \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ is a ring homomorphism of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ (Theorem 10.7) and satisfies $(F \circ G) \circ G^{(-1)\circ} = F \circ (G \circ G^{(-1)\circ}) = F$ by associativity on the admissible domain (Theorem 10.27) and $G \circ G^{(-1)\circ} = X$ (Theorem 12.7). Hence $H \circ G^{(-1)\circ} - F = (H - F \circ G) \circ G^{(-1)\circ}$, whose order equals $\nu_t(H - F \circ G)$ by Equation (490). $\square$

Remark 26.17 (Composition has one faithful certificate and several cheap unfaithful ones). Theorem 26.16 is the only single-equation certificate for composition in this volume, and it is not free: it requires the inverse of the inner map, hence an independent reversion. The cheap alternatives – the chain rule (Theorem 10.31), the multiplicativity $(F_1 F_2) \circ G = (F_1 \circ G)(F_2 \circ G)$, associativity, and the agreement of the two engines of Section 25.8 – are *consistency tests*: each of them is an identity that a wrong engine may or may not satisfy. Theorem 26.9 shows that the multiplicative tests are satisfied by a wrong engine, and the chain rule is not, so among the cheap tests the chain rule is the one to run.

Proposition 26.18 (The antiderivative equation is the one unfaithful defining equation). *Let $G \in {}^t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.*

1. *The equation $E(Y) = D_X Y - G$ does not define a single-valued operation on $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$: its solution set is a coset of $\mathbb{K}$ (Theorem 25.30), and E is not faithful with any gauge, since $E(Y + c) = E(Y)$ while $\nu_t((Y + c) - Y) = 0$ for $c \in \mathbb{K}^\times$.*
2. *Fix $c \in \mathbb{K}$ and let $\mathcal{Y}_c = \{Y \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}} : (\text{block of index 0 of } Y) \text{ takes the value } c \text{ at } L = 0\}$. Then E restricted to $\mathcal{Y}_c$ is a defining equation for a single-valued operation, and it is faithful with gauge 1.*

Proof. (i) is Theorem 25.30 together with the displayed observation. For (ii), let $Y, Y^* \in \mathcal{Y}_c$ with $D_X Y^* = G$ and put $\Delta = Y - Y^*$, so that $E(Y) = D_X \Delta$. The block of index 0 of Δ vanishes at $L = 0$, so it is not a nonzero constant. If $\nu_t \Delta = 0$, its leading block Δ_0 is a nonzero polynomial with $\Delta_0(0) = 0$, hence $\deg \Delta_0 \geq 1$ and $\partial_L \Delta_0 \neq 0$ in characteristic zero; if $\nu_t \Delta = n \geq 1$, then $(\partial_L - n)\Delta_n \neq 0$ because $\partial_L - n$ is injective on $\mathbb{K}[L]$. In both cases the block formula gives $\nu_t(D_X \Delta) = \nu_t(\Delta) + 1$. Uniqueness of the solution in $\mathcal{Y}_c$ is the case $\Delta = 0$. $\square$

Remark 26.19 (Every derivative-based certificate inherits the same defect). The Leibniz test $D_X H - (F D_X G + G D_X F)$ for a candidate product H , and the chain-rule test for a candidate composition, are residuals of the form $D_X(\text{error})$. By Theorem 25.105 they have gauge 1 *except* when the leading block of the error is a constant sitting at index 0, which they annihilate. A derivative-based test therefore cannot see an error in $\mathbb{K}$, which is exactly the error an antiderivative can commit. Pair them with one gauge-zero test.

Table 9: Defining equations, their gauges, and where they are proved. The gauge is the number of blocks by which the residual test must be run beyond the cutoff to be certified (Theorem 26.2).

Operation and candidate	Defining equation E	Gauge	Reference
Quotient Q of A/B	$BQ - A$	$\nu_t B$	Theorem 26.11
Reciprocal C of B^{-1}	$BC - 1$	$\nu_t B$	Theorem 26.11
Rational power D of $B^{p/q}$, $\nu_t B = 0$	$D^q - B^p$	0	Theorem 26.15
Logarithm Λ of $\log(1 + U)$	$\exp \Lambda - (1 + U)$	0	Theorem 26.13(i)
Logarithm Λ , differential form	$(1 + U)\partial_t \Lambda - \partial_t U$	-1	Theorem 26.13(ii)

Operation and candidate	Defining equation E	Gauge	Reference
Exponential $\mathcal{E}$ of $\exp U$	$\log \mathcal{E} - U$	0	Theorem 26.13(iii)
Composition H of $F \circ G$, $G \in \mathcal{G}_{\text{poly}, \mathbb{K}}$	$H \circ G^{(-1)\circ} - F$	0	Theorem 26.16
Reversion G of $F^{(-1)\circ}$, right	$F \circ G - X$	0	Theorem 26.5
Reversion G of $F^{(-1)\circ}$, left	$G \circ F - X$	0	Theorem 26.5
Omega truncation ω_N	$\omega_N + \log \omega_N - X$	0	Theorem 18.13
Antiderivative Y with de- clared constant	$D_X Y - G$	1 on $\mathcal{Y}_c$	Theorem 26.18
Antiderivative Y , con- stant undeclared	$D_X Y - G$	none	Theorem 26.18(i)

26.4 A specification for a self-checking implementation

The tests below are chosen so that each one fails on a specific error that this volume documents, and so that no test is checked by the code path it tests. They are stated as a specification: an implementation conforms when it runs all of them at every build, over the declared coefficient ring, with exact arithmetic.

Two of them need an object to test against. The Wright omega expansion is the canonical choice, because it is the reversion of the single map $X + L$, its blocks are the Lambert polynomials, and it satisfies four mutually independent identities: the reversion residual ([Theorem 18.13](#)), the integral recurrence (348), the Stirling formula ([Theorem 19.4](#)), and the block recurrence of the differentiated equation, which we now record because it is the only one of the four that exercises D_X and the truncated product rather than ∂_L alone.

Lemma 26.20 (The block recurrence of the omega equation, and the constant it cannot see). *Let $\mathcal{Q} = \sum_{n \geq 0} q_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $\Omega = X + \mathcal{Q} \in \mathcal{G}_{\text{poly}, \mathbb{K}}$.*

1. *If $\Omega + \log \Omega = X + c$ for some $c \in \mathbb{K}$, then*

$$(1 + t + t\mathcal{Q}) D_X \mathcal{Q} = -t. \quad (497)$$

2. *Conversely, (497) holds if and only if $\partial_L q_0 = -1$ and, for every $n \geq 1$,*

$$(\partial_L - n)q_n = -(\partial_L - n + 1)q_{n-1} - \sum_{j=1}^n q_{n-j} (\partial_L - j + 1)q_{j-1}. \quad (498)$$

In particular (497) determines q_n uniquely from $q_0, \dots, q_{n-1}$ for every $n \geq 1$, and determines q_0 only up to an additive constant.

3. *The solutions of (497) in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ are exactly the corrections $(X + L - c)^{(-1)\circ} - X$ for $c \in \mathbb{K}$, one for each value of $q_0(0) = c$; only $c = 0$ solves $\Omega + \log \Omega = X$, and the residual of [Theorem 18.13](#) detects $c \neq 0$ already in the block of index 0.*

Proof. (i) Apply D_X to $\Omega + \log \Omega = X + c$. By [Theorem 8.10](#), $D_X \log \Omega = (D_X \Omega) \Omega^{-1}$, and $D_X(X + c) = 1$, so $(D_X \Omega)(\Omega + 1) = \Omega$. Substituting $\Omega = t^{-1}(1 + t\mathcal{Q})$, $\Omega + 1 = t^{-1}(1 + t + t\mathcal{Q})$ and $D_X \Omega = 1 + D_X \mathcal{Q}$, and multiplying by t , gives $(1 + D_X \mathcal{Q})(1 + t + t\mathcal{Q}) = 1 + t\mathcal{Q}$, which rearranges to (497).

(ii) By the block formula, $D_X \mathcal{Q} = \sum_{m \geq 1} d_m t^m$ with $d_m = (\partial_L - m + 1)q_{m-1}$, and $d_0 = 0$. Extracting the coefficient of t^m in (497) gives

$$d_m + d_{m-1} + \sum_{j=1}^{m-1} q_{m-1-j} d_j = -[m = 1] \quad (m \geq 1),$$

where $[m = 1]$ is the Iverson bracket and the sum is empty for $m = 1$. The case $m = 1$ is $d_1 = -1$, that is $\partial_L q_0 = -1$; the case $m = n + 1 \geq 2$ is (498) after substituting the values of the d_j . Since $\partial_L - n$ is bijective on $\mathbb{K}[L]$ for $n \neq 0$ (Theorem 25.30), q_n is determined; and $\partial_L q_0 = -1$ has solution set $-L + \mathbb{K}$ in characteristic zero.

(iii) For $c \in \mathbb{K}$ the map $F_c = X + L - c$ lies in $\mathcal{G}_{\text{poly}, \mathbb{K}}$, so $\Omega_c = F_c^{(-1)\circ}$ exists and is unique (Theorem 14.11), and $F_c \circ \Omega_c = X$ reads $\Omega_c + \log \Omega_c = X + c$; by (i) its correction solves (497), and its block of index 0 satisfies $q_0 + L = c$, so $q_0(0) = c$. Conversely a solution of (497) has $q_0 = -L + c$ for some $c \in \mathbb{K}$ by (ii), and then agrees with the correction of Ω_c block by block, again by the uniqueness in (ii). Finally $\Omega_c + \log \Omega_c - X = c$, which is the block of index 0 of the residual. $\square$

Remark 26.21. Theorem 19.2 records the recurrence for the Lambert polynomials in its intrinsic two-term form, which is the right form for proving things about them. Theorem 26.20 is the form an implementation can run without knowing the family in advance: it is a recurrence for the *blocks of a series*, driven by D_X and one truncated product, and it therefore tests exactly the primitives of Sections 25.4 and 25.7. Its resonance at $n = 0$ is not an artefact of the derivation: it is Theorem 25.30 in the smallest instance anybody actually computes.

Example 26.22 (Hand-checkable test vectors). Every quantity below can be verified with pencil and paper, which is the point of a test vector.

1. *Lambert blocks.* $\mathbb{L}_0^{\text{Lam}} = -L$, $\mathbb{L}_1^{\text{Lam}} = L$, $\mathbb{L}_2^{\text{Lam}} = \frac{1}{2}L^2 - L$, $\mathbb{L}_3^{\text{Lam}} = \frac{1}{3}L^3 - \frac{3}{2}L^2 + L$. At $n = 3$ the recurrence (498) reads

$$(\partial_L - 3)\mathbb{L}_3^{\text{Lam}} = -L^3 + \frac{11}{2}L^2 - 6L + 1,$$

and its right-hand side,

$$-(\partial_L - 2)\mathbb{L}_2^{\text{Lam}} - [\mathbb{L}_2^{\text{Lam}} \partial_L \mathbb{L}_0^{\text{Lam}} + \mathbb{L}_1^{\text{Lam}} (\partial_L - 1)\mathbb{L}_1^{\text{Lam}} + \mathbb{L}_0^{\text{Lam}} (\partial_L - 2)\mathbb{L}_2^{\text{Lam}}],$$

is the same polynomial; both are computed from the displayed $\mathbb{L}_n^{\text{Lam}}$ in four lines.

2. *Composition through the logarithmic generator.* With $H = 1$, the Taylor engine gives $D_X L = t$, $D_X^2 L = -t^2$, $D_X^3 L = 2t^3$, hence

$$L \circ (X + 1) = L + t - \frac{1}{2}t^2 + \frac{1}{3}t^3 + \mathcal{O}_t^{\text{form}}(t^4) = L + \log(1 + t) + \mathcal{O}_t^{\text{form}}(t^4),$$

in agreement with the generator formula Theorem 10.22. An engine that returns L has substituted into the wrong generator.

3. *Guard block.* $F = X$ exactly, and G known to cutoff 1, its only stored block being the constant 1 at index 0: the sound output cutoff for FG is $\min\{+\infty, 1 - 1\} = 0$, so a routine that returns the block of index 0 of FG has returned an undetermined block (Theorem 25.13).
4. *Nonunit denominator.* $(L + t)^{-1} = \sum_{n \geq 0} (-1)^n t^n L^{-n-1}$, an element of $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ and not of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$; the identity is checked by multiplying out and telescoping. A routine that returns a polynomial answer here has violated (P4) of Theorem 25.9.
5. *Resonance.* $D_X Y = t$ has solution set $L + \mathbb{K}$; a routine that returns L without recording the choice is nondeterministic in the sense of Theorem 26.24.
6. *Sampling is not a zero test.* The residual block $\prod_{i=1}^s (L - \lambda_i)$ vanishes at every sample point $\lambda_1, \dots, \lambda_s$ and is nonzero (Theorem 26.3).

Table 10: The conformance suite. Each test is a checked identity; the third column names the error class it is there to catch, with the diagnosis in [Section 27](#).

	Checked identity	Catches
T1	Declared cutoffs of $+$, $\times$, $/$, $\circ$, D_X equal those of Theorem 25.10 on random inputs, including inputs of negative order	the “modulo t^N in, modulo t^N out” fallacy; Section 27.2
T2	$\text{Trunc}_{<N} F$ retains the block of index $N-1$ and discards the block of index N	inclusive/exclusive confusion; Section 27.1
T3	$BQ - A$ vanishes below $N - \nu_t(B)$ after a quotient at cutoff N	a gauge dropped from the division test; Theorem 26.12
T4	$\exp \log(1+U) = 1+U$ and $\log \exp U = U$ to the declared cutoff	off-by-one in the two recurrences of Theorem 25.26
T5	$D_X(F \circ G) = ((D_X F) \circ G) D_X G$ on the single input $F = L$	substitution into the wrong generator; Theorem 26.9
T6	Taylor engine and generator engine agree to the declared cutoff	an error in either engine, but not in a wrong convention shared by both
T7	$H \circ G^{(-1)\circ} = F$ to the declared cutoff	composition errors that the consistency tests miss; Theorem 26.16
T8	Both $F \circ G - X$ and $G \circ F - X$ vanish below $\min\{N_A, N_B\}$, and their leading blocks agree	a reversion certified by the residual that drove it; Theorem 26.7
T9	The four reversion strategies of Section 25.9 agree to a common cutoff	strategy-specific indexing errors, in particular the Lagrange stopping rule; Section 27.1
T10	$\omega_N + \log \omega_N - X$ has order exactly $N+1$ and leading block $-L_{N+1}^{\text{Lam}}$	any error in the Lambert blocks, and a truncation off by one, which shows as order N or $N+2$; Theorem 18.13
T11	The blocks produced by (498), by (348) and by Theorem 19.4 coincide	an error in D_X , in the truncated product, or in the Stirling coefficients; Theorem 26.20
T12	$D_X Y - G = 0$ and the recorded resonant constant is reproduced	a lost constant of integration; Section 27.6
T13	Residual blocks are tested by canonical form, and a planted residual $\prod_i (L - \lambda_i)$ is rejected	numerical sampling used as a zero test; Theorem 26.3
T14	Two runs on the same input, and runs with permuted input orderings, produce byte-identical canonical forms	nondeterminism; Section 26.5

Remark 26.23 (What a passing suite does not establish). Every test above is a formal identity in the coefficient ring. A complete pass establishes that the computed truncations satisfy the defining equations below their declared cutoffs. It establishes nothing about any function: the passage to an analytic statement needs a realization theorem ([Theorems 10.32](#) and [10.35](#)), a regime, and a derivative bound ([Theorems 14.21](#) and [18.17](#)). Nor does a pass establish that the answer is what the user wanted: the class, the branch and the coefficient ring are inputs, and a computation performed in the wrong class can be internally impeccable. [Section 27.9](#) exists for that reason.

26.5 Determinism and reproducibility

Definition 26.24 (Deterministic routine, reproducible computation). A routine is *deterministic* when its returned representative is a function of its input representatives together with the declared branch

and normalisation data — and, in particular, is independent of the order in which stored blocks are visited, of any hashing, and of any internally chosen constant. A computation is *reproducible* when its record

(series, R , cutoff, order bound, residual and its order, branch and normalisation data)

is returned in full, so that a second run, or a different implementation, can be compared with it item by item.

Proposition 26.25 (Soundness pins the answer). *Let φ be single-valued on denotations, and let R be sound for φ with declared cutoff N_{out} , returning w . Then for every admissible input tuple $(F_1, \dots, F_k)$ in the denotations,*

$$w = \text{Trunc}_{<N_{\text{out}}} \varphi(F_1, \dots, F_k).$$

Consequently any two sound routines for φ that declare the same cutoff return the same value, and if in addition both are canonical in the sense of (P2) of Theorem 25.9, they return the same stored object.

Proof. Write $z = (F_1, \dots, F_k)$. Soundness says $\varphi(z) \in \text{Den}(w, N_{\text{out}})$, which by Theorem 25.3 says $\text{Trunc}_{<N_{\text{out}}} \varphi(z) = w$. The right-hand side does not depend on the routine, and by hypothesis it does not depend on which element of the denotations was chosen either. Canonicity makes the stored representation of w unique. $\square$

Corollary 26.26 (Reproducibility is a theorem, and its failure is a diagnosis). *A conforming implementation — sound, canonical, with a declared cutoff that is a function of the declared input data — is deterministic on every operation whose φ is single-valued on denotations. Consequently, if two runs of a conforming implementation disagree, then at least one of the following holds, and these are the only possibilities:*

1. *the operation is not single-valued — an antiderivative (Theorem 25.30), a root or a fractional power (Theorem 26.15), a logarithm of a constant, or a branch choice — and the choice was made internally instead of being an input;*
2. *the declared cutoff is not a function of the declared input data, for instance because it was taken from a stale stored order bound (Theorem 25.4);*
3. *the returned object is not in canonical form, for instance because a zero block was retained or a rational coefficient was left unnormalised;*
4. *the implementation is unsound.*

Proof. The positive statement is Theorem 26.25: with (P1), (P2) and a cutoff determined by the inputs, the returned object is $\text{Trunc}_{<N_{\text{out}}} \varphi(\cdot)$ in canonical form, hence the same in both runs. For the list, negate the hypotheses one at a time: the proof used exactly single-valuedness of φ on the denotations, the fact that N_{out} is the same in both runs, canonicity, and soundness. A disagreement therefore falsifies one of them, and each of the four items is the failure of one. $\square$

A common trap

Three habits break Theorem 26.26 in practice and are worth naming. Iterating over a hash map makes the accumulation order depend on the hash seed; with exact arithmetic the mathematical value is unaffected but the canonical form is reached by a different route and any timing-dependent truncation heuristic becomes visible, which is why the ordered map of Theorem 25.1 is part of the contract and not an optimisation. Floating-point coefficients break the exact zero test of Theorem 26.3, hence break canonicity, hence break determinism. And a global “simplification” pass applied to coefficients outside the declared R can silently change the type, which is failure (P4).

Section summary

A certificate is a defining equation plus a checked valuation, and it certifies a coset up to the equation’s gauge. Every primitive of [Section 25](#) has a faithful equation with a known gauge, except the antiderivative, whose equation is blind to a constant – which is exactly the resonance. Both reversion residuals certify the same blocks; the reason to compute both is that a residual which drove the iteration certifies nothing. Zero tests are comparisons of canonical forms, never numerical samples. Reproducibility is then not a matter of discipline but a corollary of soundness, canonicity, and a cutoff that is a function of the inputs.

27 Failure modes, diagnostics, and a decision procedure

Polynomial–logarithmic computations are forgiving at leading order and unforgiving after it. Each of the eight failure modes below is one that the six source drafts either commit, warn against without a diagnosis, or warn against with the wrong diagnosis. Each is presented in the same shape: *symptom*, the observable behaviour; *diagnosis*, the theorem that was violated; *minimal reproducing example*, the smallest instance in which the error is visible, chosen so that it can be checked by hand and installed as a regression test; and *repair*, what the contract must say so that the error cannot recur silently. The section closes with a decision procedure: given a problem, which class, which algorithm, which certificate.

27.1 Truncation-convention off-by-one

Symptom. The last requested block is missing, or is present but wrong; two implementations agree in every block but the last; a residual has order one less than predicted, or one more; a loop that should run to R runs to $R - 1$ and the answer is correct except at the most expensive block, the one nobody checks by hand.

Diagnosis. Two conventions are in circulation and they differ by one block:

$$\text{Trunc}_{<N} F = \sum_{n < N} p_n(L) t^n \quad \text{against} \quad [F]_{\leq N} = \text{Trunc}_{<N+1} F,$$

the first adopted here and the second used by source 6. The conversion is harmless in a statement and fatal in a loop bound, because a loop bound is where the two conventions meet an index. The three places in this volume where the discrepancy is real are the Taylor budget ([Theorem 25.31](#)), the Lagrange stopping rule ([Theorem 25.40](#)), and the certifying residual, which for correctness *through* block N must be a $\mathcal{O}_t^{\text{form}}(t^{N+1})$ and not a $\mathcal{O}_t^{\text{form}}(t^N)$ ([Theorem 14.16](#)).

Minimal reproducing example. Reverting $F = X + L$, whose correction has $h = \nu_t(L) = 0$, by Lagrange–Bürmann ([Theorem 25.40](#)). Each summand contributes exactly one block: for $r \geq 1$,

$$\frac{(-1)^r}{r!} D_X^{r-1}(L^r) = \mathbb{L}_{r-1}^{\text{Lam}}(L) t^{r-1},$$

which is [Equation \(336\)](#). Asking for the blocks of index 0, 1, 2 – that is, for the exclusive cutoff $N = 3$ – therefore requires $r = 1, 2, 3$, which is $R = \lceil (N + 1)/(h + 1) \rceil - 1 = 3$. A loop written to the inclusive reading of “through block N ” but run against an exclusive cutoff stops at $r = 2$ and returns

$$-L + Lt + 0 \cdot t^2 \quad \text{instead of} \quad -L + Lt + \left(\frac{1}{2}L^2 - L\right)t^2,$$

losing exactly $\mathbb{L}_2^{\text{Lam}}$. The right residual then has order 2 instead of the predicted 3, which is the diagnosis printed in one line.

Repair. Carry the cutoff in the type, never in the prose; convert once, at the boundary, with $[F]_{\leq N} = \text{Trunc}_{<N+1} F$; and let the residual order be the arbiter, since by [Theorem 26.51](#) it is exactly the first index at which the answer is wrong.

27.2 Reading a block the input data does not determine

Symptom. A routine returns a block that changes when an input block *beyond* the requested cutoff is changed; two runs with different but equally admissible completions of the same input disagree; a computation is reproducible on every test case in which the inputs happen to have order ≥ 0 and wrong on the first Laurent input.

Diagnosis. The folklore rule “both inputs modulo t^N , output modulo t^N ” is the product law [Theorem 25.102](#) specialised to $m_1 = m_2 = 0$. The law is $N_{\text{out}} = \min \{N_1 + m_2, N_2 + m_1\}$, which is *better* than the folklore rule when a factor has positive order and *worse* as soon as a factor has a pole. Worse means: the routine returns a block it has no right to return, and the failure is silent, because the returned block is a perfectly good function of the stored data – it is just not a function of the mathematical object.

Minimal reproducing example. $F = X = t^{-1}$, known exactly; G known to exclusive cutoff 1, its only stored block being 1 at index 0. The sound cutoff for FG is $\min \{+\infty, 1 + (-1)\} = 0$: only the block of index -1 is determined. Indeed $G_1 = 1$ and $G_2 = 1 + t$ have the same representative and $FG_1 = t^{-1}$, $FG_2 = t^{-1} + 1$ differ at index 0. A routine that declares cutoff 1 here has invented the block 0.

Repair. Compute N_{out} from the derived order bounds ([Theorem 25.4](#)), never from the input cutoffs alone; report the relative precision ρ alongside the cutoff, since by [Theorem 25.11](#) it is ρ , not N , that the multiplicative operations conserve; and run test T1 of [Table 10](#) with at least one input of negative order, which is the only kind of input that exhibits the bug.

27.3 Substituting into the wrong generator

Symptom. The leading block is right and every later block is wrong, so the answer looks nearly correct and passes any eyeball test; the residual has order exactly one more than the leading block; ring-based consistency tests all pass.

Diagnosis. Composition moves *both* generators ([Theorem 10.22](#)):

$$t \circ G = G^{-1}, \quad L \circ G = \log G.$$

Only the second is easy to forget, and forgetting it is not an approximation but a different operation – one that is still a $\mathbb{K}[L]$ -algebra homomorphism, hence still passes every multiplicative test ([Theorem 26.9\(i\)](#)). A related error is to “substitute $t \mapsto t + H$ ”, which is not an operation on $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ at all: it is the substitution appropriate to a series in a variable that is not the reciprocal of the argument.

Minimal reproducing example. Two, at different depths.

1. One generator, one block. $L \circ (X + 1) = L + \log(1 + t) = L + t - \frac{1}{2}t^2 + \frac{1}{3}t^3 + \mathcal{O}_t^{\text{form}}(t^4)$ ([Theorem 26.22\(ii\)](#)). An engine that returns L has forgotten the logarithmic generator. The chain-rule test on this single input detects it ([Theorem 26.9\(ii\)](#)).
2. A whole reversion. The engine Ψ_G of [Theorem 26.8](#) reverts $X + L$ to $X - L$, which is the correct block of index 0 of ω and nothing else, while its own residual vanishes identically.

Repair. Implement composition only through the generator images ([Theorem 25.34](#)) or the Taylor engine ([Theorem 25.32](#)), never by textual substitution; run the chain-rule test T5 and the faithful certificate T7 of [Table 10](#); and never certify a reversion with the residual that drove it ([Theorem 26.7](#)).

27.4 Treating a formal tail as an analytic bound

Symptom. A numerical error bound of the form CX^{-N} that experiment violates; a claim that a truncation “converges”; a remainder quoted as $\mathcal{O}(X^{-3})$ that in fact contains $X^{-3}(\log X)^{100}$; an identity between formal objects presented as an identity between functions.

Diagnosis. $\mathcal{O}_t^{\text{form}}(t^N)$ is a statement about coefficient support and nothing else ([Theorems 5.7](#) and [5.8](#)). It becomes a statement about a function only through a realization ([Theorem 10.32](#)) together

with a theorem that transports it ([Theorems 10.35, 14.21 and 18.17](#)). Two further confusions travel with this one: formal summability ([Theorem 5.21](#)) is a statement about valuations and implies nothing about numerical convergence; and a bare $\mathcal{O}$ with no printed regime silently invites the reader to supply the wrong one.

Proposition 27.1 (Two ways in which the formal tail says nothing analytic). 1. Realizations are not unique. *Let f be a function on a ray $(X_0, +\infty)$ with $f \approx F$ in the sense of [Theorem 10.32](#), and let $g(X) = f(X) + e^{-X}$. Then $g \approx F$ as well, and $f \neq g$ everywhere. Consequently no formal statement about F determines any value of a realization, and the difference of two realizations is invisible to every block.*

2. Formal summability is not convergence. *The family $(n! t^n)_{n \geq 0}$ is formally summable, with sum $F = \sum_{n \geq 0} n! t^n \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, yet for every real $X > 0$ the numerical series $\sum_{n \geq 0} n! X^{-n}$ diverges.*

Proof. (i) For every fixed N , $e^{-X} = o_{X \rightarrow +\infty}(X^{-N})$, since $X^N e^{-X} \rightarrow 0$; hence g has the same asymptotic remainders as f at every order, which is the definition of $g \approx F$. And $e^{-X} > 0$, so $f \neq g$ pointwise.

(ii) Formal summability requires $\nu_t(n! t^n) = n \rightarrow \infty$ ([Theorem 5.21](#)), which holds, so the sum exists in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$; each block is a single scalar and no coefficient receives infinitely many contributions. For the divergence, fix $X > 0$; then $n! X^{-n} \rightarrow \infty$ as $n \rightarrow \infty$, because $((n+1)! X^{-n-1}) / (n! X^{-n}) = (n+1)/X \geq 2$ for $n \geq 2X$. A series whose terms do not tend to zero diverges. $\square$

Minimal reproducing example. [Theorem 27.1\(ii\)](#) with $N = 3$: the “remainder after three blocks” is formally $\mathcal{O}_t^{\text{form}}(t^3)$, while numerically, at $X = 10$, the terms $n! 10^{-n}$ decrease only until $n = 9$ and $n = 10$, where both equal $9!/10^9 = 3.63 \times 10^{-4}$, and grow without bound afterwards. The formal tail marker predicts none of this and forbids none of it: it is not a statement about numbers.

Repair. Never write a bare $\mathcal{O}$ for an asymptotic claim. Print the regime, as in

$$\mathcal{O}_{X \rightarrow +\infty}((\log X)^d X^{-3});$$

state which realization theorem is being used, and if none is, say that the statement is formal. [Section 18.6](#) is the model of the four ingredients an analytic statement needs.

27.5 Dividing by a nonunit

Symptom. Coefficients acquire denominators in L ; block degrees go negative; a routine that promised polynomial coefficients returns something else, or — worse — silently truncates an infinite series in L^{-1} and returns a polynomial that is wrong in every block; a residual test fails at the leading block for no visible reason.

Diagnosis. The units of $\mathbb{K}[L]$ are $\mathbb{K}^\times$ ([Theorem 7.1](#)); hence an element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is a unit if and only if its leading block is a nonzero scalar ([Theorem 7.3](#)). The reciprocal recurrence divides by the leading block at every step, so a leading block of positive L -degree takes the computation out of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and into $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$, or after expansion into $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$. This is not a defect of the algorithm but information about the problem, and (P4) of [Theorem 25.9](#) requires it to appear in the returned type.

Proposition 27.2 (The smallest nonunit, and the two rings it forces). 1. $B = L + t$ is not a unit of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$.

2. In $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}} = \mathbb{K}(L)((t))$ it is a unit, with $B^{-1} = \sum_{n \geq 0} (-1)^n L^{-n-1} t^n$.

3. For $B' = L + 1 + t$ the reciprocal is $\sum_{n \geq 0} (-1)^n (L + 1)^{-n-1} t^n$, and each of its blocks, when expanded in $\mathbb{K}((L^{-1}))$, is an infinite series; so the passage from $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ to $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ introduces a second, inner cutoff, which the type must carry and the certificate must quote.

Proof. (i) The leading block of B is L , which is not a unit of $\mathbb{K}[L]$ because a product $Lq(L)$ has degree ≥ 1 ; now apply [Theorem 7.3](#).

(ii) Multiply out and telescope:

$$\begin{aligned} (L+t) \sum_{n \geq 0} (-1)^n L^{-n-1} t^n &= \sum_{n \geq 0} (-1)^n L^{-n} t^n + \sum_{n \geq 0} (-1)^n L^{-n-1} t^{n+1} \\ &= \sum_{n \geq 0} (-1)^n L^{-n} t^n - \sum_{m \geq 1} (-1)^m L^{-m} t^m = 1. \end{aligned}$$

Every block lies in $\mathbb{K}(L)$ and the family is summable, the n -th term having $\nu_t = n$.

(iii) The same telescoping with L replaced by $L+1$ gives the displayed reciprocal. Expanding $(L+1)^{-n-1} = L^{-n-1}(1+L^{-1})^{-n-1} = \sum_{k \geq 0} \binom{-n-1}{k} L^{-n-1-k}$ exhibits infinitely many nonzero inner coefficients, since $\binom{-n-1}{k} = (-1)^k \binom{n+k}{k} \neq 0$ for every k . $\square$

Minimal reproducing example. [Theorem 27.2](#) with the quotient $1/(L+t)$ requested in $\mathcal{PS}_{\text{poly}, \mathbb{K}}$: a conforming routine refuses and reports the obstruction; a nonconforming one returns $L^{-1} - L^{-2}t + \dots$ typed as a polynomial-coefficient object, and every downstream degree computation is then meaningless.

Repair. Test the leading block for unit-hood before dividing, taking logarithms, or forming a general power; enlarge R explicitly and record it in the returned type; and remember that in $\mathcal{PS}_{\text{itLaur}, \mathbb{K}}$ the log-degree profile of [Theorem 6.24](#) is no longer the relevant invariant — the inner valuation is.

27.6 Losing the constant of integration at the resonant block

Symptom. Two runs differ by an additive constant; an identity that held before an integration fails after it; a derived expansion is a shift of the intended one, so that every block of positive index is right and the answer is still wrong; or, in the rational class, a routine returns nonsense where the true answer needs $\log L$.

Diagnosis. The block equation for $D_X Y = G$ is $(\partial_L - n)y_n = g_{n+1}$. The operator $\partial_L - n$ is bijective on $\mathbb{K}[L]$ for $n \neq 0$ and equals ∂_L for $n = 0$, whose kernel is $\mathbb{K}$ ([Theorem 25.30](#)). So the block of index 0 — the resonant block — is determined only up to a constant, and the defining equation $D_X Y - G$ is blind to that constant ([Theorem 26.18](#)). In $\mathcal{PS}_{\text{rat}, \mathbb{K}}$ the same block carries an obstruction as well as a freedom: the equation $\partial_L y_0 = g_1$ is solvable in $\mathbb{K}(L)$ only when g_1 has zero residue ([Theorems 8.23 and 9.23](#)).

Minimal reproducing example. Two, one for the freedom and one for the obstruction.

1. *Freedom.* Recover the ω expansion from the differentiated equation (497) instead of from $\Omega + \log \Omega = X$. By [Theorem 26.20\(ii\)–\(iii\)](#) the differentiated equation determines every block of index ≥ 1 uniquely and determines the block of index 0 only up to a constant c , and each c gives the genuine reversion of the genuinely different map $X + L - c$. Every block of positive index is then wrong as well, since it is a block of the wrong map: solving (498) at $n = 1$ with $q_0 = c - L$ gives $(\partial_L - 1)q_1 = 1 + c - L$, hence $q_1 = L - c$. For $c = 1$ the computed answer begins $1 - L + (L - 1)t$ where the intended one begins $-L + Lt$; no block is correct, and the residual $\Omega + \log \Omega - X = c$ exposes it in the block of index 0.
2. *Obstruction.* Antidifferentiate $G = t/L \in \mathcal{PS}_{\text{rat}, \mathbb{K}}$. The block equation at $n = 0$ is $\partial_L y_0 = 1/L$, which has no solution in $\mathbb{K}(L)$, because the derivative of a rational function has zero residue at 0 ([Theorem 8.23](#)) while $1/L$ has residue 1. The class must be enlarged to admit $\log L$, which is [Theorem 9.21](#).

Repair. Treat the antiderivative as an operation with an input, not as a function: the resonant constant is part of the specification, it is returned in the record of [Theorem 26.24](#), and the certificate is the pair (T12 of [Table 10](#)). Before integrating in $\mathcal{PS}_{\text{rat}, \mathbb{K}}$, test the residue.

27.7 Assuming a bound uniform in L

Symptom. A truncation that is provably correct as $X \rightarrow +\infty$ behaves badly at every argument one can actually compute with; the “next correction” is larger than the term it corrects; an error constant that was quoted once turns out to depend on the number of blocks retained, or on the log-degrees of the discarded ones.

Diagnosis. The dominance $t^{n+1}L^D \prec t^n$ is a statement about $X \rightarrow +\infty$ at *fixed* D (Theorems 2.7 and 3.13). Since $L = \log X$ grows, the crossover at which the monomial $L^D X^{-1}$ actually becomes small is at $\log X > x^*(D)$, where $x^*(D)$ is the largest root of $x = D \log x$, so $x^*(D) \sim D \log D$. A profile that grows with the block index therefore pushes the crossover out with it, and no constant covers all blocks at once.

Proposition 27.3 (No constant is uniform in the number of blocks). *Fix an integer $D \geq 1$, take $\mathbb{K} = \mathbb{R}$, and let $F = \sum_{n \geq 0} L^{Dn} t^n \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, so that $d_F(n) = Dn$. All statements below about numbers are on the positive real ray with the real logarithm.*

1. *On the ray $X > 1$ with $L = \log X$, the numerical series converges exactly when $L^D < X$, with sum $f(X) = X/(X - L^D)$, and $f \approx F$ as $X \rightarrow +\infty$.*
2. *For each fixed $N \geq 1$, the remainder is $f(X) - \sum_{n < N} L^{Dn} X^{-n} = (L^D X^{-1})^N (1 - L^D X^{-1})^{-1}$, which is $o_{X \rightarrow +\infty}(X^{-N+\varepsilon})$ for every $\varepsilon > 0$ but is not $\mathcal{O}_{X \rightarrow +\infty}(X^{-N})$.*
3. *There is no pair (C, X_0) with $|f(X) - \sum_{n < N} L^{Dn} X^{-n}| \leq C X^{-N}$ for all $N \geq 0$ and all $X \geq X_0$.*
4. *The blocks decrease in size at a given X if and only if $\log X > x^*(D)$. For $D = 10$, $x^* = 35.77 \dots$; so at $X = 10^{15}$ the block of index 1 exceeds the block of index 0 by the factor $L^{10}/X = 2.42 \dots$, and the expansion becomes numerically decreasing only for $X > 3.4 \times 10^{15}$.*

Proof. (i) The series is geometric with ratio $L^D X^{-1}$, whence the convergence criterion and the sum $1/(1 - L^D X^{-1}) = X/(X - L^D)$. For the asymptotic statement, fix $N \geq 0$; by (ii) with $N + 1$ in place of N ,

$$f(X) - \sum_{n \leq N} L^{Dn} X^{-n} = \frac{L^{D(N+1)} X^{-(N+1)}}{1 - L^D X^{-1}} = o_{X \rightarrow +\infty}(X^{-N}),$$

because $L^{D(N+1)} X^{-1} = (\log X)^{D(N+1)}/X \rightarrow 0$ and the denominator tends to 1. This is the level-wise condition of Theorem 10.32, so $f \approx F$.

(ii) Sum the geometric tail. The remainder equals $L^{DN} X^{-N} (1 - L^D X^{-1})^{-1} \geq L^{DN} X^{-N}$ for X large, and $L^{DN} \rightarrow \infty$, so the remainder divided by X^{-N} is unbounded as $X \rightarrow +\infty$; hence it is not $\mathcal{O}_{X \rightarrow +\infty}(X^{-N})$. The little-o statement is the estimate just made.

(iii) Fix any $X \geq \max\{X_0, e^2\}$, so $L = \log X \geq 2$ and in particular $L^D \geq 2$. The remainder at cutoff N is at least $L^{DN} X^{-N}$, so the asserted bound would force $C \geq L^{DN} \geq 2^{DN}$ for every N , which is impossible for a finite C .

(iv) The block of index n evaluates to $(L^D X^{-1})^n$, so the blocks decrease if and only if $L^D < X$, that is $D \log L < L$ with $L = \log X$, that is $L > x^*(D)$, where $x^*(D)$ is the largest solution of $x = D \log x$; the function $x - D \log x$ is negative at $x = D$ for $D \geq 3$ and tends to $+\infty$, and its only critical point is $x = D$, so it has exactly one root above D . For $D = 10$, $x - 10 \log x$ is -0.55 at $x = 35$ and $+0.17$ at $x = 36$, so $x^* = 35.77 \dots$ and $e^{x^*} = 3.4 \times 10^{15}$. At $X = 10^{15}$ one has $L = 34.54$ and $L^{10}/X = 2.42$. $\square$

Minimal reproducing example. Theorem 27.3(iv) with $D = 10$ at $X = 10^{15}$: a series that is a perfectly valid asymptotic expansion, whose first correction is 2.4 times its leading term.

Repair. Quote the log-degrees with the error term, as in $\mathcal{O}_{X \rightarrow +\infty}((\log X)^d X^{-N})$; state the constant’s dependence on N and on the profile; and, for numerical work, compare the actual magnitude of the first discarded block at the actual argument rather than trusting the ordering of the monomials.

27.8 Using a degree bound that was a hypothesis

Symptom. A fixed-width coefficient array overflows, or silently truncates the top logarithmic coefficients; a cost estimate is off by a factor that grows with N ; a proof of a degree law is quoted without its hypothesis.

Diagnosis. Two distinct claims are easily confused. The degree law for a reversion ([Theorem 16.21](#)) is a theorem *under the hypothesis* that the profile of the input is linearly bounded, and it bounds the output profile in terms of that hypothesis. What is not true, under any hypothesis, is that the output profile is bounded by the input profile: a single reciprocal already destroys that, and reversion destroys it spectacularly. The log-degree profile is output data ([Theorems 4.12, 6.24 and 7.15](#)), to be measured and reported, never an input constant.

Proposition 27.4 (Bounded input, unbounded output; and the exact scalar cost). 1. One reciprocal suffices. $B = 1 + Lt$ has $d_B(0) = 0$, $d_B(1) = 1$ and no other nonzero block, while $B^{-1} = \sum_{n \geq 0} (-1)^n L^n t^n$ has $d_{B^{-1}}(n) = n$ for every n .

2. Reversion. $F = X + L$ has a single nonzero correction block, of degree 1, and $F^{(-1)^\circ} = \omega$ has correction blocks $\mathbb{L}_n^{\text{Lam}}(L)$ with

$$d_{F^{(-1)^\circ} - X}(0) = 1, \quad d_{F^{(-1)^\circ} - X}(n) = n \quad (n \geq 1).$$

3. Cost. For $N \geq 1$, the schoolbook truncated product of two series with the profile of (ii) costs exactly

$$\binom{N+3}{4} + N^2 + N + 1$$

scalar multiplications, which exceeds the naive “ $\binom{N+1}{2}$ multiplications” count of [Theorem 25.17](#) by a factor $\Theta(N^2)$.

Proof. (i) The geometric series identity $(1 + Lt) \sum_{n \geq 0} (-1)^n L^n t^n = 1$ is immediate by telescoping, and $\deg((-1)^n L^n) = n$.

(ii) $(X + L)^{(-1)^\circ} = \omega = X + \sum_{n \geq 0} \mathbb{L}_n^{\text{Lam}}(L) t^n$ is [Theorem 18.10](#). For the degrees, $\mathbb{L}_0^{\text{Lam}}(u) = -u$ has degree 1, and for $n \geq 1$, $\mathbb{L}_n^{\text{Lam}} = ((-1)^{n+1}/(n+1)!) \Delta_{0,n} u^{n+1}$ by [Theorem 19.1](#), where $\Delta_{0,n} = \prod_{j=0}^{n-1} (\partial_u - j)$ contains the factor ∂_u exactly once, at $j = 0$; by [Theorem 25.29](#) the degree therefore drops by exactly one, giving $\deg \mathbb{L}_n^{\text{Lam}} = (n+1) - 1 = n$. This is the degree law of [Theorem 19.7](#).

(iii) Write $\mathcal{P} = F^{(-1)^\circ} - X$. By [Theorem 25.17](#) the scalar count is $\sum_{i+j < N} w(i)w(j)$ with $w(n) = d_{\mathcal{P}}(n) + 1$, so $w(0) = 2$ and $w(n) = n + 1$ for $n \geq 1$; no block vanishes, since $\mathbb{L}_n^{\text{Lam}} \neq 0$ for all n . Using the Iverson bracket, which is 1 when the enclosed statement holds and 0 otherwise, we have $w(i) = (i + 1) + [i = 0]$ and hence

$$w(i)w(j) = (i + 1)(j + 1) + (i + 1)[j = 0] + (j + 1)[i = 0] + [i = 0][j = 0].$$

Summing over $i, j \geq 0$ with $i + j < N$: the first term gives $\binom{N+3}{4}$ by [Theorem 25.18\(ii\)](#); the second gives $\sum_{i=0}^{N-1} (i + 1) = N(N + 1)/2$, and the third the same by symmetry; the fourth gives 1. The total is $\binom{N+3}{4} + N(N + 1) + 1 = \binom{N+3}{4} + N^2 + N + 1$. Since $\binom{N+3}{4} = N^4/24 + \mathcal{O}(N^3)$ while $\binom{N+1}{2} = N^2/2 + \mathcal{O}(N)$, the ratio is $\Theta(N^2)$. $\square$

Minimal reproducing example. [Theorem 27.4\(i\)](#): an implementation that stores coefficient polynomials in an array of width $\max_n d_B(n) + 1 = 2$, inferred from the input, loses every logarithmic coefficient above degree 1 from the third block of the reciprocal on.

Repair. Store blocks as variable-length objects ([Theorem 25.1](#)); measure the profile of the output and return it; quote costs in the coefficient-operation model and multiply by the measured profile factor, as [Theorem 25.18](#) requires; and when invoking a degree law, print its hypothesis.

27.9 A decision procedure

The eight failure modes have a common shape: a decision was made implicitly. The procedure below makes each of them explicit and in the order in which the information becomes available. Its first three steps decide the class, its next two the algorithm and the precision budget, its last three the certificate and the status of the answer. Nothing in it is optional: skipping step 3 is failure mode [Section 27.5](#), skipping step 4 is [Section 27.2](#), and skipping step 7 is [Section 27.4](#).

Algorithm 27.5 (Decision procedure: class, algorithm, certificate). **Input.** A problem: an expansion, an implicit equation, an inverse, or a composition; and a target amount of information.

Steps.

1. *Name the regime and the branch.* Which variable tends where, on which ray or sector, with which logarithm branch. For a problem at $x \rightarrow 0^+$, transport it to $X \rightarrow +\infty$ by the dictionary of [Theorem 13.18](#) before anything else; for a problem at a finite critical point, stop — the scale is ramified, not polynomial–logarithmic.
2. *Solve the leading balance and normalise.* Factor the dominant monomial. If the map is monomial-leading, $cX^\varrho(1 + \dots)$ with $\varrho > 0$ ([Theorem 17.1](#)), reduce to the near-identity case by [Theorem 17.6](#); a leading logarithmic factor is first put in normal form by [Theorem 17.7](#). If the leading derivative degenerates, the problem is not near-identity reversion at all and needs a ramified scale.
3. *Compute the generator images before anything else.* Form $t \circ G = G^{-1}$ and $L \circ G = \log G$ and read the smallest closed class off [Table 11](#). Adopt that class; if it is larger than the one the problem was posed in, say so in the output type. This is the generator-first rule, and it is much safer than expanding first and inferring the class from the mess.
4. *Fix the precision budget backwards.* From the target cutoff, propagate the required input cutoffs through [Theorem 25.10](#), remembering that the guard blocks are not a safety margin but part of the law ([Theorem 25.13](#)), and that relative precision, not absolute cutoff, is what the multiplicative operations conserve ([Theorem 25.11](#)).
5. *Choose the algorithm* from [Table 12](#), by how many blocks are wanted, whether they are wanted lazily, and whether composition is available and cheap.
6. *Choose the certificate* from [Table 9](#), compute the residual, and compare its order with the declared cutoff *plus the gauge*. If the answer was produced by an iteration, the certificate must not be the residual that drove it ([Theorem 26.7](#)).
7. *Declare the status of the answer.* Formal — an identity of coefficients, and nothing more; asymptotic — name the realization and the transfer theorem ([Theorems 10.32, 10.35, 14.21 and 18.17](#)); or convergent — name the convergence theorem, and note that this volume proves none for the ω series ([Theorem 18.20](#)).
8. *Return the record* of [Theorem 26.24](#): series, ring, cutoff, order bound, residual and its order, branch and normalisation data.

Table 11: Step 3 of [Theorem 27.5](#): what the generator images force.
The class named is the smallest one in this volume that is closed for the operation; a smaller one can be used only by refusing the operation.

Observed data	Smallest closed class	Authority
Sums, products, derivatives of blocks with polynomial coefficients	$\mathcal{PL}_{\text{poly}, \mathbb{K}} = \mathbb{K}[L]((t))$	Theorems 6.7 and 9.1

Observed data	Smallest closed class	Authority
Inner map $X + H$, $H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$; reversion of it	$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, group $\mathcal{G}_{\text{poly}, \mathbb{K}}$	Theorems 12.7 and 14.11
Inner map $cX^\varrho(1+U)$, $\varrho \in \mathbb{N}_{>0}$, $\log c \in \mathbb{K}$	$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$	Theorem 10.22
Division, logarithm or general power whose leading block is a nonconstant polynomial	$\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}} = \mathbb{K}(L)((t))$	Theorems 7.3 and 7.20
A rational coefficient expanded for large L ; a second, inner cutoff appears	$\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ $\mathbb{K}((L^{-1}))((t))$	= Theorems 7.37 and 27.2
$\varrho \in \mathbb{Q}_{>0} \setminus \mathbb{Z}$, or a formal root	Puiseux–log tower	Theorems 3.25, 6.38 and 10.17
ϱ irrational, or a non-cyclic exponent group	Hahn field of Γ	Theorem 3.19
Leading term cX^aL^b with $b \neq 0$; or an inner map with a logarithmic factor	depth two: $\log \log X$	Theorems 13.8, 17.4 and 17.5
Composition with $\log$, or with a map of logarithmic depth r	depth- r tower	Theorems 13.10 and 13.14
An exponential of a block that is not a constant; composition with $\exp$	outside every logarithmic class	Theorems 8.16 and 13.11
Antiderivative in $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ of a block with nonzero residue	adjoin $\log L$	Theorems 9.21 and 9.23

Table 12: Steps 5 and 6 of Theorem 27.5: which algorithm, and which certificate goes with it. Costs are in the coefficient-operation model of Theorem 25.15 and must be multiplied by the profile factor of Theorem 25.18 to describe a machine.

Goal	Algorithm	Certificate
Product at cutoff N	Theorem 25.19	Leibniz test, plus T1 of Table 10
Reciprocal, small N	Theorem 25.20, $\binom{\rho}{2}$ products	$BC - 1$, gauge $\nu_t B$
Reciprocal, large N	Theorem 25.21, $\lceil \log_2 \rho \rceil$ stages	same, recomputed at each stage
Quotient	Theorem 25.22	$BQ - A$, gauge $\nu_t B$
Power, α a small nonnegative integer	binary powering, Theorem 25.24	$D^q - B^p$ with $q = 1$
Power, general α ; formal roots	recurrence Equation (491)	$D^q - B^p$ and a leading-scalar check (Theorem 26.15)
$\log(1+U)$, $\exp U$, $\nu_t U \geq 1$	Theorem 25.27	Theorem 26.13, either form
Composition, inner near-identity	Theorem 25.32	T5, T6, and Theorem 26.16
Composition, inner monomial-leading	Theorem 25.34	the same three
Reversion, a few blocks, or lazily	Theorem 25.38	left residual, never the driving one
Reversion, large N , composition cheap	Theorem 25.39 with Theorem 15.16	both residuals, plus the abort test
Reversion, closed form or a map whose powers simplify	Theorem 25.40	both residuals; for $X + L$, T10
Reversion with only a composition primitive	Theorem 25.37	both residuals

Goal	Algorithm	Certificate
Antiderivative	Theorem 25.30 , blockwise	$D_X Y - G$ with the constant declared (T12)
Mixed or unknown leading data	Theorem 25.41	as at each stage

Table 13: Diagnostic index: an observed symptom and the section that explains it.

Symptom	Diagnosis
Exactly one block, the last, disagrees between two implementations	Section 27.1
A residual has order one less, or one more, than predicted	Section 27.1
An output block changes when an input block beyond the cutoff changes	Section 27.2
Everything is correct until the first input with a pole	Section 27.2
The leading block is right and every later block is wrong	Section 27.3
Multiplicative tests pass and the chain-rule test fails	Section 27.3
A numerical error bound is violated in experiment	Section 27.4 or Section 27.7
Denominators in L appear where polynomials were promised	Section 27.5
Two runs differ by an additive constant	Section 27.6
Every block of positive index is a block of a shifted problem	Section 27.6
The first correction is larger than the term it corrects	Section 27.7
A coefficient array overflows, or a cost estimate is off by a growing factor	Section 27.8
Two runs of the same input differ at all	Theorem 26.26

Section summary

The eight failure modes are eight implicit decisions. Truncation convention and guard blocks are decisions about indices; the generator images and the unit test are decisions about the class; the resonant constant and the branch are decisions about which of several correct answers is wanted; the formal tail and the uniformity in L are decisions about what kind of statement is being made; and the degree bound is a decision about which hypotheses are in force. [Theorem 27.5](#) orders them so that each is made when its information is available, and [Table 9](#) says what to compute afterwards to find out whether the decisions were kept.

Part VIII

Extensions and the wider landscape

28 Generalized power–log classes, nested logarithms, and multiserries

Parts I–IV worked inside a deliberately small ambient: integer powers of $t = X^{-1}$, polynomial (or rational, or Laurent) coefficients in the single logarithm $L = \log X$, and no exponentials. Every closure failure recorded there — the reciprocal of a nonunit ([Theorem 7.3](#)), a fractional leading exponent ([Theorem 17.3](#)), a logarithm inside the leading monomial ([Theorem 17.4](#)), the exponential of an element of negative outer order ([Theorem 8.16](#)) — names a specific enlargement. This section makes each enlargement precise and, for each, says exactly which theorems of Parts I–IV survive verbatim, which survive after a hypothesis is changed, and which become false. The failures are proved, because they are the statements a computation can actually violate; a class that is described only by what it contains is a class one will silently misuse.

28.1 Five independent axes

It is useful to separate the enlargements into five axes that can be turned independently. Nothing in this subsection is a theorem; it is the map of the territory, and each entry is made precise below or in the place indicated.

Table 14: The five axes of enlargement. The class of Parts I–IV is the corner ($\Gamma = \mathbb{Z}$, $C = \mathbb{K}[L]$, $r = 1$, height 0, well-based).

Axis	What is enlarged	Forced by	Analysed in
Exponent group in t	$\mathbb{Z} \rightsquigarrow s^{-1}\mathbb{Z} \rightsquigarrow \Gamma \subseteq \mathbb{R}$	fractional or irrational leading exponent (Theorem 17.3)	Section 28.3 , Section 28.4
Coefficient ring in L	$\mathbb{K}[L] \rightsquigarrow \mathbb{K}(L) \rightsquigarrow \mathbb{K}((L^{-1}))$	division by a nonunit (Theorem 7.3); the axis stops short of $\sqrt{L}$ (Theorem 28.17)	Section 3 , Section 28.4
Logarithmic depth r	one logarithm $\rightsquigarrow L_1, \dots, L_r$	a logarithm in the leading monomial (Theorem 17.4); an unmatched residue (Theorem 28.20)	Section 13 , Section 28.5
Exponential height	height 0 $\rightsquigarrow$ transmonomials e^p	exp of an element of negative outer order (Theorem 8.16 , Theorem 29.1)	Section 29
Support class	grid-based $\rightsquigarrow$ left-finite $\rightsquigarrow$ well-based	infinite summation; completion (Theorem 28.10)	Section 28.2

The axes are not interchangeable

An enlargement along one axis never repairs an obstruction on another. Ramifying the exponent group of t does not produce $\sqrt{L}$ ([Theorem 28.17](#)); adding logarithmic depth does not produce e^{-X} ([Theorem 29.2](#)); enlarging the coefficient ring does not remove the obstruction $\sigma \geq 1$ of [Theorem 17.3](#); and no amount of exponential height supplies the antiderivative of $1/(L_0 L_1 \cdots L_r)$ ([Theorem 28.20](#)). A closure test must therefore be run once per axis, and a diagnosis that does not name the axis is not a diagnosis.

28.2 Supports: what well-ordering does, and what it does not do

Throughout this subsection Γ is a totally ordered abelian group, written additively, and a subset $S \subseteq \Gamma$ is thought of as a set of exponents of t : a *larger* exponent means a *smaller* monomial, so the least element of a support is its dominant monomial, and the $\succ$ -greatest element of [Theorem 3.15](#) (c) is the least exponent. We keep the exponent formulation because it makes the group structure visible; the reader who prefers monomials should read every “least” as “dominant”.

Definition 28.1 (Three support classes). Let $S \subseteq \Gamma$.

- (a) S is *well based* (equivalently, well ordered) if every nonempty subset of S has a least element.
- (b) S is *left-finite* if $S \cap (-\infty, M)$ is finite for every $M \in \Gamma$.
- (c) S is *grid-based* if there are $\mathbf{n} \in \Gamma$ and finitely many $\alpha_1, \dots, \alpha_k \in \Gamma_{>0}$ with

$$S \subseteq \mathbf{n} + \{n_1\alpha_1 + \dots + n_k\alpha_k : n_1, \dots, n_k \in \mathbb{N}_0\}.$$

Each of the three classes is closed under passing to subsets, under finite unions, and under translation.

Verification of the closure claims. Subsets and translates are immediate in all three cases; for (c) a translate changes only $\mathbf{n}$. Finite unions: for (a) and (b) this is clear. For (c), let $S \subseteq \mathbf{n} + \sum \mathbb{N}_0\alpha_i$ and $T \subseteq \mathbf{m} + \sum \mathbb{N}_0\beta_j$, put $\mathbf{p} = \min(\mathbf{n}, \mathbf{m})$, and take as generating set the α_i , the β_j , and whichever of $\mathbf{n} - \mathbf{p}$, $\mathbf{m} - \mathbf{p}$ is nonzero; both of the latter are ≥ 0 , so all generators used are > 0 , and $S \cup T \subseteq \mathbf{p} + \sum \mathbb{N}_0(\dots)$. $\square$

The whole content of [Theorem 3.15](#) is that for $\Gamma = \mathbb{Z}$ the three classes coincide with “bounded below”. They separate as soon as Γ is dense, or as soon as Γ fails to be archimedean; [Theorem 28.8](#) says exactly how.

28.2.1 Well-ordering and local finiteness

The next lemma is the reason the catalogue calls the support condition “what makes multiplication locally finite”. We give the proof in full because the sources assert it and because part (iii), which is what actually makes reciprocals and geometric series legal, is not the same statement as parts (i)–(ii).

Lemma 28.2 (Convolution of well-based supports). *Let Γ be a totally ordered abelian group.*

- (i) *If $S, T \subseteq \Gamma$ are well based then so is $S + T = \{\sigma + \tau : \sigma \in S, \tau \in T\}$.*
- (ii) *If S, T are well based then for every $\gamma \in \Gamma$ the fiber $R_\gamma = \{(\sigma, \tau) \in S \times T : \sigma + \tau = \gamma\}$ is finite.*
- (iii) *Let $S \subseteq \Gamma_{>0}$ be well based and, for $k \geq 0$, let $S^{(k)} = S + \dots + S$ (k summands; $S^{(0)} = \{0\}$). Then every $S^{(k)}$ is well based, and for every k and every γ only finitely many multisets of k elements of S have sum γ . If moreover Γ is archimedean, then $S^* = \bigcup_{k \geq 0} S^{(k)}$ is well based and every $\gamma \in S^*$ has, in total, only finitely many representations as a sum of elements of S .*

Proof. We use twice the following remark: *every sequence in a well-based set has a non-decreasing subsequence*. Indeed, given $(\sigma_k)_{k \geq 1}$ in the well-based set S , the set $\{\sigma_k : k \geq 1\}$ has a least element; let k_1 attain it. Having chosen $k_1 < \dots < k_i$, let $k_{i+1} > k_i$ attain the least element of the nonempty set $\{\sigma_k : k > k_i\}$. Since $\{\sigma_k : k > k_i\} \subseteq \{\sigma_k : k > k_{i-1}\}$ while σ_{k_i} is least in the larger set, $\sigma_{k_i} \leq \sigma_{k_{i+1}}$.

(ii) Suppose R_γ is infinite. As $\tau = \gamma - \sigma$ is determined by σ , infinitely many distinct σ occur; choose pairwise distinct $\sigma_k \in S$ with $\gamma - \sigma_k \in T$. Passing to a subsequence we may assume $\sigma_{k_1} \leq \sigma_{k_2} \leq \dots$, and since they are pairwise distinct the inequalities are strict. Then $\gamma - \sigma_{k_i}$ is a strictly decreasing sequence in T , so $\{\gamma - \sigma_{k_i} : i \geq 1\}$ is a nonempty subset of T with no least element, contradicting well-basedness of T .

(i) Suppose $\gamma_1 > \gamma_2 > \dots$ is strictly decreasing in $S + T$, say $\gamma_k = \sigma_k + \tau_k$. Pass to a subsequence along which (σ_k) is non-decreasing; along it $\tau_k = \gamma_k - \sigma_k$ is strictly decreasing in T , a contradiction as

before. Hence $S + T$ has no infinite strictly decreasing sequence; a totally ordered set with that property is well based, since a nonempty subset without a least element would produce one by dependent choice.

(iii) That $S^{(k)}$ is well based follows from (i) by induction on k . For the representation count, induct on k ; the case $k \leq 1$ is trivial (the empty multiset for $\gamma = 0$, and at most one singleton otherwise). Let $k \geq 2$ and suppose infinitely many multisets $\{s_1 \leq s_2 \leq \dots \leq s_k\} \subseteq S$ have sum γ . If the value s_1 took only finitely many values, then for one of them infinitely many multisets of size $k - 1$ would sum to $\gamma - s_1$, contradicting the inductive hypothesis. So infinitely many distinct values of s_1 occur; by the remark, choose a strictly increasing sequence $s_1^{(1)} < s_1^{(2)} < \dots$ of such values. Then $\gamma - s_1^{(i)}$ is a strictly decreasing sequence in the well-based set $S^{(k-1)}$, a contradiction.

Now assume Γ archimedean. Since S is well based and $S \subseteq \Gamma_{>0}$, it has a least element $\delta > 0$ (we may assume $S \neq \emptyset$, else there is nothing to prove). A sum of k elements of S is $\geq k\delta$; by archimedeanity, for each M only finitely many k satisfy $k\delta < M$. Hence for $\emptyset \neq T \subseteq S^*$ and $\tau \in T$ we have $T \cap (-\infty, \tau] \subseteq \bigcup_{k \leq K} S^{(k)}$ for some finite K , a finite union of well-based sets and therefore well based; its least element is least in T . The total representation count of γ is the sum, over the finitely many admissible k , of the counts already bounded. $\square$

Remark 28.3 (Neumann’s lemma without archimedeanity). The last clause of [Theorem 28.2](#) (iii) is true for an arbitrary totally ordered abelian group — this is Neumann’s lemma, and it is what makes a Hahn series ring over any such group a field when the coefficients form a field [19, 13, 33]. The proof above uses archimedeanity only in the last paragraph, and $\Gamma \subseteq \mathbb{R}$ is archimedean, so every use of the clause in this volume with a *one-dimensional* exponent group is covered by the proof given. The monomial groups $\mathcal{M}_r(\Gamma)$ of [Theorem 13.1](#) are *not* archimedean for $r \geq 1$, and there we quote Neumann’s lemma rather than reprove it. Every *argument* carried out in [Section 28.5](#) uses only [Theorem 28.2](#) (i)–(ii); the one further input, used only through the word “field”, is that $\mathcal{H}_r(\mathbb{K}, \Gamma)$ is a field, which is [Theorem 13.1](#) together with Neumann’s lemma and is quoted, not reproved.

Proposition 28.4 (Well-basedness is the largest admissible support class). *Let Γ be a totally ordered abelian group and let $T \subseteq \Gamma$ fail to be well based. Then there is a well-based $S \subseteq \Gamma$, of order type ω , and a $\gamma \in \Gamma$ whose fiber $R_\gamma \subseteq S \times T$ is infinite. Consequently, if $\mathcal{F}$ is a family of subsets of Γ that contains every well-based set and for which every convolution fiber $R_\gamma \subseteq S \times T$ with $S, T \in \mathcal{F}$ is finite, then $\mathcal{F}$ is exactly the family of well-based subsets of Γ .*

Proof. If T is not well based it contains a strictly decreasing sequence $\tau_1 > \tau_2 > \dots$. Put $S = \{-\tau_k : k \geq 1\}$; then $-\tau_1 < -\tau_2 < \dots$, so S is well based of order type ω . With $\gamma = 0$, every pair $(-\tau_k, \tau_k)$ lies in R_0 , which is therefore infinite. For the second assertion, a member $T \in \mathcal{F}$ failing to be well based would, together with the well-based $S \in \mathcal{F}$ just produced, violate the finiteness hypothesis. $\square$

Remark 28.5 (What the sharp statement is, and what it is not). [Theorem 28.2](#) says well-basedness suffices, and [Theorem 28.4](#) says it cannot be relaxed *as a condition on a class of supports*. It is not necessary for a single product. Take $\Gamma = \mathbb{Q}$ and

$$T = \left\{1 + \frac{1}{k} : k \in \mathbb{N}_{>0}\right\} = \left\{2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \dots\right\},$$

which is strictly decreasing and therefore not well based. Every element of $T + T$ lies in $(2, 4]$, and a representation $\gamma = (1 + \frac{1}{k}) + (1 + \frac{1}{l})$ with $k \leq l$ forces $\frac{1}{k} + \frac{1}{l} = \gamma - 2$, hence $\frac{1}{k} \geq \frac{\gamma-2}{2}$ and $\frac{1}{k} < \gamma - 2$, so

$$(\gamma - 2)^{-1} < k \leq 2(\gamma - 2)^{-1},$$

which leaves finitely many k , and each k determines $l = (\gamma - 2 - \frac{1}{k})^{-1}$ uniquely. Thus T is a non-well-based support for which the square of the corresponding formal sum has finite coefficient fibers. What fails is *closure*: by [Theorem 28.4](#) the same T has an infinite fiber against the perfectly legitimate support $\{-1 - \frac{1}{k} : k \geq 1\}$, whose elements increase to -1 . A support condition is a condition on a class, not on an element, and this is why the axiom is stated as well-basedness.

28.2.2 Left-finite supports

Left-finiteness is the exact condition under which the finite bookkeeping of Parts II and VI – truncation, block recursion, sparse representation – survives a change of exponent group. It is [Theorem 3.15 \(d\)](#) with $\mathbb{Z}$ replaced by Γ , strengthened by the cofinality clause of [Theorem 28.6 \(i\)](#), which is automatic over $\mathbb{Z}$ and is not automatic over a dense Γ .

Proposition 28.6 (Left-finite supports). *Let Γ be a totally ordered abelian group with $\Gamma \neq \{0\}$ and let $S \subseteq \Gamma$.*

- (i) *S is left-finite if and only if it can be enumerated as a strictly increasing sequence $\gamma_0 < \gamma_1 < \dots$, finite or of order type ω , which in the infinite case is cofinal in Γ : for every $M \in \Gamma$ some $\gamma_k > M$.*
- (ii) *Left-finite $\Rightarrow$ well based, and the converse fails whenever Γ is densely ordered: for $\Gamma = \mathbb{Q}$ the set $S = \{1 - \frac{1}{k} : k \in \mathbb{N}_{>0}\}$ is well based of order type ω and is not left-finite.*
- (iii) *If S, T are left-finite then so are $S \cup T$, $S + T$ and every translate of S , and all convolution fibers are finite.*
- (iv) *Assume in addition that Γ is archimedean. If $S \subseteq \Gamma_{>0}$ is left-finite then S^* is left-finite, and every $\gamma \in S^*$ arises from only finitely many multisets of elements of S .*

Proof. (i) Let S be left-finite and nonempty. For $\mu \in S$ the set $S \cap (-\infty, \mu]$ is finite and nonempty, so S has a minimum γ_0 ; removing it leaves a left-finite set, and iterating produces a strictly increasing enumeration in which $\mu \in S$ is reached after at most $\#(S \cap (-\infty, \mu])$ steps, so every element occurs. If the enumeration is infinite and $\gamma_k \leq M$ for all k , pick $g \in \Gamma_{>0}$; then $S \cap (-\infty, M + g) = S$ is infinite, contradicting left-finiteness. Conversely, given such an enumeration and $M \in \Gamma$, choose K with $\gamma_K > M$; then $S \cap (-\infty, M) \subseteq \{\gamma_0, \dots, \gamma_{K-1}\}$.

(ii) If $\emptyset \neq T \subseteq S$ with S left-finite, pick $\tau \in T$; then $T \cap (-\infty, \tau]$ is finite and nonempty, so it has a least element, which is least in T . For the counterexample, the displayed S is enumerated by the strictly increasing sequence $1 - \frac{1}{k}$, so every nonempty subset has a least element; but $S \subseteq (-\infty, 1)$, so $S \cap (-\infty, 1) = S$ is infinite. It is exactly the cofinality clause of (i) that fails. In a general densely ordered Γ the same construction works: pick $a < b$ and, recursively, γ_{k+1} strictly between γ_k and b starting from $\gamma_1 \in (a, b)$; then $\{\gamma_k\}$ is well based of order type ω , and it is not left-finite because it is infinite and contained in $(-\infty, b)$.

(iii) Unions and translates are immediate. A nonempty left-finite set has a minimum by (i); write $\sigma_0 = \min S$, $\tau_0 = \min T$. If $\sigma + \tau < M$ with $\sigma \in S$, $\tau \in T$, then $\sigma < M - \tau_0$ and $\tau < M - \sigma_0$, so σ, τ range over finite sets; hence $(S + T) \cap (-\infty, M)$ is finite and every fiber is finite.

(iv) Put $\delta = \min S > 0$. A sum of k elements of S is at least $k\delta$, so a sum $< M$ forces $k\delta < M$, and archimedeanity leaves finitely many such k . Each summand of such a sum is $< M - (k - 1)\delta \leq M$, so lies in the finite set $S \cap (-\infty, M)$. There are therefore finitely many admissible multisets, which proves both assertions. $\square$

Remark 28.7 (Archimedean is not decoration). Part (iv) fails for non-archimedean Γ . Take $\Gamma = \mathbb{Z} \times \mathbb{Z}$ ordered lexicographically and $S = \{(0, 1)\}$, which is left-finite, indeed finite. Then $S^* = \{(0, k) : k \in \mathbb{N}_0\}$ is infinite and contained in $(-\infty, (1, 0))$, so it is not left-finite. The failure is precisely the failure of $\mathbb{Z}\delta$ to be cofinal, and it is the same phenomenon that would break separatedness of a filtration indexed by Γ . This is why [Theorem 8.11](#) fixes $\Gamma \subseteq \mathbb{R}$; the restriction cannot be dropped without redoing every summability argument of Parts II–IV. It also means that in the nested-logarithmic setting of [Section 28.5](#), where the monomial group is non-archimedean by construction, “left-finite” is the wrong notion and only well-basedness survives.

Theorem 28.8 (The support hierarchy, and where it collapses). *Let Γ be a totally ordered abelian group.*

- (i) *Grid-based $\Rightarrow$ well based, always.*
- (ii) *Left-finite $\Rightarrow$ well based, always.*

- (iii) If Γ is archimedean, grid-based $\Rightarrow$ left-finite.
- (iv) If Γ is not archimedean, grid-based $\nRightarrow$ left-finite.
- (v) If Γ is archimedean and not densely ordered — equivalently, by Hölder’s theorem, Γ is cyclic — then all three classes coincide with “bounded below”.
- (vi) For $\Gamma = \mathbb{Q}$ both inclusions in grid-based $\subsetneq$ left-finite $\subsetneq$ well based are strict.

Proof. (i) Let $S \subseteq \mathfrak{n} + M$ with $M = \{\sum_i n_i \alpha_i : n_i \in \mathbb{N}_0\}$ and $\alpha_i \in \Gamma_{>0}$. It suffices to show M is well based. Suppose $v^{(1)}, v^{(2)}, \dots \in \mathbb{N}_0^k$ give a strictly decreasing sequence $\sum_i v_i^{(m)} \alpha_i$. By Dickson’s lemma the componentwise order on $\mathbb{N}_0^k$ is a well-quasi-order, so there are $m < m'$ with $v^{(m)} \leq v^{(m')}$ componentwise; as all $\alpha_i > 0$ this gives $\sum_i v_i^{(m)} \alpha_i \leq \sum_i v_i^{(m')} \alpha_i$, contradicting strict decrease. So M has no infinite strictly decreasing sequence and is well based; well-basedness passes to subsets and translates.

(ii) [Theorem 28.6](#) (ii).

(iii) With S and M as in (i) and $M' \in \Gamma$ given, an element $\mathfrak{n} + \sum_i n_i \alpha_i < M'$ has $n_i \alpha_i < M' - \mathfrak{n}$ for each i , because the omitted summands are ≥ 0 ; archimedeanity leaves finitely many n_i for each i , hence finitely many tuples and finitely many values.

(iv) In $\mathbb{Z} \times \mathbb{Z}$ ordered lexicographically the set $M = \{(0, n) : n \in \mathbb{N}_0\}$ is grid-based with $\mathfrak{n} = (0, 0)$ and the single generator $\alpha_1 = (0, 1) > 0$, and it is not left-finite, as in [Theorem 28.7](#).

(v) An archimedean group is order-isomorphic to a subgroup of $\mathbb{R}$ (Hölder), and a subgroup of $\mathbb{R}$ is either dense or cyclic. If $\Gamma = \mathbb{Z}\delta$ is cyclic, a set bounded below is finite below every bound, hence left-finite; it is then well based by (ii); and it is grid-based with generator δ and base point any lower bound. Conversely each class consists of sets bounded below: a well-based set has a least element if nonempty. This is [Theorem 3.15](#) in the case $\Gamma = \mathbb{Z}$.

(vi) Left-finite $\subsetneq$ well based is [Theorem 28.6](#) (ii). For grid-based $\subsetneq$ left-finite, take

$$S = \{k + \frac{1}{k} : k \in \mathbb{N}_{>0}\} \subseteq \mathbb{Q}.$$

It is enumerated by a strictly increasing sequence tending to $+\infty$, hence left-finite by [Theorem 28.6](#) (i). If it were grid-based, it would lie in the subgroup of $\mathbb{Q}$ generated by $\mathfrak{n}, \alpha_1, \dots, \alpha_k$; a finitely generated subgroup of $\mathbb{Q}$ is cyclic, say $N^{-1}\mathbb{Z}$, and $k + \frac{1}{k} \in N^{-1}\mathbb{Z}$ forces $N/k \in \mathbb{Z}$, that is $k \mid N$, which holds for only finitely many k . $\square$

Structural checkpoint

Three different things are being asked of a support, and they are not the same thing. *Well based* is what makes the product defined at all ([Theorem 28.2](#)), and by [Theorem 28.4](#) it is the weakest condition that can be imposed on a whole class. *Left-finite* is what makes truncation produce finite data ([Theorem 28.10](#) (ii)). *Grid-based* is what makes the exponent arithmetic itself finitary: exponents are integer vectors on a fixed finite basis. In the integer-exponent world of Parts I–IV all three collapse into “bounded below”, which is exactly why the distinction is invisible there and unavoidable here.

28.3 Real exponents: Hahn power–log scales

[Theorem 8.11](#) already introduced the power–log scale $\mathcal{H}(C, \Gamma)$: well-based supports in an ordered group $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{R}$, blocks in $C \in \{\mathbb{K}'[L], \mathbb{K}'(L), \mathbb{K}'((L^{-1}))\}$, and the derivation (134). We now isolate inside it the subring on which the finite bookkeeping of Parts II and VI is still meaningful, and prove that outside that subring it is not.

Definition 28.9 (The left-finite power–log scale). Let $\mathcal{H} = \mathcal{H}(C, \Gamma)$ be a power–log scale in the sense of [Theorem 8.11](#). For $F = \sum_{\gamma \in S} p_\gamma t^\gamma$ put $\text{Exp}(F) = \{\gamma \in \Gamma : p_\gamma \neq 0\}$, the *exponent set* of F — the

projection to Γ of $\text{supp}_{\text{coeff}}(F)$. Define

$$\mathcal{H}_{\text{lf}}(C, \Gamma) := \{F \in \mathcal{H}(C, \Gamma) : \text{Exp}(F) \text{ is left-finite}\},$$

and write $C[t^\Gamma]$ for the subring of elements with finite exponent set.

Theorem 28.10 (Truncation, completeness, and what $\mathcal{H}_{\text{lf}}$ is). *Let $\mathcal{H} = \mathcal{H}(C, \Gamma)$ be a power–log scale.*

- (i) $\mathcal{H}_{\text{lf}}(C, \Gamma)$ is a subring of $\mathcal{H}$ containing $C[t^\Gamma]$, stable under D_X , and stable under the inversion of those of its elements that are units of $\mathcal{H}$.
- (ii) For $F \in \mathcal{H}$: every truncation $\text{Trunc}_{<N} F$, $N \in \Gamma$, has finite exponent set if and only if $F \in \mathcal{H}_{\text{lf}}(C, \Gamma)$.
- (iii) Both $\mathcal{H}$ and $\mathcal{H}_{\text{lf}}(C, \Gamma)$ are separated and complete for the filtration $\mathfrak{F}_N = \{F \in \mathcal{H} : \nu_t F \geq N\}$, $N \in \Gamma$, whose topology is metrizable because $\mathbb{Z}$ is cofinal in Γ .
- (iv) $\mathcal{H}_{\text{lf}}(C, \Gamma)$ is the closure of $C[t^\Gamma]$ in $\mathcal{H}$; equivalently it is the completion of $C[t^\Gamma]$ for the induced filtration. If Γ is dense in $\mathbb{R}$ then $\mathcal{H}_{\text{lf}}(C, \Gamma) \subsetneq \mathcal{H}(C, \Gamma)$.

Proof. (i) Closure under sums and products is Theorem 28.6 (iii), the block of a product at t^γ being the finite sum over the fiber, which is finite by Theorem 28.2 (ii). Stability under D_X holds because (134) translates the exponent set by $1 \in \Gamma$. For inverses, let $F \in \mathcal{H}_{\text{lf}}$ be a unit of $\mathcal{H}$ and write $e = \nu_t F$, $p_e = [t^e]F$. If $FG = 1$ then $\nu_t G = -e$ and, by the leading-block law (Theorem 8.12(i) and its proof, or directly: the coefficient of t^0 in FG receives exactly one contribution, from the two leading blocks), $p_e [t^{-e}]G = 1$, so $p_e \in C^\times$. Hence $F = t^e p_e (1 + v)$ with $v = t^{-e} p_e^{-1} F - 1 \in \mathcal{H}_{\text{lf}}$ and $\nu_t v > 0$; then $F^{-1} = t^{-e} p_e^{-1} \sum_{k \geq 0} (-v)^k$. Put $S = \text{Exp}(v) \subseteq \Gamma_{>0}$, which is left-finite. By Theorem 28.6 (iv), S^* is left-finite and each of its elements has finitely many representations, so the family $((-v)^k)_{k \geq 0}$ is summable and its sum has exponent set inside S^* ; translating by $-e$ preserves left-finiteness.

(ii) If $\text{Exp}(F)$ is left-finite then $\text{Exp}(F) \cap (-\infty, N)$, which is $\text{Exp}(\text{Trunc}_{<N} F)$, is finite. Conversely, if every $\text{Trunc}_{<N} F$ has finite exponent set then $\text{Exp}(F) \cap (-\infty, N)$ is finite for every $N \in \Gamma$.

(iii) *Metrizability and separatedness.* Since $1 \in \Gamma \subseteq \mathbb{R}$ and Γ is archimedean, no element of Γ exceeds every integer, so $\mathbb{Z}$ is cofinal in Γ and the countable subfamily $(\mathfrak{F}_N)_{N \in \mathbb{Z}}$ is a neighbourhood basis of 0. If $\nu_t F \geq N$ for every $N \in \mathbb{Z}$ then $F = 0$, for otherwise $\nu_t F \in \Gamma$ would exceed every integer. *Completeness.* Let (F_r) be Cauchy: for each $N \in \mathbb{Z}$ there is $r(N)$ with $\text{Trunc}_{<N} F_r = \text{Trunc}_{<N} F_{r(N)}$ for all $r \geq r(N)$, and we may take $r(N)$ non-decreasing in N . Define F blockwise by $[t^\gamma]F = [t^\gamma]F_{r(N)}$ for any integer $N > \gamma$; the definition is consistent. Then

$$\text{Exp}(F) = \bigcup_{N \in \mathbb{Z}} \left(\text{Exp}(F_{r(N)}) \cap (-\infty, N) \right),$$

which is well based: for $\emptyset \neq T \subseteq \text{Exp}(F)$ pick $\tau \in T$ and an integer $N > \tau$; then $T \cap (-\infty, N)$ is a nonempty subset of the well-based set $\text{Exp}(F_{r(N)})$ and its least element is least in T . So $F \in \mathcal{H}$ and $\nu_t(F - F_r) \geq N$ for $r \geq r(N)$. If all F_r lie in $\mathcal{H}_{\text{lf}}$ then so does F , because $\text{Exp}(F) \cap (-\infty, N) = \text{Exp}(F_{r(N)}) \cap (-\infty, N)$ is finite for every integer N , and every $M \in \Gamma$ is below some integer.

(iv) $\mathcal{H}_{\text{lf}}$ is closed in $\mathcal{H}$ by the last sentence of (iii); it contains $C[t^\Gamma]$; and every $F \in \mathcal{H}_{\text{lf}}$ is the limit of the finitely supported truncations $\text{Trunc}_{<N} F$, $N \in \mathbb{Z}$, by (ii). Conversely a limit of finitely supported elements is left-finite, since $\text{Trunc}_{<N} F = \text{Trunc}_{<N} G$ with G finitely supported makes $\text{Trunc}_{<N} F$ finitely supported. Hence $\mathcal{H}_{\text{lf}}$ is the closure of $C[t^\Gamma]$, and a closed subring containing a subring is its completion when the ambient is complete and separated. Finally, if Γ is dense in $\mathbb{R}$ then $\Gamma \cap (0, 1)$ is infinite; an infinite bounded subset of $\mathbb{R}$ contains a strictly monotone sequence, and by passing to a subsequence we obtain $\gamma_1 < \gamma_2 < \dots$ in $\Gamma \cap (0, 1)$. Put $F = 1 + t + \sum_{k \geq 1} t^{\gamma_k}$. Its exponent set $\{0\} \cup \{\gamma_k : k \geq 1\} \cup \{1\}$ is well based – a nonempty subset containing 0 has least element 0; one meeting $\{\gamma_k\}$ has as least element the γ_k of least index; and the only remaining subset is $\{1\}$ – so $F \in \mathcal{H}$; it is not left-finite, since $\text{Exp}(F) \cap (-\infty, 1)$ is infinite, so $F \notin \mathcal{H}_{\text{lf}}$. $\square$

Corollary 28.11 (The left-finite scale carries the near-identity theory). *Let $\mathcal{H} = \mathcal{H}(C, \Gamma)$ and let $\mathcal{H}_{\geq 0} = \{A \in \mathcal{H} : \nu_t A \geq 0\}$.*

- (i) *Every statement of Section 12 and of Section 14 whose proof uses only (a) that D_X raises ν_t by at least 1, (b) that a family with $\nu_t \rightarrow \infty$ may be summed termwise, (c) that a continuous $\mathbb{K}$ -algebra endomorphism is determined by the image of the monomial t and of L together with the binomial rule for $(1+u)^\gamma$, $\gamma \in \Gamma$, $\nu_t u > 0$, and (d) that ν_t is additive, holds in $\mathcal{H}$ with $\mathcal{P}\mathcal{L}^{\text{pow}, \mathbb{K}}$ replaced by $\mathcal{H}_{\geq 0}$ and the integer filtration by the Γ -filtration $\mathfrak{F}_N$. This covers the substitution operator (Theorem 12.4), the Taylor formula (Theorem 11.6), the filtered gain (Theorem 12.6), the group law (Theorem 12.7), separation and completeness of the contact filtration (Theorem 12.13), and existence and uniqueness of the near-identity inverse (Theorem 14.11).*

- (ii) *The subset*

$$\{X + A : A \in \mathcal{H}_{\text{lf}}(C, \Gamma), \nu_t A \geq 0\}$$

is a subgroup of the group of (i): it is closed under composition and under compositional inversion.

Proof. (i) Inspect the four inputs. D_X raises ν_t by 1 on monomials by (134) and hence on all of $\mathcal{H}$; summability of $(A^k D_X^k F / k!)_{k \geq 0}$ follows from $\nu_t(A^k D_X^k F) \geq \nu_t F + k$, and $\mathbb{Z}$ is cofinal in Γ , so only finitely many k affect any given block; the substitution operator is fixed on a general monomial by $t^\gamma \circ (X + A) = t^\gamma(1 + tA)^{-\gamma}$, the binomial series being summable since $\nu_t(tA) \geq 1$ (Theorem 6.37, Theorem 8.29), and on L by $L \circ (X + A) = L + \log(1 + tA)$; additivity of ν_t is Theorem 8.12(i). Each of the six listed statements uses only these.

(ii) Let $A, B \in \mathcal{H}_{\text{lf}}$ with $\nu_t A, \nu_t B \geq 0$, and put $S_A = \text{Exp}(A)$, $S_B = \text{Exp}(B)$, $S'_B = S_B \cap \Gamma_{>0}$. By the Taylor formula, $A \circ (X + B) = \sum_{k \geq 0} B^k D_X^k A / k!$ and

$$\text{Exp}(B^k D_X^k A) \subseteq (S_A + k) + (S'_B)^* \subseteq (S_A + k) + \Gamma_{\geq 0},$$

because k copies of S_B contribute a sum of elements of S'_B together with some copies of 0. Fix $M \in \Gamma$. A contribution to $(-\infty, M)$ requires $\min S_A + k < M$, so only finitely many k contribute, since $\mathbb{Z}$ is cofinal in the archimedean Γ . For each such k , the set $(S_A + k) + (S'_B)^*$ is left-finite by Theorem 28.6 (iii)–(iv). A finite union of left-finite sets is left-finite, so $A \circ (X + B) \in \mathcal{H}_{\text{lf}}$. Closure under inversion follows: the inverse of $X + A$ is $X + B$ with B the $\mathfrak{F}$ -limit of $B_0 = 0$, $B_{n+1} = -A \circ (X + B_n)$ (Theorem 14.11), every B_n lies in $\mathcal{H}_{\text{lf}}$ by what has just been proved and induction, and $\mathcal{H}_{\text{lf}}$ is closed in $\mathcal{H}$ by Theorem 28.10 (iv). $\square$

What clause (i) deliberately omits

The statements about the *associated graded* of the contact filtration – Theorem 12.14, Theorem 12.15, Theorem 12.16, Theorem 12.18 – are formulated using that $r \mapsto r + 1$ is the successor in the index group. Write $\mathcal{G}_\gamma$ for the step of the contact filtration of Theorem 12.11 indexed by $\gamma \in \Gamma$. For dense Γ the quotients $\mathcal{G}_\gamma / \mathcal{G}_{\gamma+1}$ remain defined, because the gain is the element $1 \in \Gamma$, but they are no longer single steps of the filtration, and the identification of the graded object with a Lie algebra is not the same statement. We do not claim it here. The same caution applies to any argument that enumerates blocks one at a time; see the entry for Theorem 15.2 in Table 15.

Proposition 28.12 (Truncation, and therefore all of Part VI, fails in the general Hahn scale). *Let $\Gamma \subseteq \mathbb{R}$ be dense with $\mathbb{Z} \subseteq \Gamma$, and let $\mathcal{H} = \mathcal{H}(\mathbb{K}[L], \Gamma)$. There is $F \in \mathcal{H}$ with $\nu_t F = 0$ and every block equal to 1 such that:*

- (a) $\text{Trunc}_{<N} F$ has infinite exponent set for every $N \geq 1$ in Γ ;
 (b) $\text{supp}_{\text{coeff}}(F)$ admits no enumeration as a strictly $\succ$ -decreasing sequence of order type at most ω exhausting it, so Theorem 3.15 (d) is false in $\mathcal{H}$, although Theorem 3.15 (c) remains true by definition of $\mathcal{H}$;

- (c) F has no sparse block representation in the sense of [Theorem 25.1](#) at any precision $N \geq 1$; consequently no routine of [Section 25](#) accepts F as input, and the precision calculus of [Theorem 25.10](#) and the cost model of [Theorem 25.17](#) have no meaning for it.

Proof. Take $F = 1 + t + \sum_{k \geq 1} t^{\gamma_k}$ with $\gamma_k \in \Gamma \cap (0, 1)$ strictly increasing, as constructed in the proof of [Theorem 28.10](#) (iv); then $\nu_t F = 0$ and every block is 1. (a) All the γ_k and 0 are $< 1 \leq N$, so $\text{Exp}(\text{Trunc}_{<N} F) \supseteq \{\gamma_k : k \geq 1\}$ is infinite. (b) An enumeration $s_0 \succ s_1 \succ \dots$ of order type at most ω exhausting $\text{Exp}(F)$ would be an increasing enumeration of a set whose order type is at most ω . But $\text{Exp}(F)$ has order type $\omega + 1$: the element 1 is strictly greater than each of the infinitely many elements $0 < \gamma_1 < \gamma_2 < \dots$, so in any exhausting increasing enumeration it would have to be preceded by infinitely many terms and could occupy no finite position. Hence [Theorem 3.15](#) (d) fails. By [Theorem 28.6](#) (i) the correct replacement for (d) in a general Γ is not “order type at most ω ” but left-finiteness, which additionally demands cofinality of the enumeration. (c) A sparse block representation is a *finitely supported* map v together with a cutoff N , and by [Theorem 25.3](#) its denotation is the coset $\sum_{\gamma < N} v(\gamma)t^\gamma + \mathfrak{F}_N$. If this coset contained F then $\text{Trunc}_{<N} F$ would equal the finitely supported $\sum_{\gamma < N} v(\gamma)t^\gamma$, contradicting (a). $\square$

Table 15: Transfer of Parts I–IV and VI to a Hahn power–log scale $\mathcal{H}(C, \Gamma)$ with $\Gamma \subseteq \mathbb{R}$ dense. “Verbatim” means the statement and its proof go through with $n \in \mathbb{Z}$ read as $\gamma \in \Gamma$.

Statement	Status	What changes
Domain, valuation laws (Theorem 4.2)	verbatim	—
Unit criterion (Theorem 7.3)	verbatim	—
Reciprocal of a unit (Theorem 7.7)	modified	the block-by-block recurrence presupposes a successor exponent; replace it by the geometric series, summable by Theorem 28.2 (iii)
t -adic completeness (Theorem 5.18)	verbatim	Theorem 28.10 (iii)
Support characterisation, clause (d) (Theorem 3.15)	false	Theorem 28.12 (b)
Block and repeated-derivative formulas (Theorem 9.3 , Theorem 9.7)	verbatim	the shift is $\partial_L - \gamma$ with $\gamma \in \Gamma$; the successive shifts $\gamma, \gamma + 1, \dots$ stay integrally spaced
Sharp valuation of D_X (Theorem 9.10), kernel	verbatim	—
Antiderivatives (Theorem 9.17)	verbatim	the resonant exponent is still the single value $\gamma = 0$
Exponential boundary (Theorem 8.16)	modified	m may be any element of Γ (Theorem 8.17)
Taylor formula, near-identity group, reversion (Theorem 12.7 , Theorem 14.11)	verbatim	Theorem 28.11
Triangular reversal (Theorem 15.2)	modified	“the next block” is undefined when Γ is dense; one must fix a cofinal sequence in Γ and cancel residuals along it
Lagrange–Bürmann index range $r \leq N + 1$ (Theorem 16.19)	open	the bound is an integrality statement; see Theorem 29.10
Sparse representation, precision calculus, cost model (Theorem 25.1 , Theorem 25.10 , Theorem 25.17)	false	restored verbatim on $\mathcal{H}_{\text{lf}}(C, \Gamma)$

28.4 Puiseux-log classes, and the union over denominators

The Puiseux-log ring $\mathbb{K}[L]((t^{1/s}))$ of [Theorem 3.24](#) and the ramified tower of [Theorem 3.25](#) are the minimal enlargement forced by a rational leading exponent ([Theorem 17.5](#)). At a fixed denominator nothing is lost at all, and the reason is worth stating as a theorem rather than as the folklore “the

derivative formula generalizes without change” of S1, S3 and S5: the ring is not merely similar to $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, it is $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ in a different large variable.

Proposition 28.13 (A fixed denominator is a change of variable, not an enlargement). *Fix $s \in \mathbb{N}_{>0}$ and set*

$$Y := X^{1/s}, \quad \tau := Y^{-1} = t^{1/s}, \quad \Lambda := \log Y = s^{-1}L.$$

Then the $\mathbb{K}$ -linear map sending $t^{r/s}L^j$ to $s^j\tau^r\Lambda^j$ is an isomorphism of ordered $\mathbb{K}$ -algebras

$$\Phi_s : \mathbb{K}[L]((t^{1/s})) \longrightarrow \mathbb{K}[\Lambda]((\tau)),$$

onto the polynomial–logarithmic Laurent ring of Theorem 3.3 built from the generators τ, Λ , and

$$\Phi_s \circ D_X \circ \Phi_s^{-1} = s^{-1}\tau^{s-1}D_Y, \quad (499)$$

where $D_Y = \tau\partial_\Lambda - \tau^2\partial_\tau$ is the distinguished derivation of the target. Consequently every statement of Parts I–IV that mentions only the ring structure, the order, the valuation, the truncation calculus and the derivation up to a unit factor holds verbatim in $\mathbb{K}[L]((t^{1/s}))$. In particular the unit criterion, the reciprocal recurrence, the completeness theorem, the sparse representation and the precision calculus transfer with no change whatever, and

$$D_X(t^\alpha p(L)) = t^{\alpha+1}(\partial_L p - \alpha p), \quad \alpha \in s^{-1}\mathbb{Z}. \quad (500)$$

Proof. Φ_s is the relabelling of generators $t^{1/s} \mapsto \tau$, $L \mapsto s\Lambda$; it is bijective on monomials, hence a $\mathbb{K}$ -algebra isomorphism, and it preserves the dominance order because it preserves the exponent pair (r, j) and the order is lexicographic in that pair. For (499) it suffices to compare on the two generators. On the one hand $D_X\tau = D_X t^{1/s} = -s^{-1}t^{1/s+1} = -s^{-1}\tau^{s+1}$, and $s^{-1}\tau^{s-1}D_Y\tau = -s^{-1}\tau^{s-1}\tau^2 = -s^{-1}\tau^{s+1}$. On the other hand $D_X\Lambda = s^{-1}D_X L = s^{-1}t = s^{-1}\tau^s$, and $s^{-1}\tau^{s-1}D_Y\Lambda = s^{-1}\tau^{s-1}\tau = s^{-1}\tau^s$. Both sides of (499) are derivations of $\mathbb{K}[\Lambda]((\tau))$ that raise the τ -order, hence are continuous for the τ -adic filtration; their difference is a continuous derivation vanishing on τ and Λ , so by the Leibniz rule it vanishes on $\mathbb{K}[\Lambda][\tau, \tau^{-1}]$ and by continuity and Theorem 5.18 on all of $\mathbb{K}[\Lambda]((\tau))$. Formula (500) is (134) for $\Gamma = s^{-1}\mathbb{Z}$, or equivalently (499) read backwards. $\square$

Remark 28.14 (The one thing the change of variable does *not* transport). Φ_s is an isomorphism of ordered differential rings up to a unit, but it does not carry near-identity maps in X to near-identity maps in Y identically: if $F = X + A$ with $A \in \mathbb{K}[L][[t^{1/s}]]$, then in the variable Y the same map reads

$$Y \longmapsto (Y^s + A)^{1/s} = Y(1 + \tau^s A)^{1/s} = Y + s^{-1}\tau^{s-1}A + \mathcal{O}_\tau^{\text{form}}(\tau^{2s-1}),$$

which is again near-identity but with a perturbation of outer order $s - 1$ higher. Reversion therefore transports through the conjugation $F \mapsto j_s \circ F \circ j_s^{\langle -1 \rangle \circ}$ with $j_s(X) = X^{1/s}$, not through Φ_s alone. This is the formal counterpart of the fact that tangency to the identity is a property of a map together with a coordinate.

The union over denominators behaves quite differently, and here the sources are silent about the essential point.

Theorem 28.15 (The Puiseux-log union is separated but not complete). *Let*

$$\mathfrak{P} := \mathbb{K}[L]((t^{1/\infty})) = \bigcup_{s \in \mathbb{N}_{>0}} \mathbb{K}[L]((t^{1/s}))$$

be the ramified union of Theorem 3.25, viewed inside the power–log scale $\mathcal{H}(\mathbb{K}[L], \mathbb{Q})$. Then:

- (i) $\mathfrak{P} \subseteq \mathcal{H}_{\text{lf}}(\mathbb{K}[L], \mathbb{Q})$, and $\mathfrak{P}$ contains every element with finite exponent set;

(ii) $\mathfrak{P}$ is separated for the $\mathbb{Q}$ -indexed filtration, but it is not complete: the sequence of partial sums of

$$F := \sum_{k \geq 2} t^{k-1/k} \quad (501)$$

is Cauchy in $\mathfrak{P}$ and has no limit in $\mathfrak{P}$. Hence [Theorem 5.18](#) is false for $\mathfrak{P}$;

(iii) the closure of $\mathfrak{P}$ inside $\mathcal{H}(\mathbb{K}[L], \mathbb{Q})$ is exactly $\mathcal{H}_{\text{lf}}(\mathbb{K}[L], \mathbb{Q})$; that is, the left-finite Hahn ring over $\mathbb{Q}$ is the completion of the Puiseux-log union.

Proof. (i) An element of $\mathbb{K}[L]((t^{1/s}))$ has exponent set contained in $s^{-1}\mathbb{Z}$ and bounded below, hence left-finite by [Theorem 28.8](#) (v) applied to the cyclic group $s^{-1}\mathbb{Z}$, or directly. Conversely a finite set of rationals has a common denominator s , so any element with finite exponent set lies in $\mathbb{K}[L]((t^{1/s}))$.

(ii) Separatedness is inherited from [Theorem 28.10](#) (iii). Put $F_K = \sum_{k=2}^K t^{k-1/k}$; then $F_K \in \mathbb{K}[L]((t^{1/K!})) \subseteq \mathfrak{P}$, and for $K' > K$ we have $\nu_t(F_{K'} - F_K) = (K+1) - \frac{1}{K+1} > K$, so (F_K) is Cauchy. Its limit in the complete ring $\mathcal{H}(\mathbb{K}[L], \mathbb{Q})$ is F of (501), whose exponent set $\{k - \frac{1}{k} : k \geq 2\}$ is strictly increasing and unbounded, hence left-finite, so $F \in \mathcal{H}_{\text{lf}}$. But $F \notin \mathfrak{P}$: if $F \in \mathbb{K}[L]((t^{1/s}))$ then $k - \frac{1}{k} \in s^{-1}\mathbb{Z}$ for all $k \geq 2$, that is $s/k \in \mathbb{Z}$, that is $k \mid s$, which fails for all $k > s$. By separatedness a limit is unique, so (F_K) has no limit in $\mathfrak{P}$.

(iii) By (i), $\mathbb{K}[L][t^{\mathbb{Q}}] \subseteq \mathfrak{P} \subseteq \mathcal{H}_{\text{lf}}$. Taking closures in $\mathcal{H}(\mathbb{K}[L], \mathbb{Q})$ and using [Theorem 28.10](#) (iv), which identifies the closure of $\mathbb{K}[L][t^{\mathbb{Q}}]$ with $\mathcal{H}_{\text{lf}}$, and the fact that $\mathcal{H}_{\text{lf}}$ is closed, all three closures coincide. $\square$

Remark 28.16 (Why this matters operationally). Every individual computation in the Puiseux tower happens at some finite denominator, and [Theorem 28.13](#) says that at a fixed denominator the whole apparatus of Parts I–IV applies unchanged. What [Theorem 28.15](#) adds is that a *limit* of such computations may leave the tower. An implementation that stores a single denominator per object is therefore sound for any finite sequence of operations and unsound as a model of the completed class; the completed class is $\mathcal{H}_{\text{lf}}$, on which [Theorem 28.11](#) guarantees closure under near-identity composition and reversion but no denominator bound. Bounding the denominator growth of an explicit iteration is the content of [Theorem 29.11](#).

28.4.1 An obstruction no exponent group repairs

Proposition 28.17 (No power–log scale contains a square root of L). *Let $\mathcal{H} = \mathcal{H}(C, \Gamma)$ be a power–log scale in the sense of [Theorem 8.11](#), so that $C \in \{\mathbb{K}'[L], \mathbb{K}'(L), \mathbb{K}'((L^{-1}))\}$. Then there is no $F \in \mathcal{H}$ with $F^2 = L$. This holds for every exponent group Γ , in particular for $\Gamma = \mathbb{R}$.*

Proof. Suppose $F^2 = L$. Since $\mathcal{H}$ is a domain and ν_t is additive ([Theorem 8.12](#)(i)), $2\nu_t F = \nu_t L = 0$, so $\nu_t F = 0$; as $F \neq 0$ and its exponent set is well based, the block $p_0 = [t^0]F$ is defined and nonzero, and comparing blocks at t^0 gives $p_0^2 = L$ in C . It therefore suffices to show that L is not a square in any of the three coefficient rings.

For $C = \mathbb{K}'(L)$ (and hence for the subring $\mathbb{K}'[L]$): write $p_0 = u/v$ with $u, v \in \mathbb{K}'[L]$ coprime and $v \neq 0$; then $u^2 = Lv^2$. The polynomial L is irreducible in the unique factorisation domain $\mathbb{K}'[L]$, and its multiplicity in u^2 is even while its multiplicity in Lv^2 is odd, a contradiction.

For $C = \mathbb{K}'((L^{-1}))$: write $p_0 = \sum_{j \leq d} c_j L^j$ with $c_d \neq 0$. Then $p_0^2 = c_d^2 L^{2d} + (\text{lower powers})$, so comparing the top exponent with that of L gives $2d = 1$, impossible for $d \in \mathbb{Z}$. $\square$

Remark 28.18 (Which axis repairs it). $\sqrt{L}$ does exist in the rank-two Hahn field $\mathcal{H}_1(\mathbb{K}, \Gamma)$ of [Theorem 13.1](#) whenever $\frac{1}{2} \in \Gamma$, since there $L_1^{1/2}$ is a monomial. The point of [Theorem 28.17](#) is that the four classes of Part I and all their Puiseux and Hahn refinements share one structural feature – the exponent of L inside a block runs over $\mathbb{Z}$, never over a larger group – and that no amount of refinement along the t -exponent axis touches it. The enlargement required is a ramification of the *monomial group* in which L itself carries fractional exponents; algebraically this is the step from a scale

with a coefficient *ring* to a genuine rank-two Hahn field. Concretely, $\sqrt{\log X}$ arises the moment one inverts a map with a half-integral logarithmic exponent. For $Y = X(\log X)^{1/2}$ with $X \rightarrow +\infty$, taking logarithms gives $\log Y = \log X + \frac{1}{2} \log \log X$; dividing that relation by $\log Y$ and taking logarithms gives $\log \log X - \log \log Y = \log(1 - \frac{1}{2} \log \log X / \log Y) = \mathcal{O}_{Y \rightarrow +\infty}(\log \log Y / \log Y)$, whence

$$\log X = \log Y - \frac{1}{2} \log \log Y + \mathcal{O}_{Y \rightarrow +\infty}(\log \log Y / \log Y),$$

and hence

$$X = Y (\log X)^{-1/2} = Y (\log Y)^{-1/2} \left(1 + \frac{1}{4} \frac{\log \log Y}{\log Y} + \mathcal{O}_{Y \rightarrow +\infty}((\log \log Y)^2 (\log Y)^{-2}) \right),$$

whose leading monomial $Y(\log Y)^{-1/2}$ has a half-integral exponent on the first logarithm; the expansion lives in $\mathcal{H}_2(\mathbb{K}, \frac{1}{2}\mathbb{Z})$ and in no power–log scale. The computation is the shape of [Theorem 17.7](#) at $\rho = 1, \sigma = \frac{1}{2}$; that theorem is stated for $\sigma \in \mathbb{N}_0$, and we do not widen its hypothesis. The display above is proved where it stands, and it agrees with (317) at $c = 1, \rho = 1, \sigma = \frac{1}{2}$.

28.5 Nested logarithms, and the residue that forces each new level

The nested-logarithm tower $L_0 = X, L_{j+1} = \log L_j$, its monomial groups $\mathcal{M}_r(\Gamma)$ and its Hahn ambients $\mathcal{H}_r(\mathbb{K}, \Gamma)$ were introduced in [Theorem 13.1](#); the tower derivation $\mathcal{D}_r$, the minimal Laurent coefficient ring $\mathcal{L}_r$, the near-identity group at depth r and the depth-dependent unit criterion are [Theorem 13.13](#), [Theorem 13.14](#) and [Theorem 13.15](#). Two facts remain to be proved, and they are the ones that make the tower a genuine hierarchy rather than a notational convenience: the derivation extends to the full Hahn ambient, and each level is forced by an obstruction of residue type that no enlargement along any other axis removes.

Throughout, $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{R}$ is an ordered group equipped, as in [Theorem 8.11](#), with an embedding $\Gamma \hookrightarrow \mathbb{K}$ of additive groups. We extend [Theorem 13.1](#) to $r = 0$ by the same formula, so that $\mathcal{M}_0(\Gamma) = \{X^{\alpha_0} : \alpha_0 \in \Gamma\}$ and $\mathcal{H}_0(\mathbb{K}, \Gamma)$ is the logarithm-free Hahn field of real powers of X . We write a monomial of $\mathcal{M}_r(\Gamma)$ as

$$\mathbf{m} = \prod_{j=0}^r L_j^{\alpha_j(\mathbf{m})}, \quad \alpha_j(\mathbf{m}) \in \Gamma, \quad \mathbf{n}_j := L_0 L_1 \cdots L_j \quad (0 \leq j \leq r). \quad (502)$$

Each $\alpha_j : \mathcal{M}_r(\Gamma) \rightarrow \Gamma$ is a group homomorphism, and $\mathbf{n}_j \succ 1$, so $\mathbf{n}_j^{-1} \prec 1$; moreover $\mathbf{n}_0^{-1} = t \succ \mathbf{n}_1^{-1} \succ \cdots \succ \mathbf{n}_r^{-1}$.

Lemma 28.19 (The tower derivation on the Hahn ambient). *For $0 \leq j \leq r$ let ϑ_j act on $F = \sum_{\mathbf{m} \in S} c_{\mathbf{m}} \mathbf{m} \in \mathcal{H}_r(\mathbb{K}, \Gamma)$ by $\vartheta_j F = \sum_{\mathbf{m} \in S} \alpha_j(\mathbf{m}) c_{\mathbf{m}} \mathbf{m}$, and set*

$$\partial_r := \sum_{j=0}^r \mathbf{n}_j^{-1} \vartheta_j. \quad (503)$$

Then:

- (i) each ϑ_j is a well-defined $\mathbb{K}$ -linear derivation of $\mathcal{H}_r(\mathbb{K}, \Gamma)$, and so is ∂_r ;
- (ii) $\partial_r L_j = \mathbf{n}_{j-1}^{-1}$ for $1 \leq j \leq r$ and $\partial_r L_0 = 1$, in agreement with [Theorem 13.13](#); and for $\mathbf{m} \in \mathcal{M}_r(\Gamma)$,

$$\frac{\partial_r \mathbf{m}}{\mathbf{m}} = \sum_{j=0}^r \alpha_j(\mathbf{m}) \mathbf{n}_j^{-1}; \quad (504)$$

- (iii) on $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}} \subseteq \mathcal{H}_1(\mathbb{K}, \mathbb{Z})$, ∂_1 is D_X ; on a power–log scale $\mathcal{H}(\mathbb{K}[L], \Gamma)$ it is (134); and on $\mathcal{L}_r((t))$ it is (235).

Proof. (i) $\vartheta_j F$ has support inside that of F , so it is well defined. It is a derivation: the coefficient of $\mathfrak{p}$ in $\vartheta_j(FG)$ is $\alpha_j(\mathfrak{p}) \sum_{\mathfrak{m}\mathfrak{n}=\mathfrak{p}} c_{\mathfrak{m}} d_{\mathfrak{n}}$, while the coefficient of $\mathfrak{p}$ in $(\vartheta_j F)G + F(\vartheta_j G)$ is $\sum_{\mathfrak{m}\mathfrak{n}=\mathfrak{p}} (\alpha_j(\mathfrak{m}) + \alpha_j(\mathfrak{n})) c_{\mathfrak{m}} d_{\mathfrak{n}}$; these agree because α_j is a homomorphism, and both sums are finite by Theorem 28.2 (ii). Multiplying a derivation by a fixed ring element again gives a derivation, and a finite sum of derivations is a derivation, so ∂_r is one. Its support lies in $\bigcup_j \mathfrak{n}_j^{-1} S$, a finite union of translates of a well-based set, and each monomial receives at most $r + 1$ contributions, one per j , because $\mathfrak{m} \mapsto \mathfrak{m}\mathfrak{n}_j^{-1}$ is injective.

(ii) For $\mathfrak{m} = L_j$ one has $\alpha_i(L_j) = \delta_{ij}$, so $\partial_r L_j = \mathfrak{n}_j^{-1} L_j = (L_0 \cdots L_{j-1})^{-1} = \mathfrak{n}_{j-1}^{-1}$ for $j \geq 1$, and $\partial_r L_0 = L_0^{-1} L_0 = 1$. Formula (504) is (503) applied to the single monomial $\mathfrak{m}$.

(iii) For $\mathfrak{m} = t^\gamma L^k = L_0^{-\gamma} L_1^k$, (504) gives $\partial_1 \mathfrak{m} = \mathfrak{m}(-\gamma L_0^{-1} + k(L_0 L_1)^{-1}) = t^{\gamma+1}(k L^{k-1} - \gamma L^k) = t^{\gamma+1}(\partial_L - \gamma)L^k$, which is (134), and for $\gamma = n \in \mathbb{Z}$ this is D_X . The identification with (235) is the same computation with $\mathfrak{m} = t^n L_1^{a_1} \cdots L_r^{a_r}$. $\square$

The next theorem is the exact reason the tower cannot be truncated. It is the depth- r form of Theorem 9.21, which is the case $r = 1$: there the obstruction was the residue $[t^1 L^{-1}]$, and $t^1 L^{-1} = \mathfrak{n}_1^{-1}$.

Theorem 28.20 (Each level of the tower is forced by an unmatched residue). *Let $r \geq 0$ and let $\mathcal{H}_r = \mathcal{H}_r(\mathbb{K}, \Gamma)$ with $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{R}$. Then*

$$[\mathfrak{n}_r^{-1}] \partial_r F = 0 \quad \text{for every } F \in \mathcal{H}_r, \quad (505)$$

where $[n]G$ denotes the coefficient of the monomial $\mathfrak{n}$ in G . Consequently:

- (i) $\mathfrak{n}_r^{-1} = \frac{1}{L_0 L_1 \cdots L_r}$ has no antiderivative in $\mathcal{H}_r$, so ∂_r is not surjective and Theorem 9.17 fails at every depth once the ambient is the full Hahn field;
- (ii) if $(\mathcal{T}, \partial)$ is any differential field extension of $\mathcal{H}_r$ whose derivation restricts to ∂_r , and $\Lambda \in \mathcal{T}$ satisfies $\partial \Lambda = \mathfrak{n}_r^{-1}$ — as $L_{r+1} = \log L_r$ does — then $\Lambda \notin \mathcal{H}_r$. In particular $L_{r+1} \notin \mathcal{H}_r(\mathbb{K}, \Gamma)$ for every Γ , and

$$\mathcal{H}_0(\mathbb{K}, \Gamma) \subsetneq \mathcal{H}_1(\mathbb{K}, \Gamma) \subsetneq \mathcal{H}_2(\mathbb{K}, \Gamma) \subsetneq \cdots$$

is a strictly increasing chain of differential subfields;

- (iii) no $\mathcal{H}_r(\mathbb{K}, \Gamma)$ is closed under the logarithm of its positive elements: $L_r \in \mathcal{H}_r$ but $\log L_r \notin \mathcal{H}_r$.

Proof. Fix $F = \sum_{\mathfrak{m} \in S} c_{\mathfrak{m}} \mathfrak{m} \in \mathcal{H}_r$. By (503), the monomial $\mathfrak{n}_r^{-1}$ receives, for each index $j \in \{0, \dots, r\}$, exactly one possible contribution, namely from the unique monomial $\mathfrak{m}_j$ with $\mathfrak{m}_j \mathfrak{n}_j^{-1} = \mathfrak{n}_r^{-1}$, that is

$$\mathfrak{m}_j = \mathfrak{n}_j \mathfrak{n}_r^{-1} = \frac{L_0 L_1 \cdots L_j}{L_0 L_1 \cdots L_r} = \prod_{i=j+1}^r L_i^{-1},$$

the empty product for $j = r$ being 1. Hence

$$[\mathfrak{n}_r^{-1}] \partial_r F = \sum_{j=0}^r \alpha_j(\mathfrak{m}_j) c_{\mathfrak{m}_j},$$

a finite sum, in which $c_{\mathfrak{m}_j}$ is read as 0 when $\mathfrak{m}_j \notin S$. But $\mathfrak{m}_j$ involves only the letters $L_{j+1}, \dots, L_r$, so its L_j -exponent is $\alpha_j(\mathfrak{m}_j) = 0$ for every j , and every term vanishes. This proves (505).

- (i) If $\partial_r F = \mathfrak{n}_r^{-1}$ then the left side has $\mathfrak{n}_r^{-1}$ -coefficient 0 and the right side has $\mathfrak{n}_r^{-1}$ -coefficient 1.

(ii) If $\Lambda \in \mathcal{H}_r$ then $\partial \Lambda = \partial_r \Lambda$ has vanishing $\mathfrak{n}_r^{-1}$ -coefficient by (505), contradicting $\partial \Lambda = \mathfrak{n}_r^{-1}$. That L_{r+1} has this derivative is Theorem 13.13 at index $r + 1$, rewritten with $\mathfrak{n}_r = L_0 L_1 \cdots L_r$. Inclusion $\mathcal{H}_r \subseteq \mathcal{H}_{r+1}$ holds because $\mathcal{M}_r(\Gamma)$ is the subgroup of $\mathcal{M}_{r+1}(\Gamma)$ with $\alpha_{r+1} = 0$, the induced order agrees, and well-basedness and the convolution product are inherited; strictness is the membership $L_{r+1} \in \mathcal{H}_{r+1} \setminus \mathcal{H}_r$. Each inclusion commutes with the derivations, since ∂_{r+1} restricted to monomials with $\alpha_{r+1} = 0$ is ∂_r .

- (iii) Immediate from (ii) with $\Lambda = L_{r+1}$. $\square$

Corollary 28.21 (Depth is not bounded, and the union is the smallest logarithmically closed ambient). Put $\mathcal{H}_\infty(\mathbb{K}, \Gamma) = \bigcup_{r \geq 0} \mathcal{H}_r(\mathbb{K}, \Gamma)$, a directed union of differential fields, hence a differential field. Then $\mathcal{H}_\infty$ contains L_j for every j , no proper member $\mathcal{H}_r$ of the chain does, and any subfield of an ambient differential field that contains X and is closed under $\log$ on the elements $L_0, L_1, L_2, \dots$ contains all of them. In particular a bound on logarithmic depth is never a theorem about a class; it is a hypothesis about a particular computation.

Proof. A directed union of subfields of a fixed field is a subfield, and the derivations agree on overlaps by Theorem 28.20 (ii); $L_j \in \mathcal{H}_j \subseteq \mathcal{H}_\infty$ while $L_{r+1} \notin \mathcal{H}_r$ by the same theorem. The last assertion is an induction on j starting from $L_0 = X$. $\square$

A depth bound is a hypothesis, not a conclusion

Theorem 17.4 shows one logarithm in a leading monomial forcing one new level; Theorem 28.20 shows that no level is ever redundant. Neither result bounds the depth needed by an arbitrary composite of admissible operations, and this volume proves no such bound. What is proved is that the classification of scale changes (Theorem 13.2) raises the depth of a single power–logarithm substitution by at most one; iterating it k times can therefore be guaranteed only within depth $r + k$. Whether that is sharp for every k is Theorem 29.9.

28.6 Multiseries: nesting, the fast variable, and the completion order

A *multiseries* is an expansion whose coefficients are themselves expansions, in a strictly slower scale, iterated to a finite depth. Formally this is the iterated Laurent construction of Theorem 7.36, carried up the logarithmic tower.

Definition 28.22 (Finite-depth multiseries field). For $r \geq 0$ put

$$\mathfrak{J}_r := \mathbb{K}((L_r^{-1}))((L_{r-1}^{-1})) \cdots ((L_1^{-1}))((L_0^{-1})), \quad (506)$$

the iterated field of formal Laurent series in $L_0^{-1} = t$ whose coefficients are formal Laurent series in L_1^{-1} , whose coefficients are formal Laurent series in L_2^{-1} , and so on to depth r ; the outermost construction carries the *most dominant* scale. Thus $\mathfrak{J}_0 = \mathbb{K}((t))$ and $\mathfrak{J}_1 = \mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$. An element of $\mathfrak{J}_r$ is a family $(c_m)_{m \in \mathbb{Z}^{r+1}}$ of scalars, $m = (m_0, \dots, m_r)$ representing the monomial $L_0^{-m_0} L_1^{-m_1} \cdots L_r^{-m_r}$.

Theorem 28.23 (The iterated field is the Hahn field, and only in the matching order). Let $r \geq 0$.

- (i) $\mathfrak{J}_r = \mathcal{H}_r(\mathbb{K}, \mathbb{Z})$ as $\mathbb{K}$ -algebras, the identification being the identity on monomials. Explicitly, a family $(c_m)_{m \in \mathbb{Z}^{r+1}}$ lies in $\mathfrak{J}_r$ if and only if its support is well ordered in the direction of increasing smallness for the dominance order of Theorem 13.1.
- (ii) Let π be a permutation of $\{0, 1, \dots, r\}$ and let $\mathfrak{J}_\pi$ be built as in (506) with the levels nested in the order $\pi(0)$ outermost, $\dots$, $\pi(r)$ innermost. If $\pi \neq \text{id}$ then neither of $\mathfrak{J}_\pi, \mathfrak{J}_r$ contains the other, as subsets of the space of families (c_m) . More generally, if $\pi \neq \pi'$ then neither of $\mathfrak{J}_\pi, \mathfrak{J}_{\pi'}$ contains the other; so the $(r+1)!$ nestings give $(r+1)!$ pairwise incomparable, and in particular pairwise distinct, subsets.

Proof. (i) Write $\mathfrak{m}(m) = L_0^{-m_0} \cdots L_r^{-m_r}$. By Theorem 13.1, $\mathfrak{m}(m) \succ \mathfrak{m}(m')$ exactly when $m < m'$ lexicographically, so “the support is well ordered in the direction of increasing smallness” means: every nonempty subset of the support, read as a set of vectors $m \in \mathbb{Z}^{r+1}$, has a lexicographically least element. Membership in $\mathfrak{J}_r$, unfolded, says: the set of m_0 occurring is bounded below; for each such m_0 the set of m_1 occurring with that m_0 is bounded below; and so on through m_r . These two conditions are equivalent.

$\Leftarrow$: given a nonempty subset T , let μ_0 be the least m_0 occurring in T , which exists because the set is a nonempty subset of $\mathbb{Z}$ bounded below; restrict T to that value and let μ_1 be the least m_1 occurring, and so on. The resulting vector $(\mu_0, \dots, \mu_r)$ lies in T and is lexicographically least.

$\Rightarrow$: if for some k and some fixed prefix $(\mu_0, \dots, \mu_{k-1})$ occurring in the support the set of admissible m_k were unbounded below, choose $m^{(1)}, m^{(2)}, \dots$ in the support with that prefix and with $m_k^{(i)}$ strictly decreasing; then $m^{(1)} > m^{(2)} > \dots$ lexicographically, so $\{m^{(i)} : i \geq 1\}$ has no lexicographically least element.

(ii) Since $\pi \neq \text{id}$ there are $a < b$ in $\{0, \dots, r\}$ with $\pi^{-1}(b) < \pi^{-1}(a)$: level b is nested *outside* level a in $\mathfrak{J}_\pi$, while in $\mathfrak{J}_r$ level a is outside level b . Put

$$A = \sum_{n \geq 0} L_a^{-n} L_b^n, \quad B = \sum_{n \geq 0} L_a^n L_b^{-n},$$

so that A has $m_a = n, m_b = -n$ and all other coordinates 0, and B has $m_a = -n, m_b = n$. Read in the order $0, 1, \dots, r$: for A , the coordinates $m_0, \dots, m_{a-1}$ are 0, the set of m_a is $\mathbb{N}_0$, bounded below, and once $m_a = n$ is fixed all later coordinates are determined; hence $A \in \mathfrak{J}_r$ (indeed $A = \sum_n (L_b/L_a)^n$). For B , the set of m_a occurring is $\{0, -1, -2, \dots\}$, unbounded below, so $B \notin \mathfrak{J}_r$. Read in the order π , the roles of a and b are exchanged and the same computation gives $B \in \mathfrak{J}_\pi$ and $A \notin \mathfrak{J}_\pi$.

The argument used only that the two nestings order the pair a, b oppositely, and $\mathfrak{J}_r = \mathfrak{J}_{\text{id}}$. Two distinct total orders on the finite set $\{0, \dots, r\}$ must order some pair oppositely, so for $\pi \neq \pi'$ the same witnesses – built from the two levels on which π and π' disagree – give an element of $\mathfrak{J}_\pi \setminus \mathfrak{J}_{\pi'}$ and an element of $\mathfrak{J}_{\pi'} \setminus \mathfrak{J}_\pi$. $\square$

Remark 28.24 (The completion order, in one sentence). [Theorem 28.23](#) says that iterating the Laurent construction reproduces the Hahn field *exactly when the nesting order is the reverse of the dominance order* – outermost variable most dominant – and produces an incomparable object otherwise. The case $r = 1$ is [Theorem 3.22](#) and [Theorem 7.39](#); the general case shows that the phenomenon is not an artefact of the rank-two situation but is present at every depth, and that there are $(r + 1)!$ pairwise distinct iterated fields on the same monomial group, exactly one of which is the asymptotically meaningful one. The catalogue records the same warning as “the iteration order is part of the type”.

Remark 28.25 (Nested truncation: what a finite datum is). Membership in $\mathfrak{J}_r$ does not make an element finitely representable. A single block of $\mathfrak{J}_1 = \mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ is already an infinite Laurent series in L^{-1} , so a finite datum needs $r + 1$ cutoffs, or one declared staircase in the lexicographic monomial order; an outer cutoff alone bounds nothing. This is the depth- r form of the caution attached to [Theorem 7.37](#), and it is the reason the sparse representation of [Theorem 25.1](#) fixes an inner cutoff for the coefficient ring $\mathbb{K}((L^{-1}))$. A workable contract at depth r is a monotone staircase: a bound $m_0 < N_0$, then $m_1 < N_1(m_0)$, then $m_2 < N_2(m_0, m_1)$, and so on, chosen so that the region is stable under the convolutions to be performed. The choice is part of the specification of the computation, not a detail of its implementation.

Do not expand a rational coefficient before you must

The exact block $(L^2 + 1)/(L - 1) \in \mathbb{K}(L)$ and its expansion $L + 1 + 2L^{-1} + 2L^{-2} + \dots \in \mathbb{K}((L^{-1}))$ are different data about the same object: the first is finite and exact, the second resolves a finer asymptotic scale and is infinite. By [Theorem 7.37](#) the expansion map ι is injective, so an identity proved after expansion *does* descend; but it is not surjective, so a computation carried out after expansion may leave the image and return no exact rational block, and a cancellation that is exact and finite in $\mathbb{K}(L)$ survives expansion only in the limit – at every finite truncation it is invisible. Expand when the requested scale requires it, and record that you did.

28.6.1 Changing the fast variable

The most useful practical observation about multiseries is that a nested expansion is often an ordinary polynomial–logarithmic expansion in the right variable. The mechanism is already a theorem in this volume – [Theorem 13.16](#), the level shift – and the following example shows it operating on the classical case.

Example 28.26 (A logarithmic prefactor, and where the second logarithm sits). Let $\beta \in \mathbb{K}$ and invert $Y = Z(\log Z)^\beta$ for $Z \rightarrow +\infty$. Write $S = \log Y$, $\Lambda = \log S$ and $z = \log Z$, so that the relation reads $S = z + \beta \log z$, that is $z = S - \beta \log z$. Substituting $z = S - \beta \Lambda + w$ and expanding $\log z = \Lambda + \log(1 - \beta \Lambda S^{-1} + w S^{-1})$ forces $w = \beta^2 \Lambda S^{-1} + \mathcal{O}_{S^{-1}}^{\text{form}}(S^{-2})$, the reversion being the near-identity one of $\mathbb{K}[\Lambda][[S^{-1}]]$. Since $e^S = Y$ and $e^{-\beta \Lambda} = S^{-\beta}$, exponentiating z gives

$$Z = Y S^{-\beta} \left(1 + \beta^2 \frac{\Lambda}{S} + \mathcal{O}_{S^{-1}}^{\text{form}}(S^{-2}) \right). \quad (507)$$

This is the shape of [Theorem 17.7](#) at $c = 1$, $\rho = 1$, $\sigma = \beta$, whose hypothesis $\sigma \in \mathbb{N}_0$ we do not widen; the display is proved where it stands. The right-hand side is *not* an element of any power–log scale in the variable Y , because the second logarithm Λ occurs. But the bracket is an element of $\mathbb{K}[\Lambda][[S^{-1}]]$, that is, of the polynomial–logarithmic power-series ring built on the fast variable $S = \log Y$ and its logarithm $\Lambda = \log S$. Under [Theorem 13.16](#) every statement of Parts I–IV therefore applies to the bracket verbatim, with t read as S^{-1} and L read as Λ . The only genuinely new object in (507) is the monomial prefactor $Y S^{-\beta}$, which is where the depth increase is confined.

Remark 28.27 (The pattern, and the honest limit of the pattern). [Theorem 28.26](#) is a special case of the normalisation theorem [Theorem 17.6](#): factor out a leading model whose inverse is known, and the remainder is a near-identity problem in a scale that the leading model itself selects. What the pattern does *not* do is remove the depth increase; it localises it in the prefactor. A claim of the form “after the right change of variable the problem is one-logarithmic” is true of the correction and false of the answer, and [Theorem 28.20](#) is the reason it must be false of the answer whenever $\beta \neq 0$.

29 Relation with the full transseries field

Everything in this volume happens at *exponential height zero*: no monomial of any class considered so far is an exponential of a large series. This section places the polynomial–logarithmic class inside the ambient in which exponentials are allowed, says precisely what that ambient adds, and is careful about the division of labour. Two things are proved here, because they are what a user of the smaller class needs and because they sharpen results already in the volume: an exact criterion, valid at every logarithmic depth, for an element with a prescribed logarithmic derivative to lie in the class ([Theorem 29.1](#)); and the observation that the filtered gain which makes all of Parts III–IV work is a property of the logarithmic scale and is destroyed by a single exponential ([Theorem 29.4](#)). Everything else — the construction of the transseries field, its model theory, its composition and summability theory — is quoted from the literature and explicitly *not* reproved.

29.1 The exponential boundary at every logarithmic depth

[Theorem 8.16](#) settled which elements of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ have exponentials inside $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, and [Theorem 8.17](#) extended this to a power–log scale. Both are statements at logarithmic depth one. The following theorem is the depth- r statement, and it is what makes the boundary structural rather than a feature of the particular class: it says that in $\mathcal{H}_r(\mathbb{K}, \Gamma)$ the logarithmic derivative of a nonzero element is never larger than t , and that its t -coefficient is forced to lie in the exponent group.

Theorem 29.1 (Logarithmic derivatives in a nested-logarithmic Hahn field). *Let $r \geq 0$, $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{R}$, and let $\mathcal{H}_r = \mathcal{H}_r(\mathbb{K}, \Gamma)$ with the derivation ∂_r of [Theorem 28.19](#). Let $F \in \mathcal{H}_r$ be nonzero with dominant monomial $\mathfrak{m} = \text{lm}(F)$ and leading scalar $c = \text{lc}(F)$. Then*

$$\text{lm}(\partial_r F) \preceq t \mathfrak{m}, \quad [t \mathfrak{m}] \partial_r F = \alpha_0(\mathfrak{m}) c. \quad (508)$$

Consequently, if $(\mathcal{T}, \partial)$ is a differential ring extension of $\mathcal{H}_r$ whose derivation restricts to ∂_r , if $E \in \mathcal{H}_r$ is nonzero, and if $u \in \mathcal{H}_r$ is nonzero with $\partial E = uE$, then

$$\text{lm}(u) \preceq t, \quad \text{and if } \text{lm}(u) = t \text{ then } \text{lc}(u) = \alpha_0(\text{lm } E) \in \Gamma. \quad (509)$$

Proof. Write $F = \sum_{n \in S} c_n n$ with $m = \max_{\succ} S$. By (503), every monomial occurring in $\partial_r F$ is of the form $n n_j^{-1}$ with $n \in S$ and $0 \leq j \leq r$. Since $n \prec m$ and $n_j^{-1} \prec n_0^{-1} = t$, each such monomial is $\prec tm$, which is the first half of (508). For the second half, the contributions to $tm = m n_0^{-1}$ come from the pairs (n, j) with $n = m n_0^{-1} n_j = m L_1 L_2 \cdots L_j$. For $j = 0$ this is $n = m$, contributing $\alpha_0(m)c$. For $j \geq 1$ the monomial $m L_1 \cdots L_j$ has the same L_0 -exponent as m and a strictly larger L_1 -exponent, hence $m L_1 \cdots L_j \succ m$, so it does not lie in S and contributes nothing.

For the consequence, note first that $\mathcal{H}_r$ is an integral domain with $\text{lm}(uE) = \text{lm}(u) \text{lm}(E)$ and $\text{lc}(uE) = \text{lc}(u) \text{lc}(E)$: the coefficient of $\text{lm}(u) \text{lm}(E)$ in uE is the finite sum (Theorem 28.2(ii)) over pairs $pq = \text{lm}(u) \text{lm}(E)$ with $p \in \text{supp}_{\text{coeff}}(u)$, $q \in \text{supp}_{\text{coeff}}(E)$, and any such pair other than $(\text{lm } u, \text{lm } E)$ would have $p \prec \text{lm}(u)$ or $q \prec \text{lm}(E)$ and hence $pq \prec \text{lm}(u) \text{lm}(E)$. Now put $m = \text{lm}(E)$. From $\partial E = \partial_r E = uE$ and (508), $\text{lm}(u)m = \text{lm}(uE) = \text{lm}(\partial_r E) \prec tm$, so $\text{lm}(u) \prec t$. If $\text{lm}(u) = t$, comparing the coefficients of tm gives $\text{lc}(u) \text{lc}(E) = \alpha_0(m) \text{lc}(E)$, and $\text{lc}(E) \neq 0$. $\square$

The criterion is applied by enlarging the exponent group so that the logarithmic derivative itself becomes an element of the ambient; this costs nothing, because $\mathcal{H}_r(\mathbb{K}, \Gamma) \subseteq \mathcal{H}_r(\mathbb{K}, \Gamma')$ for $\Gamma \subseteq \Gamma'$, while the conclusion $\text{lc}(u) = \alpha_0(\text{lm } E)$ still reads the exponent of an element of $\mathcal{M}_r(\Gamma)$.

Corollary 29.2 (No finite logarithmic depth admits an exponential). *Let $\mathbb{Z} \subseteq \Gamma \subseteq \Gamma' \subseteq \mathbb{R}$, let $(\mathcal{T}, \partial)$ be a differential ring extension of $\mathcal{H}_r(\mathbb{K}, \Gamma')$ whose derivation restricts to ∂_r , and let $E \in \mathcal{T}$ be nonzero with $\partial E = uE$ for some nonzero $u \in \mathcal{H}_r(\mathbb{K}, \Gamma')$. If $E \in \mathcal{H}_r(\mathbb{K}, \Gamma)$ then $\text{lm}(u) \prec t$, and if $\text{lm}(u) = t$ then $\text{lc}(u) \in \Gamma$. Consequently, for every $r \geq 0$ and every $\Gamma \subseteq \mathbb{R}$:*

- (i) $e^{cX} \notin \mathcal{H}_r(\mathbb{K}, \Gamma)$ for $c \in \mathbb{K}^\times$;
- (ii) $e^{\sqrt{X}} \notin \mathcal{H}_r(\mathbb{K}, \Gamma)$ and $X^{\log X} = e^{L^2} \notin \mathcal{H}_r(\mathbb{K}, \Gamma)$;
- (iii) no nonzero E in an ambient as above satisfies $\partial E = aX^{-1}E$ with $E \in \mathcal{H}_r(\mathbb{K}, \Gamma)$, when $a \in \mathbb{K}$ does not lie in the image of Γ ; informally, $X^a \notin \mathcal{H}_r(\mathbb{K}, \Gamma)$ whenever $a \notin \Gamma$;
- (iv) by contrast $L \in \mathcal{H}_1$ and $e^{1/X} \in \mathcal{PL}_{\text{pow}, \mathbb{K}}$, whose logarithmic derivatives are $n_1^{-1} = t/L$ and $-t^2$: the criterion is not vacuous in either direction.

Proof. The displayed criterion is (509) applied inside $\mathcal{H}_r(\mathbb{K}, \Gamma')$, which contains E and u ; the refinement is that $\text{lm}(E) \in \mathcal{M}_r(\Gamma)$, so $\alpha_0(\text{lm } E) \in \Gamma$.

(i) Take $\Gamma' = \Gamma$ and $u = c$; then $\text{lm}(u) = 1 \succ t$. (ii) For $E = e^{\sqrt{X}}$ take $\Gamma' = \Gamma + \frac{1}{2}\mathbb{Z}$, which is admissible: since $2 \cdot \frac{1}{2} = 1 \in \Gamma$ and $\mathbb{K}$ has characteristic zero, sending $\frac{1}{2}$ to half the image of 1 extends the embedding $\Gamma \hookrightarrow \mathbb{K}$ to Γ' , and the extension is injective because $\gamma + \frac{1}{2} \in \Gamma$ would force $\frac{1}{2} \in \Gamma$, in which case $\Gamma' = \Gamma$. Take $u = \frac{1}{2}X^{-1/2}$, so $\text{lm}(u) = L_0^{-1/2} \succ L_0^{-1} = t$ because $-\frac{1}{2} > -1$. For $E = e^{L^2}$ assume first $r \geq 1$; take $\Gamma' = \Gamma$ and $u = 2LX^{-1}$, so $\text{lm}(u) = L_0^{-1}L_1 \succ L_0^{-1}$: the two monomials agree in the L_0 -exponent and the first has the larger L_1 -exponent. Both contradict the criterion. For $r = 0$ the second case follows from the case $r = 1$, because $\mathcal{H}_0(\mathbb{K}, \Gamma) \subseteq \mathcal{H}_1(\mathbb{K}, \Gamma)$ by Theorem 28.20(ii). (iii) Take $\Gamma' = \Gamma$ and $u = aX^{-1} = at$, which lies in $\mathcal{H}_r(\mathbb{K}, \Gamma)$ because $t \in \mathcal{M}_r(\Gamma)$ and a is a scalar; then $\text{lm}(u) = t$ and $\text{lc}(u) = a$, which is not in the image of Γ . (iv) $\text{lm}(t/L) = L_0^{-1}L_1^{-1} \prec L_0^{-1} = t$ and $\text{lm}(-t^2) = L_0^{-2} \prec L_0^{-1}$; both are consistent with the criterion, and both elements do lie in the stated classes. $\square$

Remark 29.3 (Relation to the depth-one boundary). Theorem 29.2 does not supersede Theorem 8.16: the latter is an exact *iff*, listing the elements whose exponential stays in $\mathcal{PL}_{\text{poly}, \mathbb{K}}$, while the former is a necessary condition of a different shape — a constraint on the logarithmic derivative — whose merit is that it is uniform in the logarithmic depth and therefore rules out an element from *every* member of the tower at once. Read together they say: within the logarithmic world an exponential can only be a power of X times something already there (Theorem 8.21), and enlarging the logarithmic depth never changes that. The first genuinely new monomial is e^X , and it is new at every depth.

29.2 Where the class sits: the logarithmic–exponential field

The field $\mathbb{T}$ of logarithmic–exponential transseries is built by iterating two constructions on an ordered monomial group: adjoining exponentials e^p of *purely large* series p , and adjoining logarithms; and then closing under well-based sums. Two standard variants must be distinguished, and they correspond exactly to the support classes of [Theorem 28.1](#): the *grid-based* transseries of [\[33\]](#), whose supports lie on a finitely generated multiplicative grid, and the *well-based* field of [\[1, 13\]](#), whose supports need only be well ordered. [Theorem 28.8](#) is the elementary shadow of that distinction.

Section summary

Quoted, not proved. The construction of $\mathbb{T}$ as an ordered, exponentially closed, differential field with well-based supports and finite exponential height is [\[33, 1, 13\]](#). Its monomial group contains every $\prod_{j=0}^r L_j^{\alpha_j}$ with $\alpha_j \in \mathbb{R}$, and $\mathbb{T}$ is closed under well-based sums of its own monomials; consequently, for $\mathbb{K} = \mathbb{R}$ and every $\Gamma \subseteq \mathbb{R}$ and $r \geq 0$, the field $\mathcal{H}_r(\mathbb{R}, \Gamma)$ of [Theorem 13.1](#) is an ordered differential subfield of $\mathbb{T}$. The four classes of [Theorem 3.3](#) therefore sit inside $\mathbb{T}$ as the subrings of $\mathcal{H}_1(\mathbb{K}, \mathbb{Z})$ described in [Theorem 3.20](#). We use this placement only for orientation; no theorem of this volume depends on it.

What the exponential hierarchy adds is exactly one thing, and [Theorem 29.2](#) says it cannot be simulated: monomials whose logarithmic derivative is not $\preccurlyeq t$. Everything else – real exponents, arbitrary logarithmic depth, well-based rather than left-finite supports – lives already at height zero and was treated in [Section 28](#). The cost of the addition is severe, and it is worth stating precisely, because it explains why Parts III and IV could be elementary and the corresponding theory in $\mathbb{T}$ cannot.

Proposition 29.4 (The filtered gain is a logarithmic-scale phenomenon). *Let $(\mathcal{T}, \partial)$ be a differential ring containing a constant c and an element E with $\partial E = cE \neq 0$. Then there is no map ν from $\mathcal{T} \setminus \{0\}$ to a partially ordered set that is constant on nonzero constant multiples and satisfies $\nu(\partial G) > \nu(G)$ for every G with $G, \partial G \neq 0$. In particular the one-block gain $\nu_t(D_X A) \geq \nu_t(A) + 1$, which is the decisive hypothesis behind [Theorem 12.6](#), [Theorem 14.4](#), [Theorem 14.11](#) and the termination of every algorithm of [Section 25](#), cannot survive into any ring containing e^X .*

Proof. Apply the hypothesis to $G = E$: $\nu(\partial E) = \nu(cE) = \nu(E)$, while the assumed property gives $\nu(\partial E) > \nu(E)$. For the second sentence, take $c = 1$, $E = e^X$, and $\nu = \nu_t$, which is constant on nonzero scalar multiples. By contrast, inside $\mathcal{H}_r(\mathbb{K}, \Gamma)$ the gain does hold: [Theorem 29.1](#) gives $\text{lm}(\partial_r G) \preccurlyeq t \text{lm}(G) \prec \text{lm}(G)$ for every nonzero G . $\square$

Remark 29.5 (What replaces it). In $\mathbb{T}$ the derivation is not valuation-increasing, and the contraction arguments of Parts III–IV have no direct analogue. Their replacement is the asymptotic-differential-algebra machinery of [\[1\]](#): the notion of an H -field, the asymptotic couple attached to the valuation and the derivation, the Newton degree of a differential polynomial, and the axioms ω -freeness and newtonianity, which are what make the analogue of “the residual determines the next block” true again – now for differential equations rather than for a single composition. It is this apparatus, and not any refinement of the elementary filtration, that carries the theory across the exponential boundary.

29.3 The Hardy-field perspective

Everything above is formal. The classical counterpart is Hardy’s calculus of *orders of infinity* [\[20\]](#): the logarithmico-exponential functions – germs at $+\infty$ built from the variable and real constants by the field operations, $\exp$ and $\log$ – are eventually of constant sign, eventually monotone, and pairwise comparable, so their germs form a totally ordered differential field, a *Hardy field*. Three points deserve to be kept apart, and the sources of this consolidation do not always keep them apart.

1. A *Hardy field* consists of germs of actual real functions; its order is eventual dominance of real values.

2. A *transseries field* consists of formal objects; its order is the combinatorial dominance order on supports ([Theorem 3.12](#)).
3. An *expansion map* relates the two. Its existence for a given germ is a theorem, never a definition, and its injectivity fails in general: e^{-X} and 0 have the same polynomial–logarithmic expansion.

Within this volume, item (2) is what [Theorem 3.13](#) justifies at the level of monomials, and item (3) is what [Theorem 10.32](#) and [Theorem 10.35](#) implement, with hypotheses printed. The non-injectivity in item (3) is the exact reason the class is blind to resurgence; see [Section 29.6](#).

29.4 Model theory of the ambient, quoted

Section summary

Quoted, not proved. The following are stated as citations and used nowhere in this volume.

- Van den Dries, Macintyre and Marker [\[31\]](#) construct the field of logarithmic–exponential power series, a real closed field closed under $\exp$ and $\log$, and identify it as a model of the theory of the real field with restricted analytic functions and exponentiation; in [\[32\]](#) they develop its ordered differential and compositional structure and its relation with Hardy fields of logarithmico-exponential functions.
- Aschenbrenner, van den Dries and van der Hoeven [\[1\]](#) prove that $\mathbb{T}$, as an ordered valued differential field, is an ω -free newtonian Liouville closed H -field, that this axiomatisation is complete and model complete in their language, and develop the accompanying theory of asymptotic differential algebra.

These results are what license the phrase “the ambient field is well behaved”. None of them is needed to justify a single computation in Parts I–VI, and none of them is reproved here.

29.5 Composition, recursion, and convergence in the larger setting

Composition is the operation that degrades worst under enlargement, and the reason is visible already in [Section 28](#): substitution moves supports, and the image of a well-based support need not be well based unless the inner argument is controlled. Edgar’s treatment [\[14\]](#) isolates the admissibility conditions – in particular that the inner argument be positive and infinitely large – under which composition of transseries is defined, associative, and compatible with differentiation; the corresponding statements in this volume are [Theorem 10.6](#), [Theorem 10.7](#) and [Theorem 10.27](#), proved there in the narrow setting where admissibility is automatic. Fractional iteration and the iterative logarithm – the formal vector field V with $\exp(V)X = F$ – are developed for transseries in [\[15\]](#); in the present class they are [Theorem 12.18](#) and [Theorem 12.19](#), where the graded Lie algebra of the contact filtration makes the correspondence exact and the divisibility unique.

Two hazards that only appear after enlargement

Both are invisible in $\mathcal{PL}_{\text{poly}, \mathbb{K}}$ and must be checked in any larger class.

1. *Loss of well-basedness under substitution.* Composition substitutes an inner series into infinitely many outer monomials; the union of the resulting supports is well based only under a hypothesis, and [Theorem 28.4](#) shows the hypothesis cannot simply be dropped in favour of a weaker support class.
2. *Failure of Taylor approximation.* The formal Taylor formula [Theorem 11.6](#) is summable because D_X raises the outer order; [Theorem 29.4](#) shows this fails in an exponentially closed field, so the Taylor expansion of a transseries around a displaced argument requires a separate theorem with its own hypotheses. Such theorems exist [\[22\]](#), and they are not the trivial extension of [Theorem 11.6](#).

29.6 Borel summability and resurgence: what this class cannot see

The formal theory of Parts I–IV is indifferent to convergence, and this is not an oversight: it is a division of labour. A divergent polynomial–logarithmic series may be attached to an actual function by Borel–Laplace summation [8], and the ambiguity in that attachment — the Stokes phenomenon, and the exponentially small terms that measure it — is the subject of resurgence [12]. The following is the honest statement of where the present class stands.

Proposition 29.6 (The class is blind to exponentially small corrections). (i) *For every $r \geq 0$, every $\Gamma \subseteq \mathbb{R}$, every $\mu \in \mathbb{K}^\times$ and every ambient as in Theorem 29.2, the element $e^{-\mu X}$ does not lie in $\mathcal{H}_r(\mathbb{K}, \Gamma)$.*

(ii) *Let $S = [X_0, \infty)$ be a real ray, let $\mu > 0$ and $\lambda \in \mathbb{C}$, and let $f : S \rightarrow \mathbb{C}$ realize $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ in the sense of Theorem 10.32. Then the function $g(X) = f(X) + \lambda e^{-\mu X}$ realizes the same F , and by Theorem 10.33 no other element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$.*

Consequently no truncation, no residual certificate (Theorem 14.14) and no error estimate of Parts I–VI distinguishes f from g .

Proof. (i) This is Theorem 29.2 (i) with $c = -\mu$.

(ii) For every $N \in \mathbb{Z}$ and every $\mu > 0$ one has $e^{-\mu X} = o_{X \rightarrow +\infty}(X^{1-N})$, since $X^{N-1}e^{-\mu X} \rightarrow 0$. Hence

$$g(X) - (\text{Trunc}_{<N} F)(X) = \left(f(X) - (\text{Trunc}_{<N} F)(X) \right) + \lambda e^{-\mu X} = o_{X \rightarrow +\infty}(X^{1-N}),$$

which is the defining remainder condition of Theorem 10.32 for g . Uniqueness of the realized series is Theorem 10.33. $\square$

Remark 29.7 (Why the interesting resurgence is not here). Resurgence is a statement about the *relations* between the exponentially small sectors of a transseries and the singularities of its Borel transform. Theorem 29.6 says the polynomial–logarithmic class contains no such sector at all, at any logarithmic depth; the class is therefore too small for the phenomena to be visible, not merely inconvenient for them. What the class *can* contribute is the height-zero datum on which those phenomena act: the formal power–log series that a Borel–Laplace summation is asked to sum, together with the exact residual certificates of Theorem 14.14 and Theorem 14.15, which control the algebraic part of the error without any convergence hypothesis. Where a convergence statement is actually available in this volume it is proved and its scope is printed — Theorem 18.19 for the Lambert generating function at frozen logarithm — and where it is not, it is recorded as unproved in Section 29.8. No claim of Borel summability is made anywhere in this volume.

29.7 Symbolic asymptotics for larger classes

The algorithmic counterpart of this section is the multiseries technology of Salvy and Shackell. Their programme computes asymptotic expansions of exp–log functions by discovering the scale rather than fixing it in advance: implicit functions and functions of two variables [29], and multiseries of inverse functions [30]. The relationship with Section 25 is the following, and it is a genuine division of labour rather than a comparison.

- Their algorithms decide, at each step, *which* scale the next term lives in — an inverse power of X , an inverse power of $\log X$, a new iterated logarithm, a root, or an exponential monomial — and then recurse. Theorem 17.5, Theorem 17.3 and Theorem 28.20 are the corresponding decision content in the present class, stated as theorems about which enlargement is forced.
- Once the scale is fixed, the coefficient arithmetic is exactly the block calculus of Parts I–IV, and the cost model of Theorem 25.15 applies. The sharp scalar cost of truncated block convolution (Theorem 25.17) and the precision calculus (Theorem 25.10) are statements that a general multiseries implementation does not have available, because they use the finiteness of each block.

- The practical rule that follows is the one this volume has argued for throughout: begin in the smallest class that contains the problem, run the closure test for the next operation, and promote explicitly. Promotion hidden inside a simplifier is how a computation acquires an unproved convergence claim.

29.8 Open problems

The following are stated as open, not as theorems and not as conjectures with suppressed hypotheses. Each is precise enough to be attacked, and each is raised, and left unsettled, by material in this volume.

Remark 29.8 (Open problem: convergence of the Lambert expansion with coupled logarithm). Does there exist $x_0 > 0$ such that $\sum_{n \geq 1} |\mathbb{L}_n^{\text{Lam}}(\log x)| x^{-n} < \infty$ for every real $x > x_0$? Equivalently, is the radius of convergence $r(u)$ of $s \mapsto \sum_{n \geq 1} \mathbb{L}_n^{\text{Lam}}(u) s^n$ bounded below by κ/u for some $\kappa > 0$ and all large real u ? [Theorem 18.19](#) proves convergence for each *frozen* u with no lower bound on $r(u)$. Writing $\omega(X) = X + h$ with $h = \sum_{n \geq 0} \mathbb{L}_n^{\text{Lam}}(u) s^n$, $u = L$, $s = t$, the defining relation $\omega + \log \omega = X$ becomes $h + u + \log(1 + sh) = 0$ – this is the implicit generating equation of [Theorem 19.11](#), and the substitution just made rederives it, the value $h(u, 0) = \mathbb{L}_0^{\text{Lam}}(u) = -u$ fixing the branch. Its h -derivative $1 + s/(1 + sh)$ vanishes exactly at $1 + sh = -s$, that is at $h = -1 - s^{-1}$, and eliminating h puts the singularities of $s \mapsto h(u, s)$ on the curve $u = s^{-1} + 1 - \log(-s)$. This suggests $r(u) \asymp 1/u$; nothing in Parts I–IV proves it. A positive answer would upgrade [Theorem 18.10](#) from an asymptotic expansion to a convergent one on a half-line.

Remark 29.9 (Open problem: sharp logarithmic depth of a composite). Fix $r \geq 1$ and let $G_1, \dots, G_k$ be monomial-leading maps, each of logarithmic depth at most r in the sense that its leading monomial lies in $\mathcal{M}_r(\Gamma)$. [Theorem 13.2](#) guarantees that a single such substitution raises depth by at most one, so $G_1 \circ \dots \circ G_k$ lies at depth at most $r + k$; this volume proves no corresponding bound for its compositional inverse. Is $r + k$ attained by the composite, and is the inverse subject to any depth bound at all? More precisely: give an algorithm that, from the leading data of $G_1, \dots, G_k$, computes the least r' with $(G_1 \circ \dots \circ G_k)^{(-1)^\circ} \in \mathcal{H}_{r'}$, together with a proof of minimality. The only lower bound proved here is the single step of [Theorem 17.4](#), and the only obstruction proved here is the residue of [Theorem 28.20](#); neither is known to be complete.

Remark 29.10 (Open problem: a Lagrange–Bürmann formula at dense exponents). [Theorem 16.14](#) expresses the reversion of a near-identity map as $\sum_{r \geq 1} \frac{(-1)^r}{r!} \partial^{r-1}(A^r)$, and [Theorem 16.19](#) shows that the coefficient of t^N is determined by the indices $r \leq N + 1$ and by no fewer. Both statements use that the exponent group is $\mathbb{Z}$. In a Hahn power–log scale $\mathcal{H}(\mathbb{K}[L], \Gamma)$ with Γ dense, is there a summation index set – necessarily not $\mathbb{N}_{>0}$ with the range $r \leq N + 1$ – and an operator identity reducing to [Theorem 16.14](#) at $\Gamma = \mathbb{Z}$? The near-identity inverse exists there by [Theorem 28.11](#); what is missing is a closed formula for it.

Remark 29.11 (Open problem: denominator growth in the Puiseux tower). By [Theorem 28.15](#) the Puiseux-log union $\mathfrak{P}$ is not complete, so an infinite iteration may leave it. Give an explicit bound, or prove that none exists, for the following: if $F = X + A$ with $A \in \mathbb{K}[L][[t^{1/s}]]$, the Newton scheme of [Theorem 15.10](#) started at X produces iterates $H_0, H_1, \dots$; is there a function $s \mapsto \sigma(s)$, independent of A , with $H_k \in \mathbb{K}[L][[t^{1/\sigma(s)}]]$ for all k ? For the triangular scheme of [Theorem 15.2](#) the answer is yes with $\sigma(s) = s$, since every block of the inverse has exponent in $s^{-1}\mathbb{Z}$; the question is whether the quadratically convergent scheme, which divides by $F' \circ H_k$, preserves the denominator.

Remark 29.12 (Open problem: uniformity of the analytic transfer). [Theorem 18.17](#) and [Theorem 10.35](#) give, for each fixed truncation order N , an asymptotic error estimate on the positive ray with a constant depending on N . Is there an estimate uniform in N – for instance a bound of the form $|\omega(X) - \text{Trunc}_{<N} \omega(X)| \leq C^N N! X^{-N} (\log X)^N$ valid for all N and all $X \geq X_0$, with C, X_0 absolute? Is there a version uniform in a complex sector and in the branch index k of W_k ? None of these is claimed or proved anywhere in this volume; the literature asserts related convergence statements [5, 21, 11] whose proofs the present consolidation has not verified.

Remark 29.13 (Open problem: realisation of left-finite Hahn elements by germs). [Theorem 28.10](#) identifies $\mathcal{H}_{\text{lf}}(\mathbb{K}[L], \Gamma)$ as the completion of the finitely supported power–log series, and [Theorem 28.11](#) shows it carries the whole near-identity theory. Which of its elements are asymptotic expansions of germs at $+\infty$ lying in a common Hardy field, in the sense of [Theorem 10.32](#)? For $\Gamma = \mathbb{Z}$ the question is the classical one about power–log series; for $\Gamma = \mathbb{Q}$, where by [Theorem 28.15](#) the class strictly contains the Puiseux tower, we do not know a characterisation, nor an example of an element of $\mathcal{H}_{\text{lf}}(\mathbb{K}[L], \mathbb{Q}) \setminus \mathfrak{P}$ that is provably realised.

Remark 29.14 (Open problem: a resurgent completion with block arithmetic). [Theorem 29.6](#) shows the class cannot see exponentially small corrections. Is there a minimal enlargement – adjoining a single transmonomial $e^{-\mu X}$, $\mu > 0$, with coefficients in $\mathcal{PL}_{\text{poly}, \mathbb{K}}$, that is, the ring $\mathcal{PL}_{\text{poly}, \mathbb{K}}[[e^{-\mu X}]]$ with its natural derivation – in which the block arithmetic of Parts I–IV survives verbatim, the unit criterion and the reversion theorem included? [Theorem 29.4](#) shows the valuation gain of D_X fails on the new generator, so a positive answer would need a two-index filtration (order in t , order in $e^{-\mu X}$) and a proof that the composition operator is continuous for it. We state this as open because the naive filtration does not work and we have not verified any substitute.

Remark 29.15 (Open problem: which of these statements are machine-checkable today). The formalisation register of this volume records, for each stated result, its status in a proof assistant. The support-theoretic results of [Section 28.2](#) are finitary in character – they are statements about ordered abelian groups and well-quasi-orders – and should be reachable. [Theorem 28.20](#) is a finite computation with a monomial bookkeeping argument. [Theorem 28.10](#) needs a formal Hahn series library with a Γ -indexed filtration. Which of these is actually within reach of a current library, and at what cost, is not known to us, and the register should be read as a plan rather than as a claim.

Section summary

The polynomial–logarithmic class is a proper, well-delimited fragment of the logarithmic–exponential transseries field. What separates it from the ambient is a single structural fact, proved here at every logarithmic depth: inside the class the logarithmic derivative of a nonzero element is never larger than X^{-1} ([Theorem 29.1](#)). Everything the class does easily – triangular reversion, terminating algorithms, exact residual certificates – follows from the valuation gain that this fact expresses, and everything it cannot do – exponentially small terms, Stokes data, resurgence – follows from the same fact read as a limitation ([Theorem 29.4](#), [Theorem 29.6](#)). The right attitude is neither to inflate the class nor to abandon it, but to promote explicitly, one axis at a time, with the closure test named.

Appendices

A Bell polynomials and Faà di Bruno bookkeeping

The exponential partial Bell polynomials appear at four separate places in the body of this volume: in the coefficients of an integer power ([Theorem 6.36](#)), in the coefficients of a series substituted into a scalar power series ([Theorems 8.5 and 11.22](#)), in the shift recurrence that drives Faà di Bruno for admissible composition ([Theorems 11.24 and 11.25](#)), and in the closed form of the convolution polynomials ([Theorem 11.13 \(iv\)](#)). (The second member of that group restates the first in the part on composition, and adds a clause about a base point $z_0 \neq 0$; only the first is proved from scratch.) In each of those places the defining generating identity is quoted from [\[4\]](#). This appendix pays that debt: it defines the family, proves the generating identity and the recurrences from the definition, proves the two Stirling evaluations — one of which is exactly the arithmetic content of the Lambert polynomials of [Section 19](#) — and proves the form of Faà di Bruno that the volume needs but does not state, namely the one whose outer function is a scalar-analytic function of a polynomial–logarithmic series.

Throughout, $\mathbb{K}$ is a field of characteristic zero, so that $\mathbb{Q} \subseteq \mathbb{K}$ and every factorial occurring below is invertible. The polynomial identities of [Sections A.1 and A.2](#) are identities in $\mathbb{Q}[x_1, x_2, \dots]$ and are therefore available over every commutative $\mathbb{Q}$ -algebra, in particular over $\mathbb{K}[L]$, $\mathbb{K}(L)$, $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$.

A.1 Definition, generating identity, and the two normalizations

Definition A.1 (Exponential partial and complete Bell polynomials). Let $x_1, x_2, \dots$ be independent indeterminates. For $n, k \in \mathbb{N}_0$ put

$$\mathcal{B}_{n,k}^{\text{exp}}(x_1, x_2, \dots) := \sum_{\substack{m_1, m_2, \dots \geq 0 \\ \sum_{j \geq 1} m_j = k \\ \sum_{j \geq 1} j m_j = n}} \frac{n!}{\prod_{j \geq 1} m_j! (j!)^{m_j}} \prod_{j \geq 1} x_j^{m_j} \in \mathbb{Q}[x_1, x_2, \dots], \quad (510)$$

the sum ranging over all sequences $(m_j)_{j \geq 1}$ of nonnegative integers with finite support satisfying the two displayed constraints, and the empty product being 1. The *complete* polynomial is

$$\mathcal{B}_n^{\text{exp}}(x_1, \dots, x_n) := \sum_{k=0}^n \mathcal{B}_{n,k}^{\text{exp}}(x_1, \dots, x_{n-k+1}). \quad (511)$$

The index set in (510) is finite: the constraint $\sum_j j m_j = n$ forces $m_j = 0$ for $j > n$ and $m_j \leq n/j$. It is empty — so that $\mathcal{B}_{n,k}^{\text{exp}} = 0$ — whenever no such sequence exists. Two degenerate cases matter and are fixed once and for all by (510) itself: for $n = k = 0$ the only sequence is $m \equiv 0$, giving $\mathcal{B}_{0,0}^{\text{exp}} = 1$; and for $n > 0, k = 0$ the constraints $\sum m_j = 0$ and $\sum j m_j = n > 0$ are incompatible, so $\mathcal{B}_{n,0}^{\text{exp}} = 0$. Consequently $\mathcal{B}_0^{\text{exp}} = 1$.

Lemma A.2 (Support, degree, weight, and boundary values). Fix $n, k \in \mathbb{N}_0$ and write $B := \mathcal{B}_{n,k}^{\text{exp}}$.

- (i) $B = 0$ unless $0 \leq k \leq n$; moreover $B = 1$ when $n = k = 0$ and $B = 0$ when $n \geq 1$ and $k = 0$. When $1 \leq k \leq n$, only the variables $x_1, \dots, x_{n-k+1}$ occur in B .
- (ii) B is homogeneous of degree k in the variables and isobaric of weight n for the weighting $\text{wt}(x_j) = j$. Equivalently, for indeterminates a, b ,

$$\mathcal{B}_{n,k}^{\text{exp}}(abx_1, ab^2x_2, ab^3x_3, \dots) = a^k b^n \mathcal{B}_{n,k}^{\text{exp}}(x_1, x_2, \dots). \quad (512)$$

- (iii) The boundary rows and columns are

$$\mathcal{B}_{n,n}^{\text{exp}} = x_1^n, \quad \mathcal{B}_{n,1}^{\text{exp}} = x_n \ (n \geq 1), \quad \mathcal{B}_{n,n-1}^{\text{exp}} = \binom{n}{2} x_1^{n-2} x_2 \ (n \geq 2). \quad (513)$$

Proof. (i) A contributing sequence has $\sum_j m_j = k$ and $\sum_j jm_j = n$; since every $j \geq 1$, the second sum is at least the first, so $n \geq k$. If $k = 0$ the first sum forces $m \equiv 0$ and then $n = 0$. Suppose $1 \leq k \leq n$ and $m_i \geq 1$ for some i . The remaining $k - 1$ parts contribute at least $k - 1$ to $\sum_j jm_j$, whence $i \leq n - (k - 1) = n - k + 1$.

(ii) Each monomial $\prod_j x_j^{m_j}$ occurring in (510) has total degree $\sum_j m_j = k$ and weight $\sum_j jm_j = n$. Substituting $x_j \mapsto ab^j x_j$ multiplies that monomial by $a^{\sum_j m_j} b^{\sum_j jm_j} = a^k b^n$, which is (512).

(iii) For $k = n \geq 1$, $\sum_j m_j = \sum_j jm_j$ forces $m_j = 0$ for $j \geq 2$ and hence $m_1 = n$; the coefficient is $n!/(n! (1!)^n) = 1$. For $k = 1$ the only sequence is $m_n = 1$ with all other entries zero; the coefficient is $n!/(1! (n!)^1) = 1$. For $k = n - 1$ with $n \geq 2$ the only sequence is $m_1 = n - 2$, $m_2 = 1$; its coefficient is $n!/((n - 2)! 1! (1!)^{n-2} (2!)^1) = n(n - 1)/2 = \binom{n}{2}$. $\square$

Proposition A.3 (The defining generating identity). *Let R be a commutative $\mathbb{Q}$ -algebra, let $\xi_1, \xi_2, \dots \in R$, and put $\Xi(z) := \sum_{j \geq 1} \xi_j z^j / j! \in zR[[z]]$. Then, in $R[[z]]$,*

$$\frac{\Xi(z)^k}{k!} = \sum_{n \geq k} \mathcal{B}_{n,k}^{\exp}(\xi_1, \xi_2, \dots) \frac{z^n}{n!} \quad (k \in \mathbb{N}_0), \quad (514)$$

and the family $(\Xi^k/k!)_{k \geq 0}$ is summable in $R[[z]]$ with

$$\exp \Xi(z) := \sum_{k \geq 0} \frac{\Xi(z)^k}{k!} = \sum_{n \geq 0} \mathcal{B}_n^{\exp}(\xi_1, \dots, \xi_n) \frac{z^n}{n!}. \quad (515)$$

Conversely, (514) taken over $R = \mathbb{Q}[x_1, x_2, \dots]$ with $\xi_j = x_j$ determines (510) uniquely.

Proof. Fix $k \geq 1$ (the case $k = 0$ is $1 = 1$). By the multinomial theorem in $R[[z]]$, applied to the k -th power of a series with zero constant term,

$$\Xi(z)^k = \sum_{(j_1, \dots, j_k) \in \mathbb{N}_{>0}^k} \prod_{i=1}^k \frac{\xi_{j_i} z^{j_i}}{j_i!} = \sum_{n \geq k} z^n \sum_{\substack{j_1, \dots, j_k \geq 1 \\ j_1 + \dots + j_k = n}} \prod_{i=1}^k \frac{\xi_{j_i}}{j_i!},$$

the rearrangement being legitimate because for fixed n only the finitely many tuples with $j_1 + \dots + j_k = n$ contribute, and there are none when $n < k$. Now group the tuples by their multiplicity vector: a tuple $(j_1, \dots, j_k)$ with $m_j := \#\{i : j_i = j\}$ satisfies $\sum_j m_j = k$ and $\sum_j jm_j = n$, and the number of tuples with a prescribed multiplicity vector is the multinomial coefficient $k! / \prod_j m_j!$. Every tuple in one such class contributes the same term $\prod_j (\xi_j / j!)^{m_j}$. Hence

$$[z^n] \Xi^k = \sum_{\substack{\sum_j m_j = k \\ \sum_j jm_j = n}} \frac{k!}{\prod_j m_j!} \prod_j \left(\frac{\xi_j}{j!} \right)^{m_j} = \frac{k!}{n!} \mathcal{B}_{n,k}^{\exp}(\xi_1, \xi_2, \dots),$$

by (510). Dividing by $k!$ gives (514).

For (515): $\Xi^k/k! \in z^k R[[z]]$, so the family is summable for the z -adic filtration and its sum is computed blockwise; by (514) and Theorem A.2 (i) the coefficient of $z^n/n!$ in that sum is $\sum_{k=0}^n \mathcal{B}_{n,k}^{\exp}$, which is (511). Finally, in the universal case $R = \mathbb{Q}[x_1, x_2, \dots]$ the coefficients of a formal power series are uniquely determined, so (514) recovers (510). $\square$

Theorem A.3 is precisely the identity quoted without proof in the proofs of Theorem 6.36, Theorem 11.13 (iv) and Theorem 11.24; those citations may now be read as references to (514).

The volume uses two different normalizations for the same combinatorial data, and the conversion between them is a constant source of factorial errors. The next lemma states it once.

Lemma A.4 (Ordinary versus exponential normalization). *Let R be a commutative $\mathbb{Q}$ -algebra and $\xi_1, \xi_2, \dots \in R$. Define the ordinary partial Bell polynomial by*

$$\mathcal{B}_{n,k}^{\text{ord}}(\xi_1, \xi_2, \dots) := \sum_{\substack{m_1, m_2, \dots \geq 0 \\ \sum_j m_j = k \\ \sum_j j m_j = n}} \frac{k!}{\prod_{j \geq 1} m_j!} \prod_{j \geq 1} \xi_j^{m_j}, \quad (516)$$

so that $(\sum_{j \geq 1} \xi_j z^j)^k = \sum_{n \geq k} \mathcal{B}_{n,k}^{\text{ord}}(\xi_1, \dots) z^n$. Then for all $n, k \in \mathbb{N}_0$

$$\mathcal{B}_{n,k}^{\text{ord}}(\xi_1, \xi_2, \dots) = \frac{k!}{n!} \mathcal{B}_{n,k}^{\text{exp}}(1! \xi_1, 2! \xi_2, 3! \xi_3, \dots). \quad (517)$$

In particular, for $H = \sum_{j \geq 1} h_j(L) t^j \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ the convolution polynomials of [Theorem 11.12](#) satisfy $C_{n,k} = \mathcal{B}_{n,k}^{\text{ord}}(h_1, h_2, \dots)$, which is [Theorem 11.13](#) (iv).

Proof. Apply (512) in the substituted form: putting $\hat{\xi}_j = j! \xi_j$ in (510) gives

$$\mathcal{B}_{n,k}^{\text{exp}}(\hat{\xi}_1, \hat{\xi}_2, \dots) = \sum_{\substack{\sum_j m_j = k \\ \sum_j j m_j = n}} \frac{n!}{\prod_j m_j! (j!)^{m_j}} \prod_j (j!)^{m_j} \xi_j^{m_j} = \frac{n!}{k!} \sum_{\substack{\sum_j m_j = k \\ \sum_j j m_j = n}} \frac{k!}{\prod_j m_j!} \prod_j \xi_j^{m_j},$$

which is (517). The generating identity for (516) follows by the same multiplicity grouping as in the proof of [Theorem A.3](#), now without the factors $1/j!$. The last sentence is [Theorem 11.13](#) (i) combined with that generating identity. $\square$

Three two-index arrays, three index orders

The volume carries three arrays whose entries are indexed by a pair of integers and which are easy to confuse.

- $h_{k,m} = [t^m] H^k$ of (196): first index the exponent, second the outer order.
- $C_{m,r}$ of [Theorem 11.12](#): first index the outer order, second the exponent; $h_{k,m} = C_{m,k}$.
- $\mathcal{B}_{n,k}^{\text{exp}}$ of [Theorem A.1](#): first index the *weight* n , second the *degree* k ; it is an ordinary polynomial in abstract indeterminates, and only after the substitution $x_j \mapsto j! h_j$ and the factor $k!/n!$ of (517) does it become $C_{n,k}$.

Transferring a formula between any two of them requires transposing the indices, and between the last and either of the first two also requires the factorial conversion. A missing $k!/n!$ is the single commonest error in this circle of identities.

Proposition A.5 (Two computational recurrences). *For all $n \geq 1$ and $k \geq 1$,*

$$\mathcal{B}_{n,k}^{\text{exp}} = \sum_{j=1}^{n-k+1} \binom{n-1}{j-1} x_j \mathcal{B}_{n-j,k-1}^{\text{exp}}, \quad (518)$$

$$k \mathcal{B}_{n,k}^{\text{exp}} = \sum_{j=1}^{n-k+1} \binom{n}{j} x_j \mathcal{B}_{n-j,k-1}^{\text{exp}}. \quad (519)$$

Either recurrence, together with the boundary values $\mathcal{B}_{0,0}^{\text{exp}} = 1$, $\mathcal{B}_{n,0}^{\text{exp}} = 0$ for $n \geq 1$ and $\mathcal{B}_{0,k}^{\text{exp}} = 0$ for $k \geq 1$, determines the whole array.

Proof. Work in $R[[z]]$ with $R = \mathbb{Q}[x_1, x_2, \dots]$ and $\Xi(z) = \sum_{j \geq 1} x_j z^j / j!$.

For (519), write $\Xi^k / k! = \frac{1}{k} \Xi \cdot \Xi^{k-1} / (k-1)!$ and extract $[z^n / n!]$. For two series $P = \sum_n p_n z^n / n!$ and $Q = \sum_n q_n z^n / n!$ one has $[z^n / n!](PQ) = \sum_{j=0}^n \binom{n}{j} p_j q_{n-j}$; applying this with $P = \Xi$ (so $p_0 = 0$,

$p_j = x_j$) and $Q = \Xi^{k-1}/(k-1)!$ and using (514) gives $\frac{1}{k} \sum_{j \geq 1} \binom{n}{j} x_j \mathcal{B}_{n-j, k-1}^{\text{exp}}$. The upper limit is $n - k + 1$ because $\mathcal{B}_{n-j, k-1}^{\text{exp}} = 0$ unless $n - j \geq k - 1$ (Theorem A.2 (i)).

For (518), differentiate (514) with respect to z :

$$\frac{d}{dz} \frac{\Xi^k}{k!} = \Xi' \frac{\Xi^{k-1}}{(k-1)!}, \quad \Xi'(z) = \sum_{i \geq 0} x_{i+1} \frac{z^i}{i!}.$$

The left side is $\sum_{n \geq 0} \mathcal{B}_{n+1, k}^{\text{exp}} z^n / n!$. Applying the product rule for exponential generating functions to the right side with $p_i = x_{i+1}$ gives $\sum_{i=0}^n \binom{n}{i} x_{i+1} \mathcal{B}_{n-i, k-1}^{\text{exp}}$. Replacing n by $n - 1$ and setting $j = i + 1$ turns this into $\mathcal{B}_{n, k}^{\text{exp}} = \sum_{j \geq 1} \binom{n-1}{j-1} x_j \mathcal{B}_{n-j, k-1}^{\text{exp}}$, which is (518); the upper limit is again forced by Theorem A.2 (i).

Both recurrences express $\mathcal{B}_{n, k}^{\text{exp}}$ in terms of entries $\mathcal{B}_{n-j, k-1}^{\text{exp}}$ with $j \geq 1$, hence in terms of strictly earlier rows; with the three stated boundary values — which supply every entry of row 0 and every entry of column 0 — they determine the array by induction on n . $\square$

Remark A.6 (A third recurrence, and where each is used). Theorem 11.24 proves a third recurrence, $\mathcal{B}_{n+1, k}^{\text{exp}} = \mathcal{D} \mathcal{B}_{n, k}^{\text{exp}} + x_1 \mathcal{B}_{n, k-1}^{\text{exp}}$ with $\mathcal{D} = \sum_{i \geq 1} x_{i+1} \partial_{x_i}$, which is the one adapted to *differentiating* a Bell polynomial evaluated at a tower of derivatives; it is the engine of every Faà di Bruno induction, including Theorem 11.25 and Theorem A.12 below. By contrast (518) and (519) are the ones adapted to *tabulating* the array: each computes one entry from $n - k + 1$ entries of the previous column, so the whole triangle up to weight N costs $\mathcal{O}(N^3)$ multiplications of one indeterminate x_j into one previously computed entry. That is a count of *entry* operations, not of scalar ones: the entry $\mathcal{B}_{n, k}^{\text{exp}}$ has exactly one monomial for each partition of n into k parts, so row n carries as many monomials as there are partitions of n , and the number of scalar coefficients written while tabulating up to weight N therefore exceeds every polynomial in N . The distinction between the two levels of counting is the one drawn in Theorem 25.18.

Example A.7 (The array through weight six). Table 16 lists $\mathcal{B}_{n, k}^{\text{exp}}$ for $1 \leq k \leq n \leq 6$; the entries were generated from (518) and checked against (510). The first column is $\mathcal{B}_{n, 1}^{\text{exp}} = x_n$, the diagonal is x_1^n and the subdiagonal is $\binom{n}{2} x_1^{n-2} x_2$, as (513) predicts. Each entry is homogeneous of degree k and isobaric of weight n , which is the cheapest available check on a hand computation: in $\mathcal{B}_{6, 3}^{\text{exp}} = 15x_1^2 x_4 + 60x_1 x_2 x_3 + 15x_2^3$ every monomial has three factors and subscript sum six. A second check is that setting all $x_j = 1$ must return the Stirling number $\{n\}_k$ of Theorem A.9 (ii); for $n = 6, k = 3$ this reads $15 + 60 + 15 = 90$.

A.2 Stirling evaluations, and the link to the Lambert polynomials

The unsigned Stirling numbers of the first kind were fixed in (349) by $(z)^{\overline{n}} = \sum_k s(n, k) z^k$ and $[n]_k = (-1)^{n-k} s(n, k)$. We first record the exponential generating function that converts that definition into a Bell evaluation.

Lemma A.8 (Rising-factorial and Stirling generating identities). *In $\mathbb{Q}[z][[u]]$,*

$$\sum_{n \geq 0} (z)^{\overline{n}} \frac{u^n}{n!} = \exp\left(z \log \frac{1}{1-u}\right), \quad (z)^{\overline{n}} = z(z+1) \cdots (z+n-1), \quad (520)$$

where $\log \frac{1}{1-u} := \sum_{i \geq 1} u^i / i$ and $\exp$ is the formal exponential of a series with zero constant term. Consequently $(z)^{\overline{n}} = \sum_{k=0}^n [n]_k z^k$ and, for every $k \in \mathbb{N}_0$,

$$\frac{1}{k!} \left(\log \frac{1}{1-u} \right)^k = \sum_{n \geq k} \begin{bmatrix} n \\ k \end{bmatrix} \frac{u^n}{n!}. \quad (521)$$

Table 16: The exponential partial Bell polynomials $\mathcal{B}_{n,k}^{\text{exp}}(x_1, x_2, \dots)$ for $1 \leq k \leq n \leq 6$.

n	k	$\mathcal{B}_{n,k}^{\text{exp}}$
1	1	x_1
2	1	x_2
	2	x_1^2
3	1	x_3
	2	$3x_1x_2$
	3	x_1^3
4	1	x_4
	2	$4x_1x_3 + 3x_2^2$
	3	$6x_1^2x_2$
	4	x_1^4
5	1	x_5
	2	$5x_1x_4 + 10x_2x_3$
	3	$10x_1^2x_3 + 15x_1x_2^2$
	4	$10x_1^3x_2$
	5	x_1^5
6	1	x_6
	2	$6x_1x_5 + 15x_2x_4 + 10x_3^2$
	3	$15x_1^2x_4 + 60x_1x_2x_3 + 15x_2^3$
	4	$20x_1^3x_3 + 45x_1^2x_2^2$
	5	$15x_1^4x_2$
	6	x_1^6

Proof. The unsigned form of (349) is obtained by replacing z with $-z$: since $((-z))^n = (-z)(-z-1)\cdots(-z-n+1) = (-1)^n (z)^{\overline{n}}$, we get $(z)^{\overline{n}} = (-1)^n \sum_k s(n, k) (-z)^k = \sum_k (-1)^{n-k} s(n, k) z^k = \sum_k \begin{bmatrix} n \\ k \end{bmatrix} z^k$.

For (520) we compare two solutions of one linear differential equation in $\mathbb{Q}[z][[u]]$. A series $W = \sum_{n \geq 0} w_n u^n / n!$ satisfies

$$(1 - u) \partial_u W = zW, \quad w_0 = 1, \quad (522)$$

if and only if $w_{n+1} = (z + n)w_n$ for every $n \geq 0$; indeed $\partial_u W = \sum_{n \geq 0} w_{n+1} u^n / n!$ and $u \partial_u W = \sum_{n \geq 1} n w_n u^n / n!$, so comparing coefficients of $u^n / n!$ in (522) gives $w_{n+1} - n w_n = z w_n$. In particular (522) has exactly one solution, and $w_n = (z)^{\overline{n}}$ is it, because $(z)^{\overline{0}} = 1$ and $(z)^{\overline{n+1}} = (z + n) (z)^{\overline{n}}$.

Now put $\ell(u) := \log \frac{1}{1-u} = \sum_{i \geq 1} u^i / i$, so that $\partial_u \ell = \sum_{i \geq 1} u^{i-1} = (1 - u)^{-1}$, and $Z := \exp(z \ell(u)) = \sum_{k \geq 0} z^k \ell(u)^k / k!$, a summable family in $\mathbb{Q}[z][[u]]$ because $\ell^k \in u^k \mathbb{Q}[[u]]$. Termwise differentiation of that summable family gives $\partial_u Z = z(\partial_u \ell)Z = zZ/(1 - u)$, that is $(1 - u)\partial_u Z = zZ$, and Z has constant term 1. By the uniqueness just proved, $Z = \sum_{n \geq 0} (z)^{\overline{n}} u^n / n!$, which is (520).

Finally, compare the coefficient of z^k on the two sides of (520): on the left it is $\sum_{n \geq 0} \begin{bmatrix} n \\ k \end{bmatrix} u^n / n!$ by the expansion $(z)^{\overline{n}} = \sum_k \begin{bmatrix} n \\ k \end{bmatrix} z^k$ just established, and on the right it is $\ell(u)^k / k!$. This is (521); the sum there may be started at $n = k$ because $\begin{bmatrix} n \\ k \end{bmatrix} = 0$ for $k > n$. $\square$

Theorem A.9 (Stirling evaluations of the Bell polynomials). *For all $n, k \in \mathbb{N}_0$,*

- (i) $\mathcal{B}_{n,k}^{\text{exp}}(0!, 1!, 2!, \dots) = \begin{bmatrix} n \\ k \end{bmatrix}$, the unsigned Stirling number of the first kind; equivalently $\mathcal{B}_{n,k}^{\text{exp}}((j-1)!)_{j \geq 1} = (-1)^{n-k} s(n, k)$;
- (ii) $\mathcal{B}_{n,k}^{\text{exp}}(1, 1, 1, \dots) = \{n \}_k$, the Stirling number of the second kind;
- (iii) $\mathcal{B}_n^{\text{exp}}(0!, 1!, 2!, \dots) = n!$.

Proof. (i) Apply (514) over $R = \mathbb{Q}$ with $\xi_j = (j-1)!$. Then $\Xi(u) = \sum_{j \geq 1} (j-1)! u^j / j! = \sum_{j \geq 1} u^j / j = \log \frac{1}{1-u}$, so (514) reads

$$\frac{1}{k!} \left(\log \frac{1}{1-u} \right)^k = \sum_{n \geq k} \mathcal{B}_{n,k}^{\exp}(0!, 1!, 2!, \dots) \frac{u^n}{n!}.$$

Comparing with (521) and using uniqueness of coefficients in $\mathbb{Q}[[u]]$ gives (i). The second form of (i) is then the sign convention (349).

(ii) With $\xi_j = 1$ for all j one gets $\Xi(u) = \sum_{j \geq 1} u^j / j! = e^u - 1$, and (514) becomes $\frac{1}{k!} (e^u - 1)^k = \sum_{n \geq k} \mathcal{B}_{n,k}^{\exp}(1, 1, \dots) u^n / n!$. The left-hand side is the classical exponential generating function of the Stirling numbers of the second kind, which we verify from the definition $\{n\}_k = \frac{1}{k!} \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} i^n$. Expanding $(e^u - 1)^k$ by the binomial theorem gives

$$(e^u - 1)^k = \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} e^{iu} = \sum_{n \geq 0} \left(\sum_{i=0}^k (-1)^{k-i} \binom{k}{i} i^n \right) \frac{u^n}{n!},$$

so $[u^n / n!] \frac{1}{k!} (e^u - 1)^k = \{n\}_k$.

(iii) Sum (i) over $0 \leq k \leq n$ and use $\sum_{k=0}^n \binom{n}{k} = (1)^{\overline{n}} = 1 \cdot 2 \cdots n = n!$, which is the polynomial identity $(z)^{\overline{n}} = \sum_k \binom{n}{k} z^k$ of Theorem A.8 evaluated at $z = 1$. $\square$

Corollary A.10 (The Lambert triangle is a slice of the Bell array). *For $n \geq 1$ and $1 \leq m \leq n$, the coefficients $\lambda_{n,m} = [u^m] \mathcal{L}_n^{\text{Lam}}(u)$ of the Lambert polynomials (Theorems 19.1 and 19.4) are*

$$\begin{aligned} \lambda_{n,m} &= \frac{(-1)^{n-m}}{m!} \mathcal{B}_{n,n-m+1}^{\exp}(0!, 1!, 2!, \dots), \\ \mathcal{L}_n^{\text{Lam}}(u) &= \sum_{m=1}^n \frac{(-1)^{n-m}}{m!} \mathcal{B}_{n,n-m+1}^{\exp}(0!, 1!, 2!, \dots) u^m. \end{aligned} \tag{523}$$

Reading the array by antidiagonals, the n -th Lambert polynomial consumes exactly the n -th row of the Bell array evaluated at the factorial arguments, in reversed column order and with alternating signs.

Proof. Substitute Theorem A.9 (i) with $k = n - m + 1$ into (351). $\square$

Remark A.11 (Why the factorial arguments are the right ones). The evaluation $x_j \mapsto (j-1)!$ looks arbitrary until one notices what it encodes. By the proof of Theorem A.9 (i) it is exactly the evaluation for which the generating series Ξ of Theorem A.3 becomes a logarithm. The Lambert polynomials arise from reverting $F = X + L$, that is from a perturbation which is the logarithm (Theorem 18.10); the appearance of cycle numbers in (523) is therefore not a coincidence of two unrelated triangles but the same statement twice. The table of $\left[\begin{smallmatrix} n \\ n-m+1 \end{smallmatrix} \right]$ in Table 28 is simultaneously a table of Bell polynomials at factorial arguments. Note that the correspondence is between the *unsigned* cycle numbers and the Bell array: the signs in (523) come from the alternating prefactor of (344) and not from the Bell polynomials, which have nonnegative integer coefficients by (510).

A.3 Faà di Bruno with a scalar-analytic outer function

Theorem 11.25 computes $D_X^n(F \circ G)$ when F and G are both polynomial–logarithmic and $\circ$ is the composition of Section 10. The volume also needs the complementary case, in which the outer object is not an element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ at all but a scalar power series $\varphi \in \mathbb{K}[[z]]$ applied to a polynomial–logarithmic argument through the functional calculus of Theorem 11.22. The two statements have the same proof, and we isolate that proof once.

Lemma A.12 (Abstract Faà di Bruno for a Leibniz tower). *Let R be a commutative $\mathbb{Q}$ -algebra, ∂ a derivation of R , $\gamma \in R$, and let $(u_k)_{k \geq 0}$ be a sequence in R satisfying*

$$\partial u_k = u_{k+1} \gamma \quad (k \geq 0). \quad (524)$$

Then for every $n \geq 1$,

$$\partial^n u_0 = \sum_{k=1}^n u_k \mathcal{B}_{n,k}^{\text{exp}}(\gamma, \partial\gamma, \dots, \partial^{n-k}\gamma). \quad (525)$$

Proof. Induction on n . For $n = 1$, (525) reads $\partial u_0 = u_1 \mathcal{B}_{1,1}^{\text{exp}}(\gamma) = u_1 \gamma$, which is (524) at $k = 0$.

Assume (525) for some $n \geq 1$ and abbreviate $\beta_{n,k} := \mathcal{B}_{n,k}^{\text{exp}}(\gamma, \partial\gamma, \dots)$, the evaluation of the polynomial $\mathcal{B}_{n,k}^{\text{exp}}$ at $x_i = \partial^{i-1}\gamma$. Since ∂ is a derivation and $\mathcal{B}_{n,k}^{\text{exp}}$ is a polynomial in finitely many of the x_i , the ordinary chain rule for polynomials gives

$$\partial \beta_{n,k} = \sum_{i \geq 1} (\partial_{x_i} \mathcal{B}_{n,k}^{\text{exp}})(\gamma, \partial\gamma, \dots) \cdot \partial^i \gamma = (\mathcal{D} \mathcal{B}_{n,k}^{\text{exp}})(\gamma, \partial\gamma, \dots), \quad \mathcal{D} = \sum_{i \geq 1} x_{i+1} \partial_{x_i},$$

because $\partial^i \gamma$ is the value of x_{i+1} . Hence, using the Leibniz rule and (524),

$$\partial^{n+1} u_0 = \sum_{k \geq 1} \left(u_{k+1} \gamma \beta_{n,k} + u_k (\mathcal{D} \mathcal{B}_{n,k}^{\text{exp}})(\gamma, \partial\gamma, \dots) \right) = \sum_{k \geq 1} u_k \left[\gamma \mathcal{B}_{n,k-1}^{\text{exp}} + \mathcal{D} \mathcal{B}_{n,k}^{\text{exp}} \right](\gamma, \partial\gamma, \dots),$$

after shifting the index in the first sum (the term $k = 1$ of the shifted sum carries $\mathcal{B}_{n,0}^{\text{exp}} = 0$ for $n \geq 1$, so nothing is lost). By Theorem 11.24 evaluated at $x_1 = \gamma$, the bracket is $\mathcal{B}_{n+1,k}^{\text{exp}}$. The sum truncates at $k = n + 1$ by Theorem A.2 (i), and the arguments needed are $x_1, \dots, x_{n+1-k+1}$, that is $\gamma, \dots, \partial^{n+1-k}\gamma$. All sums are finite. $\square$

Remark A.13 (The two Faà di Bruno statements are two towers). Theorem 11.25 is Theorem A.12 in $R = \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ with $\partial = D_X$, $\gamma = D_X G$ and $u_k = (D_X^k F) \circ G$: the tower condition (524) is then exactly the chain rule Theorem 10.31. Theorem A.15 below is the same lemma with $\gamma = D_X F$ and $u_k = \varphi^{(k)}(F)$. Nothing else distinguishes the two; in particular neither is a corollary of the other, since the towers are different.

Before applying Theorem A.12 to a scalar-analytic outer function we must know that the required tower exists, that is, that the chain rule holds for the functional calculus $\varphi \mapsto \varphi(W)$ of Theorem 11.22. For a polynomial φ this is the Leibniz rule; for a power series it needs the continuity of D_X .

Lemma A.14 (Chain rule for a scalar-analytic substitution). *Let $z_0 \in \mathbb{K}$ and let the outer function be given by its expansion about z_0 : that is, let $\varphi_0, \varphi_1, \dots \in \mathbb{K}$ be arbitrary scalars and write, formally, $\varphi(z) = \sum_{k \geq 0} \frac{\varphi_k}{k!} (z - z_0)^k$, the datum being the pair $(z_0, (\varphi_k)_{k \geq 0})$. Let $F \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ satisfy $\nu_t(F - z_0) \geq 1$, write $W := F - z_0 \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and put*

$$\varphi(F) := \sum_{k \geq 0} \frac{\varphi_k}{k!} W^k, \quad \varphi^{(k)}(F) := \sum_{i \geq 0} \frac{\varphi_{k+i}}{i!} W^i \quad (k \geq 0), \quad (526)$$

each well defined by Theorem 11.22 applied to the series $\sum_{k \geq 0} (\varphi_k/k!) z^k \in \mathbb{K}[[z]]$ and to W . When $z_0 = 0$ this is the ordinary reading $\varphi(F) = \varphi(W)$; when $z_0 \neq 0$ the centre must be part of the datum, since an element of $\mathbb{K}[[z]]$ presented at the origin cannot be re-expanded about z_0 inside $\mathbb{K}[[z]]$. Then

$$D_X(\varphi(F)) = \varphi'(F) D_X F. \quad (527)$$

If $\varphi \in \mathbb{K}[z]$ is a polynomial, (527) holds for every $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, with $\varphi(F)$ the ordinary polynomial evaluation and no valuation hypothesis. The two readings of $\varphi(F)$ agree when both apply, since the Taylor expansion of a polynomial about z_0 is exact and finite.

Proof. Each $\varphi^{(k)}(F)$ is defined by (526), the family there being summable because $\nu_t(W^i) \geq i$. Write $c_k = \varphi_k/k! \in \mathbb{K}$, so that $\varphi(F) = \sum_{k \geq 0} c_k W^k$ and $\varphi'(F) = \sum_{k \geq 1} k c_k W^{k-1}$, both families being summable because $\nu_t(W^k) \geq k$ (Theorem 4.2, Theorem 8.1). Note $D_X W = D_X F$, because $D_X z_0 = 0$ for a scalar z_0 (Theorem 9.11).

The identity (527) is checked one block at a time. Fix $N \geq 0$ and split $\varphi(F) = P_N + T_N$ with $P_N = \sum_{k \leq N+1} c_k W^k$ and $\nu_t(T_N) \geq N + 2$. Since D_X is a $\mathbb{K}$ -derivation of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ with $\nu_t(D_X H) \geq \nu_t H + 1$ (Theorem 9.1, Theorem 9.10 (1)), we get $\nu_t(D_X T_N) \geq N + 3$, so

$$[t^N] D_X(\varphi(F)) = [t^N] D_X P_N = \sum_{k=1}^{N+1} k c_k [t^N] (W^{k-1} D_X W),$$

the last step being the Leibniz rule applied to the *finite* sum P_N . On the other side, $\nu_t(W^{k-1} D_X W) \geq (k-1) + 2 = k+1$, so the terms with $k \geq N+2$ do not reach the block t^N and

$$[t^N] (\varphi'(F) D_X F) = \sum_{k \geq 1} k c_k [t^N] (W^{k-1} D_X W) = \sum_{k=1}^{N+1} k c_k [t^N] (W^{k-1} D_X W).$$

The two agree for every N , and an element of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is determined by its blocks (Theorem 5.13), which proves (527). For a polynomial φ the sums are finite from the start and only the Leibniz rule is used, so no valuation hypothesis is needed and the identity holds in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. $\square$

Theorem A.15 (Faà di Bruno for a scalar-analytic outer function). *Let $z_0 \in \mathbb{K}$, let φ be given by its expansion about z_0 as in Theorem A.14, and let $F \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ satisfy $\nu_t(F - z_0) \geq 1$; or let $\varphi \in \mathbb{K}[z]$ be a polynomial and $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ arbitrary. Then for every $n \geq 1$*

$$D_X^n(\varphi(F)) = \sum_{k=1}^n \varphi^{(k)}(F) \mathcal{B}_{n,k}^{\text{exp}}(D_X F, D_X^2 F, \dots, D_X^{n-k+1} F), \quad (528)$$

an identity in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ in the first case and in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ in the second.

Proof. Apply Theorem A.12 in $R = \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ with $\partial = D_X$, $\gamma = D_X F$ and $u_k = \varphi^{(k)}(F)$. The tower condition (524) is $D_X(\varphi^{(k)}(F)) = \varphi^{(k+1)}(F) D_X F$, which is Theorem A.14 applied to $\varphi^{(k)}$ in place of φ . The arguments of the Bell polynomial are $x_i = \partial^{i-1} \gamma = D_X^i F$, $1 \leq i \leq n-k+1$. Membership in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ in the first case follows because $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is closed under D_X (Theorem 9.1 (i)) and under the substitution (Theorem 11.22). $\square$

Corollary A.16 (Derivatives of a power, and the Lagrange terms). *For every $A \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, every $r \in \mathbb{N}_0$ and every $n \geq 1$,*

$$D_X^n(A^r) = \sum_{k=1}^{\min\{n,r\}} (r)_{\underline{k}} A^{r-k} \mathcal{B}_{n,k}^{\text{exp}}(D_X A, D_X^2 A, \dots, D_X^{n-k+1} A), \quad (529)$$

where $(r)_{\underline{k}} = r(r-1) \cdots (r-k+1)$ is the falling factorial, and where the terms with $k > r$ are absent because $(r)_{\underline{k}} = 0$ there.

Proof. Take $\varphi(z) = z^r$ in the polynomial case of Theorem A.15; then $\varphi^{(k)}(z) = (r)_{\underline{k}} z^{r-k}$ for $k \leq r$ and $\varphi^{(k)} = 0$ for $k > r$. $\square$

Remark A.17 (Valuation bookkeeping). In the setting of Theorem A.15 with $\nu_t(F - z_0) \geq 1$, one has $\nu_t(D_X^i F) \geq i+1$ by Theorem 9.10, so a monomial of $\mathcal{B}_{n,k}^{\text{exp}}$ with multiplicity vector (m_j) evaluates to something of order at least $\sum_j m_j(j+1) = n+k$. The k -th summand of (528) therefore has $\nu_t \geq n+k \geq n+1$, matching the bound $\nu_t(D_X^n(\varphi(F))) \geq n+1$ that Theorem 9.10 (2) gives directly, since $\varphi(F) - \varphi(z_0) \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Consequently only $k \leq N-n$ can affect the block t^N : the combinatorial size of (528) at a fixed target block is bounded independently of any analytic consideration.

B Closed-form coefficient extraction

Parts II–IV compute coefficients by *recurrences*: the reciprocal by (100), the quotient by (101), the convolution polynomials by (201), the inverse by Theorem 15.2. A recurrence is the right tool when all coefficients up to a cutoff are wanted, and it is what an implementation should run (Section 25). It is the wrong tool when a *single* block is wanted, when the block is wanted as an explicit function of the input blocks, or when one has to prove something about the block — a degree bound, a sign, an integrality statement — for all N at once.

This section derives the non-recursive counterparts. By a *closed form* we mean a formula for $[t^N]$ or $[t^N L^d]$ of the result as a finite sum, indexed by an explicitly bounded set, of products of the input blocks and of universal combinatorial constants, with no reference to previously computed output blocks. Every such formula below is derived from the body of the volume and from Section A; none is quoted.

B.1 The multi-index bookkeeping

Two separate finiteness mechanisms are in play, and conflating them is the source of the “infinite sum” complaints that the sources occasionally raise against these formulas.

Lemma B.1 (The two finiteness mechanisms). *Let $F = \sum_{n \geq n_0} p_n(L) t^n \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and fix $N \in \mathbb{Z}$, $d \in \mathbb{N}_0$.*

- (i) (Outer, by valuation.) *If $(S_i)_{i \in I}$ is a summable family in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, then $\{i \in I : [t^N] S_i \neq 0\}$ is finite and $[t^N] \sum_i S_i = \sum_i [t^N] S_i$.*
- (ii) (Inner, by degree.) *Each block p_n lies in $\mathbb{K}[L]$, hence has finite $\deg_L$, so $[t^N L^d] F = 0$ for $d > d_F(N)$ and the block p_N is a finite sum of monomials.*
- (iii) *The two are independent: a summable family may have blocks of unbounded $\deg_L$ at a fixed level of the partial sums — for instance $S_i = t^i L^i$ is summable although $d_{\sum_{i \leq M} S_i}$ is unbounded in M — and conversely a family of bounded $\deg_L$ need not be summable. Only (i) is part of the t -adic topology; a bound on $\deg_L$ is an additional approximation and must be stated separately.*

Proof. (i) is Theorem 5.22 together with Theorem 5.13. (ii) is Theorem 4.11 and the definition of $\mathbb{K}[L]$. For the first half of (iii), $\nu_t(t^i L^i) = i \rightarrow \infty$, so the family is summable, while d of the M -th partial sum takes the value M at level M ; for the second half, the constant family $S_i = 1$ has $\deg_L S_i = 0$ and is not summable. $\square$

The formal tail does not bound a logarithmic degree

The statement $F = G + \mathcal{O}_t^{\text{form}}(t^N)$ says nothing whatever about $\deg_L$ of the discarded blocks (Theorem 5.7); truncation is exclusive in t and whole in L (Theorem 5.1). Every closed formula below therefore returns a complete polynomial block, never a polynomial block truncated in L . Discarding a high power of L inside a retained block is a second, independent approximation and requires its own error statement.

B.2 Products

Proposition B.2 (Scalar coefficients of a product). *Let $F, G \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ be nonzero with $a = \nu_t F$, $b = \nu_t G$, and write $a_{n,e} = [t^n L^e] F$, $b_{m,f} = [t^m L^f] G$ in the notation of Theorem 5.12. Then for all $N \in \mathbb{Z}$ and $d \in \mathbb{N}_0$,*

$$[t^N L^d](FG) = \sum_{n=a}^{N-b} \sum_{e=0}^d a_{n,e} b_{N-n, d-e}, \quad (530)$$

a sum of at most $(N - a - b + 1)(d + 1)$ products, empty when $N < a + b$. The nonzero terms satisfy the sharper constraints $e \leq d_F(n)$ and $d - e \leq d_G(N - n)$; in particular $[t^N L^d](FG) = 0$ whenever $d > (d_F * d_G)(N) = \max_{n+m=N} (d_F(n) + d_G(m))$.

Proof. By Theorem 6.5 and Theorem 6.6, $[t^N](FG) = \sum_{n=a}^{N-b} p_n q_{N-n}$, a finite sum in $\mathbb{K}[L]$. Extracting $[L^d]$ of a product of two polynomials is the Cauchy product in L , which gives (530); the index e is confined to $[0, d]$ because both factors are polynomials, hence have no negative powers of L . The sharper constraints and the vanishing statement are the degree bound (78) of Theorem 6.25, whose $*$ is the $(\max, +)$ -convolution displayed above. $\square$

Formula (530) is a genuine two-level convolution: outer in t by valuation, inner in L by degree. Its cost, and the fact that the resulting scalar cost is *not* $\mathcal{O}(N^2)$ once the inner level is counted, are analysed in Theorem 25.17 and Theorem 25.18.

B.3 Powers, reciprocals, and the convolution polynomials

The convolution polynomials $C_{m,r}$ of Theorem 11.12 — defined by (199) with the empty-product conventions $C_{0,0} = 1$, $C_{m,0} = 0$ for $m > 0$, and computed by the recurrence (201) — are the blocks of the powers of a series. Theorem 11.13 (iv) gives their closed form only under the hypothesis $h_0 = 0$. The general case is needed whenever the inner series has a nonzero level-zero block, which is the normal situation for a near-identity map.

Proposition B.3 (Closed form of the convolution polynomials, general case). *Let $H = \sum_{j \geq 0} h_j(L)t^j \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and let $C_{m,r}$ be as in (199). Then for all $m, r \in \mathbb{N}_0$,*

$$C_{m,r} = \sum_{k=0}^{\min\{m,r\}} \binom{r}{k} h_0^{r-k} \frac{k!}{m!} \mathcal{B}_{m,k}^{\text{exp}}(1!h_1, 2!h_2, \dots, (m-k+1)!h_{m-k+1}), \quad (531)$$

where the term $k = 0$ carries $\mathcal{B}_{m,0}^{\text{exp}}$ and is therefore h_0^r for $m = 0$ and 0 for $m > 0$. When $h_0 = 0$ only $k = r$ survives: for $r \geq 1$ formula (531) then reduces to (202), and for $r = 0$ it returns the boundary values $C_{m,0} = \delta_{m,0}$ of (199).

Proof. Split $H = h_0 + H_+$ with $H_+ = \sum_{j \geq 1} h_j t^j \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. Since h_0 and H_+ commute, the binomial theorem gives $H^r = \sum_{k=0}^r \binom{r}{k} h_0^{r-k} H_+^k$, a finite sum. By Theorem 11.13 (i) applied to H and by Theorem 6.36 applied to H_+ (whose level-zero block vanishes),

$$C_{m,r} = [t^m]H^r = \sum_{k=0}^r \binom{r}{k} h_0^{r-k} [t^m]H_+^k, \quad [t^m]H_+^k = \frac{k!}{m!} \mathcal{B}_{m,k}^{\text{exp}}(1!h_1, \dots) \quad (k \geq 1),$$

with $[t^m]H_+^k = 0$ for $m < k$ and $[t^m]H_+^0 = \delta_{m,0}$. Truncating the sum at $k = \min\{m, r\}$ is Theorem A.2 (i). If $h_0 = 0$ then, with the convention $0^0 = 1$, $h_0^{r-k} = 0$ unless $k = r$; the surviving term is (202) when $r \geq 1$, and is $(0!/m!) \mathcal{B}_{m,0}^{\text{exp}} = \delta_{m,0}$ when $r = 0$. $\square$

Theorem B.4 (Closed form for an arbitrary power of a unit). *Let $W = \sum_{j \geq 1} w_j(L)t^j \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $\sigma \in \mathbb{K}$, and let $(1+W)^\sigma \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ be the binomial power of Theorem 6.37. Then $[t^0](1+W)^\sigma = 1$ and, for $r \geq 1$,*

$$[t^r](1+W)^\sigma = \frac{1}{r!} \sum_{k=1}^r (\sigma)_k^{\overline{}} \mathcal{B}_{r,k}^{\text{exp}}(1!w_1, 2!w_2, \dots, (r-k+1)!w_{r-k+1}), \quad (532)$$

where $(\sigma)_k^{\overline{}} = \sigma(\sigma-1)\cdots(\sigma-k+1)$. The right-hand side depends only on $w_1, \dots, w_r$, and it is $\mathbb{K}$ -linear in w_r with coefficient σ .

If $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is a unit, so that $F = ct^{n_0}(1+W)$ with $c \in \mathbb{K}^\times$, $n_0 = \nu_t F$ and $W \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ by [Theorem 7.3](#), then, once a value $c^\sigma \in \mathbb{K}$ is chosen, one defines $F^\sigma := c^\sigma t^{\sigma n_0}(1+W)^\sigma$; this lies in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ when $\sigma n_0 \in \mathbb{Z}$, and otherwise in the Puiseux-log or Hahn extension of [Theorem 3.25](#). For $\sigma \in \mathbb{Z}$ the definition returns the ring power of F .

Proof. By [Theorem 6.37](#) (i)–(ii), $(1+W)^\sigma = \sum_{k \geq 0} \binom{\sigma}{k} W^k$ with the family summable, and $[t^r](1+W)^\sigma = \sum_{k=0}^r \binom{\sigma}{k} [t^r]W^k$ since $\nu_t(W^k) \geq k$. For $k \geq 1$, [Theorem 6.36](#) (formula (84), whose hypothesis $w_0 = 0$ holds) gives $[t^r]W^k = \frac{k!}{r!} \mathcal{B}_{r,k}^{\text{exp}}(1!w_1, 2!w_2, \dots)$; for $k = 0$, $[t^r]W^0 = \delta_{r,0}$, which contributes nothing when $r \geq 1$. Since $\binom{\sigma}{k} k! = (\sigma)_k$, summing gives (532).

Dependence only on $w_1, \dots, w_r$ is [Theorem A.2](#) (i). For the linear dependence on w_r : the variable x_r occurs in $\mathcal{B}_{r,k}^{\text{exp}}$ only for $k = 1$ – by [Theorem A.2](#) (i) it can occur only when $r \leq r - k + 1$, i.e. $k \leq 1$ – and $\mathcal{B}_{r,1}^{\text{exp}} = x_r$ by (513), so the coefficient of w_r is $\frac{1}{r!} (\sigma)_1 \cdot r! = \sigma$. This is (87) made explicit.

The factorization statement is [Theorem 7.3](#) (3); the exponent $t^{\sigma n_0}$ lies in the integer-exponent scale exactly when $\sigma n_0 \in \mathbb{Z}$, and otherwise requires the ramified tower of [Theorem 3.25](#). For $\sigma \in \mathbb{Z}$ the displayed definition returns the ring power, because c^σ and $t^{\sigma n_0}$ are then the ordinary powers and $(1+W)^\sigma$ agrees with the σ -fold product by the exponent law of [Theorem 6.37](#). $\square$

Corollary B.5 (Closed form for a reciprocal). *With W as in [Theorem B.4](#),*

$$[t^r](1+W)^{-1} = \frac{1}{r!} \sum_{k=1}^r (-1)^k k! \mathcal{B}_{r,k}^{\text{exp}}(1!w_1, \dots, (r-k+1)!w_{r-k+1}) \quad (r \geq 1). \quad (533)$$

Consequently, for a unit $F = ct^{n_0}(1+W)$ of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, $F^{-1} = c^{-1}t^{-n_0}(1 + \sum_{r \geq 1} [t^r](1+W)^{-1} t^r)$. In particular

$$[t^1](1+W)^{-1} = -w_1, \quad [t^2](1+W)^{-1} = w_1^2 - w_2, \quad [t^3](1+W)^{-1} = -w_1^3 + 2w_1w_2 - w_3.$$

Proof. Put $\sigma = -1$ in (532) and use $((-1))^k = (-1)(-2) \cdots (-k) = (-1)^k k!$. The factorization is [Theorem 7.7](#). The three displayed values follow from [Table 16](#): for $r = 1$, $-1!x_1/1!$ at $x_1 = w_1$; for $r = 2$, $(-1!x_2 + 2!x_1^2)/2!$ at $(x_1, x_2) = (w_1, 2w_2)$, giving $-w_2 + w_1^2$; for $r = 3$, $(-x_3 + 2 \cdot 3x_1x_2 - 6x_1^3)/6$ at $(x_1, x_2, x_3) = (w_1, 2w_2, 6w_3)$, giving $-w_3 + 2w_1w_2 - w_1^3$. $\square$

The Bell form (533) is closed but its term count grows like the partition function. A second closed form, with a term count that is a single determinant and which is the one to use for structural arguments about signs and integrality, is the following.

Proposition B.6 (Toeplitz–Hessenberg determinant for the reciprocal). *Let $B = \sum_{n \geq 0} B_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $B_0 = c \in \mathbb{K}^\times$, and let $B^{-1} = \sum_{r \geq 0} C_r(L)t^r$ be its reciprocal ([Theorem 7.7](#)). For $r \geq 1$ let M_r be the $r \times r$ matrix over $\mathbb{K}[L]$ with entries*

$$(M_r)_{ij} := B_{i-j+1}, \quad 1 \leq i, j \leq r, \quad B_m := 0 \text{ for } m < 0. \quad (534)$$

Thus M_r is lower Hessenberg, with c on the superdiagonal, B_1 on the diagonal, and B_r in the lower-left corner. Then

$$C_r = \frac{(-1)^r}{c^{r+1}} \det M_r \quad (r \geq 1), \quad C_0 = c^{-1}. \quad (535)$$

Proof. Put $D_0 := 1$ and $D_r := \det M_r$. We first show

$$D_r = - \sum_{m=1}^r (-1)^m B_m c^{m-1} D_{r-m} \quad (r \geq 1). \quad (536)$$

Expand $\det M_r$ along its last row, whose j -th entry is B_{r-j+1} : $D_r = \sum_{j=1}^r (-1)^{r+j} B_{r-j+1} \mu_j$, where μ_j is the minor obtained by deleting row r and column j . Its rows are indexed by $i \in \{1, \dots, r-1\}$ and its columns by $q \in \{1, \dots, r\} \setminus \{j\}$, with entry B_{i-q+1} . If $i \leq j-1$ and $q \geq j+1$, then $i-q+1 \leq j-1-j-1+1 < 0$, so that block vanishes; the minor is therefore block lower-triangular with diagonal blocks

- rows $1, \dots, j-1$ against columns $1, \dots, j-1$, which is M_{j-1} , of determinant D_{j-1} ; and
- rows $j, \dots, r-1$ against columns $j+1, \dots, r$. In the local indices $a = i - j + 1$ and $b = q - j$, both running over $1, \dots, r-j$, its entry is $B_{i-q+1} = B_{a-b}$; it therefore vanishes strictly above the diagonal and carries $B_0 = c$ on the diagonal, so it is lower triangular with determinant c^{r-j} .

Hence $\mu_j = D_{j-1}c^{r-j}$ and $D_r = \sum_{j=1}^r (-1)^{r+j} B_{r-j+1}c^{r-j} D_{j-1}$. Substituting $m = r - j + 1$, so that $j - 1 = r - m$, $r - j = m - 1$ and $(-1)^{r+j} = (-1)^{2r-m+1} = (-1)^{m+1}$, gives (536).

Now define $\tilde{C}_0 = c^{-1}$ and $\tilde{C}_r = (-1)^r c^{-(r+1)} D_r$ for $r \geq 1$. Multiplying (536) by $(-1)^r c^{-(r+1)}$ turns it into

$$\begin{aligned} \tilde{C}_r &= - \sum_{m=1}^r (-1)^{r+m} c^{m-1-(r+1)} B_m D_{r-m} = -c^{-1} \sum_{m=1}^r B_m (-1)^{r-m} c^{-(r-m+1)} D_{r-m} \\ &= -c^{-1} \sum_{m=1}^r B_m \tilde{C}_{r-m}, \end{aligned}$$

using $(-1)^{r+m} = (-1)^{r-m}$ and $D_0 = 1$ for the term $m = r$. This is exactly the recurrence (100) characterizing the blocks C_r , and $\tilde{C}_0 = C_0$ by (99); by the uniqueness part of Theorem 7.7, $\tilde{C}_r = C_r$ for all r . $\square$

Remark B.7 (What the determinant form is good for). Theorem B.6 makes two structural facts immediate that are awkward to see from either (100) or (533). First, $c^{r+1}C_r$ is a polynomial with integer coefficients in $B_0, \dots, B_r$, so the only denominators in a reciprocal are powers of the leading scalar: reciprocation introduces no new primes beyond those dividing c . Second, for $r \geq 1$,

$$\deg_L C_r \leq \max \left\{ \sum_{i=1}^s d_B(m_i) : s \geq 1, m_i \geq 1, m_1 + \dots + m_s = r \right\},$$

because a signed product in the expansion of $\det M_r$ is $\prod_{i=1}^r B_{i-\sigma(i)+1}$ for a permutation σ , whose subscripts sum to $\sum_i (i - \sigma(i) + 1) = r$; the factors with subscript 0 are the scalar c and contribute nothing to $\deg_L$, and the remaining factors are indexed by a composition of r into positive parts. Setting $d_B(m) \leq \alpha m$ makes the right-hand side at most αr , which is the linear bound of Theorem 7.15, now obtained without a recursion. Neither observation requires expanding the determinant.

B.4 Composition

Theorem 11.10 already gives a closed formula, in terms of the convolution polynomials. Substituting Theorem B.3 makes it closed in terms of the *input blocks* alone.

Proposition B.8 (Composition coefficients in Bell form). *Let $F = \sum_{n \geq n_0} p_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and $H = \sum_{j \geq 1} h_j(L)t^j \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ (so $h_0 = 0$). Then for every $N \in \mathbb{Z}$,*

$$[t^N](F \circ (X + H)) = \sum_{n=n_0}^N \sum_{r=0}^{\lfloor (N-n)/2 \rfloor} \frac{1}{(N-n-r)!} \mathcal{B}_{N-n-r, r}^{\text{exp}}(1! h_1, 2! h_2, \dots) \Delta_{n, r} p_n, \quad (537)$$

where $\Delta_{n, r} = \prod_{j=0}^{r-1} (\partial_L - n - j)$ is the operator of Theorem 9.6. The sum is finite, with at most $\lfloor (N - n_0 + 2)^2/4 \rfloor$ terms, and it involves only $p_{n_0}, \dots, p_N$ and $h_1, \dots, h_{N-n_0}$.

Proof. Start from (197), in which the summand indexed by (n, k, m) with $n+k+m = N$ is $\frac{h_{k, m}}{k!} \Delta_{n, k} p_n$ and $h_{k, m} = C_{m, k}$ by the index dictionary of Section 11.4. Rename k as r and use $m = N - n - r$. Since $h_0 = 0$, (202) gives $C_{m, r} = \frac{r!}{m!} \mathcal{B}_{m, r}^{\text{exp}}(1! h_1, \dots)$ for $m \geq r \geq 1$, and $C_{m, r} = 0$ for $m < r$; also $C_{0, 0} = 1 = \frac{0!}{0!} \mathcal{B}_{0, 0}^{\text{exp}}$ and $C_{m, 0} = 0 = \mathcal{B}_{m, 0}^{\text{exp}}$ for $m > 0$. Hence in every case $C_{m, r}/r! = \mathcal{B}_{m, r}^{\text{exp}}(\cdot)/m!$, which is the displayed summand. The constraint $r \leq m = N - n - r$, i.e. $2r \leq N - n$, is Theorem A.2 (i). Counting the pairs (n, r) with $n_0 \leq n \leq N$ and $0 \leq r \leq \lfloor (N - n)/2 \rfloor$ and writing $s = N - n$ gives

$\sum_{s=0}^{N-n_0} (\lfloor s/2 \rfloor + 1) = \lfloor (N - n_0 + 2)^2/4 \rfloor$. Dependence on the listed blocks only is visible from the displayed ranges: $n \leq N$, and the arguments of $\mathcal{B}_{N-n-r,r}^{\text{exp}}$ reach at most $x_{(N-n-r)-r+1}$, so h_j occurs only for $j \leq N - n \leq N - n_0$. This agrees with, and refines for this special G , the general finite determination of [Theorem 10.10](#). $\square$

Corollary B.9 (Composition with a pure level-zero shift). *Let $F = \sum_{n \geq n_0} p_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ and let $H = h_0(L) \in \mathbb{K}[L]$ have no positive t -levels, so that $X + H = X + \bar{h}_0(L)$. Then for every N ,*

$$[t^N](F \circ (X + h_0)) = \sum_{n=n_0}^N \frac{h_0^{N-n}}{(N-n)!} \Delta_{n,N-n} p_n. \quad (538)$$

In particular, for $F = X + A$ and $h_0 = aL + b$ with $a, b \in \mathbb{K}$, the whole composite is computed by a single sum with $N - n_0 + 1$ terms and no convolution at all.

Proof. Here $H^k = h_0^k$ is concentrated at level 0, so $h_{k,m} = h_0^k \delta_{m,0}$ in (196). Substituting into (197) leaves only the triples $(n, k, m) = (n, N - n, 0)$, which is (538). $\square$

Example B.10 (Composition with $X + L$). Take $F = t = X^{-1}$, so $n_0 = 1$, $p_1 = 1$ and $p_n = 0$ otherwise. Then (538) with $h_0 = L$ gives $[t^N](t \circ (X + L)) = \frac{L^{N-1}}{(N-1)!} \Delta_{1,N-1} 1$. Since $\Delta_{1,k} 1 = \prod_{j=0}^{k-1} (-1 - j) = (-1)^k k!$, this is $(-1)^{N-1} L^{N-1}$, so $t \circ (X + L) = \sum_{N \geq 1} (-1)^{N-1} L^{N-1} t^N = t/(1 + tL)$, in agreement with the generator rule (180).

B.5 Reversion

Two closed forms are available for the inverse, and they answer different questions. The first is closed in the blocks of the perturbation.

Corollary B.11 (Reversion coefficients, closed in the input blocks). *Let $A = \sum_{m \geq 0} a_m(L)t^m \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, $F = X + A$, and $F^{(-1)\circ} = X + \sum_{N \geq 0} b_N(L)t^N$. Then for every $N \in \mathbb{N}_0$,*

$$b_N = \sum_{r=1}^{N+1} \frac{(-1)^r}{r!} \left[\prod_{s=N-r+1}^{N-1} (\partial_L - s) \right] C_{N-r+1,r}(a_0, a_1, \dots), \quad (539)$$

where $C_{m,r}$ is the convolution polynomial of [Theorem 11.12](#) formed from the blocks of A , given in closed form by (531), and where a product whose lower bound exceeds its upper bound is empty. Only $a_0, \dots, a_N$ occur, and the range $r \leq N + 1$ cannot be shortened.

Proof. [Theorem 16.18](#) is (539) with $A_{r,N-r+1}$ in place of $C_{N-r+1,r}$, and [Theorem 11.13](#) (i) identifies $A_{r,m} = [t^m]A^r = C_{m,r}$. Since $C_{m,r}$ involves only $a_0, \dots, a_m$ and $m = N - r + 1 \leq N$ here, only $a_0, \dots, a_N$ occur. Sharpness of the range is [Theorem 16.19](#). $\square$

The special case in which the perturbation is a single level-zero block degenerates spectacularly: exactly one Lagrange summand survives per output block, and the whole inverse is given by one operator applied to one power.

Corollary B.12 (Reversion of a pure level-zero perturbation). *Let $P \in \mathbb{K}[L]$ and $F = X + P(L) \in \mathcal{G}_{\text{poly},\mathbb{K}}$. Then*

$$F^{(-1)\circ} = X + \sum_{N \geq 0} \frac{(-1)^{N+1}}{(N+1)!} \left[\prod_{j=0}^{N-1} (\partial_L - j) \right] P^{N+1} t^N = X + \sum_{N \geq 0} \frac{(-1)^{N+1}}{(N+1)!} \Delta_{0,N}(P^{N+1}) t^N, \quad (540)$$

the product being empty at $N = 0$, where the block is $-P$. For $P(L) = aL$ with $a \in \mathbb{K}$ this is $F^{(-1)\circ} = X + \sum_{N \geq 0} a^{N+1} \mathbb{L}_N^{\text{Lam}}(L)t^N$, which is [Theorem 18.5](#); for $a = 1$ it is the expansion of ω ([Theorem 18.10](#)). For a constant $P = c \in \mathbb{K}$ it collapses to $(X + c)^{(-1)\circ} = X - c$.

Proof. Here $a_0 = P$ and $a_m = 0$ for $m \geq 1$, so by (199) the convolution polynomials are $C_{0,r} = P^r$ and $C_{m,r} = 0$ for $m \geq 1$. In (539) the index $m = N - r + 1$ is therefore forced to 0, that is $r = N + 1$, and the operator becomes $\prod_{s=0}^{N-1} (\partial_L - s) = \Delta_{0,N}$. Substituting gives (540).

For $P = aL$, $\Delta_{0,N}((aL)^{N+1}) = a^{N+1}\Delta_{0,N}L^{N+1}$, and comparison with (344) identifies the block as $a^{N+1}\mathbb{L}_N^{\text{Lam}}(L)$. For $P = c$ constant and $N \geq 1$, the factor ∂_L at $s = 0$ annihilates the constant c^{N+1} , so every block beyond $N = 0$ vanishes. $\square$

Remark B.13 (A repair to the level-zero reversion formula). Source S3 records (540) in the equivalent but shifted form “ $[t^n](F^{(-1)\circ} - X + P) = \dots$ ”. That form is correct for $n \geq 1$, because P sits at level zero and contributes nothing to higher blocks, but it fails at $n = 0$, where the left-hand side is $-P + P = 0$ while the right-hand side is $-P$. The correction is simply to delete the $+P$; (540) is then valid for every $N \geq 0$, and the Lambert normalization $\mathbb{L}_0^{\text{Lam}}(u) = -u$ of Theorem 19.1 becomes the $N = 0$ instance rather than a separately stipulated boundary value.

The second closed form is closed in A and its D_X -derivatives. It is what one wants when A is not given blockwise — for instance when it is a symbolic expression, or when one wants a formula valid for a whole family of perturbations at once.

Theorem B.14 (The Lagrange terms as differential polynomials). *Let $A \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and $F = X + A$. Then for every $r \geq 1$*

$$\frac{(-1)^r}{r!} D_X^{r-1}(A^r) = (-1)^r \sum_{j=1}^r \frac{A^j}{j!} \mathcal{B}_{r-1, r-j}^{\text{exp}}(D_X A, D_X^2 A, \dots, D_X^j A), \quad (541)$$

and consequently

$$F^{(-1)\circ} = X + \sum_{r \geq 1} (-1)^r \sum_{j=1}^r \frac{A^j}{j!} \mathcal{B}_{r-1, r-j}^{\text{exp}}(D_X A, \dots, D_X^j A), \quad (542)$$

the outer family being summable. Writing $A' = D_X A$, $A'' = D_X^2 A$ and so on, the first five Lagrange terms are

$$\begin{aligned} F^{(-1)\circ} - X = & -A \\ & + AA' \\ & - A(A')^2 - \frac{1}{2}A^2 A'' \\ & + A(A')^3 + \frac{3}{2}A^2 A' A'' + \frac{1}{6}A^3 A''' \\ & - A(A')^4 - 3A^2(A')^2 A'' - \frac{1}{2}A^3(A'')^2 - \frac{2}{3}A^3 A' A''' - \frac{1}{24}A^4 A'''' \\ & + \mathcal{O}_t^{\text{form}}(t^5), \end{aligned} \quad (543)$$

each line being one value of r , from $r = 1$ to $r = 5$.

Proof. Apply Theorem A.16 with $n = r - 1$. For $r = 1$ both sides of (541) are $-A$ (the left side is $-D_X^0 A$; the right side has only $j = 1$, with $\mathcal{B}_{0,0}^{\text{exp}} = 1$). For $r \geq 2$, (529) gives

$$D_X^{r-1}(A^r) = \sum_{k=1}^{r-1} (r)^{\underline{k}} A^{r-k} \mathcal{B}_{r-1, k}^{\text{exp}}(D_X A, \dots, D_X^{r-k} A),$$

the index k stopping at $r-1$ because $\mathcal{B}_{r-1, k}^{\text{exp}} = 0$ for $k > r-1$. Divide by $r!$ and use $(r)^{\underline{k}}/r! = 1/(r-k)!$; putting $j = r - k$, whose range is $1 \leq j \leq r-1$, turns the sum into $\sum_j \frac{A^j}{j!} \mathcal{B}_{r-1, r-j}^{\text{exp}}(D_X A, \dots, D_X^j A)$. The missing term $j = r$ carries $\mathcal{B}_{r-1, 0}^{\text{exp}}$, which vanishes for $r \geq 2$, so it may be added without changing the value; this gives (541) in the stated uniform range $1 \leq j \leq r$. The argument list is $x_1, \dots, x_{(r-1)-(r-j)+1} = x_1, \dots, x_j$, that is $D_X A, \dots, D_X^j A$.

Formula (542) is now (281) of [Theorem 16.14](#) term by term, and the summability of the outer family is the same theorem.

For (543), evaluate (541) for $r \leq 5$ using [Table 16](#). At $r = 2$: $j = 1$ gives $A \mathcal{B}_{1,1}^{\exp}(A') = AA'$ and $j = 2$ gives $\frac{1}{2}A^2 \mathcal{B}_{1,0}^{\exp} = 0$; the sign is $(-1)^2$. At $r = 3$: $j = 1$ gives $A \mathcal{B}_{2,2}^{\exp}(A') = A(A')^2$, $j = 2$ gives $\frac{1}{2}A^2 \mathcal{B}_{2,1}^{\exp}(A', A'') = \frac{1}{2}A^2 A''$, and $j = 3$ gives 0; the sign is -1 . At $r = 4$: $\mathcal{B}_{3,3}^{\exp} = (A')^3$, $\mathcal{B}_{3,2}^{\exp} = 3A'A''$, $\mathcal{B}_{3,1}^{\exp} = A'''$, giving $A(A')^3 + \frac{1}{2}A^2 \cdot 3A'A'' + \frac{1}{6}A^3 A'''$. At $r = 5$: $\mathcal{B}_{4,4}^{\exp} = (A')^4$, $\mathcal{B}_{4,3}^{\exp} = 6(A')^2 A''$, $\mathcal{B}_{4,2}^{\exp} = 4A'A''' + 3(A'')^2$, $\mathcal{B}_{4,1}^{\exp} = A''''$, giving $A(A')^4 + \frac{1}{2}A^2 \cdot 6(A')^2 A'' + \frac{1}{6}A^3 (4A'A''' + 3(A'')^2) + \frac{1}{24}A^4 A''''$, which is the last line with the sign $(-1)^5$. Finally, [Theorem 16.8](#) gives ν_t at least $r - 1 \geq 5$ for every omitted term, which is the recorded formal tail. $\square$

Remark B.15 (Which reversion formula to run). [Theorem B.11](#) and [Theorem B.14](#) are the same series regrouped. [Theorem B.11](#) groups by output block and is the formula to use when a single b_N is wanted, since it needs only A^r for $r \leq N + 1$ and one application of a differential operator per term. [Theorem B.14](#) groups by Lagrange index and is the formula to use for structural statements, since the r -th line of (543) is a universal differential polynomial with rational coefficients, independent of A , of weight r in the grading $\text{wt}(D_X^i A) = i + 1$. Neither is competitive as an *algorithm*: for computing all blocks up to a cutoff, triangular reversal ([Theorem 15.2](#)) or Newton reversion ([Theorem 15.11](#)) is cheaper, as [Theorem 25.42](#) records.

B.6 What differs from the classical Lagrange inversion formula

The classical formula, in the form standard in enumerative combinatorics [[17](#), [16](#)], reads as follows. Let C be a commutative $\mathbb{Q}$ -algebra, let $\psi \in C[[z]]$, and let $u \in \varepsilon C[[\varepsilon]]$ be the unique solution of $u = \varepsilon\psi(u)$. Then for every $\eta \in C[[z]]$ and every $m \geq 1$,

$$[\varepsilon^m] \eta(u) = \frac{1}{m} [z^{m-1}] (\eta'(z) \psi(z)^m). \quad (544)$$

[Theorem 16.14](#) is not this statement, and the differences are worth naming precisely, because each of them is a place where a transplanted classical proof breaks.

1. *The output of one term.* In (544) the right-hand side is a single scalar, obtained by a coefficient extraction that evaluates a derivative at the origin. In (281) the r -th summand $\frac{(-1)^r}{r!} D_X^{r-1}(A^r)$ is an entire element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, contributing to every block t^N with $N \geq r - 1$; the passage to one block is a second step, performed by [Theorem 16.18](#). Accordingly the volume’s abstract ingredient [Theorem 16.11](#) states $[\varepsilon^m] \gamma_{[u]} = \frac{1}{m!} \partial_z^{m-1} (\gamma' \psi^m)$ as an identity of *series* in z , and specializes z only afterwards. Evaluating at $z = 0$ recovers (544), because $\frac{1}{m!} (\partial_z^{m-1} (\gamma' \psi^m))|_{z=0} = \frac{1}{m} [z^{m-1}] (\gamma' \psi^m)$.
2. *The derivation does not annihilate the coefficients.* In (544) the operator ∂_z kills C . Here $D_X L = t \neq 0$ ([Theorem 9.1](#)), so differentiating a Lagrange term differentiates its coefficient polynomials; this is what produces the operator $\Delta_{n,k}$ of [Theorem 9.6](#) in (539) and, ultimately, the Lambert polynomials themselves. Replacing D_X^{r-1} by “extract the coefficient of t^{r-1} ” is therefore wrong by an operator, not by a constant.
3. *What makes the family summable.* Classically the grading is by the single variable ε , and the solution is small because $u \in \varepsilon C[[\varepsilon]]$. Here A need not be small at all: $\nu_t(A) = 0$ is the normal case, since a near-identity map may have a full level-zero block $a_0(L)$. Summability comes instead from the pairing of one derivative against one extra factor of A , that is from the axiom $\nu(\partial f) \geq \nu(f) + 1$ of [Theorem 16.1](#); the resulting admissibility threshold is $\nu(A) > -1$, not $\nu(A) > 0$ ([Theorem 16.2](#), [Theorem 16.8](#)).
4. *No formal residue is available.* The usual proof of (544) runs through the formal residue on $C((z))$ and the identity $\text{Res}_z \partial_z(\cdot) = 0$. [Theorem B.16](#) shows that no such functional exists on $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. The volume’s proof of [Theorem 16.14](#) therefore goes through a weighted deformation algebra and an evaluation homomorphism ([Theorem 16.13](#)) instead of through residues.

5. *The contributing range is finite and known.* Classically every $m \geq 1$ contributes to its own coefficient. Here only $r \leq N + 1$ contributes to the block N , and that bound is attained ([Theorem 16.19](#)); this is what makes ([539](#)) a finite formula rather than a limit.
6. *No analytic content.* [Theorem 16.14](#) is an identity of formal objects. The transfer to a genuine inverse function of a real or complex variable is a separate theorem with separate hypotheses ([Theorem 14.21](#)), exactly as the catalogue’s distinction between $\mathcal{O}_t^{\text{form}}(t^N)$ and $\mathcal{O}_{X \rightarrow +\infty}(\cdot)$ requires.

Proposition B.16 (No formal residue on the polynomial–logarithmic Laurent ring). *There is no nonzero $\mathbb{K}$ -linear functional $\rho: \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} \rightarrow \mathbb{K}$ with $\rho(D_X F) = 0$ for every $F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. The same holds for any $\mathbb{K}$ -linear map from $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ to any $\mathbb{K}$ -vector space.*

Proof. By [Theorem 9.17](#) the map $D_X: \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} \rightarrow \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is surjective. If $\rho \circ D_X = 0$ then ρ vanishes on $D_X(\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}) = \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, so $\rho = 0$. The same argument applies verbatim to a linear map with any target. $\square$

Remark B.17 (The residue that does exist is a residue in L). [Theorem B.16](#) does not contradict [Theorem 8.22](#). That definition attaches a residue to a *coefficient* $h \in \mathbb{K}((L^{-1}))$, namely the coefficient of L^{-1} , and [Theorem 8.23](#) shows that it annihilates ∂_L -images. It is a residue in the inner variable, defined only after the coefficient ring has been enlarged to $\mathbb{K}((L^{-1}))$ — where ∂_L is *not* surjective ([Theorem 8.24](#)) — and it does not implement extraction of an outer block. The obstruction that a classical residue proof would need lives in the outer variable t , and there is none, because t is a unit of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ ([Theorem 5.11](#) (i)) and D_X is onto.

B.7 Which formula to use

C Operational formula sheet

This section is a navigation aid, not an exposition. Every line states a result of this volume and cites where that result is proved; nothing is asserted here that is not proved there, and no line carries a hypothesis that its citation does not carry. Read a line together with its reference before using it: the abbreviations below suppress hypotheses that are essential — which coefficient class an element lives in, whether a leading block is a unit, whether an inner argument is admissible, whether a claim is formal or analytic.

Standing conventions for the whole sheet. $\mathbb{K}$ is a field of characteristic zero; X is the large variable and the default regime is $X \rightarrow +\infty$ along the positive real ray; $t = X^{-1}$ and $L = \log X$; F, G, H denote series, p_n, q_n, a_n, b_n, h_n their blocks in $\mathbb{K}[L]$, and N, n, m, r, k integers. Truncation is *exclusive* ([Theorem 5.1](#)). Three symbols that look alike are kept apart throughout: $\mathcal{O}_t^{\text{form}}(t^N)$ is a formal valuation tail with no analytic content, $\mathcal{O}_{X \rightarrow +\infty}(A(X))$ is an analytic estimate on the printed regime, and $\mathcal{O}(N^2)$ is an operation count.

C.1 The four ambient classes

Table 18: Generators, classes, and their algebraic type.

Statement	Proved at
$t := X^{-1}$, $L := \log X$; independent generators in the formal algebra	Theorem 3.1
$\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} = \mathbb{K}[L][[t]]$, $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} = \mathbb{K}[L]((t))$,	Theorem 3.3
$\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}} = \mathbb{K}(L)((t))$, $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}} = \mathbb{K}((L^{-1}))((t))$	

Statement	Proved at
$\mathcal{PL}_{\text{pow},\mathbb{K}} \subsetneq \mathcal{PL}_{\text{poly},\mathbb{K}} \subsetneq \mathcal{PL}_{\text{rat},\mathbb{K}} \subsetneq \mathcal{PL}_{\text{itLaur},\mathbb{K}}$; the first two are domains and not fields, the last two are fields	Theorem 3.5, Theorem 3.6, Theorem 7.19
$\mathcal{PL}_{\text{itLaur},\mathbb{K}}$ is the Hahn field of the lexicographic rank-two value group	Theorem 3.19, Theorem 3.20
The coefficient extensions $\mathbb{K}[L] \rightsquigarrow \mathbb{K}(L) \rightsquigarrow \mathbb{K}((L^{-1}))$ are not localizations of $\mathcal{PL}_{\text{poly},\mathbb{K}}$	Theorem 3.8
Fractional outer exponents require the Puiseux-log tower; a general real exponent requires a Hahn exponent group	Theorem 3.24, Theorem 3.25
$\mathcal{G}_{\text{poly},\mathbb{K}} = \{X + A : A \in \mathcal{PL}_{\text{pow},\mathbb{K}}\}$ is a group under $\circ$	Theorem 12.1, Theorem 12.7
Every class is t -adically complete and separated	Theorem 5.18

C.2 Valuation, leading data, and the dominance order

Table 19: Outer valuation, rank-two valuation, leading data, dominance, log-degree profile.

Statement	Proved at
$\nu_t(F) = \min \{n : p_n \neq 0\}, \nu_t(0) = +\infty$	Theorem 4.1
$\nu_t(FG) = \nu_t F + \nu_t G; \nu_t(F + G) \geq \min \{\nu_t F, \nu_t G\}$, strict exactly when the leading blocks cancel	Theorem 4.2
Leading block $p_{\nu_t F}$, leading monomial $\text{lm}(F) = t^{\nu_t F} L^{\deg_L p_{\nu_t F}}$, leading scalar $\text{lc}(F) -$ three different objects	Theorem 4.4, Theorem 4.5
$\text{lm}(FG) = \text{lm}(F) \text{lm}(G)$ and $\text{lc}(FG) = \text{lc}(F) \text{lc}(G)$	Theorem 4.7, Theorem 6.21
The block representation of a nonzero element is unique	Theorem 4.9
$t^n L^j \succ t^m L^k$ iff $n < m$, or $n = m$ and $j > k$	Theorem 3.12
Dominance is genuine growth as $X \rightarrow +\infty$	Theorem 3.13
$d_F(n) = \deg_L p_n$, with $-\infty$ at absent blocks	Theorem 4.11
$d_{F+G} \leq \max, d_{FG} \leq$ the $(\max, +)$ -convolution, with an exact attainment criterion	Theorem 4.12, Theorem 6.25

C.3 Truncation, the formal tail, and summability

Table 20: Exclusive truncation, the formal big-O token, summability.

Statement	Proved at
$\text{Trunc}_{<N} F = \sum_{n < N} p_n(L) t^n$; every retained block is retained whole	Theorem 5.1
Inclusive-to-exclusive dictionary $[F]_{\leq N} =$	Theorem 5.2, (48)
$\text{Trunc}_{<N+1} F$	
$\text{Trunc}_{<N}(FG) = \text{Trunc}_{<N}(\text{Trunc}_{<N-\nu_t G} F \cdot$	Theorem 5.4
$\text{Trunc}_{<N-\nu_t F} G)$	
$F = G + \mathcal{O}_t^{\text{form}}(t^N)$ means $\nu_t(F - G) \geq N$, equivalently	Theorem 5.7, Theorem 5.8
$\text{Trunc}_{<N} F = \text{Trunc}_{<N} G$	

Statement	Proved at
$t^N \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ is a filtration step of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, never an ideal, because t is a unit there	Theorem 5.11
$[t^n]F = p_n(L) \in \mathbb{K}[L]$ and $[t^n L^j]F = a_{n,j} \in \mathbb{K}$; extraction is linear and filtration-continuous	Theorem 5.12 , Theorem 5.13
A family is summable iff $\nu_t \rightarrow +\infty$; sums may then be regrouped and multiplied termwise	Theorem 5.21 , Theorem 5.22 , Theorem 5.24 , Theorem 5.26
A bound on $\deg_L$ is a second, independent approximation, not part of the t -adic topology	Theorem B.1

C.4 The derivation, its powers, and antiderivatives

Table 21: The distinguished derivation and its calculus.

Statement	Proved at
$D_X = t \partial_L - t^2 \partial_t$; the unique $\mathbb{K}$ -derivation with $D_X t = -t^2$, $D_X L = t$ and $D_X(\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}) \subseteq \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$	Theorem 9.1
Block formula $D_X(t^n p(L)) = t^{n+1}(\partial_L - n)p(L)$	Theorem 9.3
Euler form $\vartheta = t^{-1} D_X$ preserves each block and acts as $\partial_L - n$ on $t^n \mathbb{K}[L]$	Theorem 9.4
$\Delta_{n,k} = \prod_{j=0}^{k-1} (\partial_L - n - j)$, $\Delta_{n,0} = 1$; the second index is never optional	Theorem 9.6
$D_X^k(t^n p(L)) = t^{n+k} \Delta_{n,k} p(L)$	Theorem 9.7
$\Delta_{0,k} = \sum_j s(k, j) \partial_L^j$, and $D_X^k t^n = (-1)^k (n)^{\bar{k}} t^{n+k}$	Theorem 9.8
$\nu_t(D_X F) \geq \nu_t F + 1$, with the sharp value and the three cases	Theorem 9.10
$\ker D_X \cap \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}} = \mathbb{K}$	Theorem 9.11
D_X is onto $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$; antiderivatives exist and are unique up to a constant of $\mathbb{K}$	Theorem 9.17
Resonance: $(\partial_L - n)p = q$ is triangular for $n \neq 0$; the block $n = 0$ integrates and adds a constant	Theorem 9.14 , Theorem 9.15
In $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$ and $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ antidifferentiation is obstructed by a residue at $L = \infty$; this is where $\log L$ is forced	Theorem 9.20 , Theorem 9.23 , Theorem 9.21

C.5 Product, reciprocal, and division

Table 22: Multiplicative calculus.

Statement	Proved at
$[t^N](FG) = \sum_{n+m=N} p_n q_m$, a finite sum for each N ; the four classes are closed under it	Theorem 6.5 , Theorem 6.6 , Theorem 6.7
$[t^N L^d](FG) = \sum_{n+m=N} \sum_{e+f=d} a_{n,e} b_{m,f}$	Theorem B.2
F is a unit of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ iff its leading block is a nonzero scalar, iff $F = c t^{n_0} (1+U)$ with $c \in \mathbb{K}^\times$, $U \in t \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$	Theorem 7.3 , Theorem 7.4
$B^{-1} = \sum_n C_n t^n$ with $C_0 = c^{-1}$ and $C_n = -c^{-1} \sum_{j=1}^n B_j C_{n-j}$, where $B_0 = c \in \mathbb{K}^\times$	Theorem 7.7

Statement	Proved at
$Q_n = B_0^{-1}(A_n - \sum_{j=1}^n B_j Q_{n-j})$, valid over $\mathbb{K}[L]$ only when $B_0 \in \mathbb{K}^\times$ – a nonzero <i>scalar</i> , not merely a nonzero polynomial	Theorem 7.8, Theorem 7.3
Newton reciprocal step $C \mapsto C(2 - BC)$ squares the residual $1 - BC$ and may be truncated at the doubled order	Theorem 7.27, Theorem 7.29
Truncated division: relative precision is exactly preserved $(1 + U)^\sigma = \sum_{r \geq 0} \binom{\sigma}{r} U^r$ for $\nu_t U \geq 1$, with finite determination at each block	Theorem 7.9, Theorem 7.11 Theorem 6.37
Closed forms: $[t^r](1 + W)^\sigma = \frac{1}{r!} \sum_{k=1}^r (\sigma)^k \mathcal{B}_{r,k}^{\exp}(1!w_1, 2!w_2, \dots)$; $\sigma = -1$ gives the reciprocal	Theorem B.4, Theorem B.5
$c^{r+1}[t^r]B^{-1}$ is a Toeplitz–Hessenberg determinant in $B_0, \dots, B_r$	Theorem B.6
$\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ is a domain but not a field; among the rings $E((t))$ with $\mathbb{K}[L] \subseteq E \subseteq \mathbb{K}(L)$, exactly one is a field, namely $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$	Theorem 6.22, Theorem 7.20, Theorem 7.23

C.6 Composition and the exact Taylor formula

Table 23: Substitution, the Taylor formula, and the coefficient formula.

Statement	Proved at
Admissible inner argument $G = cX^\rho(1 + U)$, $c \in \mathbb{K}^\times$, $\rho > 0$, $U \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$	Theorem 10.6
Generator images $\Phi_G(t) = c^{-1}t^\rho(1 + U)^{-1}$, $\Phi_G(L) = \rho L + \log c + \log(1 + U)$; Φ_G is a $\mathbb{K}$ -algebra homomorphism compatible with summable families	Theorem 10.7
Near identity: $t \circ (X + H) = t/(1 + tH)$, $L \circ (X + H) = L + \log(1 + tH)$. Substituting $t \mapsto t + H$, or updating L alone, is wrong	Theorem 10.22
$F \circ (X + H) = \sum_{k \geq 0} \frac{H^k}{k!} D_X^k F$, an exact identity, not an approximation	Theorem 11.6
$[t^r](F \circ (X + H)) = \sum_{n+k+m=r} \frac{h_{k,m}}{k!} \Delta_{n,k} p_n$, with $h_{k,m} = [t^m]H^k = C_{m,k}$	Theorem 11.10, Theorem 11.12
$C_{0,0} = 1$, $C_{m,0} = 0$ for $m > 0$, $C_{m,r+1} = \sum_{j=0}^m h_j C_{m-j,r}$; closed form in (531)	Theorem 11.13, Theorem B.3
Bell form when $h_0 = 0$, and the single-sum formula for a level-zero shift $H = h_0(L)$	Theorem B.8, Theorem B.9
Substituting into a scalar power series: $\varphi(W)$ for $W \in t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, with Bell coefficients	Theorem 8.5, Theorem 11.22

Statement	Proved at
$D_X^n(\varphi(F)) = \sum_{k=1}^n \varphi^{(k)}(F) \mathcal{B}_{n,k}^{\text{exp}}(D_X F, \dots, D_X^{n-k+1} F)$	Theorem A.15
$D_X^n(F \circ G) = \sum_{k=1}^n (D_X^k F) \circ G \mathcal{B}_{n,k}^{\text{exp}}(\gamma, \dots, D_X^{n-k} \gamma),$ $\gamma = D_X G$	Theorem 11.25, Theorem A.12
Chain rule $D_X(F \circ G) = ((D_X F) \circ G) D_X G$; associativity holds inside the admissible class	Theorem 10.31, Theorem 10.27
Finite determination and the triangular precision budget	Theorem 10.10, Theorem 10.11
Analytic transfer under composition is a separate theorem with its own hypotheses	Theorem 10.35

C.7 Reversion

Table 24: The compositional inverse: fixed point, algorithms, closed formulas.

Statement	Proved at
For $F = X + A \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ and $G = X + B$: $F \circ G = X$ iff $B = -A \circ (X + B)$	Theorem 14.7
The fixed point exists, is unique, and is a two-sided inverse; the Picard iterates gain one block per step	Theorem 14.11, Theorem 14.9, Theorem 12.7
$\text{Err}_{\text{rev}}(F, H) = F \circ H - X$; its order equals $\nu_t(H - F^{(-1)\circ})$ exactly, on both sides, with the same leading block	Theorem 14.14, Theorem 14.15, Theorem 14.16
Triangular reversal: $r_N = [t^N] \text{Err}_{\text{rev}}(F, G_{N-1})$, $G_N = G_{N-1} - r_N t^N$, and $G_N = \text{Trunc}_{<N+1} F^{(-1)\circ}$	Theorem 15.2, Theorem 14.17
At leading slope $a \in \mathbb{K}^\times$ the correction is $-r_N/a$, and the whole problem conjugates to the near-identity one	Theorem 15.4
Truncating the inputs of a reversal step is safe	Theorem 15.7, Theorem 25.8
Newton map $\mathcal{N}_F(G) = G - \frac{\text{Err}_{\text{rev}}(F, G)}{(D_X F) \circ G}$; the denominator is a unit	Theorem 15.10, Theorem 15.9
Exact arithmetic: $\nu_t \text{Err}_{\text{rev}}(F, \mathcal{N}_F(G)) \geq 2m + 2 + \alpha$ with $m = \nu_t \text{Err}_{\text{rev}}(F, G)$, $\alpha = \nu_t A$	Theorem 15.11
Guard requirement. With an inexact correction accurate only to order σ , the new residual order is $\min\{2m + 2 + \alpha, \sigma\}$; doubling is lost unless $\sigma \geq 2m + 2 + \alpha$	Theorem 15.14, Theorem 15.15, Theorem 15.16
$(X + A)^{(-1)\circ} = X + \sum_{r \geq 1} \frac{(-1)^r}{r!} D_X^{r-1}(A^r)$, the r -th summand of order $\geq r - 1$	Theorem 16.14
$b_N = \sum_{r=1}^{N+1} \frac{(-1)^r}{r!} \left[\prod_{s=N-r+1}^{N-1} (\partial_L - s) \right] A_{r, N-r+1}$, the empty product occurring exactly at $r = 1$	Theorem 16.18, Theorem B.11
The range $r \leq N + 1$ is exactly right and is attained	Theorem 16.19

Statement	Proved at
Level-zero perturbation: $(X + P)^{(-1)^\circ} = X + \sum_{N \geq 0} \frac{(-1)^{N+1}}{(N+1)!} \Delta_{0,N}(P^{N+1})t^N$ for $P \in \mathbb{K}[L]$	Theorem B.12, Theorem B.13
Transport of an observable: $\varphi \circ F^{(-1)^\circ} = \varphi + \sum_{r \geq 1} \frac{(-1)^r}{r!} \partial^{r-1}((\partial\varphi)A^r)$	Theorem 16.15, Theorem 16.16
Universal differential-polynomial form, weight-graded by $\text{wt}(D_X^i A) = i + 1$	Theorem B.14
Beyond the identity scale: reduce by the leading monomial; a leading logarithm forces a nested logarithm	Theorem 17.6, Theorem 17.3, Theorem 17.4
Analytic residual transfer needs injectivity and a derivative lower bound	Theorem 14.21

C.8 Wright omega, the Lambert polynomials, and Lambert W

Table 25: The canonical logarithmic reversion and its coefficient family.

Statement	Proved at
$y \mapsto y + \log y$ is a bijection $(0, \infty) \rightarrow \mathbb{R}$; its inverse is ω , with $\omega(x) = W_0(e^x)$	Theorem 18.1
$\omega = X + \sum_{n \geq 0} \mathbb{L}_n^{\text{Lam}}(L)t^n = X - L + \sum_{n \geq 1} \mathbb{L}_n^{\text{Lam}}(L)t^n$ in $\mathcal{G}_{\text{poly}, \mathbb{K}}$; each Lagrange–Bürmann summand is a single block	Theorem 18.10
$\mathbb{L}_n^{\text{Lam}}(u) = \frac{(-1)^{n+1}}{(n+1)!} \left[\prod_{j=0}^{n-1} (\partial_u - j) \right] u^{n+1}$; $\mathbb{L}_0^{\text{Lam}}(u) = -u$,	Theorem 19.1
$\mathbb{L}_1^{\text{Lam}}(u) = u$, $\mathbb{L}_2^{\text{Lam}}(u) = \frac{u^2}{2} - u$ $\mathbb{L}_n^{\text{Lam}}(0) = 0$; $\mathbb{L}_{n+1}^{\text{Lam}} = -\mathbb{L}_n^{\text{Lam}} + n \int_0^u \mathbb{L}_n^{\text{Lam}}$; equivalently $\mathbb{L}_{n+1}^{\text{Lam}'} = n\mathbb{L}_n^{\text{Lam}} - \mathbb{L}_n^{\text{Lam}'}$	Theorem 19.2
$\mathbb{L}_n^{\text{Lam}}(u) = \sum_{m=1}^n (-1)^{n-m} \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] \frac{u^m}{m!} \quad (n \geq 1)$	Theorem 19.4
$\deg \mathbb{L}_n^{\text{Lam}} = n$ for $n \geq 1$ only; $[u^n] \mathbb{L}_n^{\text{Lam}} = 1/n$; $\lambda_{n,1} = (-1)^{n-1}$; $\mathbb{L}_n^{\text{Lam,sc}} = n! \mathbb{L}_n^{\text{Lam}} \in \mathbb{Z}[u]$	Theorem 19.7, Theorem 19.9
$\left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] = \mathcal{B}_{n,n-m+1}^{\text{exp}}(0!, 1!, 2!, \dots)$: the Lambert triangle is a slice of the Bell array	Theorem A.9, Theorem A.10
$(X + aL)^{(-1)^\circ} = X + \sum_{n \geq 0} a^{n+1} \mathbb{L}_n^{\text{Lam}}(L)t^n$; the case $a = -1$ gives $(-1)^{n+1} \mathbb{L}_n^{\text{Lam}}$, not $(-1)^n \mathbb{L}_n^{\text{Lam}}$	Theorem 18.5, Theorem 18.6
Exact residual of the truncation: $\omega_N + \log \omega_N - X = -\mathbb{L}_{N+1}^{\text{Lam}}(L)t^{N+1} + \mathcal{O}_t^{\text{form}}(t^{N+2})$	Theorem 18.13, Theorem 18.14
Analytic form, $x \rightarrow +\infty$ on the positive ray, $K = (N+2)^2$: $\omega(x) - \omega_N(x) = \mathbb{L}_{N+1}^{\text{Lam}}(L)x^{-(N+1)} + \mathcal{O}_{x \rightarrow +\infty}(L^K x^{-(N+2)})$	Theorem 18.17

Statement	Proved at
Branch coordinates: $W_k(z) + \log W_k(z) = \xi_k$ with $\xi_k = \text{Log } z + 2\pi i k$; the formal expansion of W_k is Theorem 18.10 under $X \mapsto \xi_k$, $L \mapsto \ell_k = \log_B \xi_k$. The polynomials are branch-independent; the coordinates, sector and analytic estimates are not, and the complex analysis is not developed	Theorem 18.7
Numerical tables of L_n^{Lam} , of $\left[\begin{smallmatrix} n \\ n-m+1 \end{smallmatrix} \right]$, and of $L_n^{\text{Lam}}(1)$	Section D , Table 27 , Table 28 , Table 29

C.9 Bell polynomials and closed coefficient extraction

Table 26: The combinatorial layer of this appendix.

Statement	Proved at
$\mathcal{B}_{n,k}^{\text{exp}}$ by multiplicity sum; $\mathcal{B}_{0,0}^{\text{exp}} = 1$, $\mathcal{B}_{n,0}^{\text{exp}} = 0$ for $n \geq 1$	Theorem A.1
$\frac{\Xi^k}{k!} = \sum_{n \geq k} \mathcal{B}_{n,k}^{\text{exp}} \frac{z^n}{n!}$ for $\Xi = \sum_{j \geq 1} \xi_j z^j / j!$; and the complete version with $\exp \Xi$	Theorem A.3
Degree k , weight n , bihomogeneity, boundary rows	Theorem A.2
$\mathcal{B}_{n,k}^{\text{ord}}(\xi) = \frac{k!}{n!} \mathcal{B}_{n,k}^{\text{exp}}(1! \xi_1, 2! \xi_2, \dots)$; this is the factor that converts $C_{n,k}$ to the exponential normalization	Theorem A.4
Tabulating recurrences $\mathcal{B}_{n,k}^{\text{exp}} = \sum_j \binom{n-1}{j-1} x_j \mathcal{B}_{n-j,k-1}^{\text{exp}}$ and its $\binom{n}{j}$ variant	Theorem A.5
Differentiating recurrence $\mathcal{B}_{n+1,k}^{\text{exp}} = \mathcal{D} \mathcal{B}_{n,k}^{\text{exp}} + x_1 \mathcal{B}_{n,k-1}^{\text{exp}}$	Theorem 11.24
$\mathcal{B}_{n,k}^{\text{exp}}(0!, 1!, \dots) = \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]$; $\mathcal{B}_{n,k}^{\text{exp}}(1, 1, \dots) = \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$	Theorem A.9
Abstract Faà di Bruno for a Leibniz tower $\partial u_k = u_{k+1} \gamma$	Theorem A.12
$D_X^n(A^r) = \sum_k \binom{r}{k} A^{r-k} \mathcal{B}_{n,k}^{\text{exp}}(D_X A, \dots)$	Theorem A.16
No nonzero linear functional on $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ annihilates all D_X -images: there is no formal residue in t	Theorem B.16
Six named differences from the classical Lagrange inversion formula	Section B.6
Which closed form to use for which task	Table 17

C.10 What this sheet does not assert

Four things the sheet is silent about

1. **Analytic validity.** Every entry above whose citation is a formal statement is an identity of formal series. It becomes an assertion about a function only through an explicit transfer theorem ([Theorem 6.45](#), [Theorem 10.35](#), [Theorem 14.21](#), [Theorem 18.17](#)), each with its own hypotheses and its own printed regime. A bare $\mathcal{O}^{\text{form}}$ never implies a bare $\mathcal{O}(\cdot)$ ([Theorem 5.7](#)).
2. **Uniformity in L .** A block-level identity says nothing about the size of the coefficient polynomial at a given x . The analytic entries above therefore carry an explicit power of L in the error term – for instance $K = (N + 2)^2$ in [Theorem 18.17](#) – and none of them is uniform in N .
3. **Degree truncation.** Discarding high powers of L inside a retained block is not part of the t -adic calculus and none of the precision statements above covers it ([Theorem B.1](#), [Theorem 5.1](#)).

Table 17: Closed forms and their recursive counterparts. “Closed” means a finite sum over an explicitly bounded index set in the input blocks alone; “recursive” means a triangular recurrence in the output blocks.

Task	Closed form	Recursive alternative, and when to prefer it
Block of a product	(530) (Theorem B.2)	none needed; the Cauchy product (60) is already finite. Use Theorem 5.4 to truncate inputs safely.
Block of an integer power	(82), or (531) in Bell form	the triangular recurrence (83); preferable when all powers $r \leq R$ are wanted, since it costs $\mathcal{O}(N^2 R)$ block products.
Block of $(1 + W)^\sigma$	(532) (Theorem B.4)	the power recurrence of Theorem 25.23; preferable for a full table.
Block of a reciprocal	(533), or the determinant (535)	(100) (Theorem 7.7) for a table; Newton doubling Theorem 7.27 when N is large. Use the determinant for structural claims (Theorem B.7).
Block of a quotient	B^{-1} then (530)	(101) (Theorem 7.8); almost always preferable.
Block of a composition	(197), or (537) in Bell form	the block recurrence Theorem 10.15; preferable when one inner map is composed with many outer series.
Composition with a level-zero shift	(538) (Theorem B.9)	none needed; the closed form is already a single sum.
Block of a reversion	(539) (Theorem B.11)	triangular reversal Theorem 15.2 or Newton Theorem 15.11; both cheaper for a full table (Theorem 25.42).
Reversion of a level-zero perturbation $X + P(L)$	(540) (Theorem B.12)	none needed; one operator applied to one power gives every block. Contains the Lambert family as the case $P = aL$.
Reversion as a differential polynomial	(542), (543) (Theorem B.14)	no recursive counterpart; this is the form for symbolic A and for weight-graded structural arguments.

4. **Closure.** Membership in a class is a hypothesis, never a conclusion of a formula. Exponentials, general powers, logarithms of non-units and reversions at a non-identity leading slope leave $\mathcal{PL}_{\text{poly}, \mathbb{K}}$ under conditions recorded in [Theorem 8.16](#), [Theorem 8.26](#), [Theorem 8.31](#) and [Theorem 17.3](#), and only there.

D Tables of the Lambert polynomials

All entries below were produced from the operator definition of [Theorem 19.1](#) in exact rational arithmetic and then verified three independent ways: against the Stirling closed form of [Theorem 19.4](#), against the differential recurrence of [Theorem 19.2](#), and – the only check that does not presuppose any of the derived identities – by substituting the resulting series into the defining equation $\omega + \log \omega = X$ and confirming that every coefficient block through outer order t^8 vanishes identically.

D.1 The scaled integer family

By [Theorem 19.9](#), $n!L_n^{\text{Lam}}(u) \in \mathbb{Z}[u]$ for $n \geq 1$; the table lists that integer polynomial together with the factorial that divides it, which is the storage normalization $L_n^{\text{Lam,sc}} = n!L_n^{\text{Lam}}$. Read the row as $L_n^{\text{Lam}}(u) = (\text{entry})/n!$. The sources tabulate these polynomials only through $n = 8$; rows 9–12 are new here.

Table 27: The Lambert polynomials $L_n^{\text{Lam}}(u) = L_n^{\text{Lam,sc}}(u)/n!$.

n	$n!$	$L_n^{\text{Lam,sc}}(u) = n!L_n^{\text{Lam}}(u)$
0	1	$-u$
1	1	u
2	2	$u^2 - 2u$
3	6	$2u^3 - 9u^2 + 6u$
4	24	$6u^4 - 44u^3 + 72u^2 - 24u$
5	120	$24u^5 - 250u^4 + 700u^3 - 600u^2 + 120u$
6	720	$120u^6 - 1,644u^5 + 6,750u^4 - 10,200u^3 + 5,400u^2 - 720u$
7	5,040	$720u^7 - 12,348u^6 + 68,208u^5 - 154,350u^4 + 147,000u^3 - 52,920u^2 + 5,040u$
8	40,320	$5,040u^8 - 104,544u^7 + 735,392u^6 - 2,274,384u^5 + 3,292,800u^4 - 2,163,840u^3 + 564,480u^2 - 40,320u$
9	362,880	$40,320u^9 - 986,256u^8 + 8,504,928u^7 - 33,911,136u^6 + 67,885,776u^5 - 68,584,320u^4 + 33,022,080u^3 - 6,531,840u^2 + 362,880u$
10	3,628,800	$362,880u^{10} - 10,265,760u^9 + 105,543,000u^8 - 521,049,600u^7 + 1,357,398,000u^6 - 1,913,375,520u^5 + 1,428,840,000u^4 - 526,176,000u^3 + 81,648,000u^2 - 3,628,800u$
11	39,916,800	$3,628,800u^{11} - 116,915,040u^{10} + 1,402,893,360u^9 - 8,325,405,000u^8 + 27,062,085,600u^7 - 50,009,929,200u^6 + 52,481,610,720u^5 - 30,187,080,000u^4 + 8,781,696,000u^3 - 1,097,712,000u^2 + 39,916,800u$
12	479,001,600	$39,916,800u^{12} - 1,446,526,080u^{11} + 19,921,172,832u^{10} - 138,940,660,320u^9 + 546,429,272,400u^8 - 1,267,789,406,400u^7 + 1,754,714,586,240u^6 - 1,426,718,240,640u^5 + 652,040,928,000u^4 - 153,679,680,000u^3 + 15,807,052,800u^2 - 479,001,600u$

Remark D.1 (The $n = 0$ row is not an exception). $L_0^{\text{Lam}}(u) = -u$ has degree 1, not 0. The degree law $\deg L_n^{\text{Lam}} = n$ of [Theorem 19.7](#) is therefore stated only for $n \geq 1$, and the integrality normalization

$\mathcal{L}_n^{\text{Lam,sc}} = n! \mathcal{L}_n^{\text{Lam}}$ is likewise a statement about $n \geq 1$; the table's $n = 0$ row records the value, not an instance of either law.

D.2 The coefficient triangle

Write $\mathcal{L}_n^{\text{Lam}}(u) = \sum_{m=1}^n \lambda_{n,m} u^m$ for $n \geq 1$. By [Theorem 19.4](#),

$$\lambda_{n,m} = \frac{(-1)^{n-m}}{m!} \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right],$$

so the signs alternate along each row and the magnitudes are unsigned Stirling cycle numbers divided by a factorial. The next table lists the integers $m!(-1)^{n-m}\lambda_{n,m} = \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right]$, which is the row of the Stirling triangle read from the right.

Table 28: $\left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] = m! |\lambda_{n,m}|$, the unsigned coefficient data of $\mathcal{L}_n^{\text{Lam}}$.

$n \backslash m$	1	2	3	4	5	6	7	8	9
1	1								
2	1	1							
3	1	3	2						
4	1	6	11	6					
5	1	10	35	50	24				
6	1	15	85	225	274	120			
7	1	21	175	735	1,624	1,764	720		
8	1	28	322	1,960	6,769	13,132	13,068	5,040	
9	1	36	546	4,536	22,449	67,284	118,124	109,584	40,320

The first column is $\left[\begin{matrix} n \\ n \end{matrix} \right] = 1$, which is [Theorem 19.7\(iv\)](#) in the form $\lambda_{n,1} = (-1)^{n-1}$. The last entry of row n is $\left[\begin{matrix} n \\ 1 \end{matrix} \right] = (n-1)!$, giving the leading coefficient $\lambda_{n,n} = 1/n$. The penultimate entry is $\left[\begin{matrix} n \\ 2 \end{matrix} \right] = (n-1)!H_{n-1}$, giving $\lambda_{n,n-1} = -H_{n-1}$.

D.3 Numerical check values

The following exact values are convenient unit tests for an implementation of the recurrence; each is $\mathcal{L}_n^{\text{Lam}}(1)$, which by [Theorem 19.4](#) equals $\sum_{m=1}^n (-1)^{n-m} \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] / m!$.

Table 29: Exact values $\mathcal{L}_n^{\text{Lam}}(1)$.

n	$\mathcal{L}_n^{\text{Lam}}(1)$	n	$\mathcal{L}_n^{\text{Lam}}(1)$
0	-1	7	$\frac{15}{56}$
1	1	8	$\frac{457}{1260}$
2	$-\frac{1}{2}$	9	$-\frac{49}{90}$
3	$-\frac{1}{6}$	10	$-\frac{391}{2016}$
4	$\frac{5}{12}$	11	$\frac{32213}{36960}$
5	$-\frac{1}{20}$	12	$-\frac{299363}{1425600}$
6	$-\frac{49}{120}$		

D.4 An end-to-end numerical validation

The tables above and the identities of [Section 19](#) are algebraic. A coefficient table can be internally consistent and still be wrong in sign, because a sign error propagates through every identity that derives one normalization from another. The following check is independent of all of them: it solves the defining equation numerically and compares.

Let $\omega_N(x) = x + \sum_{n=0}^N \mathsf{L}_n^{\text{Lam}}(L)x^{-n}$ with $L = \log x$, the truncation of [Theorem 18.10](#). Solving $y + \log y = x$ to fifty significant digits by Newton’s method and comparing gives the following. The residual column is $\omega_N + \log \omega_N - x$ computed numerically; the predicted column is $-\mathsf{L}_{N+1}^{\text{Lam}}(L)x^{-(N+1)}$, which is what [Equation \(338\)](#) asserts the residual must be to leading order.

Table 30: Observed versus predicted residual of the truncated Wright-omega expansion. The ratio tends to 1 as x grows, which is the content of [Equation \(338\)](#); the departure at $x = 50$ is the omitted $\mathcal{O}_t^{\text{form}}(t^{N+2})$ block making itself felt.

x	N	observed residual	predicted	ratio
50	1	-1.534239×10^{-3}	-1.495976×10^{-3}	1.0256
50	2	-5.859441×10^{-6}	-7.300570×10^{-6}	0.8026
50	3	$+1.599261 \times 10^{-6}$	$+1.473299 \times 10^{-6}$	1.0855
50	4	$+9.405048 \times 10^{-8}$	$+8.979402 \times 10^{-8}$	1.0474
200	1	-2.211720×10^{-4}	-2.184442×10^{-4}	1.0125
200	2	-1.606059×10^{-6}	-1.596060×10^{-6}	1.0063
200	3	-1.802138×10^{-9}	-2.030332×10^{-9}	0.8876
200	4	$+2.386204 \times 10^{-10}$	$+2.316867 \times 10^{-10}$	1.0299
1000	1	-1.701321×10^{-5}	-1.695079×10^{-5}	1.0037
1000	2	-4.535166×10^{-8}	-4.520477×10^{-8}	1.0032
1000	3	$-1.013683 \times 10^{-10}$	$-1.011740 \times 10^{-10}$	1.0019
1000	4	$-9.250010 \times 10^{-14}$	$-9.319150 \times 10^{-14}$	0.9926

Two features of the table are worth reading carefully, because both are predictions of the theory rather than accidents.

First, the residual *changes sign* between $N = 2$ and $N = 3$ at $x = 50$ and between $N = 3$ and $N = 4$ at $x = 200$. That is not instability: by [Theorem 19.7](#) the coefficients of $\mathsf{L}_n^{\text{Lam}}$ alternate, so the leading residual $-\mathsf{L}_{N+1}^{\text{Lam}}(L)x^{-(N+1)}$ changes sign with N whenever L is small enough that the top coefficient does not yet dominate. A sign error anywhere in the chain would break this pattern rather than reproduce it.

Second, at $x = 1000$ the error falls by roughly three orders of magnitude per added block:

N	$ \omega_N - \omega $	N	$ \omega_N - \omega $
0	6.924751×10^{-3}	4	9.240705×10^{-14}
1	1.699609×10^{-5}	5	7.844485×10^{-16}
2	4.530604×10^{-8}	6	6.759229×10^{-18}
3	1.012664×10^{-10}		

What this table does *not* show

Improvement through $N = 6$ at one value of x is not convergence. The series of [Theorem 18.10](#) diverges for every fixed x : by [Theorem 19.7](#) the leading coefficient of $\mathsf{L}_n^{\text{Lam}}$ is $1/n$ while its degree is n , so for fixed x the terms eventually grow. What the table exhibits is asymptotic behaviour — for each fixed N , the error is $\mathcal{O}_{x \rightarrow \infty}(L^{N+2}/x^{N+2})$ — and the correct reading of the second table is a statement about N small relative to $\log x$, not a limit in N . [Section 22](#) makes the distinction precise.

E Exercises with solutions

The six source reports carry 83 exercises between them, distributed very unevenly: report 1 sets 58, three at the end of each chapter, and supplies no solutions; report 3 sets 13 and solves all of them; report 2 sets 12 and solves all of them; reports 4, 5 and 6 set none but contain worked examples that

make good problems. The overlap is heavy — five of the six ask for the reversion of $X + aL$ in one guise or another — and the quality is uneven, ranging from genuine tests of the formalism down to arithmetic drill.

What follows is a single graded set of 56 exercises, harvested from all six sources, deduplicated, corrected, and organised by the part of the volume they exercise. Every exercise carries a complete solution, not a hint. Where a source supplied a solution it was recomputed independently; the corrections are marked in the $\text{\LaTeX}$ source with % ed. : comments and are also summarised in [Section G](#). Where an important technique of the body had no exercise anywhere in the corpus — the sharp summability threshold, the guard-precision rule, the derivative certificate, the analytic transfer — one has been added.

E.1 How to use this appendix

Three conventions are worth restating, because most of the mistakes recorded below come from breaking one of them.

1. *Truncation is exclusive.* $\text{Trunc}_{<N} F$ keeps the blocks of outer order $n < N$ ([Theorem 5.1](#)). Sources 1 and 6 use the inclusive cutoff $[F]_{\leq N} = \text{Trunc}_{<N+1} F$; every index taken from them has been shifted by one block. Exercises whose title contains the word *trap* exist precisely because that shift is invisible until it is wrong.
2. *The three $\mathcal{O}(\cdot)$ -roles are distinct.* $\mathcal{O}_t^{\text{form}}(t^N)$ is a statement about a formal valuation and carries no analytic content whatsoever; $\mathcal{O}_{X \rightarrow +\infty}(A(X))$ is an inequality between functions on a stated ray; a complexity bound is neither. No exercise below asks you to pass from the first to the second without an analytic hypothesis, and three of them ask you to find where somebody did.
3. *Every statement names its coefficient field and its regime.* The default field is a field $\mathbb{K}$ of characteristic zero; the default regime is $X \rightarrow +\infty$ along the positive real ray with the real logarithm.

Exercises marked (*hard*) need either a genuinely new idea or a long exact computation. Two special kinds recur.

- *Counterexample exercises* ask you to refute a plausible strengthening of a theorem of the body, usually one obtained by deleting a hypothesis. These are [Theorems E.4, E.5, E.7, E.15, E.20, E.27, E.29, E.36, E.39, E.44](#) and [E.53](#). The hypotheses the sources kept dropping are exactly the ones these exercises defend.
- *Traps* hand you a plausible-looking computation containing one documented failure mode — a truncation convention read in the wrong direction, a formal tail treated as an analytic estimate, a division by a nonunit, a scale change applied to one generator only — and ask you to locate the error and repair it. These are [Theorems E.6, E.16, E.26, E.32, E.41, E.46, E.52](#) and [E.54](#).

E.2 The scale and its algebra

Exercise E.1 (Valuation of a sum, and how far it can jump). Work in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, using only [Theorem 4.2](#).

1. Prove $\nu_t(F + G) \geq \min\{\nu_t F, \nu_t G\}$, with equality whenever $\nu_t F \neq \nu_t G$.
2. For every $k \geq 1$ and every $d \geq 0$ exhibit $F, G \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $\nu_t F = \nu_t G = 0$, $\deg_L([t^0]F) = d$, and $\nu_t(F + G) = k$.
3. Show that strict inequality in (1) forces $\nu_t F = \nu_t G =: n$ and $[t^n]F = -[t^n]G$.

Solution. (1) Let $n = \min\{\nu_t F, \nu_t G\}$. For $m < n$ both $[t^m]F$ and $[t^m]G$ vanish, hence so does $[t^m](F + G)$; this is $\nu_t(F + G) \geq n$. If $\nu_t F < \nu_t G$ then $n = \nu_t F$ and $[t^n](F + G) = [t^n]F \neq 0$, so $\nu_t(F + G) = n$; the case $\nu_t G < \nu_t F$ is symmetric.

(2) Put

$$F = L^d + t^k, \quad G = -L^d.$$

Both lie in $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, both have outer order 0, the t^0 -block of F has L -degree d , and $F + G = t^k$ has outer order exactly k .

(3) If $\nu_t F < \nu_t G$ or $\nu_t G < \nu_t F$ then (1) gives equality, so strictness forces $\nu_t F = \nu_t G = n$. Then $0 = [t^n](F + G) = [t^n]F + [t^n]G$, which is the assertion. Note that the converse fails: cancelling leading blocks guarantee $\nu_t(F + G) \geq n + 1$ but say nothing about how much further the order jumps, which is what (2) exploits. $\square$

Exercise E.2 (Completeness of the power-series ring, block by block). Let $(F_r)_{r \geq 0}$ be a t -adically Cauchy sequence in $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$.

1. Show that for each n the block $[t^n]F_r$ is eventually independent of r ; call its value $p_n \in \mathbb{K}[L]$.
2. Show $F := \sum_{n \geq 0} p_n t^n$ lies in $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and $F_r \rightarrow F$.
3. Locate the exact step at which $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, rather than $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, was used, and show that a Cauchy sequence in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ nevertheless always satisfies $\inf_r \nu_t(F_r) > -\infty$, so that the same three steps do prove completeness of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ as well.

Solution. (1) Cauchy means: for every N there is r_N with $\nu_t(F_r - F_s) \geq N$ whenever $r, s \geq r_N$. Applying this with $N = n + 1$ gives $[t^n]F_r = [t^n]F_s$ for all $r, s \geq r_{n+1}$, which is the assertion; write p_n for the common value.

(2) Each p_n lies in $\mathbb{K}[L]$ and the index set is $\mathbb{N}_0$, so the support of F is bounded below by 0 and $F \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. Fix N and let $r \geq \max\{r_1, \dots, r_N\}$. For $n < N$ we have $[t^n]F_r = p_n = [t^n]F$, so $\nu_t(F - F_r) \geq N$. As N was arbitrary, $F_r \rightarrow F$. Uniqueness of the limit is separatedness, [Theorem 5.18](#) (i).

(3) The only step that used $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ is the claim in (2) that $\sum_n p_n t^n$ is an element of the ring: on $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ the blockwise limits p_n could be nonzero for infinitely many $n < 0$, and then the sum is not an element of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. This is not idle: the sequence of [Theorem 5.19](#) stabilises coefficientwise to exactly such a symbol. But that sequence is not Cauchy, and no Cauchy sequence can behave that way. Indeed apply the Cauchy condition with $N = 0$: there is r_0 with $\nu_t(F_r - F_{r_0}) \geq 0$ for all $r \geq r_0$, whence by part (1) of [Theorem E.1](#)

$$\nu_t(F_r) \geq \min\{\nu_t(F_{r_0}), 0\} \quad (r \geq r_0),$$

and the finitely many indices $r < r_0$ contribute a finite minimum. Hence $\mu := \inf_r \nu_t(F_r) > -\infty$, every F_r lies in $t^\mu \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, and steps (1) and (2) may be run verbatim inside that subring. This is [Theorem 5.18](#) (iii),(iv). $\square$

Exercise E.3 (Summable powers). Let $U \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. Prove that the family $(U^n)_{n \geq 0}$ is formally summable in the sense of [Theorem 5.21](#) if and only if $U = 0$ or $\nu_t(U) \geq 1$, and that in that case $\sum_{n \geq 0} U^n = (1 - U)^{-1}$.

Solution. Assume $U \neq 0$ and put $q = \nu_t(U)$. Since $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is a domain and ν_t is additive on products ([Theorem 4.2](#)), $\nu_t(U^n) = nq$ for every $n \geq 0$, with $U^0 = 1$ of order 0.

If $q \geq 1$ then $nq \rightarrow +\infty$, so for each N the set $\{n : \nu_t(U^n) < N\}$ is contained in $\{n : n < N\}$ and is finite: the family is summable. If $q \leq 0$ then $nq \leq 0$ for all n , so $\{n : \nu_t(U^n) < 1\} = \mathbb{N}_0$ is infinite and the family is not summable.

For the value, let $S = \sum_{n \geq 0} U^n$, which exists by the summability just proved. Multiplication is continuous for the t -adic filtration on a summable family ([Theorem 5.22](#)), so

$$(1 - U)S = \sum_{n \geq 0} U^n - \sum_{n \geq 0} U^{n+1} = U^0 = 1,$$

the middle step being a telescoping of two summable families. Since $\nu_t(U) \geq 1$ the element $1 - U$ is a unit of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ by [Theorem 7.3](#), and its reciprocal is unique, so $S = (1 - U)^{-1}$. This is [Theorem 5.23](#). $\square$

Exercise E.4 (Counterexample: block-local finiteness is not summability). [Theorem 5.24](#) (c) characterises summability by two conditions: every block receives finitely many contributions, and $\inf_i \nu_t(S_i) > -\infty$. Show that the second clause cannot be deleted.

1. Exhibit a family in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ that is block-locally finite but not summable.
2. Exhibit one that is block-locally finite, not summable, and whose blockwise sum is a legitimate element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. Conclude that “the blockwise sum exists in the ring” is not a substitute for summability either.
3. Show that on $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ the second clause is automatic.

Solution. (1) Take $S_i = t^{-i}$ for $i \geq 1$. The block t^n receives a contribution from S_i only for $i = -n$, so at most one contribution. But $\nu_t(S_i) = -i$, so $\{i : \nu_t(S_i) < 0\}$ is infinite and the family fails [Theorem 5.21](#). Concretely, the partial sums $T_m = \sum_{i=1}^m t^{-i}$ satisfy $\nu_t(T_{m+1} - T_m) = -(m+1)$, so they are not Cauchy, and the blockwise limits would give a symbol with infinitely many negative blocks, which is not an element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ ([Theorem 3.15](#) (b)).

(2) Take

$$S_i = t^{-i} - t^{-i-1} \quad (i \geq 1).$$

The block t^n receives contributions only from $i = -n$ and $i = -n - 1$, so at most two; the family is block-locally finite. Blockwise, $[t^{-1}] \sum_i S_i = 1$ and for $n \leq -2$ the two contributions are $+1$ (from $i = -n$) and -1 (from $i = -n - 1$), summing to 0. The blockwise sum is therefore $t^{-1} \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. Nevertheless $\nu_t(S_i) = -i - 1 \rightarrow -\infty$, so the family is not summable, and indeed the partial sums $\sum_{i=1}^m S_i = t^{-1} - t^{-m-1}$ do not converge: $\nu_t(t^{-1} - t^{-m-1})$ –differences are $-(m+2) \rightarrow -\infty$. So a blockwise sum can exist in the ring and still fail to be the limit of the partial sums in any sense the filtration recognises. Summability is a statement about the family, not about the array of coefficients it produces.

(3) If every $S_i \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ then $\nu_t(S_i) \geq 0$, so $\inf_i \nu_t(S_i) \geq 0 > -\infty$. $\square$

Exercise E.5 (Counterexample: the tail is a filtration, not an ideal). A plausible reading of “compute modulo t^N ” is that $\mathfrak{F}_N = t^N \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is an ideal of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ and that $\text{Trunc}_{<N} \cdot$ is the quotient homomorphism. Refute both halves.

1. Show $\mathfrak{F}_N$ is not an ideal of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ for any N , and that the ideal it generates is all of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$.
2. Show $\text{Trunc}_{<N} \cdot$ is not multiplicative on $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, by exhibiting F, G with $\text{Trunc}_{<N}(FG) \neq \text{Trunc}_{<N}(\text{Trunc}_{<N} F \text{Trunc}_{<N} G)$.
3. State what does survive, and prove it.

Solution. (1) t is a unit of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, with inverse t^{-1} . Hence $t^{-1} \cdot t^N = t^{N-1}$, which has outer order $N - 1 < N$ and so does not lie in $\mathfrak{F}_N$: $\mathfrak{F}_N$ is not closed under multiplication by ring elements. Moreover $t^{-N} \cdot t^N = 1$, so the ideal generated by t^N contains 1 and is all of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. In particular $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}/(t^N) = 0$ and there is no quotient ring to map onto. This is [Theorem 5.11](#) (i),(ii).

(2) Take

$$F = G = \frac{t^{-2}}{1-t} = \sum_{n \geq -2} t^n, \quad N = 1.$$

Then $FG = t^{-4}(1-t)^{-2} = \sum_{j \geq -4} (j+5)t^j$, so

$$\text{Trunc}_{<1}(FG) = t^{-4} + 2t^{-3} + 3t^{-2} + 4t^{-1} + 5.$$

On the other hand $\text{Trunc}_{<1} F = \text{Trunc}_{<1} G = t^{-2}(1 + t + t^2)$, and

$$\text{Trunc}_{<1} F \text{Trunc}_{<1} G = t^{-4}(1 + t + t^2)^2 = t^{-4} + 2t^{-3} + 3t^{-2} + 2t^{-1} + 1,$$

which is already of t -degree 0, so truncating again changes nothing. The blocks at t^{-1} and t^0 are wrong, by 2 and by 4.

(3) Three statements survive. First, $\mathfrak{F}_N$ is an additive subgroup and a $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ -submodule of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, so $\text{Trunc}_{<N} \cdot$ is $\mathbb{K}[L]$ -linear and $\text{Trunc}_{<N}(F + G) = \text{Trunc}_{<N} F + \text{Trunc}_{<N} G$. Second, on $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ the set $t^N \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ is an ideal for $N \geq 0$ and $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}/t^N \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} \cong \mathbb{K}[L][t]/(t^N)$; this is where the quotient language is legitimate. Third, the correct multiplicative law on $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is the asymmetric one of [Theorem 5.4](#): with $m = \nu_t F$ and $r = \nu_t G$,

$$\text{Trunc}_{<N}(FG) = \text{Trunc}_{<N}(\text{Trunc}_{<N-r} F \cdot \text{Trunc}_{<N-m} G).$$

In the example $m = r = -2$ and $N = 1$, so the required inputs are $\text{Trunc}_{<3} F$ and $\text{Trunc}_{<3} G$, two blocks more than were used. A direct check: $\text{Trunc}_{<3} F = t^{-2}(1 + t + t^2 + t^3 + t^4)$, whose square is $t^{-4}(1 + 2t + 3t^2 + 4t^3 + 5t^4 + \dots)$, and truncating at < 1 reproduces the correct answer. $\square$

Exercise E.6 (Trap: a symmetric truncation on the Laurent ring). A routine is asked for $\text{Trunc}_{<0}(FG)$ where

$$F = X^2 + X + \frac{L}{X} + \mathcal{O}_t^{\text{form}}(t^2), \quad G = X^3 - X^2 L + \mathcal{O}_t^{\text{form}}(t^{-1}).$$

It reasons: “both factors are known through outer order t^1 respectively t^{-2} ; truncating each at < 0 and multiplying, $FG = (X^2 + X)(X^3 - X^2 L) + \dots = X^5 - X^4 L + X^4 - X^3 L + \dots$, so $\text{Trunc}_{<0}(FG) = X^5 + (1 - L)X^4 - LX^3$.” Find the error, say which blocks of the answer are nevertheless correct, and state exactly what precision in F and G the requested output needs.

Solution. The error is that $\text{Trunc}_{<N} \cdot$ is not multiplicative on $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ ([Theorem E.5](#)), and the symmetric rule (50) is valid only when both factors have nonnegative outer order. Here $\nu_t F = -2$ and $\nu_t G = -3$, so [Theorem 5.4](#) demands

$$\begin{aligned} \text{Trunc}_{<0}(FG) &= \text{Trunc}_{<0}(\text{Trunc}_{<0-(-3)} F \cdot \text{Trunc}_{<0-(-2)} G) \\ &= \text{Trunc}_{<0}(\text{Trunc}_{<3} F \cdot \text{Trunc}_{<2} G). \end{aligned}$$

That is, F is needed through outer order t^2 and G through t^1 ; the routine used F through t^{-1} and G through t^{-1} .

Which blocks survive? Write $F_n = [t^n]F$ and $G_n = [t^n]G$, so that $F_{-2} = 1$, $F_{-1} = 1$, $F_0 = 0$, $F_1 = L$ are known and F_n is unknown for $n \geq 2$; while $G_{-3} = 1$, $G_{-2} = -L$ are known and G_n is unknown for $n \geq -1$. Convoluting,

$$\begin{aligned} [t^{-5}](FG) &= F_{-2}G_{-3} = 1, \\ [t^{-4}](FG) &= F_{-2}G_{-2} + F_{-1}G_{-3} = 1 - L, \\ [t^{-3}](FG) &= F_{-2}G_{-1} + F_{-1}G_{-2} + F_0G_{-3} = G_{-1} - L. \end{aligned}$$

The first two are determined by the data and agree with the routine’s X^5 and $(1 - L)X^4$. The third is not determined: it needs G_{-1} , which the routine both discarded and never knew. The printed $-LX^3$ is what one gets by setting $G_{-1} = 0$, an assumption nothing supports. The blocks at t^{-2} , t^{-1} , t^0 are likewise undetermined, and the routine did not print them at all.

So the data are *insufficient* for the requested output. Indeed F is known only modulo $\mathcal{O}_t^{\text{form}}(t^2)$, which is exactly $\text{Trunc}_{<2} F$, one block short of the $\text{Trunc}_{<3} F$ required; and G is known only modulo $\mathcal{O}_t^{\text{form}}(t^{-1})$, three blocks short of $\text{Trunc}_{<2} G$. The honest output is

$$\text{Trunc}_{<-3}(FG) = X^5 + (1 - L)X^4,$$

with the remaining blocks reported as unknown. This is the error rule of [Theorem 6.14](#) read backwards. Two lessons: the symmetric rule (50) is a theorem about $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, not about $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$; and a Laurent factor makes a routine need *more* input precision than the output it is asked for, never less. $\square$

Exercise E.7 (Counterexample: no monomial prefactor inside the polynomial ring). Several sources write a nonzero $A \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ in the prefactor normal form $A = cX^\rho L^\beta(1+U)$ with $c \in \mathbb{K}^\times$, $\rho \in \mathbb{Z}$, $\beta \in \mathbb{N}_0$ and $U \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, “whenever available”. Show that for $A = L + t$ it is not available, and that it becomes available exactly after passing to $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$. Deduce that no argument inside $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ may assume the normal form.

Solution. Suppose $L + t = cX^\rho L^\beta(1+U) = ct^{-\rho}L^\beta(1+U)$ with $U \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. Comparing outer orders, $\nu_t(L + t) = 0$ while $\nu_t(ct^{-\rho}L^\beta(1+U)) = -\rho$, so $\rho = 0$. Comparing the t^0 -blocks then gives $cL^\beta = L$, so $c = 1$ and $\beta = 1$. Comparing the t^1 -blocks gives

$$1 = [t^1](L(1+U)) = L \cdot u_1, \quad u_1 := [t^1]U \in \mathbb{K}[L],$$

whence $u_1 = L^{-1}$, which is not a polynomial. Contradiction.

Over $\mathbb{K}(L)$ the factorisation exists and is the obvious one, $L + t = L(1 + L^{-1}t)$ with $L^{-1}t \in t\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$. So the obstruction is not the shape of the leading monomial but the failure of $\mathbb{K}[L]$ to be a field: the general leading-block factorisation [Theorem 4.10](#) produces $A = t^{n_0}p_{n_0}(L)(1+U)$ with U in $t\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$, and the coefficient p_{n_0} can be inverted only in $\mathbb{K}(L)$. Consequently any argument that silently divides by the leading block – for instance one that defines d_A through the prefactor, or that “normalises A to be monic” – has left $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. $\square$

Exercise E.8 (A prescribed drop in the log-degree profile). Let $d \geq 1$. For each k with $1 \leq k \leq d$ construct a pair $F, G \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ with $\nu_t F = \nu_t G = 0$ and $d_F(1) = d_G(1) = d$ such that $d_{FG}(1) = d - k$; construct one more pair with $d_{FG}(1) = -\infty$; and exhibit a pair with no drop at all. Then say exactly when equality holds in the bound $d_{FG}(N) \leq (d_F * d_G)(N)$ of [Theorem 6.25](#).

Solution. For $1 \leq k \leq d$ put

$$F = 1 + (L^d + L^{d-k})t, \quad G = 1 - L^d t.$$

Both have $\nu_t = 0$, $d_F(0) = d_G(0) = 0$ and $d_F(1) = d_G(1) = d$, the term L^{d-k} being of strictly lower degree than L^d because $k \geq 1$. Now

$$[t^1](FG) = (L^d + L^{d-k}) \cdot 1 + 1 \cdot (-L^d) = L^{d-k},$$

so $d_{FG}(1) = d - k$, while the convolution bound is $(d_F * d_G)(1) = \max\{0 + d, d + 0\} = d$. For the degenerate case take $F = 1 + L^d t$, $G = 1 - L^d t$, where $[t^1](FG) = 0$ and $d_{FG}(1) = -\infty$. For no drop take $F = G = 1 + L^d t$, where $[t^1](FG) = 2L^d$ has degree d ; this is the generic situation, and it is what makes the drop worth an exercise.

Equality is decided exactly by [Theorem 6.25](#). Let $M = (d_F * d_G)(N)$ and let P_N be the set of pairs (n, m) with $n + m = N$, $[t^n]F \neq 0 \neq [t^m]G$ and $d_F(n) + d_G(m) = M$. If $P_N = \emptyset$ then $[t^N](FG) = 0$ and equality holds trivially with $M = -\infty$. If $P_N \neq \emptyset$ then the coefficient of L^M in $[t^N](FG)$ equals $\sum_{(n,m) \in P_N} \text{lc}([t^n]F) \text{lc}([t^m]G)$, and equality holds if and only if that sum is nonzero. In the dropping family $P_1 = \{(0, 1), (1, 0)\}$ and the sum is $1 \cdot (-1) + 1 \cdot 1 = 0$, which is why the degree drops; how far it drops is then decided by the next coefficients, and the parameter k tunes exactly that. In the last pair the same sum is $1 \cdot 1 + 1 \cdot 1 = 2 \neq 0$ and there is no drop. Note that condition (3) of [Theorem 6.26](#) – positive leading coefficients over an ordered field – excludes the dropping family and covers the last one, as it must. $\square$

Exercise E.9 (Which enlargement does each expression force?). For each expression name the smallest class in which it can be represented, choosing among $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$, $\mathcal{P}\mathcal{L}_{\text{itLaur},\mathbb{K}}$, a Puiseux–log ring, a ring with a second logarithmic level, and a logarithmic-exponential transseries field; and justify both membership and minimality.

$$(L + t)^{-1}, \quad \sum_{n \geq 0} n! L^{-n}, \quad X^{1/2}, \quad \log(X \log X), \quad e^{-X}.$$

Solution. $(L + t)^{-1}$. Geometric expansion in t gives

$$\frac{1}{L + t} = \frac{1}{L} \cdot \frac{1}{1 + L^{-1}t} = \sum_{n \geq 0} (-1)^n L^{-n-1} t^n, \quad (545)$$

every coefficient of which lies in $\mathbb{K}[L, L^{-1}] \subseteq \mathbb{K}(L)$. So the element lies in $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$. It does not lie in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, because the reciprocal in $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ is unique and (545) has non-polynomial coefficients; equivalently, $L + t$ is not a unit of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ by Theorem 7.3, its leading block L not being a nonzero constant. Minimality among the classes of the volume: Theorem 7.20 says $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ is the smallest field $E((t))$ with $\mathbb{K}[L] \subseteq E \subseteq \mathbb{K}(L)$ containing $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, and this example is one of the witnesses that forces the enlargement.

$\sum_{n \geq 0} n! L^{-n}$. This is a legitimate element of $\mathbb{K}((L^{-1}))$, hence of $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ (as a t^0 -block). It is not in $\mathbb{K}(L)$: a rational function of L has an eventually linearly recurrent sequence of Laurent coefficients at $L = \infty$, and $(n!)_{n \geq 0}$ satisfies no linear recurrence with constant coefficients, since $n!/(n-1)! = n$ is unbounded. So this element separates $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ from $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$, which is Theorem 3.7.

$X^{1/2} = t^{-1/2}$. A Puiseux–log ring with exponent group $\frac{1}{2}\mathbb{Z}$ contains it; no class with exponent group $\mathbb{Z}$ does, since the outer exponents of every element of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$, $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$ are integers. Enlarging the coefficient field cannot help: the exponent group is not touched by that. See Theorem 3.25.

$\log(X \log X) = L + \log L$. The symbol $\log L$ is not in $\mathbb{K}((L^{-1}))$, let alone $\mathbb{K}(L)$: if it were, it would be a Laurent series in L^{-1} , so $\partial_L \log L = L^{-1}$ would lie in the image of ∂_L on $\mathbb{K}((L^{-1}))$, contradicting Theorem 8.24 (the residue at $L = \infty$ of L^{-1} is 1). A second logarithmic level is forced; this is item (3) of Theorem 10.1.

e^{-X} . By Theorem 8.16, $\exp F$ lies in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ only if $\nu_t F \geq 0$; here $F = -X$ has $\nu_t = -1$. In fact $e^{-X} \prec t^N$ for every N , so it is not comparable with any monomial of any power–log scale (Theorem 8.15); an exponential transmonomial, hence a logarithmic-exponential transseries field, is required. $\square$

Exercise E.10 (Admissible supports, and a support that is not one). Decide, with proof, which of the following subsets of $\mathbb{Z} \times \mathbb{N}_0$ are the coefficient support $\text{supp}_{\text{coeff}}(F)$ of some $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, using Theorem 3.15:

$$\begin{aligned} S_1 &= \{(n, n) : n \geq 0\}, & S_2 &= \{(n, j) : n \geq 0, 0 \leq j \leq n\}, \\ S_3 &= \{(-n, 0) : n \geq 0\}, & S_4 &= \{(n, 0) : n \geq 0\} \cup \{(0, j) : j \geq 0\}. \end{aligned}$$

For each admissible one, exhibit an F realising it.

Solution. Criterion (b) of Theorem 3.15: the set of outer exponents occurring must be bounded below, and each fibre $S_n = \{j : (n, j) \in S\}$ must be finite.

S_1 : outer exponents $\{n \geq 0\}$ bounded below by 0; fibre $S_n = \{n\}$ finite. Admissible; realised by $F = \sum_{n \geq 0} L^n t^n$.

S_2 : exponents bounded below by 0; fibre $S_n = \{0, \dots, n\}$ finite. Admissible; realised by $F = \sum_{n \geq 0} (\sum_{j=0}^n L^j) t^n$.

S_3 : exponents $\{-n : n \geq 0\}$ are not bounded below. Not admissible. This is the support the divergent family of Theorem E.4(1) would have produced.

S_4 : exponents bounded below by 0, but the fibre at $n = 0$ is $\mathbb{N}_0$, infinite: a block would have to be a power series in L , not a polynomial. Not admissible in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. It is the support of an element of the Hahn ambient of Theorem 3.19 only if that ambient still requires reverse well-ordering, which criterion (c) shows it does; and S_4 fails (c) as well, since $\{(0, j) : j \geq 0\}$ has no $\succ$ -greatest element – larger j at fixed n means larger monomial $t^n L^j$. So S_4 is not admissible in any of the four classes. $\square$

E.3 Arithmetic: product, reciprocal, derivative

Exercise E.11 (The block Cauchy product, and what it consumed). Let

$$A = 1 + (L + 1)t + (L^2 - 1)t^2, \quad B = 1 + (2 - L)t + Lt^2,$$

with all undisplayed blocks zero. Compute AB through outer order t^3 . Then, without recomputing, say which blocks of A and of B the block $[t^3](AB)$ actually depends on, and check your answer against [Theorem 6.12](#).

Solution. The blocks convolve:

$$\begin{aligned} [t^0](AB) &= 1, \\ [t^1](AB) &= (L + 1) + (2 - L) = 3, \\ [t^2](AB) &= (L^2 - 1) + (L + 1)(2 - L) + L = L^2 - 1 + (-L^2 + L + 2) + L = 2L + 1, \\ [t^3](AB) &= (L + 1) \cdot L + (L^2 - 1)(2 - L) = L^2 + L - L^3 + 2L^2 + L - 2 \\ &= -L^3 + 3L^2 + 2L - 2. \end{aligned}$$

Hence

$$AB = 1 + 3t + (2L + 1)t^2 + (-L^3 + 3L^2 + 2L - 2)t^3 + \mathcal{O}_t^{\text{form}}(t^4),$$

and in fact the product is exact through t^4 as well, both inputs being polynomials in t .

Dependency: since $\nu_t A = \nu_t B = 0$, [Theorem 6.12](#) says $[t^N](AB)$ depends exactly on $[t^n]A$ and $[t^m]B$ for $0 \leq n, m \leq N$. For $N = 3$ that is $A_0, \dots, A_3$ and $B_0, \dots, B_3$; as $A_3 = B_3 = 0$ by hypothesis, four blocks of each are consulted and only three of each are nonzero. Note that the hypothesis “undisplayed blocks vanish” is not decoration: without it A_3 and B_3 are unknown and $[t^3](AB)$ is undetermined. $\square$

Exercise E.12 (Reciprocal of a unit, and where the unit hypothesis enters). Find $C \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $(1 + (L - 1)t + Lt^2)C = 1$, through outer order t^4 . Then explain why C has polynomial blocks, and give the one-line reason that the same recurrence applied to $L + (L - 1)t$ does not.

Solution. Write $B = 1 + B_1t + B_2t^2$ with $B_1 = L - 1$, $B_2 = L$, and $C = \sum_{n \geq 0} C_n t^n$. By [Theorem 7.7](#) with $c = 1$, $C_0 = 1$ and $C_n = -\sum_{j=1}^n B_j C_{n-j}$. Hence

$$\begin{aligned} C_1 &= -B_1 = 1 - L, \\ C_2 &= -(B_1C_1 + B_2C_0) = (L - 1)^2 - L = L^2 - 3L + 1, \\ C_3 &= -(B_1C_2 + B_2C_1) = -(L - 1)(L^2 - 3L + 1) - L(1 - L) \\ &= -(L^3 - 4L^2 + 4L - 1) - (L - L^2) = -L^3 + 5L^2 - 5L + 1, \\ C_4 &= -(B_1C_3 + B_2C_2) \\ &= -(L - 1)(-L^3 + 5L^2 - 5L + 1) - L(L^2 - 3L + 1) \\ &= -(-L^4 + 6L^3 - 10L^2 + 6L - 1) - (L^3 - 3L^2 + L) \\ &= L^4 - 7L^3 + 13L^2 - 7L + 1. \end{aligned}$$

So

$$C = 1 + (1 - L)t + (L^2 - 3L + 1)t^2 + (-L^3 + 5L^2 - 5L + 1)t^3 + (L^4 - 7L^3 + 13L^2 - 7L + 1)t^4 + \mathcal{O}_t^{\text{form}}(t^5).$$

Every C_n is a polynomial because the recurrence divides only by $c = B_0 = 1$, a unit of $\mathbb{K}$; the whole content of [Theorem 7.3](#) is that this is the only case in which the division stays inside $\mathbb{K}[L]$. For $B = L + (L - 1)t$ the recurrence starts $C_0 = L^{-1}$, which already leaves $\mathbb{K}[L]$; B is not a unit of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ because its leading block L is not a nonzero constant. $\square$

Exercise E.13 (A quotient, and the exact guard precision it needs). Let

$$A = 1 + (2L - 1)t + (L^2 + 3)t^2 + (L^3 - L)t^3 + \mathcal{O}_t^{\text{form}}(t^4), \quad B = 1 + Lt + (L^2 - 1)t^2 + \mathcal{O}_t^{\text{form}}(t^3).$$

Compute $\text{Trunc}_{<4}(A/B)$, or explain why the data do not determine it, and in either case state the sharp input requirement given by [Theorem 7.11](#).

Solution. Here $a = \nu_t A = 0$ and $b = \nu_t B = 0$, so (103) reads: $\text{Trunc}_{<N}(A/B)$ is determined by $\text{Trunc}_{<N} A$ and $\text{Trunc}_{<N} B$. With $N = 4$ we need A and B each through order t^3 . A is given to that precision; B is given only modulo $\mathcal{O}_t^{\text{form}}(t^3)$, i.e. only $\text{Trunc}_{<3} B$. So the block $[t^3](A/B)$ is not determined: changing B_3 by δ changes the quotient's third block by $-\delta$.

What is determined is $\text{Trunc}_{<3}(A/B)$. Using (101), i.e. $Q_n = A_n - \sum_{j \geq 1} B_j Q_{n-j}$ since $B_0 = 1$:

$$Q_0 = 1,$$

$$Q_1 = A_1 - B_1 Q_0 = (2L - 1) - L = L - 1,$$

$$Q_2 = A_2 - B_1 Q_1 - B_2 Q_0 = (L^2 + 3) - L(L - 1) - (L^2 - 1) = -L^2 + L + 4.$$

If in addition $B_3 = 0$ is assumed – which is what the source that supplies this example intends – then

$$Q_3 = A_3 - B_1 Q_2 - B_2 Q_1 - B_3 Q_0 = (L^3 - L) - L(-L^2 + L + 4) - (L^2 - 1)(L - 1) = L^3 - 4L - 1,$$

giving

$$\frac{A}{B} = 1 + (L - 1)t + (-L^2 + L + 4)t^2 + (L^3 - 4L - 1)t^3 + \mathcal{O}_t^{\text{form}}(t^4).$$

A useful check, and the one the same source recommends, is to multiply the computed quotient by B and compare exclusive truncations at the requested cutoff; by [Theorem 7.9](#) the truncated quotient is the unique polynomial of t -degree < 4 passing that test. $\square$

Exercise E.14 (Integer and generalised binomial powers). Let $F = 1 + Lt \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.

1. Compute $[t^n]F^k$ for every integer $k \geq 0$.
2. Compute $[t^n]F^\kappa$ for arbitrary $\kappa \in \mathbb{K}$, where $F^\kappa := \exp(\kappa \log F)$ as in [Theorem 6.38](#), and compare with (1).
3. Compute d_{F^κ} and compare with the bound of [Theorem 6.31](#).
4. Expand $(1 + Lt + (L^2 - 1)t^2)^\kappa$ through t^2 .

Solution. (1) $F^k = (1 + Lt)^k = \sum_{n=0}^k \binom{k}{n} L^n t^n$, so $[t^n]F^k = \binom{k}{n} L^n$, zero for $n > k$.

(2) $\log F = \log(1 + Lt) = \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} L^m t^m$, and exponentiating gives the generalised binomial series $F^\kappa = \sum_{n \geq 0} \binom{\kappa}{n} L^n t^n$ with $\binom{\kappa}{n} = (\kappa)_n / n!$. So the coefficient formula is *the same*, $[t^n]F^\kappa = \binom{\kappa}{n} L^n$; the only difference is that for $\kappa = k \in \mathbb{N}_0$ the falling factorial $(k)_n$ vanishes for $n > k$ and the series terminates, while for $\kappa \notin \mathbb{N}_0$ every coefficient is nonzero and the series is genuinely infinite. Both are legitimate elements of $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$; the support condition of [Theorem 3.15](#) does not care.

(3) $d_{F^\kappa}(n) = n$ for all $n \geq 0$ when $\kappa \notin \mathbb{N}_0$, and for $0 \leq n \leq k$ when $\kappa = k$. The hypothesis of [Theorem 6.31](#) holds with $\alpha = 1, \beta = 0$, and its conclusion $d_{F^k}(N) \leq N$ is attained: the bound is sharp here.

(4) Put $U = Lt + (L^2 - 1)t^2$, so $U^2 = L^2 t^2 + \mathcal{O}_t^{\text{form}}(t^3)$ and

$$(1 + U)^\kappa = 1 + \kappa U + \binom{\kappa}{2} U^2 + \mathcal{O}_t^{\text{form}}(t^3) = 1 + \kappa Lt + \left(\kappa(L^2 - 1) + \frac{\kappa(\kappa - 1)}{2} L^2 \right) t^2 + \mathcal{O}_t^{\text{form}}(t^3),$$

that is, the t^2 -block is $\frac{1}{2}\kappa(\kappa + 1)L^2 - 2$. $\square$

Exercise E.15 (Counterexample: a nonzero leading block is not enough). A plausible strengthening of [Theorem 7.3](#) is: “ F is a unit of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ if and only if its leading block is nonzero”. Refute it, locate $(L+t)^{-1}$ exactly, and deduce that $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is an integral domain that is not a field — so that the four classes of [Theorem 3.3](#) really are four different objects.

Solution. Every nonzero $F \in \mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ has, by definition, a nonzero leading block; if the proposed criterion were correct then every nonzero element would be a unit and $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ would be a field. It is not. Take $F = L + t$. Its leading block is $L \neq 0$. If F were a unit of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, with reciprocal $C = \sum_{n \geq 0} C_n t^n \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, then comparing t^0 -blocks in $FC = 1$ gives $LC_0 = 1$, impossible in $\mathbb{K}[L]$ since $\deg_L(LC_0) \geq 1$ for $C_0 \neq 0$. The reciprocal does exist one class up, in $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$, and is given by (545).

That $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ is a domain is [Theorem 6.22](#): outer orders add and leading blocks multiply in $\mathbb{K}[L]$, which is a domain, so a product of nonzero elements is nonzero. A domain that is not a field is exactly what the unit criterion describes: the units are the elements $ct^n(1+U)$ with $c \in \mathbb{K}^\times$, $n \in \mathbb{Z}$, $U \in t\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, a proper subgroup of the nonzero elements. The strengthening fails because the condition is not that the leading block be nonzero but that it be a nonzero *constant*. $\square$

Exercise E.16 (Trap: a division that leaves the ring). A student computes $(L+t)^{-1}$ inside $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ as follows. “By [Theorem 7.7](#) the reciprocal of $B = B_0 + B_1 t + \dots$ has $C_0 = B_0^{-1}$ and $C_n = -B_0^{-1} \sum_{j=1}^n B_j C_{n-j}$. Here $B_0 = L$, $B_1 = 1$, so $C_0 = L^{-1}$, $C_1 = -L^{-2}$, $C_2 = L^{-3}$, and in general $C_n = (-1)^n L^{-n-1}$. Since every C_n is a Laurent polynomial in L and Laurent polynomials are polynomials in L and L^{-1} , the answer $\sum_{n \geq 0} (-1)^n L^{-n-1} t^n$ lies in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$.” Find every error, and state precisely what is true.

Solution. There are two errors, one of hypothesis and one of vocabulary.

Hypothesis. [Theorem 7.7](#) requires $B_0 = c \in \mathbb{K}^\times$; the whole proof consists in dividing by c , and c^{-1} has to exist in the coefficient ring. Here $B_0 = L$ is not invertible in $\mathbb{K}[L]$, so the theorem does not apply, and the recurrence as written is being run over the larger ring $\mathbb{K}(L)$. That is legitimate — it computes the reciprocal in $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ — but it is no longer the theorem quoted.

Vocabulary. The blocks of an element of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ are *polynomials* in L , not Laurent polynomials. $\mathbb{K}[L]$ is the coefficient ring; $\mathbb{K}[L, L^{-1}]$ is a strictly larger ring, and $L^{-1} \notin \mathbb{K}[L]$. So the concluding sentence is false at its last step.

What is true: the computed series is correct, and it is the reciprocal of $L+t$ in $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$; all its blocks in fact lie in the localisation $\mathbb{K}[L, L^{-1}]$, so it lies in the intermediate ring $\mathbb{K}[L, L^{-1}]((t)) \subsetneq \mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$. It does not lie in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, by [Theorem E.15](#). The general statement replacing the misquoted theorem is [Theorem 7.24](#): with $\nu_t B = 0$ and $A \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, the quotient A/B lies in $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ if and only if every block of the $\mathbb{K}(L)$ -recurrence happens to be a polynomial, which happens if and only if $A \in B\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$. $\square$

Exercise E.17 (Exact divisibility conditions, block by block). Let $B = (L+1) + t$ and let $A = A_0(L) + A_1(L)t + \mathcal{O}_t^{\text{form}}(t^2)$ with $A_0, A_1 \in \mathbb{K}[L]$. Find necessary and sufficient conditions on A_0, A_1 for $\text{Trunc}_{<2}(A/B)$ to have polynomial blocks, and express them as two evaluations at a single point of $\mathbb{K}$.

Solution. Work over $E = \mathbb{K}(L)$, where B is a unit, and let $Q = A/B$ have blocks Q_n . From $BQ = A$, comparing blocks,

$$(L+1)Q_0 = A_0, \quad (L+1)Q_1 + Q_0 = A_1.$$

Thus $Q_0 \in \mathbb{K}[L]$ if and only if $(L+1) \mid A_0$; and given that, $Q_1 \in \mathbb{K}[L]$ if and only if $(L+1) \mid (A_1 - Q_0)$. This is [Theorem 7.24\(2\)](#) for the present B .

To turn the two conditions into evaluations, note that for $P \in \mathbb{K}[L]$ one has $(L+1) \mid P$ if and only if $P(-1) = 0$. The first condition is therefore $A_0(-1) = 0$. Writing $A_0 = (L+1)Q_0$ and differentiating, $A'_0 = Q_0 + (L+1)Q'_0$, so $Q_0(-1) = A'_0(-1)$. The second condition, $A_1(-1) = Q_0(-1)$, becomes

$A_0(-1) = 0 \quad \text{and} \quad A_1(-1) = A'_0(-1).$
--

Both are conditions at the single point $L = -1$, the root of the leading block. Two remarks. First, the conditions are not vacuous: for $A = 1$ the first already fails, which is another proof that B is not a unit. Second, the pattern continues: $Q_n \in \mathbb{K}[L]$ requires $(L + 1) \mid (A_n - Q_{n-1})$, so the whole quotient lies in $\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ exactly when $A \in B\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, which is [Theorem 7.24 \(3\)](#). $\square$

Exercise E.18 (Repeated derivatives of one block, three ways). 1. Differentiate $A = X^2 + L^3 + (L^2 - 1)X^{-2}$ once, using the block formula of [Theorem 9.3](#).
 2. Compute $D_X^k(L^3)$ for $k = 1, 2, 3, 4$ using [Theorem 9.7](#).
 3. Recompute $D_X^4(L^3)$ from the Stirling expansion $\Delta_{0,k} = \sum_{j=0}^k s(k, j) \partial_L^j$ of [Theorem 9.8](#), and check the two answers agree.

Solution. (1) [Theorem 9.3](#) is $D_X(t^n p(L)) = t^{n+1}(p' - np)$. Apply it to the three blocks, with $n = -2, 0, 2$:

$$\begin{aligned} D_X(t^{-2}) &= t^{-1}(0 + 2) = 2X, \\ D_X(L^3) &= t^1(3L^2 - 0) = 3L^2t, \\ D_X(t^2(L^2 - 1)) &= t^3(2L - 2(L^2 - 1)) = (-2L^2 + 2L + 2)t^3. \end{aligned}$$

Hence $D_X A = 2X + 3L^2 X^{-1} + (-2L^2 + 2L + 2)X^{-3}$.

(2) For a t^0 -block, $D_X^k p = t^k \Delta_{0,k} p$ with $\Delta_{0,k} = (\partial_L - k + 1) \cdots (\partial_L - 1) \partial_L$. With $p = L^3$:

$$\begin{aligned} \Delta_{0,1} L^3 &= 3L^2, \\ \Delta_{0,2} L^3 &= (\partial_L - 1)(3L^2) = 6L - 3L^2, \\ \Delta_{0,3} L^3 &= (\partial_L - 2)(6L - 3L^2) = 6 - 6L - 12L + 6L^2 = 6L^2 - 18L + 6, \\ \Delta_{0,4} L^3 &= (\partial_L - 3)(6L^2 - 18L + 6) = 12L - 18 - 18L^2 + 54L - 18 = -18L^2 + 66L - 36. \end{aligned}$$

So $D_X L^3 = 3L^2 t$, $D_X^2 L^3 = (6L - 3L^2)t^2$, $D_X^3 L^3 = (6L^2 - 18L + 6)t^3$, $D_X^4 L^3 = (-18L^2 + 66L - 36)t^4$.

(3) The signed Stirling numbers of the first kind in row 4 are $s(4, 0) = 0$, $s(4, 1) = -6$, $s(4, 2) = 11$, $s(4, 3) = -6$, $s(4, 4) = 1$. Applying $\sum_j s(4, j) \partial_L^j$ to L^3 , whose derivatives are $3L^2, 6L, 6, 0$,

$$-6 \cdot 3L^2 + 11 \cdot 6L - 6 \cdot 6 + 1 \cdot 0 = -18L^2 + 66L - 36,$$

which agrees with (2). Equivalently $X^4 \frac{d^4}{dX^4} (\log X)^3 = \sum_{j=0}^4 s(4, j) ((\log X)^3)^{(j)}$; note that the sum must start at $j = 0$, because $s(k, 0) = 0$ only for $k \geq 1$ and equals 1 at $k = 0$. $\square$

Exercise E.19 (Antiderivatives of a single block, resonant and not). Find polynomial-logarithmic antiderivatives of

$$F_1 = t^2(L + 1) \quad \text{and} \quad F_2 = t(L + 1),$$

and verify each by differentiating. Explain why the two computations look different, and identify the block index at which the difference occurs.

Solution. By [Theorem 9.3](#), $D_X(t^n Q) = t^{n+1}(Q' - nQ)$, so an antiderivative of $t^{n+1}P$ is $t^n Q$ with $Q' - nQ = P$.

$F_1 = t^2(L + 1)$, so $n = 1$. Solve $Q' - Q = L + 1$ in $\mathbb{K}[L]$. With $Q = \alpha L + \beta$: $\alpha - \alpha L - \beta = L + 1$, so $\alpha = -1$ and $\alpha - \beta = 1$, i.e. $\beta = -2$. Thus $Q = -L - 2$ and

$$\int F_1 dX = -t(L + 2) + c, \quad c \in \mathbb{K}.$$

Check: $D_X(-t(L + 2)) = t^2(-1 - (-(L + 2))) = t^2(L + 1)$.

$F_2 = t(L+1)$, so $n = 0$. Now the equation is $Q' = L+1$, with solution $Q = \frac{1}{2}L^2 + L$ up to a constant, and

$$\int F_2 \, dX = \frac{1}{2}L^2 + L + c.$$

Check: $D_X(\frac{1}{2}L^2 + L) = t(L+1-0) = t(L+1)$.

The difference is the operator being inverted. For $n \neq 0$ the map $Q \mapsto Q' - nQ$ is bijective on $\mathbb{K}[L]$: it is triangular with diagonal $-n \neq 0$ in the monomial basis, and the explicit inverse is the terminating sum $Q = -\sum_{j \geq 0} P^{(j)}/n^{j+1}$ of (156). This is the nonresonant case of Theorem 9.14, and it lowers the L -degree, or at worst preserves it. For $n = 0$ the map is $Q \mapsto Q'$, which is surjective on $\mathbb{K}[L]$ but has kernel $\mathbb{K}$ and raises the degree by one; this is the resonant block of Theorem 9.15, and it is the only place where the log-degree profile of an antiderivative exceeds that of its derivative, as recorded in (157). The resonant index is the one whose outer exponent in F is 1, i.e. the block t^1 of the integrand. $\square$

Exercise E.20 (Counterexample: surjectivity of D_X is not inherited). By Theorem 9.17 the derivation D_X maps $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ onto $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, with kernel $\mathbb{K}$. A plausible strengthening is that it maps each of the four classes onto itself. Refute this. Show that the element

$$F = \frac{t}{L} = \frac{1}{X \log X},$$

which lies in $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ and in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$, has no antiderivative in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$, and identify the antiderivative it does have.

Solution. Suppose $G = \sum_n g_n(L)t^n \in \mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$, so each $g_n \in \mathbb{K}((L^{-1}))$, satisfies $D_X G = t/L$. By Theorem 9.3, extended over the coefficient ring $\mathbb{K}((L^{-1}))$, the block of $D_X G$ at t^{n+1} is $g'_n - ng_n$, where $' = \partial_L$. Comparing at t^1 , i.e. $n = 0$, gives

$$g'_0 = \frac{1}{L}.$$

Now ∂_L maps $\mathbb{K}((L^{-1}))$ into itself, and for $g_0 = \sum_{k \geq k_0} c_k L^{-k}$ one has $g'_0 = \sum_{k \geq k_0} (-k)c_k L^{-k-1}$. The coefficient of L^{-1} in g'_0 is the one with $-k-1 = -1$, i.e. $k = 0$, and it equals $-0 \cdot c_0 = 0$. So L^{-1} is never in the image: the residue at $L = \infty$, $\text{res}(g) := [L^{-1}]g$, vanishes identically on derivatives (Theorem 8.23), and $\text{res}(t/L) = 1 \neq 0$. Hence no such g_0 exists, and F has no antiderivative in $\mathcal{P}\mathcal{L}_{\text{itLaur}, \mathbb{K}}$; a fortiori none in $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ (Theorem 8.24).

The obstruction is exactly one function, the residue, and it is met by adjoining one new symbol: $\Lambda = \log \log X$ satisfies $D_X \Lambda = t/L$, because $\frac{d}{dX} \log \log X = \frac{1}{X \log X}$. This is Theorem 9.21, and it is where the second logarithmic level of the whole theory comes from: not from composition, but from integration. Note that no contradiction with Theorem 9.17 arises, since $t/L \notin \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. $\square$

Exercise E.21 (Which formal exponentials stay in the class?). Decide, for each F below, whether $\exp F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$, and if not, name the smallest enlargement in which it lives:

$$L, \quad 2L+t, \quad t, \quad \frac{1}{2}L, \quad L^2, \quad t^{-1}.$$

State the criterion you used and check it in each case.

Solution. The criterion is Theorem 8.16: $\exp F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ if and only if $\nu_t F \geq 0$ and $[t^0]F = a + mL$ with $m \in \mathbb{Z}$, $a \in \mathbb{K}$ and $\exp a \in \mathbb{K}$; and then $\exp F = (\exp a)t^{-m} \exp(F - [t^0]F)$ with the last factor in $1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.

$F = L$: $\nu_t = 0$, $[t^0]F = 0 + 1 \cdot L$, $m = 1$, $a = 0$. Yes: $\exp L = t^{-1} = X$.

$F = 2L + t$: $\nu_t = 0$, $[t^0]F = 2L$, $m = 2$, $a = 0$. Yes: $\exp F = t^{-2} \exp(t) = X^2 \sum_{k \geq 0} t^k/k! \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$.

$F = t$: $\nu_t = 1 \geq 0$, $[t^0]F = 0$, $m = 0$, $a = 0$. Yes, and $\exp t \in 1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$.

$F = \frac{1}{2}L$: $m = \frac{1}{2} \notin \mathbb{Z}$. No; $\exp F = X^{1/2}$ needs the Puiseux–log ring with exponent group $\frac{1}{2}\mathbb{Z}$ ([Theorem 8.17](#)).

$F = L^2$: $[t^0]F = L^2$ is not affine in L . No. Here the failure is not repaired by any coefficient-field or exponent-group enlargement of a power–log scale: $\exp(L^2) = X^{\log X}$ grows faster than every power of X and slower than $\exp(cX)$, so it is a new transmonomial ([Theorem 8.15](#)).

$F = t^{-1} = X$: $\nu_t = -1 < 0$. No; e^X is an exponential transmonomial, and the smallest home is a logarithmic-exponential transseries field.

A remark worth extracting: the three failures are of three different kinds – a fractional slope, a nonaffine t^0 -block, a negative outer order – and [Theorem 8.15](#) says these are the only three. $\square$

E.4 Composition

Exercise E.22 (Substituting a constant shift into one monomial). Let $a \in \mathbb{K}$ and $G = X + a$. Compute $(tL) \circ G$ through outer order t^3 , and then $(tL) \circ (cX)$ exactly, for $c \in \mathbb{K}^\times$ with $\log c \in \mathbb{K}$.

Solution. By [Theorem 10.22](#), with $H = a$,

$$t \circ G = \frac{t}{1+at} = t - at^2 + a^2t^3 + \mathcal{O}_t^{\text{form}}(t^4), \quad L \circ G = L + \log(1+at) = L + at - \frac{a^2}{2}t^2 + \mathcal{O}_t^{\text{form}}(t^3).$$

Composition is a ring homomorphism ([Theorem 10.7](#)), so $(tL) \circ G = (t \circ G)(L \circ G)$, and multiplying,

$$\begin{aligned} (tL) \circ (X + a) &= (t - at^2 + a^2t^3) \left(L + at - \frac{a^2}{2}t^2 \right) + \mathcal{O}_t^{\text{form}}(t^4) \\ &= tL + a(1 - L)t^2 + a^2 \left(L - \frac{3}{2} \right) t^3 + \mathcal{O}_t^{\text{form}}(t^4). \end{aligned}$$

For the dilation $G = cX$ the substitution is exact and finite: $t \circ G = c^{-1}t$ and $L \circ G = L + \log c$, so

$$(tL) \circ (cX) = c^{-1}t(L + \log c),$$

and more generally $F \circ (cX) = \sum_n c^{-n} p_n(L + \log c) t^n$ for $F = \sum_n p_n(L) t^n$. Two things are visible here that matter later. First, a dilation changes *both* generators, the power by c^{-n} and the logarithm by the shift $\log c$; forgetting the second is the failure mode of [Theorem E.41](#). Second, the identity is a statement about $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ only if $\log c \in \mathbb{K}$; over $\mathbb{K} = \mathbb{Q}$ and $c = 2$ the right-hand side does not live in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ at all, and the field must be enlarged first. $\square$

Exercise E.23 (A composition through two blocks). Let $F(Y) = Y + (\log Y)^2$ and $G(X) = X - L$. Compute $F \circ G$ through outer order t^2 .

Solution. Write $F = t^{-1} + L^2$ in its own generators and $G = X + H$ with $H = -L$, so $H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and $G \in \mathcal{G}_{\text{poly}, \mathbb{K}}$. By [Theorem 10.22](#),

$$L \circ G = L + \log(1 - Lt) = L - Lt - \frac{1}{2}L^2t^2 + \mathcal{O}_t^{\text{form}}(t^3).$$

Therefore

$$\begin{aligned} (L \circ G)^2 &= \left(L - Lt - \frac{1}{2}L^2t^2 \right)^2 + \mathcal{O}_t^{\text{form}}(t^3) \\ &= L^2 - 2L^2t + (L^2 - L^3)t^2 + \mathcal{O}_t^{\text{form}}(t^3), \end{aligned}$$

the t^2 -block being $2 \cdot L \cdot (-\frac{1}{2}L^2) + (-L)^2$. Adding $G = X - L$,

$$F \circ G = X + (L^2 - L) - \frac{2L^2}{X} + \frac{L^2 - L^3}{X^2} + \mathcal{O}_t^{\text{form}}(t^3).$$

A check on the shape: $\nu_t(F \circ G) = \nu_t F = -1$, as [Theorem 10.22](#) requires, and no block acquired a degree in L larger than $2+$ (the number of substituted logarithms), consistent with [Theorem 10.26](#). $\square$

Exercise E.24 (Composition is multiplicative, from the generators). Let $G = X + H$ with $H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, so $G \in \mathcal{G}_{\text{poly}, \mathbb{K}}$. Prove that

$$(F\Phi) \circ G = (F \circ G)(\Phi \circ G) \quad \text{for all } F, \Phi \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}},$$

directly from the exact substitution formula (181), without invoking Theorem 10.7.

Solution. Let $F = \sum_{n \geq n_0} p_n(L)t^n$ and $\Phi = \sum_{m \geq m_0} q_m(L)t^m$, and let $G = X + H \in \mathcal{G}_{\text{poly}, \mathbb{K}}$. Write

$$\tau := t \circ G = \frac{t}{1 + tH}, \quad \lambda := L \circ G = L + \log(1 + tH),$$

so that (181) reads $F \circ G = \sum_n \tau^n p_n(\lambda)$. Two facts are needed.

(a) *The families are summable.* Since $\nu_t(\tau) = 1$ and $\nu_t(\lambda - L) \geq 1$, the summand $\tau^n p_n(\lambda)$ has outer order exactly n (Theorem 10.22), so $\{n : \nu_t < N\}$ is finite for every N and Theorem 5.21 is satisfied. The same holds for Φ and for the Cauchy product.

(b) *Multiplication distributes over summable families.* This is Theorem 5.22 (iv): if (S_i) and (T_j) are summable then so is $(S_i T_j)_{(i,j)}$ and $(\sum_i S_i)(\sum_j T_j) = \sum_{i,j} S_i T_j$.

Now compute. Let $r_N = \sum_{n+m=N} p_n q_m$ be the blocks of $F\Phi$. By (b),

$$(F \circ G)(\Phi \circ G) = \left(\sum_n \tau^n p_n(\lambda) \right) \left(\sum_m \tau^m q_m(\lambda) \right) = \sum_{n,m} \tau^{n+m} p_n(\lambda) q_m(\lambda).$$

Group by $N = n + m$; the inner sums are finite because for fixed N only finitely many pairs (n, m) with $n \geq n_0, m \geq m_0$ satisfy $n + m = N$. Since evaluation of a polynomial at the fixed element $\lambda \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ is a ring homomorphism $\mathbb{K}[L] \rightarrow \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, $\sum_{n+m=N} p_n(\lambda) q_m(\lambda) = r_N(\lambda)$. Hence

$$(F \circ G)(\Phi \circ G) = \sum_N \tau^N r_N(\lambda) = (F\Phi) \circ G,$$

the last equality being (181) applied to $F\Phi$. Additivity is the same computation with one index, and $1 \circ G = 1$; so $\Phi_G: F \mapsto F \circ G$ is a $\mathbb{K}$ -algebra endomorphism of $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. $\square$

Exercise E.25 (The sharp Taylor budget, in the exclusive convention). Let $F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$ with $\nu_t F \geq a$, let $H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $\nu_t H \geq h \geq 0$, and let $N \in \mathbb{Z}$.

1. Prove that $\text{Trunc}_{<N}(F \circ (X + H))$ is unchanged if the Taylor family $\sum_{k \geq 0} H^k D_X^k F / k!$ is truncated after the index $k = \lfloor (N - 1 - a) / (h + 1) \rfloor$.
2. Show the bound cannot be lowered, by exhibiting F, H, N for which the last retained term does change the answer.
3. A source states the same budget as $k \leq \lfloor (N - a) / (h + 1) \rfloor$. Explain the discrepancy and say whether it is harmful.

Solution. (1) By Theorem 9.9, $\nu_t(D_X^k F) \geq a + k$, and $\nu_t(H^k) \geq kh$. So the k -th Taylor term has

$$\nu_t \left(\frac{H^k}{k!} D_X^k F \right) \geq a + k + kh = a + k(h + 1).$$

Such a term is invisible to $\text{Trunc}_{<N} \cdot$ as soon as $a + k(h + 1) \geq N$, i.e. $k \geq (N - a) / (h + 1)$. The retained indices are therefore those with $a + k(h + 1) < N$, that is $k(h + 1) < N - a$, that is $k \leq \lfloor (N - 1 - a) / (h + 1) \rfloor$ for integers. Equivalently the number of retained terms is $K = \lceil (N - a) / (h + 1) \rceil$, which is Theorem 25.31.

(2) Take $F = L$ (so $a = 0$), $H = 1$ (so $h = 0$) and $N = 3$. The budget retains $k \leq 2$. Now $D_X L = t$, $D_X^2 L = D_X(t) = t^2(0 - 1) = -t^2$, $D_X^3 L = D_X(-t^2) = -t^3(0 - 2) = 2t^3$, so the Taylor sum is

$$L + t - \frac{1}{2}t^2 + \frac{1}{3}t^3 - \dots,$$

which is exactly $L \circ (X + 1) = L + \log(1 + t)$, as it must be. The term $k = 2$ contributes $-\frac{1}{2}t^2$, a block strictly below the cutoff $N = 3$; dropping it changes $\text{Trunc}_{<3}$. So the last retained index is genuinely needed, and the bound is sharp.

(3) The source is working with the inclusive cutoff $[F]_{\leq N}$. Since $[F]_{\leq N} = \text{Trunc}_{<N+1} F$, its N is our $N - 1$, and its budget $k \leq \lfloor (N - a)/(h + 1) \rfloor$ becomes, in our normalisation, exactly $k \leq \lfloor (N - 1 - a)/(h + 1) \rfloor$: the two agree. Read as an exclusive bound, however, the source’s formula retains one term too many whenever $(h + 1) \mid (N - a)$. That is harmless for *correctness* – the extra term is invisible below t^N – but not for cost, and not for a routine that uses the budget in reverse to decide how much of H to compute. This is the discrepancy recorded at [Theorem 25.33](#). $\square$

Exercise E.26 (Trap: how much of the inner argument is needed?). A routine must return $\text{Trunc}_{<1}(F \circ (X + H))$ for $F = X^2$. It argues: “the output is requested through outer order t^0 ; composition is filtration-continuous ([Theorem 10.24](#)), so H is needed only through outer order t^0 , i.e. only $\text{Trunc}_{<1} H$.” Compute $\text{Trunc}_{<1}(F \circ (X + H))$ for $H = h_0 + h_1 t + h_2 t^2 + \dots$, find the error, and state the correct input requirement for a general F .

Solution. Here $F = t^{-2}$, and composition with a near-identity map is exact:

$$F \circ (X + H) = (X + H)^2 = t^{-2} + 2t^{-1}H + H^2,$$

so, block by block,

$$[t^{-2}] = 1, \quad [t^{-1}] = 2h_0, \quad [t^0] = 2h_1 + h_0^2,$$

and $\text{Trunc}_{<1}(F \circ (X + H)) = X^2 + 2h_0X + (2h_1 + h_0^2)$. The block h_1 enters. The routine’s answer would omit it, and would be wrong by $2h_1$.

The error is a confusion of absolute with relative precision. Filtration continuity says $\nu_t(F \circ (X + B) - F \circ (X + C)) \geq \nu_t(F) + \nu_t(B - C) + 1$ ([Theorem 12.6](#)), and the summand $\nu_t(F)$ is negative here. Unwinding: a perturbation of H at outer order j moves the composite at outer order $a + j + 1$ with $a = \nu_t F$. It is therefore invisible below t^N exactly when $a + j + 1 \geq N$, so the required input is

$$\text{Trunc}_{<N-a-1} H.$$

For $F = X^2$, $a = -2$ and $N = 1$: $\text{Trunc}_{<2} H$, one block more than the routine used. Only when $\nu_t F \geq -1$ does $\text{Trunc}_{<N} H$ suffice. This is the sharp statement of [Theorem 10.10](#), and the general principle behind it is [Theorem 25.11](#): what a routine conserves is relative, not absolute, precision. $\square$

Exercise E.27 (Counterexample: the near-identity group is not abelian). Let $f, g \in \mathbb{K}[L]$, $F = X + f(L)$, $G = X + g(L)$.

1. Compute the first block of $F \circ G - G \circ F$.
2. Deduce that $\mathcal{G}_{\text{poly}, \mathbb{K}}$ is not abelian, and characterise the pairs (f, g) for which F and G commute to first order.
3. Show that vanishing of the first block does *not* imply commuting, by computing the next block for $f = L$, $g = 2L$.

Solution. (1) By [Theorem 10.23](#), $F \circ G = X + g + f \circ (X + g)$, and by [Theorem 12.6](#), (219), $f \circ (X + g) - f \equiv g D_X f \pmod{t^2 \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}}$, where $D_X f = f'(L)t$. Hence

$$F \circ G = X + f + g + g f' t + \mathcal{O}_t^{\text{form}}(t^2),$$

and by symmetry $G \circ F = X + f + g + f g' t + \mathcal{O}_t^{\text{form}}(t^2)$. Subtracting,

$$F \circ G - G \circ F = (g f' - f g') t + \mathcal{O}_t^{\text{form}}(t^2), \tag{546}$$

which is [Theorem 11.18](#).

(2) Take $f = L, g = L^2$: then $gf' - fg' = L^2 - 2L^2 = -L^2 \neq 0$, so $F \circ G \neq G \circ F$ and $\mathcal{G}_{\text{poly}, \mathbb{K}}$ is not abelian, although it is associative ([Theorem 10.27](#)). The first block vanishes if and only if $gf' = fg'$, i.e. $(f/g)' = 0$ in $\mathbb{K}(L)$ for $g \neq 0$, i.e. f and g are proportional over $\mathbb{K}$. Equivalently, the vector fields $f\partial_L$ and $g\partial_L$ commute.

(3) Take the proportional pair $f = L, g = 2L$, for which $gf' - fg' = 2L - 2L = 0$, so the first block of the commutator vanishes. The maps still do not commute. Indeed by [Theorem 10.22](#),

$$\begin{aligned}(X + L) \circ (X + 2L) &= X + 3L + 2Lt - 2L^2t^2 + \mathcal{O}_t^{\text{form}}(t^3), \\ (X + 2L) \circ (X + L) &= X + 3L + 2Lt - L^2t^2 + \mathcal{O}_t^{\text{form}}(t^3),\end{aligned}$$

the two t^2 -blocks coming from $\log(1 + 2Lt)$ and from $2\log(1 + Lt)$ respectively. Hence

$$F \circ G - G \circ F = -L^2t^2 + 2L^3t^3 - \frac{7}{2}L^4t^4 + \mathcal{O}_t^{\text{form}}(t^5).$$

So proportionality kills the first block only. Structurally this is exactly what [Theorem 12.16](#) predicts: the graded Lie algebra of the contact filtration is not abelian, and vanishing of the bracket of the leading symbols only pushes the commutator one step down the filtration ([Theorem 12.14](#)). $\square$

Exercise E.28 (A scale change that produces a second logarithm exactly). Let $F(X) = X + \log X + X^{-1}$ and $G(X) = 3X^2L^{-1}$. Compute $F \circ G$ exactly, and say which class each summand lives in. Then do the same for $F(Y) = \log Y, G(X) = X^2(\log X)^3$, and identify the smallest new symbol forced in each case.

Solution. First pair. $G = 3t^{-2}L^{-1} \in \mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$ is monomial-leading with $\rho = 2, c = 3, \sigma = -1$ and no correction. Composition acts on the generators by $t \mapsto G^{-1}$ and $L \mapsto \log G$, and here both are exact and finite:

$$G^{-1} = \frac{1}{3}Lt^2, \quad \log G = \log 3 + 2L - \log L.$$

Hence

$$F \circ G = 3X^2L^{-1} + 2L - \log L + \log 3 + \frac{1}{3}Lt^2,$$

a *finite* expression. The summands $3X^2L^{-1}$ and $\frac{1}{3}Lt^2$ lie in $\mathcal{P}\mathcal{L}_{\text{rat}, \mathbb{K}}$; $2L$ lies in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$; $\log 3$ requires $\log 3 \in \mathbb{K}$; and $\log L = \log \log X$ lies in no Puiseux–log ring over $\mathbb{K}[L]$, by the residue argument of [Theorem E.20](#). The point the source makes is worth keeping: the second logarithmic level appears here *exactly*, in a closed-form finite composite, not merely inside a remainder.

Second pair. $\log G = 2L + 3\log L$, so again the new symbol is $L_2 = \log L = \log \log X$, and the output lies in a two-level coefficient class, not in $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. This is item (3) of [Theorem 10.1](#) and [Theorem 13.8](#).

In both cases the forced symbol is read off from the images of the generators alone: G^{-1} decides whether the exponent group must grow, $\log G$ decides whether the logarithmic depth must grow. That is the whole content of the admissibility checklist of [Section 13](#). $\square$

Exercise E.29 (Counterexample: a monomial-leading inner argument is not enough). [Theorem 10.7](#) composes F with an inner argument $G = cX^\rho(1 + U)$ subject to $\rho > 0$. Show that the hypothesis $\rho > 0$ cannot be weakened to $\rho \geq 0$: prove that if $\nu_t(t \circ G) = 0$ then $F \circ G$ is undefined for *every* F whose coefficient support is infinite, whatever the coefficient field, and identify the exact failure.

Solution. Suppose G is such that $\tau := t \circ G$ has $\nu_t \tau = 0$; the simplest instance is $G = c \in \mathbb{K}^\times$, where $\tau = c^{-1}$. Let $F = \sum_{n \geq n_0} p_n(L)t^n$ have infinitely many nonzero blocks. The defining formula for the composite is the sum of the family $(\tau^n p_n(\lambda))_n$, $\lambda = L \circ G$. Since $\nu_t(\tau^n) = n\nu_t(\tau) = 0$ for every n , every member of the family has outer order $\geq \nu_t(p_n(\lambda))$, and for the constant inner argument $G = c$ — where $\lambda = \log c \in \mathbb{K}$ and $p_n(\lambda) \in \mathbb{K}$ — every member sits in the single block t^0 . The set $\{n : \nu_t < 1\}$ is then infinite, so [Theorem 5.21](#) fails; concretely the t^0 -block of the putative composite would be

$$\sum_{n \geq n_0} c^{-n} p_n(\log c),$$

an infinite sum of scalars, which is not an element of $\mathbb{K}$ unless it happens to terminate. No enlargement of the coefficient field repairs this: the obstruction is that infinitely many summands land in one block, which is a statement about the outer filtration alone. The same argument applies to any G with $\rho = 0$, since then $\nu_t(t \circ G) = 0$.

For contrast, when $\rho > 0$ one has $\nu_t(\tau) = \rho > 0$ and $\nu_t(\tau^n) = n\rho \rightarrow +\infty$, so the family is summable — in a Hahn ambient with exponent group containing $\rho\mathbb{Z}$, which is [Theorem 10.17](#). This is why $\rho > 0$ is a hypothesis and not a normalisation. $\square$

E.5 Reversion

Exercise E.30 (The first-order form of the inverse, and its sharp error). Let $H \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ with $q = \nu_t(H)$. Show

$$(X + H)^{\langle -1 \rangle \circ} = X - H + \mathcal{O}_t^{\text{form}}(t^{2q+1}),$$

and show that the exponent $2q + 1$ is attained for $H = L$.

Solution. By [Theorem 14.11](#), $(X + H)^{\langle -1 \rangle \circ} = X + B$ where B is the t -adic limit of the Picard iterates $B_0 = 0$, $B_{j+1} = -H \circ (X + B_j)$, and $\nu_t(B - B_j) \geq (j + 1)q + j$. Taking $j = 1$, so that $B_1 = -H \circ X = -H$,

$$\nu_t(B + H) \geq 2q + 1,$$

which is the claim.

Sharpness at $H = L$: here $q = 0$ and the claim is $(X + L)^{\langle -1 \rangle \circ} = X - L + \mathcal{O}_t^{\text{form}}(t)$. By [Theorem 18.10](#) the inverse is $X - L + \mathbf{L}_1^{\text{am}}(L)t + \cdots = X - L + Lt + \cdots$, so $B + L = Lt + \cdots$ has outer order exactly $1 = 2 \cdot 0 + 1$.

Two remarks. First, the estimate is a statement about ν_t only: the correction $-H$ is exact in the leading block by [Theorem 14.11](#) (4), and everything beyond it is error. Second, the same proof gives the general Picard rate: j iterations from $B_0 = 0$ are correct through outer block $(j + 1)q + j - 1$, so each further iteration gains $q + 1$ blocks. $\square$

Exercise E.31 (The universal inverse blocks through order two). Let $a, b, c \in \mathbb{K}[L]$ and $F = X + a(L) + b(L)t + c(L)t^2 + \mathcal{O}_t^{\text{form}}(t^3)$, so $F \in \mathcal{G}_{\text{poly}, \mathbb{K}}$. Write $F^{\langle -1 \rangle \circ} = X + b_0 + b_1t + b_2t^2 + \mathcal{O}_t^{\text{form}}(t^3)$. Derive b_0, b_1, b_2 as universal differential polynomials in a, b, c .

Solution. Put $G = X + B$ with $B = b_0 + b_1t + b_2t^2 + \cdots$ and impose $F \circ G = X$. By [Theorem 10.22](#),

$$L \circ G = L + \log(1 + tB) = L + b_0t + \left(b_1 - \frac{1}{2}b_0^2\right)t^2 + \mathcal{O}_t^{\text{form}}(t^3), \quad t \circ G = t(1 - b_0t + \mathcal{O}_t^{\text{form}}(t^2)).$$

Write $\delta := (L \circ G) - L$, of outer order ≥ 1 . Expanding each coefficient by [Theorem 25.33](#), a finite Taylor sum:

$$\begin{aligned} a(L \circ G) &= a + a'\delta + \frac{1}{2}a''\delta^2 + \mathcal{O}_t^{\text{form}}(t^3) \\ &= a + a'b_0t + \left(a'(b_1 - \frac{1}{2}b_0^2) + \frac{1}{2}a''b_0^2\right)t^2 + \mathcal{O}_t^{\text{form}}(t^3), \\ b(L \circ G) \cdot (t \circ G) &= (b + b'b_0t) \cdot t(1 - b_0t) + \mathcal{O}_t^{\text{form}}(t^3) \\ &= bt + (b'b_0 - bb_0)t^2 + \mathcal{O}_t^{\text{form}}(t^3), \\ c(L \circ G) \cdot (t \circ G)^2 &= ct^2 + \mathcal{O}_t^{\text{form}}(t^3). \end{aligned}$$

Adding these to $G = X + B$ and equating $F \circ G$ to X block by block:

$$\begin{aligned} t^0 : \quad & b_0 + a = 0, \\ t^1 : \quad & b_1 + a'b_0 + b = 0, \\ t^2 : \quad & b_2 + a'(b_1 - \frac{1}{2}b_0^2) + \frac{1}{2}a''b_0^2 + b'b_0 - bb_0 + c = 0. \end{aligned}$$

Solving in order, with $b_0 = -a$ and $b_1 = aa' - b$,

$$b_0 = -a, \quad b_1 = aa' - b, \quad b_2 = \frac{a^2 a'}{2} - \frac{a^2 a''}{2} - a(a')^2 + a'b + ab' - ab - c. \quad (547)$$

Three consistency checks. For $b = c = 0$ this is $b_2 = \frac{1}{2}a^2 a' - a(a')^2 - \frac{1}{2}a^2 a''$, the form used in [Theorem E.33](#). For $a = aL$ with $a \in \mathbb{K}$ constant and $b = c = 0$ it gives $a^3(L^2/2 - L) = a^3 \mathbb{L}_2^{\text{Lam}}(L)$, matching [Theorem 16.23](#). For a, b constants and $c = 0$ it gives $b_2 = -ab$, matching (266) at $\deg_L a = 0$. $\square$

Exercise E.32 (Trap: an incomplete universal formula). A source records the third universal inverse block of $F = X + a(L) + b(L)t + c(L)t^2$ as

$$b_2 \stackrel{?}{=} \frac{1}{2}a^2 a' - a(a')^2 - ab + a'b,$$

and verifies it on the example $a = 2L, b = 3, c = 0$, where it is correct. Show that the formula is nevertheless false in general, by producing three independent witnesses – one for each missing term – and state the correct formula. Explain why the source’s verification could not have detected the error.

Solution. Comparing with (547), the quoted formula omits $-\frac{1}{2}a^2 a''$, $+ab'$ and $-c$. Each is detected by a separate witness.

Missing $-\frac{1}{2}a^2 a''$. Take $a = L^2, b = c = 0$. Then $a' = 2L, a'' = 2$, and (547) gives

$$b_2 = \frac{L^4 \cdot 2L}{2} - \frac{L^4 \cdot 2}{2} - L^2 \cdot 4L^2 = L^5 - 5L^4,$$

whereas the quoted formula gives $L^5 - 4L^4$. The discrepancy is $-L^4$. The correct value is confirmed independently by [Theorem 16.23](#), which for $P = L^2$ predicts $\deg_L b_2 = d(N+1) - 1 = 5$ with leading coefficient $(d/N)a_d^{N+1} = 2/2 = 1$; both hold for $L^5 - 5L^4$.

Missing $+ab'$. Take $a = 1, b = L, c = 0$. Then (547) gives $b_2 = -ab + ab' = -L + 1$, while the quoted formula gives $-L$.

Missing $-c$. Take $a = b = 0, c = 1$. Then $F = X + t^2$, and $F^{(-1)\circ} = X - t^2 + \mathcal{O}_t^{\text{form}}(t^3)$ directly by [Theorem E.30](#); so $b_2 = -1$, while the quoted formula gives 0.

The source’s verification could not detect any of this because its single test case has $a'' = 0$ (its a is affine in L), $b' = 0$ (its b is a constant) and $c = 0$. All three omitted terms vanish simultaneously on that example. This is the standard hazard of validating a universal formula on one instance; the residual certificate of [Theorem 14.16](#) is the correct test, because it is run against F itself and is an identity in $\mathbb{K}[L][[t]]$, not a numerical sample. $\square$

Exercise E.33 (Reversion of a quadratic logarithmic perturbation). Reverse $F = X + L^2$ through outer order t^3 , and check the result against the degree law of [Theorem 16.22](#) at $N = 1, 2, 3$.

Solution. Use the Lagrange–Bürmann formula in the form [Theorem 16.23](#): for $F = X + P(L)$ only the summand $r = N + 1$ contributes to b_N , so

$$b_N = \frac{(-1)^{N+1}}{(N+1)!} \left[\prod_{s=0}^{N-1} (\partial_L - s) \right] P^{N+1}.$$

With $P = L^2$:

$$N = 0: b_0 = -P = -L^2.$$

$$N = 1: b_1 = \frac{1}{2}\partial_L(P^2) = PP' = L^2 \cdot 2L = 2L^3.$$

$$N = 2: b_2 = -\frac{1}{6}(\partial_L - 1)\partial_L(P^3). \text{ Now } \partial_L(P^3) = 3P^2 P' \text{ and } (\partial_L - 1)(3P^2 P') = 6P(P')^2 + 3P^2 P'' - 3P^2 P', \text{ so}$$

$$b_2 = \frac{1}{2}P^2 P' - P(P')^2 - \frac{1}{2}P^2 P'' = L^5 - 4L^4 - L^4 = L^5 - 5L^4.$$

$N = 3$: $b_3 = \frac{1}{24}(\partial_L - 2)(\partial_L - 1)\partial_L(P^4)$. With $P^4 = L^8$: $\partial_L L^8 = 8L^7$; $(\partial_L - 1)(8L^7) = 56L^6 - 8L^7$; $(\partial_L - 2)(56L^6 - 8L^7) = 336L^5 - 56L^6 - 112L^6 + 16L^7 = 16L^7 - 168L^6 + 336L^5$. Dividing by 24,

$$b_3 = \frac{2}{3}L^7 - 7L^6 + 14L^5.$$

Collecting,

$$(X + L^2)^{\langle -1 \rangle \circ} = X - L^2 + 2L^3t + (L^5 - 5L^4)t^2 + \left(\frac{2}{3}L^7 - 7L^6 + 14L^5\right)t^3 + \mathcal{O}_t^{\text{form}}(t^4).$$

Degree law. Here $d = \deg_L P = 2$ and $a_d = 1$. [Theorem 16.22](#) predicts, for $N \geq 1$, $\deg_L b_N = d(N+1) - 1 = 2N + 1$ and $[L^{2N+1}]b_N = (d/N)a_d^{N+1} = 2/N$. For $N = 1, 2, 3$ that is degrees 3, 5, 7 with leading coefficients 2, 1, $\frac{2}{3}$; all three agree with the computation. Note that $\deg_L b_0 = 2 \neq d \cdot 1 - 1 = 1$: the law excludes $N = 0$, exactly as the Lambert degree law excludes $n = 0$ ([Theorem E.44](#)). $\square$

Exercise E.34 (A perturbation with a genuine t -block). Reverse $F = X + L + at$ through outer order t^3 , for $a \in \mathbb{K}$, and verify the answer with the residual certificate of [Theorem 14.16](#).

Solution. Set $G = X + B$, $B = b_0 + b_1t + b_2t^2 + b_3t^3 + \dots$. Since $F = X + L + at$, [Theorem 10.22](#) gives

$$F \circ G = X + B + L + \log(1 + tB) + \frac{at}{1 + tB}.$$

Expand each piece through t^3 . Writing $u = tB = b_0t + b_1t^2 + b_2t^3$,

$$\log(1 + u) = b_0t + \left(b_1 - \frac{1}{2}b_0^2\right)t^2 + \left(b_2 - b_0b_1 + \frac{1}{3}b_0^3\right)t^3 + \mathcal{O}_t^{\text{form}}(t^4),$$

$$\frac{at}{1 + u} = at - ab_0t^2 + a(b_0^2 - b_1)t^3 + \mathcal{O}_t^{\text{form}}(t^4).$$

Equating $F \circ G = X$ block by block:

$$\begin{aligned} t^0 : b_0 + L &= 0 & \Rightarrow b_0 &= -L, \\ t^1 : b_1 + b_0 + a &= 0 & \Rightarrow b_1 &= L - a, \\ t^2 : b_2 + b_1 - \frac{1}{2}b_0^2 - ab_0 &= 0 & \Rightarrow b_2 &= \frac{1}{2}L^2 - L - aL + a, \\ t^3 : b_3 + b_2 - b_0b_1 + \frac{1}{3}b_0^3 + a(b_0^2 - b_1) &= 0. \end{aligned}$$

Substituting the earlier blocks in the last line, $b_0b_1 = -L(L - a) = -L^2 + aL$, $\frac{1}{3}b_0^3 = -\frac{1}{3}L^3$, $b_0^2 - b_1 = L^2 - L + a$, so

$$b_3 = -b_2 + b_0b_1 - \frac{1}{3}b_0^3 - a(b_0^2 - b_1) = \frac{L^3}{3} - \frac{3L^2}{2} - aL^2 + 3aL + L - a^2 - a.$$

Hence

$$\begin{aligned} (X + L + at)^{\langle -1 \rangle \circ} &= X - L + (L - a)t + \left(\frac{L^2}{2} - L - aL + a\right)t^2 \\ &\quad + \left(\frac{L^3}{3} - \frac{3L^2}{2} - aL^2 + 3aL + L - a^2 - a\right)t^3 + \mathcal{O}_t^{\text{form}}(t^4). \end{aligned}$$

Certificate. By [Theorem 15.2](#) the residual of the truncation $G_2 = X + b_0 + b_1t + b_2t^2$ is $\text{Err}_{\text{rev}}(F, G_2) = -b_3t^3 + \mathcal{O}_t^{\text{form}}(t^4)$; substituting the computed b_2 into the t^3 -equation above reproduces exactly $-b_3$, which is the check. Two sanity tests: at $a = 0$ the blocks reduce to $-L, L, \frac{1}{2}L^2 - L, \frac{1}{3}L^3 - \frac{3}{2}L^2 + L$, i.e. to $L_0^{\text{Lam}}, \dots, L_3^{\text{Lam}}$ as [Theorem 18.10](#) requires; and the whole answer agrees with (264)–(267) at $a \mapsto 1, b \mapsto a$. $\square$

Exercise E.35 (The residual measures the error, on both sides). Let $F = X + L$ and let $H = X - L + Lt + \frac{1}{2}L^2t^2$, the second Picard iterate. Compute $\text{Err}_{\text{rev}}(F, H)$ and $H \circ F - X$ through outer order t^3 , and verify [Theorem 14.15](#) on this pair, including the part that says the two residuals agree only in their *leading* block.

Solution. Write $G = F^{(-1)\circ} = \omega$, whose blocks are the Lambert polynomials ([Theorem 18.10](#)). The error is $E = H - G$, with

$$[t^0]E = [t^1]E = 0, \quad [t^2]E = \frac{1}{2}L^2 - \mathbb{L}_2^{\text{Lam}}(L) = \frac{1}{2}L^2 - \left(\frac{1}{2}L^2 - L\right) = L,$$

so $e := \nu_t(E) = 2$ and $[t^2]E = L$.

Right residual. With $B = -L + Lt + \frac{1}{2}L^2t^2$, $\text{Err}_{\text{rev}}(F, H) = B + L + \log(1 + tB)$. Expanding,

$$\text{Err}_{\text{rev}}(F, H) = Lt^2 - \frac{L^2(2L - 9)}{6}t^3 + \mathcal{O}_t^{\text{form}}(t^4).$$

Left residual. $H \circ F - X = (F - X) + B \circ F = L + B \circ F$, and expanding with (181),

$$H \circ F - X = Lt^2 - \frac{L^2(2L + 3)}{6}t^3 + \mathcal{O}_t^{\text{form}}(t^4).$$

Both residuals have outer order exactly $2 = e$, and both have leading block $L = [t^2]E$, which is [Theorem 14.15](#). Their t^3 -blocks differ, by $2L^2$: the proposition asserts equality of the leading blocks only, and this pair shows that no more is true. The practical consequence is that either residual certifies correctness through a requested block ([Theorem 14.16](#)), and either reads off the next inverse block ([Theorem 14.17](#)), but neither may be used as an approximation to the error beyond its leading block. $\square$

Exercise E.36 (Counterexample: the derivative certificate is strictly weaker). One source asserts that the derivative residual $D_X H - ((D_X F) \circ H)^{-1}$ “often detects a wrong block one order earlier” than the composition residual $\text{Err}_{\text{rev}}(F, H)$. Refute this, and quantify the true relation.

1. Show the derivative residual has outer order at least $e + 1$, where $e = \nu_t(H - F^{(-1)\circ})$: it is one level *later*, never earlier.
2. Show it is blind at $e = 0$: construct F, H with $e = 0$ for which the composition residual has order 0 while the derivative residual has order 2.

Solution. (1) Let $G = F^{(-1)\circ}$ and $E = H - G$. Differentiating $F \circ G = X$ gives $(D_X F) \circ G \cdot D_X G = 1$, so $D_X G = ((D_X F) \circ G)^{-1}$ and the derivative residual is

$$D_X H - ((D_X F) \circ H)^{-1} = D_X E + \left(((D_X F) \circ G)^{-1} - ((D_X F) \circ H)^{-1} \right).$$

The first summand has $\nu_t \geq \nu_t(E) + 1 = e + 1$ by [Theorem 9.9](#). For the second, write $F = X + A$ with $A \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, so $D_X F = 1 + D_X A$ with $\nu_t(D_X A) \geq \nu_t(A) + 1 \geq 1$. Then $(D_X F) \circ H - (D_X F) \circ G = (D_X A) \circ H - (D_X A) \circ G$ has $\nu_t \geq \nu_t(D_X A) + e + 1 \geq e + 2$ by [Theorem 12.6](#). Both $(D_X F) \circ H$ and $(D_X F) \circ G$ lie in $1 + t\mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$, and for units u, v of that form $u^{-1} - v^{-1} = (v - u)(uv)^{-1}$ with $\nu_t(uv) = 0$, so taking reciprocals preserves the order of a difference exactly; the second summand therefore has $\nu_t \geq e + 2$. Hence the derivative residual has $\nu_t \geq e + 1$, whereas $\text{Err}_{\text{rev}}(F, H)$ has ν_t exactly e ([Theorem 14.15](#)). It is therefore always one level later, and can never be earlier. This is [Theorem 14.19](#).

(2) Take $F = X + L$, $G = F^{(-1)\circ} = \omega$ and $H = G + 5$, so $E = 5 \in \mathbb{K}^\times$ and $e = 0$. The composition residual has order 0: indeed $\text{Err}_{\text{rev}}(F, H) = 5 + (\text{terms of order } \geq 1)$, whose t^0 -block is $5 \neq 0$. But $D_X E = D_X(5) = 0$, so the first summand in the display above vanishes identically, and a direct computation gives

$$D_X H - ((D_X F) \circ H)^{-1} = -5t^2 + \mathcal{O}_t^{\text{form}}(t^3),$$

of order 2. So at $e = 0$ the derivative certificate is off by two levels, and a routine that used it alone would accept an inverse that is wrong in its very first block by an additive constant. The reason is structural: D_X kills $\mathbb{K}$, so the derivative certificate is blind to exactly the kernel of D_X , which is the block that ν_t calls level 0. $\square$

Exercise E.37 (A near-identity idempotent is the identity). Let $J \in \mathcal{G}_{\text{poly}, \mathbb{K}}$ satisfy $J \circ J = J$. Prove $J = X$, first from the group structure and then, without it, by comparing the first nonzero correction block.

Solution. First proof. By [Theorem 12.7](#), $\mathcal{G}_{\text{poly}, \mathbb{K}}$ is a group under composition. Composing $J \circ J = J$ on the right with $J^{(-1)\circ}$ gives $J = X$.

Second proof, without inverses. Write $J = X + B$ with $B \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$ and suppose $B \neq 0$; let $\beta = \nu_t(B) \geq 0$. By [Theorem 10.23](#),

$$J \circ J = X + B + B \circ (X + B),$$

so $J \circ J = J$ forces $B \circ (X + B) = 0$. But [Theorem 10.22](#) says that substitution by a near-identity map preserves the outer valuation exactly: $\nu_t(B \circ (X + B)) = \nu_t(B) = \beta < \infty$. A series of finite outer order is nonzero, a contradiction. Hence $B = 0$ and $J = X$.

The second proof is the more informative one: it shows that the conclusion uses only that Φ_J is injective and valuation-preserving, not that inverses exist. Consequently the same argument works in the near-identity monoid at any logarithmic depth ([Theorem 13.14](#)), and in Puiseux–log enlargements. $\square$

Exercise E.38 (One Picard step against one Newton step). Let $F = X + \log X$ and take $G_0 = X - \log X$ as the starting approximation.

1. Compute the next Picard iterate $\Phi(G_0)$, where $\Phi(X + B) = X - A \circ (X + B)$ with $A = L$, and say how many blocks of $F^{(-1)\circ}$ it gets right.
2. Compute the Newton iterate $\mathcal{N}_F(G_0)$ of [Theorem 15.10](#) far enough to say how many blocks it gets right, and compare with [Theorem 15.11](#).

Solution. Throughout, $F^{(-1)\circ} = \omega = X - L + Lt + (\frac{1}{2}L^2 - L)t^2 + (\frac{1}{3}L^3 - \frac{3}{2}L^2 + L)t^3 + \dots$ ([Theorem 15.19](#)), and $G_0 = X - L$ is correct through block 0, so $e_0 := \nu_t(G_0 - F^{(-1)\circ}) = 1$.

(1) $\Phi(G_0) = X - L \circ (X - L) = X - (L + \log(1 - Lt))$, so

$$\Phi(G_0) = X - L + Lt + \frac{L^2}{2}t^2 + \frac{L^3}{3}t^3 + \dots$$

Blocks 0 and 1 are correct ($-L$ and L); block 2 is $\frac{1}{2}L^2$ against the true $\frac{1}{2}L^2 - L$, so the error has outer order exactly 2. One Picard step gained one block, in agreement with (247): with $\alpha = \nu_t(A) = 0$, each step gains $\alpha + 1 = 1$.

(2) $D_X F = 1 + t$, so $\mathcal{N}_F(G_0) = G_0 - \text{Err}_{\text{rev}}(F, G_0) \cdot ((1 + t) \circ G_0)^{-1}$. Here $\text{Err}_{\text{rev}}(F, G_0) = \log(1 - Lt) = -Lt - \frac{1}{2}L^2t^2 - \frac{1}{3}L^3t^3 - \dots$ and $(1 + t) \circ G_0 = 1 + t/(1 - Lt) = 1 + t + Lt^2 + \dots$, whose reciprocal is $1 - t + (1 - L)t^2 + \dots$. Multiplying and subtracting,

$$\mathcal{N}_F(G_0) = X - L + Lt + \left(\frac{1}{2}L^2 - L\right)t^2 + \left(\frac{1}{3}L^3 - \frac{3}{2}L^2 + L\right)t^3 + \left(\frac{1}{4}L^4 - \frac{11}{6}L^3 + \frac{5}{2}L^2 - L\right)t^4 + \dots,$$

whose blocks 0, 1, 2, 3 all agree with ω and whose block 4 does not (the true value is $\frac{1}{4}L^4 - \frac{11}{6}L^3 + 3L^2 - L = \mathbb{L}_4^{\text{Lam}}(L)$, differing by $\frac{1}{2}L^2$). So the error has outer order exactly 4.

[Theorem 15.11](#) predicts $\nu_t(\text{Err}_{\text{rev}}(F, \mathcal{N}_F(G_0))) \geq 2m + 2 + \alpha$ with $m = \nu_t(\text{Err}_{\text{rev}}(F, G_0)) = 1$ and $\alpha = 0$, i.e. at least 4; the computation attains it. Summary: from a start correct through block 0, one Picard step reaches block 1 and one Newton step reaches block 3. The comparison is not, however, a cost comparison: the Newton step performs a composition, a reciprocal and a multiplication, and which scheme is cheaper depends on the cost model ([Theorem 25.42](#)). $\square$

Exercise E.39 (Counterexample: “smaller than X ” is not the admissibility test (hard)). [Theorem 16.9](#) requires $\nu_t(A) > -1$, which on $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ means $A \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$. A plausible substitute is the dominance condition $A \prec X$, i.e. “ A is asymptotically smaller than the identity”. Refute it.

1. Show that $A = X/L$ satisfies $A = o_{X \rightarrow +\infty}(X)$ yet has $\nu_t(A) = -1$.
2. For $A = \alpha X$ with $\alpha \in \mathbb{K}^\times$, compute every Lagrange–Bürmann summand exactly and show they all have the same outer order, so the family is never summable.
3. Show that the formal sum of those summands is a geometric series whose *numerical* value reproduces the correct inverse $((1 + \alpha)X)^{\langle -1 \rangle^\circ} = X/(1 + \alpha)$ when $|\alpha| < 1$ and is divergent when $|\alpha| \geq 1$. Conclude that the threshold is not a technicality.

Solution. (1) $A = X/L = t^{-1}L^{-1}$ lies in $\mathcal{P}\mathcal{L}_{\text{rat},\mathbb{K}}$ with $\nu_t(A) = -1$. As a function, $A(X)/X = 1/\log X \rightarrow 0$, so $A \prec X$ in the analytic sense of [Theorem 3.13](#). So the dominance test is passed and the valuation test is failed: logarithmic gains are invisible to ν_t . This is the content of [Theorem 16.10](#).

(2) Let $A = \alpha X = \alpha t^{-1}$. By [Theorem 9.3](#), $D_X(t^n) = t^{n+1}(0 - n) = -nt^{n+1}$, so by induction

$$D_X^k(t^{-r}) = r(r-1) \cdots (r-k+1) t^{-r+k},$$

and with $k = r - 1$, $D_X^{r-1}(t^{-r}) = r! t^{-1}$. Hence the r -th Lagrange–Bürmann summand is

$$\frac{(-1)^r}{r!} D_X^{r-1}(A^r) = \frac{(-1)^r \alpha^r}{r!} r! t^{-1} = (-\alpha)^r X. \quad (548)$$

Every summand has outer order -1 , so $\{r : \nu_t < 0\}$ is infinite and the family fails [Theorem 5.21](#). This matches the exact order formula (274): with $\alpha_{\nu_t} = \nu_t(A) = -1$, $\nu_t(D_X^{r-1}(A^r)) = r(\nu_t(A) + 1) - 1 = -1$ for every r .

(3) Formally summing (548) as a geometric series,

$$X + \sum_{r \geq 1} (-\alpha)^r X = X \cdot \frac{1}{1 + \alpha} = \frac{X}{1 + \alpha},$$

which is indeed $F^{\langle -1 \rangle^\circ}$ for $F = X + \alpha X = (1 + \alpha)X$. For $|\alpha| < 1$ the scalar series converges and the manipulation is a valid computation with numbers – but not with formal series: the family is not summable, so the left-hand side names no element of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, and the agreement is a coincidence of the geometric series, not a theorem. For $\alpha = 2$, say, the same manipulation produces $X \sum_{r \geq 0} (-2)^r$, a divergent series of scalars, while the true inverse $(3X)^{\langle -1 \rangle^\circ} = X/3$ is perfectly well defined. So the recipe is not merely unjustified at the threshold; it breaks.

The moral: $\nu_t(A) > -1$ is exactly the condition under which the summands’ orders tend to $+\infty$, and no growth comparison between A and X substitutes for it. The two agree on $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$, whose value group is $\mathbb{Z}$, and separate on the Puiseux–log enlargement, where $\nu_t(A) > -1$ admits fractional orders in $(-1, 0)$ that “ $A \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ ” would exclude. $\square$

Exercise E.40 (The inverse of $X \log X$ needs two logarithms (hard)). Let $F(X) = X \log X$ and $Y = F(X) \rightarrow +\infty$. Put $S = \log Y$ and $\Lambda = \log S$.

1. Explain why no one-logarithm ansatz can represent $F^{\langle -1 \rangle^\circ}$, however many terms it carries.
2. Derive the closed form $F^{\langle -1 \rangle^\circ}(Y) = Y/W(Y)$ and expand it through S^{-4} .

Solution. (1) $F = cX^\rho L^\sigma(1 + U)$ with $c = 1$, $\rho = 1$, $\sigma = 1$, $U = 0$. [Theorem 17.3](#) says a monomial-leading inverse exists in a Puiseux–log ring if and only if ρ^{-1} is in the exponent group *and* $\sigma = 0$; here $\sigma = 1$, so no enlargement of the exponent group or of the coefficient field helps. Quantitatively, [Theorem 17.4](#) gives $\log F^{\langle -1 \rangle^\circ}(Y) = S - \Lambda + \mathcal{O}_{S \rightarrow +\infty}(\Lambda/S)$, and the coefficient -1 of $\Lambda = \log \log Y$ is nonzero, so $F^{\langle -1 \rangle^\circ}$ lies in no ring whose coefficients are polynomials in one logarithm.

(2) Put $u = \log X$. Then $Y = X \log X = ue^u$, so $u = W(Y)$ on the principal branch and

$$F^{(-1)\circ}(Y) = X = e^u = \frac{Y}{u} = \frac{Y}{W(Y)}.$$

Now $W(Y) = \omega(S)$, because $w + \log w = S$ is $we^w = Y$. By [Theorem 18.10](#),

$$W(Y) = S - \Lambda + \sum_{n \geq 1} \frac{\mathbb{L}_n^{\text{Lam}}(\Lambda)}{S^n} = S(1 + \varepsilon), \quad \varepsilon = -\frac{\Lambda}{S} + \sum_{n \geq 1} \frac{\mathbb{L}_n^{\text{Lam}}(\Lambda)}{S^{n+1}}.$$

Hence $F^{(-1)\circ}(Y) = \frac{Y}{S} \frac{1}{1 + \varepsilon}$, and expanding the reciprocal with $\mathbb{L}_1^{\text{Lam}} = \Lambda$, $\mathbb{L}_2^{\text{Lam}} = \frac{1}{2}\Lambda^2 - \Lambda$, $\mathbb{L}_3^{\text{Lam}} = \frac{1}{3}\Lambda^3 - \frac{3}{2}\Lambda^2 + \Lambda$ gives, collecting by powers of S^{-1} ,

$$F^{(-1)\circ}(Y) = \frac{Y}{S} \left[1 + \frac{\Lambda}{S} + \frac{\Lambda(\Lambda - 1)}{S^2} + \frac{\Lambda(\Lambda - 2)(2\Lambda - 1)}{2S^3} + \frac{\Lambda(6\Lambda^3 - 26\Lambda^2 + 27\Lambda - 6)}{6S^4} + \mathcal{O}_{Y \rightarrow +\infty} \left(\frac{\Lambda^5}{S^5} \right) \right]. \quad (549)$$

For the record, the coefficients arise as follows. Writing $\varepsilon = \sum_{k \geq 1} e_k S^{-k}$ with $e_1 = -\Lambda$, $e_2 = \Lambda$, $e_3 = \mathbb{L}_2^{\text{Lam}}(\Lambda)$, $e_4 = \mathbb{L}_3^{\text{Lam}}(\Lambda)$, the coefficient of S^{-4} in $(1 + \varepsilon)^{-1}$ is

$$-e_4 + (2e_1e_3 + e_2^2) - 3e_1^2e_2 + e_1^4 = -\frac{1}{3}\Lambda^3 + \frac{3}{2}\Lambda^2 - \Lambda - \Lambda^3 + 2\Lambda^2 + \Lambda^2 - 3\Lambda^3 + \Lambda^4,$$

which is $\Lambda^4 - \frac{13}{3}\Lambda^3 + \frac{9}{2}\Lambda^2 - \Lambda$, the printed S^{-4} numerator divided by 6. Every coefficient is a polynomial in Λ , which is (1) again. The body records this expansion through S^{-2} only, at [\(321\)](#); the two further blocks are computed here. $\square$

Exercise E.41 (Trap: a scale change applied to one generator only). Let $F(X) = 2X + \log X$. A computation runs: “ $F = m_2 \circ G$ with $m_2(Z) = 2Z$ and $G(X) = X + \frac{1}{2} \log X$, so $F^{(-1)\circ}(Y) = G^{(-1)\circ}(Y/2)$. By [Theorem 16.23](#) with $a = \frac{1}{2}$, $G^{(-1)\circ}(Z) = Z + \sum_{N \geq 0} 2^{-(N+1)} \mathbb{L}_N^{\text{Lam}}(\log Z) Z^{-N}$. Substituting $Z = Y/2$, that is $Z^{-N} = 2^N Y^{-N}$, gives

$$F^{(-1)\circ}(Y) \stackrel{?}{=} \frac{Y}{2} + \frac{1}{2} \sum_{N \geq 0} \mathbb{L}_N^{\text{Lam}}(\log Y) Y^{-N}.$$

” Find the error, give the correct answer, quantify the discrepancy, and say which of the volume’s checks would have caught it.

Solution. The error is that $Z = Y/2$ changes *both* generators, not one:

$$Z^{-N} = 2^N Y^{-N} \quad \text{and} \quad \log Z = \log Y - \log 2.$$

The computation substituted the first and forgot the second. The correct result is

$$F^{(-1)\circ}(Y) = \frac{Y}{2} + \frac{1}{2} \sum_{N \geq 0} \mathbb{L}_N^{\text{Lam}}(\log Y - \log 2) Y^{-N},$$

which is [\(324\)](#).

The failure is silent. The wrong answer is a perfectly well formed element of $\mathcal{PL}_{\text{poly}, \mathbb{K}}$: every block has the right outer order, the right L -degree, the right leading coefficient and the right sign pattern. No check internal to the near-identity calculation for G can see it, because the calculation for G was correct.

Size of the error. The leading discrepancy is in the $N = 0$ block: $\frac{1}{2}\mathbb{L}_0^{\text{Lam}}(\log Y - \log 2) - \frac{1}{2}\mathbb{L}_0^{\text{Lam}}(\log Y) = -\frac{1}{2}(\log Y - \log 2) + \frac{1}{2}\log Y = \frac{1}{2}\log 2 = 0.34657\dots$, an additive constant. Numerically, at $Y = 20 + \log 10 = 22.3025850929\dots$, where $F^{(-1)\circ}(Y) = 10$ exactly, the two truncations through Y^{-3} – that is, over $0 \leq N \leq 3$ – give

$$\text{correct : } 10.0000043732\dots, \quad \text{naive : } 9.6702070480\dots,$$

a gap of $0.3298\dots$, close to but not equal to $\frac{1}{2}\log 2$ because the higher blocks differ as well.

What catches it. Two things, and only two. First, the residual certificate of [Theorem 14.16](#) computed against F itself: substituting the wrong series into $2X + \log X$ leaves a residual of outer order 0, namely $\log 2$, rather than the $\mathcal{O}_t^{\text{form}}(t^{N+1})$ a correct truncation must leave. A certificate run against the normalised map G would pass, which is why step 7 of [Theorem 17.9](#) certifies against the original map. Second, the coefficient field: the correct answer does not lie in $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$ over $\mathbb{K} = \mathbb{Q}$ at all, since its blocks involve $\log 2$. A routine that tracks the field would have refused the substitution before performing it, which is the point of [Theorem 17.2](#). $\square$

E.6 Wright omega and the Lambert polynomials

Exercise E.42 (Generating the Lambert polynomials two ways). Starting from $\mathbb{L}_1^{\text{Lam}}(u) = u$, use the integral recurrence (348), $\mathbb{L}_{n+1}^{\text{Lam}} = -\mathbb{L}_n^{\text{Lam}} + n \int_0^u \mathbb{L}_n^{\text{Lam}}(v) dv$, to produce $\mathbb{L}_2^{\text{Lam}}, \mathbb{L}_3^{\text{Lam}}, \mathbb{L}_4^{\text{Lam}}$. Check each against the Stirling formula (350), using $\begin{bmatrix} 4 \\ 4 \end{bmatrix} = 1$, $\begin{bmatrix} 4 \\ 3 \end{bmatrix} = 6$, $\begin{bmatrix} 4 \\ 2 \end{bmatrix} = 11$, $\begin{bmatrix} 4 \\ 1 \end{bmatrix} = 6$.

Solution. Successive integration, the constant of integration being fixed to 0 by $\mathbb{L}_n^{\text{Lam}}(0) = 0$ ([Theorem 19.2](#)):

$$\begin{aligned} \mathbb{L}_2^{\text{Lam}} &= -u + 1 \cdot \int_0^u v dv = \frac{u^2}{2} - u, \\ \mathbb{L}_3^{\text{Lam}} &= -\left(\frac{u^2}{2} - u\right) + 2 \int_0^u \left(\frac{v^2}{2} - v\right) dv = -\frac{u^2}{2} + u + \frac{u^3}{3} - u^2 = \frac{u^3}{3} - \frac{3}{2}u^2 + u, \\ \mathbb{L}_4^{\text{Lam}} &= -\left(\frac{u^3}{3} - \frac{3}{2}u^2 + u\right) + 3 \int_0^u \left(\frac{v^3}{3} - \frac{3}{2}v^2 + v\right) dv \\ &= -\frac{u^3}{3} + \frac{3}{2}u^2 - u + \frac{u^4}{4} - \frac{3}{2}u^3 + \frac{3}{2}u^2 = \frac{u^4}{4} - \frac{11}{6}u^3 + 3u^2 - u. \end{aligned}$$

Stirling check at $n = 4$. Formula (350) reads $\mathbb{L}_n^{\text{Lam}}(u) = \sum_{m=1}^n (-1)^{n-m} \begin{bmatrix} n \\ n-m+1 \end{bmatrix} u^m / m!$. For $n = 4$:

$$-\begin{bmatrix} 4 \\ 4 \end{bmatrix} \frac{u}{1!} + \begin{bmatrix} 4 \\ 3 \end{bmatrix} \frac{u^2}{2!} - \begin{bmatrix} 4 \\ 2 \end{bmatrix} \frac{u^3}{3!} + \begin{bmatrix} 4 \\ 1 \end{bmatrix} \frac{u^4}{4!} = -u + 3u^2 - \frac{11}{6}u^3 + \frac{u^4}{4},$$

which agrees. Note that the alternation is governed by $(-1)^{n-m}$, not by $(-1)^m$: the whole row changes sign with the parity of n . A common slip is to fix the sign pattern from a single row – $n = 4$, where the coefficients run $-, +, -, +$ – and then apply it at $n = 3$, where they run $+, -, +$.

Factored forms. $\mathbb{L}_2^{\text{Lam}} = \frac{1}{2}u(u-2)$, $\mathbb{L}_3^{\text{Lam}} = \frac{1}{6}u(2u^2-9u+6)$, $\mathbb{L}_4^{\text{Lam}} = \frac{1}{12}u(3u^3-22u^2+36u-12)$, matching the scaled integer family $\mathbb{L}_n^{\text{Lam,sc}} = n!\mathbb{L}_n^{\text{Lam}}$ of [Table 27](#). $\square$

Exercise E.43 (The one-parameter family, from Lagrange–Bürmann). Let $a \in \mathbb{K}$ and $F = X + a \log X$. Prove directly from (281) that the coefficient of X^{-n} in $F^{(-1)\circ}$ is $a^{n+1}\mathbb{L}_n^{\text{Lam}}(L)$ for every $n \geq 0$, and identify which Lagrange index contributes to which block.

Solution. Apply (281) with $A = aL \in \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$, legitimate since $\nu_t(A) = 0 > -1$ ([Theorem 16.9](#)):

$$F^{(-1)\circ} = X + \sum_{r \geq 1} \frac{(-1)^r}{r!} D_X^{r-1}(a^r L^r) = X + \sum_{r \geq 1} \frac{(-1)^r a^r}{r!} D_X^{r-1}(L^r).$$

By [Theorem 9.7](#) applied to the t^0 -block L^r ,

$$D_X^{r-1}(L^r) = t^{r-1} \Delta_{0,r-1} L^r = t^{r-1} \left[\prod_{j=0}^{r-2} (\partial_L - j) \right] L^r.$$

So the r -th summand is a *single* block, sitting at outer order exactly $r - 1$. Setting $n = r - 1$, no two summands ever meet, and

$$[t^n] F^{(-1)\circ} = a^{n+1} \cdot \frac{(-1)^{n+1}}{(n+1)!} \left[\prod_{j=0}^{n-1} (\partial_L - j) \right] L^{n+1} = a^{n+1} \mathbb{L}_n^{\text{Lam}}(L),$$

by [Theorem 19.1](#). Index r contributes to block $n = r - 1$ and to no other; this is the “one Lagrange index per block” of (336), and it is special to a perturbation that is a *single* t^0 -block. At $a = 1$ one recovers ω ([Theorem 18.10](#)) and at $a = -1$ the reversion of $X - \log X$ with weights $(-1)^{n+1}$ ((329)); at $n = 0$ the value is $a \mathbb{L}_0^{\text{Lam}}(L) = -aL$, which is the leading correction predicted by [Theorem 14.11](#) (4).

A useful sanity check that does not use the formula: substituting $t \mapsto at$ in the implicit equation (333) and multiplying by a transports the unique fixed point of $X + L$ to that of $X + aL$, which is the proof of [Theorem 18.5](#). $\square$

Exercise E.44 (Counterexample: the degree law and the sign of the error at $n = 0$). The Lambert polynomials satisfy $\deg \mathbb{L}_n^{\text{Lam}} = n$, $[u^n] \mathbb{L}_n^{\text{Lam}} = 1/n$ and $[u] \mathbb{L}_n^{\text{Lam}} = (-1)^{n-1}$ – for $n \geq 1$.

1. Show that each of the three statements fails, or is meaningless, at $n = 0$, and determine which parts of [Theorem 19.7](#) do survive there.
2. A plausible consequence of $\omega - \omega_N \sim \mathbb{L}_{N+1}^{\text{Lam}}(L)x^{-(N+1)}$ is that every truncation approaches ω from below on $(1, \infty)$. Refute this, and locate the exact threshold for $N = 1$.

Solution. (1) $\mathbb{L}_0^{\text{Lam}}(u) = -u$. Its degree is 1, not 0; the assertion $[u^0] \mathbb{L}_0^{\text{Lam}} = 1/0$ is meaningless; and $[u] \mathbb{L}_0^{\text{Lam}} = -1$ whereas $(-1)^{0-1} = -1$ – so the third statement happens to hold, by accident, and should not be quoted as evidence. What does survive at $n = 0$ is exactly the part of [Theorem 19.7](#) that does not mention the degree or a coefficient indexed from the top: the constant term vanishes and $u \mid \mathbb{L}_0^{\text{Lam}}$, i.e. item (i) minus its degree clause. Every statement indexed relative to the top – (ii) on the number of nonzero coefficients, (iii), (iv), (vi) – fails or is vacuous. The integrality normalisation $\mathbb{L}_n^{\text{Lam,sc}} = n! \mathbb{L}_n^{\text{Lam}}$ is likewise an $n \geq 1$ statement; at $n = 0$ it returns $-u$, which is the value, not an instance of the law.

Why the exception is unavoidable: the zero-based normalisation is the one in which (336) says “one Lagrange index per block” and in which [Theorem 18.5](#) needs no case distinction. The price is paid entirely at $n = 0$, and it is the reason [Theorem 18.11](#) exists.

(2) The claim would follow if $\mathbb{L}_{N+1}^{\text{Lam}}(L) > 0$ for all $x > 1$, i.e. for all $L > 0$. It is false. For $N = 1$ the relevant polynomial is $\mathbb{L}_2^{\text{Lam}}(u) = \frac{1}{2}u(u-2)$, which is *negative* for $0 < u < 2$ and positive for $u > 2$. So the sign of $\omega - \omega_1$ changes at $L = 2$, that is at

$$x = e^2 = 7.389056 \dots$$

Numerically: at $x = e$ (so $L = 1$) one has $\omega(e) = 2.0167797 \dots$ while $\omega_1(e) = e - 1 + e^{-1} = 2.0861613 \dots$, an overshoot; at $x = 10$ (so $L = 2.302585 \dots$) one has $\omega(10) = 7.9294201 \dots$ and $\omega_1(10) = 7.9276734 \dots$, an undershoot. In both cases the error agrees in sign and to one digit with $\mathbb{L}_2^{\text{Lam}}(L)x^{-2}$, which is -6.77×10^{-2} and $+3.48 \times 10^{-3}$ respectively.

More generally $[u] \mathbb{L}_n^{\text{Lam}} = (-1)^{n-1}$, so near $u = 0^+$ – that is for x slightly above 1 – the sign of $\mathbb{L}_n^{\text{Lam}}(L)$ is $(-1)^{n-1}$ and the truncations alternate sides. Only for L large, where the leading term $L^n/n > 0$ dominates, do they settle on one side. The correct statement is the one the body makes, (341): an asymptotic equivalence as $x \rightarrow +\infty$, with no claim at any finite x . $\square$

Exercise E.45 (The lower real branch near zero). Let $x \in (-e^{-1}, 0)$ and write $x = -e^{-\eta}$, so $\eta \rightarrow +\infty$ as $x \rightarrow 0^-$. Put $M = \log \eta$.

1. Show that $W_{-1}(x) = -v$ where v solves $v - \log v = \eta$.
2. Derive the first four correction blocks of v , and hence of W_{-1} .
3. Some sources say “the same $\mathsf{L}_n^{\text{Lam}}$ occur, multiplied by $(-1)^n$ ”. Say precisely which quantity carries which weight.

Solution. (1) $W_{-1}(x) < -1$, so write $W_{-1}(x) = -v$ with $v > 1$. The defining equation $We^W = x$ becomes $-ve^{-v} = x = -e^{-\eta}$, i.e. $ve^{-v} = e^{-\eta}$. Taking the real logarithm of both positive sides, $\log v - v = -\eta$, that is $v - \log v = \eta$.

(2) So $v = (X - \log X)^{(-1)^\circ}(\eta)$. By (329),

$$\begin{aligned} v &= \eta + M + \sum_{n \geq 1} (-1)^{n+1} \mathsf{L}_n^{\text{Lam}}(M) \eta^{-n} \\ &= \eta + M + \frac{M}{\eta} - \frac{\frac{1}{2}M^2 - M}{\eta^2} + \frac{\frac{1}{3}M^3 - \frac{3}{2}M^2 + M}{\eta^3} - \frac{\mathsf{L}_4^{\text{Lam}}(M)}{\eta^4} + \dots, \end{aligned}$$

and therefore

$$\begin{aligned} W_{-1}(-e^{-\eta}) &= -\eta - M - \frac{M}{\eta} + \frac{M(M-2)}{2\eta^2} - \frac{M(2M^2-9M+6)}{6\eta^3} \\ &\quad + \frac{M(3M^3-22M^2+36M-12)}{12\eta^4} + \mathcal{O}_{\eta \rightarrow +\infty}\left(\frac{M^5}{\eta^5}\right), \end{aligned}$$

which is (390). Note that the Lambert part of this volume proves more than an asymptotic statement here: by Theorem 20.21 the series $\sum_{n \geq 1} (-1)^n \mathsf{L}_n^{\text{Lam}}(M) \eta^{-n}$ converges once $2(1 + |M|) < \eta$, with an explicit remainder bound. That is a theorem about this particular family, proved by a Cauchy estimate; it is not a consequence of the formal reversion, and nothing analogous is claimed for ω in Theorem E.56.

(3) The two weights belong to two different objects, and conflating them is the error the sources make. The reversion of $X - \log X$ carries $(-1)^{n+1}$: $(X - L)^{(-1)^\circ} = X + L + \sum_{n \geq 1} (-1)^{n+1} \mathsf{L}_n^{\text{Lam}}(L) t^n$, so its first correction is $+\mathsf{L}_1^{\text{Lam}}(L)/X = +L/X$, with a *plus* sign. Its negative, which is what enters W_{-1} , carries $(-1)^n$: $W_{-1} = -\eta - M + \sum_{n \geq 1} (-1)^n \mathsf{L}_n^{\text{Lam}}(M) \eta^{-n}$. Both displays above are consistent with this. $\square$

Exercise E.46 (Trap: an off-by-one in the truncation index). A source using the inclusive cutoff defines $\Omega_N = X + [\mathcal{P}]_{\leq N}$, where $\mathcal{P} = \sum_{n \geq 0} \mathsf{L}_n^{\text{Lam}}(L) t^n$, and proves

$$\Omega_N + \log \Omega_N - X = -\mathsf{L}_{N+1}^{\text{Lam}}(L) t^{N+1} + \mathcal{O}_t^{\text{form}}(t^{N+2}).$$

A reader transcribing into the normative exclusive convention writes $\Omega_N = X + \text{Trunc}_{<N} \mathcal{P}$ and keeps the conclusion. Show that the conclusion is then false, compute the correct residual, and give the correct transcription.

Solution. The two cutoffs are related by $[F]_{\leq N} = \text{Trunc}_{<N+1} F$, so the correct transcription is $\omega_N = X + \text{Trunc}_{<N+1} \mathcal{P}$, which is Theorem 18.12. Writing $\text{Trunc}_{<N} \mathcal{P}$ instead drops the block $\mathsf{L}_N^{\text{Lam}}(L) t^N$.

Compute the residual of the mistranscribed object. Put $\mathcal{P}'_N = \text{Trunc}_{<N} \mathcal{P}$ and $\Delta = \mathcal{P} - \mathcal{P}'_N = \sum_{n \geq N} \mathsf{L}_n^{\text{Lam}}(L) t^n$, so $\nu_t \Delta = N$ and $[t^N] \Delta = \mathsf{L}_N^{\text{Lam}}(L)$. Exactly the computation in the proof of Theorem 18.13 – generator substitution, then $1 + t\mathcal{P}'_N = (1 + t\mathcal{P})(1 - t\Delta/(1 + t\mathcal{P}))$ and additivity of the formal logarithm – gives

$$(X + \text{Trunc}_{<N} \mathcal{P}) + \log(X + \text{Trunc}_{<N} \mathcal{P}) - X = -\mathsf{L}_N^{\text{Lam}}(L) t^N + \mathcal{O}_t^{\text{form}}(t^{N+1}).$$

So the residual has outer order exactly N , not $N + 1$, and its leading block is $-\mathbb{L}_N^{\text{Lam}}$, not $-\mathbb{L}_{N+1}^{\text{Lam}}$. For $N = 3$, for instance, the mistranscribed truncation leaves $-(\frac{1}{3}L^3 - \frac{3}{2}L^2 + L)t^3 + \mathcal{O}_t^{\text{form}}(t^4)$, whereas the correct ω_3 leaves $-\mathbb{L}_4^{\text{Lam}}(L)t^4 + \mathcal{O}_t^{\text{form}}(t^5)$.

The practical bite: a user asking for “the expansion correct through x^{-3} ” and receiving $X + \text{Trunc}_{<3} \mathcal{P}$ gets an answer correct only through x^{-2} . The error is one block, it is systematic, and it is undetectable by inspecting the answer — every printed block is a correct Lambert polynomial. What detects it is the certificate of [Theorem 18.14](#): the residual order is the definition of how many blocks are right, and it reads N , not $N + 1$. $\square$

Exercise E.47 (The Laplace transform and its cancellations). 1. Compute the transform

$$\int_0^\infty e^{-su} \mathbb{L}_n^{\text{Lam}}(u) \, du$$

directly for $n = 1, 2, 3$ and $\Re s > 0$, and compare with (361).

2. Deduce the exponential moment cancellations

$$\int_0^\infty e^{-ru} \mathbb{L}_n^{\text{Lam}}(u) \, du = 0$$

for all integers r with $1 \leq r \leq n - 1$, and say why the range is empty at $n = 1$.

3. Explain why the identity says nothing about the convergence of $\sum_n \mathbb{L}_n^{\text{Lam}}(u)s^n$.

Solution. (1) Use $\int_0^\infty e^{-su} u^m \, du = m!/s^{m+1}$ for $\Re s > 0$.

$$\mathbb{L}_1^{\text{Lam}} = u : \quad \frac{1}{s^2}; \quad \mathbb{L}_2^{\text{Lam}} = \frac{u^2}{2} - u : \quad \frac{1}{2} \cdot \frac{2}{s^3} - \frac{1}{s^2} = \frac{1-s}{s^3};$$

$$\mathbb{L}_3^{\text{Lam}} = \frac{u^3}{3} - \frac{3}{2}u^2 + u : \quad \frac{1}{3} \cdot \frac{6}{s^4} - \frac{3}{2} \cdot \frac{2}{s^3} + \frac{1}{s^2} = \frac{2-3s+s^2}{s^4} = \frac{(1-s)(2-s)}{s^4}.$$

[Theorem 19.14](#) predicts $s^{-(n+1)} \prod_{j=1}^{n-1} (j-s)$, the product empty at $n = 1$; the three values are $1/s^2$, $(1-s)/s^3$, $(1-s)(2-s)/s^4$, matching.

(2) The right-hand side of (361) vanishes exactly at the zeros of $\prod_{j=1}^{n-1} (j-s)$, i.e. at $s = 1, 2, \dots, n-1$, all of which satisfy $\Re s > 0$ so that the integral converges absolutely. Hence $\int_0^\infty e^{-ru} \mathbb{L}_n^{\text{Lam}}(u) \, du = 0$ for $1 \leq r \leq n-1$. At $n = 1$ the product is empty and has no zeros, so the range $1 \leq r \leq 0$ is empty; the first genuine cancellation is $\int_0^\infty e^{-u} \mathbb{L}_2^{\text{Lam}}(u) \, du = 0$, which one can check by hand: $\frac{1}{2} \cdot 2 - 1 = 0$.

(3) The Laplace identity is a statement about one polynomial at a time. It integrates $\mathbb{L}_n^{\text{Lam}}$ against a fixed kernel over $u \in (0, \infty)$; it involves no sum over n , and gives no bound on $\mathbb{L}_n^{\text{Lam}}(u)$ at a fixed u as $n \rightarrow \infty$. Convergence of $\sum_n \mathbb{L}_n^{\text{Lam}}(u)s^n$ at frozen u is a separate theorem, [Theorem 18.19](#), proved by the implicit function theorem for (343); and neither statement is about the asymptotic expansion of ω , where $u = \log x$ and $s = 1/x$ move together. Keeping the three apart is the discipline [Section E.7](#) is about. $\square$

Exercise E.48 (Reciprocal and logarithm of W (hard)). Let z be large and positive, $L_1 = \log z$, $\ell = \log L_1$, $r = L_1^{-1}$, and $W = W(z)$ on the principal branch.

1. Derive $\log W = L_1 - W$ exactly, and use it to expand $\log W$ through r^4 without ever taking the logarithm of a series.
2. Expand $1/W$ through r^6 , and compute one block further than the sources tabulate.
3. Deduce a transseries for e^W with no exponentiation of a series.

Solution. (1) From $We^W = z$, taking logarithms of the positive quantities, $\log W + W = \log z = L_1$, so $\log W = L_1 - W$ exactly. Since $W = \omega(L_1)$ – because $W + \log W = L_1$ is the defining equation of ω – [Theorem 18.10](#), transported to branch coordinates by [Theorem 20.9](#), gives $W = L_1 - \ell + \sum_{n \geq 1} \mathsf{L}_n^{\text{Lam}}(\ell) r^n$. Subtracting from L_1 ,

$$\begin{aligned} \log W &= \ell - \sum_{n \geq 1} \mathsf{L}_n^{\text{Lam}}(\ell) r^n \\ &= \ell - \ell r + \left(\ell - \frac{\ell^2}{2}\right) r^2 + \left(-\frac{\ell^3}{3} + \frac{3}{2}\ell^2 - \ell\right) r^3 \\ &\quad + \left(-\frac{\ell^4}{4} + \frac{11}{6}\ell^3 - 3\ell^2 + \ell\right) r^4 + \mathcal{O}_r^{\text{form}}(r^5). \end{aligned}$$

No series logarithm was computed: the identity $\log W = L_1 - W$ did all the work. This is the same device as [Theorem 8.10](#) used in closed form.

(2) Factor $W = L_1(1 + \varepsilon)$ with

$$\begin{aligned} \varepsilon &= -\ell r + \sum_{n \geq 1} \mathsf{L}_n^{\text{Lam}}(\ell) r^{n+1} \\ &= -\ell r + \ell r^2 + \left(\frac{\ell^2}{2} - \ell\right) r^3 + \left(\frac{\ell^3}{3} - \frac{3}{2}\ell^2 + \ell\right) r^4 + \mathsf{L}_4^{\text{Lam}}(\ell) r^5 + \mathsf{L}_5^{\text{Lam}}(\ell) r^6 + \dots, \end{aligned}$$

so $\nu_t \varepsilon = 1$ in the r -adic filtration and the geometric series $\sum_k (-\varepsilon)^k$ is summable ([Theorem E.3](#)). Then $1/W = r \sum_{k \geq 0} (-\varepsilon)^k$. Collecting by powers of r , with $e_1 = -\ell$, $e_2 = \ell$, $e_3 = \mathsf{L}_2^{\text{Lam}}(\ell)$, $e_4 = \mathsf{L}_3^{\text{Lam}}(\ell)$, $e_5 = \mathsf{L}_4^{\text{Lam}}(\ell)$, $e_6 = \mathsf{L}_5^{\text{Lam}}(\ell)$ the coefficients of $\varepsilon = \sum_k e_k r^k$:

$$\begin{aligned} \frac{1}{W} &= r + \ell r^2 + (\ell^2 - \ell) r^3 + \left(\ell^3 - \frac{5}{2}\ell^2 + \ell\right) r^4 \\ &\quad + \left(\ell^4 - \frac{13}{3}\ell^3 + \frac{9}{2}\ell^2 - \ell\right) r^5 + \left(\ell^5 - \frac{77}{12}\ell^4 + \frac{71}{6}\ell^3 - 7\ell^2 + \ell\right) r^6 \\ &\quad + \left(\ell^6 - \frac{87}{10}\ell^5 + \frac{145}{6}\ell^4 - \frac{155}{6}\ell^3 + 10\ell^2 - \ell\right) r^7 + \mathcal{O}_r^{\text{form}}(r^8). \end{aligned} \quad (550)$$

The blocks through r^6 are tabulated in the sources; the r^7 block is computed here. As a sample of the bookkeeping, the coefficient of r^5 is $-e_4 + (2e_1e_3 + e_2^2) - 3e_1^2e_2 + e_1^4$, which evaluates to $-\frac{1}{3}\ell^3 + \frac{3}{2}\ell^2 - \ell - \ell^3 + 2\ell^2 + \ell^2 - 3\ell^3 + \ell^4 = \ell^4 - \frac{13}{3}\ell^3 + \frac{9}{2}\ell^2 - \ell$.

Three structural checks now follow, each provable in a line and each visible in (550). Write

$$\frac{1}{W} = r \sum_{k \geq 0} c_k(\ell) r^k,$$

so that c_k is the block of $1/W$ at r^{k+1} .

Zero constant term. Setting $\ell = 0$ kills every $\mathsf{L}_n^{\text{Lam}}$ ((346)), hence $\varepsilon|_{\ell=0} = 0$ and $1/W|_{\ell=0} = r$; so $c_k(0) = 0$ for $k \geq 1$.

Monic of degree k . In $\varepsilon = \sum_{k \geq 1} e_k r^k$ the first coefficient has ℓ -degree 1 and, for $k \geq 2$, $e_k = \mathsf{L}_{k-1}^{\text{Lam}}(\ell)$ has ℓ -degree $k-1$. A monomial $e_{i_1} \cdots e_{i_m}$ with $i_1 + \cdots + i_m = k$ therefore has degree at most k minus the number of indices $i_j \geq 2$, which equals k only when every $i_j = 1$. So the top-degree part of c_k comes from $(-e_1)^k = \ell^k$ alone, and $[\ell^k]c_k = 1$.

Linear coefficient. As ε vanishes at $\ell = 0$, the derivative of $(1 + \varepsilon)^{-1}$ there is $-\partial_\ell \varepsilon$, and by $\mathsf{L}_n^{\text{Lam}'}(0) = (-1)^{n-1}$ ([Theorem 19.7\(v\)](#)),

$$-\partial_\ell \varepsilon|_{\ell=0} = r + \sum_{n \geq 1} (-1)^n r^{n+1}.$$

Hence $[\ell]c_k = (-1)^{k-1}$ for $k \geq 1$, which is the alternation visible in (550).

Finally, (550) is, block for block, the bracket in (549): $(X \log X)^{\langle -1 \rangle \circ} (Y) = Y/W(Y)$, so the two computations are the same one.

(3) From $We^W = z$, $e^W = z/W = z \cdot (1/W)$, and (550) supplies $1/W$ already. No exponential of a series is ever formed, which matters because Theorem 8.16 shows that exponentiating a block of L -degree ≥ 2 leaves every power–log scale. $\square$

Exercise E.49 (The fifth and sixth Lambert polynomials, three ways (hard)). Compute L_5^{Lam} and L_6^{Lam} from the operator definition (344), from the integral recurrence, and from the Stirling formula, using

$$\left(\begin{bmatrix} 5 \\ 5 \end{bmatrix}, \dots, \begin{bmatrix} 5 \\ 1 \end{bmatrix}\right) = (1, 10, 35, 50, 24), \quad \left(\begin{bmatrix} 6 \\ 6 \end{bmatrix}, \dots, \begin{bmatrix} 6 \\ 1 \end{bmatrix}\right) = (1, 15, 85, 225, 274, 120).$$

Then verify the coefficient identities of Theorem 19.7(iii)–(vii) on both.

Solution. Answers.

$$\begin{aligned} L_5^{\text{Lam}}(u) &= \frac{u^5}{5} - \frac{25}{12}u^4 + \frac{35}{6}u^3 - 5u^2 + u = \frac{24u^5 - 250u^4 + 700u^3 - 600u^2 + 120u}{120}, \\ L_6^{\text{Lam}}(u) &= \frac{u^6}{6} - \frac{137}{60}u^5 + \frac{75}{8}u^4 - \frac{85}{6}u^3 + \frac{15}{2}u^2 - u \\ &= \frac{120u^6 - 1644u^5 + 6750u^4 - 10200u^3 + 5400u^2 - 720u}{720}. \end{aligned}$$

Stirling route. The two rows are the usual cycle numbers [4, 18]. By (350),

$$L_n^{\text{Lam}}(u) = \sum_{m=1}^n (-1)^{n-m} \begin{bmatrix} n \\ n-m+1 \end{bmatrix} \frac{u^m}{m!}.$$

For $n = 5$ the signs are $(-1)^{5-m}$, i.e. $+, -, +, -, +$ for $m = 1, \dots, 5$:

$$L_5^{\text{Lam}} = \begin{bmatrix} 5 \\ 5 \end{bmatrix} u - \begin{bmatrix} 5 \\ 4 \end{bmatrix} \frac{u^2}{2} + \begin{bmatrix} 5 \\ 3 \end{bmatrix} \frac{u^3}{6} - \begin{bmatrix} 5 \\ 2 \end{bmatrix} \frac{u^4}{24} + \begin{bmatrix} 5 \\ 1 \end{bmatrix} \frac{u^5}{120} = u - 5u^2 + \frac{35}{6}u^3 - \frac{25}{12}u^4 + \frac{u^5}{5}.$$

For $n = 6$ the signs are $(-1)^{6-m}$, i.e. $-, +, -, +, -, +$:

$$L_6^{\text{Lam}} = -u + \frac{15}{2}u^2 - \frac{85}{6}u^3 + \frac{225}{24}u^4 - \frac{274}{120}u^5 + \frac{120}{720}u^6,$$

which is the stated answer since $225/24 = 75/8$, $274/120 = 137/60$ and $120/720 = 1/6$.

Recurrence route. From L_4^{Lam} of Theorem E.42, $L_5^{\text{Lam}} = -L_4^{\text{Lam}} + 4 \int_0^u L_4^{\text{Lam}}$, giving $-(\frac{1}{4}u^4 - \frac{11}{6}u^3 + 3u^2 - u) + 4(\frac{1}{20}u^5 - \frac{11}{24}u^4 + u^3 - \frac{1}{2}u^2) = \frac{1}{5}u^5 - \frac{25}{12}u^4 + \frac{35}{6}u^3 - 5u^2 + u$. Then $L_6^{\text{Lam}} = -L_5^{\text{Lam}} + 5 \int_0^u L_5^{\text{Lam}}$ gives the second answer by the same arithmetic.

Operator route. $L_5^{\text{Lam}} = \frac{1}{720}(\partial_u - 4)(\partial_u - 3)(\partial_u - 2)(\partial_u - 1)\partial_u u^6$; applying the five factors in turn to $6u^5$ reproduces the same polynomial. This is the only one of the three routes that presupposes none of the derived identities.

Coefficient checks. With $H_4 = \frac{25}{12}$ and $H_5 = \frac{137}{60}$: (iii) $[u^5]L_5^{\text{Lam}} = \frac{1}{5}$ and $[u^6]L_6^{\text{Lam}} = \frac{1}{6}$, i.e. $1/n$. (iv) $[u^4]L_5^{\text{Lam}} = -\frac{25}{12} = -H_4$ and $[u^5]L_6^{\text{Lam}} = -\frac{137}{60} = -H_5$. (v) $[u]L_5^{\text{Lam}} = 1 = (-1)^4$ and $[u]L_6^{\text{Lam}} = -1 = (-1)^5$. (vii) $[u^2]L_5^{\text{Lam}} = -5 = (-1)^5 \frac{5 \cdot 4}{4}$ and $[u^2]L_6^{\text{Lam}} = \frac{15}{2} = (-1)^6 \frac{6 \cdot 5}{4}$. Finally (vi): $[u^3]L_5^{\text{Lam}} = \frac{35}{6}$ should equal $\frac{n-1}{2}(H_{n-1}^2 - H_{n-1}^{(2)}) = 2(\frac{625}{144} - \frac{205}{144}) = 2 \cdot \frac{420}{144} = \frac{35}{6}$, which it does. Multiplying by $n!$ turns every coefficient into an integer, as Theorem 19.9 requires: $L_5^{\text{Lam,sc}}$ and $L_6^{\text{Lam,sc}}$ are the numerators displayed above. $\square$

Exercise E.50 (The branch point forces a square root, not a logarithm (hard)). At $z = -e^{-1}$ the two real branches of W meet at $W = -1$.

1. Set $W = -1 + q$ and expand $1 + ez$ in powers of q through q^4 .

2. With $p = \sqrt{2(1+ez)}$, derive $W_0(z) = -1 + p - \frac{1}{3}p^2 + \frac{11}{72}p^3 - \frac{43}{540}p^4 + \dots$ and W_{-1} by $p \mapsto -p$.
3. Explain, from the local structure of we^w , why a square root rather than an iterated logarithm is forced here, and why the residual-based error estimate of [Theorem 14.21](#) degrades near this point.

Solution. (1) With $w = -1 + q$,

$$ez = e we^w = (-1 + q)e^q,$$

and multiplying $-1 + q$ into $\sum_{k \geq 0} q^k/k!$ the coefficient of q^n for $n \geq 2$ is $\frac{1}{(n-1)!} - \frac{1}{n!} = \frac{n-1}{n!}$, while the constant term is -1 and the coefficient of q is $-1 + 1 = 0$. Hence

$$1 + ez = \sum_{n \geq 2} \frac{(n-1)q^n}{n!} = \frac{q^2}{2} + \frac{q^3}{3} + \frac{q^4}{8} + \frac{q^5}{30} + \mathcal{O}_{q \rightarrow 0}(q^6), \quad (551)$$

an entire function of q ; this is [Theorem 20.23](#).

(2) Then $p^2 = 2(1 + ez) = q^2(1 + \frac{2}{3}q + \frac{1}{4}q^2 + \frac{1}{15}q^3 + \dots)$, so, choosing the branch with $p \sim q$,

$$p = q \left(1 + \frac{2}{3}q + \frac{1}{4}q^2 + \frac{1}{15}q^3 + \dots \right)^{1/2} = q + \frac{q^2}{3} + \frac{5q^3}{72} + \frac{11q^4}{1080} + \dots$$

Reverting this power series (a near-identity reversion in the ordinary one-variable sense — no logarithms are involved) gives

$$q = p - \frac{p^2}{3} + \frac{11}{72}p^3 - \frac{43}{540}p^4 + \frac{769}{17280}p^5 - \frac{221}{8505}p^6 + \dots,$$

whence $W_0 = -1 + q$ as stated; the coefficients agree with the standard tabulations [5, 11]. Since p enters only through $p^2 = 2(1 + ez)$, replacing p by $-p$ gives the second solution of $we^w = z$ near $w = -1$, namely $W_{-1}(z) = -1 - p - \frac{1}{3}p^2 - \frac{11}{72}p^3 - \frac{43}{540}p^4 - \dots$. That the reverted series has a positive radius of convergence, and that the two displayed branches are exactly the two solutions there, is [Theorem 20.24](#); the formal reversion above supplies the coefficients but not the radius.

(3) The map $w \mapsto we^w$ has derivative $(1+w)e^w$, which vanishes simply at $w = -1$; the second derivative $(2+w)e^w$ equals $e^{-1} \neq 0$ there. So $z + e^{-1}$ has a simple *quadratic* zero in w at $w = -1$, and inverting a quadratic critical point produces a two-valued square-root branch, with the two branches exchanged by $p \mapsto -p$. Nothing logarithmic occurs: the logarithmic scale of [Theorem 18.10](#) comes from the leading behaviour of we^w at $w \rightarrow +\infty$, an entirely different regime.

Near the branch point the residual-to-error transfer degrades exactly because W' blows up there. [Theorem 14.21](#) bounds $|x_N - x_*|$ by $|f(x_N) - y|/m$ with $m = \inf |f'|$ on the interval between; here $f'(w) = (1+w)e^w \rightarrow 0$ as $w \rightarrow -1$, so $m \rightarrow 0$ and the bound becomes vacuous. Concretely, a small residual no longer certifies a small error: near a quadratic critical point the error is of order the square root of the residual, not of order the residual. $\square$

E.7 Analytic transfer

Exercise E.51 (From residual to error, on a ray). Let $F(y) = y + \log y$ for $y > 0$, let $X \in \mathbb{R}$, and let $y_* > 0$ be the unique solution of $F(y) = X$.

1. Show that every $y_N > 0$ satisfies $|y_N - y_*| \leq |y_N + \log y_N - X|$, and identify the general principle at work.
2. Sharpen this to a two-sided estimate: if $y_0 > 0$ and both y_N and y_* lie in $[y_0, \infty)$, then

$$\frac{|F(y_N) - X|}{1 + 1/y_0} \leq |y_N - y_*| \leq |F(y_N) - X|.$$

Deduce that on such a ray the residual is not merely an upper bound for the error but a proxy for it, correct to within the factor $1 + 1/y_0$.

3. Explain why the same argument gives nothing for $F(y) = y - \log y$ on $(0, 1)$.

Solution. F is continuously differentiable on $(0, \infty)$ with $F'(y) = 1 + 1/y > 1$, so F is strictly increasing, hence injective, and $F((0, \infty)) = \mathbb{R}$, so y_* exists and is unique (Theorem 18.1). By the mean value theorem there is θ between y_N and y_* with

$$F(y_N) - X = F(y_N) - F(y_*) = F'(\theta)(y_N - y_*),$$

and $\theta > 0$ gives $F'(\theta) > 1$. Hence $|y_N - y_*| = |F(y_N) - X| / F'(\theta) \leq |F(y_N) - X|$, which is the claim.

The principle is Theorem 14.21: if $|f'| \geq m > 0$ between the approximation and the true point, then the error is at most the residual divided by m ; here $m = 1$ works on the whole ray. Two hypotheses are doing real work, and both are analytic: the derivative bound, and the fact that y_N and y_* lie in a common interval on which it holds.

(2) The same θ now satisfies $y_0 \leq \theta$, because θ lies between y_N and y_* and both are $\geq y_0$. Since $F'(y) = 1 + 1/y$ is decreasing, $1 < F'(\theta) \leq 1 + 1/y_0$, and dividing $F(y_N) - X = F'(\theta)(y_N - y_*)$ by $F'(\theta)$ gives both inequalities at once. So the residual determines the error to within the factor $1 + 1/y_0$, which tends to 1 as $y_0 \rightarrow \infty$. This is the two-sided enclosure of Theorem 23.2, with $m = 1$ and $M = 1 + 1/y_0$; that theorem states it on a compact interval and adds that injectivity need not be assumed, being a conclusion.

(3) For $F(y) = y - \log y$ on $(0, 1)$ one has $F'(y) = 1 - 1/y < 0$, and $F'(y) \rightarrow 0$ as $y \rightarrow 1^-$. So $m = \inf |F'| = 0$ on any interval reaching to 1, and the bound degenerates. That is not an artefact: $y = 1$ is the critical point of $y - \log y$, the two branches of its inverse meet there, and near it the error is of order the square root of the residual — the same phenomenon as at the Lambert branch point (Theorem E.50). $\square$

Exercise E.52 (Trap: a formal tail read as an analytic estimate). The following argument is offered for the asymptotic validity of the Wright omega truncations.

“By Theorem 18.13, $\text{Err}_{\text{rev}}(X + L, \omega_N) = -\mathbb{L}_{N+1}^{\text{Lam}}(L)t^{N+1} + \mathcal{O}_t^{\text{form}}(t^{N+2})$. Substituting x for X and $\log x$ for L , the residual of $\omega_N(x)$ is $-\mathbb{L}_{N+1}^{\text{Lam}}(\log x)x^{-(N+1)} + \mathcal{O}_{x \rightarrow +\infty}(x^{-(N+2)})$. Since $\left| \frac{d}{dy}(y + \log y) \right| > 1$ for $y > 0$, Theorem 14.21 converts this into

$$\omega(x) - \omega_N(x) = \mathbb{L}_{N+1}^{\text{Lam}}(\log x)x^{-(N+1)} + \mathcal{O}_{x \rightarrow +\infty}(x^{-(N+2)}).$$

Hence the truncations are asymptotically valid.”

Locate every unjustified step, say which parts of the conclusion are nevertheless true, and state what has to be proved to repair it.

Solution. There is exactly one unjustified step, and it is the second sentence.

The step. “Substituting x for X and $\log x$ for L ” in an identity of $\mathbb{K}[L][[t]]$ is not an operation on functions. The symbol $\mathcal{O}_t^{\text{form}}(t^{N+2})$ records that a formal series has outer valuation at least $N + 2$; it asserts no inequality, contains no constant, and names no domain. There is no theorem in the volume, and no theorem at all, that converts it into $\mathcal{O}_{x \rightarrow +\infty}(x^{-(N+2)})$ without further hypotheses. Two independent obstructions:

- the formal tail is an *infinite* sum of blocks, and its substitution need not converge or even be defined pointwise;
- even if a bound exists, its blocks carry powers of L , so the honest analytic remainder is of the form $\mathcal{O}_{x \rightarrow +\infty}(L^K x^{-(N+2)})$ for some K , *not* $\mathcal{O}_{x \rightarrow +\infty}(x^{-(N+2)})$. The two differ by an unbounded factor.

The third and fourth sentences are then fine *given* the second: the derivative bound and [Theorem 14.21](#) do transfer a residual estimate to an error estimate, as in [Theorem E.51](#).

What is true. The conclusion is correct in shape and wrong in detail. [Theorem 18.17](#) establishes

$$\omega(x) - \omega_N(x) = \frac{\mathbb{L}_{N+1}^{\text{Lam}}(\log x)}{x^{N+1}} + \mathcal{O}_{x \rightarrow +\infty} \left(\frac{(\log x)^K}{x^{N+2}} \right), \quad K = (N+2)^2,$$

and hence (341). The logarithmic power K cannot be removed. The next block alone contributes the term $\mathbb{L}_{N+2}^{\text{Lam}}(\log x) x^{-(N+2)}$. Its exact size is

$$\asymp (\log x)^{N+2} x^{-(N+2)}, \quad \text{which is not } \mathcal{O}_{x \rightarrow +\infty} \left(x^{-(N+2)} \right).$$

The repair. What must be supplied, and what the formal identity cannot supply, is a genuine analytic estimate for the substituted tail: an explicit finite sum of blocks plus a remainder bounded on an explicit ray, with the constant and the exponent K named. That is item (2) of the proof of [Theorem 18.17](#), and in general it is [Theorem 23.5](#), which converts a *finite* truncation’s formal residual order into an inequality $|\varphi(h(x)) - x| \leq Cx^{-N}(\log x)^M$ on a ray — note that its hypothesis makes B a finite sum of blocks, precisely so that substitution is a finite computation with numbers. The general lesson is the one printed at the head of this appendix: $\mathcal{O}_t^{\text{form}}(\cdot)$ and $\mathcal{O}_{X \rightarrow +\infty}(\cdot)$ are different assertions, and no amount of correct formal algebra produces the second. $\square$

Exercise E.53 (Counterexample: the Poincaré condition does not determine the blocks). A source proposes to define “ f has the asymptotic expansion $\sum_n p_n(\log x)x^{-n}$ ” by the minimal Poincaré requirement that, for each N , the remainder after the term $n = N$ be $\mathcal{O}_{x \rightarrow +\infty}(p_N(\log x)x^{-N})$. Show that this does not determine the blocks p_n , and that the definition actually used ([Theorem 6.43](#)) does.

Solution. Nonuniqueness. Fix N and take the two candidate blocks

$$p_N(u) = u^5, \quad q_N(u) = u^5 + u^3.$$

They differ by u^3 , and

$$(q_N(\log x) - p_N(\log x))x^{-N} = (\log x)^3 x^{-N} = \mathcal{O}_{x \rightarrow +\infty}((\log x)^5 x^{-N}).$$

So if f satisfies the Poincaré condition with block p_N it satisfies it with block q_N as well: the remainder after q_N differs from the remainder after p_N by something that is little- $\mathcal{O}$ of *either* of the two last retained terms. Hence the condition determines the block only up to anything of strictly smaller L -degree, which is to say not at all. The same construction works at every index and with u^3 replaced by any polynomial of degree < 5 .

The root cause is that the Poincaré condition compares the remainder with the last retained *term*, and on a two-generator scale a term is a whole block, whose parts have different growth. On a one-generator scale $\{x^{-n}\}$ the terms are monomials and the classical condition [25] is fine; on $\{(\log x)^m x^{-n}\}$ it is not.

What does determine them. [Theorem 6.43](#), and its extension to rational blocks in [Theorem 22.4](#), compare the remainder not with the last term but with the *next outer order*: for every N there are M_N, C_N, x_N with

$$|f(x) - (\text{Trunc}_{<N} F)(x)| \leq C_N x^{-N} (\log x)^{M_N} \quad (x \geq x_N).$$

Suppose F and G both satisfy this and $F \neq G$; let $n_0 = \nu_t(F - G)$ and $p = [t^{n_0}](F - G) \neq 0$. Applying the condition at $N = n_0 + 1$ to both and subtracting,

$$|p(\log x)x^{-n_0} + (\text{lower-order blocks})| \leq C x^{-n_0-1} (\log x)^M$$

for large x , where the omitted blocks have outer order $> n_0$. Dividing by x^{-n_0} , the left side tends to ∞ or to a nonzero constant in absolute value if $p \neq 0$ – indeed $|p(\log x)| \geq c(\log x)^{\deg_L p}$ for large x – while the right side is $Cx^{-1}(\log x)^M \rightarrow 0$. Contradiction; so $F = G$. Uniqueness holds – this is [Theorem 22.8](#), which also covers rational blocks and closed subsectors – and the expansion map of [Theorem 6.45](#) is therefore well defined. $\square$

Exercise E.54 (Trap: a guard that was not taken in the analytic estimate). Let $f(X) = X^2$ exactly, so f realises $F = t^{-2}$ with no error at all, and let g satisfy $g - (\text{Trunc}_{<N} G) = \mathcal{O}_{X \rightarrow +\infty}(X^{-N})$ for some $G \in \mathcal{P}\mathcal{L}_{\text{pow}, \mathbb{K}}$. A draft concludes:

“ $fg - (\text{Trunc}_{<N} F)(\text{Trunc}_{<N} G) = f(g - \text{Trunc}_{<N} G) = \mathcal{O}_{X \rightarrow +\infty}(X^{-N})$, since the first factor is exact and the second remainder is $\mathcal{O}_{X \rightarrow +\infty}(X^{-N})$.”

Find the error, give the correct exponent, and state the general precision rule for an analytic product.

Solution. The error is that the first factor is not $\mathcal{O}_{X \rightarrow +\infty}(1)$; it is X^2 . Multiplying a remainder of size X^{-N} by X^2 gives

$$f(g - \text{Trunc}_{<N} G) = \mathcal{O}_{X \rightarrow +\infty}(X^{-N+2}),$$

two orders worse than claimed. To obtain $\mathcal{O}_{X \rightarrow +\infty}(X^{-N})$ in the product one must know g to precision $N + 2$, i.e. $g - \text{Trunc}_{<N+2} G = \mathcal{O}_{X \rightarrow +\infty}(X^{-N-2})$.

General rule. Let $f \approx F$, $g \approx G$ with $a = \nu_t F$, $b = \nu_t G$. Suppose

$$f - \text{Trunc}_{<P} F = \mathcal{O}_{X \rightarrow +\infty}(X^{-P}(\log X)^M), \quad g - \text{Trunc}_{<Q} G = \mathcal{O}_{X \rightarrow +\infty}(X^{-Q}(\log X)^{M'}).$$

Write

$$fg - \text{Trunc}_{<P} F \text{Trunc}_{<Q} G = \text{Trunc}_{<P} F(g - \text{Trunc}_{<Q} G) + (f - \text{Trunc}_{<P} F)g.$$

The truncation $\text{Trunc}_{<P} F$ has outer order a , and g has outer order b ; so as functions on the ray they are of size X^{-a} and X^{-b} up to powers of $\log X$. The two summands are therefore of sizes

$$\mathcal{O}_{X \rightarrow +\infty}(X^{-a-Q}(\log X)^*) \quad \text{and} \quad \mathcal{O}_{X \rightarrow +\infty}(X^{-P-b}(\log X)^*)$$

respectively. So the product is known to absolute order N if and only if

$$P \geq N - b \quad \text{and} \quad Q \geq N - a,$$

which is verbatim the formal truncated-product rule (49) of [Theorem 5.4](#), and its error-tolerant version [Theorem 6.14](#). The analytic guard and the formal guard coincide; what does *not* coincide is the naive symmetric rule $P = Q = N$, which is valid only when $a, b \geq 0$. In the example $a = -2$, $b \geq 0$, so $Q \geq N + 2$, which is the two orders that went missing.

Two riders. First, the logarithmic powers accumulate: the exponent in the product’s remainder is at least $M + \deg_L$ of the retained blocks, so a statement of the form $\mathcal{O}_{X \rightarrow +\infty}(X^{-N})$ with no logarithmic factor is almost never available ([Theorem E.52](#)). Second, the same arithmetic with two *inexact* factors is where a draft of the composition transfer went wrong: bounding a product of remainders by the smaller of the two exponents is legitimate only when each remainder is multiplied by a factor of nonpositive outer order, and the hypothesis $N \geq a + \tilde{a}$ is exactly what makes that so. $\square$

Exercise E.55 (Realizations form an algebra, and the map that forgets). Let $\mathbb{K} \subseteq \mathbb{C}$, $S = [X_0, +\infty)$ with the real logarithm, and $\mathcal{A}(S) = \{f: S \rightarrow \mathbb{C} : \exists F \in \mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}, f \approx F\}$ in the sense of [Theorem 6.43](#).

1. Show F is uniquely determined by f .
2. Show $\mathcal{A}(S)$ is a $\mathbb{K}$ -algebra and $f \mapsto F$ is a $\mathbb{K}$ -algebra homomorphism into $\mathcal{P}\mathcal{L}_{\text{poly}, \mathbb{K}}$. (For surjectivity, quote [Theorem 22.13](#); do not attempt it by hand.)

3. Identify its kernel and show that it is not zero, so that the homomorphism is not injective; give an explicit nonzero element.
4. Deduce that “ $f \approx F$ ” can never be upgraded to “ f is the sum of F ” without extra hypotheses.

Solution. (1) This is the uniqueness argument of [Theorem E.53](#): if $f \approx F$ and $f \approx G$ with $F \neq G$, let $n_0 = \nu_t(F - G)$ and $p = [t^{n_0}](F - G) \neq 0$. Subtracting the two estimates at cutoff $N = n_0 + 1$ gives $|(F - G)(X)| \leq CX^{-n_0-1}(\log X)^M$ for large X , where $(F - G)(X) = p(\log X)X^{-n_0} +$ (blocks of order $> n_0$); dividing by X^{-n_0} makes the left side bounded below by a positive constant times $(\log X)^{\deg_L p}$ and the right side tend to 0. Contradiction.

(2) Linearity is immediate from the triangle inequality and linearity of $\text{Trunc}_{<N} \cdot$. For products, [Theorem 6.45](#) gives $fg \approx FG$; its proof is the guard computation of [Theorem E.54](#) run at $P = N - b$, $Q = N - a$, which is legitimate because $f \approx F$ supplies an estimate at *every* cutoff, in particular at those two. The constant function 1 realises 1, so the map is unital. Surjectivity is the Borel–Ritt theorem on this scale, [Theorem 22.13](#): every $F \in \mathcal{PL}_{\text{poly}, \mathbb{K}}$ is realised by some C^∞ function, real valued when $\mathbb{K} \subseteq \mathbb{R}$. It is worth noticing what that theorem does *not* say – it produces a smooth realisation, not an analytic one, and it produces no canonical choice, since by (3) below any two realisations differ by a flat function.

(3) The kernel is the ideal $\mathcal{N}$ of *flat* functions of [Theorem 22.11](#), $\{f : f = \mathcal{O}_{X \rightarrow +\infty}(X^{-N}) \text{ for every } N\}$, which is exactly the condition $f \approx 0$. It is an ideal, being the kernel of a ring homomorphism, and it is not zero: $f(X) = e^{-X}$ is nonzero everywhere on S and satisfies $e^{-X}X^N \rightarrow 0$ for every N , so $e^{-X} \approx 0$. Hence e^{-X} and 0 have the same expansion, and the map is not injective. A second witness, of a different character: $\sin(e^X)$ is bounded and oscillatory, but $e^{-X} \sin(e^X)$ is again flat and nonzero.

(4) Consequently the expansion F determines f only modulo flat functions, and $\mathcal{N}$ is large: by [Theorem 22.12](#) it is prime but not maximal, is not stable under differentiation, and contains no smallest nonzero scale, since $f \in \mathcal{N}$ eventually nonvanishing gives $f^2 \in \mathcal{N}$ with $f^2 = o_{X \rightarrow +\infty}(f)$. “ f is the sum of F ” would assert both that the series converges and that its sum is f ; the first is a claim about $\sum_n p_n(\log X)X^{-n}$ as a numerical series and the second would force injectivity, which (3) refutes. What $\approx$ does assert is exactly (92): a family of finite inequalities, one per cutoff, each on its own ray. Any statement stronger than that needs a summation method and its own theorem. $\square$

Exercise E.56 (Exactness at a point, and what the numbers do and do not show). 1. Show that every truncation is exact at the single point $x = 1$, that is, that $\omega_N(1) = \omega(1) = 1$ for every N ; and explain why this is not a convergence statement.

2. Take $x = 20$, so that $L = \log 20 = 2.995732 \dots$. The terms of the series, and the errors after each of them, are, to four significant figures,

n	$\mathcal{L}_n^{\text{Lam}}(L) 20^{-n}$	$\omega(20) - \omega_n(20)$
1	1.498×10^{-1}	3.507×10^{-3}
2	3.729×10^{-3}	-2.220×10^{-4}
3	-1.880×10^{-4}	-3.395×10^{-5}
4	-3.267×10^{-5}	-1.282×10^{-6}
5	-1.433×10^{-6}	1.503×10^{-7}
6	1.259×10^{-7}	2.443×10^{-8}
7	2.353×10^{-8}	8.975×10^{-10}
8	1.047×10^{-9}	-1.496×10^{-10}

Check the table against (340), and against the sign prediction of [Theorem E.44](#).

3. The terms decrease monotonically through $n = 8$ and beyond. Does that show that the series $\sum_n \mathcal{L}_n^{\text{Lam}}(\log x) x^{-n}$ converges at $x = 20$? What does this volume actually prove?

Solution. (1) At $x = 1$ one has $L = \log 1 = 0$, and $\mathbb{L}_n^{\text{Lam}}(0) = 0$ for every n by (346). Hence $\omega_N(1) = 1 - \mathbb{L}_0^{\text{Lam}}(0) \cdot (-1) + \sum_{n=1}^N \mathbb{L}_n^{\text{Lam}}(0) = 1 - 0 + 0 = 1$. And $\omega(1) = 1$ because $1 + \log 1 = 1$. So every truncation is exact at that one point.

This says nothing about convergence, and it is worth being explicit about why. The statement “ $\omega_N(1) = \omega(1)$ for all N ” is a family of identities between numbers at a single x . An asymptotic statement $\mathcal{O}_{x \rightarrow +\infty}(\cdot)$ has no content at a single point — multiplying a function by any factor that equals 1 at $x = 1$ changes nothing about it. And a convergence statement would be about the tail $\sum_{n>N}$, which here is identically zero at $x = 1$ for a trivial reason: every summand vanishes. The same phenomenon in Lambert coordinates is the exactness of the branch expansion at $z = e$, where $\xi_0 = 1$ and $\ell_0 = 0$; this is Theorem 18.16 and Theorem 20.19, and it is a consequence of $\mathbb{L}_n^{\text{Lam}}(0) = 0$, of nothing else. Indeed the convergence criterion of the Lambert part fails outright at that point.

(2) Theorem 18.17 asserts $\omega(x) - \omega_N(x) = \mathbb{L}_{N+1}^{\text{Lam}}(L)x^{-(N+1)} + \mathcal{O}_{x \rightarrow +\infty}(L^K x^{-(N+2)})$. Reading the table, the error after N should be close to the term at $n = N+1$, and it is, in every row: 3.507×10^{-3} against 3.729×10^{-3} ; -2.220×10^{-4} against -1.880×10^{-4} ; -3.395×10^{-5} against -3.267×10^{-5} ; and so on down to 8.975×10^{-10} against 1.047×10^{-9} . Both magnitude and sign agree throughout, which is (341) in action.

The signs are themselves informative. Here $L = 2.9957 > 2$, so $\mathbb{L}_2^{\text{Lam}}(L) > 0$ and ω_1 undershoots, in accordance with Theorem E.44 (2); the later signs follow the sign of $\mathbb{L}_n^{\text{Lam}}(L)$, which changes as L passes the real roots of the successive Lambert polynomials. There is no alternating pattern in n to be read off, and none should be expected: the sign is the sign of a polynomial value, not of a coefficient.

(3) No. Monotone decrease of finitely many computed terms is evidence, not proof, and in an asymptotic series the terms typically decrease for a long time before turning. What this volume proves is three separate things, and none of them is convergence of the substituted series:

- the *formal* identity (335), an equality in $\mathcal{G}_{\text{poly}, \mathbb{K}}$;
- the *asymptotic* statement (340), valid as $x \rightarrow +\infty$ with an explicit logarithmic loss $K = (N+2)^2$;
- the *frozen* convergence of Theorem 18.19: for each fixed u_0 the series $\sum_{n \geq 1} \mathbb{L}_n^{\text{Lam}}(u)s^n$ has a positive radius in s , uniformly near u_0 .

The third is the reason the numbers look convergent: at $x = 20$ the pair $(s, u) = (1/20, \log 20)$ is one point, and the frozen series at $u = \log 20$ has some positive radius which $1/20$ may well be inside. But in the asymptotic regime $u = \log x$ and $s = 1/x$ are linked and $u \rightarrow \infty$, so the frozen radius $r(u)$ is itself moving, and no convergence statement about the substituted series follows. This volume does not determine whether that series converges; the honest reading of the table is the sharp-order statement of (2), and nothing more. $\square$

Remark E.57 (Where each exercise came from). Of the 56 exercises above, 31 descend from a source problem — in most cases from several at once, since the corpus’s 83 problems are heavily duplicated, and in most cases with the statement strengthened, the hypotheses printed, or the supplied answer corrected; 9 were built from worked examples in reports 4, 5 and 6, which set no exercises at all; and 16 are new, written because a technique of the body had no exercise anywhere in the corpus. Every trap and every counterexample exercise — 8 and 11 respectively — is in the last two groups.

The source problems that left no descendant were of three kinds: duplicates of a retained problem, arithmetic drill on an operation already exercised elsewhere, and problems whose *supplied* solution is wrong. Of the last kind there are five, all corrected above and all recorded in Section G: the class in which $(L+t)^{-1}$ lives, the universal third inverse block, the sign weight in the lower Lambert branch, the range of the Stirling sum for $\Delta_{0,k}$, and the passage from a formal residual identity to an analytic error bound.

F Notation, and a concordance with the six sources

F.1 The normative source

Every mathematical symbol in this volume is defined normatively in the corpus notation catalogue, *FabiusFunction Mathematical Notation Catalogue*, whose entries polynomial-logarithmic-ambient, polynomial-logarithmic-leading-data, polynomial-logarithmic-truncation, polynomial-logarithmic-differential, polynomial-logarithmic-composition, polynomial-logarithmic-reversion, wright-omega-lambert-polynomials, and lambert-branch-expansions cover exactly the material of this volume. This document introduces no new mathematical notation. Its preamble defines short local aliases — $\mathcal{PL}_{\text{pow},\mathbb{K}}$, $\mathcal{PL}_{\text{poly},\mathbb{K}}$, $\mathcal{PL}_{\text{rat},\mathbb{K}}$, $\mathcal{PL}_{\text{itLaur},\mathbb{K}}$, $\mathcal{G}_{\text{poly},\mathbb{K}}$, ν_t , lm , lc and the rest — and every one of them expands to a catalogue command; the aliases exist to keep displayed formulas readable, not to name anything the catalogue does not already name.

F.2 Symbols used throughout

Table 31: Principal notation.

Symbol	Alias	Meaning
X	—	the large variable; the default regime is $X \rightarrow +\infty$ along the positive real ray
t	—	X^{-1} , the outer generator
L	—	$\log X$, the logarithmic generator, formally independent of t
$\mathbb{K}$	<code>\K</code>	the coefficient field, of characteristic zero
$\mathcal{PL}_{\text{pow},\mathbb{K}}$	<code>\PLpow</code>	$\mathbb{K}[L][[t]]$, the polynomial-log power-series ring
$\mathcal{PL}_{\text{poly},\mathbb{K}}$	<code>\PLlau</code>	$\mathbb{K}[L]((t))$, its Laurent extension: finitely many positive powers of X
$\mathcal{PL}_{\text{rat},\mathbb{K}}$	<code>\PLrat</code>	$\mathbb{K}(L)((t))$, the rational-log Laurent field
$\mathcal{PL}_{\text{itLaur},\mathbb{K}}$	<code>\PLit</code>	$\mathbb{K}((L^{-1}))((t))$, the iterated Laurent-log field
$\mathcal{G}_{\text{poly},\mathbb{K}}$	<code>\Gnear</code>	$\{X + A : A \in \mathcal{PL}_{\text{pow},\mathbb{K}}\}$, the near-identity group under composition
$\nu_t F$	<code>\ordt</code>	the outer valuation, with $\nu_t 0 = +\infty$
$\text{lm}(F)$, $\text{lc}(F)$	<code>\lm</code> , <code>\lc</code>	leading monomial and scalar leading coefficient; the leading <i>block</i> is the whole polynomial $p_{\nu_t F}$, a third and different object
$\deg_L p$	<code>\degL</code>	degree in L
$d_F(n)$	<code>\dprofile</code>	the log-degree profile, $-\infty$ where the block vanishes
$\text{Trunc}_{<N} F$	<code>\Trunc</code>	the <i>exclusive</i> truncation $\sum_{n < N} p_n(L)t^n$
$\mathcal{O}_t^{\text{form}}(t^N)$	—	a formal valuation tail: coefficient agreement below outer order N , and nothing analytic
$\mathcal{O}_{X \rightarrow \infty}(A(X))$	—	an analytic estimate, with the regime printed
D_X	<code>\DX</code>	the distinguished derivation $t\partial_L - t^2\partial_t$
$\Delta_{n,k}$	<code>\shiftop</code>	$\prod_{j=0}^{k-1} (\partial_L - n - j)$, empty and equal to the identity at $k = 0$
$F^{(-1)\circ}$	<code>\rev</code>	compositional inverse, $F \circ F^{(-1)\circ} = F^{(-1)\circ} \circ F = X$
F^{-1}	<code>\recip</code>	multiplicative inverse, $FF^{-1} = 1$
$\text{Err}_{\text{rev}}(F, H)$	<code>\RevErr</code>	the reversion residual $F \circ H - X$; <i>both</i> arguments are part of the object
ω	—	Wright omega, the inverse of $Y \mapsto Y + \log Y$

Symbol	Alias	Meaning
L_n^{Lam}	<code>\LamP</code>	the canonical zero-based Lambert polynomials
$L_n^{\text{Lam,sc}}$	<code>\LamPsc</code>	the scaled family $n!L_n^{\text{Lam}}$, for $n \geq 1$
$[n]_k$	<code>\stirlingone</code>	unsigned Stirling numbers of the first kind (cycle numbers)
$s(n, k)$	<code>\stirlingsigned</code>	signed Stirling numbers of the first kind
$\prec, \succ$	—	the dominance order on monomials

F.3 Concordance with the absorbed sources

The six sources used mutually incompatible local abbreviations, and in two cases the *same* abbreviation for different objects. A reader returning to a source through the Git history needs the following table.

Table 32: Local abbreviations of the absorbed sources and their meaning here. S1–S6 are as identified in Table 35.

Source	Local	This volume
S1	<code>\Plog, \Rlog, \Hlog</code>	$\mathcal{PL}_{\text{poly}, \mathbb{K}}, \mathcal{PL}_{\text{rat}, \mathbb{K}}, \mathcal{PL}_{\text{itLaur}, \mathbb{K}}$
S1	<code>\val, \trunc</code>	ν_t , exclusive $\text{Trunc}_{<N}$ ·
S1	<code>\invcomp</code>	$\langle -1 \rangle_\circ$
S2	<code>\val</code>	ν_t
S2	<code>\rev, \recip</code>	$\langle -1 \rangle_\circ, \cdot^{-1}$ (already the catalogue commands)
S2	<code>\Dop = D</code>	D_X
S2	<code>\normalexp</code>	normal-ordered shifted exponential; expanded inline here
S3	<code>\Rring, \Pring, \Fring, \Ggroup</code>	$\mathcal{PL}_{\text{pow}, \mathbb{K}}, \mathcal{PL}_{\text{poly}, \mathbb{K}}, \mathcal{PL}_{\text{itLaur}, \mathbb{K}}, \mathcal{G}_{\text{poly}, \mathbb{K}}$
S3	<code>\ordt, \D = D</code>	ν_t, D_X
S4	<code>\PL</code>	$\mathcal{PL}_{\text{poly}, \mathbb{K}}$ — <i>not</i> $\mathcal{PL}_{\text{pow}, \mathbb{K}}$; see below
S4	<code>\RL, \HL, \Gnear</code>	$\mathcal{PL}_{\text{rat}, \mathbb{K}}, \mathcal{PL}_{\text{itLaur}, \mathbb{K}}, \mathcal{G}_{\text{poly}, \mathbb{K}}$
S4	<code>\ordx, \degell</code>	$\nu_t, \deg_L$
S5	<code>\PL</code>	$\mathcal{PL}_{\text{pow}, \mathbb{K}}$ — <i>not</i> $\mathcal{PL}_{\text{poly}, \mathbb{K}}$; see below
S5	<code>\PLL, \QPL</code>	$\mathcal{PL}_{\text{poly}, \mathbb{K}}, \mathcal{PL}_{\text{rat}, \mathbb{K}}$
S5	<code>\D = D</code>	D_X
S6	<code>\PLpoly, \PLfield</code>	$\mathcal{PL}_{\text{poly}, \mathbb{K}}, \mathcal{PL}_{\text{itLaur}, \mathbb{K}}$
S6	<code>\EulerDerivation</code>	$\vartheta = XD_X$
S6	<code>\trunc{F}{N} = [F]_{\leq N}</code>	<i>inclusive</i> ; equals $\text{Trunc}_{<N+1} F$
S6	<code>\compinv</code>	$\langle -1 \rangle_\circ$

Two abbreviations that collide across sources

1. `\PL` denotes the *Laurent* ring $\mathbb{K}[L]((t))$ in S4 and the *power-series* ring $\mathbb{K}[L][[t]]$ in S5. A statement transported from one to the other without checking which ring is meant can silently acquire or lose the finitely many positive powers of X ; the unit criterion and the antiderivative obstruction both change. In S4 the power-series subring was written $[\mathcal{PL}_{\text{poly}, \mathbb{K}}]_{\geq 0}$, which is $\mathcal{PL}_{\text{pow}, \mathbb{K}}$.
2. `\trunc` is the exclusive $\text{Trunc}_{<N}$ in S1 and the inclusive $[\cdot]_{\leq N}$ in S6. Every cutoff index transported from S6 into this volume has been shifted by one block. A residual certificate stated with the wrong convention is off by exactly one order, which is precisely the amount that makes a correct algorithm look wrong and an incorrect one look right.

F.4 Conventions that are not merely notational

Three conventions are recorded here because getting them wrong changes theorems, not only glyphs.

1. **Empty products and sums.** $\Delta_{n,0} = 1$; the Lagrange–Bürmann coefficient product is empty at $r = 1$; $C_{0,0} = 1$ and $C_{m,0} = 0$ for $m > 0$; $s(0, 0) = 1$. Every statement below is written so that these conventions make the boundary case correct rather than exceptional.
2. **The three $\mathcal{O}$ roles.** A formal valuation tail, an analytic estimate, and an algorithmic cost bound are three different assertions and are printed with three different tokens. None may be inferred from another. In particular, $\mathcal{O}_t^{\text{form}}(t^N)$ carries no uniformity in L whatsoever, and a bound uniform in L is a separate theorem with separate hypotheses.
3. $t^N \mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ **is not an ideal of $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$.** In the Laurent ring t is a unit, so the principal ideal it generates is the whole ring. Quotient-algebra language is therefore available on $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}}$ and not on $\mathcal{P}\mathcal{L}_{\text{poly},\mathbb{K}}$; on the latter, use $\nu_t(F - G) \geq N$, the formal tail token, or equality of exclusive truncations.

G Ledger of editorial repairs

Every correction this consolidation made to its sources is marked at the point of repair in this document’s own $\text{\LaTeX}$ source with an `% ed. : comment`. The table below is generated from those marks, so it cannot drift out of step with the text. Repairs of pure notation — a symbol renamed to its catalogue command — are not listed; those are recorded once, in the concordance of Table 32. What is listed is every place where a source’s mathematical statement, proof, hypothesis, index range, or cost claim was wrong, incomplete, or overstated.

Three kinds of defect recur often enough to name. The first is an *omitted hypothesis*: a statement true under conditions the source did not print, most often a missing empty-product convention, a missing guard-precision requirement, or a missing restriction to $n \geq 1$. The second is *overclaiming*: an analytic conclusion drawn from formal algebra alone, typically a convergence or uniformity assertion that the argument does not support. The third is a *cost or sharpness claim* asserted without the count that would establish it.

Table 33: Editorial repairs, in document order.

Location	Repair
Theorem 2.8	source draft 5 justifies the lexicographic order only in the case $\alpha > \alpha'$, leaving the tie-breaking case $\alpha = \alpha'$ unproved. The display above proves both cases.
Theorem 2.10	the draft said $X^{\{1/2\}}$ “arises from inverting a non-integer leading power”. That is backwards, and contradicts the reading of the Fabius endpoint law in the remark on its status: a half-integer exponent is produced by inverting an “integer” leading power other than the first (X^2 inverts to $X^{\{1/2\}}$); an inner argument that is already Puiseux has left the class before the inversion begins.
Theorem 2.10	source draft 6 prints the refined scale as “..., $X^{\{-n\}}L^2$, $X^{\{-n\}}L$, $X^{\{-n\}}$, $X^{\{-n\}}L^{-1}$, ...”, which silently places negative powers of L inside the polynomial-logarithmic class. They are not there: coefficient blocks of $K[L]((t))$ are polynomials, so L -exponents are nonnegative, and L^{-1} belongs to the enlargements named above.

Location	Repair
Theorem 2.14	the draft called PLLau "the t-adic completion of the Laurent ring in (iii)". That is exactly the error item 7 of the normalizations remark warns against: t is a unit in $K[t, t^{-1}][L]$, so every power of the ideal (t) is the whole ring, the t-adic filtration is constant, and the associated completion collapses to zero. The correct construction completes the polynomial subring $K[L][t]$ and only then inverts t .
Why powers and logarithms belong in one	the draft attached the exponentiated form to the full display, where it is false — exponentiating <code>\eqref{plt:eq:mot-generic-implicit}</code> exponentiates the higher blocks as well. The equivalence holds for the two-term equation only, and is stated for it.
Theorem 2.21	the draft named the two parts of this estimate A and B, colliding with the $B = \log(1/y)$ of the corollary and the status remark that quote this proposition. The internal quantities are renamed A_1 and A_2 ; B is reserved for the inverse-endpoint variable throughout the subsection.
Why powers and logarithms belong in one	source draft 1 asserts convergence "on the principal real branch for $z \geq e$ ", supported only by a general citation. That threshold is not established anywhere in the six drafts, and the assertion also conflates a convergence claim with the separate, elementary observation that the series is exact at $z = e$. We keep the coefficient statement, proved below, and defer the convergence domain to Part VI with its own citation. the draft called the asymptotic display <code>\eqref{plt:eq:mot-W-two-orders}</code> "exact at $z = e$ ". A display whose remainder is an analytic O as $z \rightarrow +\infty$ asserts nothing at a single point, so that formulation commits precisely the conflation the box above forbids. The correct elementary statement is a coefficient identity for the underlying formal series, and it is proved here.
Reader's guide and the structure of this	the draft called "Part V may be read directly after Part IV" an exception to the rule that each part depends only on those before it. It is not an exception at all — V does come after IV — and it also contradicts reading path 1 below, which does draw two statements from Parts II and III. The intended and correct content is the lightness of that dependency.
Reader's guide and the structure of this	source draft 2 tabulates the output class of differentiation as " $t K[L]((t))$ up to the leading power". That set equals $K[L]((t))$, because t is a unit there, so the row asserts nothing. The correct statement is the valuation inequality $\nu_t(D_X F) \geq \nu_t(F) + 1$, proved in Part II, which can be strict only when a constant leading block is annihilated.
Reader's guide and the structure of this	source draft 4 prints the coefficient condition of its defining display as " p_n in $K[X]$ ", naming the large variable where the logarithmic one is meant. The intended and correct condition is p_n in $K[L]$.
The formal polynomial–logarithmic rings	the chain <code>\eqref{plt:eq:ring-chain}</code> exhibits three inclusions, not four; the count was wrong as written.
Theorem 3.6	the first inclusion is not a change of coefficient ring (the printed chain $\backslash(K[L] \subset K[L] \subset K(L))$ made it look like one); it is the inclusion of a power-series ring in the corresponding Laurent ring.
Theorem 3.6	this appeal was unsupported and mildly circular: the geometric-series corollary was stated only for coefficient rings whose list already contained $\backslash(K((L^{-1})))$, the very object being constructed here. <code>\cref{plt:cor:ring-geometric}</code> is now proved for an arbitrary commutative coefficient ring, so the instance $\backslash(E=K)$, generator $\backslash(u)$, is legitimate.

Location	Repair
Theorem 3.11	S4 calls $(tK[\ell][t])$ “the strictly small ideal” of the Laurent ring $(K[\ell](t))$; it is neither small in that ring nor an ideal of it.
Theorem 3.14	the statement imposed the standing hypothesis $(\alpha > \alpha')$ and then appended the case $(\alpha = \alpha')$, which that hypothesis excludes; the two cases are now alternatives of one hypothesis. The regime $(X > \text{EulerE})$ is printed here rather than produced inside the proof.
Theorem 3.14	the sources state this without the hypothesis $(X > \text{EulerE})$; for $(1 < X < \text{EulerE})$ one has $(0 < \log X < 1)$ and high logarithmic powers are small, so the comparison is genuinely eventual.
Valuation, leading data, and the degree	(ordt) is a discrete valuation on (R) with valuation ring $(E[[t]])$ ” overstates the case for two of the four classes: (PLpow) and (PLlau) are not fields, and on (PLpow) the image of (ordt) is (NaturalNumbers) , not (IntegerNumbers) . The claim is split accordingly.
Theorem 4.3	the domain hypothesis enters only through the reverse inequality in (i). For an arbitrary commutative ring (E) the subadditivity $(\text{ordt}(FG) \geq \text{ordt } F + \text{ordt } G)$, together with (ii) and (iii), still holds in $(E[[t]])$ and $(E((t)))$, by the first half of the computation in (i); this weaker form is what <code>\cref{plt:cor:ring-geometric}</code> uses, and stating it here is what keeps that corollary free of a circular appeal. S1 and S5 record the strictness of (ii) as happening “exactly when the leading blocks cancel”, omitting the hypothesis $(\text{ordt } F = \text{ordt } G)$; with unequal valuations no cancellation is possible and equality always holds.
Theorem 4.5	the display $(p_{\nu} = aL^d + \text{text{lower powers of } L})$ is literally a polynomial expansion, but (R) here ranges over all four classes, where a block need not be a polynomial. The reading is fixed explicitly.
Valuation, leading data, and the degree	“equivalently” was wrong: additivity of (v) is equivalent to multiplicativity of (lm) alone; multiplicativity of (lc) is a further assertion (proved below), not a restatement.
Theorem 4.8	“with value group (Γ) ” is false as stated for two of the four classes: on (PLpow) and (PLlau) a block is a polynomial, so $(\deg L \geq 0)$ and the image of (v) never meets $(\text{IntegerNumbers} \times \text{PositiveIntegers})$. The image and the group it generates are separated.
Theorem 4.11	S2 states the prefactor form $(A = cX^{\rho}L^{\beta}(1+U))$ “whenever available”; the example $(L+t)$ shows that over $(K[L])$ it is available only after passing to $(K(L))$, so no argument may assume it inside (PLlau) .
Theorem 4.15	the monotonicity clause $(\text{mathcal } D_{\sigma} \subseteq \text{mathcal } D_{\sigma'})$ was asserted without proof, and it is not the trivial inequality $(\sigma \leq \sigma')$: that fails for $(n < 0)$. The argument below uses that $(\text{dprofile}\{F\})$ is $(-\infty)$ below $(\text{ordt } F)$.
Theorem 4.15	S2 states $(\deg c_n \leq A+B+\max(\alpha, \beta)n)$ for the product. That form is correct only when the factor of smaller slope has nonnegative outer order; the counterexample of <code>\cref{plt:ex:lead-slope-fails}</code> violates it, and <code>\eqref{plt:eq:lead-slope-bound}</code> is the corrected statement, attained there with equality.
Theorem 5.3	S6 uses the inclusive marker throughout; the catalogue makes the exclusive operator normative, so every transferred formula is shifted by one block.
Theorem 5.6	S5 states <code>\eqref{plt:eq:trunc-product-pow}</code> without the hypothesis $(\text{ordt} \geq 0)$; on (PLlau) it is false, and <code>\eqref{plt:eq:trunc-product}</code> is the correct general form.

Location	Repair
Theorem 5.9	<code>\cref{plt:def:trunc-formal-o}</code> defines the token only for an integer cutoff, so (ii) and (iii) below were type-incorrect when $\backslash(H)$ or $\backslash(G)$ vanishes and the printed exponent becomes $\backslash(+\infty)$. The reading of that degenerate case is now fixed explicitly.
Theorem 5.11	the displayed analytic statement was printed with none of $\backslash(f)$, $\backslash(S)$ or $\backslash(n_0)$ introduced, contrary to the standing requirement that a statement name its regime and domain. They are named here.
Theorem 5.11	S1 offers instead the minimal Poincaré requirement that the remainder be little- o of the last retained term. That condition does not determine the blocks: with $\backslash(p_n=L^5)$ and $\backslash(q_n=L^5+L^3)$ the two candidate blocks differ by $\backslash(L^3=\text{LittleOAt}\{L^5\}\{X\text{to}+\infty\})$, so uniqueness fails. The displayed form, and S6’s $\backslash(\text{LittleOAt}\{X^{\{-N\}}\}\{X\text{to}+\infty\})$ form, both do determine them.
Theorem 5.12	the label kind "box" is not in the normative list of kinds; these boxed statements are remarks, so each label is renamed to the "rmk" kind.
Theorem 5.15	the label kind "box" is not in the normative list of kinds; these boxed statements are remarks, so each label is renamed to the "rmk" kind.
Theorem 5.19	S2’s completeness proof asserts the common lower bound because “the sequence is already Laurent-bounded at some initial precision”. That is not a hypothesis of the statement; the bound must be extracted, as above, from the Cauchy condition at cutoff $\backslash(N=0)$.
Theorem 5.19	the word “exactly” in (iv) asserts that the common lower bound is also necessary; the sources supply only sufficiency, so the converse is proved here.
Theorem 5.20	S3 asserts that $\backslash(t)$ -adic convergence is “equivalent” to coefficientwise stabilization and that both the power-series and the Laurent ring are complete “in this sense”. The equivalence is false on the Laurent ring, as follows; the completeness assertion itself is correct, but needs the proof of <code>\cref{plt:thm:trunc-completeness}</code> , (iii).
Theorem 5.23	the identification of $\backslash(\mu)$ was printed as an equality. It is only a lower bound: if every $\backslash(\text{ordt } S_{i>0})$ the right-hand side is $\backslash(0)$ while $\backslash(\mu>0)$. Finiteness is all the argument needs, and all it gives.
Theorem 5.24	the corollary was stated only for $\backslash(E\text{in}\text{SetOf}\{K[L], K(L), K((L^{\{-1\}}))\})$, yet it is invoked inside <code>\cref{plt:prop:ring-chain}</code> with $\backslash(E=K)$ and generator $\backslash(L^{\{-1\}})$, precisely in order to prove that $\backslash(K((L^{\{-1\}})))$ is a field — the fact that puts $\backslash(K((L^{\{-1\}})))$ on that list. Stating it for an arbitrary commutative coefficient ring removes the circle at no cost: the proof never needs a domain hypothesis.
Theorem 5.28	the transfer was claimed only for the four ambient classes, but <code>\cref{plt:cor:ring-geometric}</code> needs it for $\backslash(E((t)))$ with $\backslash(E)$ an arbitrary commutative ring (the case $\backslash(E=K)$, generator $\backslash(L^{\{-1\}})$, is used in <code>\cref{plt:prop:ring-chain}</code>). The scope of the transfer is widened, and the two places where the domain hypothesis entered are named.
Theorem 5.28	the label kind "box" is not in the normative list of kinds; these boxed statements are remarks, so each label is renamed to the "rmk" kind.
Theorem 5.28	the volume regime is $\backslash(X\text{to}+\infty)$ on the positive ray; the generic row printed the weaker $\backslash(X\text{to}\infty)$.
Theorem 6.8	the four classes were displayed on one line, which overfills the text block by about 45pt; set as a two-by-two array instead.

Location	Repair
Theorem 6.13	the bracketed title duplicated that of <code>\cref{plt:prop:trunc-product}</code> . The two statements are not the same result: the ring-chapter proposition is the law on $(\mathbb{P}\mathbb{L}\mathbb{a}\mathbb{u})$ at the canonical cutoffs, this one is the law over an arbitrary logarithmically graded coefficient ring at arbitrary sufficient cutoffs, together with optimality and sharpness. The titles now say so; see <code>\cref{plt:rmk:mul-leading-restated}</code> .
Multiplication: convolution logarithm	of the optimality of $(N-b, N-a)$ was asserted unconditionally. For $N \leq a+b$ both sides of <code>\eqref{plt:eq:mul-truncated-product}</code> vanish whatever P, Q are, so no pair is forced and there is nothing to be optimal about; the clause is now stated under $N > a+b$, which is what the argument actually establishes.
Theorem 6.14	the source bounds the third term by $P+Q$ and asserts $P+Q \geq N$. That inequality does not follow from $P \geq N-b$ and $Q \geq N-a$: for $a=b=0$ and $N=P=Q=-5$ one has $P+Q=-10 < N$. The repair is to bound $\text{ord } H$ by b rather than by Q , which is legitimate because every block of H is a block of G .
Theorem 6.14	S6, Theorem "Closure and finite determination", eq. (mult-error-rule). The source states the conclusion as $O(t^{\min(p+m_0, q+n_0)})$ without any hypothesis relating p, q to the valuations n_0, m_0 , and its proof silently bounds $\text{ord}(E \cdot \tilde{G})$ by $p+m_0$ and $\text{ord}(\tilde{F} \cdot H)$ by $q+n_0$, which requires $\text{ord}(\tilde{G}) \geq m_0$ and $\text{ord}(\tilde{F}) \geq n_0$. Neither follows from the stated hypotheses. The corrected statement carries the third entry $p+q$, and the hypotheses $p \geq n_0, q \geq m_0$ under which the source's form is recovered. Note that these hypotheses are sufficient but not necessary: the third entry is redundant exactly when $p \geq a$ or $q \geq b$.
Theorem 6.18	retitled to distinguish it from <code>\cref{plt:def:lead-leading-data}</code> , which is the same data on the four ambient classes; only the coefficient ring differs. See <code>\cref{plt:rmk:mul-leading-restated}</code> .
Theorem 6.19	the leading data, the rank-two valuation, the log-degree profile, the exclusive truncation law and the summability calculus are all introduced in the ring chapter. Restating them here without saying so would leave two definitions of the same symbols standing in one volume. This remark records that the restatements are verbatim generalizations over the coefficient ring and agree with the originals.
Theorem 6.21	the first equivalence of the display was left unargued. It is not a theorem of this section at all: succdom is defined by the right-hand condition in <code>\cref{plt:def:ring-dominance}</code> , so only the analytic equivalence needs an argument, and that argument restates <code>\cref{plt:lem:ring-dominance-analytic}</code> for integer exponents.
Theorem 6.23	the statement asserts that neither $\mathbb{P}\mathbb{L}\mathbb{p}\mathbb{w}$ nor $\mathbb{P}\mathbb{L}\mathbb{a}\mathbb{u}$ is a field, but the argument as given covers only $\mathbb{P}\mathbb{L}\mathbb{a}\mathbb{u}$. One line closes it.
Theorem 6.24	retitled to distinguish it from <code>\cref{plt:def:lead-profile}</code> , which is the same function on $(\mathbb{P}\mathbb{L}\mathbb{a}\mathbb{u})$, where the blocks are polynomials and the codomain is $(\mathbb{N} \cup \text{SetOf}(-\infty))$. See <code>\cref{plt:rmk:mul-leading-restated}</code> .

Location	Repair
Theorem 6.26	the criterion was stated as an unconditional “if and only if”. It fails in the degenerate case $P_N = \emptyset$ (equivalently $M = -\infty$): there $[t^N](FG) = 0$, so equality holds although the empty sum $\text{\eqref{plt:eq:mul-top-coefficient}}$ is zero. The criterion is now confined to M finite and the degenerate case is recorded separately.
Theorem 6.26	S6, Proposition “Degree and support bounds”, asserts equality “whenver the maximizing blocks are nonzero” under the hypothesis that all coefficients are nonnegative over an ordered field. That hypothesis is far stronger than necessary and the quoted sufficient condition is stated ambiguously (a zero block has degree $-\infty$ and cannot maximize at all). The criterion below is exact and the positivity hypothesis has been reduced to the leading coefficients of the maximizing blocks only.
Theorem 6.27	condition (3) as first drafted omitted $P_N \neq \emptyset$; over an ordered field its hypotheses are satisfied vacuously when P_N is empty, and the conclusion would then be the false assertion that the sum $\text{\eqref{plt:eq:mul-top-coefficient}}$ is nonzero.
Theorem 6.30	the inclusion $\mathfrak{A}_\alpha \subseteq \mathfrak{A}_{\{\alpha\}}$ for $\alpha \in \mathbb{A}$ was used without proof. On the Laurent ring it is not immediate: for $n < 0$ a larger slope gives a <i>smaller</i> bound, so the constant has to be increased, and that is possible only because the support is bounded below. The argument is supplied here.
Theorem 6.33	retitled, and its relation to $\text{\eqref{plt:def:trunc-summable}}$ made explicit: the condition is verbatim the one of the ring chapter, read over the coefficient ring R instead of $K[L]$, which $\text{\eqref{plt:rmk:trunc-summability-classes}}$ already licenses. Only the two clauses of $\text{\eqref{plt:lem:mul-summable}}$ are used below, so they are the only ones recorded here.
Theorem 6.35	the source calls the two infima “finite by (1)”; $\text{part} \sim (1)$ gives only $> -\infty$, and the value $+\infty$ does occur, namely when every member of the family is zero. That case is disposed of separately.
Theorem 6.44	retitled: “Basic properties of a scale” in the differential part ($\text{\eqref{plt:lem:dif-scale-basic}}$) is about the coefficient field of a power–log scale, not about the relation $\approx$ between a function and a series. The two lemmas share nothing but the word “basic”.
Theorem 6.46	the three-term estimate was set on one display line, overfilling the text block by about 53pt; broken across two lines.
Theorem 7.4	S5 and S6 assert the nonunit direction by citing the general power-series unit criterion; the additivity step that makes it valid on the Laurent ring is supplied here explicitly.
Theorem 7.5	the direct-product step was justified by “the three subgroups pairwise intersect trivially”, which does not imply an internal direct product for three subgroups (the Klein four-group refutes it); the isomorphism is read off instead from surjectivity plus the uniqueness just proved.
Theorem 7.8	$\text{\eqref{plt:cor:div-quotient}}$ invokes this theorem “over the coefficient ring E ”, a generality the statement did not record; the proof uses only commutativity and invertibility of B_0 , so the clause is supplied here.
Theorem 7.10	S6 states this correctness theorem with its local inclusive truncation $[F]_{\leq N}$; the statement above is the exclusive normalisation demanded by the catalogue, and the two differ by exactly one block.

Location	Repair
Theorem 7.12	S6 states a "guard precision" principle in words ("compute the quotient beyond the final requested block") but never says by how much; the exact guard is the content of the proposition above.
Theorem 7.13	the negative-guard case was explained by "the denominator has a Laurent principal part", which accounts only for the numerator guard b; the denominator guard $2b-a$ is negative under the unrelated condition $a > 2b$.
Theorem 7.15	S1 prints the block Q_3 of this example immediately after inputs given only modulo t^3 ; the hypothesis that the omitted blocks vanish is a genuine extra assumption and is stated here.
Theorem 7.19	pointwise finiteness alone does not place the sum in the field: one must also check that each block is a Laurent series in L^{-1} , i.e. that its support is bounded below. The same bound gives this at once, and no source records the step.
Theorem 7.29	the input-requirement statement (4) is absent from S2, S3, S5 and S6, each of which asserts doubling without saying which truncations suffice.
Theorem 7.29	the precondition $N \geq 1$ was not printed; with $N = 0$ the loop body never runs and the routine returns the scalar B_0^{-1} rather than $\text{Trunc}_{\leq 0}(1/B) = 0$.
Theorem 7.30	the induction covers only the exact powers of two, while <code>\cref{plt:alg:div-newton}</code> caps every step at N ; the capped step needs its own line before the "in particular" clause is justified.
Theorem 7.30	S2 justifies the doubling by "the valuation is multiplicative"; the valuation is additive on products, which is what the argument uses.
Theorem 7.32	the convention "multiplications by the scalar B_0^{-1} are not counted" was stated loosely enough to exclude also the products $B_j C_0$ and $C_0 E_j$, under which reading the two schemes cost exactly the same and the $N-1$ gap claimed below disappears. The counted set is pinned down here and the other reading is recorded at the end of <code>\cref{plt:prop:div-cost-newton}</code> .
Theorem 7.34	the $N-1$ gap is fragile: it is created entirely by charging for products against the scalar block C_0 . Recording this keeps the comparison from being read as a robust separation, which it is not.
Theorem 7.34	S3 asserts that Newton inversion "is often faster than coefficient recursion for large N even with naive multiplication"; the exact counts above show the opposite at the level of coefficient multiplications, and the honest statement is the corollary. the corollary was headed by the iff "improves ... exactly when block multiplication is subquadratic". Only the two implications below are proved (subquadratic wins asymptotically; schoolbook loses by $N-1$), and neither the converse nor the intermediate regime $M(N) = cN^2$ is covered; the claim is downgraded to what the two cost counts give.
Theorem 7.35	S6 gives the cost as $O(N^2 d^2)$ "if the typical coefficient degree is d "; under the affine log-degree profile that actually occurs there is no such constant d , and the honest count is quartic in N .
Units, reciprocals, and division	the catalogue asserts that the two iteration orders differ "in general"; no source proves it, and the witnesses above supply the proof.
Units, reciprocals, and division	"beats the recurrence only when block multiplication is subquadratic" asserts the unproved converse; only the two extreme regimes are established.

Location	Repair
Theorem 8.1	retitled. Three “Summable families” statements stood in the volume with the same bracketed title. This one is the blockwise assembly principle for a sequence in a Laurent ring over any of the coefficient rings used here; <code>\cref{plt:lem:cmp-summable}</code> is the corresponding statement for an arbitrary index set in a Hahn ring with real exponents, and <code>\cref{plt:thm:trunc-summability}</code> is the full calculus on $\mathcal{PLau}$. On $\mathcal{PLau}$ the present lemma is the sequence case of <code>\cref{plt:thm:trunc-summability}</code> , (i),(iv),(v); it is stated again because it is applied below in the $\mathcal{L}^{-1}$ -adic order and in a Hahn ambient, neither of which that theorem covers. The hypothesis $(\inf_r \text{ordt } F_r > -\infty)$ is redundant for a sequence with $(\text{ordt } F_r \rightarrow +\infty)$ and is kept only because the proof uses it.
Theorem 8.2	retitled. This is substitution of a small series into a <code>\emph{scalar}</code> power series, $\mathcal{K}[[z]] \rightarrow \mathcal{PLpow}$. The composition part has a different theorem under the same name, <code>\cref{plt:thm:cmp-substitution-hom}</code> , which substitutes a monomial-leading transseries into a transseries; neither implies the other.
Theorem 8.3	retitled: the composition part states a different theorem under the same name (<code>\cref{plt:thm:cmp-finite-determination}</code> , finite determination of $\mathcal{F} \circ G$). The present one is about $\mathcal{Phi}(U)$ with $\mathcal{Phi}$ a scalar power series.
Theorem 8.7	S3 states the analogous expansion with $U^4 = O(t^4)$ as the reason for stopping at $r = 3$; the correct reason is that $r \cdot q \geq N$ kills the term, which is the general statement proved above and matters as soon as $q > 1$.
Theorem 8.12	retitled to separate it from <code>\cref{plt:lem:multip-asymptotic-basic}</code> , which collects the basic properties of the relation $\approx$ between a function and a series.
Theorem 8.15	S4 and S6 note only that a “large part” other than an integral multiple of L may require new monomials. The decisive point, which no source states, is that a logarithmic derivative has positive L^{-1} -order, whereas the derivative of a nonaffine polynomial has order at most -1 .
Theorem 8.24	no source identifies the image of ∂_L on Laurent-log coefficients, although S5 and S6 both assert closure of the class under antiderivatives and then remark vaguely that “new logarithmic levels can be forced” once inverse powers of L are admitted. The converse of the preceding lemma is what makes that remark precise, so it is proved here and used both for the logarithm boundary and for the antiderivative obstruction.
Theorem 8.28	S6 states this example with a leading factor L^{-2} , which is not an element of $\mathcal{K}[L](t)$; the phenomenon is the same, and the general statement is that the coefficient of Λ is $-\omega$ of the leading coefficient.
Theorem 9.3	the notation catalogue and four of the six sources assert that D_X is “the unique derivation” with the two generator values, with no hypothesis. The argument above needs a bound on δ on the power-series part; without one, the passage from the dense subring to the completion is not available.
Theorem 9.7	notation reconciliation. Two sources write Δ_n for the single factor $\partial_L - n$ and never introduce the second index; the normative two-index operator is used here throughout.

Location	Repair
Theorem 9.9	S1 prints the identity as $X^k d^k/dX^k P(\log X) = \sum_j s(k,j) P^{\{j\}}$ with the sum starting at $j = 1$; that is correct only for $k \geq 1$, since $s(0,0) = 1$. The range $j = 0..k$ with the empty-product convention is uniform.
Theorem 9.16	the motivation part proves the same dichotomy in the weaker form it needs there (\cref{plt:lem:mot-block-antiderivative}), on $\backslash(K[L])$ only, with no closed form). Rather than restate it, the two lemmas above are identified with it here, so that a reader meets the resonance once as a mechanism and once in the form the antiderivative theorem consumes.
Theorem 9.18	S6 concludes only that "the polynomial-logarithmic ring is closed under termwise antiderivatives, up to an arbitrary constant", and S3 writes the weaker (and redundant) inclusion $\text{int } P \, dX \subset P + K$. The correct statement is surjectivity of D_X on $K[L]((t))$ with kernel K , which is what the block system actually proves and what is needed downstream.
Theorem 10.3	retitled. This is the summability criterion in a Hahn ring with real exponents, for an arbitrary index set, together with the notion of a map compatible with summable families that the rest of the part uses. It is not \cref{plt:lem:dif-summable} (sequences in a Laurent ring) and not \cref{plt:thm:trunc-summability} (the calculus on $\backslash(\text{PLlau})$); for $\backslash(\Gamma = \text{IntegerNumbers})$ it restricts to the latter.
Theorem 10.4	the catalogue warns that a t -adically Cauchy <i>sequence</i> in a Laurent ring may fail to converge without a common lower support bound. For <i>families</i> satisfying (plt:eq:cmp-summability-criterion) the bound is automatic, as the proof shows; the two statements are not in conflict.
Theorem 10.7	retitled to separate it from \cref{plt:prop:dif-substitution} , which substitutes a small series into a scalar power series. The present theorem substitutes a monomial-leading transseries into a transseries.
Theorem 10.8	the draft deduced "compatible with summable families" from (plt:lem:cmp-summable) directly, but that lemma only supplies the valuation half of the definition; the interchange $\Phi_G(\sum S_i) = \sum \Phi_G(S_i)$ needs the argument below. It is used by (plt:lem:tay-determined) and by the associativity proof, so it is supplied here.
Theorem 10.10	retitled: \cref{plt:thm:dif-finite-determination} carries the same name for a different statement, the finite determination of $\backslash(\Phi(U))$ with $\backslash(\Phi)$ a scalar power series.
Theorem 10.15	S2 writes the operator in (its) power-leading formula as $"e^{\{\delta(D-n)\}}"$ with D previously declared to be d/dL , while the identity requires differentiation in the <i>argument</i> of the coefficient polynomial. We print <code>partial_z</code> for the dummy variable and evaluate afterwards. an earlier draft of this warningbox described the discrepancy as "a spurious factor ρ^k ". That is false as soon as $n \neq 0$: the chain rule turns $(d_z - n)^k$ into $(\rho d_z - n)^k$, and the two are proportional only when $n = 0$. Corrected below, with a numerical witness.
Theorem 10.23	the sources state the rule in words; the numerical demonstration of the two standard mistakes is supplied here so that the size of the error is visible.
Theorem 10.33	the exclusive normalization is used throughout; a source using the inclusive cutoff states the same condition as $f - \sum_{n \leq N} = o(X^{\{-N\}})$, which is (plt:eq:cmp-realization) at $N+1$.

Location	Repair
Theorem 10.35	the draft omitted this reduction and then asserted that the product of the two remainders is $O(X^{N+\epsilon})$; that product is only $O(X^{2N+a+\tilde{\epsilon}/2})$, which is the required bound precisely when $N \geq a + \tilde{\epsilon}$. The reduction below makes the omission harmless.
The formal Taylor formula	as in (plt:thm:cmp-substitution-hom) , the earlier text checked the interchange only for the family of blocks of a single F ; the definition of compatibility quantifies over all summable families, and that is what (plt:lem:tay-determined) consumes. Argument supplied.
Theorem 11.5	this multiplicativity is exactly the step that S_3 asserts ("both sides are continuous algebra homomorphisms") and that S_6 replaces by "the equality then extends to polynomials, products, inverses"; it is not automatic, because T_H is a *normal-ordered* exponential and not $\exp(H D_X)$.
Theorem 11.10	S_4 and S_5 prove the Taylor formula by "inserting a scalar parameter s , forming $F(x+sA)$, and differentiating in s ". That argument is circular: the object $F(x+sA)$ is defined only by the composition being constructed, and "formal differentiation in s " of it presupposes the chain rule for that composition. The homomorphism argument above avoids the circle.
Theorem 11.13	the two index orders in the corpus are transposes of each other; the collision is silent because both symbols are two-index arrays of polynomials.
Theorem 11.20	the source that supplies this example specifies F only modulo $Y^{\{3\}}$ and G only modulo t^2 , yet quotes h_3 , which depends on the coefficient of $Y^{\{3\}}$ in F and on the coefficients of t^2 , t^3 in G . The data are restated here as exact finite series, which is what the quoted answer actually assumes.
Theorem 11.22	this proposition restated three results of the elementary-series part under a different normalisation of the outer coefficients: the substitution homomorphism (\cref{plt:prop:dif-substitution}), its finite determination (\cref{plt:thm:dif-finite-determination}), and the Bell-polynomial coefficient formula \eqref{plt:eq:dif-bell} of \cref{plt:prop:dif-bell} , which the displayed formula reproduces verbatim once \c{c}_k=\varphi_k/k! is substituted. The restatement and its proof are deleted in favour of the pointer below; the label is kept so that \cref{plt:ex:tay-standard-outer} and any citation from elsewhere in the volume continue to resolve. Only the translation clause was genuinely new, and it is retained with its one-line proof, as is the second route to the coefficient formula.
Theorem 12.2	S_6 writes $F/X = 1+tA = 1+o(1)$ "formally". A little- o symbol with no regime attached is not a formal statement; the three O -roles must not be conflated. The correct formal assertion is $tA \in tPLpow$, equivalently $1+tA-1=\text{FormalBigOAt}\{t\}$, equivalently $tA\prec_{\text{dom}}1$.
Theorem 12.2	this lemma was stated and proved twice in the volume, here and as \cref{plt:lem:cmp-exp-log} in the composition part, with the same transport-of-identities proof. The composition-part statement is the stronger of the two — it is set in an arbitrary polynomial-logarithmic Hahn ring, and it adds the additivity of \(\exp\) , the law \(\log(P^{\{n\}})=n\log P\) for all \(\{n\}\in\text{IntegerNumbers}\) , and the commutation with any homomorphism compatible with summable families, all of which are used below. The restatement and its proof are deleted and replaced by this pointer; the label is kept so that the citations to it elsewhere in the volume continue to resolve.

Location	Repair
Theorem 12.5	the exact formal Taylor formula was stated and proved twice, here and as \cref{plt:thm:tay-formula} in the composition part, where an entire subsection (the Taylor operator, its summability budget, its multiplicativity, and the generator-determination lemma) is built to prove it. The statements are identical, so the restatement and its proof are deleted in favour of a pointer; the label is kept because the reversion part and the algorithms part cite it. The repair recorded here — S3 concludes from agreement on the generators without proving that the normally ordered operator is multiplicative — is carried out at the surviving site, in \cref{plt:lem:tay-homomorphism} .
Theorem 12.6	the Lipschitz bound was printed as $\mathcal{O}((B-C)+1)$, dropping the summand $\mathcal{O}(A)$ that the proof below already produces. The sharper form is what the reversion part needs (it is what makes the Newton estimate of \cref{plt:thm:rev-newton-doubling} come out), and it was stated there a second time as a separate lemma. The inequality is corrected here and the restatement in the reversion part is deleted; every use of the weaker form below is unaffected, since $\mathcal{O}(A) \geq 0$ on $\mathcal{PL}^{\text{pow}}$.
Theorem 12.7	S4 proves the contraction step with a formal "mean value" identity $A \circ (X+B) - A \circ (X+C) = (B-C) \int_0^1 A' \circ (X+C+s(B-C)) \, ds$. The integrand is a composition with a series carrying an auxiliary parameter s , and neither its existence nor the legitimacy of the term-by-term integration is established there. Corollary \ref{plt:cor:grp-gain} replaces it by the exact Taylor difference, which needs nothing beyond Theorem \ref{plt:thm:grp-taylor} .
Theorem 12.8	S5 proves associativity by "introducing finite truncations mod t^N ", for which "ordinary substitution is associative". Truncation is not a ring retraction compatible with substitution, and the claim that a truncated composite equals the truncation of the composite is precisely what needs proof. S1 proposes the correct route (check on the generators) but does not carry out the verification for L ; the display above is that verification.
Theorem 12.8	the Picard construction of the inverse was carried out twice in the volume: here, and again — with the sharp rate $\mathcal{O}(B_{j+1} - B_j) \geq (j+1)\alpha + j$, the first-order form of the inverse, and the uniqueness of the two-sided inverse — as \cref{plt:thm:rev-existence} . The two constructions are the same iteration; the reversion part proves strictly more. The duplicate is deleted and item (4) is derived from the reversion theorem, which makes the dependency of the group's inverses on Part~IV explicit instead of hiding it behind two independent proofs. There is no circularity: \cref{plt:thm:rev-existence} uses items (1)–(3) of the present theorem and the gain estimate \cref{plt:cor:grp-gain} , and nothing else from this section.
Theorem 12.10	this remark previously asserted that the section is independent of Part~IV and that inverses are established here. That was true only because the same Picard construction was written out twice. With the duplicate removed, the dependency is one-directional and is stated as such.
Theorem 12.13	S6 states the two continuity relations for the contact filtration but justifies the right-composition one by " $H = 1 + \mathcal{O}(t)$ ". That is the reason for a gain in the *left* argument; the right-hand estimate is the exact isometry of Proposition \ref{plt:prop:grp-substitution} (4) and needs no derivative hypothesis at all.

Location	Repair
Theorem 12.15	S6 calls the group "pro-unipotent at finite truncation". No algebraic group structure is available here (the graded pieces are the infinite dimensional space $K[L]$), and none is needed: Propositions \ref{plt:prop:grp-limit} and \ref{plt:prop:grp-commutator} give the exact provable statement, namely that G_{near} is the inverse limit of nilpotent quotients.
Theorem 12.20	S6 introduces $\log(S_F)$ as a power series in $S_F - I$ and writes the fractional iterate as " $F^{\{s\}} = X S_F^s$ ". Three things are missing there: that $S_F - I$ raises the filtration (true: $(S_F - I)H = H \circ F - H$ has $\text{ordt} \geq \text{ordt}(H) + 1$ by Corollary \ref{plt:cor:grp-gain}); that S_F^s is again a "substitution" operator, i.e. that $F^{\{s\}}$ exists in G_{near} at all; and the notation, which should be $S_F^s(X)$, not a product. Theorem \ref{plt:thm:grp-exp} and Corollary \ref{plt:cor:grp-iteration} supply all three.
Theorem 13.4	S4's closure table records only the rational case (" $c x^{\{r/s\}}$: Puiseux-log series") and S1 says "arbitrary real beta require a Hahn support", neither separating the three regimes nor proving that each enlargement is necessary. Corollary \ref{plt:cor:grp-power-regimes} does both.
Theorem 13.14	S2 says only that "the rational field $K(L_1, \dots, L_r)$, or a Laurent-monomial field, repairs this defect". Proposition \ref{plt:prop:grp-tower-derivation} identifies the minimal repair inside the Laurent-monomial family and shows that L_r itself need not be inverted.
Theorem 14.3	report 6 states its reversion algorithms with the inclusive cutoff $[F]_{\leq N}$; every cutoff index taken from that source is shifted by one here.
Theorem 14.4	the closure of $(\backslash G_{\text{near}} \backslash)$ under composition was stated and proved a second time in this part, as a separate lemma. It is item (1) of the group theorem, proved there from the substitution proposition, so it is imported here with the other three standing facts. Note that only items (1)–(3) of that theorem are imported: its item (4) cites \cref{plt:thm:rev-existence} below, and importing it here would be circular.
Theorem 14.6	the closure lemma stated here duplicated \cref{plt:thm:grp-group} , (1), which is proved in the composition-group part with the same one-line argument. It is deleted; closure is now imported as part of (I3) and appears as \eqref{plt:eq:rev-closure} . For the record, the argument is that $(F \backslash \text{compose } G = G + A \backslash \text{compose } G \backslash)$ and that $(A \backslash \text{compose } G = \sum_{k \geq 0} B^k DX^k A/k! \backslash)$ has all summands in $(\backslash \text{PLpow} \backslash)$ by \cref{plt:lem:rev-deriv-gain} .
Theorem 14.9	this estimate was proved twice in the volume: here from the two-point Taylor expansion, and in the composition-group part from the exact Taylor formula, where it is \eqref{plt:eq:grp-gain-lipschitz} . The two arguments are the same computation, and the group-part statement was missing the summand $(\backslash \text{ordt}(A) \backslash)$ that its own proof produces; that omission is repaired at \cref{plt:cor:grp-gain} and the duplicate proof is deleted here. The inequality is recorded below as a fifth standing import, in the instantiated form used throughout this part.
Theorem 14.11	all six sources call this a contraction; none says in what sense. The statement is about nu_t , not about coefficient size, and the constant term $\alpha = \text{nu}_t(A)$ is absent from every source formulation.

Location	Repair
Theorem 14.12	this corollary restated <code>\cref{plt:thm:grp-group}</code> in full. The group statement now stands once, in the composition-group part, and its item (4) cites <code>\cref{plt:thm:rev-existence}</code> ; the restatement and its proof are replaced by the sentence below, which records what has just been supplied.
Theorem 14.19	report 1 asserts that at finite order the two residuals "need not have identical first omitted blocks"; for an exact G_N in the group that is false.
Theorem 14.21	report 1 claims the derivative residual "often detects a wrong block one order earlier"; it detects it one level later, and is blind at level 0.
Theorem 15.7	report 4 states the rule as "for a leading slope $F = ax + o(x)$ the correction is divided by a ", with no hypothesis on a and no discussion of a nonconstant leading block of $D_X F$.
Theorem 15.13	the sources give five mutually inconsistent exponents; one contradicts its own proof, and two slide from residual order to correct-block count.
Theorem 15.19	reports 1, 3, 4 and 6 assert Newton is "asymptotically superior" with no cost model stated; the claim is model-dependent and is weakened here.
Theorem 16.10	S1, S2, S3, S4 and S5 all assert only the sufficiency direction ("the r -th term has t -order at least $r-1$ "), and all state it with the hypothesis $\text{ord}_t(A) \geq 0$. The correct threshold is $\text{ord}_t(A) > -1$; the two agree on $K[L]((t))$, whose value group is $\mathbb{Z}$, but they differ on the Puiseux–log enlargement, where the weaker hypothesis is exactly what the reduction of a ramified reversion needs (<code>plt:ex:lag-puiseux</code>). None of the sources observes that the condition is also necessary, which is what makes the misuse in <code>plt:warn:lag-scale-trap</code> detectable.
Theorem 16.12	S3's appendix proves the unshifted formula by a formal residue change of variables, which is valid only when the constant term of the auxiliary series is invertible, and repairs this by a universal-coefficient localisation argument. S6 replaces that argument by the phrase "an integration by parts reduces the resulting residue". The proof above needs neither device.
Theorem 16.16	S3 states this "whenever both sides are interpreted to a fixed finite t -precision" and S6 "for any H for which the family is summable"; both hypotheses are vacuous or unverifiable. The order estimate <code>plt:eq:lag-transport-order</code> shows that no hypothesis beyond membership in the frame is needed, since $\text{nu}(\phi)$ is finite for ϕ nonzero.
Theorem 16.25	S4 gives only b_0, b_1, b_2 in this universal form; the b_3 line <code>plt:eq:lag-b3</code> is derived here and verified against the two independent worked examples <code>plt:ex:lag-mixed</code> and <code>plt:ex:lag-abc</code> .
Theorem 16.28	this proposition restated <code>\cref{plt:cor:rev-finite-certificate}</code> , and its proof re-derived <code>\cref{plt:prop:rev-residual-order}</code> along the way. The statements are identical — the hypothesis $\backslash(F=X+A\backslash)$ with $\backslash(A\text{in}\backslash\mathbb{P}\backslash\text{Lpow}\backslash)$ is membership in $\backslash(G_{\text{near}}\backslash)$, and the two candidate truncations $\backslash(G_N\backslash)$ are the same object. The restatement and its proof are deleted in favour of the pointer below; every citation of the deleted label in this part now cites the reversion part.
Theorem 17.2	the requirement $\log d$ in K is stated in the catalogue and is omitted in S1, S2 and S6, which write $\log c$ freely over an unspecified coefficient field. Over $K = \mathbb{Q}$ it fails already for $d = 2$; see <code>plt:warn:lag-scale-trap</code> .

Location	Repair
Theorem 17.8	S6 derives <code>plt:eq:lag-u-expansion</code> by inserting an undetermined ansatz $u = \rho^{-1}(S - \sigma \log S + C + (a \log S + b)/S + \dots)$ and matching three constants by hand. The derivation above obtains the same coefficients from the near-identity coefficient formula, so the pattern continues to all orders with no further ansatz, and the branch and field hypotheses ($\log c$, $\log \rho$ in K) become visible.
Theorem 17.12	S4 asserts that after the substitution $y = dx^{1/q}(1+V)$ "the formal implicit-function recursion is triangular" for any leading exponent q , which is true, but it does not record that the intermediate normalisation $\Phi = F \circ \text{rev}(F_0)$ can leave the near-identity class, as it does in <code>plt:ex:lag-puiseux</code> . The correct general statement is the converse half of <code>plt:thm:lag-rigidity</code> , where $\Phi = X(1 + U \circ \text{rev}(F_0))$ because F_0 is the exact leading monomial of F , not merely its leading power.
Theorem 18.7	S1's prose ("the same P_n occur, multiplied by $(-1)^n$ ") contradicts its own displayed expansion of the inverse of $X - \log X$, where $P_1 = L$ occurs with a plus sign. The correct weight is $(-1)^{n+1}$; the weight $(-1)^n$ belongs to the negated quantity $-\text{rev}(X-L)$, which is what enters W_{-1} .
Theorem 18.10	S5 states uniqueness only inside $-L + t K[L][[t]]$; the contraction estimate holds on all of $K[L][[t]]$, so uniqueness is unconditional there and membership in $-L + t K[L][[t]]$ is a conclusion, not a hypothesis.
Theorem 18.11	the citation named a label " <code>plt:thm:lagrange-burmann</code> " that is defined nowhere in the volume; the theorem meant is the one labelled <code>plt:thm:lag-burmann</code> in the Lagrange–Bürmann part, whose polynomial–logarithmic form is equation <code>plt:eq:lag-burmann-pl</code> .
Theorem 18.14	retitled. This is not the general finite certificate (<code>\cref{plt:cor:rev-finite-certificate}</code>), which it invokes) but its specialisation to the single map $\backslash(X+L)$, where the criterion becomes a characterisation of the Lambert polynomials themselves.
Theorem 18.17	S5 asserts that the residual-to-error principle "proves the asymptotic validity of every truncation directly from" the FORMAL residual identity, and S1 asserts an analytic O -bound for the branch residual "by construction". Neither is a proof: a formal t -adic identity supplies no inequality. The missing ingredient is item (2), supplied below.
Theorem 19.4	S3 and S6 start the recurrence at $P_1 = u$ and impose $P_n(0)=0$ as a side condition of the theorem rather than proving it; S4 derives the linear recurrence by "equating coefficients" in a NONLINEAR generating ODE, which does not produce it. The proof above is direct from the definition and proves the boundary condition instead of assuming it.
Theorem 19.12	this corollary carried its own argument inside the statement, so it read as an assertion with no proof. The claim and the argument are separated.
Theorem 20.16	S1, S3, S4 and S5 all state the branch expansion "in an admissible sector" or "in a branch-compatible sector" without ever defining one, and none proves that the constructed solution is the branch conventionally labelled W_k . Repaired: the estimate is proved on an explicit region with no sector at all (Theorem <code>lw-uniform-remainder</code>), and the labelling is a separate proposition with a printed distance-to-the-cut hypothesis (Proposition <code>lw-branch-label</code>).

Location	Repair
Theorem 20.17	S6 deduces convergence of the Lambert series from (a) convergence at frozen u and (b) $ s u \rightarrow 0$ along the Lambert path. That inference is invalid: the radius of convergence in s depends on u and shrinks as $ u \rightarrow \infty$. Repaired by proving the two-variable statement (Lemma lw-two-variable), which is what the argument actually needs.
Theorem 20.18	S1, S3 and S5 each assert, as though it were established here, that the principal-branch series converges for every real $z \geq e$. Downgraded to an attributed literature report: nothing in this volume proves it, and S5's supporting observation (all terms vanish at $z = e$) is vacuous.
Theorem 20.23	every source displays the lower-branch expansion with the factor $(-1)^n$ and explains it only by analogy ("the same polynomials reappear with alternating block signs"). Repaired: the alternation is derived as the single value $a = -1$ of the scalar a^{n+1} in Theorem lw-affine-log, and the box states explicitly that no second polynomial family exists.
Theorem 20.24	S1 and S3 assert the Puiseux expansion at the branch point without constructing the local inverse; S1 says only "ordinary series reversion in p produces" the coefficients, and neither source proves that a power-logarithmic expansion is impossible there. Repaired: the local biholomorphism is constructed, the coefficients are rederived (and carried one order further than any source, to $p^6 = -221/8505$, verified against a high-precision evaluation), and the impossibility is proved from the order of the local monodromy in Corollary lw-branch-point-monodromy.
Theorem 20.32	the four source tables are individually correct but use two incompatible column conventions (S1 and S4 index by $x = \log z$, S3 and S5 by z), which makes them look contradictory when read side by side. All four were recomputed to 60 digits and merged into one table indexed by the intrinsic variable $\xi_0 = \log z$; the concordance is recorded below, together with the observation that the S3/S5 columns at $z = 10$ and $z = 100$ lie outside the region where any convergence statement is proved here.
Theorem 21.3	S3, S4, S5 and S6 all give the closed form $y = (\alpha/\beta) W_k((\beta/\alpha) x^{1/\alpha})$ with the disclaimer "branches chosen consistently", which hides a genuine ambiguity: for non-integer α the symbol $x^{1/\alpha}$ is not determined by x . Repaired by Proposition lw-power-exp, which makes the three choices explicit and states the correspondence as a bijection.
Theorem 21.13	S6 states that "the coefficient of Y^{-2} is no longer a polynomial in L and M alone". That is true but not sharp: expanding the Y^{-1} block gives $A^{\text{varpi}} = \alpha^2 L + \alpha^{\text{gamma}} M + \alpha^{\text{gamma}} + \text{gamma}^2 M/L$, so a negative power of L is already present one block earlier whenever gamma is nonzero. The corrected statement and its proof are below.
Theorem 22.2	S2 states that "the monomials $X^{-n}L^m$ form an asymptotic scale under the lexicographic order", and S6 opens its analytic chapter by letting (ϕ_{nu}) be "a totally ordered asymptotic scale, so that $\phi_{\text{nu}+1} = o(\phi_{\text{nu}})$ in the chosen enumeration" and then applying this to the polynomial-logarithmic monomials. The pairwise dominance comparisons those drafts display are all correct; what fails is the existence of the enumeration, by the proposition above. The repair is to fix a degree bound first, which is what the block formulation does automatically.

Location	Repair
Theorem 22.4	this definition carried the same title as <code>\creff{plt:def:mul-asymptotic}</code> , which it extends rather than repeats: the blocks here are rational in L and may have poles, so the evaluation needs the large- X caveat below. Retitled so the two are distinguishable in the list of results.
Theorem 22.8	S6 writes the blockwise condition as $f - \sum_{n \leq N} p_n(\log X) X^{-n} = o(X^{-N}(\log X)^{d_N})$ and remarks that "one must specify that inner ordering" of the monomials inside the last block. Form (v) of the theorem shows the logarithmic weight is harmless once all N are quantified, and form (vi) shows the inner ordering never has to be specified: the blockwise and monomialwise conditions coincide for every admissible profile.
Theorem 22.17	S5 states products and quotients together with a "proof sketch" and, in its part (ii), adds the hypothesis "and g is eventually nonzero". That hypothesis is redundant: it is a consequence of the leading block being nonzero, as <code>(plt:eq:an-g-lower)</code> shows. S5's part (iii) requires instead that " $Q(\log x)$ stays nonzero eventually", which is likewise automatic for a nonzero rational function. The full proof above replaces the sketch, and in particular exhibits the logarithmic degree k of the leading block explicitly, which the sketch suppresses; it is what makes the remainder exponent $M'-k$, not M' .
Theorem 22.21	S6 writes, in the chapter that this part consolidates, "For $f(X) = X + O(\log X)$, one has $f'(X) = 1 + o(1)$ under the relevant differentiability hypotheses". No such hypotheses are named, and the implication is false without them, by the display in the box above. S6 uses the conclusion to justify choosing m close to 1 in its residual-transfer theorem. The repair is <code>(plt:thm:an-omega-computable)</code> and <code>(plt:thm:an-inverse-transfer)</code> , which take the derivative bound as a hypothesis and verify it directly for the map at hand instead of deducing it from an expansion.
Theorem 23.5	S5 writes, immediately after its residual-to-error principle: "For $f(x) = x + \log x$, this proves the asymptotic validity of every truncation ω_N directly from [the formal residual identity]." It does not: the principle converts an analytic residual into an analytic error, and the formal identity provides no analytic residual. The missing step is the one carried out in <code>(plt:thm:om-analytic)</code> for this map and in <code>(plt:thm:an-residual-analytic)</code> in general. The same gap is present in S1 ("by construction") and is repaired in the same way.
Theorem 23.8	S2's transfer theorem lists the hypotheses " $F(y) = y + o(y)$ " and " $F'(y) = 1 + o(1)$ " and then assumes an ANALYTIC residual bound $F(G_N(X)) - X = O(X^{-N-1}L^M)$ outright. Assuming it is legitimate; the surrounding text, however, obtains it from the formal reversion, so the theorem as used is circular. The first hypothesis is never used in S2's proof. (H2) above is the honest replacement, and <code>(plt:thm:an-residual-analytic)</code> discharges it. S2 also writes the analytic inverse as <code>\rev{F}</code> ; in this volume that symbol denotes the FORMAL compositional inverse, and the two must not share a glyph in a statement whose whole content is that they agree.

Location	Repair
Theorem 23.14	S1 asserts, of its own error table at $x = 10$, that "the $N = 2$ truncation is slightly worse than $N = 1$ ". Its table gives $1.7466787e-3$ at $N = 1$ and $1.7369609e-3$ at $N = 2$, which is smaller, and recomputation at 40 digits confirms both table entries. The prose contradicts the table it is describing. What is true at $x = 10$ is that the $N = 2$ correction almost exactly cancels the $N = 1$ error and flips its sign, improving it by only 0.6 per cent, and that genuine non-monotonicity occurs later, at $N = 6 \rightarrow 7$ and $N = 11 \rightarrow 12$, as recorded above. S3's table at $\log z = 2.3026$ makes the analogous claim correctly, for the pair $N = 2, 3$.
Theorem 24.2	S2 sets $\text{ell} = \log(1/x)$; S1 and S6, following the normal-form literature, set $\text{ell} = -1/\log x$. The two are reciprocal, and no single displayed formula distinguishes them. We fix Lambda for the large log and never write a bare ell for it.
Power–log series near a finite point an	S6 displays the expansion with a bare remainder $O(M^3/S^3)$ and no argument. The estimate below is proved; the sharp form of S6's remainder is recovered in the last sentence.
Theorem 24.12	S2 gives the product form of (13.7) but never names the operator; the shift index $n-k+1$ is the trap, since the naive mirror of $\text{plt:eq:dif-repeated}$ would be $\Delta_{\{n,k\}}$, which is wrong for $k \geq 2$.
Theorem 24.14	S2's $\text{eq:Dulac-repeated-derivative}$ is correct as printed; the repair is that it is stated without proof, without the empty-product convention, and without identifying the operator, so its $k \geq 2$ index shift is invisible.
Theorem 24.16	S2 (eq:small-Lagrange) states the near-zero Lagrange-Burmam formula as having "exactly the same form" and gives no proof; S1 and S6 do the same for the fixed-point construction. The formula is true (Theorem $\text{plt:thm:du-lagrange-zero}$) but it is not the transport of $\text{plt:thm:lag-burmam}$: the transported frame has a different derivation and a different distinguished element, and no rescaling of nu_x repairs (F3).
Theorem 24.22	S1's $\text{eq:dulac-inverse-first}$ is correct, but it is written in the reciprocal convention $\text{ell} = -1/\log x$, so its exponents of the logarithm are the negatives of ours; transporting it without noting this produces the wrong sign on every logarithmic exponent.
Theorem 25.4	S4 stores the least allowed exponent $n_{\min}$ as an independent field and then warns that a stale zero at that exponent corrupts order and unit tests. Lemma $\text{plt:lem:alg-denotation-coset}$ shows the field is redundant: the order bound is a function of the canonical form, so the failure is eliminated by deriving it rather than by documenting it.
Data structures and finite-order algorit	the previous version of (v) asserted that sharpness fails at $N_1=0$ because DX annihilates the unknown constant block. That is an artifact of scalar power series and is false over $R=K[L]$: the unknown block of index N_1 is an arbitrary polynomial, and $a=t^{\{N_1\}}L$ already has $\text{DX } a = t^{\{N_1+1\}}(1-N_1 L) \neq 0$ for every N_1 , including $N_1=0$. What is true, and is what the sources were reaching for, is that the output ORDER bound can fail to be attained – exactly when the leading block is a nonzero constant sitting at index 0 – and that this is a loss of relative precision, not of absolute precision.

Location	Repair
Data structures and finite-order algorit	the sharpness witnesses below were originally exhibited only at the zero representative (sum) or for one of the two competing bounds (product, quotient), which proves sharpness of the LAW but not of each instance. Since Definition plt:def:alg-sound quantifies sharpness per instance, the witnesses are now chosen so that the block that differs is exactly the declared cutoff, for every instance.
Data structures and finite-order algorit	the witness that perturbs the first argument produces a difference in the block N_1+m_2 , which is the declared cutoff only when that bound is the smaller of the two. The previous version used it unconditionally and so proved sharpness of the law, not of each instance as Definition plt:def:alg-sound requires. The witness must be selected by which bound realises the minimum.
Data structures and finite-order algorit	"follows from the two witnesses used in (ii)" was too quick: as in (ii) the witness must be the one realising the minimum. Here, unlike (ii), both witnesses exist for every instance, because the numerator is allowed to be zero and may then be replaced by $t^{\{N_1\}}$.
Theorem 25.11	the proof still carried the discarded claim that sharpness fails at $N_1=0$, contradicting the corrected statement above. The witness $a=t^{\{N_1\}}$ is the wrong one: it is annihilated exactly at $N_1=0$. Over $R=K[L]$ the undetermined block is an arbitrary polynomial, and $a=t^{\{N_1\}}L$ works for every N_1 .
Theorem 25.11	the witness was a single pair ($F=t^{\{m_1\}}$ with m_1 nonzero), which proves sharpness of the law but not the per-instance claim actually stated. Under the printed hypothesis $\text{ordt}(DX\ v_1)=m_1+1$ the generic witness works, and the hypothesis is exactly what makes the $k=1$ term dominate.
Theorem 25.12	the inequality was stated as $\rho(DX\ F) \geq \rho_1$. It runs the other way: the absolute cutoff gains one block, but in the exceptional case of part (v) the output ORDER gains more than one, so the difference – which is ρ – can only fall. The extreme case $v_1 = c$ in K^x returns the zero representative, whose relative precision is 0.
Theorem 25.13	"only addition can destroy it" overlooked the annihilating case of the derivation, which the corrected corollary now records.
Theorem 25.15	the source sentence asked for G "through block $N-1$ ", which is the datum already assumed and yields the product only through block $N-2$. Since multiplication by $t^{\{-1\}}$ shifts indices down by one, the block of index n of the product is read off the block of index $n+1$ of G .
Theorem 25.15	S2 states only that inputs known modulo t^N with nonnegative valuation give a sum and product known modulo t^N . That is sound but not sharp: by part (ii) the product is known modulo $t^{\{N+\min(m_1,m_2)\}}$, which is strictly better whenever a factor has positive order, and strictly worse – the case S2's formulation hides – as soon as a factor is Laurent.
Data structures and finite-order algorit	"four times the naive quadratic guess in the exponent" is not a statement: the exponent is four rather than two. Also, the profile of the inverse of $X+L$ is $d(0)=1$, $d(n)=n$ for $n \geq 1$, which differs from case (ii) in the block of index 0; the exact count is corrected below.
Theorem 25.29	the two recurrences do not cost the same: the logarithm sums $j \leq n-1$ and the exponential sums $j \leq n$, so the exact counts differ by $N-1$. The single figure $\text{binom}(N,2)$ was right only for the exponential.

Location	Repair
Theorem 25.30	the claim was " $\text{shiftopt}\{n\}\{k\}p = 0$ as soon as $k > \deg p$ and $n = 0$ ", presumably by analogy with iterated <code>partial_L</code> . It is false: only the single factor with index $j = -n$ is <code>partial_L</code> , all the others are bijective, so the operator drops the degree by exactly one and no further. Already $\text{shiftopt}\{0\}\{2\}L = (\text{partial_L} - 1)(1) = -1$, nonzero with $k = 2 > 1 = \deg L$. The corrected kernel statement is what the annihilation clause of the precision calculus needs.
Data structures and finite-order algorit	the witness $F = t^a$ fails whenever $a < 0$ and $K - 1 > a $, because then $\text{shiftopt}\{a\}\{K - 1\}$ annihilates the constant 1 – e.g. $a = -1$, $h = 0$, $N = 10$ needs $k = 9$ while $DX^k t^{-1} = 0$ already for $k = 2$. Taking the block to be L instead of 1 removes the exception, by the kernel description in <code>plt:prop:alg-iterated-derivative</code> .
Theorem 25.33	$S1$ records the budget as $k \leq \text{floor}((N - r)/(h + 1))$. Under the exclusive cutoff adopted here that admits one Taylor term too many; the sharp exclusive bound is $k \leq \text{floor}((N - 1 - a)/(h + 1))$. The discrepancy is exactly the inclusive/exclusive one-block conversion of the catalogue and is harmless for correctness but not for cost.
Theorem 25.33	the running power was truncated at the output cutoff N . That is wrong as soon as $m_F < 0$: the term $P D$ contributes to a block of index below N through every block of P of index below $N - \text{ordt}(D)$, and $\text{ordt}(D) \geq m_F + k$ can be negative. Truncating P at $N - \text{ordt}(D)$, with the CURRENT D , repairs it and never demands a block of H beyond N_H , since $N - \text{ordt}(D) \leq N - m_F - k \leq N_H + 1 - k$.
Theorem 25.42	$S4$'s driver hard-codes the schedule as $q := \min(N, \max(1, 2p + 2))$. The additive $+2$ is a fudge for an unstated guard; the guard actually required is the one computed by <code>plt:thm:rev-newton-guard</code> together with the composition law of <code>plt:thm:alg-precision-calculus(vi)</code> , and the abort test above makes any residual shortfall visible instead of silently absorbing it.
Data structures and finite-order algorit	the product count was quoted as the ceiling itself; the stopping rule of <code>plt:alg:reverse-lagrange</code> runs $r = 1..R$ with $R = \text{ceil}((N + 1)/(h + 1)) - 1$.
Theorem 26.1	"only addition destroys it, and only differentiation improves it" is wrong twice: differentiation improves the ABSOLUTE cutoff by one block while leaving the relative precision unchanged, and in the annihilating case it destroys the relative precision outright.
Theorem 26.12	$S2$ (eq. <code>recip-residual-test</code>) and $S3$ (eq. <code>divisionresidual</code>) both state the residual test as " $AB_N - 1 = O(t^N)$ " or " $BC_{\{<N\}} - A = O(t^N)$ " and read off correctness through block $N - 1$. That reading is the gauge-zero one and is correct only for a denominator of order 0. With a denominator of order m the same residual certifies through block $N - m - 1$, which is fewer blocks when $m > 0$ and more when the denominator has a pole.
Theorem 28.4	the draft claimed that no argument of <code>\cref{plt:sec:ext-depth}</code> needs more than parts (i)–(ii). That is not accurate: the statements of <code>\cref{plt:thm:ext-tower-strict}</code> and <code>\cref{plt:cor:ext-depth-unbounded}</code> call H_r a differential FIELD, and the field property of a Hahn series ring over a non-archimedean monomial group is exactly Neumann's lemma. What is true is that every argument carried out here needs only (i)–(ii); the field property is imported, not reproved.

Location	Repair
Theorem 28.6	S1 ("What changes is the proof that every target coefficient receives only finitely many contributions ... generalized supports make it a central axiom") and S3 ("The well-order condition guarantees that each coefficient receives only an admissible family of contributions") both present well-ordering as if it were equivalent to local finiteness of a single product. It is sufficient, and it is necessary for the class, but it is not necessary for an individual product; the witness below shows this.
Theorem 28.6	the naive bound " $k, l \leq 2/(\gamma-2)$ " is false – for $\gamma-2 = 1/2$ the pair (3,6) is a representation with $l = 6 > 4$. Only the smaller index is bounded above; the larger is then determined.
Theorem 28.6	the draft said left-finiteness "is literally $\text{\cref{plt:thm:ring-support}}(d)$ with Z replaced by Γ ". That is false and contradicts $\text{\cref{plt:prop:ext-hahn-trunc-fails}}(b)$ below, which observes that clause (d) read verbatim over a general Γ is strictly weaker: it asks only for an exhausting enumeration of order type at most ω and does not demand that the enumeration be cofinal. Over Z cofinality is automatic, which is why the two conditions are indistinguishable in Parts I–IV.
Generalized power–log classes, nested I	the draft cited a nonexistent label $\text{\plt:thm:grp-taylor}$; the formal Taylor formula of this volume is $\text{\cref{plt:thm:tay-formula}}$.
Theorem 28.12	the draft used an undefined macro $\text{\Gfilt}$ here; the volume preamble provides no such macro, so the filtration steps are written out.
Generalized power–log classes, nested I	S1 ("The arithmetic formulas from Part II therefore survive almost unchanged"), S3 ("The differential rule is again ... Near-identity composition remains controlled") and S5 ("Near-identity composition and reversion continue to work when the support conditions ensure coefficientwise finiteness") all extend Parts I–IV to Hahn supports without recording that the finite bookkeeping of Part VI – truncation, sparse representation, precision calculus – requires strictly more than well-basedness. $\text{\cref{plt:prop:ext-hahn-trunc-fails}}$ isolates what is lost, and $\text{\cref{plt:thm:ext-lf-completion}}$ isolates the class where nothing is.
Generalized power–log classes, nested I	the draft recorded only " $+ o(1)$ " here, which is too weak to produce the error term displayed below: an additive $o(1)$ inside $\log X$ contributes a relative error $o(1/\log Y)$, larger than $(\log \log Y)^2 (\log Y)^{-2}$. The sharper form is what the two-line computation actually gives, and it is what the displayed remainder needs.
Theorem 28.19	the draft concluded that the display "is exactly the $\sigma = 1/2$ instance" of a theorem stated only for σ in N . That is a claim about a proof belonging to another part, and it is not made here: the display above is a self-contained computation, and the agreement with the theorem is recorded as a check, not used as an input.
Theorem 28.19	$\text{\cref{plt:thm:lag-log-normal-form}}$ restricts to σ in N although its proof uses only that σ is a scalar of K ; the half-integral instance is needed here and is verified directly above rather than by widening the hypothesis of a theorem belonging to another part. The displayed coefficient $1/4$ agrees with $\text{\eqref{plt:eq:lag-X-expansion}}$ at $c=1, \rho=1, \sigma=1/2$.
Theorem 28.20	the draft pointed at an equation label $\text{\plt:eq:grp-tower-derivative}$ that the volume does not carry; the statement it needs is the enclosing proposition, which does exist.

Location	Repair
Theorem 28.24	the draft proved incomparability only against the identity nesting, while \cref{plt:rmk:ext-completion-order} asserted $(r+1)!$ pairwise distinct iterated fields. The argument already gives the general statement, because it uses nothing about the identity beyond the relative order of the two chosen levels; the last paragraph records this.
Theorem 28.26	the draft cited a label plt:rmk:div-embedding-warning that the volume does not carry; the statement being invoked is the expansion map itself.
Theorem 28.26	the draft wrote "an identity proved after expansion need not descend". That contradicts the injectivity it cites in the same sentence: if $\iota(a)=\iota(b)$ then $a=b$. What non-surjectivity actually costs is that a computation carried out after expansion may leave the image, and that an exact cancellation in $K(L)$ is, after expansion, visible only in the limit and at no finite truncation. Both claims are restated correctly.
Theorem 28.27	the draft obtained the display by asserting that the proof of \cref{plt:thm:lag-log-normal-form}, which is stated only for σ in $\mathbb{N}$, works verbatim for every scalar β . That is a claim about a proof belonging to another part and is not made here; the two-line derivation is given instead, and the agreement with that theorem is recorded as a check.
Theorem 29.3	"a $\notin \Gamma$ " is a statement about the image of Γ in K under the embedding of \cref{plt:def:dif-scale}; the exponent a is read as a scalar of K , which is what the proof needs and what makes the enlargement of the exponent group in the draft's proof unnecessary.
Theorem 29.3	the draft passed to $\Gamma' = \Gamma + (1/2)\mathbb{Z}$ without checking that Γ' is admissible, i.e. that the embedding $\Gamma \hookrightarrow K$ of \cref{plt:def:dif-scale} extends to it injectively. It does, and the verification is one line, given here.
Theorem 29.3	the draft took $\Gamma' = \Gamma + a\mathbb{Z}$ here. That enlargement is neither needed nor in general admissible – for $K = \mathbb{Q}$, $\Gamma = \mathbb{Z}$ and a irrational the group $\mathbb{Z} + a\mathbb{Z}$ has rank two and admits no injective homomorphism into $\mathbb{Q}$. No enlargement is required: a is a coefficient, not an exponent, and $u = at$ already lies in $H_r(K, \Gamma)$.
Theorem 29.4	the draft pointed at an equation label plt:eq:dif-exp-value that the volume does not carry; the statement meant is the classification of which elements are exponentials.
Theorem 29.5	the draft assumed only " E nonzero and c a nonzero constant". In a ring with zero divisors that does not force $cE \neq 0$, and the proof needs $\partial E = cE \neq 0$ in order to apply the hypothesis on ν at $G = E$. The hypothesis is stated in the form the proof uses; it is satisfied whenever $\mathcal{T}$ is a domain.
Relation with the full transseries field	the draft opened this list with a Dahn–Göhring item citing the key dahn-goering-1987, which is not in the merged bibliography of this volume. Rather than cite a key that does not resolve, the item is dropped; it carried no content used anywhere else.
Theorem 29.7	the draft referred to an equation label plt:eq:cmp-realization that the volume does not carry; the condition invoked is the defining one of the realization relation.

Location	Repair
Theorem 29.8	three appeals to a label <code>plt:rmk:om-open</code> , which the volume does not carry, are replaced by the statement they were standing in for; see also <code>\cref{plt:rmk:ext-open-convergence}</code> and <code>\cref{plt:rmk:ext-open-uniformity}</code> .
Theorem 29.8	the draft also cited <code>salvy-shackell-1992</code> , a key that is not in the merged bibliography; the item it carried (functional inverses) is covered by <code>salvy-shackell-1999</code> , which is.
Theorem 29.9	the draft attributed the singularity locus to a label <code>plt:rmk:om-open</code> that the volume does not carry. The locus is recomputed here from the generating equation, and it does come out as the draft stated it.
Theorem A.1	the second group is a restatement inside the volume, not an error, but the draft called its two members "the same statement" and declared them interchangeable. They are not: <code>plt:prop:tay-scalar-outer</code> restates <code>plt:prop:dif-bell</code> and then adds a clause about expanding the outer function about a base point $z_0 \neq 0$. Nothing below leans on that clause – see the editorial note at <code>plt:lem:bell-chain</code> , where the base point is made part of the datum instead – but the two statements are not interchangeable.
Theorem A.7	the draft wrote " $\mathcal{O}(N^3)$ coefficient multiplications" without saying what a coefficient is here. The count $\mathcal{O}(N^3)$ is correct only for the operation "multiply one indeterminate into one stored entry"; the entries are not scalars, and the scalar count is genuinely superpolynomial. Left unqualified this is exactly the conflation that <code>plt:cor:alg-cost-not-quadratic</code> warns against, so the two levels are now separated.
Theorem A.14	the draft wrote $\varphi = \sum (\varphi_k/k!) z^k$ – coefficients at the origin – and then set $\varphi(F) = \sum (\varphi_k/k!) W^k$ with $W = F - z_0$. That is φ evaluated at F only when $z_0 = 0$; in general the coefficients of the k -th power of W are $\varphi^{(k)}(z_0)/k!$, not $\varphi_k/k!$. (Take $\varphi(z) = z$, $z_0 = 1$, $F = 1 + t$: the displayed sum returns t , not F .) The justification offered, "apply the translate $z \mapsto \varphi(z_0 + z)$ ", is also unavailable: re-expanding a formal element of $\mathbb{K}[[z]]$ about a centre $z_0 \neq 0$ requires an infinite sum in $\mathbb{K}$ for every coefficient, and $\mathbb{K}$ carries no topology here. Both defects are repaired at once by making the centre part of the datum – the outer function IS its expansion at z_0 . Nothing downstream changes: the proof only ever uses $c_k = \varphi_k/k!$ and the relation $\varphi'(F) = \sum_{k \geq 1} k c_k W^{k-1}$.
Theorem B.4	the draft said that for $h_0 = 0$ the formula "reduces to <code>plt:eq:tay-C-bell</code> ", which is stated only for $m \geq r \geq 1$; at $r = 0$ it reduces instead to the boundary convention $C_{\{m,0\}} = \delta_{\{m,0\}}$. Range added.
Theorem B.5	the draft asserted "then $F^\sigma = c^\sigma t^{\sigma n_0} (1+W)^\sigma$ " as though it were an identity. For non-integer σ the symbol F^σ has no prior meaning in the volume, and the right-hand side depends on the chosen value of c^σ ; it is a definition, and is now printed as one.
Theorem B.13	S3's formula sheet displays this identity as $[t^n](\text{rev } F - X + P) = ((-1)^{n+1}/(n+1)!) D(D-1)\dots(D-n+1) P^{n+1}$. The added "+P" is a slip: it is correct for $n \geq 1$, where P (a level-zero block) contributes nothing, but false at $n = 0$, where the left side is $-P + P = 0$ while the right side is $-P$. Removing the "+P", as in <code>plt:eq:cext-level-zero-reversion</code> , makes the formula correct for every $n \geq 0$ and makes the Lambert case a specialization rather than an exception.

Location	Repair
Operational formula sheet	this row printed the division recurrence with no hypothesis on B_0 , which repeats the defect of S1's sheet (" $B_0 \neq 0$ "). In $K[L]((t))$ a nonzero polynomial block is not enough: B_0 must be a unit of $K[L]$, i.e. a nonzero scalar, or the quotient blocks leave $K[L]$. Hypothesis printed.
Operational formula sheet	this row read "\PLrat is the smallest field of the tower with polynomial-log coefficients", which inverts the content of <code>plt:thm:div-minimal-field</code> : \PLrat has RATIONAL-log coefficients, and the theorem says $E((t))$ is a field iff $E = K(L)$. Restated as proved.
Exercises with solutions	source 1 asks only for a jump "by at least two", i.e. the case $k = 2$ of part (2), and does not ask for the converse. The stronger form is proved above because part (3) is what the sum rule of <code>\cref{plt:thm:ring-valuation-laws}</code> actually asserts.
Exercises with solutions	source 2 proves the common lower bound by asserting that "the sequence is already Laurent-bounded at some initial precision". That is not a hypothesis of completeness; the bound has to be extracted from the Cauchy condition at cutoff $N = 0$, as in part (3).
Exercises with solutions	source 2 states the prefactor form "whenever available" without an instance of unavailability; $L + t$ is one, and it is the reason the log-degree profile of <code>\cref{plt:def:lead-profile}</code> is defined blockwise rather than through a normal form.
Exercises with solutions	source 3 answers this item with "a natural home is $K((L^{-1}))((t))$ ", i.e. $\text{\PLit}$. That overshoots by a whole class: the coefficients in <code>\eqref{plt:eq:exr-recip-L-plus-t}</code> are Laurent *monomials* in L , so even the localisation $K[L, L^{-1}]((t))$ suffices, and among the four ambient classes the answer is $\text{\PLrat}$. $\text{\PLit}$ is strictly larger (<code>\cref{plt:prop:ring-PLit-strict}</code>), and the next item is what actually needs it.
Exercises with solutions	source 3 poses this product only through t^2 and source 1 poses a different pair only through t^3 without saying that the undisplayed blocks must be assumed zero. The two are merged here and the hypothesis is printed, since <code>\cref{plt:thm:mul-finite-determination}</code> is the point of the exercise.
Exercises with solutions	source 5 prints the block Q_3 of this example immediately after inputs given only modulo t^3 . The hypothesis $B_3 = 0$ is a genuine extra assumption, and <code>\cref{plt:prop:div-relative-precision}</code> shows exactly how much is missing; it is stated here rather than assumed silently.
Exercises with solutions	source 1 prints the Stirling identity with the sum starting at $j = 1$. That is correct only for $k \geq 1$; the uniform range $j = 0..k$ with the empty-product convention covers $k = 0$ as well, and is used above.
Exercises with solutions	source 1 records the budget as $k \leq \text{floor}((N - r)/(h + 1))$ without saying which truncation convention its N refers to. Under the exclusive cutoff, which the catalogue makes normative, that form admits one Taylor term too many. Part (3) prints the conversion rather than silently shifting the index, because the same formula is what a routine would apply in reverse to size its inputs, where the extra term is not free.
Exercises with solutions	source 1 states this with the hypothesis $q \geq 1$. The hypothesis is unnecessary: the Picard estimate <code>\eqref{plt:eq:rev-picard-accuracy}</code> holds for every $\alpha = q \geq 0$, and the case $q = 0$ – the Wright omega case – is exactly the one in which the statement is sharp.

Location	Repair
Exercises with solutions	sources 2 and 3 both quote a general third-order inverse block; source 3's version (the $b = c = 0$ case above) is correct, source 2's is not (<code>\cref{plt:exer:exr-trap-universal-formula}</code>). Source 3 checks the degree law only at $N = 1, 2$; the block <code>b_3</code> above is computed here and extends the check to $N = 3$.
Exercises with solutions	the two numbers in <code>\cref{plt:warn:lag-scale-trap}</code> were computed at two different cutoffs. Recomputed in exact arithmetic at $Y = 20 + \log 10$: $N \leq 2 \rightarrow$ correct 10.0000781540, naive 9.6702691781 $N \leq 3 \rightarrow$ correct 10.0000043732, naive 9.6702070480 The warningbox says "through Y^{-3} " and then quotes 10.000004 (which is the $N \leq 3$ value) alongside 9.67026 (which is the $N \leq 2$ value). Both truncations are quoted at $N \leq 3$ above. The conclusion of the warningbox is unaffected; only the second numeral changes.
Exercises with solutions	source 1's prose ("the same P_n occur, multiplied by $(-1)^n$ ") contradicts its own displayed expansion of the inverse of $X - \log X$, in which $P_1 = L$ occurs with a plus sign. The weight $(-1)^{n+1}$ belongs to $\text{rev}(X - L)$; the weight $(-1)^n$ belongs to its negative, which is what $W_{\{-1\}}$ is. Part (3) separates them.
Exercises with solutions	source 5 asserts that the residual-to-error principle "proves the asymptotic validity of every truncation directly from" the formal residual identity, and source 1 asserts an analytic O -bound for the branch residual "by construction". Neither is a proof; the missing ingredient is the substituted-tail estimate, which is why this exercise exists.
(front matter)	the six sources cited this work inconsistently as a 2010 preprint and as a 2013 journal article; the published version is given, with the preprint retained as a secondary pointer.

Remark G.1 (What is not in this ledger). Three classes of change are deliberately excluded. Deduplication is not a repair: where several sources proved the same result, the strongest statement and the best proof were kept and the rest collapsed, which the provenance section records in aggregate rather than line by line. Restructuring is not a repair. And a source claim that this volume could neither prove nor refute has not been silently dropped: it is stated here as an explicitly labelled open problem or conjecture, with the source's assertion quoted as an assertion.

H Formalization register

H.1 What this register claims

No proof in this volume is machine-checked. The corpus that hosts it does contain a large Lean development, and the honest question is not whether these results are formalized — they are not — but which of them have a formalized *neighbour*, and what a formalization campaign would have to build first. This register answers that. Every Lean name below was checked to resolve in the corpus; a name that no longer resolves is a defect, and the corpus audit `scripts/audit_crosswalk_names.py` enforces it.

Three statuses are used. **Crosswalked** means a Lean declaration states the same mathematical fact, possibly in a different but faithfully translated setting. **Neighbouring** means a Lean declaration states a genuinely different theorem that a reader might mistake for this one; the entry exists to prevent that mistake. **Absent** means nothing in the corpus is close.

H.2 The formal algebra: absent

Nothing in Parts I–IV corresponds to a Lean declaration. The corpus has no polynomial–logarithmic ring, no two-generator scale, no outer valuation, no distinguished derivation D_X , no composition of transseries, and no reversion at infinity. Concretely, a formalization campaign would have to build, in order:

1. the ring $\mathcal{P}\mathcal{L}_{\text{pow},\mathbb{K}} = \mathbb{K}[L][[t]]$ with t and L as independent generators, and its Laurent extension (Section 3);
2. the outer valuation and the leading-data trio, which are three different objects and are easy to conflate in a formal setting (Section 4);
3. the derivation D_X characterized by $D_X t = -t^2$, $D_X L = t$, and the repeated-derivative operator $\Delta_{n,k}$ (Section 9);
4. composition by generator substitution, which is where an informal treatment most often goes wrong, since $t \mapsto t + H$ is *not* the substitution meant (Section 10);
5. the exact formal Taylor formula (Theorem 11.6), whose proof here is an algebra-homomorphism argument on the two generators and should formalize without the coefficient bookkeeping;
6. the reversion fixed point and its uniqueness (Theorem 14.11).

Only after those is the Lagrange–Bürmann theorem at infinity (Theorem 16.14) even statable.

H.3 Lagrange–Bürmann: neighbouring, not crosswalked

The corpus *does* formalize Lagrange–Bürmann inversion — for formal power series at the origin, over a commutative $\mathbb{Q}$ -algebra. The relevant declarations are `Fabius.Lagrange.solution`, `Fabius.Lagrange.coeff_solution`, and the division-free identity `Fabius.Lagrange.coeff_subst_mul_derivative`.

That is a different theorem from Theorem 16.14, and the difference is not cosmetic:

- the formalized version inverts $g = z\varphi(g)$ in $R[[z]]$, a one-generator power-series ring with the ordinary derivative; the version here inverts $X + A$ in the two-generator polynomial–logarithmic scale with the derivation D_X ;
- the formalized version is a statement about coefficients of z^n ; the version here is a statement about coefficient *blocks* $p_n(L)$, each a polynomial whose degree is unbounded across n ;
- summability is automatic at the origin and is a theorem here (Theorem 16.9).

So the corpus’s Lagrange inversion cannot be cited as proving the at-infinity formula, and this volume does not do so. It is, however, the natural model for a future formalization, and its proof strategy — purely algebraic, avoiding formal residues, which Mathlib does not have — is the same strategy the proof here uses.

H.4 The Lambert W series: neighbouring

`Fabius.lambertW` and `Fabius.coeff_lambertW` formalize the *small-argument* expansion $W(z) = \sum_{n \geq 1} (-n)^{n-1} z^n / n!$, obtained from the formalized Lagrange inversion above. This volume expands W at *infinity* (Section 20). The two expansions share a name, a function, and nothing else: one is a convergent Taylor series at the origin with radius e^{-1} , the other an asymptotic polynomial–logarithmic expansion at infinity that converges nowhere. Confusing them is the single most likely misreading of this volume’s Part V, which is why the entry is here.

H.5 The real branches: crosswalked

The analytic facts about the two real branches that this volume *uses* — as opposed to *derives* — are kernel-verified:

Table 34: Analytic branch facts used in Part V and their Lean declarations.

Fact	Declaration
The two real branches exist and are characterized by the defining equation	<code>Fabius.principalLambertW,</code> <code>Fabius.lowerLambertW,</code> <code>Fabius.principalLambertW_unique,</code> <code>Fabius.lowerLambertW_unique</code>
Every real solution of $we^w = z$ is one of the two branches	<code>Fabius.eq_principalLambertW_or_eq_lowerLambertW</code>
Continuity on the full closed domains, branch point included	<code>Fabius.principalLambertW_continuousOn_Ici,</code> <code>Fabius.lowerLambertW_continuousOn_Ico</code>
The derivative of the principal branch, and $W_0''(0) = -2$	<code>Fabius.deriv_principalLambertW,</code> <code>Fabius.deriv_deriv_principalLambertW_zero</code>
Strict concavity of the principal branch	<code>Fabius.strictConcaveOn_principalLambertW</code>
The square-root scale at the branch point, and the signed leading laws	<code>Fabius.lambertWBranchPointScale,</code> <code>Fabius.principalLambertW_add_one_isEquivalent_branchPoint,</code> <code>Fabius.lowerLambertW_add_one_isEquivalent_branchPoint</code>

The branch-point entry is the sharpest of these: it confirms, in the kernel, the assertion of [Section 20](#) that the inverse near $z = -e^{-1}$ is *not* a power–logarithmic series but carries a square-root scale, which is exactly the boundary of this volume’s class.

H.6 The lower-branch expansion near zero: partially crosswalked

One expansion statement of Part V has a genuine Lean counterpart. `Fabius.tendsto_lowerLambertW_expansion` proves

$$W_{-1}(-\varepsilon) = \log \varepsilon - \log |\log \varepsilon| + o_{\varepsilon \downarrow 0}(1),$$

which in the coordinates of [Section 20](#), with $S = \log(1/\varepsilon)$ and $L = \log S$, is $W_{-1}(-\varepsilon) = -S - L + o_{\varepsilon \downarrow 0}(1)$: precisely the first two blocks of the lower-branch expansion derived there.

A second, sharper declaration pins the next block as well. `Fabius.lowerLambertW_near_zero_bounds` proves, for $x \in (-e^{-1}, 0)$ and $\eta = \log(1/(-x))$,

$$-\eta - \frac{\eta \log \eta}{\eta - 1} \leq W_{-1}(x) < -\eta - \log \eta.$$

In the coordinates above, $\eta = S$ and $\log \eta = L$, so the bracket says that the deviation of W_{-1} from the two-block truncation $-S - L$ lies in $(-L/(S-1), 0]$. That is a two-sided, non-asymptotic statement: it fixes the sign of the third block and its order L/S , which is exactly what [Section 20](#) derives formally. The strictness of the upper bound is the formal statement that the third block is nonzero and negative.

The supporting declaration is worth naming separately, because it is the piece most likely to be reusable: `Fabius.SubLog.bracket_of_sub_log_eq` brackets the inverse of $y \mapsto y - \log y$ for

$y > 1$ with no Lambert content at all, and `Fabius.SubLog.strictMonoOn_sub_log` supplies the monotonicity that makes that inverse well defined. Since this volume’s lower-branch treatment *runs* on the inverse of $u \mapsto u - \log u$ rather than on W_{-1} directly, that lemma – not the branch-specific one – is the natural target for a future crosswalk of the whole lower-branch section. The side condition $\eta > 1$ is itself a lemma (`Fabius.one_lt_log_one_div_neg`) rather than a hypothesis the caller must discharge, so the $(\eta - 1)$ denominator cannot be reached at zero.

Two elementary bounds on the principal branch are also available and are used here only as sanity constraints: `Fabius.principalLambertW_elementary_bounds` gives $x/(1+x) \leq W_0(x) \leq \log(1+x) \leq x$ for $x \geq 0$, resting on the Lambert-free inequality `Fabius.exp_1e_one_add_mul_exp_self`, valid for every real argument.

Beyond the third block – the L_n^{Lam} coefficients for $n \geq 2$, the remainder order at every order, and any uniformity – everything is human-proved here and unformalized.

H.7 Summary

Of the results in this volume, none is crosswalked in the strong sense; one two-term asymptotic statement is; six analytic facts that the volume cites rather than proves are. The formal algebra of Parts I–IV is absent from the corpus in its entirety, and the ordered list at the head of this register is what a campaign would need to build. This register is deliberately conservative: where the corpus proves a neighbouring theorem, that is recorded as a neighbour and not as coverage.

I Provenance of the consolidation

I.1 What was merged

This volume is an editorial consolidation, carried out on 2 September 2026, of six independently written article packages that arrived together as bare directories in the direct-arrival commit `730e1763291099cd50ca1e20ed2c62c38d95ab4f` and were filed on 1 September 2026 as standalone quick-gate receipts. No archive or checksum ledger accompanied them. All six treat the same subject – the algebra, division, composition, and asymptotic reversion of polynomial–logarithmic transseries, with Lambert W as the guiding example – from independent starting points, with substantial overlap and no containment relation among them.

Editorial merge means: every result is stated once, in the strongest form that the six sources jointly support and that we can prove; the best available proof is retained and gaps are filled; conflicting conventions are resolved against the corpus notation catalogue; errors are corrected and the corrections are marked in the source with `% ed. : comments`; and every source-specific layer that is not redundant is preserved. Nothing was discarded merely for being repeated: repetition was collapsed, content was not.

I.2 Absorbed sources and their receipts

Each source is recorded twice: as filed at intake, and as actually absorbed. The two differ by exactly three bytes per file because a regrouping on 2 September 2026 moved the packages one directory level deeper and the shared notation input path was repaired from four to five levels ([Table 35](#)). No mathematical content changed between the two states.

The six retained arrival PDFs, which are historical artefacts and are not renders of the absorbed sources, had SHA-256 values `a4fc4af0 . . . 69e886` (S1, 119 custom 522-by-738-point pages), `5e9ff596 . . . bfc68e` (S2, 102 custom pages), `3f7c4bc1 . . . 58a4af` (S3, 87 Letter pages), `c2d75b35 . . . 45a3eb` (S4, 47 A4 pages), `189e95ab . . . 58a2db` (S5, 44 Letter pages), and `b5142bad . . . 467aa` (S6, 100 A4 pages). Git history is the archive for all twelve files.

Table 35: The six absorbed sources. “Intake” is the receipt recorded in the group manifest on 1 September 2026; “absorbed” is the state merged into this volume. The retained arrival PDFs were historical publication checkpoints that already predated their own sources and were not rebuilt for this merge.

Source	Lines	Bytes	SHA-256
S1 = Polynomial-Logarithmic-Transseries-1.tex			
intake	4,020	187,071	01a03e0934d05b8b94369da9656fc0da4dde9b7c559b4e0738075484e14f4e61
absorbed	4,020	187,074	4fd42b7e344c485f8d878279dc3b9a6d5dc326a9c739114e443fd9c07a5eb2b8
S2 = polynomial_logarithmic_transseries-2.tex			
intake	5,006	173,396	aa25baa03099472df765d35c03a2bb265dbe9af5aab500ec0a4a04a378ef83e5
absorbed	5,006	173,399	d172b81453a805366f056f6114bb6df6fdf781146641320185530940754bdefe
S3 = Polynomial_Logarithmic_Transseries-3.tex			
intake	4,249	150,182	0962c15683cb4610d45961c227f6e9b102d66ee37073432d1943d31a1e6ad348
absorbed	4,249	150,185	d2a8bd25d71cc41fc81c1c75fb66ca75496b8425ac496ce105d3220e1ea07e3f
S4 = Polynomial-Logarithmic-Transseries-4.tex			
intake	3,132	120,607	387ca51f94292a996fb339ec6dfb3477f68575255b2aaaf539fcffb1c32ff564
absorbed	3,132	120,610	37026b625cf7b2fd106005b8c89f720b8b724f99a5446ed84ca52531c6e2b18f
S5 = Polynomial_Logarithmic_Transseries-5.tex			
intake	2,443	106,141	1149ae6884e63dc77b747c786d0455419b6c79cdd3b33dc915bf3875cc8d26cc
absorbed	2,443	106,144	187bd510f33ea4296f0247b286d0fcae6200ce0b5bd175c6c8706a37652dc17a
S6 = polynomial_logarithmic_transseries-6.tex			
intake	4,388	155,846	df4e4bc5932a59e4ed265b32d8463edc65f67cbd2735b34902acf479647f8491
absorbed	4,388	155,849	16978aae6fb1e458e236cac9807ab81bb5314fdb087a2504a416867a4d43b4c4

I.3 What each source contributed

No source was a superset of the others, and each contributed at least one layer that no other supplied.

- **S1** contributed the block-algebra presentation of the scale, the systematic comparison of four reversal algorithms, the residual-certificate and reproducible-verification discipline, the Dulac and power–log normal-form literature, and the largest exercise set.
- **S2** contributed the t -adic completeness treatment, the normal-ordered composition formalism, the power–log series near a finite point with the Dulac conventions, the derivation of the universal composition formulas, and the pitfalls-and-diagnostics decision procedure.
- **S3** contributed the formal summability criterion, the degree law for polynomial perturbations, the Lagrange–Bürmann transport formula, the fully formal proof of Lagrange–Bürmann inversion at infinity, and the reusable inverse templates beyond Lambert.
- **S4** contributed the sharpest statement of the unit criterion and the field property, the finite reversal certificate, and closed-form coefficient extraction.
- **S5** contributed the tightest and most completely proved core – it is the spine of Parts I–IV – together with the Laplace transform of the Lambert polynomials, the resulting exponential-moment cancellations, and the generalized Lambert expansion.
- **S6** contributed the filtered Lipschitz gain underlying the fixed-point construction, the composed-observable form of Lagrange–Bürmann, the systematic change of variable between infinity and a finite point, and the failure-mode catalogue.

I.4 Conventions reconciled

The six sources used mutually incompatible local conventions. All are resolved here in favour of the corpus notation catalogue; the concordance in [Section F](#) records the mapping in full. The reconciliations that change a formula rather than only a glyph are:

- **Truncation.** S6 used the inclusive cutoff $[F]_{\leq N}$; every other source and the catalogue use the exclusive $\text{Trunc}_{<N} F$. The relation is $[F]_{\leq N} = \text{Trunc}_{<N+1} F$, so every S6 statement transported here has had its cutoff index shifted by one.
- **Lambert polynomial indexing.** The sources split between a zero-based family with $L_0^{\text{Lam}}(u) = -u$ and a one-based family that separates the logarithmic block. The catalogue’s zero-based normalization is used throughout; the degree law $\deg L_n^{\text{Lam}} = n$ is therefore stated only for $n \geq 1$, since $L_0^{\text{Lam}} = -u$ has degree 1.
- **Coefficient extraction.** Two sources printed $[M]F$ and two printed $[F]_M$ for the same operation. This volume writes the series to the right of the bracket throughout.
- **The three $\mathcal{O}$ roles.** Formal valuation tails, analytic estimates, and algorithmic cost bounds are printed with three different tokens and are never interchanged. Several source statements that used one glyph for all three have been split accordingly.

I.5 Status of the claims

Every theorem, proposition, lemma, and corollary in this volume carries a proof. Where a source asserted a result without proof, the proof is supplied here or the assertion is demoted to an explicitly labelled conjecture or open problem. Where a source claimed analytic validity on the strength of formal algebra alone, the claim has been weakened to what the algebra establishes and the missing analytic hypothesis is named. Corrections to the sources are marked in this document’s own source with % ed. : comments at the point of repair, and are collected in [Section G](#).

None of the proofs in this volume is a machine-checked proof. The corpus formalizes the real Lambert pair and its branch-point geometry in Lean, and the concordance in [Section F](#) records which statements here correspond to declarations that exist there; the remaining results are human-proved frontier mathematics.

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